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HEAD MOUNTABLE DISPLAY

Abstract

A head-mountable display device includes a housing defining a front opening and a rear opening, a display screen disposed in the front opening, a display assembly disposed in the rear opening, a first securement strap coupled to the housing, the first securement strap including a first electronic component, a second securement strap coupled to the housing, the second securement strap including a second electronic component, and a securement band extending between and coupled to the first securement strap and the second securement strap.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This is a continuation of U.S. application Ser. No. 18/662,906, filed 13 May 2024, and entitled “HEAD MOUNTABLE DISPLAY,” which claims priority to U.S. Provisional Patent Application No. 63/586,403, filed 28 Sep. 2023, and entitled “HEAD MOUNTABLE DISPLAY,” to U.S. Provisional Patent Application No. 63/506,020, filed 2 Jun. 2023, and entitled “HEAD MOUNTABLE DISPLAY,” and to U.S. Provisional Patent Application No. 63/502,408, filed 15 May 2023, and entitled “HEAD MOUNTABLE DISPLAY,” the entire disclosures of which are hereby incorporated by reference in their entirety.

FIELD

[0002] The present disclosure relates generally to head mountable computer systems that provide computer-generated experiences, including, but not limited to, electronic devices that provide virtual reality and mixed reality experiences via a display.

BACKGROUND

[0003] The development of computer systems for augmented reality, including head mountable computer systems, has increased significantly in recent years. Example augmented reality environments include at least some virtual elements that replace or augment the physical world. Input devices, such as cameras, controllers, joysticks, touch-sensitive surfaces, and touch-screen displays for computer systems and other electronic computing devices are used to interact with virtual/augmented reality environments. Example virtual elements include virtual objects, such as digital images, video, text, icons, and control elements such as buttons and other graphics.

SUMMARY

[0004] In at least one example of the present disclosure, a head-mountable display device includes a housing defining a front opening and a rear opening, a display screen disposed in the front opening, a display assembly disposed in the rear opening, a first securement strap coupled to the housing, the first securement strap including a first electronic component, a second securement strap coupled to the housing, the second securement strap including a second electronic component,

and a securement band extending between and coupled to the first securement strap and the second securement strap.

[0005] In one example of the present disclosure, the display assembly is a first display assembly; and the head-mountable display device further includes a second display assembly disposed in the rear opening and including a second display screen and a third display screen

[0006] In one example of the present disclosure, the first display screen is oriented to project light in a first direction and the second display screen and the third display screen are oriented to direct light in a second direction opposite the first direction.

[0007] In one example of the present disclosure, the first electronic component includes a speaker.

[0008] In one example of the present disclosure, the second electronic component includes a computing component.

[0009] In one example of the present disclosure, the display screen has a curvature.

[0010] In one example of the present disclosure, the curvature follows contours of a user's face.

[0011] In one example of the present disclosure, the securement band includes a flexible textile material.

[0012] In at least one example of the present disclosure, a display device includes a housing defining a first opening, a second opening opposite the first opening, an internal volume, a first aperture between the first opening and the second opening, and a second aperture between the first opening and the second opening. The display device includes a front-facing cover assembly disposed in the first opening, a rear facing display assembly disposed in the internal volume, an elastic curtain occluding the second opening between the housing and the rear-facing display assembly, a dial disposed in the first aperture, and a button disposed in the second aperture.

[0013] In one example of the present disclosure, the rear-facing display assembly includes a display screen and the display device further includes an adjustment mechanism configured to adjust a position of the display screen.

[0014] In one example of the present disclosure, the dial is electrically coupled to the adjustment mechanism and a manipulation of the dial causes the adjustment mechanism to adjust the position of the display screen.

[0015] In one example of the present disclosure, the front facing cover assembly includes a first display screen configured to project light in a first direction and the rear-facing display assembly includes a second display screen configured to project light in a second direction different than the first direction.

[0016] In one example of the present disclosure, the first display screen is curved.

[0017] In one example of the present disclosure, the second direction is opposite the first direction.

[0018] In one example of the present disclosure, the display device further includes a light seal coupled to the housing around the second opening. The light seal is configured to press against a face of a user around eyes of the user to block light external to the device from reaching the user's eyes, including light from the first display screen.

[0019] In at least one example of the present disclosure, a head-mountable electronic device includes a housing defining an internal volume and a front opening, a display assembly disposed in the internal volume, a curved front cover assembly disposed in the front opening, and a securement mechanism extending rearward from the housing. The securement mechanism includes a first electronic strap including a first proximal end coupled to the housing and a first distal end opposite the first proximal end, a second electronic strap including a second proximal end coupled to the housing and a second distal end opposite the second proximal end, a first band and a second band. The first band includes a first end coupled to the first distal end and a second end coupled to the second distal end. The second band extends between the first electronic strap and the second electronic strap.

[0020] In one example of the present disclosure, the second band includes a first end coupled to the first electronic strap between the first proximal end and the first distal end and a second end

coupled to the second electronic strap between the second proximal end and the second distal end.
[0021] In one example of the present disclosure, the first electronic strap and the second electronic strap include plastic material and the first band and the second band include flexible material.
[0022] In one example of the present disclosure, the flexible material includes a woven textile material.
[0023] In one example of the present disclosure, the first electronic strap defines an internal strap volume and includes an electronic component disposed in the internal strap volume.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like structural elements, and in which:

I: Overall System

[0025] FIG. 1-1A illustrates a front, perspective view of an example of a head-mountable device (HMD).

[0026] FIG. 1-1B illustrates rear, perspective view of an example of an HMD.

[0027] FIG. 1-2 illustrates an example of an HMD.

[0028] FIG. 1-3 illustrates a display module of an HMD.

[0029] FIG. 1-4 illustrates a display module of an HMD.

II: Cover Glass

[0030] FIGS. 2.0-1 illustrates a view of an example of an HMD.

2.1: Systems with Transparent Layers

[0031] FIGS. 2.1-1 is a perspective view of an illustrative system with a transparent layer in accordance with an embodiment.

[0032] FIGS. 2.1-2 is a cross-sectional side view of an illustrative transparent layer overlapping optical components that operate through the transparent layer.

[0033] FIGS. 2.1-3 is a cross-sectional side view of an illustrative transparent layer in accordance with an embodiment.

2.2: Systems with Displays and Sensors

[0034] FIGS. 2.2-1 is a side view of an illustrative electronic device such as a head-mounted device in accordance with an embodiment.

[0035] FIGS. 2.2-2 is schematic diagram of an illustrative system with an electronic device in accordance with an embodiment.

[0036] FIGS. 2.2-3 is a front view of an illustrative head-mounted device in accordance with an embodiment.

[0037] FIGS. 2.2-4 is a cross-sectional top view of an illustrative head-mounted device in accordance with an embodiment.

[0038] FIGS. 2.2-5A is a cross-sectional side view of an illustrative head-mounted device in accordance with an embodiment.

[0039] FIGS. 2.2-5B is a cross-sectional side view of another illustrative head-mounted device in accordance with an embodiment.

[0040] FIGS. 2.2-6 is a front view of an upper left portion of an illustrative head-mounted device with a publicly viewable display in accordance with an embodiment.

[0041] FIGS. 2.2-7, 2.2-8, 2.2-9, 2.2-10, 2.2-11, and 2.2-12 are front views of portions of an illustrative head-mounted device in accordance with embodiments.

[0042] FIGS. 2.2-13 is a cross-sectional top view of a portion of an illustrative head-mounted device in accordance with an embodiment.

[0043] FIGS. **2.2-14** is a cross-sectional side view of a portion of an illustrative head-mounted device with a display in accordance with an embodiment.

[0044] FIGS. **2.2-15**, **2.2-16**, and **2.2-17** are cross-sectional side views of illustrative display cover layers overlapping illustrative optical components in accordance with embodiments.

2.3: Systems with Supplemental Illumination

[0045] FIGS. **2.3-1** is a cross-sectional side view of a portion of an illustrative electronic device with an environmental illumination system in accordance with an embodiment.

[0046] FIGS. **2.3-2** is a top view of an illustrative electronic device with an environmental illumination system in accordance with an embodiment.

[0047] FIGS. **2.3-3**, **2.3-4**, **2.3-5**, and **2.3-6** are cross-sectional side view of illustrative light sources for a supplemental illumination system in accordance with an embodiment.

[0048] FIGS. **2.3-7**, **2.3-8**, and **2.3-9** are graphs showing illustrative illumination patterns that may be produced by a supplemental illumination system in accordance with an embodiment.

[0049] FIGS. **2.3-10** is a flow chart of illustrative operations involved in using an electronic device such as a head-mounted device with a supplemental illumination system in accordance with an embodiment.

2.4: Systems with Displays and Sensor-Hiding Structures

[0050] FIGS. **2.4-1** is a front view of an illustrative head-mounted device in accordance with an embodiment.

2.5: Systems with Cover Layer Sealing Structures

[0052] FIGS. **2.5-1** is a side view of an illustrative electronic device such as a head-mounted device in accordance with an embodiment.

[0053] FIGS. **2.5-2** is schematic diagram of an illustrative system with an electronic device in accordance with an embodiment.

[0054] FIGS. **2.5-3** is a front view of an illustrative head-mounted device in accordance with an embodiment.

[0055] FIGS. **2.5-4** is a front view of an illustrative shroud in accordance with an embodiment.

[0056] FIGS. **2.5-5** is a top view of a portion of an illustrative head-mounted device with a display, cover layer, and shroud in accordance with an embodiment.

[0057] FIGS. **2.5-6** is a side view of an illustrative cover layer with encapsulation material that seals an edge surface of the cover layer and overlaps a laminate on the cover layer in accordance with an embodiment.

[0058] FIGS. **2.5-7** is a side view of an illustrative cover layer with encapsulation material that seals an edge surface of the cover layer in accordance with an embodiment.

[0059] FIGS. **2.5-8** is a side view of an illustrative cover layer having an edge surface that is spaced apart from a head-mounted device housing in accordance with an embodiment.

[0060] FIGS. **2.5-9** is a side view of an illustrative cover layer with a bumper ring or overmold structure that seals an edge surface of the cover layer in accordance with an embodiment.

[0061] FIGS. **2.5-10** is a side view of an illustrative cover layer with an upper laminate that wraps an edge surface of the cover layer in accordance with an embodiment.

[0062] FIGS. **2.5-11** is a side view of an illustrative cover layer with a lower laminate that wraps an edge surface of the cover layer in accordance with an embodiment.

[0063] FIGS. **2.5-12** is a side view of an illustrative cover layer and glue that fills a gap between an edge surface of the cover layer and a housing structure in accordance with an embodiment.

[0064] FIGS. **2.5-13** is a side view of an illustrative cover layer with an upper laminate that extends over the cover layer to a housing structure to separate an edge surface of the cover layer from an exterior of the device in accordance with an embodiment.

[0065] FIGS. **2.5-14** is a side view of an illustrative cover layer and a lip formed from a shroud or housing member that overlaps an edge portion of the cover layer in accordance with an embodiment.

[0066] FIGS. **2.5-15** is a side view of an illustrative cover layer and a lip formed from a shroud or housing member that overlaps an edge portion of the cover layer, along with an upper laminate that wraps around the edge portion in accordance with an embodiment.

2.6: Electronic Devices with Antennas and Optical Components

[0067] FIGS. **2.6-1** is a top view of a head-mounted device.

[0068] FIGS. **2.6-2** is a rear view of a head-mounted device.

[0069] FIGS. **2.6-3** is a schematic diagram of a head-mounted device.

[0070] FIGS. **2.6-4** is a view of a portion of a head-mounted device with a head-mounted housing frame and a camera support member.

[0071] FIGS. **2.6-5** is a front view of a portion of a head-mounted device with a camera support structure.

[0072] FIGS. **2.6-6** is a cross-sectional side view of a portion of a head-mounted device with a camera support structure.

[0073] FIGS. **2.6-7** is a schematic diagram of wireless communications circuitry.

[0074] FIGS. **2.6-8** is a diagram of an antenna.

[0075] FIGS. **2.6-9, 2.6-10, 2.6-11, and 2.6-12** are cross-sectional side views of portions of a support structure, such as a camera support structure with antennas.

[0076] FIGS. **2.6-13** is a top view of a camera support structure.

[0077] FIGS. **2.6-14** is a cross-sectional side view of a camera support structure.

[0078] FIGS. **2.6-15** is a cross-sectional side view of a portion of a camera support structure with a bend sensor to detect camera misalignment.

[0079] FIGS. **2.6-16** is a cross-sectional side view of a portion of a camera support structure with an adjustable-orientation camera.

III: Display Integration Assembly

[0080] FIG. **3-1** is a view of a display and front-cover assembly of an HMD.

[0081] FIG. **3-2** is a cross-sectional view of a portion of a display assembly of an example HMD.

[0082] FIG. **3-3** is a side view of an example of a display assembly of an HMD.

[0083] FIG. **3-4** is a side cross-sectional view of a portion of a display assembly of an example HMD.

[0084] FIG. **3-4A** is a side cross-sectional view of a portion of a display assembly of an example HMD.

[0085] FIG. **3-4B** is a perspective cross-sectional view of a portion of a display assembly of an example HMD.

[0086] FIG. **3-4C** is a perspective cross-sectional view of a portion of a display assembly of an example HMD.

[0087] FIG. **3-5** is a perspective cross-sectional view of a portion of a display assembly of an example HMD.

[0088] FIG. **3-6** is a perspective cross-sectional view of a portion of a display assembly of an example HMD.

IV: Shroud

4.0: Systems with Displays and Sensor-Hiding Structures

[0089] FIG. **4-1** is a front view of an illustrative shroud in accordance with an embodiment.

[0090] FIG. **4-2** is a front view of a portion of an illustrative shroud with a curved periphery in accordance with an embodiment.

[0091] FIG. **4-3** is a front view of a portion of an illustrative forward-facing display in accordance with an embodiment.

[0092] FIG. **4-4** is a cross-sectional top view of a portion of an illustrative display in accordance with an embodiment.

[0093] FIG. **4-5** is a cross-sectional top view of a portion of an illustrative head-mounted device with a display and shroud in accordance with an embodiment.

[0094] FIG. **4-6** is a cross-sectional side view of a portion of an illustrative shroud with a through-hole opening to accommodate an optical component in accordance with an embodiment.

[0095] FIG. **4-7** is a cross-sectional side view of a portion of an illustrative shroud with a window member in a through-hole opening in accordance with an embodiment.

[0096] FIG. **4-8** is a cross-sectional side view of a portion of a head-mounted device with a shroud covering a display in accordance with an embodiment.

[0097] FIG. **4-9** is a cross-sectional side view of an illustrative head-mounted device optical component mounting arrangement with an optical component window coating in accordance with an embodiment.

[0098] FIG. **4-10** is a cross-sectional side view of an illustrative head-mounted device optical component mounting arrangement using shroud through-hole openings in accordance with an embodiment.

[0099] FIG. **4-11** is a cross-sectional side view of an illustrative head-mounted device optical component mounting arrangement with a window formed from a transparent window member such as a layer of glass or clear polymer with a coating in accordance with an embodiment.

4.1: System with Cover Layer Sealing Structures

[0100] FIGS. **4.1-1** is a side view of an illustrative electronic device such as a head-mounted device in accordance with an embodiment.

[0101] FIGS. **4.1-2** is schematic diagram of an illustrative system with an electronic device in accordance with an embodiment.

[0102] FIGS. **4.1-3** is a front view of an illustrative head-mounted device in accordance with an embodiment.

[0103] FIGS. **4.1-4** is a front view of an illustrative shroud in accordance with an embodiment.

[0104] FIGS. **4.1-5** is a top view of a portion of an illustrative head-mounted device with a display, cover layer, and shroud in accordance with an embodiment.

[0105] FIGS. **4.1-6** is a side view of an illustrative cover layer with encapsulation material that seals an edge surface of the cover layer and overlaps a laminate on the cover layer in accordance with an embodiment.

[0106] FIGS. **4.1-7** is a side view of an illustrative cover layer with encapsulation material that seals an edge surface of the cover layer in accordance with an embodiment.

[0107] FIGS. **4.1-8** is a side view of an illustrative cover layer having an edge surface that is spaced apart from a head-mounted device housing in accordance with an embodiment.

[0108] FIGS. **4.1-9** is a side view of an illustrative cover layer with a bumper ring or overmold structure that seals an edge surface of the cover layer in accordance with an embodiment.

[0109] FIGS. **4.1-10** is a side view of an illustrative cover layer with an upper laminate that wraps an edge surface of the cover layer in accordance with an embodiment.

[0110] FIGS. **4.1-11** is a side view of an illustrative cover layer with a lower laminate that wraps an edge surface of the cover layer in accordance with an embodiment.

[0111] FIGS. **4.1-12** is a side view of an illustrative cover layer and glue that fills a gap between an edge surface of the cover layer and a housing structure in accordance with an embodiment.

[0112] FIGS. **4.1-13** is a side view of an illustrative cover layer with an upper laminate that extends over the cover layer to a housing structure to separate an edge surface of the cover layer from an exterior of the device in accordance with an embodiment.

[0113] FIGS. **4.1-14** is a side view of an illustrative cover layer and a lip formed from a shroud or housing member that overlaps an edge portion of the cover layer in accordance with an embodiment.

[0114] FIGS. **4.1-15** is a side view of an illustrative cover layer and a lip formed from a shroud or housing member that overlaps an edge portion of the cover layer, along with an upper laminate that wraps around the edge portion in accordance with an embodiment.

[0115] FIGS. **4.1-16** and **4.1-17** are side views of illustrative cover layers with upper and lower

laminates in accordance with some embodiments.

[0116] FIGS. **4.1-18** is a side view of an illustrative cover layer with a laminate and a seal that covers an edge of the laminate in accordance with some embodiments.

V: Dust Seal

5.1: Seal for an Electronic Device

[0117] FIG. **5-1** shows a cross-sectional view of a portion of an electronic device, according to an example;

[0118] FIG. **5-2** shows a cross-sectional view of a seal, according to an example;

[0119] FIG. **5-3** shows a cross-sectional view of an electronic device, according to an example;

[0120] FIG. **5-4A** shows a top perspective view of an electronic component and a seal, according to an example;

[0121] FIG. **5-4B** shows a cross-sectional view of a portion of an electronic device, according to an example; and

[0122] FIG. **5-4C** shows a cross-sectional view of a portion of an electronic device, according to an example.

VI: Sensor System

[0123] FIG. **6-0** illustrates a view of an example of an HMD.

[0124] FIG. **6-1** illustrates a front perspective view of an example of a sensor system for an HMD.

[0125] FIG. **6-2** illustrates a lower perspective view of an example of a sensor system for and HMD.

[0126] FIG. **6-3** illustrates a lower perspective view of an example of a sensor system for and HMD without a front cover assembly.

[0127] FIG. **6-4** illustrates a lower perspective view of an example of a sensor system of and HMD.

VII: Antennas

[0128] FIGS. **7.0-1** illustrates a view of an example of a display unit of and HMD.

7.1: Electronic Devices with Antenna Mounting Structures

[0129] FIGS. **7.1-1** is a top view of an illustrative electronic device such as a head-mounted device in accordance with an embodiment.

[0130] FIGS. **7.1-2** is a diagram of an illustrative antenna for an electronic device in accordance with an embodiment.

[0131] FIGS. **7.1-3** is a perspective view of an illustrative antenna on an illustrative unidirectional structured foam antenna biasing structure in accordance with an embodiment.

[0132] FIGS. **7.1-4** is a top view of an illustrative structured foam member in accordance with an embodiment.

[0133] FIGS. **7.1-5** is a diagram illustrating how a structured foam member may exhibit preferential unidirectional compression and expansion characteristics in accordance with an embodiment.

[0134] FIGS. **7.1-6** is a cross-sectional top view of a right-hand edge portion of an illustrative head-mounted device in which a unidirectional structured foam antenna biasing member (antenna biasing structure) is being used to mount an antenna against a surface of an overlapping layer such as a display cover layer in accordance with an embodiment.

7.2: Electronic Devices with Millimeter Wave Antennas

[0135] FIGS. **7.2-1** is a top view of an illustrative electronic device with an antenna in accordance with an embodiment.

[0136] FIGS. **7.2-2** is a front view of an illustrative antenna for an electronic device in accordance with an embodiment.

[0137] FIGS. **7.2-3** is a side view of an illustrative millimeter wave antenna having an array of patch antenna elements in accordance with an embodiment.

[0138] FIGS. **7.2-4** is a cross-sectional side view of a corner portion of an illustrative head-mounted device with an antenna in accordance with an embodiment.

[0139] FIGS. **7.2-5** is a cross-sectional view of a front portion of an illustrative head-mounted device with an antenna in accordance with an embodiment.

7.3: Electronic Devices with Antennas Having Compound Curvature

[0140] FIGS. **7.3-1** is a top view of an illustrative electronic device with an antenna in accordance with an embodiment.

[0141] FIGS. **7.3-2** is a diagram of an illustrative antenna for an electronic device in accordance with an embodiment.

[0142] FIGS. **7.3-3** is a perspective view of an illustrative flexible printed circuit antenna with compound curvature in accordance with an embodiment.

[0143] FIGS. **7.3-4** is a side view of illustrative equipment for laminating a flexible printed circuit antenna to a dielectric member such as a polymer layer in accordance with an embodiment.

[0144] FIGS. **7.3-5** is a side view of an illustrative printed circuit antenna with compound curvature attached to a compound curvature surface of dielectric member with compound curvature in accordance with an embodiment.

[0145] FIGS. **7.3-6** is a perspective view of an illustrative printed circuit antenna with compound curvature laminated to the inner surface of a dielectric member with compound curvature in accordance with an embodiment.

VIII: Bent MLB

[0146] FIG. **8-0** illustrates a view of an HMD including a logic board.

[0147] FIG. **8-1** illustrates a plan view of an example of a logic board.

[0148] FIG. **8-2** illustrates a top view of an example of a logic board.

[0149] FIG. **8-3** illustrates a close up view of the logic board shown in FIG. **8-2**.

[0150] FIG. **8-4** illustrates an example of a logic board.

[0151] FIG. **8-5** illustrates a perspective view of a logic board coupled with a fan assembly of an HMD.

IX: Thermals

[0152] FIGS. **9.0-1** illustrates a view of an HMD.

9.1: Air Deflector for a Cooling System in a Head-Mounted Device

[0153] FIGS. **9.1-1** illustrates a schematic diagram of an example of a head-mounted device.

[0154] FIGS. **9.1-2** illustrates a front view of an example of a head-mounted device.

[0155] FIGS. **9.1-3** illustrates a side view of an example of a cooling system.

[0156] FIGS. **9.1-4** illustrates a side view of an example of a cooling system having an air deflector.

[0157] FIGS. **9.1-5** illustrates a side view of an example of a cooling system having an air deflector.

[0158] FIGS. **9.1-6** illustrates a side view of an example of a cooling system having an air deflector.

[0159] FIGS. **9.1-7** illustrates a side view of an example of air flow in a cooling system.

[0160] FIGS. **9.1-8** illustrates a side view of an example of air flow in a cooling system.

[0161] FIGS. **9.1-9** illustrates a block diagram of an example of a head-mounted device.

9.2: Fan with Debris Mitigation

[0162] FIGS. **9.2-1** illustrates a side view of a head-mountable device, according to some embodiments of the present disclosure.

[0163] FIGS. **9.2-2** illustrates a perspective view of a fan for a head-mountable device, according to some embodiments of the present disclosure.

[0164] FIGS. **9.2-3** illustrates a sectional view of an assembly of the head-mountable device of FIGS. **9.2-1** including the fan of FIGS. **9.2-2** in operation to generate a flow, according to some embodiments of the present disclosure.

[0165] FIGS. **9.2-4** illustrates another sectional view of the assembly of FIGS. **9.2-3** with the fan of FIGS. **9.2-2** in stasis and particles entering through the outlet, according to some embodiments of

the present disclosure.

[0166] FIGS. **9.2-5** illustrates a perspective and sectional view of a fan having an annular ring to direct incoming particles, according to some embodiments of the present disclosure.

[0167] FIGS. **9.2-6** illustrates a perspective and sectional view of a fan having an annular ring to direct incoming particles, according to some embodiments of the present disclosure.

[0168] FIGS. **9.2-7** illustrates a sectional view of a fan having a base plate with variable thickness, according to some embodiments of the present disclosure.

[0169] FIGS. **9.2-8** illustrates a view of a fan having a base plate forming openings, according to some embodiments of the present disclosure.

[0170] FIGS. **9.2-9** illustrates a bottom view of a fan having a base plate forming openings and an adhesive pad, according to some embodiments of the present disclosure.

[0171] FIGS. **9.2-10** illustrates a sectional view of the fan of FIGS. **9.2-9**, according to some embodiments of the present disclosure.

[0172] FIGS. **9.2-11** illustrates a view of a fan having a base plate forming openings, according to some embodiments of the present disclosure.

[0173] FIGS. **9.2-12** illustrates a perspective and sectional view of the fan of FIGS. **9.2-11**, according to some embodiments of the present disclosure.

[0174] FIGS. **9.2-13** illustrates a block diagram of a head-mountable device, in accordance with some embodiments of the present disclosure.

9.3: Ventilation

[0175] FIGS. **9.3-1** illustrates a view of an example of an HMD.

[0176] FIGS. **9.3-2** illustrates a rear perspective view of an example of a ventilation assembly of and HMD.

[0177] FIGS. **9.3-3** illustrates a perspective cross-sectional view of an example of a fan assembly of an HMD.

[0178] FIGS. **9.3-4** illustrates a cross-sectional view of an example of a fan assembly of an HMD.

[0179] FIGS. **9.3-5** illustrates a top plan view of an example of a fan for an HMD.

[0180] FIGS. **9.3-6** illustrates a bottom plan view of an example of a fan for an HMD.

[0181] FIGS. **9.3-7** illustrates an exploded view of an example of a fan for and HMD.

[0182] FIGS. **9.3-8** illustrates a rear perspective view of an example of a fan and circuit board assembly of an HMD.

[0183] FIGS. **9.3-9** illustrates a perspective view of an example of a fan and circuit board assembly of an HMD.

[0184] FIGS. **9.3-10** illustrates a close up perspective view of an example of a fan and circuit board assembly of an HMD.

[0185] FIGS. **9.3-11** illustrates a side cross-sectional view of an example of a fan and circuit board assembly of an HMD.

X: Chassis

[0186] FIG. **10-0** illustrates a view of an example of an HMD.

[0187] FIG. **10-1** illustrates a view of an example of an HMD.

[0188] FIG. **10-2** illustrates a rear perspective view of an example of an HMD.

[0189] FIG. **10-3** illustrates a front perspective view of an example of a frame assembly of an HMD.

[0190] FIG. **10-4** illustrates a front plan view of an example of a frame assembly of an HMD.

[0191] FIG. **10-5** illustrates a front plan view of an example of a frame assembly of an HMD.

[0192] FIG. **10-6** illustrates a close-up cross-sectional view of a portion of an example of an HMD.

XI: Optical Module

[0193] FIG. **11-1** illustrates a view of an example of an HMD.

11.1: IPD Adjust

[0194] FIGS. **11.1-1** illustrates a partial perspective view of an example of an HMD including an

optical module adjustment system.

11.1.1.1: Crown

[0195] FIG. **11.1.1-1** illustrates a partial perspective view of an example of an HMD including an optical module adjustment system.

11.1.1.1.1: Adjustment Mechanism for Head-Mounted Display

[0196] FIG. **11.1.1.1-1** is a top view of a head-mounted display.

[0197] FIG. **11.1.1.1-2A** is a detail view of an actuator disposed within a head-mounted display similar to the head-mounted display of FIG. **11.1.1.1-1**.

[0198] FIG. **11.1.1.1-2B** is a partially-exploded sectional view of the actuator of FIG. **11.1.1.1-2A**.

[0199] FIG. **11.1.1.1-3A** is a detail view of another actuator disposed within a head-mounted display similar to the head-mounted display of FIG. **11.1.1.1-1**.

[0200] FIG. **11.1.1.1-3B** is a partially-exploded sectional view of the actuator of FIG. **11.1.1.1-3A**.

[0201] FIG. **11.1.1.1-4A** is a detail view of another actuator disposed within a head-mounted display similar to the head-mounted display of FIG. **11.1.1.1-1**.

[0202] FIG. **11.1.1.1-4B** is a partially-exploded sectional view of the actuator of FIG. **11.1.1.1-4A**.

[0203] FIG. **11.1.1.1-5A** is a detail view of an electromagnetic dampening mechanism for an actuator similar to the actuators of FIGS. **11.1.1.1-2A**, **11.1.1.1-2B**, **11.1.1.1-3A**, **11.1.1.1-3B**, **11.1.1.1-4A**, and **11.1.1.1-4B**.

[0204] FIG. **11.1.1.1-5B** is a detail view of another electromagnetic dampening mechanism for an actuator similar to the actuator of FIGS. **11.1.1.1-2A**, **11.1.1.1-2B**, **11.1.1.1-3A**, **11.1.1.1-3B**, **11.1.1.1-4A**, and **11.1.1.1-4B**.

[0205] FIG. **11.1.1.1-6A** is a detail view of a mechanical dampening mechanism for an actuator similar to the actuator of FIGS. **11.1.1.1-2A**, **11.1.1.1-2B**, **11.1.1.1-3A**, **11.1.1.1-3B**, **11.1.1.1-4A**, and **11.1.1.1-4B**.

[0206] FIG. **11.1.1.1-6B** is a detail view of another mechanical dampening mechanism for an actuator similar to the actuator of FIGS. **11.1.1.1-2A**, **11.1.1.1-2B**, **11.1.1.1-3A**, **11.1.1.1-3B**, **11.1.1.1-4A**, and **11.1.1.1-4B**.

[0207] FIG. **11.1.1.1-7** is a flowchart depicting a process of operation for an actuator disposed within a head-mounted display similar to the head-mounted display of FIG. **11.1.1.1-1**.

[0208] FIG. **11.1.1.1-8** is a schematic hardware configuration for a controller in the head-mounted display of FIG. **11.1.1.1-1**.

11.1.1.2: Crown Input and Feedback for Head-Mountable Devices

[0209] FIG. **11.1.1.2-1** illustrates a top view of a head-mountable device, according to some embodiments of the present disclosure.

[0210] FIG. **11.1.1.2-2** illustrates a top exploded view of a head-mountable device, according to some embodiments of the present disclosure.

[0211] FIG. **11.1.1.2-3** illustrates a sectional view of a crown module of the head-mountable device of FIG. **11.1.1.2-2**, according to some embodiments of the present disclosure.

[0212] FIG. **11.1.1.2-4** illustrates a partial sectional view of a crown module of the head-mountable device of FIG. **11.1.1.2-2**, according to some embodiments of the present disclosure.

[0213] FIG. **11.1.1.2-5** illustrates a sectional view of the crown module of FIG. **11.1.1.2-4** taken along line A-A, according to some embodiments of the present disclosure.

[0214] FIG. **11.1.1.2-6** illustrates a side view of the crown module of FIG. **11.1.1.2-4**, according to some embodiments of the present disclosure.

[0215] FIG. **11.1.1.2-7** illustrates a circuit diagram of a sensor of the crown module of FIG. **11.1.1.2-4**, according to some embodiments of the present disclosure.

[0216] FIG. **11.1.1.2-8** illustrates a block diagram of a head-mountable device, in accordance with some embodiments of the present disclosure.

11.1.2: Wishbone and Mustache

[0217] FIG. **11.1.2-1** illustrates a front perspective view of an HMD with the front cover and

display assembly omitted to show an example of a sensor system.

[0218] FIG. **11.1.2-2** illustrates a perspective view of a portion of a sensor system including sensors coupled to a bracket.

[0219] FIG. **11.1.2-3** illustrates a rear perspective view of a portion of an example HMD including a display module bracket.

[0220] FIG. **11.1.2-4** illustrates a top view of a portion of a display assembly of an example HMD.

[0221] FIG. **11.1.2-5** illustrates a side cross-sectional view of an example HMD.

11.1.3: Upper Guide Rod System

[0222] FIG. **11.1.3-1** illustrates a rear perspective view of an example HMD including a display adjustment system.

[0223] FIG. **11.1.3-2** illustrates a close-up view thereof with the display module omitted.

[0224] FIG. **11.1.3-3** illustrates a close-up view of the system shown in FIG. **11.1.3-1** with the display module omitted.

11.1.3.1: Motors

[0225] FIG. **11.1.3.1-1** illustrates a rear perspective view of an example HMD including a display adjustment system.

[0226] FIG. **11.1.3.1-2** illustrates a perspective view of an example of a motor of a display adjustment system of an example HMD.

[0227] FIG. **11.1.3.1-3** illustrates a cross-sectional view of an example of a motor of a display adjustment system of an example HMD.

11.1.3.1.1: Electronic Devices with Optical Module Positioning Systems

[0228] FIG. **11.1.3.1.1-1** is a top view of an illustrative head-mounted device in accordance with an embodiment.

[0229] FIG. **11.1.3.1.1-2** is a rear view of an illustrative head-mounted device in accordance with an embodiment.

[0230] FIG. **11.1.3.1.1-3** is a schematic diagram of an illustrative head-mounted device in accordance with an embodiment.

[0231] FIG. **11.1.3.1.1-4** is a rear view of an interior portion of an illustrative head-mounted device in accordance with an embodiment.

[0232] FIG. **11.1.3.1.1-5** is a side view of an illustrative portion of an optical module that is configured to receive a guide rail and a threaded actuator rod in accordance with an embodiment.

[0233] FIG. **11.1.3.1.1-6** is an exploded cross-sectional view of an illustrative guide rod and end cap in accordance with an embodiment.

[0234] FIG. **11.1.3.1.1-7** is a side view of the illustrative guide rod of FIG. **11.1.3.1.1-6** following attachment of the end cap in accordance with an embodiment.

[0235] FIG. **11.1.3.1.1-8** is a cross-sectional top view of the illustrative guide rod of FIGS.

11.1.3.1.1-6 and **11.1.3.1.1-7** showing how the guide rod may be mounted to a housing structure such as a frame in head-mounted support structures in accordance with an embodiment.

[0236] FIGS. **11.1.3.1.1-9**, **11.1.3.1.1-10**, **11.1.3.1.1-11**, and **11.1.3.1.1-12** are views of illustrative guide rods in accordance with embodiments.

[0237] FIG. **11.1.3.1.1-13** is a cross-sectional side view of an illustrative guide rod tube that has been partially filled with a core in accordance with an embodiment.

[0238] FIG. **11.1.3.1.1-14** is a top view of a portion of an illustrative guide rod formed from fiber-composite material in accordance with an embodiment.

[0239] FIG. **11.1.3.1.1-15** is a cross-sectional end view of an illustrative portion of a guide rod formed from fiber-composite material in accordance with an embodiment.

[0240] FIG. **11.1.3.1.1-16** is a cross-sectional side view of an illustrative end portion of a guide rod in accordance with an embodiment.

[0241] FIG. **11.1.3.1.1-17** is a cross-sectional side view of an illustrative tapered end portion of a guide rod in accordance with an embodiment.

11.1.3.1.2: Electronic Device with Lens Positioning Sensing

[0242] FIG. **11.1.3.1.2-1** is a schematic diagram of an illustrative electronic device such as a head-mounted display device in accordance with an embodiment.

[0243] FIG. **11.1.3.1.2-2** is a top view of an illustrative head-mounted device in accordance with an embodiment.

[0244] FIG. **11.1.3.1.2-3** is a front view of an illustrative lens assembly having a force or position sensor in accordance with an embodiment.

[0245] FIG. **11.1.3.1.2-4A** is a front view of an illustrative direct force sensor in accordance with an embodiment.

[0246] FIG. **11.1.3.1.2-4B** is a top view of an illustrative sensor woven into a fabric in accordance with an embodiment.

[0247] FIG. **11.1.3.1.2-4C** is a cross-sectional side view of an illustrative nasal flap with an air bladder sensor in accordance with an embodiment.

[0248] FIG. **11.1.3.1.2-5** is a front view of an illustrative lens assembly having a proximity sensor in accordance with an embodiment.

[0249] FIG. **11.1.3.1.2-6** is a front view of an illustrative lens assembly having movable components that block a light-emitting component to indicate a position of the lens assembly in accordance with an embodiment.

[0250] FIG. **11.1.3.1.2-7** is a circuit diagram of an illustrative control circuit for controlling a positioner motor while monitoring for feedback from the motor in accordance with an embodiment.

[0251] FIG. **11.1.3.1.2-8** is a flow chart of illustrative steps involved operating a head-mounted device in accordance with an embodiment.

11.1.3.2: Sensors/Encoders

[0252] FIG. **11.1.3.2-1** illustrates a perspective view of an example encoder for an HMD display adjustment system.

[0253] FIG. **11.1.3.2-2** illustrates a top perspective view of an example display adjustment system for an HMD.

[0254] FIG. **11.1.3.2-3** illustrates a top view of an example encoder assembly for an HMD display adjustment system.

11.1.3.2.1: Sensor Assembly

[0255] FIG. **11.1.3.2.1-1** illustrates a side view of a head-mountable device, according to some embodiments of the present disclosure.

[0256] FIG. **11.1.3.2.1-2** illustrates an exploded perspective view of a sensor assembly of the head-mountable device of FIG. **11.1.3.2.1-1**, according to some embodiments of the present disclosure.

[0257] FIG. **11.1.3.2.1-3** illustrates a side sectional view of a sensor assembly, according to some embodiments of the present disclosure.

[0258] FIG. **11.1.3.2.1-4** illustrates a side sectional view of a sensor assembly, according to some embodiments of the present disclosure.

[0259] FIG. **11.1.3.2.1-5** illustrates a block diagram of a head-mountable device, in accordance with some embodiments of the present disclosure.

11.1.3.2.2: Electronic Devices with Movable Optical Assemblies

[0260] FIG. **11.1.3.2.2-1** is a diagram of an illustrative head-mounted device in accordance with an embodiment.

[0261] FIGS. **11.1.3.2.2-2** and **11.1.3.2.2-3** are rear views of portions of illustrative head-mounted devices in accordance with embodiments.

[0262] FIG. **11.1.3.2.2-4** is a graph in which illustrative optical assembly adjustment values are plotted as a function of measured eye relief for multiple different illustrative measured interpupillary distances in accordance with an embodiment.

[0263] FIG. **11.1.3.2.2-5** is a flow chart of illustrative operations involved in using a head-mounted device in accordance with an embodiment.

11.1.3.2.3: Electronic Devices with Movable Optical Assemblies

[0264] FIG. **11.1.3.2.3-1** is a diagram of an illustrative head-mounted device in accordance with an embodiment.

[0265] FIGS. **11.1.3.2.3-2** and **11.1.3.2.3-3** are flow charts of illustrative operations involved in using a head-mounted device with movable optical assemblies in accordance with embodiments.

[0266] FIG. **11.1.3.2.3-4** is a cross-sectional end view of an illustrative clutch based on a split nut may be used in limiting how much force is applied to an optical assembly in accordance with an embodiment.

[0267] FIGS. **11.1.3.2.3-5** and **11.1.3.2.3-6** are diagrams showing how magnetic clutches may be used in limiting the force applied to optical assemblies in accordance with embodiments.

[0268] FIGS. **11.1.3.2.3-7**, **11.1.3.2.3-8**, **11.1.3.2.3-9**, and **11.1.3.2.3-10** are diagrams of illustrative mechanical clutch mechanisms that may be used in moving optical assemblies in accordance with embodiments.

[0269] FIG. **11.1.3.2.3-11** is a diagram showing how force-sensitive switches may be used in coupling a nut to an optical assembly in accordance with an embodiment.

[0270] FIG. **11.1.3.2.3-12** is a diagram showing how torque-sensitive switches may be coupled between a rotating motor and a portion of a rotating shaft in accordance with an embodiment.

[0271] FIG. **11.1.3.2.3-13** is a circuit diagram showing how motor load may be measured electrically while moving optical assemblies in accordance with an embodiment.

[0272] FIG. **11.1.3.2.3-14** is a diagram of an illustrative motor with a rotary encoder in accordance with an embodiment

[0273] FIG. **11.1.3.2.3-15** is a diagram of an illustrative motor, movable optical assembly, and associated linear magnetic encoder in accordance with an embodiment.

[0274] FIG. **11.1.3.2.3-16** is a graph showing how motor stalling may be detected while controlling a motor to move an optical assembly in accordance with an embodiment.

[0275] FIG. **11.1.3.2.3-17** is a flow chart of illustrative operations involved in using a head-mounted device with motors to move optical assemblies in accordance with an embodiment.

11.1.3.3: Hard Stops

[0276] FIG. **11.1.3.3-1** illustrates a perspective view of a portion of an example HMD including a hard stop.

[0277] FIG. **11.1.3.3-2** illustrates a perspective view of a portion of an example HMD including a hard stop.

11.1.3.4: Upper Biasing Members

[0278] FIG. **11.1.3.4-1** illustrates a perspective view of a portion of a display adjustment system of an example HMD.

[0279] FIG. **11.1.3.4-2** illustrates a perspective view of a portion of a display adjustment system of an example HMD.

11.1.4: Lower Guide Rod System

11.1.4.1: Electronic Devices with Biased Guide Rails

[0280] FIG. **11.1.4.1-1** is a top view of an illustrative electronic device in accordance with an embodiment.

[0281] FIG. **11.1.4.1-2** is a schematic diagram of an illustrative electronic device in accordance with an embodiment.

[0282] FIG. **11.1.4.1-3** is top view of an illustrative electronic device with optical module guide rails in accordance with an embodiment.

[0283] FIG. **11.1.4.1-4** is a rear view of illustrative electronic device with optical module guide rails in accordance with an embodiment.

[0284] FIG. **11.1.4.1-5** is a side view of an illustrative optical module with guide rails in accordance with an embodiment.

[0285] FIGS. **11.1.4.1-6A**, **11.1.4.1-6B**, and **11.1.4.1-7** are cross-sectional side views of illustrative

guide rail biasing mechanisms in accordance with embodiments.

[0286] FIG. **11.1.4.1-8** is a cross-sectional side view of a portion of a kinematic guide rail mounting system in accordance with an embodiment.

[0287] FIG. **11.1.4.1-9** is a side view of a kinematic optical module guide rail mounting system in accordance with an embodiment.

[0288] FIG. **11.1.4.1-10** is a perspective view of an illustrative guide rail sensor based on a strain gauge in accordance with an embodiment.

[0289] FIG. **11.1.4.1-11** is a cross-sectional side view of an illustrative optical module with a guide rail sensor in accordance with an embodiment.

11.1.4.2: Lower Guide Rods

[0290] FIG. **11.1.4.2-1** illustrates a plan view of a portion of an example HMD including a guide system for an adjustable display.

11.1.4.2.1: Electrical Contacts

[0291] FIG. **11.1.4.2.1-1** illustrates a perspective view of a portion of an example HMD.

11.1.4.2.2: BIASING MEMBERS

[0292] FIG. **11.1.4.2.2-1** illustrates a perspective view of a portion of an example HMD.

11.2: Barrels and Baskets

11.2.1: Lens Mounting Systems

[0293] FIG. **11.2.1-1** is a diagram of an illustrative head-mounted device in accordance with an embodiment.

[0294] FIG. **11.2.1-2** is a front view of an illustrative lens in accordance with embodiments.

[0295] FIGS. **11.2.1-3** and **11.2.1-4** are cross-sectional side view of peripheral portions of illustrative lenses and associated mounting structures in accordance with embodiments.

[0296] FIGS. **11.2.1-5** and **11.2.1-6** are top views of illustrative flexures for mounting a lens in accordance with embodiments.

[0297] FIGS. **11.2.1-7**, **11.2.1-8**, **11.2.1-9**, and **11.2.1-10** are cross-sectional side views of additional illustrative flexure arrangements for mounting a lens in accordance with embodiments.

[0298] FIG. **11.2.1-11** is a diagram showing how adhesive may be introduced into a gap between an illustrative flexure and a lens in accordance with an embodiment.

11.3: Rear-Facing Cameras

11.3.1: Optical Module for Head-Mounted Device

[0299] FIG. **11.3.1-1** is a block diagram that shows an example of a hardware configuration for a head-mounted device.

[0300] FIG. **11.3.1-2** is a top view illustration that shows the head-mounted device, including a device housing and a support structure.

[0301] FIG. **11.3.1-3** is a rear view illustration taken along line A-A of FIG. **11.3.1-2** that shows the device housing.

[0302] FIG. **11.3.1-4** is a perspective view illustration that shows an optical module of the head-mounted device.

[0303] FIG. **11.3.1-5** is an exploded side view diagram showing components of an optical module according to an example.

[0304] FIG. **11.3.1-6** is a front view that shows the lens according to an example.

[0305] FIG. **11.3.1-7** is a cross-section view taken along line B-B of FIG. **11.3.1-6** showing the lens.

[0306] FIG. **11.3.1-8** is a front view illustration that shows a housing body of an optical module housing assembly

[0307] FIG. **11.3.1-9** is a cross-section view illustration taken along line C-C of FIG. **11.3.1-8** showing the housing body.

[0308] FIG. **11.3.1-10** is a front view illustration that shows a retainer of the optical module housing assembly.

[0309] FIG. **11.3.1-11** is a cross-section view illustration taken along line D-D of FIG. **11.3.1-10** showing the retainer.

[0310] FIG. **11.3.1-12** is a front view illustration that shows an infrared emitter.

[0311] FIG. **11.3.1-13** is a cross-section view illustration showing a portion of the infrared emitter and a peripheral wall of the housing body.

[0312] FIG. **11.3.1-14** is a cross-section view illustration that shows the optical module.

[0313] FIG. **11.3.1-15** is a cross-section view illustration that shows the optical module according to an alternative implementation in which an optical axis of the eye camera is angled toward an optical axis of the optical module.

[0314] FIG. **11.3.1-16** is a cross-section view illustration that shows the optical module according to an alternative implementation in which the infrared emitter is located outside of the housing body of the optical module housing assembly.

[0315] FIG. **11.3.1-17** is a side-view illustration that shows a display module according to an implementation.

[0316] FIG. **11.3.1-18** is a top-view illustration that shows interpupillary adjustment mechanisms that each support one of the optical modules.

[0317] FIG. **11.3.1-19** is a side view illustration that shows one of the interpupillary adjustment mechanisms.

[0318] FIG. **11.3.1-20** is a top-view cross-section illustration that shows front-facing cameras that are supported by each of the optical modules.

[0319] FIG. **11.3.1-21** is an illustration that shows connection of the eye camera and the infrared emitter to a computing device by an optical module jumper board.

11.3.2: Cameras and LEDs

[0320] FIG. **11.3.2-1** illustrates a perspective view of a portion of an example of an optical module of an HMD.

[0321] FIG. **11.3.2-2** illustrates a top view of a portion of an example of an optical module of an HMD.

[0322] FIG. **11.3.2-3** illustrates a perspective cutaway view of a portion of an example of an optical module of an HMD.

[0323] FIG. **11.3.2-4** illustrates a plan view of a portion of an example of an optical module of and HMD.

[0324] FIG. **11.3.2-5** illustrates a cutaway view of a portion of an example of an optical module of an HMD.

11.4: Display

11.4.1: Display System with Interchangeable Lens

[0325] FIG. **11.4.1-0** illustrates a view of an HMD.

[0326] FIG. **11.4.1-1** is a side view of a display system with hidden components illustrated in dashed lines.

[0327] FIG. **11.4.1-2** is a cross-sectional view of the display system of FIG. **11.4.1-1** taken along line 2-2 in FIG. **11.4.1-1**.

[0328] FIG. **11.4.1-3A** is a cross-sectional view of a display unit and interchangeable lens assembly of the display system of FIG. **11.4.1-1** taken along line 3-3 in FIG. **11.4.1-2** and shown in an assembled state.

[0329] FIG. **11.4.1-3B** is a cross-sectional view of the display unit and the interchangeable lens assembly of FIG. **11.4.1-3A** shown in a disassembled state.

[0330] FIG. **11.4.1-4** is a rear view of a removable lens of the display system of FIG. **11.4.1-1** with light emission, entry, and exit points illustrated by dashed lines (i.e., dash-dot lines).

[0331] FIG. **11.4.1-5** is a rear view of another embodiment of a removable lens.

[0332] FIG. **11.4.1-6** is a rear view of another embodiment of a removable lens.

[0333] FIG. **11.4.1-7** is a cross-sectional view another embodiment of a removable lens.

[0334] FIG. **11.4.1-8** is a cross-sectional view another embodiment of a removable lens.

[0335] FIG. **11.4.1-9** is a cross-sectional view another embodiment of a removable lens.

[0336] FIG. **11.4.1-10A** is a cross-sectional view of another display unit and another interchangeable lens assembly for the display system of FIG. **11.4.1-1** shown in a disassembled state.

[0337] FIG. **11.4.1-10B** is a cross-sectional view of the display unit and the interchangeable lens assembly of FIG. **11.4.1-10A** shown in an assembled state.

[0338] FIG. **11.4.1-11A** is a cross-sectional view of another display unit and another interchangeable lens assembly for the display system of FIG. **11.4.1-1** shown in a disassembled state.

[0339] FIG. **11.4.1-11B** is a cross-sectional view of the display unit and the interchangeable lens assembly of FIG. **11.4.1-10A** shown in an assembled state.

[0340] FIG. **11.4.1-12A** is a side view of a display module for use in the display system.

[0341] FIG. **11.4.1-12B** is a front view of a display module for use in the display system.

[0342] FIG. **11.4.1-12C** is a front view of a display module for use in the display system.

[0343] FIG. **11.4.1-12D** is a front view of a display module for use in the display system.

[0344] FIG. **11.4.1-13A** is a front view of a display module for use in the display system.

[0345] FIG. **11.4.1-13B** is a front view of a removable lens assembly for use with the display module of FIG. **11.4.1-13A**.

[0346] FIG. **11.4.1-13C** is a cross-sectional view of the display module of FIG. **11.4.1-13A** taken along line **11.4.1-13A-11.4.1-13A**.

[0347] FIG. **11.4.1-13D** is a cross-sectional view of the removable lens assembly of FIG. **11.4.1-13B** taken along line **11.4.1-13B-11.4.1-13B**.

[0348] FIG. **11.4.1-13E** is a cross-sectional view of the display module of FIG. **11.4.1-13A** and the removable lens assembly of FIG. **11.4.1-13B** in a partially coupled state.

[0349] FIG. **11.4.1-13F** is a cross-sectional view of the display module of FIG. **11.4.1-13A** and the removable lens assembly of FIG. **11.4.1-13B** in a coupled state.

[0350] FIG. **11.4.1-14A** is a schematic view of a display system.

[0351] FIG. **11.4.1-14B** is a flow chart of a method for operating the display system.

[0352] FIG. **11.4.1-15** is a flow chart of a process for determining compatibility of a removable lens and a user.

[0353] FIG. **11.4.1-16** is a flow chart of a method for determining compatibility of the removable lens and the user.

[0354] FIG. **11.4.1-17** is a schematic of an example hardware configuration of a controller of the display system.

11.4.2: Electronic Device System with Supplemental Lenses

[0355] FIG. **11.4.2-1** is a schematic diagram of an illustrative electronic device such as a head-mounted display device in accordance with an embodiment.

[0356] FIG. **11.4.2-2** is a top view of an illustrative head-mounted device in accordance with an embodiment.

[0357] FIG. **11.4.2-3** is a diagram of an illustrative removable supplemental lens in accordance with an embodiment.

[0358] FIG. **11.4.2-4** is a flow chart of illustrative operations associated with using a head-mounted device in accordance with an embodiment.

11.4.3: Rx Lenses

[0359] FIG. **11.4.3-1** illustrates a perspective view of a portion of an optical assembly of an example HMD.

[0360] FIG. **11.4.3-2** illustrates a perspective view of a portion of an optical assembly of an example HMD.

[0361] FIG. **11.4.3-3** illustrates a perspective view of a portion of an optical assembly of an

example HMD.

[0362] FIG. **11.4.3-4** illustrates a plan and exploded view of a portion of an optical assembly of an example HMD.

[0363] FIG. **11.4.3-5** illustrates a magnet array for an example display module of an HMD.

[0364] FIG. **11.4.3-6** illustrates a perspective view of an example lens of an HMD.

[0365] FIG. **11.4.3-7** illustrates a side view of an example lens of an HMD.

[0366] FIG. **11.4.3-8** illustrates a side view of an example lens of an HMD.

[0367] FIG. **11.4.3-9** illustrates a side view of an example lens of an HMD.

[0368] FIG. **11.4.3-10** illustrates a side view of an example lens of an HMD.

[0369] FIG. **11.4.3-11** illustrates a side view of an example lens of an HMD.

XII: Curtain

[0370] FIGS. **12.0-1** illustrates a view of an HMD.

12.1: Electronic Devices with Stretchable Fabric Covers

[0371] FIGS. **12.1-1** is a top view of an illustrative head-mounted device in accordance with an embodiment.

[0372] FIGS. **12.1-2** is a rear view of an illustrative head-mounted device in accordance with an embodiment.

[0373] FIGS. **12.1-3** is a schematic diagram of an illustrative head-mounted device in accordance with an embodiment.

[0374] FIGS. **12.1-4** is a top view of an illustrative head-mounted device in which left-eye and right-eye optical modules have been placed close to each other to accommodate a user with a small interpupillary distance in accordance with an embodiment.

[0375] FIGS. **12.1-5** is a top view of the illustrative head-mounted device of FIGS. **12.1-4** in which the optical modules have been moved away from each other to accommodate a user with a large interpupillary distance in accordance with an embodiment.

[0376] FIGS. **12.1-6** is a front view of an illustrative cover layer with stretchable fabric in an unstretched state in accordance with an embodiment.

[0377] FIGS. **12.1-7** is a front view of the illustrative cover layer of FIGS. **12.1-6** with stretchable fabric in a stretched state in accordance with an embodiment.

[0378] FIGS. **12.1-8** is a side view of an illustrative first strand that may be used in a cover layer of the type shown in FIGS. **12.1-6** and **12.1-7** in accordance with an embodiment.

[0379] FIGS. **12.1-9** is a side view of an illustrative second strand that may be used in a cover layer of the type shown in FIGS. **12.1-6** and **12.1-7** in accordance with an embodiment.

[0380] FIGS. **12.1-10** is a front view of an illustrative cover layer having regions with different levels of stretch and opacity in accordance with an embodiment.

[0381] FIGS. **12.1-11** is a perspective view of an illustrative cover layer formed from a three-dimensional fabric in accordance with an embodiment.

12.2: Curtain Assembly

[0382] FIGS. **12.2-1** illustrates a view of an example of an HMD.

[0383] FIGS. **12.2-2** illustrates a rear perspective view of an example HMD including a curtain assembly.

[0384] FIGS. **12.2-3** illustrates a rear view of an example HMD including a curtain assembly.

[0385] FIGS. **12.2-4** illustrates a side cross-sectional view of an example HMD including a curtain assembly.

[0386] FIGS. **12.2-5** illustrates a perspective view of an example of a curtain assembly of an HMD.

[0387] FIGS. **12.2-6** illustrates an exploded view of an example of a curtain assembly of an HMD.

[0388] FIGS. **12.2-7** illustrates a rear view of an example of a curtain assembly of an HMD.

[0389] FIGS. **12.2-8** illustrates a partial view of an example curtain assembly.

[0390] FIGS. **12.2-9** illustrates a partial view of an example curtain assembly.

[0391] FIGS. **12.2-10** illustrates a partial view of an example curtain assembly.

[0392] FIGS. **12.2-11** illustrates a partial view of an example curtain assembly.

[0393] FIGS. **12.2-12** illustrates a partial view of an example curtain assembly.

[0394] FIGS. **12.2-13** illustrates a partial view of an example curtain assembly.

XIII: Light Seal

[0395] FIGS. **13.0-1** illustrates a view of an HMD.

[0396] FIGS. **13.0-2A** illustrates a front perspective view of a device seal according to one embodiment.

[0397] FIGS. **13.0-2B** illustrates a bottom-rear perspective view of the device seal of FIG. **13.0-2A**.

[0398] FIGS. **13.0-2C** illustrates a rear view of the device seal of FIGS. **13.0-2A**.

13.1: Electronic Devices with Covering Structures

[0399] FIGS. **13.1-1** is a top view of a head-mounted device.

[0400] FIGS. **13.1-2** is a rear view of a head-mounted device.

[0401] FIGS. **13.1-3** is a schematic diagram of a head-mounted device.

[0402] FIGS. **13.1-4** is a top view of a head-mounted device with left-eye and right-eye optical modules.

[0403] FIGS. **13.1-5** is a top view of the head-mounted device of FIGS. **13.1-4** with the optical modules spaced further apart.

[0404] FIGS. **13.1-6** is a cross-sectional side view of a head-mounted device with a fan.

[0405] FIGS. **13.1-7** is an exploded perspective view of a curtain having a frame and a cover layer supported on the frame.

[0406] FIGS. **13.1-8** is a top view of an optical module and a cover layer.

[0407] FIGS. **13.1-9** is a view of a cover layer with a peripheral elastic band.

[0408] FIGS. **13.1-10** is a view of a cover layer with woven elastic strands forming a peripheral elastic band.

[0409] FIGS. **13.1-11** is a diagram of a cover layer formed from a material that stretches.

[0410] FIGS. **13.1-12** is a view of a frame for a curtain.

[0411] FIGS. **13.1-13** is a cross-sectional side view of a cover layer with a peripheral elastic band that moves relative to a rigid frame.

[0412] FIGS. **13.1-14** is a cross-sectional top view of a head-mounted device with a floating curtain.

[0413] FIGS. **13.1-15** is a rear view of a curtain with locations for attaching the curtain to a head-mounted device housing member.

[0414] FIGS. **13.1-16** is a cross-sectional side view of a portion of a head-mounted device showing with a curtain attached to a head-mounted device housing member.

[0415] FIGS. **13.1-17** is a top view of an apparatus with a movable member surrounded by a curtain.

13.2: Device with a Removable Cushion

[0416] FIGS. **13.2-1** is a top view of an electronic device such as a head-mounted device.

[0417] FIGS. **13.2-2** is top view of an optical module for an electronic device.

[0418] FIGS. **13.2-3A** is a cross-sectional top view of a head-mounted device with a removable cushion in an unattached state.

[0419] FIGS. **13.2-3B** is a cross-sectional top view of a head-mounted device with a removable cushion in an attached state.

[0420] FIGS. **13.2-4** is a perspective view of a head-mounted support structure.

[0421] FIGS. **13.2-5A** is a rear view of a flexible structure of a head-mounted support structure attached to support posts.

[0422] FIGS. **13.2-5B** is a rear view of a removable cushion with high-rigidity portions configured to overlap support posts in a corresponding head-mounted support structure.

[0423] FIGS. **13.2-6A** is a rear view of a flexible structure with primary attachment structures and auxiliary attachment structures.

[0424] FIGS. **13.2-6B** is a rear view of a removable cushion with primary attachment structures and auxiliary attachment structures.

[0425] FIGS. **13.2-7** is a cross-sectional top view of a head-mounted device with a removable cushion with magnets and recesses.

[0426] FIGS. **13.2-8** is a rear view of a removable cushion with hinge structures.

[0427] FIGS. **13.2-9** is a schematic diagram of a system that includes a head-mounted support structure and multiple removable cushions.

13.3: Electronic Devices with Light-Blocking Fabrics

[0428] FIGS. **13.3-1** is a top view of a head-mounted device.

[0429] FIGS. **13.3-2** is a rear view of a head-mounted device.

[0430] FIGS. **13.3-3** is a schematic diagram of a head-mounted device.

[0431] FIGS. **13.3-4** is a perspective view of a head-mounted device with a fabric-covered face frame.

[0432] FIGS. **13.3-5A** is a schematic diagram of a knitting system.

[0433] FIGS. **13.3-5B** is a schematic diagram of a knitting system.

[0434] FIGS. **13.3-6** is a diagram of a portion of a weft knit fabric layer.

[0435] FIGS. **13.3-7** is a cross-sectional side view of a light seal.

[0436] FIGS. **13.3-8** is a perspective view of an inner fabric layer for a light seal.

[0437] FIGS. **13.3-9** is a cross-sectional side view of a light seal.

13.4: Electronic Devices with Stretchable Fabrics

[0438] FIGS. **13.4-6** is a diagram of a portion of a fabric layer with knit stitches.

[0439] FIGS. **13.4-7** is a diagram of a portion of a fabric layer with knit stitches and miss stitches.

[0440] FIGS. **13.4-8** is a diagram of a portion of a fabric layer with knit stitches and tuck stitches.

[0441] FIGS. **13.4-9** is a knitting chart of a fabric layer may have a four-row-repeat-pattern with knit stitches and tuck stitches.

13.5: Contactless Sensors for a Head-Mountable Device

[0442] FIGS. **13.5-1** shows a top view profile of a head-mountable device including a facial interface.

[0443] FIGS. **13.5-2A** shows a side view of a head-mountable device including a facial interface.

[0444] FIGS. **13.5-2B** shows a front view of a head-mountable device including a facial interface.

[0445] FIGS. **13.5-3** shows a top view of a facial interface with a sensor.

[0446] FIGS. **13.5-4** shows a top view of a facial interface with multiple sensors at various positions.

[0447] FIGS. **13.5-5** shows yet another top view of a facial interface with multiple sensors at various positions.

[0448] FIGS. **13.5-6A** shows a top view of a facial interface with various components, including a sensor.

[0449] FIGS. **13.5-6B** shows a top view of a facial interface with various components, including a sensor.

[0450] FIGS. **13.5-7A-13.5-7B** shows non-exploded and exploded perspective views of a facial interface with sensors.

13.6: Integrated Health Sensors

[0451] FIGS. **13.6-1** shows a block diagram of a head-mountable device.

[0452] FIGS. **13.6-2** shows a top view of an example head-mountable device.

[0453] FIGS. **13.6-3** shows a rear perspective view of an example head-mountable device including a facial interface incorporated with sensors.

[0454] FIGS. **13.6-4** shows a cross-sectional view of a facial interface with sensors disposed at various locations.

[0455] FIGS. **13.6-5** shows a perspective view of a head-mountable device including sensors.

[0456] FIGS. **13.6-6** shows a perspective view of a head-mountable device including a facial

interface, a frame, and a plurality of electronic components.

13.7: Health Sensing Retention Band

[0457] FIGS. **13.7-1** shows a schematic block diagram of a head-mountable device.

[0458] FIGS. **13.7-2** shows a top view of a head-mountable device.

[0459] FIGS. **13.7-3** shows a cross-sectional side view of a head-mountable device.

[0460] FIGS. **13.7-4A** shows a rear perspective view of a retention band.

[0461] FIGS. **13.7-4B** shows a side view of the retention band of FIGS. **13.7-4A** in an articulated position.

[0462] FIGS. **13.7-4C** shows a side view of the retention band of FIGS. **13.7-4A** in an articulated position.

[0463] FIGS. **13.7-5** shows a perspective exploded view of a head-mountable device.

[0464] FIGS. **13.7-6** shows a side view of a retention band having sensors.

13.8: Conductive Fabric Architecture

[0465] FIGS. **13.8-1A** shows a schematic block diagram of a head-mountable device.

[0466] FIGS. **13.8-1B** shows a top view of a head-mountable device.

[0467] FIGS. **13.8-2** shows a bottom perspective view of a light seal.

[0468] FIGS. **13.8-3** shows a top view of a head-mountable device.

[0469] FIGS. **13.8-4A** shows a conductive fabric in a neutral state.

[0470] FIGS. **13.8-4B** shows the conductive fabric of FIGS. **13.8-4A** in a compressed state.

[0471] FIGS. **13.8-4C** shows the conductive fabric of FIGS. **13.8-4A** in a stretched state.

[0472] FIGS. **13.8-5A** shows a conductive component on an exterior of a cover.

[0473] FIGS. **13.8-5B** shows a conductive component interwoven into a cover.

[0474] FIGS. **13.8-5C** shows a conductive component on an interior of a cover.

[0475] FIGS. **13.8-5D** shows free-floating conductive components.

[0476] FIGS. **13.8-6** shows a side perspective view of a light seal.

[0477] FIGS. **13.8-7** shows a bottom perspective view of a light seal.

13.9: Facial Interface Having Integrated Health Sensors

[0478] FIGS. **13.9-1** shows a block diagram of a head-mountable device.

[0479] FIGS. **13.9-2A** shows a top view of a head-mountable device.

[0480] FIGS. **13.9-2B** shows a rear view of a facial interface for a head-mountable device.

[0481] FIGS. **13.9-3** shows a rear perspective view of a facial interface with sensors disposed near a nasal region of a head-mountable device.

[0482] FIGS. **13.9-4A** shows an exploded perspective view of pressure sensor assembly of a head-mountable device.

[0483] FIGS. **13.9-4B** shows an assembled perspective view of a pressure sensor assembly of a head-mountable device.

[0484] FIGS. **13.9-5A** shows sensors disposed on a forehead region of a facial interface of a head-mountable device.

[0485] FIGS. **13.9-5B** shows sensors disposed on a forehead region of a facial interface of a head-mountable device.

[0486] FIGS. **13.9-6** shows a cross-sectional view of a pressure sensors assembly of a head-mountable device.

13.10: Touch Sensitive Input Surface

[0487] FIGS. **13.10-1A** shows a schematic block diagram of a head-mountable device.

[0488] FIGS. **13.10-1B** shows a top view of a head-mountable device.

[0489] FIGS. **13.10-2** shows a bottom perspective view of a light seal.

[0490] FIGS. **13.10-3A** shows a top view of a head-mountable device with conductive fabric in a light seal of the head-mountable device.

[0491] FIGS. **13.10-3B** shows a top view of a head-mountable device with a user engaging a touch sensitive surface of a light seal of the head-mountable device.

[0492] FIGS. **13.10-4** shows a touch sensitive surface of a head-mountable device.

[0493] FIGS. **13.10-5** shows a touch sensitive surface of a light seal of a head-mountable device.

[0494] FIGS. **13.10-6** shows a touch sensitive surface of a light seal of a head-mountable device.

[0495] FIGS. **13.10-7** shows a head-mountable device with a sensor incorporated with a frame of the head-mountable device.

[0496] FIGS. **13.10-8A** shows a head-mountable device with a sensor incorporated with a frame of the head-mountable device.

[0497] FIGS. **13.10-8B** shows the head-mountable device of FIGS. **13.10-8A** with a user mechanically deflecting the frame of the head-mountable device.

13.11: Face Engaging Structures

[0498] FIGS. **13.11-1** shows a top view of an example head-mountable device.

[0499] FIGS. **13.11-2A** shows a side view of an example head-mountable device.

[0500] FIGS. **13.11-2B** shows a front view of an example head-mountable device.

[0501] FIGS. **13.11-3A** shows a perspective view of a head-mountable device including a connector positioned at a forehead location.

[0502] FIGS. **13.11-3B-3E** show various connector types.

[0503] FIGS. **13.11-4A** shows a perspective view of a head-mountable device including a connector positioned at the zygoma location.

[0504] FIGS. **13.11-4A-4H** show various connector types.

[0505] FIGS. **13.11-5A** shows a perspective view of a head-mountable device including a facial interface.

[0506] FIGS. **13.11-5B-5G** show various facial interfaces.

[0507] FIGS. **13.11-6A-6B** show another variation of a facial interface.

[0508] FIGS. **13.11-7A** shows a perspective view of a display including a display frame.

[0509] FIGS. **13.11-7B** shows an exploded perspective view of a display including display frame.

[0510] FIGS. **13.11-8A-8B** show a display frame with a relief cutout.

[0511] FIGS. **13.11-9A** shows a head-mountable device with no relief cutout.

[0512] FIGS. **13.11-9B** shows a head-mountable device with a relief cutout.

[0513] FIGS. **13.11-10A-10B** show a head-mountable device with a relief cutout at various locations.

[0514] FIGS. **13.11-11A-11C** show a display frame with a relief cutout.

[0515] FIGS. **13.11-12** shows a display frame with thru-holes.

[0516] FIGS. **13.11-13** shows a display frame with stiffeners.

[0517] FIGS. **13.11-14A** is a top view of a frame for a device seal including stiffeners.

[0518] FIGS. **13.11-14B** is a cross-sectional view of the frame of FIGS. **13.11-14A**.

[0519] FIGS. **13.11-14C** is a bottom view of the frame of FIGS. **13.11-14A**.

[0520] FIGS. **13.11-14D** is a top view of the frame of FIGS. **13.11-14A**.

[0521] FIGS. **13.11-14** shows a perspective view of an example connector.

[0522] FIGS. **13.11-15A** shows a side view of an example connector positioned between a display frame and a facial interface.

[0523] FIGS. **13.11-15B** shows an example facial interface.

[0524] FIGS. **13.11-15C-13.11-15D** show example cross-sections of the facial interface shown in FIGS. **13.11-15B**.

[0525] FIGS. **13.11-16** shows a cross-sectional view of an example connector with a connector frame and post.

[0526] FIGS. **13.11-17** shows a top view of an example connector.

[0527] FIGS. **13.11-18** shows a side perspective view of a base of an example connector where attached to an example display frame.

[0528] FIGS. **13.11-19** shows another cross-sectional view of an example connector.

[0529] FIGS. **13.11-20-13.11-21** respectively show perspective and top views of example adhesives

in an example head-mountable device.

13.12: Face Engaging Structure

[0530] FIGS. **13.12-1** shows a top view of a head-mountable device including a facial interface.

[0531] FIGS. **13.12-2A** shows a side view of a head-mountable device including a facial interface connected to a display.

[0532] FIGS. **13.12-2B** shows a top view of a head-mountable device including a facial interface connected to a display.

[0533] FIGS. **13.12-3** shows a perspective view of a head-mountable device including a facial interface and an example connector.

[0534] FIGS. **13.12-4A** shows a perspective view of a head-mountable device with an example connector between a display and a facial interface.

[0535] FIGS. **13.12-4B** shows a front view of an example connector.

[0536] FIGS. **13.12-4C** shows a side view of an example connector portion.

[0537] FIGS. **13.12-5A-13.12-5B** show views of a connector in example positional states.

[0538] FIGS. **13.12-6A** shows a perspective view of a head-mountable device including a facial interface and another example connector.

[0539] FIGS. **13.12-6B** shows a top view of an example connector.

[0540] FIGS. **13.12-7A-13.12-7B** show side views of another connector in example positional states.

[0541] FIGS. **13.12-8A-13.12-8B** show schematic views of an example sliding connector.

[0542] FIGS. **13.12-9A** shows a bottom view of another example head-mountable device.

[0543] FIGS. **13.12-9B-13.12-9F** show various positions of a connector of a head-mountable device.

[0544] FIGS. **13.12-10** shows a cutaway view of an example connector.

[0545] FIGS. **13.12-11** shows a perspective view of another example connector.

[0546] FIGS. **13.12-12** shows a side view of yet another example connector.

13.13: Adjustment Mechanism

[0547] FIGS. **13.13-1** shows a top view profile of a head-mountable device including a facial interface.

[0548] FIGS. **13.13-2A** shows a side view profile of a head-mountable device including a facial interface.

[0549] FIGS. **13.13-2B** shows a top view profile of a head-mountable device including a facial interface.

[0550] FIGS. **13.13-3A-13.13-3D** show exemplary locations of adjustment mechanism of a head-mountable device.

[0551] FIGS. **13.13-4A-13.13-4C** show exemplary translatable positions of an adjustment mechanism.

[0552] FIGS. **13.13-5A-13.13-5C** show exemplary rotatable positions of an adjustment mechanism of a head-mountable device.

[0553] FIGS. **13.13-6A-13.13-6B** show an exemplary adjustment mechanism.

[0554] FIGS. **13.13-7A-13.13-7B** show an exemplary rotatable adjustment mechanism.

[0555] FIGS. **13.13-8** shows another exemplary adjustment mechanism.

[0556] FIGS. **13.13-9A-13.13-24B** respectively show example head-mountable devices with an actuator control.

[0557] FIGS. **13.13-25A-13.13-25D** show an example head-mountable device with an example connection and corresponding actuator control.

[0558] FIGS. **13.13-26** shows an example connection of a head-mountable device.

[0559] FIGS. **13.13-27** shows another example connection of a head-mountable device.

[0560] FIGS. **13.13-28-13.13-30** respectively show top, front, and side views of another example head-mountable device.

[0561] FIGS. **13.13-31-13.13-33** respectively show a perspective view of lock-slider disengagement, a front view of lock-slider disengagement, and a front view of lock-slider engagement of a portion of a linear adjustment connection.

[0562] FIGS. **13.13-34** illustrates a perspective view of a portion of a head-mountable device having multiple linear adjustment connections, according to one exemplary embodiment.

13.14: Nosepiece

[0563] FIGS. **13.14-1** is a diagram of an illustrative electronic device in accordance with an embodiment.

[0564] FIGS. **13.14-2** is a front view of an illustrative electronic device with a light-shielding structure in accordance with an embodiment.

[0565] FIGS. **13.14-3** is a diagram of an illustrative light-shielding structure having a fabric cover in accordance with an embodiment.

[0566] FIGS. **13.14-4** is a front view of an illustrative light-shielding structure having a structural frame in accordance with an embodiment.

[0567] FIGS. **13.14-5** is a side view of an illustrative light-shielding structure having a fabric and elastomer layers in accordance with an embodiment.

[0568] FIGS. **13.14-6** is a front view of an illustrative light-shielding structure having an extension in accordance with an embodiment.

[0569] FIGS. **13.14-7** is a side view of an illustrative light-shielding structure having an embedded service loop in accordance with an embodiment.

[0570] FIGS. **13.14-8** is a side view of an illustrative light-shielding structure having an embedded deformable stiffener in accordance with an embodiment.

[0571] FIGS. **13.14-9A** is a side view of an illustrative light-shielding structure having a rolled edge in accordance with an embodiment.

[0572] FIGS. **13.14-9B** is a side view of an illustrative light-shielding structure having embedded foam in accordance with an embodiment.

[0573] FIGS. **13.14-9C** is a top view of an illustrative light-shielding structure having a crumple zone in accordance with an embodiment.

[0574] FIGS. **13.14-9D** is a side view of an illustrative light-shielding structure having a hemmed edge in accordance with an embodiment.

[0575] FIGS. **13.14-9E** is a top view of an illustrative light-shielding structure having a foam in corner regions in accordance with an embodiment.

[0576] FIGS. **13.14-9F** is a side view of an illustrative light-shielding structure having segmented foam or elastomeric regions in accordance with an embodiment.

[0577] FIGS. **13.14-9G** is a side view of an illustrative light-shielding structure having a stiffener and a foam layer in accordance with an embodiment.

[0578] FIGS. **13.14-10** is a front view of an illustrative light-shielding structure having a semi-rigid stiffener in accordance with an embodiment.

13.15: Removable Facial Interface

[0579] FIGS. **13.15-1A** is a schematic block diagram of an example of a head-mountable device.

[0580] FIGS. **13.15-1B** is a top view of an example of a head-mountable device.

[0581] FIGS. **13.15-2A** is a perspective view of an example of a device seal.

[0582] FIGS. **13.15-2B** is a perspective view of an example of a facial interface frame.

[0583] FIGS. **13.15-2C** is a perspective view of an example of a facial interface frame and a removable facial interface.

[0584] FIGS. **13.15-2D** is a cross-sectional view of an example of a facial interface shim.

[0585] FIGS. **13.15-3A** is a perspective view of an example of a device seal.

[0586] FIGS. **13.15-3B** is a plan view of an example of a removable facial interface.

[0587] FIGS. **13.15-4** is a cross-sectional view of an example of a magnetic attachment mechanism.

[0588] FIGS. **13.15-5A** is a cross-sectional view of an example of an interlocking attachment mechanism.

[0589] FIGS. **13.15-5B** is a cross-sectional view of an example of an interlocking attachment mechanism.

[0590] FIGS. **13.15-6** is a cross-sectional view of an example of a magnetic slide attachment mechanism.

[0591] FIGS. **13.15-7** is a cross-sectional view of an example of a hook-and-loop attachment mechanism.

[0592] FIGS. **13.15-8** is a cross-sectional view of an example of a magnetic attachment mechanism.

[0593] FIGS. **13.15-9** is a cross-sectional view of an example of a spring snap attachment mechanism.

[0594] FIGS. **13.15-10** is a cross-sectional view of an example of an interlocking attachment mechanism.

[0595] FIGS. **13.15-11** is a cross-sectional view of an example of a suction attachment mechanism.

[0596] FIGS. **13.15-12** is a cross-sectional view of an example of a bi-stable attachment mechanism.

[0597] FIGS. **13.15-13A** is a plan view of an example of a removable facial interface.

[0598] FIGS. **13.15-13B** is a plan view of an example of a removable facial interface.

[0599] FIGS. **13.15-14** is a cross-sectional view of an example facial interface.

[0600] FIGS. **13.15-15A** is a cross-sectional view of a compressible portion.

[0601] FIGS. **13.15-15B** is a cross-sectional view of a compressible portion.

[0602] FIGS. **13.15-15C** is a cross-sectional view of a compressible portion.

13.16: Electronic Devices with Light Blocking Structures

[0603] FIGS. **13.16-1** is a diagram of an illustrative electronic device in accordance with an embodiment.

[0604] FIGS. **13.16-2** is a front view of an illustrative electronic device with a light-shielding structure in accordance with an embodiment.

[0605] FIGS. **13.16-3** is a diagram of an illustrative light-shielding structure having a fabric cover in accordance with an embodiment.

[0606] FIGS. **13.16-4A** and **13.16-4B** are front views of illustrative elastomeric layers that may be used in a nosepiece in accordance with some embodiments.

[0607] FIGS. **13.16-5** is a front view of an illustrative light-shielding structure having a structural frame in accordance with an embodiment.

[0608] FIGS. **13.16-6** is a side view of an illustrative light-shielding structure having a fabric and elastomer layers in accordance with an embodiment.

[0609] FIGS. **13.16-7** is a front view of an illustrative light-shielding structure having an extension in accordance with an embodiment.

[0610] FIGS. **13.16-8** is a side view of an illustrative light-shielding structure having an embedded service loop in accordance with an embodiment.

[0611] FIGS. **13.16-9** is a side view of an illustrative light-shielding structure having an embedded deformable stiffener in accordance with an embodiment.

[0612] FIGS. **13.16-10A** is a side view of an illustrative light-shielding structure having a rolled edge in accordance with an embodiment.

[0613] FIGS. **13.16-10B** is a side view of an illustrative light-shielding structure having embedded foam in accordance with an embodiment.

[0614] FIGS. **13.16-10C** is a top view of an illustrative light-shielding structure having a crumple zone in accordance with an embodiment.

[0615] FIGS. **13.16-10D** is a side view of an illustrative light-shielding structure having a hemmed edge in accordance with an embodiment.

[0616] FIGS. **13.16-10E** is a top view of an illustrative light-shielding structure having a foam in corner regions in accordance with an embodiment.

[0617] FIGS. **13.16-10F** is a side view of an illustrative light-shielding structure having segmented foam or elastomeric regions in accordance with an embodiment.

[0618] FIGS. **13.16-10G** is a side view of an illustrative light-shielding structure having a stiffener and a foam layer in accordance with an embodiment.

[0619] FIGS. **13.16-11** is a front view of an illustrative light-shielding structure having a semi-rigid stiffener in accordance with an embodiment.

[0620] FIGS. **13.16-12** is a perspective view of an illustrative light-shield structure formed from multiple fabric layers in accordance with an embodiment.

XIV: Powerstraps and Securement Band

[0621] FIGS. **14.0-1** illustrates a view of an HMD.

14.1: Electrical Connectors

[0622] FIGS. **14.1-1A** shows a perspective side view of an electronic device.

[0623] FIGS. **14.1-1B** shows a perspective side view of the electronic device of FIGS. **14.1-1A**.

[0624] FIGS. **14.1-2** shows a perspective view of a display, a support, and a plug connector.

[0625] FIGS. **14.1-3A** shows a perspective view of a receptacle connector.

[0626] FIGS. **14.1-3B** shows a perspective view of a plug connector.

[0627] FIGS. **14.1-4** shows an exploded view of a receptacle connector.

[0628] FIGS. **14.1-5A** shows a front view of a receptacle connector.

[0629] FIGS. **14.1-5B** shows a partial cut-away front view of the receptacle connector of FIGS. **14.1-5A**.

[0630] FIGS. **14.1-6A** shows a side sectional view of a receptacle connector.

[0631] FIGS. **14.1-6B** shows a side sectional view of a receptacle connector.

[0632] FIGS. **14.1-7A** shows a detail perspective view of a receptacle connector.

[0633] FIGS. **14.1-7B** shows a detail perspective view of a plug connector.

[0634] FIGS. **14.1-8A** shows a sectional view of a plug connector inserted into a receptacle connector.

[0635] FIGS. **14.1-8B** shows a detail sectional view of the plug connector inserted into the receptacle connector of FIGS. **14.1-8A**.

[0636] FIGS. **14.1-9A** shows a detail sectional view of a plug connector inserted into a receptacle connector.

[0637] FIGS. **14.1-9B** shows a detail sectional view of the plug connector inserted into the receptacle connector of FIGS. **14.1-9A**.

[0638] FIGS. **14.1-9C** shows a detail sectional view of a plug connector inserted into a receptacle connector.

[0639] FIGS. **14.1-9D** shows a detail sectional view of the plug connector inserted into the receptacle connector of FIGS. **14.1-9C**.

[0640] FIGS. **14.1-9E** shows a detail sectional view of a plug connector inserted into a receptacle connector.

[0641] FIGS. **14.1-9F** shows a detail sectional view of the plug connector inserted into the receptacle connector of FIGS. **14.1-9E**.

[0642] FIGS. **14.1-10** shows a sectional view of a tool ejecting a plug connector from a receptacle connector.

[0643] FIGS. **14.1-11A** through **14.1-11D** show perspective views of tools for ejecting plug connectors from receptacle connectors.

[0644] FIGS. **14.1-12A** shows a front view of a receptacle connector.

[0645] FIGS. **14.1-12B** shows a front view of a plug connector.

[0646] FIGS. **14.1-13A** shows a sectional view of a receptacle connector.

[0647] FIGS. **14.1-13B** through **14.1-13D** show detail sectional views of seals of the receptacle

connector of FIGS. **14.1-13A**.

[0648] FIGS. **14.1-14A** shows a perspective view of a receptacle connector.

[0649] FIGS. **14.1-14B** shows a detail sectional view of a fastener of the receptacle connector of FIGS. **14.1-14A**.

[0650] FIGS. **14.1-15** shows a perspective view of a receptacle connector, a plug connector, and a housing.

[0651] FIGS. **14.1-16A** shows a side sectional view of a receptacle connector and a plug connector.

[0652] FIGS. **14.1-16B** shows a side view of the plug connector of FIGS. **14.1-16A**.

[0653] FIGS. **14.1-17A** shows a side view of a plug connector.

[0654] FIGS. **14.1-17B** shows a bottom-up view of the plug connector of FIGS. **14.1-17A**.

[0655] FIGS. **14.1-18A** shows a perspective view of a receptacle connector.

[0656] FIGS. **14.1-18B** shows a top-down view of the receptacle connector of FIGS. **14.1-18A**.

[0657] FIGS. **14.1-18C** shows a side sectional view of the receptacle connector of FIG. **14.1-18A**.

[0658] FIGS. **14.1-18D** shows a side sectional view of the receptacle connector of FIG. **14.1-18A** and a plug connector.

[0659] FIGS. **14.1-18E** shows a side sectional view of the receptacle connector of FIG. **14.1-18A** and the plug connector of FIGS. **14.1-18D**.

[0660] FIGS. **14.1-19A** shows a detail top-down view of a receptacle connector.

[0661] FIGS. **14.1-19B** shows a perspective view of a detent of the receptacle connector of FIGS. **14.1-19A**.

[0662] FIGS. **14.1-19C** shows a sectional view of the detent of FIGS. **14.1-19B** and a plug connector.

[0663] FIGS. **14.1-19D** shows a detail top-down view of a receptacle connector.

[0664] FIGS. **14.1-19E** shows a perspective view of a detent of the receptacle connector of FIGS. **14.1-19D**.

[0665] FIGS. **14.1-19F** shows a sectional view of the detent of FIGS. **14.1-19E** and a plug connector.

[0666] FIGS. **14.1-19G** shows a detail top-down view of a receptacle connector.

[0667] FIGS. **14.1-19H** shows a perspective view of a detent of the receptacle connector of FIGS. **14.1-19G**.

[0668] FIGS. **14.1-19I** shows a detail top-down view of a receptacle connector.

[0669] FIGS. **14.1-19J** shows a perspective view of a detent of the receptacle connector of FIGS. **14.1-19I**.

[0670] FIGS. **14.1-19K** shows a sectional view of the detent of FIGS. **14.1-19J** and a plug connector.

[0671] FIGS. **14.1-19L** shows a detail top-down view of a receptacle connector.

[0672] FIGS. **14.1-19M** shows a perspective view of a detent of the receptacle connector of FIGS. **14.1-19L**.

[0673] FIGS. **14.1-20A** shows a bottom view of a receptacle connector and a plug connector.

[0674] FIGS. **14.1-20B** shows a bottom view of a receptacle connector and a plug connector.

[0675] FIGS. **14.1-21A** shows a bottom view of a receptacle connector and a plug connector.

[0676] FIGS. **14.1-21B** shows a bottom view of a receptacle connector and a plug connector.

[0677] FIGS. **14.1-22A** through **14.1-22E** show sectional side views of a receptacle connector, a plug connector, and seals between the receptacle connector and the plug connector.

[0678] FIGS. **14.1-23A** through **14.1-23G** show sectional side views of a receptacle connector, a plug connector, and seals between the receptacle connector and the plug connector.

[0679] FIGS. **14.1-24A** and **14.1-24B** show exploded views of receptacle connectors.

[0680] FIGS. **14.1-25A** and **14.1-25B** show exploded views of receptacle connectors.

[0681] FIGS. **14.1-26A** shows a perspective view of an electronic device.

[0682] FIGS. **14.1-26B** shows a perspective view of the electronic device of FIGS. **14.1-26A** and a

plug connector.

[0683] FIGS. **14.1-27A** shows a perspective view of a plug connector inserted into an electronic device.

[0684] FIGS. **14.1-27B** shows a partially exploded view of the plug connector of FIG. **14.1-27A**, a trim ring, and a receptacle connector.

[0685] FIGS. **14.1-28A** shows a sectional view of a plug connector being inserted into a trim ring and a receptacle connector.

[0686] FIGS. **14.1-28B** shows a detail sectional view of the plug connector and a latch of the trim ring of FIGS. **14.1-28A**.

[0687] FIGS. **14.1-28C** shows a sectional view of the plug connector being inserted into the trim ring and the receptacle connector of FIGS. **14.1-28A**.

[0688] FIGS. **14.1-28D** shows a sectional view of the plug connector inserted into the trim ring and the receptacle connector of FIGS. **14.1-28A**.

[0689] FIGS. **14.1-28E** shows a sectional view of the plug connector being un-latched from the trim ring and the receptacle connector of FIGS. **14.1-28A**.

[0690] FIGS. **14.1-28F** shows a perspective view of a lever arm of the trim ring of FIG. **14.1-28A**.

[0691] FIGS. **14.1-29A** shows a sectional view of a plug connector in a trim ring and a receptacle connector.

[0692] FIGS. **14.1-29B** shows a sectional view of a tool used to un-latch the plug connector from the trim ring and the receptacle connector of FIGS. **14.1-29A**.

[0693] FIGS. **14.1-29C** shows a perspective view of the plug connector and the trim ring of FIGS. **14.1-29A**.

[0694] FIGS. **14.1-29D** shows a sectional view of the trim ring of FIGS. **14.1-29A**.

[0695] FIGS. **14.1-29E** shows a perspective view of a lever arm of the trim ring of FIG. **14.1-29A**.

[0696] FIGS. **14.1-30A** shows a perspective view of a plug connector and a trim ring.

[0697] FIGS. **14.1-30B** shows a sectional view of the plug connector inserted in the trim ring of FIGS. **14.1-30A** and a receptacle connector.

[0698] FIGS. **14.1-30C** shows a sectional view of the plug connector inserted in the trim ring and the receptacle connector of FIGS. **14.1-30B**.

[0699] FIGS. **14.1-31A** shows a sectional view of a plug connector being inserted into a trim ring and a receptacle connector.

[0700] FIGS. **14.1-31B** shows a sectional view of the plug connector being un-latched from into the trim ring and the receptacle connector of FIGS. **14.1-31A**.

[0701] FIGS. **14.1-32A** through **14.1-32C** show perspective views of a trim ring and a receptacle connector being assembled in a housing.

[0702] FIGS. **14.1-33A** shows a top-down view of a plug connector being inserted into a trim ring and a receptacle connector.

[0703] FIGS. **14.1-33B** shows a detail view of the trim ring and the plug connector of FIGS. **14.1-33A** prior to the trim ring latching with the receptacle connector.

[0704] FIGS. **14.1-33C** shows a top-down view of the plug connector latched in the trim ring and the receptacle connector of FIGS. **14.1-33A**.

[0705] FIGS. **14.1-33D** shows a detail view of the trim ring and the plug connector of FIGS. **14.1-33C** with the plug connector latched in the trim ring.

[0706] FIGS. **14.1-33E** shows a top-down view of the plug connector being un-latched from the trim ring and the receptacle connector of FIGS. **14.1-33A**.

[0707] FIGS. **14.1-33F** shows a detail view of the trim ring and the plug connector of FIGS. **14.1-33E** with the plug connector un-latched from the trim ring.

[0708] FIGS. **14.1-34A** and **14.1-34B** show perspective views of a plug connector inserted into a trim ring and a receptacle connector.

[0709] FIGS. **14.1-35A** shows an exploded view of a receptacle connector.

[0710] FIGS. **14.1-35B** shows a side sectional view of the receptacle connector of FIG. **14.1-35A**.
[0711] FIGS. **14.1-36A** and **14.1-36B** respectively illustrate a semi-transparent view and a solid view of an electrical connector portion.
[0712] FIGS. **14.1-37** through **14.1-38** illustrate respective top and bottom views of an electrical connector portion.
[0713] FIGS. **14.1-39** illustrates example method steps for manufacturing an electrical connector portion.
[0714] FIGS. **14.1-40** depicts example method steps of providing an interface connector to an electrical connector portion.
[0715] FIGS. **14.1-41A** and **14.1-41B** show side schematic views of the assembling an interface connector to an electrical connector portion.

14.2: Modular Components for Wearable Electronic Devices

[0716] FIGS. **14.2-1A** shows a wearable electronic device being worn by a user.
[0717] FIGS. **14.2-1B** shows a top view of the wearable electronic device of FIGS. **14.2-1A**.
[0718] FIGS. **14.2-1C** shows an exploded view of the wearable electronic device of FIG. **14.2-1A**.
[0719] FIGS. **14.2-2A** shows an exploded view of a wearable electronic device.
[0720] FIGS. **14.2-2B** shows a side view of a component of the wearable electronic device of FIGS. **14.2-2A**.
[0721] FIGS. **14.2-2C** shows a side view of a component of the wearable electronic device of FIGS. **14.2-2A**.
[0722] FIGS. **14.2-2D** shows a cross-sectional view of the component of FIGS. **14.2-2C**.
[0723] FIGS. **14.2-3** shows a side view of a component of a wearable electronic device.
[0724] FIGS. **14.2-4** shows a side view of a component of a wearable electronic device.
[0725] FIGS. **14.2-5A** shows a top view of a component of a wearable electronic device.
[0726] FIGS. **14.2-5B** shows a side view of the component of FIGS. **14.2-5A**.
[0727] FIGS. **14.2-5C** shows a cross-sectional view of the component of FIGS. **14.2-5A**.
[0728] FIGS. **14.2-6A** shows a top view of a component of a wearable electronic device.
[0729] FIGS. **14.2-6B** shows a side view of the component of FIGS. **14.2-6A**.
[0730] FIGS. **14.2-6C** shows a cross-sectional view of the component of FIGS. **14.2-6A**.
[0731] FIGS. **14.2-7A** shows a top view of a component of a wearable electronic device.
[0732] FIGS. **14.2-7B** shows a side view of the component of FIGS. **14.2-7A**.
[0733] FIGS. **14.2-7C** shows a cross-sectional view of the component of FIGS. **14.2-7A**.
[0734] FIGS. **14.2-8A** shows a top view of a component of a wearable electronic device.
[0735] FIGS. **14.2-8B** shows a side view of the component of **14.2-FIG. 8A**.
[0736] FIGS. **14.2-8C** shows a cross-sectional view of the component of FIGS. **14.2-8A**.
[0737] FIGS. **14.2-9A** shows a top view of a component of a wearable electronic device.
[0738] FIGS. **14.2-9B** shows a side view of the component of FIGS. **14.2-9A**.
[0739] FIGS. **14.2-9C** shows a cross-sectional view of the component of FIGS. **14.2-9A**.
[0740] FIGS. **14.2-10A** shows a top view of a component of a wearable electronic device.
[0741] FIGS. **14.2-10B** shows a side view of the component of FIGS. **14.2-10A**.
[0742] FIGS. **14.2-10C** shows a cross-sectional view of the component of FIGS. **14.2-10A**.
[0743] FIGS. **14.2-11** shows an exploded view of a wearable electronic device.
[0744] FIGS. **14.2-12** shows an exploded view of a wearable electronic device.
[0745] FIGS. **14.2-13** exploded view of a wearable electronic device.

14.3: Modular Strap for an Electronic Device

[0746] FIGS. **14.3-1** shows a top view of an example of an electronic device donned by a user.
[0747] FIGS. **14.3-2** shows a perspective view of an example of an electronic device.
[0748] FIGS. **14.3-3** shows an exploded perspective view of an example of an electronic device.
[0749] FIGS. **14.3-4** shows a side profile view of an example removable strap of an HMD system.
[0750] FIGS. **14.3-5** shows a top cross-sectional profile view of an example electronics pod.

[0751] FIGS. **14.3-6** shows a top view of another example of an electronic device donned by a user.
[0752] FIGS. **14.3-7** and **14.3-8** show example cable management mechanisms of an example HMD system.

14.4: Devices with Detachable Headbands

[0753] FIGS. **14.4-1** is a side view of an electronic device with a detachable headband.
[0754] FIGS. **14.4-2** is a view of a detachable headband.
[0755] FIGS. **14.4-3** is a cross-sectional side view of a portion of a detachable headband.
[0756] FIGS. **14.4-4** is a top view of a spring.
[0757] FIGS. **14.4-5** is a diagram of a detachable headband having a latch with a release tab.
[0758] FIGS. **14.4-6** is a cross-sectional side view of a detachable headband with a release tab.
[0759] FIGS. **14.4-7** is a top view of a magnet arrangement.
[0760] FIGS. **14.4-8**, **14.4-9**, and **14.4-10** are diagrams showing latch biasing mechanisms.
[0761] FIGS. **14.4-11** is a cross-sectional side view of a latch biasing mechanism.
[0762] FIGS. **14.4-12** and **14.4-13** are cross-sectional side views of detachable headbands.
[0763] FIGS. **14.4-14** is a perspective view of a detachable headband with recesses.
[0764] FIGS. **14.4-15** is a top view of a headband attachment post.
[0765] FIGS. **14.4-16** is a cross-sectional side view of a headband attachment post.
[0766] FIGS. **14.4-17** is a cross-sectional side view of a headband attachment post with a recess and a corresponding detachable headband.

14.5: Cable Tensioning System and Dial

[0767] FIGS. **14.5-1** is a side view of an example of a head-mountable display device with an adjustable headband;
[0768] FIGS. **14.5-2** is a plan view of an example of an adjustable headband;
[0769] FIGS. **14.5-3** is a perspective view of an example of a tensioning system for an adjustable headband;
[0770] FIGS. **14.5-4** is a partially exploded view of an example of a tensioning system for an adjustable headband;
[0771] FIGS. **14.5-5** is a partial cross-sectional view of an example of a tensioning system for an adjustable headband;
[0772] FIGS. **14.5-6** is a partially exploded view of an example of a dial cap for a tensioning system;
[0773] FIGS. **14.5-7A** and **14.5-7B** are partial cross-sectional views of an example of a disc-type angular restraint system;
[0774] FIGS. **14.5-8** is partial cross-sectional view of an example of a dial cap including a spring detent mechanism; and
[0775] FIGS. **14.5-9A** through **14.5-9C** are perspective views of an example of angular restraint systems.

14.6: Two-Part Speaker System

[0776] FIGS. **14.6-1A** shows a side view of an electronic device.
[0777] FIGS. **14.6-1B** shows a perspective view of an electronic device.
[0778] FIGS. **14.6-1C** shows a perspective view of an electronic device.
[0779] FIGS. **14.6-1D** shows a perspective view of an electronic device.
[0780] FIGS. **14.6-2** shows a cross-sectional side view of a speaker assembly.
[0781] FIGS. **14.6-3A** shows a perspective view of a speaker assembly.
[0782] FIGS. **14.6-3B** shows a cross-sectional side view of a speaker assembly.
[0783] FIGS. **14.6-3C** shows a cross-sectional perspective view of a speaker assembly.
[0784] FIGS. **14.6-3D** shows a top perspective view of a speaker assembly.
[0785] FIGS. **14.6-3E** shows a bottom perspective view of a speaker assembly.
[0786] FIGS. **14.6-4** shows a perspective exploded view of a port barrier.

14.7: Bifurcated Band

[0787] FIGS. **14.7-1** is a side view of an illustrative electronic device such as a head-mounted display device with an adjustable headband in accordance with some embodiments.

[0788] FIGS. **14.7-2A** and **14.7-2B** are side views of opposing sides of an illustrative headband in accordance with some embodiments.

[0789] FIGS. **14.7-3** is an illustrative front view of an edge of a headband in accordance with some embodiments.

[0790] FIGS. **14.7-4** is a side view of an illustrative headband with a seam that is invisible to a naked eye in accordance with some embodiments.

[0791] FIGS. **14.7-5** is a side view of an illustrative headband having stiffeners on a surface of the headband in accordance with some embodiments.

[0792] FIGS. **14.7-6A-14.7-6C** are side views of illustrative stiffeners that may be incorporated onto a surface of a headband in accordance with some embodiments.

[0793] FIGS. **14.7-7** is a side view of an illustrative headband having embedded stiffeners in accordance with some embodiments.

[0794] FIGS. **14.7-8** is a perspective view of an illustrative stiffener in a channel of a headband in accordance with some embodiments.

[0795] FIGS. **14.7-9A** and **14.7-9B** are side views of illustrative headbands with local stiffeners that change the curvature of the headband when under tension in accordance with some embodiments.

14.8: Over the Head Strap

[0796] FIGS. **14.8-1** is a side view of an illustrative electronic device such as a head-mounted display device with a detachable headband in accordance with some embodiments.

[0797] FIGS. **14.8-2** is a perspective view of an illustrative headband with a post that couples to a post on a head-mounted structure in accordance with some embodiments.

[0798] FIGS. **14.8-3** is a cross-sectional side view of an illustrative headband with a post that couples to a post on a head-mounted structure in accordance with some embodiments.

[0799] FIGS. **14.8-4** is a cross-sectional side view of an illustrative detachable headband with a release tab in accordance with some embodiments.

[0800] FIGS. **14.8-5** is a perspective view of an illustrative headband with a magnet that couples to a post on a head-mounted structure in accordance with some embodiments.

[0801] FIGS. **14.8-6** is a cross-sectional side view of an illustrative headband with a magnet that couples to a post on a head-mounted structure in accordance with some embodiments.

[0802] FIGS. **14.8-7** is a cross-sectional side view of an illustrative headband with a magnet and a protrusion that couples to a post with a recess on a head-mounted structure in accordance with some embodiments.

[0803] FIGS. **14.8-8** is a perspective view of an illustrative headband that has portions that wrap around a head-mounted support structure to attach to the support structure in accordance with some embodiments.

[0804] FIGS. **14.8-9** is a perspective view of an illustrative headband that attaches to a head-mounted support structure with a lug and socket system in accordance with some embodiments.

[0805] FIGS. **14.8-10** is a cross-sectional side view of two illustrative headbands that attach to a head-mounted support structure with latches in accordance with some embodiments.

[0806] FIGS. **14.8-11** is a cross-sectional side view of two illustrative headbands, one of which attaches to a head-mounted support structure with a latch, and one of which attaches to the head-mounted support structure with a protrusion, in accordance with some embodiments.

[0807] FIGS. **14.8-12** is a diagram of an illustrative headband that attaches to a head-mounted support structure with a twist-to-lock system in accordance with some embodiments.

[0808] FIGS. **14.8-13** is a perspective view of an illustrative headband that has an opening to surround a post of a head-mounted support structure in accordance with some embodiments.

[0809] FIGS. **14.8-14** is a cross-sectional side view of an illustrative headband that has an opening

to surround a post of a head-mounted support structure in accordance with some embodiments.

[0810] FIGS. **14.8-15A** and **14.8-15B** are perspective views of an illustrative post with an extendable magnet in accordance with some embodiments.

[0811] FIGS. **14.8-16** is a perspective view of an illustrative headband with an opening to receive a magnet and couple to a head-mounted support structure in accordance with some embodiments.

[0812] FIGS. **14.8-17A** and **14.8-17B** are cross-sectional side views of an illustrative headband engaging with an extendable magnet of a post in accordance with some embodiments.

XV: User Interface

[0813] FIG. **15-1** shows an example user interface displayed by a display module of an HMD.

[0814] FIG. **15-2A** shows an example user interface displayed by a display module of an HMD.

[0815] FIG. **15-2B** shows an example user interface displayed by a display module of an HMD.

[0816] FIG. **15-3A** shows an example of a user interface of a display module of an electronic device.

[0817] FIG. **15-3B** shows an example of a user interface of a display module of an electronic device.

[0818] FIG. **15-4A** shows an example of users interacting with a user interface of two display modules of an electronic device.

[0819] FIG. **15-4B** shows an example of users interacting with a user interface of two display modules of an electronic device.

[0820] FIG. **15-5A** shows an example user interface displayed by a display module of an HMD.

[0821] FIG. **15-5B** shows an example user interface displayed by a display module of an HMD.

[0822] FIG. **15-5C** shows an example user interface displayed by a display module of an HMD.

[0823] FIG. **15-6A** shows an example user interface displayed by a display module of an HMD.

[0824] FIG. **15-6B** shows an example user interface displayed by a display module of an HMD.

DETAILED DESCRIPTION

I: Overall System

[0825] FIG. **1-1A** illustrates a front, top, perspective view of an example of a head-mountable display (HMD) device **1-100** configured to be donned by a user and provide virtual and altered/mixed reality (VR/AR) experiences. The HMD **1-100** can include a display unit **1-102** or assembly, an electronic strap assembly **1-104** connected to and extending from the display unit **1-102**, and a band assembly **1-106** secured at either end to the electronic strap assembly **1-104**. The electronic strap assembly **1-104** and the band **1-106** can be part of a retention assembly configured to wrap around a user's head to hold the display unit **1-102** against the face of the user.

[0826] In at least one example, the band assembly **1-106** can include a first band **1-116** configured to wrap around the rear side of a user's head and a second band **1-117** configured to extend over the top of a user's head. The second strap can extend between first and second electronic straps **1-105a**, **1-105b** of the electronic strap assembly **1-104** as shown. The strap assembly **1-104** and the band assembly **1-106** can be part of a securement mechanism extending rearward from the display unit **1-102** and configured to hold the display unit **1-102** against a face of a user.

[0827] In at least one example, the securement mechanism includes a first electronic strap **1-105a** including a first proximal end **1-134** coupled to the display unit **1-102**, for example a housing **1-150** of the display unit **1-102**, and a first distal end **1-136** opposite the first proximal end **1-134**. The securement mechanism can also include a second electronic strap **1-105b** including a second proximal end **1-138** coupled to the housing **1-150** of the display unit **1-102** and a second distal end **1-140** opposite the second proximal end **1-138**. The securement mechanism can also include the first band **1-116** including a first end **1-142** coupled to the first distal end **1-136** and a second end **1-144** coupled to the second distal end **1-140** and the second band **1-117** extending between the first electronic strap **1-105a** and the second electronic strap **1-105b**. The straps **1-105a-b** and band **1-116** can be coupled via connection mechanisms or assemblies **1-114**. In at least one example, the second band **1-117** includes a first end **1-146** coupled to the first electronic strap **1-105a** between

the first proximal end **1-134** and the first distal end **1-136** and a second end **1-148** coupled to the second electronic strap **1-105b** between the second proximal end **1-138** and the second distal end **1-140**.

[0828] In at least one example, the first and second electronic straps **1-105a-b** include plastic, metal, or other structural materials forming the shape the substantially rigid straps **1-105a-b**. In at least one example, the first and second bands **1-116**, **1-117** are formed of elastic, flexible materials including woven textiles, rubbers, and the like. The first and second bands **1-116**, **1-117** can be flexible to conform to the shape of the user's head when donning the HMD **1-100**.

[0829] In at least one example, one or more of the first and second electronic straps **1-105a-b** can define internal strap volumes and include one or more electronic components disposed in the internal strap volumes. In one example, as shown in FIG. **1-1A**, the first electronic strap **1-105a** can include an electronic component **1-112**. In one example, the electronic component **1-112** can include a speaker. In one example, the electronic component **1-112** can include a computing component such as a processor.

[0830] In at least one example, the housing **1-150** defines a first, front-facing opening **1-152**. The front-facing opening is labeled in dotted lines at **1-152** in FIG. **1-1A** because the front display assembly **1-108** is disposed to occlude the first opening **1-152** from view when the HMD **1-100** is assembled. The housing **1-150** can also define a rear-facing second opening **1-154**. The housing **1-150** also defines an internal volume between the first and second openings **1-152**, **1-154**. In at least one example, the HMD **1-100** includes the display assembly **1-108**, which can include a front cover and display screen (shown in other figures) disposed in or across the front opening **1-152** to occlude the front opening **1-152**. In at least one example, the display screen of the display assembly **1-108**, as well as the display assembly **1-108** in general, has a curvature configured to follow the curvature of a user's face. The display screen of the display assembly **1-108** can be curved as shown to compliment the user's facial features and general curvature from one side of the face to the other, for example from left to right and/or from top to bottom where the display unit **1-102** is pressed.

[0831] In at least one example, the housing **1-150** can define a first aperture **1-126** between the first and second openings **1-152**, **1-154** and a second aperture **1-130** between the first and second openings **1-152**, **1-154**. The HMD **1-100** can also include a first button **1-128** disposed in the first aperture **1-126** and a second button **1-132** disposed in the second aperture **1-130**. The first and second buttons **1-128**, **1-132** can be depressible through the respective apertures **1-126**, **1-132**. In at least one example, the first button **1-126** and/or second button **1-130** can be twistable dials as well as depressible buttons. In at least one example, the first button **1-126** is a depressible and twistable dial button and the second button **1-132** is a depressible button.

[0832] FIG. **1-1B** illustrates a rear, perspective view of the HMD **1-100**. The HMD **1-100** can include a light seal **1-110** extending rearward from the housing **1-150** of the display assembly **1-108** around a perimeter of the housing **1-150** as shown. The light seal **1-110** can be configured to extend from the housing **1-150** to the user's face around the user's eyes to block external light from being visible. In one example, the HMD **1-100** can include first and second display assemblies **1-120a**, **1-120b** disposed at or in the rearward facing second opening **1-154** defined by the housing **1-150** and/or disposed in the internal volume of the housing **1-150** and configured to project light through the second opening **1-154**. In at least one example, each display assembly **1-120a-b** can include respective display screens **1-122a**, **1-122b** configured to project light in a rearward direction through the second opening **1-154** toward the user's eyes.

[0833] In at least one example, referring to both FIGS. **1-1A** and **1-1B**, the display assembly **1-108** can be a front-facing, forward display assembly including a display screen configured to project light in a first, forward direction and the rear facing display screens **1-122a-b** can be configured to project light in a second, rearward direction opposite the first direction. As noted above, the light seal **1-110** can be configured to block light external to the HMD **1-100** from reaching the user's eyes, including light projected by the forward facing display screen of the display assembly **1-108**.

shown in the front perspective view of FIG. 1-1A. In at least one example, the HMD 1-100 can also include a curtain 1-124 occluding the second opening 1-154 between the housing 1-150 and the rear-facing display assemblies 1-120a-b. In at least one example, the curtain 1-124 can be elastic or at least partially elastic.

[0834] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 1-1A and 1-1B can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. 1-2-1-4 and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. 2-4 can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 1-1A and 1-1B.

[0835] FIG. 1-2 illustrates a view of an example of an HMD 1-200 including various portions or parts thereof separated according to the modularity and selective coupling of those parts. For example, the HMD 1-200 can include a band 1-216 which can be selectively coupled to first and second electronic straps 1-205a, 1-205b. The first securement strap 1-205a can include a first electronic component 1-212a and the second securement strap 1-205b can include a second electronic component 1-212b. In at least one example, the first and second straps 1-205a-b can be removably coupled to the display unit 1-202.

[0836] In addition, the HMD 1-200 can include a light seal 1-210 configured to be removably coupled to the display unit 1-202. The HMD 1-200 can also include lenses 1-218 which can be removably coupled to the display unit 1-202, for example over first and second display assemblies including display screens. The lenses 1-218 can include customized prescription lenses configured for corrective vision. As noted, each part shown in the view of FIG. 1-2 and described above can be removably coupled, attached, re-attached, and changed out to update parts or swap out parts for different users. For example, bands such as the band 1-216, light seals such as the light seal 1-210, lenses such as the lenses 1-218, and electronic straps such as the straps 1-205a-b can be swapped out depending on the user such that these parts are customized to fit and correspond to the individual user of the HMD 1-200.

[0837] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 1-2 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. 1-1A, 1-1B, and 1-3-1-4 and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. 1-1A, 1-1B, and 1-3-1-4 can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 1-2.

[0838] FIG. 1-3 illustrates a view of an example of a display unit 1-306 of an HMD. The display unit 1-306 can include a front display assembly 1-308, a frame/housing assembly 1-350, and a curtain assembly 1-324. The display unit 1-306 can also include a sensor assembly 1-356, logic board assembly 1-358, and cooling assembly 1-360 disposed between the frame assembly 1-350 and the front display assembly 1-308. In at least one example, the display unit 1-306 can also include a rear-facing display assembly 1-320 including first and second rear-facing display screens 1-322a, 1-322b disposed between the frame 1-350 and the curtain assembly 1-324.

[0839] In at least one example, the display unit 1-306 can also include a motor assembly 1-362 configured as an adjustment mechanism for adjusting the positions of the display screens 1-322a-b of the display assembly 1-320 relative to the frame 1-350. In at least one example, the display assembly 1-320 is mechanically coupled to the motor assembly 1-362, with at least one motor for each display screen 1-322a-b, such that the motors can translate the display screens 1-322a-b to match an interpupillary distance of the user's eyes.

[0840] In at least one example, the display unit 1-306 can include a dial or button 1-328 depressible relative to the frame 1-350 and accessible to the user outside the frame 1-350. The button 1-328 can

be electronically connected to the motor assembly **1-362** via a controller such that the button **1-328** can be manipulated by the user to cause the motors of the motor assembly **1-362** to adjust the positions of the display screens **1-322a-b**.

[0841] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **1-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **1-1A-1-2** and **1-4** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **1-1A-1-2** and **1-4** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **1-3**.

[0842] FIG. **1-4** illustrates a view of another example of a display unit **1-406** of an HMD device similar to other HMD devices described herein. The display unit **1-406** can include a front display assembly **1-402**, a sensor assembly **1-456**, a logic board assembly **1-458**, a cooling assembly **1-460**, a frame assembly **1-450**, a rear-facing display assembly **1-421**, and a curtain assembly **1-424**. The display unit **1-406** can also include a motor assembly **1-462** for adjusting the positions of first and second display sub-assemblies **1-420a**, **1-420b** of the rear-facing display assembly **1-421**, including first and second respective display screens for interpupillary adjustments, as described above.

[0843] The various parts, systems, and assemblies shown in the view of FIG. **1-4** are described in greater detail herein with reference to FIGS. **1-1A-1-3** as well as subsequent figures referenced in the present disclosure. The display unit **1-406** shown in FIG. **1-4** can be assembled and integrated with the securement mechanisms shown in FIGS. **1-1A-1-3**, including the electronic straps, bands, and other components including light seals, connection assemblies, and so forth.

[0844] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **1-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **1-1A-1-3** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **1-1A-1-3** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **1-4**.

II: Cover Glass

[0845] FIGS. **2.0-1** illustrates a view of an HMD **2.0-100** including a front cover and display assembly **2.0-102**, including one or more transparent layers, display integration assemblies, shroud, and dust seal. The transparent layers, display assemblies, shroud, and dust seal are described below in sections II, III, IV, and V.

2.1: Systems with Transparent Layers

[0846] Transparent layers may be used to form windows in buildings, vehicles, and/or other systems. Transparent layers may also be used in forming protective cover layers such as cover layers for optical components.

[0847] FIGS. **2.1-1** is a cross-sectional side view of an illustrative system containing a transparent layer. System **2.1-10** of FIGS. **2.1-1** has a support such as support **2.1-12** in which one or more transparent layers such as transparent layer **2.1-14** may be mounted. System **2.1-10** may be a building (e.g., support **2.1-12** may include building walls), may be a vehicle (e.g., support **2.1-12** may be vehicle body), may be an electronic device (e.g., support **2.1-12** may be an electronic device housing such as a head-mounted housing for a head-mounted device), and/or may be any other suitable system. In arrangement in which system **2.1-10** is a building or vehicle, layer **2.1-14** may serve as a window. In arrangements in which system **2.1-10** is an electronic device, layer **2.1-14** may overlap and protect components in the device. For example, layer **2.1-14** may serve as a protective cover layer that overlaps optical components. In an illustrative configuration, system **2.1-10** is a portable electronic device (e.g., a cellular telephone, head-mounted device, tablet computer, laptop computer, wristwatch, etc.).

[0848] Transparent layer **2.1-14** and support **2.1-12** may separate interior region **2.1-16** of system **2.1-10** from exterior region **2.1-18**. System components may be mounted in interior region **2.1-16**. Layer **2.1-14** may have opposing inner and outer surfaces. The outer surface of layer **2.1-14** may face exterior region **2.1-18** and the inner surface of layer **2.1-14** may face interior region **2.1-16**. The surfaces of layer **2.1-14** may include planar portions and/or portions that are curved. For example, layer **2.1-14** may have a shape with a curved cross-sectional profile such as shape **2.1-20**. In arrangements in which layer **2.1-14** is curved, the inner and outer surfaces may be parallel to each other (e.g., the thickness of layer **2.1-14** may be constant across layer **2.1-14**). If desired, some or all of the surfaces of layer **2.1-14** may have compound curvature (surfaces that can only be flattened into a plane with distortion). Surface areas of compound curvature may be bent about both the X and Y axes of FIGS. **2.1-1**.

[0849] FIGS. **2.1-2** shows how layer **2.1-14** may overlap components in interior region **2.1-16** such as illustrative components **2.1-20** and **2.1-22**. Components **2.1-20** and **2.1-22** may include optical components that emit and/or detect light. As an example, component **2.1-22** may be a display that emits visible light that passes through layer **2.1-14**. This allows a viewer in exterior region **2.1-18** to view an image on the display through layer **2.1-14** (e.g., layer **2.1-14** may serve as a display cover layer). Components such as component **2.1-20** may include, for example, visible and/or infrared cameras and/or other optical sensors that receive light through layer **2.1-14**. By overlapping components **2.1-20** and **2.1-22** as shown in FIGS. **2.1-2**, layer **2.1-14** may serve as a protective cover layer for components **2.1-20** and **2.1-22**.

[0850] During events such as drop events in which system **2.1-10** abruptly contacts the ground or other hard surface, layer **2.1-14** may be subjected to undesirably large amounts of stress. To help enhance durability, layer **2.1-14** may be provided with one or more layers of polymer. As an example, a polymer layer may be used to laminate multiple layers of transparent material together and/or polymer layers may be formed on exposed inner and/or outer surfaces of layer **2.1-14**.

[0851] FIGS. **2.1-3** is a cross-sectional side view of layer **2.1-14**. As shown in FIGS. **2.1-3**, layer **2.1-14** may include multiple layers of transparent material such as layers **2.1-40**, **2.1-34**, **2.1-32**, and **2.1-30**. In an illustrative configuration, layer **2.1-14** includes two layers of hard transparent material and one or more softer layers that are attached to the harder layers. The softer layers may be, for example, polymer layers that help enhance durability.

[0852] In the example of FIGS. **2.1-3**, layer **2.1-34** may be a hard layer such as a layer of glass (including glass ceramic) or sapphire or other crystalline material. Illustrative configurations in which layer **2.1-34** is a layer of glass may sometimes be described herein as an example. Layer **2.1-34** may be formed from alumina silicate glass or other glass materials and may optionally be chemically strengthened using an ion-exchange chemical strengthening process that places the surfaces of layer **2.1-34** in compression relative to the core of layer **2.1-34**. Layer **2.1-34** may have a thickness that is sufficient to provide layer **2.1-14** with some or all of its structural strength, so layer **2.1-34** may sometimes be referred to as a structural layer, structural transparent layer, or structural glass layer. Layer **2.1-34** may, as an example, have a thickness of 700 microns, at least 400 microns, at least 500 microns, at least 600 microns, less than 1200 microns, less than 1000 microns, less than 900 microns, less than 800 microns, 400-1200 microns, 400-1100 microns, 400-1000 microns, 400-800 microns, and/or other suitable thickness.

[0853] One or more polymer layers may be attached to layer **2.1-34**. In an illustrative configuration, polymer layer **2.1-40** is attached to inner surface **2.1-42** of layer **2.1-34**. Layer **2.1-40** may include a first layer such as layer **2.1-38** and a second layer such as layer **2.1-36**. Layer **2.1-38** may be a polymer film (e.g., a film of polycarbonate, polyethylene terephthalate, or other polymer film) and may have a thickness of 50 microns, 10-250 microns, 25-100 microns, at least 20 microns, less than 200 microns, less than 150 microns, or other suitable thickness. Layer **2.1-36** may be a polymer layer such as a layer of polymer adhesive (e.g., epoxy, acrylic adhesive, cured liquid adhesive, pressure sensitive adhesive, and/or other adhesive) that attaches layer **2.1-38** to

layer **2.1-34** and may have a thickness of 100 microns, 20-500 microns, at least 30 microns, less than 250 microns, less than 300 microns, or other suitable thickness.

[0854] If desired, an additional polymer layer such as polymer layer **2.1-32** may be attached to upper surface outer surface **1.3-44** of layer **2.1-34**. Layer **2.1-32** may be formed from an elastomeric polymer or other soft polymer material. Examples of materials that may be used in forming polymer layer **2.1-32** include polyvinyl butyral and ethylene vinyl acetate. Other polymers may be used in forming layer **2.1-32**, if desired. Layer **2.1-32** may be the outermost layer of material of layer **2.1-14** (e.g., the outer surface of layer **2.1-32** may be exposed to region **2.1-18**) or layer **2.1-32** may be covered with a harder outer layer.

[0855] As shown in FIGS. **2.1-3**, for example, layer **2.1-32**, which may sometimes be referred to as an elastomeric polymer layer or polymer interlayer, may be used to attach a thin hard layer such as outer layer **2.1-30** to layer **2.1-34**. The thickness of layer **2.1-32** may be 50 microns, 25-100 microns, at least 20 microns, at least 40 microns, at least 50 microns, less than 400 microns, 25-400 microns, less than 300 microns, less than 200 microns, 20-200 microns, 50-400 microns, or other suitable thickness. Layer **2.1-30** be formed from glass (including glass ceramic), a crystalline material such as sapphire, or hard polymer (e.g., hardened acrylic). The thickness of layer **2.1-30** is preferably less than the thickness of layer **2.1-34** to help minimize the weight of layer **2.1-14**.

[0856] In an illustrative arrangement, layer **2.1-30** is formed as a separate layer (e.g., a separate glass layer from layer **2.1-34**) that is attached to layer **2.1-34** by laminating layers **2.1-30** and **2.1-34** together using polymer layer **2.1-32**. The thickness of layer **2.1-30** in this type of arrangement may be at least 50 microns, at least 75 microns, at least 100 microns, less than 300 microns, less than 250 microns, less than 200 microns, less than 150 microns, less than 100 microns, 50-200 microns, 25-300 microns, 50-150 microns, or other suitable thickness (e.g., a thickness that provides the outermost surface of layer **2.1-14** with sufficient hardness to resist scratches). In addition to resisting scratches, the inclusion of a hard outer layer such as layer **2.1-30** to layer **2.1-14** may help enhance the strength of layer **2.1-14** and thereby allow the thickness of layer **34** to be reduced. To help match the curvature of layers **2.1-30** and **2.1-34** in this type of arrangement, layers **2.1-30** and **2.1-34** may be formed into desired shapes using molding operations (e.g., glass molding), machining and/or polishing operations, etching (wet and/or dry chemical etching), and/or other suitable shaping operations.

[0857] In some embodiments, layer **2.1-30** may be deposited as a coating on layer **2.1-32**. As an example, deposition techniques such as physical vapor deposition and sol-gel deposition may be used to deposit an inorganic dielectric layer of a hard material (e.g., a glass coating formed of silicon nitride, silicon oxynitride, zirconia, alumina, and/or other hard dielectric coating deposited by physical vapor deposition or a glass coating formed of an inorganic dielectric based on silicon oxide deposited by sol-gel deposition techniques). The thickness of this coating may be sufficient to allow the coating to enhance durability (e.g., to help prevent scratches in layer **2.1-32**). As an example, layer **2.1-30** may have a thickness of at least 20 microns, at least 25 microns, at least 35 microns, and/or other suitable thickness). If desired, a liquid polymer (e.g., liquid acrylic) may be deposited and cured to form an acrylic-based hard coat (e.g., layer **2.1-30** may be a polymer hard coat that is harder than layer **2.1-32** and that therefore helps resist scratching).

2.2: Systems with Displays and Sensors

[0858] FIGS. **2.2-1** is a side view of an illustrative head-mounted electronic device. As shown in FIGS. **2.2-1**, head-mounted device **2.2-10** may include head-mounted support structure **2.2-26**.

Support structure **2.2-26** may have walls or other structures that separate an interior region of device **2.2-10** such as interior region **2.2-42** from an exterior region surrounding device **2.2-10** such as exterior region **2.2-44**. Electrical components **2.2-40** (e.g., integrated circuits, sensors, control circuitry, light-emitting diodes, lasers, and other light-emitting devices, other control circuits and input-output devices, etc.) may be mounted on printed circuits and/or other structures within device **2.2-10** (e.g., in interior region **2.2-42**).

[0859] To present a user with images for viewing from eye boxes such as eye box 2.2-34, device 2.2-10 may include rear-facing displays such as display 2.2-14R and lenses such as lens 2.2-38. These components may be mounted in optical modules such as optical module 2.2-36 (e.g., a lens barrel) to form respective left and right optical systems. There may be, for example, a left rear-facing display for presenting an image through a left lens to a user's left eye in a left eye box and a right rear-facing display for presenting an image to a user's right eye in a right eye box. The user's eyes are located in eye boxes 34 at rear side R of device 2.2-10 when structure 2.2-26 rests against the outer surface (face surface 2.2-30) of the user's face.

[0860] Support structure 2.2-26 may include a main support structure such as main housing portion 2.2-26M (sometimes referred to as a main portion). Main housing portion 2.2-26M may extend from front side F of device 2.2-10 to opposing rear side R of device 2.2-10. On rear side R, main housing portion 2.2-26M may have cushioned structures to enhance user comfort as portion 2.2-26M rests against face surface 2.2-30. If desired, support structure 2.2-26 may include optional head straps such as strap 2.2-26B and/or other structures that allow device 2.2-10 to be worn on a head of a user.

[0861] Device 2.2-10 may have a publicly viewable front-facing display such as display 2.2-14F that is mounted on front side F of main housing portion 2.2-26M. Display 2.2-14F may be viewable to the user when the user is not wearing device 2.2-10 and/or may be viewable by others in the vicinity of device 2.2-10. Display 2.2-14F may, as an example, be visible on front side F of device 2.2-10 by an external viewer such as viewer 2.2-50 who is viewing device 2.2-10 in direction 2.2-52.

[0862] A schematic diagram of an illustrative system that may include a head-mounted device is shown in FIGS. 2.2-2. As shown in FIGS. 2.2-2, system 2.2-8 may have one or more electronic devices 2.2-10. Devices 2.2-10 may include a head-mounted device (e.g., device 2.2-10 of FIGS. 2.2-1), accessories such as controllers and headphones, computing equipment (e.g., a cellular telephone, tablet computer, laptop computer, desktop computer, and/or remote computing equipment that supplies content to a head-mounted device), and/or other devices that communicate with each other.

[0863] Each electronic device 2.2-10 may have control circuitry 2.2-12. Control circuitry 2.2-12 may include storage and processing circuitry for controlling the operation of device 2.2-10. Circuitry 2.2-12 may include storage such as hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry 2.2-12 may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, and other integrated circuits. Software code may be stored on storage in circuitry 2.2-12 and run on processing circuitry in circuitry 2.2-12 to implement control operations for device 2.2-10 (e.g., data gathering operations, operations involving the adjustment of the components of device 2.2-10 using control signals, etc.). Control circuitry 2.2-12 may include wired and wireless communications circuitry. For example, control circuitry 2.2-12 may include radio-frequency transceiver circuitry such as cellular telephone transceiver circuitry, wireless local area network transceiver circuitry (e.g., WiFi® circuitry), millimeter wave transceiver circuitry, and/or other wireless communications circuitry.

[0864] During operation, the communications circuitry of the devices in system 2.2-8 (e.g., the communications circuitry of control circuitry 2.2-12 of device 2.2-10) may be used to support communication between the electronic devices. For example, one electronic device may transmit video data, audio data, control signals, and/or other data to another electronic device in system 2.2-8. Electronic devices in system 2.2-8 may use wired and/or wireless communications circuitry to communicate through one or more communications networks (e.g., the internet, local area networks, etc.). The communications circuitry may be used to allow data to be received by device

2.2-10 from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, online computing equipment such as a remote server or other remote computing equipment, or other electrical equipment) and/or to provide data to external equipment. [0865] Each device **2.2-10** in system **2.2-8** may include input-output devices **2.2-22**. Input-output devices **2.2-22** may be used to allow a user to provide device **2.2-10** with user input. Input-output devices **2.2-22** may also be used to gather information on the environment in which device **2.2-10** is operating. Output components in devices **2.2-22** may allow device **2.2-10** to provide a user with output and may be used to communicate with external electrical equipment.

[0866] As shown in FIGS. **2.2-2**, input-output devices **2.2-22** may include one or more displays such as displays **2.2-14**. Displays **2.2-14** may include rear facing displays such as display **2.2-14R** of FIGS. **2.2-1**. Device **2.2-10** may, for example, include left and right components such as left and right scanning mirror display devices or other image projectors, liquid-crystal-on-silicon display devices, digital mirror devices, or other reflective display devices, left and right display panels based on light-emitting diode pixel arrays (e.g., organic light-emitting displays with polymer or semiconductor substrates or display devices based on pixel arrays formed from crystalline semiconductor light-emitting diode dies), liquid crystal display panels, and/or or other left and right display devices that provide images to left and right eye boxes for viewing by the user's left and right eyes, respectively. Display components such as these (e.g., an organic light-emitting display with a flexible polymer substrate or a display based on a pixel array formed from crystalline semiconductor light-emitting diode dies on a flexible substrate) may also be used in forming a forward-facing display for device **2.2-10** such as forward-facing display **2.2-14F** of FIGS. **2.2-1** (sometimes referred to as a front-facing display, front display, or publicly viewable display).

[0867] During operation, displays **2.2-14** (e.g., displays **2.2-14R** and/or **2.2-14F**) may be used to display visual content for a user of device **2.2-10** (e.g., still and/or moving images including pictures and pass-through video from camera sensors, text, graphics, movies, games, and/or other visual content). The content that is presented on displays **2.2-14** may, for example, include virtual objects and other content that is provided to displays **2.2-14** by control circuitry **2.2-12**. This virtual content may sometimes be referred to as computer-generated content. Computer-generated content may be displayed in the absence of real-world content or may be combined with real-world content. In some configurations, a real-world image may be captured by a camera (e.g., a forward-facing camera, sometimes referred to as a front-facing camera) and computer-generated content may be electronically overlaid on portions of the real-world image (e.g., when device **2.2-10** is a pair of virtual reality goggles).

[0868] Input-output circuitry **2.2-22** may include sensors **2.2-16**. Sensors **2.2-16** may include, for example, three-dimensional sensors (e.g., three-dimensional image sensors such as structured light sensors that emit beams of light and that use two-dimensional digital image sensors to gather image data for three-dimensional images from dots or other light spots that are produced when a target is illuminated by the beams of light, binocular three-dimensional image sensors that gather three-dimensional images using two or more cameras in a binocular imaging arrangement, three-dimensional LIDAR (light detection and ranging) sensors, sometimes referred to as time-of-flight cameras or three-dimensional time-of-flight cameras, three-dimensional radio-frequency sensors, or other sensors that gather three-dimensional image data), cameras (e.g., two-dimensional infrared and/or visible digital image sensors), gaze tracking sensors (e.g., a gaze tracking system based on an image sensor and, if desired, a light source that emits one or more beams of light that are tracked using the image sensor after reflecting from a user's eyes), touch sensors, capacitive proximity sensors, light-based (optical) proximity sensors, other proximity sensors, force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), sensors such as contact sensors based on switches, gas sensors, pressure sensors, moisture sensors, magnetic sensors, audio sensors (microphones), ambient light sensors, flicker sensors that gather temporal information on ambient lighting conditions such as the presence of a time-varying ambient light intensity associated with

artificial lighting, microphones for gathering voice commands and other audio input, sensors that are configured to gather information on motion, position, and/or orientation (e.g., accelerometers, gyroscopes, compasses, and/or inertial measurement units that include all of these sensors or a subset of one or two of these sensors), and/or other sensors.

[0869] User input and other information may be gathered using sensors and other input devices in input-output devices 2.2-22. If desired, input-output devices 2.2-22 may include other devices 2.2-24 such as haptic output devices (e.g., vibrating components), light-emitting diodes, lasers, and other light sources (e.g., light-emitting devices that emit light that illuminates the environment surrounding device 2.2-10 when ambient light levels are low), speakers such as ear speakers for producing audio output, circuits for receiving wireless power, circuits for transmitting power wirelessly to other devices, batteries and other energy storage devices (e.g., capacitors), joysticks, buttons, and/or other components.

[0870] As described in connection with FIGS. 2.2-1, electronic device 2.2-10 may have head-mounted support structures such as head-mounted support structure 2.2-26 (e.g., head-mounted housing structures such as housing walls, straps, etc.). The head-mounted support structure may be configured to be worn on a head of a user (e.g., against the user's face covering the user's eyes) during operation of device 2.2-10 and may support displays 2.2-14, sensors 2.2-16, other components 2.2-24, other input-output devices 2.2-22, and control circuitry 2.2-12 (see, e.g., components 2.2-40 and optical module 2.2-36 of FIGS. 2.2-1).

[0871] FIGS. 2.2-3 is a front view of device 2.2-10 in an illustrative configuration in which device 2.2-10 has a publicly viewable display such as forward-facing display 2.2-14F. As shown in FIGS. 2.2-3, support structure 2.2-26M of device 2.2-10 may have right and left portions such as portions 2.2-26R and 2.2-26L that are coupled by an interposed nose bridge portion such as portion 2.2-26NB. Portion 2.2-26NB may have a curved exterior surface such as nose bridge surface 2.2-90 that is configured to receive and rest upon a user's nose to help support main housing portion 2.2-26M on the head of the user.

[0872] Display 2.2-14F may have an active area such as active area AA that is configured to display images and an inactive area IA that does not display images. The outline of active area AA may be rectangular, rectangular with rounded corners, may have teardrop shaped portions on the left and right sides of device 2.2-10, may have a shape with straight edges, a shape with curved edges, a shape with a peripheral edge that has both straight and curved portions, and/or other suitable outlines. As shown in FIGS. 2.2-3, active area AA may have a curved recessed portion at nose bridge portion 2.2-26NB of main housing portion 2.2-26. The presence of the nose-shaped recess in active area AA may help fit active area AA within the available space of housing portion 2.2-26M without overly limiting the size of active area AA.

[0873] Active area AA contains an array of pixels. The pixels may be, for example, light-emitting diode pixels formed from thin-film organic light-emitting diodes or crystalline semiconductor light-emitting diode dies (sometimes referred to as micro-light-emitting diodes) on a flexible display panel substrate. Configurations in which display 2.2-14F uses other display technologies may also be used, if desired. Illustrative arrangements in which display 2.2-14 is formed from a light-emitting diode display such as an organic light-emitting diode display that is formed on a flexible substrate (e.g., a substrate formed from a bendable layer of polyimide or a sheet of other flexible polymer) may sometimes be described herein as an example. The pixels of active area AA may be formed on a display device such as display panel 2.2-14P of FIGS. 2.2-3 (e.g., a flexible organic light-emitting diode display panel). In some configurations, the outline of panel 2.2-14P may have a peripheral edge that contains straight segments or a combination of straight and curved segments. Configurations in which the entire outline of panel 2.2-14P is characterized by a curved peripheral edge may also be used.

[0874] Display 2.2-14F may have an inactive area such as inactive area IA that is free of pixels and that does not display images. Inactive area IA may form an inactive border region that runs along

one more portions of the peripheral edge of active area AA. In the illustrative configuration of FIGS. 2.2-3, inactive area IA has a ring shape that surrounds active area AA. In this type of arrangement, the width of inactive area IA may be relatively constant and the inner and outer edges of area IA may be characterized by straight and/or curved segments or may be curved along their entire lengths. For example, the outer edge of area IA (e.g., the periphery of display 2.2-14F) may have a curved outline that runs parallel to the curved edge of active area AA.

[0875] In some configurations, device 2.2-10 may operate with other devices in system 2.2-8 (e.g., wireless controllers and other accessories). These accessories may have magnetic sensors that sense the direction and intensity of magnetic fields. Device 2.2-10 may have one or more electromagnets configured to emit a magnetic field. The magnetic field can be measured by the wireless accessories near device 2.2-10, so that the accessories can determine their orientation and position relative to device 2.2-10. This allows the accessories to wirelessly provide device 2.2-10 with real-time information on their current position, orientation, and movement so that the accessories can serve as wireless controllers. The accessories may include wearable devices, handled devices, and other input devices.

[0876] In an illustrative configuration, device 2.2-10 may have a coil such as illustrative coil 2.2-54 that runs around the perimeter of display 2.2-14F (e.g., under inactive area IA or other portion of display 2.2-14F). Coil 2.2-54 may have any suitable number of turns (e.g., 1-10, at least 2, at least 5, at least 10, 10-50, fewer than 100, fewer than 25, fewer than 6, etc.). These turns may be formed from metal traces on a substrate, may be formed from wire, and/or may be formed from other conductive lines. During operation, control circuitry 2.2-12 may supply coil 2.2-54 with an alternating-current (AC) drive signal. The drive signal may have a frequency of at least 1 kHz, at least 10 kHz, at least 100 kHz, at least 1 MHz, less than 10 MHz, less than 3 MHz, less than 300 kHz, or less than 30 kHz (as examples). As AC current flows through coil 2.2-54, a corresponding magnetic field is produced in the vicinity of device 2.2-10. Electronic devices such as wireless controllers with magnetic sensors that are in the vicinity of device 2.2-10 may use the magnetic field as a reference so that the wireless controllers can determine their orientation, position, and/or movement while being moved relative to device 2.2-10 to provide device 2.2-10 with input.

[0877] Consider, as an example, a handheld wireless controller that is used in controlling the operation of device 2.2-10. During operation, device 2.2-10 uses coil 2.2-54 to emit a magnetic field. As the handheld wireless controller is moved, the magnetic sensors of the controller can monitor the location of the controller and the movement of the controller relative to device 2.2-10 by monitoring the strength, orientation, and change to the strength and/or orientation of the magnetic field emitted by coil 2.2-54 as the controller is moved through the air by the user. The electronic device can then wirelessly transmit information on the location and orientation of the controller to device 2.2-10. In this way, a handheld controller, wearable controller, or other external accessory can be manipulated by a user to provide device 2.2-10 with air gestures, pointing input, steering input, and/or other user input.

[0878] Device 2.2-10 may have components such as optical components (e.g., optical sensors among sensors 2.2-16 of FIGS. 2.2-2). These components may be mounted in any suitable location on head-mounted support structure 2.2-26 (e.g. on head strap 2.2-26B, on main housing portion 2.2-26M, etc.). Optical components and other components may face rearwardly (e.g., when mounted on the rear face of device 2.2-10), may face to the side (e.g. to the left or right), may face downwardly or upwardly, may face to the front of device 2.2-10 (e.g., when mounted on the front face of device 2.2-10), may be mounted so as to point in any combination of these directions (e.g., to the front, to the right, and downward) and/or may be mounted in other suitable orientations. In an illustrative configuration, at least some of the components of device 2.2-10 are mounted so as to face outwardly to the front (and optionally to the sides and/or up and down). For example, forward-facing cameras for pass-through video may be mounted on the left and right sides of the front of device 2.2-10 in a configuration in which the cameras diverge slightly along the horizontal

dimension so that the fields of view of these cameras overlap somewhat while capturing a wide-angle image of the environment in front of device **2.2-10**. The captured image may, if desired, include portions of the user's surroundings that are below, above, and to the sides of the area directly in front of device **2.2-10**.

[0879] To help hide components such as optical components from view from the exterior of device **2.2-10**, it may be desirable to cover some or all of the components with cosmetic covering structures. The covering structures may include transparent portions (e.g., optical component windows) that are characterized by sufficient optical transparency to allow overlapped optical components to operate satisfactorily. For example, an ambient light sensor may be covered with a layer that appears opaque to an external viewer to help hide the ambient light sensor from view, but that allows sufficient ambient light to pass to the ambient light sensor for the ambient light sensor to make a satisfactory ambient light measurement. As another example, an optical component that emits infrared light may be overlapped with a visibly opaque material that is transparent to infrared light.

[0880] In an illustrative configuration, optical components for device **2.2-10** may be mounted in inactive area IA of FIGS. **2.2-3** and cosmetic covering structures may be formed in a ring shape overlapping the optical components in inactive area IA. Cosmetic covering structures may be formed from ink, polymer structures, structures that include metal, other materials, and/or combinations of these materials. In an illustrative configuration, a cosmetic covering structure may be formed from a ring-shaped member having a footprint that matches the footprint of inactive area IA. If, for example, active area AA has left and right portions with teardrop shapes, the ring-shaped member may have curved edges that follow the curved periphery of the teardrop-shaped portions of active area AA. The ring-shaped member may be formed from one or more polymer structures (e.g., the ring-shaped member may be formed from a polymer ring). Because the ring-shaped member can help hide overlapped components from view, the ring-shaped member may sometimes be referred to as a shroud or ring-shaped shroud member. The outward appearance of the shroud or other cosmetic covering structures may be characterized by a neutral color (white, black, or gray) or a non-neutral color (e.g., blue, red, green, gold, rose gold, etc.).

[0881] Display **2.2-14F** may, if desired, have a protective display cover layer. The cover layer may overlap active area AA and inactive area IA (e.g., the entire front surface of device **2.2-10** as viewed from direction **2.2-52** of FIGS. **2.2-1** may be covered by the cover layer). The cover layer, which may sometimes be referred to as a housing wall or transparent housing wall, may have a rectangular outline, an outline with teardrop portions, an oval outline, or other shape with curved and/or straight edges.

[0882] The cover layer may be formed from a transparent material such as glass, polymer, transparent crystalline material such as sapphire, clear ceramic, other transparent materials, and/or combinations of these materials. As an example, a protective display cover layer for display **2.2-14F** may be formed from safety glass (e.g., laminated glass that includes a clear glass layer with a laminated polymer film). Optional coating layers may be applied to the surfaces of the display cover layer. If desired, the display cover layer may be chemically strengthened (e.g., using an ion-exchange process to create an outer layer of material under compressive stress that resists scratching). In some configurations, the display cover layer may be formed from a stack of two or more layers of material (e.g., first and second structural glass layers, a rigid polymer layer coupled to a glass layer or another rigid polymer layer, etc.) to enhance the performance of the cover layer.

[0883] In active area AA, the display cover layer may overlap the pixels of display panel **2.2-14P**. The display cover layer in active area AA is preferably transparent to allow viewing of images presented on display panel **2.2-14P**. In inactive area IA, the display cover layer may overlap the ring-shaped shroud or other cosmetic covering structure. The shroud and/or other covering structures (e.g., opaque ink coatings on the inner surface of the display cover layer and/or structures) may be sufficiently opaque to help hide some or all of the optical components in

inactive area IA from view. Windows may be provided in the shroud or other cosmetic covering structures to help ensure that the optical components that are overlapped by these structures operate satisfactorily. Windows may be formed from holes, may be formed from areas of the shroud or other cosmetic covering structures that have been locally thinned to enhance light transmission, may be formed from window members with desired light transmission properties that have been inserted into mating openings in the shroud, and/or may be formed from other shroud window structures.

[0884] In the example of FIGS. 2.2-3, device 2.2-10 includes optical components such as optical components 2.2-60, 2.2-62, 2.2-64, 2.2-66, 2.2-68, 2.2-70, 2.2-72, 2.2-74, 2.2-76, 2.2-78, and 2.2-80 (as an example). Each of these optical components (e.g., optical sensors selected from among sensors 2.2-16 of FIGS. 2.2-2, light-emitting devices, etc.) may be configured to detect light and, if desired to emit light (e.g., ultraviolet light, visible light, and/or infrared light).

[0885] In an illustrative configuration, optical component 2.2-60 may sense ambient light (e.g., visible ambient light). In particular, optical component 2.2-60 may have a photodetector that senses variations in ambient light intensity as a function of time. If, as an example, a user is operating in an environment with an artificial light source, the light source may emit light at a frequency associated with its source of wall power (e.g., alternating-current mains power at 60 Hz). The photodetector of component 2.2-60 may sense that the artificial light from the artificial light source is characterized by 60 Hz fluctuations in intensity. Control circuitry 2.2-12 can use this information to adjust a clock or other timing signal associated with the operation of image sensors in device 2.2-10 to help avoid undesired interference between the light source frequency and the frame rate or other frequency associated with image capture operations. Control circuitry 2.2-12 can also use measurements from component 2.2-60 to help identify the presence of artificial lighting and the type of artificial lighting that is present. In this way, control circuitry 2.2-12 can detect the presence of lights such as fluorescent lights or other lights with known non-ideal color characteristics and can make compensating color cast adjustments (e.g., white point adjustments) to color-sensitive components such as cameras and displays. Because optical component 2.2-60 may measure fluctuations in light intensity, component 2.2-60 may sometimes be referred to as a flicker sensor or ambient light frequency sensor.

[0886] Optical component 2.2-62 may be an ambient light sensor. The ambient light sensor may include one or more photodetectors. In a single-photodetector configuration, the ambient light sensor may be a monochrome sensor that measures ambient light intensity. In a multi-photodetector configuration, each photodetector may be overlapped by an optical filter that passes a different band of wavelengths (e.g. different visible and/or infrared passbands). The optical filter passbands may overlap at their edges. This allows component 2.2-62 to serve as a color ambient light sensor that measures both ambient light intensity and ambient light color (e.g., by measuring color coordinates for the ambient light). During operation of device 2.2-10, control circuitry 2.2-12 can take action based on measured ambient light intensity and color. As an example, the white point of a display or image sensor may be adjusted or other display or image sensor color adjustments may be made based on measured ambient light color. The intensity of a display may be adjusted based on light intensity. For example, the brightness of display 2.2-14F may be increased in bright ambient lighting conditions to enhance the visibility of the image on the display and the brightness of display 2.2-14F may be decreased in dim lighting conditions to conserve power. Image sensor operations and/or light source operations may also be adjusted based on ambient light readings.

[0887] The optical components in active area IA may also include components along the sides of device 2.2-10 such as components 2.2-80 and 2.2-64. Optical components 2.2-80 and 2.2-64 may be pose-tracking cameras that are used to help monitor the orientation and movement of device 2.2-10. Components 2.2-80 and 2.2-64 may be visible light cameras (and/or cameras that are sensitive at visible and infrared wavelengths) and may, in conjunction with an inertial measurement unit, form a visual inertial odometry (VIO) system.

[0888] Optical components **2.2-78** and **2.2-66** may be visible-light cameras that capture real-time images of the environment surrounding device **2.2-10**. These cameras, which may sometimes be referred to as scene cameras or pass-through-video cameras, may capture moving images that are displayed in real time to displays **2.2-14R** for viewing by the user when the user's eyes are located in eye boxes **2.2-34** at the rear of device **2.2-10**. By displaying pass-through images (pass-through video) to the user in this way, the user may be provided with real-time information on the user's surroundings. If desired, virtual content (e.g. computer-generated images) may be overlaid over some of the pass-through video. Device **2.2-10** may also operate in a non-pass-through-video mode in which components **2.2-78** and **2.2-66** are turned off and the user is provided only with movie content, game content, and/or other virtual content that does not contain real-time real-world images.

[0889] Input-output devices **2.2-22** of device **2.2-10** may gather user input that is used in controlling the operation of device **2.2-10**. As an example, a microphone in device **2.2-10** may gather voice commands. Buttons, touch sensors, force sensors, and other input devices may gather user input from a user's finger or other external object that is contacting device **2.2-10**. In some configurations, it may be desirable to monitor a user's hand gestures or the motion of other user body parts. This allows the user's hand locations or other body part locations to be replicated in a game or other virtual environment and allows the user's hand motions to serve as hand gestures (air gestures) that control the operation of device **2.2-10**. User input such as hand gesture input can be captured using cameras that operate at visible and infrared wavelengths such as tracking cameras (e.g., optical components **2.2-76** and **2.2-68**). Tracking cameras such as these may also track fiducials and other recognizable features on controllers and other external accessories (additional devices **2.2-10** of system **2.2-8**) during use of these controllers in controlling the operation of device **2.2-10**. If desired, tracking cameras can help determine the position and orientation of a handheld controller or wearable controller that senses its location and orientation by measuring the magnetic field produced by coil **2.2-54**. The use of tracking cameras may therefore help track hand motions and controller motions that are used in moving pointers and other virtual objects being displayed for a user and can otherwise assist in controlling the operation of device **2.2-10**.

[0890] Tracking cameras may operate satisfactorily in the presence of sufficient ambient light (e.g., bright visible ambient lighting conditions). In dim environments, supplemental illumination may be provided by supplemental light sources such as supplemental infrared light sources (e.g., optical components **2.2-82** and **2.2-84**). The infrared light sources may each include one or more light-emitting devices (light-emitting diodes or lasers) and may each be configured to provide fixed and/or steerable beams of infrared light that serve as supplemental illumination for the tracking cameras. If desired, the infrared light sources may be turned off in bright ambient lighting conditions and may be turned on in response to detection of dim ambient lighting (e.g., using the ambient light sensing capabilities of optical component **2.2-62**).

[0891] Three-dimensional sensors in device **2.2-10** may be used to perform biometric identification operations (e.g., facial identification for authentication), may be used to determine the three-dimensional shapes of objects in the user's environment (e.g., to map the user's environment so that a matching virtual environment can be created for the user), and/or to otherwise gather three-dimensional content during operation of device **2.2-10**. As an example, optical components **2.2-74** and **2.2-70** may be three-dimensional structured light image sensors. Each three-dimensional structured light image sensor may have one or more light sources that provide structured light (e.g., a dot projector that projects an array of infrared dots onto the environment, a structured light source that produces a grid of lines, or other structured light component that emits structured light). Each of the three-dimensional structured light image sensors may also include a flood illuminator (e.g., a light-emitting diode or laser that emits a wide beam of infrared light). Using flood illumination and structured light illumination, optical components **2.2-74** and **2.2-70** may capture facial images, images of objects in the environment surrounding device **2.2-10**, etc.

[0892] Optical component **2.2-72** may be an infrared three-dimensional time-of-flight camera that uses time-of-flight measurements on emitted light to gather three-dimensional images of objects in the environment surrounding device **2.2-10**. Component **2.2-72** may have a longer range and a narrower field of view than the three-dimensional structured light cameras of optical components **2.2-74** and **2.2-70**. The operating range of component **2.2-72** may be 30 cm to 7 m, 60 cm to 6 m, 70 cm to 5 m, or other suitable operating range (as examples).

[0893] FIGS. **2.2-4** is a top view of device **2.2-10** in an illustrative arrangement in which display **2.2-14F** and main housing portion **2.2-26M** have been configured to curve about the curved surface of a user's face (curved face surface **2.2-30**). In particular, rear surface **2.2-96** of housing portion **2.2-26M** on rear side R of device **2.2-10** may have a curved shape that is bent about axis **2.2-98** (e.g., an axis parallel to the vertical Z axis in the example of FIGS. **2.2-4**). By wrapping housing portion **2.2-26M** smoothly about the curved surface of the user's head, comfort may be enhanced when wearing device **2.2-10**.

[0894] As shown in FIGS. **2.2-4**, display **2.2-14F** and other structures on the front of device **2.2-10** may have a protective cover layer such as display cover layer **2.2-92** (e.g., a front portion of housing portion **2.2-26M**, which may sometimes be referred to as a front housing wall, transparent dielectric housing wall, or dielectric housing member). In some embodiments, display cover layer **2.2-92** may include areas that are characterized by curved surfaces that can be flattened into a plane without distortion (sometimes referred to as developable surfaces or curved surfaces without compound curvature). Display cover layer **2.2-92** may also include areas that are characterized by compound curvature (e.g., surfaces that can only be flattened into a plane with distortion, sometimes referred to as non-developable surfaces).

[0895] In active area AA of display **2.2-14F**, cover layer **2.2-92** overlaps an array of pixels P in display panel **2.2-14P**. In inactive area IA, cover layer **2.2-92** does not overlap any pixels, but may overlap optical components such as the optical components shown in FIGS. **2.2-3**. To help reduce the size and weight of device **2.2-10**, display **2.2-14F** may have a curved shape that wraps around the front of the user's head parallel to face surface **2.2-30** and parallel to curved rear surface **2.2-96** of housing portion **2.2-26M**. For example, display panel **2.2-14P** may have a flexible substrate that allows panel **2.2-14P** to bend about bend axis **2.2-94** (e.g., a bend axis that is parallel to the Z axis in the example of FIGS. **2.2-4**). In active area AA of display **2.2-14F**, display cover layer **2.2-92** may have an inner surface with a curved cross-sectional profile that conforms to bent display panel **2.2-14P** and a corresponding curved outer surface. In inactive area IA, display cover layer **2.2-92** may also be curved (e.g., with a tighter bend radius and more curvature than in active area AA). If desired, a polymer layer (sometimes referred to as a shroud canopy or polymer member) may be interposed between display cover layer **2.2-92** and display panel **2.2-14P**. The polymer layer may be separated from the pixels of panel **2.2-14P** by an air gap and may be separated from the inner surface of display cover layer **2.2-92** by an air gap (as an example).

[0896] FIGS. **2.2-5A** is a cross-sectional side view of display **2.2-14F** viewed in the X direction. As shown in FIGS. **2.2-5A**, the cross-sectional profile of display panel **2.2-14P** (in planes parallel to the YZ plane) may, in an illustrative configuration, be straight rather than curved. This may help prevent wrinkling or other distortion to the flexible substrate material of display panel **2.2-14P** as display panel **2.2-14P** is bent about bend axis **2.2-94** to wrap around the curved surface of the user's face. Display panel **2.2-14P** may, in this example, have a developable surface (e.g., a surface that has a curved cross-sectional profile but that does not have any compound curvature). Panel **2.2-14P** of FIGS. **2.2-5A** may be attached to the inner surface of layer **2.2-92** (e.g., with adhesive). In this scenario, the inner surface of layer **2.2-92** may be a developable surface that mates with the outwardly facing developable surface of panel **2.2-14P**. The corresponding outer surface of layer **2.2-92** in active area AA may be a developable surface or may be a surface of compound curvature. In inactive area IA, layer **2.2-92** may have inner and/or outer surfaces of compound curvature and/or the inner and/or outer surfaces may be developable surfaces. If desired, the entire outer

surface of layer **2.2-92** may have compound curvature (both in active area AA and in inactive area IA), the inner surface of layer **2.2-92** in active area AA may be a developable surface to which panel **2.2-14P** is laminated with adhesive, and the inner surface of layer **2.2-92** in inactive area IA may have compound curvature and/or may be a developable surface.

[0897] Another illustrative configuration for display **2.2-14F** is shown in FIGS. **2.2-5B**. As shown in the cross-sectional side view of FIGS. **2.2-5B**, display cover layer **2.2-92** may, if desired, have a cross-sectional profile that is curved across all of layer **2.2-92**. With this type of arrangement, the surface of inactive area IA of display cover layer **2.2-92** may have compound curvature and active area AA of display cover layer **2.2-92** may have compound curvature (e.g., layer **2.2-92** may be free of any areas with developable surfaces). A polymer layer such as polymer layer **2.2-130**, which may sometimes be referred to as a shroud or shroud canopy, may be interposed between the inner surface of display cover layer **2.2-92** and the opposing outer surface of display panel **2.2-14P**. The outer surface of display panel **2.2-14P** may be a developable surface (e.g., display panel **2.2-14P** may be bent about axis **2.2-94**). In active area AA, where polymer layer **2.2-130** overlaps the pixels of panel **2.2-14P**, polymer layer **2.2-130** may also be bent about axis **2.2-94** (e.g., the inner and outer surfaces of polymer layer **2.2-130** in active area AA may be developable surfaces). In inactive area IA, the inner and outer surfaces of polymer layer **2.2-130** may have compound curvature. Air gaps may separate panel **2.2-14P** from the inner surface of layer **2.2-130** and may separate the outer surface of layer **2.2-130** from the inner surface of layer **2.2-92**.

[0898] If desired, other arrangements for layer **2.2-130** may be used. For example, the side of layer **2.2-130** facing display panel **2.2-14P** may have a developable surface in active area AA, whereas the side of layer **2.2-130** facing layer **2.2-92** may have compound curvature in active area AA (e.g., layer **2.2-130** may have a non-uniform thickness). Layer **2.2-92** may also have different configurations. For example, the outer surface of layer **2.2-92** may have compound curvature, whereas the inner surface of layer **2.2-92** in active area AA and/or in area IA may be a developable surface. Other arrangements in which layer **2.2-92** and/or layer **2.2-130** have variable thicknesses may also be used. In inactive area IA, multiple polymer structures may be joined. For example, in area IA, a ring-shaped polymer member, sometimes referred to as a shroud trim, may be joined to layer **2.2-130**, which may form a shroud canopy member that extends across the entire front face of device **2.2-10**. The shroud trim and shroud canopy may, if desired, sometimes be referred to individually or collectively as forming a shroud, shroud member(s), etc. Tinting (e.g., dye, pigment, and/or other colorant) may be included in layer **2.2-130**. For example, layer **2.2-130** may be tinted to exhibit a visible light transmission of 30-80% to help obscure internal structures in device **2.2-10** such as display panel **2.2-14P** from view when not in use.

[0899] FIGS. **2.2-6** is a front view of a portion of display **2.2-14F** and display cover layer **2.2-92**. The inner and outer surfaces of display cover layer **2.2-92** that directly overlap active area AA and display panel **2.2-14P** may be developable surfaces and/or may include areas of compound curvature. In an illustrative configuration, the inner surface of cover layer **2.2-92** in area AA may, as described in connection with FIGS. **4** and **5A**, bend about bend axis **2.2-94** without exhibiting curvature about any axis orthogonal to axis **2.2-94**. The outer surface of layer **2.2-92** in area AA may be a developable surface or a surface of compound curvature. The use of a developable surface for the inwardly facing side of display cover layer **2.2-92** (and, if desired, the use of a developable surface for the inwardly facing side of optional layer **2.2-130** of FIGS. **2.2-5B**) may help ensure that display panel **2.2-14P** is not wrinkled or otherwise damaged during the bending of panel **2.2-14P** to form a curved display shape that conforms to the shape of the user's head.

[0900] Display panel **2.2-14P** may have an outwardly facing surface in active area AA that is a developable surface. This display panel surface may be adhered to the corresponding inner developable surface of layer **2.2-130** or a corresponding inner developable surface of layer **2.2-92** or may be spaced apart from the layer **2.2-130** and/or the inner surface of layer **2.2-92** by an air gap (as examples).

[0901] Some or all portions of the inner and outer surfaces of display cover layer **2.2-92** in inactive area **IA** may, if desired, be characterized by compound curvature. This allows the periphery of display **2.2-14F** to smoothly transition away from the active area and provides an attractive appearance and compact shape for device **2.2-10**. The compound curvature of display cover layer **2.2-92** in inactive area **IA** may also facilitate placement of the optical components under inactive area **IA** in desired orientations. If desired, all areas of layer **2.2-92** may have compound curvature (e.g., the inner and outer surfaces of layer **2.2-92** may have compound curvature in both area **IA** and area **AA**).

[0902] In the illustrative configuration of FIGS. **2.2-6**, in which display cover layer **2.2-92** has a curved peripheral edge and in which the inwardly facing and outwardly facing surfaces of display cover layer **2.2-92** have compound curvature in inactive area **IA**, the cross-sectional profiles of display cover layer **2.2-92** taken along each of illustrative lines **2.2-100** of FIGS. **2.2-6** are curved (e.g., the entire peripheral ring-shaped inactive area of display **2.2-14F** in the FIGS. **2.2-6** example is covered by a portion of display cover layer **2.2-92** with inner and outer surfaces of compound curvature). This type of shape for display cover layer **2.2-92** may be produced by glass forming, polymer molding, machining, and/or other display cover layer fabrication techniques. Other arrangements (e.g., configurations in which display cover layer **2.2-92** has at least some developable surfaces (inner and/or outer surfaces) in inactive area **IA**) may also be used. The arrangement of FIGS. **2.2-6** is illustrative.

[0903] FIGS. **2.2-7**, **2.2-8**, and **2.2-9** are front views of illustrative upper left portions of display cover layer **2.2-92**. Device **2.2-10** may have symmetrical right-hand cover layer portions. The example of FIGS. **2.2-7** shows how the peripheral edge of display cover layer **2.2-92** may have straight edges (e.g., a generally rectangular shape with straight edges) and rounded corners. In the example of FIGS. **2.2-8**, display cover layer **2.2-92** has teardrop shapes on the upper left and right sides. FIGS. **2.2-9** shows how the upper corners of display cover layer **2.2-92** may have sweeping curves (e.g., to help soften the visual appearance of device **2.2-10** when viewed from the front).

[0904] FIGS. **10**, **11**, and **12** are front views of illustrative lower left portions of display cover layer **2.2-92**. As shown in FIGS. **2.2-10**, the lower half of cover layer **2.2-92** may be characterized by a rectangular shape with rounded corners. Cover layer **2.2-92** of FIGS. **2.2-10** may have an upper portion with a shape of the type shown in FIGS. **2.2-7** (as an example). In the nose bridge portion of device **2.2-10**, cover layer **2.2-92** may have a recessed curved nose-bridge edge shape (see, e.g., curved edge surface **2.2-90**). In the illustrative arrangement of FIGS. **2.2-11**, display cover layer **2.2-92** has lower left and right sides with teardrop shapes (e.g., shapes that may be used with a display cover layer having upper left and right teardrop shapes of the type shown in FIGS. **2.2-8**). FIGS. **2.2-12** shows how the lower portion of display cover layer **2.2-92** may have a more gradually curved outline.

[0905] In general, the upper and lower portions of cover layer **2.2-92** may have any suitable outlines when viewed from the front of device **2.2-10**. The shape used for cover layer **2.2-92** may be determined by factors such as aesthetics, size, the ability to facilitate suitable placement for optical components in inactive area **IA**, the ability to provide desired active area coverage (overlap over active area **AA**), etc. Any of the illustrative shapes for the upper portion of device **2.2-10** shown in FIGS. **7**, **8**, and/or **9** may be used in combination with any of the illustrative shapes for the lower portion of device **2.2-10** shown in FIGS. **10**, **11**, and **12**. The overall shape for cover layer **2.2-92** may be symmetric about the nose bridge (e.g., left and right halves of layer **2.2-92** may exhibit mirror symmetry). The shapes of FIGS. **7**, **8**, **9**, **10**, **11**, and **12** are illustrative. Other shapes may be used, if desired.

[0906] FIGS. **2.2-13** is an exploded cross-sectional top view of a portion of device **2.2-10** showing how display cover layer **2.2-92** may have a portion overlapping display panel **2.2-14P** and a portion overlapping a cosmetic covering structure such as shroud **2.2-130** (e.g., a ring-shaped shroud portion sometimes referred to as a shroud trim or shroud trim member, which may optionally be

attached in area IA to a shroud canopy that covers display 2.2-14F such as optional polymer layer 2.2-130). Cosmetic covering structures in inactive area IA may be formed from opaque masking layers (e.g., black ink layers) and/or other coatings on the inner surface of display cover layer 2.2-92 and/or on the shroud, from separate structures formed from metal, polymer, glass, or other materials, and/or other structures that can help hide overlapped components 2.2-104. Components 2.2-104 may include sensors 16 and other input-output devices 2.2-22 of FIGS. 2.2-2. For example, components 2.2-104 may be optical components such as components 2.2-60, 2.2-62, 2.2-64, 2.2-84, 2.2-66, 2.2-68, 2.2-70, 2.2-72, 2.2-74, 2.2-76, 2.2-78, 2.2-82, and 2.2-80 of FIGS. 2.2-3. In inactive area IA, cover layer 2.2-92 may have curved inner and outer surfaces (e.g., surfaces with compound curvature). Shroud 2.2-102 (and, if desired, layer 2.2-130 in area IA) may optionally have corresponding inner and outer surfaces (e.g., surfaces with compound curvature). Components 2.2-104 may operate through optical component windows in shroud 2.2-102 (and optionally in layer 2.2-130 in area IA) and corresponding areas in layer 2.2-92. These windows may be formed by recesses and/or through-hole openings in shroud 2.2-102 (and optionally in layer 2.2-130) and/or layer 2.2-92, by window members that are installed within openings in shroud 2.2-102 (and optionally in layer 2.2-130) and/or layer 2.2-92, by portions of shroud 2.2-102 (and optionally portions of layer 2.2-130) and/or layer 2.2-92 that exhibit optical transparency sufficient for satisfactory operation of overlapped components, and/or by other structures in shroud 2.2-102 (and optionally in layer 2.2-130) and/or window 2.2-92.

[0907] If desired, components 2.2-104 may include components such as cameras (e.g., visible and/or infrared image sensors, time-of-flight sensors, structured light three-dimensional sensors, etc.) that are sensitive to optical distortion imposed by the curved shapes of the curved inner and/or outer surface of cover layer 2.2-92. For example, a camera or other optical component 104 may operate through a portion of cover layer 2.2-92 in inactive area IA that is characterized by an outer surface that has compound curvature and an inner surface with compound curvature or a developable inner surface. In this type of situation, the control circuitry of device 2.2-10 may be configured to digitally compensate for the optical distortion introduced as light (e.g., real-world image light) passes through layer 2.2-92 to the camera or other optical sensor. As an example, the amount of image distortion imposed by layer 2.2-92 (e.g., stretching, shifting, keystoneing, barrel distortion, pincushion distortion, and/or other optical distortion) may be measured and characterized for each optical component that operates through layer 2.2-92 (e.g., through a portion of layer 2.2-92 in inactive area IA that has inner and/or outer surfaces of compound curvature). During operation of device 2.2-10, the image data captured by a camera and/or other sensor data that is gathered by an optical component overlapped by layer 2.2-92 may be compensated accordingly (e.g., an equal and opposite amount of digital image warping may be applied to the captured image data, thereby removing the known distortion effects of layer 2.2-92). In this way, high quality (undistorted) images and/or other sensor data may be gathered by cameras and/or other optical components that operate through curved portions of layer 2.2-92. This allows layer 2.2-92 to be provided with an attractive shape (e.g., a shape with one or more surfaces characterized by compound curvature).

[0908] When assembled into device 2.2-10, display cover layer 2.2-92 and shroud 2.2-102 (and optionally layer 2.2-130) may be mounted to an exposed edge portion of a polymer housing structure, a metal housing wall, or other housing structure in main housing portion 2.2-26M. As an example, main housing portion 2.2-26M may have a polymer sidewall member that runs around the periphery of display cover layer 2.2-92 and that supports the peripheral edge of display cover layer 2.2-92. Shroud 2.2-102 may have a ring shape that runs along the edge of display cover layer 2.2-92 in inactive area IA. In an illustrative configuration, adhesive is used to attach display cover layer 2.2-92 to shroud 2.2-102 (and/or layer 2.2-130) and adhesive is used to attach shroud 2.2-102 (and/or layer 2.2-130) to the exposed front edge of the sidewall in main housing portion 2.2-26M. Components 2.2-104 may be attached to shroud 2.2-102 (and/or layer 2.2-130) and/or may be

supported on internal housing structures (e.g., brackets, frame members, etc.) in alignment with optical windows in shroud **2.2-102** (and/or layer **2.2-130**) and corresponding portions of layer **2.2-92**.

[0909] FIGS. **2.2-14** is a cross-sectional side view of a portion of display **2.2-14F**. In the example of FIGS. **2.2-14**, display panel **2.2-14P** is a three-dimensional display panel having an array of pixels **P** overlapped by lenticular lenses **2.2-106** (e.g., display panel **2.2-14P** is an autostereoscopic display that produces glasses-free three-dimensional images for viewers such as viewer **2.2-50** of FIGS. **2.2-1**). Lenses **2.2-106** may, as an example, be formed from semicylindrical lens elements that are elongated along columns of pixels (e.g., lens elements that extend parallel to the **Z** dimension in the example of FIGS. **2.2-14**). If desired, lenses **2.2-106** may be omitted (e.g., display panel **2.2-14P** may have an array of pixels **P** that are not overlapped by lenses **2.2-106** to form a two-dimensional display).

[0910] An air gap such as gap **2.2-114** may separate display panel **2.2-14P** of display **2.2-14F** from display cover layer **2.2-92**. Optional layer **2.2-130** may be formed within gap **2.2-114** of FIG. **2.2-14**, so that layer **2.2-130** has an outer surface that is separated from layer **2.2-92** by a first air gap and an opposing inner surface that is separated from lenses **2.2-106** and pixels **P** of display panel **2.2-14P** by a second air gap. In arrangements in which lenses **2.2-106** are present, air gap **2.2-114** (and the resulting absence of direct contact between the inner surface of layer **2.2-130** and lenses **2.2-106**) may allow lenses **2.2-106** to operate satisfactorily. Display cover layer **2.2-92** and optional layer **2.2-130** may be formed from transparent material such as glass, polymer, clear ceramic, crystalline material such as sapphire, one or more sublayers of these materials and/or other materials that have been laminated together (e.g., using adhesive, etc.), etc. Configurations in which layer **2.2-92** is a glass layer and layer **2.2-130** is a polymer layer may sometimes be described herein as an example.

[0911] Coatings may be provided on one or more of the layers in display cover layer **2.2-92**. As shown in the illustrative configuration of FIGS. **2.2-14**, display cover layer **2.2-92** may include, for example, a layer such as layer **2.2-108** that is formed from one or more sublayers (e.g., layer(s) of glass and/or polymer), a polymer layer that helps provide layer **2.2-92** with safety glass functionality (see, e.g., illustrative polymer film **112**, which has been attached to the inner surface of glass layer **2.2-108** to form a layer of laminated glass), and coating **2.2-110** on the front (outwardly facing) surface of layer **2.2-92** (e.g., the outer surface of glass layer **2.2-108**). Coating **2.2-110** may be, for example, an antireflection coating formed from one or more inorganic dielectric layers and/or other layers with thicknesses and refractive index values selected to minimize visible light reflections from the outermost surface of layer **2.2-92** and help maintain a desired appearance (e.g., a neutral tint) for layer **2.2-92**. If desired, display panel **2.2-14P** may be a touch sensitive display (e.g., a display that is overlapped by or incorporates capacitive touch sensor circuitry). In configurations in which display **2.2-14F** is touch sensitive, the outermost surface of layer **2.2-92** may be coated with an oleophobic coating layer (e.g., a fluoropolymer layer).

[0912] To help strengthen layer **2.2-92**, layer **2.2-108** may be formed from chemically strengthened glass (e.g., a glass layer that has been treated in an ion-exchange bath to place the exterior surfaces of the glass layer under compression relative to the interior of the glass layer). This may help layer **2.2-108** resist scratching and cracks. Layer **2.2-108** may be formed from a single glass layer, a single polymer layer, a stack of two laminated glass layers (e.g., first and second glass layers laminated together with a layer of polymer), a stack of two polymer layers, three or more polymer and/or glass layers, etc. If desired, layer **2.2-108** may be formed from a hybrid stack of layers that includes one or more glass layers attached to one or more polymer layers. As an example, layer **2.2-92** may include a rigid structural polymer layer that is covered with a thin glass layer (e.g., a glass layer attached to the structural polymer layer using heat and/or pressure or a glass layer attached to the structural polymer layer using a layer of polymer adhesive). The thin glass layer in this type of arrangement may help protect the structural polymer layer from scratches.

[0913] One or more of the structures in layer **2.2-92** (e.g., coating **2.2-110**, the layer(s) forming layer **2.2-108**, layer **2.2-112**, optional layer **2.2-130**, etc.) may, if desired, be provided with a dye, pigment, or other colorant that creates a desired neutral tint (e.g., gray or black) or non-neutral tint (e.g., red). Thin metal coatings, polarizers, and/or other structures may also be incorporated into layer **2.2-92** to help provide layer **2.2-92** with desired optical properties and/or to provide layer **2.2-92** with a desired external appearance.

[0914] If desired, the portion of layer **2.2-92** that overlaps optical components **2.2-104** and/or other portions of layer **2.2-92** may be provided with a coating that helps prevent scratches that could adversely affect optical quality for components **2.2-104**. As shown in FIGS. **2.2-15**, for example, display cover layer **2.2-92** may have a transparent layer such as transparent layer **2.2-116** (e.g., one or more layers of polymer, glass, and/or other transparent layers such as layer **2.2-108** of FIGS. **2.2-14**). Transparent layer **2.2-116** may be covered with one or more coating layers such as coating layer **2.2-118**. Layer **2.2-118** may be a thin-film layer formed from an inorganic material (e.g., an oxide, nitride, diamond-like carbon etc.) that helps resist scratches. This type of approach may be used, for example, to ensure that the portion of display cover layer **2.2-92** that overlaps optical component **2.2-104** does not become hazy from scratches when layer **2.2-116** is formed from a material such as polymer that may be prone to scratching when exposed to excessive rubbing from sharp external objects. Layer **2.2-118** may sometimes be referred to as a hard coat and may have a higher hardness (e.g., a higher Mohs hardness) than layer **2.2-116**. Layer **2.2-118** may be a thin-film coating with a thickness of less than 3 microns, less than 2 microns, less than 1 micron, less than 0.5 microns, or other suitable thickness.

[0915] Another way in which to help prevent undesired scratches on the surface of display cover layer **2.2-92** where layer **2.2-92** overlaps optical components **2.2-104** is illustrated in the cross-sectional side view of display cover layer **2.2-92** of FIGS. **2.2-16**. As this example demonstrates, the outer surface of display cover layer **2.2-92** may be provided with a recess such as recess **2.2-120** (e.g., a shallow circular depression or a depression with a rectangular shape or other footprint). This places recessed display cover layer surface **2.2-124** of recess **2.2-120** below surrounding external surfaces **2.2-122** of layer **2.2-92**. When device **2.2-10** is laid on a tabletop or other surface, the unrecessed portion of the surface of layer **2.2-92** (external surface **122**) will contact the tabletop surface and will thereby help prevent the tabletop surface from contacting the recessed portion of the surface of layer **2.2-92** (surface **2.2-124**). As a result, recessed surface **2.2-124**, which overlaps component **104**, will remain free of scratches. Haze will therefore not generally develop in the area of layer **2.2-92** that overlaps component **104**, even when layer **2.2-92** is exposed to excessive wear.

[0916] Layer **2.2-92** may be formed from materials having optical properties that are compatible with overlapped optical components **2.2-104**. For example, if an optical component that is overlapped by a portion of layer **2.2-92** in inactive area **IA** is configured to operate at visible and infrared wavelengths, that portion of layer **2.2-92** may be provided with sufficient visible light and infrared light transparency to allow the overlapped component to operate satisfactorily at visible and infrared wavelengths. In arrangements in which the material from the bulk of layer **2.2-92** does not have desired optical properties for an optical component, an optical component window member (e.g., a disk of window material such as a disk of infrared-transparent and, if desired, visible-transparent glass or other inserted window member) may be mounted within an opening in layer **2.2-92** overlapping the optical component.

[0917] Consider, as an example, an arrangement in which layer **2.2-92** is transparent to visible light but has low transmission at infrared wavelengths. An optical component in this type of arrangement may operate at infrared wavelengths. To ensure that the optical component can transmit and/or receive infrared light through layer **2.2-92**, layer **2.2-92** may be provided with a through-hole opening and an infrared-transparent optical component window member such as an infrared-transparent disk. The infrared-transparent window member may be formed from a different material than the material forming layer **2.2-92** and may be mounted within the through-hole

opening in layer **2.2-92**. This type of arrangement is shown in the cross-sectional side view of FIGS. **2.2-17** in which display cover layer **2.2-92** has been provided with optical component window member **2.2-92W** in a through-hole opening in layer **2.2-92**. Member **2.2-92W** may be a glass optical component window member that is transparent to infrared light (and optionally transparent to visible light), whereas surrounding portions of layer **2.2-92** may be formed from different material (e.g., polymer, different glass material, etc.). By providing an infrared-transparent window in layer **2.2-92**, the infrared optical component (e.g., optical component **2.2-102** of FIGS. **2.2-17**) can transmit and/or received infrared light through display cover layer **2.2-92** (e.g., through the window in the display cover layer), even when layer **2.2-92** has been formed from materials that are not infrared-transparent. This approach may be used to provide an optical component window with any suitable optical properties that are different than those of the rest of layer **2.2-92** (e.g., desired amounts of opacity, light transmission, reflection, absorption, and/or haze level, desired polarization properties, etc.).

2.3: Systems with Supplemental Illumination

[0918] FIGS. **2.3-1** is a cross-sectional side view of a head-mounted device in an illustrative configuration in which the device includes an illumination system for providing environmental illumination. Head-mounted device **2.3-10** of FIGS. **2.3-1** may have optical sensors. These sensors may include cameras. The cameras of device **2.3-10** may have lenses and image sensors that are configured to capture images at ultraviolet light wavelengths, visible light wavelengths, and/or infrared wavelengths.

[0919] Some cameras (e.g., cameras of the type that may sometimes be referred to as scene cameras) may be used for capturing images of a user's environment that are displayed on displays **2.3-14** in real time (e.g., real-time pass-through video). Cameras in device **2.3-10** may also be used in tracking the positions and movements of external objects. As an example, tracking cameras may track a user's hand (see, e.g., hand **2.3-30H**) or the user's torso or other body part (see, e.g., user body part **2.3-30B**). Hand gesture input may, as an example, be used in controlling operation of device **2.3-10**. Body part monitoring may be used to allow a user's body motions to be replicated by content displayed in a virtual environment. If desired, cameras may also be used in tracking the position of external accessories (e.g., the position and movement of controllers that are moved by a user to control device **2.3-10**). In some scenario, visual inertial odometry (VIO) systems or other systems that determine the position, movement, and/or orientation of device **2.3-10** relative to the environment surrounded by device **2.3-10** may be formed by combining data from one or more cameras in device **2.3-10** with additional sensor data (e.g., data from an inertial measurement unit). Cameras may perform dedicated functions (tracking, visual inertial odometry functions, scene capture, ranging, three-dimensional image capture for facial recognition and environment mapping, etc.) or two or more of these operations may be performed by a shared camera.

[0920] It may be desirable to allow a user of device **2.3-10** to operate device **2.3-10** in low lighting conditions. As an example, a user may be viewing content on displays **14** while in a dark room or dark vehicle interior. To ensure that camera tracking functions such as hand tracking, body tracking, accessory tracking, and optionally other camera-based functions (e.g., visual inertial odometry, etc.) can be performed satisfactorily, device **2.3-10** may provide supplemental illumination. The supplemental illumination may be provided by light sources that produce supplemental ultraviolet light, supplemental visible light, and/or supplemental infrared light to augment any ambient light that is available. In an illustrative configuration, supplemental illumination is provided at infrared wavelengths, as this light is detectable by tracking cameras or other cameras with infrared sensing capabilities and is invisible to the human eye. Because supplemental infrared illumination is invisible, people in the vicinity of the user of device **2.3-10** (e.g., people in the same room or vehicle as the user) will not be disturbed by the presence of the supplemental illumination.

[0921] Any suitable light sources may be used in forming the supplemental illumination system for

device **2.3-10** (e.g., light-emitting diodes, lasers, etc.). In an illustrative configuration, these light-emitting devices are laser diodes or light-emitting diodes that emit infrared light at a wavelength of 940 nm or other infrared wavelength (e.g., one or more wavelengths such as 740-1500 nm, at least 800 nm, 940 nm, at least 900 nm, 800-1200 nm, 900-1000 nm, 750-1100 nm, 800-1100 nm, less than 1500 nm, etc.). There may be N cameras that use supplemental illumination in device **2.3-10** and M supplemental light sources. The values of N and M may be 1-10, at least 2, at least 3, at least 4, at least 6, at least 8, 2-10, 4-6, 2-4, less than 10, less than 5, less than 4, or other suitable values. The value of N may be larger than the value of M, the value of N may be equal to the value of M, or the value of N may be less than the value of M. As one example, there may be four cameras that use supplemental infrared illumination and there may be two light sources that emit supplemental illumination.

[0922] The cameras that use the supplemental infrared illumination may be configured to be sensitive at the wavelengths illuminated by the supplemental illumination system (e.g., the infrared light wavelengths associated with the M supplemental light sources). The cameras may also be sensitive at visible light wavelengths so that when ample visible ambient light illumination is present, the cameras can operate without any supplemental illumination. To help avoid infrared interference during normal ambient lighting conditions, the supplemental illumination system may, as an example, be configured to emit light in a narrow infrared band (e.g., 940 nm) and the cameras may be provided with filters that pass visible light while blocking all infrared light except light in the narrow infrared band. In another illustrative configuration, the cameras are sensitive across the visible spectrum (e.g., 380 to 740 nm) and into the infrared spectrum (e.g., 740-1000 nm, or other suitable broader infrared wavelength band in which the infrared supplemental illumination is produced). If desired, switchable filters may be used to block infrared light from the cameras when supplemental infrared illumination is not being used and that pass infrared light when supplemental infrared illumination is being used.

[0923] As shown in FIGS. **2.3-1**, the right-hand side of device **2.3-10** may contain a first camera such as camera **2.3-50** that faces in a direction such as direction **2.3-54** (e.g., in the -Z direction and slightly in the +Y direction as an example) and may contain a second camera such as camera **2.3-52** (sometimes referred to as a forward-facing camera) that faces in a forward direction such as direction **2.3-56** (e.g., in the +Y direction and slightly in the -Z direction as an example). The left-hand side of device **2.3-10** may have a corresponding pair of cameras that are oriented in the same way. The angles of view of the cameras on the left and right sides may be configured to overlap in front of device **2.3-10**, so that there are no gaps in coverage in front of the user. If desired, cameras **2.3-50** and **2.3-52** may be replaced by a single camera (e.g., a camera in the position of camera **2.3-52**, a camera in the position of camera **2.3-50**, or a camera in another suitable forward-facing and/or downward-facing orientation that captures images while viewing outwardly from a location on front side F of device **2.3-10**). There may be, for example, a single tracking camera (e.g., camera **2.3-52**) on the right side of device **2.3-10** and a corresponding single tracking camera on the left side of device **2.3-10**.

[0924] Regardless of the number of tracking cameras provided on each side of device **2.3-10**, there may be a right-hand infrared light source such as light source **2.3-58** that provides supplemental illumination (infrared light) in direction **2.3-60** to illuminate objects such as hand **2.3-30H**, body **2.3-30B**, and other external objects for the tracking camera(s) on the right-hand side of device **2.3-10** and there may be a corresponding left-hand infrared light source that provides supplemental infrared light for the tracking camera(s) on the left side of device **2.3-10**. The use of a single supplemental infrared light source on each side of device **2.3-10** to provide supplemental illumination for the tracking camera(s) on that side of device **2.3-10** may help to conserve space within the tight confines of housing **2.3-26**.

[0925] The supplemental illumination system of device **2.3-10** may provide infrared illumination in an area (range of angles) that is larger than the area (range of angles) covered by the tracking

camera(s) of device **2.3-10**, that is equal in area to the area covered by the camera(s), or that is smaller than the area covered by the camera(s).

[0926] Consider, as an example the coverage of the supplemental illumination system of device **2.3-10** of FIGS. **2.3-1** within the YZ plane. As shown in the side view of FIGS. **2.3-1**, downward-facing camera **2.3-50** may be characterized by an angle of view **A1** in the YZ plane and forward-facing camera **2.3-52** may be characterized by an angle of view **A3** in the YZ plane. These angles of view may overlap to provide continuous tracking coverage in the YZ plane. If desired, the same amount of coverage in the YZ plane or another suitable amount of coverage may be provided using a single tracking camera. The example of FIGS. **2.3-1** is illustrative.

[0927] Supplemental illumination from light source **2.3-58** may be characterized by an illumination angle **A2** in the YZ plane. The value of **A2** may be larger than, equal to, or smaller than the combined angle-of-view of cameras **2.3-50** and **2.3-52** or may be larger than, equal to, or smaller than the angle-of-view of a single tracking camera being used in place of cameras **2.3-50** and **2.3-52**. In an illustrative configuration, **A2** is smaller than the overall angle of view of the tracking camera(s) and is directed outwardly in a forward and downward direction in front of device **2.3-10** (where hand and body tracking is most likely to take place). The use of a somewhat reduced illumination area for the supplemental illumination system (e.g., an area of illumination that is less than the area covered by the tracking camera system) may help to conserve power when operating for extended periods of time in dark operating environments while preserving the ability to track objects in all but peripheral areas.

[0928] FIGS. **2.3-2** is a top view of device **2.3-10** showing how device **2.3-10** may contain cameras on both the left and right sides of support structure **2.3-26**. The center of housing portion **2.3-26M** may contain nose bridge portion **2.3-26NB**. Nose bridge portion **2.3-26NM** may have a lower edge with a curved shape configured to rest on a user's nose while device **2.3-10** is worn on a user's face. Nose bridge portion **2.3-26NB** may couple right housing portion **2.3-26R** to left housing portion **2.3-26L**. Optical components **2.3-62** may include side-facing visible light cameras, forward-facing visible light cameras, a time-of-flight camera (e.g., a time-of-flight sensor in nose bridge portion **2.3-26NM** that faces forward), three-dimensional structured light cameras (e.g., left and right structured light cameras adjacent to nose bridge portion **2.3-26NB**), a flicker sensor for detecting ambient light fluctuations (e.g., 60 Hz fluctuations associated with indoor artificial lighting), an ambient light sensor, etc.

[0929] Right camera **2.3-52** may be supported in right housing portion **2.3-26R** and corresponding left camera **2.3-52'** may be supported in left housing portion **2.3-26L**. Similarly, an optional additional right camera such as camera **2.3-50** of FIGS. **2.3-1** may be supported in right housing portion **2.3-26R** and a corresponding optional additional left camera may be supported in left housing portion **2.3-26L**. In this type of configuration, supplemental illumination for the single right-side tracking camera or the pair of right side tracking cameras may be provided by right supplemental light source **2.3-58** and supplemental illumination for the left side camera(s) may be provided by left supplemental light source **2.3-58'**.

[0930] During supplemental illumination operations, light sources **2.3-58** and **2.3-58'** produce supplemental illumination in directions **2.3-60** and **2.3-60'**, respectively. As described in connection with the relative coverage areas of the cameras and light source of FIGS. **2.3-1**, it is not necessary for the illumination coverage area of the supplemental illumination system to exactly match the coverage area of the cameras. For example, the tracking cameras on each side of device **2.3-10** may be characterized by an angle of view that is larger in the XY plane than the angle of coverage of the associated light source. Arrangements in which the illumination from the supplemental light source on each side of device **2.3-10** is provided over the same range of angles as the angle of view of the cameras or in which the illumination is provided over a wider range of angles than the cameras' angle-of-view may also be used.

[0931] Supplemental illumination may be provided over a relatively large fixed area in a global

fashion or a desired area may be covered by activating or moving a narrower beam of illumination towards or across the desired area. If desired, a dynamic illumination system with a steered or addressable beam of supplemental illumination may steer or activate the illumination beam so that the beam follows a user's hand or other object of interest. In this way, power is not needlessly expended illuminating areas that do not contain objects to track.

[0932] FIGS. 2.3-5 and 2.3-6 are side views of illustrative fixed-area supplemental illumination light sources. Illustrative light source 2.3-58 of FIGS. 2.3-3 has a semiconductor light-emitting device 2.3-70. Device 2.3-70 may be a solid state light-emitting device such as a light-emitting diode, a superluminous light-emitting diode, a resonant cavity light-emitting diode, an edge-emitting light-emitting diode, or a vertical-cavity-surface-emitting diode, may be a diode-pumped laser such as a diode-pumped fiber laser or other diode-pumped laser, etc. As shown in FIG. 2.3-3, device 2.3-70 may be mounted on optional interposer 2.3-72 (e.g., using solder). Interposer 2.3-72 may be mounted to package substrate 2.3-74 (e.g., a printed circuit). During operation, device 2.3-70 may emit infrared light that is spread over a desired illumination area by one or more optical structures that overlap device 2.3-70. In the example of FIGS. 2.3-3, these optical structures include optional overmolded polymer lens 3.3-76 and optional secondary optical structures such as peanut lens 3.3-78. It is also possible to form curved reflective optical structures on interposer 2.3-76 or substrate 2.3-74 to enhance side and/or back light recollection. The optical structures that overlap device 2.3-70 may be used to shape the light intensity to produce a desirable far-field distribution different than the native light source intensity distribution (e.g., a light-emitting diode with a Lambertian intensity distribution). If desired, safety enhancement structures such as resistive safety traces or capacitive traces may be embedded in optics or may overlap optics or a photodetector may be used to form a closed loop with a safety interlock on light-source drivers (e.g., in connection with module architectures of the types shown in connection with FIGS. 2.3-5 through 2.3-8).

[0933] In the illustrative configuration of FIGS. 2.3-4, light-emitting device 2.3-70 (e.g., a laser) has been mounted under a light spreading structure such as beam shaping layer 2.3-82. Layer 2.3-82 may be supported in light source package 2.3-80. Device 2.3-70 may be mounted in package 2.3-80 on optional interposer 2.3-72 on a printed circuit or other substrate. During operation, device 2.3-70 of FIGS. 2.3-4 may emit infrared light in an upward direction that is spread out laterally by beam shaping layer 2.3-82 to cover a desired illumination area (e.g., $\pm 60^\circ$ or other suitable range of angles).

[0934] In general, any suitable optical components that serve as light spreading structures may overlap device 2.3-70 of FIGS. 2.3-5 and 2.3-6. These optical components may include optical components such as refractive beam shaping optical components, diffractive optics, diffusive optics, optical nanostructures (e.g., thin two-dimensional metamaterial layers such as patterned structures of clear dielectric with subwavelength dimensions that form metasurfaces that are configured to spread the emitted beam), curved reflectors, etc. Multiple devices 2.3-70 may be mounted in a common package and/or multiple packaged devices 2.3-70 may be mounted on a printed circuit adjacent to each other when forming light source 2.3-58. The use of a single light-emitting device 2.3-70 in forming light source 58 in the examples of FIGS. 2.3-5 and 2.3-6 is illustrative.

[0935] FIGS. 2.3-7 and 2.3-8 are side views of illustrative dynamic pattern illuminators that may be used in an illumination system for device 2.3-10. Using light source of the types shown in FIGS. 2.3-7 and 2.3-8, control circuitry 2.3-12 can selectively activate or steer an emitted beam of infrared light so that one or more objects of interest can be provided with targeted supplemental illumination.

[0936] In the example of FIGS. 2.3-5, light source 2.3-58 has an array of light-emitting devices 2.3-70. Devices 2.3-70 may include multiple semiconductor dies mounted on a substrate such as printed circuit 2.3-84 in package 2.3-86, may include multiple individually addressable emitters, or

may include multiple individually addressable segments of emitters mounted on a substrate such as silicon, ceramic, printed circuit board **2.3-84**, or other substrate in package **2.3-86**. A zoned beam shaper layer or other optical component such as layer **2.3-88** may overlap devices **2.3-70**.

[0937] Layer **2.3-88** may have multiple zones each with a respective beam steering and beam shaping optical structure. These structures may be refractive structures, diffractive structures, nanostructures, etc. Structures on both surfaces of layer **2.3-88** and/or multiple layers of layer **2.3-88** with vertically aligned or misaligned zones may be employed. Each zone may be used to steer and shape a beam of light emitted from a respective light-emitting device in a different respective direction. For example, a first zone may direct a beam of light that has been emitted vertically from a first device **2.3-70** to the left, whereas a second zone may direct a beam of light that has been emitted vertically from a second device **2.3-70** to the right. By overlapping an array of individually controlled devices **70** with a corresponding array of individualized beam steering structures, each device **2.3-70** can be configured to emit a beam of light in a different respective direction (see, e.g., illustrative beams **2.3-90**), providing light source **2.3-58** of FIGS. **2.3-5** with the ability to emit a steered beam of light. The emission area of each beam may overlap with adjacent beams to avoid potential gaps in coverage. Beams **2.3-90** may all be emitted simultaneously or one or more selected beams **2.3-90** may be emitted at a time. If desired, beams **2.3-90** may be emitted in sequence (e.g., to scan the emitted beam from light source **58** across an area of interest).

[0938] Another illustrative light source that may be used in forming a dynamic pattern illuminator for the supplemental illumination system of device **2.3-10** is shown in FIGS. **2.3-6**. Light source **2.3-58** of FIGS. **2.3-6** may have one or more light-emitting devices such as device **2.3-70** that emit one or more beams of light such as light beam **2.3-92** (e.g., an infrared light beam). Device **2.3-70** may be mounted on a printed circuit or other substrate **2.3-94** in package **2.3-96**. Electrically controlled beam steerer **2.3-98** may have one or more beam steerers such as steerable microelectromechanical systems mirror **2.3-100** or other electrically adjustable beam steering element(s) controlled by control signals from control circuitry **2.3-12**. When it is desired to emit light in a first direction, mirror **2.3-100** may be placed in a first orientation that reflects beam **2.3-92** to create first emitted beam **2.3-102**. When it is desired to emit light in a second direction, mirror **2.3-100** may be placed in a second orientation that is different than the first orientation, thereby reflecting beam **2.3-92** to create second emitted beam **2.3-104**. Mirror **2.3-100** may be placed in any suitable number of different orientations (e.g., at least 2, at least 10, at least 25, at least 100, less than 5000, less than 1000, less than 500, or other suitable number). Mirror **2.3-100** may be rotated about a single axis (to change the angle of emitted light beams along a single dimension) or may be rotated about two axes (e.g., to change the angle of emitted light beams arbitrarily in two dimensions). If desired, beam shaping optics (e.g., beam collimating lenses, etc.) may be incorporated into beam steerer **2.3-98** to help ensure that the steered beam has a desired intensity profile.

[0939] If desired, a hybrid illuminator architecture may be employed, such that multiple channels of device **2.3-70** or multiple device **2.3-70** as described in connection with FIGS. **2.3-5** can be selectively activated to provide one or more additional dimensions of dynamic illumination to beam steering optics such as mirror **2.3-100** of FIGS. **2.3-6**.

[0940] Light sources that emit static wide-area beams (see, e.g., illustrative light sources **2.3-58** of FIGS. **2.3-5** and **2.3-6**) may be configured to emit light beams of any suitable shape to help provide supplemental illumination for the tracking cameras of device **2.3-10**. FIGS. **2.3-7** is a graph showing how light source **2.3-58** may be configured to emit a circular beam field of regards (FoG) such as circular beam **2.3-110** (e.g., a beam of infrared light with full width half maximum (FWHM) intensity characterized by an angular spread of $\pm 60^\circ$ or other suitable coverage area) or may be configured to emit a rectangular beam FoG such as rectangular beam **2.3-112** with a similar angular spread vertically and a smaller angular spread horizontally. Two rectangular beams such as beam **2.3-112** may be produced side by side to provide sufficient horizontal illumination coverage

for both the left and right cameras in device **2.3-10** (as an example).

[0941] In most general use cases, a goal of the illumination system is to provide a uniform signal-to-noise ratio for the illuminated scene captured by one or more cameras. Within the desired FWHM 2-D FOG, a uniform far-field intensity at each instantaneous FOG (iFoG) can be achieved to provide uniform illumination and working range for the cameras. However, there are cases when non-uniform far-field intensity distributions may be desired. For example, when a target of the illumination is flat or when camera vignetting is significant, a symmetric “bat-wing” intensity distribution may be used to compensate for the relative intensity fall-off of the camera image sensor. Further examples include asymmetric intensity distribution for cameras that are aligned with a non-co-axial orientation relative to the illumination system, for targets such as hands that have asymmetric occurrence/residence across FoGs, for multiple illuminators with overlapping FoGs, for multiple non-co-axial cameras, for irregular occlusions at certain FOG regions, etc.

[0942] The graphs of FIGS. **2.3-10** and **2.3-11** show illustrative beam outputs (angular beam distributions) associated with a dynamically adjustable illumination system. In the example of FIGS. **2.3-8**, a light source such as light source **58** of FIGS. **2.3-5** or light source **58** of FIGS. **2.3-6** has been configured to produce a beam with an elongated rectangular shape (e.g., a rectangle having a larger horizontal spread than vertical spread). Using beam steering, light source **2.3-58** may emit this elongated rectangular beam in one or more vertical locations, such as illustrative location **2.3-114** of FIGS. **2.3-8**. In an arrangement of the type shown in FIGS. **2.3-5**, each light-emitting device **2.3-70** may produce a different respective elongated rectangular beam each of which is associated with a different vertical position in the output of light source **2.3-58**. One or more of these beams may be emitted at the same time by turning on one or more respective light-emitting devices **2.3-70**. In an arrangement of the type shown in FIGS. **2.3-6**, light-emitting device **2.3-70** may produce a beam such as beam **2.3-92** of FIGS. **2.3-6** that is steered to a desired location (e.g., illustrative location **2.3-114** of FIGS. **2.3-8**) and/or to other locations by beam steerer **2.3-98**, thereby providing a desired coverage for light source **2.3-58**.

[0943] In the illustrative example of FIGS. **2.3-8**, light is output over a larger vertical angular range than horizontal range. Additional horizontal coverage may be supplied using an additional light source (e.g., a light source on an opposing side of device **2.3-10**). In this way, a desired angular output range (e.g., $\pm 60^\circ$ in both horizontal and vertical dimensions or other suitable angular output range) may be covered.

[0944] In the illustrative configuration of FIGS. **2.3-9**, light source **2.3-58** (e.g., a dynamically configured light source such as light source **58** of FIGS. **2.3-5** or FIGS. **2.3-6**) is configured to supply a relatively small circular or square output beam that can be steered in both horizontal and vertical dimensions so that a desired overall amount of coverage is produced.

[0945] In both those light sources that are static and do not have steerable beams and in those light sources with dynamically patterned output, beam power can be controlled in a binary fashion (on/off) or in an analog fashion (e.g., by adjusting output power continuously or in a stepwise fashion between more than two different output levels). As shown in FIGS. **2.3-9**, for example, no light may be output in certain portions of a coverage area such as areas **2.3-116** (e.g., beam power may be zero for these areas), full power light may be output in areas such as areas **2.3-118** (e.g., beam power may be maximized for these areas), and an intermediate power level may be used when supplying output light to other areas such as areas **2.3-120** that are immediately adjacent to areas **2.3-118**.

[0946] Arrangements in which full-power light is only output in a subset of the total coverage area for light source **58** may help device **2.3-10** use power efficiently. As shown in the diagram of FIGS. **2.3-9**, for example, there may be one or more external objects of interest such as objects **2.3-122** within the coverage area of a given light source. Device **2.3-10** may, as an example, be tracking a user's hands or other external objects. When these objects are relatively small compared to the overall angle-of-view of the cameras in device **2.3-10**, power can be conserved by restricting the

output of supplemental illumination (or at least restricting the output of full-power supplemental illumination) to only those regions that overlap the tracked external objects.

[0947] In the FIGS. 2.3-9 example, objects 2.3-122 (e.g., the user's hands or other body part or other objects in the user's environment) are being actively tracked by device 2.3-10. As a result, the supplemental illumination system of device 2.3-10 is being used to provide full-power illumination to areas 2.3-118 that overlap objects 2.3-122. Elsewhere in the coverage area of light-emitting device 58, beam power is reduced (see, e.g., intermediate power areas 2.3-120) or shut off entirely (see, e.g., unilluminated areas 2.3-116). This type of approach may be used for either scanned beam arrangements (e.g., using a scanning mirror device or other beam steerer as described in connection with FIGS. 2.3-6) or using light sources with addressable arrays of devices 70 each of which can provide output in different directions (e.g., light source 58 of FIGS. 2.3-5).

[0948] In areas such as areas 2.3-116 of FIGS. 2.3-9, no supplemental illumination is present, so items in those areas will not receive supplemental illumination. Nevertheless, once objects such as objects 2.3-122 are being tracked, device 2.3-10 can monitor the position and direction of movement of objects 2.3-122 in real time. This allows device 2.3-10 to provide supplemental illumination of full power to the areas overlapping objects 2.3-122 and intermediate power (or, if desired, full power), to portions of the output area of light source 58 that are immediately adjacent to objects 2.3-122 (e.g., areas where objects 2.3-122 may possibly move and/or are predicted to occupy in the near future based on tracked movements). In the event that the positions of objects 2.3-122 move into one of those adjacent areas, device 2.3-10 can increase the supplemental illumination on those areas to full power and can update the beam powers so that adjacent areas again have intermediate power level coverage.

[0949] Although the multi-power-level beam scheme of FIGS. 2.3-9 has been described in connection with a two-dimensional scanning light beam from light sources 2.3-58 of FIGS. 7 and 8, such adjustable power output schemes may also be used with light sources 2.3-58 that provide one-dimensional adjustable direction light sources (e.g., light sources that produce slices of supplemental illumination of the type shown in FIGS. 2.3-8) and/or may be used with fixed-area light sources. In a fixed-area light source scheme, for example, a right-hand light source 58 of the type shown in FIGS. 2.3-3 or FIGS. 2.3-4 may be used to supply supplemental illumination for tracking objects 2.3-122 that are in front of the right-hand camera(s) of device 2.3-10, whereas a left-hand light source 58 of the type shown in FIGS. 2.3-3 or FIGS. 2.3-4 may be used to supply supplemental illumination for tracking objects 2.3-122 that are in front of the left-hand camera(s) of device 2.3-10. Device 2.3-10 can activate either the right-hand light-source or the left-hand light source or both depending on the current and expected locations of objects 2.3-122.

[0950] Another way in which to help use power efficiently for the supplemental illumination system involves using light sources 2.3-58 to produce supplemental illumination only when the cameras for which the supplemental illumination is being provided will benefit from the supplemental illumination. For example, in bright lighting conditions, ambient visible light will provide sufficient illumination, so supplemental infrared light beams can be turned off (or at least reduced in power to a lower level than otherwise used) to help conserve power. The activation of supplemental lighting may take place when dim ambient lighting conditions are detected or when other suitable conditions are detected to trigger the production of supplemental lighting.

[0951] FIGS. 2.3-10 is a flow chart of illustrative operations involved in using electronic device 2.3-10. During the operations of block 2.3-150, device 2.3-10 may be used to provide a user with content such as visual content, audio content, and other output. Device 2.3-10 may, as an example be worn on a user's head while images are presented for viewing. The operations of block 2.3-150 may be performed while device 2.3-10 is in a normal operating environment with satisfactory visible ambient light levels.

[0952] Visual content may be presented for the user on displays 2.3-14. This visual content may include camera images from cameras in device 2.3-10 (e.g., pass-through video) and/or other

content. In some scenarios, computer-generated content (sometimes referred to as virtual content) may be overlaid on top of real-world content from cameras in device **2.3-10**. In this type of mixed reality environment, camera data may be used to help track the locations of the user's hands and other real-world objects and thereby help register the overlaying of virtual content on real-world images. For example, by tracking the location of a user's hand, a computer-generated image of a glove may be accurately overlaid on top of a real-world image of the user's hand. By tracking the location of a table surface, a computer-generated image may be placed on top of the table surface. Camera data can be used to track the motion of a user's hands, fingers, and/or other body parts in real time. In this way, hand gestures, finger gestures, and/or other body part motions that serve as user input (sometimes referred to as air gestures) can be used in controlling the operation of device **2.3-10** (e.g., in a mixed-reality or completely virtual environment).

[0953] Device **2.3-10** may have any suitable number of cameras including three-dimensional cameras (e.g., structured light cameras, time-of flight cameras, etc.), cameras for capturing real-world visible-light images (e.g., for video passthrough), and/or cameras that perform tracking operations, that serve as parts of visual inertial odometry systems, and/or that otherwise support the operation of device **2.3-10**. The cameras of device **2.3-10** may face forward, down, to the side, up, to the rear, and/or in multiple directions. Some cameras may operate only at visible wavelengths. Other cameras may operate at visible and infrared wavelengths.

[0954] As described in connection with FIGS. **3** and **4**, device **2.3-10** may, as an example, have a one or more tracking cameras on each side of device **2.3-10**. These cameras may be sensitive at visible and infrared wavelengths and may be used for tracking operations (e.g., hand and body tracking, air gesture input tracking, accessory tracking) and optionally additional functions such as imaging structures in the user's environment for a visual inertial odometry system). Tracking cameras may be sensitive at visible and infrared wavelengths such as wavelengths from 400-1000 nm, 400-740 nm and 940 nm, or other suitable visible and infrared wavelengths. The infrared sensitivity of the tracking cameras preferably coincides with the wavelength or wavelengths emitted by light sources **2.3-58** in the supplemental illumination system, allowing these cameras to operate when most or all available illumination is being provided by light sources **2.3-58** rather than ambient light sources.

[0955] Supplemental illumination may, if desired, be provided continuously. Arrangements in which power is conserved by at least occasionally depowering the supplemental illumination system are described herein as an example. In configurations for device **2.3-10** in which supplemental illumination is turned on and off, device **2.3-10** may, during the operations of block **2.3-150**, monitor for the occurrence of conditions indicating that supplemental illumination should be switched on for satisfactory operation of the cameras (e.g., the tracking cameras). These monitoring activities may take place while the cameras of device **2.3-10** (e.g., the tracking cameras) are operating normally in the absence of supplemental illumination from the supplemental illumination system.

[0956] Any suitable trigger criteria may be used to determine when to activate the supplemental illumination system by turning on light sources **2.3-58**. As an example, device **2.3-10** may contain an ambient light sensor. The ambient light sensor may measure the amount of visible ambient light that is present in the environment surrounding device **2.3-10**. A threshold or other criteria may be applied to ambient light readings from the ambient light sensor. In response to determining that ambient light levels are below a predetermined ambient light threshold or are otherwise too dim for satisfactory operation of the tracking cameras, control circuitry **12** can turn on light sources **2.3-58** to provide supplemental illumination (e.g., infrared light).

[0957] Another illustrative criteria that may be used in determining when to activate supplemental illumination involves evaluating an image processing algorithm quality metric. During the operations of block **2.3-150**, captured images may be processed by one or more image processing algorithms. These algorithms may include, as an example, a hand tracking algorithm. The hand

tracking algorithm may produce a quality factor or other metric that is indicative of the ability of the hand tracking algorithm to satisfactorily track the user's hands. In response to detecting that the tracking algorithm quality metric is below a desired threshold value, control circuitry **12** can turn on light sources **2.3-58** to provide supplemental illumination for the cameras.

[0958] If desired, the tracking cameras or other image sensor hardware may supply information indicating that performance is being adversely affected by low ambient lighting levels. As an example, frames of image data may be evaluated to determine whether lighting levels are low. The output of the tracking camera hardware of device **2.3-10** may also indicate whether signal-to-noise levels are satisfactory. If the tracking cameras are producing only dark and/or noisy image data, control circuitry **12** can determine that light sources **2.3-58** should be turned on.

[0959] In some arrangements, device **2.3-10** may be configured to determine the location of a user relative to walls and other obstructions in the user's environment. As an example, device **2.3-10** may contain a map of known wall locations (e.g., a map obtained from an external source or a map based on previous map-building operations performed by device **2.3-10** when a user wore device **2.3-10** while walking throughout a building or other environment). Satellite navigation system circuitry (e.g., Global Positioning System circuitry) may use satellite signals to determine the location of device **2.3-10** (e.g. the location of device **2.3-10** relative to building walls and other obstructions). From the user's known location and movement and using information on the locations of known obstructions such as walls, device **2.3-10** can predict when a user is likely to approach a wall or other obstruction. Sensors **16** in device **2.3-10** (such as proximity sensors, time of flight sensors, radar, LIDAR, etc.) may also be used in monitoring the user's movements relative to walls and other obstructions. By using some or all of this information in combination with additional information on the operating environment for device **2.3-10** (e.g., ambient light readings indicating that ambient lighting is dim), device **2.3-10** can determine when light sources **2.3-58** should be turned on to provide supplemental illumination to help ensure that the tracking cameras of device **2.3-10** will operate satisfactorily. This may help ensure that the cameras of device **2.3-10** can track the locations of obstructions in the user's environment using the infrared illumination of light sources **2.3-58**. By tracking the locations of obstructions accurately, these obstructions or alerts regarding the presence of the obstructions can be displayed on displays **2.3-14** to help the user avoid undesired collisions with the obstructions.

[0960] If desired, multiple electronic devices **2.3-10** in system **2.3-8** may monitor for conditions indicating that supplemental illumination is needed. For example, multiple users may be wearing head-mounted devices and one device may detect low levels of ambient lighting before another. In this type of system, any of the devices that detect a low level of ambient lighting can signal the other devices in the system to request that supplemental illumination be provided. In response, one or more of the other devices may provide supplemental illumination to assist the cameras of the requesting device in gathering images. The supplemental illumination systems of different devices may therefore assist each other by contributing shared supplemental illumination. This may allow a wall-powered device to help provide supplemental illumination for a battery powered device or may allow an electronic device that is close to a tracked object to provide supplemental illumination to that object more efficiently than an electronic device that is farther from the tracked object (as examples).

[0961] So long as conditions for triggering supplemental illumination are not detected, device **2.3-10** (e.g., control circuitry **12**) may continue to monitor for conditions that satisfy supplemental illumination trigger criteria (e.g., dim ambient lighting, reduction of tracking camera image processing quality, reduction of camera hardware performance, criteria based on obstruction proximity, requests from other devices, etc.) during the operations of block **2.3-150**.

[0962] In the event that the trigger criteria are satisfied, processing may proceed to block **2.3-152**. During the operations of block **2.3-152**, control circuitry **2.3-14** can use the supplemental illumination system to provide supplemental illumination for the cameras (e.g., infrared light

emitted by light sources **2.3-58** that illuminates exterior objects in the field of view of the tracking cameras). In providing the supplemental illumination, the power of the infrared light emitted by each light source **2.3-58** and/or the direction of the light beam(s) emitted by each light source **2.3-58** may be adjusted. For example, some devices **2.3-70** may be turned on while other devices **2.3-70** remain off, beams of emitted light may be directed to areas containing tracked objects (e.g., the known locations of the user's hands or other external objects of interest being tracked by the tracking cameras) and/or adjacent areas, emitted power levels may be adjusted in a stepwise fashion or continuously (e.g., so that sufficient supplemental illumination is provided to ensure satisfactory tracking camera operation without providing excess illumination), etc.

[0963] Light sources such as light sources **2.3-58** of FIGS. **2.3-5** and **2.3-6** that are configured to provide illumination over a fixed area may be turned on to ensure that objects in those fixed areas are illuminated. Light sources that emit steerable beams such as light sources **2.3-58** of FIGS. **2.3-5** and **2.3-8** may be used to emit supplemental illumination over a relatively large area (e.g. by scanning a beam across the large area or by simultaneously using multiple smaller beams to illuminate different respective parts of the larger area) or may be used to emit supplemental illumination to particular locations such as the location(s) containing the user's hands or other objects being tracked.

[0964] Supplemental illumination may be provided for cameras that track user body parts, cameras that track the locations of accessories, cameras that capture pass-through video, cameras that form part of a visual inertial odometry system, and/or other optical components that gather light from objects in the vicinity of device **2.3-10**. If desired, light sources **2.3-58** may be configured to emit structured light (e.g., lines, dots, features distributed in pseudorandom patterns, etc.). Structured light may be used, for example, in scenarios in which the tracking cameras gather three-dimensional images.

[0965] During the operations of block **2.3-152**, device **2.3-10** may monitor for conditions that indicate that supplemental illumination is no longer needed. Control circuitry **2.3-12** may, for example, monitor to determine whether supplemental illumination trigger conditions cease to be satisfied. So long as dim ambient lighting conditions or other conditions indicating that supplemental illumination should be provided continue to be present, device **2.3-10** can continue to use light sources **2.3-58** to provide supplemental illumination. In the event that dim lighting conditions cease or that other conditions in which supplemental illumination is desired are determined to no longer be present, device **2.3-10** can turn off the supplemental illumination system. In particular, control circuitry **2.3-12** can turn off light sources **2.3-58** during the operations of block **156**. As indicated by line **2.3-152**, operations may then return to block **2.3-150**.

2.4: Systems with Displays and Sensor-Hiding Structures

[0966] FIGS. **2.4-1** is a front view of device **2.4-10** in an illustrative configuration in which device **2.4-10** has a publicly viewable display such as forward-facing display **2.4-14F**. As shown in FIGS. **2.4-1**, support structure **2.4-16M** of device **2.4-10** may have right and left portions such as portions **2.4-16R** and **2.4-16L** that are coupled by an interposed nose bridge portion such as portion **2.4-16NB**. Portion **2.4-16NB** may have a curved exterior surface such as nose bridge surface **2.4-90** that is configured to receive and rest upon a user's nose to help support main housing portion **2.4-16M** on the head of the user.

[0967] Display **2.4-14F** may have an active area such as active area **AA** that is configured to display images and an inactive area **IA** that does not display images. The outline of active area **AA** may be rectangular, rectangular with rounded corners, may have teardrop shaped portions on the left and right sides of device **2.4-10**, may have a shape with straight edges, a shape with curved edges, a shape with a peripheral edge that has both straight and curved portions, and/or other suitable outlines. As shown in FIGS. **2.4-1**, active area **AA** may have a curved recessed portion at nose bridge portion **2.4-16NB** of main housing portion **2.4-16**. The presence of the nose-shaped recess in active area **AA** may help fit active area **AA** within the available space of housing portion

2.4-16M without overly limiting the size of active area **AA**.

[0968] Active area **AA** contains an array of pixels. The pixels may be, for example, light-emitting diode pixels formed from thin-film organic light-emitting diodes or crystalline semiconductor light-emitting diode dies (sometimes referred to as micro-light-emitting diodes) on a flexible display panel substrate. Configurations in which display **2.4-14F** uses other display technologies may also be used, if desired. Illustrative arrangements in which display **14** is formed from a light-emitting diode display such as an organic light-emitting diode display that is formed on a flexible substrate (e.g., a substrate formed from a bendable layer of polyimide or a sheet of other flexible polymer) may sometimes be described herein as an example. The pixels of active area **AA** may be formed on a display device such as display panel **2.4-14P** of FIGS. **2.4-1** (e.g., a flexible organic light-emitting diode display panel). In some configurations, the outline of active area **AA** (and, if desired, panel **2.4-14P**) may have a peripheral edge that contains straight segments or a combination of straight and curved segments. Configurations in which the entire outline of active area **AA** (and optionally panel **2.4-14P**) is characterized by a curved peripheral edge may also be used.

[0969] Display **2.4-14F** may have an inactive area such as inactive area **IA** that is free of pixels and that does not display images. Inactive area **IA** may form an inactive border region that runs along one or more portions of the peripheral edge of active area **AA**. In the illustrative configuration of FIGS. **2.4-1**, inactive area **IA** has a ring shape that surrounds active area **AA** and forms an inactive border. In this type of arrangement, the width of inactive area **IA** may be relatively constant and the inner and outer edges of area **IA** may be characterized by straight and/or curved segments or may be curved along their entire lengths. For example, the outer edge of area **IA** (e.g., the periphery of display **2.4-14F**) may have a curved outline that runs parallel to the curved edge of active area **AA**.

[0970] In some configurations, device **2.4-10** may operate with other devices in system **2.4-8** (e.g., wireless controllers and other accessories). These accessories may have magnetic sensors that sense the direction and intensity of magnetic fields. Device **2.4-10** may have one or more electromagnets configured to emit a magnetic field. The magnetic field can be measured by the wireless accessories near device **2.4-10**, so that the accessories can determine their orientation and position relative to device **2.4-10**. This allows the accessories to wirelessly provide device **2.4-10** with real-time information on their current position, orientation, and movement so that the accessories can serve as wireless controllers. The accessories may include wearable devices, handheld devices, and other input devices.

[0971] In an illustrative configuration, device **2.4-10** may have a coil such as illustrative coil **2.4-54** that runs around the perimeter of display **2.4-14F** (e.g., under inactive area **IA** or other portion of display **2.4-14F**). Coil **2.4-54** may have any suitable number of turns (e.g., 1-10, at least 2, at least 5, at least 10, 10-50, fewer than 100, fewer than 25, fewer than 6, etc.). These turns may be formed from metal traces on a substrate, may be formed from wire, and/or may be formed from other conductive lines. During operation, control circuitry **2.4-12** may supply coil **2.4-54** with an alternating-current (AC) drive signal. The drive signal may have a frequency of at least 1 kHz, at least 10 kHz, at least 100 kHz, at least 1 MHz, less than 10 MHz, less than 3 MHz, less than 300 kHz, or less than 30 kHz (as examples). As AC current flows through coil **2.4-54**, a corresponding magnetic field is produced in the vicinity of device **2.4-10**. Electronic devices such as wireless controllers with magnetic sensors that are in the vicinity of device **2.4-10** may use the magnetic field as a reference so that the wireless controllers can determine their orientation, position, and/or movement while being moved relative to device **2.4-10** to provide device **2.4-10** with input.

[0972] Consider, as an example, a handheld wireless controller that is used in controlling the operation of device **2.4-10**. During operation, device **2.4-10** uses coil **2.4-54** to emit a magnetic field. As the handheld wireless controller is moved, the magnetic sensors of the controller can monitor the location of the controller and the movement of the controller relative to device **2.4-10** by monitoring the strength, orientation, and change to the strength and/or orientation of the magnetic field emitted by coil **2.4-54** as the controller is moved through the air by the user. The

electronic device can then wirelessly transmit information on the location and orientation of the controller to device **2.4-10**. In this way, a handheld controller, wearable controller, or other external accessory can be manipulated by a user to provide device **2.4-10** with air gestures, pointing input, steering input, and/or other user input.

[0973] Device **2.4-10** may have components such as optical components (e.g., optical sensors among sensors **2.4-16** of FIGS. **2.4-1**). These components may be mounted in any suitable location on head-mounted support structure **2.4-16** (e.g., on head strap **2.4-16B**, on main housing portion **2.4-16M**, etc.). Optical components and other components may face rearwardly (e.g., when mounted on the rear face of device **2.4-10**), may face to the side (e.g., to the left or right), may face downwardly or upwardly, may face to the front of device **2.4-10** (e.g., when mounted on the front face of device **2.4-10**), may be mounted so as to point in any combination of these directions (e.g., to the front, to the right, and downward) and/or may be mounted in other suitable orientations. In an illustrative configuration, at least some of the components of device **2.4-10** are mounted so as to face outwardly to the front (and optionally to the sides and/or up and down). For example, forward-facing cameras for pass-through video may be mounted on the left and right sides of the front of device **2.4-10** in a configuration in which the cameras diverge slightly along the horizontal dimension so that the fields of view of these cameras overlap somewhat while capturing a wide-angle image of the environment in front of device **2.4-10**. The captured image may, if desired, include portions of the user's surroundings that are below, above, and to the sides of the area directly in front of device **2.4-10**.

[0974] To help hide components such as optical components from view from the exterior of device **2.4-10**, it may be desirable to cover some or all of the components with cosmetic covering structures. The covering structures may include transparent portions (e.g., optical component windows) that are characterized by sufficient optical transparency to allow overlapped optical components to operate satisfactorily. For example, an ambient light sensor may be covered with a layer that appears opaque to an external viewer to help hide the ambient light sensor from view, but that allows sufficient ambient light to pass to the ambient light sensor for the ambient light sensor to make a satisfactory ambient light measurement. As another example, an optical component that emits infrared light may be overlapped with a visibly opaque material that is transparent to infrared light.

[0975] In an illustrative configuration, optical components for device **2.4-10** may be mounted in inactive area IA of FIGS. **2.4-1** and cosmetic covering structures may be formed in a ring shape overlapping the optical components in inactive area IA. Cosmetic covering structures may be formed from ink, polymer structures, structures that include metal, glass, other materials, and/or combinations of these materials. In an illustrative configuration, a cosmetic covering structure may be formed from a ring-shaped member having a footprint that matches the footprint of inactive area IA. If, for example, active area AA has left and right portions with teardrop shapes, the ring-shaped member may have curved edges that follow the curved periphery of the teardrop-shaped portions of active area AA. The ring-shaped member may be formed from one or more polymer structures (e.g., the ring-shaped member may be formed from a polymer ring). Because the ring-shaped member can help hide overlapped components from view, the ring-shaped member may sometimes be referred to as a shroud or ring-shaped shroud member. The outward appearance of the shroud or other cosmetic covering structures may be characterized by a neutral color (white, black, or gray) or a non-neutral color (e.g., blue, red, green, gold, rose gold, etc.).

[0976] Display **2.4-14F** may, if desired, have a protective display cover layer. The cover layer may overlap active area AA and inactive area IA (e.g., the entire front surface of device **2.4-10** as viewed from direction **2.4-52** of FIGS. **2.4-1** may be covered by the cover layer). The cover layer, which may sometimes be referred to as a housing wall or transparent housing wall, may have a rectangular outline, an outline with teardrop portions, an oval outline, or other shape with curved and/or straight edges.

[0977] The cover layer may be formed from a transparent material such as glass, polymer, transparent crystalline material such as sapphire, clear ceramic, other transparent materials, and/or combinations of these materials. As an example, a protective display cover layer for display **2.4-14F** may be formed from safety glass (e.g., laminated glass that includes a clear glass layer with a laminated polymer film). Optional coating layers may be applied to the surfaces of the display cover layer. If desired, the display cover layer may be chemically strengthened (e.g., using an ion-exchange process to create an outer layer of material under compressive stress that resists scratching). In some configurations, the display cover layer may be formed from a stack of two or more layers of material (e.g., first and second structural glass layers, a rigid polymer layer coupled to a glass layer or another rigid polymer layer, etc.) to enhance the performance of the cover layer. [0978] In active area AA, the display cover layer may overlap the pixels of display panel **2.4-14P**. The display cover layer in active area AA is preferably transparent to allow viewing of images presented on display panel **2.4-14P**. In inactive area IA, the display cover layer may overlap the ring-shaped shroud or other cosmetic covering structure. The shroud and/or other covering structures (e.g., opaque ink coatings on the inner surface of the display cover layer and/or structures) may be sufficiently opaque to help hide some or all of the optical components in inactive area IA from view. Windows may be provided in the shroud or other cosmetic covering structures to help ensure that the optical components that are overlapped by these structures operate satisfactorily. Windows may be formed from holes, may be formed from areas of the shroud or other cosmetic covering structures that have been locally thinned to enhance light transmission, may be formed from window members with desired light transmission properties that have been inserted into mating openings in the shroud, and/or may be formed from other shroud window structures.

[0979] In the example of FIGS. **2.4-1**, device **2.4-10** includes optical components such as optical components **2.4-60**, **2.4-62**, **2.4-64**, **2.4-66**, **2.4-68**, **2.4-70**, **2.4-72**, **2.4-74**, **2.4-76**, **2.4-78**, and **2.4-80** (as an example). Each of these optical components (e.g., optical sensors selected from among sensors **2.4-16** of FIGS. **2.4-1**, light-emitting devices, etc.) may be configured to detect light and, if desired to emit light (e.g., ultraviolet light, visible light, and/or infrared light).

[0980] In an illustrative configuration, optical component **2.4-60** may sense ambient light (e.g., visible ambient light). In particular, optical component **2.4-60** may have a photodetector that senses variations in ambient light intensity as a function of time. If, as an example, a user is operating in an environment with an artificial light source, the light source may emit light at a frequency associated with its source of wall power (e.g., alternating-current mains power at 60 Hz). The photodetector of component **2.4-60** may sense that the artificial light from the artificial light source is characterized by 60 Hz fluctuations in intensity. Control circuitry **2.4-12** can use this information to adjust a clock or other timing signal associated with the operation of image sensors in device **2.4-10** to help avoid undesired interference between the light source frequency and the frame rate or other frequency associated with image capture operations. Control circuitry **2.4-12** can also use measurements from component **2.4-60** to help identify the presence of artificial lighting and the type of artificial lighting that is present. In this way, control circuitry **2.4-12** can detect the presence of lights such as fluorescent lights or other lights with known non-ideal color characteristics and can make compensating color cast adjustments (e.g., white point adjustments) to color-sensitive components such as cameras and displays. Because optical component **2.4-60** may measure fluctuations in light intensity, component **2.4-60** may sometimes be referred to as a flicker sensor or ambient light frequency sensor.

[0981] Optical component **2.4-62** may be an ambient light sensor. The ambient light sensor may include one or more photodetectors. In a single-photodetector configuration, the ambient light sensor may be a monochrome sensor that measures ambient light intensity. In a multi-photodetector configuration, each photodetector may be overlapped by an optical filter that passes a different band of wavelengths (e.g., different visible and/or infrared passbands). The optical filter passbands

may overlap at their edges. This allows component **2.4-62** to serve as a color ambient light sensor that measures both ambient light intensity and ambient light color (e.g., by measuring color coordinates for the ambient light). During operation of device **2.4-10**, control circuitry **2.4-12** can take action based on measured ambient light intensity and color. As an example, the white point of a display or image sensor may be adjusted or other display or image sensor color adjustments may be made based on measured ambient light color. The intensity of a display may be adjusted based on light intensity. For example, the brightness of display **2.4-14F** may be increased in bright ambient lighting conditions to enhance the visibility of the image on the display and the brightness of display **2.4-14F** may be decreased in dim lighting conditions to conserve power. Image sensor operations and/or light source operations may also be adjusted based on ambient light readings. [0982] The optical components in active area IA may also include components along the sides of device **2.4-10** such as components **2.4-80** and **2.4-64**. Optical components **2.4-80** and **2.4-64** may be pose-tracking cameras that are used to help monitor the orientation and movement of device **2.4-10**. Components **2.4-80** and **2.4-64** may be visible light cameras (and/or cameras that are sensitive at visible and infrared wavelengths) and may, in conjunction with an inertial measurement unit, form a visual inertial odometry (VIO) system.

[0983] Optical components **2.4-78** and **2.4-66** may be visible-light cameras that capture real-time images of the environment surrounding device **2.4-10**. These cameras, which may sometimes be referred to as scene cameras or pass-through-video cameras, may capture moving images that are displayed in real time to displays **2.4-14R** for viewing by the user when the user's eyes are located in eye boxes **2.4-24** at the rear of device **2.4-10**. By displaying pass-through images (pass-through video) to the user in this way, the user may be provided with real-time information on the user's surroundings. If desired, virtual content (e.g., computer-generated images) may be overlaid over some of the pass-through video. Device **2.4-10** may also operate in a non-pass-through-video mode in which components **2.4-78** and **2.4-66** are turned off and the user is provided only with movie content, game content, and/or other virtual content that does not contain real-time real-world images.

[0984] Input-output devices **2.4-12** of device **2.4-10** may gather user input that is used in controlling the operation of device **2.4-10**. As an example, a microphone in device **2.4-10** may gather voice commands. Buttons, touch sensors, force sensors, and other input devices may gather user input from a user's finger or other external object that is contacting device **2.4-10**. In some configurations, it may be desirable to monitor a user's hand gestures or the motion of other user body parts. This allows the user's hand locations or other body part locations to be replicated in a game or other virtual environment and allows the user's hand motions to serve as hand gestures (air gestures) that control the operation of device **2.4-10**. User input such as hand gesture input can be captured using cameras that operate at visible and infrared wavelengths such as tracking cameras (e.g., optical components **2.4-76** and **2.4-68**). Tracking cameras such as these may also track fiducials and other recognizable features on controllers and other external accessories (additional devices **2.4-10** of system **2.4-8**) during use of these controllers in controlling the operation of device **2.4-10**. If desired, tracking cameras can help determine the position and orientation of a handheld controller or wearable controller that senses its location and orientation by measuring the magnetic field produced by coil **2.4-54**. The use of tracking cameras may therefore help track hand motions and controller motions that are used in moving pointers and other virtual objects being displayed for a user and can otherwise assist in controlling the operation of device **2.4-10**.

[0985] Tracking cameras may operate satisfactorily in the presence of sufficient ambient light (e.g., bright visible ambient lighting conditions). In dim environments, supplemental illumination may be provided by supplemental light sources such as supplemental infrared light sources (e.g., optical components **2.4-82** and **2.4-84**). The infrared light sources may each include one or more light-emitting devices (light-emitting diodes or lasers) and may each be configured to provide fixed and/or steerable beams of infrared light that serve as supplemental illumination for the tracking

cameras. If desired, the infrared light sources may be turned off in bright ambient lighting conditions and may be turned on in response to detection of dim ambient lighting (e.g., using the ambient light sensing capabilities of optical component **2.4-62**).

[0986] Three-dimensional sensors in device **2.4-10** may be used to perform biometric identification operations (e.g., facial identification for authentication), may be used to determine the three-dimensional shapes of objects in the user's environment (e.g., to map the user's environment so that a matching virtual environment can be created for the user), and/or to otherwise gather three-dimensional content during operation of device **2.4-10**. As an example, optical components **2.4-74** and **2.4-70** may be three-dimensional structured light image sensors. Each three-dimensional structured light image sensor may have one or more light sources that provide structured light (e.g., a dot projector that projects an array of infrared dots onto the environment, a structured light source that produces a grid of lines, or other structured light component that emits structured light). Each of the three-dimensional structured light image sensors may also include a flood illuminator (e.g., a light-emitting diode or laser that emits a wide beam of infrared light). Using flood illumination and structured light illumination, optical components **2.4-74** and **2.4-70** may capture facial images, images of objects in the environment surrounding device **2.4-10**, etc.

[0987] Optical component **2.4-72** may be an infrared three-dimensional time-of-flight camera that uses time-of-flight measurements on emitted light to gather three-dimensional images of objects in the environment surrounding device **2.4-10**. Component **2.4-72** may have a longer range and a narrower field of view than the three-dimensional structured light cameras of optical components **2.4-74** and **2.4-70**. The operating range of component **2.4-72** may be 30 cm to 7 m, 2.4-60 cm to 6 m, 70 cm to 5 m, or other suitable operating range (as examples).

2.5: Systems with Cover Layer Sealing Structures

[0988] A head-mounted device may include a head-mounted support structure that allows the device to be worn on the head of a user. The head-mounted device may have displays that are supported by the head-mounted support structure for presenting a user with visual content. The displays may include rear-facing displays that present images to eye boxes at the rear of the head-mounted support structure. The displays may also include a forward-facing display. The forward-facing display may be mounted to the front of the head-mounted support structure and may be viewed by the user when the head-mounted device is not being worn on the user's head. The forward-facing display, which may sometimes be referred to as a publicly viewable display, may also be viewable by other people in the vicinity of the head-mounted device.

[0989] Optical components such as image sensors and other light sensors may be provided in the head-mounted device. In an illustrative configuration, optical components are mounted under peripheral portions of a display cover layer that protects the forward-facing display. The display cover layer, or other layers within the head-mounted device, may be formed from materials, such as glass, that are prone to shattering. Because the head-mounted device is near a user's eyes during operation, it may be desirable to reduce the likelihood that these layers will shatter into the user's eyes. Therefore, laminates, such as plastic laminates, may be formed on top and bottom surfaces of the cover layer. To protect the edges of the cover layer, encapsulation material may be coupled to the edge surface, or the head-mounted device housing structures may be modified to reduce the likelihood that glass from the cover layer exits the device.

[0990] FIGS. **2.5-1** is a side view of an illustrative head-mounted electronic device. As shown in FIGS. **2.5-1**, head-mounted device **2.5-10** may include head-mounted support structure **2.5-26**. Support structure **2.5-26** may have walls or other structures that separate an interior region of device **2.5-10** such as interior region **2.5-42** from an exterior region surrounding device **2.5-10** such as exterior region **2.5-44**. Electrical components **2.5-40** (e.g., integrated circuits, sensors, control circuitry, light-emitting diodes, lasers, and other light-emitting devices, other control circuits and input-output devices, etc.) may be mounted on printed circuits and/or other structures within device **2.5-10** (e.g., in interior region **2.5-42**).

[0991] To present a user with images for viewing from eye boxes such as eye boxes **2.5-34**, device **2.5-10** may include rear-facing displays such as displays **2.5-14R**, which may have associated lenses that focus images for viewing in the eye boxes. These components may be mounted in optical modules (e.g., a lens barrel) to form respective left and right optical systems. There may be, for example, a left rear-facing display for presenting an image through a left lens to a user's left eye in a left eye box and a right rear-facing display for presenting an image to a user's right eye in a right eye box. The user's eyes are located in eye boxes **2.5-34** at rear side R of device **2.5-10** when structure **2.5-26** rests against the outer surface of the user's face.

[0992] Support structure **2.5-26** may include a main support structure (sometimes referred to as a main portion or housing). The main housing support structure may extend from front side F of device **2.5-10** to opposing rear side R of device **2.5-10**. On rear side R, support structure **2.5-26** may have cushioned structures to enhance user comfort as support structure **2.5-26** rests against the user's face. If desired, support structure **2.5-26** may include optional head straps and/or other structures that allow device **2.5-10** to be worn on a head of a user.

[0993] Device **2.5-10** may have a publicly viewable front-facing display such as display **2.5-14F** that is mounted on front side F of support structure **2.5-26**. Display **2.5-14F** may be viewable to the user when the user is not wearing device **2.5-10** and/or may be viewable by others in the vicinity of device **2.5-10**. Display **2.5-14F** may, as an example, be visible on front side F of device **2.5-10** by an external viewer who is viewing device **2.5-10** from front side F.

[0994] A schematic diagram of an illustrative system that may include a head-mounted device is shown in FIGS. **2.5-2**. As shown in FIGS. **2.5-2**, system **2.5-8** may have one or more electronic devices **2.5-10**. Devices **2.5-10** may include a head-mounted device (e.g., device **2.5-10** of FIGS. **2.5-1**), accessories such as controllers and headphones, computing equipment (e.g., a cellular telephone, tablet computer, laptop computer, desktop computer, and/or remote computing equipment that supplies content to a head-mounted device), and/or other devices that communicate with each other.

[0995] Each electronic device **2.5-10** may have control circuitry **2.5-12**. Control circuitry **2.5-12** may include storage and processing circuitry for controlling the operation of device **2.5-10**. Circuitry **2.5-12** may include storage such as hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **2.5-12** may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, and other integrated circuits. Software code may be stored on storage in circuitry **2.5-12** and run on processing circuitry in circuitry **2.5-12** to implement control operations for device **2.5-10** (e.g., data gathering operations, operations involving the adjustment of the components of device **2.5-10** using control signals, etc.). Control circuitry **2.5-12** may include wired and wireless communications circuitry. For example, control circuitry **2.5-12** may include radio-frequency transceiver circuitry such as cellular telephone transceiver circuitry, wireless local area network transceiver circuitry (e.g., WiFi® circuitry), millimeter wave transceiver circuitry, and/or other wireless communications circuitry.

[0996] During operation, the communications circuitry of the devices in system **2.5-8** (e.g., the communications circuitry of control circuitry **2.5-12** of device **2.5-10**) may be used to support communication between the electronic devices. For example, one electronic device may transmit video data, audio data, control signals, and/or other data to another electronic device in system **2.5-8**. Electronic devices in system **2.5-8** may use wired and/or wireless communications circuitry to communicate through one or more communications networks (e.g., the internet, local area networks, etc.). The communications circuitry may be used to allow data to be received by device **2.5-10** from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, online computing equipment such as a remote server or other remote

computing equipment, or other electrical equipment) and/or to provide data to external equipment. [0997] Each device 2.5-10 in system 2.5-8 may include input-output devices 2.5-22. Input-output devices 2.5-22 may be used to allow a user to provide device 2.5-10 with user input. Input-output devices 2.5-22 may also be used to gather information on the environment in which device 2.5-10 is operating. Output components in devices 2.5-22 may allow device 2.5-10 to provide a user with output and may be used to communicate with external electrical equipment.

[0998] As shown in FIGS. 2.5-2, input-output devices 2.5-22 may include one or more displays such as displays 2.5-14. Displays 2.5-14 may include rear facing displays such as display 2.5-14R of FIGS. 2.5-1. Device 2.5-10 may, for example, include left and right components such as left and right scanning mirror display devices or other image projectors, liquid-crystal-on-silicon display devices, digital mirror devices, or other reflective display devices, left and right display panels based on light-emitting diode pixel arrays (e.g., thin-film organic light-emitting displays with polymer or semiconductor substrates such as silicon substrates or display devices based on pixel arrays formed from crystalline semiconductor light-emitting diode dies), liquid crystal display panels, and/or or other left and right display devices that provide images to left and right eye boxes for viewing by the user's left and right eyes, respectively. Display components such as these (e.g., a thin-film organic light-emitting display with a flexible polymer substrate or a display based on a pixel array formed from crystalline semiconductor light-emitting diode dies on a flexible substrate) may also be used in forming a forward-facing display for device 2.5-10 such as forward-facing display 2.5-14F of FIGS. 2.5-1 (sometimes referred to as a front-facing display, front display, or publicly viewable display).

[0999] During operation, displays 2.5-14 (e.g., displays 2.5-14R and/or 2.5-14F) may be used to display visual content for a user of device 2.5-10 (e.g., still and/or moving images including pictures and pass-through video from camera sensors, text, graphics, movies, games, and/or other visual content). The content that is presented on displays 2.5-14 may, for example, include virtual objects and other content that is provided to displays 2.5-14 by control circuitry 2.5-12. This virtual content may sometimes be referred to as computer-generated content. Computer-generated content may be displayed in the absence of real-world content or may be combined with real-world content. In some configurations, a real-world image may be captured by a camera (e.g., a forward-facing camera, sometimes referred to as a front-facing camera) and computer-generated content may be electronically overlaid on portions of the real-world image (e.g., when device 2.5-10 is a pair of virtual reality goggles).

[1000] Input-output circuitry 2.5-22 may include sensors 2.5-16. Sensors 2.5-16 may include, for example, three-dimensional sensors (e.g., three-dimensional image sensors such as structured light sensors that emit beams of light and that use two-dimensional digital image sensors to gather image data for three-dimensional images from dots or other light spots that are produced when a target is illuminated by the beams of light, binocular three-dimensional image sensors that gather three-dimensional images using two or more cameras in a binocular imaging arrangement, three-dimensional LIDAR (light detection and ranging) sensors, sometimes referred to as time-of-flight cameras or three-dimensional time-of-flight cameras, three-dimensional radio-frequency sensors, or other sensors that gather three-dimensional image data), cameras (e.g., two-dimensional infrared and/or visible digital image sensors), gaze tracking sensors (e.g., a gaze tracking system based on an image sensor and, if desired, a light source that emits one or more beams of light that are tracked using the image sensor after reflecting from a user's eyes), touch sensors, capacitive proximity sensors, light-based (optical) proximity sensors, other proximity sensors, force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), sensors such as contact sensors based on switches, gas sensors, pressure sensors, moisture sensors, magnetic sensors, audio sensors (microphones), ambient light sensors, flicker sensors that gather temporal information on ambient lighting conditions such as the presence of a time-varying ambient light intensity associated with artificial lighting, microphones for gathering voice commands and other audio input, sensors that

are configured to gather information on motion, position, and/or orientation (e.g., accelerometers, gyroscopes, compasses, and/or inertial measurement units that include all of these sensors or a subset of one or two of these sensors), and/or other sensors.

[1001] User input and other information may be gathered using sensors and other input devices in input-output devices 2.5-22. If desired, input-output devices 2.5-22 may include other devices 2.5-24 such as haptic output devices (e.g., vibrating components), light-emitting diodes, lasers, and other light sources (e.g., light-emitting devices that emit light that illuminates the environment surrounding device 2.5-10 when ambient light levels are low), speakers such as ear speakers for producing audio output, circuits for receiving wireless power, circuits for transmitting power wirelessly to other devices, batteries and other energy storage devices (e.g., capacitors), joysticks, buttons, and/or other components.

[1002] As described in connection with FIGS. 2.5-1, electronic device 2.5-10 may have head-mounted support structures such as head-mounted support structure 2.5-26 (e.g., head-mounted housing structures such as housing walls, straps, etc.). The head-mounted support structure may be configured to be worn on a head of a user (e.g., against the user's face covering the user's eyes) during operation of device 2.5-10 and may support displays 2.5-14, sensors 2.5-16, other components 2.5-24, other input-output devices 2.5-22, and control circuitry 2.5-12 (see, e.g., components 2.5-40 and displays 2.5-14R and 2.5-14F of FIGS. 2.5-1, which may include associated optical modules).

[1003] FIGS. 2.5-3 is a front view of device 2.5-10 in an illustrative configuration in which device 2.5-10 has a publicly viewable display such as forward-facing display 2.5-14F. As shown in FIGS. 2.5-3, support structure 2.5-26 of device 2.5-10 may have right and left portions on either side of nose bridge 2.5-90. Nose bridge 2.5-90 may be a curved exterior surface that is configured to receive and rest upon a user's nose to help support housing 2.5-26 on the head of the user.

[1004] Display 2.5-14F may have an active area such as active area AA that is configured to display images and an inactive area IA that does not display images. The outline of active area AA may be rectangular, rectangular with rounded corners, may have teardrop shaped portions on the left and right sides of device 2.5-10, may have a shape with straight edges, a shape with curved edges, a shape with a peripheral edge that has both straight and curved portions, and/or other suitable outlines. As shown in FIGS. 2.5-3, active area AA may have a curved recessed portion at nose bridge 2.5-90. The presence of the nose-shaped recess in active area AA may help fit active area AA within the available space of housing 2.5-26 without overly limiting the size of active area AA.

[1005] Active area AA contains an array of pixels. The pixels may be, for example, light-emitting diode pixels formed from thin-film organic light-emitting diodes or crystalline semiconductor light-emitting diode dies (sometimes referred to as micro-light-emitting diodes) on a flexible display panel substrate. Configurations in which display 2.5-14F uses other display technologies may also be used, if desired. Illustrative arrangements in which display 2.5-14 is formed from a light-emitting diode display such as an organic light-emitting diode display that is formed on a flexible substrate (e.g., a substrate formed from a bendable layer of polyimide or a sheet of other flexible polymer) may sometimes be described herein as an example. The pixels of active area AA may be formed on a display device such as a display panel (e.g., a flexible organic light-emitting diode display panel). In some configurations, the outline of active area AA may have a peripheral edge that contains straight segments or a combination of straight and curved segments. Configurations in which the entire outline of active area AA is characterized by a curved peripheral edge may also be used.

[1006] Display 2.5-14F may have an inactive area such as inactive area IA that is free of pixels and that does not display images. Inactive area IA may form an inactive border region that runs along one more portions of the peripheral edge of active area AA. In the illustrative configuration of FIGS. 2.5-3, inactive area IA has a ring shape that surrounds active area AA and forms an inactive

border. In this type of arrangement, the width of inactive area IA may be relatively constant and the inner and outer edges of area IA may be characterized by straight and/or curved segments or may be curved along their entire lengths. For example, the outer edge of area IA (e.g., the periphery of display 2.5-14F) may have a curved outline that runs parallel to the curved edge of active area AA. [1007] In some configurations, device 2.5-10 may operate with other devices in system 2.5-8 (e.g., wireless controllers and other accessories). These accessories may have magnetic sensors that sense the direction and intensity of magnetic fields. Device 2.5-10 may have one or more electromagnets configured to emit a magnetic field. The magnetic field can be measured by the wireless accessories near device 2.5-10, so that the accessories can determine their orientation and position relative to device 2.5-10. This allows the accessories to wirelessly provide device 2.5-10 with real-time information on their current position, orientation, and movement so that the accessories can serve as wireless controllers. The accessories may include wearable devices, handled devices, and other input devices.

[1008] In an illustrative configuration, device 2.5-10 may have a coil that runs around the perimeter of display 2.5-14F (e.g., under inactive area IA along the periphery of active area AA). The coil may have any suitable number of turns (e.g., 1-10, at least 2, at least 5, at least 10, 10-50, fewer than 100, fewer than 25, fewer than 6, etc.). These turns may be formed from metal traces on a substrate, may be formed from wire, and/or may be formed from other conductive lines. During operation, control circuitry 2.5-12 may supply the coil with an alternating-current (AC) drive signal. The drive signal may have a frequency of at least 1 kHz, at least 10 kHz, at least 100 kHz, at least 1 MHz, less than 10 MHz, less than 3 MHz, less than 300 kHz, or less than 30 kHz (as examples). As AC current flows through the coil a corresponding magnetic field is produced in the vicinity of device 2.5-10. Electronic devices such as wireless controllers with magnetic sensors that are in the vicinity of device 2.5-10 may use the magnetic field as a reference so that the wireless controllers can determine their orientation, position, and/or movement while being moved relative to device 2.5-10 to provide device 2.5-10 with input.

[1009] Consider, as an example, a handheld wireless controller that is used in controlling the operation of device 2.5-10. During operation, device 2.5-10 uses the coil to emit a magnetic field. As the handheld wireless controller is moved, the magnetic sensors of the controller can monitor the location of the controller and the movement of the controller relative to device 2.5-10 by monitoring the strength, orientation, and change to the strength and/or orientation of the magnetic field emitted by the coil as the controller is moved through the air by the user. The electronic device can then wirelessly transmit information on the location and orientation of the controller to device 2.5-10. In this way, a handheld controller, wearable controller, or other external accessory can be manipulated by a user to provide device 2.5-10 with air gestures, pointing input, steering input, and/or other user input.

[1010] Device 2.5-10 may have components such as optical components (e.g., optical sensors among sensors 2.5-16 of FIGS. 2.5-2). These components may be mounted in any suitable location on head-mounted support structure 2.5-26 (e.g. on a head strap, on housing 2.5-26, etc.). Optical components and other components may face rearwardly (e.g., when mounted on the rear face of device 2.5-10), may face to the side (e.g. to the left or right), may face downwardly or upwardly, may face to the front of device 2.5-10 (e.g., when mounted on the front face of device 2.5-10), may be mounted so as to point in any combination of these directions (e.g., to the front, to the right, and downward) and/or may be mounted in other suitable orientations. In an illustrative configuration, at least some of the components of device 2.5-10 are mounted so as to face outwardly to the front (and optionally to the sides and/or up and down). For example, forward-facing cameras for pass-through video may be mounted on the left and right sides of the front of device 2.5-10 in a configuration in which the cameras diverge slightly along the horizontal dimension so that the fields of view of these cameras overlap somewhat while capturing a wide-angle image of the environment in front of device 2.5-10. The captured image may, if desired, include portions of the

user's surroundings that are below, above, and to the sides of the area directly in front of device **2.5-10**.

[1011] To help hide components such as optical components from view from the exterior of device **2.5-10**, it may be desirable to cover some or all of the components with cosmetic covering structures. The covering structures may include transparent portions (e.g., optical component windows) that are characterized by sufficient optical transparency to allow overlapped optical components to operate satisfactorily. For example, an ambient light sensor may be covered with a layer that appears opaque to an external viewer to help hide the ambient light sensor from view, but that allows sufficient ambient light to pass to the ambient light sensor for the ambient light sensor to make a satisfactory ambient light measurement. As another example, an optical component that emits infrared light may be overlapped with a visibly opaque material that is transparent to infrared light.

[1012] In an illustrative configuration, optical components for device **2.5-10** may be mounted in inactive area IA of FIGS. **2.5-3** and cosmetic covering structures may be formed in a ring shape overlapping the optical components in inactive area IA. Cosmetic covering structures may be formed from ink, polymer structures, structures that include metal, glass, other materials, and/or combinations of these materials. In an illustrative configuration, a cosmetic covering structure may be formed from a ring-shaped member having a footprint that matches the footprint of inactive area IA. If, for example, active area AA has left and right portions with teardrop shapes, the ring-shaped member may have curved edges that follow the curved periphery of the teardrop-shaped portions of active area AA. The ring-shaped member may be formed from one or more polymer structures (e.g., the ring-shaped member may be formed from a polymer ring). Because the ring-shaped member can help hide overlapped components from view, the ring-shaped member may sometimes be referred to as a shroud or ring-shaped shroud member. The outward appearance of the shroud or other cosmetic covering structures may be characterized by a neutral color (white, black, or gray) or a non-neutral color (e.g., blue, red, green, gold, rose gold, etc.).

[1013] Display **2.5-14F** may, if desired, have a protective display cover layer. The cover layer may overlap active area AA and inactive area IA (e.g., the entire front surface of device **2.5-10** as viewed from front F of FIGS. **2.5-1** may be covered by the cover layer). The cover layer, which may sometimes be referred to as a housing wall or transparent housing wall, may have a rectangular outline, an outline with teardrop portions, an oval outline, or other shape with curved and/or straight edges.

[1014] The cover layer may be formed from a transparent material such as glass, polymer, transparent crystalline material such as sapphire, clear ceramic, other transparent materials, and/or combinations of these materials. As an example, a protective display cover layer for display **2.5-14F** may be formed from safety glass (e.g., laminated glass that includes a clear glass layer with a laminated polymer film). Optional coating layers may be applied to the surfaces of the display cover layer. If desired, the display cover layer may be chemically strengthened (e.g., using an ion-exchange process to create an outer layer of material under compressive stress that resists scratching). In some configurations, the display cover layer may be formed from a stack of two or more layers of material (e.g., first and second structural glass layers, a rigid polymer layer coupled to a glass layer or another rigid polymer layer, etc.) to enhance the performance of the cover layer.

[1015] In active area AA, the display cover layer may overlap the pixels of display panel **2.5-14P**. The display cover layer in active area AA is preferably transparent to allow viewing of images presented on display panel **2.5-14P**. In inactive area IA, the display cover layer may overlap the ring-shaped shroud or other cosmetic covering structure. The shroud and/or other covering structures (e.g., opaque ink coatings on the inner surface of the display cover layer and/or structures) may be sufficiently opaque to help hide some or all of the optical components in inactive area IA from view. Windows may be provided in the shroud or other cosmetic covering structures to help ensure that the optical components that are overlapped by these structures operate

satisfactorily. Windows may be formed from holes, may be formed from areas of the shroud or other cosmetic covering structures that have been locally thinned to enhance light transmission, may be formed from window members with desired light transmission properties that have been inserted into mating openings in the shroud, and/or may be formed from other shroud window structures.

[1016] In the example of FIGS. 2.5-3, device 2.5-10 includes optical components such as optical components 2.5-60, 2.5-62, 2.5-64, 2.5-66, 2.5-68, 2.5-70, 2.5-72, 2.5-74, 2.5-76, 2.5-78, and 2.5-80 (as an example). Each of these optical components (e.g., optical sensors selected from among sensors 2.5-16 of FIGS. 2.5-2, light-emitting devices, etc.) may be configured to detect light and, if desired to emit light (e.g., ultraviolet light, visible light, and/or infrared light).

[1017] In an illustrative configuration, optical component 2.5-60 may sense ambient light (e.g., visible ambient light). In particular, optical component 2.5-60 may have a photodetector that senses variations in ambient light intensity as a function of time. If, as an example, a user is operating in an environment with an artificial light source, the light source may emit light at a frequency associated with its source of wall power (e.g., alternating-current mains power at 60 Hz). The photodetector of component 2.5-60 may sense that the artificial light from the artificial light source is characterized by 60 Hz fluctuations in intensity. Control circuitry 2.5-12 can use this information to adjust a clock or other timing signal associated with the operation of image sensors in device 2.5-10 to help avoid undesired interference between the light source frequency and the frame rate or other frequency associated with image capture operations. Control circuitry 2.5-12 can also use measurements from component 2.5-60 to help identify the presence of artificial lighting and the type of artificial lighting that is present. In this way, control circuitry 2.5-12 can detect the presence of lights such as fluorescent lights or other lights with known non-ideal color characteristics and can make compensating color cast adjustments (e.g., white point adjustments) to color-sensitive components such as cameras and displays. Because optical component 2.5-60 may measure fluctuations in light intensity, component 2.5-60 may sometimes be referred to as a flicker sensor or ambient light frequency sensor.

[1018] Optical component 2.5-62 may be an ambient light sensor. The ambient light sensor may include one or more photodetectors. In a single-photodetector configuration, the ambient light sensor may be a monochrome sensor that measures ambient light intensity. In a multi-photodetector configuration, each photodetector may be overlapped by an optical filter that passes a different band of wavelengths (e.g. different visible and/or infrared passbands). The optical filter passbands may overlap at their edges. This allows component 2.5-62 to serve as a color ambient light sensor that measures both ambient light intensity and ambient light color (e.g., by measuring color coordinates for the ambient light). During operation of device 2.5-10, control circuitry 2.5-12 can take action based on measured ambient light intensity and color. As an example, the white point of a display or image sensor may be adjusted or other display or image sensor color adjustments may be made based on measured ambient light color. The intensity of a display may be adjusted based on light intensity. For example, the brightness of display 2.5-14F may be increased in bright ambient lighting conditions to enhance the visibility of the image on the display and the brightness of display 2.5-14F may be decreased in dim lighting conditions to conserve power. Image sensor operations and/or light source operations may also be adjusted based on ambient light readings.

[1019] The optical components in active area IA may also include components along the sides of device 2.5-10 such as components 2.5-80 and 2.5-64. Optical components 2.5-80 and 2.5-64 may be pose-tracking cameras that are used to help monitor the orientation and movement of device 2.5-10. Components 2.5-80 and 2.5-64 may be visible light cameras (and/or cameras that are sensitive at visible and infrared wavelengths) and may, in conjunction with an inertial measurement unit, form a visual inertial odometry (VIO) system.

[1020] Optical components 2.5-78 and 2.5-66 may be visible-light cameras that capture real-time images of the environment surrounding device 2.5-10. These cameras, which may sometimes be

referred to as scene cameras or pass-through-video cameras, may capture moving images that are displayed in real time to displays **2.5-14R** for viewing by the user when the user's eyes are located in eye boxes **2.5-34** at the rear of device **2.5-10**. By displaying pass-through images (pass-through video) to the user in this way, the user may be provided with real-time information on the user's surroundings. If desired, virtual content (e.g. computer-generated images) may be overlaid over some of the pass-through video. Device **2.5-10** may also operate in a non-pass-through-video mode in which components **2.5-78** and **2.5-66** are turned off and the user is provided only with movie content, game content, and/or other virtual content that does not contain real-time real-world images.

[1021] Input-output devices **2.5-22** of device **2.5-10** may gather user input that is used in controlling the operation of device **2.5-10**. As an example, a microphone in device **2.5-10** may gather voice commands. Buttons, touch sensors, force sensors, and other input devices may gather user input from a user's finger or other external object that is contacting device **2.5-10**. In some configurations, it may be desirable to monitor a user's hand gestures or the motion of other user body parts. This allows the user's hand locations or other body part locations to be replicated in a game or other virtual environment and allows the user's hand motions to serve as hand gestures (air gestures) that control the operation of device **2.5-10**. User input such as hand gesture input can be captured using cameras that operate at visible and infrared wavelengths such as tracking cameras (e.g., optical components **2.5-76** and **2.5-68**). Tracking cameras such as these may also track fiducials and other recognizable features on controllers and other external accessories (additional devices **2.5-10** of system **2.5-8**) during use of these controllers in controlling the operation of device **2.5-10**. If desired, tracking cameras can help determine the position and orientation of a handheld controller or wearable controller that senses its location and orientation by measuring the magnetic field produced by coil **2.5-54**. The use of tracking cameras may therefore help track hand motions and controller motions that are used in moving pointers and other virtual objects being displayed for a user and can otherwise assist in controlling the operation of device **2.5-10**.

[1022] Tracking cameras may operate satisfactorily in the presence of sufficient ambient light (e.g., bright visible ambient lighting conditions). In dim environments, supplemental illumination may be provided by supplemental light sources such as supplemental infrared light sources (e.g., optical components **2.5-82** and **2.5-84**). The infrared light sources may each include one or more light-emitting devices (light-emitting diodes or lasers) and may each be configured to provide fixed and/or steerable beams of infrared light that serve as supplemental illumination for the tracking cameras. If desired, the infrared light sources may be turned off in bright ambient lighting conditions and may be turned on in response to detection of dim ambient lighting (e.g., using the ambient light sensing capabilities of optical component **2.5-62**).

[1023] Three-dimensional sensors in device **2.5-10** may be used to perform biometric identification operations (e.g., facial identification for authentication), may be used to determine the three-dimensional shapes of objects in the user's environment (e.g., to map the user's environment so that a matching virtual environment can be created for the user), and/or to otherwise gather three-dimensional content during operation of device **2.5-10**. As an example, optical components **2.5-74** and **2.5-70** may be three-dimensional structured light image sensors. Each three-dimensional structured light image sensor may have one or more light sources that provide structured light (e.g., a dot projector that projects an array of infrared dots onto the environment, a structured light source that produces a grid of lines, or other structured light component that emits structured light). Each of the three-dimensional structured light image sensors may also include a flood illuminator (e.g., a light-emitting diode or laser that emits a wide beam of infrared light). Using flood illumination and structured light illumination, optical components **2.5-74** and **2.5-70** may capture facial images, images of objects in the environment surrounding device **2.5-10**, etc.

[1024] Optical component **2.5-72** may be an infrared three-dimensional time-of-flight camera that uses time-of-flight measurements on emitted light to gather three-dimensional images of objects in

the environment surrounding device **2.5-10**. Component **2.5-72** may have a longer range and a narrower field of view than the three-dimensional structured light cameras of optical components **2.5-74** and **2.5-70**. The operating range of component **2.5-72** may be 30 cm to 7 m, 60 cm to 6 m, 70 cm to 5 m, or other suitable operating range (as examples).

[1025] FIGS. **2.5-4** is a front view of an illustrative ring-shaped cosmetic covering structure for device **2.5-10**. Illustrative ring-shaped shroud **2.5-100** of FIGS. **2.5-4** may be mounted under the inner surface of the display cover layer for display **2.5-14F** in inactive area IA. This may help hide the optical components and other internal portions of device **2.5-10** from view from the exterior of device **2.5-10**. Shroud **2.5-100** may be formed from one or more unbroken ring-shaped members and/or may be formed from multiple shroud segments that are attached using adhesive, fasteners, or other attachment structures. If desired, shroud **2.5-100** may be formed from multiple members that are sandwiched together along some or all of their lengths. In an illustrative configuration, which may sometimes be described herein as an example, shroud **2.5-100** may be formed from an inner piece (e.g., an inner full or partial ring), which may sometimes be referred to as an inner shroud member, shroud trim, or shroud trim member and may be formed from an outer piece or pieces (e.g., one or more strips of material or covering members, an full ring, one or more partial rings, etc.), which may sometimes be referred to as a shroud cover, canopy, or shroud canopy.

[1026] As shown in FIGS. **2.5-4**, shroud **2.5-100** may have optical component windows to accommodate components **2.5-60**, **2.5-62**, **2.5-64**, **2.5-84**, **2.5-66**, **2.5-68**, **2.5-70**, **2.5-72**, **2.5-74**, **2.5-76**, **2.5-78**, **2.5-82**, and **2.5-80**. The optical component windows may be formed from through-hole openings in shroud **2.5-100**, from recesses or other partial openings that do not pass entirely through shroud **2.5-100**, from inserted optical window members in shroud through-hole openings, and/or from other shroud optical component window structures. Display **2.5-14F** may have a display cover layer that has corresponding optical component windows (through-hole openings, recessed areas, inserted window members in through-hole openings, etc.) and/or that is formed from bulk material that has desired optical properties (e.g., a display cover layer formed from one or more layers of material such as glass and/or polymer with sufficient transparency at the operating wavelength range of the overlapped optical component to allow the optical component to operate satisfactorily through the cover layer without forming openings or other window structures in the cover layer).

[1027] Shroud **2.5-100** may have any suitable shape. For example, the outline of shroud **2.5-100** may be rectangular with rounded corners as shown in FIGS. **2.5-4**, may have teardrop shapes on the left and right sides of device **2.5-10**, may have an oval outline, and/or may have other outlines with curved and/or straight edge segments. For example, the inner and outer edges of shroud **2.5-100** may be curved (e.g., to follow a teardrop shape). Shroud **2.5-100** may, if desired, have a peripheral edge that is curved along most or all of its length.

[1028] The width of shroud **2.5-100** may be constant along its length or shroud **2.5-100** may have portions that are wider than others. The thickness of shroud **2.5-100** (e.g., the dimension of shroud **2.5-100** into the page in the orientation of FIGS. **2.5-4**) may be smaller than the width of shroud **2.5-100** (the lateral dimension of shroud **2.5-100** within the page in the orientation of FIGS. **2.5-4**) or the thickness of the shroud may be equal to or greater than the width of the shroud. The shroud may have a two-dimensional shape (e.g., shroud **2.5-100** may have a planar shape) or may have a three-dimensional shape (e.g., a shape with a curved cross-sectional profile and/or a shape characterized by inner and/or outer surfaces of compound curvature). In an illustrative configuration, most or all of the inner and outer surfaces of shroud have a compound-curvature surface.

[1029] The optical components under inactive area IA may include components on the left and right sides of device **2.5-10** that operate in conjunction with each other. For example, scene cameras, tracking cameras, and/or structured light cameras in device **2.5-10** may be formed in pairs, each of which includes a left camera and a corresponding right camera. A left scene camera

and a right scene camera may, as an example, operate together to capture overlapping images that provide device **2.5-10** with a wide field of view for gathering pass-through video. Left and right tracking cameras may operate together to track a user's hands or other external objects. Left and right structured light cameras or other three-dimensional cameras may be used together to capture three-dimensional images of the user's environment. To enhance performance of the left and right optical components in these types of paired component arrangements, it may be desirable to maintain accurate alignment between the left and right optical components. To help maintain left and right optical components on the respective left and right sides of device **2.5-10** in alignment with each other, device **2.5-10** may be provided with one or more housing structures that help support the optical components. An illustrative example of device **2.5-10** having housing structures that support the optical components and a cover layer that overlaps the optical components is shown in FIGS. **2.5-5**.

[1030] As shown in FIGS. **2.5-5**, shroud **2.5-100** and display cover layer **2.5-92** may be attached to housing **2.5-26** using adhesive, screws and other fasteners, press-fit connections, and/or other attachment mechanisms. An illustrative configuration in which shroud **2.5-100** and cover layer **2.5-92** are attached to forward-facing edge of a housing wall in the main housing portion of structure **2.5-26** using adhesive is shown in FIGS. **2.5-5**. In the example of FIGS. **2.5-5**, shroud **2.5-100** has an inner shroud member such as shroud trim **2.5-100A** and has a corresponding outer shroud member such as shroud canopy **2.5-100B**. Shroud trim **2.5-100A** and shroud canopy **2.5-100B** may be formed from metal, polymer, ceramic, glass, other materials, and/or combinations of these materials. In an illustrative example, shroud trim **2.5-100A** is formed from black polymer or other dark material and shroud canopy **2.5-100B** is formed from clear polymer. The outer surface of shroud canopy **2.5-100B** may be smooth to provide shroud **2.5-100** with a cosmetically attractive appearance.

[1031] A layer of pressure sensitive adhesive may be used in attaching canopy **2.5-100B** to trim **2.5-100A**, or canopy **2.5-100B** may be formed integrally with trim **2.5-100A**. Adhesive may also be used in attaching cover layer **2.5-92** and shroud **2.5-100** to housing portion **2.5-26**. As shown in FIGS. **2.5-5**, for example, a first adhesive such as adhesive **2.5-122** may be used to attach display cover layer **2.5-92** to shroud **2.5-100** (e.g., to a ledge in shroud trim **2.5-100A**). A second adhesive such as adhesive **2.5-124** may, in turn, be used to attach shroud **2.5-100** (e.g., shroud trim **2.5-100A**) to an adjacent lip of a wall in housing **2.5-26**.

[1032] In some configurations, adhesives **2.5-122** and **2.5-124** may be formed from the same type of material. In an illustrative configuration, adhesives **2.5-122** and **2.5-124** are different. Housing portion **2.5-26** may have a wall with a lip shape that creates a shearing force on adhesive **2.5-124** as display **2.5-14F** is attached to housing **2.5-26** by pressing display **2.5-14F** against housing **2.5-26**. In this type of scenario, it may be desirable to form adhesive **2.5-124** from an adhesive that can bond satisfactorily in the presence of shear forces such as a molten hot melt glue (thermoplastic adhesive) or other liquid adhesive rather than pressure sensitive adhesive. Adhesive **2.5-124** may, if desired, be exposed to a curing agent (ultraviolet light, moisture, etc.) before display **2.5-14F** is assembled into housing **2.5-26**.

[1033] It may be desirable to repair device **2.5-10**. For example, if a user exposes display **2.5-14F** to excessive force during a drop event, it may be desirable to replace display **2.5-14F** with a new display. This can be accomplished by heating adhesive **2.5-124** to loosen the adhesive bond formed by adhesive **2.5-124**. To help prevent display cover layer **2.5-92** from detaching from shroud **2.5-100** while softening adhesive **2.5-124** with heat, adhesive **2.5-122** may be provided with a higher-temperature softening point than adhesive **2.5-124** (e.g., adhesive **2.5-122** may be a two-part hot melt glue with a higher melting point than adhesive **2.5-124**).

[1034] Optical components that are overlapped by display cover layer **2.5-92** and shroud **2.5-100** in inactive area **IA** may transmit and/or receive light through shroud **2.5-100** and display cover layer **2.5-92**. Layer **2.5-92** may be formed from a single layer of glass, laminated glass, or other clear

material that allows light for each overlapped optical component **2.5-104** to pass through layer **2.5-92**. If desired, a partial recess or a through-hole opening may be formed in the portion of layer **2.5-92**. An optional optical component window member may then be inserted within layer **2.5-92** (e.g., a window that overlaps component **2.5-104**). As an example, layer **2.5-92** may be formed from one or more layers of glass and/or polymer and may be characterized by a first level of light transmission at operating wavelength(s) for component **2.5-104**. A window member in layer **2.5-92** may be formed from polymer, glass, and/or other materials that are characterized by a second level of light transmission at the operating wavelength(s) that is greater than the first level of light transmission. In other illustrative arrangements, no window member is inserted in layer **2.5-92** (e.g., when layer **2.5-92** alone is sufficiently transparent to pass light for component **2.5-104**).

[1035] Shroud **2.5-100** may be provided with an optical component window that overlaps optical component to help accommodate overlapped optical component **2.5-104**. Component **2.5-104** may operate at ultraviolet light wavelengths, visible light wavelengths, and/or infrared light wavelengths. To accommodate component **2.5-104**, shroud trim **2.5-100A** has been provided with a through-hole opening, whereas shroud canopy **2.5-100B** has no openings overlapping component **2.5-104**. This effectively forms a window recess in shroud **2.5-100** in alignment with components **2.5-104**. Trim **2.5-100A** may be formed from black polymer or other light-absorbing material, so the formation of opening **120** in trim **2.5-100A** may help ensure that sufficiently light may pass through to allow component **2.5-104** to operate satisfactorily. The portion of canopy **2.5-100B** that overlaps component **2.5-104** may be transparent (e.g., clear polymer). Alternatively, canopy **2.5-100B** may be formed from light-absorbing material, and a portion of canopy **2.5-100B** overlapping component **2.5-104** may be removed.

[1036] To help hide component **2.5-104** from view, the inner surface of shroud canopy **2.5-100B** may be covered with one or more coatings, which may be used to provide region the region overlapping component **2.5-104** with a desired outward appearance and optical properties that ensure that component **2.5-104** can operate satisfactorily. The coatings may include a thin-film-interference filter formed from a stack of thin-film dielectric layers of alternating refractive index values (with indices and thicknesses selected to create a desired transmission spectrum and a desired reflection spectrum for the filter), may include a layer of ink (e.g., a polymer layer including dye, pigment, and/or other colorant), and/or may include any other suitable coating with desired optical properties.

[1037] Consider, as an example, a scenario in which component **2.5-104** transmits and/or receives infrared light. In this type of arrangement, canopy **2.5-100B** may be coated with a coating that is opaque at visible wavelengths and transparent at infrared wavelengths. This helps to hide component **2.5-104** from view from the exterior of device **2.5-10** while allowing infrared light associated with the operation of component **2.5-104** to pass through shroud **2.5-100** and layer **2.5-92**.

[1038] As another example, consider a scenario in which component **2.5-104** is an ambient light sensor. In this configuration, canopy **2.5-100B** may be coated with a coating that exhibits a visible light transmission of 1-8% (as an example). This may allow sufficient visible ambient light to reach the ambient light sensor for the ambient light sensor to make an ambient light reading. At the same time, the transmission of the coating may be sufficiently low to reduce the visibility of component **2.5-104** from the exterior of device **2.5-10**.

[1039] As these examples demonstrate, regions of display **2.5-14F** that overlap optical components such as component **2.5-104** of FIGS. **2.5-5** may be provided with optical component window structures in layer **2.5-92** and/or shroud **2.5-100** that help accommodate the optical component.

[1040] As described in connection with FIGS. **2.5-3** and **2.5-4**, there may be numerous optical components such as component **2.5-104** in inactive area **IA**. Each optical component may potentially have a different type of optical component window structure in shroud **2.5-100** and/or layer **2.5-92** to accommodate that component. For example, some areas of shroud **2.5-100** may

have openings that receive components, other areas of shroud **2.5-100** may have inserted optical window member, and/or other areas of shroud **2.5-100** may have partial shroud openings (e.g., non-through-hole recesses) such as the opening of FIGS. **2.5-8** (which may optionally be coated to modify the optical properties of shroud **2.5-100**).

[1041] Because of the proximity of cover layer **2.5-92** to a user's eyes, it may be desirable to reduce the likelihood of the cover layer material (e.g., glass) from shattering and injuring the user's eyes. An illustrative example of cover layer **2.5-92** with encapsulation to reduce the likelihood of such an event is shown in FIGS. **2.5-6**.

[1042] As shown in FIGS. **2.5-6**, cover layer **2.5-92** may be coupled to shroud **2.5-100**. Cover layer **2.5-92** may include glass layer **2.5-126**, front laminate **2.5-128** and rear laminate **2.5-130**. Front laminate **2.5-128** and rear laminate **2.5-128** may be for example, layers of plastic that are laminated to cover layer **2.5-92**, layers of plastic that are adhesively attached to cover layer **2.5-92**, or other shatter-resistant material that is attached to the front and rear surfaces of glass layer **2.5-126**.

Although not shown in FIGS. **2.5-6**, multiple layers, such as antireflection coatings, antismudge coatings, acrylic layers, or other desired layers, may be included as part of cover layer **2.5-92**.

[1043] Although front laminate **2.5-128** and rear laminate **2.5-130** may reduce the chances of glass layer **2.5-126** shattering toward the front or rear of device **2.5-10**, an edge surface of glass layer **2.5-126** may still be exposed. In the case of a shatter event, such as if device **2.5-10** were dropped, glass layer **2.5-126** could shatter and glass could exit device **2.5-10** from the edge surface of glass layer **2.5-126**. To mitigate this risk, encapsulation material **2.5-132** may be attached to the edge surface of glass layer **2.5-126**. Encapsulation material **2.5-132** may be an epoxy material, such as a ductile epoxy, that seals the edge surface of glass layer **2.5-126** and prevents glass layer **2.5-126** from shattering at the edge surface. Alternatively, acrylate, polyvinyl butyral (PVB), polyurethane, or moisture cure materials may be used for encapsulation material **2.5-132**. In general, encapsulation material **2.5-132** may be formed from material that adheres to glass layer **2.5-126**, while preventing glass layer **2.5-126** from shattering.

[1044] Encapsulation material **2.5-132** may substantially fill the opening between the edge surface of glass layer **2.5-126** and shroud **2.5-100**. For example, encapsulation material **2.5-132** may extend approximately 150 microns from the edge surface. In general, however, any amount of encapsulation material **2.5-132** may be applied to the edge surface.

[1045] As shown in FIGS. **2.5-6**, encapsulation material **2.5-132** may cover the edge surface and may also cover an edge portion of laminate **2.5-128**. However, this is merely illustrative. If desired, encapsulation material **2.5-132** may cover the edge surface of glass layer **2.5-126** without covering an edge portion of laminate **2.5-128**. For example, as shown in FIGS. **2.5-7**, encapsulation material may cover only the edge surface of glass layer **2.5-126**. In the example of FIG. **7**, laminate **2.5-128** may extend over encapsulation material **2.5-132**. However, this is merely illustrative. Laminate **2.5-128** may be flush with the edge surface of glass layer **2.5-126**, if desired.

[1046] In some embodiments, it may be determined that the shattering risk of the edge surface of glass layer **2.5-126** can be mitigated by modifying the position of glass layer **2.5-126** relative to shroud **2.5-100** (or support structure **2.5-126**). For example, as shown in the illustrative embodiment of FIGS. **2.5-8**, the edge surface of glass layer **2.5-126** may be left unencapsulated, but the size of opening **2.5-134** between the edge surface and shroud **2.5-100** may be adjusted to reduce the risk of glass escaping through opening **2.5-134** in the case of a shatter event. For example, if glass layer **2.5-126** is sufficiently close to shroud **2.5-100** (e.g., if opening **2.5-134** is sufficiently small), glass from layer **2.5-126** may not escape through opening **2.5-134** if layer **2.5-126** shatters.

[1047] Instead of, or in addition to, adding material to the edge surface of layer **2.5-126**, it may be desirable to add material in the gap between layer **2.5-126** and the shroud/support structure. An illustrative example of adding material in this gap is shown in FIGS. **2.5-9**.

[1048] As shown in FIGS. **2.5-9**, material **2.5-136** may be included between the edge surface of

glass layer **2.5-126** and shroud **2.5-100**. Material **2.5-136** may be, for example, a bumper ring. The bumper ring may be formed from elastomer, rigid plastic, or another material that helps protect the edge surface of layer **2.5-126**.

[1049] As an alternative to material **2.5-136** being a bumper ring between layer **2.5-126** and shroud **2.5-100**, material **2.5-136** may be an overmolded structure on layer **2.5-126**, on shroud **2.5-100**, or on a chassis that is coupled to support structure **26**. In general, the overmolded structure may fill the gap between layer **2.5-126** and the support structure, shroud, and/or chassis and help to prevent the edge surface of layer **2.5-126** from shattering out of device **2.5-10**.

[1050] Although not shown in FIGS. **2.5-9**, a portion of material **2.5-136** may extend underneath layer **2.5-126** if desired. In particular, there may be a portion of material **2.5-136** between the bottom surface of layer **2.5-126** and shroud **2.5-100**.

[1051] Instead of, or in addition to, adding material between layer **2.5-126** and shroud **2.5-100**, upper laminate **2.5-128** and/or lower laminate **2.5-130** may wrap around the edge surface of layer **2.5-126**. Illustrative examples of the laminates wrapping the edge surface are shown in FIGS. **2.5-10** and **2.5-11**.

[1052] As shown in FIGS. **2.5-10**, upper laminate **2.5-128** may wrap around the edge surface of layer **2.5-126**. In particular, upper laminate **2.5-128** may have portion **2.5-128A** that extends around and covers the edge surface of layer **2.5-126**. Layer **2.5-126** may have a rounded edge surface to allow upper laminate **2.5-128** to wrap the edge surface and sufficiently adhere to the surface, as shown in FIGS. **2.5-10**. By forming layer **2.5-126** with a rounded edge, the curve of laminate **2.5-128** around the edge may be reduced, thereby reducing stress on laminate **2.5-128**. However, layer **2.5-126** may have a planar edge surface, or a surface with any other desired profile, around which upper laminate **2.5-128** wraps, if desired. By wrapping upper laminate around the edge surface of layer **2.5-126**, glass may be prevented from shattering out of the edge surface.

[1053] As shown in FIGS. **2.5-11**, lower laminate **2.5-130** may wrap around the edge surface of layer **2.5-126**. In particular, lower laminate may have portion **2.5-130A** that extends around and covers the edge surface of layer **2.5-126**. Layer **2.5-126** may have a rounded edge surface to allow lower laminate **2.5-130** to wrap the edge surface and sufficiently adhere to the surface, may have a planar edge surface, or may have a surface with any other desired profile, around which lower laminate **2.5-130** wraps. By wrapping lower laminate **2.5-130** around the edge surface of layer **2.5-126**, glass may be prevented from shattering out of the edge surface.

[1054] In the example of FIGS. **2.5-11**, lower laminate portion **2.5-130A** wraps entirely around the edge surface of layer **2.5-126** and partially overlaps upper laminate **2.5-128**. This arrangement may ensure that no glass can escape if layer **2.5-126** shatters. However, the arrangement is merely illustrative. If desired, lower laminate portion **2.5-130A** may wrap around only the edge surface of layer **2.5-126** without overlapping or extending over upper laminate **2.5-128**.

[1055] Another example of material that may be used to prevent layer **2.5-126** from shattering and glass from exiting through the edge surface is shown in FIGS. **2.5-12**. In the example of FIGS. **2.5-12**, glue (or another similar material) **2.5-138** may be used to completely fill the gap between layer **2.5-126** and shroud **2.5-100**. For example, the glue may be inserted into the gap after cover layer **2.5-92** has been assembled into the head-mounted device. Glue **2.5-138** may help prevent glass from exiting through the edge surface of layer **2.5-126** if layer **2.5-126** shatters.

[1056] Rather than wrapping the upper or lower laminate around the edge surface of layer **2.5-126**, upper laminate **2.5-128** may extend to shroud **2.5-100** to cover the gap between the edge surface and shroud **2.5-100**. For example, as shown in FIGS. **2.5-13**, upper laminate **2.5-128** may have portion **2.5-128B** that extends to shroud **2.5-100** (or support structure **2.5-26** or another portion of device **2.5-10**). By covering the gap between the edge surface of layer **2.5-126** and shroud **2.5-100**, glass may be prevented from exiting device **2.5-10** during a shatter event.

[1057] Instead of, or in addition to, adding material or extending the laminates to prevent glass from shattering out of device **2.5-10**, shroud **2.5-100** or a chassis attached to support structure **2.5-**

26 may be modified to reduce the risk of layer **2.5-126** shattering. Illustrative examples of modifying these components to reduce the risk of layer **2.5-126** shattering are shown in FIGS. **2.5-14** and **2.5-15**.

[1058] As shown in FIGS. **2.5-14**, structure **2.5-140** may have a lip that covers the gap between layer **2.5-126** and shroud **2.5-100**/support structure **2.5-126**. Structure **2.5-140** may be formed from a portion of shroud **2.5-100** or from a portion of support structure **2.5-126** (e.g., a chassis of support structure **2.5-126**). The lip of structure **2.5-140** may help prevent glass that would otherwise shatter from layer **2.5-126** and exit device **2.5-10** from exiting device **2.5-10**, thereby protecting users of device **2.5-10**.

[1059] If desired, the lip of structure **2.5-140** may be combined with the extension of laminate around the edge surface of layer **2.5-126**. For example, as shown in FIGS. **2.5-15**, upper laminate **2.5-128A** may wrap the edge surface of layer **2.5-126** to prevent glass from exiting through the edge surface, and the lip of structure **2.5-140** may provide additional protection should some glass get through the laminate.

[1060] Although cover layer **2.5-92** has been described as being coupled to shroud **2.5-100**, this is merely illustrative. In some embodiments, cover layer **2.5-92** may be coupled directly to support structure **2.5-26**. In other embodiments, device **2.5-10** may include a chassis attached to support structure **2.5-26** (e.g., a chassis to support various components in device **2.5-10**), and cover layer **2.5-92** may be coupled to the chassis.

[1061] Moreover, although cover layer **2.5-92** has been described as including a glass layer that may shatter, this material is merely illustrative. Layer **2.5-126** may be formed from ceramic, sapphire, or any other desired material.

2.6: Electronic Devices with Antennas and Optical Components

[1062] An electronic device such as a head-mounted device may have a front face that faces away from a user's head and may have an opposing rear face that faces the user's head. Optical modules may be used to provide images to a user's eyes. The positions of the optical modules may be adjusted to accommodate different user interpupillary distances. The head-mounted device may have actuators and optical module guide structures to allow the optical module positions to be adjusted.

[1063] The head-mounted device may have wireless communications circuitry to communicate with external equipment such as a computer, cellular telephone, or other computing device. This allows the external equipment to provide the head-mounted device with content for viewing on the head-mounted device and/or allows the head-mounted device to otherwise interact with the remote equipment. The wireless communications circuitry may include multiple antennas.

[1064] The head-mounted device may have one or more cameras. For example, forward-facing (front-facing) cameras may allow the head-mounted device to monitor movement of the head-mounted device relative to the environment surrounding the head-mounted device (e.g., the cameras may be used in forming a visual odometry system or part of a visual inertial odometry system). Forward-facing cameras may also be used to capture images of the environment that are displayed to a user of the head-mounted device. If desired, images from multiple forward-facing cameras may be merged with each other and/or forward-facing camera content can be merged with computer-generated content for a user.

[1065] A top view of an illustrative head-mounted device is shown in FIGS. **2.6-1**. As shown in FIGS. **2.6-1**, head-mounted devices such as electronic device **2.6-10** may have head-mounted support structures such as housing **2.6-12**. Housing **2.6-12** may include portions (e.g., head-mounted support structures **2.6-12T**) to allow device **2.6-10** to be worn on a user's head. Support structures **2.6-12T** may be formed from fabric, polymer, metal, and/or other material. Support structures **2.6-12T** may form a strap or other head-mounted support structures to help support device **2.6-10** on a user's head. A main support structure (e.g., a head-mounted housing such as main housing portion **2.6-12M**) of housing **2.6-12** may support electronic components such as

displays **2.6-14**.

[1066] Main housing portion **2.6-12M** may include housing structures formed from metal, polymer, glass, ceramic, and/or other material. For example, housing portion **2.6-12M** may have housing walls on front face **F** and housing walls on adjacent top, bottom, left, and right side faces that are formed from rigid polymer or other rigid support structures and these rigid walls may optionally be covered with electrical components, fabric, leather, or other soft materials, etc. Housing portion **2.6-12M** may also have internal support structures such as a frame and/or structures that perform multiple functions such as controlling airflow while providing structural support. The walls of housing portion **2.6-12M** may enclose internal components **2.6-38** in interior region **2.6-34** of device **2.6-10** and may separate interior region **2.6-34** from the environment surrounding device **2.6-10** (exterior region **2.6-36**). Internal components **2.6-38** may include integrated circuits, actuators, batteries, sensors, and/or other circuits and structures for device **2.6-10**. Housing **2.6-12** may be configured to be worn on a head of a user and may form glasses, a hat, a helmet, goggles, and/or other head-mounted device. Configurations in which housing **2.6-12** forms goggles may sometimes be described herein as an example.

[1067] Front face **F** of housing **2.6-12** may face outwardly away from a user's head and face. Opposing rear face **R** of housing **2.6-12** may face the user. Portions of housing **2.6-12** (e.g., portions of main housing **2.6-12M**) on rear face **R** may form a cover such as cover **2.6-12C** (sometimes referred to as a curtain). The presence of cover **2.6-12C** on rear face **R** may help hide internal housing structures, internal components **2.6-38**, and other structures in interior region **2.6-34** from view by a user.

[1068] Device **2.6-10** may have one more cameras such as cameras **2.6-46**. For example, device **2.6-10** may have **K** cameras, where the value of **K** is at least one, at least two, at least four, at least six, at least eight, at least ten, at least 12, less than 20, less than 14, less than 12, less than 10, from 4 to 10, or other suitable value. Cameras **2.6-46** may be sensitive at infrared wavelengths (e.g., cameras **2.6-46** may be infrared cameras), may be sensitive at visible wavelengths (e.g., cameras **2.6-46** may be visible cameras), and/or cameras **2.6-46** may be sensitive at other wavelengths. If desired, cameras **2.6-46** may be sensitive at both visible and infrared wavelengths.

[1069] Cameras **2.6-46** that are mounted on front face **F** and that face outwardly (towards the front of device **2.6-10** and away from the user) may sometimes be referred to herein as forward-facing or front-facing cameras. Forward-facing cameras (e.g., cameras **2.6-46** of FIGS. **2.6-1**) may include a first set of two or more front-facing cameras on the left side of front face **F** of device **2.6-10** and/or may include a second set of two or more front-facing cameras on the right side of front face **F** of device **2.6-10**. Cameras **2.6-46** may also be provided elsewhere in housing portion **2.6-12M**. Cameras **2.6-46** may, if desired, include cameras that are oriented at a slight angle relative to the $-Z$ axis of FIGS. **2.6-1**. For example, some of cameras **2.6-46** may be oriented directly ahead, whereas some cameras **2.6-46** along the left and right edges of front face **F** may be respectively angled slightly to the left and right of the $-Z$ axis to capture peripheral images on the left and right. Cameras **2.6-46** may capture visual odometry information, image information that is processed to locate objects in the user's field of view (e.g., so that virtual content can be registered appropriately relative to real-world objects), image content that is displayed in real time for a user of device **2.6-10**, and/or other suitable image data.

[1070] Device **2.6-10** may have left and right optical modules **2.6-40**. Optical modules **2.6-40** support electrical and optical components such as light-emitting components and lenses and may therefore sometimes be referred to as optical assemblies, optical systems, optical component support structures, lens and display support structures, electrical component support structures, or housing structures. Each optical module may include a respective display **2.6-14**, lens **2.6-30**, and support structure such as support structure **2.6-32**. Support structure **2.6-32**, which may sometimes be referred to as a lens support structure, optical component support structure, optical module support structure, or optical module portion, or lens barrel, may include hollow cylindrical

structures with open ends or other supporting structures to house displays **2.6-14** and lenses **2.6-30**. Support structures **2.6-32** may, for example, include a left lens barrel that supports a left display **2.6-14** and left lens **2.6-30** and a right lens barrel that supports a right display **2.6-14** and right lens **2.6-30**.

[1071] Displays **2.6-14** may include arrays of pixels or other display devices to produce images. Displays **2.6-14** may, for example, include organic light-emitting diode pixels formed on substrates with thin-film circuitry and/or formed on semiconductor substrates, pixels formed from crystalline semiconductor dies, liquid crystal display pixels, scanning display devices, and/or other display devices for producing images.

[1072] Lenses **2.6-30** may include one or more lens elements for providing image light from displays **2.6-14** to respective eyes boxes **2.6-13**. Lenses may be implemented using refractive glass lens elements, using mirror lens structures (catadioptric lenses), using Fresnel lenses, using holographic lenses, and/or other lens systems.

[1073] When a user's eyes are located in eye boxes **2.6-13**, displays (display panels) **2.6-14** operate together to form a display for device **2.6-10** (e.g., the images provided by respective left and right optical modules **2.6-40** may be viewed by the user's eyes in eye boxes **2.6-13** so that a stereoscopic image is created for the user). The left image from the left optical module fuses with the right image from a right optical module while the display is viewed by the user.

[1074] It may be desirable to monitor the user's eyes while the user's eyes are located in eye boxes **2.6-13**. For example, it may be desirable to use a camera to capture images of the user's irises (or other portions of the user's eyes) for user authentication. It may also be desirable to monitor the direction of the user's gaze. Gaze tracking information may be used as a form of user input and/or may be used to determine where, within an image, image content resolution should be locally enhanced in a foveated imaging system. To ensure that device **2.6-10** can capture satisfactory eye images while a user's eyes are located in eye boxes **2.6-13**, each optical module **2.6-40** may be provided with a camera such as camera **2.6-42** and one or more light sources such as light-emitting diodes **2.6-44** or other light-emitting devices such as lasers, lamps, etc. Cameras **2.6-42** and light-emitting diodes **2.6-44** may operate at any suitable wavelengths (visible, infrared, and/or ultraviolet). As an example, diodes **2.6-44** may emit infrared light that is invisible (or nearly invisible) to the user. This allows eye monitoring operations to be performed continuously without interfering with the user's ability to view images on displays **2.6-14**.

[1075] Not all users have the same interpupillary distance IPD. To provide device **2.6-10** with the ability to adjust the interpupillary spacing between modules **2.6-40** along lateral dimension X and thereby adjust the spacing IPD between eye boxes **2.6-13** to accommodate different user interpupillary distances, device **2.6-10** may be provided with optical module positioning systems in housing **2.6-12**. The positioning systems may have guide members and actuators **2.6-43** that are used to position optical modules **2.6-40** with respect to each other.

[1076] Actuators **2.6-43** can be manually controlled and/or computer-controlled actuators (e.g., computer-controlled motors) for moving support structures (lens barrels) **2.6-32** relative to each other. Information on the locations of the user's eyes may be gathered using, for example, cameras **2.6-42**. The locations of eye boxes **2.6-13** can then be adjusted accordingly.

[1077] As shown in the rear view of device **2.6-10** of FIGS. **2.6-2**, cover **2.6-12C** may cover rear face R while leaving lenses **2.6-30** of optical modules **2.6-40** uncovered (e.g., cover **2.6-12C** may have openings that are aligned with and receive modules **2.6-40**). As modules **2.6-40** are moved relative to each other along dimension X to accommodate different interpupillary distances for different users, modules **2.6-40** move relative to fixed housing structures such as the walls of main portion **2.6-12M** and move relative to each other.

[1078] A schematic diagram of an illustrative electronic device such as a head-mounted device or other wearable device is shown in FIGS. **2.6-3**. Device **2.6-10** of FIGS. **2.6-3** may be operated as a stand-alone device and/or the resources of device **2.6-10** may be used to communicate with

external electronic equipment. As an example, communications circuitry in device **2.6-10** may be used to transmit user input information, sensor information, and/or other information to external electronic devices (e.g., wirelessly or via wired connections). Each of these external devices may include components of the type shown by device **2.6-10** of FIGS. **2.6-3**.

[1079] As shown in FIGS. **2.6-3**, a head-mounted device such as device **2.6-10** may include control circuitry **2.6-20**. Control circuitry **2.6-20** may include storage and processing circuitry for supporting the operation of device **2.6-10**. The storage and processing circuitry may include storage such as nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **2.6-20** may be used to gather input from sensors and other input devices and may be used to control output devices. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors and other wireless communications circuits, power management units, audio chips, application specific integrated circuits, etc. During operation, control circuitry **2.6-20** may use display(s) **2.6-14** and other output devices in providing a user with visual output and other output.

[1080] To support communications between device **2.6-10** and external equipment, control circuitry **2.6-20** may communicate using communications circuitry **2.6-22**. Circuitry **2.6-22** may include antennas, radio-frequency transceiver circuitry, and other wireless communications circuitry and/or wired communications circuitry. Circuitry **2.6-22**, which may sometimes be referred to as control circuitry and/or control and communications circuitry, may support bidirectional wireless communications between device **2.6-10** and external equipment (e.g., a companion device such as a computer, cellular telephone, or other electronic device, an accessory such as a point device, computer stylus, or other input device, speakers or other output devices, etc.) over a wireless link. For example, circuitry **2.6-22** may include radio-frequency transceiver circuitry such as wireless local area network transceiver circuitry configured to support communications over a wireless local area network link, near-field communications transceiver circuitry configured to support communications over a near-field communications link, cellular telephone transceiver circuitry configured to support communications over a cellular telephone link, or transceiver circuitry configured to support communications over any other suitable wired or wireless communications link. Wireless communications may, for example, be supported over a Bluetooth® link, a WiFi® link, a wireless link operating at a frequency between 10 GHz and 400 GHz, a 60 GHz link, or other millimeter wave link, a cellular telephone link, or other wireless communications link. Device **2.6-10** may, if desired, include power circuits for transmitting and/or receiving wired and/or wireless power and may include batteries or other energy storage devices. For example, device **2.6-10** may include a coil and rectifier to receive wireless power that is provided to circuitry in device **2.6-10**.

[1081] Device **2.6-10** may include input-output devices such as devices **2.6-24**. Input-output devices **2.6-24** may be used in gathering user input, in gathering information on the environment surrounding the user, and/or in providing a user with output. Devices **2.6-24** may include one or more displays such as display(s) **2.6-14**. Display(s) **2.6-14** may include one or more display devices such as organic light-emitting diode display panels (panels with organic light-emitting diode pixels formed on polymer substrates or silicon substrates that contain pixel control circuitry), liquid crystal display panels, microelectromechanical systems displays (e.g., two-dimensional mirror arrays or scanning mirror display devices), display panels having pixel arrays formed from crystalline semiconductor light-emitting diode dies (sometimes referred to as microLEDs), and/or other display devices.

[1082] Sensors **2.6-16** in input-output devices **2.6-24** may include force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors such as a touch sensor that forms a button,

trackpad, or other input device), and other sensors. If desired, sensors **2.6-16** may include optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, optical touch sensors, optical proximity sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, image sensors (e.g., cameras), fingerprint sensors, iris scanning sensors, retinal scanning sensors, and other biometric sensors, temperature sensors, sensors for measuring three-dimensional non-contact gestures (“air gestures”), pressure sensors, sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or inertial measurement units that contain some or all of these sensors), health sensors such as blood oxygen sensors, heart rate sensors, blood flow sensors, and/or other health sensors, radio-frequency sensors, three-dimensional camera systems such as depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices that capture three-dimensional images) and/or optical sensors such as self-mixing sensors and light detection and ranging (LIDAR) sensors that gather time-of-flight measurements (e.g., time-of-flight cameras), humidity sensors, moisture sensors, gaze tracking sensors, electromyography sensors to sense muscle activation, facial sensors, and/or other sensors. In some arrangements, device **2.6-10** may use sensors **2.6-16** and/or other input-output devices to gather user input. For example, buttons may be used to gather button press input, touch sensors overlapping displays can be used for gathering user touch screen input, touch pads may be used in gathering touch input, microphones may be used for gathering audio input (e.g., voice commands), accelerometers may be used in monitoring when a finger contacts an input surface and may therefore be used to gather finger press input, etc.

[1083] If desired, electronic device **2.6-10** may include additional components (see, e.g., other devices **2.6-18** in input-output devices **2.6-24**). The additional components may include haptic output devices, actuators for moving movable housing structures, audio output devices such as speakers, light-emitting diodes for status indicators, light sources such as light-emitting diodes that illuminate portions of a housing and/or display structure, other optical output devices, and/or other circuitry for gathering input and/or providing output. Device **2.6-10** may also include a battery or other energy storage device, connector ports for supporting wired communication with ancillary equipment and for receiving wired power, and other circuitry.

[1084] Housing **2.6-12** may include support structures for optical modules **2.6-40** and other components of device **2.6-10**. In an illustrative configuration, housing **2.6-12** may include a head-mounted support structure such as frame **2.6-12I** of FIGS. **2.6-4**. Frame **2.6-12I** may have support structures that run vertically (e.g., frame portion **2.6-12I-M** in the middle of device **2.6-10** that are aligned with the user's nose bridge) and may have support structures that run horizontally across the top edge of housing **2.6-12**, along the bottom of edge of housing **2.6-12**, and along the left and right edges of housing **2.6-12** (see, e.g., peripheral edge portion **2.6-12I-E**). This forms left and right openings in frame **2.6-12I** that receive, respectively, left and right optical modules **2.6-40**. There may, in general, be one or more supporting members in device housing **2.6-12** that help create housing portion **2.6-12M** and that support the components in housing portion **2.6-12M**. The frame **2.6-12I** of FIGS. **2.6-4** is illustrative.

[1085] As shown in FIGS. **2.6-4**, one or more component support structures such as camera support structure **2.6-50** may be coupled to frame **2.6-12I** (e.g., left and right camera support structures **2.6-50** may be attached to respective left and right peripheral edges such as edge portions **2.6-12I-E** of frame **2.6-12I**). Support structures for device **2.6-10** such as frame **2.6-12I** and camera support structure **2.6-50** may be formed from polymer, glass, ceramic, metal, carbon-fiber composite material or other fiber-composite material, other materials, and/or combinations of these materials (e.g., sheets of rigid polymer or other material, and/or other structural members).

[1086] There may be multiple component support structures coupled to frame **2.6-12I**. For example, there may be a right-hand camera support structure **2.6-50** coupled to a right side of frame **2.6-12I** and a left-hand camera support structure **2.6-50** coupled to a left side of frame **2.6-12I**. A

single side of frame **2.6-12I** and corresponding camera support structure **2.6-50** is shown in the example of FIGS. **2.6-4**.

[1087] Camera support structure **2.6-50** may be coupled to frame **2.6-12I** using adhesive, welds, screws or other fasteners, mating engagement structures (e.g., recesses and protrusions for forming a snap fit), press-fit connections, and/or other coupling arrangements. In the example of FIGS. **2.6-4**, fasteners **2.6-56** (e.g., threaded fasteners such as screws) pass through through-hole openings **2.6-54** of camera support structure **2.6-50** and are received in corresponding openings **2.6-52** of frame **2.6-12I**. Openings **2.6-52** may be threaded openings or may be unthreaded through-hole openings in configurations in which fasteners **2.6-56** are supplied with corresponding threaded nuts (as examples).

[1088] Camera support structure **2.6-50** may be configured to receive cameras **2.6-46** (e.g., structure may have recesses, openings, and/or other structures configured to receive front-facing cameras). As an example, camera support structures **2.6-50** may have at least two openings **2.6-58** (e.g., through-hole openings), each of which is configured to receive an associated camera. Each camera **2.6-46**, which may sometimes be referred to as a camera module, may have a camera module housing and may have a lens and image sensor coupled to the camera module housing. Cameras **2.6-46** may be sensitive to any suitable wavelengths of light (e.g., infrared, visible, both infrared and visible, and/or other wavelengths), may be stereoscopic (three-dimensional) cameras or two-dimensional cameras, may be time-of-flight cameras, may be structured light three-dimensional cameras may be cameras that gather information for use in placing virtual objects in a scene containing real-world and virtual content, may be cameras that are used as part of a visual odometry system, and/or may be other imaging systems. If desired, other optical components may be mounted to camera mounting structure **2.6-50**. For example, ambient light sensors, proximity sensors, and/or other components that emit and/or detect light may be mounted to structure **2.6-50**. Configurations in which two or more cameras **2.6-46** are attached to each camera mounting structure **2.6-50** may sometimes be described herein as an example.

[1089] When cameras **2.6-46** are received within respective openings **2.6-58** of a rigid unitary camera support structure **2.6-50** and/or are otherwise mounted to camera support structure **2.6-50**, the relative position of these cameras becomes fixed. This ensures that the direction in which each camera is pointing (e.g., the orientation of the camera's field of view) is fixed relative to the other, thereby helping to avoid misalignment issues arising from cameras orientations that vary during use of device **2.6-10**. By attaching camera support structure **2.6-50** to frame **2.6-12I**, the rigidity and strength of frame **2.6-12I** may be enhanced. This helps ensure that housing portion **2.6-12M** is sturdy and able to maintain sensitive components such as optical modules **2.6-40** in alignment with each other in the event that device **2.6-10** is subjected to an undesired drop event.

[1090] Camera support structure **2.6-50** may be formed from a layer of polymer or other material with optional ribs and/or other features to help strengthen structure **2.6-50** without adding excessive weight. To help maintain the rigidity and strength of camera support structure **2.6-50**, support structure **2.6-50** may be partly or completely free of large notches along the periphery of structure **2.6-50**. This may help ensure that there are no portions with locally narrowed widths along the length of structure **2.6-50** that could compromise the rigidity of structure **2.6-50**. The width of support structure may be relatively large near the middle of structure **2.6-50**. For example, support structure **2.6-50** may have a maximum width across its shorter lateral dimension that is at least 2 mm, at least 4 mm, at least 8 mm, at least 16 mm, at least 32 mm, less than 40 mm, less than 25 mm, less than 18 mm, less than 15 mm, less than 10 mm, less than 7 mm, or other suitable value. The longitudinal dimension (length) of support structure **2.6-50** may be at least 2 cm, at least 4 cm at least 8 cm, at least 16 cm, less than 20 cm, less than 14 cm, less than 10 cm, less than 6 cm, less than 4 cm, or other suitable value. The minimum thickness of support **2.6-50** may be at least 0.3 mm, at least 0.6 mm, at least 1.2 mm, at least 2.4 mm, less than 5 mm, less than 2.5 mm, less than 1.3 mm, less than 0.8 mm, less than 0.5 mm, or other suitable value.

[1091] In addition to supporting cameras **2.6-54** and/or other optical components, camera support structure **2.6-50** may serve as a support for wireless communications components such as antennas **2.6-60**. In the example of FIGS. **2.6-4**, camera support structure **2.6-50** serves as a support member for a pair of antennas **2.6-60**. In general, camera support structure **2.6-50** may support at least one antenna, at least two antennas, at least three antennas, fewer than ten antennas, from 2 to 5 antennas, or other suitable number of antennas. Antennas **2.6-60** may be formed using any suitable antenna types. For example, antennas **2.6-60** may include antennas with resonating elements that are formed from loop antenna structures, patch antenna structures, inverted-F antenna structures, slot antenna structures, planar inverted-F antenna structures, monopoles, dipoles, helical antenna structures, Yagi (Yagi-Uda) antenna structures, hybrids of these designs, etc. If desired, one or more of antennas **2.6-60** may be cavity-backed antennas. Different types of antennas may be used for different bands and combinations of bands. For example, one type of antenna may be used in forming a local wireless link antenna and another type of antenna may be used in forming a remote wireless link antenna. Dedicated antennas may be used for receiving satellite navigation system signals or, if desired, antennas **2.6-60** can be configured to receive both satellite navigation system signals and signals for other communications bands (e.g., wireless local area network signals and/or cellular telephone). Antennas **2.6-60** may be formed from metal members, patterned thin-film metal layers, and/or other conductive structures.

[1092] The front face of device **2.6-10** may be covered with an inactive housing wall (e.g., a polymer layer). In the example of FIGS. **2.6-5**, front face F of device **2.6-10** is covered by display **2.6-14F** (e.g., an organic light-emitting diode display, a microLED display, an electrophoretic display, liquid crystal display, etc.). The pixels of display **2.6-14F** may be covered with an outer protective display cover layer (e.g., a layer of glass, a layer of clear polymer, etc.).

[1093] Optical windows such as camera windows **2.6-62** may be provided in the display cover layer. Camera windows **2.6-62** may be formed from portions of the display cover layer or from clear window structures that are mounted in openings in the display cover layer. Each optical window may overlap a corresponding optical component and may allow light from the component to be emitted through the optical window and/or may allow ambient light from the environment to pass to the optical component. Camera windows **2.6-62** (e.g., camera windows in the display cover layer for display **2.6-14F** and/or optical windows formed in other portions of housing **2.6-12**) may have optical characteristics that allow an associated optical component to operate satisfactorily. Consider, as an example, a camera window **2.6-62** that overlaps one of forward-facing cameras **2.6-46**. As shown in FIGS. **2.6-5**, camera support structure **2.6-50** may be mounted in the interior of device **2.6-10** so that cameras **2.6-46** are aligned with camera windows **2.6-62** and so that antennas **2.6-60** are overlapped by the display cover layer for display **2.6-14F**. Each camera window **2.6-62** may have a visible-light and/or infrared-light transparency level sufficient to allow the forward-facing camera **2.6-46** that is overlapped by that window to capture images of real-world objects in the user's environment and/or to gather other image data. The transmission of camera window **2.6-62** may be, as an example, at least 50%, at least 90%, at least 95%, or other suitable value (at visible and/or infrared wavelengths). Non-camera components (e.g., an ambient light sensor, an optical proximity sensor, etc.) may have optical windows with other transmission values.

[1094] FIGS. **2.6-6** is a cross-sectional view of a portion of device **2.6-10** in an illustrative configuration in which display **2.6-14F** is formed on front face F of device **2.6-10**. As shown in FIGS. **2.6-6**, display **2.6-14F** includes pixel array **2.6-14M** (e.g., a display layer such as an organic light-emitting diode display layer, an array of crystalline light-emitting diodes, an electrophoretic display layer, a liquid crystal display layer, etc.). Display cover layer CG of display **2.6-14F** may cover and protect pixel array **2.6-14M**. During operation, display **2.6-14F** may present images to a user (while device **2.6-10** is or is not being worn on a user's head). If desired, display **2.6-14F** may have touch screen functionality, so that a user may supply touch input to front face F of device **2.6-10**.

[1095] Camera **2.6-46** may be located at the edge of display **2.6-14F** (e.g., outside of the active area of the display), camera **2.6-46** may operate through an opening in pixel array **2.6-14M**, and/or camera **2.6-46** may sense light that passes through gaps in the opaque structures of pixel array **2.6-14M**. In the illustrative configuration of FIGS. **2.6-6**, camera **2.6-46** is located in an inactive display border region that is free of pixels. As shown in FIGS. **2.6-6**, camera window **2.6-62** may be formed from an opening in opaque masking layer **2.6-64** that allows light to pass through display cover layer CG. Opaque masking layer **2.6-64** may be, as an example, a layer of black ink that is formed on the inner surface of display cover layer CG.

[1096] Camera **2.6-46** may be mounted to an opening in camera support structure **2.6-50** using bonds **2.6-66** (e.g., adhesive bonds, welds, etc.), using screws or other fasteners such as illustrative fastener **2.6-68**, or using other attachment mechanisms (press-fit connections, mating engagement structures, etc.). In turn, camera support structure **2.6-50** may be attached to frame **2.6-12I** by heat stakes (e.g., heat staked protrusions extending from camera support structure **2.6-50** into mating openings in frame **2.6-12I** and/or heat staked protrusions extending from frame **2.6-12I** into openings), adhesive, welds (e.g., laser welds joining a metal camera support structure to a metal frame, laser welds joining polymer camera support structure to a polymer frame, and/or other welds), press-fit connections, mating engagement structures (e.g., snaps), or other attachment structures **2.6-70** and/or screws or other fasteners **2.6-56** (e.g., screws that are received within threaded openings in camera support structure **2.6-50** and/or frame **2.6-12I**, screws that are received within insert nuts, etc.).

[1097] As shown in FIGS. **2.6-6**, antenna **2.6-60** may be formed from conductive antenna structures (e.g., metal traces, stamped metal foil, etc.) supported by camera support structure **2.6-50**. During operation, antenna **2.6-60** may transmit and/or receive wireless signals that pass through display cover layer CG and other portions of housing **2.6-12M**.

[1098] A schematic diagram of an illustrative antenna (antenna **2.6-60**) coupled to illustrative radio-frequency transceiver circuitry **2.6-90** is shown in FIGS. **2.6-7**. Communications circuitry **2.6-22** of FIG. **3** may include transceiver circuitry **2.6-90** (FIGS. **2.6-7**) and/or other wireless circuitry formed from one or more integrated circuits, power amplifier circuitry, low-noise input amplifiers, passive radio-frequency (RF) components, one or more antennas **2.6-60**, transmission lines, and other circuitry for handling RF wireless signals.

[1099] Radio-frequency transceiver circuitry **2.6-90** of FIGS. **2.6-7** may use antenna **2.6-60** for handling various radio-frequency communications bands. For example, circuitry **2.6-90** may include wireless local area network transceiver circuitry (e.g., circuitry **2.6-90** may handle 2.4 GHz and 5 GHz bands for WiFi® (IEEE 802.11) communications) and may handle the 2.4 GHz Bluetooth® communications band. If desired, circuitry **2.6-90** may use cellular telephone transceiver circuitry or other circuitry for handling cellular telephone wireless communications and/or other wireless communications in frequency ranges such as a communications band from 700 to 2700 MHz, from 3.4 to 3.6 GHz, from 450 MHz to 6 GHz, from 24 to 53 GHz, from 5 to 8 GHz, from 60 to 90 GHz, and/or other communications bands. Circuitry **2.6-90** may handle voice data and non-voice data.

[1100] Transceiver circuitry **2.6-90** may include satellite navigation system circuitry such as Global Positioning System (GPS) receiver circuitry for receiving GPS signals at 1575 MHz or for handling other satellite positioning data (e.g., GLONASS signals at 1609 MHz). Satellite navigation system signals are received from a constellation of satellites orbiting the earth.

[1101] In satellite navigation system links, cellular telephone links, and other long-range links, wireless signals are typically used to convey data over thousands of feet or miles. In WiFi® and Bluetooth® links at 2.4 and 5 GHz and other short-range wireless links, wireless signals are typically used to convey data over tens or hundreds of feet. If desired, device **2.6-10** may include millimeter wave wireless transceiver circuitry. To enhance signal reception for millimeter wave communications, phased antenna arrays and beam steering techniques may be used (e.g., schemes

in which antenna signal phase and/or magnitude for each antenna in an array is adjusted to perform beam steering). Antenna diversity schemes may also be used to ensure that the antennas that have become blocked or that are otherwise degraded due to the operating environment of device **2.6-10** can be switched out of use and higher-performing antennas used in their place. Circuitry **2.6-90** can include circuitry for other short-range and long-range wireless links if desired. For example, circuitry **2.6-90** may include circuitry for receiving television and radio signals, paging system transceivers, near field communications (NFC) circuitry, etc. If desired, circuitry **2.6-90** and/or other wireless circuitry may use antennas such as antenna **2.6-60** for radio-frequency sensing (e.g., to determine the orientation and/or distance between device **2.6-10** and other wireless equipment, to form radar-based sensors, etc.).

[1102] As shown in FIGS. **2.6-7**, radio-frequency transceiver circuitry **2.6-90** may be coupled to antenna feed **2.6-102** of antenna **2.6-60** using transmission line **2.6-92**. Antenna feed **2.6-102** may include a positive antenna feed terminal such as positive antenna feed terminal **2.6-98** and may have a ground antenna feed terminal such as ground antenna feed terminal **2.6-100**. Transmission line **2.6-92** may be formed from metal traces on a printed circuit or other conductive structures and may have a positive transmission line signal path such as path **2.6-94** that is coupled to terminal **2.6-98** and a ground transmission line signal path such as path **2.6-96** that is coupled to terminal **2.6-100**. Transmission line paths such as path **2.6-92** may be used to route antenna signals within device **2.6-10**. For example, transmission line paths may be used to couple antenna structures such as one or more antennas in an array of antennas to transceiver circuitry **2.6-90**. Transmission lines in device **2.6-10** may include coaxial cable paths, microstrip transmission lines, stripline transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, transmission lines formed from combinations of transmission lines of these types, etc. Filter circuitry, switching circuitry, impedance matching circuitry, and other circuitry may be interposed within transmission line **2.6-92** and/or circuits such as these may be incorporated into antenna **2.6-60** (e.g., to support antenna tuning, to support operation in desired frequency bands, etc.).

[1103] Device **2.6-10** may contain multiple antennas **2.6-60**. The antennas may be used together or one of the antennas may be switched into use while other antenna(s) are switched out of use. If desired, control circuitry **2.6-20** may be used to select an optimum antenna to use in device **2.6-10** in real time and/or to select an optimum setting for adjustable wireless circuitry associated with one or more of antennas **2.6-60**. Antenna adjustments may be made to tune antennas to perform in desired frequency ranges, to perform beam steering with a phased antenna array, and to otherwise optimize antenna performance. Sensors may be incorporated into antennas **2.6-60** to gather sensor data in real time that is used in adjusting antennas **2.6-60**.

[1104] FIGS. **2.6-8** is a diagram of an illustrative antenna that may be used in device **2.6-10**. In the example of FIGS. **2.6-8**, antenna **2.6-60** is an inverted-F antenna. As shown in FIGS. **2.6-8**, antenna **2.6-60** may include an antenna resonating element such as antenna resonating element **2.6-110** and an antenna ground such as antenna ground **2.6-112**. Antenna resonating element **2.6-110** may have one or more branches such as antenna resonating element arm **2.6-116** and optional antenna resonating element arm **2.6-116'**. Return path **2.6-118** (sometimes referred to as a short circuit path) may be coupled between resonating element arm **2.6-116** and ground **2.6-112**. Antenna feed **2.6-102** may include positive antenna feed terminal **2.6-98** and ground antenna feed terminal **2.6-100** and may be coupled between element **2.6-110** (e.g., arm **2.6-116**) and ground **2.6-112** in parallel with return path **2.6-118**. One or more optional components (switches, tunable circuits such as tunable capacitors, tunable inductors, etc.) may be coupled between antenna ground **2.6-112** and resonating element arm **2.6-116** and may be adjusted to tune antenna **2.6-60**. The configuration of FIGS. **2.6-8** in which no tunable components are coupled between arm **2.6-116** and ground **2.6-112** is merely illustrative.

[1105] Antenna resonating element arm **2.6-116** may be separated from ground **2.6-112** by dielectric opening **2.6-122**. If desired, opening **2.6-122** may form a slot antenna element that

contributes to the antenna response of antenna **2.6-60**. In the example of FIGS. **2.6-8**, antenna **2.6-40** is an inverted-F antenna that does not include a slot antenna element.

[1106] Optional parasitic antenna elements such as optional parasitic element **2.6-124** may be included in antenna **2.6-60** to adjust the frequency response of antenna **2.6-60**.

[1107] Antennas such as antenna **2.6-60** of FIGS. **2.6-8** (e.g., inverted-F antennas, slot antennas, hybrid inverted-F slot antennas, etc.) and/or other types of antenna **2.6-60** (e.g., patch antennas, loop antennas, etc.) may be used in supporting any suitable operations involving transmission and/or reception of wireless signals.

[1108] Antennas (e.g., antenna resonating elements, parasitic elements, antenna ground structures, feed structures, and/or other structures for each antenna **2.6-60**) may be formed from conductive structures such as metal members (e.g., metal structures formed from wireless, machined metal parts, stamped sheet metal, etc.), metal traces (e.g., patterned metal deposited by physical vapor deposition or laser-assisted deposition techniques), other conductive materials (e.g., carbon nanowires, etc.), and/or other conductive antenna structures. These conductive structures may be supported by substrates such as rigid and/or flexible printed circuit substrates, by polymer housing structures (e.g., by portions of camera support structure **2.6-50**), dielectric members formed from glass, ceramic, and/or other dielectric, and/or other antenna support structures.

[1109] FIGS. **2.6-9**, **2.6-10**, **2.6-11**, and **2.6-12** are cross-sectional side views of illustrative conductive antenna structures **2.6-126** for use in forming antennas **2.6-60**.

[1110] In the illustrative configuration of FIGS. **2.6-9**, laser direct structuring (LDS) techniques are being used to form antenna structures **2.6-126**. Laser beam **2.6-128** is used to selectively illuminate area **2.6-130** on the surface of a dielectric antenna support structure such as structure **2.6-50**.

Structure **2.6-50** in the example of FIGS. **2.6-9** may be formed from polymer with additives to help sensitize structure **2.6-50** to laser light exposure. After laser light exposure with beam **2.6-128**, electroplating operations are used to selectively electrodeposit conductive structures **2.6-126** on area **2.6-130** without depositing the conductive structures elsewhere on the exposed surface of structure **2.6-50**, thereby forming structures **2.6-126** with a desired antenna shape (e.g., to form an antenna resonating element, parasitic element, ground, and/or other patterned antenna structures as shown in FIGS. **2.6-8**).

[1111] In the example of FIGS. **2.6-10**, conductive antenna structures **2.6-126** are metal traces deposited on printed circuit **2.6-132**. These metal traces may be deposited by physical vapor deposition and patterned using photolithography, and/or may be formed using other deposition and patterning techniques. Metal traces **2.6-134** of printed circuit **2.6-132** may help convey radio-frequency signals to and/or from antenna structures **2.6-126**. Adhesive **2.6-136** may be used to attach printed circuit **2.6-132** to a surface of support structure **2.6-50**.

[1112] If desired, conductive antenna structures **2.6-126** can be formed from metal structures embedded in support structure **2.6-50**. For example, metal antenna structures (wire, metal foil, structural metal members, sheet metal parts, and/or other conductive antenna structures forming antenna structures **2.6-126**) can be embedded in polymer that forms support structure **2.6-50**, as shown in FIGS. **2.6-11** (e.g., one or more shots of polymer for support structure **2.6-50** may be molded over conductive antenna structures **2.6-126**).

[1113] In the illustrative example of FIGS. **2.6-12**, printed circuit **2.6-132** has metal traces forming conductive antenna structures **2.6-126** and metal traces **2.6-134** forming signal paths such as transmission lines. As shown in FIGS. **2.6-12**, printed circuit **2.6-132** may be embedded within support structure **2.6-50** (e.g., polymer forming support structure **2.6-50** may be molded over printed circuit **2.6-132**).

[1114] The arrangements of FIGS. **2.6-9**, **2.6-10**, **2.6-11**, and/or **2.6-12** and/or other arrangements may be used in forming antennas **2.6-60** on camera support structure **2.6-50**, while camera support structure **2.6-50** simultaneously serves as a support and alignment member for cameras **2.6-46**.

[1115] FIGS. **2.6-13** is a top view of an illustrative camera support structure formed using multiple

shots of polymer. One shot of polymer forms portion **2.6-50-1** of camera support structure **2.6-50** and another shot of polymer forms portion **2.6-50-2** of camera support structure **2.6-50**. Portion **2.6-50-1** may, as an example, include fibers or other filler embedded in the shot of polymer forming portion **2.6-50-1** or portion **2.6-50-1** may have an embedded fiber-composite member (e.g., a stiffening member formed from a rod, strip, or other elongated member of carbon-fiber material or other stiffening member). This may help to locally stiffen and strengthen portion **2.6-50-1** (e.g., to enhance the stiffness of portion **2.6-50-1** relative to portion **2.6-50-2**). As shown in FIGS. **2.6-13**, stiffening member **2.6-50M** may extend between openings **2.6-58** (and therefore cameras **2.6-46**) to prevent bending of the intervening portion of structure **2.6-50** (e.g., to prevent bending of structure **2.6-50** out of the X-Y plane of FIGS. **2.6-13**) and thereby prevent undesired bending-induced camera misalignment. Portion **2.6-50-1** may, if desired, be free of conductive material such as conductive carbon fibers (e.g., to reduce the presence of conductive material that could interfere with the operation of overlapping antennas).

[1116] FIGS. **2.6-14** is a cross-sectional side view of camera support structure **2.6-50** taken along line **2.6-140** of FIGS. **2.6-13** and viewed in direction **2.6-142** of FIGS. **2.6-13**. As shown in FIGS. **2.6-14**, camera support structure **2.6-50** may include an embedded stiffening structure such as fiber-composite stiffening member **2.6-50M** (e.g., an elongated strip-shaped carbon-fiber stiffening member). Member **2.6-50M** may be embedded within portion **2.6-50-2**. Portions **2.6-50-1** and **2.6-50-2** may be formed from first and second shots of molded polymer material or may be formed using other techniques.

[1117] It may be desirable to detect misalignment of cameras **2.6-46** due to deformation of camera support structure **2.6-50**. As shown in the cross-sectional side view of structure **2.6-50** of FIGS. **2.6-15**, a bend sensor such as sensor **2.6-16B** may be mounted to camera support structure **2.6-50** between cameras **2.6-46**. Sensor **2.6-16B** may be a strain gauge or other sensor that is configured to detect bending of structure **2.6-50** (e.g., bending about bend axis **2.6-150**). Flexible printed circuit **2.6-152** may have signal lines that carry bending measurements to control circuitry **2.6-20** (FIG. **3**). In response to measuring bending in structure **2.6-50**, control circuitry **2.6-20** can take corrective action to compensate for any predicted misalignment between cameras **2.6-46**. For example, if cameras **2.6-46** are detected as being misaligned by 10 from data gathered by sensor **2.6-16B**, control circuitry **2.6-20** can digitally compensate for the measured misalignment (e.g., by shifting and/or warping the camera image data gathered by cameras **2.6-46** to ensure that the images from cameras **2.6-46** can be stitched together as desired or otherwise used as desired in operating device **2.6-10**).

[1118] If desired, device **2.6-10** may have one or more camera positioning devices such as actuator **2.6-160** of FIGS. **2.6-16**. Actuator **2.6-160** can change the angular orientation of camera **2.6-46** relative to structure **2.6-50**. In response to detecting with sensor **2.6-16B** that structure **2.6-50** has bent about axis **150** of FIGS. **2.6-15** by **20**, for example, control circuitry **2.6-20** may direct actuator **2.6-160** to move camera **2.6-46** to compensate. For example, camera **2.6-46** may be tilted in an opposing direction by a compensating amount (e.g., **-20**), thereby ensuring that cameras **2.6-46** remain aligned even if structure **2.6-50** experiences deformation during operation of device **2.6-10**.

[1119] The use of a strain gauge to detect bending is illustrative. Any suitable sensor **2.6-16** may be used to detect camera misalignment due to deformation of support structure **2.6-50**. The effects of camera misalignment may be compensated by physically steering optical components such as cameras **2.6-46** (as described in connection with FIGS. **2.6-16**), by processing the image data from cameras **2.6-46** (e.g., image warping, etc.), and/or by otherwise compensating for detected misalignment. The examples of FIGS. **2.6-13**, **2.6-14**, **2.6-15**, and **2.6-16** are illustrative.

III: Display Integration Assembly

[1120] FIG. **3-1** illustrates a perspective view of a front cover assembly **3-100** of an HMD device described herein, for example the front cover assembly **3-1** of the HMD **3-100** shown in FIG. **3-1**

or any other HMD device shown and described herein. The front cover assembly **3-100** shown in FIG. **1** can include a transparent or semi-transparent cover **3-102**, shroud **3-104** (or “canopy”), adhesive layers **3-106**, display assembly **3-108** including a lenticular lens panel or array **3-110**, and a structural trim **3-112**. The adhesive layer **3-106** can secure the shroud **3-104** and/or transparent cover **3-102** to the display assembly **3-108** and/or the trim **3-112**. The trim **3-112** can secure the various components of the front cover assembly **3-100** to a frame or chassis of the HMD device. [1121] In at least one example, as shown in FIG. **3-1**, the transparent cover **3-102**, shroud **3-104**, and display assembly **3-108**, including the lenticular lens array **3-110**, can be curved to accommodate the curvature of a user's face. The transparent cover **3-102** and the shroud **3-104** can be curved in two or three dimensions, e.g. vertically curved in the Z-direction in and out of the Z-X plane and horizontally curved in the X-direction in and out of the Z-X plane. In at least one example, the display assembly **3-108** can include the lenticular lens array **3-110** as well as a display panel having pixels configured to project light through the shroud **3-104** and the transparent cover **3-102**. The display assembly **3-108** can be curved in at least one direction, for example the horizontal direction, to accommodate the curvature of a user's face from one side (e.g., left side) of the face to the other (e.g., right side). In at least one example, each layer or component of the display assembly **3-108**, which will be shown in subsequent figures and described in more detail, but which can include the lenticular lens array **3-110** and a display layer, can be similarly or concentrically curved in the horizontal direction to accommodate the curvature of the user's face. [1122] In at least one example, the shroud **3-104** can include a transparent or semi-transparent material through which the display assembly **3-108** projects light. In one example, the shroud **3-104** can include one or more opaque portions, for example opaque ink-printed portions or other opaque film portions on the rear surface of the shroud **3-104**. The rear surface can be the surface of the shroud **3-104** facing the user's eyes when the HMD device is donned. In at least one example, opaque portions can be on the front surface of the shroud **3-104** opposite the rear surface. In at least one example, the opaque portion or portions of the shroud **3-104** can include perimeter portions visually hiding any components around an outside perimeter of the display screen of the display assembly **3-108**. In this way, the opaque portions of the shroud hide any other components, including electronic components, structural components, and so forth, of the HMD device that would otherwise be visible through the transparent or semi-transparent cover **3-102** and/or shroud **3-104**.

[1123] In at least one example, the shroud **3-104** can define one or more apertures transparent portions **3-120** through which sensors can send and receive signals. In one example, the portions **3-120** are apertures through which the sensors can extend or send and receive signals. In one example, the portions **3-120** are transparent portions, or portions more transparent than surrounding semi-transparent or opaque portions of the shroud, through which sensors can send and receive signals through the shroud and through the transparent cover **3-102**. In one example, the sensors can include cameras, IR sensors, LUX sensors, or any other visual or non-visual environmental sensors of the HMD device.

[1124] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **3-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **3-2-3-6** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **3-2-3-6** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **3-1**.

[1125] FIG. **3-2** illustrates a partial, cross-sectional view of an example of a front cover assembly **3-200**, including a shroud **3-204** and a display assembly **3-208** coupled to the shroud **3-204** via a display bracket **3-214**. The shroud **3-204** can be coupled to a shroud bracket **3-216**, which can couple the front cover assembly **3-200** to one or more structural frame members **3-218** of an HMD

device. The display bracket **3-214** can extend behind the display assembly **3-208** and fix the display assembly **3-208** adjacent to the shroud **3-204** such that pixels of the display screen of the display assembly **3-208** project light outward through the shroud **3-204**.

[1126] In at least one example, the display assembly **3-208** is curved in one direction, e.g., the horizontal direction, but not the vertical direction, e.g. up and down in the orientation shown in FIG. **3-2**. In such an example, the shroud **3-204** can be curved in both vertical and horizontal directions, as shown in FIG. **1** and described above.

[1127] FIG. **3-3** shows a side view of another front cover assembly **3-300** similar to the front cover assembly **3-200** shown in FIG. **3-2**. In FIG. **3-3**, the front cover assembly **3-300** is shown without an exterior transparent cover but includes a shroud **3-304** defining mechanical or visual apertures **3-320** through which sensors can send and receive signals as well as a display assembly **3-308** secured to a back or rear side of the shroud **3-304**.

[1128] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **3-2** and **3-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **3-1** and **3-4** through **3-6** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **3-1** and **3-4** through **3-6** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **3-2** and **3-3**.

[1129] FIG. **3-4** illustrates a partial, cross-sectional view of an example of a front cover assembly **3-400**, including an outer transparent or semi-transparent cover **3-402**, a shroud **3-404**, a blinker film **3-422** secured to a back/rear side of the shroud **3-404**, a display assembly **3-408** including a lenticular lens array **3-410**, a display bracket **3-414** securing the display assembly relative to the shroud **3-404** such that an air gap **3-424** is defined between the lenticular lens array **3-410** and the blinker film **3-422**, and a graphite layer **3-426** disposed on a back/rear side of the display bracket **3-414**. The air gap **3-424** can define an open space or volume between the lenticular lens array **3-410** and the blinker film **3-422** such that no other components are disposed within the gap **3-424** between the lenticular lens array **3-410** and the blinker film **3-422**. In at least one example, the blinker film **3-422** can be configured to hide the display from view when the display of the display assembly **3-408** is not on or projecting light and/or when someone views the front display assembly from certain predetermined angles (e.g., wide angles instead of narrow, more “straight on” viewing angles in front of the display **3-408**). In at least one example, the lenticular lens array **3-410** can be used to generate a three-dimensional effect of those not donning the HMD and viewing the outward facing display **3-408** through the cover **3-402** and shroud **3-404**. The gap **3-424** can be present to allow light transmitted through the cover **3-402**, shroud **3-404**, and blinker film **3-422** to pass appropriately through the lenses of the lenticular lens array **3-410** to create the desired three-dimensional effect.

[1130] In at least one example, the blinker film **3-422** can be adhered to the shroud **3-404** via an optically clear adhesive **3-423** shown in FIG. **3-4A**. In one example, the lenticular lens array **3-410** can be adhered to the display assembly **3-408** via an optically clear adhesive **3-411** shown in FIG. **3-4A**. In at least one example, the blinker film **3-422** can diffuse and darken light, including specular highlights from the lenticular lens array **3-410**, passing through from the display assembly **3-408**. In at least one example, diffusion particles can be formed in the optically clear adhesive layer **3-423**, which can be referred to as a haze film **3-432**, between the blinker film **3-422** and the shroud **3-404**. In at least one example, the diffusion particles can be formed within the blinker film **3-422** itself. In at least one example, the diffusion particles can be formed within the shroud **3-404**. In at least one example, the diffusion particles can include titanium dioxide. In at least one example, the display assembly **3-408** can include an LCD and/or OLED display layer rather than the lenticular lens array **3-410**, e.g., with a flat display assembly (as opposed to the curved display assembly **3-408** shown) to provide 3-d visual effects. The front cover assembly of the HMD can

also include a dust seal **3-409** disposed between the shroud **3-404** and the lenticular lens array **3-422** and/or other components of the assembly, including the display assembly **3-408**, to prevent dust and other particles/pollutants from entering the air gap **3-424** between the lenticular lens array **3-410** and the blinker film **3-422**.

[1131] In at least one example, the graphite layer **3-426** can be one of multiple graphite layers, for example two layers, three layers, four layers, or more than four layers. The graphite layers **3-426** can be configured to spread heat from the display assembly **3-408** coupled to the display bracket **3-414**. In at least one example, the display bracket **3-414** is a metal material that acts as a heat sink to dissipate heat from the display assembly **3-408**. In one example, the display bracket **3-414** includes a thermally conductive material. In one example, the display bracket **3-414** includes magnesium.

[1132] In at least one example, the cover **3-402** includes a clear core material **3-427**. In one example, the core material **3-427** can include polycarbonate. In one example, the core material includes glass or other ceramic materials. In at least one example, optical clear adhesive (OCA) layers **3-428a**, **3-428b** can be disposed on either side, respectively, of the core material **3-427**. In at least one example, first and second protective coating layers/films **3-430a**, **3-430b** can be disposed on either side, respectively, of the core material **3-427**, with the OCA layers **3-428a**, **3-428b** disposed between respective protective coating films **3-430a**, **3-430b** and the core material **3-427**. In at least one example, an additional hard-coat layer and or an additional anti-reflective coating, which can be part of a hard-coat layer, can be disposed on top of the first polycarbonate film **3-430a** to define an external surface of the transparent cover **3-402**.

[1133] In at least one example, the cover assembly **3-400** can include a haze film **3-432** disposed on the back/rear side of the shroud **3-404**. In one example, an optical bonding material, such as an optically clear bonding material, can be disposed between the blinker film **3-422** and the shroud **3-404**. In one example, the bonding material can include or form the haze film **3-432** from particles in the bonding material. In at least one example, the front cover assembly **3-400** can also include a bracket **3-434** disposed against a rear/back side of the shroud **3-404** outside a perimeter of the display assembly **3-408**. The display bracket **3-414** can be mounted to the shroud **3-404** via the bracket **3-434** as shown in FIG. **3-4**. In at least one example, one or more opaque painted portions **3-436** can be applied or formed on the rear/back side of the shroud **3-404**, for example, between the bracket **3-434** and the shroud **3-404** to visually hide the bracket **3-434** from being seen from outside or in front of the front cover assembly **3-400**. The shroud can also include an opaque portion or multiple portions disposed generally around an outer perimeter area of the shroud to hide components behind the perimeter portion, for example cameras, sensors, brackets, circuitry, frame and structural components of the HMD, and so forth. The opaque portions can be applied to the shroud **3-404**.

[1134] In at least one example, the bracket **3-434** can be referred to as a shroud canopy. The shroud **3-404** can include a layer of ink, paint, film, or other opaque layer **3-435** disposed between the bracket **3-434** and the shroud **3-404** to visually hide any glue or other connection mechanisms securing the bracket **3-434** to the shroud **3-404** from someone looking in through the front of the device. The bracket **3-434** can also be opaque and provide a visual barrier to components behind/within the HMD device. The film or paint layer **3-435** can be disposed on the shroud **3-404** around a peripheral edge or edge area surrounding the display assembly **3-408** or around a more opaque area of the shroud **3-404** through which the display assembly **3-408** is configured to project light.

[1135] FIG. **3-4B** illustrates an exploded view of a front cover assembly **3-400** of an HMD device, including a transparent cover **3-402**, shroud **3-404**, and a frame **3-448** to which the cover glass **3-402** and shroud **3-404** can be secured. The shroud **3-404** can include or define an aperture **3-440** through which one or more cameras or other sensors of the HMD can send and receive signals. The shroud **3-404** can also include a visually opaque and infrared-transparent window **3-438** through which an infrared sensor and/or emitter can send and receive infrared light. The shroud **3-404** can

also include another window or opening **3-442** through which another sensor of the HMD can send and receive signals.

[1136] In at least one example, the cover **3-402** can include back-painted, opaque trim layer **3-444** disposed on a back side of the cover **3-402** between the cover **3-402** and the shroud **3-404**. The trim layer **3-444** can include opaque ink, paint, film, or other opaque layers to reduce stray signals, for example light signals, from the various emitters of the HMD from undesirably bouncing between the cover **3-402** and the shroud **3-404**, thus reducing cross-talk between sensors and emitters. The trim layer **3-444** can include a sensor peripheral trim portion **3-402** on a back side of the cover **3-402** as well to limit stray light directly around a sensor disposed behind the window **3-438**. The trim portion **3-402** can include the same or different layers and materials as that of the trim layer **3-444** but be aligned with the window **3-438**, and thus disposed directly around a periphery of the emitter or sensor of the HMD sending and/or receiving signals through the window **3-438**, to prevent stray signals, including stray light signals, from cross-talking with other emitters/sensors. In this way, stray cross-talk between the cover **3-402** and the shroud **3-404** can be reduced.

[1137] FIG. **3-4C** illustrates another exploded, perspective view of a front cover assembly **3-400**, including the shroud **3-404**, cover **3-402**, trim layer **3-444**, trim portion **3-446** defining a sensor window **3-445**, windows **3-440**, and sensor/emitter window **3-438**. In at least one example, the assembly **3-400** can also include a light seal **3-450** disposed around the window **3-438** on a front side of the shroud **3-404** between the shroud **3-404** and the cover **3-402**. The light seal **3-450** can be a physical barrier extending between the shroud **3-404** and the cover **3-402**, for example physically contacting both the shroud **3-404** and the cover **3-402**, and around the window **3-438**, such that light or other signals sent by an emitter or received by a sensor of the HMD through the window **3-438** is directed through the window **3-445** of the cover **3-402** to reduce stray signals and cross-talk. The seal **3-450** can be aligned with the trim portion **3-446** of the cover **3-402**.

[1138] In at least one example, the shroud **3-404** and cover **3-402** can include radio-frequency transparent materials and the HMD can include antennas and emitters configured to pass radio-frequency signals through shroud **3-404** and cover **3-402**.

[1139] In at least one example, the shroud **3-404** can include a notch or cutout recessed into or disposed on a back side of the shroud **3-404** to accommodate one or more antennas. The cutouts can be shaped and positioned to accommodate the placement of antennas adjacent to or against the shroud to allow more distance between the antennas and one or more grounding planes relative to the antennas. The increased resonating distance of the antennas provided by the shroud cutouts can improve the performance of the antennas. In one example, an increase of about 0.8 mm of distance between an antenna disposed in the cutout and the grounding plane can improve the antenna performance up to about 3 dB.

[1140] In at least one example, the backside of shroud **3-404** (e.g., the side of the shroud **3-404** facing inward toward an internal volume of the HMD device) can include posts, brackets, or other datum features to align cameras within the HMD device when assembled.

[1141] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **3-4-3-4C** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **3-1-3-3** and **3-5-3-6** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **3-1-3-3** and **3-5-3-6** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **3-4-3-4C**.

[1142] FIG. **3-5** shows another cutaway view of an example of a front cover display **3-500** without the shroud or external transparent cover shown. In the illustrated example of FIG. **3-5**, the assembly **3-500** includes a blinker film **3-522** separated from the lenticular lens array **3-510** by the gap **3-524**, the lenticular lens array **3-510** being part of or disposed on/against the display assembly **3-508**. FIG. **3-5** also shows the display bracket **3-514** coupling the display assembly **3-508** to the

bracket **3-534**. FIG. **3-5** illustrates the curvature of the display assembly **3-508**, including the lenticular lens **3-510**, and the curvature of the blinker film **3-522**.

[1143] FIG. **3-6** illustrates a close-up view of a lenticular lens array **3-610** similar to the array **3-510** shown in FIG. **3-5**, with individual lens or array portions **3-611** shown. In at least one example, because the lens array **3-610** can be curved, as well as the display screen **3-609** of the display assembly **3-608**, as noted above in other examples and as shown in FIG. **3-6**, the angle of the array portions **3-611** can vary across a horizontal width or length of the lenticular lens array **3-610**. The variation of the angles of the array portions **3-611** can be such that they do not block light from the display screen as the display screen **3-609** curves from the point of view of someone external to and in front of the front cover assembly **3-600**, to which the light from the display screen **3-609** is projected. In one example, the display screen **3-609** can include a panel of pixels. In one example, the display screen **3-609** includes an OLED panel.

[1144] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **3-5** and **3-6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **3-1-3-4** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **3-1-3-4** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **3-5** and **3-6**.

IV: Shroud

4.0: Systems with Displays and Sensor-Hiding Structures

[1145] FIG. **4-1** is a front view of an illustrative ring-shaped cosmetic covering structure for device **4-10**. Illustrative ring-shaped shroud **4-100** of FIG. **4-1** may be mounted under the inner surface of the display cover layer for display **4-14F** in inactive area **IA**. This may help hide the optical components and other internal portions of device **4-10** from view from the exterior of device **4-10**. Shroud **4-100** may be formed from one or more unbroken ring-shaped members and/or may be formed from multiple shroud segments that are attached using adhesive, fasteners, or other attachment structures. If desired, shroud **4-100** may be formed from multiple members that are sandwiched together along some or all of their lengths. In an illustrative configuration, which may sometimes be described herein as an example, shroud **4-100** may be formed from an inner piece (e.g., an inner full or partial ring), which may sometimes be referred to as an inner shroud member, shroud trim, or shroud trim member and may be formed from an outer piece or pieces (e.g., one or more strips of material or covering members, an full ring, one or more partial rings, etc.), which may sometimes be referred to as a shroud cover, canopy, or shroud canopy.

[1146] As shown in FIG. **4-1**, shroud **4-100** may have optical component windows to accommodate components **4-60**, **4-62**, **4-64**, **4-84**, **4-66**, **4-68**, **4-70**, **4-72**, **4-74**, **4-76**, **4-78**, **4-82**, and **4-80**. The optical component windows may be formed from through-hole openings in shroud **4-100**, from recesses or other partial openings that do not pass entirely through shroud **4-100**, from inserted optical window members in shroud through-hole openings, and/or from other shroud optical component window structures. Display **4-14F** may have a display cover layer that has corresponding optical component windows (through-hole openings, recessed areas, inserted window members in through-hole openings, etc.) and/or that is formed from bulk material that has desired optical properties (e.g., a display cover layer formed from one or more layers of material such as glass and/or polymer with sufficient transparency at the operating wavelength range of the overlapped optical component to allow the optical component to operate satisfactorily through the cover layer without forming openings or other window structures in the cover layer).

[1147] Shroud **4-100** may have any suitable shape. For example, the outline of shroud **4-100** may be rectangular with rounded corners as shown in FIG. **4-1**, may have teardrop shapes on the left and right sides of device **4-10**, may have an oval outline, and/or may have other outlines with curved and/or straight edge segments. FIG. **4-2** is a front view of a portion of shroud **4-100**

showing how the inner and outer edges of shroud **4-100** may be curved (e.g., to follow a teardrop shape). Shroud **4-100** may, if desired, have a peripheral edge that is curved along most or all of its length.

[1148] The width of shroud **4-100** may be constant along its length or shroud **4-100** may have portions that are wider than others. The thickness of shroud **4-100** (e.g., the dimension of shroud **4-100** into the page in the orientation of FIG. **4-1**) may be smaller than the width of shroud **4-100** (the lateral dimension of shroud **4-100** within the page in the orientation of FIG. **4-1**) or the thickness of the shroud may be equal to or greater than the width of the shroud. The shroud may have a two-dimensional shape (e.g., shroud **4-100** may have a planar shape that lies in the XZ plane in the example of FIG. **4-1**) or may have a three-dimensional shape (e.g., a shape with a curved cross-sectional profile and/or a shape characterized by inner and/or outer surfaces of compound curvature). In an illustrative configuration, most or all of the inner and outer surfaces of shroud have a compound-curvature surface.

[1149] The optical components under inactive area IA may include components on the left and right sides of device **4-10** that operate in conjunction with each other. For example, scene cameras, tracking cameras, and/or structured light cameras in device **4-10** may be formed in pairs, each of which includes a left camera and a corresponding right camera. A left scene camera and a right scene camera may, as an example, operate together to capture overlapping images that provide device **4-10** with a wide field of view for gathering pass-through video. Left and right tracking cameras may operate together to track a user's hands or other external objects. Left and right structured light cameras or other three-dimensional cameras may be used together to capture three-dimensional images of the user's environment. To enhance performance of the left and right optical components in these types of paired component arrangements, it may be desirable to maintain accurate alignment between the left and right optical components. To help maintain left and right optical components on the respective left and right sides of device **4-10** in alignment with each other, device **4-10** may be provided with one or more housing structures that help support the optical components.

[1150] As shown in FIG. **4-3**, for example, device **4-10** may be provided with an internal support structure such as bracket **4-102** that helps support optical components **4-104** on the left and right sides of device **4-10**. Components **4-104** may be, for example, optical components of the type shown under inactive area IA of FIG. **4-1**. Bracket **4-102** may be formed from stiff metal and/or other rigid materials (e.g., rigid polymer, carbon fiber composite material or other fiber-composite material, etc.). A nose-bridge recess in bracket **4-102** (e.g., in the portion of bracket **4-102** near nose-bridge portion **4-16NB**) may help bracket **4-102** conform to the shape of the user's face. Bracket **4-102** may have an elongated strip shape that runs along a portion of the length of inactive area IA (e.g., on the lower edge of device **4-10**).

[1151] Bracket **4-102** may be coupled to device **4-10** with attachment structures (adhesive, fasteners, press-fit connections, and/or other attachment mechanism) that allow bracket **4-102** to float with respect to the rest of housing portion **4-16M** during a drop event. The stiffness of bracket **4-102** and the ability of bracket **4-102** to shift in position somewhat relative to other housing structures without deforming the shape of bracket **4-102** significantly may help hold components on the left and right sides of device **4-10** in alignment with each other during periods of excessive stress such as when device **4-10** experiences high stress during an unexpected drop event.

[1152] In the example of FIG. **4-3**, bracket **4-102** is mounted under inactive area IA and has a nose bridge recess with a curved edge that is configured to accommodate a user's nose when device **4-10** is worn on a user's head. Bracket **4-102** may have other shapes, if desired. Components **4-104** may be attached to respective left and right sides of bracket **4-102** and/or other supporting structures in device **4-10** (e.g., shroud **4-100**) using adhesive, fasteners, press fit connections, and/or other attachment structures.

[1153] FIG. **4-4** is a cross-sectional top view of a portion of device **4-10**. As shown in FIG. **4-4**,

shroud **4-100** may overlap one or more optical components **4-104** in inactive area IA. Inactive area IA may form a ring-shaped border that surrounds active area AA. Display **4-14F** may have a display cover layer such as display cover layer **4-92**. Layer **4-92** may be formed from glass, polymer, ceramic, crystalline material such as sapphire, other materials, and/or combinations of these materials. Layer **4-92** may include a single layer of material or multiple stacked layers of material. In active area AA, pixels P in display panel **4-14P** display images that are viewable through display cover layer **92**. Shroud **4-100** may be absent from active area AA (e.g., shroud may have a ring shape that surrounds an opening over panel **4-14P** as shown in FIG. **4-4**) or shroud **4-100** may optionally have a portion (sometimes referred to as a canopy or shroud structure) that overlaps display panel **4-14P**. The canopy may be fully or partly transparent. In inactive area IA, shroud **4-100** overlaps components **4-104**. Components **4-104** may be optical components that emit and/or detect light that passes through transparent portions of layer **92** and shroud **4-100** and/or through optical component windows formed from recesses, through-hole openings, window members, and/or other window structures in layer **4-92** and shroud **4-100**.

[1154] Display cover layer **4-92** may include planar surfaces and/or curved surfaces. In an illustrative configuration, most or all of the inner and outer surfaces of display cover layer **4-92** have curvature.

[1155] The curved surfaces of display cover layer **4-92** may include curved surfaces that can be flattened into a plane without distortion (sometimes referred to as developable surfaces or curved surfaces without compound curvature). Surfaces such as these may, as an example overlap active area AA. The curved surfaces of display cover layer **4-92** may also include curved surfaces that are characterized by compound curvature (e.g., surfaces that can only be flattened into a plane with distortion, sometimes referred to as non-developable surfaces). Some or all portions of the inner and outer surfaces of display cover layer **4-92** in inactive area IA may, as an example, be characterized by compound curvature. This allows the periphery of display **4-14F** to smoothly transition away from the active area and provides an attractive appearance and compact shape for device **4-10**. The compound curvature of display cover layer **4-92** in inactive area IA may also facilitate placement of the optical components under inactive area IA in desired orientations. The inner and outer surfaces of display cover layer **4-92** in active area AA may have compound curvature, may be developable surfaces, or may include both developable surface areas and compound curvature areas.

[1156] Image data and other data gathered by optical components can be warped digitally to compensate for optical distortion associated with display cover layer **4-92**. To help minimize optical distortion, one or more of the optical components may optionally be oriented in a direction that is parallel or close to parallel to the surface normal of the portion of the display cover layer surface that is overlapping the optical component.

[1157] Consider, as an example, optical components **4-104** of FIG. **4-4**. As shown in FIG. **4-4**, some optical components such as illustrative optical component **4-104B**, which operates in direction **4-112**, may face forward (e.g., direction **4-112** may be parallel to or nearly parallel to the Y axis of FIG. **4-4**) in portions of display cover layer **4-92** where the surface normal of layer **4-92** is oriented parallel to the Y axis or close to parallel to the Y axis. Other optical components such as illustrative optical component **4-104A**, which operates in direction **4-110**, may be angled away from the forward direction by a non-zero angle (e.g., by an angle of at least 10°, at least 20°, less than 90°, less than 50°, or other suitable amount). Direction **4-110** may be parallel or closely parallel (e.g., aligned within 30°, within 20°, within 10° or other suitable amount) to the surface normal of the overlapping surface of display cover layer **92** and may lie in the XY plane of FIG. **4-4** or be angled out of the XY plane (e.g., by orienting component **4-104A** so that direction **4-110** is angled upwards in the +Z direction or downwards in the -Z direction in addition to angling direction **4-110** away from the +Y direction as shown in FIG. **4-4**).

[1158] In this type of arrangement, display cover layer **4-92** may have compound curvature in

inactive area IA and shroud **4-100** may have a shape with a cross-sectional profile that mirrors that of display cover layer **92** in inactive area IA (e.g., the outer and/or inner surfaces of shroud **4-100** in inactive area IA may be compound-curvature surfaces). When components such as components **4-104A** and **4-104B** are mounted to shroud **4-100** and/or are otherwise supported by the support structures of device **4-10** to operate through shroud **4-100** and display cover layer **4-92**, the curved shape of display cover layer **4-92** and shroud **4-100** may help allow these components to face in desired orientations (e.g., in a forward direction for components such as component **4-104B** or angled away from the forward direction for components such as component **4-104A**).

[1159] As an example, optical components that are mounted to the left and right sides of nose bridge portion **4-16NB** may be oriented respectively somewhat to the left and somewhat to the right of the +Y forward direction (e.g., to ensure an adequate angle-of-view for a pair of cameras). As another example, the curved shape of display cover layer **4-92** and shroud **4-100** along the lower edge of device **4-10** may allow the components in this portion to point somewhat downward out of the XY plane, which may help orient cameras such as tracking cameras towards the user's hands.

[1160] Display panel **4-14P** may be a flexible display such as a flexible organic light-emitting diode display with a flexible substrate or a light-emitting diode display formed from crystalline semiconductor light-emitting diode dies mounted on a flexible substrate. This allows display panel **4-14P** and the pixels of panel **4-14P** that form active area AA to be bent about a bend axis that runs parallel to vertical axis Z, thereby helping to wrap display **4-14F** and housing portion **4-16M** about the curved surface of the user's face. If desired, display panel **4-14P** may be a lenticular display configured to display three-dimensional images (e.g., an autostereoscopic display having a series of parallel lenticular lenses, each of which overlaps a respective group of multiple columns of pixels).

[1161] The outer and inner surfaces of display cover layer **4-92** may have the same shape (e.g., these surfaces may be parallel to each other) or the outer surface and inner surfaces may have different shapes. In arrangements in which display panel **4-14P** of display **4-14F** is flexible, it may be desirable to configure the inner surface of display cover layer **4-92** in active area AA to exhibit a bent surface shape that matches the bent outwardly-facing surface of display panel **4-14P** (e.g., the inner and, if desired, the outer surface of display cover layer **4-92** in active area AA may be developable surfaces without compound curvature to match the developable outward-facing surface of display panel **4-14P**).

[1162] Shroud **4-100** and display cover layer **4-92** may be attached to main housing portion **4-16M** using adhesive, screws and other fasteners, press-fit connections, and/or other attachment mechanisms. An illustrative configuration in which shroud **4-100** and cover layer **4-92** are attached to forward-facing edge of a housing wall in main housing portion **4-16M** using adhesive is shown in FIG. **4-4** through **4-6**. In the example of FIG. **4-5**, shroud **4-100** has an inner shroud member such as shroud trim **4-100A** and has a corresponding outer shroud member such as shroud canopy **4-100B**. Shroud trim **4-100A** and shroud canopy **4-100B** may be formed from metal, polymer, ceramic, glass, other materials, and/or combinations of these materials. In an illustrative example, shroud trim **4-100A** is formed from black polymer or other dark material and shroud canopy **4-100B** is formed from clear polymer. The outer surface of shroud canopy **4-100B** may be smooth to provide shroud **4-100** with a cosmetically attractive appearance.

[1163] A layer of pressure sensitive adhesive (see, e.g., adhesive **114**) may be used in attaching canopy **4-100B** to trim **4-100A**. Adhesive may also be used in attaching cover layer **92** and shroud **4-100** to housing portion **4-16M**. As shown in FIG. **4-5**, for example, a first adhesive such as adhesive **4-122** may be used to attach display cover layer **92** to shroud **4-100** (e.g., to a ledge in shroud trim **4-100A**). A second adhesive such as adhesive **4-124** may, in turn, be used to attach shroud **4-100** (e.g., shroud trim **4-100A**) to an adjacent lip of a wall in main housing portion **4-16M**.

[1164] In some configurations, adhesives **4-122** and **4-124** may be formed from the same type of

material. In an illustrative configuration, adhesives **4-122** and **4-124** are different. Housing portion **4-16M** may have a wall with a lip shape that creates a shearing force on adhesive **4-124** as display **4-14F** is attached to housing portion **4-16M** by pressing display **4-14F** against housing portion **4-16M** in the $-Y$ direction. In this type of scenario, it may be desirable to form adhesive **4-124** from an adhesive that can bond satisfactorily in the presence of shear forces such as a molten hot melt glue (thermoplastic adhesive) or other liquid adhesive rather than pressure sensitive adhesive. Adhesive **4-124** may, if desired, be exposed to a curing agent (ultraviolet light, moisture, etc.) before display **4-14F** is assembled into housing **4-16M**.

[1165] It may be desirable to repair device **4-10**. For example, if a user exposes display **4-14F** to excessive force during a drop event, it may be desirable to replace display **4-14F** with a new display. This can be accomplished by heating adhesive **4-124** to loosen the adhesive bond formed by adhesive **4-124**. To help prevent display cover layer **4-92** from detaching from shroud **4-100** while softening adhesive **4-124** with heat, adhesive **4-122** may be provided with a higher-temperature softening point than adhesive **4-124** (e.g., adhesive **4-122** may be a two-part hot melt glue with a higher melting point than adhesive **4-124**).

[1166] Optical components that are overlapped by display cover layer **4-92** and shroud **4-100** in inactive area **IA** may transmit and/or receive light through shroud **4-100** and display cover layer **4-92**. Layer **4-92** may be formed from laminated glass or other clear material that allows light for each overlapped optical component **4-104** to pass through layer **4-92**. If desired, a partial recess or a through-hole opening may be formed in the portion of layer **4-92**. An optional optical component window member **4-116** may then be inserted within layer **4-92** (e.g., in window region **4-118**). As an example, layer **4-92** may be formed from one or more layers of glass and/or polymer and may be characterized by a first level of light transmission at operating wavelength(s) for component **4-104**, whereas window member **4-116** may be formed from polymer, glass, and/or other materials that are characterized by a second level of light transmission at the operating wavelength(s) that is greater than the first level of light transmission. In other illustrative arrangements, no window member is inserted in layer **4-92** (e.g., optional window member **4-116** of FIG. 4-5 can be omitted when layer **4-92** alone is sufficiently transparent to pass light for component **4-104**).

[1167] Shroud **4-100** may be provided with an optical component window in region **4-118** to accommodate overlapped optical component **4-104**. Component **4-104** may operate at ultraviolet light wavelengths, visible light wavelengths, and/or infrared light wavelengths. To accommodate component **4-104** in the example of FIG. 4-5, shroud trim **4-100A** has been provided with a through-hole opening such as opening **120**, whereas shroud canopy **4-100B** has no openings in region **4-118**. This effectively forms a window recess in shroud **4-100** in alignment with components **4-104**. Trim **4-100A** may be formed from black polymer or other light-absorbing material, so the formation of opening **120** in trim **4-100A** may help ensure that sufficiently light may pass through region **4-118** to allow component **4-104** to operate satisfactorily. The portion of canopy **4-100B** that overlaps opening **4-120** may be transparent (e.g., clear polymer).

[1168] To help hide component **4-104** from view, the inner surface of shroud canopy **4-100B** of FIG. 4-5 has been covered with coating **4-126**. Coating **4-126** may be used to provide region **4-118** with a desired outward appearance and optical properties that ensure that component **104** can operate satisfactorily. Coating **4-126** may be a thin-film-interference filter formed from a stack of thin-film dielectric layers of alternating refractive index values (with indices and thicknesses selected to create a desired transmission spectrum and a desired reflection spectrum for the filter), may be a layer of ink (e.g., a polymer layer including dye, pigment, and/or other colorant), and/or may be any other suitable coating with desired optical properties.

[1169] Consider, as an example, a scenario in which component **4-104** transmits and/or receives infrared light. In this type of arrangement, coating **4-126** may be opaque at visible wavelengths and transparent at infrared wavelengths. This helps to hide component **4-104** from view from the exterior of device **4-10** while allowing infrared light associated with the operation of component **4-**

104 to pass through shroud **4-100** and layer **4-92**.

[1170] As another example, consider a scenario in which component **4-104** is an ambient light sensor. In this configuration, coating **4-126** may exhibit a visible light transmission of 1-8% (as an example). This may allow sufficient visible ambient light to reach the ambient light sensor for the ambient light sensor to make an ambient light reading. At the same time, the transmission of coating **4-126** may be sufficiently low that coating **4-126** helps reduce the visibility of component **4-104** from the exterior of device **4-10**.

[1171] As these examples demonstrate, regions of display **4-14F** that overlap optical components such as component **4-104** of FIG. **4-5** may be provided with optical component window structures in layer **4-92** and/or shroud **4-100** that help accommodate the optical component.

[1172] If desired, shroud **4-100** may be provided with a through-hole opening to accommodate an overlapped optical component. As shown in FIG. **4-6**, for example, shroud **4-100** may contain one or more sublayers (e.g., a trim, a canopy, and/or other layers). Through-hole opening **4-130** may pass from the inner surface of shroud **4-100** to the outer surface of shroud **4-100**. Opening **4-130** may be aligned with optical component **104**. Component **104** may be mounted behind opening **4-130** and/or may be partly or fully receive within opening **4-130** as shown in FIG. **4-6**. This allows light to be emitted and/or received by component **104** without being blocked by shroud **4-100**.

[1173] In the illustrative configuration of FIG. **4-7**, shroud **4-100** also contains one more sublayers (e.g., a trim, a canopy, and/or other layers). As shown in FIG. **4-7**, a through-hole opening may formed in shroud **4-100** in alignment with optical component **104** and may be filled with optical component window member **4-132** (e.g., a glass or polymer member or a window structure formed from other material and/or combinations of these materials). Optical component window member **4-132** has optical characteristics (e.g., light transmission, reflection, absorption, haze, etc.) that allow component **4-104** to transmit and/or receive light satisfactorily through region **4-118**. As an example, member **4-130** may be formed from glass that is transparent to infrared light and that is opaque or transparent to visible light.

[1174] As described in connection with FIGS. **4-2** and **4-3**, there may be numerous optical components such as component **4-104** in inactive area **IA**. Each optical component may potentially have a different type of optical component window structure in shroud **4-100** and/or layer **4-92** to accommodate that component. For example, some areas of shroud **4-100** may have openings that receive components as described in connection with FIG. **4-6**, other areas of shroud **4-100** may have inserted optical window member such as member **4-132** of FIG. **4-7**, and/or other areas of shroud **4-100** may have partial shroud openings (e.g., non-through-hole recesses) such as opening **4-120** of FIG. **4-5** (which may optionally be covered with a layer such as coating **4-126** to modify the optical properties of shroud **4-100**).

[1175] FIG. **4-8** is a cross-sectional side view of a portion of a head-mounted device with a fully or partly transparent shroud covering the front face of the device. As shown in FIG. **4-8**, head-mounted device **4-10** may include display panel **4-14P** for forward-facing display **4-14**. Panel **4-14P** may be a lenticular display (e.g., an autostereoscopic display with lenticular lenses **4-14P'** configured to display three-dimensional images for a user).

[1176] In the arrangement of FIG. **4-8**, display cover layer **92** has inner and outer surfaces with compound curvature in inactive area **IA** (e.g., a ring-shaped area running along the periphery of layer **4-92**). The inner and outer surfaces of display cover layer **4-92** in active area **AA** may also have compound curvature or one or both of these surfaces may be developable surfaces. In the example of FIG. **4-8**, the inner and outer surfaces of layer **4-92** have compound curvature in both inactive area **IA** and active area **AA** (e.g., these surfaces may be free of any developable surfaces), which may help provide device **4-10** with an attractive appearance.

[1177] The shroud of device **4-10** of FIG. **4-8** includes a shroud trim **4-100A** and shroud canopy **4-100B**. Trim **4-100A** may have a ring shape and may extend around the periphery of display **14**. Canopy **4-100B**, which may be formed from a material such as polymer, may have an outline equal

to or nearly equal to that of display cover layer **4-92** and may cover substantially the entire front face of device **4-10**. With this type of arrangement, shroud canopy **4-100B** overlaps all of display panel **4-14P**. The polymer that makes up canopy **4-100B** may have a bulk tint (e.g., a colorant such as dye and/or pigment that provides canopy **4-100B** with a desired optical transmission characteristic). For example, canopy **4-100B** may be tinted so that canopy **4-100B** exhibits a visible light transmission of 30-80%, at least 20%, at least 40%, less than 95%, less than 90%, less than 85%, less than 75%, 4-60%, or other suitable amount. By configuring canopy **10B** to exhibit partial light transmission (e.g., 30-80% or other suitable value), canopy **4-100B** may help visually hide internal components such as lenses **4-14P'** and other structures of display **4-14P** from view (e.g., when display **4-14P** is not in use).

[1178] The inner surface of canopy **4-100B** may also be provided with an optical layer such as optical layer (optical film) **4-146**. Layer **4-146** may have texture and/or light-scattering particles that create haze. The haze may help hide the structures of display panel **4-14P** from view from the exterior of device **4-10**. Layer **4-146** may also have microlouvers or other features that help suppress off-axis light transmission (e.g., layer **4-146** may have privacy structures that reduce light transmission for light rays that are not parallel to the Y axis). Because layer **4-146** may contain haze and/or privacy structures, layer **4-146** may sometimes be referred to as a privacy layer, a haze layer, and/or a privacy and haze layer.

[1179] In an illustrative configuration, layer **4-146** may have a flexible substrate layer covered with a hazy coating. The hazy coating may be a pad-printed polymer coating that contains embedded light-scattering particles (e.g., inorganic light-scattering particles such as titanium oxide particles, etc.). The flexible substrate layer may be a privacy film such as a microlouver film or other privacy layer that prevents off-axis (away from the Y axis) viewing of display panel **4-14P**.

[1180] Haze for layer **4-146** may be provided using any suitable haze structures (e.g., a coating of hazy polymer having a thickness of 3-10 microns on a flexible privacy film or other substrate, a laminated hazy film, or other layer that exhibits 3%-40% haze or other suitable value, sometimes referred to as a haze coating). Haze may be provided by embedded light-scattering particles and/or surface texture (e.g., texture in layer **4-146** or optionally texture on the surface of canopy **4-100B**). The haze provided by the hazy coating of layer **4-146** and/or other haze structures is preferably provided sufficiently close to display **4-14P** that the resolution of display **4-14P** is not significantly affected. At the same time, the presence of the haze (e.g., the hazy coating of layer **4-146**) may help hide lenses and other structures in layer **4-14P** from view when not in use.

[1181] Device **4-10** may have an air gap between display panel **4-14P** and canopy **4-100B** (e.g., an air gap such as air gap **4-144** may be present between the inwardly facing side of canopy **4-100B** and any coatings and/or films on this side of canopy **4-100B** such as haze layer **4-146** and the opposing upper surface of display panel **4-14P** (and lenses **4-14P'** and the pixels on panel **4-14P**). The presence of air gap **4-144** may help ensure that lenses **4-14P'** operate satisfactorily. Bracket **4-156** may help support display panel **4-14P**.

[1182] To help hide internal components from view, an opaque masking layer such as layer **BM-1** may be formed on the inner surface of display cover layer **4-92** in inactive area **IA**. Adhesive **4-122** may attach layer **4-92** to the edge of canopy **4-100B**. Additional opaque masking material (see, e.g., canopy opaque masking layer **BM-2**) may be formed on the inner surface of canopy **4-100B** in inactive area **IA**. Adhesive **4-114** may be used to attach shroud trim **4-100A** to shroud canopy **4-100B**. Adhesive **4-124** may be used to attach shroud trim **4-100A** to housing portion **4-16M**. Adhesive **14-60** may be used to attach bracket **4-156** (which is attached with adhesive to the rear of panel **4-14P**) to canopy **4-100B**.

[1183] In the example of FIG. **4-8**, outer surface **4-148** and inner surface **4-150** of display cover layer **4-92** have compound curvature in inactive area **IA** and in active area **AA**. Outer surface **4-152** and opposing inner surface coil **4-54** of shroud canopy **4-100B** may have matching compound curvature in inactive area **IA**. In active area **AA**, outer surface **4-152** and inner surface 1coil **4-54** of

shroud canopy **4-100B** may be developable surfaces (e.g., surfaces without compound curvature that exhibit a curved cross-sectional profile that bends about a single bend axis such as axis **4-142**). Axis **4-142** is an axis that runs parallel to the Z axis in this example. Display panel **4-14P** may exhibit the same amount of bending about axis **4-142** and may also be characterized by a developable surface (e.g., the pixel array on the outer surface of panel **4-14P** may have a developable surface).

[1184] The amount of bending of canopy **4-100B** and the corresponding amount of bending of display panel **4-14P** about axis **4-142** may be selected to help device **4-10** conform to the curved shape of a user's face.

[1185] In the illustrative configuration of FIG. **4-8**, canopy **4-100B** does not have any areas of compound curvature that overlap display panel **4-14P**. Rather, the portion of canopy **4-100B** that overlaps panel **4-14P** has inner and outer developable surfaces. If desired, one or both of surfaces **4-152** and **1coil 4-54** may have compound curvature. For example, outer surface **4-152** may have compound curvature and may be configured to establish a uniform thickness for air gap **4-140** under some or all of inner surface **4-150** of layer **4-92**. In the example of FIG. **4-9**, there is an air gap **4-140** of uneven thickness between layer **92** and canopy **4-100B**.

[1186] Bracket **4-156** may be formed from a metal sheet or other support structure and may be characterized by inner and outer surfaces that are developable surfaces (e.g., surfaces that bend about axis **4-142** and that do not contain areas of compound curvature). By avoiding compound curvature in the structures that support and immediately overlap display panel **4-14P**, display panel **4-14P** may be formed from a bent flexible substrate such as a polyimide substrate that bends about axis **4-142** without risk of creating wrinkles or other artifacts of the type that might be introduced if panel **4-14P** had areas of compound curvature.

[1187] The shroud and other structures of device **4-10** of FIG. **4-8** (e.g., the opaque masking layer coatings such as layers **BM-1** and **BM-2** which may be, for example, black ink layers) may be configured to form optical windows for optical components **4-104**.

[1188] FIG. **4-9** shows how opaque masking layer **BM-2** on canopy **4-100B** may have a window opening that is filled with a coating layer such as coating **4-170**. Optical component **4-104** (e.g., a flicker sensor, an ambient light sensor, and/or other photodetector) may be aligned with the window opening. A transparent canopy portion may overlap this window opening or a canopy opening may overlap this window opening. Layer **BM-2** may be opaque, which helps prevent internal components in device **4-10** from being viewed from the exterior of device **4-10**. The presence of the opening in layer **BM-2** allows optical component **4-104** to operate satisfactorily (e.g., to receive and measure ambient light). Coating **4-170** may be configured to allow component **104** to operate, while helping to visually hide component **4-104**. As an example, coating **4-170** may be formed from a layer of ink with a visible light transmission of 2-25%, at least 1%, at least 2%, at least 4%, less than 80%, less than 30%, or other suitable amount, whereas layer **BM-2** may have a visible transmission of less than 2%, less than 1%, or less than 0.5% (as examples).

[1189] FIG. **4-10** is a cross-sectional side view of another illustrative head-mounted device optical component mounting arrangement. The arrangement of FIG. **4-10** uses shroud through-hole openings in trim **4-100A** and canopy **4-100B**. These through-hole openings are aligned with an opening in display opaque masking layer **BM-1** (and are optionally aligned with a corresponding opening in canopy opaque masking layer **BM-2**). An optional coating layer such as layer **4-164** may cover the optical window formed from these openings. Layer **4-164** and the other openings of FIG. **4-11** may be aligned with optical component **4-104**, which may be mounted behind the shroud and/or which may have portions protruding into the through-hole openings of the shroud. In a first illustrative configuration, component **4-104** of FIG. **4-10** is an infrared illuminator (e.g., an infrared light-emitting diode). In this type of arrangement, coating layer **4-164** may be formed from a layer of ink, a thin-film interference filter, or other filter layer that blocks visible light and that is transparent to infrared light (e.g., a visible-light-blocking-and-infrared-light-transmitting filter

layer). In a second illustrative configuration, component **4-104** of FIG. **4-10** is a camera (e.g., a visible pass-through camera, an infrared camera, and/or other camera operating at visible and/or infrared wavelengths). In this arrangement, coating **4-164** may be omitted (to pass visible and/or infrared light), may be configured to form an antireflection coating, and/or may otherwise be configured to operate with the camera.

[1190] FIG. **4-11** is a cross-sectional side view of an illustrative head-mounted device optical component mounting arrangement with an optical component window formed from a transparent window member. Transparent window member **4-166** (e.g., a layer of glass or polymer) may be mounted in through-hole openings in trim **4-100A** and canopy **4-100B** and may be aligned with optical component **4-104** and an opening in opaque masking layer **BM-1** on layer **4-92** (and, if desired, may be aligned with an opening in opaque masking layer **BM-2** on canopy **4-100B**). Filter coating **4-168** may be provided on window member **4-166**. In an illustrative configuration, component **4-104** of FIG. **4-11** is a three-dimensional camera such as a time-of-flight camera or a structured light camera and may operate at infrared wavelengths. Filter **4-168** in this type of arrangement may be transparent to infrared light and may be transparent to visible light or may be opaque to visible light (e.g., filter **4-168** may be an infrared-light-transparent-and-visible-light-blocking filter). Filter coating **4-168** may be formed from ink, from a thin-film interference filter, or other filter structures.

[1191] The presence of window member **4-166**, which may be configured to exhibit relatively small amounts of optical distortion, may help enhance the optical performance of component **4-104**. If desired, optical-component-compatible surface areas for an optical component window for component **4-104** may be formed directly in canopy **4-100B** (e.g., so that canopy **4-100B** may overlap component **4-104** without forming a through-hole opening in canopy **4-100B**).

4.1: System with Cover Layer Sealing Structures

[1192] A head-mounted device may include a head-mounted support structure that allows the device to be worn on the head of a user. The head-mounted device may have displays that are supported by the head-mounted support structure for presenting a user with visual content. The displays may include rear-facing displays that present images to eye boxes at the rear of the head-mounted support structure. The displays may also include a forward-facing display. The forward-facing display may be mounted to the front of the head-mounted support structure and may be viewed by the user when the head-mounted device is not being worn on the user's head. The forward-facing display, which may sometimes be referred to as a publicly viewable display, may also be viewable by other people in the vicinity of the head-mounted device.

[1193] Optical components such as image sensors and other light sensors may be provided in the head-mounted device. In an illustrative configuration, optical components are mounted under peripheral portions of a display cover layer that protects the forward-facing display. The display cover layer, or other layers within the head-mounted device, may be formed from materials, such as glass, and, laminates, such as plastic laminates, may be formed on top and bottom surfaces of the cover layer. To protect the edges of the cover layer, encapsulation material may be coupled to the edge surface, or the head-mounted device housing structures may be modified.

[1194] FIGS. **4.1-1** is a side view of an illustrative head-mounted electronic device. As shown in FIGS. **4.1-1**, head-mounted device **4.1-10** may include head-mounted support structure **4.1-26**. Support structure **4.1-26** may have walls or other structures that separate an interior region of device **4.1-10** such as interior region **4.1-42** from an exterior region surrounding device **4.1-10** such as exterior region **4.1-44**. Electrical components **4.1-40** (e.g., integrated circuits, sensors, control circuitry, light-emitting diodes, lasers, and other light-emitting devices, other control circuits and input-output devices, etc.) may be mounted on printed circuits and/or other structures within device **4.1-10** (e.g., in interior region **4.1-42**).

[1195] To present a user with images for viewing from eye boxes such as eye boxes **4.1-34**, device **4.1-10** may include rear-facing displays such as displays **4.1-14R**, which may have associated

lenses that focus images for viewing in the eye boxes. These components may be mounted in optical modules (e.g., a lens barrel) to form respective left and right optical systems. There may be, for example, a left rear-facing display for presenting an image through a left lens to a user's left eye in a left eye box and a right rear-facing display for presenting an image to a user's right eye in a right eye box. The user's eyes are located in eye boxes **4.1-34** at rear side R of device **4.1-10** when structure **4.1-26** rests against the outer surface of the user's face.

[1196] Support structure **4.1-26** may include a main support structure (sometimes referred to as a main portion or housing). The main housing support structure may extend from front side F of device **4.1-10** to opposing rear side R of device **4.1-10**. On rear side R, support structure **4.1-26** may have cushioned structures to enhance user comfort as support structure **4.1-26** rests against the user's face. If desired, support structure **4.1-26** may include optional head straps and/or other structures that allow device **4.1-10** to be worn on a head of a user.

[1197] Device **4.1-10** may have a publicly viewable front-facing display such as display **4.1-14F** that is mounted on front side F of support structure **4.1-26**. Display **4.1-14F** may be viewable to the user when the user is not wearing device **4.1-10** and/or may be viewable by others in the vicinity of device **4.1-10**. Display **4.1-14F** may, as an example, be visible on front side F of device **4.1-10** by an external viewer who is viewing device **4.1-10** from front side F.

[1198] A schematic diagram of an illustrative system that may include a head-mounted device is shown in FIGS. **4.1-2**. As shown in FIGS. **4.1-2**, system **4.1-8** may have one or more electronic devices **4.1-10**. Devices **4.1-10** may include a head-mounted device (e.g., device **4.1-10** of FIGS. **4.1-1**), accessories such as controllers and headphones, computing equipment (e.g., a cellular telephone, tablet computer, laptop computer, desktop computer, and/or remote computing equipment that supplies content to a head-mounted device), and/or other devices that communicate with each other.

[1199] Each electronic device **4.1-10** may have control circuitry **4.1-12**. Control circuitry **4.1-12** may include storage and processing circuitry for controlling the operation of device **4.1-10**. Circuitry **4.1-12** may include storage such as hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **4.1-12** may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, and other integrated circuits. Software code may be stored on storage in circuitry **4.1-12** and run on processing circuitry in circuitry **4.1-12** to implement control operations for device **4.1-10** (e.g., data gathering operations, operations involving the adjustment of the components of device **4.1-10** using control signals, etc.). Control circuitry **4.1-12** may include wired and wireless communications circuitry. For example, control circuitry **4.1-12** may include radio-frequency transceiver circuitry such as cellular telephone transceiver circuitry, wireless local area network transceiver circuitry (e.g., WiFi® circuitry), millimeter wave transceiver circuitry, and/or other wireless communications circuitry.

[1200] During operation, the communications circuitry of the devices in system **4.1-8** (e.g., the communications circuitry of control circuitry **4.1-12** of device **4.1-10**) may be used to support communication between the electronic devices. For example, one electronic device may transmit video data, audio data, control signals, and/or other data to another electronic device in system **4.1-8**. Electronic devices in system **4.1-8** may use wired and/or wireless communications circuitry to communicate through one or more communications networks (e.g., the internet, local area networks, etc.). The communications circuitry may be used to allow data to be received by device **4.1-10** from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, online computing equipment such as a remote server or other remote computing equipment, or other electrical equipment) and/or to provide data to external equipment.

[1201] Each device **4.1-10** in system **4.1-8** may include input-output devices **4.1-22**. Input-output

devices **4.1-22** may be used to allow a user to provide device **4.1-10** with user input. Input-output devices **4.1-22** may also be used to gather information on the environment in which device **4.1-10** is operating. Output components in devices **4.1-22** may allow device **4.1-10** to provide a user with output and may be used to communicate with external electrical equipment.

[1202] As shown in FIGS. **4.1-2**, input-output devices **4.1-22** may include one or more displays such as displays **4.1-14**. Displays **4.1-14** may include rear facing displays such as display **4.1-14R** of FIGS. **4.1-1**. Device **4.1-10** may, for example, include left and right components such as left and right scanning mirror display devices or other image projectors, liquid-crystal-on-silicon display devices, digital mirror devices, or other reflective display devices, left and right display panels based on light-emitting diode pixel arrays (e.g., thin-film organic light-emitting displays with polymer or semiconductor substrates such as silicon substrates or display devices based on pixel arrays formed from crystalline semiconductor light-emitting diode dies), liquid crystal display panels, and/or or other left and right display devices that provide images to left and right eye boxes for viewing by the user's left and right eyes, respectively. Display components such as these (e.g., a thin-film organic light-emitting display with a flexible polymer substrate or a display based on a pixel array formed from crystalline semiconductor light-emitting diode dies on a flexible substrate) may also be used in forming a forward-facing display for device **4.1-10** such as forward-facing display **4.1-14F** of FIGS. **4.1-1** (sometimes referred to as a front-facing display, front display, or publicly viewable display).

[1203] During operation, displays **4.1-14** (e.g., displays **4.1-14R** and/or **4.1-14F**) may be used to display visual content for a user of device **4.1-10** (e.g., still and/or moving images including pictures and pass-through video from camera sensors, text, graphics, movies, games, and/or other visual content). The content that is presented on displays **4.1-14** may, for example, include virtual objects and other content that is provided to displays **4.1-14** by control circuitry **4.1-12**. This virtual content may sometimes be referred to as computer-generated content. Computer-generated content may be displayed in the absence of real-world content or may be combined with real-world content. In some configurations, a real-world image may be captured by a camera (e.g., a forward-facing camera, sometimes referred to as a front-facing camera) and computer-generated content may be electronically overlaid on portions of the real-world image (e.g., when device **4.1-10** is a pair of virtual reality goggles).

[1204] Input-output circuitry **4.1-22** may include sensors **4.1-16**. Sensors **4.1-16** may include, for example, three-dimensional sensors (e.g., three-dimensional image sensors such as structured light sensors that emit beams of light and that use two-dimensional digital image sensors to gather image data for three-dimensional images from dots or other light spots that are produced when a target is illuminated by the beams of light, binocular three-dimensional image sensors that gather three-dimensional images using two or more cameras in a binocular imaging arrangement, three-dimensional lidar (light detection and ranging) sensors, sometimes referred to as time-of-flight cameras or three-dimensional time-of-flight cameras, three-dimensional radio-frequency sensors, or other sensors that gather three-dimensional image data), cameras (e.g., two-dimensional infrared and/or visible digital image sensors), gaze tracking sensors (e.g., a gaze tracking system based on an image sensor and, if desired, a light source that emits one or more beams of light that are tracked using the image sensor after reflecting from a user's eyes), touch sensors, capacitive proximity sensors, light-based (optical) proximity sensors, other proximity sensors, force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), sensors such as contact sensors based on switches, gas sensors, pressure sensors, moisture sensors, magnetic sensors, audio sensors (microphones), ambient light sensors, flicker sensors that gather temporal information on ambient lighting conditions such as the presence of a time-varying ambient light intensity associated with artificial lighting, microphones for gathering voice commands and other audio input, sensors that are configured to gather information on motion, position, and/or orientation (e.g., accelerometers, gyroscopes, compasses, and/or inertial measurement units that include all of these sensors or a

subset of one or two of these sensors), and/or other sensors.

[1205] User input and other information may be gathered using sensors and other input devices in input-output devices **4.1-22**. If desired, input-output devices **4.1-22** may include other devices **4.1-24** such as haptic output devices (e.g., vibrating components), light-emitting diodes, lasers, and other light sources (e.g., light-emitting devices that emit light that illuminates the environment surrounding device **4.1-10** when ambient light levels are low), speakers such as ear speakers for producing audio output, circuits for receiving wireless power, circuits for transmitting power wirelessly to other devices, batteries and other energy storage devices (e.g., capacitors), joysticks, buttons, and/or other components.

[1206] As described in connection with FIGS. **4.1-1**, electronic device **4.1-10** may have head-mounted support structures such as head-mounted support structure **4.1-26** (e.g., head-mounted housing structures such as housing walls, straps, etc.). The head-mounted support structure may be configured to be worn on a head of a user (e.g., against the user's face covering the user's eyes) during operation of device **4.1-10** and may support displays **4.1-14**, sensors **4.1-16**, other components **4.1-24**, other input-output devices **4.1-22**, and control circuitry **4.1-12** (see, e.g., components **4.1-40** and displays **4.1-14R** and **4.1-14F** of FIGS. **4.1-1**, which may include associated optical modules).

[1207] FIGS. **4.1-3** is a front view of device **4.1-10** in an illustrative configuration in which device **4.1-10** has a publicly viewable display such as forward-facing display **4.1-14F**. As shown in FIGS. **4.1-3**, support structure **4.1-26** of device **4.1-10** may have right and left portions on either side of nose bridge **4.1-90**. Nose bridge **4.1-90** may be a curved exterior surface that is configured to receive and rest upon a user's nose to help support housing **4.1-26** on the head of the user.

[1208] Display **4.1-14F** may have an active area such as active area AA that is configured to display images and an inactive area IA that does not display images. The outline of active area AA may be rectangular, rectangular with rounded corners, may have teardrop shaped portions on the left and right sides of device **4.1-10**, may have a shape with straight edges, a shape with curved edges, a shape with a peripheral edge that has both straight and curved portions, and/or other suitable outlines. As shown in FIGS. **4.1-3**, active area AA may have a curved recessed portion at nose bridge **4.1-90**. The presence of the nose-shaped recess in active area AA may help fit active area AA within the available space of housing **4.1-26** without overly limiting the size of active area AA.

[1209] Active area AA contains an array of pixels. The pixels may be, for example, light-emitting diode pixels formed from thin-film organic light-emitting diodes or crystalline semiconductor light-emitting diode dies (sometimes referred to as micro-light-emitting diodes) on a flexible display panel substrate. Configurations in which display **4.1-14F** uses other display technologies may also be used, if desired. Illustrative arrangements in which display **4.1-14** is formed from a light-emitting diode display such as an organic light-emitting diode display that is formed on a flexible substrate (e.g., a substrate formed from a bendable layer of polyimide or a sheet of other flexible polymer) may sometimes be described herein as an example. The pixels of active area AA may be formed on a display device such as a display panel (e.g., a flexible organic light-emitting diode display panel). In some configurations, the outline of active area AA may have a peripheral edge that contains straight segments or a combination of straight and curved segments. Configurations in which the entire outline of active area AA is characterized by a curved peripheral edge may also be used.

[1210] Display **4.1-14F** may have an inactive area such as inactive area IA that is free of pixels and that does not display images. Inactive area IA may form an inactive border region that runs along one more portions of the peripheral edge of active area AA. In the illustrative configuration of FIGS. **4.1-3**, inactive area IA has a ring shape that surrounds active area AA and forms an inactive border. In this type of arrangement, the width of inactive area IA may be relatively constant and the inner and outer edges of area IA may be characterized by straight and/or curved segments or may

be curved along their entire lengths. For example, the outer edge of area IA (e.g., the periphery of display **4.1-14F**) may have a curved outline that runs parallel to the curved edge of active area AA. [1211] In some configurations, device **4.1-10** may operate with other devices in system **4.1-8** (e.g., wireless controllers and other accessories). These accessories may have magnetic sensors that sense the direction and intensity of magnetic fields. Device **4.1-10** may have one or more electromagnets configured to emit a magnetic field. The magnetic field can be measured by the wireless accessories near device **4.1-10**, so that the accessories can determine their orientation and position relative to device **4.1-10**. This allows the accessories to wirelessly provide device **4.1-10** with real-time information on their current position, orientation, and movement so that the accessories can serve as wireless controllers. The accessories may include wearable devices, handled devices, and other input devices.

[1212] In an illustrative configuration, device **4.1-10** may have a coil that runs around the perimeter of display **4.1-14F** (e.g., under inactive area IA along the periphery of active area AA). The coil may have any suitable number of turns (e.g., 1-10, at least 2, at least 5, at least 10, 10-50, fewer than 100, fewer than 25, fewer than 6, etc.). These turns may be formed from metal traces on a substrate, may be formed from wire, and/or may be formed from other conductive lines. During operation, control circuitry **4.1-12** may supply the coil with an alternating-current (AC) drive signal. The drive signal may have a frequency of at least 1 kHz, at least 10 kHz, at least 100 kHz, at least 1 MHz, less than 10 MHz, less than 3 MHz, less than 300 kHz, or less than 30 kHz (as examples). As AC current flows through the coil a corresponding magnetic field is produced in the vicinity of device **4.1-10**. Electronic devices such as wireless controllers with magnetic sensors that are in the vicinity of device **4.1-10** may use the magnetic field as a reference so that the wireless controllers can determine their orientation, position, and/or movement while being moved relative to device **4.1-10** to provide device **4.1-10** with input.

[1213] Consider, as an example, a handheld wireless controller that is used in controlling the operation of device **4.1-10**. During operation, device **4.1-10** uses the coil to emit a magnetic field. As the handheld wireless controller is moved, the magnetic sensors of the controller can monitor the location of the controller and the movement of the controller relative to device **4.1-10** by monitoring the strength, orientation, and change to the strength and/or orientation of the magnetic field emitted by the coil as the controller is moved through the air by the user. The electronic device can then wirelessly transmit information on the location and orientation of the controller to device **4.1-10**. In this way, a handheld controller, wearable controller, or other external accessory can be manipulated by a user to provide device **4.1-10** with air gestures, pointing input, steering input, and/or other user input.

[1214] Device **4.1-10** may have components such as optical components (e.g., optical sensors among sensors **4.1-16** of FIGS. **4.1-2**). These components may be mounted in any suitable location on head-mounted support structure **4.1-26** (e.g. on a head strap, on housing **4.1-26**, etc.). Optical components and other components may face rearwardly (e.g., when mounted on the rear face of device **4.1-10**), may face to the side (e.g. to the left or right), may face downwardly or upwardly, may face to the front of device **4.1-10** (e.g., when mounted on the front face of device **4.1-10**), may be mounted so as to point in any combination of these directions (e.g., to the front, to the right, and downward) and/or may be mounted in other suitable orientations. In an illustrative configuration, at least some of the components of device **4.1-10** are mounted so as to face outwardly to the front (and optionally to the sides and/or up and down). For example, forward-facing cameras for pass-through video may be mounted on the left and right sides of the front of device **4.1-10** in a configuration in which the cameras diverge slightly along the horizontal dimension so that the fields of view of these cameras overlap somewhat while capturing a wide-angle image of the environment in front of device **4.1-10**. The captured image may, if desired, include portions of the user's surroundings that are below, above, and to the sides of the area directly in front of device **4.1-10**.

[1215] To help hide components such as optical components from view from the exterior of device **4.1-10**, it may be desirable to cover some or all of the components with cosmetic covering structures. The covering structures may include transparent portions (e.g., optical component windows) that are characterized by sufficient optical transparency to allow overlapped optical components to operate satisfactorily. For example, an ambient light sensor may be covered with a layer that appears opaque to an external viewer to help hide the ambient light sensor from view, but that allows sufficient ambient light to pass to the ambient light sensor for the ambient light sensor to make a satisfactory ambient light measurement. As another example, an optical component that emits infrared light may be overlapped with a visibly opaque material that is transparent to infrared light.

[1216] In an illustrative configuration, optical components for device **4.1-10** may be mounted in inactive area IA of FIGS. **4.1-3** and cosmetic covering structures may be formed in a ring shape overlapping the optical components in inactive area IA. Cosmetic covering structures may be formed from ink, polymer structures, structures that include metal, glass, other materials, and/or combinations of these materials. In an illustrative configuration, a cosmetic covering structure may be formed from a ring-shaped member having a footprint that matches the footprint of inactive area IA. If, for example, active area AA has left and right portions with teardrop shapes, the ring-shaped member may have curved edges that follow the curved periphery of the teardrop-shaped portions of active area AA. The ring-shaped member may be formed from one or more polymer structures (e.g., the ring-shaped member may be formed from a polymer ring). Because the ring-shaped member can help hide overlapped components from view, the ring-shaped member may sometimes be referred to as a shroud or ring-shaped shroud member. The outward appearance of the shroud or other cosmetic covering structures may be characterized by a neutral color (white, black, or gray) or a non-neutral color (e.g., blue, red, green, gold, rose gold, etc.).

[1217] Display **4.1-14F** may, if desired, have a protective display cover layer. The cover layer may overlap active area AA and inactive area IA (e.g., the entire front surface of device **4.1-10** as viewed from front F of FIGS. **4.1-1** may be covered by the cover layer). The cover layer, which may sometimes be referred to as a housing wall or transparent housing wall, may have a rectangular outline, an outline with teardrop portions, an oval outline, or other shape with curved and/or straight edges.

[1218] The cover layer may be formed from a transparent material such as glass, polymer, transparent crystalline material such as sapphire, clear ceramic, other transparent materials, and/or combinations of these materials. As an example, a protective display cover layer for display **4.1-14F** may be formed from safety glass (e.g., laminated glass that includes a clear glass layer with a laminated polymer film). Optional coating layers may be applied to the surfaces of the display cover layer. If desired, the display cover layer may be chemically strengthened (e.g., using an ion-exchange process to create an outer layer of material under compressive stress that resists scratching). In some configurations, the display cover layer may be formed from a stack of two or more layers of material (e.g., first and second structural glass layers, a rigid polymer layer coupled to a glass layer or another rigid polymer layer, etc.) to enhance the performance of the cover layer.

[1219] In active area AA, the display cover layer may overlap the pixels of display panel **4.1-14P**. The display cover layer in active area AA is preferably transparent to allow viewing of images presented on display panel **4.1-14P**. In inactive area IA, the display cover layer may overlap the ring-shaped shroud or other cosmetic covering structure. The shroud and/or other covering structures (e.g., opaque ink coatings on the inner surface of the display cover layer and/or structures) may be sufficiently opaque to help hide some or all of the optical components in inactive area IA from view. Windows may be provided in the shroud or other cosmetic covering structures to help ensure that the optical components that are overlapped by these structures operate satisfactorily. Windows may be formed from holes, may be formed from areas of the shroud or other cosmetic covering structures that have been locally thinned to enhance light transmission,

may be formed from window members with desired light transmission properties that have been inserted into mating openings in the shroud, and/or may be formed from other shroud window structures.

[1220] In the example of FIGS. **4.1-3**, device **4.1-10** includes optical components such as optical components **4.1-60**, **4.1-62**, **4.1-64**, **4.1-66**, **4.1-68**, **4.1-70**, **4.1-72**, **4.1-74**, **4.1-76**, **4.1-78**, and **4.1-80** (as an example). Each of these optical components (e.g., optical sensors selected from among sensors **4.1-16** of FIGS. **4.1-2**, light-emitting devices, etc.) may be configured to detect light and, if desired to emit light (e.g., ultraviolet light, visible light, and/or infrared light).

[1221] In an illustrative configuration, optical component **4.1-60** may sense ambient light (e.g., visible ambient light). In particular, optical component **4.1-60** may have a photodetector that senses variations in ambient light intensity as a function of time. If, as an example, a user is operating in an environment with an artificial light source, the light source may emit light at a frequency associated with its source of wall power (e.g., alternating-current mains power at 4.1-60 Hz). The photodetector of component **4.1-60** may sense that the artificial light from the artificial light source is characterized by 60 Hz fluctuations in intensity. Control circuitry **4.1-12** can use this information to adjust a clock or other timing signal associated with the operation of image sensors in device **4.1-10** to help avoid undesired interference between the light source frequency and the frame rate or other frequency associated with image capture operations. Control circuitry **4.1-12** can also use measurements from component **4.1-60** to help identify the presence of artificial lighting and the type of artificial lighting that is present. In this way, control circuitry **4.1-12** can detect the presence of lights such as fluorescent lights or other lights with known non-ideal color characteristics and can make compensating color cast adjustments (e.g., white point adjustments) to color-sensitive components such as cameras and displays. Because optical component **4.1-60** may measure fluctuations in light intensity, component **4.1-60** may sometimes be referred to as a flicker sensor or ambient light frequency sensor.

[1222] Optical component **4.1-62** may be an ambient light sensor. The ambient light sensor may include one or more photodetectors. In a single-photodetector configuration, the ambient light sensor may be a monochrome sensor that measures ambient light intensity. In a multi-photodetector configuration, each photodetector may be overlapped by an optical filter that passes a different band of wavelengths (e.g. different visible and/or infrared passbands). The optical filter passbands may overlap at their edges. This allows component **4.1-62** to serve as a color ambient light sensor that measures both ambient light intensity and ambient light color (e.g., by measuring color coordinates for the ambient light). During operation of device **4.1-10**, control circuitry **4.1-12** can take action based on measured ambient light intensity and color. As an example, the white point of a display or image sensor may be adjusted or other display or image sensor color adjustments may be made based on measured ambient light color. The intensity of a display may be adjusted based on light intensity. For example, the brightness of display **4.1-14F** may be increased in bright ambient lighting conditions to enhance the visibility of the image on the display and the brightness of display **4.1-14F** may be decreased in dim lighting conditions to conserve power. Image sensor operations and/or light source operations may also be adjusted based on ambient light readings.

[1223] The optical components in active area IA may also include components along the sides of device **4.1-10** such as components **4.1-80** and **4.1-64**. Optical components **4.1-80** and **4.1-64** may be pose-tracking cameras that are used to help monitor the orientation and movement of device **4.1-10**. Components **4.1-80** and **4.1-64** may be visible light cameras (and/or cameras that are sensitive at visible and infrared wavelengths) and may, in conjunction with an inertial measurement unit, form a visual inertial odometry (VIO) system.

[1224] Optical components **4.1-78** and **4.1-66** may be visible-light cameras that capture real-time images of the environment surrounding device **4.1-10**. These cameras, which may sometimes be referred to as scene cameras or pass-through-video cameras, may capture moving images that are displayed in real time to displays **4.1-14R** for viewing by the user when the user's eyes are located

in eye boxes **4.1-34** at the rear of device **4.1-10**. By displaying pass-through images (pass-through video) to the user in this way, the user may be provided with real-time information on the user's surroundings. If desired, virtual content (e.g. computer-generated images) may be overlaid over some of the pass-through video. Device **4.1-10** may also operate in a non-pass-through-video mode in which components **4.1-78** and **4.1-66** are turned off and the user is provided only with movie content, game content, and/or other virtual content that does not contain real-time real-world images.

[1225] Input-output devices **4.1-22** of device **4.1-10** may gather user input that is used in controlling the operation of device **4.1-10**. As an example, a microphone in device **4.1-10** may gather voice commands. Buttons, touch sensors, force sensors, and other input devices may gather user input from a user's finger or other external object that is contacting device **4.1-10**. In some configurations, it may be desirable to monitor a user's hand gestures or the motion of other user body parts. This allows the user's hand locations or other body part locations to be replicated in a game or other virtual environment and allows the user's hand motions to serve as hand gestures (air gestures) that control the operation of device **4.1-10**. User input such as hand gesture input can be captured using cameras that operate at visible and infrared wavelengths such as tracking cameras (e.g., optical components **4.1-76** and **4.1-68**). Tracking cameras such as these may also track fiducials and other recognizable features on controllers and other external accessories (additional devices **4.1-10** of system **4.1-8**) during use of these controllers in controlling the operation of device **4.1-10**. If desired, tracking cameras can help determine the position and orientation of a handheld controller or wearable controller that senses its location and orientation by measuring the magnetic field produced by coil **4.1-54**. The use of tracking cameras may therefore help track hand motions and controller motions that are used in moving pointers and other virtual objects being displayed for a user and can otherwise assist in controlling the operation of device **4.1-10**.

[1226] Tracking cameras may operate satisfactorily in the presence of sufficient ambient light (e.g., bright visible ambient lighting conditions). In dim environments, supplemental illumination may be provided by supplemental light sources such as supplemental infrared light sources (e.g., optical components **4.1-82** and **4.1-84**). The infrared light sources may each include one or more light-emitting devices (light-emitting diodes or lasers) and may each be configured to provide fixed and/or steerable beams of infrared light that serve as supplemental illumination for the tracking cameras. If desired, the infrared light sources may be turned off in bright ambient lighting conditions and may be turned on in response to detection of dim ambient lighting (e.g., using the ambient light sensing capabilities of optical component **4.1-62**).

[1227] Three-dimensional sensors in device **4.1-10** may be used to perform biometric identification operations (e.g., facial identification for authentication), may be used to determine the three-dimensional shapes of objects in the user's environment (e.g., to map the user's environment so that a matching virtual environment can be created for the user), and/or to otherwise gather three-dimensional content during operation of device **4.1-10**. As an example, optical components **4.1-74** and **4.1-70** may be three-dimensional structured light image sensors. Each three-dimensional structured light image sensor may have one or more light sources that provide structured light (e.g., a dot projector that projects an array of infrared dots onto the environment, a structured light source that produces a grid of lines, or other structured light component that emits structured light). Each of the three-dimensional structured light image sensors may also include a flood illuminator (e.g., a light-emitting diode or laser that emits a wide beam of infrared light). Using flood illumination and structured light illumination, optical components **4.1-74** and **4.1-70** may capture facial images, images of objects in the environment surrounding device **4.1-10**, etc.

[1228] Optical component **4.1-72** may be an infrared three-dimensional time-of-flight camera that uses time-of-flight measurements on emitted light to gather three-dimensional images of objects in the environment surrounding device **4.1-10**. Component **4.1-72** may have a longer range and a narrower field of view than the three-dimensional structured light cameras of optical components

4.1-74 and 4.1-70. The operating range of component **4.1-72** may be 30 cm to 7 m, 60 cm to 6 m, 70 cm to 5 m, or other suitable operating range (as examples).

[1229] FIGS. **4.1-4** is a front view of an illustrative ring-shaped cosmetic covering structure for device **4.1-10**. Illustrative ring-shaped shroud **4.1-100** of FIGS. **4.1-4** may be mounted under the inner surface of the display cover layer for display **4.1-14F** in inactive area IA. This may help hide the optical components and other internal portions of device **4.1-10** from view from the exterior of device **4.1-10**. Shroud **4.1-100** may be formed from one or more unbroken ring-shaped members and/or may be formed from multiple shroud segments that are attached using adhesive, fasteners, or other attachment structures. If desired, shroud **4.1-100** may be formed from multiple members that are sandwiched together along some or all of their lengths. In an illustrative configuration, which may sometimes be described herein as an example, shroud **4.1-100** may be formed from an inner piece (e.g., an inner full or partial ring), which may sometimes be referred to as an inner shroud member, shroud trim, or shroud trim member and may be formed from an outer piece or pieces (e.g., one or more strips of material or covering members, an full ring, one or more partial rings, etc.), which may sometimes be referred to as a shroud cover, canopy, or shroud canopy.

[1230] As shown in FIGS. **4.1-4**, shroud **4.1-100** may have optical component windows to accommodate components **4.1-60**, **4.1-62**, **4.1-64**, **4.1-84**, **4.1-66**, **4.1-68**, **4.1-70**, **4.1-72**, **4.1-74**, **4.1-76**, **4.1-78**, **4.1-82**, and **4.1-80**. The optical component windows may be formed from through-hole openings in shroud **4.1-100**, from recesses or other partial openings that do not pass entirely through shroud **4.1-100**, from inserted optical window members in shroud through-hole openings, and/or from other shroud optical component window structures. Display **4.1-14F** may have a display cover layer that has corresponding optical component windows (through-hole openings, recessed areas, inserted window members in through-hole openings, etc.) and/or that is formed from bulk material that has desired optical properties (e.g., a display cover layer formed from one or more layers of material such as glass and/or polymer with sufficient transparency at the operating wavelength range of the overlapped optical component to allow the optical component to operate satisfactorily through the cover layer without forming openings or other window structures in the cover layer).

[1231] Shroud **4.1-100** may have any suitable shape. For example, the outline of shroud **4.1-100** may be rectangular with rounded corners as shown in FIGS. **4.1-4**, may have teardrop shapes on the left and right sides of device **4.1-10**, may have an oval outline, and/or may have other outlines with curved and/or straight edge segments. For example, the inner and outer edges of shroud **4.1-100** may be curved (e.g., to follow a teardrop shape). Shroud **4.1-100** may, if desired, have a peripheral edge that is curved along most or all of its length.

[1232] The width of shroud **4.1-100** may be constant along its length or shroud **4.1-100** may have portions that are wider than others. The thickness of shroud **4.1-100** (e.g., the dimension of shroud **4.1-100** into the page in the orientation of FIGS. **4.1-4**) may be smaller than the width of shroud **4.1-100** (the lateral dimension of shroud **4.1-100** within the page in the orientation of FIGS. **4.1-4**) or the thickness of the shroud may be equal to or greater than the width of the shroud. The shroud may have a two-dimensional shape (e.g., shroud **4.1-100** may have a planar shape) or may have a three-dimensional shape (e.g., a shape with a curved cross-sectional profile and/or a shape characterized by inner and/or outer surfaces of compound curvature). In an illustrative configuration, most or all of the inner and outer surfaces of shroud have a compound-curvature surface.

[1233] The optical components under inactive area IA may include components on the left and right sides of device **4.1-10** that operate in conjunction with each other. for example, scene cameras, tracking cameras, and/or structured light cameras in device **4.1-10** may be formed in pairs, each of which includes a left camera and a corresponding right camera. A left scene camera and a right scene camera may, as an example, operate together to capture overlapping images that provide device **4.1-10** with a wide field of view for gathering pass-through video. Left and right

tracking cameras may operate together to track a user's hands or other external objects. Left and right structured light cameras or other three-dimensional cameras may be used together to capture three-dimensional images of the user's environment. To enhance performance of the left and right optical components in these types of paired component arrangements, it may be desirable to maintain accurate alignment between the left and right optical components. To help maintain left and right optical components on the respective left and right sides of device **4.1-10** in alignment with each other, device **4.1-10** may be provided with one or more housing structures that help support the optical components. An illustrative example of device **4.1-10** having housing structures that support the optical components and a cover layer that overlaps the optical components is shown in FIGS. **4.1-5**.

[1234] As shown in FIGS. **4.1-5**, shroud **4.1-100** and display cover layer **4.1-92** may be attached to housing **4.1-26** using adhesive, screws and other fasteners, press-fit connections, and/or other attachment mechanisms. An illustrative configuration in which shroud **4.1-100** and cover layer **4.1-92** are attached to forward-facing edge of a housing wall in the main housing portion of structure **4.1-26** using adhesive is shown in FIGS. **4.1-5**. In the example of FIGS. **4.1-5**, shroud **4.1-100** has an inner shroud member such as shroud trim **4.1-100A** and has a corresponding outer shroud member such as shroud canopy **4.1-100B**. Shroud trim **4.1-100A** and shroud canopy **4.1-100B** may be formed from metal, polymer, ceramic, glass, other materials, and/or combinations of these materials. In an illustrative example, shroud trim **4.1-100A** is formed from black polymer or other dark material and shroud canopy **4.1-100B** is formed from clear polymer. The outer surface of shroud canopy **4.1-100B** may be smooth to provide shroud **4.1-100** with a cosmetically attractive appearance.

[1235] A layer of pressure sensitive adhesive may be used in attaching canopy **4.1-100B** to trim **4.1-100A**, or canopy **4.1-100B** may be formed integrally with trim **4.1-100A**. Adhesive may also be used in attaching cover layer **4.1-92** and shroud **4.1-100** to housing portion **4.1-26**. As shown in FIGS. **4.1-5**, for example, a first adhesive such as adhesive **4.1-122** may be used to attach display cover layer **4.1-92** to shroud **4.1-100** (e.g., to a ledge in shroud trim **4.1-100A**). A second adhesive such as adhesive **4.1-124** may, in turn, be used to attach shroud **4.1-100** (e.g., shroud trim **4.1-100A**) to an adjacent lip of a wall in housing **4.1-26**.

[1236] In some configurations, adhesives **4.1-122** and **4.1-124** may be formed from the same type of material. In an illustrative configuration, adhesives **4.1-122** and **4.1-124** are different. Housing portion **4.1-26** may have a wall with a lip shape that creates a shearing force on adhesive **4.1-124** as display **4.1-14F** is attached to housing **4.1-26** by pressing display **4.1-14F** against housing **4.1-26**. In this type of scenario, it may be desirable to form adhesive **4.1-124** from an adhesive that can bond satisfactorily in the presence of shear forces such as a molten hot melt glue (thermoplastic adhesive) or other liquid adhesive rather than pressure sensitive adhesive. Adhesive **4.1-124** may, if desired, be exposed to a curing agent (ultraviolet light, moisture, etc.) before display **4.1-14F** is assembled into housing **4.1-26**.

[1237] Adhesive **4.1-124** may be heated to loosen the adhesive bond formed by adhesive **4.1-124**, if desired. To help prevent display cover layer **4.1-92** from detaching from shroud **4.1-100** while softening adhesive **4.1-124** with heat, adhesive **4.1-122** may be provided with a higher-temperature softening point than adhesive **4.1-124** (e.g., adhesive **4.1-122** may be a two-part hot melt glue with a higher melting point than adhesive **4.1-124**).

[1238] Optical components that are overlapped by display cover layer **4.1-92** and shroud **4.1-100** in inactive area **IA** may transmit and/or receive light through shroud **4.1-100** and display cover layer **4.1-92**. Layer **4.1-92** may be formed from a single layer of glass, laminated glass, or other clear material that allows light for each overlapped optical component **4.1-104** to pass through layer **4.1-92**. If desired, a partial recess or a through-hole opening may be formed in the portion of layer **4.1-92**. An optional optical component window member may then be inserted within layer **4.1-92** (e.g., a window that overlaps component **4.1-104**). As an example, layer **4.1-92** may be formed from one

or more layers of glass and/or polymer and may be characterized by a first level of light transmission at operating wavelength(s) for component **4.1-104**. A window member in layer **4.1-92** may be formed from polymer, glass, and/or other materials that are characterized by a second level of light transmission at the operating wavelength(s) that is greater than the first level of light transmission. In other illustrative arrangements, no window member is inserted in layer **4.1-92** (e.g., when layer **4.1-92** alone is sufficiently transparent to pass light for component **4.1-104**). [1239] Shroud **4.1-100** may be provided with an optical component window that overlaps optical component to help accommodate overlapped optical component **4.1-104**. Component **4.1-104** may operate at ultraviolet light wavelengths, visible light wavelengths, and/or infrared light wavelengths. To accommodate component **4.1-104**, shroud trim **4.1-100A** has been provided with a through-hole opening, whereas shroud canopy **4.1-100B** has no openings overlapping component **4.1-104**. This effectively forms a window recess in shroud **4.1-100** in alignment with components **4.1-104**. Trim **4.1-100A** may be formed from black polymer or other light-absorbing material, so the formation of opening **4.1-120** in trim **4.1-100A** may help ensure that sufficiently light may pass through to allow component **4.1-104** to operate satisfactorily. The portion of canopy **4.1-100B** that overlaps component **4.1-104** may be transparent (e.g., clear polymer). Alternatively, canopy **4.1-100B** may be formed from light-absorbing material, and a portion of canopy **4.1-100B** overlapping component **4.1-104** may be removed.

[1240] To help hide component **4.1-104** from view, the inner surface of shroud canopy **4.1-100B** may be covered with one or more coatings, which may be used to provide region the region overlapping component **4.1-104** with a desired outward appearance and optical properties that ensure that component **4.1-104** can operate satisfactorily. The coatings may include a thin-film-interference filter formed from a stack of thin-film dielectric layers of alternating refractive index values (with indices and thicknesses selected to create a desired transmission spectrum and a desired reflection spectrum for the filter), may include a layer of ink (e.g., a polymer layer including dye, pigment, and/or other colorant), and/or may include any other suitable coating with desired optical properties.

[1241] Consider, as an example, a scenario in which component **4.1-104** transmits and/or receives infrared light. In this type of arrangement, canopy **4.1-100B** may be coated with a coating that is opaque at visible wavelengths and transparent at infrared wavelengths. This helps to hide component **4.1-104** from view from the exterior of device **4.1-10** while allowing infrared light associated with the operation of component **4.1-104** to pass through shroud **4.1-100** and layer **4.1-92**.

[1242] As another example, consider a scenario in which component **4.1-104** is an ambient light sensor. In this configuration, canopy **4.1-100B** may be coated with a coating that exhibits a visible light transmission of 1-8% (as an example). This may allow sufficient visible ambient light to reach the ambient light sensor for the ambient light sensor to make an ambient light reading. At the same time, the transmission of the coating may be sufficiently low to reduce the visibility of component **4.1-104** from the exterior of device **4.1-10**.

[1243] As these examples demonstrate, regions of display **4.1-14F** that overlap optical components such as component **4.1-104** of FIGS. **4.1-5** may be provided with optical component window structures in layer **4.1-92** and/or shroud **4.1-100** that help accommodate the optical component.

[1244] As described in connection with FIGS. **4.1-3** and **4.1-4**, there may be numerous optical components such as component **4.1-104** in inactive area IA. Each optical component may potentially have a different type of optical component window structure in shroud **4.1-100** and/or layer **4.1-92** to accommodate that component. For example, some areas of shroud **4.1-100** may have openings that receive components, other areas of shroud **4.1-100** may have inserted optical window member, and/or other areas of shroud **4.1-100** may have partial shroud openings (e.g., non-through-hole recesses) such as the opening of FIGS. **4.1-8** (which may optionally be coated to modify the optical properties of shroud **4.1-100**).

[1245] In some embodiments, it may be desirable to provide encapsulation material over cover layer **4.1-92**. An illustrative example of cover layer **4.1-92** with encapsulation is shown in FIGS. **4.1-6**.

[1246] As shown in FIGS. **4.1-6**, cover layer **4.1-92** may be coupled to shroud **4.1-100**. Cover layer **4.1-92** may include glass layer **4.1-126**, front laminate **4.1-128** and rear laminate **4.1-130**. Front laminate **4.1-128** and rear laminate **4.1-128** may be for example, layers of plastic that are laminated to cover layer **4.1-92**, layers of plastic that are adhesively attached to cover layer **4.1-92**, or other protective material that is attached to the front and rear surfaces of glass layer **4.1-126**. Although not shown in FIGS. **4.1-6**, multiple layers, such as antireflection coatings, antismudge coatings, acrylic layers, or other desired layers, may be included as part of cover layer **4.1-92**.

[1247] Although front laminate **4.1-128** and rear laminate **4.1-130** may protect the front and rear of glass layer **4.1-126**, an edge surface of glass layer **4.1-126** may still be exposed. To further protect the edge surface of glass layer **4.1-126**, encapsulation material **4.1-132** may be attached to the edge surface of glass layer **4.1-126**. Encapsulation material **4.1-132** may be an epoxy material, such as a ductile epoxy, that seals the edge surface of glass layer **4.1-126** and protects glass layer **4.1-126** at the edge surface. Alternatively, acrylate, polyvinyl butyral (PVB), polyurethane, or moisture cure materials may be used for encapsulation material **4.1-132**.

[1248] In some embodiments, encapsulation material **4.1-132** may be an epoxy that adheres to glass layer **4.1-126** without a primer (e.g., encapsulation material **4.1-132** may be a primer-less adhesive). Moreover, encapsulation material **4.1-132** may have adequate ductility to elongate and not fracture, such as a Young's modulus of less than 3 GPa, less than 4 GPa, less than 2.5 GPa, or other suitable modulus. Additionally, encapsulation material **4.1-132** may be chemically resistant to chemicals and resistant to degradation due to solar exposure. It may also be desirable for encapsulation material **4.1-132** to match an appearance of the shroud. For example, encapsulation material **4.1-132** may have a black appearance, a white appearance, a gray appearance, a shiny appearance, and/or a matte appearance. In general, encapsulation material **4.1-132** may be formed from material that adheres to glass layer **4.1-126**, while protecting the edge surface of glass layer **4.1-126**.

[1249] Encapsulation material **4.1-132** may substantially fill the opening between the edge surface of glass layer **4.1-126** and shroud **4.1-100**. For example, encapsulation material **4.1-132** may extend approximately 150 microns from the edge surface. In general, however, any amount of encapsulation material **4.1-132** may be applied to the edge surface.

[1250] As shown in FIGS. **4.1-6**, encapsulation material **4.1-132** may cover the edge surface and may also cover an edge portion of laminate **4.1-128**. However, this is merely illustrative. If desired, encapsulation material **4.1-132** may cover the edge surface of glass layer **4.1-126** without covering an edge portion of laminate **4.1-128**. For example, as shown in FIGS. **4.1-7**, encapsulation material may cover only the edge surface of glass layer **4.1-126**. In the example of FIGS. **4.1-7**, laminate **4.1-128** may extend over encapsulation material **4.1-132**. However, this is merely illustrative. Laminate **4.1-128** may be flush with the edge surface of glass layer **4.1-126**, if desired.

[1251] In some embodiments, it may be determined that glass layer **4.1-126** can be protected by modifying the position of glass layer **4.1-126** relative to shroud **4.1-100** (or support structure **4.1-126**). For example, as shown in the illustrative embodiment of FIGS. **4.1-8**, the edge surface of glass layer **4.1-126** may be left unencapsulated, but the size of opening **4.1-134** between the edge surface and shroud **4.1-100** may be adjusted. By increasing or decreasing the size of opening **4.1-134**, glass layer **4.1-126** may be protected.

[1252] Instead of, or in addition to, adding material to the edge surface of layer **4.1-126**, it may be desirable to add material in the gap between layer **4.1-126** and the shroud/support structure. An illustrative example of adding material in this gap is shown in FIGS. **4.1-9**.

[1253] As shown in FIGS. **4.1-9**, material **4.1-136** may be included between the edge surface of glass layer **4.1-126** and shroud **4.1-100**. Material **4.1-136** may be, for example, a bumper ring. The

bumper ring may be formed from elastomer, rigid plastic, or another material that helps protect the edge surface of layer **4.1-126**.

[1254] As an alternative to material **4.1-136** being a bumper ring between layer **4.1-126** and shroud **4.1-100**, material **4.1-136** may be an overmolded structure on layer **4.1-126**, on shroud **4.1-100**, or on a chassis that is coupled to support structure **4.1-26**. In general, the overmolded structure may fill the gap between layer **4.1-126** and the support structure, shroud, and/or chassis and help protect the edge surface of layer **4.1-126**.

[1255] Although not shown in FIGS. **4.1-9**, a portion of material **4.1-136** may extend underneath layer **4.1-126** if desired. In particular, there may be a portion of material **4.1-136** between the bottom surface of layer **4.1-126** and shroud **4.1-100**.

[1256] Instead of, or in addition to, adding material between layer **4.1-126** and shroud **4.1-100**, upper laminate **4.1-128** and/or lower laminate **4.1-130** may wrap around the edge surface of layer **4.1-126**. Illustrative examples of the laminates wrapping the edge surface are shown in FIGS. **4.1-10** and **4.1-11**.

[1257] As shown in FIGS. **4.1-10**, upper laminate **4.1-128** may wrap around the edge surface of layer **4.1-126**. In particular, upper laminate **4.1-128** may have portion **4.1-128A** that extends around and covers the edge surface of layer **4.1-126**. Layer **4.1-126** may have a rounded edge surface to allow upper laminate **4.1-128** to wrap the edge surface and sufficiently adhere to the surface, as shown in FIGS. **4.1-10**. By forming layer **4.1-126** with a rounded edge, the curve of laminate **4.1-128** around the edge may be reduced, thereby reducing stress on laminate **4.1-128**. However, layer **4.1-126** may have a planar edge surface, or a surface with any other desired profile, around which upper laminate **4.1-128** wraps, if desired. By wrapping upper laminate around the edge surface of layer **4.1-126**, the edge surface may be protected.

[1258] As shown in FIGS. **4.1-11**, lower laminate **4.1-130** may wrap around the edge surface of layer **4.1-126**. In particular, lower laminate may have portion **4.1-130A** that extends around and covers the edge surface of layer **4.1-126**. Layer **4.1-126** may have a rounded edge surface to allow lower laminate **4.1-130** to wrap the edge surface and sufficiently adhere to the surface, may have a planar edge surface, or may have a surface with any other desired profile, around which lower laminate **4.1-130** wraps. By wrapping lower laminate **4.1-130** around the edge surface of layer **4.1-126**, the edge surface may be protected.

[1259] In the example of FIGS. **4.1-11**, lower laminate portion **4.1-130A** wraps entirely around the edge surface of layer **4.1-126** and partially overlaps upper laminate **4.1-128**. However, the arrangement is merely illustrative. If desired, lower laminate portion **4.1-130A** may wrap around only the edge surface of layer **4.1-126** without overlapping or extending over upper laminate **4.1-128**.

[1260] Another example of material that may be used to protect layer **4.1-126** is shown in FIGS. **4.1-12**. In the example of FIGS. **4.1-12**, glue (or another similar material) **4.1-138** may be used to completely fill the gap between layer **4.1-126** and shroud **4.1-100**. For example, the glue may be inserted into the gap after cover layer **4.1-92** has been assembled into the head-mounted device. Glue **4.1-138** may help protect the edge surface of layer **4.1-126**.

[1261] Rather than wrapping the upper or lower laminate around the edge surface of layer **4.1-126**, upper laminate **4.1-128** may extend to shroud **4.1-100** to cover the gap between the edge surface and shroud **4.1-100**. For example, as shown in FIGS. **4.1-13**, upper laminate **4.1-128** may have portion **4.1-128B** that extends to shroud **4.1-100** (or support structure **4.1-26** or another portion of device **4.1-10**). By covering the gap between the edge surface of layer **4.1-126** and shroud **4.1-100**, the edge surface may be protected.

[1262] Instead of, or in addition to, adding material or extending the laminates to protect the edge surface of layer **4.1-126**, shroud **4.1-100** or a chassis attached to support structure **4.1-26** may be modified to protect the edge surface. Illustrative examples of modifying these components to protect layer **4.1-126** are shown in FIGS. **4.1-14** and **4.1-15**.

[1263] As shown in FIGS. **4.1-14**, structure **4.1-140** may have a lip that covers the gap between layer **4.1-126** and shroud **4.1-100**/support structure **4.1-26**. Structure **4.1-140** may be formed from a portion of shroud **4.1-100** or from a portion of support structure **4.1-26** (e.g., a chassis of support structure **4.1-26**). The lip of structure **4.1-140** may help protect the edge surface of layer **4.1-126**.

[1264] If desired, the lip of structure **4.1-140** may be combined with the extension of laminate around the edge surface of layer **4.1-126**. For example, as shown in FIGS. **4.1-15**, upper laminate **4.1-128A** may wrap the edge surface of layer **4.1-126** to protect the edge surface, and the lip of structure **4.1-140** may provide additional protection.

[1265] Although cover layer **4.1-92** has been described as being coupled to shroud **4.1-100**, this is merely illustrative. In some embodiments, cover layer **4.1-92** may be coupled directly to support structure **4.1-26**. In other embodiments, device **4.1-10** may include a chassis attached to support structure **4.1-26** (e.g., a chassis to support various components in device **4.1-10**), and cover layer **4.1-92** may be coupled to the chassis.

[1266] Moreover, although cover layer **4.1-92** has been described as including a glass layer, this material is merely illustrative. Layer **4.1-126** may be formed from ceramic, sapphire, or any other desired material.

[1267] In general, laminates **4.1-128** and **4.1-130** may protect layer **4.1-126**. An illustrative stackup of cover layer **4.1-92** including detailed laminates **4.1-128** and **4.1-130** is shown in FIGS. **4.1-16**.

[1268] As shown in FIGS. **4.1-16**, cover layer **4.1-92** may include layer **4.1-126**, which may be glass, sapphire, or other material, and laminates **4.1-128** and **4.1-130**. Although laminates **4.1-128** and **4.1-130** are shown as planar in FIGS. **4.1-16** for illustrative purposes, laminate **4.1-128** may be a convex laminate, and laminate **4.1-130** may be a concave laminate (as shown in FIGS. **4.1-6-4.1-15**), if desired.

[1269] Laminate **4.1-128** may include polymer layer **4.1-134** coupled to layer **4.1-126** with adhesive **4.1-132**. Polymer layer **4.1-134** may be a polycarbonate (PC) layer, a polymethyl methacrylate (PMMA) layer, or other suitable polymer layer. Adhesive **4.1-132** may be a pressure-sensitive adhesive (PSA), an optically clear adhesive (OCA), or other suitable adhesive. In some illustrative embodiments, adhesive **4.1-132** may be an ultraviolet-curable OCA. Adhesive **4.1-132** may have a thickness of at least 100 microns, at least 200 microns, 150-250 microns, or other suitable thickness.

[1270] Hard coat **4.1-136** may be formed on polymer layer **4.1-134**. Hard coat **4.1-136** may be an acrylic layer, a thin glass layer, a sapphire layer, or other material. In some embodiments, if polymer layer **4.1-134** is a PMMA layer, hard coat **4.1-136** may be a polycarbonate layer or a blend of polycarbonate and acrylic material. For example, hard coat **4.1-136** may be formed from an ultraviolet-curable film. Hard coat **4.1-136** may be at least 2 microns thick, three microns thick, 3-5 microns thick, or other suitable thickness.

[1271] Coating layers **4.1-138** may be formed on hard coat **4.1-136**. Coating layers **4.1-138** may include an antireflection coating (e.g., a layer that matches an index of refraction of glass **4.1-126** to the air outside of glass **4.1-126**) and an antismudge coating (e.g., a fluoropolymer or other oleophobic material), as examples.

[1272] By including adhesive **4.1-132** between glass **4.1-126** and polymer layer **4.1-134**, polymer **4.1-134** and glass **4.1-126** may be decoupled. As a result, less stress may be applied to glass **4.1-126** from polymer **4.1-134**. To reduce the amount of stress further, a UV-curable OCA may be used, as such an OCA may be soft when applied. However, this is merely illustrative. Any suitable adhesive may be used to decouple polymer **4.1-134** from glass **4.1-126** and reduce the stress applied to glass **4.1-126**.

[1273] In some illustrative embodiments, polymer layer **4.1-134** may be formed from PMMA, which may match the optical properties of glass **4.1-126**, particularly when cover layer **4.1-92** is curved. However, any suitable material may be used to form polymer layer **4.1-134** and match the appearance of polymer layer **4.1-134** to glass **4.1-126**.

[1274] Although hard coat **4.1-136** has been described as being overlapped by coating layers **4.1-138**, this is merely illustrative. In some embodiments, hard coat **4.1-136** may be formed as the outermost layer of cover layer **4.1-92**. For example, as shown in FIGS. **4.1-17**, hard coat **4.1-136** may be formed as the outermost layer.

[1275] If desired, hard coat **4.1-136** may include antismudge properties. For example, hard coat **4.1-136** may have fluoropolymer or other oleophobic material incorporated into the hard coat material. Additionally, if an antireflection coating is desired, antireflection coating **4.1-137** may be formed as the lowermost layer of laminate **4.1-128**, directly on glass **4.1-126**.

[1276] Returning to FIGS. **4.1-16**, laminate **4.1-130** may include adhesive **4.1-140** that couples polymer **4.1-142** to glass **4.1-126**. Adhesive **4.1-140** and polymer **4.1-142** may correspond with adhesive **4.1-132** and polymer layer **4.1-132**. For example, adhesive **4.1-140** may be a PSA or OCA (e.g., a UV-curable OCA), and polymer layer **4.1-132** may be polycarbonate or PMMA. Polymer layer **4.1-142** may be acrylic, sapphire, glass, UV-curable material, or other suitable material.

[1277] Antireflection coating **4.1-146** may be formed on polymer layer **4.1-142**. Although not shown, a UV stabilizer material, such as alumina, may be formed between polymer layer **4.1-142** and antireflection coating **4.1-146**. The UV stabilizer material may have a thickness of at least 10 nm, at least 20 nm, or other suitable thickness, and may stabilize the surrounding layers when adhesives **4.1-140** and **4.1-132**, and/or polymer layers **4.1-134** and **4.1-142** are cured.

[1278] Laminate **4.1-130** may also include ink **4.1-148**. Ink **4.1-148** may be formed over an optical component in a head-mounted device, such as one of components **4.1-60**, **4.1-62**, **4.1-64**, **4.1-66**, **4.1-68**, **4.1-70**, **4.1-72**, **4.1-74**, **4.1-76**, **4.1-78**, and **4.1-80** of FIGS. **4.1-3**. In an illustrative embodiment, ink **4.1-148** may overlap a flood illuminator, and ink **4.1-148** may be an infrared-transparent-visible-light-blocking ink, such as an ink with 10% visible light transmission or less. However, this is merely illustrative. In general, ink **4.1-148** may overlap any component in a head-mounted device and may have a corresponding transmission spectrum.

[1279] Although FIGS. **4.1-16** and **4.1-17** show laminate **4.1-130** without a hard coat layer, this is merely illustrative. If desired, laminate **4.1-130** may have a hard coat layer, such as hard coat layer **4.1-136**, either within laminate **4.1-130** or as an outermost layer of laminate **4.1-130**.

[1280] An edge seal, such as the seals of FIGS. **4.1-6-4.1-9**, may protect one or more layers of laminate **4.1-128** and/or laminate **4.1-130**. An illustrative example of a seal overlapping the edges of layers in laminate **4.1-128** is shown in FIGS. **4.1-18**.

[1281] As shown in FIGS. **4.1-18**, seal **4.1-150** may overlap the edges of laminate **4.1-128**. In particular, seal **4.1-150** may overlap and bond to a PMMA layer, a polycarbonate layer, and/or a hard coat layer in laminate **4.1-128**. In some embodiments, seal **4.1-150** may form chemical bonds with an upper surface of laminate **4.1-128**. Moreover, in the illustrative example of FIGS. **4.1-18**, seal **4.1-150** may extend to glass layer **4.1-126**. Seal **4.1-150** may be formed from polyurethane, epoxy, acrylate, PVB, or other suitable material to protect the edges of laminate **4.1-128**.

[1282] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

V: Dust Seal

5.1: Seal for an Electronic Device

[1283] The electronic devices described in the present disclosure include components and arrangements configured to protect the LEDs, antennas, sensors, and/or other components communicating with the external environment noted above, from light and debris that may otherwise harm or negatively affect the components. More specifically, the present disclosure describes seals and electronic components that ensure internal components can sufficiently function without interference from an environment external to the electronic device, as needed, without the risk of damage to the internal component. Such interferences can include light, debris, or moisture. Such internal components can include, but are not limited to, cameras, LEDs, ambient light

sensors, flicker sensors, and other sensors, and the like. These and other internal components can be disposed adjacent to, or aligned below, one or more apertures/ports of the electronic device, with a seal disposed between the aperture/vent and the internal electronic component. In this way, the seals described herein can act as a barrier between the internal component and the external environment or the internal component and other internal components while also enabling the proper functioning of the internal component.

[1284] In a particular example, an electronic device includes a housing that defines an internal volume and an aperture defined by the housing. An electronic component can be disposed within the internal volume and a seal can be disposed between the aperture and the electronic component, the seal being configured to protect the electronic component within the housing.

[1285] In a particular example, an electronic component can include a light emitting component coupled to a base support. The electronic device can include other electronic components that are light sensitive. The seal can be disposed and configured between the light emitting component and light sensitive component to prevent and/or minimize light interference.

[1286] The seals described herein obstruct light having wavelengths in the visible range, as well as those having wavelengths extending into to ultraviolet and infrared ranges. The seals also inhibit dust and/or debris from traversing into the housing from apertures and/or any vents disposed on the housing of the electronic device. The seals described herein are also compressible to aid in assembly of the electronic device. In addition, the seals and assemblies described herein are durable and resistant to damage. The seal can be die cut and the components of the electronic device can be made of materials intended to minimize the weight of the components and/or the electronic device.

[1287] For example, a seal described herein can include an open or closed cell foam. The seal can engage a bracket or base support of the electronic device in an interference fit. In some examples, the seal can include a seal configured to surround the light emitting component. Because the light emitting component is mounted to an end of the base support in some examples, the seal can be die cut to have a precise interference fit with the base support to separate the light emitting component from other electronic components disposed in the electronic device and to facilitate a top-down assembly orientation.

[1288] FIG. 5-1 illustrates a cross-section of an electronic device 5-100. The electronic device can include any of the electronic devices described above. The electronic device can include a housing 5-105. The housing can include a first wall 5-110. In some examples, the first wall 5-110 can include an exterior wall of the electronic device 5-100 or can include a wall that separate chambers or compartments within electronic device 5-100. The housing 5-105 can include a base support 5-115 coupled to the first wall 5-110. The base support 5-115 can include a base bracket or support piece for the electronic device 5-100 and/or electronic components disposed within the housing 5-105. In some examples, the base support 5-115 can include one or more stiff metal materials, including steel, stainless steel, magnesium, and/or titanium. In some examples, the base support 5-115 can include plastic and/or plastic components to reduce weight of the electronic device 5-100. The base support 5-115 can be cylindrical shaped or can include a cubic shape and may be a solid material or hollow to reduce weight of the electronic device 5-100. The base support 5-115 can include a first surface 5-120 coupled to the first wall 5-110 by adhesive or any non-limiting fastening method and/or device. An electronic component can be coupled to a second surface 5-125 of the base support 5-115. The second surface 5-125 of the base support 5-115 can be configured opposite the first surface 5-120. The electronic component can include a light emitting component 5-130 (e.g., a light emitting diode or LED).

[1289] The electronic device 5-100 can further include a second wall 5-135 disposed over the base support 5-115. The first wall 5-110 and the second wall 5-135 can form an enclosure 5-140 within the housing 5-105 and the base support 5-115 can be disposed within the enclosure 5-140. In some examples, an electronic component 5-145 can also be disposed within the enclosure 5-140. The electronic component 5-145 can include a light sensitive component, in some examples, subject to

light interference from the light emitting component **5-130**. The electronic component **5-145** can be mounted to a second base support or can be coupled to either the first wall **5-110** and/or the second wall **5-135** directly.

[1290] The electronic device **5-100** can further include a seal **5-150** surrounding the light emitting component **5-130**. The seal **5-150** can be disposed between the base support **5-115** and the second wall **5-135**. In some examples, the seal **5-150** separates the electronic component **5-145** from the light emitting component **5-130**. The seal **5-150** can be configured to obstruct light having wavelengths between about 10⁻⁸ m and about 10⁻³ m. In other words, the seal **5-150** can obstruct light in the visible region, the ultraviolet region, and the infrared region. The seal **5-150** can also inhibit dust and/or debris from traversing between the base support **5-115** and the second wall **5-135**. In some examples, the seal **5-150** surrounds the light emitting component **5-130** on all sides. As such, the seal **5-150** engages the base support **5-115** either around a perimeter of an upper portion of the base support **5-115** or the second surface **5-125** of the base support **5-115**. In some examples, the second surface **5-125** includes the same surface having the light emitting component **5-130** disposed thereon.

[1291] In some examples, the base support **5-115** defines a ledge **5-155**. The seal **5-150** can surround an upper portion of the base support **5-115** and contact or engage an upper surface of the ledge **5-155**. As such, the seal **5-150** and the base support **5-115** can be engaged in an interference fit. For example, the radial seal **5-150** can be disposed in a press fit or friction fit between the base support **5-115** and the second wall **5-135**. The seal **5-150** and the base support **5-115** can be held together by friction after the housing **5-105** and other components are assembled. For example, the seal **5-150** can be compressed between the second wall **5-135** and the second surface **5-125** or the ledge **5-155** of the base support **5-115** to separate or seal the light emitting component **5-130** from the electronic component **5-145**. In some examples, the seal **5-150** can separate and/or seal off the enclosure **5-140** from an area or environment outside the housing **5-105** or external to the electronic device **5-100**.

[1292] In some examples, the seal **5-150** can be die cut. The seal **5-150** can be cut into a precise shape by a metal “die” during production. Because of the precise shape (e.g., radial) and dimensions of the seal **5-150** to properly seal against light, dust, and other debris and to fit properly against the second wall **5-135** and the base support **5-115**, including the ledge **5-155** defined thereby, the die cutting process can provide the exact shape and properties of the seal **5-150** for proper assembly of the electronic device **5-100**.

[1293] In some examples, the seal **5-150** can include a silicone material. The silicone seal **5-150** can be manufactured inexpensively and when shaped appropriately, provide the required compressibility and interference fit to the base support **5-115**. In some examples, the silicone seal **5-150** can include a foaming agent and/or be dyed to obstruct light from passing through from either the external environment and/or the light emitting component **5-130** to the electronic component **5-145** within the enclosure **5-140**. The seal **5-150** can include portions formed of different materials. In some examples, the seal **5-150** can include a plastic shim **5-160**. The plastic shim **5-160** can be included as a component of the seal **5-150** to reduce weight of the seal **5-150**, and thus, can reduce the weight of the electronic device **5-100**. In some examples, the plastic shim **5-160** can contact the base support **5-115**. In some examples, the plastic shim **5-160** can contact the ledge **5-155** and/or the upper portion of the base support **5-115**. The plastic shim **5-160** can also be impenetrable or resistant to light, dust, and/or debris. The seal can further include a foam material **5-165** coupled to the plastic shim **5-160**.

[1294] Referring now to FIG. 5-2, a cross-section of the seal **5-150** illustrated in FIG. 5-1 is shown. In some examples, the plastic shim **5-160** can be coupled to the foam material **5-165** with a pressure sensitive adhesive **5-170**, or any suitable adhesive. The plastic shim **5-160** can form a precision seal with the base support **5-115** and can be less expensive to manufacture than the silicone and/or a seal including only foam material. The foam material **5-165** can be compressed

between the plastic shim 5-160 and the second wall 5-135, as shown in FIG. 5-1. The plastic shim 5-160 can include a bottom surface 5-162 coupled to the base support 5-115. The plastic shim 5-160 can be joined to the base support 5-115 with the pressure sensitive adhesive 5-170. A top surface of the plastic shim 5-160 can also include a pressure sensitive adhesive 5-170 to couple the plastic shim 5-160 to the foam material 5-165. In some examples, the foam material 5-165 can include a compressible open or closed cell foam to inhibit light, dust, and/or debris.

[1295] FIG. 5-3 illustrates an electronic device 5-200 having a housing 5-205 and electronic components therein. The electronic device 5-200 can include a head mountable device (e.g., device 30). In some examples, the electronic device 5-200 can include a first enclosure 5-210 having a light emitting component 5-222 disposed therein. The first enclosure can be defined, at least in part, by the housing 5-205, an internal wall 5-255, and an external wall 5-240 defining an external surface 5-242. The housing 5-205 can include a first wall 5-215. In some examples, the first wall 5-215 can include an exterior wall of the electronic device 5-200 or can include a wall that separate chambers or compartments within the electronic device 5-200. The housing 5-205 can include a base support 5-220 coupled to the first wall 5-215. The base support 5-220 can include a first surface 5-225 coupled to the first wall 5-215 by an adhesive or any non-limiting fastening method and/or device. An electronic component can be coupled to a second surface 5-230 of the base support 5-220. The second surface 5-230 of the base support 5-220 can be opposite the first surface 5-225. The light emitting component 5-222 can include a light emitting diode (LED).

[1296] The housing 5-205 can further include a second enclosure 5-230. The second enclosure 5-230 can include at least one light sensitive component 5-235 disposed therein. The second enclosure 5-230 can be defined by the housing 5-205, including various internal and external walls. In some examples, the light sensitive component 5-235 can include an optically sensitive device. In some examples, the light sensitive component 5-235 can include at least one of a camera, an ambient light sensor, or a flicker sensor. In some examples, the second enclosure 5-230 can include an exterior surface 5-240 of the housing 5-205. In other words, one of the walls of the second enclosure 5-230 includes the exterior surface 5-240 of the housing 5-205.

[1297] In some examples, the housing 5-205 can further define a third enclosure 5-245. The base support 5-220 can be disposed within the third enclosure 5-245. In some examples, the base support 5-220 can include a metal or plastic material. In some examples, the third enclosure 5-245 can include an interior surface 5-250 of the housing 5-205. In other words, one of the walls of the third enclosure 5-245 includes the interior surface 5-250 of the housing 5-205. In some examples, the housing 5-205 of the electronic device 5-200 includes an internal wall 5-255 separating at least a portion of the second enclosure 5-230 and the third enclosure 5-245. In some examples, a seal 5-260 is included. The seal 5-260 can be coupled to a top portion or the second surface 5-230 of the base support 5-220. The seal 5-260 can separate the first enclosure 5-210, the second enclosure 5-230, and the third enclosure 5-245. In other words, the seal 5-260 includes at least a portion of a wall or divider that separates the first enclosure 5-210, the second enclosure 5-230, and the third enclosure 5-245 from each other. In some examples, the seal 5-260 is impenetrable to dust, debris, and/or light. The seal can extend from the second surface 5-230 of the base support 5-220 past the internal wall 5-255 and press fit to the exterior surface 5-240 of the housing 5-205. In such a configuration as shown in FIG. 6, the seal 5-260 functions to both protect the light sensitive component 5-235 from the light emitting component 5-222 and protects both the light sensitive component 5-235 and the light emitting component 5-222 from debris and/or dust 5-265 present either outside the housing 5-205 or within the third enclosure 5-245.

[1298] In some examples, the dust 5-265 can ingress into the third enclosure 5-245 through an air vent 5-270 or other aperture. The seal 5-260 prevents or substantially prevents the dust 5-265 or other debris, such as, but in no way limited to, water, dirt, and smoke, from passing through to the first enclosure 5-210 or second enclosure 5-230, which can include sensitive electronic components. The air vent 5-270 can be included on the electronic device 5-200 for temperature

control and/or heat dissipation. In some examples, the air vent **5-270** can be included for comfort and/or safety for the user of the electronic device **5-200**. According to one example, the air vent **5-270** can include a perforated material to cover the air vent **5-270** or other apertures in the electronic device. The size, locations, and number of perforations extending through such a material can vary from one example to another. Such perforations can include machined, laser cut, or otherwise manufactured openings defined by and extending through the material. Such openings can be sized and arranged to prevent a certain size particle from the external environment from passing through the air vent **5-270**. Such openings can also be sized and include other types of filters to prevent moisture or other harmful debris from passing through the air vent **5-270**.

[1299] In some examples, the seal **5-260** can be cut and/or shaped to seal around the internal wall **5-255** and between the second surface **5-230** of the base support **5-220** and the exterior surface **5-240** of the housing **5-205**. The seal **5-260** can include a compressible foam and can compress in an interference or press fit between the exterior surface **5-240** of the housing **5-205** and a top portion or the second surface **5-230** of the base support **5-220**. The seal **5-260** can include an open cell polyurethane foam, which is lightweight and more compressible compared to a closed cell foam. In other examples, the seal **5-260** can include a silicone material. The seal **5-260** can include a reticulated Polyurethane, PVC/Nitrile, Ethylene Propylene Diene Monomer (EPDM) rubber, or other suitable material. The materials which can be used to produce the closed cell foam for the seal **5-260** can vary greatly from ethylene-vinyl acetate (EVA), polyethylene, polystyrene, rubber to polypropylene etc. The closed cell foam can include trapped gas bubbles which are formed during the expansion and cure of the foam. The bubbles are permanently locked to a place, as the trapped gas is very efficient in increasing the insulation capability of the foam. The foam which is formed is strong and is usually of a greater density than open cell foam, which enables the gas bubbles to lock into place. The nature of the foam enables it to be vapor retardant and resist liquid water. In some examples, the materials for the seal can be open celled, but can include a sufficiently tortuous path so as to restrict or eliminate the passage of contaminants between enclosures.

[1300] In some examples, the top portion of the base support **5-220** can include a ledge (e.g., ledge **5-155** shown in FIG. **5-1**). The seal **5-260** can surround an upper portion of the base support **5-220** and contact the ledge. In some examples, the seal **5-260** can further include a plastic shim (e.g., plastic shim **5-160**) that contacts the ledge. The seal **5-260** can fit tightly to the base support **5-220**. The tight fit can function not only to seal the enclosures within the housing, but also to control and/or retain electronic components in place, which can improve the design of the electronic device **5-200**, minimize noise and/or vibrations, and reduce manufacturing costs.

[1301] FIGS. **5-4A-5-4C** illustrate an electronic device **5-300**. FIG. **5-4A** shows a top perspective view of an electronic component **5-305** and a dust seal **5-310** surrounding the electronic component **5-305** within the electronic device **5-300**, according to one example. In some examples, the dust seal **5-310** can be coupled to a base support **5-315**. The dust seal **5-310** can couple to the base support **5-315** via an interference fit. In some examples, the dust seal **5-310** can be adhered to the base support **5-315** with an adhesive (e.g., pressure sensitive adhesive **5-170** shown in FIG. **5-2**). In some examples, the electronic component **5-305** can include a light emitting component (e.g., light emitting component **5-222** shown in FIG. **5-3**). The electronic device **5-300** can further include a flexible printed circuit board or flexible printed circuit (flex PCB) **5-320**. At least a portion of the flex PCB **5-320** can be disposed between the dust seal **5-310** and the base support **5-315**. In some examples, a flex PCB can include an arrangement of printed circuitry and/or components that utilize flexible materials with a flexible overlay. A flexible printed circuit board can include a metallic layer of traces, usually copper, bonded to a dielectric layer. Thickness of the metallic layer can be very thin ($<.0001''$) to very thick ($>0.010''$) and the dielectric thickness can also vary. The flex PCB can be included in the electronic device **5-300** to reduce the assembly process and improve reliability. The flex PCB can be used as a connector, power supply, and also as full circuits assembled with components such as electronic component **5-305**. A benefit of including the flex

PCB 5-320 in the electronic device 5-300 is connecting the electronic component 5-305 to other components and/or a power source (not shown) and having the flex PCB 5-320 disposed between the dust seal 5-310 and the base support 5-315. In some examples, the dust seal 5-310 retains the flex PCB adjacent to the base support 5-315.

[1302] FIG. 5-4B shows a cross-sectional view of a portion of electronic device 5-300. The flex PCB 5-320 can be coupled to the electronic component 5-305. In some examples, electronic component 5-305 can include a light emitting device mounted to the base support 5-315. In some examples, the light emitting device can be coupled to a top surface 5-325 of the flex PCB 5-320 and a bottom surface 5-330 of the flex PCB 5-320 can be coupled to the base support 5-315. The dust seal 5-310 can encircle the light emitting device. In some examples, the dust seal 5-310 is impenetrable or resistant to light, dust, and/or debris. FIG. 5-4B shows the base support 5-315 having a ledge 5-335. The dust seal 5-310 can extend from the upper portion of the base support 5-315 adjacent to the base support 5-315 and past the dust seal 5-310. In some examples, the ledge 5-335 can include a cut-out portion. The cut-out portion can have the width of the flex PCB to minimize folding of the PCB and retain the flex PCB 5-320 adjacent to the base support 5-315. Referring to FIG. 5-4C, the flex PCB 5-320 can further include an adhesive 5-340 on the bottom surface 5-330 to retain the flex PCB 5-320 adjacent to the base support 5-315. FIG. 7C shows components of the dust seal, each component configured to retain the flex PCB 5-320 and/or prevent dust and/or debris ingress into the electronic device.

[1303] In some examples, the dust seal 5-310 can act as a radial seal to apply a radial force 5-345 to the flex PCB 5-320 to retain the flex PCB 5-320 adjacent the base support 5-315. In other words, the dust seal 5-310 prevents ingress path and aids adhesive 5-340 in flex management by applying compressive normal force that keeps the flex PCB 5-320 from delaminating from the base support 5-315. The dust seal 5-310 can also act as a compressive seal to apply a compressive force 5-350 to compress the dust seal 5-310 between a housing wall 5-355 and the base support 5-315. As such, the dust seal 5-310 can include a compressible foam. The dust seal 5-310 can be coupled to the base support 5-315 and engaged to the base support and the housing wall 5-355 in an interference fit. As described in other examples above, the housing wall 5-355 of the electronic device 5-300 can include a vent 5-360 and/or other apertures. FIG. 7C includes the dust ingress route 5-365 that the dust seal 5-310 is configured to prevent. Because the dust seal 5-310 includes a compressible foam and is engaged in a compressive seal and a radial seal, the dust ingress route 5-365 is closed off, the dust seal 5-310 being impenetrable to dust and/or debris. In some examples, the dust seal 5-310 is also impenetrable to light.

[1304] In some examples, the components described above for the electronic device examples (e.g., electronic device 5-100, 5-200, and/or 5-300) can aid in the assembly of the electronic device housing (e.g. housing 5-305). The assembly process can, in some examples, demand the use of a minimal amount of metal material for the base support 5-315 to maximize the available distance that can be used between the components. The assembly methods can also demand that the ledge 5-335 be minimized so that the base support 5-315 does not interfere with the inclusion/installation of other electronic components (e.g., electronic component 5-145) during assembly. In other words, due to the materials used for the base support 5-315 and/or the shape of the base support 5-315, the base support 5-315 can be installed first, then the other electronic components (e.g., electronic component 5-145) can be installed after the base support 5-315, followed by the dust seal 5-310 being placed on or around the base support, and finally the housing wall (e.g., housing wall 5-355) can be installed to compress the dust seal 5-310 in an interference fit between the housing wall 5-355 and the base support 5-315. In other words, the present exemplary sealing systems and methods facilitate a top or front assembly of the device, while ensuring a secure light and/or dust seal between sections.

VI: Sensor System

[1305] FIG. 6-0 illustrates a view of an example of an HMD device 6-100. The HMD device 6-100

can include a sensor array or system **6-102** including one or more sensors, cameras, projectors, and so forth mounted to one or more components of the HMD **6-100**. In at least one example, the sensor system **6-102** can include a bracket **1-338** on which one or more sensors of the sensor system **6-102** can be fixed/secured.

[1306] FIG. **6-1** illustrates a portion of an HMD device **6-100** including a front transparent cover **6-104** and a sensor system **6-102**. The sensor system **6-102** can include a number of different sensors, emitters, receivers, including cameras, IR sensors, projectors, and so forth. The transparent cover **6-104** is illustrated in front of the sensor system **6-102** to illustrate relative positions of the various sensors and emitters as well as the orientation of each sensor/emitter of the system **6-102**. As referenced herein, “sideways,” “side,” “lateral,” “horizontal,” and other similar terms refer to orientations or directions as indicated by the X-axis shown in FIG. **6-2**. Terms such as “vertical,” “up,” “down,” and similar terms refer to orientations or directions as indicated by the Z-axis shown in FIG. **6-2**. Terms such as “frontward,” “rearward,” “forward,” “backward,” and similar terms refer to orientations or directions as indicated by the Y-axis shown in FIG. **6-2**.

[1307] In at least one example, the transparent cover **6-104** can define a front, external surface of the HMD device **6-100** and the sensor system **6-102**, including the various sensors and components thereof, can be disposed behind the cover **6-104** in the Y-axis/direction. The cover **6-104** can be transparent or semi-transparent to allow light to pass through the cover **6-104**, both light detected by the sensor system **6-102** and light emitted thereby.

[1308] As noted elsewhere herein, the HMD device **6-100** can include one or more controllers including processors for electrically coupling the various sensors and emitters of the sensor system **6-102** with one or more mother boards, processing units, and other electronic devices such as display screens and the like. In addition, as will be shown in more detail below with reference to other figures, the various sensors, emitters, and other components of the sensor system **6-102** can be coupled to various structural frame members, brackets, and so forth of the HMD device **6-100** not shown in FIG. **6-1**. FIG. **6-1** shows the components of the sensor system **6-102** unattached and un-coupled electrically from other components for illustrative clarity sake.

[1309] In at least one example, the device can include one or more controllers having processors configured to execute instructions stored on memory components electrically coupled to the processors. The instructions can include, or cause the processor to execute, one or more algorithms for self-correcting angles and positions of the various cameras described herein overtime with use as the initial positions, angles, or orientations of the cameras get bumped or deformed due to unintended drop events or other events.

[1310] In at least one example, the sensor system **6-102** can include one or more scene cameras **6-106**. The system **6-102** can include two scene cameras **6-106** disposed on either side of the nasal bridge or arch of the HMD device **6-100** such that each of the two cameras **6-106** correspond generally in position with left and right eyes of the user behind the cover **6-103**. In at least one example, the scene cameras **6-106** are oriented generally forward in the Y-direction to capture images in front of the user during use of the HMD **6-100**. In at least one example, the scene cameras are color cameras and provide images and content for MR video pass through to the display screens facing the user's eyes when using the HMD device **6-100**. The scene cameras **6-106** can also be used for environment and object reconstruction.

[1311] In at least one example, the HMD **6-100** can include a controller electrically coupled to the various sensors and displays of the HMD **6-100**. In one example, the controller is configured to cause mixed-reality video passthrough from the first and second scene cameras **6-106** to the rearward facing displays, respectively, including one or more images captured by the first and second scene cameras **6-106**.

[1312] In at least one example, the sensor system **6-102** can include a first depth sensor **6-108** pointed generally forward in the Y-direction. In at least one example, the first depth sensor **6-108** can be used for environment and object reconstruction as well as user hand and body tracking. In at

least one example, the sensor system **6-102** can include a second depth sensor **6-110** disposed centrally along the width (i.e., along the X-axis) of the HMD device **6-100**. For example, the second depth sensor **6-110** can be disposed above the central nasal bridge or accommodating features over the nose of the user when donning the HMD **6-100**. In at least one example, the second depth sensor **6-110** can be used for environment and object reconstruction as well as hand and body tracking. In at least one example, the second depth sensor can include a LIDAR sensor. [1313] In at least one example, the sensor system **6-102** can include a depth projector **6-112** facing generally forward to project electromagnetic waves, for example in the form of a predetermined pattern of light dots, out into and within a field of view of the user and/or the scene cameras **6-106** or a field of view including and beyond the field of view of the user and/or scene cameras **6-106**. In at least one example, the depth projector can project electromagnetic waves of light in the form of a dotted light pattern to be reflected off objects and back into the depth sensors noted above, including the depth sensors **6-108**, **6-110**. In at least one example, the depth projector **6-112** can be used for environment and object reconstruction as well as hand and body tracking.

[1314] In at least one example, the sensor system **6-102** can include downward facing cameras **6-114** with a field of view pointed generally downward relative to the HMD device **6-100** in the Z-axis. In at least one example, the downward cameras **6-114** can be disposed on left and right sides of the HMD device **6-100** as shown and used for hand and body tracking, headset tracking, and facial avatar detection and creation for display a user avatar on the forward facing display screen of the HMD device **6-100** described elsewhere herein. The downward cameras **6-114**, for example, can be used to capture facial expressions and movements for the face of the user below the HMD device **6-100**, including the cheeks, mouth, and chin.

[1315] In at least one example, the sensor system **6-102** can include jaw cameras **6-116**. In at least one example, the jaw cameras **6-116** can be disposed on left and right sides of the HMD device **6-100** as shown and used for hand and body tracking, headset tracking, and facial avatar detection and creation for display a user avatar on the forward facing display screen of the HMD device **6-100** described elsewhere herein. The jaw cameras **6-116**, for example, can be used to capture facial expressions and movements for the face of the user below the HMD device **6-100**, including the user's jaw, cheeks, mouth, and chin. for hand and body tracking, headset tracking, and facial avatar

[1316] In at least one example, the sensor system **6-102** can include side cameras **6-118**. The side cameras **6-118** can be oriented to capture side views left and right in the X-axis or direction relative to the HMD device **6-100**. In at least one example, the side cameras **6-118** can be used for hand and body tracking, headset tracking, and facial avatar detection and re-creation.

[1317] In at least one example, the sensor system **6-102** can include a plurality of eye tracking and gaze tracking sensors for determining an identity, status, and gaze direction of a user's eyes during and/or before use. In at least one example, the eye/gaze tracking sensors can include nasal eye cameras **6-120** disposed on either side of the user's nose and adjacent the user's nose when donning the HMD device **6-100**. The eye/gaze sensors can also include bottom eye cameras **6-122** disposed below respective user eyes for capturing images of the eyes for facial avatar detection and creation, gaze tracking, and iris identification functions.

[1318] In at least one example, the sensor system **6-102** can include infrared illuminators **6-124** pointed outward from the HMD device **6-100** to illuminate the external environment and any object therein with IR light for IR detection with one or more IR sensors of the sensor system **6-102**. In at least one example, the sensor system **6-102** can include a flicker sensor **6-126** and an ambient light sensor **6-128**. In at least one example, the flicker sensor **6-126** can detect overhead light refresh rates to avoid display flicker. In one example, the infrared illuminators **6-124** can include light emitting diodes and can be used especially for low light environments for illuminating user hands and other objects in low light for detection by infrared sensors of the sensor system **6-102**.

[1319] In at least one example, multiple sensors, including the scene cameras **6-106**, the downward cameras **6-114**, the jaw cameras **6-116**, the side cameras **6-118**, the depth projector **6-112**, and the

depth sensors **6-108**, **6-110** can be used in combination with an electrically coupled controller to combine depth data with camera data for hand tracking and for size determination for better hand tracking and object recognition and tracking functions of the HMD device **6-100**. In at least one example, the downward cameras **6-114**, jaw cameras **6-116**, and side cameras **6-118** described above and shown in FIG. **6-1** can be wide angle cameras operable in the visible and infrared spectrums. In at least one example, these cameras **6-114**, **6-116**, **6-118** can operate only in black and white light detection to simplify image processing and gain sensitivity.

[1320] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **6-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **6-2-6-4** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **6-2-6-4** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **6-1**.

[1321] FIG. **6-2** illustrates a lower perspective view of an example of an HMD **6-200** including a cover or shroud **6-204** secured to a frame **6-230**. In at least one example, the sensors **6-203** of the sensor system **6-202** can be disposed around a perimeter of the HMD **6-200** such that the sensors **6-203** are outwardly disposed around a perimeter of a display region or area **6-232** so as not to obstruct a view of the displayed light. In at least one example, the sensors can be disposed behind the shroud **6-204** and aligned with transparent portions of the shroud allowing sensors and projectors to allow light back and forth through the shroud **6-204**. In at least one example, opaque ink or other opaque material or films/layers can be disposed on the shroud **6-204** around the display area **6-232** to hide components of the HMD **6-200** outside the display area **6-232** other than the transparent portions defined by the opaque portions, through which the sensors and projectors send and receive light and electromagnetic signals during operation. In at least one example, the shroud **6-204** allows light to pass therethrough from the display (e.g., within the display region **6-232**) but not radially outward from the display region around the perimeter of the display and shroud **6-204**.

[1322] In some examples, the shroud **6-204** includes a transparent portion **6-205** and an opaque portion **6-207**, as described above and elsewhere herein. In at least one example, the opaque portion **6-207** of the shroud **6-204** can define one or more transparent regions **6-209** through which the sensors **6-203** of the sensor system **6-202** can send and receive signals. In the illustrated example, the sensors **6-203** of the sensor system **6-202** sending and receiving signals through the shroud **6-204**, or more specifically through the transparent regions **6-209** of the (or defined by) the opaque portion **6-207** of the shroud **6-204** can include the same or similar sensors as those shown in the example of FIG. **6-1**, for example depth sensors **6-108** and **6-110**, depth projector **6-112**, first and second scene cameras **6-106**, first and second downward cameras **6-114**, first and second side cameras **6-118**, and first and second infrared illuminators **6-124**. These sensors are also shown in the examples of FIGS. **6-3** and **6-4**. Other sensors, sensor types, number of sensors, and relative positions thereof can be included in one or more other examples of HMDs.

[1323] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **6-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **6-1** and **6-3-6-4** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **6-1** and **6-3-6-4** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **6-2**.

[1324] FIG. **6-3** illustrates a front view of a portion of an example of an HMD device **6-300** including a display **6-334**, brackets **6-336**, **6-338**, and frame or housing **6-330**. The example shown in FIG. **6-3** does not include a front cover or shroud in order to illustrate the brackets **6-336**, **6-338**. For example, the shroud **6-204** shown in FIG. **6-2** includes the opaque portion **6-207** that would

visually cover/block a view of anything outside (e.g., radially/peripherally outside) the display/display region **6-334**, including the sensors **6-303** and bracket **6-338**.

[1325] In at least one example, the various sensors of the sensor system **6-302** are coupled to the brackets **6-336**, **6-338**. In at least one example, the scene cameras **6-306** include tight tolerances of angles relative to one another. For example, the tolerance of mounting angles between the two scene cameras **6-306** can be 0.5 degrees or less, for example 0.3 degrees or less. In order to achieve and maintain such a tight tolerance, in one example, the scene cameras **6-306** can be mounted to the bracket **6-338** and not the shroud. The bracket can include cantilevered arms on which the scene cameras **6-306** and other sensors of the sensor system **6-302** can be mounted to remain undeformed in position and orientation in the case of a drop event by a user resulting in any deformation of the other bracket **6-226**, housing **6-330**, and/or shroud.

[1326] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **6-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **6-1-6-2** and **6-4** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **6-1-6-2** and **6-4** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **6-3**.

[1327] FIG. **6-4** illustrates a bottom view of an example of an HMD **6-400** including a front display/cover assembly **6-404** and a sensor system **6-402**. The sensor system **6-402** can be similar to other sensor systems described above and elsewhere herein, including in reference to FIGS. **6-1-6-3**. In at least one example, the jaw cameras **6-416** can be facing downward to capture images of the user's lower facial features. In one example, the jaw cameras **6-416** can be coupled directly to the frame or housing **6-430** or one or more internal brackets directly coupled to the frame or housing **6-430** shown. The frame or housing **6-430** can include one or more apertures/openings **6-415** through which the jaw cameras **6-416** can send and receive signals.

[1328] In at least one example, an outward facing sensor assembly configured to capture images from an external environment in front of the head mountable device can include the first scene camera **6-106** pointed in a first direction and the second scene camera **6-106** pointed in a second direction, and an inward facing sensor assembly configured to capture images of the user when the user dons the head mountable device **6-100**. The inward facing sensor assembly can include the various eye tracking camera **6-120** and **6-122** pointed in the rearward direction. In at least one example, the device **6-100** can include a controller electrically coupled to the outward facing sensor assembly and the inward facing sensor assembly. The controller can be configured to cause the rearward facing display to project first images captured by the outward facing sensor assembly and cause the forward facing display to project second images captured by the various eye tracking cameras **6-120**, **6-122**.

[1329] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **6-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **6-1-6-3** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **6-1-6-3** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **6-4**.

VII: Antennas

[1330] FIGS. **7.0-1** illustrates a view of an example of an HMD display unit **100** including an antenna assembly **7.0-102**. Section VII of the present application describes the antennas and various components, assemblies, and systems associated with the antenna system **7.0-102** of the HMD display unit **100**.

7.1: Electronic Devices with Antenna Mounting Structures

[1331] Electronic devices may be provided with component mounting structures. The electronic devices may include portable electronic devices, wearable devices, desktop devices, embedded systems, and other electronic equipment. Illustrative configurations in which the electronic devices include a head-mounted device with a component mounting system may sometimes be described herein as an example.

[1332] Component mounting systems may be used to help mount electrical components in a device. As an example, a biasing member (sometimes referred to as a basing structure) may be used to help mount an antenna in the housing of a device. The biasing member, the antenna, and additional structures such as an antenna support member (sometimes referred to as an antenna support structure or antenna support) may form an antenna assembly that helps mount the antenna in a location in the device in which antenna signals are not blocked by conductive housing structures. If desired, biasing members may be used in mounting components other than antennas in a device. The use of biasing members to help mount antennas in head-mounted devices is illustrative.

[1333] A biasing member for an antenna assembly may be formed from a material such as structural foam or other compressible material that exhibits unidirectional compression and expansion characteristics. With this type of arrangement, the foam preferentially compresses and expands along a particular direction and exhibits little or no expansion and compression along orthogonal directions. The direction of preferential compression and expansion, which may sometimes be referred to as a preferential compression direction, unidirectional compression direction, preferential compression/expansion axis, axis of preferential compression, etc., need not be parallel with the surface normals of the biasing member. For example, a biasing member may be formed from a layer of foam with first and second opposing surfaces that are characterized by surface normals and the preferential compression direction may be oriented at a non-zero angle (e.g., an angle of at least 10°, at least 20°, less than 80°, less than 70°, or other suitable angle) with respect to the surface normals.

[1334] FIGS. 7.1-1 is a top view of an illustrative electronic device that may include a component biasing member. In the example of FIGS. 7.1-1, device 7.1-10 is a head-mounted device. As shown in FIGS. 7.1-1, head-mounted device 7.1-10 may include housing 7.1-12. Housing 7.1-12 is configured to be worn on a user's head and may sometimes be referred to as a head-mounted housing or head-mounted support structure. Housing 7.1-12 may have curved head-shaped surfaces, a nose-bridge portion such as portion NB that is configured to rest on a user's nose when device 7.1-10 is on a user's head, may have a headband such as strap 7.1-12T for supporting device 7.1-10 on the user's head, and/or may have other features that allow device 7.1-10 to be worn by a user. Housing 7.1-12 may have walls or other structures that separate an interior region of device 7.1-10 such as interior region 7.1-42 from an exterior region surrounding device 7.1-10 such as exterior region 7.1-44. As an example, housing 7.1-12 may include a transparent layer that forms a housing wall on front F of device 7.1-10 such as display cover layer 7.1-12CG. Display cover layer 7.1-12CG may overlap a forward-facing display such as display 7.1-20 (e.g., a pixel array based on organic light-emitting diodes or other display panel). Electrical components 7.1-36 (e.g., integrated circuits, sensors, control circuitry, light-emitting diodes, lasers, and other light-emitting devices, other control circuits and input-output devices, etc.) may be mounted on printed circuits and/or other structures within device 7.1-10 (e.g., in interior region 7.1-42).

[1335] To present a user with images for viewing from eye boxes such as eye box 7.1-34, device 7.1-10 may include rear-facing displays in optical modules 7.1-16. There may be, for example, a left rear-facing display in a left optical module 7.1-16 for presenting an image through a left lens to a user's left eye in a left eye box 7.1-34 and a right rear-facing display in right optical module 7.1-16 for presenting an image through a right lens to a user's right eye in a right eye box 7.1-34.

[1336] The user's eyes are located in eye boxes 7.1-34 at rear R of device 7.1-10 when inwardly facing surface 7.1-18 of housing 7.1-12 rests against the outer surface of the user's face. On rear R,

housing **7.1-12** may have cushioned structures (sometimes referred to as light seal structures) to enhance user comfort as surface **7.1-18** rests against the user's face. Device **7.1-10** may have forward-facing components such as forward-facing cameras and other sensors on front **F** that face outwardly away from the user. These components may generally be oriented in the +Y (forward) direction of FIGS. **7.1-1**.

[1337] During operation, device **7.1-10** may receive image data (e.g., image data for video, still images, etc.) and may present this information on the displays of optical modules **7.1-16**. Device **7.1-10** may also receive other data, control commands, user input, etc. Device **7.1-10** may transmit data to accessories and other electronic equipment. For example, image data from a forward-facing camera may be provided to an associated device, audio output may be provided to a device with speakers such as a headphone device, user input and sensor readings may be transmitted to remote equipment, etc.

[1338] Communications such as these may be supported using wired and/or wireless communications. In an illustrative configuration, components **7.1-36** may include wireless communications circuitry for supporting wireless communications between device **7.1-10** and remote wireless equipment (e.g., a cellular telephone, a wireless base station, a computer, headphones or other accessories, a remote control, peer devices, internet servers, and/or other equipment). Wireless communications may be supported using one or more antennas operating at one or more wireless communications frequencies. In an illustrative configuration, one or more antennas may be coupled to wireless transceiver circuitry. The wireless transceiver circuitry may include transmitter circuitry configured to transmit wireless communications signals using the antenna(s) and receiver circuitry configured to receive wireless communications signals using the antenna(s).

[1339] The wireless circuitry of device **7.1-10** may be formed from one or more integrated circuits, power amplifier circuitry, low-noise input amplifiers, passive RF components, one or more antennas, transmission lines, and other circuitry for handling RF wireless signals. The wireless circuitry may include radio-frequency transceiver circuitry for handling various radio-frequency communications bands. For example, the wireless circuitry of device **7.1-10** may include wireless local area network (WLAN) and wireless personal area network (WPAN) transceiver circuitry. This transceiver circuitry may handle 2.4 GHz and 5 GHz bands for WiFi® (IEEE 802.11) communications and other WLAN communications and the 2.4 GHz Bluetooth® communications band or other WPAN bands and may sometimes be referred to herein as WLAN/WPAN transceiver circuitry or local transceiver circuitry.

[1340] The wireless circuitry of device **7.1-10** may use remote wireless circuitry such as cellular telephone transceiver circuitry for handling wireless communications in frequency ranges (communications bands) such as a cellular low band (LB) from 600 to 960 MHz, a cellular low-midband (LMB) from 1410 to 1510 MHz, a cellular midband (MB) from 1710 to 2170 MHz, a cellular high band (HB) from 2300 to 2700 MHz, a cellular ultra-high band (UHB) from 3300 to 5000 MHz, or other communications bands between 600 MHz and 5000 MHz. If desired, the cellular telephone transceiver circuitry may support 5G communications using a low band at 600-850 MHz, a mid-band at 2.5-3.7 GHz, and a high band at 25-39 GHz. Wireless communications may also be provided using other frequency ranges (e.g., frequencies above 100 MHz, above 1 GHz, 1-30 GHz, 100 Mhz-300 GHz, 24 GHz, less than 300 GHz, less than 100 GHz, 10-300 GHz or other mm-wave frequencies, and/or other suitable frequencies). WLAN/WPAN transceiver circuitry and/or cellular transceiver circuitry may handle voice data and non-voice data.

[1341] If desired, the antennas and other wireless circuitry of device **7.1-10** may include satellite navigation system circuitry such as Global Positioning System (GPS) receiver circuitry for receiving GPS signals at 1575 MHz or for handling other satellite positioning data (e.g., GLONASS signals at 1609 MHz). Satellite navigation system signals are received from a constellation of satellites orbiting the earth. Wireless circuitry in device **7.1-10** can include circuitry

for other short-range (local) and long-range (remote) wireless links if desired. For example, wireless circuitry in device **7.1-10** may be provided to receive television and radio signals, paging signals, near field communications (NFC) signals at 13.56 MHz or other suitable NFC frequencies, ultrawideband (UWB) signals (e.g., UWB signals from 6-8.5 GHz, UWB signals from 3.5-9 GHz, etc.). Wireless circuitry in device **7.1-10** may also include antennas and transceiver for handling sensing applications (e.g., radar). If desired, antennas may be provided in arrays (e.g., phased antenna arrays) that support beam steering. These arrangements and other arrangements may be used in supporting wireless communications, wireless sensing, wireless location services, wireless power, and other wireless operations.

[1342] The wireless circuitry of device **7.1-10** may include antennas that are formed using any suitable antenna types. For example, the antennas of device **7.1-10** may include antennas with resonating elements that are formed from slot antenna structures, loop antenna structures, patch antenna structures, stacked patch antenna structures, antenna structures having parasitic elements, inverted-F antenna structures, planar inverted-F antenna structures, helical antenna structures, monopole antennas, dipole antenna structures, Yagi (Yagi-Uda) antenna structures, surface integrated waveguide structures, coils, hybrids of these designs, etc. If desired, one or more of the antennas may be cavity-backed antennas.

[1343] Different types of antennas may be used for different bands and combinations of bands. For example, one type of antenna may be used in forming a local wireless link antenna whereas another type of antenna is used in forming a remote wireless link antenna. If desired, space may be conserved within device **7.1-10** by using a single antenna to handle two or more different communications bands. For example, a single antenna in device **7.1-10** may be used to handle communications in a WiFi® or Bluetooth® communication band while also handling communications at one or more cellular telephone frequencies. In some configurations, some cellular telephone communications (e.g., low-band and mid-band communications) may be handled using a first antenna (e.g., an inverted-F antenna), whereas other communications (e.g., high-band communications) may be handled using one or more phased antenna arrays (e.g., multiple linear patch antenna arrays each of which is mounted in a different orientation and each of which has a different angle of view so that a desired amount of angular coverage is achieved).

[1344] To provide antenna structures in device **7.1-10** with the ability to cover different frequencies of interest, one or more of the antennas of device **7.1-10** may be provided with circuitry such as filter circuitry (e.g., one or more passive filters and/or one or more tunable filter circuits). Discrete components such as capacitors, inductors, and resistors may be incorporated into the filter circuitry. Capacitive structures, inductive structures, and resistive structures may also be formed from patterned metal structures (e.g., part of an antenna). If desired, antenna(s) in device **7.1-10** may be provided with adjustable circuits such as tunable components that tune the antenna over communications (frequency) bands of interest. The tunable components may be part of a tunable filter or tunable impedance matching network, may be part of an antenna resonating element, may span a gap between an antenna resonating element and antenna ground, etc.

[1345] Radio-frequency transmission line paths may be used to convey antenna signals between the radio-frequency transceiver circuitry of device **7.1-10** and the antenna(s) of device **7.1-10**. These paths may include one or more radio-frequency transmission lines (sometimes referred to herein simply as transmission lines). Radio-frequency transmission line paths may each include a positive signal conductor and a ground signal conductor. Transmission lines in device **7.1-10** may include coaxial cable transmission lines, stripline transmission lines, microstrip transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, transmission lines formed from waveguide structures (e.g., coplanar waveguides or grounded coplanar waveguides), and combinations of these types of transmission lines and/or other transmission line structures.

[1346] If desired, matching networks may be used to help match impedances in the wireless circuitry of device **7.1-10**. A matching network may, for example, include components such as

inductors, resistors, and capacitors configured to match the impedance of an antenna to the impedance of an associated radio-frequency transmission line path that is used in coupling the antenna to a transceiver. Matching network components may be provided as discrete components (e.g., surface mount technology components) or may be formed from housing structures, printed circuit board structures, traces on plastic supports, etc. Components such as these may also be used in forming antenna filter circuitry and may be tunable and/or fixed components.

[1347] Radio-frequency transmission line paths may be coupled to antenna feed structures associated with antennas in device **7.1-10**. As an example, an antenna in device **7.1-10** such as an inverted-F antenna, a planar inverted-F antenna, a patch antenna, a loop antenna, or other antenna may have an antenna feed with a positive antenna feed terminal and a ground antenna feed terminal. The positive antenna feed terminal may be coupled to an antenna resonating (radiating) element within the antenna. The ground antenna feed terminal may be coupled to an antenna ground in the antenna. The positive feed terminal may be coupled to a positive signal line in a transmission line and the ground feed terminal may be coupled to a ground signal line in the transmission line.

[1348] Other types of antenna feed arrangements may be used if desired. For example, an antenna may be fed using multiple feeds each coupled to a respective port of a transceiver over a corresponding transmission line. If desired, a given transmission line signal conductor may be coupled to multiple locations on an antenna and/or switches may be interposed within the paths between a transceiver and the feed terminals of an antenna.

[1349] FIGS. **7.1-2** is a diagram of illustrative wireless communications circuitry for device **7.1-10**. As shown in FIGS. **7.1-2**, the wireless circuitry includes radio-frequency transceiver **7.1-60**, which is coupled to antenna **7.1-40** by transmission line **7.1-62**. Antenna **7.1-40** may have an antenna resonating element **7.1-52** and antenna ground **7.1-50**. Antenna resonating element **7.1-52** may be formed from any suitable antenna resonating element structures. In the example of FIG. **2**, antenna resonating element **7.1-52** is an inverted-F antenna resonating element having resonating element arm **7.1-56**, which is coupled to ground **7.1-50** by return path **7.1-54** and which is fed using antenna feed **7.1-58**. Feed **7.1-58** has positive and ground feed terminals coupled respectively to positive and ground signal lines in transmission line **7.1-62**. Conductive structures making up antenna **7.1-40** may be formed from thin-film metal traces on printed circuits (e.g., rigid printed circuit boards formed from fiberglass-filled epoxy and other rigid printed circuit board substrate material and/or flexible printed circuits formed from sheets of polyimide or other flexible polymer substrates), metal traces on molded polymer antenna substrates, metal traces on other dielectric substrates, metal foil, conductive structural members such as portions of a housing for device **7.1-10** (e.g., a metal chassis and/or other internal and/or external frame structures, metal housing walls, metal component support brackets, and/or other conductive housing structures), and/or other structures in device **7.1-10** that are formed from metal and/or other conductive material.

[1350] Antennas may be mounted within device **7.1-10** using mounting brackets, using biasing structures that press antenna components against housing structures, using adhesive, using screws and other fasteners, using press-fit connections, using solder, welds, conductive adhesive, and/or other conductive attachment mechanisms, and/or other mounting arrangements. Due to the order of assembly of device components (e.g., due to the desire to assemble some components such as a display cover layer and/or other housing structures after other structures such as the rear of housing **7.1-12** and optical modules **7.1-16** have been assembled), it may be desirable to mount an antenna using a compressible structure such as a layer of foam or other biasing member that helps bias the antenna towards a known (reference) location in the device when the display cover layer or other housing structure is attached to other device structures.

[1351] Consider, as an example, the illustrative antenna mounting structures of FIGS. **7.1-3**. As shown in FIGS. **7.1-3**, antenna **7.1-40** may be formed from antenna substrate **7.1-64** (e.g., a flexible printed circuit or other substrate, which may be planar, as shown in FIGS. **7.1-3**, or which may

have a curved developable surface and/or a surface of compound curvature). Antenna substrate **7.1-64** contains metal traces **7.1-66** that have been patterned to form antenna resonating element **7.1-52** and/or other antenna structures.

[1352] Antenna **7.1-40** may be mounted in interior **7.1-42** using an antenna support member such as support member **7.1-68** (sometimes referred to as a support structure or support). Support member **7.1-68** may be formed from polymer, glass, ceramic, or other dielectric and/or other materials (e.g., metal, etc.) and/or combinations of these materials. A biasing member such as member **7.1-70** (sometimes referred to as a biasing structure) may be located between antenna **7.1-40** and support member **7.1-68**. Member **7.1-70** may be formed from a compressible structure (e.g., one or more springs, foam layers, elastomeric polymer layers, and/or other structures that can exhibit a restoring force when compressed). This allows member **7.1-70** to provide a biasing force that helps hold antenna **7.1-40** (e.g., substrate **7.1-64**) against the inner surface of cover layer **7.1-12CG** and/or other portions of device **7.1-10** such as other housing structures. As an example, member **7.1-68** may be mounted in device **7.1-10** so that antenna **7.1-40** faces outwardly towards front **F** (e.g., in the +Y direction away from eye boxes **7.1-34**). After mounting member **7.1-68** in this way, one or more additional layers of material may be installed in device **7.1-10** (e.g., by attaching such layer(s) to housing sidewalls and/or other housing structures). The layers of material that are installed in this way (which may sometimes be referred to as housing layers) may include cosmetic covering layers (e.g., a ring-shaped cover, sometimes referred to as a shroud or shroud trim, that runs around the border of display **7.1-20**, a tinted polymer layer that covers display **7.1-20** on the front of device **7.1-10** (sometimes referred to as a shroud canopy), display cover layer **7.1-12CG**, and/or other layer(s) of material. During installation of one or more of these housing layers (e.g., layer **7.1-12CG**, shroud layer(s), etc.), a curved inner surface or other inner surface of layer **7.1-12CG** and/or other housing layer(s) may contact antenna **7.1-40** and may press antenna **7.1-40** towards member **7.1-68**. This compresses biasing member **7.1-70**, which generates a restoring force (biasing force) that helps hold antenna **7.1-40** in place against the inner surface of layer **7.1-12CG** and/or other overlapping housing layers. In this way, the location of antenna **7.1-40** relative to layer **7.1-12CG** and/or other housing layers may be reliably established.

[1353] Substrate **7.1-64** may be formed from a layer with opposing first and second sides characterized by respective surface normals (see, e.g., surface normal **n**). In configurations in which antenna **7.1-40** (e.g., substrate **7.1-64**) and member **7.1-70** are compressed in a direction parallel that is parallel to the surface normal **n** of antenna **7.1-40** and substrate **7.1-64**, member **7.1-70** will compress in the absence of sheering forces (off-axis forces with respect to surface normal **n**). In some configurations, however, the inner surface of layer **7.1-12CG** (or other structure against which antenna **7.1-40** is being placed) may exert force on antenna **7.1-40** and member **7.1-70** in a direction that is orientated at a non-zero angle (see, e.g., angle **A2** of FIGS. **7.1-4**, which may be between 10° and 80°, between 20° and 70°, etc.) with respect to surface normal **n**. For example, the inner surface of layer **7.1-12CG** may be angled with respect to the Y axis. As a result, sheering forces may be generated that give rise to uncertainty in the lateral placement (location along the X axis) of antenna **7.1-40** after an overlapping layer such as layer **7.1-12CG** and/or other overlapping housing layer(s) has been assembled into device **7.1-10**.

[1354] To overcome undesired lateral movement of antenna **7.1-40** in response to installation of layer **7.1-12CG** or other overlapping layer(s) and associated application of force from layer **7.1-12CG** or other overlapping layer(s) onto antenna **7.1-40** and member **7.1-70**, member **7.1-70** may be formed from a compressible structure that exhibits preferential compression along an axis (sometimes referred to as a compression axis, unidirectional compression axis, axis of compression, axis of preferential compression, preferential axis of compression, or unidirectional axis of compression) that is not parallel to surface normals such as surface normal **n** that are associated with the inner surface of the display cover layer, the surfaces of the antenna substrate, and the adjacent surface of member **7.1-70**.

[1355] The use of a biasing member with a preferential axis of compression that is not parallel to surface normal n shown in FIGS. 7.1-4. In the example of FIGS. 7.1-4, member 7.1-70 has been formed from a polymer such as polymer foam (e.g., elastomeric open-cell and/or closed-cell foam). Fibers 7.1-72 (e.g., strands of polymer, glass, carbon, or other materials) have been embedded into member 7.1-70. Most or all of fibers 7.1-72 extend parallel to the X axis of FIGS. 7.1-4, which is perpendicular to the Y axis of FIGS. 7.1-4, and/or the lengths of fibers 7.1-72 that run parallel to the X axis are longer than those of fibers 7.1-72 that run parallel to the Y axis. The presence of fibers 7.1-72 helps preferentially stiffen and reduce compressibility in member 7.1-70 along the direction in which fibers 7.1-72 are aligned). As a result, the foam material of member 7.1-70 of FIGS. 7.1-4 exhibits elevated resistance to compression along the X axis (oriented at a non-zero angle $A1$ with respect to surface normal n in the example of FIG. 4) and exhibits lowered (e.g., less) resistance to compression along the Y axis (oriented at a non-zero angle $A2$ with respect to surface normal n in the example of FIGS. 7.1-4). The Y axis in this example serves as the preferential axis of compression for member 7.1-70. Because member 7.1-70 compresses and stretches along the Y axis but does not significantly compress or stretch along the X axis, which is orthogonal to the Y axis, member 7.1-70 may sometimes be referred to as unidirectional structural foam antenna biasing member (unidirectional structural foam antenna biasing structure) or unidirectional antenna biasing member (unidirectional antenna biasing structure).

[1356] FIGS. 7.1-5 is a diagram illustrating the behavior of member 7.1-70 of FIGS. 7.1-4 when exposed to applied force along the Y axis (e.g., a direction that is at a non-zero angle $A2$ with respect to surface normal n). Initially, member 7.1-70 is uncompressed and has thickness T . After compression, member 7.1-70 has reduced thickness TR . Due to the presence of fibers 7.1-72 (FIGS. 7.1-4) and/or other structures that promote uniaxial expansion and contraction, when force is applied to the surfaces of member 7.1-70 along the Y axis, member 7.1-70 compresses in direction 7.1-74 along the Y axis from its initially uncompressed state to a compressed state (see, e.g., compressed member shape 7.1-70' of FIGS. 7.1-5). During these compression activities, compression occurs along the Y axis and not the orthogonal X axis. As a result, the opposing outwardly facing and inwardly facing surfaces of member 7.1-70 do not shift laterally (e.g., there is no movement of these surfaces relative to each other along the X axis). This helps ensure that antenna 7.1-40 is mounted in a desired location in device 7.1-10 and does not experience undesirable lateral movement during assembly.

[1357] FIGS. 7.1-6 shows how the uniaxial compression properties of member 7.1-70 may be used to help ensure satisfactory placement of antenna 7.1-40 within device 7.1-10. As shown in FIGS. 7.1-6, forward-facing display 7.1-20 may be mounted under display cover layer 7.1-12CG, so that images on display 7.1-20 may be viewed on front F of device 7.1-10. If desired, an air gap may separate display 7.1-20 from display cover layer 7.1-12CG. One or more additional structures (e.g., a shroud having a ring-shaped trim portion that surrounds the pixels of display 7.1-20 and having a canopy portion that covers the pixels of display 7.1-20) may optionally be located between display cover layer 7.1-12CG and antenna 7.1-40, as shown by illustrative dielectric layer(s) 7.1-85. As an example, device 7.1-10 may include an internal housing structure such as a polymer structure forming a shroud (e.g., layer 7.1-85 may be a polymer should layer) and this layer may be interposed between layer 7.1-12CG and antenna 7.1-40. The shroud in this type of arrangement may have a ring shape that extends around the periphery of display 7.1-20 and/or may have portions that overlap display 7.1-20. In general, antenna 7.1-40 may be overlapped by any structure having a surface (e.g., an inner surface) against which antenna 7.1-40 is mounted. The overlapping structure, which may be a housing structure such as display cover layer 7.1-12CG, a shroud trim member, a shroud canopy, or other polymer layer, a dielectric housing wall, and/or any other dielectric member (sometimes referred to as a housing structure or housing layer), may have an inwardly facing surface against which antenna 7.1-40 is mounted. In the illustrative configuration of FIGS. 7.1-6, the housing layer that overlaps antenna 7.1-40 is display cover layer 7.1-12CG (or,

in the situation where optional interposed layer 7.1-85 is present, the layer that overlaps antenna 7.1-40 is a polymer layer that is between layer 7.1-12CG and antenna 7.1-40). These are illustrative examples. In general, any suitable polymer layer or other dielectric structure may overlap antenna 7.1-40 and may have a surface against which antenna 7.1-40 may be mounted during assembly of device 7.1-10.

[1358] In the illustrative configuration of FIGS. 7.1-6, antenna 7.1-40 is mounted in device 7.1-10 in peripheral (edge) region 7.1-80 (on the right side of device 7.1-10 in the example of FIG. 7.1-6) under layer 7.1-12CG. Layer 7.1-12CG (and/or other housing structures overlapping antenna 7.1-40) may be planar or may be curved and may be tilted to the right sufficiently to create a region of surface 7.1-84 with a surface normal that is angled with respect to the X and Y axes of FIGS. 7.1-6 (see, e.g., surface normal n of antenna 7.1-40, which is parallel to the surface normal of surface 7.1-84 in region 7.1-80). Because region 7.1-80 is angled away from the central portion of layer 7.1-12CG (or other overlapping dielectric layer) and may have a curved cross-sectional profile, region 7.1-80 may sometimes be referred to as forming a curved edge portion of layer 7.1-12CG or a curved edge portion of other overlapping housing layer.

[1359] Antenna 7.1-40 is attached (e.g., with adhesive) to biasing member 7.1-70 (e.g., a layer of unidirectional structural foam of the type described in connection with FIGS. 7.1-4 and 7.1-5 that is configured to serve as an antenna biasing member). Member 7.1-70 has a first side that faces antenna 7.1-40 and an opposing second side that faces support member 7.1-68. Member 7.1-70 is attached (e.g., with adhesive) to support member 7.1-68. Support member 7.1-68, in turn, is mounted to housing 7.1-12 (e.g., fastener 7.1-82 may be used to attach member 7.1-68 to portion 7.1-12P of housing 7.1-12 and/or other attachment mechanisms may be used to secure member 7.1-68 relative to housing 7.1-12 on rear R and/or elsewhere in device 7.1-10). When device 7.1-10 is assembled, there is generally slight compression of member 7.1-70 (e.g., along the Y axis), which creates a restoring force outward against inner surface 7.1-84. The arrangement of FIGS. 7.1-6 therefore places antenna 7.1-40 in a known spatial relationship with overlapping structures such as display cover layer 7.1-12CG (e.g., in direct contact with surface 7.1-84), thereby eliminating uncertainty in the distance between antenna 7.1-40 and layer 7.1-12CG. This may help avoid the possibility of forming variable-size air gaps between antenna 7.1-40 and surface 7.1-84, which could have varying impacts on antenna performance. Layer 7.1-12CG and/or other overlapping housing structures (e.g., a polymer shroud layer or other polymer layer) is preferably formed from a dielectric such as glass or polymer, so radio-frequency antenna signals for antenna 7.1-40 may pass through portion 7.1-80 of layer 7.1-12CG or other overlapping layer.

[1360] Antenna performance, which is affected by the distance between antenna 7.1-40 and the structures of device 7.1-10, and the reliability of the mounting arrangement shown in FIGS. 7.1-6 could potentially be adversely affected by undesired lateral movement of antenna 7.1-40 relative to its nominal position under portion 7.1-80. This lateral movement is prevented by using uniaxial foam in forming member 7.1-70. During assembly of device 7.1-10, antenna 7.1-40, member 7.1-70, and member 7.1-68 are initially mounted to housing 7.1-12. In this initial state, the front wall of housing 7.1-12 (e.g., cover layer 7.1-12CG in the current example) may not be present. After components 7.1-36 have been installed within housing 7.1-12, cover layer 7.1-12CG and/or other overlapping housing layer(s) may be moved in the $-Y$ direction and mounted to housing 7.1-12 (e.g., using adhesive layer 7.1-86, using fasteners, and/or using other attachment structures). Because surface normal n of antenna 7.1-40 and the opposing surface normal of surface 7.1-84 in edge region 7.1-80 are at a non-zero angle with respect to the Y axis (see, e.g., angle $A2$ of FIGS. 7.1-4), movement of layer 7.1-12CG in the $-Y$ direction creates a lateral force on antenna 7.1-40 and member 7.1-70 along the X axis that has the potential to compress member 7.1-70 laterally. Nevertheless, because member 7.1-70 is formed from unidirectional structural foam that preferentially compresses along the Y axis, this lateral movement (movement parallel to the X axis) is prevented. Rather, as layer 7.1-12CG (or other overlapping dielectric housing structure) is

mounted to housing 7.1-12 and presses inwardly on antenna 7.1-40, member 7.1-70 compresses only along the Y axis. As a result, antenna 7.1-40 moves slightly in the -Y direction when pressed by surface 7.1-84, but does not shift position relative to the X axis. This ensures that antenna 7.1-40 is located satisfactorily in device 7.1-10 relative to housing 7.1-12 (which may include, for example, metal structures such as a metal chassis that forms some or all of antenna ground 7.1-50 of FIGS. 7.1-2). Satisfactory performance of antenna 7.1-40 may therefore be achieved.

[1361] Although sometimes described herein in the context of an antenna biasing member in a head-mounted device, unidirectional structural foam or other compressible structures with a unidirectional axis of compression may be used in other contexts and/or devices. As an example, unidirectional structural foam may be used in device 7.1-10 for sound management (e.g., to exhibit preferential sound channeling along a particular direction), may be used for directional vibration absorption (perpendicular to the axis of fibers 7.1-72), may be used to exhibit directional electrical conductivity (e.g., for sensing and signal routing applications), and/or may otherwise be used in electronic devices that can benefit from elevated control of antenna placement and/or elevated control of the placement of other device components.

7.2: Systems with Transparent Layers

[1362] Electronic devices may be provided with components such as antennas. The electronic devices may include portable electronic devices, wearable devices, desktop devices, embedded systems, and other electronic equipment. Illustrative configurations in which the electronic devices include a head-mounted device may sometimes be described herein as an example.

[1363] The antennas in an electronic device may be configured to cover communications bands of interest (e.g., local area network bands, cellular telephone bands, etc.). To handle some communications such as 5G cellular communications, the antennas may include millimeter wave antennas (e.g., antennas operating at one or more frequencies between 20 GHz and 300 GHz, as an example). A millimeter wave antenna may use a phased-antenna array architecture in which multiple antenna elements such as patch antenna elements are arranged in an array (e.g., multiple patches in a row). During operation, the relative phases of each of the elements may be adjusted (e.g., so that the phased-antenna array performs beam steering).

[1364] Electronic device housing structures and other parts of an electronic device may include areas that are characterized by curved surfaces that can be flattened into a plane without distortion (sometimes referred to as developable surfaces or curved surfaces without compound curvature). Electronic device housing structures and other parts of an electronic device may also include areas that are characterized by compound curvature (surfaces that can only be flattened into a plane with distortion, sometimes referred to as non-developable surfaces). Mounting millimeter wave antennas and/or other antennas in an electronic device with curved surfaces can be challenging, because the presence of a curved surface adjacent to an antenna may give rise to different amounts of loading for different antenna elements in the antenna.

[1365] To help ensure satisfactory antenna operation when integrating a millimeter wave antenna into an electronic device with a curved structure, a dielectric member (sometimes referred to as a dielectric structure or dielectric layer) may be provided between the curved structure and the antenna. The dielectric member may have a planar surface facing the antenna. As an example, in a head-mounted device with a curved display cover layer, a polymer layer with a planar surface facing an antenna may be placed between the antenna and the curved display cover layer. In this way, the dielectric structure may help even out the amount of loading experienced by each antenna element in the antenna. The dielectric structure may also assist in impedance matching.

[1366] FIGS. 7.2-1 is a top view of an illustrative electronic device having antennas such as one or more millimeter wave antennas. In the example of FIGS. 7.2-1, device 7.2-10 is a head-mounted device. In general, device 7.2-10 may be any suitable electronic equipment.

[1367] As shown in FIGS. 7.2-1, head-mounted device 7.2-10 may include housing 7.2-12.

Housing 7.2-12 is configured to be worn on a user's head and may sometimes be referred to as a

head-mounted housing or head-mounted support structure. Housing 7.2-12 may have curved head-shaped surfaces, a nose-bridge portion such as portion NB that is configured to rest on a user's nose when device 7.2-10 is on a user's head, may have a headband such as strap 7.2-12T for supporting device 7.2-10 on the user's head, and/or may have other features that allow device 7.2-10 to be worn by a user.

[1368] Housing 7.2-12 may have walls or other structures that separate an interior region of device 7.2-10 such as interior region 7.2-42 from an exterior region surrounding device 7.2-10 such as exterior region 7.2-44. As an example, housing 7.2-12 may include a transparent layer that forms a housing wall on front F of device 7.2-10 such as display cover layer 7.2-12CG. Housing 7.2-12 may also include internal frame structures (e.g., a metal chassis), cosmetic covering members, polymer layers (e.g., fully or partly transparent polymer layers), housing walls formed from polymer and/or other materials, and/or other housing structures. In an illustrative configuration, housing 7.2-12 includes a dielectric structure such as dielectric member 7.2-13 that is overlapped by display cover layer 7.2-12CG. Dielectric member 7.2-13, which may sometimes be referred to as a polymer layer, shroud, dielectric layer, or dielectric structure, may be formed from one or more individual dielectric structures (e.g., structures formed from polymer, glass, ceramic, and/or other dielectric). Member 7.2-13 may be formed in a ring shape that runs along the periphery of cover layer 7.2-12CG (e.g., under peripheral edge portion E of cover layer 7.2-12CG) or may, as shown in FIGS. 7.2-1, overlap substantially all of display cover layer 7.2-12CG.

[1369] Display cover layer 7.2-12CG and member 7.2-13 may overlap a forward-facing display such as display 7.2-20 (e.g., a flexible display panel formed from a pixel array based on organic light-emitting diodes or other display panel). The portion of member 7.2-13 that overlaps display 7.2-20 may be formed from fully transparent polymer or partly transparent polymer that helps hide display 7.2-20 from view. The portion of member 7.2-13 in edge portion E may be opaque or transparent. Display cover layer 7.2-12CG may be formed from transparent polymer or glass (as examples).

[1370] Portions of display cover layer 7.2-12CG and member 7.2-13 such as edge portions of display cover layer 7.2-12CG and member 7.2-13 that surround display 7.2-20 may have curved cross-sectional profiles. As an example, edge portion E of cover layer 7.2-12CG and the underlying edge portion of member 7.2-13 may have one or more surfaces characterized by compound curvature (e.g., non-developable surfaces). The central portions of display cover layer 7.2-12CG and member 7.2-13 that overlap pixels of display 7.2-20 may have compound curvature and/or may have developable surfaces. In an illustrative arrangement, cover layer 7.2-12CG has inner and outer surfaces with compound curvature and member 7.2-13 has an outer surface of compound curvature around the edges of device 7.2-10 (e.g., the portion of member 7.2-13 surrounding display 7.2-20) and has developable inner and outer surfaces overlapping display 7.2-20. In the areas of compound curvature, at least some portions of the curved surface of layer 7.2-12CG and/or member 7.2-13 may be characterized by a radius of curvature R of 4 mm to 250 mm, 8 mm to 200 mm, 10 mm to 150 mm, at least 5 mm, at least 12 mm, at least 16 mm, at least 20 mm, at least 30 mm, less than 200 mm, less than 100 mm, less than 75 mm, less than 55 mm, less than 35 mm, and/or other suitable amount of curvature. In this illustrative configuration, display 7.2-20 may be a flexible display panel that is bent into a curved shape (e.g., a curved shape following the curved face of a user) and that is characterized by inner and outer developable surfaces. The portion of member 7.2-13 overlapping display 7.2-20 may have corresponding inner and outer developable surfaces. The innermost surface of member 7.2-13 in edge portion E may be planar to accommodate millimeter wave antennas. Other arrangements for the shapes of display cover layer 7.2-12CG and member 7.2-13 may be used in device 7.2-10, if desired.

[1371] Device 7.2-10 may include millimeter wave antennas and other antennas. Millimeter wave antennas may use phased-antenna arrays to implement beam steering. Each millimeter wave antenna may have an associated angle of view. To help provide satisfactory antenna coverage for

device **7.2-10** at millimeter wave frequencies, it may be desired to provide device **7.2-10** with multiple millimeter wave antennas and to orient each of these antennas in a different direction so that the angular coverage of each of the antennas overlaps.

[1372] Consider, as an example, illustrative device **7.2-10** of FIGS. **7.2-1**, which has three millimeter wave antennas, each of which is pointed in a different direction. A first of the three antennas (millimeter wave antenna **7.2-40-1**) is located under edge portion E of member **7.2-13** and cover layer **7.2-12CG** on the left side of device **7.2-10**. Antenna **7.2-40-1** is oriented in direction **7.2-72**, which is rotated counter clockwise by angle AX relative to the Y axis (where the Y axis is oriented in the forward direction facing outwardly in front of device **7.2-10**). A second of the three antennas (millimeter wave antenna **7.2-40-2**) is located in the center of device **7.2-10**, and, in this illustrative example, points straight ahead (in direction **7.2-74**, along the Y axis). A third of the three antennas (millimeter wave antenna **7.2-40-3**) is oriented in direction **7.2-76**, which is rotated clockwise by an angle AX relative to the Y axis. Directions **7.2-72**, **7.2-74**, and **7.2-76** may each lie in the XY plane of FIG. **1** or may be angled above or below the XY plane. With this type of arrangement, each antenna has a respective angle of view VA (e.g., a value within the range of 15°-90°, as an example). By overlapping the angle-of-view coverage of each antenna (e.g., by pointing antenna **7.2-40-1** slightly to the left of center, by pointing antenna **7.2-40-2** straight ahead, and by pointing antenna **7.2-40-3** slightly to the right of center), device **7.2-10** can be provided with a larger angular coverage at millimeter wave frequencies than if only one of these antennas were to be used.

[1373] During operation, device **7.2-10** may receive image data (e.g., image data for video, still images, etc.) and may present this information on the displays of optical modules **7.2-16**. Device **7.2-10** may also receive other data, control commands, user input, etc. Device **7.2-10** may transmit data to accessories and other electronic equipment. For example, image data from a forward-facing camera may be provided to an associated device, audio output may be provided to a device with speakers such as a headphone device, user input and sensor readings may be transmitted to remote equipment, etc.

[1374] Communications such as these may be supported using wired and/or wireless communications. In an illustrative configuration, components **7.2-36** may include wireless communications circuitry for supporting wireless communications between device **7.2-10** and remote wireless equipment (e.g., a cellular telephone, a wireless base station, a computer, headphones or other accessories, a remote control, peer devices, internet servers, and/or other equipment). Wireless communications may be supported using one or more antennas operating at one or more wireless communications frequencies (see, e.g., antennas **7.2-40-1**, **7.2-40-2**, and **7.2-40-3** of FIGS. **7.2-1**). In an illustrative configuration, one or more antennas may be coupled to wireless transceiver circuitry. The wireless transceiver circuitry may include transmitter circuitry configured to transmit wireless communications signals using the antenna(s) and receiver circuitry configured to receive wireless communications signals using the antenna(s).

[1375] The wireless circuitry of device **7.2-10** may be formed from one or more integrated circuits, power amplifier circuitry, low-noise input amplifiers, passive RF components, one or more antennas, transmission lines, and other circuitry for handling RF wireless signals. The wireless circuitry may include radio-frequency transceiver circuitry for handling various radio-frequency communications bands. For example, the wireless circuitry of device **7.2-10** may include wireless local area network (WLAN) and wireless personal area network (WPAN) transceiver circuitry. This transceiver circuitry may handle 2.4 GHz and 5 GHz bands for WiFi® (IEEE 802.11) communications and other WLAN communications and the 2.4 GHz Bluetooth® communications band or other WPAN bands and may sometimes be referred to herein as WLAN/WPAN transceiver circuitry or local transceiver circuitry.

[1376] The wireless circuitry of device **7.2-10** may use remote wireless circuitry such as cellular telephone transceiver circuitry for handling wireless communications in frequency ranges

(communications bands) such as a cellular low band (LB) from 600 to 960 MHz, a cellular low-midband (LMB) from 1410 to 1510 MHz, a cellular midband (MB) from 1710 to 2170 MHz, a cellular high band (HB) from 2300 to 2700 MHz, a cellular ultra-high band (UHB) from 3300 to 5000 MHz, or other communications bands between 600 MHz and 5000 MHz. If desired, the cellular telephone transceiver circuitry may support 5G communications using a low band at 600-850 MHz, a mid-band at 2.5-3.7 GHz, and a high band at 25-39 GHz. Wireless communications may also be provided using other frequency ranges (e.g., frequencies above 100 MHz, above 1 GHz, 1-30 GHz, 100 Mhz-300 GHz, 24 GHz, less than 300 GHz, less than 100 GHz, 10-300 GHz or other millimeter wave frequencies, and/or other suitable frequencies). WLAN/WPAN transceiver circuitry and/or cellular transceiver circuitry may handle voice data and non-voice data.

[1377] If desired, the antennas and other wireless circuitry of device **7.2-10** may include satellite navigation system circuitry such as Global Positioning System (GPS) receiver circuitry for receiving GPS signals at 1575 MHz or for handling other satellite positioning data (e.g., GLONASS signals at 1609 MHz). Satellite navigation system signals are received from a constellation of satellites orbiting the earth. Wireless circuitry in device **7.2-10** can include circuitry for other short-range (local) and long-range (remote) wireless links if desired. For example, wireless circuitry in device **7.2-10** may be provided to receive television and radio signals, paging signals, near field communications (NFC) signals at 13.56 MHz or other suitable NFC frequencies, ultrawideband (UWB) signals (e.g., UWB signals from 6-8.5 GHz, UWB signals from 3.5-9 GHz, etc.). Wireless circuitry in device **7.2-10** may also include antennas and transceiver for handling sensing applications (e.g., radar). If desired, antennas may be provided in arrays (e.g., phased antenna arrays) that support beam steering. These arrangements and other arrangements may be used in supporting wireless communications, wireless sensing, wireless location services, wireless power, and other wireless operations.

[1378] The wireless circuitry of device **7.2-10** may include antennas that are formed using any suitable antenna types. For example, the antennas of device **7.2-10** may include antennas with resonating elements that are formed from slot antenna structures, loop antenna structures, patch antenna structures, stacked patch antenna structures, antenna structures having parasitic elements, inverted-F antenna structures, planar inverted-F antenna structures, helical antenna structures, monopole antennas, dipole antenna structures, Yagi (Yagi-Uda) antenna structures, surface integrated waveguide structures, coils, hybrids of these designs, etc. If desired, one or more of the antennas may be cavity-backed antennas.

[1379] Different types of antennas may be used for different bands and combinations of bands. For example, one type of antenna may be used in forming a local wireless link antenna whereas another type of antenna is used in forming a remote wireless link antenna. If desired, space may be conserved within device **7.2-10** by using a single antenna to handle two or more different communications bands. For example, a single antenna in device **7.2-10** may be used to handle communications in a WiFi® or Bluetooth® communication band while also handling communications at one or more cellular telephone frequencies. In some configurations, some cellular telephone communications (e.g., low-band and mid-band communications) may be handled using a first antenna (e.g., an inverted-F antenna), whereas other communications (e.g., high-band cellular communications) may be handled using one or more phased antenna arrays (e.g., multiple linear patch antenna arrays each of which is mounted in a different orientation and each of which has a different angle of view so that a desired amount of angular coverage is achieved).

[1380] To provide antenna structures in device **7.2-10** with the ability to cover different frequencies of interest, one or more of the antennas of device **7.2-10** may be provided with circuitry such as filter circuitry (e.g., one or more passive filters and/or one or more tunable filter circuits). Discrete components such as capacitors, inductors, and resistors may be incorporated into the filter circuitry. Capacitive structures, inductive structures, and resistive structures may also be formed from patterned metal structures (e.g., part of an antenna). If desired, antenna(s) in device **7.2-10** may be

provided with adjustable circuits such as tunable components that tune the antenna over communications (frequency) bands of interest. The tunable components may be part of a tunable filter or tunable impedance matching network, may be part of an antenna resonating element, may span a gap between an antenna resonating element and antenna ground, etc.

[1381] Radio-frequency transmission line paths may be used to convey antenna signals between the radio-frequency transceiver circuitry of device **7.2-10** and the antenna(s) of device **7.2-10**. These paths may include one or more radio-frequency transmission lines (sometimes referred to herein as transmission lines). Radio-frequency transmission line paths may each include a positive signal conductor and a ground signal conductor. Transmission lines in device **7.2-10** may include coaxial cable transmission lines, stripline transmission lines, microstrip transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, transmission lines formed from waveguide structures (e.g., coplanar waveguides or grounded coplanar waveguides), and combinations of these types of transmission lines and/or other transmission line structures.

[1382] If desired, matching networks may be used to help match impedances in the wireless circuitry of device **7.2-10**. A matching network may, for example, include components such as inductors, resistors, and capacitors configured to match the impedance of an antenna to the impedance of an associated radio-frequency transmission line path that is used in coupling the antenna to a transceiver. Matching network components may be provided as discrete components (e.g., surface mount technology components) or may be formed from housing structures, printed circuit board structures, traces on plastic supports, etc. Components such as these may also be used in forming antenna filter circuitry and may be tunable and/or fixed components.

[1383] Radio-frequency transmission line paths may be coupled to antenna feed structures associated with antennas in device **7.2-10**. As an example, an antenna in device **7.2-10** such as an inverted-F antenna, a planar inverted-F antenna, a patch antenna, a loop antenna, or other antenna may have an antenna feed with a positive antenna feed terminal and a ground antenna feed terminal. The positive antenna feed terminal may be coupled to an antenna resonating (radiating) element within the antenna. The ground antenna feed terminal may be coupled to an antenna ground in the antenna. The positive feed terminal may be coupled to a positive signal line in a transmission line and the ground feed terminal may be coupled to a ground signal line in the transmission line.

[1384] Other types of antenna feed arrangements may be used if desired. For example, an antenna may be fed using multiple feeds each coupled to a respective port of a transceiver over a corresponding transmission line. If desired, a given transmission line signal conductor may be coupled to multiple locations on an antenna and/or switches may be interposed within the paths between a transceiver and the feed terminals of an antenna.

[1385] FIGS. **7.2-2** is a front view of device **7.2-10** showing illustrative locations for millimeter wave antennas **7.2-40-1**, **7.2-40-2**, and **7.2-40-3**. As shown in FIGS. **7.2-2**, front-facing display **7.2-20** may be surrounded by peripheral edge portions E of display cover layer **7.2-12CG** (which may be formed from dielectric materials such as glass and/or polymer) and dielectric member **7.2-13**, and these dielectric structures may overlap antennas **7.2-40-1**, **7.2-40-2**, and **7.2-40-3**. During operation, transmitted antenna signals from the millimeter wave antennas and received antenna signals for the millimeter wave antennas may pass through display cover layer **7.2-12CG** and member **7.2-13**. To enhance antenna efficiency, conductive structures such as conductive pixel structures and other conductive structures associated with display **7.2-20** may be present only in the center of device **7.2-10** (e.g., edge portions E may be free of any conductive display structures overlapping the millimeter wave antennas).

[1386] FIGS. **7.2-3** is a cross-sectional side view of an illustrative millimeter wave antenna. As shown in FIGS. **7.2-3**, the wireless circuitry of device **7.2-10** includes radio-frequency transceiver **7.2-60**. Transceiver **7.2-60** may be coupled to antenna **7.2-40** by signal path **7.2-62** (e.g., one or more transmission lines). The configuration of millimeter wave antenna **7.2-40** of FIGS. **7.2-3** may

be used for antenna 7.2-40-1, may be used for antenna 7.2-40-2, and may be used for antenna 7.2-40-3. As shown in FIGS. 7.2-3, millimeter wave antenna 7.2-40 may have multiple antenna elements 7.2-40E. Elements 7.2-40E may be formed from millimeter wave antenna resonating elements such as patch antenna elements (e.g., patch antennas formed from thin-film metal structures). The patch antennas may be arranged in a linear array (e.g., a line) on antenna substrate 7.2-40B (e.g., a printed circuit substrate, a ceramic or glass layer, or other dielectric substrate). A ground for antenna 7.2-40 may be formed from a ground antenna trace in substrate 7.2-40B and/or other conductive structures in device 7.2-10 (e.g., a metal chassis in device 7.2-10, a heat sink in device 7.2-10, a support bracket in device 7.2-10, etc.). Antenna 7.2-40 (e.g., elements 7.2-40E and substrate 7.2-40B) may have a planar surface that is characterized by a surface normal (see, e.g., surface normal nb of FIGS. 7.2-3). During operation, the control circuitry of device 7.2-10 may perform beam steering operations by adjusting the relative phases of the signals for each respective element 7.2-40E. In this way, angle BA of antenna beam direction 7.2-92 relative to antenna surface normal nb (e.g., the direction in which antenna 7.2-40 is nominally pointed) may be adjusted (e.g., to ensure that antenna signals are transmitted and received along a direction that provides satisfactory antenna performance).

[1387] Antennas may be mounted within device 7.2-10 using mounting brackets, using biasing structures that press antenna components against housing structures, using adhesive, using screws and other fasteners, using press-fit connections, using solder, welds, conductive adhesive, and/or other conductive attachment mechanisms, using one or more frames, carriers, and/or other internal support structures, and/or other mounting arrangements.

[1388] To ensure even loading for each of elements 7.2-40E, antenna 7.2-40 may be mounted adjacent to a planar dielectric structure that is evenly spaced from each of elements 7.2-40E. In arrangements such as the arrangement of FIGS. 7.2-1 and 7.2-2, in which edge portion E of display cover layer 7.2-12CG has an inner surface with a curved cross-sectional profile (e.g., an inwardly facing concave surface of compound curvature), a dielectric structure such as dielectric member 7.2-13 that is located between display cover layer 7.2-12CG and antenna 7.2-40 may be used to form the planar dielectric structure. (Arrangements in which the planar inner surface is formed directly on the inner side of layer 7.2-12CG and in which member 7.2-13 is omitted may also be used, if desired).

[1389] Consider, as an example, the top cross-sectional view of the left front corner of device 7.2-10 that is shown in FIGS. 7.2-4. As shown in FIGS. 7.2-4, the inner and outer surfaces of display cover layer 7.2-12CG (e.g., the surfaces of edge portion E of layer 7.2-12CG) may have curved cross-sectional profiles. These surfaces may have compound curvature. For example, inner surface 7.2-83 of display cover layer 7.2-12CG may be a concave surface of compound curvature. Antenna elements 7.2-40E of antenna 7.2-40 are supported on planar outer surface 7.2-82 of substrate 7.2-40B. To equalize antenna loading for each of the elements 7.2-40E in antenna 7.2-40 and thereby facilitate beam forming by antenna 7.2-40, dielectric member 7.2-13 may be provided with planar inner surface 7.2-70 that is parallel to planar surface 7.2-82 (e.g., surface normal na of surface 7.2-70 may be parallel to surface normal nb of surface 7.2-82). This ensures that each antenna element 7.2-40E will be separated from surface 7.2-70 by an air gap of equal size, thereby ensuring equal loading on each element 7.2-40E.

[1390] The outer surface of member 7.2-13 may be curved (e.g., the outer surface of member 7.2-13 that is attached to or adjacent to inner surface 7.2-83 of layer 7.2-12CG may have a convex shape such as a convex shape with compound curvature that matches the concave shape of surface 7.2-83). The permittivity of member 7.2-13, the thickness of member 7.2-13, and the size of the air gap between antenna 7.2-40 and member 7.2-13 may be selected to help match the impedance of antenna 7.2-40 to the impedance of layer 7.2-12CG, thereby reducing antenna signal reflections. In an illustrative configuration, the permittivity of layer 7.2-12CG has a first permittivity value, the permittivity of air has a second permittivity value that is lower than the first permittivity value, and

the permittivity of the polymer that makes up member 7.2-13 has a third permittivity value that is between the first and second values. This configuration may help match the impedance of antenna 7.2-40 to that of layer 7.2-12CG. The presence of the air gap between antenna 7.2-40 and member 7.2-13 may help reduce surface waves and may facilitate assembly of device 7.2-10 (e.g., by physically decoupling antenna 7.2-40 from overlapping structures such as member 7.2-13). [1391] Antenna 7.2-40 may be supported on internal housing structures and/or other support structures (see, e.g., illustrative support structures 7.2-86 of FIGS. 7.2-4). The support structures under antenna 7.2-40 may include a metal plate and/or other heatsink structure (see, e.g., heat sink 7.2-84). In the illustrative configuration of FIGS. 7.2-4, antenna 7.2-40 is located on the left front corner of device 7.2-10 (e.g., antenna 7.2-40 of FIGS. 7.2-4 may serve as antenna 7.2-40-1 of FIGS. 7.2-1). If desired, the approach of FIGS. 7.2-4 may be used to form antenna 7.2-40-3 on the right front corner of device.

[1392] In the center of device 7.2-10, antenna 7.2-40-2 may be installed using an arrangement of the type shown in the top view of FIGS. 7.2-5. As shown in FIGS. 7.2-5, antenna 7.2-40 (e.g., antenna 7.2-40-2 of FIGS. 7.2-1) may be mounted on heatsink 7.2-84 and support structure 7.2-86 so that the planar outer surface of antenna 7.2-40 faces an opposing inwardly facing planar surface of member 7.2-13. The adjacent surfaces of member 7.2-13 and antenna 7.2-40 may be parallel to each other (e.g., surface normal n_a may be parallel to surface normal n_b).

7.3: Systems with Transparent Layers

[1393] Electronic devices may be provided with components such as antennas. The electronic devices may include portable electronic devices, wearable devices, desktop devices, embedded systems, and other electronic equipment. Illustrative configurations in which the electronic devices include a head-mounted device may sometimes be described herein as an example.

[1394] Antennas may be formed from thin flexible substrates such as flexible printed circuits. A flexible printed circuit antenna may have metal traces that are patterned to form an antenna resonating element (sometimes referred to as an antenna resonating structure or antenna resonator). The metal traces may be supported by a flexible printed circuit substrate layer. The flexible printed circuit substrate layer may be formed from one or more sheets of polyimide or layers of other polymer.

[1395] Electronic device housing structures and other parts of an electronic device may include areas that are characterized by curved surfaces that can be flattened into a plane without distortion (sometimes referred to as developable surfaces or curved surfaces without compound curvature). Electronic device housing structures and other parts of an electronic device may also include areas that are characterized by compound curvature (surfaces that can only be flattened into a plane with distortion, sometimes referred to as non-developable surfaces).

[1396] To help conform a flexible printed circuit antenna to a surface of an electronic device housing structure or other dielectric member in an electronic device and/or to otherwise provide the antenna with a shape that facilitates installation and use of the antenna in a device with potentially complex shapes such as surfaces with compound curvature, a flexible printed circuit antenna may be formed into a three-dimensional shape (e.g., an unwrinkled shape characterized by surfaces of compound curvature). A flexible printed circuit antenna that has been provided with compound curvature surfaces in this way may then be attached to a supporting housing structure with compound curvature. For example, a flexible printed circuit antenna with compound curvature may be laminated to a dielectric member having matching compound curvature using a layer of adhesive.

[1397] FIGS. 7.3-1 is a top view of an illustrative electronic device that may include a flexible printed circuit antenna with compound curvature. In the example of FIGS. 7.3-1, device 7.3-10 is a head-mounted device. In general, device 7.3-10 may be any suitable electronic equipment.

[1398] As shown in FIGS. 7.3-1, head-mounted device 7.3-10 may include housing 7.3-12. Housing 7.3-12 is configured to be worn on a user's head and may sometimes be referred to as a

head-mounted housing or head-mounted support structure. Housing 7.3-12 may have curved head-shaped surfaces, a nose-bridge portion such as portion NB that is configured to rest on a user's nose when device 7.3-10 is on a user's head, may have a headband such as strap 7.3-12T for supporting device 7.3-10 on the user's head, and/or may have other features that allow device 7.3-10 to be worn by a user.

[1399] Housing 7.3-12 may have walls or other structures that separate an interior region of device 7.3-10 such as interior region 7.3-42 from an exterior region surrounding device 7.3-10 such as exterior region 7.3-44. As an example, housing 7.3-12 may include a transparent layer that forms a housing wall on front F of device 7.3-10 such as display cover layer 7.3-12CG. Housing 7.3-12 may also include internal frame structures (e.g., a metal chassis), cosmetic covering members, polymer layers (e.g., fully or partly transparent polymer layers), housing walls formed from polymer and/or other materials, and/or other housing structures. In an illustrative configuration, housing 7.3-12 includes a dielectric structure such as dielectric member 7.3-13 that is overlapped by display cover layer 7.3-12CG. Dielectric member 7.3-13, which may sometimes be referred to as a polymer layer, shroud, dielectric layer, or dielectric structure, may be formed from one or more individual dielectric structures (e.g., structures formed from polymer, glass, ceramic, and/or other dielectric). In the example of FIGS. 7.3-1, dielectric member 7.3-13, includes a first dielectric layer such as polymer layer 7.3-13-1 that extends across substantially all of front F of device 7.3-10 (e.g., layer 7.3-13-1 of FIGS. 7.3-1 has a footprint similar to or the same as that of layer 7.3-12CG). With this arrangement, layer 7.3-13-1, which may sometimes be referred to as a shroud canopy or shroud, has a central portion that overlaps display 7.3-20 and has a peripheral portion (e.g., the portion under edge portion E of display cover layer 7.3-12CG) with a ring shaped footprint that surrounds display 7.3-20. Dielectric member 7.3-13 of FIGS. 7.3-1 also has a second polymer layer such as layer 7.3-13-2. Layer 7.3-13-2, which may sometimes be referred to as a shroud trim or shroud, may have a ring shape that surrounds display 7.3-20. In the peripheral portion of member 7.3-13, layers 7.3-13-1 and 7.3-13-2 may be attached to each other using adhesive, press-fit connections, screws or other fasteners, and/or other attachment mechanisms.

[1400] Display cover layer 7.3-12CG and member 7.3-13 (e.g., layer 7.3-13-1) may overlap a forward-facing display such as display 7.3-20 (e.g., a flexible display panel formed from a pixel array based on organic light-emitting diodes or other display panel). Layer 7.3-13-1 may be formed from fully transparent polymer or partly transparent polymer that helps hide display 7.3-20 from view. Display cover layer 7.3-12CG may be formed from transparent polymer or glass (as examples).

[1401] Portions of display cover layer 7.3-12CG and member 7.3-13 such as edge portions of display cover layer 7.3-12CG and member 7.3-13 that surround display 7.3-20 may have curved cross-sectional profiles. As an example, edge portion E of cover layer 7.3-12CG and the underlying edge portion of member 7.3-13 may have inner and/or outer surfaces characterized by compound curvature (e.g., non-developable surfaces). The central portions of display cover layer 7.3-12CG and member 7.3-13 may have compound curvature and/or may have developable surfaces. In an illustrative arrangement, cover layer 7.3-12CG has inner and outer surfaces with compound curvature and member 7.3-13 has surfaces of compound curvature around the edges of device 7.3-10 (e.g., the portion of member 7.3-13 surrounding display 7.3-20) and has developable surfaces overlapping display 7.3-20. In this illustrative configuration, display 7.3-20 may be a flexible display panel that is bent into a curved shape (e.g., a curved shape following the curved face of a user) and that is characterized by inner and outer developable surfaces. The portion of member 7.3-13 overlapping display 7.3-20 may have corresponding inner and outer developable surfaces. Other arrangements for the shapes of display cover layer 7.3-12CG and member 7.3-13 may be used in device 7.3-10, if desired.

[1402] Device 7.3-10 may have one or more antennas. As an example, antenna 7.3-40 may be mounted in device 7.3-10 along the edge of display 7.3-20. As shown in FIGS. 7.3-1, antenna 7.3-

40 may, as an example, be mounted to the inner surface of dielectric member **7.3-13** under edge portion E of display cover layer **7.3-12CG**. During operation, antenna signals may pass through these overlapping dielectric structures.

[1403] Antenna **7.3-40** may be attached to the surface of member **7.3-13** (e.g., the inner surface of layer **7.3-13-2** in the example of FIGS. **7.3-1**) using adhesive **7.3-15**. The portion of the inner surface of member **7.3-13** to which antenna **7.3-40** is mounted in this way may have compound curvature. Antenna **7.3-40** may be formed from a flexible printed circuit with matching compound curvature.

[1404] Device **7.3-10** may include electrical components **7.3-36** (e.g., integrated circuits, sensors, control circuitry, light-emitting diodes, lasers, and other light-emitting devices, other control circuits and input-output devices, etc.). Components **7.3-36** may be mounted on printed circuits and/or other structures within device **7.3-10** (e.g., in interior region **7.3-42**).

[1405] To present a user with images for viewing from eye boxes such as eye box **7.3-34**, device **7.3-10** may include rear-facing displays in optical modules **7.3-16**. There may be, for example, a left rear-facing display in a left optical module **7.3-16** for presenting an image through a left lens to a user's left eye in a left eye box **7.3-34** and a right rear-facing display in right optical module **7.3-16** for presenting an image through a right lens to a user's right eye in a right eye box **7.3-34**.

[1406] The user's eyes are located in eye boxes **7.3-34** at rear R of device **7.3-10** when inwardly facing surface **7.3-18** of housing **7.3-12** rests against the outer surface of the user's face. On rear R, housing **7.3-12** may have cushioned structures (sometimes referred to as light seal structures) to enhance user comfort as surface **7.3-18** rests against the user's face. Device **7.3-10** may have forward-facing components such as forward-facing cameras and other sensors on front F that face outwardly away from the user. These components may generally be oriented in the +Y (forward) direction of FIGS. **7.3-1**.

[1407] During operation, device **7.3-10** may receive image data (e.g., image data for video, still images, etc.) and may present this information on the displays of optical modules **7.3-16**. Device **7.3-10** may also receive other data, control commands, user input, etc. Device **7.3-10** may transmit data to accessories and other electronic equipment. For example, image data from a forward-facing camera may be provided to an associated device, audio output may be provided to a device with speakers such as a headphone device, user input and sensor readings may be transmitted to remote equipment, etc.

[1408] Communications such as these may be supported using wired and/or wireless communications. In an illustrative configuration, components **7.3-36** may include wireless communications circuitry for supporting wireless communications between device **7.3-10** and remote wireless equipment (e.g., a cellular telephone, a wireless base station, a computer, headphones or other accessories, a remote control, peer devices, internet servers, and/or other equipment). Wireless communications may be supported using one or more antennas operating at one or more wireless communications frequencies (see, e.g., antenna **7.3-40** of FIGS. **7.3-1**). In an illustrative configuration, one or more antennas may be coupled to wireless transceiver circuitry. The wireless transceiver circuitry may include transmitter circuitry configured to transmit wireless communications signals using the antenna(s) and receiver circuitry configured to receive wireless communications signals using the antenna(s).

[1409] The wireless circuitry of device **7.3-10** may be formed from one or more integrated circuits, power amplifier circuitry, low-noise input amplifiers, passive RF components, one or more antennas, transmission lines, and other circuitry for handling RF wireless signals. The wireless circuitry may include radio-frequency transceiver circuitry for handling various radio-frequency communications bands. For example, the wireless circuitry of device **7.3-10** may include wireless local area network (WLAN) and wireless personal area network (WPAN) transceiver circuitry. This transceiver circuitry may handle 2.4 GHz and 5 GHz bands for WiFi® (IEEE 802.11) communications and other WLAN communications and the 2.4 GHz Bluetooth® communications

band or other WPAN bands and may sometimes be referred to herein as WLAN/WPAN transceiver circuitry or local transceiver circuitry.

[1410] The wireless circuitry of device **7.3-10** may use remote wireless circuitry such as cellular telephone transceiver circuitry for handling wireless communications in frequency ranges (communications bands) such as a cellular low band (LB) from 600 to 960 MHz, a cellular low-midband (LMB) from 1410 to 1510 MHz, a cellular midband (MB) from 1710 to 2170 MHz, a cellular high band (HB) from 2300 to 2700 MHz, a cellular ultra-high band (UHB) from 3300 to 5000 MHz, or other communications bands between 600 MHz and 5000 MHz. If desired, the cellular telephone transceiver circuitry may support 5G communications using a low band at 600-850 MHz, a mid-band at 2.5-3.7 GHz, and a high band at 25-39 GHz. Wireless communications may also be provided using other frequency ranges (e.g., frequencies above 100 MHz, above 1 GHz, 1-30 GHz, 100 Mhz-300 GHz, 24 GHz, less than 300 GHz, less than 100 GHz, 10-300 GHz or other mm-wave frequencies, and/or other suitable frequencies). WLAN/WPAN transceiver circuitry and/or cellular transceiver circuitry may handle voice data and non-voice data.

[1411] If desired, the antennas and other wireless circuitry of device **7.3-10** may include satellite navigation system circuitry such as Global Positioning System (GPS) receiver circuitry for receiving GPS signals at 1575 MHz or for handling other satellite positioning data (e.g., GLONASS signals at 1609 MHz). Satellite navigation system signals are received from a constellation of satellites orbiting the earth. Wireless circuitry in device **7.3-10** can include circuitry for other short-range (local) and long-range (remote) wireless links if desired. For example, wireless circuitry in device **7.3-10** may be provided to receive television and radio signals, paging signals, near field communications (NFC) signals at 13.56 MHz or other suitable NFC frequencies, ultrawideband (UWB) signals (e.g., UWB signals from 6-8.5 GHz, UWB signals from 3.5-9 GHz, etc.). Wireless circuitry in device **7.3-10** may also include antennas and transceiver for handling sensing applications (e.g., radar). If desired, antennas may be provided in arrays (e.g., phased antenna arrays) that support beam steering. These arrangements and other arrangements may be used in supporting wireless communications, wireless sensing, wireless location services, wireless power, and other wireless operations.

[1412] The wireless circuitry of device **7.3-10** may include antennas that are formed using any suitable antenna types. For example, the antennas of device **7.3-10** may include antennas with resonating elements that are formed from slot antenna structures, loop antenna structures, patch antenna structures, stacked patch antenna structures, antenna structures having parasitic elements, inverted-F antenna structures, planar inverted-F antenna structures, helical antenna structures, monopole antennas, dipole antenna structures, Yagi (Yagi-Uda) antenna structures, surface integrated waveguide structures, coils, hybrids of these designs, etc. If desired, one or more of the antennas may be cavity-backed antennas.

[1413] Different types of antennas may be used for different bands and combinations of bands. For example, one type of antenna may be used in forming a local wireless link antenna whereas another type of antenna is used in forming a remote wireless link antenna. If desired, space may be conserved within device **7.3-10** by using a single antenna to handle two or more different communications bands. For example, a single antenna in device **7.3-10** may be used to handle communications in a WiFi® or Bluetooth® communication band while also handling communications at one or more cellular telephone frequencies. In some configurations, some cellular telephone communications (e.g., low-band and mid-band communications) may be handled using a first antenna (e.g., an inverted-F antenna), whereas other communications (e.g., high-band cellular communications) may be handled using one or more phased antenna arrays (e.g., multiple linear patch antenna arrays each of which is mounted in a different orientation and each of which has a different angle of view so that a desired amount of angular coverage is achieved).

[1414] To provide antenna structures in device **7.3-10** with the ability to cover different frequencies of interest, one or more of the antennas of device **7.3-10** may be provided with circuitry such as

filter circuitry (e.g., one or more passive filters and/or one or more tunable filter circuits). Discrete components such as capacitors, inductors, and resistors may be incorporated into the filter circuitry. Capacitive structures, inductive structures, and resistive structures may also be formed from patterned metal structures (e.g., part of an antenna). If desired, antenna(s) in device **7.3-10** may be provided with adjustable circuits such as tunable components that tune the antenna over communications (frequency) bands of interest. The tunable components may be part of a tunable filter or tunable impedance matching network, may be part of an antenna resonating element, may span a gap between an antenna resonating element and antenna ground, etc.

[1415] Radio-frequency transmission line paths may be used to convey antenna signals between the radio-frequency transceiver circuitry of device **7.3-10** and the antenna(s) of device **7.3-10**. These paths may include one or more radio-frequency transmission lines (sometimes referred to herein as transmission lines). Radio-frequency transmission line paths may each include a positive signal conductor and a ground signal conductor. Transmission lines in device **7.3-10** may include coaxial cable transmission lines, stripline transmission lines, microstrip transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, transmission lines formed from waveguide structures (e.g., coplanar waveguides or grounded coplanar waveguides), combinations of these types of transmission lines and/or other transmission line structures.

[1416] If desired, matching networks may be used to help match impedances in the wireless circuitry of device **7.3-10**. A matching network may, for example, include components such as inductors, resistors, and capacitors configured to match the impedance of an antenna to the impedance of an associated radio-frequency transmission line path that is used in coupling the antenna to a transceiver. Matching network components may be provided as discrete components (e.g., surface mount technology components) or may be formed from housing structures, printed circuit board structures, traces on plastic supports, etc. Components such as these may also be used in forming antenna filter circuitry and may be tunable and/or fixed components.

[1417] Radio-frequency transmission line paths may be coupled to antenna feed structures associated with antennas in device **7.3-10**. As an example, an antenna in device **7.3-10** such as an inverted-F antenna, a planar inverted-F antenna, a patch antenna, a loop antenna, or other antenna may have an antenna feed with a positive antenna feed terminal and a ground antenna feed terminal. The positive antenna feed terminal may be coupled to an antenna resonating (radiating) element within the antenna. The ground antenna feed terminal may be coupled to an antenna ground in the antenna. The positive feed terminal may be coupled to a positive signal line in a transmission line and the ground feed terminal may be coupled to a ground signal line in the transmission line.

[1418] Other types of antenna feed arrangements may be used if desired. For example, an antenna may be fed using multiple feeds each coupled to a respective port of a transceiver over a corresponding transmission line. If desired, a given transmission line signal conductor may be coupled to multiple locations on an antenna and/or switches may be interposed within the paths between a transceiver and the feed terminals of an antenna.

[1419] FIGS. **7.3-2** is a diagram of illustrative wireless communications circuitry for device **7.3-10**. As shown in FIGS. **7.3-2**, the wireless circuitry includes radio-frequency transceiver **7.3-60**, which is coupled to antenna **7.3-40** by transmission line **7.3-62**. Antenna **7.3-40** may have an antenna resonating element **52** (sometimes referred to as an antenna resonating structure or antenna resonator) and antenna ground **7.3-50**. Antenna resonating element **52** may be formed from any suitable antenna resonating element structures. In the example of FIGS. **7.3-2**, antenna resonating element **52** is an inverted-F antenna resonating element having resonating element arm **7.3-56**, which is coupled to ground **7.3-50** by return path **7.3-54** and which is fed using antenna feed **7.3-58**. Feed **7.3-58** has positive and ground feed terminals coupled respectively to positive and ground signal lines in transmission line **7.3-62**. Conductive structures making up antenna **7.3-40** may be formed from thin-film metal traces on printed circuits (e.g., flexible printed circuits formed from

sheets of polyimide or other flexible polymer substrates). If desired, the conductive structures making up antenna **7.3-40** (e.g., ground structures for antenna **7.3-40**) may include conductive structural members such as portions of a housing for device **7.3-10** (e.g., a metal chassis and/or other internal and/or external frame structures, metal housing walls, metal component support brackets, and/or other conductive housing structures), and/or other structures in device **7.3-10** that are formed from metal and/or other conductive material.

[1420] Antennas may be mounted within device **7.3-10** using mounting brackets, using biasing structures that press antenna components against housing structures, using adhesive, using screws and other fasteners, using press-fit connections, using solder, welds, conductive adhesive, and/or other conductive attachment mechanisms, using one or more frames, carriers, and/or other internal support structures, and/or other mounting arrangements. In an illustrative configuration, flexible printed circuit antenna **7.3-40** has compound curvature and is attached to an overlapping dielectric member such as display cover layer **7.3-12CG** and/or member **7.3-13** that has an opposing surface of matching compound curvature. By matching the compound curvature of the substrate of antenna **7.3-40** to the compound curvature of an associated overlapping dielectric layer, antenna **7.3-40** may be configured to fit within the potentially tight confines of device **7.3-10** without adversely affecting the shape and appearance of device **7.3-10**. As an example, by matching the compound curvature of the substrate of antenna **7.3-40** to the compound curvature of an overlapping dielectric structure such as member **7.3-13**, antenna **7.3-40** can be attached to the inner surface or outer surface of member **7.3-13** with adhesive. Display cover layer **7.3-12CG** may then be mounted on device **7.3-10** so that edge portion E of layer **7.3-12CG** overlaps member **7.3-13** and antenna **7.3-40**.

[1421] Consider, as an example, the illustrative antenna structures of FIGS. **7.3-3**. As shown in FIGS. **7.3-3**, antenna **7.3-40** may be formed from antenna substrate **7.3-64**. Substrate **7.3-64** may be formed from a flexible printed circuit with a surface of compound curvature (sometimes referred to as a non-developable surface). Antenna substrate **7.3-64** contains metal traces such as metal trace **7.3-66** (e.g., a patterned thin-film metal layer). In an illustrative configuration, substrate **7.3-64** has an upper layer (e.g., an upper polyimide layer or other sheet of polymer) and a lower layer (e.g., a lower polyimide layer or other sheet of polymer) and trace **7.3-66** is formed from a patterned thin-film metal layer that is between the upper and lower layers. Trace **7.3-66** is patterned to form antenna resonating element **52** (FIGS. **7.3-2**) and/or other antenna structures. Antenna **7.3-40** of FIGS. **7.3-3** is formed from a planar sheet of printed circuit substrate material that was contoured in a contouring tool to produce a desired three-dimensional shape with compound curvature. At least some portions of the curved surface of substrate **7.3-64** may be characterized by a radius of curvature R of 4 mm to 250 mm, 8 mm to 200 mm, 10 mm to 150 mm, at least 5 mm, at least 12 mm, at least 16 mm, at least 20 mm, at least 30 mm, less than 200 mm, less than 100 mm, less than 75 mm, less than 55 mm, less than 35 mm, and/or other suitable amount of curvature.

[1422] After forming a flexible printed circuit antenna with compound curvature of the type shown in FIGS. **7.3-3**, this compound curvature flexible printed circuit antenna may be attached to the surface of a dielectric support structure in device **7.3-10**. In an illustrative arrangement, the compound curvature flexible printed circuit antenna is attached to the inner surface of dielectric member **7.3-13** using a layer of adhesive. FIGS. **7.3-4** is a cross-sectional side view of an illustrative vacuum lamination tool that may be used in attaching antenna **7.3-40** to member **7.3-13**. As shown in FIGS. **7.3-4**, tool **7.3-70** may have movable upper and lower dies such as upper die **7.3-82** with concave surface **7.3-72** and lower die **7.3-74** with convex surface **7.3-76**.

[1423] Surfaces **7.3-72** and **7.3-76** may be characterized by compound curvature (e.g., compound curvature that matches the compound curvature of the inner and outer surfaces of member **7.3-13** and that matches the compound curvature of the inner and outer surfaces of flexible printed circuit antenna **7.3-40**). Prior to lamination, a layer of adhesive such as adhesive **7.3-15** may be suspended between member **7.3-13** and antenna **7.3-40**. During lamination, tool **7.3-70** may use vacuum

enclosure 7.3-80 to produce a vacuum while member 30 is pressed against antenna 7.3-40 by moving die 7.3-82 towards die 7.3-74. While pressure is applied between member 7.3-13 and antenna 7.3-40 in this way, dies 7.3-82 and 7.3-74 may optionally apply heat to facilitate lamination. The presence of vacuum helps prevent air bubbles from forming as adhesive 7.3-15 is compressed between member 30 and antenna 7.3-40.

[1424] FIGS. 7.3-5 is a cross-sectional side view of member 7.3-13 (e.g., layer 7.3-13-2 of FIGS. 7.3-1 or other dielectric antenna support structure) following lamination in tool 7.3-70 to attach antenna 7.3-40 to member 7.3-13 with adhesive. In general, antenna 7.3-40 may be attached to an inner or outer surface of a supporting member and this attachment surface may have convex or concave curvature. In the example of FIGS. 7.3-5, member 7.3-13 has an inwardly facing concave surface of compound curvature and the outwardly facing surface of antenna 7.3-40 has matching compound curvature.

[1425] After antenna 7.3-40 is attached to a shroud or other dielectric member (e.g., member 7.3-13 of FIGS. 7.3-1 or other member) using adhesive 7.3-15, the shroud or other dielectric member may be attached to other portions of housing 7.3-12 (e.g., using screws or other fasteners, using adhesive, etc.). In arrangements in which member 7.3-13 is separate from cover layer 7.3-12CG, the attachment of antenna 7.3-40 to member 7.3-13 may help preserve the ability of cover layer 7.3-12CG to be removed (e.g., to permit rework or repair of device 7.3-10).

[1426] FIGS. 7.3-6 is a perspective view of antenna 7.3-40 on member 7.3-13 taken from the outside of device 7.3-10 with cover layer 7.3-12CG removed. As shown in the example of FIGS. 7.3-6, antenna 7.3-40 may be mounted to the underside (inner surface) of member 7.3-13. The outwardly facing surface of member 7.3-13 in FIGS. 7.3-6 is convex. The opposing inwardly facing surface of member 7.3-13 in FIGS. 7.3-6 is concave (e.g., the surface of member 7.3-13 on the far side of member 7.3-13 of FIGS. 7.3-6 is concave). Antenna 7.3-40 may have an outwardly facing convex surface that is attached to the concave inwardly facing surface of member 7.3-13.

[1427] One or more metal structures in device 7.3-10 such as metal structure 7.3-90 (e.g., a metal chassis or other metal housing structure) may serve as antenna ground 7.3-50 of FIG. 7.3-2. Member 7.3-13 may have openings 7.3-92 through which leg portions or other protruding portions of antenna 7.3-40 may pass. In the example of FIGS. 7.3-6, protruding portion 7.3-40-1 of antenna 7.3-40 has a return path metal trace (forming return path 7.3-54 of FIGS. 7.3-2) that is shorted to metal structure 7.3-90 using metal fastener 7.3-94. Protruding portion 7.3-40-2 of antenna 7.3-40 may include a metal trace forming a positive feed terminal. A cable or other transmission line (see, e.g., transmission line 7.3-62 of FIGS. 7.3-2) may be coupled to connector 7.3-96. Connector 7.3-96 may have a positive terminal coupled to the positive feed terminal and may have a negative terminal that is shorted to metal structure 7.3-90 (e.g., via metal fastener 7.3-98). Conductive adhesive, solder, welded connections, and/or other conductive connections may be used in attaching the metal trace of antenna 7.3-40 to metal structure 7.3-80 and connector 7.3-96, if desired. The use of fasteners 7.3-94 and 7.3-98 (e.g., screws) is illustrative. Following installation of member 7.3-13 and antenna 7.3-40 into device 7.3-10 (e.g., by attaching member 7.3-13 to housing 7.3-12 and attaching antenna 7.3-40 to structure 7.3-90), cover layer 7.3-12CG may be mounted to the front of housing 7.3-12, thereby covering member 7.3-13 and antenna 7.3-40 as shown in FIGS. 7.3-1.

VIII: Bent MLB

[1428] FIG. 8-0 illustrates a view of an example of an HMD device 8-100. The HMD device 8-100 can include a logic board system 8-102 including one or more printed circuit boards, processors, other computing components, memory components, circuitry, and so forth mounted to one or more components of the HMD 8-100.

[1429] FIG. 8-1 illustrates a plan view of an example of a logic board 8-102, which can include a main logic board of the HMD 8-100 shown in FIG. 8-0. In at least one example, the logic board 8-102 includes a first portion 8-104 and a second portion 8-106 joined together at a transition portion

8-108. As shown in the top view of FIG. **8-23**, in at least one example, the first portion **8-104** of the MLB **8-102** can be disposed at an angle θ relative to the second portion **8-106** of the MLB **8-102**. In at least one example, the angle θ can disposed the first and second portions **8-104**, **8-106** of the MLB **8-102** to accommodate the curvature of a user's face. For example, the lateral (left-to-right) direction of the user's face is curved and the HMD **8-100** can be generally curved to accommodate the user's face. This curvature can form a thin profile against the user's face and minimize the moment arm and thus the torque applied to the user's face from the HMD **8-100**, including components within the HMD **8-100**.

[1430] In order to accommodate the curvature of the user's face, not only is the outside or external surfaces of the HMD **8-100** curved, as shown in FIG. **8-0**, but internal components of the HMD **8-100** can be similarly curved to fit tightly and compactly within the HMD **8-100** to minimize bulk and size of the device. For example, in at least one example, the angle θ of the first portion **8-104** relative to the second portion **8-106** can be between about 5-degrees and about 60-degrees to form a bent MLB **8-102**. In one example, the angle θ can be between about 5-degrees and about 60-degrees. In one example, the angle θ can be between about 10-degrees and about 50-degrees. In one example, the angle θ can be between about 12-degrees and about 45-degrees. In one example, the angle θ can be about 15-degrees.

[1431] As shown in the close up top view of FIG. **8-24**, the transition portion **8-108**, which can be referred to as the bend portion **8-108**, can be thinner than the first and second portion **8-104**, **8-106**, as shown. In at least one example, the MLB **8-102** can be manufactured and assembled, including the application of adhesive tapes and other components, while the MLB **8-102** is not bent. That is, in its resting state, the MLB **8-102** can be planar such that the first portion **8-104** is not disposed at an angle relative to the second portion **8-106**. Then, when the MLB **8-102** is disposed within the HMD **8-100** during assembly/manufacture of the HMD **8-100**, the MLB **8-102** can be bent at the transition/bend portion **8-108** as shown and fixed in position within the HMD **8-100**.

[1432] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **8-1-8-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **8-4-8-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **8-4-8-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **8-1-8-3**.

[1433] FIG. **8-4** illustrates a cross-sectional view of another example of a MLB **8-202** including a first portion **8-204** and a second portion **8-206** with a transition portion **8-208** disposed between the first portion **8-204** and the second portion **8-206**. In the illustrated example of FIG. **8-4**, the MLB **8-202** is not bent at the transition portion **8-208**, for example when the MLB **8-202** has not yet been bent for assembly within the HMD **8-100**. In one example, the transition portion **8-208** is thinner than the first and second portion **8-204**, **8-206** such that when the first and second portions **8-204**, **8-206** are urged toward each other, the MLB **8-202** naturally bends at the transition portion **8-208**.

[1434] In at least one example, the transition portion **8-208** includes a number of layers **8-201** including conductive layers, such as copper conductive layers, and insulation layers, including pre-preg insulation layers disposed between the conductive layers. The thicknesses of the layers for the first, second, and transition portions **8-204**, **8-206**, **8-208** of the MLB **8-202** can be between about 0.006 mm and about 0.053 mm. The number of layers, types of layers, and thicknesses of each of those layers can vary in other examples.

[1435] In at least one example, the transition portion **8-208** can include an upper combined coverlay-adhesive layer **8-203**. In at least one example, the transition portion **8-208** can include a lower sliver shield layer **8-205** below a lower combined coverlay-adhesive layer **8-207**. The copper and pre-preg layers can be disposed between the coverlay-adhesive layers **8-203**, **8-207**. The MLB **8-202** can include a core layer **8-209** disposed between adjacent copper layers and extending from

the first portion **8-204**, through the transition portion **8-208**, on through the second portion **8-206** as a single core layer **8-209**. Likewise, any of the other layers noted herein can extend from the first portion **8-204**, through the transition portion **8-208**, on through the second portion **8-206** as single continuous layers that cross the transition portion **8-208**. In one example, the transition portion **8-208** is not a separate MLB portion disposed between separate, unconnected other portions. Rather, the transition portion **8-208** can be an integral part of the MLB **8-202** and include the same or similar layers of material electronically connecting or otherwise connecting the first and second portion **8-204**, **8-206** thereof.

[1436] In at least one example, one or more core and/or conductive material layers extends continuously from the first portion **8-204** to the second portion **8-206** through the transition portion **8-208**. In at least one example, one or more of these layers can form a thermal/heat connection between the first portion **8-204** and the second portion **8-206**. In at least one example, there is no need for connectors between the first portion **8-204** and the second portion **8-208**. In at least one example, the MLB **8-202** can include aluminum plating for solder-ability.

[1437] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **8-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **8-1-8-3** and **8-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **8-1-8-3** and **8-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **8-4**.

[1438] FIG. **8-5** illustrates a top perspective view of a subassembly of the HMD **8-100** including an MLB **8-302** and first and second fans **8-310a**, **8-310b** secured to the MLB **8-302**. In at least one example the first fan **8-310a** is secured to or adjacent the first portion **8-304** of the MLB **8-302** and the second fan **8-310b** is secured to or adjacent the second portion **8-306** of the MLB **8-302**. In such an example, the fans **8-310a-b** can be disposed at the same angle relative to one another as the first and second portions **8-304**, **8-306** of the MLB **8-302** are disposed relative to one another. In this way, the MLB **8-302** as well as the fans **310a-b** can be disposed compactly within the curved HMD **8-100** device to minimize device volume.

[1439] In addition, the first and second fans **8-310a-b** can be secured to the MLB **8-302** such that thermally conductive housings of the fans **8-310a-b** can be thermally coupled to one or more heat-generating components of the MLB **8-302**. In one example, the fans **8-310a-b** can be secured to the first and second portions **8-304**, **8-306** of the MLB **8-302** such that the housings of the fans **8-310a-b** form at least part of one or more electromagnetic interference shields for components of the MLB **8-302**. More details regarding the EMI shielding and thermal conductivity of the fans **8-310a-b** with relation to the MLB are given elsewhere herein.

[1440] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **8-5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **8-1-8-4** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **8-1-8-5** and **8-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **8-5**.

IX: Thermals

[1441] FIGS. **9.0-1** illustrates a view of an HMD **100** including a thermal management system **102**. The thermal management system **102** is described in more detail here in section IX.

9.1: Air Deflector for a Cooling System in a Head-Mounted Device

[1442] Head-mounted devices are an attractive technology for providing an immersive user experience. For example, head-mounted devices are gaining increased popularity for providing VR, AR, and MR experiences for applications such as gaming, movies, or simulations for professional

training, among other potential applications.

[1443] Head-mounted devices can employ a wearable device housing that is secured to a user's head, and various electronic components within the housing, such as displays, integrated circuits, memory, audio devices, or electronic circuitry. As with other electronic devices, head-mounted devices can employ a cooling system based on circulation of air to maintain electronic components at desirable operating temperatures. The cooling system can also be used to cool the user's face from heat build-up inside the head-mounted device.

[1444] Maintaining efficient operation without unduly detracting from the user experience is a challenging task for head-mounted devices. The shape of the head-mounted device or layout of internal components can lead to a tortuous flow path for the cooling system. The proximity of the air flow path to the user's head can create undesired effects that detract from the user experience, such as excessive noise that interferes with the audio of the device in a noticeable manner. Some head-mounted devices may employ movable components that can interrupt the air flow path, such as adjustable optics that can be moved to account for a given user's interpupillary distance (IPD). IPD is defined as the distance between the centers of the pupils of a user's eyes. This adjustability can in turn make it difficult to design a cooling system in a given device that is suitable for different users.

[1445] According to some embodiments disclosed herein, a cooling system for a head-mounted device may employ an air deflector designed to affect a flow of air within the head-mounted device. The air deflector may be positioned in an air flow path extending through the housing of the head-mounted device, and can be designed to reduce the turbulence of air in the cooling system. For example, the air deflector can be positioned between a surface of an internal component and an incoming stream of air, at a reduced angle relative to the surface of the internal component so as to create smooth or more laminar flow over or across the component. The air deflector can be mounted to a movable component, such as an adjustable display assembly, so as to affect the flow of air as the moveable component is adjusted for particular users in a manner that results in a partial occlusion of the air flow path by the movable component. The air deflector can be configured to pivot or otherwise move to account for changes in the incident angle of air resulting from changes in the position of movable components. The air deflector can include or be coupled to additional thermal structures to enhance heat transfer affects resulting from the flow of air over the air deflector. For example, the air deflector can include an integrated heat sink and/or be coupled to heat-generating components via a thermally conductive interface material to enhance dissipation of heat from such components.

[1446] These and other embodiments are discussed below with reference to FIGS. **9.1-1** through FIGS. **9.1-9**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting.

[1447] FIGS. **9.1-1** illustrates an example of a head-mounted device **9.1-100** secured to a head **9.1-20** of a user **9.1-10**. As seen in FIGS. **9.1-1**, the head-mounted device **9.1-100** can include a housing **9.1-110** that is securable to the user's head **9.1-20** via a securement element **9.1-150**. The securement element **9.1-150** can include a band, a strap, a rim, temples of a glasses frame, or any other suitable mechanism that serves to secure and retain the housing **9.1-110** on the head **9.1-20** of the user **9.1-10**. The securement element **9.1-150** can be an integral part of the housing **9.1-110** or be implemented as a separate component attached thereto. The housing **9.1-110** can further include or be coupled to one or more nose pads that serve to rest the housing **9.1-110** on the nose of the user **9.1-10**.

[1448] The housing **9.1-110** can enclose and support various functional components therein, such as integrated circuits, memory devices, processors, electronic circuitry, input/output devices, or other electronic components. In FIGS. **9.1-1**, housing **9.1-110** is shown as containing a display **9.1-120**, a controller **9.1-130**, and an air circulation device **9.1-140** therein. The display **9.1-120** can be

positioned in front of the eyes of the user **9.1-10** to provide information within the user's field of view. The air circulation device **9.1-140** can urge air through the housing **9.1-110** and over components such as the display **9.1-120** to cool such components. The controller **9.1-130** can be configured to control operation of one or more components, such as the display **9.1-120** and/or air circulation device **9.1-140**.

[1449] The display **9.1-120** can transmit light from a physical environment for viewing by the user **9.1-10**. For example, the display **9.1-120** can include optical elements, such as lenses for vision correction. The display **9.1-120** can be configured to present information in addition to (e.g., overlaid with) the physical environment viewed by the user. Alternatively, the display **9.1-120** can be configured to provide information to the exclusion of the physical environment. In either case, the display **9.1-120** can be configured to present graphics to, for example, present a computer-generated reality environment to the user **9.1-10**.

[1450] A physical environment refers to a physical world that people can sense and/or interact with without aid of electronic systems. Physical environments, such as a physical park, include physical articles, such as physical trees, physical buildings, and physical people. People can directly sense and/or interact with the physical environment, such as through sight, touch, hearing, taste, and smell.

[1451] In contrast, a computer-generated reality (CGR) environment refers to a wholly or partially simulated environment that people sense and/or interact with via an electronic system. In CGR, a subset of a person's physical motions, or representations thereof, are tracked, and, in response, one or more characteristics of one or more virtual objects simulated in the CGR environment are adjusted in a manner that comports with at least one law of physics. For example, a CGR system may detect a person's head turning and, in response, adjust graphical content and an acoustic field presented to the person in a manner similar to how such views and sounds would change in a physical environment. In some situations, (e.g., for accessibility reasons), adjustments to characteristic(s) of virtual object(s) in a CGR environment may be made in response to representations of physical motions (e.g., vocal commands).

[1452] A person may sense and/or interact with a CGR object using any one of their senses, including sight, sound, touch, taste, and smell. For example, a person may sense and/or interact with audio objects that create 3D or spatial audio environment that provides the perception of point audio sources in 3D space. In another example, audio objects may enable audio transparency, which selectively incorporates ambient sounds from the physical environment with or without computer-generated audio. In some CGR environments, a person may sense and/or interact only with audio objects.

[1453] Examples of CGR include virtual reality and mixed reality.

[1454] A virtual reality (VR) environment refers to a simulated environment that is designed to be based entirely on computer-generated sensory inputs for one or more senses. A VR environment comprises a plurality of virtual objects with which a person may sense and/or interact. For example, computer-generated imagery of trees, buildings, and avatars representing people are examples of virtual objects. A person may sense and/or interact with virtual objects in the VR environment through a simulation of the person's presence within the computer-generated environment, and/or through a simulation of a subset of the person's physical movements within the computer-generated environment.

[1455] In contrast to a VR environment, which is designed to be based entirely on computer-generated sensory inputs, a mixed reality (MR) environment refers to a simulated environment that is designed to incorporate sensory inputs from the physical environment, or a representation thereof, in addition to including computer-generated sensory inputs (e.g., virtual objects). On a virtuality continuum, a mixed reality environment is anywhere between, but not including, a wholly physical environment at one end and virtual reality environment at the other end.

[1456] In some MR environments, computer-generated sensory inputs may respond to changes in

sensory inputs from the physical environment. Also, some electronic systems for presenting an MR environment may track location and/or orientation with respect to the physical environment to enable virtual objects to interact with real objects (that is, physical articles from the physical environment or representations thereof). For example, a system may account for movements so that a virtual tree appears stationery with respect to the physical ground.

[1457] Different forms of head-mounted devices that enable a person to sense and/or interact with various CGR environments. Examples include smart glasses, helmets, visors, or goggles. A head-mounted device may have one or more speaker(s) and an integrated opaque display. Alternatively, a head-mounted system may be configured to accept an external opaque display (e.g., a smartphone). The head-mounted device may incorporate one or more imaging sensors to capture images or video of the physical environment, and/or one or more microphones to capture audio of the physical environment. Rather than an opaque display, a head-mounted device may have a transparent or translucent display. The transparent or translucent display may have a medium through which light representative of images is directed to a person's eyes. The display may utilize digital light projection, OLEDs, LEDs, uLEDs, liquid crystal on silicon, laser scanning light source, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In one embodiment, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface.

[1458] FIGS. 9.1-2 shows an example of the head-mounted device 9.1-100 in front view. As seen in FIGS. 9.1-2, the display 9.1-120 (FIGS. 9.1-1) can include a first display assembly 9.1-121a and a second display assembly 9.1-121b, which collectively form a pair of display assemblies corresponding to the two eyes of a user. Each of the display assemblies may include any appropriate combination of electronic and optical elements to present graphical information to the user. For example, each display assembly may include a display layer having an array of electronically controlled pixels that can provide a visual output. The display assembly may further include optical elements, such as lenses, mirrors, etc., and/or a gaze tracking device, to facilitate generation of an enhanced computer generated reality that is responsive to a gaze and/or pose of the user.

[1459] The pair of display assemblies can be mounted to the housing 9.1-110 and separated by a distance 9.1-215. The distance 9.1-215 between the pair of display assemblies can be designed to correspond to the IPD of a user. The distance 9.1-215 can be adjustable to account for different IPDs of different users that may wear the head-mounted device 9.1-100. For example, either or both of the display assemblies may be movably mounted to the housing 9.1-110 to permit the display assemblies to move or translate laterally to make the distance 9.1-215 larger or smaller. Any type of manual or automatic mechanism may be used to permit the distance 9.1-215 between the display assemblies to be an adjustable distance. For example, the display assemblies can be mounted to the housing via slidable tracks or guides that permit manual or electronically actuated movement of one or more of the display assemblies to adjust the distance 9.1-215.

[1460] As seen in FIGS. 9.1-2, the air circulation device 9.1-140 can be positioned in or otherwise mounted to the housing 9.1-110 so as to urge a flow of air through an interior space 9.1-225 of the housing 9.1-110. The housing 9.1-110 can include a port that permits fluid communication between the interior space 9.1-225 and an environment external to the housing 9.1-110, to create a flow path of air in the housing 9.1-110. In FIGS. 9.1-2, the housing is shown with a pair of inlet ports 9.1-240 at a bottom side thereof, and an outlet port 9.1-250 at a top side thereof, which creates air flow paths 9.1-275 extending from the inlet ports 9.1-240 to the outlet port 9.1-250. Each of the ports can include a vent, screen, hole, porous membrane, and/or other fluidic opening that permits fluid communication thereacross. However, it is contemplated that the housing 9.1-110 can generally

include any suitable number inlet ports and outlet ports in any suitable locations with respect to the housing to permit a flow of air therein. The air circulation device **9.1-140** can be implemented as a fan that is configured to draw air into the inlet port(s) **9.1-240** and urge air out of the outlet port **9.1-250**. However, any suitable number of fans or other air circulation devices can be included to urge movement of air.

[1461] The air flow path **9.1-275** can extend over or across components such as heat-generating electronic components mounted within the housing. For example, the pair of display assemblies may include heat-generating display layers, and the air circulation device **9.1-140** may be configured to generate a flow of air so that the air flow path **9.1-275** extends over each of the display assemblies **9.1-121a** and **9.1-121b** to cool the heat-generating layers by dissipating heat therefrom. Alternatively, or in combination, the air circulation device **9.1-140** may be configured to circulate air over other electronic components, such as integrated circuit chips, other input/output devices, or the like, or across the user's face.

[1462] FIGS. **9.1-3** shows an example of a display assembly **9.1-121**, and an air flow path **9.1-275** extending over surfaces of the display assembly. The display assembly **9.1-121** can be one of a pair of display assemblies like in the example shown in FIGS. **9.1-2**, where each of the first and second display assemblies **9.1-121a** and **9.1-121b** can be configured similarly to the display assembly **9.1-121**. As seen in FIGS. **9.1-3**, the display assembly **9.1-121** can include a display layer **9.1-314**, a heat sink **9.1-342**, a circuit board **9.1-370**, one or more components **9.1-386** on the circuit board, and an enclosure **9.1-393** that serves to enclose and support the foregoing components.

[1463] As shown in FIGS. **9.1-3**, the display assembly **9.1-121** can have a front side for viewing images and a back side opposite the front side. The display layer **9.1-314** can include operative components of the display that form images capable of being viewed by the user from the front side thereof. The display layer **9.1-314** can, for example, include any suitable operational display panel having an array of electronically controlled pixels that can provide a visual output, such as an OLED, uLED, or LCD panel. The display assembly **9.1-121** can further include other optional components, which can support specialized display functions for providing an immersive head-mounted display. For example, the display assembly **9.1-121** can include as gaze-tracking devices or eye trackers (e.g., positioned beside the display layer **9.1-314**), and/or optics (e.g., positioned in front of the display layer). The optics can be configured to help optically adjust and correctly project the image based content being displayed by the display layer **9.1-314** for close up viewing. The optics can include one or more lenses, mirrors, or other optical elements.

[1464] In the example shown in FIGS. **9.1-3**, the air flow path **9.1-275** passes over the display assembly **9.1-121** across a back side thereof, so as to dissipate heat generated from the display layer **9.1-314** through the back side. To facilitate dissipation of heat, a heat sink **9.1-342** can be positioned behind the back side of the display layer **9.1-314**. The heat sink **9.1-342** can include a plurality of fins **9.1-377** positioned in the air flow path **9.1-275** so as to increase a surface area of the back surface that is exposed to the flow of air. The heat sink **9.1-342** can be thermally coupled to the back side or back surface of the display layer **9.1-314** via a thermal interface **9.1-361**, such as a thermally conductive adhesive or other appropriate thermally conductive material, to enhance heat transfer (e.g., conduction) from the display layer through the heat sink and to the stream of air.

[1465] As seen in FIGS. **9.1-3**, the display assembly **9.1-121** can further include other structures, such as a circuit board **9.1-370** (e.g., a flexible or rigid printed circuit board) on the back side of the display assembly. The circuit board **9.1-370** can have one or more components **9.1-386** mounted thereon. The components **9.1-386** can be, for example, passive or active electronic components surface mounted to the circuit board **9.1-370**, such as integrated circuit chips, resistors, capacitors, or other structures that can protrude from the surface of the circuit board **9.1-370**.

[1466] The component(s) **9.1-386**, and/or or other structures of the display assembly, can partially impede or obstruct the free flow of air and have a tendency to increase a turbulence of the air in the flow path. For example, FIGS. **9.1-3** shows an example in which increased impedance caused by

the presence of components **9.1-386** causes more turbulent flow, which can degrade the efficiency or user experience as described above.

[1467] FIGS. **9.1-4** shows another example of the display assembly **9.1-121**. The example shown in FIGS. **9.1-4** employs a similar structure to FIGS. **9.1-3**, but additionally includes an air deflector **9.1-400** positioned in the air flow path **9.1-275**. The air deflector **9.1-400** is a structure that can be mounted to surfaces within the head-mounted device to reduce a turbulence of air passing through the head-mounted device and across the air deflector. The air deflector **9.1-400** can have a surface designed to create less turbulent, more laminar flow for air that is incident on the surface. For example, the air deflector **9.1-400** can provide a smoother surface or lower angle with respect to an incoming stream of air generated by the air circulation device **9.1-140** (e.g., FIGS. **9.1-2**), compared to structures in the head-mounted device that the incoming stream of air would otherwise contact if the air deflector were not present. The air deflector **9.1-400** can be a rigid component made of any appropriate material, such as plastic, ceramic, or metal. In some embodiments, the air deflector **9.1-400** can be configured as a dedicated wall structure that is mounted to internal structures or inserted into an interior space of the housing solely to affect the properties of the flow of air incident through the interior space, without providing other mechanical or electrical functions.

[1468] In the example shown in FIGS. **9.1-4**, the air deflector **9.1-400** is mounted to the back side of the display assembly **9.1-121**. The air deflector **9.1-400** is mounted on and attached to the circuit board **9.1-370**, and extends at least partially over the component(s) **9.1-386** so as to at least partially shield the components **9.1-386** from the incoming air in the air flow path **9.1-275**. The air deflector **9.1-400** can, for example, be positioned down-stream from the heat sink **9.1-342** with respect to the air flow path **9.1-275**. Compared to the surfaces of the components **9.1-386** in the absence of the air deflector **9.1-400**, the surface of the air deflector that is in the flow path and positioned to receive an incident stream of air can have a smoother surface with fewer bends or steps. Accordingly, the air deflector **9.1-400** can be configured to make the air flow path **9.1-275** less tortuous.

[1469] Although the air deflector **9.1-400** is shown mounted to the circuit board **9.1-370** on the back side of the display assembly **9.1-121**, it is contemplated that the air deflector **9.1-400** can be mounted in any other desired location within the housing of the head-mounted device in which reduced turbulence is desired. For example, the air deflector **9.1-400** can be mounted to the heat-sink or another surface on the back side of the display assembly **9.1-121**, another non-back side surface of the display assembly, or another internal component within the housing of the head-mounted device.

[1470] FIGS. **9.1-5** illustrates another example of the air deflector **9.1-400**. In the example shown in FIGS. **9.1-5**, the air deflector **9.1-400** is configured similar to the example shown in FIGS. **9.1-4**, but also includes an integral heat sink so that the air deflector **9.1-400** can further dissipate heat from the components **9.1-386** that are shielded by the air deflector **9.1-400**, where such components can be heat-generating electronic components. The surface of the air deflector **9.1-400** that receives the incident air can include multiple fins **9.1-477**, which increase a surface area of the surface receiving incident air. To maintain sufficiently laminar flow, the fins **9.1-477** can, for example, be configured as longitudinal fins that extend in the direction of air flow, or as a series of aligned pins that are arranged in rows extending along the direction of air flow, among other possible structural arrangements. To facilitate heat transfer, the air deflector **9.1-400** can be made of a material having a sufficiently high thermal conductivity, such as copper or aluminum. The air deflector **9.1-400** can be coupled to one or several of the components **9.1-386**. To further enhance the ability of the air deflector **9.1-400** to dissipate heat, the air deflector **9.1-400** can be thermally coupled to such components via a thermally conductive interface **9.1-461**, such as a conductive adhesive or other appropriate thermally conductive material.

[1471] FIGS. **9.1-6** illustrates another example of the air deflector **9.1-400**. The air deflector **9.1-**

400 as shown in FIGS. **9.1-6** can be configured similar to the examples of FIGS. **9.1-4** or FIGS. **9.1-5**, except that in FIGS. **9.1-6**, the air deflector **9.1-400** is movably mounted to a surface (in this case, a surface of the circuit board **9.1-370**), rather than non-movably or fixedly mounted to the surface like in the previous examples. The movable mounting can permit the air deflector **9.1-400** to have an adjustable angle with respect to the incoming stream of air in the air flow path **9.1-275**. This can be useful to, for example, allow the adjustable angle to be optimized for reduced turbulence in various positions of the air deflector when the position of the air deflector with respect to the flow path or housing is otherwise moved. When the air deflector **9.1-400** is mounted to a display assembly that is movable adjust the distance **9.1-215** (e.g., FIGS. **9.1-2**), the air deflector **9.1-400** can be configured to compensate for its changed position by also moving with respect to the display assembly. For example, the air deflector **9.1-400** can be configured to move or rotate relative to the display assembly in response to or otherwise in accordance with movement of the display assembly relative to the housing. The movement of the air deflector **9.1-400** can be achieved using, for example, a piezo electric actuator or other actuator, and/or a mechanical linkage that synchronizes movement of the display assembly to rotation of the air deflector **9.1-400**. In the example shown in FIGS. **9.1-6**, the air deflector **9.1-400** is pivotally mounted to the surface of the display assembly **9.1-121**. The actuator can be configured to rotate the air deflector **9.1-400** about pivot point **9.1-603**, based on movement of the display assembly **9.1-121** relative to the housing or based on changes to the distance **9.1-215** between the pair of display assemblies, so as to adjust an angle of incidence of air onto the surface of the air deflector **9.1-400** to account for a new position of the air deflector with respect to the flow path extending through the housing.

[1472] FIGS. **9.1-7** through **9.1-8** show an example of how an air deflector **9.1-400** can reduce turbulence of air in a head-mounted device. FIGS. **9.1-7** through **9.1-8** show examples of arrangement without and with an air deflector, respectively.

[1473] FIGS. **9.1-7** shows an arrangement having a component **9.1-386** mounted in a flow path and positioned to receive an incoming stream **9.1-735** of air in the flow path (e.g., similar to FIGS. **9.1-3**). The component **9.1-386** has a surface **9.1-759** positioned in the flow path to receive the incoming stream **9.1-735** of air thereon. The incoming stream **9.1-735** is incident on the surface **9.1-759**, and forms an angle $9.1-\theta$ with respect to the surface **9.1-759**. In this example, the angle θ between the incoming stream **9.1-735** and the surface of the component **9.1-386** is approximately 90 degrees. Stated another way, the angle of incidence of the incoming stream **9.1-735** is approximately zero, wherein the angle of incidence is defined by the angle between the incoming stream and the normal to the incident surface. The large angle between the incoming stream and the incidence surface, or equivalently the low incident angle, causes a dramatic change in the current of air that tends to create turbulent patterns as the incoming stream impinges on the component, and then continues to flow along a tortuous flow path around the component **9.1-386**.

[1474] FIGS. **9.1-8** shows an identical arrangement to FIGS. **9.1-7**, except that air deflector **9.1-400** is mounted in the flow path to deflect, at least partially, the incoming stream **9.1-735** of air away from incidence onto the surface **9.1-759** of the component **9.1-386** (e.g., similar to FIGS. **9.1-4**). The incoming stream **9.1-735** is directed towards the surface **9.1-759** of the component **9.1-386** at the same angle $9.1-\theta$ as shown in FIGS. **9.1-7**. This is represented in FIGS. **9.1-8** by the dashed arrow, which illustrates what the path of the incoming stream **9.1-735** would be if the air deflector **9.1-400** were not present. However, due to the presence of air deflector **9.1-400**, the incoming stream **9.1-735** is wholly or partially deflected away from incidence onto the surface **9.1-759** of the component **9.1-386**. The air deflector **9.1-400** has a surface **9.1-859** positioned in the flow path to receive the incoming stream **9.1-735** of air thereon, and the surface **9.1-859** of the air deflector **9.1-400** forms an angle $9.1-\phi$ relative to the incoming stream **9.1-735** that is less than the angle $9.1-\theta$. Stated another way, the angle of incidence of the incoming stream **9.1-735** onto the surface **9.1-859** of the air deflector is greater than what the angle of incidence onto the surface **9.1-759** of the component **9.1-386** would be if the air deflector were not present. As a result of such configuration,

the air that further propagates downstream after incidence onto the air deflector **9.1-400** is deflected to a less drastic degree compared to if the air deflector **9.1-400** were not present and the air were to be incident unobstructed onto the component **9.1-386**. As a result, the air follows a less tortuous flow path, which can beneficially reduce noise within the device for a given flow velocity and/or improve efficiency of the cooling system.

[1475] The air deflector **9.1-400** can be configured as any suitable wall structure that forms the desired, turbulence-reducing angle, with respect to the incoming stream **9.1-735** of air. Although the wall is shown in FIGS. **9.1-8** as having a straight geometry, in various embodiments the wall can, for example, have a straight, a bent, or curved geometry. As shown in FIGS. **9.1-8**, the air deflector **9.1-400** and the component **9.1-386** can both be mounted to the same common surface **9.1-801**, which can be, for example, any suitable surface of the display assembly. As shown in FIGS. **9.1-8**, the surface **9.1-859** (air-receiving surface) of the air deflector can further form an obtuse angle with respect to the surface **9.1-801** onto which the air deflector **9.1-400** is mounted, which can be useful for reducing turbulence in cases where the incoming stream **9.1-735** of air propagates in a direction parallel to the surface **9.1-801**.

[1476] Components of the head-mounted device can be operably connected to provide the performance described herein. FIGS. **9.1-9** shows a simplified block diagram of an example of the head-mounted device **9.1-100**.

[1477] As shown in FIGS. **9.1-9**, the head-mounted device **9.1-100** can include a controller **9.1-130** with one or more processing units that include or are configured to access a memory **9.1-918** having instructions stored thereon. The instructions or computer programs may be configured to perform one or more of the operations or functions described with respect to the head-mounted device **9.1-100**. The controller **9.1-130** can be implemented as any electronic device capable of processing, receiving, or transmitting data or instructions. For example, the controller **9.1-130** may include one or more of: a microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a digital signal processor (DSP), or combinations of such devices. As described herein, the term “processor” is meant to encompass a single processor or processing unit, multiple processors, multiple processing units, or other suitably configured computing element or elements.

[1478] The memory **9.1-918** can store electronic data that can be used by the head-mounted device **9.1-100**. For example, the memory **9.1-918** can store electrical data or content such as, for example, audio and video files, documents and applications, device settings and user preferences, timing and control signals or data for the various modules, data structures or databases, and so on. The memory **9.1-918** can be configured as any type of memory. By way of example only, the memory **9.1-918** can be implemented as random access memory, read-only memory, Flash memory, removable memory, or other types of storage elements, or combinations of such devices.

[1479] The head-mounted device **9.1-100** can further include a display **9.1-120** for displaying visual information for a user. The display **9.1-120** can provide visual (e.g., image or video) output, and can include a pair of display assemblies as described herein. The display **9.1-120** can be or include an opaque, transparent, and/or translucent display. The display **9.1-120** may have a transparent or translucent medium through which light representative of images is directed to a user's eyes. The display **9.1-120** may utilize digital light projection, OLEDs, LEDs, uLEDs, liquid crystal on silicon, laser scanning light source, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In some embodiments, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface. The head-mounted device **9.1-100** can include optics configured to help optically adjust and correctly project the image based content being displayed by the

display **9.1-120** for close up viewing. The optics can include one or more lenses, mirrors, or other optical devices.

[1480] In some embodiments, controller **9.1-130** can receive user inputs from controls **9.1-908** and execute operations in response to the inputs. For example, controller **9.1-130** can be configured to receive sound from the microphone **9.1-930**. In response to receiving the sound, controller **9.1-130** can run the voice recognition module to identify voice commands.

[1481] Head-mounted device **9.1-100** can include a battery **9.1-920**, which can charge and/or power components of the head-mounted device **9.1-100**. The battery **9.1-920** can also charge and/or power components connected to the head-mounted device **9.1-100**, such as a portable electronic device **9.1-902**.

[1482] Head-mounted device **9.1-100** can include the air circulation device **9.1-140** for cooling down components of the head-mounted device **9.1-100**. The head-mounted device **9.1-100** can further include an air deflector **9.1-400** disposed in an air flow path and configured to receive a stream of air generated by the air circulation device **9.1-140**, as further described herein. The air deflector **9.1-400** can optionally be movable by an actuator **9.1-949**, as further described herein. The controller **9.1-130** can be configured to operate the actuator **9.1-949** to move or rotate the air deflector based on inputs from a user and/or adjustments to assemblies of the display **9.1-120**.

[1483] The head-mounted device **9.1-100** can include an input/output component **9.1-926**, which can include any suitable component for connecting head-mounted device **9.1-100** to other devices. Suitable components can include, for example, audio/video jacks, data connectors, or any additional or alternative input/output components.

[1484] The head-mounted device **9.1-100** can include communications circuitry **9.1-928** for communicating with one or more servers or other external devices **9.1-90** using any suitable communications protocol. For example, communications circuitry **9.1-928** can support Wi-Fi (e.g., a 802.11 protocol), Ethernet, Bluetooth, high frequency systems (e.g., 900 MHZ, 2.4 GHZ, and 5.6 GHz communication systems), infrared, TCP/IP (e.g., any of the protocols used in each of the TCP/IP layers), HTTP, BitTorrent, FTP, RTP, RTSP, SSH, any other communications protocol, or any combination thereof. Communications circuitry **9.1-928** can also include an antenna for transmitting and receiving electromagnetic signals.

[1485] The head-mounted device **9.1-100** can include audio devices such as a microphone **9.1-930** and/or speaker **9.1-912**. The microphone **9.1-930** can be configured to detect sounds from a user and/or environment. The microphone **9.1-930** can be operably connected to the controller **9.1-130** for detection of sound levels and communication of detections for further processing. The speaker **9.1-212** can be configured to emit sounds to a user and/or environment. The speaker **9.1-212** can be operably connected to the controller **9.1-130** for control of speaker output, including sound levels and/or other sound characteristics.

[1486] The head-mounted device **9.1-100** can optionally connect to a portable electronic device **9.1-902**, which can provide certain functions. For the sake of brevity, the portable electronic device **9.1-902** will not be described in detail in FIGS. **9.1-9**. It should be appreciated, however, that the portable electronic device **9.1-902** may be embodied in a variety of forms including a variety of features, all or some of which can be utilized by the head-mounted device **9.1-100** (e.g., input/output, controls, processing, battery, etc.). The portable electronic device **9.1-902** can be configured to receive cooling from operation of air circulation device **9.1-140**. The portable electronic device **9.1-902** can provide a handheld form factor (e.g., small portable electronic device which is light weight, fits in a pocket, etc.). Although not limited to these, examples include media players, phones (including smart phones), PDAs, computers, and the like. The portable electronic device **9.1-902** may include a screen **9.1-913** for presenting the graphical portion of the media to the user. The screen **9.1-913** can be utilized as the primary screen of the head-mounted device **9.1-100**.

[1487] The head-mounted device **9.1-100** can include a dock **9.1-906** operative to receive the

portable electronic device **9.1-902**. The dock **9.1-906** can include a connector (e.g., Lightning, USB, Fire Wire, power, DVI, etc.), which can be plugged into a complementary connector of the portable electronic device **9.1-902**. The dock **9.1-906** may include features for helping to align the connectors during engagement and for physically coupling the portable electronic device **9.1-902** to the head-mounted device **9.1-100**. For example, the dock **9.1-906** may define a cavity for placement of the portable electronic device **9.1-902**. The dock **9.1-906** may also include retaining features for securing portable electronic device **9.1-902** within the cavity. The connector on the dock **9.1-906** can function as a communication interface between the portable electronic device **9.1-902** and the head-mounted device **9.1-100**.

[1488] The head-mounted device **9.1-100** can include one or more other sensors. Such sensors can be configured to sense substantially any type of characteristic such as, but not limited to, images, pressure, light, touch, force, temperature, position, motion, and so on. For example, the sensor can be a photodetector, a temperature sensor, a light or optical sensor, an atmospheric pressure sensor, a humidity sensor, a magnet, a gyroscope, an accelerometer, a chemical sensor, an ozone sensor, a particulate count sensor, and so on. By further example, the sensor can be a bio-sensor for tracking biometric characteristics, such as health and activity metrics. Other user sensors can perform facial feature detection, facial movement detection, facial recognition, eye tracking, user mood detection, user emotion detection, voice detection, etc. Sensors can include a camera which can capture image based content of the outside world.

9.2: Fan with Debris Mitigation

[1489] Head-mountable devices, such as head-mountable displays, headsets, visors, smartglasses, head-up display, etc., can perform a range of functions that are managed by the components (e.g., sensors, circuitry, and other hardware) included with the wearable device. The head-mountable device can provide a user experience that is immersive or otherwise natural so the user can easily focus on enjoying the experience without being distracted by the mechanisms of the head-mountable device.

[1490] Components of a head-mountable device can generate heat during operation. The performance of electronic devices are often limited by their ability to effectively dissipate the heat generated by computing and other workloads. Excessive heat for long durations of time can damage the components of the head-mountable device and cause discomfort to the user. Heat can be mitigated in a number of ways, including with active mechanisms (e.g., fans, blowers, air movers, and the like) that are integrated into the head-mountable device. In order to dissipate heat effectively within a small form factor, active cooling is sometimes used. Active cooling refers to a thermal architecture wherein heat is dissipated via forced convection.

[1491] An active cooling device (e.g., fans, blowers, air movers, and the like) can generate a flow of air from an inlet to an outlet thereof. Such inlets and outlets can define a pathway that receives air from and/or delivers air to an environment external to the head-mountable device. However, by providing such exposure to an external environment, the fan is susceptible to intrusion by particles or other debris from the external environment. In particular, when the fan is not in operation, such particles can collect and become lodged between the parts of the fan. In such a condition, the fan may encounter difficulty resuming operation due to the introduction of particles between moving parts of the fan. In particular, particles lodged between the stationary parts (i.e., fan housing) and moving parts (i.e., rotor, impeller, etc.) can cause the fan to stall.

[1492] Systems of the present disclosure can provide fans that mitigate the intrusion of particles and other debris. Fans can include a protrusion that creates a tortuous pathway to direct incoming particles away from sensitive regions. Fans can include openings to allow particles to exit the fan. Fans can include a variable spacing between the stationary parts (i.e., fan housing) and moving parts (i.e., rotor, impeller, etc.) to avoid collection of particles. Fans can include an adhesive pad that collects and retains particles at a location that does not interfere with operation of the impeller.

[1493] These and other embodiments are discussed below with reference to FIGS. **9.2-1** through

9.2-13. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting.

[1494] According to some embodiments, for example as shown in FIGS. **9.2-1**, a head-mountable device **9.2-100** includes a frame **9.2-110** that is worn on a head of a user. The frame **9.2-110** can be positioned in front of the eyes of a user to provide information within a field of view of the user. The frame **9.2-110** can provide nose pads or another feature to rest on a user's nose. The frame **9.2-110** can be supported on a user's head with the head engager **9.2-120**. The head engager **9.2-120** can wrap or extend along opposing sides of a user's head. The head engager **9.2-120** can include earpieces for wrapping around or otherwise engaging or resting on a user's ears. It will be appreciated that other configurations can be applied for securing the head-mountable device **9.2-100** to a user's head. For example, one or more bands, straps, belts, caps, hats, or other components can be used in addition to or in place of the illustrated components of the head-mountable device **9.2-100**. By further example, the head engager **9.2-120** can include multiple components to engage a user's head.

[1495] The frame **9.2-110** can provide structure around a peripheral region thereof to support any internal components of the frame **9.2-110** in their assembled position. For example, the frame **9.2-110** can enclose and support various internal components (including for example integrated circuit chips, processors, memory devices, cameras, displays, lenses, and other circuitry) to provide computing and functional operations for the head-mountable device **9.2-100**, as discussed further herein. Any number of components can be included within and/or on the frame **9.2-110** and/or the head engager **9.2-120**.

[1496] The frame **9.2-110** can include and/or support one or more cameras **9.2-130**. The cameras **9.2-130** can be positioned on or near an outer side of the frame **9.2-110** to capture images of views external to the head-mountable device **9.2-100**. The captured images can be used for display to the user or stored for any other purpose.

[1497] The head-mountable device can be provided with displays that provide visual output for viewing by a user wearing the head-mountable device. As further shown in FIGS. **9.2-1**, one or more displays **9.2-140** can be positioned on an inner side **9.2-124** of the head-mountable device **9.2-100**, for example within an eye chamber **9.2-126**. For example, a pair of displays **9.2-140** can be provided, where each display **9.2-140** is movably positioned to be within the field of view of each of a user's two eyes. Each display **9.2-140** can be adjusted to align with a corresponding eye of the user. For example, each display **9.2-140** can be moved along one or more axes until a center of each display **9.2-140** is aligned with a center of the corresponding eye.

[1498] A display **9.2-140** can transmit light from a physical environment (e.g., as captured by a camera) for viewing by the user. Such a display **9.2-140** can include optical properties, such as lenses for vision correction based on incoming light from the physical environment. Additionally or alternatively, a display **9.2-140** can provide information as a display within a field of view of the user. Such information can be provided to the exclusion of a view of a physical environment or in addition to (e.g., overlaid with) a physical environment.

[1499] A physical environment relates to a physical world that people can sense and/or interact with without necessarily requiring the aid of an electronic device. A computer-generated reality environment relates to a wholly or partially simulated environment that people sense and/or interact with the assistance of an electronic device. Examples of computer-generated reality include mixed reality and virtual reality. Examples of mixed realities can include augmented reality and augmented virtuality. Some examples of electronic devices that enable a person to sense and/or interact with various computer-generated reality environments include head-mountable systems, projection-based systems, heads-up displays (HUDs), vehicle windshields having integrated display capability, windows having integrated display capability, displays formed as lenses designed to be placed on a person's eyes (e.g., similar to contact lenses), headphones/earphones,

speaker arrays, input systems (e.g., wearable or handheld controllers with or without haptic feedback), smartphones, tablets, and desktop/laptop computers. A head-mountable device can have an integrated opaque display, have a transparent or translucent display, or be configured to accept an external opaque display (e.g., smartphone).

[1500] Referring again to FIGS. **9.2-1**, the head-mountable device can be provided with one or more flow channels that extend through at least a portion of the frame thereof to provide cooling to components of the head-mountable device. As shown in FIGS. **9.2-1**, the flow channels can include and/or be connected to a system inlet **9.2-112**. The system inlet **9.2-112** can provide airflow directly to the fan **9.2-200** and/or one or more components of the head-mountable device **9.2-100**, such as a circuit component **9.2-150**. The air received by the system inlet **9.2-112** can be directed to a system outlet **9.2-116** and/or other outlets for exhaust out of the head-mountable device **9.2-100** by the fan **9.2-200**.

[1501] While the system inlet **9.2-112** is depicted at a bottom portion of the frame **9.2-110** and the system outlet **9.2-116** is depicted at a top portion of the frame **9.2-110**, it will be recognized that inlets, outlets, and flow channels there between can be positioned at any portion of the head-mountable device **9.2-100**. The system outlet **9.2-116** can be provided at a location that will allow exiting air to exhaust to an environment that is not disruptive to the user. For example, the system outlet **9.2-116** can be provided at a location and in an orientation that directs hot air away from the user. Multiple flow channels can be interconnected, such that multiple inlets and/or multiple outlets are connected to each other.

[1502] One or more fans **9.2-200** can be operated to provide cooling to one or more circuit components **9.2-150** of the head-mountable device **9.2-100**. The circuit component **9.2-150** can be an electrical component that generates heat during operation. The circuit component **9.2-150** can be a component of a circuit board **9.2-152**. The circuit component **9.2-150** can be operably and structurally coupled to the circuit board **9.2-152**. A portion of the fan **9.2-200** can be thermally connected to the circuit component **9.2-150**.

[1503] The fan **9.2-200** can receive a flow of air from a system inlet **9.2-112** and direct the flow of air to a system outlet **9.2-116**. For example, the fan **9.2-200**, the circuit component **9.2-150**, and/or the circuit board **9.2-152** can be positioned within an internal chamber **9.2-118** (e.g., plenum chamber) of the head-mountable device **9.2-100**.

[1504] While several components are shown within the frame **9.2-110**, it will be understood that some or all of these components can be located anywhere within or on the head-mountable device **9.2-100**. For example, one or more of these components can be positioned within the head engager **9.2-120** of the head-mountable device **9.2-100**.

[1505] Referring now to FIGS. **9.2-2**, the fan can provide features for moving air across surfaces for dissipating heat. As shown in FIGS. **9.2-2**, the fan **9.2-200** can include a fan housing **9.2-208** that includes a cover **9.2-210** over an impeller **9.2-220**. The cover **9.2-210** can form a fan inlet **9.2-212**, for example through a top surface thereof. The cover **9.2-210** can include or be connected to an exhaust duct **9.2-290** that forms a fan outlet **9.2-292** of the fan **9.2-200**.

[1506] The impeller **9.2-220** of the fan **9.2-200** can be positioned to receive air through the fan inlet **9.2-212** and direct the air to the fan outlet **9.2-292**. The fan **9.2-200** can include a motor to drive rotation of the impeller **9.2-220**. For example, the fan **9.2-200** can include a stator and an impeller **9.2-220** configured to rotate about the stator. The impeller **9.2-220** can include multiple blades **9.2-224** that extend radially outwardly away from the central hub of the impeller. As the impeller **9.2-220** rotates, the blades **9.2-224** receive air from the fan inlet **9.2-212** and direct the air radially outwardly and toward the fan outlet **9.2-292**. The impeller **9.2-220** can be stabilized by one or more bearings between the impeller **9.2-220** and the stator. The bearing can include a fluid hydrodynamic bearing, for example with oil or another fluid between the impeller **9.2-220** and the stator. It will be understood that other types of bearings are contemplated, including mechanical bearings, journal bearings, plain bearings, ball bearings, and the like. The bearing can provide radial and/or axial

support to the impeller **9.2-220** as it rotates about the stator.

[1507] The impeller **9.2-220** can direct air or another gas within, against, or across one or more components of the fan **9.2-200**. The fan **9.2-200** can be operated based on one or more operating parameters that are controllable during use. The operating parameters can be determined, at least in part, based on a demand for cooling (e.g., based on a temperature of one or more components). The operating parameters can be further determined based on acceptable sound levels and characteristics to be produced by the fan **9.2-200** and along the flow channel.

[1508] Referring now to FIGS. **9.2-3** and **9.2-4**, the fan can be thermally connected to components to be cooled. As shown in FIGS. **9.2-3**, the fan housing **9.2-208** can further include a base plate **9.2-230**. The cover **9.2-210** and the base plate **9.2-230** of the fan housing **9.2-208** can, together, define an interior space **9.2-280** there between. The impeller **9.2-220** can be positioned within the interior space **9.2-280** and mounted to the base plate **9.2-230**. As such, the base plate **9.2-230** provides structural support to the impeller **9.2-220** for stable operation (e.g., during rotation).

[1509] A heat sink **9.2-240** can provide one or more fins **9.2-242** between the interior space **9.2-280** and the fan outlet **9.2-292** formed by the exhaust duct **9.2-290**. The heat sink **9.2-240** is thermally connected to the circuit component **9.2-150**, which is operably and structurally coupled to the circuit board **9.2-152**. The heat sink **9.2-240** can be thermally connected to the circuit component **9.2-150** by a direct connection (e.g., no intervening structure) or by a thermal interface **9.2-160**. For example, a thermal paste or other thermally conductive material can be provided to thermally and/or structurally connect the circuit component **9.2-150** to the heat sink **9.2-240**.

[1510] The base plate **9.2-230**, the heat sink **9.2-240**, and/or the fins **9.2-242** can be of a metal or other material having high thermal conductivity. The material can provide high rigidity and strength to provide support to components mounted to the base plate **9.2-230** and to securely mount to other components (e.g., the frame) of the head-mountable device.

[1511] The cover **9.2-210** can be of a material that provides protection to the impeller **9.2-220** and any other components in the interior space. The material can be plastic, metal, and/or another material. The cover **9.2-210** can be a monolithic, unitary, and/or unibody structure, rather than an assembly of parts.

[1512] As shown in FIGS. **9.2-3** and FIGS. **9.2-4**, the fan **9.2-200** can generate a flow of air from an inlet (e.g., system inlet **9.2-112** and/or fan inlet **9.2-212**) to an outlet (e.g., system outlet **9.2-116** and/or fan outlet **9.2-292**). It will be understood that the system outlet **9.2-116** can be directly connected to the fan outlet **9.2-292**, such that all air directed through the fan outlet **9.2-292** can be further directed to the system outlet **9.2-116**. Such inlets and outlets can define a pathway that receives air from and/or delivers air to an environment external to the head-mountable device. During operation of the fan **9.2-200**, the air flow can be maintained such that any particles or other debris travelling with the air flow moves through the fan **9.2-200** without remaining therein. Furthermore, the movement of the impeller **9.2-220** can disrupt and propel any resting particles until they are ejected from the fan **9.2-200**.

[1513] However, as shown in FIGS. **9.2-5**, by providing such exposure to an external environment, the fan **9.2-200** is susceptible to intrusion by particles or other debris from the external environment. In particular, when the fan **9.2-200** is not in operation (e.g., when the impeller **9.2-220** is not rotating), particles **9.2-2** can collect and become lodged between the moving and stationary parts of the fan **9.2-200**. For example, the particles **9.2-2** can enter through the system inlet **9.2-112** and into the fan inlet **9.2-212** and/or through the system outlet **9.2-116** and into the fan outlet **9.2-292** and become lodged between the stationary parts (i.e., fan housing, base plate, etc.) and moving parts (i.e., rotor, impeller, etc.) of the fan **9.2-200**. In such a condition, the fan **9.2-200** may encounter difficulty resuming operation due to the introduction of particles between moving and stationary parts of the fan **9.2-200**. In particular, particles **9.2-2** lodged between the impeller **9.2-220** and the base plate **9.2-230** can cause the impeller **9.2-220** to stall.

[1514] It can be desirable to provide features that mitigate the collection of such particles while the

fan **9.2-200** is not in operation without reducing the effectiveness of the fan while it is in operation. [1515] Referring now to FIGS. **9.2-5**, a fan can include features that create a tortuous pathway to prevent the incoming particles from entering the sensitive regions. For example, as shown in FIGS. **9.2-5**, a protrusion **9.2-232** can extend from the base plate **9.2-230** to block and/or redirect a pathway of an incoming particle **9.2-2**. For example, the protrusion **9.2-232** can be positioned between at least a portion of the impeller **9.2-220** and the fan outlet **9.2-292**. The protrusion **9.2-232** can extend toward the cover **9.2-210** while allowing an air flow pathway over the protrusion between the impeller **9.2-220** and the fan outlet **9.2-292**.

[1516] The impeller **9.2-220** can be formed with a support disk **9.2-228** on one side of the blades **9.2-224**. Where a support disk **9.2-228** is provided, such as is shown in FIGS. **9.2-5**, the protrusion **9.2-232** can extend from the base plate **9.2-230** to at least a height of a portion of the support disk **9.2-228**. Where blades **9.2-224** are provided, the protrusion **9.2-232** can extend from the base plate **9.2-230** to at least a height of one or more of the blades **9.2-224**. As such, the protrusion **9.2-232** can shield and/or occupy a space between the base plate **9.2-230** and the support disk **9.2-228**. It will be understood that a tortuous pathway around the protrusion **9.2-232** can connect the fan outlet **9.2-292** to the space between the base plate **9.2-230** and the support disk **9.2-228**. However, the protrusion can be formed such that incoming particles **9.2-2** are instead deflected to above the support disk **9.2-228**, so that they do not lodge between the base plate **9.2-230** and the support disk **9.2-228**.

[1517] In some embodiments, as further shown in FIGS. **9.2-5**, the protrusion **9.2-232** can be formed as an annular ring or closed loop. Such a closed loop can surround a periphery of the impeller **9.2-220**. For example, the annular ring can encircle the support disk **9.2-228** and/or a portion of each of the blades **9.2-224**.

[1518] In some embodiments, as further shown in FIGS. **9.2-5**, the protrusion **9.2-232** defines a surface **9.2-286** facing away from the impeller **9.2-220**. This surface **9.2-286** can be that on which the particle **9.2-2** is incident upon entry through the fan outlet **9.2-292**. The surface **9.2-286** can be formed to direct the particle away from the space between the base plate **9.2-230** and the impeller **9.2-220**. For example, the surface can form an angle that is oblique with respect to the base plate **9.2-230** to form a ramp.

[1519] Referring now to FIGS. **9.2-6**, the protrusion **9.2-232** can overlap with at least a portion of the impeller **9.2-220**. For example, the blades **9.2-224** can be positioned between the cover **9.2-210** and the base plate **9.2-230**, and each of the blades can form a notch **9.2-288**. The protrusion **9.2-232** extends from the base plate **9.2-230** and into at least some of the notches **9.2-288**. As the impeller **9.2-220** rotates, the notches **9.2-288** pass over the protrusion **9.2-232** without physical contact.

[1520] Where a support disk **9.2-228** of the impeller **9.2-220** is provided, such as is shown in FIGS. **9.2-6**, the protrusion **9.2-232** can extend from the base plate **9.2-230** to at least a height of a portion of the support disk **9.2-228**. In some embodiments, as further shown in FIGS. **9.2-6**, the protrusion **9.2-232** can be formed as an annular ring. For example, the annular ring can encircle the support disk **9.2-228** without encircling the blades **9.2-224**, but instead protruding towards the blades **9.2-224** and into the notches **9.2-288**. As such, the protrusion **9.2-232** can block a space between the base plate **9.2-230** and the support disk **9.2-228**. It will be understood that a tortuous pathway around the protrusion **9.2-232** can connect the fan outlet **9.2-292** to the space between the base plate **9.2-230** and the support disk **9.2-228**. However, the protrusion can be formed such that incoming particles are instead deflected to above the support disk **9.2-228**, so that they do not lodge between the base plate **9.2-230** and the support disk **9.2-228**.

[1521] It will be understood that any protrusion **9.2-232** disclosed herein need not form an annular ring and that a variety of other shapes are contemplated, such as an arc, a flat wall, and the like. It will be understood that any number of protrusions **9.2-232** can be provided. For example, both the protrusion of FIGS. **9.2-5** (i.e., overlapping the impeller **9.2-220**) and the protrusion of FIGS. **9.2-6** (i.e., surrounding the impeller **9.2-220**) can be provided in combination.

[1522] In some embodiments, a distance between any protrusion **9.2-232** disclosed herein and the impeller **9.2-220** (e.g., the support disk **9.2-228** and/or the blades **9.2-224**) can be smaller than a distance between the base plate **9.2-230** and the impeller **9.2-220** (e.g., the support disk **9.2-228**). The smaller distance between the protrusion **9.2-232** and the impeller **9.2-220** can serve to block particles having a dimension that is greater than such a distance. As such, only particles smaller than such a distance can pass the protrusion **9.2-232**. When such small particles are between the base plate **9.2-230** and the impeller **9.2-220**, the particles will not be large enough to extend to both the base plate **9.2-230** and the impeller **9.2-220** and will thereby be unable to lodge therein or arrest operation of the impeller **9.2-220**.

[1523] Referring now to FIGS. **9.2-7**, the base plate can have features that facilitate particle blocking and promote impeller operation. For example, as shown in FIGS. **9.2-7**, the base plate **9.2-230** can have a variable profile at different portions thereof and with respect to the impeller **9.2-220**. For example, the base plate can form a base plate central portion **9.2-262** and one or more base plate peripheral portions **9.2-260**. The impeller **9.2-220** can form an impeller central portion **9.2-252** and one or more impeller peripheral portions **9.2-250**. A distance **9.2-266** between the base plate central portion **9.2-262** and the impeller central portion **9.2-252** is greater than a distance **9.2-264** between the base plate peripheral portion **9.2-260** and the impeller peripheral portion **9.2-250**. The smaller distance **9.2-264** between the base plate central portion **9.2-262** and the impeller central portion **9.2-252** can serve to block particles having a dimension that is greater than the distance **9.2-264**. As such, only particles smaller than such a distance can pass the gap defined by the distance **9.2-264**. When such small particles are between the base plate central portion **9.2-262** and the impeller central portion **9.2-252**, the particles will not be large enough to span the distance **9.2-266** to contact both the base plate central portion **9.2-262** and the impeller central portion **9.2-252** and will thereby be unable to lodge therein or arrest operation of the impeller **9.2-220**. It will be understood that a number of variations can be provided, including any number of different distances and/or transitions there between.

[1524] Referring now to FIGS. **9.2-8**, the base plate can have features that facilitate particle evacuation and/or capture. For example, as shown in FIGS. **9.2-8**, the base plate **9.2-230** can define one or more openings **9.2-270** that extend entirely through the base plate **9.2-230**. The impeller **9.2-220** can be positioned to overlap at least some of the openings **9.2-270** of the base plate **9.2-230**. When particles are introduced between the base plate **9.2-230** and the impeller **9.2-220**, the particles can be provided with an exit from such a region through the openings **9.2-270**. The openings **9.2-270** can be connected to an external environment or a collection chamber for releasing and/or capturing the particles. It will be understood that any number of openings **9.2-270** can be provided in any arrangement. It will be further understood that the openings **9.2-270** can have any size and/or dimension to facilitate particle removal. It will be understood that openings can be provided at any location, including other than the base plate **9.2-230**.

[1525] Referring now to FIGS. **9.2-9** and **9.2-10**, a fan can include a retention mechanism that captures particles. For example, as shown in FIGS. **9.2-9**, a pad **9.2-272** can be provided on a side of the base plate **9.2-230** facing and/or covering the openings **9.2-270**. In such an embodiment, the openings **9.2-270** can be the same or similar to those described elsewhere herein. As shown in FIGS. **9.2-10**, the pad **9.2-272** can be positioned on a side of the openings that is opposite the impeller **9.2-220**, such as on an outer side of the base plate **9.2-230**. Rather than releasing the particles that pass through the openings **9.2-270**, the pad **9.2-272** can capture the particles and retain them away from the impeller **9.2-220**.

[1526] Once the particles are removed from the interior space **9.2-280** containing the impeller **9.2-220**, they can be retained at a location that does not interfere with the operation of the impeller **9.2-220**. For example, the pad **9.2-272** can provide a surface that retains particles upon contact of the particle with the pad **9.2-272**. For example, the pad **9.2-272** can include an adhesive (e.g., pressure-sensitive adhesive). As used herein, an adhesive can include any material that has adhesion

properties and/or stickiness, such as a polymer, glue, cement, paste, laminate, and/or other material that bonds to particles upon contact therewith. By further example, the pad **9.2-272** can include an uncured or partially cured substance that is exposed to the interior space **9.2-280** and/or the openings **9.2-270**. In some embodiments, the fan **9.2-200** can include one or more other retention mechanisms, such as an operable electrode or other surface that is configured to be electrically charged to attract particles. Additionally or alternatively, the fan **9.2-200** can include a filter and/or a mechanically actuated container for selectively containing particles that migrate out of the interior space **9.2-280** and/or through the openings **9.2-270**.

[1527] Referring now to FIGS. **9.2-11** and **9.2-12**, a fan can provide multiple features that provide particle mitigation properties. For example, the fan **9.2-200** of FIGS. **9.2-11** and **9.2-12** includes one or more protrusions **9.2-232** extending from the base plate **9.2-230** to block and/or redirect a pathway of an incoming particle. Such protrusion(s) **9.2-232** can include any one or more features described herein with respect to the fan of FIGS. **9.2-5** or **9.2-6**.

[1528] By further example, the fan **9.2-200** of FIGS. **9.2-11** and **9.2-12** includes a base plate **9.2-230** having a variable profile at different portions thereof and with respect to the impeller **9.2-220**, such as with a base plate central portion **9.2-262** and one or more base plate peripheral portions **9.2-260**. Along with features of the impeller **9.2-220**, the base plate **9.2-230** can define different distances there between, as described with respect to the fan of FIGS. **9.2-7**.

[1529] By further example, the fan **9.2-200** of FIGS. **9.2-11** and **9.2-12** includes a base plate **9.2-230** that defines one or more openings **9.2-270** extending entirely through the base plate **9.2-230**. Such openings **9.2-270** can include any one or more features described herein with respect to the fan of FIGS. **9.2-8**.

[1530] By further example, the fan **9.2-200** of FIGS. **9.2-11** and **9.2-12** includes a pad **9.2-272** on an outer side of the base plate **9.2-230** to capture particles from the interior space **9.2-280**. The pad **9.2-272** can include any one or more features described herein with respect to the fan of FIGS. **9.2-9** and **9.2-10**.

[1531] It will be understood that any number of features described herein can be combined in a single fan to provide particle mitigation. It will be further understood that no one feature is necessary to provide effective particle mitigation to a fan.

[1532] Referring now to FIGS. **9.2-13**, components of the head-mountable device can be operably connected to provide the performance described herein. FIGS. **9.2-13** shows a simplified block diagram of an illustrative head-mountable device **9.2-100** in accordance with one embodiment of the invention. It will be appreciated that components described herein can be provided on either or both of a frame and/or a head engager of the head-mountable device **9.2-100**. It will be understood that additional components, different components, or fewer components than those illustrated may be utilized within the scope of the subject disclosure.

[1533] As shown in FIGS. **9.2-13**, the head-mountable device **9.2-100** can include a controller **9.2-180** with one or more processing units that include or are configured to access a memory **9.2-182** having instructions stored thereon. The instructions or computer programs may be configured to perform one or more of the operations or functions described with respect to the head-mountable device **9.2-100**. The controller **9.2-180** can be implemented as any electronic device capable of processing, receiving, or transmitting data or instructions. For example, the controller **9.2-180** may include one or more of: a microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a digital signal processor (DSP), or combinations of such devices. As described herein, the term “processor” is meant to encompass a single processor or processing unit, multiple processors, multiple processing units, or other suitably configured computing element or elements.

[1534] The memory **9.2-182** can store electronic data that can be used by the head-mountable device **9.2-100**. For example, the memory **9.2-182** can store electrical data or content such as, for example, audio and video files, documents and applications, device settings and user preferences,

timing and control signals or data for the various modules, data structures or databases, and so on. The memory **9.2-182** can be configured as any type of memory. By way of example only, the memory **9.2-182** can be implemented as random access memory, read-only memory, Flash memory, removable memory, or other types of storage elements, or combinations of such devices.

[1535] The head-mountable device **9.2-100** can further include a display **9.2-140** for displaying visual information for a user. The display **9.2-140** can provide visual (e.g., image or video) output. The display **9.2-140** can be or include an opaque, transparent, and/or translucent display. The display **9.2-140** may have a transparent or translucent medium through which light representative of images is directed to a user's eyes. The display **9.2-140** may utilize digital light projection, OLEDs, LEDs, uLEDs, liquid crystal on silicon, laser scanning light source, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In one embodiment, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface. The head-mountable device **9.2-100** can include an optical subassembly **9.2-214** configured to help optically adjust and correctly project the image-based content being displayed by the display **9.2-140** for close up viewing. The optical subassembly **9.2-214** can include one or more lenses, mirrors, or other optical devices.

[1536] The head-mountable device **9.2-100** can include the fan **9.2-200** and/or any other suitable component for cooling down components of the head-mountable device **9.2-100**. Suitable components can include, for example, impellers, pipes for transferring heat, vents, apertures, holes, any other component suitable for distributing and diffusing heat, or any combination thereof. The fan **9.2-200** may also or instead be manufactured from materials selected for heat dissipation properties. For example, a housing of the head-mountable device **9.2-100** may be configured to distribute heat away from components thereof and/or the user.

[1537] The head-mountable device **9.2-100** can include a battery **9.2-184**, which can charge and/or power components of the head-mountable device **9.2-100**. The battery **9.2-184** can also charge and/or power components connected to the head-mountable device **9.2-100**.

[1538] The head-mountable device **9.2-100** can include an input/output component **9.2-186**, which can include any suitable component for connecting head-mountable device **9.2-100** to other devices. Suitable components can include, for example, audio/video jacks, data connectors, or any additional or alternative input/output components. The input/output component **9.2-186** can include buttons, keys, or another feature that can act as a keyboard for operation by the user.

[1539] The head-mountable device **9.2-100** can include the microphone **9.2-188** as described herein. The microphone **9.2-188** can be operably connected to the controller **9.2-180** for detection of sound levels and communication of detections for further processing, as described further herein.

[1540] The head-mountable device **9.2-100** can include the speakers **9.2-190** as described herein. The speakers **9.2-190** can be operably connected to the controller **9.2-180** for control of speaker output, including sound levels, as described further herein.

[1541] The head-mountable device **9.2-100** can include one or more other sensors. Such sensors can be configured to sense substantially any type of characteristic such as, but not limited to, images, pressure, light, touch, force, temperature, position, motion, and so on. For example, the sensor can be a photodetector, a temperature sensor, a light or optical sensor, an atmospheric pressure sensor, a humidity sensor, a magnet, a gyroscope, an accelerometer, a chemical sensor, an ozone sensor, a particulate count sensor, and so on. By further example, the sensor can be a bio-sensor for tracking biometric characteristics, such as health and activity metrics. Other user sensors can perform facial feature detection, facial movement detection, facial recognition, eye tracking, user mood detection, user emotion detection, voice detection, etc. Sensors can include a camera which can capture image based content of the outside world.

[1542] The head-mountable device **9.2-100** can include communications circuitry **9.2-192** for communicating with one or more servers or other devices using any suitable communications protocol. For example, communications circuitry **9.2-192** can support Wi-Fi (e.g., a 802.11 protocol), Ethernet, Bluetooth, high frequency systems (e.g., 900 MHz, 2.4 GHz, and 5.6 GHz communication systems), infrared, TCP/IP (e.g., any of the protocols used in each of the TCP/IP layers), HTTP, BitTorrent, FTP, RTP, RTSP, SSH, any other communications protocol, or any combination thereof. Communications circuitry **9.2-192** can also include an antenna for transmitting and receiving electromagnetic signals.

[1543] While various embodiments and aspects of the present disclosure are illustrated with respect to a head-mountable device, it will be appreciated that the subject technology can encompass and be applied to other devices. For example, a noise mitigation system in accordance with embodiments disclosed herein can be included with an electronic device that generates heat during operation. Such an electronic device can be or include a desktop computing device, a laptop-computing device, a display, a television, a portable device, a phone, a tablet computing device, a mobile computing device, a wearable device, a watch, and/or a digital media player. Such devices can include an impeller and flow channels to facilitate cooling as described herein.

[1544] Accordingly, embodiments of the present disclosure provide a head-mountable device that provides a fan that effectively manages heat while also mitigating the intrusion of particles and other debris. Fans can include a protrusion that creates a tortuous pathway to direct incoming particles away from sensitive regions. Fans can include openings to allow particles to exit the impeller. Fans can include a variable spacing between the impeller and the base plate to avoid collection of particles. Fans can include an adhesive pad that collects and retains particles at a location that does not interfere with operation of the impeller.

9.3: Ventilation

[1545] FIGS. **9.3-1** illustrates an example of a head-mountable display (HMD) **9.3-100**. The HMD **9.3-100** can include a ventilation system **9.3-102** for controlling airflow through the HMD **9.3-100** and cooling various components of the HMD **9.3-100** described herein. In at least one example, the ventilation system **9.3-102** can be configured to cool one or more heat generating components or subsystems of the HMD **9.3-100**, including, for example, the optical modules **9.3-108a**, **9.3-108b** and/or one or more processors or other heat generating components of the logic board **9.3-110**. The ventilation system **9.3-102** can include a frame assembly **9.3-106**, which can include a multi-part chassis, and a fan assembly **9.3-104**. The fan assembly **9.3-104** can be coupled to the frame assembly **9.3-106** and the frame assembly **9.3-106** such that the fan assembly **9.3-104** is disposed between the frame assembly **9.3-106** and the logic board **9.3-110**.

[1546] In at least one example, when the HMD **9.3-100** is assembled, the optical modules **9.3-108a**, **9.3-108b** can be coupled to the frame assembly **9.3-106** such that the optical modules **9.3-108a**, **9.3-108b** are each aligned with an aperture defined by the frame assembly **9.3-106** (illustrated in subsequent figures, including FIGS. **9.3-2** and **9.3-11**). In this way, in at least one example, the optical modules **9.3-108a**, **9.3-108b** can be in fluid communication with the fan assembly **9.3-104** through the frame assembly **9.3-106** disposed between the optical modules **9.3-108a**, **9.3-108b** and the fan assembly **9.3-104**. In at least one example, one or the optical modules **9.3-108a**, **9.3-108b** can each include one or more display screens (illustrated in subsequent figures, for example FIGS. **9.3-11**) or other heat-generating electronic components. For example, the first optical module **9.3-108a** can include a first display screen and the second optical module **9.3-108b** can include a second display screen, both of which can be oriented toward the user's eyes when donning the HMD **9.3-100**.

[1547] FIGS. **9.3-2** illustrates an example of a ventilation system **9.3-202** in an assembled configuration. In at least one example, the ventilation system **9.3-202** includes a first fan **9.3-212a** and a second fan **9.3-212b** coupled with a frame assembly **9.3-206** (e.g., the chassis). In at least one example, the frame assembly **9.3-206** can include an outer frame **9.3-215** and an inner frame **9.3-**

214 coupled to the outer frame **9.3-215**. The frame assembly **9.3-206** can be a structural frame assembly. The outer frame **9.3-215** can be a structural outer frame and define an exterior surface of the HMD **9.3-100** and the inner frame **9.3-214** can be an inner structural frame disposed within an internal volume **9.3-226** defined by the outer frame **9.3-215**. As noted above with reference to FIGS. **9.3-1**, an optical module including a display screen (not shown in FIGS. **9.3-2**) can be coupled to the frame assembly **9.3-206**, for example coupled to the inner frame **9.3-214**, and aligned with a first aperture **9.3-216a** or the second aperture **9.3-216b**. In one example, a first optical module with a first screen can be aligned with the first aperture **9.3-216a** and a second optical module with a second screen can be aligned with the second aperture **9.3-216b**.

[1548] In at least one example, the inner frame **9.3-214** of the frame assembly **9.3-206** of the head mountable display device can be coupled to the outer frame **9.3-215** at discrete touch points **9.3-218** to minimize contact and thermal transfer between the inner frame **9.3-214** and the outer frame **9.3-215**. Two discrete touch points **9.3-218** are labeled and shown in FIGS. **9.3-2** and are given as non-limiting examples. More or less than the touch points **9.3-218** shown can be included at various positions or locations of the inner frame **9.3-214** and outer frame **9.3-215**. The discrete touch points **9.3-218** where the inner frame **9.3-214** is coupled to the outer frame **9.3-215** can be a part of a first set of discrete touch points such that the inner frame **9.3-214** is thermally isolated from the outer frame **9.3-215** via the first set of discrete touch points. The inner frame **9.3-214** can be referred to as an internal frame and the outer frame **9.3-215** can be referred to as an external frame.

[1549] In at least one example, each touch-point **9.3-218** of the first set of discrete touch points can include an insulating member **9.3-220** disposed between the inner frame **9.3-214** and the outer frame **9.3-215**. In this way, the inner frame **9.3-214** can be thermally isolated from the outer frame **9.3-215** via a first set of insulating members, including the insulating members **9.3-220** shown in FIGS. **9.3-2**, disposed between the inner frame **9.3-214** and the outer frame **9.3-215** at the first set of discrete touch points, with at least one insulating member **9.3-220** of the first set of insulating members disposed at each discrete touch point **9.3-218** of the first set of discrete touch points.

[1550] In at least one example, the outer frame **9.3-215** can include a first material and the inner frame **9.3-214** can include a second material different than the first material. The first material of the outer frame **9.3-215** can include aesthetically pleasing materials, including aluminum, other metals, plastics, composite materials, and so forth. In at least one example, the second material of the inner frame **9.3-214** can include a material stronger and/or lighter than the first material of the outer frame **9.3-215**. In one example, the second material can include a magnesium alloy. Other examples of the second material can include other metals stiffer and/or stronger than the first material, including, metal alloys, carbon fiber, composite materials and alloys, and so forth.

[1551] The external/outer frame **9.3-215** can define the internal volume **9.3-226** as well as a first opening **9.3-122** (shown in the front, perspective view of the frame assembly **9.3-106** of FIGS. **9.3-1**) and a second opening **9.3-224** opposite the first opening **9.3-122**. In at least one example, the outer frame **9.3-215** can define one or more intake ports **9.3-228** between the first opening **9.3-122** and the second opening **9.3-224**. The intake ports **9.3-228** can include one or more apertures defined by the outer frame **9.3-215** through which the internal volume **9.3-226** can be in fluid communication with an external environment.

[1552] In at least one example, the external/outer frame **9.3-215** can define one or more exhaust ports shown on the top surface of the ventilation system **9.3-102** adjacent the input ports **9.3-230** between the first opening **9.3-122** and the second opening **9.3-224**. The exhaust ports can include one or more apertures defined by the outer frame **9.3-215** through which the internal volume **9.3-226** can be in fluid communication with the external environment. In at least one example, one or more of the exhaust ports can be centrally located between the first and second apertures **9.3-216a-b** of the internal frame **9.3-214**.

[1553] In at least one example, the fan assembly **9.3-204** of the HMD **9.3-100** can include a first

fan **9.3-212a** disposed in the internal volume **9.3-226** and a second fan **9.3-212b** disposed in the internal volume **9.3-226**. The first fan **9.3-212a** can be aligned with the first aperture **9.3-216a** and the second fan **9.3-212b** can be aligned with the second aperture **9.3-216b**. In general, the term “aligned” as used herein can refer to a relative position with two components adjacent or near one another. For example, the first fan **9.3-212a** can be aligned with the first aperture **9.3-216a** such that an air intake feature or opening of the first fan **9.3-212a** is positioned or configured to receive air into the first fan **9.3-212a** through the first aperture **9.3-216a**. In at least one example, the first fan **9.3-212a** can be “aligned” with the first aperture **9.3-216a** such that an axis of rotation of a fan blade assembly **9.3-217a** of the first fan **9.3-212a** can pass through the first aperture **9.3-216a**. In one example, the axis of rotation of the fan blade assembly **9.3-217a** can be parallel, aligned with, or close in proximity and angle with a central, normal axis of the first aperture **9.3-216a** defined by the inner frame **9.3-214**. Thus, the term “aligned” is used herein to denote the arrangement of the fans **9.3-212a-b** and respective apertures **9.3-216a-b** as shown in FIGS. **9.3-2**. The same use of the term “aligned” can be used to describe the relative position of the second fan **9.3-212b** with the second aperture **9.3-216a**, as illustrated in FIG. **9.3-2**.

[1554] In at least one example, the first fan **9.3-212a** and the second fan **9.3-212b** are configured to draw air into the internal volume **9.3-226** through the intake port **9.3-228** and push the air out from the internal volume **9.3-226** through the exhaust port. As noted above, the first and second fans **9.3-212a**, **9.3-212b** can be aligned with the first and second apertures **9.3-216a**, **9.3-216b** of the inner frame **9.3-214**, respectively, to draw air through the apertures **9.3-216a-b** and into the fans **9.3-212a-b** via fan inlets **9.3-232a**, **9.3-232b**, respectively. The first fan **9.3-212a** can include a housing **9.3-234a** defining the fan inlet **9.3-232a** through which air can enter the first fan **9.3-212a** before being blown through the exhaust port. Likewise, the second fan **9.3-212b** can include a housing **9.3-234b** defining the fan inlet **9.3-232b** through which air can enter the second fan **9.3-212b** before being blown through the exhaust port. The fan inlets **9.3-232a-b** can be adjacent to and/or aligned with the apertures **9.3-216a-b**, respectively, such that air flows into the internal volume **9.3-226** through the intake ports **9.3-228** then through the apertures **9.3-216a-b** and then through the fan inlets **9.3-232a-b**. The fans **9.3-212a-b** are configured to blow the air out from the internal volume **9.3-226** through the exhaust ports.

[1555] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **9.3-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **9.3-3-9.3-11** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **9.3-3-9.3-11** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **9.3-2**.

[1556] FIGS. **9.3-3** illustrates an example of a fan **9.3-312** including a housing **9.3-334**. The housing **9.3-334** can include a first housing shell **9.3-336** and a second housing shell **9.3-338** coupled to the first housing shell **9.3-336**. The first and second housing shells **9.3-336**, **9.3-338** can define an internal fan volume **9.3-340**. The first and second housing shells **9.3-336**, **9.3-338** can also be referred to as first and second housings, respectively. In at least one example, a plurality of fan fins or blades **9.3-342** are disposed in the internal fan volume **9.3-340** and configured or assembled to rotate about an axis of rotation **9.3-344**. This axis of rotation **9.3-344** can be the same axis of rotations discussed above with reference to FIGS. **9.3-2** and which is aligned to pass through the aperture **9.3-216a** or **9.3-216b** shown in FIGS. **9.3-2**. In at least one example, the first housing shell **9.3-336** and the second housing shell **9.3-338** can define a fan outlet **9.3-346** through which air is pushed by the blades **9.3-342** when rotated. In at least one example, the second housing shell **9.3-338** can define a fan inlet **9.3-332** through which the blades **9.3-342** are configured to draw air into the internal fan volume **9.3-340**.

[1557] In at least one example, the fan **9.3-312** can be coupled to a logic board assembly **9.3-310**

including one or more printed circuit boards (PCBs) and electronic components. In particular, the first housing shell **9.3-336** can be thermally coupled with or in direct thermal or mechanical contact with a heat generating component **9.3-348** of the logic board assembly **9.3-310**. In at least one example, the first housing shell **9.3-336** can be formed of or include a thermally conductive material, such as a metal, such that heat generated by the heat generating component **9.3-348** is conductively transferred through the first housing shell **9.3-336** and air blowing through the internal fan volume **9.3-340** can convectively cool the first housing shell **9.3-336**. In this way, heat generated by the heat generating component **9.3-348** is carried out through the fan outlet **9.3-346**.

[1558] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **9.3-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **9.3-2** and **9.3-4-9.3-11** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **9.3-2** and **9.3-4-9.3-11** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **9.3-3**.

[1559] FIGS. **9.3-4** illustrates a side, cross-sectional view of a fan **9.3-412** including a housing **9.3-434**. The housing **9.3-434** can include a first housing shell **9.3-436** and a second housing shell **9.3-438**, which can also be referred to as first and second housings, respectively, defining an internal fan volume **9.3-440**. In at least one example, a plurality of fan fins or blades **9.3-442** are disposed in the internal fan volume **9.3-440** and configured or assembled to rotate about an axis of rotation **9.3-444**. This axis of rotation **9.3-444** can be the same axis of rotations discussed above with reference to FIGS. **9.3-2** and which is aligned to pass through the aperture **9.3-216a** or **9.3-216b** shown in FIGS. **9.3-2**. In at least one example, the first housing shell **9.3-436** and the second housing shell **9.3-438** can define a fan outlet **9.3-446** through which air is pushed by the blades **9.3-442** when rotated. In at least one example, the second housing shell **9.3-438** can define a fan inlet (not shown in the side view of FIGS. **9.3-4**) through which the blades **9.3-442** are configured to draw air into the internal fan volume **9.3-440**.

[1560] In at least one example, the fan **9.3-412** can be coupled to a logic board assembly or printed circuit board (PCB) **9.3-410** including one or more electronic components. In particular, the first housing shell **9.3-436** can be thermally coupled with or in direct thermal or mechanical contact with a heat generating component **9.3-448** of the PCB **9.3-410**. In at least one example, the first housing shell **9.3-436** can be formed of or include a thermally conductive material, such as a metal, such that heat generated by the heat generating component **9.3-448** is conductively transferred through the first housing shell **9.3-436** and air blowing through the internal fan volume **9.3-440** can convectively cool the first housing shell **9.3-436**. In this way, heat generated by the heat generating component **9.3-448** is carried out through the fan outlet **9.3-446**.

[1561] In at least one example, the first housing shell **9.3-436** can be thermally coupled directly with the heat generating component **9.3-448** via a thermal paste or adhesive **9.3-450** disposed between the heat generating component **9.3-448** and the first housing shell **9.3-436**. In this way, the first housing shell **9.3-436** can be a heat sink disposed in the internal volume of an electronic device, such as the HMD **9.3-100** shown in FIGS. **9.3-1**. The housing **9.3-434** of the fan **9.3-412** can include the metal heat sink. As noted above and shown in FIGS. **9.3-2**, multiple fans **9.3-212a-b** can be disposed in the HMD **9.3-100** such that the fan **9.3-412** shown in FIGS. **9.3-4**, which can also serve as and thus be referred to as a heat sink for the heat generating component **9.3-448**, can be one of multiple fans including a second fan and heat sink of an HMD or other electronic device.

[1562] In at least one example, the system or device shown in FIGS. **9.3-4** can include an electrically conductive fence **9.3-452** coupled to the PCB **9.3-410** and surrounding a perimeter of the electronic/heat generating component **9.3-448**. As noted above, the fan **9.3-412** or fan assembly, can include the housing **9.3-434** having the first housing shell **9.3-436** coupled to the fence **9.3-452** and disposed over the electronic/heat generating component **9.3-448**. The first housing shell **9.3-**

436 and the fence **9.3-452** can form an electromagnetic interference (EMI) shield around the electronic/heat generating component **9.3-448** or any other electronic component coupled to or part of the PCB **9.3-410**.

[1563] In at least one embodiment, the fence **9.3-452**, which can also be referred to as an EMI fence **9.3-452**, can include an EMI foam **9.3-456** and a metal portion **9.3-454** disposed between the PCB **9.3-410** and the first housing shell **9.3-436** as shown, with the EMI foam **9.3-456** disposed between the metal fence **9.3-452** or fence portion/sidewall **9.3-454** and the second housing shell **9.3-436**. The metal fence or fence portion **9.3-454** can be disposed between the EMI foam **9.3-456** and the PCB **9.3-410**. In at least one example, housing **9.3-434**, including the second housing shell **9.3-436**, can be coupled to the metal EMI fence **9.3-452** or metal portion/sidewall **9.3-454** via the EMI foam **9.3-456**. In at least one example, the fence **9.3-452** or sidewall **9.3-454** can extend from the second housing shell **9.3-436** of the housing **9.3-434** of the fan **9.3-412**.

[1564] In at least one example, the first housing shell **9.3-436** defines or is disposed in a first major plane **9.3-458** parallel to the PCB **9.3-410**, or parallel to a second major plane **9.3-460** in which the PCB **9.3-410** is disposed or which the PCB **9.3-410** defines. In at least one example, the second housing shell **9.3-436** defines or includes a planar portion **9.3-464** disposed parallel to the PCB **9.3-410**, for example parallel to the second major plane **9.3-460**, and the heat generating component **9.3-448** is coupled to the planar portion **9.3-464**.

[1565] The fan **9.3-412** shown in FIGS. **9.3-4** can thus be part of an EMI shield assembly protecting the various component of the PCB or electronic components coupled thereto, including the heat generating component **9.3-448**. The metal housing **9.3-434**, and in particular the second housing shell **9.3-436** thereof, can form a shield volume **9.3-462** in which the heat generating component and other components associated with the PCB **9.3-410** can be disposed. The components within the shield assembly can include one or more antennas or other components sensitive to electromagnetic waves. In at least one example, the heat generating component **9.3-448** can include one or more processors or processing units. The EMI shielding assembly can also be referred to as a shield can and the fan **9.3-412** can be a part of a convective cooling EMI shield can that simultaneously acts to cool various components associated with the PCB **9.3-410** and shield those components from electromagnetic interference.

[1566] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **9.3-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **9.3-2-9.3-3** and **9.3-5-9.3-11** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **9.3-2-9.3-3** and **9.3-5-9.3-11** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **9.3-4**.

[1567] FIGS. **9.3-5** illustrates a plan view of a fan **9.3-512** having a first side or housing shell **9.3-536**. The housing shell **9.3-536** can define a fan inlet **9.3-532** and a plurality of fan blades **9.3-542** can be arranged and configured to draw air into the fan **9.3-512** when rotated. The housing shell **9.3-536** can also at least partially define a fan outlet **9.3-546** through which air is expelled from the fan when the blades **9.3-542** rotate.

[1568] In at least one example, the housing or housing shell **9.3-536** can include one or more discrete connection features or points **9.3-566**. These discrete connection features or points **9.3-566** can be used to couple the fan **9.3-512** to one or more other components of an electronic device, including the HMD **9.3-100** shown in **9.3-100**, by providing material features through which connectors or connector assemblies can extend or couple to one or more frames, chassis, or other structural components of the HMD **9.3-100**. In at least one example, the discrete connection features or points **9.3-566** can provide reduce surface area contacting other structural components, relative to a total area or perimeter of the housing shell **9.3-536**, such that heat transfer from the fan **9.3-512** (as pulled from the heat-generating components with which the fan can be contacting as

shown in FIGS. **9.3-4** and described above) to structural components with which a user may come into contact is limited.

[1569] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **9.3-5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **9.3-2-9.3-4** and **9.3-6-9.3-11** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **9.3-2-9.3-4** and **9.3-6-9.3-11** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **9.3-2**.

[1570] FIGS. **9.3-6** illustrates a plan view of a fan **9.3-612** having a second side or housing shell **9.3-638**. The housing shell **9.3-638** can at least partially define a fan outlet **9.3-646** through which air is expelled from the fan during operation.

[1571] In at least one example, the housing or housing shell **9.3-638** can include one or more discrete connection features or points **9.3-666**. These discrete connection features or points **9.3-666** can be used to couple the fan **9.3-612** to one or more other components of an electronic device, including the HMD **9.3-100** shown in **9.3-100**, by providing material features through which connectors or connector assemblies can extend or couple to one or more frames, chassis, or other structural components of the HMD **9.3-100**. In at least one example, the discrete connection features or points **9.3-666** can provide reduce surface area contacting other structural components, relative to a total area or perimeter of the housing shell **9.3-638**, such that heat transfer from the fan **9.3-612** (as pulled from the heat-generating components with which the fan can be contacting as shown in FIGS. **9.3-4** and described above) to structural components with which a user may come into contact is limited.

[1572] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **9.3-6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **9.3-2-9.3-5** and **9.3-7-9.3-11** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **9.3-2-9.3-5** and **9.3-7-9.3-11** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **9.3-6**.

[1573] FIGS. **9.3-7** is an exploded perspective view of a fan **9.3-712** that can be similar to the fans **9.3-512**, **9.3-612** shown in FIGS. **9.3-5** and **9.3-6**, respectively. The fan **9.3-712** can include a housing including a first housing shell **9.3-736** and an opposing second housing shell **9.3-738** defining an internal volume in which a fan blade assembly including a plurality of fan blades **9.3-742** are disposed. The first or second housing shells **9.3-736**, **9.3-738** can include or be coupled to one or more outlet fins **9.3-768** to direct air out through an outlet of the fan **9.3-712** defined by the first and/or second housing shells **9.3-736**, **9.3-738**.

[1574] In at least one example, the first housing shell **9.3-736** can include or be coupled to one or more discrete connection features or points **9.3-766** similar to those discrete connection features or points **9.3-566**, **9.3-666** described above with reference to FIGS. **9.3-5** and **9.3-6**, respectively. In at least one example, the second housing shell **9.3-738** can define a fan inlet **9.3-732** through which air is pulled into an internal volume of the fan **9.3-712**.

[1575] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **9.3-7** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **9.3-2-9.3-6** and **9.3-8-9.3-11** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **9.3-2-9.3-6** and **9.3-8-9.3-11** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **9.3-7**.

[1576] FIGS. **9.3-8** illustrates a perspective view of a fan assembly **9.3-804** including a first fan

9.3-812a and a second fan **9.3-812b**. Each fan **9.3-812a-b** can include respective housings **9.3-836a-b**. The housings **9.3-836a-b** can be similar to the first housing shell **9.3-736** shown in FIGS. **9.3-7** and the first housing shell **9.3-436** shown in FIGS. **9.3-4** and contacting the heat generating component **9.3-448**. In at least one example, the housings **9.3-836a-b**, or housing shells, can include respective etched portions **9.3-870a** and **9.3-870b**. The etched portions **9.3-870a** and **9.3-870b** can correspond in position and shape to the EMI fence **9.3-452** shown in FIGS. **9.3-4** such that the EMI fence **9.3-452** is coupled or in contact with the housings **9.3-836a-b** of the fans **9.3-812a-b** at the etched portions **9.3-870a** and **9.3-870b**, respectively. The etched portions **9.3-870a-b** can be formed using chemical etching, laser etching, or other etching processes used to prepare the surface of the housings **9.3-836a-b** for conductive contact with the EMI fence **9.3-452**. In at least one example, the etched portions **9.3-870a-b** are better suited than other portions or surfaces of the housings **9.3-836a-b** to make contact with the EMI fence **9.3-452** so that an effective EMI shield is formed as noted above with reference to at least FIGS. **9.3-4**.

[1577] In at least one example, as shown in FIGS. **9.3-8**, the first fan **9.3-812a** can be disposed in or define a first major plane **9.3-872a** and the second fan **9.3-812a** can be disposed in or define a second major plane **9.3-872b**. The term “major plane” when referring to major planes **9.3-872a-b** can include a plane that intersects or define a cross-sectional area of each fan **9.3-812a-b** parallel to a planar surface of the first or second housing shells of the respective housings **9.3-836a-b**. A cross-sectional areas of the fans **9.3-812a-b** disposed in the major planes **9.3-872a-b** are greater than cross-sectional areas of the fans **9.3-812a-b** disposed in planes orthogonal to the major planes **9.3-872a-b**. In at least one example, the fans **9.3-812a-b** include blades rotating about an axis or rotation normal to the major planes **9.3-872a-b**. These axes of rotation can be similar to the axis of rotation **9.3-444** shown in FIGS. **9.3-4** and the major planes **9.3-872a-b** can be similar to and/or parallel to the plane **9.3-458** shown in FIGS. **9.3-4**.

[1578] In at least one example, the first major plane **9.3-872a** and the second major plane **9.3-872b** are disposed at an angle θ relative to one another such that the first and second major planes **9.3-872a-b** are non-parallel with one another. The angle θ can accommodate a curvature of one or more other components of the HMD **9.3-100** shown in FIGS. **9.3-1**, including the frame, front cover and display screen, and so forth. This curvature, and the angle θ of the major planes **9.3-872a-b** relative to one another can be designed to accommodate and complements the curvature of a user's face, including relative angles of left and right facial features and brow/forehead/cheek curvatures. The angle θ can dispose the fans **9.3-812a-b** such that the fans **9.3-812a-b** can fit compactly within the HMD **9.3-100** and a general curvature thereof as the fans are aligned with apertures of the internal frame and various MLBs and other components disposed relative to one another at the same or similar angle θ .

[1579] In at least one example, the fan assembly **9.3-804** can include a graphite bridge between the fans **9.3-812a-b**. In at least one example, the fan assembly **9.3-804** can include tape disposed between bodies of the fans **9.3-812a-b** for heat transfer equalization between the fans **9.3-812a-b**.

[1580] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **9.3-8** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **9.3-2-9.3-7** and **9.3-9-9.3-11** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **9.3-2-9.3-7** and **9.3-9-9.3-11** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **9.3-8**.

[1581] FIGS. **9.3-9** illustrates a partially assembled view of an MLB **9.3-910** including one or more heat generating components **9.3-948a-b** surrounded by respective EMI fences **9.3-952a-b**. A fan **9.3-912** is illustrated in dotted lines to indicate a position of the fan overlaying the MLB **9.3-910** and coupled with the first EMI fence **9.3-952** forming a first EMI fence about the component **9.3-948** and other components illustrated on the MLB **9.3-910**. The fan **9.3-912** is transparent to

illustrate component otherwise visually blocked in the view of FIGS. 9.3-9.

[1582] In at least one example, a barrier **9.3-974** can be coupled to the MLB **9.3-910** surrounding the heat generating component **9.3-948b** with which the housing of a fan can be thermally coupled, as shown and described with reference to FIGS. 9.3-4. The barrier can extend from the MLB **9.3-910** to contact a housing or other surface of the fan **9.3-912** disposed against the heat generating component **9.3-948** to define a containment volume in which the heat generating component **9.3-948b** is disposed. In at least one example, the barrier **9.3-974** is configured to contain a thermal paste, such as the thermal paste **9.3-450** shown in FIGS. 9.3-4, thermally coupling the heat generating component **9.3-948** and the fan contacting the heat generating component **9.3-948b**, which is not shown but can be similar to the fan **9.3-912** shown in contact with the heat generating component **9.3-948a**. That is, during assembly, as the fan is pressed toward or against the heat generating component **9.3-948b**, the thermal paste disposed there between can expand outward but the barrier **9.3-974** contains the thermal paste and prevents it from contacting or contaminating other component of or on the MLB **9.3-910** outside the barrier **9.3-974**. In at least one example, the barrier **9.3-974** can include foam, silicone gel, other compressible materials, or other materials suitable for forming the barrier **9.3-974** as described above.

[1583] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 9.3-9 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. 9.3-2-9.3-8 and 9.3-10-9.3-11 and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. 9.3-2-9.3-8 and 9.3-10-9.3-11 can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 9.3-9.

[1584] FIGS. 9.3-10 illustrates a partial perspective view of an example of a fan **9.3-1012** coupled to an inner frame **9.3-1014** of an HMD device, which can be similar to the HMD **9.3-100** shown in FIGS. 9.3-1. In the example of FIGS. 9.3-10, the housing or housing shell **9.3-1036** of the fan **9.3-1012** can be coupled to the inner frame **9.3-1014** at a discrete touch point **9.3-1018** of a set of discrete touch points that minimize or reduce thermal transfer from the fan **9.3-1012** to the inner frame **9.3-1014**. In at least one example, the touch point **9.3-1018** can be an assembly including one or more insulating members or insulators **9.3-1020a** and **9.3-1020b** disposed between the housing shell **9.3-1036** of the fan **9.3-1012** and the inner frame **9.3-1014**.

[1585] In such an example, the internal/inner frame **9.3-1014** can be thermally isolated from the housing/housing shell **9.3-1036**. In at least one example, the touch point **9.3-1018** or touch point assembly can include an interface between the housing or housing shell **9.3-1036** of the fan **9.3-1012** including the insulator(s) **9.3-1020a-b** thermally isolating the fan **9.3-1012** from the internal frame **9.3-1014**.

[1586] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 9.3-10 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. 9.3-2-9.3-9 and 9.3-11 and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. 9.3-2-9.3-9 and 9.3-11 can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 9.3-10.

[1587] FIGS. 9.3-11 illustrates a side, cross-sectional view of an example of an HMD **9.3-1100** including an outer frame **9.3-1115** defining a first opening **9.3-1122** and a second opening **9.3-1124** and an internal volume **9.3-1126**. The outer frame **9.3-1115** can define an external surface of the HMD **9.3-1100** and be thus referred to as an external frame. In at least one example, a front cover assembly **9.3-1176** can be disposed in or across the first opening **9.3-1122** and a curtain assembly **9.3-1178** can occlude the second opening **9.3-1124**. In at least one example, the curtain assembly **9.3-1178** can include an elastic layer **9.3-1180** defining an external surface and an air-impermeable

layer **9.3-1182** defining the internal volume **9.3-1126**.

[1588] In at least one example, the external frame **9.3-1115** can define an intake port **9.3-1128** disposed between the first and second openings **9.3-1122**, **9.3-1124** and an exhaust port **9.3-1130** disposed between the first and second openings **9.3-1122**, **9.3-1124**. In particular, the external frame can include an upper sidewall **9.3-1113** defining the exhaust port **9.3-1130** and a lower sidewall **9.3-1119** defining the inlet port **9.3-1128**. The lower sidewall **9.3-1119** can be disposed opposite the upper sidewall **9.3-1113**.

[1589] In at least one example, the HMD **9.3-1100** includes an internal or inner frame **9.3-1114** disposed in the internal volume **9.3-1126** and coupled to the external frame **9.3-1115**. The inner frame **9.3-1114** can define an aperture **9.3-1116**. The HMD **9.3-1100** can also include a fan **9.3-1112** disposed in the internal volume **9.3-1126**. The fan can include a housing defining an internal fan volume in which a fan blade assembly **9.3-1117** is disposed. In at least one example, the housing of the fan **9.3-1112** can include a first housing shell **9.3-1136** and a second housing shell **9.3-1138**. In one example, the second housing shell **9.3-1138** can define a fan inlet **9.3-1132** and the first and second housing shells **9.3-1136**, **9.3-1138** can together define a fan outlet **9.3-1146**.

[1590] In at least one example, the HMD **9.3-1100** can include an MLB or PCB **9.3-1110** disposed in the internal volume **9.3-1126**. The HMD **9.3-1100** can also include one or more heat generating components **9.3-1148** coupled to the MLB **9.3-1110**. In at least one example, the first housing shell **9.3-1136** can be thermally coupled or in direct contact with the PCB **9.3-1110** and in particular with the heat generating component **9.3-1148**. In one or more examples, the second housing shell **9.3-1138** can be coupled to the inner frame **9.3-1114** and the fan inlet **9.3-1132** can be aligned with the aperture **9.3-1116** defined by the inner frame **9.3-1114** as shown. The second housing shell **9.3-1138** defining the fan inlet **9.3-1132** can face the display assembly, including the display screen **9.3-1188** of the optical module **9.3-1108**.

[1591] In at least one example, the HMD **9.3-1100** can include an optical module **9.3-1108** disposed in the internal volume **9.3-1126** and connected to the inner frame **9.3-1114**. In one example, the optical module **9.3-1108** is coupled to the inner frame **9.3-1114** via a bracket **9.3-1184**. In one example, the optical module **9.3-1108** can include a display assembly having a housing **9.3-1186**, a display screen **9.3-1188**, and a lens **9.3-1190** aligned with the display screen **9.3-1188**. The fan **9.3-1112** can be disposed between the display assembly of the optical module **9.3-1108** and the PCB **9.3-1110**. As shown, in at least one example, the curtain assembly **9.3-1178** can be coupled to the outer frame **9.3-1115** and occlude the second opening **9.3-1124** between the outer frame **9.3-1115** and the lens **9.3-1190**.

[1592] In at least one example, the fan **9.3-1112** is configured to draw air (represented by dotted arrows in FIGS. **9.3-11**) from an external environment into the internal volume **9.3-1126** through the intake port **9.3-1128** and push the air out from the internal volume through the exhaust port **9.3-1130**. In at least one example, the fan **9.3-1112** can be an air mover configured to draw air into the internal volume **9.3-1126** through the intake port **9.3-1128** and push the air out of the internal volume **9.3-1126** through the exhaust port **9.3-1130**.

[1593] In at least one example, the air inlet port **9.3-1128**, the fan inlet **9.3-1132**, the fan outlet **9.3-1146**, and the air exhaust port **9.3-1130** can define an airflow path (as indicated by the dotted arrows in FIGS. **9.3-11**) with the air inlet port **9.3-1128** upstream from the fan inlet **9.3-1132**, the fan inlet **9.3-1132** upstream from the fan outlet **9.3-1146** and the fan outlet **9.3-1146** upstream from the air exhaust port **9.3-1130**. The airflow path can thus be defined between the display assembly of the optical module **9.3-1108** and the fan inlet **9.3-1132**. In at least one example, the internal frame **9.3-1114** can be coupled to the external frame **9.3-1115** such that the internal frame **9.3-1114** is disposed between the display assembly, including the display screen **9.3-1188**, and the fan **9.3-1112** defining the fan inlet **9.3-1132** aligned with the aperture **9.3-1116** (or “opening”). In at least one example, the fan **9.3-1112** is coupled to the internal frame **9.3-1114** such that the internal frame **9.3-1114** is disposed between the fan **9.3-1112** and the optical module **9.3-1108** and the fan **9.3-1112** is

disposed between the internal frame **9.3-1114** and the PCB **9.3-1110**, including the heat generating component **9.3-1148** thermally coupled to the fan **9.3-1112**. In at least one example, the heat generating component **9.3-1148** can include a processor or multiple processors and components of a processing unit.

[1594] In this way, the fan **9.3-1112** is configured to convectively cool the display assembly, including the display screen **9.3-1188** of the optical module **9.3-1108** via the air flowing in the airflow path. The fan **9.3-1112** can also be configured to conductively cool the processor **9.3-1148** via direct contact with the housing or first housing shell **9.3-1136** of the housing, as shown. The air moving through the fan **9.3-1112** via the fan blade assembly **9.3-1117** can convectively cool the first housing shell **9.3-1136** such that the processor **9.3-1148** is indirectly convectively cooled by the fan **9.3-1112**. In at least one example, the processor **9.3-1148** is disposed outside the airflow path. In one example, the metal housing shell **9.3-1136** is disposed between the processor **9.3-1148** and the airflow path.

[1595] The external/outer frame **9.3-1115** can be referred to as a housing or an outer housing. In at least one example, the fan **9.3-1112** is configured to draw air through the inlet port **9.3-1128**, through the aperture **9.3-1116** of the inner frame **9.3-1114** downstream from the inlet port **9.3-1128**, through the fan inlet **9.3-1132** downstream from the aperture **9.3-1116**, and through the exhaust port **9.3-1130** downstream from the fan inlet **9.3-1132**. The aperture/opening **9.3-1116** defined by the internal frame **9.3-1114** can be aligned with the display screen **9.3-1188** and the fan **9.3-1112** can be disposed between the display screen **9.3-1188** and the processor **9.3-1148**. The fan **9.3-1112** can be aligned with the aperture. In at least one example, the fan outlet **9.3-1146** is positioned, oriented, and otherwise shaped or configured to direct the air away from the processor **9.3-1148** and the PCB **9.3-1110** in general as the fan **9.3-1112** pushed the air out through the exhaust port **9.3-1130** from the internal volume **9.3-1126** to the external environment. The fan outlet **9.3-1146** is configured to direct air toward the exhaust port **9.3-1130**.

[1596] In at least one example, the inner frame **9.3-1114** can define a first side **9.3-1192** and a second side **9.3-1194** opposite the first side **9.3-1192**. In at least one example, the optical module **9.3-1108**, including the display screen **9.3-1188** of the display assembly can be coupled to the first side **9.3-1192** of the inner frame **9.3-1114** and be aligned with the aperture **9.3-1116** defined by the inner frame **9.3-1114**. In at least one example, the fan **9.3-1112** can be coupled to the second side **9.3-1194** of the inner frame **9.3-1114** and be aligned with the aperture **9.3-1116**.

[1597] In the example of the HMD **9.3-1100** shown in FIGS. **9.3-11**, only one of the fan **9.3-1112**, optical module **9.3-1108**, aperture **9.3-1116**, MLB/PCB **9.3-1110**, and heat generating component **9.3-1148** are shown based on the cross-sectional side view of the HMD **9.3-1100**. However, it will be understood that examples of the HMD **9.3-1100** can include multiple of these components, including two fans, two optical modules, two apertures, two MLB/PCB components, and two heat generating components as illustrated in examples of other HMD devices and components illustrated in FIGS. **9.3-1-9.3-10**. Any of the features described with reference to these components in FIGS. **9.3-11** can be applied to the second of any of these components included in the HMD **9.3-1100**.

[1598] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **9.3-11** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **9.3-2-9.3-10** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **9.3-2-9.3-10** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **9.3-11**.

X: Chassis

[1599] FIGS. **10.0-1** illustrates a view of an example of an HMD **10-100** including a structural frame assembly **10-102**. The structural frame assembly **10-102** is described in more detail here in section X.

[1600] FIG. **10-1** illustrates a view of a head mountable device (HMD) **10-100** similar to examples described herein. The HMD **10-100** can include a structural frame assembly **10-102** referred to as a chassis or frame assembly including one or more frame components or sub-frames coupled together. The structural frame assembly **10-102** can be configured to provide mechanical strength and support to one or more other components of the HMD **10-100** coupled thereto. Other components are described herein and can include optical modules, fan assemblies, printed circuit boards, processors and other electronic computing components, sensor, displays, screens, and any other component disclosed herein.

[1601] The term “structural” as used herein can include components adding to the structural rigidity, durability, shape, and strength of the HMD **10-100**. As shown and noted elsewhere with reference to other figures, the frame assembly **10-102** can define two or more apertures, for example first and second apertures **10-104a**, **10-104b**, which align in position with corresponding fan assemblies **10-106a**, **10-106b**, which each include a fan. The apertures **10-104a**, **10-104b** can also correspond in position, when assembled, to corresponding optical modules **10-108a**, **10-108b**, which can each include a display screen.

[1602] FIG. **10-2** shows a partially assembled view of the HMD **10-100** shown in FIG. **10-1**, including a frame assembly **10-202** defining first and second apertures **10-204a**, **10-204b**. In at least one example, the frame assembly **10-202** can include an intermediate frame **10-210** defining the apertures **10-204a-b**. In at least one example, an optical mounting bracket **10-211** can be coupled to the intermediate frame **10-210** between the apertures **10-204a**, **10-204b**. The display screens of the optical modules **10-108a-b** shown in FIG. **10-1** can be coupled to first and second cantilevered guide rods **10-212a**, **10-212b** of the mounting bracket **10-211**. The optical modules **10-108a-b** are not shown in FIG. **10-2** in order to illustrate the intermediate frame **10-210**, the mounting bracket **10-211**, and the cantilevered guide rods **10-212a-b**, but when assembled, the optical modules **10-108a-b** can be coupled to respective guide rods **10-212a-b** such that the first optical module **10-108a** is aligned in position with the first aperture **10-204a** and the second optical module **10-108b** is aligned with the second aperture **10-204b**.

[1603] The optical mounting bracket **10-211** can also be referred to as a mounting frame or an optical mounting frame where respective display assemblies of the optical modules **10-108a-b** can slidably engage the guide rods **10-212a-b**. In at least one example, the mounting bracket **10-211** can include first and second cantilever arms **10-214a**, **10-214b** from which the respective guide rods **10-212a**, **10-212b** extend. In this way, one or more optical modules, assemblies, display assemblies, display screens, and so forth can be coupled to the mounting bracket **10-211** to align with respective apertures **10-204a**, **10-204b**. In this way, the fans of the fan assemblies **10-216a**, **10-216b** coupled to the intermediate frame **10-210** and aligned with the apertures **10-204a-b** as shown, can convectively cool the displays or display assemblies coupled to the guide rods **10-212a-b** through the apertures **10-204a-b**. The intermediate frame **10-210** can be disposed between the mounting bracket **10-211** and any display assemblies coupled thereto such that the display assemblies are mounted to a first side of the intermediate frame **10-210** and the fan assemblies **10-216a-b** are mounted to a second side of the intermediate frame **10-210** opposite the first side.

[1604] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **10-1** and **10-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **10-3-10-6** described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **10-3-10-6** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **10-1** and **10-2**.

[1605] FIG. **10-3** shows a perspective view of a sub-assembly of an HMD similar to the HMD **10-100** of FIG. **10-1**. The sub-assembly shown in FIGS. **9.3-3** includes a frame assembly **10-302**. The frame assembly **10-302** can be referred to as a chassis or a chassis assembly and include an outer

frame **10-318** and an intermediate frame **10-310**. The outer frame **10-318** can include a first material and define an internal volume **10-322**. In at least one example, the intermediate frame **10-310** can be disposed in the internal volume **10-322** and be coupled to the external frame **10-318**. The intermediate frame **10-310** can include a second material different than the first material. The intermediate frame **10-310** can define a first aperture **10-304a** and a second aperture **10-304b**. [1606] In at least one example, the frame assembly **10-302** can also include an inner frame **10-324** coupled to the intermediate frame **10-310**. In at least one example, the inner frame **10-324** can include a third material. In one example, the third material can be different than the first and second materials of the outer/external frame **10-318** and the intermediate frame **10-310**, respectively. In at least one example, the second material of the intermediate frame **10-310** is stronger than the first material of the outer frame **10-318**. In at least one example, the third material of the inner frame **10-324** is stiffer than the first material of the outer frame **10-318** and the second material of the intermediate frame **10-310**. In at least one example, the inner frame **10-324** is stiffer due to its geometry and/or size.

[1607] In at least one example, the outer frame **10-318** can define an outer visible surface of the HMD **10-100** and the internal volume **10-322**. In at least one example, the intermediate frame **10-310** can be coupled to the outer frame **10-318** and define a first aperture **10-304a** and a second aperture **10-304b**. In at least one example, the inner frame **10-324** can be coupled to the intermediate frame **10-310** between the first aperture **10-304a** and the second aperture **10-304b** on a first side **10-326** of the intermediate frame **10-324**.

[1608] The first material of the outer frame **10-318** can include any number of materials providing structure to the HMD **10-100** and can include aesthetically pleasing materials defining the external, visible surface of the HMD **10-100**. In at least one example, the first material can include aluminum, titanium, other metals, ceramics, plastics, composites, carbon fibers, and so forth. In at least one example, the second material of the intermediate frame **10-318** can include materials stronger than the first material. Examples of the second material can include magnesium and magnesium alloys. In at least one example the third material of the inner frame can include carbon fiber.

[1609] Accordingly, the first material of the outer frame **10-318** can be aesthetically pleasing and light-weight as well as color-customizable and so forth while the second material of the intermediate frame **10-310** can include a stronger, machine-able material for structural integrity and durability. The intermediate frame **10-310** can also thus be machined to include features compatible with other components of the HMD **10-100** coupling thereto. Also, the third material of the inner frame **10-324** can be stiffer to provide added structural strength and integrity in the center of the structure between the apertures **10-304a-b** to reduce deformation of the frame assembly **10-302** as a whole in case of user drop events and so forth.

[1610] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **10-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **10-1-10-2** and **10-4-10-6** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **10-1-10-2** and **10-4-10-6** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **10-3**.

[1611] FIG. **10-4** in at least one example, such as the frame assembly **10-402** shown in FIG. **10-4**, the frame assembly **10-402** can include an outer frame **10-418** and an inner frame **10-410**. The frame assembly **10-402** can be referred to as a chassis or a chassis assembly and include an outer frame **10-418** and an intermediate frame **10-410** without any inner frame coupled to the intermediate frame **10-410** as shown in the example of FIG. **10-3** with the inner frame **10-324** coupled to the intermediate frame **10-324**. Referring back to FIG. **10-4**, the outer frame **10-418** can include a first material and define an internal volume **10-422**. In at least one example, the

intermediate frame **10-410** can be disposed in the internal volume **10-422** and be coupled to the external frame **10-418**. The intermediate frame **10-410** can include a second material different than the first material. The intermediate frame **10-410** can define a first aperture **10-404a** and a second aperture **10-404b**.

[1612] In at least one example, the outer frame **10-418** can define an outer visible surface of the HMD **100** and the internal volume **10-422**. In at least one example, the intermediate frame **10-410** can be coupled to the outer frame **10-418** and define a first aperture **10-404a** and a second aperture **10-404b**.

[1613] The first material of the outer frame **10-418** can include any number of materials providing structure to the HMD **10-100** and can include aesthetically pleasing materials defining the external, visible surface of the HMD **10-100**. In at least one example, the first material can include aluminum, titanium, other metals, ceramics, plastics, composites, carbon fibers, and so forth. In at least one example, the second material of the intermediate frame **10-418** can include materials stronger than the first material. Examples of the second material can include magnesium and magnesium alloys.

[1614] Accordingly, the first material of the outer frame **10-418** can be aesthetically pleasing and light-weight as well as color-customizable and so forth while the second material of the intermediate frame **10-410** can include a stronger, machine-able material for structural integrity and durability. The intermediate frame **10-410** can also thus be machined to include features compatible with other components of the HMD **10-100** coupling thereto.

[1615] In at least one example, the intermediate frame **10-410** can be coupled to the outer frame **10-418** at discrete touch points **10-428** which limit the heat transfer between the intermediate frame **10-410** and the external/outer frame **10-418**. In at least one example, the intermediate frame **10-410** is not coupled continuously around an outer edge thereof to the external/outer frame **10-418**.

Rather, a set of discrete touch points **10-428** are the only physical contact between the intermediate frame **10-410** and the outer frame **10-418**. In this way, heat generated by components of the HMD **10-100** and transferred to the intermediate frame **10-410**, for example by heat generating components coupled to the intermediate frame **10-410**, does not substantially transfer to the user-facing, external surface defining outer frame **10-418** with which the user can make contact. In this way, the external/outer frame **10-418** can act as a thermal barrier between the user and the intermediate frame **10-410** and any heat generating components coupled thereto.

[1616] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **10-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **10-1-10-3** and **10-5-10-6** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **10-1-10-3** and **10-5-10-6** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **10-4**.

[1617] FIG. **10-5** illustrates a plan view of an example of a frame assembly **10-402** for an HMD device including an outer frame **10-518**, an intermediate frame **10-510** coupled to the outer frame **10-518** and an inner frame **10-524** coupled to the intermediate frame **10-510**. In at least one example, the intermediate frame **10-510** can define two apertures **10-504a**, **10-504b** and the inner frame **10-524** can be coupled to the inner frame **10-510** centrally between the apertures **10-504a-b**.

[1618] In at least one example, the inner frame **10-524** can be coupled mechanically to the intermediate frame **10-510** via one or more fasteners or fastener assemblies. In at least one example, the inner frame **10-524** can be coupled to the inner frame **10-510** via an adhesive such as glue or epoxy. In at least one example, the inner frame **10-524** is glued to the intermediate frame **10-510** along substantially an entire surface area of the inner frame **10-524** facing the intermediate frame **10-510**. In this way, mechanical stresses are transferred from the intermediate frame **10-510** to substantially all of the inner frame **10-524** to structurally support and stiffen the intermediate

frame **10-510** and thus the whole frame assembly **10-502**. In at least one example, the outer frame **10-518** surrounds a periphery of the intermediate frame **10-510**.

[1619] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **10-5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **10-1-10-4** and **10-6** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **10-1-10-4** and **10-6** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **10-5**.

[1620] FIG. **10-6** shows a close-up, cutaway view of an example of a touch point **10-628** where an intermediate frame **10-610** is coupled to an outer frame **10-618**. In at least one example, the intermediate frame **10-610** can be coupled to the outer frame **10-618** at discrete connection or touch points **10-628** which limit the heat transfer between the intermediate frame **10-610** and the external/outer frame **10-618**. In at least one example, the intermediate frame **10-610** is not coupled continuously around an outer edge thereof to the external/outer frame **10-618**. Rather, a set of discrete touch points **10-628** are the only physical contact between the intermediate frame **10-610** and the outer frame **9.3-618**. In this way, heat generated by components of the HMD **10-100** and transferred to the intermediate frame **10-610**, for example by heat generating components coupled to the intermediate frame **10-610**, does not substantially transfer to the user-facing, external surface defining outer frame **10-618** with which the user can make contact. In this way, the external/outer frame **10-618** can act as a thermal barrier between the user and the intermediate frame **10-610** and any heat generating components coupled thereto.

[1621] In at least one example, each of the discrete touch points **10-628** where the intermediate frame **10-610** couples to the external/outer frame **10-618** includes a fastener **10-630** and one or more insulating members, for example a first insulating member **10-632** disposed between the fastener **10-630** and the intermediate frame **10-610** and a second insulating member **10-634** disposed between the fastener **10-630** and the outer frame **10-618**. These insulating members **10-632**, **10-634** can be referred to as insulating interfaces or can be a part of an insulating interface to reduce heat transfer between the intermediate frame **10-610** and the outer frame **10-618**. In at least one example, the insulating members **10-632**, **10-634** can be O-rings. In at least one example, the insulating members **10-632**, **10-634** of the insulating interfaces at the touch points **10-628** can include insulating materials such as plastics, rubbers, foams, or other insulating, non-conductive materials. In at least one example, the fastener **10-630** can include an insulating material, such as a plastic, rubber, foam, or other non-conductive material preventing heat transfer between the intermediate frame **10-610** and the outer frame **10-618**.

[1622] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **10-6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **10-1-10-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **10-1-10-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **10-6**.

XI: Optical Module

[1623] FIG. **11-1** illustrates a view of an example of an HMD **11-100** including an optical module system **11-102**. The optical module system **11-102** is described in more detail here in section XI.

11.1: IPD Adjust

[1624] FIGS. **11.1-1** illustrates a rear perspective view of a display assembly **11.1-102** of an HMD **11.1-100** with one or more components removed, including a curtain assembly, in order to illustrate various internal components of the HMD **11.1-100**. FIG. **1** illustrates the display assembly **11.1-102**, which can include a first display module **11.1-104a** and a second display module **11.1-104b**.

The first display module **11.1-104a** can be slidably engaged/coupled with a first adjustment mechanism **11.1-106a** and the second optical module **11.1-104b** can be slidably engaged/coupled with a second adjustment mechanism **11.1-106b**. The first and second adjustment mechanisms **11.1-106a-b** can include first and second respective guide-rods **11.1-108a-b** and motors **11.1-110a-b**. The first and second guide-rods can be similar such that the description of one applies to the other, with the first guide-rod **11.1-108a** extending a first direction away from a bracket **11.1-112** and the second guide-rod **11.1-108b** extending a second direction away from the bracket **11.1-112** opposite the first direction.

[1625] Likewise, the first and second motors **11.1-110a-b** can be similar such that the description of one applies to the other, with the first motor **110a** mechanically engaging the first optical module **11.1-104a** to adjust the position thereof via the first guide-rod **11.1-108a** and the second motor **11.1-110b** mechanically engaging the second optical module **11.1-104b** to adjust the position thereof via the second guide-rod **11.1-108b**. In at least one example, the HMD **11.1-100** also includes a depressible and/or rotatable button **11.1-114** electrically coupled to the first and second motors **11.1-110a-b**. The button **11.1-114** can electrically communicate with the first and second motors **11.1-110a-b** via a processor or other circuitry components to cause the first and second motors **11.1-110a-b** to activate and cause the first and second optical modules **11.1-104a-b**, respectively, to change position relative to one another.

[1626] In at least one example, the first and second optical modules **11.1-104a-b** can include respective display screens configured to project light toward the user's eyes when donning the HMD **11.1-100**. In at least one example, the user can manipulate (i.e., depress and/or rotate) the button **11.1-114** to activate a positional adjustment of the optical modules **11.1-104a-b** to match the inter-pupillary distance of the user's eyes. The optical modules **11.1-104a-b** can also include one or more cameras or other sensors/sensor systems for imaging and measuring the IPD of the user such that the optical modules **11.1-104a-b** can be adjusted to match the IPD.

[1627] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **11.1-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **11.1-1**.

11.1.1: Crown

[1628] FIG. **11.1.1-1** illustrates a rear perspective view of an inter-pupillary distance (IPD) adjustment system **11.1.1-102** including first and second optical modules **11.1.1-104a-b** slidably engaging/coupled to respective guide-rods **11.1.1-108a-b** and motors **11.1.1-110a-b** of left and right adjustment subsystems **11.1.1-106a-b**. The IPD adjustment system **11.1.1-102** can be coupled to a bracket **11.1.1-112** and include a button **11.1.1-114** in electrical communication with the motors **11.1.1-110a-b**. In at least one example, the button **11.1.1-114** can electrically communicate with the first and second motors **11.1.1-110a-b** via a processor or other circuitry components to cause the first and second motors **11.1.1-110a-b** to activate and cause the first and second optical modules **11.1.1-104a-b**, respectively, to change position relative to one another.

[1629] In at least one example, the first and second optical modules **11.1.1-104a-b** can include respective display screens configured to project light toward the user's eyes when donning the HMD **11.1.1-100**. In at least one example, the user can manipulate (i.e., depress and/or rotate) the button **11.1.1-114** to activate a positional adjustment of the optical modules **11.1.1-104a-b** to match the inter-pupillary distance of the user's eyes. The optical modules **11.1.1-104a-b** can also include one or more cameras or other sensors/sensor systems for imaging and measuring the IPD of the user such that the optical modules **11.1.1-104a-b** can be adjusted to match the IPD.

[1630] In one example, the user can manipulate the button **11.1.1-114** to cause an automatic

positional adjustment of the first and second optical modules **11.1.1-104a-b**. In one example, the user can manipulate the button **11.1.1-114** to cause a manual adjustment such that the optical modules **11.1.1-104a-b** move further or closer away, for example when the user rotates the button **11.1.1-114** one way or the other, until the user visually matches her/his own IPD. In one example, the manual adjustment is electronically communicated via one or more circuits and power for the movements of the optical modules **11.1.1-104a-b** via the motors **11.1.1-110a-b** is provided by an electrical power source. In one example, the adjustment and movement of the optical modules **11.1.1-104a-b** via a manipulation of the button **11.1.1-114** is mechanically actuated via the movement of the button **11.1.1-114**.

[1631] In at least one example, the IPD adjustment system, including the button **11.1.1-114**, motors, guide-rods, optical modules **11.1.1-104a-b**, encoders, sensors, controllers, and/or other IPD adjustment components described herein, can include an automatic back-off functionality. In such an example, when the user pushes the button **11.1.1-104a-b** to adjust the optical modules **11.1.1-104a-b**, the optical modules **11.1.1-104a-b** can move closer together, for example toward a user's nose. At a predetermined/set position of the optical modules **11.1.1-104a-b** during IPD adjustment, the predetermined position being detected by the encoders or otherwise determined by the controller controlling the IPD adjustment motors, the controller can cause the optical modules **11.1.1-104a-b** to reverse direction and move away from one another, either after the user has released the button **11.1.1-114** and/or even if the user has not released the button **11.1.1-114**. In this way, the automatic back-off operation of the optical modules **11.1.1-104a-b** during IPD adjustments can prevent the optical modules **11.1.1-104a-b** from contacting or running into any facial features of the user or unintended contact against any other components of the HMD.

[1632] In at least one example, the automatic back-off function of the IPD adjust system can cause the optical modules **11.1.1-104a-b** to back off or reverse direction so the optical modules **11.1.1-104a-b** move away from one another a set distance away from where the unintended contact or facial feature interference position occurs. In at least one example, the automatic back-off function of the IPD adjust system can cause the optical modules **11.1.1-104a-b** to back off or reverse direction so the optical modules **11.1.1-104a-b** move away from one another to a set position predetermined to be a position where the optical modules **11.1.1-104a-b** are not in any contact with user facial features or any unintended contact with other HMD components.

[1633] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.1-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.1-1**.

11.1.1.1: Adjustment Mechanism for Head-Mounted Display

[1634] For improved performance, head-mounted displays require the alignment of optical components such as lenses, projection units, displays, screens, etc. to unique facial features among a variety of users in order to promote comfort, fit, performance, etc. During the alignment process, optical components can be physically repositioned in respect to other components within a head-mounted display. As packaging space is constrained within head-mounted displays, physical repositioning mechanisms should be small in size with tightly controlled motion, for example, small electro-mechanical actuators. Though small in size, the actuators should be able to withstand dynamic events and properly function after experiencing these dynamic events. Dynamic events can include shaking, jostling, dropping, impacting, or otherwise handling the head-mounted display in a manner that leads to forces being exerted upon optical components or subcomponents within the electro-mechanical actuators.

[1635] In an example where an actuator used to reposition optical components employs a lead

screw, locking-release mechanisms that allow disengagement or slip along the lead screw can be used in conjunction with dampening mechanisms that allow controlled sliding and/or stopping along a guide rail to prevent damage both to the actuator and to the optical components that undergo motion through use of the actuator. The locking-release and dampening mechanisms can include motor-driven, lead-screw system designs that leverage linear spring-loaded nut assemblies, torsional spring-loaded plates, drop-detection-based lock engagement or disengagement, and optical-component motion dampening during dynamic events. Detailed descriptions of these and other applicable mechanisms that improve performance of physical repositioning within a head-mounted display are described herein.

[1636] In a first aspect, a head-mounted display includes an optical assembly and an actuator. The actuator includes a movement mechanism configured to adjust a position of an optical component within the optical assembly, a locking-release mechanism configured to modify operation of the movement mechanism upon detection of a dynamic event, and a dampening mechanism configured to control positional changes of the optical component during the dynamic event.

[1637] In the first aspect, the movement mechanism can be configured to adjust the position of the optical component within the optical assembly to a pre-dynamic event position in accordance with detection of an end of the dynamic event. The movement mechanism can comprise a lead screw and a nut assembly, the nut assembly configured to translate along the lead screw based on rotation of the lead screw. The nut assembly can comprise separable first and second portions, and the locking-release mechanism can include a spring that linearly biases one of the first and second portions of the nut assembly against the lead screw. The modified operation of the movement mechanism can include one of the first and second portions of the nut assembly being disengaged from the lead screw. The dampening mechanism can comprise an electromagnetic component configured to generate attraction between the nut assembly and the lead screw during the dynamic event. The movement mechanism can comprise a lead screw and a plate configured to translate along the lead screw based on rotation of the lead screw. The locking-release mechanism can include a spring that torsionally biases the plate against the lead screw. Modified operation of the movement mechanism can include the plate being disengaged from the lead screw. The dampening mechanism can comprise an electromagnetic component configured to generate attraction between the plate and the lead screw during the dynamic event. The dampening mechanism can comprise an electromagnetic component configured to repel the movement mechanism from end portions of the actuator during the dynamic event. The dampening mechanism can comprise a spring or cable configured to restrict motion of the movement mechanism during the dynamic event. The first aspect may include any combination of the features described in this paragraph.

[1638] In a second aspect, a method includes detecting, using a sensor in a head-mounted display, a start of a dynamic event. In accordance with detection of the start of the dynamic event and using a locking-release mechanism, the method includes modifying operation of a movement mechanism. The movement mechanism is configured to adjust a position of an optical component within an optical assembly of the head-mounted display. The method includes detecting, using the sensor in the head-mounted display, an end of the dynamic event. In accordance with detection of the end of the dynamic event and using the movement mechanism, the method includes adjusting the position of the optical component within the optical assembly to a pre-dynamic event position.

[1639] In the second aspect, and in accordance with detection of the start of the dynamic event and using a dampening mechanism, the method can include controlling positional changes of the movement mechanism during the dynamic event. The dampening mechanism can comprise an electromagnetic component configured to repel components within the movement mechanism during the dynamic event. The dampening mechanism can comprise an electromagnetic component configured to cause attraction between components within the movement mechanism during the dynamic event. The dampening mechanism can comprise a spring or cable configured to restrict motion of the movement mechanism during the dynamic event. The movement mechanism can

comprise a lead screw and a nut assembly configured to translate along the lead screw based on rotation of the lead screw. The nut assembly can comprise separable first and second portions. The locking-release mechanism can include a spring that linearly biases one of the first and second portions of the nut assembly against the lead screw. Modified operation of the movement mechanism can include one of the first and second portions of the nut assembly being disengaged from the lead screw. The movement mechanism can comprise a lead screw and a plate configured to translate along the lead screw based on rotation of the lead screw. The locking-release mechanism can include a spring that torsionally biases the plate against the lead screw. Modified operation of the movement mechanism can include the plate being disengaged from the lead screw. The second aspect may include any combination of the features described in this paragraph.

[1640] In a third aspect, an actuator includes a movement mechanism configured to adjust a position of an optical component within an optical assembly of a head-mounted display. The movement mechanism includes a lead screw and a threaded component. The threaded component is configured to translate along the lead screw based on rotation of the lead screw. The actuator also includes a locking-release mechanism configured to modify operation of the movement mechanism upon detection of a dynamic event. The locking-release mechanism includes a spring configured to bias the threaded component against the lead screw. Modified operation of the movement mechanism includes the threaded component being disengaged from the lead screw. The actuator also includes a dampening mechanism configured to control positional changes of the movement mechanism during the dynamic event.

[1641] In the third aspect, the dampening mechanism can comprise an electromagnetic component configured to generate attraction between the threaded component and the lead screw during the dynamic event. The dampening mechanism can comprise an electromagnetic component configured to repel the movement mechanism from end portions of the actuator during the dynamic event. The third aspect may include any combination of the features described in this paragraph.

[1642] FIG. **11.1.1.1-1** is a top view of a head-mounted display **11.1.1.1-100** that includes a display assembly **11.1.1.1-102** and a head support **11.1.1.1-104**. The display assembly **11.1.1.1-102** can include a housing **11.1.1.1-106** and various sub-assemblies and electronics therein, such as one or more optical assemblies **11.1.1.1-108**, **11.1.1.1-110**, sensors **11.1.1.1-112** (e.g., detecting gaze and/or user status), power electronics **11.1.1.1-114**, and actuators **11.1.1.1-116**. Various portions or subcomponents of the optical assemblies **11.1.1.1-108**, **11.1.1.1-110** (e.g., such as lenses, projection units, screens, or displays) can be repositionable using the one or more actuators **11.1.1.1-116**, details of which are described further herein. The optical assemblies **11.1.1.1-108**, **11.1.1.1-110**, sensors **11.1.1.1-112**, power electronics **11.1.1.1-114**, and actuators **11.1.1.1-116** are depicted schematically and in dashed lines to illustrate being hidden from view and disposed within the display assembly **11.1.1.1-102**.

[1643] The head support **11.1.1.1-104** is coupled to left and right sides of the display assembly **11.1.1.1-102** (e.g., to the housing **11.1.1.1-106**), extending rearward of the display assembly **11.1.1.1-102** and between the left and right sides thereof. When the head-mounted display **11.1.1.1-100** is worn on a head of a user, the display assembly **11.1.1.1-102** extends across a front of the head of the user (i.e., the face of the user), while the head support **11.1.1.1-104** extends rearward along left and right sides of the head of the user and across a rear of the head of the user. Thus, the display assembly **11.1.1.1-102** and the head support **11.1.1.1-104** cooperatively extend around the head of the user. The head support **11.1.1.1-104** includes a band **11.1.1.1-118** and one or more circumferential adjustment mechanisms **11.1.1.1-120** (e.g., two as shown on the left and right sides)

[1644] The band **11.1.1.1-118** forms a primary portion of the head support **11.1.1.1-104**, which engages the head of the user to support the head-mounted display **11.1.1.1-100** worn on the head of the user. Each of the circumferential adjustment mechanisms **11.1.1.1-120** removably couples the head support **11.1.1.1-104** to the display assembly **11.1.1.1-102** and allows the head support **11.1.1.1-104** to change length overall by changing the length between the band **11.1.1.1-118** and

the circumferential adjustment mechanisms **11.1.1.1-120**. Though two circumferential adjustment mechanisms **11.1.1.1-120** are shown as acting from opposing sides of the head support **11.1.1.1-104**, the head support **11.1.1.1-104** may alternatively employ a single adjustment mechanism. [1645] The band **11.1.1.1-118** can be elastic, flexible, or otherwise deformable to accommodate different shapes of heads of the users. The band **11.1.1.1-118** can be formed of or otherwise include an elastic or silicone material. The band **11.1.1.1-118** may also be configured to transfer electrical power from a power source (e.g., an external battery) to the display assembly **11.1.1.1-102**. For example, the band **11.1.1.1-118** may include a flexible circuit **11.1.1.1-122** extending from a power connector **11.1.1.1-124**, through the circumferential adjustment mechanism **11.1.1.1-120**, to a corresponding power connector (not shown) of the display assembly **11.1.1.1-102**. The flexible circuit **11.1.1.1-122** (hidden and illustrated in dash-dot lines) may be embedded in the elastic or silicone material of the band **11.1.1.1-118** or otherwise hidden from view by the material of the band **11.1.1.1-118**. In this manner, the band **11.1.1.1-118** may supply electrical current to the display assembly **11.1.1.1-102** inclusive of the optical assemblies **11.1.1.1-108**, **11.1.1.1-110**, sensors **11.1.1.1-112**, power electronics **11.1.1.1-114**, and actuators **11.1.1.1-116**.

[1646] FIGS. **11.1.1.1-2A** and **11.1.1.1-2B** are detail and partially-exploded sectional views of an actuator **11.1.1.1-200**. The partially-exploded sectional view of FIG. **11.1.1.1-2B** is representative of the location indicated in FIG. **11.1.1.1-2A**. The actuator **11.1.1.1-200** may serve as the actuator **11.1.1.1-116** disposed in the head-mounted display **11.1.1.1-100** of FIG. **11.1.1.1-1**.

[1647] The actuator **11.1.1.1-200** includes a movement mechanism formed by a motor **11.1.1.1-202**, a lead screw **11.1.1.1-204**, a guide rail **11.1.1.1-206**, and a nut assembly **11.1.1.1-208**. The nut assembly **11.1.1.1-208** is configured to translate along the lead screw **11.1.1.1-204** based on rotation of the lead screw **11.1.1.1-204** and thread engagement between the nut assembly **11.1.1.1-208** and the lead screw **11.1.1.1-204**. For example, the lead screw **11.1.1.1-204** can be rotationally driven by the motor **11.1.1.1-202** to cause translation of the nut assembly **11.1.1.1-208**. In this manner, the movement mechanism is configured to adjust a position of an optical component (e.g., a lens, a display, a projector, a screen, a lens and a display, etc., not shown) within an optical assembly (e.g., within the optical assemblies **11.1.1.1-108**, **11.1.1.1-110** within the head-mounted display **11.1.1.1-100**). The threads on the lead screw **11.1.1.1-204**, the threads on the nut assembly **11.1.1.1-208**, or both, can be coated to provide a longer life for the actuator **11.1.1.1-200** and to reduce friction and noise associated with operation of the actuator **11.1.1.1-200**.

[1648] Positional adjustment of optical component(s) can be achieved using the described movement mechanism, for example, by coupling the optical component(s) that require(s) repositioning to a portion of the nut assembly **11.1.1.1-208**, though the coupling is not shown. The movement mechanism can be used to modify optical component spacing along a focal axis or modify interpupillary distance, for example, between optical assemblies within the same head-mounted display. The nut assembly **11.1.1.1-208** can be designed to avoid unintended rotation, for example, by use of the guide rail **11.1.1.1-206** extending parallel to the lead screw **11.1.1.1-204**, though other anti-rotation features in the actuator **11.1.1.1-200** are also possible. When the guide rail **11.1.1.1-206** is used as shown in FIGS. **11.1.1.1-2A** and **11.1.1.1-2B**, the nut assembly **11.1.1.1-208** translates along both the lead screw **11.1.1.1-204** and the guide rail **11.1.1.1-206** based on rotation of the lead screw **11.1.1.1-204** as driven by the motor **11.1.1.1-202**.

[1649] The actuator **11.1.1.1-200** also includes a locking-release mechanism formed by separable halves or portions **11.1.1.1-210**, **11.1.1.1-212** of the nut assembly **11.1.1.1-208** and one or more springs **11.1.1.1-214** that can linearly bias the portion **11.1.1.1-212** against the lead screw **11.1.1.1-204**, against the guide rail **11.1.1.1-206**, and against the portion **11.1.1.1-210** of the nut assembly **11.1.1.1-208** as is best seen in the partially-exploded sectional view of FIG. **11.1.1.1-2B**. In this example, the nut assembly **11.1.1.1-208** has a split-nut configuration with the portion **11.1.1.1-212** of the nut assembly **11.1.1.1-208** configured to translate away from the portion **11.1.1.1-210** of the nut assembly **11.1.1.1-208** when a load experienced is sufficient to overcome a load generated by

the springs **11.1.1.1-214** against the portion **11.1.1.1-212** of the nut assembly **11.1.1.1-208**.

Sufficient loads of this type can occur during dynamic events (e.g., shaking, jostling, dropping, or impacting a head-mounted display that includes the actuator **11.1.1.1-200**).

[1650] During a dynamic event that generates sufficient force, the threads on the lead screw **11.1.1.1-204** and the threads on the portions **11.1.1.1-210**, **11.1.1.1-212** can be disengaged, and the nut assembly **11.1.1.1-208** becomes free to slide along the lead screw **11.1.1.1-204** and the guide rail **11.1.1.1-206**, thus limiting or avoiding damage to the nut assembly **11.1.1.1-208** and/or to the lead screw **11.1.1.1-204** during the dynamic event based on operation of the described locking-release mechanism. After a dynamic event, when sufficient loads are no longer experienced by the actuator **11.1.1.1-200**, the threads on the lead screw **11.1.1.1-204** and the threads on the portions **11.1.1.1-210**, **11.1.1.1-212** of the nut assembly **11.1.1.1-208** can become reengaged, and the nut assembly **11.1.1.1-208** can once again be driven along the lead screw **11.1.1.1-204** by the motor.

[1651] Upon reengagement, the nut assembly **11.1.1.1-208** may be in a different position along the lead screw **11.1.1.1-204** than prior to or during the dynamic event. In this case, the movement mechanism may be configured to adjust the position of the nut assembly **11.1.1.1-208**, and in turn, any optical components coupled or otherwise connected to the nut assembly **11.1.1.1-208**, to a pre-dynamic event position, that is, to a position that was documented for the nut assembly **11.1.1.1-208** in respect to the lead screw **11.1.1.1-204** prior to the dynamic event. Though both portions **11.1.1.1-210**, **11.1.1.1-212** of the nut assembly **11.1.1.1-208** are shown as being threaded at the interface to the lead screw **11.1.1.1-204**, in some examples, only the portion **11.1.1.1-212** may include threads without impacting operation of the actuator **11.1.1.1-200**.

[1652] To effectively implement the locking-release mechanism of the actuator **11.1.1.1-200**, a position sensor (not shown) can be used to determine a position of the nut assembly **11.1.1.1-208** in respect to the lead screw **11.1.1.1-204** before, during, and after a dynamic event. An accelerometer, inertial measurement unit, or any other sensor (not shown) can be used to determine a beginning, duration, and end of the dynamic event. In another example, a magnetic encoder system can include a sensor that determines a landing position of the portions **11.1.1.1-210**, **11.1.1.1-212** of the nut assembly **11.1.1.1-208** in respect to the lead screw **11.1.1.1-204** (or other portion of the actuator **11.1.1.1-200**), and a controller can perform a recalibration, repositioning the portions **11.1.1.1-210**, **11.1.1.1-212** of the nut assembly **11.1.1.1-208** to pre-dynamic event positions along the lead screw **11.1.1.1-204** after the dynamic event.

[1653] FIGS. **11.1.1.1-3A** and **11.1.1.1-3B** are detail and partially-exploded sectional views of another actuator **11.1.1.1-300**. The partially-exploded sectional view of FIG. **11.1.1.1-3B** is representative of the location indicated in FIG. **11.1.1.1-3A**. The actuator **11.1.1.1-300** may serve as the actuator **11.1.1.1-116** disposed in the head-mounted display **11.1.1.1-100** of FIG. **11.1.1.1-1** and may be similar to the actuator **11.1.1.1-200** described in respect to FIGS. **11.1.1.1-2A** and **11.1.1.1-2B**. Differences between the actuator **11.1.1.1-300** and the actuator **11.1.1.1-200** are highlighted herein.

[1654] The actuator **11.1.1.1-300** includes a movement mechanism formed by a motor **11.1.1.1-302**, a lead screw **11.1.1.1-304**, a guide rail **11.1.1.1-306**, and a nut assembly **11.1.1.1-308**. The nut assembly **11.1.1.1-308** is configured to translate along the lead screw **11.1.1.1-304** and the guide rail **11.1.1.1-306** based on rotation of the lead screw **11.1.1.1-304** and thread engagement between the nut assembly **11.1.1.1-308** and the lead screw **11.1.1.1-304**. The nut assembly **11.1.1.1-308** can be designed to avoid unintended rotation, for example, by use of the guide rail **11.1.1.1-306** extending parallel to the lead screw **11.1.1.1-304**, though other anti-rotation features in the actuator **11.1.1.1-300** are also possible.

[1655] The actuator **11.1.1.1-300** also includes a locking-release mechanism formed by separable portions **11.1.1.1-310**, **11.1.1.1-311**, **11.1.1.1-312** of the nut assembly **11.1.1.1-308** and one or more springs **11.1.1.1-314** that can linearly bias the portion **11.1.1.1-312** against the guide rail **11.1.1.1-306** and the central-most portion **11.1.1.1-311** of the nut assembly **11.1.1.1-308**. In turn, the portion

11.1.1.1-311 can be linearly biased against the lead screw **11.1.1.1-304** and the portion **11.1.1.1-310** of the nut assembly **11.1.1.1-308** as is best seen in the partially-exploded sectional view of FIG. **11.1.1.1-3B**. In this example, the nut assembly **11.1.1.1-308** has three portions **11.1.1.1-310**, **11.1.1.1-311**, **11.1.1.1-312**, with the portions **11.1.1.1-311**, **11.1.1.1-312** of the nut assembly **11.1.1.1-208** configured to translate away and become disengaged from the portion **11.1.1.1-310** of the nut assembly **11.1.1.1-308** when a load experienced is sufficient to overcome a load generated by the springs **11.1.1.1-314** against the portion **11.1.1.1-312** of the nut assembly **11.1.1.1-208**. Sufficient loads of this type can occur during dynamic events (e.g., shaking, jostling, dropping, or impacting a head-mounted display that includes the actuator **11.1.1.1-300**).

[1656] The actuator **11.1.1.1-300** of FIGS. **11.1.1.1-3A** and **11.1.1.1-3B** differs from the actuator **11.1.1.1-200** of FIGS. **11.1.1.1-2A** and **11.1.1.1-2B** in that the nut assembly **11.1.1.1-308** is used in place of the nut assembly **11.1.1.1-208**. The actuator **11.1.1.1-300** of FIGS. **11.1.1.1-3A** and **11.1.1.1-3B** may be useful in vertically tight packaging spaces, that is, the actuator **11.1.1.1-300** may be shorter in height than the actuator **11.1.1.1-200** due to the construction of the nut assemblies **11.1.1.1-208**, **11.1.1.1-308**. In comparison, the actuator **11.1.1.1-200** of FIGS. **11.1.1.1-2A** and **11.1.1.1-2B** may be useful in horizontally tight packaging spaces, that is, the actuator **11.1.1.1-200** may be narrower in width than the actuator **11.1.1.1-300** due to the construction of the nut assemblies **11.1.1.1-208**, **11.1.1.1-308**.

[1657] FIGS. **11.1.1.1-4A** and **11.1.1.1-4B** are detail and partially-exploded sectional views of another actuator **11.1.1.1-400**. The partially-exploded sectional view of FIG. **11.1.1.1-4B** is representative of the location indicated in FIG. **11.1.1.1-4A**. The actuator **11.1.1.1-400** may serve as the actuator **11.1.1.1-116** disposed in the head-mounted display **11.1.1.1-100** of FIG. **11.1.1.1-1** and may be similar to the actuators **11.1.1.1-200**, **11.1.1.1-300** described in respect to FIGS. **11.1.1.1-2A**, **11.1.1.1-2B**, **11.1.1.1-3A**, and **11.1.1.1-3B**. Differences between the actuator **11.1.1.1-400** and the actuators **11.1.1.1-200**, **11.1.1.1-300** are highlighted herein.

[1658] The actuator **11.1.1.1-400** includes a movement mechanism formed by a motor **11.1.1.1-402**, a lead screw **11.1.1.1-404**, a guide rail **11.1.1.1-406**, and a torsion assembly **11.1.1.1-408**. The torsion assembly **11.1.1.1-408** is configured to translate along the lead screw **11.1.1.1-404** and the guide rail **11.1.1.1-406** based on rotation of the lead screw **11.1.1.1-404** and thread engagement between the torsion assembly **11.1.1.1-408** and the lead screw **11.1.1.1-404**. The torsion assembly **11.1.1.1-408** can be designed to avoid unintended rotation, for example, by use of the guide rail **11.1.1.1-406** extending through the torsion assembly **11.1.1.1-408**, though other anti-rotation features in the actuator **11.1.1.1-400** are also possible.

[1659] The torsion assembly **11.1.1.1-408** includes a plate **11.1.1.1-410**, a support **11.1.1.1-412**, and a torsional spring **11.1.1.1-414** that can torsionally bias the plate **11.1.1.1-410** against the lead screw **11.1.1.1-404** to form a locking-release mechanism of the actuator **11.1.1.1-400**. In this example, the torsional spring **11.1.1.1-414** acts against both the plate **11.1.1.1-410** and the support **11.1.1.1-412** such that the plate **11.1.1.1-410** is configured to rotate away and become disengaged from the lead screw **11.1.1.1-404** when a load experienced is sufficient to overcome a load generated by the spring **11.1.1.1-414** against the plate **11.1.1.1-410** and the support **11.1.1.1-412**. Sufficient loads of this type can occur during dynamic events (e.g., shaking, jostling, dropping, or impacting a head-mounted display that includes the actuator **11.1.1.1-400**).

[1660] The actuator **11.1.1.1-400** of FIGS. **11.1.1.1-4A** and **11.1.1.1-4B** differs from the actuators **11.1.1.1-200**, **11.1.1.1-300** of FIGS. **11.1.1.1-2A**, **11.1.1.1-2B**, **11.1.1.1-3A**, and **11.1.1.1-3B** in that the torsion assembly **11.1.1.1-408** with the rotatable plate **11.1.1.1-410** is used in place of the nut assemblies **11.1.1.1-208**, **11.1.1.1-308**. Use of the actuator **11.1.1.1-400** of FIGS. **11.1.1.1-4A** and **11.1.1.1-4B** can support unique positional relationships between the lead screw **11.1.1.1-404** and the guide rail **11.1.1.1-406** since these components can be both vertically and horizontally offset while remaining generally parallel. Further, there is less physical engagement between the plate **11.1.1.1-410** and the lead screw **11.1.1.1-404** than is present between the nut assemblies **11.1.1.1-**

208, 11.1.1.1-308 and the lead screws **11.1.1.1-204, 11.1.1.1-304**, reducing for example, friction-based and wear-based failure potential.

[1661] The actuators **11.1.1.1-200, 11.1.1.1-300, 11.1.1.1-400** described in respect to FIGS.

11.1.1.1-2A, 11.1.1.1-2B, 11.1.1.1-3A, 11.1.1.1-3B, 11.1.1.1-4A, and 11.1.1.1-4B include split-nut and torsional-type movement mechanisms using lead screws **11.1.1.1-204, 11.1.1.1-304, 11.1.1.1-404**. Additional designs for movement mechanisms can include living-hinge nut assemblies and back-driveable lead screws (not shown). For example, a living-hinge nut assembly can be separated along a central axis with coupled top and bottom edges such that the sides of the nut assembly disengage from the lead screw under sufficient force while the top and bottom edges remain coupled. In another example, a back-driveable lead screw can be controlled to reduce forces experienced during a dynamic event by enabling back-driving during the dynamic event. In addition, movement mechanisms may use rack-and-pinion, pneumatic, hydraulic, piezo-actuated, and electromagnetic designs (not shown) to achieve positioning and repositioning of optical components within optical assemblies.

[1662] A consequence of using the locking-release mechanisms described in respect to FIGS.

11.1.1.1-2A, 11.1.1.1-2B, 11.1.1.1-3A, 11.1.1.1-3B, 11.1.1.1-4A, and 11.1.1.1-4B is that the nut assemblies **11.1.1.1-208, 11.1.1.1-308** and torsion assembly **11.1.1.1-408** can become generally free-sliding along the guide rails **11.1.1.1-206, 11.1.1.1-306, 11.1.1.1-406** during a dynamic event sufficient to disengage the nut assemblies **11.1.1.1-208, 11.1.1.1-308** and torsion assembly **11.1.1.1-408** from the lead screws **11.1.1.1-204, 11.1.1.1-304, 11.1.1.1-404**. To avoid damage to optical components having motion controlled by the actuators **11.1.1.1-200, 11.1.1.1-300, 11.1.1.1-400**, various dampening mechanisms can be implemented to control positional changes of the otherwise free-sliding optical components during dynamic events.

[1663] FIGS. **11.1.1.1-5A** and **11.1.1.1-5B** are detail views of magnetic dampening mechanisms for an actuator **11.1.1.1-500** similar to the actuators **11.1.1.1-200, 11.1.1.1-300, 11.1.1.1-400** of FIGS.

11.1.1.1-2A, 11.1.1.1-2B, 11.1.1.1-3A, 11.1.1.1-3B, 11.1.1.1-4A, and 11.1.1.1-4B. The actuator **11.1.1.1-500** may serve as the actuator **11.1.1.1-116** disposed in the head-mounted display **11.1.1.1-100** of FIG. **11.1.1.1-1**. The actuator **11.1.1.1-500** includes a movement mechanism formed by a motor **11.1.1.1-502**, a lead screw **11.1.1.1-504**, a guide rail **11.1.1.1-506**, and a nut assembly **11.1.1.1-508**. The nut assembly **11.1.1.1-508** is configured to translate along the lead screw **11.1.1.1-504** and the guide rail **11.1.1.1-506** based on rotation of the lead screw **11.1.1.1-504** and thread engagement between the nut assembly **11.1.1.1-508** and the lead screw **11.1.1.1-504**.

[1664] Locking-release mechanisms similar to those described in reference to FIGS. **11.1.1.1-2A, 11.1.1.1-2B, 11.1.1.1-3A, and 11.1.1.1-3B**, can cause the nut assembly **11.1.1.1-508** to become disengaged from the lead screw **11.1.1.1-504**, for example, during a dynamic event. To minimize free sliding of the nut assembly **11.1.1.1-508**, the actuator **11.1.1.1-500** can include an electromagnetic dampening mechanism formed by an electromagnetic component **11.1.1.1-516** disposed within or forming a portion of the nut assembly **11.1.1.1-508** as shown in dotted line in FIGS. **11.1.1.1-5A** and **11.1.1.1-5B**.

[1665] Referring to FIG. **11.1.1.1-5A**, the electromagnetic dampening mechanism can include electromagnetic components **11.1.1.1-518, 11.1.1.1-520** disposed at ends of the guide rail **11.1.1.1-506** as shown in dotted line for hidden representation. In this embodiment, the electromagnetic components **11.1.1.1-518, 11.1.1.1-520** can be controlled to actuate and repel the electromagnetic component **11.1.1.1-516** during a dynamic event, thus serving as cushions or bumpers to slow or prohibit the nut assembly **11.1.1.1-508** from impacting the ends of the guide rail **11.1.1.1-506** when sliding. In other embodiments, the nut assembly **11.1.1.1-508** may include portions formed of ferritic or other material susceptible to magnetic repulsion, and the nut assembly **11.1.1.1-508** may be repelled by the electromagnetic components **11.1.1.1-518, 11.1.1.1-520** during dynamic events.

[1666] Referring to FIG. **11.1.1.1-5B**, the electromagnetic dampening mechanism can include electromagnetic component **11.1.1.1-522** that runs the length of the guide rail **11.1.1.1-506** as

shown in dotted line. In this embodiment, the electromagnetic component **11.1.1.1-522** can be controlled to actuate and attract the electromagnetic component **11.1.1.1-516** during a dynamic event, thus serving to slow motion of the nut assembly **11.1.1.1-508** during the dynamic event due to attraction between the electromagnetic components **11.1.1.1-516**, **11.1.1.1-522**. In other embodiments, the guide rail **11.1.1.1-506** may be formed of ferritic or other material susceptible to magnetic attraction, and the electromagnetic component **11.1.1.1-516** may be attracted to the guide rail **11.1.1.1-506** during dynamic events.

[1667] Other magnetic dampening mechanisms are also possible. For example, the electromagnetic components **11.1.1.1-516**, **11.1.1.1-518**, **11.1.1.1-520**, **11.1.1.1-522** may be designed to repel in non-powered states and attract in powered states (or vice-versa), and some of the electromagnetic components **11.1.1.1-516**, **11.1.1.1-518**, **11.1.1.1-520**, **11.1.1.1-522** may be replaced with non-powered ferritic or other magnetically-susceptible materials without changing the overall function of the dampening mechanism.

[1668] FIG. **11.1.1.1-6A** and **11.1.1.1-6B** are detail views of mechanical dampening mechanisms for an actuator **11.1.1.1-600** similar to the actuators **11.1.1.1-200**, **11.1.1.1-300**, **11.1.1.1-500** of FIGS. **11.1.1.1-2A**, **11.1.1.1-2B**, **11.1.1.1-3A**, **11.1.1.1-3B**, **11.1.1.1-5A**, and **11.1.1.1-5B**. The actuator **11.1.1.1-600** may serve as the actuator **11.1.1.1-116** disposed in the head-mounted display **11.1.1.1-100** of FIG. **11.1.1.1-1**. The actuator **11.1.1.1-600** includes a movement mechanism formed by a motor **11.1.1.1-602**, a lead screw **11.1.1.1-604**, a guide rail **11.1.1.1-606**, and a nut assembly **11.1.1.1-608**. The nut assembly **11.1.1.1-608** is configured to translate along the lead screw **11.1.1.1-604** and the guide rail **11.1.1.1-606** based on rotation of the lead screw **11.1.1.1-604** and thread engagement between the nut assembly **11.1.1.1-608** and the lead screw **11.1.1.1-604**.

[1669] Locking-release mechanisms similar to those described in reference to FIGS. **11.1.1.1-2A**, **11.1.1.1-2B**, **11.1.1.1-3A**, and **11.1.1.1-3B** can cause the nut assembly **11.1.1.1-608** to become disengaged from the lead screw **11.1.1.1-604**, for example, during a dynamic event. To minimize free sliding of the nut assembly **11.1.1.1-608**, the actuator **11.1.1.1-600** can include a mechanical dampening mechanism such as those described in reference to FIGS. **11.1.1.1-6A** and **11.1.1.1-6B**.

[1670] Referring to FIG. **11.1.1.1-6A**, the mechanical dampening mechanism can include one or more cables **11.1.1.1-626**, **11.1.1.1-628** (two are shown in solid and dotted line) disposed proximate to ends of the lead screw **11.1.1.1-604** and the guide rail **11.1.1.1-606**. In this embodiment, the cables **11.1.1.1-626**, **11.1.1.1-628** can either wind or unwind during a dynamic event, serving to slow or counteract motion of the nut assembly **11.1.1.1-608** to avoid having the nut assembly **11.1.1.1-608** impact the ends of the lead screw **11.1.1.1-604** and guide rail **11.1.1.1-606** when sliding. For example, the cables **11.1.1.1-626**, **11.1.1.1-628** can be engaged under high torque conditions, or the cables **11.1.1.1-626**, **11.1.1.1-628** can be designed with a predetermined amount of unwinding or stretching that occurs before stopping movement of the nut assembly **11.1.1.1-608**.

[1671] Referring to FIG. **11.1.1.1-6B**, the mechanical dampening mechanism can include one or more springs **11.1.1.1-630**, **11.1.1.1-632** (two are shown) disposed proximate to ends of the lead screw **11.1.1.1-604** and the guide rail **11.1.1.1-606**. In this embodiment, the springs **11.1.1.1-630**, **11.1.1.1-632** can compress or expand during a dynamic event, serving to slow or counteract motion of the nut assembly **11.1.1.1-608** to avoid having the nut assembly **11.1.1.1-608** impact the ends of the lead screw **11.1.1.1-604** and guide rail **11.1.1.1-606** when sliding. In another example, the springs **11.1.1.1-630**, **11.1.1.1-632** can be coupled to bumpers or electromagnetic components in order to combine mechanical and electromagnetic dampening means (not shown).

[1672] FIG. **11.1.1.1-7** is a flowchart depicting a process **11.1.1.1-700** of operation for an actuator similar to the actuators **11.1.1.1-200**, **11.1.1.1-300**, **11.1.1.1-400**, **11.1.1.1-500**, **11.1.1.1-600** of FIGS. **11.1.1.1-2A**, **11.1.1.1-2B**, **11.1.1.1-3A**, **11.1.1.1-3B**, **11.1.1.1-4A**, **11.1.1.1-4B**, **11.1.1.1-5A**, **11.1.1.1-5B**, **11.1.1.1-6A**, and **11.1.1.1-6B** disposed within a head-mounted display similar to the head-mounted display **11.1.1.1-100** of FIG. **11.1.1.1-1**.

[1673] At **11.1.1.1-710**, the process **11.1.1.1-700** includes detecting a start of a dynamic event, for

example, using a sensor such as a position sensor, an accelerometer, an inertial measurement unit, or other sensor within or associated with the head-mounted display. The dynamic event can include events such as shaking, jostling, dropping, impacting, or otherwise handling the head-mounted display in a manner that leads to forces above a disengagement threshold being exerted upon components of the actuator and optical components of an optical assembly.

[1674] At **11.1.1.1-720**, the process **11.1.1.1-700** includes activating a locking-release mechanism such as the locking-release mechanisms described in reference to the actuators **11.1.1.1-200**, **11.1.1.1-300**, **11.1.1.1-400**. The locking-release mechanism can be activated in accordance with detection of the start of the dynamic event, for example, based on or associated with detection of the start of the dynamic event. The locking-release mechanism can modify operation of the movement mechanism within the associated actuator. The movement mechanism can include a lead screw and a nut assembly or a torsion assembly. Modifying operation of the movement mechanism can include disengaging the nut assembly (e.g., the nut assemblies **11.1.1.1-208**, **11.1.1.1-308**) or the torsion assembly (e.g., the torsion assembly **11.1.1.1-408**) from the lead screw (e.g., the lead screws **11.1.1.1-204**, **11.1.1.1-304**, **11.1.1.1-404**).

[1675] At **11.1.1.1-730**, shown in dotted line to represent its optional nature, the process **11.1.1.1-700** includes activating a dampening mechanism such as the dampening mechanisms described in reference to the actuators **11.1.1.1-500**, **11.1.1.1-600**. The dampening mechanism can be activated in accordance with detection of the start of the dynamic event. In another example, the dampening mechanism can be activated in accordance with detected motion of the movement mechanism that surpasses a predetermined threshold speed or distance, indicating that a dynamic event is ongoing. The dampening mechanism can include electromagnetic components susceptible to attraction or repulsion or mechanical components such as springs and cables.

[1676] At **11.1.1.1-740**, the process **11.1.1.1-700** includes detecting an end of the dynamic event, for example, using a sensor such as a position sensor, an accelerometer, an inertial measurement unit, or other sensor within or associated with the head-mounted display. The end of the dynamic event can also be detected based on detecting arrested motion of components within a movement mechanism that were previously in motion during the dynamic event.

[1677] At **11.1.1.1-750**, shown in dotted line to represent its optional nature, the process **11.1.1.1-700** includes adjusting a position of an optical component within an optical assembly to a pre-dynamic event position. The movement mechanism can be used to move, for example, a nut assembly or a torsion assembly within an actuator from its current post-dynamic event position to the pre-dynamic event position. Movement of the optical component is tied to movement of the movement mechanism in this example. Repositioning supports improved function of the overall head-mounted display as the user will not be required to readjust optical components within optical assemblies after a dynamic event.

[1678] FIG. **11.1.1.1-8** shows an example of a hardware configuration for a controller **11.1.1.1-800** that may be used to implement portions of the head-mounted display **11.1.1.1-100**. In the illustrated example, the controller **11.1.1.1-800** includes a processor **11.1.1.1-802**, a memory device **11.1.1.1-804**, a storage device **11.1.1.1-806**, one or more input devices **11.1.1.1-808**, and one or more output devices **11.1.1.1-810**. These components may be interconnected by hardware such as a bus **11.1.1.1-812** that allows communication between the components.

[1679] The processor **11.1.1.1-802** may be a conventional device such as a central processing unit and is operable to execute computer program instructions and perform operations described by the computer program instructions. The memory device **11.1.1.1-804** may be a volatile, high-speed, short-term information storage device such as a random-access memory module. The storage device **11.1.1.1-806** may be a non-volatile information storage device such as a hard drive or a solid-state drive. The input devices **11.1.1.1-808** may include sensors and/or any type of human-machine interface, such as buttons, switches, a keyboard, a mouse, a touchscreen input device, a gestural input device, or an audio input device. The output devices **11.1.1.1-810** may include any

type of device operable to provide an indication to a user regarding an operating state, such as a display screen, a light-control panel, or an audio output.

11.1.1.2: Crown Input and Feedback for Head-Mountable Devices

[1680] It can be desirable to provide a mechanism for a user to provide inputs to a head-mountable device to facilitate user interaction with the head-mountable device. It can be further desirable to provide a mechanism for providing feedback to the user. Such feedback can be provided in the form of haptic feedback delivered to the user. However, haptic feedback can feel unpleasant when applied across an entire device that is mounted on a head of the user. Where the user is providing tactile inputs by contacting an input member with another portion of the body, such as a finger or hand, the haptic feedback can be locally applied to that portion of the user's body, so that the haptic feedback is delivered in a way that is effective and pleasant to the user.

[1681] Systems of the present disclosure can provide a head-mountable device with a crown module with an input system that allows a user to provide inputs by rotating or otherwise applying torque to a crown of the crown module. The head-mountable device can interpret the rotation and/or torque as a user input. The crown module can further include a feedback system that provides localized haptic feedback at the crown. The haptic feedback can be effectively perceived by the user at the crown without causing the entire head-mountable device to vibrate against the head and/or face of the user.

[1682] According to some embodiments, for example as shown in FIG. **11.1.1.2-1**, a head-mountable device **11.1.1.2-100** includes a frame **11.1.1.2-110** that is worn on a head of a user. The frame **11.1.1.2-110** can be positioned in front of the eyes of a user to provide information within a field of view of the user. The frame **11.1.1.2-110** can provide nose pads or another feature to rest on a user's nose. The frame **11.1.1.2-110** can be supported on a user's head with the securement element **11.1.1.2-120**. The securement element **11.1.1.2-120** can wrap or extend along opposing sides of a user's head. The securement element **11.1.1.2-120** can include earpieces for wrapping around or otherwise engaging or resting on a user's ears. It will be appreciated that other configurations can be applied for securing the head-mountable device **11.1.1.2-100** to a user's head. For example, one or more bands, straps, belts, caps, hats, or other components can be used in addition to or in place of the illustrated components of the head-mountable device **11.1.1.2-100**. By further example, the securement element **11.1.1.2-120** can include multiple components to engage a user's head.

[1683] The frame **11.1.1.2-110** can provide structure around a peripheral region thereof to support any internal components of the frame **11.1.1.2-110** in their assembled position. For example, the frame **11.1.1.2-110** can enclose and support various internal components (including for example integrated circuit chips, processors, memory devices and other circuitry) to provide computing and functional operations for the head-mountable device **11.1.1.2-100**, as discussed further herein. Any number of components can be included within and/or on the frame **11.1.1.2-110** and/or the securement element **11.1.1.2-120**.

[1684] The frame **11.1.1.2-110** can include and/or support one or more cameras **11.1.1.2-50**. The cameras **11.1.1.2-50** can be positioned on or near an outer side **11.1.1.2-112** of the frame **11.1.1.2-110** to capture images of views external to the head-mountable device **11.1.1.2-100**. As used herein, an outer side **11.1.1.2-112** of a portion of a head-mountable device is a side that faces away from the user and/or towards an external environment. The captured images can be used for display to the user or stored for any other purpose.

[1685] The head-mountable device can be provided with one or more display screens **11.1.1.2-90** that provide visual output for viewing by a user wearing the head-mountable device. As shown in FIG. **11.1.1.2-1**, one or more optical modules containing display screens **11.1.1.2-90** can be positioned on an inner side **11.1.1.2-114** of the frame **11.1.1.2-110**. As used herein, an inner side of a portion of a head-mountable device is a side that faces toward the user and/or away from the external environment. For example, a pair of optical modules can be provided, where each optical

module is movably positioned to be within the field of view of each of a user's two eyes. Each optical module can be adjusted to align with a corresponding eye of the user. Movement of each of the optical modules can match movement of a corresponding camera **11.1.1.2-50**. Accordingly, the optical module is able to accurately reproduce, simulate, or augment a view based on a view captured by the camera **11.1.1.2-50** with an alignment that corresponds to the view that the user would have naturally without the head-mountable device **11.1.1.2-100**.

[1686] A display screen **11.1.1.2-90** can transmit light from a physical environment (e.g., as captured by a camera) for viewing by the user. Such a display screen can include optical properties, such as lenses for vision correction based on incoming light from the physical environment.

Additionally or alternatively, a display screen **11.1.1.2-90** can provide information as a display within a field of view of the user. Such information can be provided to the exclusion of a view of a physical environment or in addition to (e.g., overlaid with) a physical environment.

[1687] As further shown in FIG. **11.1.1.2-1**, the head-mountable device **11.1.1.2-100** can include a crown module **11.1.1.2-200** that received input from a user and provides feedback to the user. The crown module **11.1.1.2-200** can be provided on exterior surface of the head-mountable device **11.1.1.2-100**, such as on the frame **11.1.1.2-110**. As shown in FIG. **11.1.1.2-1**, the crown module **11.1.1.2-200** can be provided on a lateral side **11.1.1.2-116** that is defined by and outwardly facing surface between the outer side **11.1.1.2-112** and the inner side **11.1.1.2-114** of the frame **11.1.1.2-110**. It will be understood that the crown module **11.1.1.2-200** can be provided at any portion of the head-mountable device **11.1.1.2-100**, including any portion of the frame **11.1.1.2-110** (e.g., outer side **11.1.1.2-112** or inner side **11.1.1.2-114**) and/or on the securement element **11.1.1.2-120**. It will be further understood that multiple crown modules **11.1.1.2-200** can be provided by the head-mountable device **11.1.1.2-100**. For example, separate crown modules **11.1.1.2-200** can be provided on a same side or opposing sides of the head-mountable device **11.1.1.2-100**.

[1688] Referring now to FIG. **11.1.1.2-2**, the crown module **11.1.1.2-200** can be provided as a self-contained component that is connected both to other portions of the head-mountable device **11.1.1.2-100**. For example, the frame **11.1.1.2-110** or another portion of the head-mountable device **11.1.1.2-100** can provide a recess **11.1.1.2-118** into which the crown module **11.1.1.2-200** can be inserted. By providing the crown module **11.1.1.2-200** as a self-contained component, the crown module **11.1.1.2-200** can be sealed so that its internal components are protected from an external environment.

[1689] The crown module **11.1.1.2-200** can include one or more attachment elements configured to facilitate mechanical coupling or connection of the crown module **11.1.1.2-200** and the frame **11.1.1.2-110** by engaging complementary attachment elements of the frame **11.1.1.2-110** (e.g., within the recess **11.1.1.2-118**). The attachment elements can include protrusions, grooves, locks, latches, snaps, screws, clasps, threads, magnets, and/or pins can be included on the crown module **11.1.1.2-200** and/or the frame **11.1.1.2-110** for securely attaching the crown module **11.1.1.2-200** to the frame **11.1.1.2-110**.

[1690] The crown module **11.1.1.2-200** and the frame **11.1.1.2-110** can each include one or more communication interfaces that facilitate a communication link between the crown module **11.1.1.2-200** and the frame **11.1.1.2-110** (e.g., a controller within the frame **11.1.1.2-110**). The communication interfaces can include one or more of a variety of features, such as electrical connectors, pogo pins, conductive surfaces, wireless receivers/transmitters, and/or inductive coupling features (e.g., coils) for communicably coupling the components of the frame **11.1.1.2-110** and the crown module **11.1.1.2-200**.

[1691] Referring now to FIG. **11.1.1.2-3**, the crown module **11.1.1.2-200** can provide an input system **11.1.1.2-230** that facilitates receiving input by a user and a feedback system **11.1.1.2-240** that provides feedback to the user. For the purposes of the following description, the described crown module **11.1.1.2-200** is one example of that shown and discussed above with respect to FIGS. **11.1.1.2-1** and **11.1.1.2-2**. However, certain features of the crown module **11.1.1.2-200**,

including the external surface geometry, may be simplified or vary with respect to aspects of the crown module **11.1.1.2-200** discussed above.

[1692] As shown in FIG. **11.1.1.2-3**, the crown module **11.1.1.2-200** can include a housing **11.1.1.2-210** that defines at least a portion of an outer periphery of the crown module **11.1.1.2-200** and contains internal components thereof. A crown **11.1.1.2-222** can be provided at an exterior portion of the housing **11.1.1.2-210**. For example, the crown **11.1.1.2-222** can protrude from a surface of the housing **11.1.1.2-210** to be accessible by user. The crown **11.1.1.2-222** can be connected to a shaft **11.1.1.2-220** that extends within the housing **11.1.1.2-210**. The crown **11.1.1.2-222** and/or the shaft **11.1.1.2-220** can be supported relative to the housing **11.1.1.2-210** by one or more bearings **11.1.1.2-236** that facilitates rotation and/or translation of the crown **11.1.1.2-222** and/or the shaft **11.1.1.2-220** relative to the housing **11.1.1.2-210**.

[1693] The housing **11.1.1.2-210** can define a first chamber **11.1.1.2-232** that is sealed from an external environment. Components of the input system **11.1.1.2-230** can be positioned within the first chamber **11.1.1.2-232**. As such, the components of the input system **11.1.1.2-230** can be protected from ingress of fluids and/or particles that would interfere with operation of the input system **11.1.1.2-230**. The first chamber **11.1.1.2-232** can be defined at least in part by one or more seal members (e.g., O-rings) that move with the shaft **11.1.1.2-220** within the first chamber **11.1.1.2-232** of the housing **11.1.1.2-210**.

[1694] The housing **11.1.1.2-210** can define a second chamber **11.1.1.2-242** that is also sealed from an external environment. Components of the feedback system **11.1.1.2-240** can be positioned within the second chamber **11.1.1.2-242**. As such, the components of the feedback system **11.1.1.2-240** can be protected from ingress of fluids and/or particles that would interfere with operation of the feedback system **11.1.1.2-240**. At least one component (e.g., connector **11.1.1.2-250**) of the feedback system **11.1.1.2-240** (e.g., in the second chamber **11.1.1.2-242**) can be connected to the shaft **11.1.1.2-220** or another component of the input system **11.1.1.2-230** (e.g., in the first chamber **11.1.1.2-232**). The second chamber **11.1.1.2-242** and the first chamber **11.1.1.2-232** can optionally be connected to form a single continuous chamber.

[1695] The crown **11.1.1.2-222** and/or the shaft **11.1.1.2-220** can be supported by one or more bearings **11.1.1.2-236** that facilitates rotation and/or translation of the crown **11.1.1.2-222** and/or the shaft **11.1.1.2-220** relative to the housing **11.1.1.2-210**. The connector **11.1.1.2-250** can be supported by one or more bearings **11.1.1.2-246** that facilitates translation of the connector **11.1.1.2-250** with the crown **11.1.1.2-222** and/or the shaft **11.1.1.2-220**. The bearings **11.1.1.2-246** can optionally allow rotation of the connector **11.1.1.2-250** relative to the crown **11.1.1.2-222** and/or the shaft **11.1.1.2-220** while coupling the connector **11.1.1.2-250** to the shaft **11.1.1.2-220** for translation in unison.

[1696] In some embodiments, the crown **11.1.1.2-222** may be used to accept rotary input from the user, which may be used to control aspects of the head-mountable device. The crown **11.1.1.2-222** may be knurled or otherwise textured to improve grip with the user's finger and/or thumb. In some embodiments, a crown **11.1.1.2-222** may be turned by the user to scroll a display or select from a range of values. In other embodiments, the crown **11.1.1.2-222** may be rotated to move a cursor or other type of selection mechanism from a first displayed location to a second displayed location in order to select an icon or move the selection mechanism between various icons that are output on the display. The crown may also be used to control the volume of a speaker, the brightness of the display screen, visual output of the head-mountable device, or control other hardware settings.

[1697] In some embodiments, an optical encoder may be used to detect the rotational motion of the crown about an axis. More specifically, the example provided below with respect to FIG. **11.1.1.2-3** may use an optical encoder to detect rotational movement, rotational direction and/or rotational speed of a component of the electronic device. Once the rotational movement, rotational direction and/or rotational speed have been determined, this information may be used to output or change information and images that are presented on a display or user interface of the head-mountable

device.

[1698] As shown in the example embodiment of FIG. **11.1.1.2-3**, the optical encoder of the present disclosure includes a light source **11.1.1.2-270**, an optical sensor **11.1.1.2-272** (e.g., photodiode and/or photodiode array), and a shaft **11.1.1.2-220**. In some embodiments, the optical encoder of the present disclosure can utilize an encoding pattern **11.1.1.2-262** disposed directly on the shaft **11.1.1.2-220**. For example, the encoding pattern **11.1.1.2-262** can include a number of light and dark markings or stripes that are axially disposed along the shaft **11.1.1.2-220**. Each stripe or combination of stripes on the shaft **11.1.1.2-220** may be used to identify a position of the shaft **11.1.1.2-220**. For example, as light is emitted from the light source **11.1.1.2-270** and reflected off of the shaft **11.1.1.2-220** into the optical sensor **11.1.1.2-272**, a position, rotation, rotation direction and rotation speed of the shaft **11.1.1.2-220** may be determined. Once the rotation direction and speed are determined, this information may be used to output or change information or images that are presented on the display or user interface of the head-mountable device.

[1699] In other embodiments, the shape or form of the shaft **11.1.1.2-220** of the encoder may be used to determine a position, rotation, rotation direction and rotation speed of the shaft **11.1.1.2-220**. For example, the shaft **11.1.1.2-220** may be fluted or have a number of channels that cause the light to be reflected in a number of different directions. Accordingly, a diffractive pattern may be used to determine the rotation, rotation direction and rotation speed of the shaft **11.1.1.2-220**.

[1700] As shown in FIG. **11.1.1.2-3**, a crown assembly may be provided partially within the housing **11.1.1.2-210** of the crown module **11.1.1.2-200** and may be formed from a crown **11.1.1.2-222** disposed at the end of a shaft **11.1.1.2-220**. As discussed above, the crown module **11.1.1.2-200** includes an optical encoder that includes a shaft **11.1.1.2-220**, a light source **11.1.1.2-270**, and an optical sensor **11.1.1.2-272**. Although an optical sensor is specifically mentioned, embodiments disclosed herein may use various types of sensors that are arranged in various configurations for detecting the movement described herein. For example, the movement of the shaft **11.1.1.2-220** may be detected by an image sensor, a light sensor such as a CMOS light sensor or imager, a photovoltaic cell or system, photo resistive component, a laser scanner and the like.

[1701] The optical encoder may produce an encoder output that is used to determine positional data of the crown **11.1.1.2-222**. In particular, the optical encoder may produce an output that is used to detect that movement of the crown **11.1.1.2-222** including the direction of the movement, speed of the movement and so on. The movement may be rotational movement (e.g., about the axis **11.1.1.2-290**), translational movement (e.g., along or parallel to the axis **11.1.1.2-290**), angular movement (e.g., tilt relative to the axis **11.1.1.2-290**), and so on. The optical encoder may also be used to detect the degree of the change of rotation of the crown **11.1.1.2-222** and/or the angle of rotation of the crown **11.1.1.2-222** as well as the speed and the direction of the rotation of the crown **11.1.1.2-222**.

[1702] The crown **11.1.1.2-222** can be coupled to and/or monolithically formed with the shaft **11.1.1.2-220**. In some cases, the shaft **11.1.1.2-220** and crown **11.1.1.2-222** may be formed as a single piece. As the shaft **11.1.1.2-220** is coupled to, or is otherwise a part of the crown **11.1.1.2-222**, as the crown **11.1.1.2-222** rotates or moves in a particular direction and at a particular speed, the shaft **11.1.1.2-220** also rotates or moves in the same direction and with the same speed.

[1703] As further shown in FIG. **11.1.1.2-3**, the feedback system **11.1.1.2-240** can include mechanisms that facilitate haptic feedback. A feedback system can be implemented as any suitable device configured to provide force feedback, vibratory feedback, tactile sensations, and the like. For example, in one embodiment, the feedback system may be implemented as a linear actuator configured to provide a punctuated haptic feedback, such as a tap or a knock.

[1704] According to some embodiments, the feedback system **11.1.1.2-240** can include a magnetic element **11.1.1.2-252**. The magnetic element **11.1.1.2-252** can be coupled to the shaft **11.1.1.2-220** (e.g., via a connector **11.1.1.2-250**) such that the magnetic element **11.1.1.2-252** moves along the axis **11.1.1.2-290** relative to the housing **11.1.1.2-210** along with the shaft **11.1.1.2-220**. For

example, the magnetic element **11.1.1.2-252** can be directly or indirectly (e.g., via the connector **11.1.1.2-250** and/or the intermediate bearing **11.1.1.2-246**) connected to the shaft **11.1.1.2-220**.

[1705] The magnetic element **11.1.1.2-252** can include a temporary magnet of a soft magnetic material or a permanent magnet of a hard magnetic material. As used herein, “magnet” can include a magnet of a hard magnetic material and/or a magnet of a soft magnetic material. Hard magnetic materials include materials that retain their magnetism even after the removal of an applied magnetic field. Magnets that include hard magnetic material can form permanent magnets. Hard magnetic materials include neodymium (NdFeB), iron-neodymium, iron-boron, cobalt-samarium, iron-chromium-cobalt, and combinations or alloys thereof. Soft magnetic materials include materials that are responsive to magnetic fields, but do not retain their magnetism after removal of an applied magnetic field. Magnets that include soft magnetic material can form temporary magnets. Soft magnetic materials include iron, iron-cobalt, iron-silicon, steel, stainless steel, iron-aluminum-silicon, nickel-iron, ferrites, and combinations or alloys thereof. It will be recognized that “hard magnetic” and “soft magnetic” does not necessarily relate to the rigidity of the materials.

[1706] The feedback system **11.1.1.2-240** can further include a magnetic field generator to induce a magnetic field in the magnetic element **11.1.1.2-252**. For example, one or more coils **11.1.1.2-244** can be positioned on one or more sides of the magnetic element **11.1.1.2-252**. The coils **11.1.1.2-244** can include one or more helical windings in one or more layers. It will be recognized that any number of windings and arrangements of the coil can be provided to induce a magnetic field.

[1707] It will be recognized that various arrangements and alterations to the above description can be implemented to provide haptic feedback. For example, the magnetic element **11.1.1.2-252** can be exchanged with the coils **11.1.1.2-244** such that the magnetic element **11.1.1.2-252** is moveable with the housing **11.1.1.2-210** and the coils **11.1.1.2-244** are moveable with the shaft **11.1.1.2-220**. The magnetic element **11.1.1.2-252** can have a variety of shapes and sizes. Multiple magnetic elements can be provided. These and other designs can be implemented to facilitate an induced magnetic field and magnetic forces between the magnetic elements.

[1708] As shown in FIG. **11.1.1.2-3**, the coils **11.1.1.2-244** are operated to induce a magnetic field near the magnetic element **11.1.1.2-252**. When the coils **11.1.1.2-244** are activated with an electric current, the causes the magnetic element **11.1.1.2-252** to move under the influence of a magnetic force. For example, where the magnetic element **11.1.1.2-252** is a temporary magnet of a soft magnetic material, the magnetic field can cause the magnetic domains of the magnetic element **11.1.1.2-252** to align with the magnetic field. The magnetic element **11.1.1.2-252** will then be attracted toward a direction based on the activated coils **11.1.1.2-244**. Additionally or alternatively, the magnetic element **11.1.1.2-252** can be a permanent magnet of a hard magnetic material. Based on the alignment (i.e., polarity) of such a permanent magnet, the magnetic field causes the magnetic element **11.1.1.2-252** to attract toward or repel away from one or more coils **11.1.1.2-244** when activated.

[1709] As the magnetic element **11.1.1.2-252** moves (e.g., along the axis **11.1.1.2-290**), the shaft **11.1.1.2-220** moves relative to the housing **11.1.1.2-210** (e.g., along the axis **11.1.1.2-290**). As described above, the magnetic element **11.1.1.2-252** is connected to the shaft **11.1.1.2-220** to move with the shaft **11.1.1.2-220** and the coils **11.1.1.2-244** are connected to the housing **11.1.1.2-210** to move with the housing **11.1.1.2-210**. As such, magnetic forces between the magnetic element **11.1.1.2-252** and the coils **11.1.1.2-244** are transmitted to the shaft **11.1.1.2-220** and the housing **11.1.1.2-210** to cause relative movement there between.

[1710] The haptic feedback can include movement of the shaft **11.1.1.2-220** relative to the housing and along the axis **11.1.1.2-290** of the crown module **11.1.1.2-200**. For example, the magnetic element **11.1.1.2-252** can be aligned along the axis **11.1.1.2-290** of the crown module **11.1.1.2-200**. Movement from haptic feedback can be along the same axis **11.1.1.2-290** about which the crown **11.1.1.2-222** and the shaft **11.1.1.2-220** rotate. Additionally or alternatively, movement from haptic feedback can be along another axis or in multiple axes and directions.

[1711] In use, the coils **11.1.1.2-244** can be operated to provide haptic feedback while the user is operating (e.g., contacting and/or rotating) the crown **11.1.1.2-222**. The haptic feedback can be provided based on a variety of conditions and parameters. For example, the haptic feedback can be controlled by providing an electric current to the coils **11.1.1.2-244**. The induced and corresponding magnetic force between the magnetic element **11.1.1.2-252** and the coils **11.1.1.2-244** is based on the current in the coils **11.1.1.2-244**. As such, the current can have a duration, amplitude, frequency, waveform, duty cycle, or other parameters as desired for a desired and corresponding haptic feedback.

[1712] For example, the shaft **11.1.1.2-220** can be made to vibrate by applying a control signal to the coils **11.1.1.2-244**. The control signal may be a wave having a predetermined amplitude and/or frequency. When the control signal is applied, the induced magnetic field causes the shaft **11.1.1.2-220** to vibrate at the frequency of the control signal. The frequency can be in a range between 10 Hz and 5,000 Hz, 50 Hz and 1,000 Hz, or 100 Hz and 500 Hz. The frequency of the control signal may be adjusted to alter the rate of movement of the shaft **11.1.1.2-220** if a certain vibration is desired. The amplitude of the control signal may be correlated to the magnitude of movement of the shaft **11.1.1.2-220**, and may be adjusted to alter the intensity of the vibration.

[1713] The feedback system **11.1.1.2-240** can provide haptic feedback to a user by moving the shaft **11.1.1.2-220** of the crown module **11.1.1.2-200** relative to the housing **11.1.1.2-210**. In contrast to haptic feedback applied directly to the housing **11.1.1.2-210** and/or other portions of the head-mountable device, haptic feedback provided at the shaft **11.1.1.2-220** more directly provides sensations relating to the shaft **11.1.1.2-220**. For example, haptic feedback can be provided while the user is operating (e.g., contacting and/or rotating) the crown **11.1.1.2-222**. As the shaft **11.1.1.2-220** and the crown **11.1.1.2-222** are moved (e.g., vibrated) relative to the housing **11.1.1.2-210**, the rest of the head-mountable device can remain stationary, so that the haptic feedback is not felt by the user at other locations of contact. By further example, while the user is wearing the head-mountable device, the haptic feedback can nonetheless be localized to the crown **11.1.1.2-222** so that the user feels the haptic feedback only at that location.

[1714] The feedback system **11.1.1.2-240** can provide haptic feedback based on operation of the crown module **11.1.1.2-200**. For example, haptic feedback can be provided while the crown **11.1.1.2-222** and/or the shaft **11.1.1.2-220** are rotated by the user. Incremental and/or periodic haptic feedback can be provided based on the rotation performed by the user. By further example, the haptic feedback can be provided at a speed that corresponds to the speed of rotation performed by the user. As such, the haptic feedback can provide confirmation to the user relating to the input that is received by the user.

[1715] The feedback system **11.1.1.2-240** can provide haptic feedback based on activities performed by the head-mountable device. For example, the haptic feedback can correspond to visual information that is output to the user by the head-mountable device. By further example, visual information can be modified by use or operation of the crown, and haptic feedback can be provided to indicate how the user can interact with the visual information. For example, the user can rotate the crown in one or both of two directions to cause the head-mountable device to perform certain actions. Such rotation be performed to control the volume of a speaker, the brightness of the display screen, visual output of the head-mountable device, optical settings of an optical subassembly, or control other hardware settings. Rotation can be performed to scroll through a list or other set of items visually displayed by the head-mountable device.

[1716] While a first type of haptic feedback can be provided as the user rotates the crown, a second type of haptic feedback can be provided to indicate how the user can interact with the head-mountable device and/or limitations regarding the user input. For example, as the user scrolls through a list displayed by the head-mountable device, a first type of haptic feedback can be provided based on the user input (e.g., speed of rotation, etc.). By further example, as the user reaches the end of a list, a second type of haptic feedback can be provided to indicate that the user

has reached the end of the list. Additionally or alternatively, different types of feedback can be provided in this way for other actions, such as zooming in on or out from an image, changing volume settings, changing display brightness, and the like.

[1717] The feedback system **11.1.1.2-240** can provide haptic feedback for one or more other purposes. According to some embodiments, the haptic feedback can notify the user based on a message, alert, or alarm. Such notifications can be accompanied by other feedback, including tactile, auditory, and/or visual feedback on the crown module **11.1.1.2-200** and/or the external device. According to some embodiments, the haptic feedback can provide confirmation that a user selection (e.g., made with the crown module **11.1.1.2-200**) has been received by the head-mountable device and/or an external device. According to some embodiments, the haptic feedback can inform the user regarding status or operation of the head-mountable device and/or an external device.

[1718] Referring now to FIGS. **11.1.1.2-4-11.1.1.2-7**, a crown module **11.1.1.2-300** can be provided for receiving input from a user and providing feedback to the user. The crown module **11.1.1.2-300** can be provided with the head-mountable device as an alternative to and/or in addition to the crown module **11.1.1.2-200**. For example, the crown module **11.1.1.2-300** can be a self-contained module that is assembled with other components of the head-mountable device as described herein with respect to the crown module **11.1.1.2-200**.

[1719] As shown in FIG. **11.1.1.2-4**, the crown module **11.1.1.2-300** can include a housing **11.1.1.2-310** that defines at least a portion of an outer periphery of the crown module **11.1.1.2-300** and contains internal components thereof. A crown **11.1.1.2-322** can be provided at an exterior portion of the housing **11.1.1.2-310**. For example, the crown **11.1.1.2-322** can protrude from a surface of the housing **11.1.1.2-310** to be accessible by user. The crown **11.1.1.2-322** can be connected to a shaft **11.1.1.2-320** that extends within the housing **11.1.1.2-310**. The crown **11.1.1.2-322** and/or the shaft **11.1.1.2-320** can be supported relative to the housing **11.1.1.2-310** by one or more bearings **11.1.1.2-336**.

[1720] The housing **11.1.1.2-310** can define a chamber **11.1.1.2-332** that is sealed from an external environment. As such, the components of the crown module **11.1.1.2-300** can be protected from ingress of fluids and/or particles that would interfere with operation thereof.

[1721] The crown module **11.1.1.2-300** can include a torque sensor **11.1.1.2-330** configured to detect torque that is applied by a user to the crown **11.1.1.2-322** and transferred to the shaft **11.1.1.2-320**. For example, the user can apply a torque to the crown **11.1.1.2-322** by urging the crown **11.1.1.2-322** to rotate about the axis **11.1.1.2-390**. It will be understood that such a torque may not result in significant rotation about the axis **11.1.1.2-390**. For example, the crown **11.1.1.2-322** and/or the shaft **11.1.1.2-320** can be coupled to the housing **11.1.1.2-310** such that no significant rotation is achieved. Despite this coupling, a torque can be applied to the crown **11.1.1.2-322** and transferred to the shaft **11.1.1.2-320**. As the shaft **11.1.1.2-320** is subjected to such a torque, the torque sensor **11.1.1.2-330** can detect the torque and interpret the torque as an input from the user. Mechanisms for detecting torque are described further herein.

[1722] In some embodiments, the crown **11.1.1.2-322** may be used to accept torque input from the user, which may be used to control aspects of the head-mountable device. The crown **11.1.1.2-322** may be knurled or otherwise textured to improve grip with the user's finger and/or thumb. In some embodiments, the crown **11.1.1.2-322** may be operated to provide inputs such as those described with respect to the crown module **11.1.1.2-200**. For example, the crown **11.1.1.2-322** may be torqued by the user to scroll a display or select from a range of values. In other embodiments, the crown **11.1.1.2-322** may be torqued to move a cursor or other type of selection mechanism from a first displayed location to a second displayed location in order to select an icon or move the selection mechanism between various icons that are output on the display. The crown may also be used to control the volume of a speaker, the brightness of the display screen, visual output of the head-mountable device, or control other hardware settings.

[1723] As further shown in FIG. **11.1.1.2-4**, the crown module **11.1.1.2-300** can be provided with a feedback system **11.1.1.2-350** that includes mechanisms that facilitate haptic feedback. According to some embodiments, the feedback system **11.1.1.2-350** can be the same or similar to the feedback system **11.1.1.2-240** of the crown module **11.1.1.2-200**. For example, the feedback system **11.1.1.2-350** can be coupled to the shaft **11.1.1.2-320** such that a component of the feedback system **11.1.1.2-350** moves along the axis **11.1.1.2-390** relative to the housing **11.1.1.2-310** along with the shaft **11.1.1.2-320**. Additionally or alternatively, the feedback system **11.1.1.2-350** can include or be connected to motors, hydraulic actuators, pneumatic actuators, magnetic actuators, piezoelectric actuators, electroactive materials (e.g., polymers), stepper motors, shape-memory alloys, and/or the like for providing mechanical movement as haptic feedback.

[1724] As shown in FIGS. **11.1.1.2-5** and **11.1.1.2-6**, the torque sensor **11.1.1.2-330** can include multiple strain gauges **11.1.1.2-330A**, **11.1.1.2-330B**, **11.1.1.2-330C**, and **11.1.1.2-330D**. The strain gauges of the torque sensor **11.1.1.2-330** can operate as a resistive sensor formed from a material that exhibits a change in electrical resistance (e.g., conductance) in response to a dimensional change such as compression, tension, or force. The strain gauges can each be a compliant material that exhibits at least one electrical property that is variable in response to deformation, deflection, or shearing of the electrode. The strain gauges may be formed from a piezoelectric, piezoresistive, resistive, or other strain-sensitive materials.

[1725] As further shown in FIGS. **11.1.1.2-5** and **11.1.1.2-6**, the strain gauges **11.1.1.2-330A**, **11.1.1.2-330B**, **11.1.1.2-330C**, and **11.1.1.2-330D** can be distributed at different locations on the shaft **11.1.1.2-320**. For example, as shown in FIG. **11.1.1.2-5**, some of the strain gauges can be placed on a first radial side of the shaft **11.1.1.2-320**, and other strain gauges can be placed on a second radial side of the shaft **11.1.1.2-320**. By further example, as shown in FIG. **11.1.1.2-6**, some of the strain gauges can be placed in a first orientation relative to an axis of the shaft **11.1.1.2-320**, and other strain gauges can be placed in a second orientation relative to the axis of the shaft **11.1.1.2-320**. In such different orientations, an applied torque will cause some of the strain gauges to be placed in compression whereas another strain gauge is placed into tension (e.g., tensile strain) in response to the torque.

[1726] As shown in FIG. **11.1.1.2-7**, the strain gauges **11.1.1.2-330A**, **11.1.1.2-330B**, **11.1.1.2-330C**, and **11.1.1.2-330D** can be coupled to each other in an electrical circuit **11.1.1.2-360**. The electrical circuit can be configured to monitor one or more electrical properties (e.g., resistance, capacitance, accumulated charge, inductance, and so on) of the strain gauges for changes. The electrical circuit **11.1.1.2-360** then quantifies these changes, which may be used to estimate the applied torque. For example, the electrical circuit **11.1.1.2-360** can include a Wheatstone bridge arrangement to measure an electrical resistance by balancing two legs of a bridge circuit with opposing pairs of strain gauges. An applied voltage can be compared to a measured voltage within the Wheatstone bridge of the electrical circuit **11.1.1.2-360** to determine a combined effect of torque on the multiple strain gauges **11.1.1.2-330A**, **11.1.1.2-330B**, **11.1.1.2-330C**, and **11.1.1.2-330D**.

[1727] The multiple strain gauges **11.1.1.2-330A**, **11.1.1.2-330B**, **11.1.1.2-330C**, and **11.1.1.2-330D** can be connected to approximate a magnitude of torsional strain (e.g., torque) experienced by the shaft **11.1.1.2-320**. In some embodiments, a magnitude of strain can be obtained by measuring a common property (e.g., parallel and/or series resistance) and/or a differential property (e.g., voltage division) of the multiple strain gauges. For example, differential property estimates can be combined with or compared to common property estimates. By further example, the differential property estimate and common property estimate can be combined by unweighted or weighted averaging. In some embodiments, the maximum or minimum of the two estimates can be used. In some embodiments, other methods of combining or deciding between the two estimates can be used.

[1728] Once the resistance of each strain gauge is obtained via calculation or measurement, each

can be compared to a known baseline resistance value in order to determine whether the strain gauges are experiencing tension or compression. In other words, when the force-sensitive structure experiences a reaction force, it may deform, causing one or more strain gauges to either expand (e.g., tension) or contract (e.g., compression), which can cause the resistance thereof to change in a mathematically predictable manner. In some cases, the resistance or other electrical property of a strain gauge is measured as a relative value, which may factor environmental effects, such as temperature and/or residual or static strain.

[1729] For certain materials, resistance can change linearly with compression or tension. For other materials, resistance can change following a known curve in response to compression or tension. Accordingly, depending upon the material selected for the strain gauges, and the position of the strain gauges on the shaft **11.1.1.2-320**, a particular resistance and/or measured voltage can be correlated to a particular amount of strain experienced by a particular strain gauge, which in turn can itself be correlated to an amount of force applied to the force-sensitive structure, which in turn can be correlated to an amount of torque applied to the crown.

[1730] Referring now to FIG. **11.1.1.2-8**, components of the head-mountable device can be operably connected to provide the performance described herein. FIG. **11.1.1.2-8** shows a simplified block diagram of an illustrative head-mountable device **11.1.1.2-100** in accordance with one embodiment of the invention. It will be appreciated that components described herein can be provided on either or both of a frame and/or a securement element of the head-mountable device **11.1.1.2-100**.

[1731] As shown in FIG. **11.1.1.2-8**, the head-mountable device **11.1.1.2-100** can include a controller **11.1.1.2-70** with one or more processing units that include or are configured to access a memory **11.1.1.2-18** having instructions stored thereon. The instructions or computer programs may be configured to perform one or more of the operations or functions described with respect to the head-mountable device **11.1.1.2-100**. The controller **11.1.1.2-70** can be implemented as any electronic device capable of processing, receiving, or transmitting data or instructions. For example, the controller **11.1.1.2-70** may include one or more of: a microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a digital signal processor (DSP), or combinations of such devices. As described herein, the term “processor” is meant to encompass a single processor or processing unit, multiple processors, multiple processing units, or other suitably configured computing element or elements.

[1732] The memory **11.1.1.2-18** can store electronic data that can be used by the head-mountable device **11.1.1.2-100**. For example, the memory **11.1.1.2-18** can store electrical data or content such as, for example, audio and video files, documents and applications, device settings and user preferences, timing and control signals or data for the various modules, data structures or databases, and so on. The memory **11.1.1.2-18** can be configured as any type of memory. By way of example only, the memory **11.1.1.2-18** can be implemented as random access memory, read-only memory, Flash memory, removable memory, or other types of storage elements, or combinations of such devices.

[1733] The head-mountable device **11.1.1.2-100** can include the camera **11.1.1.2-50** for capturing a view of an environment external to the head-mountable device **11.1.1.2-100**. The camera **11.1.1.2-50** can include an optical sensor, such as a photodiode or a photodiode array. Additionally or alternatively, the camera **11.1.1.2-50** can include one or more of various types of optical sensors that are arranged in various configurations for detecting user inputs described herein. The camera **11.1.1.2-50** may be configured to capture an image of a scene or subject located within a field of view of the camera **11.1.1.2-50**. The image may be stored in a digital file in accordance with any one of a number of digital formats. In some embodiments, the head-mountable device **11.1.1.2-100** includes a camera, which includes an image sensor formed from a charge-coupled device (CCD) and/or a complementary metal-oxide-semiconductor (CMOS) device, a photovoltaic cell, a photo resistive component, a laser scanner, and the like. It will be recognized that a camera can include

other motion sensing devices.

[1734] The head-mountable device **11.1.1.2-100** can include a battery **11.1.1.2-20**, which can charge and/or power components of the head-mountable device **11.1.1.2-100**.

[1735] The head-mountable device **11.1.1.2-100** can include one or more other input/output components **11.1.1.2-26**, which can include any suitable component for connecting head-mountable device **11.1.1.2-100** to other devices. Suitable components can include, for example, audio/video jacks, data connectors, and/or any additional or alternative input/output components. The input/output components **11.1.1.2-26** can include a microphone. The microphone can be operably connected to the controller **11.1.1.2-70** for receiving audio input, including voice commands from the user. The input/output components **11.1.1.2-26** can include one or more speakers. The speakers can be operably connected to the controller **11.1.1.2-70** for control of speaker output, including sound levels.

[1736] The head-mountable device **11.1.1.2-100** can include communications circuitry **11.1.1.2-28** for communicating with one or more servers or other devices using any suitable communications protocol. For example, the communications circuitry **11.1.1.2-28** can support Wi-Fi (e.g., a 802.11 protocol), Ethernet, Bluetooth, high frequency systems (e.g., 900 MHz, 2.4 GHz, and 5.6 GHz communication systems), infrared, TCP/IP (e.g., any of the protocols used in each of the TCP/IP layers), HTTP, BitTorrent, FTP, RTP, RTSP, SSH, any other communications protocol, or any combination thereof. The communications circuitry **11.1.1.2-28** can also include an antenna for transmitting and receiving electromagnetic signals.

[1737] The head-mountable device **11.1.1.2-100** can further include an optical module for displaying visual information for a user, including the display screen **11.1.1.2-90** and/or an optical subassembly **11.1.1.2-14**. The head-mountable device **11.1.1.2-100** can include an optical subassembly **11.1.1.2-14** configured to help optically adjust and correctly project the image based content being displayed by the display **11.1.1.2-90** for close up viewing. The optical subassembly **11.1.1.2-14** can include one or more lenses, mirrors, or other optical devices.

[1738] The head-mountable device **11.1.1.2-100** can optionally connect to a portable electronic device **11.1.1.2-22**, which can provide certain functions. The portable electronic device **11.1.1.2-22** can provide a handheld form factor (e.g., small portable electronic device which is light weight, fits in a pocket, etc.). Although not limited to these, examples include media players, phones (including smart phones), PDAs, watches, computers, and the like. The portable electronic device may include a screen for presenting the graphical portion of the media to the user. The portable electronic device can provide processing capabilities by communicating with the controller **11.1.1.2-70** of the head-mountable device **11.1.1.2-100**. The portable electronic device can provide actuation to dislodge particles from components of the head-mountable device **11.1.1.2-100** by operating a haptic feedback element of the portable electronic device.

[1739] The head-mountable device **11.1.1.2-100** can include a dock **11.1.1.2-32** operative to receive the portable electronic device **11.1.1.2-22**. The dock **11.1.1.2-32** can include a connector (e.g., Lightning, USB, FireWire, power, DVI, etc.), which can be plugged into a complementary connector of the portable electronic device **11.1.1.2-22**. The dock **11.1.1.2-32** may include features for helping to align the connectors during engagement and for physically coupling the portable electronic device **11.1.1.2-22** to the head-mountable device **11.1.1.2-100**. For example, the dock **11.1.1.2-32** may define a cavity for placement of the portable electronic device **11.1.1.2-22**. The dock **11.1.1.2-32** may also include retaining features for securing portable electronic device **11.1.1.2-22** within the cavity. The connector on the dock **11.1.1.2-32** can function as a communication interface between the portable electronic device **11.1.1.2-22** and the head-mountable device **11.1.1.2-100**.

[1740] The head-mountable device **11.1.1.2-100** can include one or more other sensors **11.1.1.2-40**. Such sensors can be configured to sense substantially any type of characteristic such as, but not limited to, images, pressure, light, touch, force, temperature, position, motion, and so on. For

example, the sensor can be a photodetector, a temperature sensor, a light or optical sensor, an atmospheric pressure sensor, a humidity sensor, a magnet, a gyroscope, an accelerometer, a chemical sensor, an ozone sensor, a particulate count sensor, and so on. By further example, the sensor can be a bio-sensor for tracking biometric characteristics, such as health and activity metrics. Other user sensors can perform facial feature detection, facial movement detection, facial recognition, eye tracking, user mood detection, user emotion detection, voice detection, etc.

[1741] The head-mountable device **11.1.1.2-100** can include one or more sensors (e.g., eye sensor) for tracking features of the user wearing the head-mountable device **11.1.1.2-100**. For example, such sensors can perform facial feature detection, facial movement detection, facial recognition, eye tracking, user mood detection, user emotion detection, voice detection, etc. For example, an eye sensor can optically capture a view of an eye (e.g., pupil) and determine a direction of a gaze of the user. Such eye tracking may be used to determine a location and/or direction of interest. Detection and/or amplification of sound can then be focused if it is received from sources at such a location and/or along such a direction.

[1742] While some embodiments of touch-based input devices disclosed herein relate to head-mountable devices, it will be appreciated that the subject technology can encompass and be applied to other devices. For example, an input device (e.g., crown module) in accordance with embodiments disclosed herein can include a phone, a tablet computing device, a mobile computing device, a watch, a laptop computing device, a mouse, a game controller, a remote control, a digital media player, a stylus, and/or any other electronic device. Further, the external device can be any device that interacts with a touch-based input device. For example, an external device in accordance with embodiments disclosed herein can include a tablet, a phone, a laptop computing device, a desktop computing device, a wearable device, a mobile computing device, a tablet computing device, a display, a television, a phone, a digital media player, and/or any other electronic device.

[1743] Accordingly, embodiments of the present disclosure provide a head-mountable device with a crown module having an input system that allows a user to provide inputs by rotating or otherwise applying torque to a crown of the crown module. The head-mountable device can interpret the rotation and/or torque as a user input. The crown module can further include a feedback system that provides localized haptic feedback at the crown. The haptic feedback can be effectively perceived by the user at the crown without causing the entire head-mountable device to vibrate against the head and/or face of the user.

[1744] Various examples of aspects of the disclosure are described below as clauses for convenience. These are provided as examples, and do not limit the subject technology.

[1745] Clause A: a head-mountable device comprising: a housing; a crown positioned at least partially outside the housing; a shaft positioned within the housing and connected to the crown such that the shaft rotates with the crown and about an axis; a sensor for detecting rotation of the shaft; a magnetic element coupled to the shaft such that the magnetic element moves along the axis with the shaft; and a coil coupled to the housing and configured to induce a magnetic field in the magnetic element, such that, when the coil is activated, the magnetic element provides haptic feedback by moving the shaft and the crown relative to the housing.

[1746] Clause B: a head-mountable device comprising: a housing; a crown positioned at least partially outside the housing; a shaft positioned within the housing and connected to the crown such that a torque applied to the crown is transferred to the shaft; a sensor for detecting the torque transferred to the shaft; and a haptic feedback device coupled to the housing and configured to provide haptic feedback by moving the shaft and the crown relative to the housing.

[1747] Clause C: a crown module for a head-mountable device, the crown module comprising: a housing configured to be coupled to a frame of the head-mountable device; a crown positioned at least partially outside the housing; a shaft positioned within a sealed chamber of the housing and connected to the crown such that the shaft rotates with the crown and about an axis; a sensor for

detecting rotation of the shaft and positioned within the sealed chamber; and a haptic feedback device positioned within the sealed chamber of the housing and configured to provide haptic feedback by moving the shaft and the crown along the axis and relative to the housing.

[1748] One or more of the above clauses can include one or more of the features described below. It is noted that any of the following clauses may be combined in any combination with each other, and placed into a respective independent clause, e.g., clause A, B, or C.

[1749] Clause 1: a frame; a display on an inner side of the frame; a camera on an outer side of the frame; a speaker; and a microphone.

[1750] Clause 2: the shaft is positioned within a sealed chamber of the housing that contains the sensor.

[1751] Clause 3: a pair of seal members each sealingly engaging an inner surface of the housing and an outer surface of the shaft, wherein the sensor is positioned axially between the pair of seal members.

[1752] Clause 4: the sensor is an optical sensor and the shaft comprises a visual feature for detection by the optical sensor.

[1753] Clause 5: the magnetic element and the coil are positioned within a sealed chamber of the housing.

[1754] Clause 6: the magnetic element is coupled to the shaft such that the magnetic element is configured to rotate independently of the shaft.

[1755] Clause 7: the coil is a first coil on a first axial side of the magnetic element; and the head-mountable device comprises a second coil on a second axial side of the magnetic element.

[1756] Clause 8: a controller configured to operate the coil in response to a detection by the sensor that the shaft is rotating.

[1757] Clause 9: the torque is about an axis and the haptic feedback comprises movement along the axis.

[1758] Clause 10: the sensor comprises multiple strain gauges.

[1759] Clause 11: the sensor comprises multiple strain gauges electrically connected to each other in a Wheatstone bridge arrangement.

[1760] Clause 12: at least two of the multiple strain gauges have different orientations relative to a longitudinal axis of the shaft.

[1761] Clause 13: the haptic feedback device comprises an actuator configured to expand toward and contract away from the shaft.

[1762] Clause 14: a controller configured to operate the haptic feedback device in response to a detection by the sensor that the torque is applied to the crown.

[1763] Clause 15: a pair of seal members each sealingly engaging an inner surface of the housing and an outer surface of the shaft, wherein the sensor is positioned axially between the pair of seal members.

[1764] Clause 16: the sensor is an optical sensor and the shaft comprises a visual feature for detection by the optical sensor.

[1765] Clause 17: the haptic feedback device comprises: a magnetic element coupled to the shaft such that the magnetic element moves along the axis with the shaft; and a coil coupled to the housing and configured to induce a magnetic field in the magnetic element, such that, when the coil is activated, the magnetic element provides haptic feedback by moving the shaft and the crown relative to the housing.

11.1.2: Wishbone and Mustache

[1766] FIG. **11.1.2-1** illustrates a front perspective view of a portion of an HMD **11.1.2-100**, including an outer structural frame **11.1.2-102** and an inner or intermediate structural frame **11.1.2-104** defining first and second apertures **11.1.2-106a**, **11.1.2-106b**. The apertures **11.1.2-106a-b** are shown in dotted lines in FIG. **11.1.2-1** because a view of the apertures **11.1.2-106a-b** can be blocked by one or more other components of the HMD **11.1.2-100** coupled to the inner frame

11.1.2-104 and/or the outer frame **11.1.2-102**, as shown. In at least one example, the HMD **11.1.2-100** can include a first mounting bracket **11.1.2-108** coupled to the inner frame **11.1.2-104**. In at least one example, the mounting bracket **11.1.2-108** is coupled to the inner frame **11.1.2-104** between the first and second apertures **11.1.2-106a-b**.

[1767] The mounting bracket **11.1.2-108** can include a middle or central portion **11.1.2-109** coupled to the inner frame **11.1.2-104**. In some examples, the middle or central portion **11.1.2-109** may not be the geometric middle or center of the bracket **11.1.2-108**. Rather, the middle/central portion **11.1.2-109** can be disposed between first and second cantilevered extension arms extending away from the middle portion **11.1.2-109**. In at least one example, the mounting bracket **108** includes a first cantilever arm **11.1.2-112** and a second cantilever arm **11.1.2-114** extending away from the middle portion **11.1.2-109** of the mount bracket **11.1.2-108** coupled to the inner frame **11.1.2-104**.

[1768] As shown in FIG. **11.1.2-1**, the outer frame **11.1.2-102** can define a curved geometry on a lower side thereof to accommodate a user's nose when the user dons the HMD **11.1.2-100**. The curved geometry can be referred to as a nose bridge **11.1.2-111** and be centrally located on a lower side of the HMD **11.1.2-100** as shown. In at least one example, the mounting bracket **11.1.2-108** can be connected to the inner frame **11.1.2-104** between the apertures **11.1.2-106a-b** such that the cantilevered arms **11.1.2-112**, **11.1.2-114** extend downward and laterally outward away from the middle portion **11.1.2-109** to compliment the nose bridge **11.1.2-111** geometry of the outer frame **11.1.2-102**. In this way, the mounting bracket **11.1.2-108** is configured to accommodate the user's nose as noted above. The nose bridge **11.1.2-111** geometry accommodates the nose in that the nose bridge **11.1.2-111** provides a curvature that curves with, above, over, and around the user's nose for comfort and fit.

[1769] The first cantilever arm **11.1.2-112** can extend away from the middle portion **11.1.2-109** of the mounting bracket **11.1.2-108** in a first direction and the second cantilever arm **11.1.2-114** can extend away from the middle portion **11.1.2-109** of the mounting bracket **11.1.2-10** in a second direction opposite the first direction. The first and second cantilever arms **11.1.2-112**, **11.1.2-114** are referred to as “cantilevered” or “cantilever” arms because each arm **11.1.2-112**, **11.1.2-114**, includes a distal free end **11.1.2-116**, **11.1.2-118**, respectively, which are free of affixation from the inner and outer frames **11.1.2-102**, **11.1.2-104**. In this way, the arms **11.1.2-112**, **11.1.2-114** are cantilevered from the middle portion **11.1.2-109**, which can be connected to the inner frame **11.1.2-104**, with distal ends **11.1.2-102**, **11.1.2-104** unattached.

[1770] In at least one example, the HMD **11.1.2-100** can include one or more components coupled to the mounting bracket **11.1.2-108**. In one example, the components include a plurality of sensors **11.1.2-110a-f**. Each sensor of the plurality of sensors **11.1.2-110a-f** can include various types of sensors, including cameras, IR sensors, and so forth. In some examples, one or more of the sensors **11.1.2-110a-f** can be used for object recognition in three-dimensional space such that it is important to maintain a precise relative position of two or more of the plurality of sensors **11.1.2-110a-f**. The cantilevered nature of the mounting bracket **11.1.2-108** can protect the sensors **11.1.2-110a-f** from damage and altered positioning in the case of accidental drops by the user. Because the sensors **11.1.2-110a-f** are cantilevered on the arms **11.1.2-112**, **11.1.2-114** of the mounting bracket **11.1.2-108**, stresses and deformations of the inner and/or outer frames **11.1.2-104**, **11.1.2-102** are not transferred to the cantilevered arms **11.1.2-112**, **11.1.2-114** and thus do not affect the relative positioning of the sensors **11.1.2-110a-f** coupled/mounted to the mounting bracket **11.1.2-108**.

[1771] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.2-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.1.2-2-11.1.2-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **11.1.2-2-11.1.2-5** can be included, either alone or in any combination, in the example of the devices,

features, components, and parts shown in FIG. **11.1.2-1**.

[1772] FIG. **11.1.2-2** illustrates a perspective view of an example of a mounting bracket **11.1.2-208** including a middle portion **11.1.2-209** and first and second arms **11.1.2-212**, **11.1.2-214** extending from the middle portion **11.1.2-209** to form a nose bridge **11.1.2-211** geometry. One or more components including sensors **11.1.2-210a-f** can be coupled to the mounting bracket **11.1.2-208**, including to the cantilever arms **11.1.2-212**, **11.1.2-214**. In at least one example, the material of the mounting bracket **11.1.2-208** can include a strong, durable, and stiff material and/or geometry to maintain a relative position between sensors **11.1.2-210a-f** during drop events or other forces causing deformations to the frame to which the mounting bracket **11.1.2-209** is coupled. The material of the mounting bracket **11.1.2-208** can include metals such as aluminum, steel including stainless steel, magnesium or magnesium alloys, hard plastics, ceramics, composite materials, carbon fiber, or any combination thereof.

[1773] In at least one example, a first sensor, including any of the sensors **11.1.2-210a-c** can be coupled to the first cantilever arm **11.1.2-212** and a second sensor, including any of the sensors **11.1.2-210e-f** can be coupled to the second cantilever arm **11.1.2-214**. In at least one example, the first sensor can include a visual sensor such as a camera or an infrared sensor. In at least one example, the second sensor can include a visual sensor such as a camera or an infrared sensor.

[1774] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.2-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.1.2-1** and **11.1.2-3-11.1.2-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **11.1.2-1** and **11.1.2-3-11.1.2-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.2-2**.

[1775] FIG. **11.1.2-3** illustrates a perspective view of a partially assembled HMD **11.1.2-300** including an outer frame **11.1.2-302**, an inner frame **11.1.2-304** coupled to the outer frame **11.1.2-302**, and a second mounting bracket **11.1.2-320** coupled to the inner frame **11.1.2-304** between first and second apertures **11.1.2-306a-b** defined by the inner frame **11.1.2-304**. In at least one example, the second mounting bracket **11.1.2-320** can include first and second cantilevered mounting arms **11.1.2-322**, **11.1.2-324** extending in opposite directions from one another from a middle portion **11.1.2-326** of the mounting bracket **11.1.2-320**. The first and second cantilever arms **11.1.2-322**, **11.1.2-324** can be free of affixation to any other component or frame of the HMD **11.1.2-300** similar to the cantilever arms **11.1.2-112**, **11.1.2-114** of the mounting bracket **11.1.2-108** shown in FIG. **11.1.2-1**. In at least one example, the mounting bracket **11.1.2-320** can be coupled to the inner frame **11.1.2-304** at the middle portion **11.1.2-326** of the mounting bracket **11.1.2-320**.

[1776] In at least one example, the first cantilever arm **11.1.2-322** can extend from the middle portion **11.1.2-326** in a first direction and the second cantilever arm **11.1.2-324** can extend from the middle portion **11.1.2-326** in a second direction opposite the first direction. In at least one example, a first guide rod **11.1.2-328** can be coupled to the first cantilever arm **11.1.2-322** and extend in the first direction and a second guide rod **11.1.2-330** can be coupled to the second cantilever arm **11.1.2-324** and extend in the second direction opposite the first direction.

[1777] In at least one example, first and second optical modules (not shown in FIG. **11.1.2-3**) can be slidably/adjustably coupled to the mounting bracket **11.1.2-320** via the first and second guide rods **11.1.2-328**, **11.1.2-330**. The guide rods **11.1.2-328**, **11.1.2-330** can include distal ends **11.1.2-332**, **11.1.2-334**, respectively, which are opposite respective proximal ends **11.1.2-333**, **11.1.2-335**. The distal ends **11.1.2-332**, **11.1.2-334** can be free of affixation to any other component of frame **11.1.2-302**, **11.1.2-304** of the HMD **11.1.2-300**. In this way, the guide rods **11.1.2-328**, **11.1.2-330** are cantilevered with free ends floating. In one or more examples, as noted above, one or more optical modules, including display screens, sensors, structural components, and electronic components can be slidably coupled to the inner frame **11.1.2-304** via the guide rods **11.1.2-328**,

11.1.2-330 and cantilever arms **11.1.2-322**, **11.1.2-324** of the mounting bracket **11.1.2-320**. The sensors, display screens, and other components of the optical module may be such that they operate most effectively when a relative distance between and position of those coupled to the first guide rod **11.1.2-328** and the second guide rod **11.1.2-330** are maintained, even during drop events and when other structural components, including the outer and/or inner frames **11.1.2-302**, **11.1.2-304** are inadvertently deformed or bent.

[1778] Along these lines, the cantilevered nature of the mounting bracket **11.1.2-320** can protect the optical modules and components thereof from damage and altered positioning in the case of accidental drops by the user. Because the optical modules cantilevered on the guide rods **11.1.2-328**, **11.1.2-330** and the cantilever arms **11.1.2-322**, **11.1.2-324** of the mounting bracket **11.1.2-320**, stresses and deformations of the inner and/or outer frames **11.1.2-304**, **11.1.2-302** are not transferred to the cantilevered arms **11.1.2-322**, **11.1.2-323** and/or the guide rods **11.1.2-328**, **11.1.2-330** and thus do not affect the relative positioning of the optical modules slidably coupled/mounted to the mounting bracket **11.1.2-320**. Along these lines, the material of the mounting bracket **11.1.2-320** can include metals such as aluminum, steel including stainless steel, magnesium or magnesium alloys, hard plastics, ceramics, composite materials, carbon fiber, or any combination thereof. In at least one example, the guide rods **11.1.2-328**, **11.1.2-330** can include low weight, low friction materials. For example, the guide rods **11.1.2-328**, **11.1.2-330** can include carbon fiber laminate materials. The guide rods **11.1.2-328**, **11.1.2-330** can be formed as hollow tubes to decrease weight.

[1779] In at least one example, the first and second cantilever arms **11.1.2-322** and **11.1.2-324** and/or the first and second guide-rods **11.1.2-328** and **11.1.2-330** extending therefrom, respectively, can be disposed at an angle relative to one another. The angle can be such that the bracket **11.1.2-326** and associated components, including guide-rods **11.1.2-328** and **11.1.2-330**, cantilever arms **11.1.2-322** and **11.1.2-324**, and display units connected to the guide-rods **11.1.2-328** and **11.1.2-330** are curved and/or oriented to complement the curvature of the user's face. In at least one example, the angle can be between about 5-degrees and about 25-degrees. In one example, the angle can be between about 10-degrees and about 20-degrees, for example about 15-degrees. In at least one example, the bracket **11.1.2-326** can be made of magnesium and carbon fiber panels on a rear side thereof to add stiffness.

[1780] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.2-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.1.2-1-11.1.2-2** and **11.1.2-4-11.1.2-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **11.1.2-1-11.1.2-2** and **11.1.2-4-11.1.2-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.2-3**.

[1781] FIG. **11.1.2-4** illustrates a portion of an HMD **11.1.2-400** including an optical module **11.1.2-436** slidably engaged to a mounting bracket via a guide rod **11.1.2-428**, which can be similar to the guide rod **11.1.2-328** shown in FIG. **11.1.2-3** and described above. The optical module **11.1.2-436** can include a display assembly **11.1.2-438** including a display screen **11.1.2-440**, all of which can be slidably engaged with the guide rod **11.1.2-428** as shown. As noted above, the components and parts shown and described in FIGS. **11.1.2-1-11.1.2-3** can be one of a set of components, including a set of two optical modules as part of a single HMD **11.1.2-400** such that the optical module **11.1.2-428** shown in FIG. **11.1.2-4** is one of two optical modules with the second optical module slidably engaged/coupled with a guide rod extending in an opposite direction to that of the guide rod **11.1.2-428** shown from a mounting bracket. Likewise, the display assembly **11.1.2-438** and the display screen **11.1.2-440** can be one of a set of two display assemblies and display screens within the single HMD **11.1.2-400**.

[1782] In at least one example, the optical module **11.1.2-436** can include a micro-OLED display coupled to a heat sink via an adhesive, such as a pressure sensitive adhesive or other adhesive. In at least one example, the optical module **11.1.2-436** can include a silicone backplane. The heat sink can include magnesium and have no fins. In one example, a drop or more of structural adhesive can be applied during assembly/manufacturing to the center of the back of the display of the optical module **11.1.2-436** to lock the display in with the heat sink. The adhesive can include epoxy.

[1783] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.2-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.1.2-1-11.1.2-3** and **11.1.2-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **11.1.2-1-11.1.2-3** and **11.1.2-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.2-4**.

[1784] FIG. **11.1.2-5** illustrates an example of an HMD **11.1.2-500** including a first, outer structural frame **11.1.2-502** defining an internal volume **11.1.2-542**. A front cover **11.1.2-541** and a rear cover **11.1.2-543** disposed opposite the front cover **11.1.2-541** can also define the internal volume **11.1.2-542**. In at least one example, the HMD **11.1.2-500** can also include a second, inner structural frame **11.1.2-504** disposed in the internal volume **11.1.2-542** and coupled to the outer frame **11.1.2-502**. The inner frame **11.1.2-502** can define a first side **11.1.2-503** and a second side **11.1.2-505** opposite the first side **11.1.2-503**.

[1785] In at least one example, the HMD **11.1.2-500** can include a first mounting bracket **11.1.2-508** coupled to the first side **11.1.2-503** of the inner frame **11.1.2-504** and a second mounting bracket **11.1.2-520** coupled to the second side **11.1.2-505** of the inner frame **11.1.2-504**. In at least one example, the first mounting bracket **11.1.2-508** can be similar to the first mounting brackets **11.1.2-108**, **11.1.2-208** shown in FIGS. **11.1.2-1** and **11.1.2-2** and described above. In at least one example, the second mounting bracket **11.1.2-520** can be similar to the other second mounting bracket **11.1.2-320** described above and shown in FIG. **11.1.2-3**.

[1786] In at least one example, one or more sensors **11.1.2-510a**, **11.1.2-510b** can be mounted or coupled to a cantilever arm or middle portion of the first mounting bracket **11.1.2-508**. In at least one example, an optical module **11.1.2-536** can be slidably engaged/coupled to the second mounting bracket **11.1.2-520**. The optical module **11.1.2-536** can include a display assembly including a display screen **11.1.2-540** and one or more sensors **11.1.2-544a**, **11.1.2-544b**. In at least one example, at least one of the sensors **11.1.2-544a-b** can include a visual sensor such as a camera. In at least one example, the first and second sensors **11.1.2-544a-b** are both cameras.

[1787] As oriented in the view of FIG. **11.1.2-5**, the sensors **11.1.2-510a-b** coupled to the first mounting bracket **11.1.2-508** can be pointed in a forward direction and the sensors **11.1.2-544a-b** and/or display screen **11.1.2-540** of the optical module **11.1.2-536** can be pointed in a rearward direction opposite the forward direction. In at least one example, the sensors **11.1.2-510a-b** coupled to the first mounting bracket **11.1.2-508** can include image capturing sensors or other sensors configured to object recognition and detection in front of or around the HMD **11.1.2-500**. The detected, sensed, or captured images and objects can be displayed by the display screen **11.1.2-536**, which can be configured to project the images and light rearward toward a user's eyes when the user dons the HMD **11.1.2-500**.

[1788] In at least one example, the HMD **11.1.2-500** can also include a controller electronically coupled to the sensors **11.1.2-510a-b** and optical module **11.1.2-536**, including the display screen **11.1.2-540** and sensors **11.1.2-544a-b**. The controller can be configured to capture an image via one or more of the sensors **11.1.2-510a-b** coupled to the first mounting bracket **11.1.2-508** and display the image on the display screen **11.1.2-540** of the optical module **11.1.2-536** coupled to the second mounting bracket **11.1.2-520**. In order to project an accurate representation of the captured objects in real space to the user on the display screen **11.1.2-540**, the relative position, distance, angles, and

orientation between the sensors **11.1.2-510a-b** and the display screen **11.1.2-540** can be maintained even during drop events or when the outer/inner frames **11.1.2-502**, **11.1.2-504** are deformed or bumped. The first mounting bracket **11.1.2-508** can be cantilevered and centrally coupled to the inner frame **11.1.2-504** at the first side **11.1.2-503** thereof as noted above and the second mounting bracket **11.1.2-520** can be cantilevered and centrally coupled to the inner frame **11.1.2-504** at the second side **11.1.2-505** thereof. In this way, the positions, orientations, angles, and distances between sensors **11.1.2-510a-b** of the first mounting bracket **11.1.2-508** and the sensors **11.1.2-544a-b** and display screen **11.1.2-540** of the second mounting bracket **11.1.2-520** can be maintained.

[1789] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.2-5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.1.2-1-11.1.2-4** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **11.1.2-1-11.1.2-4** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.2-5**.

11.1.3: Upper Guide Rod System

[1790] FIG. **11.1.3-1** illustrates a rear perspective view of a display assembly **11.1.3-102** of an HMD **11.1.3-100** with one or more components removed, including a curtain assembly, in order to illustrate various internal components of the HMD **11.1.3-100**. FIG. **11.1.3-1** illustrates the display assembly **11.1.3-102**, which can include a first display module **11.1.3-104a** and a second display module **11.1.3-104b**. The first display module **11.1.3-104a** can be slidably engaged/coupled with a first adjustment mechanism **11.1.3-106a** and the second optical module **11.1.3-104b** can be slidably engaged/coupled with a second adjustment mechanism **11.1.3-106b**. The first and second adjustment mechanisms **11.1.3-106a-b** can include first and second respective guide-rods **11.1.3-108a-b** and motors **11.1.3-110a-b**. The first and second guide-rods can be similar such that the description of one applies to the other, with the first guide-rod **11.1.3-108a** extending a first direction away from a bracket **11.1.3-112** and the second guide-rod **11.1.3-108b** extending a second direction away from the bracket **11.1.3-112** opposite the first direction.

[1791] Likewise, the first and second motors **11.1.3-110a-b** can be similar such that the description of one applies to the other, with the first motor **11.1.3-110a** mechanically engaging the first optical module **11.1.3-104a** to adjust the position thereof via the first guide-rod **11.1.3-108a** and the second motor **11.1.3-110b** mechanically engaging the second optical module **11.1.3-104b** to adjust the position thereof via the second guide-rod **11.1.3-108b**. In at least one example, the HMD **11.1.3-100** also includes a depressible and/or rotatable button **11.1.3-114** electrically coupled to the first and second motors **11.1.3-110a-b**. The button **11.1.3-114** can electrically communicate with the first and second motors **11.1.3-110a-b** via a processor or other circuitry components to cause the first and second motors **11.1.3-110a-b** to activate and cause the first and second optical modules **11.1.3-104a-b**, respectively, to change position relative to one another.

[1792] In at least one example, the first and second optical modules **11.1.3-104a-b** can include respective display screens configured to project light toward the user's eyes when donning the HMD **11.1.3-100**. In at least one example, the user can manipulate (i.e., depress and/or rotate) the button **11.1.3-114** to activate a positional adjustment of the optical modules **11.1.3-104a-b** to match the inter-pupillary distance of the user's eyes. The optical modules **11.1.3-104a-b** can also include one or more cameras or other sensors/sensor systems for imaging and measuring the IPD of the user such that the optical modules **11.1.3-104a-b** can be adjusted to match the IPD.

[1793] The first and second optical modules **11.1.3-104a-b** can be mechanically and slidably coupled to the first and second guide-rods **11.1.3-108a-b**, respectively. In at least one example, the first optical module **11.1.3-104a** can include an aperture, through-hole, or channel engaged with the guide-rod **11.1.3-108a** to hold the optical module **11.1.3-104a** in position axially with the guide-rod

11.1.3-108a as the optical module **11.1.3-104a** slides along the guide-rod **11.1.3-108a** during IPD adjustments. The same can be included in the second optical module **11.1.3-104b**. In some examples, the optical modules **11.1.3-104a-b** can include a guide-rod slidably engaged with respective channels of the adjustment mechanism **11.1.3-102**.

[1794] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.3-1**.

[1795] FIG. **11.1.3-2** illustrates a close up view of an example of an HMD **11.1.3-200** and an adjustment mechanism **11.1.3-202** similar to those shown in FIG. **11.1.3-1** but with the optical module removed to more clearly illustrate the upper guide-rod **11.1.3-208**. The guide rod **11.1.3-208** can extend at or near or above a top portion or side of the optical module and guide the optical module back and forth axially along the guide-rod **11.1.3-208** during IPD adjustments described above. In at least one example, the guide-rod **11.1.3-208** extends from a central bracket **11.1.3-212** secured to a frame or chassis **11.1.3-216** of the HMD **11.1.3-200**. In one example, the guide-rod **11.1.3-208** is cantilevered such that a first, proximal end **11.1.3-220** of the guide-rod **11.1.3-208** is secured to the bracket **11.1.3-212** and a second, distal end **11.1.3-222** opposite the first end **11.1.3-220** is free of affixation to any other structure. In this way, the guide-rod **11.1.3-208** can be cantilevered away from the bracket **11.1.3-212**.

[1796] In at least one example, the adjustment mechanism **11.1.3-202** also includes a stop **11.1.3-218** positioned at or near the distal end **11.1.3-222** of the guide-rod **11.1.3-208**. In at least one example, the stop **11.1.3-218** can include a cavity or channel in which the distal end of the guide-rod **11.1.3-208** extends. However, in at least one example, no contact is made between the guide-rod **11.1.3-208** and the stop **11.1.3-218** such that the guide-rod **11.1.3-208** is cantilevered as noted above. The stop **11.1.3-218** can be configured to prevent the optical module (not shown) from travelling too far distally along the guide-rod **11.1.3-208** and beyond the distal end **11.1.3-222** thereof during IPD adjustment. The stop **11.1.3-218** can also serve to prevent too much deformation of the cantilevered guide-rod **11.1.3-208** during unintended drop events or other forces that would cause a deformation of the guide-rod **11.1.3-208** and a movement of the distal end **11.1.3-222** of the guide-rod **11.1.3-208**. In at least one example, the distal end **11.1.3-222** of the guide-rod **11.1.3-208** extends into or is at least partially surrounded by the stop **11.1.3-218** without making contact during normal use of the HMD **11.1.3-200**. If unintentionally moved or deformed elastically, the distal end **11.1.3-222** or portion of the guide-rod **11.1.3-208** can make contact with the stop **11.1.3-218** such that the stop **11.1.3-218** prevents too much deformation of the guide-rod **11.1.3-208**, thus preventing plastic deformation and damage to the guide-rod **11.1.3-208**.

[1797] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3-1-11.1.3-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **11.1.3-1-11.1.3-2**.

[1798] FIG. **11.1.3-3** shows another view of an example of an adjustment system **11.1.3-302** similar to those shown in FIGS. **11.1.3-1** and **11.1.3-2** but with a first guide-rod **11.1.3-308a** and a second guide-rod **11.1.3-308b** shown extending in opposite directions from a central bracket **11.1.3-312**. In the illustrated example, the adjustment mechanism **11.1.3-302** also includes a second stop **11.1.3-318a** associated with the distal end of the first guide-rod **11.1.3-308a**, which can be similar in form

and function to the stop **11.1.3-218** described above with reference to FIG. **11.1.3-2**.

[1799] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **11.1.3-3**.

11.1.3.1: Motors

[1800] FIG. **1** illustrates a rear perspective view of a display assembly **11.1.3.1-102** of an HMD **11.1.3.1-100** with one or more components removed, including a curtain assembly, in order to illustrate various internal components of the HMD **11.1.3.1-100**. FIG. **11.1.3.1-1** illustrates the display assembly **11.1.3.1-102**, which can include a first display module **11.1.3.1-104a** and a second display module **11.1.3.1-104b**. The first display module **11.1.3.1-104a** can be slidably engaged/coupled with a first adjustment mechanism **11.1.3.1-106a** and the second optical module **11.1.3.1-104b** can be slidably engaged/coupled with a second adjustment mechanism **11.1.3.1-106b**. The first and second adjustment mechanisms **11.1.3.1-106a-b** can include first and second respective guide-rods **11.1.3.1-108a-b** and motors **11.1.3.1-110a-b**. The first and second guide-rods can be similar such that the description of one applies to the other, with the first guide-rod **11.1.3.1-108a** extending a first direction away from a bracket **11.1.3.1-112** and the second guide-rod **11.1.3.1-108b** extending a second direction away from the bracket **11.1.3.1-112** opposite the first direction.

[1801] Likewise, the first and second motors **11.1.3.1-110a-b** can be similar such that the description of one applies to the other, with the first motor **11.1.3.1-110a** mechanically engaging the first optical module **11.1.3.1-104a** to adjust the position thereof via the first guide-rod **11.1.3.1-108a** and the second motor **11.1.3.1-110b** mechanically engaging the second optical module **11.1.3.1-104b** to adjust the position thereof via the second guide-rod **11.1.3.1-108b**. In at least one example, the HMD **11.1.3.1-100** also includes a depressible and/or rotatable button **11.1.3.1-114** electrically coupled to the first and second motors **11.1.3.1-110a-b**. The button **11.1.3.1-114** can electrically communicate with the first and second motors **11.1.3.1-110a-b** via a processor or other circuitry components to cause the first and second motors **11.1.3.1-110a-b** to activate and cause the first and second optical modules **11.1.3.1-104a-b**, respectively, to change position relative to one another.

[1802] In at least one example, the motors **11.1.3.1-110a-b** are configured to operate the optical modules **11.1.3.1-104a-b** inward and outward relative to the bracket **11.1.3.1-112** and thus relative to the user's nose. As such, in at least one example, the motors **11.1.3.1-110a-b** operate under MS1 safety standards and the button **11.1.3.1-114** operating/instigating the motors **11.1.3.1-110a-b** includes a “dead-man's button” so the motors **11.1.3.1-110a-b** are only operable when the button **11.1.3.1-110a-b** is manipulated, e.g. depressed and/or rotated.

[1803] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3.1-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.3.1-1**.

[1804] FIG. **11.1.3.1-2** illustrates a perspective view of an example of a motor **11.1.3.1-210** similar to those **11.1.3.1-110a-b** shown in FIG. **11.1.3.1-1**. FIG. **11.1.3.1-3** illustrates a cross-sectional view thereof to illustrate internal components of the motor **11.1.3.1-210**. In at least one example, the motor **11.1.3.1-210** can include a gear box and breakaway, spring-loaded clutches in a nut (e.g.,

“split nut”). The split nut can be configured to prevent forces in a drop event from transferring through the motor **11.1.3.1-210**. In at least one example, during use, the motor **11.1.3.1-210** can be operated in resonant valleys to reduce noise.

[1805] In at least one example, the motor can include one or more dust covers (not shown) surrounding the housing to prevent dust and other debris from contaminating the internal volume of the motor **11.1.3.1-210** where the lead screw and other components are disposed. In at least one example, the dust covers can include tape wraps. In at least one example, the motor **11.1.3.1-210** can include a split screw/nut, as noted above, to reduce noise and form a breakaway component to release the optical module driven by the motor **11.1.3.1-210** such that forces transferred from the motor **11.1.3.1-210** to the optical module are minimized during high impact scenarios such as a drop event.

[1806] In at least one example, the motor **11.1.3.1-210** can include a gear reduction mechanism or assembly, such as a planetary gear reduction assembly. In at least one example, the motor **11.1.3.1-210** can include a clutch to tune the force to a certain level to release down length. In at least one example, the motor **11.1.3.1-210** can include pre-loaded ball bearings to reduce play in the motor **11.1.3.1-210**.

[1807] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **11.1.3.1-2** and **11.1.3.1-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **11.1.3.1-2** and **11.1.3.1-3**.

11.1.3.1.1: Electronic Devices with Optical Module Positioning Systems

[1808] A head-mounted device may have optical modules that present images to a user's eyes. Each optical module may have a lens barrel with a display and a lens that presents an image from the display to a corresponding eye box.

[1809] To accommodate users with different interpupillary distances, the optical modules may be slidably coupled to guide members such as guide rods. Actuators may slide the optical modules towards or away from each other along the guide rods, thereby accommodating different interpupillary distances.

[1810] The guide rods may be formed from fiber-reinforced composite tubes with one or more end caps that are fastened to a frame in the head-mounted device. A common end cap may, if desired, be used to join a pair of guide rods. End caps may be formed as separate pieces that are attached to the ends of the fiber composite tubes or other guide rod structures and/or may be integral portions of the fiber composite tubes or other guide rod structures.

[1811] The guide rods may include a left guide rod or left pair of guide rods slidably engaged with a left optical module and a right guide rod or right pair of guide rods slidably engaged with a right optical module. Left and right guide rods may be angled at a non-zero angle with respect to each other to help guide the optical modules parallel to the surface of a user's face.

[1812] The tubes of the guide rods may be partly or completely filled with cores to add strength. Low-friction coatings such as metal coatings may be applied to the tubes and on corresponding inner surfaces of the optical module structures that receive the tubes.

[1813] An electronic device such as a head-mounted device may have a front face that faces away from a user's head and may have an opposing rear face that faces the user's head. Optical modules at the rear face may be used to provide images to a user's eyes. The positions of the optical modules may be adjusted to accommodate different user interpupillary distances. The head-mounted device may have actuators and optical module guide structures to allow the optical module positions to be adjusted.

[1814] A top view of an illustrative head-mounted device is shown in FIG. **11.1.3.1.1-1**. As shown

in FIG. **11.1.3.1.1-1**, head-mounted devices such as electronic device **11.1.3.1.1-10** may have head-mounted support structures such as housing **11.1.3.1.1-12**. Housing **11.1.3.1.1-12** may include portions (e.g., head-mounted support structures **11.1.3.1.1-12T**) to allow device **11.1.3.1.1-10** to be worn on a user's head. Support structures **11.1.3.1.1-12T** may be formed from fabric, polymer, metal, and/or other material. Support structures **11.1.3.1.1-12T** may form a strap or other head-mounted support structures to help support device **11.1.3.1.1-10** on a user's head. A main support structure (e.g., main housing portion **11.1.3.1.1-12M**) of housing **11.1.3.1.1-12** may support electronic components such as displays **11.1.3.1.1-14**.

[1815] Main housing portion **11.1.3.1.1-12M** may include housing structures formed from metal, polymer, glass, ceramic, and/or other material. For example, housing portion **11.1.3.1.1-12M** may have housing walls on front face **F** and housing walls on adjacent top, bottom, left, and right side faces that are formed from rigid polymer or other rigid support structures and these rigid walls may optionally be covered with electrical components, fabric, leather, or other soft materials, etc. Housing portion **11.1.3.1.1-12M** may also have internal support structures such as a frame and/or structures that perform multiple functions such as controlling airflow while providing structural support. The walls of housing portion **11.1.3.1.1-12M** may enclose internal components **11.1.3.1.1-38** in interior region **11.1.3.1.1-34** of device **11.1.3.1.1-10** and may separate interior region **11.1.3.1.1-34** from the environment surrounding device **11.1.3.1.1-10** (exterior region **11.1.3.1.1-36**). Internal components **11.1.3.1.1-38** may include integrated circuits, actuators, batteries, sensors, and/or other circuits and structures for device **11.1.3.1.1-10**. Housing **11.1.3.1.1-12** may be configured to be worn on a head of a user and may form glasses, a hat, a helmet, goggles, and/or other head-mounted device. Configurations in which housing **11.1.3.1.1-12** forms goggles may sometimes be described herein as an example.

[1816] Front face **F** of housing **11.1.3.1.1-12** may face outwardly away from a user's head and face. Opposing rear face **R** of housing **11.1.3.1.1-12** may face the user. Portions of housing **11.1.3.1.1-12** (e.g., portions of main housing **11.1.3.1.1-12M**) on rear face **R** may form a cover such as cover **11.1.3.1.1-12C** (sometimes referred to as a curtain). The presence of cover **11.1.3.1.1-12C** on rear face **R** may help hide internal housing structures, internal components **11.1.3.1.1-38**, and other structures in interior region **11.1.3.1.1-34** from view by a user.

[1817] Device **11.1.3.1.1-10** may have left and right optical modules **11.1.3.1.1-40**. Optical modules **11.1.3.1.1-40** support electrical and optical components such as light-emitting components and lenses and may therefore sometimes be referred to as optical assemblies, optical systems, optical component support structures, lens and display support structures, electrical component support structures, or housing structures. Each optical module may include a respective display **11.1.3.1.1-14**, lens **11.1.3.1.1-30**, and support structure such as lens barrel **11.1.3.1.1-32**. Lens barrel **11.1.3.1.1-32**, which may sometimes be referred to as lens support structures, optical component support structures, optical module support structures, or optical module portions, may include hollow cylindrical structures with open ends or other supporting structures to house displays **11.1.3.1.1-14** and lenses **11.1.3.1.1-30**. Lens barrels **11.1.3.1.1-32** may, for example, include a left lens barrel that supports a left display **11.1.3.1.1-14** and left lens **11.1.3.1.1-30** and a right lens barrel that supports a right display **11.1.3.1.1-14** and right lens **11.1.3.1.1-30**.

[1818] Displays **11.1.3.1.1-14** may include arrays of pixels or other display devices to produce images. Displays **11.1.3.1.1-14** may, for example, include organic light-emitting diode pixels formed on substrates with thin-film circuitry and/or formed on semiconductor substrates, pixels formed from crystalline semiconductor dies, liquid crystal display pixels, scanning display devices, and/or other display devices for producing images.

[1819] Lenses **11.1.3.1.1-30** may include one or more lens elements for providing image light from displays **11.1.3.1.1-14** to respective eyes boxes **11.1.3.1.1-13**. Lenses may be implemented using refractive glass lens elements, using mirror lens structures (catadioptric lenses), using Fresnel lenses, using holographic lenses, and/or other lens systems.

[1820] When a user's eyes are located in eye boxes **11.1.3.1.1-13**, displays (display panels) **11.1.3.1.1-14** operate together to form a display for device **11.1.3.1.1-10** (e.g., the images provided by respective left and right optical modules **11.1.3.1.1-40** may be viewed by the user's eyes in eye boxes **11.1.3.1.1-13** so that a stereoscopic image is created for the user). The left image from the left optical module fuses with the right image from a right optical module while the display is viewed by the user.

[1821] It may be desirable to monitor the user's eyes while the user's eyes are located in eye boxes **11.1.3.1.1-13**. For example, it may be desirable to use a camera to capture images of the user's irises (or other portions of the user's eyes) for user authentication. It may also be desirable to monitor the direction of the user's gaze. Gaze tracking information may be used as a form of user input and/or may be used to determine where, within an image, image content resolution should be locally enhanced in a foveated imaging system. To ensure that device **11.1.3.1.1-10** can capture satisfactory eye images while a user's eyes are located in eye boxes **11.1.3.1.1-13**, each optical module **11.1.3.1.1-40** may be provided with a camera such as camera **11.1.3.1.1-42** and one or more light sources such as light-emitting diodes **11.1.3.1.1-44** or other light-emitting devices such as lasers, lamps, etc. Cameras **11.1.3.1.1-42** and light-emitting diodes **11.1.3.1.1-44** may operate at any suitable wavelengths (visible, infrared, and/or ultraviolet). As an example, diodes **11.1.3.1.1-44** may emit infrared light that is invisible (or nearly invisible) to the user. This allows eye monitoring operations to be performed continuously without interfering with the user's ability to view images on displays **11.1.3.1.1-14**.

[1822] Not all users have the same interpupillary distance IPD. To provide device **11.1.3.1.1-10** with the ability to adjust the interpupillary spacing between modules **11.1.3.1.1-40** along lateral dimension X and thereby adjust the spacing IPD between eye boxes **11.1.3.1.1-13** to accommodate different user interpupillary distances, device **11.1.3.1.1-10** may be provided with optical module positioning systems in housing **11.1.3.1.1-12**. The positioning systems may have guide members and actuators **11.1.3.1.1-43** that are used to position optical modules **11.1.3.1.1-40** with respect to each other.

[1823] Actuators **11.1.3.1.1-43** can be manually controlled and/or computer-controlled actuators (e.g., computer-controlled motors) for moving lens barrels **11.1.3.1.1-32** relative to each other. Information on the locations of the user's eyes may be gathered using, for example, cameras **11.1.3.1.1-42**. The locations of eye boxes **11.1.3.1.1-13** can then be adjusted accordingly.

[1824] As shown in the rear view of device **11.1.3.1.1-10** of FIG. **11.1.3.1.1-2**, cover **11.1.3.1.1-12C** may cover rear face R while leaving lenses **11.1.3.1.1-30** of optical modules **11.1.3.1.1-40** uncovered (e.g., cover **11.1.3.1.1-12C** may have openings that are aligned with and receive modules **11.1.3.1.1-40**). As modules **11.1.3.1.1-40** are moved relative to each other along dimension X to accommodate different interpupillary distances for different users, modules **11.1.3.1.1-40** move relative to fixed housing structures such as the walls of main portion **11.1.3.1.1-12M** and move relative to each other.

[1825] A schematic diagram of an illustrative electronic device such as a head-mounted device or other wearable device is shown in FIG. **11.1.3.1.1-3**. Device **11.1.3.1.1-10** of FIG. **11.1.3.1.1-3** may be operated as a stand-alone device and/or the resources of device **11.1.3.1.1-10** may be used to communicate with external electronic equipment. As an example, communications circuitry in device **11.1.3.1.1-10** may be used to transmit user input information, sensor information, and/or other information to external electronic devices (e.g., wirelessly or via wired connections). Each of these external devices may include components of the type shown by device **11.1.3.1.1-10** of FIG. **11.1.3.1.1-3**.

[1826] As shown in FIG. **11.1.3.1.1-3**, a head-mounted device such as device **11.1.3.1.1-10** may include control circuitry **11.1.3.1.1-20**. Control circuitry **11.1.3.1.1-20** may include storage and processing circuitry for supporting the operation of device **11.1.3.1.1-10**. The storage and processing circuitry may include storage such as nonvolatile memory (e.g., flash memory or other

electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **11.1.3.1.1-20** may be used to gather input from sensors and other input devices and may be used to control output devices. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors and other wireless communications circuits, power management units, audio chips, application specific integrated circuits, etc. During operation, control circuitry **11.1.3.1.1-20** may use display(s) **11.1.3.1.1-14** and other output devices in providing a user with visual output and other output. [1827] To support communications between device **11.1.3.1.1-10** and external equipment, control circuitry **11.1.3.1.1-20** may communicate using communications circuitry **11.1.3.1.1-22**. Circuitry **11.1.3.1.1-22** may include antennas, radio-frequency transceiver circuitry, and other wireless communications circuitry and/or wired communications circuitry. Circuitry **11.1.3.1.1-22**, which may sometimes be referred to as control circuitry and/or control and communications circuitry, may support bidirectional wireless communications between device **11.1.3.1.1-10** and external equipment (e.g., a companion device such as a computer, cellular telephone, or other electronic device, an accessory such as a point device, computer stylus, or other input device, speakers or other output devices, etc.) over a wireless link. For example, circuitry **11.1.3.1.1-22** may include radio-frequency transceiver circuitry such as wireless local area network transceiver circuitry configured to support communications over a wireless local area network link, near-field communications transceiver circuitry configured to support communications over a near-field communications link, cellular telephone transceiver circuitry configured to support communications over a cellular telephone link, or transceiver circuitry configured to support communications over any other suitable wired or wireless communications link. Wireless communications may, for example, be supported over a Bluetooth® link, a WiFi® link, a wireless link operating at a frequency between 10 GHz and 400 GHz, a 60 GHz link, or other millimeter wave link, a cellular telephone link, or other wireless communications link. Device **11.1.3.1.1-10** may, if desired, include power circuits for transmitting and/or receiving wired and/or wireless power and may include batteries or other energy storage devices. For example, device **11.1.3.1.1-10** may include a coil and rectifier to receive wireless power that is provided to circuitry in device **11.1.3.1.1-10**.

[1828] Device **11.1.3.1.1-10** may include input-output devices such as devices **11.1.3.1.1-24**. Input-output devices **11.1.3.1.1-24** may be used in gathering user input, in gathering information on the environment surrounding the user, and/or in providing a user with output. Devices **11.1.3.1.1-24** may include one or more displays such as display(s) **11.1.3.1.1-14**. Display(s) **11.1.3.1.1-14** may include one or more display devices such as organic light-emitting diode display panels (panels with organic light-emitting diode pixels formed on polymer substrates or silicon substrates that contain pixel control circuitry), liquid crystal display panels, microelectromechanical systems displays (e.g., two-dimensional mirror arrays or scanning mirror display devices), display panels having pixel arrays formed from crystalline semiconductor light-emitting diode dies (sometimes referred to as microLEDs), and/or other display devices.

[1829] Sensors **11.1.3.1.1-16** in input-output devices **11.1.3.1.1-24** may include force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors such as a touch sensor that forms a button, trackpad, or other input device), and other sensors. If desired, sensors **11.1.3.1.1-16** may include optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, optical touch sensors, optical proximity sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, image sensors, fingerprint sensors, iris scanning sensors, retinal scanning sensors, and other biometric sensors, temperature sensors, sensors for measuring three-dimensional non-contact gestures (“air gestures”), pressure sensors, sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors

such as compass sensors, gyroscopes, and/or inertial measurement units that contain some or all of these sensors), health sensors such as blood oxygen sensors, heart rate sensors, blood flow sensors, and/or other health sensors, radio-frequency sensors, depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices that capture three-dimensional images), optical sensors such as self-mixing sensors and light detection and ranging (LIDAR) sensors that gather time-of-flight measurements, humidity sensors, moisture sensors, gaze tracking sensors, electromyography sensors to sense muscle activation, facial sensors, and/or other sensors. In some arrangements, device **11.1.3.1.1-10** may use sensors **11.1.3.1.1-16** and/or other input-output devices to gather user input. For example, buttons may be used to gather button press input, touch sensors overlapping displays can be used for gathering user touch screen input, touch pads may be used in gathering touch input, microphones may be used for gathering audio input (e.g., voice commands), accelerometers may be used in monitoring when a finger contacts an input surface and may therefore be used to gather finger press input, etc.

[1830] If desired, electronic device **11.1.3.1.1-10** may include additional components (see, e.g., other devices **11.1.3.1.1-18** in input-output devices **11.1.3.1.1-24**). The additional components may include haptic output devices, actuators for moving movable housing structures, audio output devices such as speakers, light-emitting diodes for status indicators, light sources such as light-emitting diodes that illuminate portions of a housing and/or display structure, other optical output devices, and/or other circuitry for gathering input and/or providing output. Device **11.1.3.1.1-10** may also include a battery or other energy storage device, connector ports for supporting wired communication with ancillary equipment and for receiving wired power, and other circuitry.

[1831] A rear view of device **11.1.3.1.1-10** is shown in FIG. **11.1.3.1.1-4**. In the example of FIG. **11.1.3.1.1-4**, cover **11.1.3.1.1-12C** has been removed to expose internal housing structures such as frame **11.1.3.1.1-12FC**. Frame **11.1.3.1.1-12FC** may be formed from polymer support structures, support structures formed from carbon-fiber composite and/or other fiber-composite materials, metal support structures, glass housing structures, and/or other support structures for main housing portion **11.1.3.1.1-12M**. Portions of frame **11.1.3.1.1-12FC** that run horizontally (along the upper and lower edges of device **11.1.3.1.1-10** parallel to the X axis of FIG. **11.1.3.1.1-4**) may be joined by vertically extending edge portions of frame **11.1.3.1.1-12FC** (see, e.g., frame edge portions **11.1.3.1.1-12FC-E**) along the left and right sides of device **11.1.3.1.1-10** and may be joined in the center of device **11.1.3.1.1-10** by a vertically extending nose bridge portion (see, e.g., frame center portion **11.1.3.1.1-12FC-M**).

[1832] Optical modules **11.1.3.1.1-40** may be guided using optical module guide structures such as optical module guide rods **11.1.3.1.1-50**. Guide rods **11.1.3.1.1-50** may extend horizontally across device **11.1.3.1.1-10** (e.g., parallel to the X axis of FIG. **11.1.3.1.1-4**). In the example of FIG. **11.1.3.1.1-4**, guide rods **11.1.3.1.1-50** include an upper set of left and right guide rods at upper position **11.1.3.1.1-52** and a lower set of left and right guide rods at lower position **11.1.3.1.1-54**. There may be more guide rods or fewer guide rods in device **11.1.3.1.1-10**, if desired. Fasteners **11.1.3.1.1-56** (e.g., screws) or other attachment structures (e.g., adhesive, welds, etc.) may be used to attach guide rods **11.1.3.1.1-50** to frame **11.1.3.1.1-12FC**. The fasteners may, as an example, attach the ends of rods **11.1.3.1.1-50** that are located on the left of device **11.1.3.1.1-10** between left frame edge portion **11.1.3.1.1-12FC-E** and frame center portion **11.1.3.1.1-12FC-M** and may attach the ends of rods **11.1.3.1.1-50** that are located on the right of device **11.1.3.1.1-10** between right frame edge portion **11.1.3.1.1-12FC-E** and frame center portion **11.1.3.1.1-12FC-M**. Guide rods **11.1.3.1.1-50** may also be attached to the frame or to an intermediary subframe. Guide rods may be fixed to a frame or subframe at both ends (as illustrated) or each guide rod may only be attached to the frame at a single end. For example, each guide rod may only be fixed to the frame at a central portion of the frame.

[1833] Each optical module may have portions that slidably engage guide rods **11.1.3.1.1-50**. For example, each optical module **11.1.3.1.1-40** may have an upper guide rod engagement portion

(sometimes referred to as a hanger or hanger portion) such as optical module portion **11.1.3.1.1-40H** that receives and engages a respective guide rod **11.1.3.1.1-50** at upper guide rod position **11.1.3.1.1-52**. Each optical module **11.1.3.1.1-40** may also have a lower guide rod engagement portion (sometimes referred to as a toe or toe portion) such as optical module portion **11.1.3.1.1-40T** that receives and engages a respective guide rod **11.1.3.1.1-50** at lower guide rod position **11.1.3.1.1-54**. Portion **11.1.3.1.1-40T** may be an integral portion of lens barrel **11.1.3.1.1-32** or other support structure for optical module **11.1.3.1.1-40** or may be formed from one or more separate structures attached to lens barrel **11.1.3.1.1-32**. Portion **11.1.3.1.1-40H** may be an integral portion of lens barrel **11.1.3.1.1-32** or may, as shown in FIG. **11.1.3.1.1-4**, be formed from one or more separate optical module structures that are attached to lens barrel **11.1.3.1.1-32** by fasteners **11.1.3.1.1-60** (e.g., screws). If desired, guide rods **11.1.3.1.1-50** may be provided with structures that serve as stops to prevent over-travel of optical modules **11.1.3.1.1-40**. As an example, each of guide rods **11.1.3.1.1-50** may have one or more stop structures such as plug **11.1.3.1.1-57**. During sliding movement of optical module **11.1.3.1.1-40**, excessive movement will be prevented when portion **11.1.3.1.1-40H** contacts plug **11.1.3.1.1-57** and is thereby stopped by plug **11.1.3.1.1-57**. Optical module sliding motion stop structures such as plug **11.1.3.1.1-57** may be attached to guide rods **11.1.3.1.1-50** by screwing, by welding, using adhesive, or using other attachment structures. Plug **11.1.3.1.1-57** may be a separate structure from rods **11.1.3.1.1-50** or such structures may be formed as part of guide rods **11.1.3.1.1-50** (e.g., a metal guide rod or other guide rod may be machined or otherwise formed into a shape that includes an integral sliding motion stop structure).

[1834] Guide rods **11.1.3.1.1-50** may have circular cross-sectional shapes (when viewed in the Y-Z plane of FIG. **11.1.3.1.1-4**) or may have other suitable cross-sectional shape. The portions of optical module **11.1.3.1.1-40** that receive guide rods **11.1.3.1.1-50** may have corresponding mating shapes (e.g., full or partial circular openings with inner diameters corresponding to the outer diameters of guide rods **11.1.3.1.1-50**). To prevent sticking, the inner surfaces of the optical module guide rod openings and/or the outer surfaces of guide rods **11.1.3.1.1-50** may be provided with low stick surfaces (e.g., using low-stick coatings, lubricant such as grease, etc.).

[1835] Actuators **11.1.3.1.1-43** may have associated threaded members such as threaded actuator rods **11.1.3.1.1-62**. Optical module portions **11.1.3.1.1-40H** may have corresponding threaded nuts **11.1.3.1.1-64** or other threaded portions that receive threaded actuator rods **11.1.3.1.1-62**. During operation, actuators **11.1.3.1.1-43** may rotate threaded actuator rods **11.1.3.1.1-62** about actuator rod rotational axes **11.1.3.1.1-66**, thereby moving optical modules **11.1.3.1.1-40** outwardly (away from each other) in directions **11.1.3.1.1-68** or inwardly (towards each other) in directions **11.1.3.1.1-70** as desired to adjust the positions of optical modules **11.1.3.1.1-40** relative to each other (e.g., to adjust the lens-center-to-lens-center spacing of the left and right lenses in device **11.1.3.1.1-10** to accommodate different interpupillary distances for different users).

[1836] FIG. **11.1.3.1.1-5** is an end view of an illustrative portion of optical module **11.1.3.1.1-40** such as portion **11.1.3.1.1-40H** of FIG. **11.1.3.1.1-4**. As shown in FIG. **11.1.3.1.1-5**, optical module portion **11.1.3.1.1-40H** may have a first opening such as opening **11.1.3.1.1-72** (e.g., a threaded opening in nut **11.1.3.1.1-64** or other portion of optical module portion **11.1.3.1.1-40H**). Opening **72** may receive threaded rod **11.1.3.1.1-62**.

[1837] Optical module portion **11.1.3.1.1-40H** may also have a second opening such as opening **11.1.3.1.1-76**. Opening **11.1.3.1.1-76** may receive a guide rod (e.g., a guide rod located at upper position **11.1.3.1.1-52**). If desired, optical module portion **11.1.3.1.1-40T** may have an opening such as opening **11.1.3.1.1-76** to receive a guide rod (e.g., a guide rod located at lower position **11.1.3.1.1-54**). As shown in FIG. **11.1.3.1.1-5**, the inner surfaces of opening **11.1.3.1.1-76** may have one or more layers of material that form low-friction coating **11.1.3.1.1-78**. In an illustrative configuration, optical module **11.1.3.1.1-40** (e.g., optical module portion **11.1.3.1.1-40H**, lens barrel **11.1.3.1.1-32**, and/or optical module portion **11.1.3.1.1-40T**) are formed from metal (e.g., aluminum) and coating **11.1.3.1.1-78** is formed from one or more deposited (e.g., electrodeposited)

metal layers (e.g., nickel, etc.). Configurations in which modules **11.1.3.1.1-40** are formed from polymer coated with metal and/or in which modules **11.1.3.1.1-40** are formed from other materials may be used, if desired.

[1838] Guide rods **11.1.3.1.1-52** may be formed from elongated guide member structures such as tubes. The tubes may be cylindrical tubes or may be tubes of other suitable shapes (e.g., tubes with rectangular cross-sectional shapes, etc.). Guide rods **11.1.3.1.1-52** may be formed from tubes that are completely hollow, that are partly hollow and partly filled with cores, or tubes that are completely filled with filler material (e.g., rods that are formed from tubes entirely filled with a core material that differs in composition, density, method of fabrication, or other respects from the material of the tubes such as a composite core material formed from fibers or other structures embedded in polymer, rods that are formed from a single material such as solid cylindrical rods, solid composite rods of fiber-composite material such as solid cylindrical composite rods, rods formed from other composites, solid rods of metal or polymer, etc.). The use of tubes that are at least partly hollow may help save weight and thereby help enhance the comfort of a user wearing device **11.1.3.1.1-10**.

[1839] FIG. **11.1.3.1.1-6** is a cross-sectional view of an illustrative guide rod. As shown in FIG. **11.1.3.1.1-6**, guide rod **11.1.3.1.1-50** may include guide rod tube **11.1.3.1.1-82** and a guide rod end member such as end cap **11.1.3.1.1-86**. Guide rod tube **11.1.3.1.1-82** may be hollow and may be characterized by cylindrical walls surrounding interior **84**. In the example of FIG. **11.1.3.1.1-6**, tube **11.1.3.1.1-82** extends along longitudinal axis **11.1.3.1.1-80** and is rotationally symmetric around axis **11.1.3.1.1-80** (e.g., tube **11.1.3.1.1-82** is cylindrical). End cap **11.1.3.1.1-86** may be attached to the end of tube **11.1.3.1.1-82** using adhesive, fasteners, welds, by molding polymer end cap material over the end of tube **11.1.3.1.1-82**, and/or other attachment mechanisms. In an illustrative configuration, end cap **11.1.3.1.1-86** may have a first opening such as opening **11.1.3.1.1-88** (e.g., a through-hole opening) to receive a first fastener (e.g., a first screw) and may have a second opening such as opening **11.1.3.1.1-90** (e.g., a through-hole opening) to receive a second fastener (e.g., a second screw). The first screw may be used to attach end cap **11.1.3.1.1-86** to the end of tube **11.1.3.1.1-82** and the second screw may be used to attach end cap **11.1.3.1.1-86** to main housing portion **11.1.3.1.1-12M** (e.g., frame **11.1.3.1.1-12FC**). Other numbers of end cap openings and/or different features (e.g., threads) in end cap **11.1.3.1.1-86** for attaching end cap **11.1.3.1.1-86** to the end of tube **11.1.3.1.1-82** may be used, if desired.

[1840] FIG. **11.1.3.1.1-7** is a side view of guide rod **11.1.3.1.1-50** of FIG. **11.1.3.1.1-7** following insertion of end cap **11.1.3.1.1-86** into the end of tube **11.1.3.1.1-82**. The portion of end cap **11.1.3.1.1-86** that is inserted into tube **11.1.3.1.1-82** may have an outer diameter that corresponds to the inner diameter of tube **11.1.3.1.1-82**.

[1841] As shown in the cross-sectional view of FIG. **11.1.3.1.1-8**, fasteners **11.1.3.1.1-92** (e.g., screws) may be received by opening **11.1.3.1.1-88** (to attach end cap **11.1.3.1.1-86** to tube **11.1.3.1.1-82**) and opening **11.1.3.1.1-90** (to attach end cap **11.1.3.1.1-86** and the rest of guide rod **11.1.3.1.1-50** to a housing structure such as frame **11.1.3.1.1-12FC** (e.g., by screwing a threaded end of the fastener passing through opening **11.1.3.1.1-90** into threaded frame opening **11.1.3.1.1-94**). If desired, end cap **11.1.3.1.1-86** may have a flattened surface portion such as planar surface **96** that rests against a corresponding planar portion of frame **11.1.3.1.1-12FC** when end cap **11.1.3.1.1-86** is attached to frame **11.1.3.1.1-12FC**. The presence of mating planar surfaces may help prevent undesired rotation of guide rod **11.1.3.1.1-50** about axis **11.1.3.1.1-80**.

[1842] As shown in FIG. **11.1.3.1.1-4**, device **11.1.3.1.1-10** may have left and right guide rods (e.g., upper left and right guide rods and lower left and right guide rods). Left and right guide rods **11.1.3.1.1-50** may have separate end caps **11.1.3.1.1-86** each with a respective opening **90**, as shown in FIG. **11.1.3.1.1-9**, or left and right guide rods **11.1.3.1.1-50** may share a common end cap **11.1.3.1.1-86** with one or more openings **90**, as shown in FIG. **11.1.3.1.1-10**. A shared end cap may be straight (e.g., so that axes **11.1.3.1.1-80** of the left and right guide rods are parallel and aligned

with each other as shown by straight end cap **11.1.3.1.1-86** of FIG. **11.1.3.1.1-11**) or a shared end cap may be bent (e.g., so that axes **11.1.3.1.1-80** of the left and right guide rods are not parallel and instead are angled at a non-zero angle A with respect to each other as shown by bent end cap **11.1.3.1.1-86** of FIG. **11.1.3.1.1-12**). Bent end cap structures such as the bent end cap of FIG. **11.1.3.1.1-12** may be formed from single pieces of metal (or other material) or may be formed from two members (e.g., two metal members) joined by a welded joint (e.g., a laser weld), an adhesive joint, or other bond **11.1.3.1.1-98**. The use of separate or common end cap structures to orient the left and right guide rods **11.1.3.1.1-50** at a non-zero angle A with respect to each other (e.g., an angle A of at least 3°, at least 6°, at least 9°, at least 15°, at least 20°, at least 30°, at least 40°, less than 50°, less than 45°, less than 35°, less than 25°, and/or less than) 15° may help orient the left and right optical modules of device **11.1.3.1.1-10** in front of the left and right sides of the user's face (which tend to angle away from each other slightly). Left and right guide rods may also be made from a single tube. The single tube may be straight (e.g., left and right guide rods may be formed from respective left and right portions of a single straight tube) or the left and right guide rods may be formed from respective left and right portions of a single tube that has been molded or bent so that the left and right guide rods are oriented at a non-zero angle A with respect to each other.

[1843] If desired, rods **11.1.3.1.1-50** may be attached to housing **11.1.3.1.1-12M** (e.g., frame **11.1.3.1.1-12FC**) using press-fit connections between rods **11.1.3.1.1-50** and housing **11.1.3.1.1-12M**, using shrink-fit connections between rods **11.1.3.1.1-50** and housing **11.1.3.1.1-12M**, and/or using other attachment mechanisms such as gluing (e.g., gluing rods **11.1.3.1.1-50** to housing **11.1.3.1.1-12M**). In some arrangements, some or all of end caps **11.1.3.1.1-86** may be omitted (e.g., to help reduce weight). For example, housing **11.1.3.1.1-12M** (e.g., frame **11.1.3.1.1-12FC**) may include portions that are configured to be received within the cylindrical hollow interior of rods **11.1.3.1.1-50** at the ends of rods **11.1.3.1.1-50** (e.g., housing **11.1.3.1.1-12M** may have integral housing portions with the shapes of end caps **11.1.3.1.1-84** that attach to the insides of rods **11.1.3.1.1-50** using a friction fit from a press-fit or shrink-fit connection and/or using adhesive), housing **11.1.3.1.1-12M** may include portions that are configured to form cylindrical openings or other shapes (e.g., clamp shapes) that receive the ends of rods **11.1.3.1.1-50** (e.g., rods **11.1.3.1.1-50** may be inserted within openings in housing **11.1.3.1.1-12M** to attach to the outer surfaces of the ends of rods **11.1.3.1.1-50** to housing **11.1.3.1.1-12M** with a friction fit and/or adhesive), and/or other housing structures (e.g., portions of housing **11.1.3.1.1-12M** such as portions of frame **11.1.3.1.1-12FC**) that mount rods **11.1.3.1.1-50** to housing **11.1.3.1.1-12M** directly.

[1844] If desired, some or all of the interior of each guide rod tube may be filled with supporting material. Consider, as an example, guide rod **11.1.3.1.1-50** of FIG. **11.1.3.1.1-13**. As shown in FIG. **11.1.3.1.1-13**, each guide rod **11.1.3.1.1-50** in device **11.1.3.1.1-10** may have a pair of end caps **11.1.3.1.1-86** (one at each of the opposing ends of guide rod tube **11.1.3.1.1-82**). At one or more locations along the length of tube **11.1.3.1.1-82** such as illustrative location **11.1.3.1.1-100**, tube **11.1.3.1.1-82** may be provided with a solid supportive core such as core **11.1.3.1.1-102** (e.g., a polymer core such as a cylindrical rod-shaped polymer foam core, a gel core, low-density thermoplastic polymer, or other low-density core of polymer, a metal core, a wood core, or a core formed from other supportive structures). Location **11.1.3.1.1-100** may correspond to the center portion of tube **11.1.3.1.1-82** over which openings in optical module **11.1.3.1.1-40** such as opening **11.1.3.1.1-76** travel (as an example). Core **11.1.3.1.1-102** may be shear-coupled to the inner surface of tube **11.1.3.1.1-82** and may help provide guide tube **11.1.3.1.1-50** with bending strength and torsional strength.

[1845] Tube **11.1.3.1.1-82** may be formed from metal, polymer, and/or fiber-composite material such as carbon-fiber material, fiberglass material (e.g., glass-fiber-reinforced structural polymer), other fiber-reinforced polymer, etc. The use of fiber-composite tubes may help reduce the weight of rods **11.1.3.1.1-50**.

[1846] Consider, as an example, tube **11.1.3.1.1-82** of FIGS. **11.1.3.1.1-14** and **11.1.3.1.1-15**. As shown in the side view of FIG. **11.1.3.1.1-14** and the cross-sectional end view of FIG. **11.1.3.1.1-15**, fiber-composite tube **11.1.3.1.1-82** of FIGS. **11.1.3.1.1-14** and **11.1.3.1.1-15** may have fibers **11.1.3.1.1-104** embedded in polymer **11.1.3.1.1-106**. Fibers **11.1.3.1.1-104** may be carbon fibers, glass fibers, or other strands of material for enhancing the strength of polymer **11.1.3.1.1-106**. Polymer **11.1.3.1.1-106**, which may sometimes be referred to as binder or resin, may be epoxy, polyether ether ketone (PEEK), a thermoset polymer, a thermoplastic polymer, and/or other polymer material. Materials such as PEEK may exhibit satisfactory wear properties and a low coefficient of friction. Other polymers may be used for forming binder for tube **11.1.3.1.1-82** if desired.

[1847] Fibers **11.1.3.1.1-104** may extend in one or more different directions. For example, fibers **11.1.3.1.1-104** may include fibers that extend longitudinally (parallel to tube longitudinal axis **11.1.3.1.1-80**), that wrap around the circumference of tube **84** (e.g., about axis **11.1.3.1.1-80**), and/or that have angled orientations (e.g., $\pm 45^\circ$) relative to axis **11.1.3.1.1-80**. These different types of fiber may be formed in a single layer of fibers or multiple layers of fiber may overlap in a stack. The stack of fiber layers may wrap around tube **11.1.3.1.1-82** and may optionally be covered with a low-friction coating.

[1848] Consider, as an example, the fiber-composite guide rod tube in the cross-sectional side view of FIG. **11.1.3.1.1-16**. In the example of FIG. **11.1.3.1.1-16**, guide rod **11.1.3.1.1-50** includes tube **11.1.3.1.1-82** and end cap **11.1.3.1.1-86**. End cap **11.1.3.1.1-86** has a hollow tube shape with hollow interior region **11.1.3.1.1-110**. The outer diameter of end cap **11.1.3.1.1-86** (which includes a hollow portion in the example of FIG. **11.1.3.1.1-16**) corresponds to the inner diameter of tube **11.1.3.1.1-82**. A layer of optional adhesive **11.1.3.1.1-112** may be used to help attach end cap **11.1.3.1.1-86** to tube **11.1.3.1.1-82**.

[1849] Tube **11.1.3.1.1-82** is hollow and surrounds interior region **11.1.3.1.1-84**. Tube **11.1.3.1.1-82** has a hollow cylindrical fiber-composite tube portion formed (in the example of FIG. **11.1.3.1.1-16**) from fiber-composite layers **11.1.3.1.1-114** covered with low-friction coating **11.1.3.1.1-116**. Layers **11.1.3.1.1-114** form a hollow cylindrical fiber-composite tube surrounding interior **11.1.3.1.1-84**. There may be any suitable number N of fiber layers in the tube formed from layers **11.1.3.1.1-114** (e.g., N may be at least one, at least three, at least four, at least six, less than ten, less than eight, less than seven, 3-8, etc.). Fiber composite layers **11.1.3.1.1-114** in the example of FIG. **11.1.3.1.1-16** include six fiber-composite layers: **L1**, **L2**, **L3**, **L4**, **L5**, and **L6**. These layers may have uniaxially aligned fibers and may have fibers that are oriented at 0° , 90° , 0° , 0° , 90° , and 0° , respectively relative to axis **11.1.3.1.1-80**. Configurations in which layers **11.1.3.1.1-114** include fibers oriented at $\pm 45^\circ$ or other angles may also be used. Fibers oriented along the length of tube **11.1.3.1.1-82** may enhance bend strength. Fibers that wrap around tube **11.1.3.1.1-82** perpendicular to axis **11.1.3.1.1-80** may enhance tube strength. Fibers oriented at $\pm 45^\circ$ may enhance torsional rigidity. If desired, different portions along the length of tube **11.1.3.1.1-82** may have different fiber orientations and/or different number of fiber layers.

[1850] Coating layer **11.1.3.1.1-116** may, if desired, be formed from one or more metal layers. As an example, first (inner) metal layer **C1** may be a nickel cobalt layer that has a thickness of 50 microns or other suitable thickness and second (outer) metal layer **C2** may be an electroless nickel layer that is deposited on top of the nickel cobalt layer and has a thickness of 50-400 microns thick, 100-200 microns thick, or other suitable thickness. Layer **C1** may serve as an adhesion promotion layer. The outer surface of layers **11.1.3.1.1-114** may be etched prior to coating layers **11.1.3.1.1-114** with layer **C1** to enhance adhesion of layer **C1** to layers **11.1.3.1.1-114**. Layer **C1** may interlock with epoxy (or other polymer) in layers **11.1.3.1.1-114** and may enhance adhesion of layer **C2**. Layer **C2** may help provide tube **11.1.3.1.1-82** with low friction as tube **11.1.3.1.1-82** moves back and forth within an opening in portion **11.1.3.1.1-40H** or **11.1.3.1.1-40T** of optical module **11.1.3.1.1-40** (see, e.g., opening **11.1.3.1.1-76** of FIG. **11.1.3.1.1-5**, which may have a nickel

coating or other low-friction coating **11.1.3.1.1-78**). The thickness of layer C2 may help enhance the strength of tube **11.1.3.1.1-82** (e.g., bending strength).

[1851] By using guide rods **11.1.3.1.1-50**, the lateral positions of modules **11.1.3.1.1-40** in device **11.1.3.1.1-10** may be adjusted to accommodate different user interpupillary distances. Optical module position adjustments may be automated or manual. The optical positioning system(s) of device **11.1.3.1.1-10** may use guide structures such as guide rods **11.1.3.1.1-50** to allow optical modules **11.1.3.1.1-40** to move along a desired axis. The guide rods and the mounting structures used to attach the guide rods to housing **11.1.3.1.1-12M** (e.g., frame **11.1.3.1.1-12FC**) may be sufficiently rigid and strong to resist deformation and misalignment in the event that device **11.1.3.1.1-10** is inadvertently dropped.

[1852] To reduce the burden on actuators **11.1.3.1.1-43** as actuators **11.1.3.1.1-43** rotate threaded rods **11.1.3.1.1-62** to move modules **11.1.3.1.1-40** along guide rods **11.1.3.1.1-50**, guide rods **11.1.3.1.1-50** may be slidably coupled to modules **11.1.3.1.1-40** using low-friction structures. These low friction structures may include using low-friction coating materials such as nickel. The coating layer(s) may be polished (e.g., using a centerless grinding tool) and/or otherwise finished to help reduce friction. If desired, the use of nickel coating material may be omitted (e.g., when finishing rods **11.1.3.1.1-50** using burnishing, grinding, or polishing to provide a low-friction surface). The weight of guide rods **11.1.3.1.1-50**, which may affect user comfort, may be reduced by using fiber-composite materials or other light materials in forming guide rod tubes **11.1.3.1.1-82**.

[1853] An example of a finishing process that may be used to help reduce friction between modules **11.1.3.1.1-40** and rods **11.1.3.1.1-50** is superfinishing. Superfinishing is a microfinishing technique that can be used to enhance the surface finish of an item while also enhancing the accuracy of the contours of the item (e.g., enhancing the accuracy of the desired shapes of rods **11.1.3.1.1-50** such as enhancing the cylindricity of rods **11.1.3.1.1-50** and/or the accuracy of the desired shapes for the mating portions of modules **11.1.3.1.1-40**). With superfinishing, small amounts of surface material (e.g., 1-2 microns) are removed by superfinishing equipment using abrasive. The surface of a superfinished item may be less smoothly polished than when the item is finished using smooth polishing equipment (e.g., there may be residual cross-hatched microscratches on the surface of a superfinished item due to oscillations and/or other movements of the abrasive and rotations of the item during finishing). By superfinishing or otherwise treating (e.g., by burnishing, grinding, polishing, etc.) one or more surfaces of parts that slide relative to each other (e.g., the surface of rods **11.1.3.1.1-50**), wear may be decreased and smooth sliding operations may be ensured (with or without using coatings such as nickel coating layers).

[1854] Fiber-composite tubes may include multiple layers of fiber-composite material (e.g., carbon fiber layers with different fiber orientations). The fiber orientations used in the fiber-composite layers may be selected to enhance bending strength, hoop strength (resistance to tube crushing), and/or torsional strength. End caps **11.1.3.1.1-86** may have solid portions and/or hollow portions and may be formed from one or more metals, polymer, fiber-composite material, etc.

[1855] Low-friction coatings for tubes **11.1.3.1.1-82** (see, e.g., coating layers **11.1.3.1.1-116** of FIG. **11.1.3.1.1-16**) may be formed from metals such as nickel, nickel cobalt, nickel iron, cobalt, chrome (e.g., a top coat of chrome to serve as a hard coat that reduces friction), and/or other low-friction durable (low-wear) coatings.

[1856] If desired, the fibers in tubes **11.1.3.1.1-82** may have different fiber orientations (layups) at different portions of tubes **11.1.3.1.1-82** (e.g., bending strength may be enhanced with fibers that run the length of tubes **11.1.3.1.1-82**, torsional rigidity may be enhanced by fibers oriented $\pm 45^\circ$ with respect to axis **11.1.3.1.1-80**, and these fibers may be present along the entire length of tubes **11.1.3.1.1-82** or only parts of tubes **11.1.3.1.1-82**), and/or crush/hoop strength may be enhanced using fibers that wrap around axis **11.1.3.1.1-80** (e.g., particularly at the ends of tubes **11.1.3.1.1-82** where tubes **11.1.3.1.1-82** are being attached to end caps **11.1.3.1.1-86**).

[1857] Tubes **11.1.3.1.1-82** may, if desired, be provided with strength-enhancing members such as

overmolded polymer strengthening members, bonded polymer and/or metal pieces, etc. The surface of tube **11.1.3.1.1-82** may be treated using acid, laser ablation, primer, sand blasting, and/or other treatments to enhance adhesion prior to overmolding operations.

[1858] If desired, tubes **11.1.3.1.1-82** may be provided with tapered portions. As shown in FIG. **11.1.3.1.1-17**, for example, tube **11.1.3.1.1-82** may have tapered portions **11.1.3.1.1-82T** that are formed by grinding and/or machining the end of tube **11.1.3.1.1-82**. By forming a tapered outer surface formed at the end of tube **11.1.3.1.1-82**, tube **11.1.3.1.1-82** may be precisely aligned with a mating tapered portion of housing **11.1.3.1.1-12** (e.g., a tapered opening in a portion of main housing portion **11.1.3.1.1-12M**, a portion of frame **11.1.3.1.1-12FC**, or other tapered portion of housing **11.1.3.1.1-12** that is machined or otherwise formed in the housing). Tapered tube **11.1.3.1.1-82** of FIG. **11.1.3.1.1-17** may be retained within the corresponding tapered opening of housing **11.1.3.1.1-12** using a press-fit connection, using adhesive, using fasteners, and/or using other suitable attachment mechanisms. In the example of FIG. **11.1.3.1.1-17**, tube **11.1.3.1.1-82** has been provided with insert **11.1.3.1.1-120**. Insert **11.1.3.1.1-120** may be a solid cylindrical member or a tube that is configured to be received within the end of the hollow center of tube **11.1.3.1.1-82**. Adhesive **11.1.3.1.1-22** or other attachment mechanisms may be used to help attach insert **11.1.3.1.1-120** to tube **11.1.3.1.1-82**. Insert **11.1.3.1.1-120** may be formed from metal or other suitable materials and may have threads **11.1.3.1.1-124** that mate with corresponding threads in housing **11.1.3.1.1-12**. These threads provide axial retention for tube **11.1.3.1.1-82** (e.g., threads **11.1.3.1.1-124** help hold the tapered end of tube **11.1.3.1.1-82** into the corresponding tapered opening of housing **11.1.3.1.1-12**). Adhesive may, if desired, be used to lock threads **11.1.3.1.1-124** in place, may be used to help attach the outer surface of tapered portion **11.1.3.1.1-82T** to housing **11.1.3.1.1-12**, and/or may otherwise be used to help reinforce the joint between tube **11.1.3.1.1-82** and housing **11.1.3.1.1-12**.

11.1.3.1.2: Electronic Device with Lens Positioning Sensing

[1859] To accommodate users with different interpupillary distances, the left and right lens assemblies may be moved towards or away from each other. A user may supply the interpupillary distance of the user to the head-mounted device, an image sensor or other device may be used in measuring the interpupillary distance to provide to the head-mounted device, and/or gaze tracking sensors in the head-mounted device may measure the interpupillary distance of the user while the head-mounted device is being worn on the head of the user. Other sensing arrangements may be used to measure lens assembly positions relative to the user's nose, if desired.

[1860] To prevent excessive pressure on the surface of the user's nose, force sensors can be used determine how much pressure is applied to the user's nose with the lenses as the lens-to-lens spacing is changed. Control circuitry in the head-mounted device may adjust the left and right lenses to match the user's interpupillary distance, unless the lenses apply too much pressure to the user's nose (e.g., the pressure measured by the force sensors exceeds a threshold). In some situations, the left and right lenses may be spaced so that the lens-to-lens spacing between the left and right lenses matches the user's interpupillary distance. In other situations, the lens-to-lens spacing between the left and right lenses will be slightly larger than the user's interpupillary distance to ensure that the lenses do not press excessively against the user's nose. Sensor circuitry such as force sensing circuitry may be used to provide the control circuitry with real-time feedback on the pressure applied by the lenses to the user's nose, thereby ensuring that the positions of the left and right lenses are adjusted satisfactorily.

[1861] A schematic diagram of an illustrative system having an electronic device with sensor circuitry that ensures satisfactory placement of lenses relative to a user's facial features is shown in FIG. **11.1.3.1.2-1**. As shown in FIG. **11.1.3.1.2-1**, system **11.1.3.1.2-8** may include one or more electronic devices such as electronic device **11.1.3.1.2-10**. The electronic devices of system **11.1.3.1.2-8** may include computers, cellular telephones, head-mounted devices, wristwatch devices, and other electronic devices. Configurations in which electronic device **11.1.3.1.2-10** is a

head-mounted device are sometimes described herein as an example.

[1862] As shown in FIG. **11.1.3.1.2-1**, electronic devices such as electronic device **11.1.3.1.2-10** may have control circuitry **11.1.3.1.2-12**. Control circuitry **11.1.3.1.2-12** may include storage and processing circuitry for controlling the operation of device **11.1.3.1.2-10**. Circuitry **11.1.3.1.2-12** may include storage such as hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **11.1.3.1.2-12** may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, and other integrated circuits. Software code may be stored on storage in circuitry **11.1.3.1.2-12** and run on processing circuitry in circuitry **11.1.3.1.2-12** to implement control operations for device **11.1.3.1.2-10** (e.g., data gathering operations, operations involved in processing three-dimensional facial image data, operations involving the adjustment of components using control signals, etc.). Control circuitry **11.1.3.1.2-12** may include wired and wireless communications circuitry. For example, control circuitry **11.1.3.1.2-12** may include radio-frequency transceiver circuitry such as cellular telephone transceiver circuitry, wireless local area network (WiFi®) transceiver circuitry, millimeter wave transceiver circuitry, and/or other wireless communications circuitry.

[1863] During operation, the communications circuitry of the devices in system **11.1.3.1.2-8** (e.g., the communications circuitry of control circuitry **11.1.3.1.2-12** of device **11.1.3.1.2-10**), may be used to support communication between the electronic devices. For example, one electronic device may transmit video and/or audio data to another electronic device in system **11.1.3.1.2-8**.

Electronic devices in system **11.1.3.1.2-8** may use wired and/or wireless communications circuitry to communicate through one or more communications networks (e.g., the internet, local area networks, etc.). The communications circuitry may be used to allow data to be received by device **11.1.3.1.2-10** from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, online computing equipment such as a remote server or other remote computing equipment, or other electrical equipment) and/or to provide data to external equipment.

[1864] Device **11.1.3.1.2-10** may include input-output devices **11.1.3.1.2-22**. Input-output devices **11.1.3.1.2-22** may be used to allow a user to provide device **11.1.3.1.2-10** with user input. Input-output devices **11.1.3.1.2-22** may also be used to gather information on the environment in which device **11.1.3.1.2-10** is operating. Output components in devices **11.1.3.1.2-22** may allow device **11.1.3.1.2-10** to provide a user with output and may be used to communicate with external electrical equipment.

[1865] As shown in FIG. **11.1.3.1.2-1**, input-output devices **11.1.3.1.2-22** may include one or more displays such as display **11.1.3.1.2-14**. In some configurations, display **11.1.3.1.2-14** of device **11.1.3.1.2-10** includes left and right display panels (sometimes referred to as left and right portions of display **11.1.3.1.2-14** and/or left and right displays) that are in alignment with the user's left and right eyes and are viewable through left and right lens assemblies, respectively. In other configurations, display **11.1.3.1.2-14** includes a single display panel that extends across both eyes.

[1866] Display **11.1.3.1.2-14** may be used to display images. The visual content that is displayed on display **11.1.3.1.2-14** may be viewed by a user of device **11.1.3.1.2-10**. Displays in device **11.1.3.1.2-10** such as display **11.1.3.1.2-14** may be organic light-emitting diode displays or other displays based on arrays of light-emitting diodes, liquid crystal displays, liquid-crystal-on-silicon displays, projectors or displays based on projecting light beams on a surface directly or indirectly through specialized optics (e.g., digital micromirror devices), electrophoretic displays, plasma displays, electrowetting displays, microLED displays, or any other suitable displays.

[1867] Display **11.1.3.1.2-14** may present computer-generated content such as virtual reality content and mixed reality content to a user. Virtual reality content may be displayed in the absence

of real-world content. Mixed reality content, which may sometimes be referred to as augmented reality content, may include computer-generated images that are overlaid on real-world images. The real-world images may be captured by a camera (e.g., a forward-facing camera) and merged with overlaid computer-generated content or an optical coupling system may be used to allow computer-generated content to be overlaid on top of real-world images. As an example, a pair of mixed reality glasses or other augmented reality head-mounted display may include a display device that provides images to a user through a beam splitter, prism, holographic coupler, or other optical coupler. Configurations in which display **11.1.3.1.2-14** is used to display virtual reality content to a user through lenses are described herein as an example.

[1868] Input-output devices **11.1.3.1.2-22** may include sensors **11.1.3.1.2-16**. Sensors **11.1.3.1.2-16** may include, for example, three-dimensional sensors (e.g., three-dimensional image sensors such as structured light sensors that emit beams of light and that use two-dimensional digital image sensors to gather image data for three-dimensional images from light spots that are produced when a target is illuminated by the beams of light, binocular three-dimensional image sensors that gather three-dimensional images using two or more cameras in a binocular imaging arrangement, three-dimensional LIDAR (light detection and ranging) sensors, three-dimensional radio-frequency sensors, or other sensors that gather three-dimensional image data), cameras (e.g., infrared and/or visible digital image sensors), gaze tracking sensors (e.g., a gaze tracking system based on an image sensor and, if desired, a light source that emits one or more beams of light that are tracked using the image sensor after reflecting from a user's eyes), touch sensors, buttons, force sensors, sensors such as contact sensors based on switches, gas sensors, pressure sensors, moisture sensors, magnetic sensors, audio sensors (microphones), ambient light sensors, microphones for gathering voice commands and other audio input, sensors that are configured to gather information on motion, position, and/or orientation (e.g., accelerometers, gyroscopes, compasses, and/or inertial measurement units that include all of these sensors or a subset of one or two of these sensors), fingerprint sensors and other biometric sensors, optical position sensors (optical encoders), and/or other position sensors such as linear position sensors, and/or other sensors. As shown in FIG.

11.1.3.1.2-1, sensors **11.1.3.1.2-16** may include sensing circuitry (sensor circuitry) that is configured to measure the pressure applied between objects in system **11.1.3.1.2-8**. The sensing circuitry may include one or more sensors such as one or more force sensors **11.1.3.1.2-20**. Sensing circuitry such as force sensors **11.1.3.1.2-20** may, for example, be used to sense an amount of pressure applied by lens assemblies in device **11.1.3.1.2-10** to a user's nose.

[1869] User input and other information may be gathered using sensors and other input devices in input-output devices **11.1.3.1.2-22**. If desired, input-output devices **11.1.3.1.2-22** may include other devices **11.1.3.1.2-24** such as haptic output devices (e.g., vibrating components), light-emitting diodes and other light sources, speakers such as ear speakers for producing audio output, and other electrical components. Device **11.1.3.1.2-10** may include circuits for receiving wireless power, circuits for transmitting power wirelessly to other devices, batteries and other energy storage devices (e.g., capacitors), joysticks, buttons, and/or other components.

[1870] Electronic device **11.1.3.1.2-10** may have housing structures (e.g., housing walls, straps, etc.), as shown by illustrative support structures **11.1.3.1.2-26** of FIG. **11.1.3.1.2-1**. In configurations in which electronic device **11.1.3.1.2-10** is a head-mounted device (e.g., a pair of glasses, goggles, a helmet, a hat, a headband, etc.), support structures **11.1.3.1.2-26** may include head-mounted support structures (e.g., a helmet housing, head straps, temples in a pair of eyeglasses, goggle housing structures, and/or other head-mounted structures). The head-mounted support structures may be configured to be worn on a head of a user during operation of device **11.1.3.1.2-10** and may support display(s) **11.1.3.1.2-14**, sensors **11.1.3.1.2-16**, other components **11.1.3.1.2-24**, other input-output devices **11.1.3.1.2-22**, and control circuitry **11.1.3.1.2-12**.

[1871] FIG. **11.1.3.1.2-2** is a top view of electronic device **11.1.3.1.2-10** in an illustrative configuration in which electronic device **11.1.3.1.2-10** is a head-mounted device. As shown in FIG.

11.1.3.1.2-2, electronic device **11.1.3.1.2-10** may include support structures (see, e.g., support structures **11.1.3.1.2-26** of FIG. **11.1.3.1.2-1**) that are used in housing the components of device **11.1.3.1.2-10** and mounting device **11.1.3.1.2-10** onto a user's head. These support structures may include, for example, structures that form housing walls and other structures for main unit **11.1.3.1.2-26-2** (e.g., exterior housing walls, lens assembly structures, etc.) and straps or other supplemental support structures such as structures **11.1.3.1.2-26-1** that help to hold main unit **11.1.3.1.2-26-2** on a user's face so that the user's eyes are located within eye boxes **11.1.3.1.2-60**. [1872] Display **11.1.3.1.2-14** may include left and right display panels (e.g., left and right pixel arrays, sometimes referred to as left and right displays or left and right display portions) that are mounted respectively in left and right display modules **11.1.3.1.2-70** corresponding respectively to a user's left eye (and left eye box **11.1.3.1.2-60**) and right eye (and right eye box). Modules **11.1.3.1.2-70**, which may sometimes be referred to as lens support structures, lens assemblies, lens housings, or lens and display housings, may be individually positioned relative to the housing wall structures of main unit **11.1.3.1.2-26-2** and relative to the user's eyes using positioning circuitry such as respective left and right positioners **11.1.3.1.2-58**. Positioners **11.1.3.1.2-58** may include stepper motors, piezoelectric actuators, motors, linear electromagnetic actuators, and/or other electronic components for adjusting lens assembly positions. Positioners **11.1.3.1.2-58** may be controlled by control circuitry **11.1.3.1.2-12** during operation of device **11.1.3.1.2-10**. For example, positioners **11.1.3.1.2-58** may be used to adjust the spacing between modules **11.1.3.1.2-70** (and therefore the lens-to-lens spacing between the left and right lenses of modules **11.1.3.1.2-70**) to match the interpupillary distance (IPD) of a user's eyes. This allows the user to view the left and right display portions of display **11.1.3.1.2-14** in the left and right lens modules. In some cases, however, the lenses may apply excess pressure to the user's nose if adjusted to the user's IPD. Therefore, sensors may be incorporated into device **11.1.3.1.2-10** to monitor the pressure on the user's nose. An illustrative arrangement that includes force sensors to monitor the pressure applied to the user's nose is shown in FIG. **11.1.3.1.2-3**.

[1873] As shown in FIG. **11.1.3.1.2-3**, lens assembly **11.1.3.1.2-70** (e.g., one of the two lens modules **11.1.3.1.2-70** shown in FIG. **11.1.3.1.2-2**) may be adjacent to the user's nose **11.1.3.1.2-40** when device **11.1.3.1.2-10** is being worn on a user's head. Only one of the two lens modules is shown in FIG. **11.1.3.1.2-3** for simplicity, but the other lens assembly **11.1.3.1.2-70** may have the same structure.

[1874] To ensure that display **11.1.3.1.2-14** is viewable by the user when the user's eyes are located in eye boxes **11.1.3.1.2-60** (FIG. **11.1.3.1.2-2**), control circuitry **11.1.3.1.2-12** may attempt to align lens centers LC with the centers PC of the user's eyes. At the same time, control circuitry **11.1.3.1.2-12** may use sensor circuitry such as force sensors **11.1.3.1.2-20** to monitor the pressure applied to nose **11.1.3.1.2-40** by lens modules **11.1.3.1.2-70** to ensure that lens modules **11.1.3.1.2-70** do not press excessively on nose **11.1.3.1.2-40** and cause discomfort.

[1875] In scenarios in which the user's nose is small, there may be ample room available to align lens centers LC with eye centers PC. In scenarios in which the user's nose is larger, control circuitry **11.1.3.1.2-12** may position modules **11.1.3.1.2-70** as shown in FIG. **11.1.3.1.2-3**. For example, a distance between lens modules **11.1.3.1.2-70** (the lens-to-lens spacing) may be larger than would be desired for perfect alignment of lens centers LC with eye centers PC. The use of this wider lens-to-lens spacing helps ensure that lens modules **11.1.3.1.2-70** will not exert more inward force on nose **11.1.3.1.2-40** than would be comfortable to a user, while still allowing satisfactory viewing of content on display **11.1.3.1.2-14** through lenses **11.1.3.1.2-72**. Lens modules **11.1.3.1.2-70** may be placed at a non-zero distance (gap) from the side surfaces of nose **11.1.3.1.2-40** or may be spaced apart from the side surfaces of nose **11.1.3.1.2-40** by a predetermined gap. A user may select which of these options is most comfortable to the user and/or a default setting may be supplied to control circuitry **11.1.3.1.2-12**.

[1876] In operation, positioners **11.1.3.1.2-58** may move lens modules **11.1.3.1.2-70** (FIG.

11.1.3.1.2-2) toward nose **11.1.3.1.2-40**, attempting to align the lens center LC of each lens with eye centers PC. Sensors may be incorporated into device **11.1.3.1.2-10** to ensure that excess pressure is not applied to nose **11.1.3.1.2-40** by lens modules **11.1.3.1.2-70**. In general, any desired sensor circuitry may be used to measure the pressure on nose **11.1.3.1.2-40**. In one example, each lens module **11.1.3.1.2-70** may have one or more force sensors **11.1.3.1.2-20**.

[1877] Force sensor **11.1.3.1.2-20** may be incorporated between nasal flap **11.1.3.1.2-29** and each lens assembly **11.1.3.1.2-70**. Nasal flap **11.1.3.1.2-29** may be a fabric, polymer, or other material component that allows for a comfortable fit for a user of device **11.1.3.1.2-10**. For example, nasal flap **11.1.3.1.2-29** may be interposed between each lens assembly **11.1.3.1.2-70** and nose **11.1.3.1.2-40**. In some embodiments, nasal flap **11.1.3.1.2-29** may extend along both sides and over the top of nose **11.1.3.1.2-40** (e.g., over at least a portion of a bridge of nose **11.1.3.1.2-40**). However, this is merely illustrative. Nasal flap **11.1.3.1.2-29** may have two separate portion, one between left lens assembly **11.1.3.1.2-70** and nose **11.1.3.1.2-40** and another between right lens assembly **11.1.3.1.2-70** and nose **11.1.3.1.2-40**, or may be omitted from device **11.1.3.1.2-10** if desired.

[1878] In embodiments in which nasal flap **11.1.3.1.2-29** is included in device **11.1.3.1.2-10**, force sensors **11.1.3.1.2-20** may be incorporated between each lens assembly **11.1.3.1.2-70** (i.e., left lens assembly **11.1.3.1.2-70** and right lens assembly **11.1.3.1.2-70**) and nasal flap **11.1.3.1.2-29**. However, this location of force sensors **11.1.3.1.2-20** is merely illustrative. In general, force sensors **11.1.3.1.2-20** may be included in any desired location within device **11.1.3.1.2-10**. For example, force sensors **11.1.3.1.2-20** may be formed between nasal flap **11.1.3.1.2-29** and nose **11.1.3.1.2-40** as shown by position **11.1.3.1.2-20'**, may be formed within nasal flap **11.1.3.1.2-29** or be integral with nasal flap **11.1.3.1.2-29** as shown by position **11.1.3.1.2-20''**, or may be formed within lens assembly **11.1.3.1.2-70** as shown by position **11.1.3.1.2-20'''**.

[1879] Although FIG. **11.1.3.1.2-3** shows a single force sensor **11.1.3.1.2-20** between nose **11.1.3.1.2-40** and lens assembly **11.1.3.1.2-70**, this is merely illustrative. Device **11.1.3.1.2-10** may have one force sensor **11.1.3.1.2-20** between nose **11.1.3.1.2-40** and each lens assembly **11.1.3.1.2-70**, may have multiple force sensors between nose **11.1.3.1.2-40** and each lens assembly **11.1.3.1.2-70**, may have one single force sensor between nose **11.1.3.1.2-40** and one lens assembly **11.1.3.1.2-70**, or may have any other desired arrangement of force sensor(s) **11.1.3.1.2-20**.

[1880] Regardless of where force sensors **11.1.3.1.2-20** are formed within device **11.1.3.1.2-10**, force sensors **11.1.3.1.2-20** may monitor the amount of force applied to nose **11.1.3.1.2-40** by lens modules **11.1.3.1.2-70** to ensure that excess pressure is not applied to nose **11.1.3.1.2-40**. Force sensors **11.1.3.1.2-20** may continuously monitor the applied force, or may flag control circuitry **11.1.3.1.2-12** when the applied force surpasses a threshold. Force sensors **11.1.3.1.2-20** may be any desired type of sensor that monitors the amount of force/pressure applied to nose **11.1.3.1.2-40**. Some examples of force sensors **11.1.3.1.2-20** are shown in FIGS. **11.1.3.1.2-4A-11.1.3.1.2-4C**.

[1881] In some examples, force sensor(s) **11.1.3.1.2-20** may be direct force sensors, such as force sensor **11.1.3.1.2-20** of FIG. **11.1.3.1.2-4A**. Direct force sensor **11.1.3.1.2-20** of FIG. **11.1.3.1.2-4A** may be a pressure sensitive resistor and may include electrodes formed from interdigitated finger traces **11.1.3.1.2-27**. In particular, when pressure is applied to direct force sensor **11.1.3.1.2-20**, the resistance may be determined due to a change in distance between portions of interdigitated finger traces **11.1.3.1.2-27**, and circuitry, such as control circuitry **11.1.3.1.2-12**, may determine the force applied to direct force sensor **11.1.3.1.2-20**. If direct force sensor **11.1.3.1.2-20** is placed in any of the possible locations shown in FIG. **11.1.3.1.2-3**, or otherwise placed between a user's nose and lens assembly **11.1.3.1.2-70**, direct force sensor **11.1.3.1.2-20** may be used to determine the force that lens assembly **11.1.3.1.2-70** applies to nose **11.1.3.1.2-40** as it is moved toward nose **11.1.3.1.2-40**. Direct force sensor **11.1.3.1.2-20** may be used to continuously measure the pressure against nose **11.1.3.1.2-40**, or may be used as a thresholding sensor (i.e., control circuitry **11.1.3.1.2-12** may determine that the resistance of direct force sensor **11.1.3.1.2-20** is over a

threshold and that the force on nose **11.1.3.1.2-40** is therefore too great).

[1882] In other examples, force sensor(s) **11.1.3.1.2-20** may be formed from conductive strands, yarn, or fibers and incorporated into a fabric in device **11.1.3.1.2-10**. For example, force sensor(s) **11.1.3.1.2-20** may be formed from smart fabrics or other fabrics that include conductive strands. The conductive strands may be arranged to form a force sensor. An example of this arrangement is shown in FIG. **11.1.3.1.2-4B**.

[1883] As shown in FIG. **11.1.3.1.2-4B**, force sensor **11.1.3.1.2-20** may be incorporated into fabric **11.1.3.1.2-31**. In particular, fabric **11.1.3.1.2-31** may include conductive strands **11.1.3.1.2-28** and non-conductive strands **11.1.3.1.2-33**. Conductive strands **11.1.3.1.2-28** may be arranged to form a force sensor. For example, control circuitry, such as control circuitry **11.1.3.1.2-12**, may measure capacitance or resistance changes between conductive strands **11.1.3.1.2-28**. Because the capacitance/resistance will change as the distance between conductive strands **11.1.3.1.2-28** changes, the capacitance/resistance measurements are indicative of the amount of force on fabric **11.1.3.1.2-31**. Conductive strands may be formed from a metal, such as silver or copper, which may optionally be coated onto a polymer or fabric strand.

[1884] Although conductive strands **11.1.3.1.2-28** are shown as straight strands in FIG. **11.1.3.1.2-4B**, this is merely illustrative. In some embodiments, conductive strands **11.1.3.1.2-28** may be serpentine or have other desired shapes, which may allow fabric **11.1.3.1.2-31** to have improved flexibility.

[1885] Non-conductive strands **11.1.3.1.2-33** may be interspersed between at least some of conductive strands **11.1.3.1.2-28**, if desired. As shown in FIG. **11.1.3.1.2-4B**, every other strand may be a non-conductive strand. However, this is merely illustrative. Any desired number of conductive and non-conductive strands may be incorporated into fabric **11.1.3.1.2-31** to form force sensor **11.1.3.1.2-20**.

[1886] Fabric **11.1.3.1.2-31** may form some or all of nasal flap **11.1.3.1.2-29**, or may otherwise cover nasal flap **11.1.3.1.2-29**. Because nasal flap **11.1.3.1.2-29** is between nose **11.1.3.1.2-40** and lens assembly **11.1.3.1.2-70** when device **11.1.3.1.2-10** is worn by a user (FIG. **11.1.3.1.2-2**), force sensor **11.1.3.1.2-20** may indicate the amount of force applied to nose **11.1.3.1.2-40** by lens assembly **11.1.3.1.2-70**. Any desired number of force sensors **11.1.3.1.2-20** may be incorporated into nasal flap **11.1.3.1.2-29** in this way.

[1887] Alternatively or additionally, fabric **11.1.3.1.2-31** may be otherwise incorporated into device **11.1.3.1.2-10**. For example, fabric **11.1.3.1.2-31** may form a curtain fabric that is incorporated into device **11.1.3.1.2-10** between support structure **11.1.3.1.2-26-2** and lenses **11.1.3.1.2-70** (FIG. **11.1.3.1.2-2**) to hide internal components. At least a portion of the curtain fabric having force sensor(s) **11.1.3.1.2-20** may be between lens assembly **11.1.3.1.2-70** and nose **11.1.3.1.2-40** (e.g., between lens assembly **11.1.3.1.2-70** and nasal flap **11.1.3.1.2-29**) and may therefore measure the force applied to nose **11.1.3.1.2-40** by lens assembly **11.1.3.1.2-70**. In general, however, fabric **11.1.3.1.2-31** may be incorporated into device **11.1.3.1.2-10** in any desired manner. Any desired number of force sensors **11.1.3.1.2-20** may be incorporated into fabric **11.1.3.1.2-31**.

[1888] In other examples, force sensor(s) **11.1.3.1.2-20** may be incorporated into an interior region of nasal flap **11.1.3.1.2-29**. An example of an arrangement in which a force sensor is located in the interior of nasal flap **11.1.3.1.2-29** is shown in FIG. **11.1.3.1.2-4C**.

[1889] As shown in FIG. **11.1.3.1.2-4C**, nasal flap **11.1.3.1.2-29** may have peripheral region **11.1.3.1.2-29A** that surrounds inner region **11.1.3.1.2-29B** (also referred to as a cavity herein). Peripheral region **11.1.3.1.2-29A** may be formed from polymer, rubber, or any other desired material. Inner region **11.1.3.1.2-29B** may be filled with air, gas, liquid, or any other desired substance. Force sensor **11.1.3.1.2-20** may be located in inner region **11.1.3.1.2-29B** and may produce a force measurement in response to increased pressure in inner region **11.1.3.1.2-29B**. For example, as lens assembly **11.1.3.1.2-70** squeezes nasal flap **11.1.3.1.2-29** against nose **11.1.3.1.2-40** (FIG. **11.1.3.1.2-2**), the pressure of the air, gas, liquid, or other material within nasal flap

11.1.3.1.2-29 may increase, increasing the pressure on force sensor **11.1.3.1.2-20**. In some examples, force sensor **11.1.3.1.2-20** may be a barometric pressure sensor that measures the increased pressure within inner region **11.1.3.1.2-29B** when lens assembly **11.1.3.1.2-70** is moved against nose **11.1.3.1.2-40**. Therefore, force sensor **11.1.3.1.2-20** may produce a force measurement indicative of the force applied to the user's nose by lens assembly **11.1.3.1.2-70**.

[1890] Although FIG. **11.1.3.1.2-4C** shows one side or portion of nasal flap **11.1.3.1.2-29**, this is merely illustrative. Any desired number of force sensors **11.1.3.1.2-20** may be incorporated into inner region **11.1.3.1.2-29B** of nasal flap **11.1.3.1.2-29** on one or both sides of the user's nose. Additionally, although FIG. **11.1.3.1.2-4C** shows force sensor **11.1.3.1.2-20** implemented as a pressure sensor in nasal flap **11.1.3.1.2-29**, force sensor **11.1.3.1.2-20** may be a pressure sensor anywhere between nose **11.1.3.1.2-40** and lens assembly **11.1.3.1.2-70** when worn by a user. For example, force sensor **11.1.3.1.2-20** may be implemented as a pressure sensor within an edge portion of lens assembly **11.1.3.1.2-70** (i.e., in position **11.1.3.1.2-20'''** of FIG. **11.1.3.1.2-3**), if desired.

[1891] As an alternative to the sensors shown in FIGS. **11.1.3.1.2-4A-11.1.3.1.2-4C**, force sensor **11.1.3.1.2-20** may be implemented to measure the deflection of nasal flap **11.1.3.1.2-29** relative to lens assembly **11.1.3.1.2-70**. An example of this type of force sensor is shown in FIG. **11.1.3.1.2-5**.

[1892] As shown in FIG. **11.1.3.1.2-5**, force sensor **11.1.3.1.2-20** may be mounted to (or within) a portion of lens assembly **11.1.3.1.2-70**. Force sensor **11.1.3.1.2-20** may be a sensor that measures the proximity of nasal flap **11.1.3.1.2-29**, such as a capacitive proximity sensor, a resistive proximity sensor, an optical proximity sensor, an ultrasonic proximity sensor, or any other desired type of proximity sensor. Optionally, magnet **11.1.3.1.2-32** may be embedded within (or mounted to) nasal flap **11.1.3.1.2-29**, and force sensor **11.1.3.1.2-20** may be implemented as a Hall Effect sensor. In this way, the Hall Effect sensor may determine the proximity of magnet **11.1.3.1.2-32** and therefore the proximity of nasal flap **11.1.3.1.2-29**. By measuring the proximity of nasal flap **11.1.3.1.2-29** (i.e., the amount by which nasal flap **11.1.3.1.2-29** has moved), the proximity sensor may provide an output that is indicative of the amount of force applied to the user's nose by lens assembly **11.1.3.1.2-70**.

[1893] Although the previous embodiments have included a dedicated force sensor **11.1.3.1.2-20** to ensure that excessive force is not applied to a user's nose, this is merely illustrative. Device **11.1.3.1.2-10** may use other sensors (such as sensors **11.1.3.1.2-16**) to determine if the force applied to nose **11.1.3.1.2-40** exceeds a threshold. An example of this arrangement is shown in FIG. **11.1.3.1.2-6**.

[1894] As shown in FIG. **11.1.3.1.2-6**, one or more flexible members **11.1.3.1.2-34** may extend from lens assembly **11.1.3.1.2-70**. Flexible members **11.1.3.1.2-34** may be formed from rubber, polymer, fabric, or any other desired flexible material. Device **11.1.3.1.2-10** may also include light-emitting component **11.1.3.1.2-36** and light-detecting component **11.1.3.1.2-42**, which may be used in gaze tracking or other desired operations. For example, light-emitting component **11.1.3.1.2-36** may be an infrared light-emitting component and light-detecting component **11.1.3.1.2-42** may be an infrared light-detecting component. To determine the gaze of a user of device **11.1.3.1.2-10**, infrared light-emitting component **11.1.3.1.2-36** may emit light toward the eye of the user, and the reflections from the user's eye may be detected by infrared light-detecting component **11.1.3.1.2-42**. These reflections may indicate the direction of the user's gaze. In general, however, light-emitting component **11.1.3.1.2-36** and light-detecting component **11.1.3.1.2-42** may be any desired components and may operate in any desired wavelength.

[1895] When assembly **11.1.3.1.2-70** moves toward nasal flap **11.1.3.1.2-29** (and therefore nose **11.1.3.1.2-40**), it will eventually contact and push against nasal flap **11.1.3.1.2-29** and nose **11.1.3.1.2-40**, causing flexible members **11.1.3.1.2-34** to be moved in direction **11.1.3.1.2-38**. If flexible members **11.1.3.1.2-34** move far enough (i.e., the amount of force applied to nose **11.1.3.1.2-40** exceeds or meets a threshold), flexible members **11.1.3.1.2-34** may block light-

emitting component **11.1.3.1.2-36**. As a result, light-detecting component **11.1.3.1.2-42** may stop detecting light emitted by light-emitting component **11.1.3.1.2-36**. Based on the changed signal of light-detecting component **11.1.3.1.2-42**, control circuitry **11.1.3.1.2-12** may stop positioners **11.1.3.1.2-58** from moving lens assembly **11.1.3.1.2-70** further toward the user's nose.

[1896] Although flexible members **11.1.3.1.2-34** are shown as covering light-emitting component **11.1.3.1.2-36**, this is merely illustrative. Flexible members **11.1.3.1.2-34** may cover light-detecting component **11.1.3.1.2-42**, or may merely move between light-emitting component **11.1.3.1.2-36** and light-detecting component **11.1.3.1.2-42**. Additionally, any desired number of flexible members **11.1.3.1.2-34** may be used.

[1897] If desired, the position of lens modules **11.1.3.1.2-70** relative to the corresponding surfaces of nose **11.1.3.1.2-40** may be measured using feedback from motors in positioners **11.1.3.1.2-58** as lens modules **11.1.3.1.2-70** are moved into contact with the surfaces of nose **11.1.3.1.2-40**. An illustrative control circuit for a positioner such as positioner **11.1.3.1.2-58** is shown in FIG.

11.1.3.1.2-7. Control circuitry **11.1.3.1.2-12** (FIG. **11.1.3.1.2-1**) may include a motor controller such as controller **11.1.3.1.2-80**. Controller **11.1.3.1.2-80** may drive motor **11.1.3.1.2-86** in a positioner **11.1.3.1.2-58** to move an associated lens module **11.1.3.1.2-70** by supplying a power supply voltage V_{in} to motor **11.1.3.1.2-86** using path **11.1.3.1.2-84**. While voltage V_{in} is being supplied to motor **11.1.3.1.2-86**, controller **11.1.3.1.2-80** of control circuitry **11.1.3.1.2-12** monitors the resulting current flow (current I) through path **11.1.3.1.2-84** using sensor circuit **82** (e.g., a current sensing resistor with a corresponding analog-to-digital converter circuit, etc.). Power supply voltage V_{in} may remain relatively constant while motor **11.1.3.1.2-86** moves lens assembly **11.1.3.1.2-70**. Positioner **11.1.3.1.2-58** may initially be used to position an edge of lens assembly **11.1.3.1.2-70** at a location that is distant from nose **11.1.3.1.2-40**. Control circuitry **11.1.3.1.2-12** may then direct positioner **11.1.3.1.2-58** to move lens assembly **11.1.3.1.2-70** toward nose **11.1.3.1.2-40**. Controller **11.1.3.1.2-80** of control circuitry **11.1.3.1.2-12** may monitor the current I that flows through path **11.1.3.1.2-84** and I sensed by sensor **82**. When lens assembly **11.1.3.1.2-70** is pressed against the side of nose **11.1.3.1.2-40**, current I will increase. When current I surpasses a desired threshold (that is related to the force applied to nose **11.1.3.1.2-40**), control circuitry **11.1.3.1.2-12** may stop positioner **11.1.3.1.2-58** from applying more force to nose **11.1.3.1.2-40** with lens assembly **11.1.3.1.2-70**.

[1898] Illustrative operations involved in operating device **11.1.3.1.2-10** in system **11.1.3.1.2-8** are shown in FIG. **11.1.3.1.2-8**.

[1899] During the operations of block **11.1.3.1.2-100**, information on the distance between the user's eyes (interpupillary distance IPD, sometimes referred to as pupillary distance) may be gathered. With one illustrative arrangement, device **11.1.3.1.2-10** or other equipment in system **11.1.3.1.2-8** gathers the user's interpupillary distance from the user by prompting the user to type the interpupillary distance into a data entry box on display **11.1.3.1.2-14** or a display in other equipment in system **11.1.3.1.2-8**. The user may also supply the user's interpupillary distance using voice input or other user input arrangements. With another illustrative arrangement, a sensor in device **11.1.3.1.2-10** or other a sensor in a stand-alone computer, portable device, or other equipment in system **11.1.3.1.2-8** may measure the user's interpupillary distance. For example, a sensor such as a two-dimensional or three-dimensional image sensor may gather an image of the user's face to measure the value of interpupillary distance IPD. After the measurement of the interpupillary distance has been made, the interpupillary distance may be provided to device **11.1.3.1.2-10** (e.g., over a wired or wireless communications paths). If desired, gaze trackers may measure the locations of the centers of the user's eyes PD and thereby determine IPD from direct measurement as a user is wearing device **11.1.3.1.2-10** on the user's head.

[1900] After gathering interpupillary distance IPD, control circuitry **11.1.3.1.2-12** of device **11.1.3.1.2-10** may, during the operations of block **11.1.3.1.2-102**, use positioners **11.1.3.1.2-58** to adjust the lens-to-lens spacing between lens centers LC so that this distance matches interpupillary

distance IPD and so that the centers of lenses **11.1.3.1.2-72** are aligned with respective eye centers PC. While positioners **11.1.3.1.2-58** are moving lens modules **11.1.3.1.2-70** and lenses **11.1.3.1.2-72** (e.g., while the lens-to-lens spacing is being reduced to move modules **11.1.3.1.2-70** towards adjacent surfaces of the user's nose), control circuitry **11.1.3.1.2-12** uses force sensing circuitry (e.g., force sensor(s) **11.1.3.1.2-20**) to monitor the force applied by lens modules **11.1.3.1.2-70** on nose **11.1.3.1.2-40**. In some situations, the user's nose **11.1.3.1.2-40** may prevent lenses **11.1.3.1.2-72** from being brought sufficiently close to each other to allow the lens-to-lens spacing to exactly match IPD without creating a risk of discomfort for the user. In other words, force sensor(s) **11.1.3.1.2-20** may indicate that too much force is being applied to nose **11.1.3.1.2-40** by lens modules **11.1.3.1.2-70**, or that the force being applied to nose **11.1.3.1.2-40** by lens modules **11.1.3.1.2-70** has reached a threshold. Control circuitry **11.1.3.1.2-12** may then stop positioners **11.1.3.1.2-58** from moving lens modules **11.1.3.1.2-70** further toward nose **11.1.3.1.2-40**. If desired, positioners **11.1.3.1.2-58** may move lens modules **11.1.3.1.2-70** off of nose **11.1.3.1.2-40** by a desired gap (e.g., a gap G of at least 0.1 mm, at least 0.2 mm, at least 1 mm, at least 2 mm, less than 5 mm, or other suitable spacing).

[1901] Following the positioning of modules **11.1.3.1.2-70** at desired locations relative to nose **11.1.3.1.2-40** to ensure user comfort while wearing device **11.1.3.1.2-10**, control circuitry **11.1.3.1.2-12** may use display **11.1.3.1.2-14** to present visual content to the user through lenses **11.1.3.1.2-72** (block **11.1.3.1.2-104**).

11.1.3.2: Sensors/Encoders

[1902] FIG. **11.1.3.2-1** illustrates a front, right, perspective view of a portion of an HMD **11.1.3.2-100** including first and second optical modules **11.1.3.2-102a**, **11.1.3.2-102b**. The first and second optical modules **11.1.3.2-102a-b** can be secured to an adjustment mechanism configured to alter the position of the optical modules **11.1.3.2-102a-b** relative to other components of the HMD **11.1.3.2-100**, for example relative to the bracket **11.1.3.2-108**. For example, the first optical module **11.1.3.2-102a** can include follower **11.1.3.2-104** slidably engaged with a guide-rod **11.1.3.2-106** via a channel of the follower **11.1.3.2-104** such that the optical module **11.1.3.2-102a** can be adjusted laterally in the axial direction of the guide-rod **11.1.3.2-106**. In some examples, the optical modules **11.1.3.2-102a-b** can include a guide-rod slidably engaged with a follower mechanism or some other sliding track system. The illustrated example is not limiting.

[1903] In at least one example, a motor (not shown) can be activated to move and adjust the position of the optical module **11.1.3.2-102a** along the guide rod **11.1.3.2-106**. In order to determine the position of the optical module **11.1.3.2-102a** relative to the second optical module **11.1.3.2-102b** or relative to other components of the HMD **11.1.3.2-100**, the HMD **11.1.3.2-100** can include an encoder **11.1.3.2-109**. In one example, the encoder **11.1.3.2-109** is not on or disposed with the motor. Rather, in at least one example, such as the example shown in FIG. **11.1.3.2-1**, the encoder **11.1.3.2-109** can be disposed outside the motor and secured in to the frame or bracket **11.1.3.2-108** of the HMD **11.1.3.2-100**. In one example, the encoder **109** can be secured to the frame and/or bracket **108** of the HMD **11.1.3.2-100** directly. In one example, the encoder assembly **11.1.3.2-109** can be secured to the frame and/or bracket **11.1.3.2-108** of the HMD **11.1.3.2-100** indirectly via an encoder bracket.

[1904] In one example, the encoder **11.1.3.2-109** assembly can include a magnet **11.1.3.2-110** having a master strip **11.1.3.2-113** and a Nonius strip **11.1.3.2-111**, each including magnetic pole strips **11.1.3.2-112**, forming adjacent magnetic pole pairs, for interacting with a ferromagnetic or magnetic chip component translatable with the optical module when the optical module **11.1.3.2-102a** is moved along the guide-rod **11.1.3.2-106**. In at least one example, the magnetic pole pairs **11.1.3.2-112** of the Nonius strip **11.1.3.2-111** are phase shifted relative to the master strip **11.1.3.2-113** such that the master strip **11.1.3.2-113** can be used for local position detection and the Nonius strip **11.1.3.2-111** can be used for determining which pole pair is correlated to the detected position. In this way, the magnet **11.1.3.2-110** of the encoder **11.1.3.2-109** can serve as part of an absolute

encoder for detecting position across the whole range of movement of the optical module **11.1.3.2-102a**.

[1905] In at least one example, the encoder **11.1.3.2-109** is a non-contact encoder. In one example, the encoder **11.1.3.2-109** is a Hall-effect encoder.

[1906] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3.2-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.3.2-1**.

[1907] FIG. **11.1.3.2-2** illustrates a top, left, perspective view of a portion of an HMD **11.1.3.2-200** including an optical module **11.1.3.2-202** having a follower **11.1.3.2-204** slidably engaged with a guide-rod **11.1.3.2-206**. The optical module **11.1.3.2-202** can be slid along the guide-rod **11.1.3.2-206** via a motor (not shown). In at least one example, the HMD **11.1.3.2-200** can include an encoder assembly **11.1.3.2-209**. The encoder assembly **11.1.3.2-209** can include a magnet **11.1.3.2-210** fixed in position to the HMD **11.1.3.2-200**, for example to the chassis, frame, or bracket **11.1.3.2-208** of the HMD **11.1.3.2-200**. In at least one example, the encoder assembly **11.1.3.2-209** can include a chip **11.1.3.2-216** or chip assembly including a housing and a chip configured to slide relative to the magnet **11.1.3.2-210** along with the optical module **11.1.3.2-202**. The chip **11.1.3.2-216** can be secured with the optical module **11.1.3.2-202** via a flexure bracket **11.1.3.2-214** secured to the chip **11.1.3.2-216** (or chip assembly **11.1.3.2-216**). In at least one example, the flexure bracket **11.1.3.2-214** is fixed to the optical module **11.1.3.2-202** at a first end **11.1.3.2-218** and biased against the magnet **11.1.3.2-210** at a second end **11.1.3.2-222**. In one example, the flexure bracket **11.1.3.2-214** biases the chip **11.1.3.2-216** against the magnet **11.1.3.2-210** at the second end **11.1.3.2-222** thereof so the chip **11.1.3.2-216** is disposed adjacent to the magnet **11.1.3.2-210** and configured to slide relative to the magnet **11.1.3.2-210** when the optical module **11.1.3.2-202** is moved via the motor (not shown).

[1908] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3.2-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.3.2-2**.

[1909] FIG. **11.1.3.2-3** illustrates an example of an encoder assembly **11.1.3.2-309** including a bracket **11.1.3.2-314** biasing a chip **11.1.3.2-316** against a magnet **11.1.3.2-310**. In at least one example, the bracket **11.1.3.2-314** includes a first end/portion **11.1.3.2-318** secured to an optical module of an HMD device and a second end **11.1.3.2-322** opposite the first. In at least one example, the flexure bracket **11.1.3.2-314** is fixed to the chip **11.1.3.2-316** via a secondary flexure bracket **11.1.3.2-320** to correct for the angle of the magnet **11.1.3.2-310** during assembly of the encoder assembly **11.1.3.2-309**. The secondary flexure bracket **11.1.3.2-320** can extend from a distal end **11.1.3.2-322** of a primary flexure bracket **11.1.3.2-317** which is coupled to a proximal end/portion **11.1.3.2-318** configured to be fixed to an optical module of an HMD device.

[1910] In at least one example, the primary and/or secondary flexure brackets **11.1.3.2-317**, **11.1.3.2-320**, as well as the proximal portion **11.1.3.2-318**, can be formed of a plastic such as Ultem. In one example, the encoder magnet **11.1.3.2-310** can include a low friction tape, for example a Teflon tape.

[1911] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3.2-3** can be included, either alone or in any

combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.3.2-3**.

11.1.3.2.1: Sensor Assembly

[1912] Systems of the present disclosure can provide a head-mountable device that securely supports sensors that are isolated from an external environment within a sealed container and orientations over time and avoids applying excessive strain on the sensors, thereby protecting them from harm. The sensor can be mounted on a flex circuit or circuit board, which is rigidly secured within an enclosure, including for example a plate and a case. The case and the plate can be sealed together with the flex circuit extending there through (e.g., at an opening formed by sealing glue) to operably connected to another component.

[1913] Accordingly, the sensor assemblies described herein facilitate effective performance of sensors mounted therein. The sensor assemblies provides high robustness during the entirety of the service life of the head-mountable device without placing strain on the sensors themselves. The seal of the enclosure can isolate the sensor from external influences by preventing ingress of elements through the case and plate. The sensor assemblies further enable the mounting of sensors on a flex circuit, where desired.

[1914] These and other embodiments are discussed below with reference to FIGS. **11.1.3.2.1-1** through **11.1.3.2.1-5**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting.

[1915] According to some embodiments, for example as shown in FIG. **11.1.3.2.1-1**, a head-mountable device **11.1.3.2.1-100** includes a frame **11.1.3.2.1-110** that is worn on a head of a user. The frame **11.1.3.2.1-110** can be positioned in front of the eyes of a user to provide information within a field of view of the user. The frame **11.1.3.2.1-110** can provide nose pads or another feature to rest on a user's nose. The frame **11.1.3.2.1-110** can be supported on a user's head with the head engager **11.1.3.2.1-120**. The head engager **11.1.3.2.1-120** can wrap or extend along opposing sides of a user's head. The head engager **11.1.3.2.1-120** can include earpieces for wrapping around or otherwise engaging or resting on a user's ears. It will be appreciated that other configurations can be applied for securing the head-mountable device **11.1.3.2.1-100** to a user's head. For example, one or more bands, straps, belts, caps, hats, or other components can be used in addition to or in place of the illustrated components of the head-mountable device **11.1.3.2.1-100**. By further example, the head engager **11.1.3.2.1-120** can include multiple components to engage a user's head.

[1916] The frame **11.1.3.2.1-110** can provide structure around a peripheral region thereof to support any internal components of the frame **11.1.3.2.1-110** in their assembled position. For example, the frame **11.1.3.2.1-110** can enclose and support various internal components (including for example integrated circuit chips, processors, memory devices and other circuitry) to provide computing and functional operations for the head-mountable device **11.1.3.2.1-100**, as discussed further herein. While several components are shown within the frame **11.1.3.2.1-110**, it will be understood that some or all of these components can be located anywhere within or on the head-mountable device **11.1.3.2.1-100**. For example, one or more of these components can be positioned within the head engager **11.1.3.2.1-120** of the head-mountable device **11.1.3.2.1-100**.

[1917] The frame **11.1.3.2.1-110** can include and/or support one or more cameras **11.1.3.2.1-130**. The cameras **11.1.3.2.1-130** can be positioned on or near an outer side **11.1.3.2.1-122** of the frame **11.1.3.2.1-110** to capture images of views external to the head-mountable device **11.1.3.2.1-100**. As used herein, an outer side of a portion of a head-mountable device is a side that faces away from the user and/or towards an external environment. The captured images can be used for display to the user or stored for any other purpose. Each of the cameras **11.1.3.2.1-130** can be movable along

the outer side **11.1.3.2.1-122**. For example, a track or other guide can be provided for facilitating movement of the camera **11.1.3.2.1-130** therein.

[1918] The head-mountable device **11.1.3.2.1-100** can include displays **11.1.3.2.1-140** that provide visual output for viewing by a user wearing the head-mountable device **11.1.3.2.1-100**. One or more displays **11.1.3.2.1-140** can be positioned on or near an inner side **11.1.3.2.1-124** of the frame **11.1.3.2.1-110**. As used herein, an inner side **11.1.3.2.1-124** of a portion of a head-mountable device is a side that faces toward the user and/or away from the external environment.

[1919] A display **11.1.3.2.1-140** can transmit light from a physical environment (e.g., as captured by a camera) for viewing by the user. Such a display **11.1.3.2.1-140** can include optical properties, such as lenses for vision correction based on incoming light from the physical environment. Additionally or alternatively, a display **11.1.3.2.1-140** can provide information as a display within a field of view of the user. Such information can be provided to the exclusion of a view of a physical environment or in addition to (e.g., overlaid with) a physical environment.

[1920] The head-mountable device **11.1.3.2.1-100** can include one or more sensors **11.1.3.2.1-70**. The sensors **11.1.3.2.1-70** can be mounted to the frame **11.1.3.2.1-110** in a secure manner, as described further herein. Any number and type of sensors can be provided. For example, one or more of the sensors **11.1.3.2.1-70** can be or include an inertial measurement unit (“IMU”) that provides information regarding a characteristic of the head-mountable device **11.1.3.2.1-100**, such as inertial angles thereof. For example, the IMU can include a six-degrees of freedom IMU that calculates the position, velocity, and/or acceleration of the head-mountable device **11.1.3.2.1-100** based on six degrees of freedom (x, y, z, $\theta_{\text{sub.x}}$, $\theta_{\text{sub.y}}$, and $\theta_{\text{sub.z}}$). The IMU can include one or more of an accelerometer, a gyroscope, and/or a magnetometer. Additionally or alternatively, the sensors **11.1.3.2.1-70** can detect motion characteristics of the head-mountable device **11.1.3.2.1-100** with one or more other motion sensors, such as an accelerometer, a gyroscope, a global positioning sensor, a tilt sensor, and so on for detecting movement and acceleration of the head-mountable device **11.1.3.2.1-100**. The sensors **11.1.3.2.1-70** can provide data to a controller for processing. Such data can influence other operations of the head-mountable device **11.1.3.2.1-100**, such as the information provided on the display **11.1.3.2.1-140**.

[1921] It will be understood that the sensors **11.1.3.2.1-70** (e.g., IMUs) and associated assemblies can be coupled to and/or integrated with one or more other components of the head-mountable device **11.1.3.2.1-100**. For example, one or more sensors **11.1.3.2.1-70** can be coupled to components that move relative to the frame **11.1.3.2.1-110** or other components of the head-mountable device **11.1.3.2.1-100**. By further example, one or more sensors **11.1.3.2.1-70** can be coupled to one or more displays **11.1.3.2.1-140**, the frame **11.1.3.2.1-110**, and/or other components **11.1.3.2.1-160** of the head-mountable device **11.1.3.2.1-100** (e.g., controllers, input/output devices, and the like). Accordingly, by operating multiple sensors **11.1.3.2.1-70**, the absolute and/or relative position and/or orientation of each of the associated components can be determined. For example, one or more (e.g., a pair of) displays **11.1.3.2.1-140** can move relative to each other and/or the frame **11.1.3.2.1-110** to be aligned with the eyes of the user, and sensors **11.1.3.2.1-70** coupled to each of the displays **11.1.3.2.1-140** can track such movement. By further example, the sensors **11.1.3.2.1-70** of separate head-mountable devices **11.1.3.2.1-100** (e.g., worn by different users) can share information so that each of the head-mountable devices **11.1.3.2.1-100** can determine the absolute and/or relative position and/or orientation of the other and optionally output corresponding information (e.g., visual information via displays **11.1.3.2.1-140**) so the users can perceive each other's presence, for example in a shared CGR environment.

[1922] Solid state motion sensors, such as IMUs, can be affected by strain, temperature, and humidity. Over time, exposure to these influences can result in a wider and less predictable operating range over the life of the IMU. For example, devices relying on absolute measurement of orientation and/or acceleration using MEMS IMUs may require bounded changes in bias, axis orthogonality, and cross-axis sensitivity. Typically, these sensors are surface mounted on a rigid

PCB and care is taken to choose PCB locations that minimize strain and temperature shift.

However, for some more demanding applications this is not sufficient to extract all performance from the sensor.

[1923] Referring now to FIGS. **11.1.3.2.1-2-11.1.3.2.1-4**, a sensor assembly can be provided to securely support one or more sensors of a head-mountable device while simultaneously protecting the sensors from exposure to an external environment as well as forces applied to the assembly. It will be understood that the assembly described herein can be one of multiple assemblies and can include one or more other components of the head-mountable device. It will be further understood that a sensor assembly can be integrated into the head-mountable device and/or provided as a module thereto.

[1924] As shown in FIG. **11.1.3.2.1-2**, a sensor assembly **11.1.3.2.1-10** can include a sensor **11.1.3.2.1-70** (e.g., IMU) housed within a protective case. A top case **11.1.3.2.1-20** of the sensor assembly **11.1.3.2.1-10** can be securely coupled to the bottom plate **11.1.3.2.1-112** or another component of the head-mountable device. For example, the bottom plate **11.1.3.2.1-112** can be part of, integrated with, and/or coupled to a frame and/or another component of the head-mountable device.

[1925] The sensor **11.1.3.2.1-70** can be coupled to a flex circuit **11.1.3.2.1-60**. For example, the sensor **11.1.3.2.1-70** can be directly mounted to a surface of the flex circuit **11.1.3.2.1-60**. Additionally or alternatively, the sensor **11.1.3.2.1-70** can be mounted to board that is directly or indirectly coupled to the flex circuit **11.1.3.2.1-60**. The flex circuit **11.1.3.2.1-60** can, in turn, directly or indirectly couple the sensor **11.1.3.2.1-70** and/or the board to other components of the sensor assembly **11.1.3.2.1-10**, such as the top case **11.1.3.2.1-20** and/or the bottom plate **11.1.3.2.1-112**.

[1926] The sensor **11.1.3.2.1-70** can be implemented as an integrated circuit, such as one or more of an industry standard integrated circuit, an application-specific integrated circuit (ASIC), an application-specific standard product (ASSP), and the like. The sensor **11.1.3.2.1-70** can have a size and shape that is accommodated by the housing structure of the assembly and provides the desired performance characteristics. For example, the sensor **11.1.3.2.1-70** can be integrated with a wafer and/or board.

[1927] The one or more sensors **11.1.3.2.1-70** of the sensor assembly **11.1.3.2.1-10** can be mounted to the flex circuit **11.1.3.2.1-60** that operably connects the sensors **11.1.3.2.1-70** of the sensor assembly **11.1.3.2.1-10** to each other and/or other components. As used herein, “flexible circuit” or “flex circuit” is a structure that includes a conductive layer, an insulation layer, and optionally a substrate layer. A flex circuit can be provided in electrical communication with at least one electrode, terminal, and/or connector. A flex circuit forms circuitry that includes a pattern of conductors of the conductive layer typically in the form of pads, which are typically formed on a surface of an insulating material of the insulation layer. Such circuitry is typically metallic, such as of a copper or copper alloy. In general, a flex circuit is thin, having a total thickness of from about 1 mm to about 30 mm. A flex circuit is generally flexible, such that it can conform to contours of other components. A flex circuit may be any suitable size and constructed in any suitable shape. For example, the size of a flex circuit may be determined by the power requirements of the components connected thereto, the conductivity of the flex circuit, the distance between operably connected components, or any other suitable criteria. It will be understood that, additionally or alternatively, a cable or other conductive material surrounded by an insulating layer can be provided. Such a cable can be connected via a connector and/or a hot bar and/or wire bond to one or more boards and/or other electronic components.

[1928] Providing operable connections to and from the sensors **11.1.3.2.1-70** via a flex circuit **11.1.3.2.1-60** can facilitate such connections while occupying little space within the head-mountable device. Additionally, a flex circuit **11.1.3.2.1-60** can conform and bend around other components of the head-mountable device. These features can help the head-mountable device

maintain a low weight and small size.

[1929] The flex circuit **11.1.3.2.1-60** can include multiple segments that have different characteristics. For example, a first segment **11.1.3.2.1-62** of the flex circuit **11.1.3.2.1-60** can support the sensor **11.1.3.2.1-70**. The first segment **11.1.3.2.1-62** can define a terminal and portion of the flex circuit **11.1.3.2.1-60** for residing within the enclosure of the sensor assembly **11.1.3.2.1-10**. The flex circuit **11.1.3.2.1-60** can further include a second segment **11.1.3.2.1-64** that extends from the first segment **11.1.3.2.1-62** and/or within the enclosure. The second segment **11.1.3.2.1-64** can provide strain relief features, such as a shape and/or size that promotes flexibility, bending, and/or stretching. For example, as shown in FIG. **11.1.3.2.1-2**, the second segment **11.1.3.2.1-64** can include an S-curve, a serpentine shape, an undulating shape, and/or another non-linear shape. With such a shape, the second segment **11.1.3.2.1-64** can easily alter its shape upon application of tension and/or other forces along the length of the flex circuit **11.1.3.2.1-60**. For example, tension and/or other forces applied at a third segment **11.1.3.2.1-66** of the flex circuit **11.1.3.2.1-60** that is away from the sensor **11.1.3.2.1-70** (e.g., within or outside the enclosure) can be absorbed by adjustments along the second segment **11.1.3.2.1-64**, such that such tension and/or other forces are not transmitted to the sensor **11.1.3.2.1-70**. Such adjustments can include an adjustment to the effective length of the second segment **11.1.3.2.1-64** (e.g., between the first segment **11.1.3.2.1-62** and the third segment **11.1.3.2.1-66**), adjustments to the curvature along the second segment **11.1.3.2.1-64**, and the like. By further example, the second segment **11.1.3.2.1-64** can alter its shape along and/or about one or more axes extending through the second segment **11.1.3.2.1-64**. As such, the sensor **11.1.3.2.1-70** can maintain a consistent position and/or orientation despite the application of external forces.

[1930] A base plate **11.1.3.2.1-80** can provide support to the sensor **11.1.3.2.1-70** and the flex circuit **11.1.3.2.1-60**. For example, the base plate **11.1.3.2.1-80** can be disposed adjacent to the sensor **11.1.3.2.1-70** on a side of the flex circuit **11.1.3.2.1-60** that is opposite the sensor **11.1.3.2.1-70**. Additionally or alternatively, the flex circuit **11.1.3.2.1-60** can be or include a circuit board that provides a rigid substrate with electrical connections between components and/or to other components. Where the flex circuit **11.1.3.2.1-60** includes or forms a rigid circuit board, the base plate **11.1.3.2.1-80** can optionally be omitted.

[1931] The sensor assembly **11.1.3.2.1-10** can further include a mounting adhesive **11.1.3.2.1-50** disposed between the base plate **11.1.3.2.1-80** and the bottom plate **11.1.3.2.1-112**. The mounting adhesive **11.1.3.2.1-50** can bond opposing surfaces of the base plate **11.1.3.2.1-80** and the bottom plate **11.1.3.2.1-112** to each other. The mounting adhesive **11.1.3.2.1-50** can have a coefficient of thermal expansion that is complementary to the coefficients of thermal expansion of the base plate **11.1.3.2.1-80** and/or the bottom plate **11.1.3.2.1-112** to minimize strain on the sensor **11.1.3.2.1-70**. As used herein, a coefficient of thermal expansion can refer to either a linear coefficient of thermal expansion and/or a volumetric coefficient of thermal expansion. As used herein, any two coefficients of thermal expansion are “complementary” when one is within $\pm 20\%$, $\pm 15\%$, $\pm 10\%$, $\pm 9\%$, $\pm 8\%$, $\pm 7\%$, $\pm 6\%$, $\pm 5\%$, $\pm 4\%$, $\pm 3\%$, $\pm 2\%$, or $\pm 1\%$ of the other, inclusive of any intermediate value there between. For example, the mounting adhesive **11.1.3.2.1-50** can have a coefficient of thermal expansion that is within $\pm 10\%$, $\pm 9\%$, $\pm 8\%$, $\pm 7\%$, $\pm 6\%$, $\pm 5\%$, $\pm 4\%$, $\pm 3\%$, $\pm 2\%$, or $\pm 1\%$ of the coefficient of thermal expansion of the base plate **11.1.3.2.1-80** and/or the bottom plate **11.1.3.2.1-112**.

[1932] The top case **11.1.3.2.1-20** defines an opening **11.1.3.2.1-28** to permit electrical communication with the sensor **11.1.3.2.1-70**, for example, at a side portion **11.1.3.2.1-22** of the top case **11.1.3.2.1-20**. For example, the flex circuit **11.1.3.2.1-60** (e.g., at the third segment **11.1.3.2.1-66** thereof) can extend through the opening **11.1.3.2.1-28**, so that a connector **11.1.3.2.1-68** at a terminal end of the flex circuit **11.1.3.2.1-60** (e.g., at an end opposite the sensor **11.1.3.2.1-70** and the first segment **11.1.3.2.1-62**) can connect to another component. The connector **11.1.3.2.1-68** coupled to the flex circuit **11.1.3.2.1-60** can be in electrical or other operable communication with

the sensor **11.1.3.2.1-70**. As such, the connector **11.1.3.2.1-68** can operably connect the sensor **11.1.3.2.1-70** to another component, such as a controller of the head-mountable device.

[1933] The sensor assembly **11.1.3.2.1-10** further includes a sealing member **11.1.3.2.1-52** disposed within the opening **11.1.3.2.1-28** to seal the enclosure from an external environment and permit electrical communication with the sensor **11.1.3.2.1-70** along the flex circuit **11.1.3.2.1-60**. For example, the sealing member **11.1.3.2.1-52** can entirely surround a portion of the flex circuit **11.1.3.2.1-60** (e.g., along the third segment **11.1.3.2.1-66**). The flex circuit **11.1.3.2.1-60** can extend entirely through the sealing member **11.1.3.2.1-52** to extend to either side thereof, such as to the interior and exterior of the enclosure. The sealing member **11.1.3.2.1-52** can include an adhesive, silicone, and/or other sealing member. The sealing member **11.1.3.2.1-52** can be provided as a liquid that solidifies (e.g., cures) during assembly.

[1934] As shown in FIG. **11.1.3.2.1-3**, the sensor assembly **11.1.3.2.1-10** can form an enclosure **11.1.3.2.1-12** to house the sensor **11.1.3.2.1-70** (e.g., IMU). For example, the top case **11.1.3.2.1-20**, the bottom plate **11.1.3.2.1-112**, and the sealing member **11.1.3.2.1-52** can form a sealed enclosure **11.1.3.2.1-12** that surrounds a sealed volume. The sensor **11.1.3.2.1-70**, the base plate **11.1.3.2.1-80**, and a portion of the flex circuit **11.1.3.2.1-60** can be contained within the sealed volume.

[1935] To form the enclosure **11.1.3.2.1-12**, the top case **11.1.3.2.1-20** can be coupled to the bottom plate **11.1.3.2.1-112** in a sealing manner. For example, the top case **11.1.3.2.1-20** can be welded (e.g., laser welded) to the bottom plate **11.1.3.2.1-112**. Other types of couplings are contemplated, such as bonding, adhesion, and the like. The coupling can maintain the opening **11.1.3.2.1-28**, for example at a side portion **11.1.3.2.1-22** of the top case **11.1.3.2.1-20**.

[1936] The flex circuit **11.1.3.2.1-60**, including the connector **11.1.3.2.1-68**, can be provided through the opening **11.1.3.2.1-28**. The sealing member **11.1.3.2.1-52** can be formed about a portion of the flex circuit **11.1.3.2.1-60** that extends through the opening **11.1.3.2.1-28**. The sealing member **11.1.3.2.1-52** can extend from the top case **11.1.3.2.1-20** to the bottom plate **11.1.3.2.1-112**. As such, the sealing member **11.1.3.2.1-52** can fill the only opening **11.1.3.2.1-28** that provides fluid communication between the interior of the enclosure **11.1.3.2.1-12** and an external environment (e.g., exterior to the enclosure **11.1.3.2.1-12**).

[1937] As further shown in FIG. **11.1.3.2.1-3**, the opening **11.1.3.2.1-28** can be defined on a side portion **11.1.3.2.1-22** of the top case **11.1.3.2.1-20**. It will be understood that the opening **11.1.3.2.1-28** can be provided at any portion of the top case **11.1.3.2.1-20**, as described further herein. Regardless of its location, the opening **11.1.3.2.1-28** can be filled with the sealing member **11.1.3.2.1-52** and around the flex circuit **11.1.3.2.1-60**.

[1938] By coupling the top case **11.1.3.2.1-20** to the bottom plate **11.1.3.2.1-112** and by filling the opening **11.1.3.2.1-28** with the sealing member **11.1.3.2.1-52**, the enclosure **11.1.3.2.1-12** can provide a sealed volume for the sensor **11.1.3.2.1-70**. As used herein, and enclosure, volume, or other space that is “sealed” is at least partially and/or selectively isolated from an external environment. For example, the top case **11.1.3.2.1-20** and the bottom plate **11.1.3.2.1-112** can be impermeable, such that they form a barrier against ingress and/or egress of matter there through. The sealing member **11.1.3.2.1-52** can also form a barrier. It will be understood that one or more barriers can be selectively permeable while still maintaining a sealed volume and/or enclosure. For example, the sealing member **11.1.3.2.1-52** can be permeable to one or more gases and/or substances on an atomic level. However, the sealing member **11.1.3.2.1-52** can be impermeable to (e.g., preventing ingress or egress of) molecules, compounds, and/or liquids. In some embodiments, the enclosure can be hermetically or “airtight” sealed, however other seals are contemplated.

[1939] As shown in FIG. **11.1.3.2.1-4**, the connector **11.1.3.2.1-68** can extend through other portions of the top case **11.1.3.2.1-20**, such as an upper portion **11.1.3.2.1-24** of the top case **11.1.3.2.1-20**. The upper portion **11.1.3.2.1-24** can be a portion of the top case that is opposite the bottom plate **11.1.3.2.1-112**. One or more openings **11.1.3.2.1-28** can be defined on the upper

portion **11.1.3.2.1-24** of the top case **11.1.3.2.1-20**, and corresponding connectors **11.1.3.2.1-68** can extend through the openings **11.1.3.2.1-28**. Outside of the enclosure **11.1.3.2.1-12**, a mating connector **11.1.3.2.1-90** can be operably coupled to the connectors **11.1.3.2.1-68**. For example, the interface between the mating connector **11.1.3.2.1-90** and the connector(s) **11.1.3.2.1-68** can include one or more electrodes, contact plates, wires, pogo pins, and the like.

[1940] Where the connectors **11.1.3.2.1-68** extend through the opening **11.1.3.2.1-28**, one or more sealing members **11.1.3.2.1-52** can be formed to fill the openings **11.1.3.2.1-28** and surround corresponding portions of the connectors **11.1.3.2.1-68**. Where the openings are formed in the upper portion **11.1.3.2.1-24** of the top case **11.1.3.2.1-20**, the sealing members **11.1.3.2.1-52** need not engage the bottom plate **11.1.3.2.1-112**. As such, the top case **11.1.3.2.1-20** and the bottom plate **11.1.3.2.1-112** can be coupled with a continuous seal, such as a weld.

[1941] It will be understood that the opening(s) **11.1.3.2.1-28** can be provided at any portion of the top case **11.1.3.2.1-20** and/or the bottom plate **11.1.3.2.1-112**. Regardless of its location, the opening **11.1.3.2.1-28** can be filled with the sealing member **11.1.3.2.1-52** and around the flex circuit **11.1.3.2.1-60** and/or connector(s) **11.1.3.2.1-68**.

[1942] Sensor assemblies of the present disclosure can provide predictable performance over the life of the head-mountable device by isolating the sensor from strain and humidity changes. This can enables absolute orientation measurements between multiple sensors over the life of the head-mountable device, without requiring repeated calibration. By providing each sensor (e.g., IMU) in a consistent position and orientation, any one or multiple sensors can be dedicated as a reference sensor for the entire head-mountable device.

[1943] Referring now to FIG. **11.1.3.2.1-5**, components of the head-mountable device can be operably connected to provide the performance described herein. FIG. **11.1.3.2.1-5** shows a simplified block diagram of an illustrative head-mountable device **11.1.3.2.1-100** in accordance with one embodiment of the invention. It will be appreciated that components described herein can be provided on either or both of a frame and/or a head engager of the head-mountable device **11.1.3.2.1-100**. It will be understood that additional components, different components, or fewer components than those illustrated may be utilized within the scope of the subject disclosure.

[1944] As shown in FIG. **11.1.3.2.1-5**, the head-mountable device **11.1.3.2.1-100** can include a controller **11.1.3.2.1-180** (e.g., control circuitry) with one or more processing units that include or are configured to access a memory **11.1.3.2.1-182** having instructions stored thereon. The instructions or computer programs may be configured to perform one or more of the operations or functions described with respect to the head-mountable device **11.1.3.2.1-100**. The controller **11.1.3.2.1-180** can be implemented as any electronic device capable of processing, receiving, or transmitting data or instructions. For example, the controller **11.1.3.2.1-180** may include one or more of: a microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a digital signal processor (DSP), or combinations of such devices. As described herein, the term “processor” is meant to encompass a single processor or processing unit, multiple processors, multiple processing units, or other suitably configured computing element or elements.

[1945] The memory **11.1.3.2.1-182** can store electronic data that can be used by the head-mountable device **11.1.3.2.1-100**. The memory **11.1.3.2.1-182** can include the memory of the flex circuit **11.1.3.2.1-170** described herein. For example, the memory **11.1.3.2.1-182** can store electrical data or content such as, for example, audio and video files, documents and applications, device settings and user preferences, timing and control signals or data for the various modules, data structures or databases, and so on. The memory **11.1.3.2.1-182** can be configured as any type of memory. By way of example only, the memory **11.1.3.2.1-182** can be implemented as random access memory, read-only memory, Flash memory, removable memory, or other types of storage elements, or combinations of such devices.

[1946] The head-mountable device **11.1.3.2.1-100** can further include a display **11.1.3.2.1-140** for displaying visual information for a user. The display **11.1.3.2.1-140** can provide visual (e.g., image

or video) output. The display **11.1.3.2.1-140** can be or include an opaque, transparent, and/or translucent display. The display **11.1.3.2.1-140** may have a transparent or translucent medium through which light representative of images is directed to a user's eyes. The display **11.1.3.2.1-140** may utilize digital light projection, OLEDs, LEDs, uLEDs, liquid crystal on silicon, laser scanning light source, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In one embodiment, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface. The head-mountable device **11.1.3.2.1-100** can include an optical subassembly configured to help optically adjust and correctly project the image-based content being displayed by the display **11.1.3.2.1-140** for close up viewing. The optical subassembly can include one or more lenses, mirrors, or other optical devices.

[1947] The head-mountable device **11.1.3.2.1-100** can include one or more sensors **11.1.3.2.1-70**, such as the sensors of a sensor assembly, as described herein. The head-mountable device **11.1.3.2.1-100** can include one or more other sensors, including sensors of a sealed enclosure assembly and/or other sensors. Such sensors can be configured to sense substantially any type of characteristic such as, but not limited to, images, pressure, light, touch, force, temperature, position, motion, and so on. For example, the sensor can be a photodetector, a temperature sensor, a light or optical sensor, an atmospheric pressure sensor, a humidity sensor, a magnet, a gyroscope, an accelerometer, a chemical sensor, an ozone sensor, a particulate count sensor, and so on. By further example, the sensor can be a bio-sensor for tracking biometric characteristics, such as health and activity metrics. Other user sensors can perform facial feature detection, facial movement detection, facial recognition, eye tracking, user mood detection, user emotion detection, voice detection, etc. Sensors can include a camera which can capture image based content of the outside world.

[1948] The head-mountable device **11.1.3.2.1-100** can include an input/output component **11.1.3.2.1-186**, which can include any suitable component for connecting head-mountable device **11.1.3.2.1-100** to other devices. Suitable components can include, for example, audio/video jacks, data connectors, or any additional or alternative input/output components. The input/output component **11.1.3.2.1-186** can include buttons, keys, or another feature that can act as a keyboard for operation by the user.

[1949] The head-mountable device **11.1.3.2.1-100** can include the microphone **11.1.3.2.1-188** as described herein. The microphone **11.1.3.2.1-188** can be operably connected to the controller **11.1.3.2.1-180** for detection of sound levels and communication of detections for further processing, as described further herein.

[1950] The head-mountable device **11.1.3.2.1-100** can include the speakers **11.1.3.2.1-190** as described herein. The speakers **11.1.3.2.1-190** can be operably connected to the controller **11.1.3.2.1-180** for control of speaker output, including sound levels, as described further herein.

[1951] The head-mountable device **11.1.3.2.1-100** can include communications circuitry **11.1.3.2.1-192** for communicating with one or more servers or other devices using any suitable communications protocol. For example, communications circuitry **11.1.3.2.1-192** can support Wi-Fi (e.g., a 802.11 protocol), Ethernet, Bluetooth, high frequency systems (e.g., 900 MHz, 2.4 GHz, and 5.6 GHz communication systems), infrared, TCP/IP (e.g., any of the protocols used in each of the TCP/IP layers), HTTP, BitTorrent, FTP, RTP, RTSP, SSH, any other communications protocol, or any combination thereof. Communications circuitry **11.1.3.2.1-192** can also include an antenna for transmitting and receiving electromagnetic signals.

[1952] The head-mountable device **11.1.3.2.1-100** can include a battery, which can charge and/or power components of the head-mountable device **11.1.3.2.1-100**. The battery can also charge

and/or power components connected to the head-mountable device **11.1.3.2.1-100**.

[1953] While various embodiments and aspects of the present disclosure are illustrated with respect to a head-mountable device, it will be appreciated that the subject technology can encompass and be applied to other devices. For example, a sensor assembly in accordance with embodiments disclosed herein can be included with an electronic device that is moved while in use. Such an electronic device can be or include a desktop computing device, a laptop-computing device, a display, a television, a portable device, a phone, a tablet computing device, a mobile computing device, a wearable device, a watch, and/or a digital media player.

[1954] Accordingly, embodiments of the present disclosure provide a head-mountable device that securely supports sensors in a manner that maintains their positions and orientations over time and avoids applying excessive strain on the sensors, thereby protecting them from harm. Accordingly, the sensor assemblies described herein facilitate effective performance of sensors mounted therein. The sensor assemblies provides high robustness during the entirety of the service life of the head-mountable device without placing strain on the sensors themselves. The seal of the enclosure can isolate the sensor from external influences by preventing ingress of elements through the case and plate. The sensor assemblies further enable the mounting of sensors on a flex circuit, where desired.

[1955] Various examples of aspects of the disclosure are described below as clauses for convenience. These are provided as examples, and do not limit the subject technology.

[1956] Clause A: a sensor assembly for a head-mountable device, the sensor assembly comprising: a flex circuit; an inertial measurement unit coupled to the flex circuit; a base plate disposed adjacent to the inertial measurement unit on a side of the flex circuit that is opposite the inertial measurement unit; and an enclosure disposed around the inertial measurement unit, the base plate, and a portion of the flex circuit, wherein the enclosure comprises: a top case; and a bottom plate coupled to the top case, wherein the bottom plate and the top case are positioned to define a sealed volume containing the inertial measurement unit, the base plate, and the portion of the flex circuit.

[1957] Clause B: a sensor assembly for a head-mountable device, the sensor assembly comprising: a flex circuit; an inertial measurement unit coupled to the flex circuit; an enclosure disposed around the inertial measurement unit and a first portion of the flex circuit, wherein the enclosure defines an opening, wherein a second portion of the flex circuit extends through the opening to permit electrical communication with the inertial measurement unit; and a sealing member disposed within the opening and around the second portion of the flex circuit to seal the enclosure from an external environment.

[1958] Clause C: a sensor assembly for a head-mountable device, the sensor assembly comprising: a flex circuit; an inertial measurement unit coupled to the flex circuit; a connector in electrical communication with the inertial measurement unit and coupled to the flex circuit; an enclosure disposed around the inertial measurement unit and the flex circuit, wherein the enclosure defines an opening, wherein the connector extends through the opening to permit electrical communication with the inertial measurement unit; and a sealing member disposed within the opening and around the connector to seal the enclosure from an external environment.

[1959] One or more of the above clauses can include one or more of the features described below. It is noted that any of the following clauses may be combined in any combination with each other, and placed into a respective independent clause, e.g., clause A, B, or C.

[1960] Clause 1: the inertial measurement unit comprises an accelerometer, a gyroscope, or a magnetometer.

[1961] Clause 2: the top case defines an opening to permit electrical communication with the inertial measurement unit.

[1962] Clause 3: the flex circuit extends through the opening in the top case.

[1963] Clause 4: a connector coupled to the flex circuit and in electrical communication with the inertial measurement unit, wherein the connector extends through the opening in the top case.

[1964] Clause 5: the opening is defined on a side portion of the top case.

[1965] Clause 6: the opening is defined on an upper portion of the top case.

[1966] Clause 7: a sealing member disposed within the opening to seal the enclosure from an external environment and permit electrical communication with the inertial measurement unit.

[1967] Clause 8: the sealing member comprises a sealing adhesive or silicone.

[1968] Clause 9: a mounting adhesive disposed between the base plate and the bottom plate, wherein the mounting adhesive has a coefficient of thermal expansion that is complementary to a coefficient of thermal expansion of the top case and the bottom plate to minimize strain on the inertial measurement unit.

[1969] Clause 10: the inertial measurement unit is mounted to the flex circuit extending entirely through the enclosure.

[1970] Clause 11: the first portion of the flex circuit within the enclosure defines an S-curve portion.

[1971] Clause 12: the connector comprises a plurality of pins extending through the sealing member.

11.1.3.2.2: Electronic Devices with Movable Optical Assemblies

[1972] The optical assemblies may have gaze trackers. The gaze trackers may be used to make interpupillary distance measurements and eye relief measurements. Adjustments to the positions of the optical assemblies may be made by the motors based on the interpupillary distance measurements and eye relief measurements. For example, no position adjustments may be made unless a measured eye relief measurement exceeds an eye relief threshold and the measured interpupillary distance is lower than an interpupillary distance threshold. If the measured eye relief measurement is sufficiently large and the measured interpupillary distance is sufficiently small, adjustments to the positions of the optical assemblies may be made in which, for a given measured eye relief, a smaller measured interpupillary distance results in a larger adjustment to an optical assembly position than a larger measured interpupillary distance.

[1973] FIG. **11.1.3.2.2-1** is a schematic diagram of an illustrative electronic device of the type that may include movable components. Device **11.1.3.2.2-10** of FIG. **11.1.3.2.2-1** may be a head-mounted device (e.g., goggles, glasses, a helmet, and/or other head-mounted device), a cellular telephone, a tablet computer, a laptop computer, a wristwatch, a peripheral device (sometimes referred to as a peripheral) such as a pair of headphones, or other electronic equipment. In an illustrative configuration, device **11.1.3.2.2-10** is a head-mounted device such as a pair of goggles (sometimes referred to as virtual reality goggles, mixed reality goggles, augmented reality glasses, etc.).

[1974] As shown in the illustrative top view of device **11.1.3.2.2-10** of FIG. **11.1.3.2.2-1**, device **11.1.3.2.2-10** may have a housing such as housing **11.1.3.2.2-12** (sometimes referred to as a head-mounted support structure or head-mounted support). Housing **11.1.3.2.2-12** may include a main portion such as portion **11.1.3.2.2-12M** (sometimes referred to as a main unit or head-mounted unit) and other head-mounted support structures such as head strap **11.1.3.2.2-12T**. When housing **11.1.3.2.2-12** is being worn on the head of a user, the front of housing **11.1.3.2.2-12** may face outwardly away from the user, the rear of housing **11.1.3.2.2-12** may face towards the user, and the user's eyes may be located in eye boxes **11.1.3.2.2-36**.

[1975] Device **11.1.3.2.2-10** may have electrical and optical components that are used in displaying images to eye boxes **11.1.3.2.2-36** when device **11.1.3.2.2-10** is being worn. These components may include left and right optical assemblies **11.1.3.2.2-20** (sometimes referred to as optical modules). Each optical assembly **11.1.3.2.2-20** may have an optical assembly support **11.1.3.2.2-38** (sometimes referred to as a lens barrel or optical module support) and guide rails **11.1.3.2.2-22** along which optical assemblies **11.1.3.2.2-20** may slide to adjust optical-assembly-to-optical-assembly separation to accommodate different user interpupillary distances.

[1976] Each assembly **11.1.3.2.2-20** may have a display **11.1.3.2.2-32** that has an array of pixels for displaying images and a lens **11.1.3.2.2-34**. Display **11.1.3.2.2-32** and lens **11.1.3.2.2-34** of each

assembly **11.1.3.2.2-20** may be coupled to and supported by support **11.1.3.2.2-38**. During operation, images displayed by displays **11.1.3.2.2-32** may be presented to eye boxes **11.1.3.2.2-36** through lenses **11.1.3.2.2-34** for viewing by a user. Each optical assembly **11.1.3.2.2-20** may also have a gaze tracker **11.1.3.2.2-50**. Gaze trackers **11.1.3.2.2-50** may each include one or more light sources (e.g., infrared light-emitting diodes that provide flood illumination and glints for eye tracking) and an associated camera (e.g., an infrared camera). Using gaze trackers **11.1.3.2.2-50**, which may sometimes be referred to as gaze tracking systems or gaze tracking sensors, device **11.1.3.2.2-10** can gather data on a user's eyes located in eye boxes **11.1.3.2.2-36**. As an example, the direction in which a user's eyes are pointing (sometimes referred to as a user's point of gaze or direction of view) may be measured. Biometric information such as iris scan information may also be gathered. In addition, gaze trackers **11.1.3.2.2-50**, may be used to measure the location of a user's eyes relative to device **11.1.3.2.2-10** and thereby measure the eye relief of the user's eyes (e.g., the distance between the lenses of device **11.1.3.2.2-10** and the eyes) and the separation between the user's left and right eyes (sometimes referred to as the user's interpupillary distance). If desired, gaze trackers **11.1.3.2.2-50** (e.g., the cameras of trackers **11.1.3.2.2-50**) may capture images of the skin of the user's face surrounding the user's eyes (e.g., to measure whether this skin is loose or taut).

[1977] Each optical assembly may have magnets, clips, and/or other engagement features to allow removable vision correction lenses (sometimes referred to as prescription lenses) to be removably attached to assemblies **11.1.3.2.2-20** in alignment with lenses **11.1.3.2.2-34** (see, e.g., illustrative optional vision correction lenses **11.1.3.2.2-51**). Lenses **11.1.3.2.2-51** may have magnets that are sensed by sensors **11.1.3.2.2-53** (e.g., magnetic sensors in assemblies **11.1.3.2.2-20**) or sensors **11.1.3.2.2-53** may be optical sensors, switches, or other sensors configured to gather other information indicating when lenses **11.1.3.2.2-51** are present.

[1978] Housing **11.1.3.2.2-12** may have a flexible curtain (sometimes referred to as a flexible rear housing wall or fabric housing wall) such as curtain **11.1.3.2.2-12R** on the rear of device **11.1.3.2.2-10** facing eye boxes **11.1.3.2.2-36**. Curtain **11.1.3.2.2-12R** has openings that receive assemblies **11.1.3.2.2-20**. The edges of curtain **11.1.3.2.2-12R** that surround each support **11.1.3.2.2-38** may be coupled to that support **11.1.3.2.2-38**. The outer peripheral edge of curtain **11.1.3.2.2-12R** may be attached to rigid housing walls forming an outer shell portion of main housing **11.1.3.2.2-12M**.

[1979] The walls of housing **11.1.3.2.2-12** may separate interior region **11.1.3.2.2-28** within device **11.1.3.2.2-10** from exterior region **11.1.3.2.2-30** surrounding device **11.1.3.2.2-10**.

[1980] Inner ends **11.1.3.2.2-24** of guide rails **11.1.3.2.2-22** may be attached to central housing portion **11.1.3.2.2-12C**. Opposing outer ends **11.1.3.2.2-26** may, in an illustrative configuration, be unsupported (e.g., the outer end portions of rails **11.1.3.2.2-22** may not directly contact housing **11.1.3.2.2-12**, so that these ends float in interior region **11.1.3.2.2-28** with respect to housing **11.1.3.2.2-12**).

[1981] Device **11.1.3.2.2-10** may include control circuitry and other components such as component **11.1.3.2.2-40**. The control circuitry may include storage, processing circuitry formed from one or more microprocessors and/or other circuits. To support communications between device **11.1.3.2.2-10** and external equipment, the control circuitry may include wireless communications circuitry. Components **11.1.3.2.2-40** may include sensors such as force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors, optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, image sensors, sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or sensors such as inertial measurement units that contain some or all of these sensors), radio-frequency sensors, depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices), optical sensors such as self-mixing sensors and

light detection and ranging (LIDAR) sensors that gather time-of-flight measurements, humidity sensors, moisture sensors, visual inertial odometry sensors, gaze tracking sensors, and/or other sensors. In some arrangements, devices **11.1.3.2.2-10** may use sensors to gather user input (e.g., button press input, touch input, etc.). Sensors may also be used in gathering environmental motion (e.g., device motion measurements, temperature measurements, ambient light readings, etc.) and/or may be used in measuring user activities and/or attributes (e.g., point-of-gaze, eye relief, interpupillary distance, etc.). If desired, position sensors such as encoders (e.g., optical encoders, magnetic encoders, etc.) may measure the position and therefore the movement (e.g., the velocity, acceleration, etc.) of optical assemblies **11.1.3.2.2-20** along rails **11.1.3.2.2-22**.

[1982] FIG. **11.1.3.2.2-2** is a rear view of an illustrative portion of device **11.1.3.2.2-10** (e.g., an inside left portion). Device **11.1.3.2.2-10** may have left and right actuators (e.g., motors) such as motor **11.1.3.2.2-48** that are used to rotate an elongated threaded shaft such as screw **11.1.3.2.2-44**. Nut **11.1.3.2.2-46** has threads that engage the threads on screw **11.1.3.2.2-44**. As the motor **11.1.3.2.2-48** on each side of device **11.1.3.2.2-10** is turned, a corresponding nut **11.1.3.2.2-46** is driven in the +X or -X direction (in accordance with whether screw **11.1.3.2.2-44** is being rotated clockwise or counterclockwise). In turn, this moves the optical assembly **11.1.3.2.2-20** on that side of device **11.1.3.2.2-10** in the +X or -X direction along its optical assembly guide rail **11.1.3.2.2-22**. If desired, the left and right motors **11.1.3.2.2-48** may be adjusted independently, so that optical assemblies **11.1.3.2.2-20** on the left and right of device **11.1.3.2.2-10** may be moved independently.

[1983] Each assembly **11.1.3.2.2-20** (e.g., each support **11.1.3.2.2-38** of FIG. **11.1.3.2.2-1**) may have portions that receive a corresponding rail **11.1.3.2.2-22** and that guide that assembly **11.1.3.2.2-20** along that rail **11.1.3.2.2-22**. By controlling the activity of motors **11.1.3.2.2-48** in tandem or individually, the spacing between the left and right optical assemblies of device **11.1.3.2.2-10** can be adjusted to accommodate the interpupillary distance of different users. For example, if gaze trackers **11.1.3.2.2-50** determine that a user has closely spaced eyes, assemblies **11.1.3.2.2-20** may be moved inwardly (towards each other) and if a user has widely spaced eyes, assemblies **11.1.3.2.2-20** may be moved outwardly (away from each other). By matching the spacing between optical assemblies **11.1.3.2.2-20** to the measured interpupillary distance of a user, the user may satisfactorily view visual content being presented by the displays of the optical assemblies.

[1984] The position and therefore the movement of each optical assembly may be monitored using one or more sensors. In the illustrative configuration of FIG. **11.1.3.2.2-2**, motor **11.1.3.2.2-48** (e.g., the motor on the left side of device **11.1.3.2.2-10**) and optical module **11.1.3.2.2-20** have been provided with an optical assembly position sensor based on a magnetic encoder. The encoder includes magnetic strip **11.1.3.2.2-54** and magnetic sensor **11.1.3.2.2-52**. Magnetic sensor **11.1.3.2.2-52** may be a Hall Effect sensor or other suitable magnetic sensor that is configured to move with optical assembly **11.1.3.2.2-20**. Magnetic strip **11.1.3.2.2-54**, which may be affixed to housing **11.1.3.2.2-12**, has a series of magnet poles (e.g., north and south poles) that extend along the X dimension parallel to guide rail **11.1.3.2.2-22** and parallel to the X-axis position adjustment direction associated with optical assembly **11.1.3.2.2-20**. As optical assembly **11.1.3.2.2-20** is moved along guide rail **11.1.3.2.2-22**, the magnetic encoder (e.g., sensor **11.1.3.2.2-52**) measures the change in magnetic field resulting as magnetic sensor **11.1.3.2.2-52** passes by different magnets in strip **11.1.3.2.2-54**. In this way, optical assembly **11.1.3.2.2-20** is provided with a position reference (e.g., by counting north and south magnetic poles in strip **11.1.3.2.2-54**). The position measurements made with the position sensor may reveal attributes of the motion of optical assembly **11.1.3.2.2-20** such as the velocity of optical assembly **11.1.3.2.2-20** and, if desired, the acceleration and deceleration of assembly **11.1.3.2.2-20**. Accordingly, the magnetic encoder (or other position sensor) associated with each optical assembly **11.1.3.2.2-30** may be used by device **11.1.3.2.2-10** to measure the location of that optical assembly so that the spacing between optical assemblies **11.1.3.2.2-20** can be satisfactorily adjusted (e.g., to help ensure that the spacing

between optical assemblies **11.1.3.2.2-20** has been adjusted to match or nearly match the user's interpupillary distance). The position readings from the optical assembly position sensors (e.g., the magnetic encoders) may also be used to determine the velocities of the optical assemblies as they are moved.

[1985] Optical assembly velocity information and/or other information from the optical assembly position sensors (e.g., linear position sensors formed from the magnetic encoders) may be used in monitoring whether the optical assemblies have slowed down their movement due to contact between the optical assemblies and the nose of a user. Consider, as an example, the arrangement of FIG. **11.1.3.2.2-3**. FIG. **11.1.3.2.2-3** is a rear view of a central nose-bridge portion NB of device **11.1.3.2.2-10**. As shown in FIG. **11.1.3.2.2-3**, nose bridge portion NB of housing portion **11.1.3.2.2-12C** has a nose-shaped recess **11.1.3.2.2-60**, which is configured to receive the user's nose (see, e.g., illustrative nose surface **11.1.3.2.2-62**). In some situations (e.g., when a user has a wide interpupillary distance), optical assemblies **11.1.3.2.2-20** reside at a non-zero distance G from surface **11.1.3.2.2-62** (sometimes referred to as gap G) after motors **11.1.3.2.2-48** adjusted the positions of assemblies **11.1.3.2.2-20** to match the user's interpupillary distance. In other situations, such as when a user has a large eye relief and small interpupillary distance, motors **11.1.3.2.2-48** may move optical assemblies inward until curtain **11.1.3.2.2-12R** (FIG. **11.1.3.2.2-1**) and optical assemblies **11.1.3.2.2-20** contact the left and right sides of nose surface **11.1.3.2.2-62**.

[1986] When optical assemblies **11.1.3.2.2-20** contact the left and right sides of nose surface of **11.1.3.2.2-62**, motors **11.1.3.2.2-48** will encounter resistance to further lateral movement of the optical assemblies along the X axis. This will cause optical assemblies **11.1.3.2.2-20** to move more slowly. The optical assembly position sensors will sense the reduction in the velocity of the optical assemblies. In this way, device **11.1.3.2.2-10** is informed that optical assemblies **11.1.3.2.2-20** are contacting and pressing against nose surface **11.1.3.2.2-62**. To ensure that device **11.1.3.2.2-10** is comfortable as device **11.1.3.2.2-10** is being worn on the head of the user, the position of the optical assemblies may, in response to this detected nose contact, be adjusted outward, away from nose surface **11.1.3.2.2-62** (e.g., by 1-3 mm or other suitable amount). This outward nudge in the positions of the optical assemblies may be made even if the final separation between the optical assemblies is slightly larger than the user's measured interpupillary distance.

[1987] Following outward adjustment of optical assemblies **11.1.3.2.2-20**, a non-zero gap G may be created between optical assemblies **11.1.3.2.2-20** and corresponding side portions of the user's nose (e.g., adjacent portions of nose surface **11.1.3.2.2-62**) and/or inward pressure imposed on the sides of the user's nose by optical assemblies **11.1.3.2.2-20** can be reduced to enhance comfort. By monitoring the optical assembly position sensors during optical assembly position adjustment with motors **11.1.3.2.2-48**, device **11.1.3.2.2-10** can identify a location for assemblies **11.1.3.2.2-20** in which the left and right assemblies **11.1.3.2.2-20** are separated by distance that is matched as closely as possible to the user's measured interpupillary distance while ensuring satisfactory comfort for the user.

[1988] Another way in which the position of optical assemblies **11.1.3.2.2-20** can be adjusted satisfactorily involves the use of eye relief measurements from gaze trackers **11.1.3.2.2-50**. When device **11.1.3.2.2-10** is placed onto the head of the user, gaze trackers **11.1.3.2.2-50** may measure the user's interpupillary distance and may measure the user's eye relief. Motors **11.1.3.2.2-48** may then adjust the positions of optical assemblies **11.1.3.2.2-20** based on the measured eye relief and measured interpupillary distance of the user. In an illustrative configuration, the positions of optical assemblies **11.1.3.2.2-20** maybe offset (e.g., nudged outwards from the location where the spacing between assemblies **11.1.3.2.2-20** matches the measured interpupillary distance of the user) by an amount that varies depending on both measured interpupillary distance and measured eye relief.

[1989] This type of approach is illustrated in the graph of FIG. **11.1.3.2.2-4**. In the graph of FIG. **11.1.3.2.2-4**, lateral offset ΔX represents the distance by which an optical assembly is adjusted outwardly (beyond the nominal position in which optical assembly separation matches measured

interpupillary distance) to enhance fit. There are three illustrative curves in the graph of FIG.

11.1.3.2.2-4. Curve **11.1.3.2.2-64** corresponds to a measured interpupillary distance IPDA, curve **11.1.3.2.2-66** corresponds to a measured interpupillary distance IPDB, which is larger than interpupillary distance IPDA, and curve **11.1.3.2.2-68** corresponds to a measured interpupillary distance IPDC, which is larger than interpupillary distance IPDA and larger than interpupillary distance IPDB. The values of IPDA, IPDB, and IPDC may be in the range of 52 to 74 mm or any other suitable range. In practice, device **11.1.3.2.2-10** may store a family of curves (e.g., empirically determined curves) for any suitable number of measured interpupillary distances and/or may represent the relationships embodied in curves such as curves **11.1.3.2.2-64**, **11.1.3.2.2-66**, and **11.1.3.2.2-68** using tables, functions, and/or other data structures.

[1990] As the graph of a FIG. **11.1.3.2.2-4** illustrates, if the measured eye relief a user falls below a predetermined threshold (e.g., ERT_H in the example of FIG. **11.1.3.2.2-4**), the user is unlikely to experience pressure from optical assemblies **11.1.3.2.2-20** on the sides of the user's nose.

Accordingly, for measured user eye relief values below eye relief threshold ERT_H, optical assemblies **11.1.3.2.2-20** may be separated along the X axis by a distance that matches the measured interpupillary distance of the user (e.g., there is no need to use a non-zero value of ΔX to adjust the position of optical assemblies **11.1.3.2.2-20** outwardly after they have reached the appropriate spacing to match the user's interpupillary distance). The value of ΔX is therefore zero for users with measured eye relief values of less than ERT_H (which may be, for example, a value between 18 mm and 22 mm or other suitable eye relief threshold value).

[1991] If however, the measured eye relief of a user exceeds eye relief threshold ERT_H, it may be beneficial for certain users to increase the spacing between optical assemblies **11.1.3.2.2-20** by nudging each optical assembly outwardly by an amount ΔX . As an example, consider users with measured interpupillary distances of IPDA. For this class of user, it may be beneficial to adjust the spacing of optical assemblies **11.1.3.2.2-20** outwardly by ΔX values that follow curve **11.1.3.2.2-64**. As shown by curve **11.1.3.2.2-64**, ΔX may be zero for measured eye relief values of less than ERT_H, whereas for measured eye relief values exceeding ERT_H, ΔX may rise progressively as a function of measured eye relief (e.g., up to a maximum ΔX value at a maximum measured eye relief value of ERM_X, which may be, for example, 25 mm or other suitable value). Users with larger interpupillary distances, such as a measured interpupillary distance of IPDB, may benefit from a less aggressive outward increase in optical assembly spacing, as shown by curve **11.1.3.2.2-66** in which the value of ΔX rises more slowly as a function of increasing measured eye relief over ERT_H than curve **11.1.3.2.2-64**.

[1992] An interpupillary distance threshold may be present above which it may not be desirable to make any ΔX adjustments for a user, regardless of their measured eye relief. When, for example, a user has a large measured interpupillary distance (e.g., a measured interpupillary distance of IPDC, which exceeds the interpupillary distance threshold), there will generally not be a comfort benefit in increasing optical assembly spacing. As a result, the recommended outward adjustment in optical assembly position ΔX as a function of measured eye relief for users with these larger measured interpupillary distances follows curve **11.1.3.2.2-68** (e.g., ΔX remains at zero, even for a measured eye relief of ERM_X). With this approach, only at measured interpupillary distances below the interpupillary distance threshold will outward adjustments in optical assembly position be used.

[1993] The graph of FIG. **11.1.3.2.2-4** illustrates how motors **11.1.3.2.2-48** may place optical assemblies **11.1.3.2.2-20** at positions that vary from the measured positions of the user's eyes. In the example of FIG. **11.1.3.2.2-4**, measured eye relief and interpupillary distance values were used in determining satisfactory positions for assemblies **11.1.3.2.2-20**. In particular, outward adjustments ΔX in the positions of optical assemblies **11.1.3.2.2-20** were recommended as a function of the interpupillary distance and eye relief values measured for a user with gaze trackers **11.1.3.2.2-50**. If desired, additional factors may be taken into consideration in making optical assembly adjustments such as these (e.g., additional factors that may be used in combination with or as an alternative to

using interpupillary distance measurements and eye relief measurements) and/or other suitable action may be taken based on data gathered with gaze trackers **11.1.3.2.2-50** and/or other sensors in device **11.1.3.2.2-10**.

[1994] FIG. **11.1.3.2.2-5** is a flow chart illustrating operations involved in using motors **11.1.3.2.2-48** and other components in device **11.1.3.2.2-10** to make adjustments to device **11.1.3.2.2-10** as a function of data gathered with gaze trackers **11.1.3.2.2-50** and/or other sensors.

[1995] During the operations of block **11.1.3.2.2-70**, device **11.1.3.2.2-10** may gather data. The gathered data may include, for example, measurements obtained by gaze trackers **11.1.3.2.2-50**. These measurements may include, for example, the measured interpupillary distance of a user wearing device **11.1.3.2.2-10**, the measured eye relief of a user wearing device **11.1.3.2.2-10**, the measured skin tautness around the eyes of a user wearing device **11.1.3.2.2-10**, and/or other gaze tracker measurements. The measurements of block **11.1.3.2.2-70** may also include measurements with the position sensors (e.g., the magnetic encoders) of optical assemblies **11.1.3.2.2-20**. For example, when a user dons device **11.1.3.2.2-10**, motors **11.1.3.2.2-48** may automatically start to move optical assemblies **11.1.3.2.2-20** to positions associated with the user's measured interpupillary distance from gaze trackers **11.1.3.2.2-50**. During this initial movement or during movement in response to a user input command or other movement, the position sensors may be used to monitor the velocities of assemblies **11.1.3.2.2-20**. In response to detected slowing of the speed of inward movement of assemblies **11.1.3.2.2-20**, device **11.1.3.2.2-10** can conclude that assemblies **11.1.3.2.2-20** are beginning to exert pressure on the sides of the user's nose (e.g., nose surface **11.1.3.2.2-62**). The locations associated with the measured reduction in optical assembly velocity are another form of measurement data that may be gathered during block **11.1.3.2.2-70**.

[1996] Further information that can be gathered during the operations of block **11.1.3.2.2-70** relates to the status of vision correction lenses **11.1.3.2.2-51** on device **11.1.3.2.2-10**. Vision correction lenses **11.1.3.2.2-51** may contain magnets that produce a magnetic field. Device **11.1.3.2.2-10** (e.g., assemblies **11.1.3.2.2-20**) may have vision correction lens sensors such as magnetic sensors **11.1.3.2.2-53** that determine whether or not lenses **11.1.3.2.2-51** are present by monitoring for the presence of the magnetic fields produced by lenses **11.1.3.2.2-51**. In response to detection of the magnetic fields from lenses **11.1.3.2.2-51** with sensors **11.1.3.2.2-53**, it can be concluded that lenses **11.1.3.2.2-51** are present.

[1997] During the operations of block **11.1.3.2.2-72**, device **11.1.3.2.2-10** can take action based on the data gathered at block **11.1.3.2.2-70**. As an example, the positions of optical assemblies **11.1.3.2.2-20** may be adjusted using motors **11.1.3.2.2-48**. In some configurations, the positions of optical assemblies **11.1.3.2.2-20** may be nudged outwards by an amount ΔX determined from measured interpupillary distance and eye relief values, as described in connection with FIG. **11.1.3.2.2-4**. If desired, optical assemblies **11.1.3.2.2-20** may, in some scenarios, be moved inward slightly (e.g., assemblies **11.1.3.2.2-20** may be moved towards each other such that assemblies **11.1.3.2.2-20** are closer than dictated by the measured value of a user's interpupillary distance to increase nasal field-of-view overlap when doing so presents a low risk of optical assembly nose surface contact or when any potential nose contact can be detected with an optical module position sensor or other sensor). To reduce wrinkles in rear curtain **11.1.3.2.2-12R**, optical assemblies **11.1.3.2.2-20** may be moved inwards or outwards slightly to tighten curtain **11.1.3.2.2-12R**. Optical assembly movement with motors **11.1.3.2.2-48** may also be used to provide a user with an alert (e.g., a haptic alert indicating that an incoming message has been received and/or another condition has been determined to be present). In some arrangements, gaze sensor measurements from trackers **11.1.3.2.2-50** during block **11.1.3.2.2-70** can be used to determine whether the user's skin surrounding the user's eyes is taut. In response to detecting an increase in skin tautness as assemblies **11.1.3.2.2-20** are moved (e.g., as assemblies **11.1.3.2.2-20** are being moved towards each other), it can be concluded that assemblies **11.1.3.2.2-20** are pressing or are about to press on nose surface **11.1.3.2.2-62**, so further movement inward of assemblies **11.1.3.2.2-20** can be halted

and/or assemblies **11.1.3.2.2-20** may be nudged outwardly to compensate. The positions of optical assemblies **11.1.3.2.2-20** may also be adjusted in response to detection that vision correction lenses **11.1.3.2.2-51** are present or are not present. The presence of lenses **11.1.3.2.2-51** may, as an example, increase or decrease the risk of nose contact by assemblies **11.1.3.2.2-20**, so information on the presence of lenses **11.1.3.2.2-51** (and, if desired, the prescription associated with lenses **11.1.3.2.2-51**) may be taken into account when using motors **11.1.3.2.2-48** to adjust the position of assemblies **11.1.3.2.2-20**. In general, these types of adjustments and/or other suitable actions may be taken during the operations of block **11.1.3.2.2-72** in response to gaze tracker measurement data and/or other data measured during the operations of block **11.1.3.2.2-70**.

11.1.3.2.3: Electronic Devices with Movable Optical Assemblies

[1998] The force with which the motors move the optical assemblies towards a central nose bridge portion of the device and therefore towards nose surfaces located at the nose bridge portion may be limited. The force may be limited using a clutch such as a magnetic clutch or a physical clutch based on structures that decouple from each other to limit the force. The force may also be limited by monitoring the force and halting the motors in response to detection of a given amount of force. Sensor measurements and electrical motor load measurements may be used in measuring the force. If desired, motor operation may be controlled by a user-operated button. The direction of permitted optical assembly movement when accommodating different interpupillary distances may also be controlled.

[1999] FIG. **11.1.3.2.3-1** is a schematic diagram of an illustrative electronic device of the type that may include movable optical assemblies to accommodate different interpupillary distances. Device **11.1.3.2.3-10** of FIG. **11.1.3.2.3-1** may be a head-mounted device (e.g., goggles, glasses, a helmet, and/or other head-mounted device). In an illustrative configuration, device **11.1.3.2.3-10** is a head-mounted device such as a pair of goggles (sometimes referred to as virtual reality goggles, mixed reality goggles, augmented reality glasses, etc.).

[2000] As shown in the illustrative cross-sectional top view of device **11.1.3.2.3-10** of FIG. **11.1.3.2.3-1**, device **11.1.3.2.3-10** may have a housing such as housing **11.1.3.2.3-12** (sometimes referred to as a head-mounted support structure, head-mounted housing, or head-mounted support). Housing **11.1.3.2.3-12** may include a front portion such as front portion **11.1.3.2.3-12F** and a rear portion such as rear portion **11.1.3.2.3-12R**. When device **11.1.3.2.3-10** is worn on the head of a user, rear portion **11.1.3.2.3-12R** rests against the face of the user and helps block stray light from reaching the eyes of the user and nose bridge portion NB of housing **11.1.3.2.3-12** rests on the nose of the user.

[2001] Main portion **11.1.3.2.3-12M** of housing **11.1.3.2.3-12** may be attached to head strap **11.1.3.2.3-12T**. Head strap **11.1.3.2.3-12T** may be used to help mount main portion **11.1.3.2.3-12** on the head and face of a user. Main portion **11.1.3.2.3-12M** may have a rigid shell formed from housing walls of polymer, glass, metal, and/or other materials. When housing **11.1.3.2.3-12** is being worn on the head of a user, the front of housing **11.1.3.2.3-12** may face outwardly away from the user, the rear of housing **11.1.3.2.3-12** (and rear portion **11.1.3.2.3-12R**) may face towards the user. In this configuration, rear portion **11.1.3.2.3-12R** may face the user's eyes located in eye boxes **11.1.3.2.3-36**.

[2002] Device **11.1.3.2.3-10** may have electrical and optical components that are used in displaying images to eye boxes **11.1.3.2.3-36** when device **11.1.3.2.3-10** is being worn. These components may include left and right optical assemblies **11.1.3.2.3-20** (sometimes referred to as optical modules). Each optical assembly **11.1.3.2.3-20** may have an optical assembly support **11.1.3.2.3-38** (sometimes referred to as a lens barrel, optical module support, or support structure) and guide rails **11.1.3.2.3-22** along which optical assemblies **11.1.3.2.3-20** may slide to adjust optical-assembly-to-optical-assembly separation to accommodate different user interpupillary distances.

[2003] Each assembly **11.1.3.2.3-20** may have a display **11.1.3.2.3-32** that has an array of pixels for displaying images and a lens **11.1.3.2.3-34**. Lens **11.1.3.2.3-34** may optionally have a removable

vision correction lens for correcting user vision defects (e.g., refractive errors such as nearsightedness, farsightedness, and/or astigmatism). In each assembly **11.1.3.2.3-20**, display **11.1.3.2.3-32** and lens **11.1.3.2.3-34** may be coupled to and supported by support **11.1.3.2.3-38**. During operation, images displayed by displays **11.1.3.2.3-32** may be presented to eye boxes **11.1.3.2.3-36** through lenses **11.1.3.2.3-34** for viewing by the user.

[2004] Rear portion **11.1.3.2.3-12R** may include flexible structures (e.g., a flexible polymer layer, a flexible fabric layer, etc.) so that portion **11.1.3.2.3-12R** can stretch to accommodate movement of supports **11.1.3.2.3-38** toward and away from each other to accommodate different user interpupillary distances.

[2005] The walls of housing **11.1.3.2.3-12** may separate interior region **11.1.3.2.3-28** within device **11.1.3.2.3-10** from exterior region **11.1.3.2.3-30** surrounding device **11.1.3.2.3-10**. In interior region **11.1.3.2.3-28**, optical assemblies **11.1.3.2.3-20** may be mounted on guide rails **11.1.3.2.3-22**. Guide rails **11.1.3.2.3-22** may be attached to central housing portion **11.1.3.2.3-12C**. If desired, the outer ends of guide rails **11.1.3.2.3-22** may be unsupported (e.g., the outer end portions of rails **11.1.3.2.3-22** may not directly contact housing **11.1.3.2.3-12**, so that these ends float in interior region **11.1.3.2.3-28** with respect to housing **11.1.3.2.3-12**).

[2006] Device **11.1.3.2.3-10** may include control circuitry and other components such as components **11.1.3.2.3-40**. The control circuitry may include storage, processing circuitry formed from one or more microprocessors and/or other circuits. To support communications between device **11.1.3.2.3-10** and external equipment, the control circuitry may include wireless communications circuitry. Components **11.1.3.2.3-40** may include sensors such as force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors, optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, image sensors, sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or sensors such as inertial measurement units that contain some or all of these sensors), radio-frequency sensors, depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices), optical sensors such as self-mixing sensors and light detection and ranging (LIDAR) sensors that gather time-of-flight measurements, humidity sensors, moisture sensors, visual inertial odometry sensors, current sensors, voltage sensors, and/or other sensors. In some arrangements, devices **11.1.3.2.3-10** may use sensors to gather user input (e.g., button press input, touch input, etc.). Sensors may also be used in gathering environmental motion (e.g., device motion measurements, temperature measurements, ambient light readings, etc.).

[2007] Optical assemblies **11.1.3.2.3-20** may have gaze trackers **11.1.3.2.3-62** (sometimes referred to as gaze tracker sensors). Gaze trackers **11.1.3.2.3-62**, which may operate through lenses **11.1.3.2.3-34**, may include one or more light sources such as infrared light-emitting diodes that emit infrared light to illuminate the eyes of a user in eye boxes **11.1.3.2.3-36**. Gaze trackers **11.1.3.2.3-62** also include infrared cameras for capturing images of the user's eyes and measuring reflections (glints) of infrared light from each of the infrared light sources. By processing these eye images, gaze trackers **11.1.3.2.3-62** may track the user's eyes and determine the point-of-gaze of the user. Gaze trackers **11.1.3.2.3-62** may also measure the locations of the user's eyes (e.g., the user's eye relief and the user's interpupillary distance).

[2008] To accommodate users with different interpupillary distances (eye-to-eye spacings), the spacing between the left and right optical assemblies **11.1.3.2.3-20** in device **11.1.3.2.3-10** can be adjusted (e.g., to match or nearly match the user's measured interpupillary distance). Device **11.1.3.2.3-10** may have left and right actuators (e.g., motors) such as motors **11.1.3.2.3-48**. Each motor **11.1.3.2.3-48** may be used to rotate an elongated threaded shaft (screw) such as shaft **11.1.3.2.3-44**. A nut **11.1.3.2.3-46** is provided on each shaft **11.1.3.2.3-44**. The nut has threads that

engage the threads on that shaft **11.1.3.2.3-44**. When a shaft is rotated, the nut on the shaft is driven in the +X or -X direction (in accordance with whether the shaft is being rotated clockwise or counterclockwise). In turn, this moves the optical assembly **11.1.3.2.3-20** that is attached to the nut in the +X or -X direction along its optical assembly guide rail **11.1.3.2.3-22**. Each assembly **11.1.3.2.3-20** (e.g., support **11.1.3.2.3-38**) may have portions that receive one of guide rails **11.1.3.2.3-22** so that the assembly is guided along the guide rail. By controlling the activity of motors **11.1.3.2.3-48**, the spacing between the left and right optical assemblies of device **11.1.3.2.3-10** can be adjusted to accommodate the interpupillary distance of different users. For example, if a user has closely spaced eyes, assemblies **11.1.3.2.3-20** may be moved inwardly (towards each other and towards nose bridge portion NB of housing **11.1.3.2.3-12**) and if a user has widely spaced eyes, assemblies **11.1.3.2.3-20** may be moved outwardly (away from each other).

[2009] When device **11.1.3.2.3-10** is being worn by a user, the user's head is located in region **11.1.3.2.3-68**. The presence of the user's head (and therefore a determination of whether device **11.1.3.2.3-10** is being worn or is unworn) may be made using one or more sensors (e.g., gaze trackers **11.1.3.2.3-62**, which may detect the presence of the eyes of the user in eye boxes **11.1.3.2.3-36**, rear-facing sensors such as sensor **11.1.3.2.3-66** on main housing **11.1.3.2.3-12M**, head-facing sensors mounted on strap **11.1.3.2.3-12T** such as sensor **11.1.3.2.3-64**, and/or other head presence sensors. These sensors may include cameras, light sensors (e.g., visible light or infrared sensors that measure when ambient light levels have dropped due to shadowing by the head of a user), proximity sensors (e.g., sensors that emit light such as infrared light and that measure corresponding reflected light from a user's head with an infrared light sensor, capacitive proximity sensors, ultrasonic acoustic proximity sensors, etc.), switches and/or other force-sensing sensors that detect head pressure when a user's head is present, and/or other head presence sensors.

[2010] When device **11.1.3.2.3-10** is being worn and a user's head is present in region **11.1.3.2.3-68**, the nose of the user will be present under nose bridge portion NB of housing **11.1.3.2.3-12**. When optical assemblies **11.1.3.2.3-20** are moved towards each other so that assemblies **11.1.3.2.3-20** are spaced apart by an amount that matches or nearly matches the user's interpupillary distance, inner side surfaces **11.1.3.2.3-60** of support structures **11.1.3.2.3-38** in assemblies **11.1.3.2.3-20** will move toward opposing outer side surfaces **11.1.3.2.3-61** of the user's nose. With sufficient inward movement of assemblies **11.1.3.2.3-20**, surfaces **11.1.3.2.3-60** may contact and press against nose surfaces **11.1.3.2.3-61**. As a result, an outward force on assemblies **11.1.3.2.3-20** is created by nose surfaces **11.1.3.2.3-61**. To avoid discomfort that might arise if the user's nose is pressed against by more than a desired amount, device **11.1.3.2.3-10** may be provided with features to limit inward nose pressure (e.g., to limit inward force by assemblies **11.1.3.2.3-20**).

[2011] With an illustrative embodiment, whenever device **11.1.3.2.3-10** is mounted on the head of a user, motors **11.1.3.2.3-48** may only be permitted to move optical assemblies **11.1.3.2.3-20** away from each other and not towards each other. This ensures that surfaces **11.1.3.2.3-60** will never move towards each other while the user's nose is present, so that the user's nose will never be pressed excessively by moving surfaces **11.1.3.2.3-60**.

[2012] The operation of device **11.1.3.2.3-10** in this type of arrangement is illustrated in the flow chart of FIG. **11.1.3.2.3-2**.

[2013] During the operations of block **11.1.3.2.3-80**, device **11.1.3.2.3-10** may be powered up. Device **11.1.3.2.3-10** may, for example, be powered up in response to detection of a user button press on button **11.1.3.2.3-70**. During power up operations, a power supply supplies power (e.g., a power supply voltage) to the control circuits, sensors, displays, and other components **11.1.3.2.3-40** of device **11.1.3.2.3-10**.

[2014] During the operations of block **11.1.3.2.3-82**, motors **11.1.3.2.3-48**, in response to detection of the power-up condition (e.g., in response to detecting the presence of the power supply voltage), may move optical assemblies **11.1.3.2.3-20** away from each other. Gaze trackers **11.1.3.2.3-62** may, in response to detection of the power up condition (e.g., in response to the power supply voltage),

capture images of the user's eyes in eye boxes **11.1.3.2.3-36** and may use this information in determining a target separation between optical assemblies **11.1.3.2.3-20** (e.g., gaze trackers **11.1.3.2.3-62** may measure the user's interpupillary distance and/or other eye characteristics such as the user's eye relief and may use the user's measured interpupillary distance and/or other eye characteristics in establishing a target separation for optical assemblies **11.1.3.2.3-20**). During the operations of block **11.1.3.2.3-82**, motors **11.1.3.2.3-48** may move optical assemblies **11.1.3.2.3-20** apart until the target separation (target positions) for optical assemblies **11.1.3.2.3-20** is reached. [2015] During the operations of block **11.1.3.2.3-84**, after motors **11.1.3.2.3-48** have placed optical assemblies **11.1.3.2.3-20** into their desired positions, further movement of assemblies **11.1.3.2.3-20** may be halted and device **11.1.3.2.3-10** may be used to present images to eye boxes **11.1.3.2.3-36** for viewing by the user.

[2016] After the user has finished viewing content with device **11.1.3.2.3-10**, device **11.1.3.2.3-10** may be powered down for storage. In an illustrative scenario, a button press on button **11.1.3.2.3-70** or other input is used to instruct device **11.1.3.2.3-10** to shut down. Optical assemblies **11.1.3.2.3-20** may be maintained in their current position while device **11.1.3.2.3-10** is powered down or can be moved towards each other in preparation for subsequent outward movements (see, e.g., block **11.1.3.2.3-82**). When moving optical assemblies **11.1.3.2.3-20** towards each other (at block **11.1.3.2.3-86** or another time such as during initial power up operations), a head-presence sensor may be used to detect whether the user's head is present in region **11.1.3.2.3-68**. The head-presence sensor may, as an example, be used to confirm that the user's head is not present whenever optical assemblies **11.1.3.2.3-82** are being moved towards each other. For example, during the operations of block **11.1.3.2.3-86**, motors **11.1.3.2.3-48** may move optical assemblies **11.1.3.2.3-20** towards each other in response to the detected user power down command (e.g., the button press input on button **11.1.3.2.3-70**) provided that no head is being detected by the head-presence sensor.

[2017] With another illustrative embodiment, which is illustrated in the flow chart of FIG. **11.1.3.2.3-3**, motors **11.1.3.2.3-48** may be configured to only move optical assemblies toward each other when a user actively permits movement of optical assemblies **11.1.3.2.3-20**. As an example, a user input device such as button **11.1.3.2.3-70** may be mounted on housing **11.1.3.2.3-12** (e.g., on an external surface of housing **11.1.3.2.3-12M** or other external surface of housing **11.1.3.2.3-12**). Motors **11.1.3.2.3-48** may be configured to inhibit movement of assemblies **11.1.3.2.3-20** (e.g., motors **11.1.3.2.3-48** may remain stationary) whenever button **11.1.3.2.3-70** is not being pressed. In response to detecting the pressing and continual holding of button **11.1.3.2.3-70** by a user, motors **11.1.3.2.3-48** may move optical assemblies **11.1.3.2.3-20** to adjust the spacing between assemblies **11.1.3.2.3-20**. Motors **11.1.3.2.3-48** may, for example, move optical assemblies **11.1.3.2.3-20** from an initial configuration in which optical assemblies **11.1.3.2.3-20** are spaced apart by their maximum spacing or other wide spacing to a target configuration in which optical assemblies **11.1.3.2.3-20** are separated by a distance equal to or nearly equal to the measured interpupillary distance associated with the user's eyes (e.g., assemblies **11.1.3.2.3-20** may be moved toward their target positions as measured by gaze trackers **11.1.3.2.3-62**).

[2018] As shown in FIG. **11.1.3.2.3-3**, device **11.1.3.2.3-10** may be powered up during the operations of block **11.1.3.2.3-90**. Following power-up operations, when device **11.1.3.2.3-10** is being worn on the head of the user, gaze trackers **11.1.3.2.3-62** may optionally measure the user's interpupillary distance and/or other eye characteristics of the user's eyes. These measurements may establish a desired separation (e.g., target positions) for optical assemblies **11.1.3.2.3-20**.

[2019] During the operations of block **11.1.3.2.3-92**, a user has an opportunity to press button **11.1.3.2.3-70**. When no button press input is detected, motors **11.1.3.2.3-48** remain stationary, so that optical assemblies **11.1.3.2.3-20** do not move. When button press input is detected, motors **11.1.3.2.3-48** are allowed to move to adjust the separation between assemblies **11.1.3.2.3-20**. As an example, motors **11.1.3.2.3-48** may move assemblies **11.1.3.2.3-20** inwardly towards their target positions. During movement of assemblies **11.1.3.2.3-20**, motors **11.1.3.2.3-48** may be halted in

response to any detected user release of button **11.1.3.2.3-70**. In this way, the positions of assemblies **11.1.3.2.3-20** will be adjusted so long as button **11.1.3.2.3-70** is being pressed, but will stop in response to release of button **11.1.3.2.3-70** (e.g., when the user desires to prevent movement of assemblies **11.1.3.2.3-20** that could press against nose surfaces **11.1.3.2.3-62**). The user in this scenario remains in continuous control of assemblies **11.1.3.2.3-20**.

[2020] Once the desired positions of assemblies **11.1.3.2.3-20** have been reached, device **11.1.3.2.3-10** may be used to view images while being worn by the user. Motors **11.1.3.2.3-48** may stop automatically when the target positions measured by gaze trackers **11.1.3.2.3-62** are reached or motors **11.1.3.2.3-48** may stop when the user releases button **11.1.3.2.3-70**. If desired, the direction of movement of assemblies **11.1.3.2.3-20** may be controlled by providing device **11.1.3.2.3-10** with two of buttons **11.1.3.2.3-70** (e.g., an inward movement button and an outward movement button) or by providing a first button **11.1.3.2.3-70** to control movement (e.g., a go/stop button) and a second button (e.g., a slider with two positions) that is used to choose between inward and outward movement settings. Arrangements in which device **11.1.3.2.3-10** has non-button user input devices such as microphones for gather voice commands, touch screen displays, and/or other user input devices may also be used in controlling movement of motors **11.1.3.2.3-48**.

[2021] Clutches may be used to limit the amount of inward force that is applied by optical assemblies **11.1.3.2.3-20** when assemblies **11.1.3.2.3-20** are moved towards nose surfaces **11.1.3.2.3-61** by motors **11.1.3.2.3-48**.

[2022] In the example of FIG. **11.1.3.2.3-4**, a split nut arrangement is used for nut **11.1.3.2.3-46**, forming a mechanical clutch (sometimes referred to as a force limiter). As shown in FIG. **11.1.3.2.3-4**, nut **11.1.3.2.3-46** may be moved (parallel to the X axis) by rotating threaded shaft **11.1.3.2.3-44**. Nut **11.1.3.2.3-46** may have a split such as split **11.1.3.2.3-91**. When excess force is experienced (e.g., when assembly **11.1.3.2.3-20** starts to press against nose surface **11.1.3.2.3-61**), nut **11.1.3.2.3-46** will be forced apart in directions **11.1.3.2.3-93** and will slide over the threads on shaft **11.1.3.2.3-44** rather than being driven inwardly by the interaction between the threads on shaft **11.1.3.2.3-44** and the corresponding threads in nut **11.1.3.2.3-46**. In this way, excess force is released and the clutch provided by split **11.1.3.2.3-91** helps prevent more than a desired amount of force from being applied to the user's nose. The split nut configuration of FIG. **11.1.3.2.3-4** therefore serves as a release mechanism that limits pressure from assembly **11.1.3.2.3-20** on nose surface **11.1.3.2.3-61**.

[2023] Illustrative magnetic clutches are shown in FIGS. **11.1.3.2.3-5** and **11.1.3.2.3-6**. In the magnetic clutch arrangement of FIG. **11.1.3.2.3-5**, a magnet **11.1.3.2.3-95** is attached to nut **11.1.3.2.3-46** and a magnet **11.1.3.2.3-96**, which is attracted to magnet **11.1.3.2.3-95**, is attached to support **11.1.3.2.3-38**. Optional structures **11.1.3.2.3-98** (e.g., non-magnetic coatings and/or protruding structures that establish an air gap between magnets **11.1.3.2.3-95** and **11.1.3.2.3-96**) may be used to adjust the holding force established by the attraction between magnets **11.1.3.2.3-95** and **11.1.3.2.3-96**. During normal operations, magnets **11.1.3.2.3-95** and **11.1.3.2.3-96** are magnetically coupled to each other, so that optical assembly support **11.1.3.2.3-38** is moved back and forth along the length of shaft **11.1.3.2.3-44** (e.g., parallel to the X axis in the example of FIG. **11.1.3.2.3-5**), thereby adjusting the spacing between assemblies **11.1.3.2.3-20**. If, however, inward movement of assemblies **11.1.3.2.3-20** is resisted due to contact between surfaces **11.1.3.2.3-60** and nose surfaces **11.1.3.2.3-61**, the holding force of the magnetic clutch will be exceeded, magnets **11.1.3.2.3-95** and **11.1.3.2.3-96** will separate, and this decoupling of magnets **11.1.3.2.3-95** and **11.1.3.2.3-96** will prevent further inward movement of assemblies **11.1.3.2.3-20**.

[2024] In the example of FIG. **11.1.3.2.3-6**, two rows of magnets are provided that are configured so that their exposed poles oppose each other. With this type of arrangement, the poles of magnets **11.1.3.2.3-95'** attract the poles of magnets **11.1.3.2.3-96'**, so that during normal operations, nut **11.1.3.2.3-46** is magnetically coupled to support **11.1.3.2.3-38** and is able to move support **11.1.3.2.3-38** and thereby adjust the position of the optical assembly associated with support

11.1.3.2.3-38. In the event that surface **11.1.3.2.3-60** of support **11.1.3.2.3-38** presses against nose surface **11.1.3.2.3-61**, the magnetic holding ability of the magnetic clutch of FIG. **11.1.3.2.3-6** will be exceeded, magnets **11.1.3.2.3-95'** and **11.1.3.2.3-96'** will decouple and will slide past each other, and additional inward movement of assemblies **11.1.3.2.3-20** towards the nose of the user will be prevented.

[2025] FIG. **11.1.3.2.3-7** is a diagram showing how interlocking physical features may be used to implement a mechanical version of the magnetic clutch of FIG. **11.1.3.2.3-6**. In the example of FIG. **11.1.3.2.3-7**, pin **11.1.3.2.3-100** is coupled to nut **11.1.3.2.3-46** and protrusions **11.1.3.2.3-102** (sometimes referred to as snap features, locking structures, or a releasable pin lock), which are coupled to support **11.1.3.2.3-38**. During normal operations, pin **11.1.3.2.3-100** will not bend and will transfer force along the X direction from nut **11.1.3.2.3-46** to support **11.1.3.2.3-38** to move assembly **11.1.3.2.3-20**. If excess force is generated (e.g., when assembly **11.1.3.2.3-20** contacts nose surface **11.1.3.2.3-61**), pin **11.1.3.2.3-100** will bend and slip past protrusions **11.1.3.2.3-102** in the pin lock, thereby decoupling nut **11.1.3.2.3-46** from support **11.1.3.2.3-38** and preventing further movement of assemblies **11.1.3.2.3-20** towards the user's nose.

[2026] FIGS. **11.1.3.2.3-8**, **11.1.3.2.3-9**, and **11.1.3.2.3-10** show illustrative mechanical clutch designs that may be used to implement mechanical version of the magnetic clutch of FIG.

11.1.3.2.3-5. In the arrangement of FIG. **11.1.3.2.3-8**, pin **11.1.3.2.3-108** is coupled to nut **11.1.3.2.3-46** and interlocks with a releasable pin lock formed from snap **11.1.3.2.3-110** on support **11.1.3.2.3-38**. When excess force is applied, snap **11.1.3.2.3-110** deforms and pin **11.1.3.2.3-108** escapes from snap **11.1.3.2.3-110** to release the clutch. In the arrangement of FIG. **11.1.3.2.3-9**, the downward protrusion of snap **11.1.3.2.3-110** of FIG. **11.1.3.2.3-8** has been replaced with protrusion **11.1.3.2.3-112** to form a releasable pin lock. In the example of FIG. **11.1.3.2.3-10**, a releasable pin lock is formed from spring **11.1.3.2.3-114** and movable ball **11.1.3.2.3-116** instead of protrusion **11.1.3.2.3-112** of FIG. **11.1.3.2.3-9**.

[2027] As shown in FIG. **11.1.3.2.3-11**, force-sensitive components (sometimes referred to as switches or force sensors) may be used in detecting when more than a desired amount of force is being applied by nut **11.1.3.2.3-46** to support **11.1.3.2.3-38**. Motors **11.1.3.2.3-48** are configured to deactivate in response to detecting this high amount of force (e.g., a force exceeding a predetermined threshold), thereby preventing excess nose pressure from assemblies **11.1.3.2.3-20**. In the illustrative arrangement of FIG. **11.1.3.2.3-11**, a protruding structure such as pin **11.1.3.2.3-122** is attached to nut **11.1.3.2.3-46**. Pin **11.1.3.2.3-122** extends between first and second switches **11.1.3.2.3-120** (or other force sensors such as strain gauges, etc.). Each switch **11.1.3.2.3-120** in the example of FIG. **11.1.3.2.3-11** has a stationary switch body **11.1.3.2.3-124** within which a movable plunger **11.1.3.2.3-126** is mounted. As nut **11.1.3.2.3-46** is moved back and forth along the X axis by rotation of shaft **11.1.3.2.3-44** from motors **11.1.3.2.3-48**, pin **11.1.3.2.3-122** pushes against plungers **11.1.3.2.3-126** (e.g., the plunger in the right-hand switch **11.1.3.2.3-120** when pushing support **11.1.3.2.3-38** to the right and the plunger in the left-hand switch **11.1.3.2.3-120** when pushing support **11.1.3.2.3-38** to the left). Plungers **11.1.3.2.3-126** are not pressed significantly into bodies **11.1.3.2.3-124** so long as a threshold amount of force on the plungers **11.1.3.2.3-126** is not exceeded. The interaction between pin **11.1.3.2.3-122** and switches **11.1.3.2.3-120** therefore allows lateral force to be transferred from nut **11.1.3.2.3-46** to support **11.1.3.2.3-38** to slide support **11.1.3.2.3-38** along rails **11.1.3.2.3-22** parallel to the X axis. If, however, assembly **11.1.3.2.3-20** contacts nose surface **11.1.3.2.3-61**, support **11.1.3.2.3-38** will resist further movement and pin **11.1.3.2.3-122** will push against the plunger **11.1.3.2.3-126** that is in contact with pin **11.1.3.2.3-122** with more than the threshold amount of force. In response to detecting a change in the state of switch **11.1.3.2.3-120** due to this elevated force, motors **11.1.3.2.3-48** are halted, thereby preventing more than a desired amount of force from being applied to nose surface **11.1.3.2.3-61**.

[2028] In the example of FIG. **11.1.3.2.3-12**, a torque-sensitive switch mechanism (switch **11.1.3.2.3-130**, sometimes referred to as a torque sensor or torque-sensitive switch) is used to

couple rotating shaft portion **11.1.3.2.3-44-2** of shaft **11.1.3.2.3-44** to rotating shaft portion **11.1.3.2.3-44-1** of shaft **11.1.3.2.3-44**. Nut **11.1.3.2.3-46** may be coupled to portion **11.1.3.2.3-44-1** and may be used to move support **11.1.3.2.3-38** parallel to shaft **11.1.3.2.3-44**. Motor **11.1.3.2.3-48** may rotate shaft portion **11.1.3.2.3-44-2** to move nut **11.1.3.2.3-46**. Switch **11.1.3.2.3-130** may have a first switch portion **11.1.3.2.3-130-1** and a second switch portion **11.1.3.2.3-130-2**. When less than a threshold amount of torque is applied to portion **11.1.3.2.3-130-2** by shaft portion **11.1.3.2.3-44-2**, portion **11.1.3.2.3-130-2** transfers this applied torque to portion **11.1.3.2.3-130-1**, which, in turn, transfers this applied torque to portion **11.1.3.2.3-44-1**. The rotation of portion **11.1.3.2.3-44-1** moves nut **11.1.3.2.3-46** laterally to adjust the lateral position of support **11.1.3.2.3-38** and assembly **11.1.3.2.3-20**. In the event that support **11.1.3.2.3-38** contacts nose surface **11.1.3.2.3-61**, support **11.1.3.2.3-38** will resist further lateral motion by nut **11.1.3.2.3-46**. This will create a rise in torque that is detected by switch **11.1.3.2.3-130**. In response, motor **11.1.3.2.3-48** halts its motion. By stopping motors **11.1.3.2.3-48** in response to detecting an amount of torque exceeding a predetermined torque threshold, motors **11.1.3.2.3-48** can be prevented from applying more force than desired to nose surfaces **11.1.3.2.3-61**.

[2029] If desired, the load on motors **11.1.3.2.3-48** can be monitored electronically, so that motors **11.1.3.2.3-48** can be halted if more than a desired motor load is encountered. An illustrative motor control circuit is shown in FIG. **11.1.3.2.3-13**. As shown in FIG. **11.1.3.2.3-13**, motor **11.1.3.2.3-48** may be driven using a drive circuit such as driver **11.1.3.2.3-140**. Driver **11.1.3.2.3-140** may apply any suitable drive signal to motor **11.1.3.2.3-48** (e.g., a sinusoidal motor drive current, etc.). Current sensor **11.1.3.2.3-144** may measure the current I_{meas} that flows through motor **11.1.3.2.3-48**. Voltage sensor **11.1.3.2.3-142** may measure the voltage V_{meas} across the terminals of motor **11.1.3.2.3-48**. The counterelectromotive force (“back EMF”) that is produced when operating motor **11.1.3.2.3-48** is equal to $(I_{meas} \cdot R_m - V_{meas})$, where R_m is the known motor resistance associated with motor **11.1.3.2.3-48**. During operation of motor **11.1.3.2.3-48**, back EMF may be calculated in real time and the phase difference between I_{meas} and the back EMF may be monitored to determine the motor load being experienced by motor **11.1.3.2.3-48** (e.g., motor **11.1.3.2.3-48** may be determined to be unloaded when the measured phase difference is 90° , may be determined to be fully loaded when the measured phase difference is 0° , and may be determined to be experiencing an intermediate amount of load when the measured phase difference is between 0° and 90°). Motors **11.1.3.2.3-48** may be configured to halt operation in response to a determination that the motor load has exceeded a desired amount (e.g., a determination that the measured phase difference between I_{meas} and back EMF is less than a predetermined threshold phase shift value such as less than 85° or less than 40°).

[2030] If desired, motor **11.1.3.2.3-48** may be supplied with a rotatory encoder, as shown in the example of FIG. **11.1.3.2.3-14**. In this example, magnet **11.1.3.2.3-150** is attached to shaft **11.1.3.2.3-44** and magnetic sensor **11.1.3.2.3-152** is used to monitor the magnetic field produced by magnet **11.1.3.2.3-150**. Sensor **11.1.3.2.3-152** and magnet **11.1.3.2.3-152** therefore serve as an encoder that measures the rotation of shaft **11.1.3.2.3-44** and therefore the rotation of the rotor in motor **11.1.3.2.3-48**. With this arrangement, the phase shift between the angle of the rotor and the drive current (e.g., I_{meas}) may be used as a measure of motor load or (in an arrangement in which motor **11.1.3.2.3-48** is being controlled by establishing a phase of 0° between rotor angle and drive current) motor load may be determined by monitoring I_{meas} and comparing I_{meas} to a threshold (as examples). When motor load is not elevated (e.g., below a predetermined threshold), motors **11.1.3.2.3-48** may rotate normally to move nut **11.1.3.2.3-46** along shaft **11.1.3.2.3-44** and thereby move optical assemblies **11.1.3.2.3-20**. When optical assemblies **11.1.3.2.3-20** contact nose surfaces **11.1.3.2.3-61**, motor load will increase beyond a threshold amount. In response, motors **11.1.3.2.3-48** may be halted, thereby preventing excess pressure on nose surfaces **11.1.3.2.3-61**.

[2031] Another illustrative technique for electronically monitoring motor load involves the use of a linear encoder of the type shown in FIG. **11.1.3.2.3-15**. As shown in FIG. **11.1.3.2.3-15**, encoder

11.1.3.2.3-156 may include a strip of magnets **11.1.3.2.3-158** and a corresponding magnetic sensor **11.1.3.2.3-160**. Sensor **11.1.3.2.3-160** may be attached to nut **11.1.3.2.3-46**. Magnets **11.1.3.2.3-158** may be attached to a stationary support surface (e.g., a portion of housing **11.1.3.2.3-12**). When motor **11.1.3.2.3-48** rotates shaft **11.1.3.2.3-44**, nut **11.1.3.2.3-46** is moved parallel to shaft **11.1.3.2.3-44**. This causes sensor **11.1.3.2.3-160** to move past magnets **11.1.3.2.3-158** while measuring the magnetic fields from magnets **11.1.3.2.3-158**. By monitoring the measured changes in magnetic field strength due to movement of sensor **11.1.3.2.3-160** past magnets **11.1.3.2.3-158**, motors **11.1.3.2.3-48** can determine the speed of nut **11.1.3.2.3-46** and can determine when nut **11.1.3.2.3-46** has stopped and motors **11.1.3.2.3-48** have stalled (e.g., when applied current to motors **11.1.3.2.3-48** is failing to cause lateral motion in nuts **11.1.3.2.3-46**).

[2032] The graph of FIG. **11.1.3.2.3-16** illustrates how stall detection may be used in measuring the force experienced by assemblies **11.1.3.2.3-20** when motors **11.1.3.2.3-48** contact nose surfaces **11.1.3.2.3-61**. Solid line **11.1.3.2.3-160** represent an increasing amount of force F of the type that may be experienced by optical assemblies **11.1.3.2.3-20** (e.g., support **11.1.3.2.3-38** and nut **11.1.3.2.3-46**) as optical assemblies **11.1.3.2.3-20** come into contact with nose surfaces **11.1.3.2.3-61** and begin to feel resistance (back pressure) from nose surfaces **11.1.3.2.3-61**. Current I_m may be applied to motors **11.1.3.2.3-48** by driver **11.1.3.2.3-140** (FIG. **11.1.3.2.3-13**) in a pattern that follows dashed line **11.1.3.2.3-162**, which represents the amount of resulting motor force produced by current I_m .

[2033] In the example of FIG. **11.1.3.2.3-16**, motor **11.1.3.2.3-48** periodically stalls at stall points **11.1.3.2.3-170**. This is because the amount of forward force produced by the current I_m applied to the motor is equaled by the back pressure (reverse force F) produced as nose surface **11.1.3.2.3-61** resists further forward motion by optical assembly **11.1.3.2.3-20**. By using an encoder such as encoder **11.1.3.2.3-156** of FIG. **11.1.3.2.3-15**, the motion of nut **11.1.3.2.3-46** and therefore the movement of support **11.1.3.2.3-38** and optical assembly **11.1.3.2.3-20** can be monitored.

Whenever no motion is detected in response to a particular value of applied current I_m , it can be concluded that optical assembly **11.1.3.2.3-20** has halted (e.g., motor **11.1.3.2.3-48** has stalled at one of stall points **11.1.3.2.3-170**). Motor **11.1.3.2.3-48** may be a stepper motor that is controlled by application of sets of pulses (e.g., 10 pulses at a time). Following detection of a stall (e.g., when it is detected that motor **11.1.3.2.3-48** moved assembly **11.1.3.2.3-20** less than expected after a given set of current pulses was applied), the magnitude of applied current I_m may, as shown in FIG. **11.1.3.2.3-16**, initially increase (to ensure further movement) and then decrease until the force being produced by motor **11.1.3.2.3-48** matches the reverse force due to nose surface contact. Each time a stall point **11.1.3.2.3-170** is reached in this way, the amount of current I_m may be evaluated. With this approach, the magnitude of the current applied to motor **11.1.3.2.3-48** closely tracks the increase in force F due to nose contact so that the magnitude of the current can be used as a measure of pressure on nose surface **11.1.3.2.3-61**. Motors **11.1.3.2.3-48** may be configured to halt (e.g., no more current will be applied) in response to the value of I_m exceeding a predetermined threshold value.

[2034] The flow chart of FIG. **11.1.3.2.3-17** shows illustrative operations that may be involved in operating device **11.1.3.2.3-10** in an arrangement in which electrical monitoring of motor load is used to determine when to halt motor operation and thereby prevent more than a desired amount of force being applied by optical assemblies **11.1.3.2.3-20** on nose surfaces **11.1.3.2.3-61**.

[2035] During the operations of block **11.1.3.2.3-200**, device **11.1.3.2.3-10** may be powered up (e.g., in response to a detected button press or other activity).

[2036] Once device **11.1.3.2.3-10** has powered up, gaze trackers **11.1.3.2.3-62** may measure the separation between the user's eyes (user interpupillary distance) and other eye characteristics to determine target positions for optical assemblies **11.1.3.2.3-20**. Motors **11.1.3.2.3-48** may then rotate shafts **11.1.3.2.3-44** to move optical assemblies **11.1.3.2.3-20** towards the target positions (e.g., by moving assemblies **11.1.3.2.3-20** towards nose bridge portion NB of housing **11.1.3.2.3-**

12). During motor operation, motor load may be electrically monitored (e.g., using back EMF measurements, using encoder output, using measurements of applied current, and/or using other measurements of the types described in connection with FIGS. **11.1.3.2.3-13**, **11.1.3.2.3-14**, **11.1.3.2.3-15**, and **11.1.3.2.3-16**).

[2037] If optical assemblies **11.1.3.2.3-20** contact nose surfaces **11.1.3.2.3-61**, nose surfaces **11.1.3.2.3-61** will produce a force against optical assemblies **11.1.3.2.3-20** that tends to resist further movement. Motors **11.1.3.2.3-48** are configured to halt operation in response to detection of more than a desired amount of motor load (see, e.g., block **11.1.3.2.3-204**). At this point, device **11.1.3.2.3-10** may be operated normally and used in presenting images to the user's eyes in eye boxes **11.1.3.2.3-36**.

[2038] Following use of device **11.1.3.2.3-10**, device **11.1.3.2.3-10** can be powered down (see, e.g., the operations of block **11.1.3.2.3-206**). Device **11.1.3.2.3-10** may, as an example, be powered down in response to detection of a user button press or other activities.

11.1.3.3: Hard Stops

[2039] FIG. **11.1.3.3-1** illustrates a portion of an example of an HMD **11.1.3.3-100** including a frame **11.1.3.3-102** and an optical module **11.1.3.3-104** adjustably connected to the frame **11.1.3.3-102**. The HMD **11.1.3.3-100** can also include a stop bracket **11.1.3.3-106** coupled to the frame adjacent the optical module **11.1.3.3-104**. In at least one example, the stop bracket **11.1.3.3-106** can include a first hard-stop feature **11.1.3.3-108a** and a second hard stop feature **11.1.3.3-108b** extending/protruding therefrom to form a stop surface against which the optical module **11.1.3.3-104** can contact in cases where the optical module **11.1.3.3-104** travels or is adjusted too far laterally during IPD adjustments described elsewhere herein.

[2040] In particular, the optical module **11.1.3.3-104** can include a housing **11.1.3.3-110** configured to contact the stop bracket **11.1.3.3-106**, or the tabs **11.1.3.3-108a-b** thereof, in a drop event or, for example, when a user adjusts or causes the optical module **11.1.3.3-104** to move too far laterally during the IPD adjustment.

[2041] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3.3-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.3.3-1**.

[2042] FIG. **11.1.3.3-2** illustrates another view of an example of an HMD **11.1.3.3-200** including an optical module **11.1.3.3-204** having a housing **11.1.3.3-210** and a stop bracket **11.1.3.3-206**, including a tab **11.1.3.3-208**, secured to a frame **11.1.3.3-202**. The HMD **11.1.3.3-200** can also include an adjustment mechanism **11.1.3.3-203** for the optical module **11.1.3.3-204** of an HMD **11.1.3.3-200**. In at least one example, the adjustment mechanism **11.1.3.3-203** also includes a stop **11.1.3.3-218** positioned at or near a distal end of a guide-rod **11.1.3.3-212**. The guide rod **11.1.3.3-212** can be disposed at least partially within a follower **11.1.3.3-214** of the optical module **11.1.3.3-204** such that the optical module **11.1.3.3-204** can be translated laterally along the guide rod **11.1.3.3-212** during adjustment for user IPD.

[2043] In at least one example, the stop **11.1.3.3-218** can include a cavity or channel in which the distal end of the guide-rod **11.1.3.3-212** extends. However, in at least one example, no contact is made between the guide-rod **11.1.3.3-212** and the stop **11.1.3.3-218** such that the guide-rod **11.1.3.3-212** is cantilevered. The stop **11.1.3.3-218** can be configured to prevent the optical module **11.1.3.3-204** from travelling too far distally along the guide-rod **11.1.3.3-212** and beyond the distal end thereof during IPD adjustment. The stop **11.1.3.3-218** can also serve to prevent too much deformation of the cantilevered guide-rod **11.1.3.3-212** during unintended drop events or other forces that would cause a deformation of the guide-rod **11.1.3.3-212** and a movement of the distal

end of the guide-rod **11.1.3.3-212**. In at least one example, the distal end of the guide-rod **11.1.3.3-212** extends into or is at least partially surrounded by the stop **11.1.3.3-218** without making contact during normal use of the HMD **11.1.3.3-200**. If unintentionally moved or deformed elastically, the distal end or portion of the guide-rod **11.1.3.3-212** can make contact with the stop **11.1.3.3-218** such that the stop **11.1.3.3-218** prevents too much deformation of the guide-rod **11.1.3.3-212**, thus preventing plastic deformation and damage to the guide-rod **11.1.3.3-212**.

[2044] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3.3-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **11.1.3.3-2**.

11.1.3.4: Upper Biasing Members

[2045] FIG. **11.1.3.4-1** illustrates an example of an adjustment mechanism **11.1.3.4-102** of an HMD **11.1.3.3-100** including a follower **11.1.3.3-104** of an optical module slidably engaged with a guide-rod **11.1.3.4-106**. The adjustment mechanism **11.1.3.4-102** can also include an encoder **11.1.3.4-110** for position sensing as the optical module is adjusted/moved along the guide-rod **11.1.3.4-106** for IPD adjustments.

[2046] In at least one example, the adjustment mechanism **11.1.3.4-102** can also include a biasing element **11.1.3.4-108** biasing the guide-rod **11.1.3.4-106** within the follower **11.1.3.4-104** to reduce slack and play between the follower **11.1.3.3-104** and the guide-rod **11.1.3.4-106**, either during adjustments or when stationary. In at least one example, the biasing element **11.1.3.4-108** includes an elongate spring finger fixed to the follower **11.1.3.3-104** or other optical module component at fixation points **11.1.3.4-112** and a distal end **11.1.3.4-114** biased toward the guide-rod **11.1.3.3-106** and extending through an aperture **11.1.3.4-116** of the follower **11.1.3.4-104** to contact the guide-rod **11.1.3.4-106** through the aperture **11.1.3.4-116**.

[2047] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3.4-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.3.4-1**.

[2048] FIG. **11.1.3.4-2** illustrates a portion of another example of an adjustment mechanism **11.1.3.4-202** including a guide-rod **11.1.3.4-206** and a biasing member **11.1.3.4-208**. The follower is not illustrated in FIG. **11.1.3.4-2** for illustrative purposes but the biasing element **11.1.3.4-208** can be fixed to a follower at the fixation point **11.1.3.4-212**. The distal end **11.1.3.4-214** of the biasing element **11.1.3.4-208** can include a distal spring finger **11.1.3.4-218** or other feature biased toward and contacting the guide-rod in order to bias the guide-rod **11.1.3.4-206** within the follower (to which the biasing element **11.1.3.4-208** is fixed) to remove slack and play between the follower and the guide-rod **11.1.3.4-206**.

[2049] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.3.4-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **11.1.3.4-2**.

11.1.4: Lower Guide Rod System

11.1.4.1: Electronic Devices with Biased Guide Rails

[2050] An electronic device may have input-output devices for gathering input and providing output. These devices may include optical components such as cameras, displays, and lenses.

[2051] A top view of an illustrative electronic device is shown in FIG. **11.1.4.1-1**. Electronic device **11.1.4.1-10** of FIG. **11.1.4.1-1** may be a head-mounted device or other suitable device. As shown in FIG. **11.1.4.1-1**, device **11.1.4.1-10** may have a housing such as housing **11.1.4.1-12**. Housing **11.1.4.1-12**, which may sometimes be referred to as a housing wall, external housing, housing structures, enclosure, or case, may be formed from materials such as polymer, glass, metal, crystalline materials such as sapphire, ceramic, fabric, foam, wood, other materials, and/or combinations of these materials.

[2052] Device **11.1.4.1-10** may have any suitable shape. Housing **11.1.4.1-12** may, for example, be configured to form a head-mounted housing in the shape of a pair of goggles (e.g., goggles having optional head straps such as straps **11.1.4.1-12T**, a nose bridge portion in nose bridge region NB that is configured to fit over a user's nose and help support housing **11.1.4.1-12** on the user's nose, etc.) and/or other head-mounted structures. Housing **11.1.4.1-12** may separate interior region **11.1.4.1-26** from exterior region **11.1.4.1-28**. Housing **11.1.4.1-12** may include portions such as front portion (front wall) **11.1.4.1-12F** on front face F of device **11.1.4.1-10**, rear portion (rear wall) **11.1.4.1-12R** on opposing rear face R of device **11.1.4.1-10**, and sidewall portions such as sidewall portions **11.1.4.1-12W** on sides W that extend between front portion **11.1.4.1-12F** and rear portion **11.1.4.1-12R**, so that housing **11.1.4.1-12** encloses interior region **11.1.4.1-26**.

[2053] Electrical and optical components may be mounted within housing **11.1.4.1-12** (e.g., in interior region **11.1.4.1-26**). As an example, housing **11.1.4.1-12** may have optical components in interior region **11.1.4.1-26** such as displays **11.1.4.1-14** and lenses **11.1.4.1-38**. Displays **11.1.4.1-14** and lenses **11.1.4.1-38** may be mounted in optical modules **11.1.4.1-30** (sometimes referred to as lens barrels, display and lens support structures, etc.). Images from displays **11.1.4.1-14** may be viewable from eye boxes **11.1.4.1-36** through lenses **11.1.4.1-38**. A left display and left lens in a left optical module **11.1.4.1-30** may be used to present a left-eye image to a user's left eye in a left eye box **11.1.4.1-36** and a right display and right lens in a right optical module **11.1.4.1-30** may be used to present a right-eye image to a user's right eye in right eye box **11.1.4.1-36**. Manual adjustment mechanisms and/or electrically adjustable actuators **11.1.4.1-13** (e.g., motors or other electrically adjustable positioners) may be used to move optical modules **11.1.4.1-30** horizontally across the front of the user's face (e.g., to adjust distance D between modules **11.1.4.1-30** along a direction parallel to the X-axis or nearly parallel to the X-axis of FIG. **11.1.4.1-1**). Optical modules **11.1.4.1-30** may, for example, be moved closer to each other or farther apart from each other as needed to accommodate different user interpupillary distances.

[2054] A schematic diagram of an illustrative electronic device is shown in FIG. **11.1.4.1-2**. Device **11.1.4.1-10** of FIG. **11.1.4.1-2** may be operated as a stand-alone device and/or the resources of device **11.1.4.1-10** may be used to communicate with external electronic equipment. As an example, communications circuitry in device **11.1.4.1-10** may be used to transmit user input information, sensor information, and/or other information to external electronic devices (e.g., wirelessly or via wired connections) and/or may be used to receive such information from external electronic devices. Each of these external devices may include components of the type shown by device **11.1.4.1-10** of FIG. **11.1.4.1-2**.

[2055] As shown in FIG. **11.1.4.1-2**, electronic device **11.1.4.1-10** may include control circuitry **11.1.4.1-20**. Control circuitry **11.1.4.1-20** may include storage and processing circuitry for supporting the operation of device **11.1.4.1-10**. The storage and processing circuitry may include storage such as nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **11.1.4.1-20** may be used to gather input from sensors (e.g., cameras) and other input devices and may be used to control output

devices. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors and other wireless communications circuits, power management units, audio chips, application specific integrated circuits, etc. During operation, control circuitry **11.1.4.1-20** may use display(s) **11.1.4.1-14** and other output devices in providing a user with visual output and other output.

[2056] To support communications between device **11.1.4.1-10** and external equipment, control circuitry **11.1.4.1-20** may communicate using communications circuitry **11.1.4.1-22**. Circuitry **11.1.4.1-22** may include antennas, radio-frequency transceiver circuitry, and other wireless communications circuitry and/or wired communications circuitry. Circuitry **11.1.4.1-22**, which may sometimes be referred to as control circuitry and/or control and communications circuitry, may support bidirectional wireless communications between device **11.1.4.1-10** and external equipment (e.g., a companion device such as a computer, cellular telephone, or other electronic device, an accessory such as a point device, computer stylus, or other input device, speakers or other output devices, etc.) over a wireless link. For example, circuitry **11.1.4.1-22** may include radio-frequency transceiver circuitry such as wireless local area network transceiver circuitry configured to support communications over a wireless local area network link, near-field communications transceiver circuitry configured to support communications over a near-field communications link, cellular telephone transceiver circuitry configured to support communications over a cellular telephone link, or transceiver circuitry configured to support communications over any other suitable wired or wireless communications link. Wireless communications may, for example, be supported over a Bluetooth® link, a WiFi® link, a wireless link operating at a frequency between 10 GHz and 400 GHz, a 60 GHz link, or other millimeter wave link, a cellular telephone link, or other wireless communications link. Device **11.1.4.1-10** may, if desired, include power circuits for transmitting and/or receiving wired and/or wireless power and may include batteries or other energy storage devices. For example, device **11.1.4.1-10** may include a coil and rectifier to receive wireless power that is provided to circuitry in device **11.1.4.1-10**.

[2057] Device **11.1.4.1-10** may include input-output devices such as devices **11.1.4.1-24**. Electronic components such as input-output devices **11.1.4.1-24** may be used in gathering user input, in gathering information on the environment surrounding the user, and/or in providing a user with output.

[2058] Devices **11.1.4.1-24** may include one or more displays such as display(s) **11.1.4.1-14**. Display(s) **11.1.4.1-14** may include one or more display devices such as organic light-emitting diode display panels (panels with organic light-emitting diode pixels formed on polymer substrates or silicon substrates that contain pixel control circuitry), liquid crystal display panels, microelectromechanical systems displays (e.g., two-dimensional mirror arrays or scanning mirror display devices), display panels having pixel arrays formed from crystalline semiconductor light-emitting diode dies (sometimes referred to as microLEDs), and/or other display devices.

[2059] Devices **11.1.4.1-24** may also include cameras **11.1.4.1-34**. Cameras **11.1.4.1-34** may include visible light cameras, infrared cameras, and/or cameras that are sensitive at multiple wavelengths, may include three-dimensional camera systems such as depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices that capture three-dimensional images), may include time-of-flight cameras, and/or may include other cameras. Cameras **11.1.4.1-34** may face toward the user of device **11.1.4.1-10** and/or away from the user of device **11.1.4.1-10**.

[2060] Sensors **11.1.4.1-16** in input-output devices **11.1.4.1-24** may include force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors such as a touch sensor that forms a button, trackpad, or other input device), and other sensors. If desired, sensors **11.1.4.1-16** may include optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, optical touch sensors, optical proximity sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, fingerprint sensors, iris scanning sensors,

retinal scanning sensors, and other biometric sensors, temperature sensors, sensors for measuring three-dimensional non-contact gestures (“air gestures”), pressure sensors, sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or inertial measurement units that contain some or all of these sensors), health sensors such as blood oxygen sensors, heart rate sensors, blood flow sensors, and/or other health sensors, radio-frequency sensors, optical sensors such as self-mixing sensors and light detection and ranging (LIDAR) sensors, humidity sensors, moisture sensors, gaze tracking sensors, electromyography sensors to sense muscle activation, facial sensors, interferometric sensors, time-of-flight sensors, magnetic sensors, resistive sensors, distance sensors, angle sensors, and/or other sensors. In some arrangements, device **11.1.4.1-10** may use sensors **11.1.4.1-16** and/or other input-output devices **11.1.4.1-24** to gather user input. For example, input-output devices **11.1.4.1-24** such as buttons may be used to gather button press input, touch sensors overlapping displays can be used for gathering user touch screen input, touch pads may be used in gathering touch input, microphones may be used for gathering audio input (e.g., voice commands), accelerometers may be used in monitoring when a finger contacts an input surface and may therefore be used to gather finger press input, etc.

[2061] Input-output devices **11.1.4.1-24** may include optical components such as depth sensors (e.g., structured light sensors or other sensors that gather three-dimensional image data), optical proximity sensors, ambient light sensors (e.g., color ambient light sensors), optical time-of-flight sensors and other sensors **11.1.4.1-16** that are sensitive to visible and/or infrared light and that may emit visible and/or infrared light (e.g., devices **11.1.4.1-24** may contain optical sensors that emit and/or detect light). For example, a visible-light image sensor in a camera may have a visible light flash or an associated infrared flood illuminator to provide illumination while the image sensor captures a two-dimensional and/or three-dimensional image. An infrared camera such as an infrared structured light camera that captures three-dimensional infrared images may have an infrared flood illuminator that emits infrared flood illumination and/or may have a dot projector the emits an array of infrared light beams. Infrared proximity sensors may emit infrared light and detect the infrared light after the infrared light has reflected from a target object.

[2062] If desired, electronic device **11.1.4.1-10** may include additional components (see, e.g., other devices **11.1.4.1-18** in input-output devices **11.1.4.1-24**). The additional components may include haptic output devices, actuators for moving movable structures in device **11.1.4.1-10**, audio output devices such as speakers, light-emitting diodes for status indicators, light sources such as light-emitting diodes that illuminate portions of a housing and/or display structure, other optical output devices, and/or other circuitry for gathering input and/or providing output. Device **11.1.4.1-10** may also include a battery or other energy storage device, connector ports for supporting wired communication with ancillary equipment and for receiving wired power, and other circuitry.

[2063] To help maintain desired alignment between optical modules **11.1.4.1-30** as optical modules **11.1.4.1-30** are moved by actuators **11.1.4.1-13** (FIG. **11.1.4.1-1**), optical modules **11.1.4.1-30** may be mounted on optical module guiding structures such as guide rails or other elongated support members. This type of arrangement is shown in the top view of device **11.1.4.1-10** of FIG. **11.1.4.1-3**. As shown in FIG. **11.1.4.1-3**, optical modules **11.1.4.1-30** may be slidably coupled to guide rails **11.1.4.1-44** to allow modules **11.1.4.1-30** to move horizontally (e.g., laterally along the X-axis to accommodate different user interpupillary distances).

[2064] Guide rails **11.1.4.1-44** may have circular cross-sectional shapes (e.g., guide rails **11.1.4.1-44** may be cylindrical rods) or may have other cross-sectional shapes. Guide rods **11.1.4.1-44** may be formed from metal, polymer, and/or other materials. Hollow and/or solid members may be used in forming guide rods **11.1.4.1-44**. To help reduce friction between guide rods **11.1.4.1-44** and optical modules **11.1.4.1-30**, guide rods **11.1.4.1-44** and/or mating portions of modules **11.1.4.1-30** may, if desired, be provided with a low-friction coating (e.g., nickel, etc.).

[2065] Guide rails **11.1.4.1-44** may span the width of housing **11.1.4.1-12**. There may be left and

right guide rails **11.1.4.1-44** in device **11.1.4.1-10** that are joined at a housing support structure aligned with nose bridge portion NB or left and right guide rails **11.1.4.1-44** may be formed as integral portions of a single guide rail member that extends across housing **11.1.4.1-12**. Rails **11.1.4.1-44** may be straight or may, if desired, have a slight bend at nose bridge portion NB (e.g., to rake the left and right sides of the guide rails backwards slightly to conform to the shape of a user's face). As shown in the rear view of device **11.1.4.1-10** of FIG. **11.1.4.1-4**, there may be upper and lower guide rails **11.1.4.1-44** on the left and right of device **11.1.4.1-10** such as upper guide rail **11.1.4.1-44T** and lower guide rail **11.1.4.1-44B**. Arrangements with fewer guide rails or more guide rails may be used, if desired.

[2066] FIG. **11.1.4.1-5** is a side view of an illustrative optical module **11.1.4.1-30** mounted on guide rails **11.1.4.1-44**. In the example of FIG. **11.1.4.1-5**, optical module **11.1.4.1-30** has an upper portion such as portion **11.1.4.1-30T** and a lower portion such as portion **11.1.4.1-30B**. Portions **11.1.4.1-30T** and/or **11.1.4.1-30B** may be integrally formed with main supporting structure **11.1.4.1-30M** of the lens barrel structures and/or other support structures of optical module **11.1.4.1-30** and/or may be separate members that are coupled (e.g., using welds, fasteners, adhesive, etc.) to main supporting structure **11.1.4.1-30M**. Lens **11.1.4.1-38** may be aligned with display **11.1.4.1-14** so that an image on display **11.1.4.1-14** may be viewed through lens **11.1.4.1-38** from eye box **11.1.4.1-36**.

[2067] As shown in FIG. **11.1.4.1-5**, optical module **11.1.4.1-30** may have portions that receive and couple to guide rails **11.1.4.1-44** while allowing optical module **11.1.4.1-30** to slide along guide rails **11.1.4.1-44**. For example, upper portion **11.1.4.1-30T** may have a guide rail opening (optical module opening) **11.1.4.1-50** such as opening **11.1.4.1-50T** that receives upper guide rail **11.1.4.1-44T** and lower portion **11.1.4.1-30B** may have a guide rail opening such as opening **11.1.4.1-50B** that receives lower guide rail **11.1.4.1-44B**. Openings **11.1.4.1-50T** and **11.1.4.1-50B** may be cylindrical openings with circular cross-sectional shapes that receive the cylindrical members forming rails **11.1.4.1-44T** and **11.1.4.1-44B** and/or may have other shapes that partly or fully surround rails **11.1.4.1-44T** and **11.1.4.1-44B**.

[2068] To prevent rails **11.1.4.1-44** from becoming stuck in guide rail openings **11.1.4.1-50** of optical module **11.1.4.1-30**, the inner diameter of optical module openings may be slightly larger (e.g., by 2-50 microns, at least 5 microns, less than 100 microns, or other suitable amount) than the outer diameter of rails **11.1.4.1-44**. To prevent excess motion, which could lead to misalignment of optical modules, device **11.1.4.1-10** may be provided with guide rail biasing systems. The guide rail biasing systems may have movable biasing members (e.g., pins, plates, spherical members, hemispherical members, etc.) and biasing elements that apply force to the biasing members. The biasing elements may be, for example, springs such as coil springs, leaf springs, and/or other spring members, may be compressed polymer (e.g., elastomeric material, foam, etc.), may be magnets, and/or may be other biasing components that can be used to bias the biasing members in a desired direction.

[2069] Using the guide rail biasing systems, guide rails **11.1.4.1-44** may be pushed towards desired positions within openings **11.1.4.1-50** to help remove undesired play between guide rails **11.1.4.1-44** and openings **11.1.4.1-50**. As an example, guide rails **11.1.4.1-44** may be pressed against the inner surfaces of openings **11.1.4.1-50** (e.g., at a location on the side of openings **11.1.4.1-50** that faces eye boxes **11.1.4.1-36** rather than the opposing side of openings **11.1.4.1-50** that faces outwardly away from the user) and/or may be pressed against a structure mounted within openings **11.1.4.1-50** such as a pin or other support member.

[2070] Consider, as an example, the cross-sectional side view of optical module portion **11.1.4.1-30B** of FIG. **11.1.4.1-6A**. As shown in FIG. **11.1.4.1-6A**, portion **11.1.4.1-30B** may have an opening such as opening **11.1.4.1-50B** that receives lower guide rail **11.1.4.1-44B**. Lower guide rail biasing system **11.1.4.1-52B** may have a lower guide rail biasing element such as lower guide rail biasing element **11.1.4.1-54B** (e.g., a spring) and a corresponding movable biasing system member

such as biasing member **11.1.4.1-56B** (e.g., a pin with a hemispherical head) and/or system **11.1.4.1-52B** may be formed from other biasing structures (e.g., a leaf spring, compressed foam, etc.). Under pressure from biasing element **11.1.4.1-54B**, member **11.1.4.1-56B** may press in direction **11.1.4.1-58** against an adjacent surface of guide rail **11.1.4.1-44B** (e.g., the side of rail **11.1.4.1-44B** facing biasing system **11.1.4.1-52B** and facing away from the user). Pin **11.1.4.1-60** may be mounted on an opposing side of opening **11.1.4.1-50B**. As a result of the force imposed on guide rail **11.1.4.1-44B** in direction **11.1.4.1-58**, guide rail **11.1.4.1-44B** may bear against member **11.1.4.1-56** (or, if desired, the inner surface of opening **11.1.4.1-50B**) at contact location **11.1.4.1-62** (e.g., a location on the surface of opening **11.1.4.1-50B** facing the user and eye boxes). This creates a well-defined location for guide rail **11.1.4.1-44B** relative to the structures of optical module **11.1.4.1-30** and helps prevent rail **11.1.4.1-44B** from moving excessively in the Y and/or Z dimensions within opening **11.1.4.1-50B**, thereby helping to ensure that optical module **11.1.4.1-30** is aligned satisfactorily with respect to eye box **11.1.4.1-30**. Additional friction and resistance to sliding of optical module **11.1.4.1-30** along the X axis is created by the use of biasing systems such as biasing system **11.1.4.1-52B** of FIG. **11.1.4.1-6A**, but when it is desired to move optical modules **11.1.4.1-30** with respect to each other to adjust their spacing to accommodate a user interpupillary distance, sufficient force along the X dimension can be applied to overcome this friction.

[2071] If desired, spring **11.1.4.1-54B** and biasing element **11.1.4.1-56B** may be located on opposite sides of guide rail **11.1.4.1-44B**. Consider, as an example, the illustrative configuration of biasing system **11.1.4.1-52B** that is shown in FIG. **11.1.4.1-6B**. In this configuration, biasing system **11.1.4.1-52B** has a biasing member **11.1.4.1-57** with a portion that forms biasing element **11.1.4.1-56B**. Biasing member **11.1.4.1-57** may be attached to portion **11.1.4.1-30B** of optical module **11.1.4.1-30** using an attachment mechanism such as screw **11.1.4.1-67** or other fastener. Screw **11.1.4.1-67** may have a threaded shaft or other structure that is fixedly attached to portion **11.1.4.1-30B**. The shaft of screw **11.1.4.1-67** may be received within a slot in member **11.1.4.1-57**. The slot may extend parallel to the Y axis of FIG. **11.1.4.1-6B** to allow biasing member **11.1.4.1-57** to slide back and forth parallel to the Y axis.

[2072] A biasing element such as spring **11.1.4.1-54B** (e.g., a compression spring or other biasing spring) may be used to bias member **11.1.4.1-57** in the -Y direction. Spring **11.1.4.1-54B** may have a first end that presses against a surface of member **11.1.4.1-57** such as surface **11.1.4.1-61** and an opposing second end that presses against a surface of portion **11.1.4.1-30B** such as surface **11.1.4.1-63**. When member **11.1.4.1-57** is moved in direction **11.1.4.1-65**, spring **11.1.4.1-59** is compressed. Spring **11.1.4.1-59** thereafter presses against surface **11.1.4.1-61** and biases the portion of member **11.1.4.1-57** that forms element **11.1.4.1-54B** in direction **11.1.4.1-58** relative to portion **11.1.4.1-30B**. This causes member **11.1.4.1-57** to contact rail **11.1.4.1-44B** at contact location **11.1.4.1-69** and to thereby bias the opposing side of rail **11.1.4.1-44B** against pin **11.1.4.1-60** at contact location **11.1.4.1-62**.

[2073] Member **11.1.4.1-57** may be formed from one or more materials. For example, member **11.1.4.1-57** may be formed from a metal (e.g., aluminum, stainless steel, etc.). The metal may be covered with a hard low-friction coating such as an electroless nickel coating to enhance wear resistance. As another example, member **11.1.4.1-57** may be formed from a polymer (e.g., a low-friction composite polymer filled with carbon fibers, fiberglass fibers, or other fibers). The low-friction composite polymer may be formed from a polymer such as PEEK (polyether ether ketone) or other polymer (e.g., a polymer that may be shaped by a molding process such as injection molding).

[2074] The examples of FIG. **11.1.4.1-6A** and **11.1.4.1-6B** demonstrate how lower guide rail **11.1.4.1-44B** may be provided with an associated guide rail biasing system (illustrative system **11.1.4.1-52B**). If desired, upper guide rails **11.1.4.1-44T** may also be provided with guide rail biasing systems. As shown in the cross-sectional side view of upper portion **11.1.4.1-30T** of optical module **11.1.4.1-30**, optical module **11.1.4.1-30** may have biasing systems **11.1.4.1-52T-1** and

11.1.4.1-52T-2. System **11.1.4.1-52T-1** may have biasing element **11.1.4.1-54T-1** for forcing movable biasing member **11.1.4.1-56T-1** against an adjacent surface of guide rail **11.1.4.1-44T** in direction **11.1.4.1-66** (or may have other biasing structures such as a leaf spring, compressed foam, etc.). System **11.1.4.1-52T-2** may have biasing element **11.1.4.1-54T-2** for forcing movable biasing member **11.1.4.1-56T-2** against an adjacent surface of guide rail **11.1.4.1-44** (or may have other biasing structures such as a leaf spring, compressed foam, etc.). Using systems **11.1.4.1-52T-1** and **11.1.4.1-52T-2**, guide rail **11.1.4.1-44T** may be biased leftward so that the surface of guide rail **11.1.4.1-44T** contacts the inner surface of opening **11.1.4.1-50T** at nominal biasing location **11.1.4.1-68** (e.g., a location on the surface of opening **11.1.4.1-50T** that faces the user and the eye boxes associated with device **11.1.4.1-10**). The force of gravity tends to pull downwardly (in the $-Z$ direction of FIG. **11.1.4.1-7**) on guide rail **11.1.4.1-44T**. To compensate for the force of gravity and thereby ensure that location **11.1.4.1-68** is located on the left side of guide rail **11.1.4.1-44T** (facing the user and eye box **11.1.4.1-36**) as shown in FIG. **11.1.4.1-7**, biasing element **11.1.4.1-54T-1** may be stronger than biasing element **11.1.4.1-54T-2**.

[2075] During operation of device **11.1.4.1-10**, a user's face and/or other external objects may impose forces on optical modules **11.1.4.1-30**. This gives rise to a potential for three different types of misalignment between the left and right optical modules in device **11.1.4.1-10**.

[2076] Rotation of optical modules **11.1.4.1-30** about the X axis, which may sometimes be referred to as splay, may cause a first type of misalignment in which one image appears at a different height than the other image. For example, splay may rotate the left eye image from the left optical module up relative to the left eye box while rotating the right eye image from the right optical module down relative to the right eye box.

[2077] Rotation of optical modules **11.1.4.1-30** about the Y axis, which may sometimes be referred to as image rotation, may produce a second type of misalignment in which the left and/or right images from modules **11.1.4.1-30** to appear tilted relative to the horizon. For example, the left eye image may be rotated counterclockwise and the right eye image may be rotated clockwise.

[2078] Another type of misalignment that may arise between optical modules **11.1.4.1-30** relates to rotation of one or both of optical modules **11.1.4.1-30** about the Z axis. This type of misalignment, which may sometimes be referred to as vergence, may be characterized by situations in which the left and/or right optical module points too far to the left or right in the X-Y plane.

[2079] All of these types of misalignment are preferably maintained at low levels (e.g., below $+/-0.5^\circ$, below $+/-0.4^\circ$, below $+/-0.3^\circ$, or below $+/-0.2^\circ$, as examples). In some situations, splay is the most undesired type of misalignment for users, so minimization of splay may be particular helpful in enhancing user comfort.

[2080] The use of guide rail biasing systems such as the illustrative biasing systems of FIGS. **11.1.4.1-6A**, **11.1.4.1-6B**, and **11.1.4.1-7** may help minimize optical module misalignment. When biasing systems **11.1.4.1-52T-1** and **11.1.4.1-52T-2** perform satisfactorily, the left optical module will be biased against upper rail **11.1.4.1-44T** at a location such as location **11.1.4.1-68** of FIG. **11.1.4.1-7** and will be biased against lower rail **11.1.4.1-44B** at a location such as location **11.1.4.1-62** of FIG. **11.1.4.1-6A** or FIG. **11.1.4.1-6B**, whereas the right optical module will likewise be biased against upper rail **11.1.4.1-44T** at a location such as location **11.1.4.1-68** of FIG. **11.1.4.1-7** and will be biased against lower rail **11.1.4.1-44B** at a location such as location **11.1.4.1-62** of FIG. **11.1.4.1-6A** or FIG. **11.1.4.1-6B**. In this case, both the left and right optical modules will have the same position relative to guide rails **11.1.4.1-44T** and **11.1.4.1-44B** and there will be no splay. A potential for splay may arise when stress from a drop event or other unexpected excessive force causes a biasing system to fail.

[2081] Consider, as an example, a worse case splay scenario in which biasing system **11.1.4.1-52T-2** in the left optical module fails due to excessive force and in which biasing system **11.1.4.1-52T-1** in the right optical module fails due to excessive force. Although this type of asymmetric failure changes the biasing locations of the left and right optical modules, the biasing arrangement of FIG.

11.1.4.1-7 helps prevent undesired splay from arising. Failure of biasing system **11.1.4.1-52T-2** in the left optical module will cause guide rail **11.1.4.1-44T** in the left optical module to be positioned against the inner surface of opening **11.1.4.1-50T** at location **11.1.4.1-72**, whereas failure of biasing system **11.1.4.1-52T-1** in the right optical module will cause guide rail **11.1.4.1-44T** in the right optical module to be positioned against the inner surface of opening **11.1.4.1-50T** at a different location such as location **11.1.4.1-70**.

[2082] In the example of FIG. **11.1.4.1-7**, system **11.1.4.1-52T-2** is located at an angle $A=+45^\circ$ relative to the Y axis and system **11.1.4.1-52T-1** is located at an angle $A=-45^\circ$ relative to the Y axis. When systems **11.1.4.1-52T-2** and **11.1.4.1-52T-1** are operating satisfactorily, guide rail **11.1.4.1-44T** will therefore press against the inner side wall of opening **11.1.4.1-50T** at location **11.1.4.1-68**. When opening **11.1.4.1-50T** in the left optical module contacts rail **11.1.4.1-44T** at location **11.1.4.1-72** while opening **11.1.4.1-50T** in the right optical module contacts rail **11.1.4.1-44T** at location **11.1.4.1-70**, both the left and right optical modules will be positioned more to the right (in the orientation of FIG. **11.1.4.1-7**) than when in their desired nominal biased position. Lower guide rod **11.1.4.1-44B** is biased satisfactorily at location **11.1.4.1-62** with respect to lower optical module portion **11.1.4.1-30B** for both the left and right optical modules (in this example). As a result, both the left and right optical modules will tip forward slightly (rotating a small amount about the X axis), under outward pressure from the user's face. Although both the left and right optical modules tilt forward in this way, the amount of tilt of the left optical module, which is dictated by the lateral displacement of rail **11.1.4.1-44T** experienced when rail **11.1.4.1-44T** contacts opening **11.1.4.1-50T** at position **11.1.4.1-72**, is equal to the amount of tilt of the right optical module, which is dictated by the lateral displacement of rail **11.1.4.1-44T** experienced when rail **11.1.4.1-44T** contacts opening **11.1.4.1-50T** at position **11.1.4.1-70**. No splay will therefore result. Because the configuration of the biasing systems of FIG. **11.1.4.1-7** tends to prevent splay from arising, this configuration may sometimes be referred to as a splay-optimized or splay-immune biasing scheme. Other biasing schemes may be used, if desired (e.g., schemes for the upper guide rail that have more biasing systems per optical module, schemes with fewer biasing systems per optical module, schemes in which the biasing systems press in different directions such as the +Z direction, -Y direction, etc.). The configuration of FIG. **11.1.4.1-7** is illustrative.

[2083] FIGS. **11.1.4.1-8** and **11.1.4.1-9** show how a kinematic mounting scheme may be used to couple guide rails **11.1.4.1-44** and optical modules **11.1.4.1-30**. As shown in FIG. **11.1.4.1-8**, upper guide rails **11.1.4.1-44T** may be biased in the -Y direction by biasing system **11.1.4.1-70**. Biasing system **11.1.4.1-70** may include biasing element **11.1.4.1-72** and movable biasing member **11.1.4.1-74** that contacts an adjacent portion of guide rail **11.1.4.1-44T** and pushes guide rail **11.1.4.1-44T** in direction **11.1.4.1-78** or may include other biasing structures (e.g., a leaf spring, compressed foam, etc.). Biasing system **11.1.4.1-80** may have a biasing element such as biasing element **11.1.4.1-82** that pushes biasing member **11.1.4.1-84** (e.g., a cylindrical pin) in direction **11.1.4.1-86** until motion of member **11.1.4.1-84** is stopped by stop structures **11.1.4.1-88**. Biasing system **11.1.4.1-90** may have a biasing element such as biasing element **11.1.4.1-92** that pushes biasing member **11.1.4.1-94** in direction **11.1.4.1-96** until motion of member **11.1.4.1-94** is stopped by stop structures **11.1.4.1-88**. In this configuration, guide rail **11.1.4.1-44T** is biased into contact with member **11.1.4.1-84** at known location **11.1.4.1-100** and into contact with member **11.1.4.1-94** at known location **11.1.4.1-102**. Lower guide rail **11.1.4.1-44B** may be biased to known location **11.1.4.1-62** using a biasing system such as system **11.1.4.1-52B** of FIG. **11.1.4.1-6A** or FIG. **11.1.4.1-6B**.

[2084] FIG. **11.1.4.1-9** is a rear view of optical module **11.1.4.1-30** with this type of kinematic rail mounting arrangement. As shown in FIG. **11.1.4.1-9**, there may be two of biasing systems **11.1.4.1-80** in portion **11.1.4.1-30T** of module **11.1.4.1-30** and two of biasing systems **11.1.4.1-90** in portion **11.1.4.1-30T** of module **11.1.4.1-30** in addition to two of biasing systems **11.1.4.1-70**. This establishes a first pair of known contact locations towards the right end of rail **11.1.4.1-44T** (e.g., a

first location **11.1.4.1-100** and a first location **11.1.4.1-102**) and, establishes a second pair of known contact locations at a different location along the length of rail **11.1.4.1-44T** such as towards the left end of rail **11.1.4.1-44T** (e.g., a second location **11.1.4.1-100** and a second location **11.1.4.1-102**). In lower portion **11.1.4.1-30B**, the location of optical module relative to guide rail **11.1.4.1-44B** is established at known location **11.1.4.1-62**. Determining these five known locations on optical module **11.1.4.1-30** where rails **11.1.4.1-44** contact module **11.1.4.1-30** (e.g., two locations **11.1.4.1-100** in upper portion **11.1.4.1-30T**, two locations **11.1.4.1-102** in upper portion **11.1.4.1-30T**, and one location **11.1.4.1-62** in lower portion **11.1.4.1-30B**), helps constrain five of the six degrees of freedom for motion of module **11.1.4.1-30** relative to rails **11.1.4.1-44** and the other support structures of device **11.1.4.1-10**. The sixth degree of freedom (motion along the X axis) is not constrained except by operation of actuator **11.1.4.1-13**, so that actuator **11.1.4.1-13** may be used to adjust the X-axis location of module **11.1.4.1-30** to accommodate different user interpupillary distances.

[2085] If desired, control circuitry **11.1.4.1-20** may use one or more sensors **11.1.4.1-16** to monitor the location of optical modules **11.1.4.1-30**. If misalignment is detected, corresponding action can be taken. For example, positioners may be adjusted to correct for the misalignment, image data for a display and/or camera may be warped to compensate for misalignment, alerts may be provided to the user and/or others, etc.

[2086] An illustrative optical module sensor system is shown in FIGS. **11.1.4.1-10** and **11.1.4.1-11**. FIG. **11.1.4.1-10** is a perspective view of an illustrative guide rail position sensor. Sensor **11.1.4.1-120** of FIG. **11.1.4.1-10** has a biasing member such as leaf spring member **11.1.4.1-122** with a protruding portion such as portion **11.1.4.1-124**. Strain gauge **11.1.4.1-126** may be coupled to member **11.1.4.1-122** and may be monitored by control circuitry **11.1.4.1-20** to detect bending in member **11.1.4.1-122**. As shown in FIG. **11.1.4.1-11**, sensor **11.1.4.1-120** of FIG. **11.1.4.1-10** may be attached to optical module **11.1.4.1-30** with a fastener such as screw **11.1.4.1-128** or other attachment mechanism so that protruding portion **11.1.4.1-124** protrudes into opening **11.1.4.1-50** and into contact an adjacent surface of guide rail **11.1.4.1-44**. Changes in the position of guide rail **11.1.4.1-44** will result in changes in the detected strain by sensor **11.1.4.1-120**, so sensor **11.1.4.1-120** can monitor the position of guide rail **11.1.4.1-44** relative to optical module **11.1.4.1-30** along axis **11.1.4.1-130**. If desired, additional sensors such as sensor **11.1.4.1-120** may be located at additional positions about rail **11.1.4.1-44** (see, e.g., illustrative orthogonal position **11.1.4.1-120'** of FIG. **11.1.4.1-11**, which allows position measurements along axis **11.1.4.1-132**).

11.1.4.2: Lower Guide Rods

[2087] FIG. **11.1.4.2-1** illustrates a subassembly of an HMD **11.1.4.2-100** including a display bracket **11.1.4.2-106** having upper arms **11.1.4.2-110a**, **11.1.4.2-110b** extending outward in opposite directions and lower arms **11.1.4.2-112a**, **11.1.4.2-112b** extending in opposite directions, respectively. The subassembly can include a first optical module **11.1.4.2-108a** slidably coupled to the display bracket **11.1.4.2-106** at a first upper guide-rod **11.1.4.2-102a** and a first lower guide-rod **11.1.4.2-104a** as well as a second optical module **11.1.4.2-108b** slidably coupled to the display bracket **11.1.4.2-106** at a second upper guide rod **11.1.4.2-102b** and a second lower guide-rod **11.1.4.2-104b**. The first and second optical modules **11.1.4.2-108a-b** can be adjusted laterally relative to one another and/or the display bracket **11.1.4.2-106** along the guide-rods **11.1.4.2-102a-b**, **11.1.4.2-104a-b** during IPD adjustments. The first and second upper guide-rods **11.1.4.2-102a-b** can be part of an upper guide-rod assembly **11.1.4.2-102** and the first and second lower guide-rods **11.1.4.2-104a-b** can be part of a lower guide-rod assembly **11.1.4.2-104**.

[2088] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.4.2-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any

other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.4.2-1**.

11.1.4.2.1: Electrical Contacts

[2089] FIG. **11.1.4.2.1-1** illustrates a lower guide-rod system **11.1.4.2.1-102** of an HMD **11.1.4.2.1-100** including a lower guide-rod **11.1.4.2.1-106** configured to guide the movement of an optical module **11.1.4.2.1-104**. In at least one example, the lower guide-rod system **11.1.4.2.1-102** also includes a stop **11.1.4.2.1-108** similar to other stops described herein, for protecting the cantilevered guide-rod **11.1.4.2.1-106**, which does not make contact with the stop **11.1.4.2.1-108** during normal use but through which the distal end of the guide-rod **11.1.4.2.1-106** may extend to contact the stop **11.1.4.2.1-108** in a drop event or other circumstance causing a deflection of the distal end of the guide-rod **11.1.4.2.1-106**. In at least one example, stop **11.1.4.2.1-108** can also prevent the optical module **11.1.4.2.1-104** from travelling too far along the guide-rod **11.1.4.2.1-106**.

[2090] In at least one example, the lower guide-rod system **11.1.4.2.1-102** can also include an electrical contact **11.1.4.2.1-110** biased against the guide-rod **11.1.4.2.1-106**. The biased electrical contact **11.1.4.2.1-110** can thus maintain contact with the guide-rod **11.1.4.2.1-106** during operation, including movements, vibrations, and other disturbances in the position of the guide-rod **11.1.4.2.1-106** during use. The biased electrical contact **11.1.4.2.1-110** can maintain electrical contact for electrical grounding purposes and for forming other electrical pathways between the guide-rod **11.1.4.2.1-106** and other components of the HMD **11.1.4.2.1-100**.

[2091] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.4.2.1-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.4.2.1-1**.

11.1.4.2.2: Biasing Members

[2092] FIG. **11.1.4.2.2-1** illustrates a perspective view of an example of an adjustment mechanism or assembly **11.1.4.2.2-100** including a lower guide-rod **11.1.4.2.2-102** and a stop **11.1.4.2.2-104**. The adjustment assembly **11.1.4.2.2-100** can also include a spring **11.1.4.2.2-106** biasing the guide-rod **11.1.4.2.2-106** passing through the stop **11.1.4.2.2-104**. The spring **11.1.4.2.2-106** can press the guide-rod **11.1.4.2.2-106** against the stop **11.1.4.2.2-104** to remove any slack or play between the guide-rod **11.1.4.2.2-102** and the stop **11.1.4.2.2-104** during use while allowing some flexure of the guide-rod **11.1.4.2.2-102**. In at least one example, the spring **11.1.4.2.2-106** can be a conductive biasing member maintaining contact with the guide-rod **11.1.4.2.2-102** for electrical grounding or providing other electrical pathways between the guide-rod **11.1.4.2.2-102** and one or more other components of the HMD.

[2093] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.1.4.2.2-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.1.4.2.2-1**.

11.2: Barrels and Baskets

11.2.1: Lens Mounting Systems

[2094] The lenses may be mounted in a head-mounted device using support structures that help minimize lens stress. In an illustrative embodiment, the lenses are mounted in rigid lens supports such as metal lens barrels using lens mounts based on flexures. The flexures help prevent stress

from being applied to the lenses even when the electronic devices are subjected to changes in operating temperature that cause the lenses to expand and contract.

[2095] FIG. **11.2.1-1** is a schematic diagram of an illustrative electronic device of the type that may include lenses mounted with flexures. Device **11.2.1-10** of FIG. **11.2.1-1** may be a head-mounted device (e.g., goggles, glasses, a helmet, and/or other head-mounted device. In an illustrative configuration, device **11.2.1-10** is a head-mounted device such as a pair of goggles (sometimes referred to as virtual reality goggles, mixed reality goggles, augmented reality glasses, etc.).

[2096] As shown in the illustrative cross-sectional top view of device **11.2.1-10** of FIG. **11.2.1-1**, device **11.2.1-10** may have a housing such as housing **11.2.1-12** (sometimes referred to as a head-mounted support structure, head-mounted housing, or head-mounted support). Housing **11.2.1-12** may include a front portion such as front portion **11.2.1-12F** and a rear portion such as rear portion **11.2.1-12R**. When device **11.2.1-10** is worn on the head of a user, rear portion **11.2.1-12R** rests against the face of the user and helps block stray light from reaching the eyes of the user.

[2097] Main portion **11.2.1-12M** of housing **11.2.1-12** may be attached to head strap **11.2.1-12T**. Head strap **11.2.1-12T** may be used to help secure main portion **11.2.1-12** on the head and face of a user. Main portion **11.2.1-12M** may have a rigid shell formed from housing walls of polymer, glass, metal, and/or other materials. When housing **11.2.1-12** is being worn on the head of a user, the front of housing **11.2.1-12** may face outwardly away from the user and the rear of housing **11.2.1-12** (and rear portion **11.2.1-12R**) may face towards the user. In this configuration, rear portion **11.2.1-12R** may face the user's eyes located in eye boxes **11.2.1-36**.

[2098] Device **11.2.1-10** may have electrical and optical components that are used in displaying images to eye boxes **11.2.1-36** when device **11.2.1-10** is being worn. These components may include left and right optical assemblies **11.2.1-20** (sometimes referred to as optical modules). Each optical assembly **11.2.1-20** may have an optical assembly support **11.2.1-38** (sometimes referred to as a lens barrel, optical module support, or support structure). Supports **11.2.1-38** may have hollow tubular shapes or other suitable shapes. Optical assemblies **11.2.1-20** may slide laterally along guide rails to adjust optical-assembly-to-optical-assembly separation to accommodate different user interpupillary distances. Rear portion **11.2.1-12R** may include flexible structures (e.g., a flexible polymer layer, a flexible fabric layer, etc.) so that portion **11.2.1-12R** can stretch to accommodate movement of assemblies **11.2.1-20**.

[2099] Each assembly **11.2.1-20** may have a display **11.2.1-32** coupled to one end of support **11.2.1-38** and a lens mounted to an opposing end of support **11.2.1-38**. Display **11.2.1-32** has an array of pixels for displaying images. Lens **11.2.1-34** may optionally have a removable vision correction lens for correcting user vision defects (e.g., refractive errors such as nearsightedness, farsightedness, and/or astigmatism). During operation, images displayed by displays **11.2.1-32** may be presented to eye boxes **11.2.1-36** through lenses **11.2.1-34** for viewing by the user.

[2100] To help satisfy design constraints (e.g., low weight, compact size, wide field of view, high resolution, etc.), lenses **11.2.1-34** may be catadioptric lenses. A catadioptric lens incorporates optical components such as polarizers and wave plates to create a folded optical path that can help reduce lens size. The use of these polarization-sensitive optical components may, however, make lenses **11.2.1-34** sensitive to performance degradation due to stress-induced birefringence effects. Lenses **11.2.1-34** may, as an example, include lens elements formed from polymers (for example, COC polymer (cyclic olefin copolymer) or other suitable polymers) that help minimize lens weight, but these polymers may exhibit birefringence when subjected to excess stress.

[2101] Satisfactory operation for lenses **11.2.1-34** may be achieved by mounting lenses **11.2.1-34** using lens mounts that help isolate lenses **11.2.1-34** from sources of stress. In an illustrative configuration, supports **11.2.1-38** may be formed from strong rigid materials such as metal (e.g., aluminum, etc.). When operating device **11.2.1-10** over wide temperature ranges (e.g., 0-50° C.), there is a risk that expansion and contraction of materials in assemblies **11.2.1-20** and, more particularly, mismatches in the coefficient of thermal expansion between the polymer of lenses

11.2.1-34 and the metal of supports **11.2.1-38**, can give rise to unwanted stress in lenses **11.2.1-34**, leading to unwanted stress-induced birefringence in lenses **11.2.1-34**. Unwanted temperature-change-induced stress may be avoided by mounting lenses **11.2.1-34** within supports **11.2.1-38** using lens mounts based on flexures.

[2102] FIG. **11.2.1-2** is a front view of an illustrative lens for device **11.2.1-10**. As shown in FIG. **11.2.1-2**, lens **11.2.1-34** may be surrounded by flexible lens mount **11.2.1-40**. Lens mount **11.2.1-40** may include one or more flexures. In an illustrative embodiment, lens mount **11.2.1-40** contains a ring-shaped flexure that runs around the entire perimeter of lens **11.2.1-34**. Arrangements in which mount **11.2.1-40** includes a number of separate segments (see, e.g., illustrative segment SG) each of which has a separate strip-shaped flexure segment or in which mount **11.2.1-40** includes discrete flexures such as illustrative discrete flexure DL (e.g., a set of three discrete flexures attached to the periphery of lens **11.2.1-34** at three evenly spaced locations along the periphery) may also be used. Support **11.2.1-38** may be attached to mount **11.2.1-40** along the outer peripheral edge of mount **11.2.1-40** or support **11.2.1-38** may be attached to mount **11.2.1-40** where support **11.2.1-38** overlaps mount **11.2.1-40** (see, e.g., FIG. **11.2.1-3**).

[2103] FIG. **11.2.1-3** is a cross-sectional side view of an illustrative peripheral portion of lens **11.2.1-38** and mount **11.2.1-40** of FIG. **11.2.1-2** taken along line **11.2.1-42** of FIG. **11.2.1-2** and viewed in direction **11.2.1-44**. As shown in FIG. **11.2.1-3**, support (lens barrel) **11.2.1-38** may be overlapped by mount **11.2.1-40**. Mount **11.2.1-40** may be formed from polymer, metal, or other material that is shaped to form a flexure. In the example of FIG. **11.2.1-3**, mount **11.2.1-40** is a ring-shaped flexure with a U-shaped cross section that runs along the ring-shaped peripheral edge of lens **11.2.1-34**. First portion **11.2.1-40-1** of the flexure is attached to the peripheral edge of lens **11.2.1-34** with a first ring of adhesive **11.2.1-50** and second portion **11.2.1-40-2** of the flexure is attached to support **11.2.1-38** with a second ring of adhesive **11.2.1-52**. Bend **11.2.1-40-3** runs along mount **11.2.1-40** between portions **11.2.1-40-1** and **11.2.1-40-2** and allows mount **11.2.1-40** to flex. For example, portion **11.2.1-40-1** may move radially outwardly towards portion **11.2.1-40-2** when lens **11.2.1-34** expands radially due to an increase in operating temperature. The metal of support **11.2.1-38** has a significantly lower coefficient of thermal expansion than the polymer of lens **11.2.1-34**, but is separated by an air gap from the edge of lens **11.2.1-34**, so that lens **11.2.1-34** can expand radially outwardly without contacting support **11.2.1-38** and is therefore not stressed due to pressure from contact with support **11.2.1-38**.

[2104] Mount **11.2.1-40** may, if desired, be formed from a material such as polyetherimide (PEI) or other rigid polymer that exhibits a high yield strength and a high elongation-to-failure value. The elastic modulus of the material forming mount **11.2.1-40** may be, for example, at least 1.5 GPa, at least 2 GPa, or at least 3 GPa, as examples. The high strength and high rigidity of the material forming mount **11.2.1-40** helps mount **11.2.1-40** accurately hold lens **11.2.1-34** in place. The material of mount **11.2.1-40** may exhibit a yield strength (e.g., at least 30 MPa, at least 70 MPa, or at least 100 MPa), so that mount **11.2.1-40** exhibits a good yield-strength-to-stiffness ratio. To provide mount **11.2.1-40** with satisfactory durability, it may be desirable for the material of mount to have an elongation-to-failure value of at least 25%, at least 35%, or at least 40%. To ensure satisfactory attachment of mount **11.2.1-40** with adhesive **11.2.1-52** and **11.2.1-50**, the material of mount **11.2.1-40** preferably also exhibits satisfactory compatibility (bond strength) with adhesives **11.2.1-52** and **11.2.1-50**. Adhesives **11.2.1-52** and **11.2.1-50** may be, for example, epoxy, so the material of mount **11.2.1-40** preferably forms satisfactory bonds with epoxy. The coefficient of thermal expansion of the material of mount **11.2.1-40** may be matched (e.g., within 20%, within 10%, within 5%, or within 2%) to the coefficient of thermal expansion of the material of lens **11.2.1-34**. In this way, lens stress can be avoided that might otherwise be imposed by mount **11.2.1-40** on lens **11.2.1-34** if, for example, the perimeter of mount **11.2.1-40** were to expand less quickly than the perimeter of lens **11.2.1-34** as a function of increasing operating temperature and thereby constrain the expansion of lens **11.2.1-34**. Other types of materials may be used in forming mount

11.2.1-40, if desired. The use of polyetherimide that is matched to the polymer of lens **11.2.1-34** in its coefficient of thermal expansion and that exhibits high yield strength and high elongation-to-failure values is illustrative.

[2105] If desired, thermally induced lens stress may be minimized using a reverse flange adhesive attachment arrangement of the type shown in FIG. **11.2.1-4**. With this type of arrangement, lens **11.2.1-34** may be provided with a milled flange that is bonded to a surface of support **11.2.1-38** that faces radially outward (away from lens **11.2.1-34**). As shown in FIG. **11.2.1-4**, adhesive **11.2.1-50** (e.g., a soft adhesive) may be used to attach outwardly facing surface **11.2.1-53** of support **11.2.1-38** to opposing inwardly faced milled flange surface **11.2.1-54** of lens **11.2.1-34**. An air gap such as air gap **11.2.1-56** is preferably present between outer peripheral edge surface **11.2.1-58** of lens **11.2.1-34** and opposing inwardly facing surface **11.2.1-60** of support **11.2.1-38**, thereby ensuring that the periphery of lens **11.2.1-34** (surface **11.2.1-58**) will not contact support **11.2.1-38** and will therefore not receive stress from support **11.2.1-38** even as lens **11.2.1-34** expands radially outward as a function of increasing operating temperature.

[2106] If desired a set of discretely located flexures such as the illustrative flexures of FIGS. **11.2.1-5** and **11.2.1-6** may also be used in supporting lens **11.2.1-34** (e.g., at three points along the perimeter of lens **11.2.1-34**). In the examples of FIGS. **11.2.1-5** and **11.2.1-6**, lens **11.2.1-34** is viewed from the rear of device **11.2.1-10** (e.g., along the Z axis). One side of illustrative flexures **11.2.1-64** and **11.2.1-66** may be attached to the periphery of lens **11.2.1-34** using adhesive **11.2.1-68** and another side of flexures **11.2.1-64** and **11.2.1-66** may be attached to support **11.2.1-38** (e.g., using adhesive, screws or other fasteners, etc.). During fluctuations in operating temperature, lens **11.2.1-34** may expand radially outward and may contract radially inward. With the flexure arrangements of FIGS. **11.2.1-5** and **11.2.1-6**, flexures **11.2.1-64** and **11.2.1-66** may radially flex to accommodate the lens expansion and contraction, while accurately maintaining a desired mounting location for lens **11.2.1-34** within support **11.2.1-38**. In arrangements in which flexures are located at discrete locations around the periphery of lens **11.2.1-34** (e.g., three discrete locations), sealing structures (e.g., ring-shaped elastomeric boots) may be used to help seal the periphery of lens **11.2.1-34** against support **11.2.1-38** and thereby prevent dust and moisture ingress into assembly **11.2.1-20**. As with the ring-shaped flexure of FIGS. **11.2.1-2** and **11.2.1-3**, the discrete flexures of FIGS. **11.2.1-5** and **11.2.1-6** may be used to maintain lens **11.2.1-34** in a satisfactory position (without significant lateral shifting in X and Y position and without significant tilt away from the Z axis) while flexing to accommodate expansion and contraction due to temperature changes. The strength of mount **11.2.1-40** may help assemblies **11.2.1-20** resist plastic deformation and other damage during drop events and other undesired impacts leading to excessive stress.

[2107] FIGS. **11.2.1-7**, **11.2.1-8**, **11.2.1-9**, and **11.2.1-10** are illustrative cross-sectional side views of illustrative flexures for mount **11.2.1-40**. The flexures of FIGS. **11.2.1-7**, **11.2.1-8**, **11.2.1-9**, and **11.2.1-10** may extend in a continuous ring around the periphery of lens **11.2.1-34**, may be formed from a number of segments that extend around lens **11.2.1-34**, or may be formed from discretely located flexures (as shown, for example, in FIGS. **11.2.1-5** and **11.2.1-6**). In the arrangements of FIGS. **11.2.1-7**, **11.2.1-8**, and **11.2.1-9**, mount **11.2.1-40** has a U-shaped cross-sectional shape.

[2108] In the example of FIG. **11.2.1-7**, screw **11.2.1-70** attaches mount (flexure) **11.2.1-40** to support **11.2.1-38** and adhesive **11.2.1-78** attaches mount **11.2.1-40** to lens **11.2.1-34**. When using a screw such as screw **11.2.1-70** to attach mount **11.2.1-40** to support **11.2.1-38** instead of adhesive, a separate seal such as seal **11.2.1-72** may be used to seal lens **11.2.1-34** to support **11.2.1-38** and thereby help prevent moisture and dust ingress into the interior of assembly **11.2.1-20**. Seal **11.2.1-72** may be an O-ring formed from an elastomeric gasket material, may be a ring of adhesive (e.g., pressure sensitive adhesive or other soft sealing adhesive), or may be other sealant. If desired, adhesive **11.2.1-76** may be provided at the interface between mount **11.2.1-40** and support **11.2.1-38** (e.g., epoxy or other rigid adhesive) and may be used both to attach mount **11.2.1-40** to support **11.2.1-38** and to seal mount **11.2.1-40** to support **11.2.1-38**. The use of screws such as screw **11.2.1-**

70 of FIG. 11.2.1-7 to attach mount 11.2.1-40 to support 11.2.1-38 is illustrative.

[2109] The illustrative configurations of FIGS. 11.2.1-8, 11.2.1-9, and 11.2.1-10 may also involve the use of screws 11.2.1-70, adhesive 11.2.1-76, and/or seal 11.2.1-72 to attach and seal mount 11.2.1-40 to support 11.2.1-38. In the example of FIG. 11.2.1-8, mount 11.2.1-40 protrudes inwardly under the outer edge of lens 11.2.1-34. In the example of FIG. 11.2.1-9, screw 11.2.1-70 extends radially inward towards the center of lens 11.2.1-34. In the example of FIG. 11.2.1-10, the flexure formed by mount 11.2.1-40 does not have a central bend, but rather relies on its length to ensure that sufficient flexure bending is available to accommodate radial outward expansion of lens 11.2.1-34 as operating temperatures rise.

[2110] The diagram of FIG. 11.2.1-11 shows how adhesive 11.2.1-78 may be dispensed to attach mount 11.2.1-40 to the outer edge of lens 11.2.1-34. In the example of FIG. 11.2.1-11, lens 11.2.1-34 and mount 11.2.1-40 have been flipped upside down relative to the orientation of FIG. 11.2.1-3. As shown in FIG. 11.2.1-11, mount 11.2.1-40 may have a chamfer such as chamfer 11.2.1-84 that locally widens the gap between lens 11.2.1-34 and mount 11.2.1-40 to facilitate the introduction of liquid adhesive into this gap. Adhesive may be dispensed in liquid form in direction 11.2.1-82 from dispenser 11.2.1-80 and may then be drawn into the gap between mount 11.2.1-40 and lens 11.2.1-34 by capillary action. After the liquid adhesive material has been drawn into the gap, the adhesive may be cured to form adhesive 11.2.1-78 (e.g., using ultraviolet light curing, curing using a two-part adhesive arrangement, etc.). To help control the downward progress of adhesive 11.2.1-78 while adhesive 11.2.1-78 is liquid, mount 11.2.1-40 may be provided with a stop feature such as step 11.2.1-86. The presence of step 11.2.1-86 on the inwardly facing surface of mount 11.2.1-40 creates a surface tension break that stops the downward flow of adhesive 11.2.1-78 when flowing into the joint between lens 11.2.1-34 and mount 11.2.1-40. In this way, the dispensing of adhesive 11.2.1-78 may be limited to a desired bonding area between mount 11.2.1-40 and lens 11.2.1-34.

11.3: Rear-Facing Cameras

11.3.1: Optical Module for Head-Mounted Device

[2111] One aspect of the disclosure is an optical module for a head-mounted device that is configured to present content to a user. The optical module includes an optical module housing assembly, a display assembly, and an eye camera. The optical module housing assembly has a first end and a second end. The lens is connected to the optical module housing assembly and positioned at the first end of the optical module housing assembly. The display assembly is connected to the optical module housing assembly and is positioned at the second end of the optical module housing assembly. The display assembly is configured to cause the content to be displayed to the user through the lens. The eye camera is connected to the optical module housing assembly and is positioned at the second end of the optical module housing assembly. The eye camera is configured to obtain images through the lens.

[2112] In some implementations of the optical module, the optical module housing assembly includes a first portion that is connected to a second portion, and the lens is retained between the first portion and the second portion. In some implementations of the optical module, projections are defined on the lens and channels are defined on the first portion of the optical module housing assembly such that the projections are located in the channels and engage the first portion of the optical module housing assembly within the channels to secure the lens relative to the optical module housing assembly and restrain movement of the lens relative to the optical module housing assembly. In some implementations of the optical module, the lens and the display assembly are connected to the optical module housing assembly in a side-by-side arrangement. In some implementations of the optical module, the optical module housing assembly defines an internal space between the lens and the display assembly.

[2113] In some implementations of the optical module, the optical module also includes a vent port that allows air to travel between the internal space and an outside environment, and a filter element that restrains foreign particles from entering the internal space. In some implementations of the

optical module, the optical module also includes a dust trap that is located in the internal space and is configured to retain foreign particles.

[2114] In some implementations of the optical module, the optical module also includes a fiducial marker that is formed on the lens and is visible in images obtained by the eye camera for use in calibration. In some implementations of the optical module, the lens is a catadioptric lens. In some implementations of the optical module, the lens is a part of a catadioptric optical system.

[2115] Another aspect of the disclosure is an optical module for a head-mounted device that is configured to present content to a user. The optical module includes an optical module housing assembly that defines an internal space, a lens that is connected to the optical module housing assembly, a display assembly that is connected to the optical module housing assembly. The display assembly is configured to cause the content to be displayed to the user through the lens. An infrared emitter is located between the lens and the display assembly in the internal space of the optical module housing assembly. The infrared emitter is configured to emit infrared radiation through the lens.

[2116] In some implementations of the optical module, the infrared emitter includes a flexible circuit and emissive components that are connected to the flexible circuit and are configured to emit infrared radiation. In some implementations of the optical module, wherein the emissive components are arranged in an array around an optical axis of the optical module housing assembly. In some implementations of the optical module, the flexible circuit extends through an electrical port that is formed through the optical module housing assembly and a sealing element is formed on the flexible circuit and is engaged with the optical module housing assembly at the electrical port. In some implementations of the optical module, the optical module housing assembly defines an optical pathway opening that is adjacent to the display assembly and is configured to allow light to pass from the display assembly to the lens, a base surface that extends around the optical pathway opening, wherein the infrared emitter is located on the base surface, and a peripheral wall that is located outward from the base surface.

[2117] In some implementations of the optical module, the optical module also includes an eye camera that is configured to obtain images that show reflected portions of the infrared radiation that is emitted by the infrared emitter. In some implementations of the optical module, the eye camera is connected to the optical module housing assembly and is configured to obtain the images through the lens. In some implementations of the optical module, the lens is a catadioptric lens. In some implementations of the optical module, the lens is a part of a catadioptric optical system.

[2118] Another aspect of the disclosure is a head-mounted device that is configured to present content to a user. The head-mounted device includes a housing, a first optical module that is located in the housing, and a second optical module that is located in the housing. An interpupillary distance adjustment assembly supports the first optical module and the second optical module with respect to the housing to allow adjustment of a distance between the first optical module and the second optical module. The head-mounted device also includes a first front-facing camera that is connected to the first optical module and is movable in unison with the first optical module by the interpupillary distance adjustment assembly, and a second front-facing camera that is connected to the second optical module and is movable in unison with the second optical module by the interpupillary distance adjustment assembly. Adjustment of the distance between the first optical module and the second optical module by the interpupillary distance adjustment assembly also adjusts a distance between the first front-facing camera and the second front-facing camera.

[2119] In some implementations of the head-mounted device, the housing includes one or more optically transmissive panels through which the first front-facing camera and the second front-facing camera may obtain images of an environment.

[2120] In some implementations of the head-mounted device, an optical axis of the first front-facing camera is aligned with an optical axis of the first optical module and an optical axis of the second front-facing camera is aligned with an optical axis of the second optical module.

[2121] In some implementations of the head-mounted device, the first front-facing camera is connected in a fixed relationship with respect to the first optical module, and the second front-facing camera is connected in a fixed relationship with respect to the second optical module.

[2122] In some implementations of the head-mounted device, the interpupillary distance adjustment assembly maintains a first spacing between an optical axis of the first optical module and an optical axis of the second optical module generally equal to a second spacing between an optical axis of the first front-facing camera and an optical axis of the second front facing camera during adjustment of the distance between the first optical module and the second optical module.

[2123] FIG. **11.3.1-1** is a block diagram that shows an example of a hardware configuration for a head-mounted device **11.3.1-100**. The head-mounted device **11.3.1-100** is intended to be worn on the head of a user and includes components that are configured to display content to the user. Components that are included in the head-mounted device **11.3.1-100** may be configured to track motion of parts of the user's body, such as the user's head and hands. Motion tracking information that is obtained by components of the head-mounted device can be utilized as inputs that control aspects of the generation and display of the content to the user, so that the content displayed to the user can be part of a CGR experience in which the user is able to view and interact with virtual environments and virtual objects. In the illustrated example, the head-mounted device **11.3.1-100** includes a device housing **11.3.1-102**, a face seal **11.3.1-104**, a support structure **11.3.1-106**, a processor **11.3.1-108**, a memory **11.3.1-110**, a storage device **11.3.1-112**, a communications device **11.3.1-114**, sensors **11.3.1-116**, a power source **11.3.1-118**, and optical modules **11.3.1-120**. The head-mounted device **11.3.1-100** includes two of the optical modules **11.3.1-120**, to display content to the user's eyes. The optical modules **11.3.1-120** may each include an optical module housing **11.3.1-122**, a display assembly **11.3.1-124**, and a lens assembly **11.3.1-126**.

[2124] The device housing **11.3.1-102** is a structure that supports various other components that are included in the head-mounted device. The device housing **11.3.1-102** may be an enclosed structure such that certain components of the head-mounted device **11.3.1-100** are contained within the device housing **11.3.1-102** and thereby protected from damage.

[2125] The face seal **11.3.1-104** is connected to the device housing **11.3.1-102** and is located at areas around a periphery of the device housing **11.3.1-102** where contact with the user's face is likely. The face seal **11.3.1-104** functions to conform to portions of the user's face to allow the support structure **11.3.1-106** to be tensioned to an extent that will restrain motion of the device housing **11.3.1-102** with respect to the user's head. The face seal **11.3.1-104** may also function to reduce the amount of light from the physical environment around the user that reaches the user's eyes. The face seal **11.3.1-104** may contact areas of the user's face, such as the user's forehead, temples, and cheeks. The face seal **11.3.1-104** may be formed from a compressible material, such as open-cell foam or closed cell foam.

[2126] The support structure **11.3.1-106** is connected to the device housing **11.3.1-102**. The support structure **11.3.1-106** is a component or collection of components that function to secure the device housing **11.3.1-102** in place with respect to the user's head so that the device housing **11.3.1-102** is restrained from moving with respect to the user's head and maintains a comfortable position during use. The support structure **11.3.1-106** can be implemented using rigid structures, elastic flexible straps, or inelastic flexible straps.

[2127] The processor **11.3.1-108** is a device that is operable to execute computer program instructions and is operable to perform operations that are described by the computer program instructions. The processor **11.3.1-108** may be implemented using a conventional device, such as a central processing unit, and provided with computer-executable instructions that cause the processor **11.3.1-108** to perform specific functions. The processor **11.3.1-108** may be a special-purpose processor (e.g., an application-specific integrated circuit or a field-programmable gate array) that implements a limited set of functions. The memory **11.3.1-110** may be a volatile, high-speed, short-term information storage device such as a random-access memory module.

[2128] The storage device **11.3.1-112** is intended to allow for long term storage of computer program instructions and other data. Examples of suitable devices for use as the storage device **11.3.1-112** include non-volatile information storage devices of various types, such as a flash memory module, a hard drive, or a solid-state drive.

[2129] The communications device **11.3.1-114** supports wired or wireless communications with other devices. Any suitable wired or wireless communications protocol may be used.

[2130] The sensors **11.3.1-116** are components that are incorporated in the head-mounted device **11.3.1-100** to provide inputs to the processor **11.3.1-108** for use in generating CGR content. The sensors **11.3.1-116** include components that facilitate motion tracking (e.g., head tracking and optionally handheld controller tracking in six degrees of freedom). The sensors **11.3.1-116** may also include additional sensors that are used by the device to generate and/or enhance the user's experience in any way. The sensors **11.3.1-116** may include conventional components such as cameras, infrared cameras, infrared emitters, depth cameras, structured-light sensing devices, accelerometers, gyroscopes, and magnetometers. The sensors **11.3.1-116** may also include biometric sensors that are operable to physical or physiological features of a person, for example, for use in user identification and authorization. Biometric sensors may include fingerprint scanners, retinal scanners, and face scanners (e.g., two-dimensional and three-dimensional scanning components operable to obtain image and/or three-dimensional surface representations). Other types of devices can be incorporated in the sensors **11.3.1-116**. The information that is generated by the sensors **11.3.1-116** is provided to other components of the head-mounted device **11.3.1-100**, such as the processor **11.3.1-108**, as inputs.

[2131] The power source **11.3.1-118** supplies electrical power to components of the head-mounted device **11.3.1-100**. In some implementations, the power source **11.3.1-118** is a wired connection to electrical power. In some implementations, the power source **11.3.1-118** may include a battery of any suitable type, such as a rechargeable battery. In implementations that include a battery, the head-mounted device **11.3.1-100** may include components that facilitate wired or wireless recharging.

[2132] In some implementations of the head-mounted device **11.3.1-100**, some or all of these components may be included in a separate device that is removable. For example, any or all of the processor **11.3.1-108**, the memory **11.3.1-110**, and/or the storage device **11.3.1-112**, the communications device **11.3.1-114**, and the sensors **11.3.1-116** may be incorporated in a device such as a smart phone that is connected (e.g., by docking) to the other portions of the head-mounted device **11.3.1-100**.

[2133] In some implementations of the head-mounted device **11.3.1-100**, the processor **11.3.1-108**, the memory **11.3.1-110**, and/or the storage device **11.3.1-112** are omitted, and the corresponding functions are performed by an external device that communicates with the head-mounted device **11.3.1-100**. In such an implementation, the head-mounted device **11.3.1-100** may include components that support a data transfer connection with the external device using a wired connection or a wireless connection that is established using the communications device **11.3.1-114**.

[2134] The components that are included in the optical modules support the function of displaying content to the user in a manner that supports CGR experiences. The optical modules **11.3.1-120** are each assemblies that include multiple components, which include the optical module housing **11.3.1-122**, the display assembly **11.3.1-124**, and the lens assembly **11.3.1-126**, as will be described further herein.

[2135] Other components may also be included in each of the optical modules. Although not illustrated in FIGS. **11.3.1-2-3**, the optical modules **11.3.1-120** may be supported by adjustment assemblies that allow the position of the optical modules **11.3.1-120** to be adjusted. As an example, the optical modules **11.3.1-120** may each be supported by an interpupillary distance adjustment mechanism that allows the optical modules **11.3.1-120** to slide laterally toward or away from each

other. As another example, the optical modules **11.3.1-120** may be supported by an eye relief distance adjustment mechanism that allows adjustment of the distance between the optical modules **11.3.1-120** and the user's eyes.

[2136] FIG. **11.3.1-2** is a top view illustration that shows the head-mounted device **11.3.1-100**, including the device housing **11.3.1-102**, the face seal **11.3.1-104**, and the support structure **11.3.1-106**. FIG. **11.3.1-3** is a rear view illustration taken along line A-A of FIG. **11.3.1-2**. In the illustrated example, the device housing **11.3.1-102** is a generally rectangular structure having a width that is selected to be similar to the width of the head of a typical person, and a height selected so as to extend approximately from the forehead to the base of the nose of a typical person. This configuration is an example, and other shapes and sizes may be used.

[2137] An eye chamber **11.3.1-328** is defined by the device housing **11.3.1-102** and is bordered by the face seal **11.3.1-104** at its outer periphery. The eye chamber **11.3.1-328** is open to the exterior of the head-mounted device **11.3.1-100** to allow the user's face to be positioned adjacent to the eye chamber **11.3.1-328**, which is otherwise enclosed by the device housing **11.3.1-102**. The face seal **11.3.1-104** may extend around part of all of the periphery of the device housing **11.3.1-102** adjacent to the eye chamber **11.3.1-328**. The face seal **11.3.1-104** may function to exclude some of the light from the environment around the head-mounted device **11.3.1-100** from entering the eye-chamber **11.3.1-328** and reaching the user's eyes.

[2138] In the illustrated example, the support structure **11.3.1-106** is a headband type device that is connected to left and right lateral sides of the device housing **11.3.1-102** and is intended to extend around the user's head. Other configurations may be used for the support structure **11.3.1-106**, such as a halo-type configuration in which the device housing **11.3.1-102** is supported by a structure that is connected to a top portion of the device housing **11.3.1-102**, engages the user's forehead above the device housing **11.3.1-102**, and extends around the user's head, or a mohawk-type configuration in which a structure extends over the user's head. Although not illustrated, the support structure **11.3.1-106** may include passive or active adjustment components, which may be mechanical or electromechanical, that allow portions of the support structure **11.3.1-106** to expand and contract to adjust the fit of the support structure **11.3.1-106** with respect to the user's head.

[2139] The optical modules **11.3.1-120** are located in the device housing **11.3.1-102** and extend outward into the eye chamber **11.3.1-328**. Portions of the optical modules **11.3.1-120** are located in the eye chamber **11.3.1-328** so that the user can see the content that is displayed by the optical modules **11.3.1-120**. The optical modules **11.3.1-120** are located within the eye chamber **11.3.1-328** at locations that are intended to be adjacent to the user's eyes. As an example, the head-mounted device **11.3.1-100** may be configured to position portions of the lens assemblies **11.3.1-126** of the optical modules **11.3.1-120** approximately 15 millimeters from the user's eyes.

[2140] FIG. **11.3.1-4** is a perspective view illustration that shows one of the optical modules **11.3.1-120**, including the optical module housing **11.3.1-122**, the display assembly **11.3.1-124**, and the lens assembly **11.3.1-126**. The display assembly **11.3.1-124** and the lens assembly **11.3.1-126** are each connected to the optical module housing **11.3.1-122**. In the illustrated example, the lens assembly **11.3.1-126** is positioned at a front end of the optical module **11.3.1-120**, and the display assembly **11.3.1-124** is positioned at a rear end of the optical module **11.3.1-120**. The optical module housing **11.3.1-122** defines an internal space between the display assembly **11.3.1-124** and the lens assembly **11.3.1-126** to allow light to travel from the display assembly **11.3.1-124** to the lens assembly **11.3.1-126** within an environment that is sealed and protected from external contaminants while protecting sensitive components from damage.

[2141] The display assembly **11.3.1-124** includes a display screen that is configured to display content, such as images, according to signals received from the processor **11.3.1-108** and/or from external devices using the communications device **11.3.1-114** in order to output CGR content to the user. As an example, the display assembly **11.3.1-124** may output still images and/or video images in response to received signals. The display assembly **11.3.1-124** may include, as examples, an

LED screen, an LCD screen, an OLED screen, a micro LED screen, or a micro OLED screen.

[2142] The lens assembly **11.3.1-126** includes one or more lenses that direct light to the user's eyes in a manner that allows viewing of CGR content. In some implementations, the lens assembly **11.3.1-126** is a catadioptric optical system that utilizes both reflection and refraction in order to achieve desired optical properties in a small package size. Reflection, in some implementations, may be achieved by internal reflection at boundaries between material layers of a single lens. Thus, in some implementations, the lens assembly **11.3.1-126** may be implemented using a single multi-layered catadioptric lens.

[2143] The lens assembly **11.3.1-126** may be positioned partially within the optical module housing **11.3.1-122**. As will be explained further herein, the optical module housing **11.3.1-122** may include two or more components that are configured to retain the lens assembly in a desired position and orientation.

[2144] FIG. **11.3.1-5** is an exploded side view diagram showing components of an optical module **11.3.1-520** according to a first example. FIG. **11.3.1-5** is a schematic view intended to show the positional relationships between various features and does not include specific structural details of the components of the optical module **11.3.1-520**. The optical module **11.3.1-520** can be implemented in the context of a head-mounted display (e.g., the head-mounted device **11.3.1-100**) and may be implemented according to the description of the optical module **11.3.1-120** and the further description herein. The optical module **11.3.1-520** includes an optical module housing assembly **11.3.1-522**, a display assembly **11.3.1-524**, a lens **11.3.1-526**, an eye camera **11.3.1-530**, and an infrared emitter **11.3.1-532**. As will be described further herein, these components are arranged along an optical axis **11.3.1-521** of the optical module **11.3.1-520** such that images generated using the display assembly are projected to the user along the optical axis **11.3.1-521**.

[2145] Although the lens **11.3.1-526** is described as a single element herein, it should be understood that the lens **11.3.1-526** may be part of an assembly of optical elements or may be an assembly of optical elements, as described with respect to the lens assembly **11.3.1-126**. Thus, for example the lens **11.3.1-526** may be a catadioptric lens or the lens **11.3.1-526** may be part of a catadioptric optical system.

[2146] The optical module housing assembly **11.3.1-522** may include multiple parts that are connected to each other. In the illustrated example, the optical module housing assembly **11.3.1-522** includes a housing body **11.3.1-534** and a retainer **11.3.1-536**. The housing body **11.3.1-534** is configured to be connected to other structures within the housing of a head-mounted display (e.g., in the device housing **11.3.1-102** of the head-mounted device **11.3.1-100**). The housing body **11.3.1-534** is also provides a structure to which other components of the optical module **11.3.1-520** may be attached, including the display assembly **11.3.1-524**, the eye camera **11.3.1-530** and the infrared emitter **11.3.1-532**. The primary portions of the optical module housing assembly **11.3.1-522**, such as the housing body **11.3.1-534** and the retainer **11.3.1-536**, may be made from a rigid material, such as plastic or aluminum. The optical module housing assembly **11.3.1-522** is arranged around the optical axis **11.3.1-521**, and both visible light and infrared radiation may be incident on surfaces of the optical module housing assembly **11.3.1-522**. For this reason, portions of the optical module housing assembly **11.3.1-522** may be coated with materials (e.g., paints or other coating materials) that exhibit low reflectance of both visible and infrared wavelengths of electromagnetic radiation.

[2147] The retainer **11.3.1-536** is connected to an outer (e.g., user-facing) end of the housing body **11.3.1-534** of the optical module **11.3.1-520**. As examples, the retainer **11.3.1-536** may be connected to the housing body **11.3.1-534** by fasteners or by an adhesive. The retainer **11.3.1-536** and the housing body **11.3.1-534** of the optical module housing assembly **11.3.1-522** are configured such that the lens **11.3.1-526** is retained between the retainer **11.3.1-536** and the housing body **11.3.1-534**, as will be explained further herein. The retainer **11.3.1-536** and the housing body **11.3.1-534** have ring-like configurations along the optical axis **11.3.1-521** to allow light from the

display assembly **11.3.1-524** to pass through the lens **11.3.1-526** and toward the user.

[2148] The display assembly **11.3.1-524** includes a seal **11.3.1-538**, a bezel **11.3.1-540**, a display module **11.3.1-542**, a thermal interface **11.3.1-544**, and a heat sink **11.3.1-546**. The display assembly **11.3.1-524** is connected to the optical module housing assembly **11.3.1-522**. As an example, the display assembly **11.3.1-524** may be connected to the optical module housing assembly **11.3.1-522** by screws or other fasteners that allow disassembly of the display assembly **11.3.1-524** from the optical module housing assembly **11.3.1-522** (e.g., to allow for inspection and/or repair). The seal **11.3.1-538** is a sealing material of any suitable type that is configured to prevent foreign particle (e.g., dust) intrusion at the interface of the display assembly **11.3.1-524** with the optical module housing assembly **11.3.1-522**. The bezel **11.3.1-540** is a structural component that supports the display module **11.3.1-542** and protects it from damage. As an example, bezel **11.3.1-540** may be connected to the heat sink **11.3.1-546** (e.g., by screws or other fasteners) to capture the display module **11.3.1-542** and the heat sink **11.3.1-546**. The seal **11.3.1-538** may be engaged with the bezel **11.3.1-540** and the optical module housing assembly **11.3.1-522** to seal the interface between them.

[2149] The seal **11.3.1-538** and the bezel **11.3.1-540** have a ring-like configuration with central openings along the optical axis **11.3.1-521** in order to avoid blocking light emission from the display module **11.3.1-542** toward the lens **11.3.1-526**.

[2150] The display module **11.3.1-542** includes a display screen that displays images (e.g., by emitting light using a grid of light-emitting elements to define a picture). The display module **11.3.1-542** may be implemented using any suitable display technology, including light-emitting diode-based display technologies, organic light-emitting diode-based display technologies, and micro light-emitting diode-based display technologies. In some implementations, a layer of cover glass is attached (e.g., by laminating) to the display surface of the display module **11.3.1-542** to provide strength, to serve as a mounting feature, and to serve as a sealing interface.

[2151] The thermal interface **11.3.1-544** is a thermally conductive and electrically non-conductive material that is located between the display module **11.3.1-542** and the heat sink **11.3.1-546** to promote heat transfer from the display module **11.3.1-542** to the heat sink **11.3.1-546**. The thermal interface **11.3.1-544** is a compliant material that is able to fill in gaps that would otherwise be present between the display module **11.3.1-542** and the heat sink **11.3.1-546**, and which would reduce the efficiency of heat transfer. As an example, the thermal interface may be dispensable thermal gel that is applied to the display module **11.3.1-542** or the heat sink **11.3.1-546**. A reworkable material may be used for the thermal interface **11.3.1-544**, such as a material that is applied by room-temperature vulcanization.

[2152] The heat sink **11.3.1-546** is a rigid structure (e.g., formed from metal) that readily conducts heat and is configured to release heat to the ambient environment. As an example, the heat sink **11.3.1-546** may incorporate structures that increase surface area, such as fins, to promote heat dissipation, and/or may include features that conduct heat away from heat-generating components (e.g., the display module **11.3.1-542**), such as a heat pipe.

[2153] FIG. **11.3.1-6** is a front view illustration that shows the lens **11.3.1-526** according to an example, and FIG. **11.3.1-7** is a cross-section view illustration taken along line B-B of FIG. **11.3.1-6** showing the lens **11.3.1-526**. The lens **11.3.1-526** is an optical element (or combination of multiple optical elements, e.g., multiple lenses) that is configured to refract and/or reflect light that is incident on the lens **11.3.1-526**. In the illustrated example, the lens **11.3.1-526** is formed from molded transparent plastic, by glass may be used. Surface configurations that cause refraction and/or reflection of light (e.g., convexity and concavity) are not shown in the figures for simplicity and clarity, and these features may be defined as needed for desired performance of the optical system.

[2154] The lens **11.3.1-526** includes a lens body **11.3.1-648** and projections **11.3.1-649** that extend outward from the lens body **11.3.1-648**. The lens body **11.3.1-648** extends from an outer surface

11.3.1-750 (oriented toward the user) to an inner surface **11.3.1-751** (oriented toward the display assembly **11.3.1-524**). The lens body **11.3.1-648** will typically have a width (or range of widths) that is greater than the height of the lens body **11.3.1-648** as measured along the optical axis **11.3.1-521** of the optical module **11.3.1-520**. The lens body **11.3.1-648** may be formed in any shape (as viewed from an end along the optical axis **11.3.1-521**), such as generally cylindrical, oval, rounded rectangle, or irregular. The projections **11.3.1-649** may have a height (in the direction of the optical axis **11.3.1-521**) that is less than the height of the lens body **11.3.1-648**, such as 10 percent to 50 percent of the height of the lens body **11.3.1-648**. As will be explained herein, the projections **11.3.1-649** facilitate alignment and retention of the lens **11.3.1-526** relative to the optical module housing assembly **11.3.1-522**.

[2155] In the illustrated example, a peripheral wall of the lens body **11.3.1-648** extends from the outer surface **11.3.1-750** to the inner surface **11.3.1-751** without tapering, so that the peripheral wall is generally in alignment with the optical axis **11.3.1-521** and the outer surface **11.3.1-750** and the inner surface **11.3.1-751** are generally the same in shape and size (e.g., except for minor deviations such as the projections **11.3.1-649**). In other implementations, the peripheral wall of the lens body **11.3.1-648** may be tapered. For example, the peripheral wall of the lens body **11.3.1-648** may be tapered progressively away from the optical axis **11.3.1-521** in a direction of travel extending from the outer surface **11.3.1-750** to the inner surface **11.3.1-751**, so that the size of the outer surface **11.3.1-750** is smaller than the size of the inner surface **11.3.1-751**.

[2156] FIG. **11.3.1-8** is a front view illustration that shows the housing body **11.3.1-534** of the optical module housing assembly **11.3.1-522**, and FIG. **11.3.1-9** is a cross-section view illustration taken along line C-C of FIG. **11.3.1-8** showing the housing body **11.3.1-534**. The housing body **11.3.1-534** includes a base portion **11.3.1-852**, an optical pathway opening **11.3.1-853** that is formed through the base portion **11.3.1-852**, a peripheral wall **11.3.1-854** that extends around the optical pathway opening **11.3.1-853**.

[2157] The base portion **11.3.1-852** extends generally perpendicular to the optical axis **11.3.1-521** of the optical module **11.3.1-520**. The base portion **11.3.1-852** may incorporate features that allow attachment of other components to the optical module housing assembly **11.3.1-522**. As one example, the display assembly **11.3.1-524** may be attached to the base portion **11.3.1-852** (e.g., by fasteners or adhesives). As another example, the eye camera **11.3.1-530** may be attached to the base portion **11.3.1-852** (e.g., by fasteners or adhesives).

[2158] The peripheral wall **11.3.1-854** extends outward from the base portion **11.3.1-852** in a direction that is generally toward the user and generally aligned with the optical axis **11.3.1-521** of the optical module **11.3.1-520**. As viewed along the optical axis **11.3.1-521**, the shape and size of the peripheral wall **11.3.1-854** is similar to that of the outer periphery of the lens **11.3.1-526**, since the peripheral wall **11.3.1-854** is part of the structure that supports and retains the lens **11.3.1-526**, as will be described further herein. A vent port **11.3.1-855** is formed through the peripheral wall **11.3.1-854** and may extend, for example, between inner and outer surfaces of the peripheral wall **11.3.1-854** in a direction that is generally perpendicular to the optical axis **11.3.1-521** of the optical module **11.3.1-520**. An electrical port **11.3.1-856** is formed through the peripheral wall **11.3.1-854** and may extend, for example, between inner and outer surfaces of the peripheral wall **11.3.1-854** in a direction that is generally perpendicular to the optical axis **11.3.1-521** of the optical module **11.3.1-520**.

[2159] A base surface **11.3.1-857** is defined on the base portion **11.3.1-852** and is located inward from the peripheral wall **11.3.1-854**. The base surface **11.3.1-857** is adjacent to and extends around the optical pathway opening **11.3.1-853**, which is an opening that is defined by the housing body **11.3.1-534** to allow light to travel from the display assembly **11.3.1-524** to the lens **11.3.1-526**. A camera opening **11.3.1-858** is formed through the base surface **11.3.1-857** and is adjacent to, but separate from, the optical pathway opening **11.3.1-853**. The camera opening **11.3.1-858** extends through the base surface **11.3.1-857** in a direction that is generally toward the user. As examples the

camera opening **11.3.1-858** may extend through the base surface **11.3.1-857** in a direction that is generally aligned with the optical axis **11.3.1-521** of the optical module **11.3.1-520**, or within 45 degrees of parallel to the optical axis **11.3.1-521** of the optical module **11.3.1-520**.

[2160] FIG. **11.3.1-10** is a front view illustration that shows the retainer **11.3.1-536** of the optical module housing assembly **11.3.1-522**, and FIG. **11.3.1-11** is a cross-section view illustration taken along line D-D of FIG. **11.3.1-10** showing the retainer **11.3.1-536**. The retainer **11.3.1-536** includes a peripheral wall **11.3.1-1060** that extends around an optical pathway opening **11.3.1-1061**. The peripheral wall **11.3.1-1060** includes an upper inner periphery portion **11.3.1-1062** that borders and extends around the optical pathway opening **11.3.1-1061**. The upper inner periphery portion **11.3.1-1062** is configured to receive the lens body **11.3.1-648** of the lens **11.3.1-526**. Channels **11.3.1-1063** are formed in the upper inner periphery portion **11.3.1-1062** and are open to the optical pathway opening **11.3.1-1061**. The size and position of the channels **11.3.1-1063** corresponds to the size and position of the projections **11.3.1-649** of the lens **11.3.1-526** such that the projections **11.3.1-649** can be received in the channels **11.3.1-1063** to secure the lens **11.3.1-526** relative to the housing body **11.3.1-534** and restrain relative movement. The peripheral wall **11.3.1-1060** includes a lower inner periphery portion **11.3.1-1064** that borders and extends around the optical pathway opening **11.3.1-1061**. The lower inner periphery portion **11.3.1-1064** is configured for connection to the peripheral wall **11.3.1-854** of the housing body **11.3.1-534**.

[2161] FIG. **11.3.1-12** is a front view illustration that shows the infrared emitter **11.3.1-532**. The infrared emitter **11.3.1-532** includes a flexible circuit **11.3.1-1266**, emissive components **11.3.1-1267**, an electrical connector **11.3.1-1268**, and a sealing element **11.3.1-1269**. The flexible circuit **11.3.1-1266** is a flexible substrate that has electrical conductors formed on it. The flexible substrate may be nonconductive polymer film. The electrical conductors may be conductive traces formed from copper. As an example, the flexible circuit **11.3.1-1266** may be formed by multiple layers of nonconductive polymer film with conductive traces formed between adjacent layers of the film. As will be explained further herein, the shape of the flexible circuit **11.3.1-1266** may be arranged such that it conforms to the shape of a portion of the optical module housing assembly **11.3.1-522** such that the infrared emitter may be located in or connected to the optical module housing assembly **11.3.1-522**. In the illustrated example, the flexible circuit **11.3.1-1266** has a c-shaped configuration that allows the flexible circuit **11.3.1-1266** to extend around the optical axis **11.3.1-521** of the optical module **11.3.1-520** so that the emissive components **11.3.1-1267** may be arranged around the optical axis **11.3.1-521** in an array without blocking the optical path (pathway along which light may travel) between the display assembly **11.3.1-524** and the lens **11.3.1-526** of the optical module **11.3.1-520**.

[2162] The emissive components **11.3.1-1267** are components that are configured to emit infrared radiation within one or more wavelength bands. The infrared radiation that is emitted by the emissive components **11.3.1-1267** and reflected by the user's eye may be imaged by the eye camera **11.3.1-530** for use in imaging tasks.

[2163] The emissive components **11.3.1-1267** may be for example, infrared light emitting diodes. In one implementation, the emissive components **11.3.1-1267** include a first group of components that are configured to emit infrared radiation in a first wavelength band and a second group of components that are configured to emit infrared radiation in a second wavelength band. The first and second wavelength bands may correspond to different imaging tasks. As an example, the first wavelength band may be configured for use in biometric identification by iris scanner (e.g., a wavelength band including 850 nanometers), and the second wavelength band may be configured for use in eye gaze direction tracking (e.g., a wavelength band including 940 nanometers).

[2164] The electrical connector **11.3.1-1268** of the infrared emitter **11.3.1-532** is a standard component of any suitable type that allows connection to other components to provide electrical power and, optionally, operating commands, to the infrared emitter **11.3.1-532**. The sealing element **11.3.1-1269** is formed on the flexible circuit **11.3.1-1266** between the electrical connector **11.3.1-**

1268 and the emissive components **11.3.1-1267**. As best seen in FIG. **11.3.1-13**, which is a cross-section view illustration showing the flexible circuit and a portion of the peripheral wall **11.3.1-854** of the housing body **11.3.1-534** of the optical module housing assembly **11.3.1-522**, the flexible circuit **11.3.1-1266** extends through and is surrounded by the sealing element **11.3.1-1269**. The sealing element **11.3.1-1269** is formed from a resilient flexible material that is configured to engage a portion of the optical module housing assembly **11.3.1-522** to allow the flexible circuit **11.3.1-1266** to exit the interior of the optical module housing assembly **11.3.1-522** without providing a pathway along which foreign particles (e.g., dust particles) may enter the interior of the optical module housing assembly **11.3.1-522**. As an example, the sealing element **11.3.1-1269** may be formed from silicone that is overmolded onto the flexible circuit **11.3.1-1266** such that the flexible circuit extends through the sealing element **11.3.1-1269**. In the illustrated example, the flexible circuit **11.3.1-1266** extends through the electrical port **11.3.1-856** of the housing body **11.3.1-534** such that the sealing element **11.3.1-1269** is located in the electrical port **11.3.1-856** and is engaged with the housing body **11.3.1-534** and the electrical port **11.3.1-856** to define a seal and occupy the electrical port **11.3.1-856** to prevent entry of foreign particles.

[2165] FIG. **11.3.1-14** is a cross section view illustration that shows the optical module **11.3.1-520**.

[2166] The lens **11.3.1-526** is disposed between the housing body **11.3.1-534** and the retainer **11.3.1-536**. The housing body **11.3.1-534** is connected to the retainer **11.3.1-536** such that the lens **11.3.1-526** is located between the housing body **11.3.1-534** and the retainer **11.3.1-536**. Thus, the housing body **11.3.1-534** and the retainer **11.3.1-536** engage the lens **11.3.1-526** such that the lens **11.3.1-526** is restrained from moving relative to the housing body **11.3.1-534** and the retainer **11.3.1-536**. To protect the lens **11.3.1-526** from damage (e.g., if the head-mounted device **11.3.1-100** is dropped), a layer of adhesive may be present between the lens **11.3.1-526** and portions of the housing body **11.3.1-534** and/or the retainer **11.3.1-536**. The adhesive that is used for this purposes is strong to secure the lens **11.3.1-526** in a desired alignment and is flexible and elastic to cushion the lens **11.3.1-526** in the event of vibration or impact shock, and to allow the lens **11.3.1-526** to return to its original position.

[2167] The vent port **11.3.1-855** is formed through the peripheral wall **11.3.1-854** of the housing body **11.3.1-534** and allows air to enter and exit an internal space **11.3.1-1470** of the optical module **11.3.1-520**. The internal space **11.3.1-1470** is defined within the optical module housing assembly **11.3.1-522** by the housing body **11.3.1-534** and the retainer **11.3.1-536** and between the lens **11.3.1-526** and the display assembly **11.3.1-524**. The internal space **11.3.1-1470** is sealed from the outside environment except at the vent port **11.3.1-855**. The vent port **11.3.1-885** is a passage that allows air to travel between the internal space **11.3.1-1470** and the outside environment that is located around the optical module **11.3.1-520**. By allowing air to enter and exit the internal space **11.3.1-1470**, air pressure within the internal space **11.3.1-1470** remains at or near ambient (e.g., outside the optical module **11.3.1-520**) air pressure. To exclude foreign particles from the internal space **11.3.1-1470**, a filter element **11.3.1-1471** is connected to the vent port **11.3.1-885** such that any air that passes through the vent port **11.3.1-855** must pass through filter element **11.3.1-1471**. The filter element **11.3.1-1471** is configured to restrain foreign particles from entering the internal space through the vent port **11.3.1-885** (e.g., by preventing entry of foreign particles that are larger than a pore size of the filter material). As examples, the filter element **11.3.1-1471** may be located in or on the vent port **11.3.1-855**. The filter element **11.3.1-1471** has a small pore size that is intended to exclude small particles (e.g., dust particles) from the internal space **11.3.1-1470**. As one example, the filter element **11.3.1-1471** may be formed from a polytetrafluoroethylene (PTFE) filter material. To capture particles that are present inside the internal space **11.3.1-1470**, a dust trap **11.3.1-1472** may be located in the internal space **11.3.1-1470**, for example, connected to an internal surface of the housing body **11.3.1-534**. The dust trap **11.3.1-1472** is configured to retain foreign particles on its surface, so that the foreign particles do not instead settle on surfaces where they may cause an optical aberration. As an example, the dust trap **11.3.1-1472** may be an adhesive element, such as a

sheet coated in adhesive material, to which airborne particles that are inside the internal space

11.3.1-1470 may become affixed, which prevents the particles from attaching to the display assembly **11.3.1-524**, or the lens **11.3.1-526**, which could cause optical aberrations that are perceptible to the user (e.g., a visual artifact similar to a dead pixel).

[2168] The eye camera **11.3.1-530** is a still image camera or video camera that is configured to obtain images. When in use, the images that are obtained by the eye camera **11.3.1-530** include a visual representation of part of or all of the user's eye, so that the obtained images may be used for biometric identification (e.g., verifying the identity of the user based on an image of the user's eye) and gaze tracking. In the implementations that are discussed herein, the eye camera **11.3.1-530** is sensitive to infrared light (i.e., electromagnetic radiation in the infrared portion of the electromagnetic spectrum). Thus, the eye camera **11.3.1-530** may be configured to obtain images that show reflected portions of the infrared radiation that is emitted by the infrared emitter **11.3.1-532**, and these reflected portions of infrared radiation, as represented in the images, are useful for observing and identifying features of the user's eye, which may be done using a machine vision-based system that is implemented in software that is executed by the head-mounted device **11.3.1-100**. In alternative implementations, the eye camera **11.3.1-530** may instead be implemented using a visible spectrum camera or may be supplemented using the visible spectrum camera in addition to an infrared spectrum camera.

[2169] The eye camera **11.3.1-530** is connected to the housing body **11.3.1-534** of the optical module housing assembly **11.3.1-522** adjacent to the camera opening **11.3.1-858** of the housing body **11.3.1-534** such that an optical axis **11.3.1-1431** of the eye camera **11.3.1-530** extends through the camera opening **11.3.1-858**. In the illustrated example, the eye camera **11.3.1-530** is oriented such that an optical axis **11.3.1-1431** of the eye camera **11.3.1-530** is substantially aligned with the optical axis **11.3.1-521** of the optical module **11.3.1-520**. However, the eye camera **11.3.1-530** is positioned near an outer periphery of the lens **11.3.1-526** and is therefore offset and outward from the optical axis **11.3.1-521** of the optical module **11.3.1-520**. Thus, the housing body **11.3.1-534** and/or the eye camera **11.3.1-530** may be configured (e.g., by an inclined mounting surface) such that the optical axis **11.3.1-1431** of the eye camera **11.3.1-530** is angled toward the optical axis **11.3.1-521** of the optical module **11.3.1-520**, as shown in FIG. **11.3.1-15**, which is a cross-section view illustration that shows the optical module **11.3.1-520** according to an alternative implementation.

[2170] Returning to FIG. **11.3.1-14**, in some implementations, a fiducial marker **11.3.1-1465** may be formed on the lens **11.3.1-526**. The fiducial marker **11.3.1-1465** is any manner of marking that can be perceived and located in images obtained by the eye camera **11.3.1-530**. The fiducial marker **11.3.1-1465** is visible in images obtained by the eye camera **11.3.1-530** for use in calibration. The head-mounted device **11.3.1-100** is calibrated to account for manufacturing conditions, user attributes, and/or other factors that may cause visual aberrations. During an initial calibration, the position of the fiducial marker **11.3.1-1465** is determined and stored. The lens **11.3.1-526** may shift with respect to other components, such as the optical module housing assembly **11.3.1-522**, for example, if the head-mounted device **11.3.1-100** is dropped. The changed position of the lens **11.3.1-526** can be identified by comparing the position of the lens **11.3.1-526** in images obtained by the eye camera **11.3.1-530** with the position of the lens **11.3.1-526** in the images that was obtained at the time of calibration. In response to determining that the lens position has changed, calibration is performed again to address any visual aberrations that may have resulted from the shift in position of the lens **11.3.1-526**.

[2171] The infrared emitter **11.3.1-532** is located on the base surface **11.3.1-857** of the housing body **11.3.1-534** and extends around the optical axis **11.3.1-521** within the internal space **11.3.1-1470** that is defined within the optical module housing assembly **11.3.1-522** by the housing body **11.3.1-534** and the retainer **11.3.1-536** and between the lens **11.3.1-526** and the display assembly **11.3.1-524**. The display assembly **11.3.1-524** is connected to the housing body **11.3.1-534** of optical

module housing assembly **11.3.1-522** adjacent to the optical pathway opening **11.3.1-853** of the housing body **11.3.1-534**.

[2172] In one implementation, the optical module **11.3.1-520** includes the optical module housing assembly **11.3.1-522**, the display assembly **11.3.1-524**, the lens **11.3.1-526**, and the eye camera **11.3.1-530**. The lens **11.3.1-526** is positioned at a first end of the optical module housing assembly **11.3.1-522**, the display assembly and the eye camera **11.3.1-530** are positioned at a second end of the optical module housing assembly **11.3.1-522**, and the internal space **11.3.1-1470** is defined within the optical module housing assembly **11.3.1-522** between the first end and the second end. The lens **11.3.1-526** is positioned such that it is able to obtain images of the user's eye through the lens **11.3.1-526**. The lens **11.3.1-526** may be connected to the optical module housing assembly **11.3.1-522** such that it is positioned adjacent to the display assembly **11.3.1-524**, such as in a side-by-side arrangement with respect to the display assembly **11.3.1-524**.

[2173] In one implementation, the optical module **11.3.1-520** includes the optical module housing assembly **11.3.1-522**, the display assembly **11.3.1-524**, the lens **11.3.1-526**, and the infrared emitter **11.3.1-532**. The lens **11.3.1-526** is positioned at a first end of the optical module housing assembly **11.3.1-522**, the display assembly is positioned at a second end of the optical module housing assembly **11.3.1-522**, and the internal space **11.3.1-1470** is defined within the optical module housing assembly **11.3.1-522** between the first end and the second end. The infrared emitter **11.3.1-532** is positioned in the internal space **11.3.1-1470** between the lens **11.3.1-526** and the display assembly **11.3.1-524**. The infrared emitter **11.3.1-532** is positioned such that is able to project infrared radiation onto the user's eye through the lens **11.3.1-526**. The optical module **11.3.1-520** also includes the eye camera **11.3.1-530**, which is connected to the optical module housing assembly **11.3.1-522** such that the infrared emitter **11.3.1-532** is positioned between (e.g., along the optical axis **11.3.1-521**) the eye camera **11.3.1-530** and the lens **11.3.1-526**.

[2174] The lens **11.3.1-526** may be connected to the optical module housing assembly **11.3.1-522** such that it is positioned adjacent to the display assembly **11.3.1-524**, such as in a side-by-side arrangement with respect to the display assembly **11.3.1-524**.

[2175] In the implementation shown in FIG. **11.3.1-14**, the infrared emitter **11.3.1-532** is located on the base surface **11.3.1-857** of the housing body **11.3.1-534** in the internal space **11.3.1-1470**. FIG. **11.3.1-16** is a cross-section view illustration that shows the optical module **11.3.1-520** according to an alternative implementation in which the infrared emitter **11.3.1-532** is located outside of the housing body **11.3.1-534** of the optical module housing assembly **11.3.1-522**. In this implementation, the infrared emitter **11.3.1-532** is connected (e.g., by an adhesive) to an exterior surface of the housing body **11.3.1-534** such that it is positioned adjacent to and extends around the display assembly **11.3.1-524**.

[2176] An infrared-transmissive panel **11.3.1-1673** is formed in the housing body **11.3.1-534** to allow infrared radiation that is emitted by the infrared emitter **11.3.1-532** to travel through the optical module housing assembly **11.3.1-522** and the lens **11.3.1-526**. The infrared-transmissive panel **11.3.1-1673** is formed from a material that allows infrared radiation to pass through it without significant losses. As examples, the infrared-transmissive panel **11.3.1-1673** may be formed from glass or from an infrared transmissive plastic. In the illustrated example, the infrared-transmissive panel **11.3.1-1673** extends through an aperture that is formed through the base surface **11.3.1-857**. The infrared-transmissive panel **11.3.1-1673** may be a single panel that extends along the base surface **11.3.1-857** adjacent to all of the emissive components **11.3.1-1267** of the infrared emitter **11.3.1-532**, or may be multiple panels that extend through separate apertures that are formed through the base surface **11.3.1-857** adjacent to individual ones of the emissive components **11.3.1-1267**. In some implementations, the infrared-transmissive panel **11.3.1-1673** may be omitted in favor of forming part or all of the optical module housing assembly **11.3.1-522** (e.g., the housing body **11.3.1-534**) from an infrared-transmissive material.

[2177] In the examples shown in FIGS. **11.3.1-14-16**, the optical module **11.3.1-120** is shown as

including a single eye camera, which is represented by the eye camera **11.3.1-530**. The optical module **11.3.1-120** could instead include more than one eye camera (e.g., two eye cameras), with each of the eye cameras being configured to obtain images showing infrared radiation that is reflected from the eye of the user. The eye cameras are located at different locations (e.g., opposite lateral sides of the eye of the user) and may be oriented at different angular orientations. The images output by multiple eye cameras may provide a more complete view of the eye of the user. [2178] FIG. **11.3.1-17** is a side-view illustration that shows the display module **11.3.1-542** according to an implementation. The display module **11.3.1-542** includes a silicon wafer **11.3.1-1775** and a display element layer **11.3.1-1776** (e.g., an organic light-emitting diode layer) that is located on the silicon wafer **11.3.1-1775**. The display element layer **11.3.1-1776** may be covered by a glass layer **11.3.1-1777**. A display connector **11.3.1-1778** includes a first portion **11.3.1-1779** and a second portion **11.3.1-1780**. The first portion **11.3.1-1779** of the display connector **11.3.1-1778** is a flexible connector (e.g., a two-layer flexible connector) that is connected to silicon wafer **11.3.1-1775** by an electrical connection **11.3.1-1781** that connects individual conductors formed on the silicon wafer **11.3.1-1775** with individual conductors formed on the first portion **11.3.1-1779** of the display connector **11.3.1-1778**. As an example, the electrical connection **11.3.1-1781** may include an anisotropic film that bonds the display connector **11.3.1-1778** to the silicon wafer **11.3.1-1775** while allowing electrical communication.

[2179] The second portion **11.3.1-1780** of the display connector **11.3.1-1778** is a multi-layer (e.g., six layer) flexible connector, of the type commonly referred to as a “rigid flex” connector. The second portion **11.3.1-1780** may include a cavity **11.3.1-1782** that is defined by removal of one or more of the layers of the multi-layer structure of the second portion **11.3.1-1780**. A driver integrated circuit **11.3.1-1783** is located in the cavity **11.3.1-1782** in order to protect the driver integrated circuit **11.3.1-1783**. The function of the driver integrated circuit **11.3.1-1783** is to receive display signals in a first format (e.g., encoded or multiplexed) and interpret the signals into a second format that is usable by the display element layer **11.3.1-1776** of the display module **11.3.1-542** to output images. A connector **11.3.1-1784** (e.g., a micro-coaxial connector) may be located on and electrically connected to the second portion **11.3.1-1780** of the display connector **11.3.1-1778** in order to connect the display module **11.3.1-542** to other components (e.g., to a computing device that provides content to be displayed).

[2180] FIG. **11.3.1-18** is a top-view illustration that shows interpupillary distance adjustment mechanisms **11.3.1-1885** that each support one of the optical modules **11.3.1-520** (i.e., left and right optical modules) with respect to the device housing **11.3.1-102**. The interpupillary distance adjustment mechanisms **11.3.1-1885** are an example of an interpupillary distance adjustment assembly that is configured to adjust a distance between the optical modules **11.3.1-520** that display content to the left eye and the right eye of the user, in order to match the spacing between the optical modules **11.3.1-520** with the spacing between the user's eyes.

[2181] The optical modules **11.3.1-520** may be supported such that the optical axis **11.3.1-521** of each of the optical modules **11.3.1-520** extends generally in a front-to-back direction of the device housing **11.3.1-102**. The interpupillary distance adjustment mechanisms **11.3.1-1885** include support rods **11.3.1-1886** and actuator assemblies **11.3.1-1887** that are configured to cause movement of the optical modules **11.3.1-520** along the support rods **11.3.1-1886** in response to a control signal. The actuator assemblies **11.3.1-1887** may include conventional motion control components such as electric motors that are connected to the optical modules **11.3.1-520** by components such as lead screws or belts to cause movement. Mounting brackets **11.3.1-1888** may be connected to the optical modules **11.3.1-520** such that the support rods **11.3.1-1886** are connected to the mounting brackets **11.3.1-1888**, such as by extending through apertures **11.3.1-1889** that are formed through the mounting brackets **11.3.1-1888**. The interpupillary distance adjustment mechanisms **11.3.1-1885** may also include biasing elements such as springs that are engaged with the mounting brackets **11.3.1-1888** to reduce or eliminate unintended motion of the

mounting brackets **11.3.1-1888** and/or the optical modules **11.3.1-520** with respect to the support rods **11.3.1-1886**. The support rods **11.3.1-1886** may be angled relative to a lateral (e.g., side-to-side) dimension of the device housing **11.3.1-102** such that they move toward the user as they move outward. As an example, the support rods may be angled by five degrees relative to the lateral dimension.

[2182] FIG. **11.3.1-19** is a side view illustration that shows one of the interpupillary distance adjustment mechanisms **11.3.1-1885**. The support rods **11.3.1-1886** may include upper and lower support rods for each of the optical modules that support the optical modules **11.3.1-520** such that the optical axis **11.3.1-521** of each optical module **11.3.1-520** is angled slightly downward, such as by five degrees. Springs **11.3.1-1990** (e.g., leaf springs) may be seated in the apertures **11.3.1-1889** of the mounting brackets **11.3.1-1888** and located forward from the support rods **11.3.1-1886** to bias the optical modules **11.3.1-520** toward the user.

[2183] FIG. **11.3.1-20** is a top-view cross-section illustration that shows front-facing cameras **11.3.1-2091** that are supported by each of the optical modules **11.3.1-520**. Openings or optically-transmissive panels **11.3.1-2092** (e.g., clear plastic) are included in the device housing **11.3.1-102** such that the front-facing cameras **11.3.1-2091** are able to obtain images of the surrounding environment through the optically-transmissive panels **11.3.1-2092**. A single panel or separate panels may be used for the optically-transmissive panels **11.3.1-2092**, and as such the device housing **11.3.1-102** may include one or more of the optically-transmissive panels **11.3.1-2092**. Thus, the device housing **11.3.1-102** may include one or more of the optically-transmissive panels **11.3.1-2092** through which the front-facing cameras may obtain images of an environment from a point of view that simulates the point of view of the user.

[2184] The front-facing cameras **11.3.1-2091** may be connected to and supported by a corresponding one of the optical modules **11.3.1-520**. The front-facing cameras **11.3.1-2091** may be positioned such that they are located on and substantially aligned with the optical axis **11.3.1-521** of a corresponding one of the optical modules **11.3.1-520** (e.g., the optical axes of the front-facing cameras **11.3.1-2091** may be substantially aligned with the optical axes of the optical modules **11.3.1-520**). The front-facing cameras **11.3.1-2091** are oriented away from the user and are supported such that they are moved by the interpupillary distance adjustment mechanisms **11.3.1-1885**. Accordingly, when the user adjusts the interpupillary distance between the optical modules **11.3.1-520**, the distance between the front-facing cameras **11.3.1-2091** is also adjusted. Thus, images from the front-facing cameras **11.3.1-2091**, when displayed to the user, have been captured at the user's own interpupillary distance and therefore are presented more accurately in stereo vision. Thus, in some implementations, the optical axis of a first one of the front-facing cameras **11.3.1-2091** is aligned with an optical axis of a first one of the optical modules **11.3.1-520** and an optical axis of a second one of the front-facing cameras **11.3.1-2091** is aligned with an optical axis of a second one of the optical modules **11.3.1-520**. Thus, in some implementations, a first one of the front-facing cameras **11.3.1-2091** is connected in a fixed relationship with respect to a first one of the optical modules **11.3.1-520**, and a second one of the front-facing cameras **11.3.1-2091** is connected in a fixed relationship with respect to a second one of the optical modules **11.3.1-520**. Thus, in some implementations, the interpupillary distance adjustment mechanisms a first spacing between an optical axis of a first one of the optical modules **11.3.1-520** and an optical axis of a second one of the optical modules **11.3.1-520** generally equal to a second spacing between an optical axis of a first one of the front-facing cameras **11.3.1-2091** and an optical axis of a second one of the front facing cameras **11.3.1-2091** during adjustment of the distance between the optical modules **11.3.1-520**.

[2185] FIG. **11.3.1-21** is an illustration that shows connection of the eye camera **11.3.1-530** and the infrared emitter **11.3.1-532** to a computing device **11.3.1-2193** by an optical module jumper board **11.3.1-2194**. The computing device **11.3.1-2193** may be, for example, a computing device that incorporates the processor **11.3.1-108** of the head-mounted device **11.3.1-100**. The optical module

jumper board **11.3.1-2194** has a data connection to the computing device **11.3.1-2193** over which signals and data to and from the eye camera **11.3.1-530** and the infrared emitter **11.3.1-532** are transmitted. The optical module jumper board **11.3.1-2194** also has separate data connections to each of the eye camera **11.3.1-530** and the infrared emitter **11.3.1-532**. Additional components could be included in the optical module **11.3.1-520** and connected to the optical module jumper board **11.3.1-2194** by additional separate connections. The optical module jumper board **11.3.1-2194** may be mounted to the optical module **11.3.1-520**, and therefore, moves in unison with the optical module **11.3.1-520** during interpupillary distance adjustment. As a result, the number and size of electrical connections that are made to components that are not mounted to the optical module (e.g., the computing device **11.3.1-2193**) is decreased. The optical module jumper board **11.3.1-2194** may be, as examples, a rigid flex circuit board, a flexible circuit board, or a printed component board.

11.3.2: Cameras and Leds

[2186] FIG. **11.3.2-1** illustrates an example of an optical module **11.3.2-100** for use in an electronic device such as an HMD, including HMD devices described herein. As shown in one or more other examples described herein, the optical module **11.3.2-100** can be one of two optical modules within an HMD, with each optical module aligned to project light toward a user's eye. In this way, a first optical module can project light via a display screen toward a user's first eye and a second optical module of the same device can project light via another display screen toward the user's second eye.

[2187] In at least one example, the optical module **11.3.2-100** can include an optical frame or housing **11.3.2-102**, which can also be referred to as a barrel or optical module barrel. The optical module **11.3.2-100** can also include a display **11.3.2-104**, including a display screen or multiple display screens, coupled to the housing **11.3.2-102**. The display **11.3.2-104** can be coupled to the housing **11.3.2-102** such that the display **11.3.2-104** is configured to project light toward the eye of a user when the HMD of which the display module **11.3.2-100** is a part is donned during use. In at least one example, the housing **11.3.2-102** can surround the display **11.3.2-104** and provide connection features for coupling other components of optical modules described herein.

[2188] In one example, the optical module **11.3.2-100** can include one or more cameras **11.3.2-106** coupled to the housing **11.3.2-102**. The camera **11.3.2-106** can be positioned relative to the display **11.3.2-104** and housing **11.3.2-102** such that the camera **11.3.2-106** is configured to capture one or more images of the user's eye during use. In at least one example, the optical module **11.3.2-100** can also include a light strip **11.3.2-108** surrounding the display **11.3.2-104**. In one example, the light strip **11.3.2-108** is disposed between the display **11.3.2-104** and the camera **11.3.2-106**. The light strip **11.3.2-108** can include a plurality of lights **11.3.2-110**. The plurality of lights can include one or more light emitting diodes (LEDs) or other lights configured to project light toward the user's eye when the HMD is donned. The individual lights **11.3.2-110** of the light strip **11.3.2-108** can be spaced about the strip **11.3.2-108** and thus spaced about the display **11.3.2-104** uniformly or non-uniformly at various locations on the strip **11.3.2-108** and around the display **11.3.2-104**.

[2189] In at least one example, the housing **11.3.2-102** defines a viewing opening **11.3.2-101** through which the user can view the display **11.3.2-104** when the HMD device is donned. In at least one example, the LEDs are configured and arranged to emit light through the viewing opening **11.3.2-101** and onto the user's eye. In one example, the camera **11.3.2-106** is configured to capture one or more images of the user's eye through the viewing opening **11.3.2-101**.

[2190] As noted above, each of the components and features of the optical module **11.3.2-100** shown in FIG. **11.3.2-1** can be replicated in another (e.g., second) optical module disposed with the HMD to interact (e.g., project light and capture images) of another eye of the user.

[2191] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.3.2-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.3.2-2-**

11.3.2-5 and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **11.3.2-2-11.3.2-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.3.2-1**.

[2192] FIG. **11.3.2-2** illustrates a cross-sectional view of an example of an optical module **11.3.2-200** including a housing **11.3.2-202**, display assembly **11.3.2-204** coupled to the housing **11.3.2-202**, and a lens **11.3.2-216** coupled to the housing **11.3.2-202**. In at least one example, the housing **11.3.2-202** defines a first aperture or channel **11.3.2-212** and a second aperture or channel **11.3.2-214**. The channels **11.3.2-212**, **11.3.2-214** can be configured to slidably engage respective rails or guide rods of an HMD device to allow the optical module **11.3.2-200** to adjust in position relative to the user's eyes for match the user's interpupillary distance (IPD). The housing **11.3.2-202** can slidably engage the guide rods to secure the optical module **11.3.2-200** in place within the HMD.

[2193] In at least one example, the optical module **11.3.2-200** can also include a lens **11.3.2-216** coupled to the housing **11.3.2-202** and disposed between the display assembly **11.3.2-204** and the user's eyes when the HMD is donned. The lens **11.3.2-216** can be configured to direct light from the display assembly **11.3.2-204** to the user's eye. In at least one example, the lens **11.3.2-216** can be a part of a lens assembly including a corrective lens removably attached to the optical module **11.3.2-200**. In at least one example, the lens **11.3.2-216** is disposed over the light strip **11.3.2-208** and the one or more eye-tracking cameras **11.3.2-206** such that the camera **11.3.2-206** is configured to capture images of the user's eye through the lens **11.3.2-216** and the light strip **11.3.2-208** includes lights configured to project light through the lens **11.3.2-216** to the users' eye during use.

[2194] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.3.2-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.3.2-1** and **11.3.2-3-11.3.2-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **11.3.2-1** and **11.3.2-3-11.3.2-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.3.2-2**.

[2195] FIG. **11.3.2-3** illustrates a close-up cutaway view of an example of an optical module **11.3.2-304** for use in HMD devices described herein. The illustrated example includes a display assembly **11.3.2-304**, a lens **11.3.2-316**, and a light strip **11.3.2-308** including a plurality of lights **11.3.2-310** coupled to the optical module barrel/housing **11.3.2-302**. In at least one example, the light strip **11.3.2-308** is secured to the housing **11.3.2-302** via an adhesive **11.3.2-318** strip or portion. The adhesive **11.3.2-318** can include a pressure sensitive adhesive, epoxy, or other adhesive. The display assembly **11.3.2-304** can include a display housing **11.3.2-319** coupling various layers, including display layers and covers of the display assembly **11.3.2-304** to the housing **11.3.2-302**. In at least one example, the display housing **11.3.2-319** of the display assembly **11.3.2-304** can be coupled to the optical module housing **11.3.2-302** via a dust seal component **11.3.2-320**. The dust seal component **11.3.2-320** can be configured to prevent dust and debris from the air and/or external environment from infiltrating the display assembly **11.3.2-304** and lens **11.3.2-316** and spaces there between. In at least one example, the dust seal component **11.3.2-320** can include an adhesive. In one example, the dust seal component **11.3.2-320** can include a foam. In one example, the dust seal component **11.3.2-320** can include a material block or portion disposed between the display assembly housing **11.3.2-319** and the optical module housing **11.3.2-302**.

[2196] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.3.2-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.3.2-1-11.3.2-2** and **11.3.2-4-11.3.2-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with

reference to FIGS. **11.3.2-1-11.3.2-2** and **11.3.2-4-11.3.2-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.3.2-3**.

[2197] FIG. **11.3.2-4** illustrates a plan view of a portion of an example of an optical module **11.3.2-400** for use in HMD devices described herein. The optical module **11.3.2-400** can include a structural frame or housing **11.3.2-402** defining a viewing opening **11.3.2-401** and a display screen **11.3.2-404** coupled to the housing **11.3.2-402**. The display screen **11.3.2-404** can be a part of a multi-component/layered display assembly. The display screen **11.3.2-404** can include or define an inner edge **11.3.2-422**, an outer edge **11.3.2-424** opposite the inner edge **11.3.2-422**, a lower edge **11.3.2-426** extending between the inner edge **11.3.2-422** and the outer edge **11.3.2-424** and an upper edge **11.3.2-428** opposite the lower edge **11.3.2-426** and also extending between the inner edge **11.3.2-422** and the outer edge **11.3.2-424**. The optical module **11.3.2-400** can also include a first camera **11.3.2-406a** and a second camera **11.3.2-406b** coupled to the housing **11.3.2-402**.

[2198] In at least one example, the first and second cameras **11.3.2-406a-b** can be positioned to capture images and visualize the user's pupil without being obstructed in view by other facial features of the user, including eyelids and eyelashes, cheeks protrusion, and the like. Accordingly, in at least one example the first camera **11.3.2-406a** can be disposed adjacent the inner edge **11.3.2-422** and the second camera **11.3.2-406b** can be disposed adjacent the lower edge **11.3.2-426**. In one example, the first camera **11.3.2-406a** can be disposed closer to the lower edge **11.3.2-426** than the upper edge **11.3.2-428**. In one example, the second camera **11.3.2-406b** can be disposed closer to the outer edge **11.3.2-424** than the inner edge **11.3.2-422**.

[2199] In at least one example, the optical module **11.3.2-400** can include an illumination strip **11.3.2-408** disposed on the housing **11.3.2-402** between the first and second cameras **11.3.2-406a-b** and the display screen **11.3.2-404**. The illumination strip **11.3.2-408** can be disposed around a periphery of the display screen **11.3.2-404**. In at least one example, the illumination strip **11.3.2-408** can include a plurality of lights **11.3.2-410**, which can include one or more types of lights or light sources, such as LEDs.

[2200] As noted above with reference to FIG. **11.3.2-1**, the optical modules described herein can be one of two optical modules of an HMD, each optical module configured to align and project light onto one eye of the user. Accordingly, the inner edge **11.3.2-422**, outer edge **11.3.2-424**, lower edge **11.3.2-426**, and upper edge **11.3.2-428** of the display screen **11.3.2-404** shown in FIG. **11.3.2-4** can each be one of a set of inner edges, outer edges, lower edges, and upper edges of two adjacent display screens within an HMD. In such an example, the inner edges of the adjacent display screens of adjacent optical modules within a single HMD device can be disposed between respective outer edges.

[2201] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.3.2-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.3.2-1-11.3.2-3** and **11.3.2-5** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **11.3.2-1-11.3.2-3** and **11.3.2-5** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.3.2-4**.

[2202] In the examples of optical modules shown and described with reference to FIGS. **11.3.2-1-511.3.2**—and elsewhere herein, a controller of the HMD can be electrically coupled to the first and second cameras **11.3.2-406a**, **11.3.2-406b** as well as to the plurality of lights **11.3.2-410** of the illumination strip **11.3.2-408**. The controller can be configured to determine a gaze direction of an eye of the user by analyzing a pattern of light reflected off the eye and projected by the plurality of lights **11.3.2-410**. In at least one example, determining the gaze direction can include assigning a first weight to a first pattern of reflected light from the plurality of lights **11.3.2-410** and captured by the first camera **11.3.2-410a** and assigning a second weight to a second pattern of reflected light

from the plurality of lights **11.3.2-410** and captured by the second camera **11.3.2-410b**. Then, the controller can determine a weighted average pattern from the first pattern and the second pattern. The first weight and the second weight can be set based on a position of the display screen relative to the user's eye.

[2203] The position of the display screen **11.3.2-404** relative to the eye of the user can be determined by a first image capture by the first camera **11.3.2-406a** and a second image captured by the second camera **11.3.2-406b**. In at least one example, the controller can be configured to change the first weight and the second weight as the position of the display screen **11.3.2-404** relative to the eye changes. That is, as the user moves the gaze direction of the eye such that the position of the pupil relative to the display screen **11.3.2-404** and the first and second cameras **11.3.2-406a-b** changes, the weight given to images taken by the first and second cameras **11.3.2-406a-b** can be changed to rely more heavily on the camera **11.3.2-406a-b** positioned for the best determination of gaze direction based on the new position of the pupil. In addition, changes to the overall position of the cameras **11.3.2-406a-b** and screen **11.3.2-404** relative to the user's eye, for example when donning and using the HMD device as the device slightly shifts around or is adjusted for comfort by the user over time during use, the weights can be changed to optimize gaze direction measurement and determination. Other factors, such as optical module adjustment for IPD adjustments and so forth can also affect the weight given to each camera **11.3.2-406a-b** during gaze detection determination.

[2204] FIG. **11.3.2-5** illustrates a partial cutaway view of an example of an optical module **11.3.2-500** including a display assembly **11.3.2-504** and a camera **11.3.2-506** mounted on or coupled with a housing **11.3.2-502**. The display assembly **11.3.2-504** can include a display screen disposed within, parallel to, or defining a first major plane **11.3.2-530**. The major plane **11.3.2-530** can define a first normal axis **11.3.2-532** perpendicular to the major plane **11.3.2-530**. The display screen of the display assembly **11.3.2-504** is configured to direct light in a direction parallel to the first normal axis **11.3.2-532**. The camera **11.3.2-506** can include a lens **11.3.2-534** disposed within, parallel to, or defining a second major plane **11.3.2-536** defining a second normal axis **11.3.2-538** perpendicular to the second major plane **11.3.2-536**.

[2205] In at least one example, the first normal axis **11.3.2-532** is disposed at a first angle relative to the second normal axis **11.3.2-538** such that the first normal axis **11.3.2-532** and the second normal axis **11.3.2-528** are non-parallel. The first camera **406a** of FIG. **11.3.2-4** illustrates a similar first lens **11.3.2-434** and the second camera **11.3.2-406b** includes a second lens **11.3.2-440**. In at least one example, the second lens **11.3.2-440** can define a normal axis disposed at a second angle relative to the first normal axis **11.3.2-532**. The first angle can be different than the second angle. The normal axis defined by the second lens **11.3.2-440** of the second camera **11.3.2-406b** can also be non-parallel, or disposed at an angle relative to, the second normal axis **11.3.2-538** of the other camera **11.3.2-406a** (or **11.3.2-506** illustrated in FIG. **11.3.2-5**). The first angle can be different than the second angle relative to the first major plane **530** defined by the display screen of the display assembly **11.3.2-504**. The angles of the cameras and lenses thereof can be configured to provide unobstructed and direct visualization of the user's pupil to capture images and track the user's gaze.

[2206] In at least one example, the cameras, including the camera **11.3.2-506** illustrated in FIG. **11.3.2-5** are disposed between the display assembly **11.3.2-504** and the viewing opening **11.3.2-501** defined by the housing **11.3.2-502** and/or the HMD frame.

[2207] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.3.2-5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **11.3.2-1-11.3.2-4** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **11.3.2-1-11.3.2-4** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.3.2-5**.

11.4: Display

11.4.1: Display System with Interchangeable Lens

[2208] FIG. **11.4.1-0** illustrates a view of an HMD **11.4.1-100** including a display assembly **11.4.1-102**. The display assembly **11.4.1-102** is described here in more detail in sections **11.4.1-11.4.3**.

[2209] Different users may have different refractive errors of their eyes, such that different users require different corrective lenses. Disclosed herein are implementations of display systems. In one implementation, a display system includes a display, a removable lens assembly, and a lens detection sensor. The removable lens is removably coupleable to the display. The lens detection sensor detects the removable lens assembly coupled to the display. The display system may further include a head-mounted display unit that includes the display and the lens detection sensor. The display system may determine lens information from the removable lens with the lens detection sensor, and may provide an indicator of the removable lens according to the lens information.

[2210] In another implementation, a method is provided for operating a head-mounted display unit, which includes identifying a removable lens coupled to a display module of the head-mounted display unit, and providing an indication according to the identifying of the removable lens. The indication may be a configuration indication that differs between the removable lens and another removable lens coupleable to the display module. The method may further include identifying a user, while the indication may be a compatibility indication of compatibility between the removable lens and the user.

[2211] In another implementation, a display system includes a head-mounted display unit and a removable lens assembly. The head-mounted display unit includes a display module. The removable lens assembly includes a lens element and a frame coupled to the lens element. The removable lens assembly may be removably to the display module in a single orientation with a magnetic attachment features coupled to the frame and surrounding the lens element. The removable lens assembly may be removably coupleable to the display module with an interference fit, for example, with the removably lens include a compliant annular protrusion that receives a lens mount of the display module axially therein. The removable lens assembly may be removably coupleable to the display module with sprung latch mechanisms.

[2212] In an implementation, a display system includes a head-mounted display unit and a removable lens assembly. The head mounted display unit includes a display module and a corresponding mechanical coupling feature. The removable lens assembly includes a corrective lens element, a frame coupled to the corrective lens element, and a mechanical coupling feature coupled to the frame. The mechanical coupling feature is receivable by the corresponding mechanical coupling feature to removably couple the removable lens assembly to the head-mounted display unit.

[2213] The mechanical coupling feature may engage the corresponding mechanical coupling feature to prevent movement of the removable lens assembly relative to the head-mounted display unit transverse to, along, and/or about an optical axis of the corrective lens element. The removable lens assembly may further include a magnetic coupling feature spaced part from the mechanical coupling feature and by which the removable lens assembly is magnetically coupleable to the head-mounted display unit. The display module may include a lens mount that includes the corresponding mechanical coupling feature, and the removable lens assembly may be removably coupleable to the display module.

[2214] Disclosed herein are embodiments of display systems that include a head-mounted display unit and one or more interchangeable lenses. The interchangeable lenses may be configured according to characteristics of different eyes of different users (e.g., according to an eyeglass prescription). For example, one pair of interchangeable lenses may be associated with (e.g., prescribed for) one user, while another pair of interchangeable. Lenses may be associated with another user. The interchangeable lenses are mountable to the head-mounted display unit in an interchangeable manner such that different lenses may be mounted to the head-mounted display

unit to accommodate different users with different eye characteristics. The interchangeable lenses may be mounted to the head-mounted display unit in various different manners. The display system may also identify the interchangeable lenses mounted to the head-mounted display unit and/or identify the user and perform various operations in response thereto.

[2215] Referring to FIGS. **11.4.1-1** and **11.4.1-2**, a display system **11.4.1-100** generally includes a head-mounted display unit **11.4.1-110** and one or more removable lens assemblies **11.4.1-120**. The head-mounted display unit **11.4.1-110** generally includes a housing **11.4.1-112**, a head support **11.4.1-114**, one or more display modules **11.4.1-116** (e.g., two; one for each eye), and various other electronics (e.g., controller or computing device, communications interface, audio input and/or output devices, sensors, battery or other power electronics; see also FIG. **11.4.1-12**). The display system **11.4.1-100** is configured to provide a computer-generated reality (e.g., virtual reality or mixed reality) in which case the display system **11.4.1-100** may be considered a computer-generated reality system or computer-generated reality display system, a virtual reality system or a virtual reality display system, or a mixed reality display system or mixed reality system. The terms computer-generated reality, virtual reality, and mixed reality are discussed in further detail below.

[2216] The housing **11.4.1-112** contains or is otherwise coupled to the one or more display modules **11.4.1-116** and the other electronics. The housing **11.4.1-112** may also include various compliant components coupled thereto for engaging the face of the user to be supported thereon (not shown). The head support **11.4.1-114** is coupled to the housing **11.4.1-112** to support the head-mounted display unit **11.4.1-110** on a head of a user with the one or more display modules **11.4.1-116** in suitable positions relative to eyes of the user. As shown, the head support **11.4.1-114** may be configured as a strap or band that is configured to extend around the head of the user. The display module **11.4.1-116** is coupled to the housing **11.4.1-112** in a fixed position (as shown).

[2217] Referring to FIGS. **11.4.1-3A-3B**, the display module **11.4.1-116** generally includes a display **11.4.1-316a**, a primary lens **11.4.1-316b**, and a lens mount **11.4.1-316c**. The display **11.4.1-316a** is configured to display graphics to the user. The display **11.4.1-316a** may, for example, be a display screen, such as a liquid crystal display (LCD), organic light-emitting diode (OLED) display, or other suitable display. The display **11.4.1-316a** may be coupled to a circuit board or other support.

[2218] The primary lens **11.4.1-316b** is coupled to the display **11.4.1-316a**. The primary lens **11.4.1-316b** is positioned between the display **11.4.1-316a** and the eye of the user and refracts light emitted from the display **11.4.1-316a** before reaching the eye of the user. The primary lens **11.4.1-316b** may be omitted depending on configurations of the display **11.4.1-316a** and/or the removable lens assemblies **11.4.1-120**.

[2219] The lens mount **11.4.1-316c** is configured to couple to and support the removable lens assembly **11.4.1-120** between the display **11.4.1-316a** and the eye of the user (i.e., with the primary lens **11.4.1-316b** being between the display **11.4.1-316a** and the removable lens assembly **11.4.1-120**). As shown, the lens mount **11.4.1-316c** may additionally couple to and support the primary lens **11.4.1-316b** relative to the display **11.4.1-316a**, such as by being coupled to the circuit board or other support of the display **11.4.1-316a**. The lens mount **11.4.1-316c** may surround the primary lens **11.4.1-316b** (e.g., forming a bezel or barrel structure of the primary lens **11.4.1-316b**).

[2220] The removable lens assembly **11.4.1-120** generally includes a lens element **11.4.1-330** and a lens frame **11.4.1-340**. The lens frame **11.4.1-340** is coupled to the lens element **11.4.1-330** and is configured to removably couple to the lens mount **11.4.1-316c** for supporting the lens element **11.4.1-330** in a predetermined position on to the display module **11.4.1-116** (e.g., relative to the display **11.4.1-316a** and/or the primary lens **11.4.1-316b**). The lens frame **11.4.1-340** is, for example, coupled to an outer peripheral surface of the lens element **11.4.1-330**. Variations of the lens mount **11.4.1-316c** and interaction with the lens frame **11.4.1-340** are discussed in further detail below. The removable lens assembly **11.4.1-120** may also be referred to as an interchangeable lens assembly, an interchangeable lens, a removable lens, or a lens.

[2221] The lens element **11.4.1-330** may be a corrective lens, for example, to address refractive errors of the eye of the user, such as myopia, hypermetropia, and/or astigmatism. The removable lens assembly **11.4.1-120** and the lens element **11.4.1-330** may be associated with a particular user by having particular lens characteristics to address the particular refractive errors of the eye of the particular user (e.g., eye characteristics). Such lens characteristics and eye characteristics may each include sphere, cylinder, axis, and/or other parameters conventionally used to define eyeglass prescriptions or refractive errors of eyes of one or more users.

[2222] The lens characteristics of the lens element **11.4.1-330** and, thereby the removable lens assembly **11.4.1-120**, may also include a center (e.g., optical axis) that is to be aligned with a pupil of the user. Therefore, the removable lens assembly **11.4.1-120** may be mounted to the lens mount **11.4.1-316c** in a predetermined position (i.e., up/down, left/right, and rotational position) to ensure proper spatial positioning of the lens element **11.4.1-330** relative to the eye of the user.

[2223] The lens element **11.4.1-330** may, instead, be a non-corrective lens that protects the primary lens **11.4.1-316b** from contact by the user and/or debris, which might otherwise interfere with the user viewing the display **11.4.1-316a** (e.g., scratching or obstructing the view). For example, one pair of removable lens assemblies **11.4.1-120** may include lens elements **11.4.1-330** that are corrective for use by a particular user, while another pair of removable lens assemblies **11.4.1-120** may include lens elements **11.4.1-330** that are non-corrective for use by other users (e.g., users not requiring corrective lenses).

[2224] Referring to FIGS. **11.4.1-4-11.4.1-6**, an outer periphery of the removable lens assembly **11.4.1-120** may have different shapes. As shown in FIG. **11.4.1-4**, the removable lens assembly **11.4.1-120** has an outer periphery that is formed by the lens frame **11.4.1-340** and that is ovular (i.e., having curved left and right ends; as shown). The outer periphery may instead be circular (see lens frame **11.4.1-540** of lens assembly **11.4.1-520** in FIG. **11.4.1-5**) or may be rectangular (see lens frame **11.4.1-640** of lens assembly **11.4.1-620** in FIG. **11.4.1-6**). As discussed in further detail below, those lens assemblies having outer peripheries that are rotationally symmetric (e.g., being ovular, circular, rectangular, or other suitable shape) may be further configured to couple to the lens mount **11.4.1-316c** in different rotational positions about the optical axis of the lens element **11.4.1-330** for use with the display module **11.4.1-116**. Alternatively, the removable lens assemblies **11.4.1-120** may have outer peripheries that are rotationally asymmetric, such that the lens mount **11.4.1-316c** is coupleable in only one position for use with the display module **11.4.1-116**.

[2225] Referring again to FIG. **11.4.1-3A** and FIG. **11.4.1-4**, light is emitted by the display **11.4.1-316a** and passes through the primary lens **11.4.1-316b** and then through the lens element **11.4.1-330** before reaching the eye of the user. For example, light emitted from outer edges **11.4.1-360a** of the display **11.4.1-316a** may generally follow an outer light path **11.4.1-360** (e.g., a ray trace), whereby the light emitted from the display **11.4.1-316a** is refracted by the primary lens **11.4.1-316b** and is subsequently refracted by the lens element **11.4.1-330**. The outer light path **11.4.1-360** passes through the primary lens **11.4.1-316b** generally a passage location **11.4.1-360b**. The outer light path **11.4.1-360** enters an entry side **11.4.1-330a** of the lens element **11.4.1-330** at entry point **11.4.1-360c** and exits an exit side **11.4.1-330b** of the light path at an entry point **11.4.1-360c**. Moving through the lens element **11.4.1-330** from the entry side **11.4.1-330a** to the exit side **11.4.1-330b**, the outer light path **11.4.1-360** narrows, so as to focus toward the eye of the user.

[2226] Areas of the lens element **11.4.1-330** outside of the entry point **11.4.1-360c** and the exit point **11.4.1-360d** (i.e., radially outward toward edges of the lens element **11.4.1-330**) do not function to refract light toward the eye of the user and may be considered non-functional (e.g., non-refractive). Such non-functional areas of the lens element **11.4.1-330** may be removed and/or blocked by the lens frame **11.4.1-340**, thereby permitting different structural configurations of the lens frame **11.4.1-340**. As shown in FIG. **11.4.1-4**, the functional area of the lens element **11.4.1-330** may, as shown, be smaller than a functional area of the primary lens **11.4.1-316b** (e.g., smaller in area, width, and/or height than the functional the passage location **11.4.1-360b** on the entry side

11.4.1-330a at the entry point **11.4.1-360c** and/or on the exit side **11.4.1-330b** at the exit point **11.4.1-360d**). The functional area of the lens element **11.4.1-330** may also, as shown, be smaller than the display **11.4.1-316a** (e.g., smaller in area, width and/or height than the light emitting of the display **11.4.1-316a** on the entry side **11.4.1-330a** at the entry point **11.4.1-360c** and/or on the exit side **11.4.1-330b** at the exit point **11.4.1-360d**). The functional area of the primary lens **11.4.1-316b** may, as shown, be larger than the display **11.4.1-316a** (e.g., larger in area, width, and/or height than a light emitting area of the display **11.4.1-316a**). Additionally, the functional areas of the primary lens **11.4.1-316b** and the lens element **11.4.1-330** may allow for any eye camera **11.4.1-319** of the head-mounted display unit **11.4.1-110** (e.g., of the display module **11.4.1-116**) to observe the eye of the user (e.g., light moving in a direction opposite that emitted by the display **11.4.1-316a**).

[2227] Still referring to FIGS. **11.4.1-3A-3B**, an outer periphery of the lens element **11.4.1-330** is coupled to an inner periphery of the lens frame **11.4.1-340**, for example, being engaged thereby and/or adhered thereto. The outer periphery of the lens element **11.4.1-330** and the inner periphery of the lens frame **11.4.1-340** have corresponding shapes allowing coupling therebetween, such as being straight (e.g., having the same cross-sectional shape moving axially) or tapering (e.g., widening moving toward the display module **11.4.1-116**). The outer periphery of the lens element **11.4.1-330** may have generally the same size as the primary lens **11.4.1-316b** (e.g., having generally the same area, width, and/or height), which defines non-functional area of the lens element **11.4.1-330**.

[2228] Referring to FIGS. **11.4.1-7-11.4.1-9** and as referenced above, the non-functional area of the lens element **11.4.1-330** may be reduced with the lens element **11.4.1-330** being smaller than the primary lens **11.4.1-316b**.

[2229] As shown in FIG. **11.4.1-7**, a removable lens assembly **11.4.1-720** is a variation of the removable lens assembly **11.4.1-120** and generally includes a lens element **11.4.1-730** and a lens frame **11.4.1-740**. The lens element **11.4.1-730** is a variation of the lens element **11.4.1-330** differing by being smaller, such as by being smaller than the primary lens **11.4.1-316b** (e.g., smaller in area, width, and/or height). The lens frame **11.4.1-740** is a variation of the lens frame **11.4.1-340** differing by extending radially inward further than the lens frame **11.4.1-740**, such as by having an inner periphery that is smaller than the primary lens **11.4.1-316b** and/or the lens mount **11.4.1-316c** (e.g., being smaller in area, width, and/or height).

[2230] Referring to FIG. **11.4.1-8**, a removable lens assembly **11.4.1-820** is a variation of the removable lens assembly **11.4.1-120** and generally includes a lens element **11.4.1-830** and a lens frame **11.4.1-840**. The lens element **11.4.1-830** is a variation of the lens element **11.4.1-330** differing by having edges that are beveled (e.g., tapered) at an angle or curvature that more closely conforms to the angle at which the outer light path **11.4.1-360** (e.g., the angle of refraction) passes through the lens element **11.4.1-830**. For example, an outer peripheral surface of the lens element **11.4.1-830** may extend at an angle measured relative to the optical axis of the lens element that is greater than zero, such as between 15 and 75 degrees, such as between 30 and 60 degrees). Edges of the outer peripheral surface on entry and exit sides of the lens element **11.4.1-830** may be smaller than peripheral dimensions of the primary lens **11.4.1-316b** (e.g., being smaller in area, width, and/or height). The edge of the peripheral surface on the exit side of the lens element **11.4.1-830** may be smaller than the edge of the peripheral surface on the entry side thereof (e.g., reducing in size moving away from the primary lens **11.4.1-316b** and toward the eye of the user). The lens frame **11.4.1-840** is a variation of the lens frame **11.4.1-340** by extending radially inward further than the lens frame **11.4.1-340**, such as by having an inner periphery that is smaller than the primary lens **11.4.1-316b** on one or both of entry and exit sides of the lens element **11.4.1-830**. The lens frame **11.4.1-840** is coupled to the outer peripheral surface of the lens element **11.4.1-830** and may follow or otherwise accommodate the shape of the peripheral surface of the lens element **11.4.1-830** (e.g., having a matching bevel or contour).

[2231] Referring to FIG. **11.4.1-9** a removable lens assembly **11.4.1-920** is a variation of the

removable lens assembly **11.4.1-120** and generally includes a lens element **11.4.1-930** and a lens frame **11.4.1-940**. The lens element **11.4.1-930** is a variation of the lens element **11.4.1-330** differing by having non-functional areas of the exit side (e.g., or other concave side) forming a forward surface, which may be planar, that is coupled to the lens frame **11.4.1-940** and surrounds functional areas of the lens element **11.4.1-930**. The lens frame **11.4.1-940** is a variation of the lens frame **11.4.1-340** by being coupled to the forward surface of the exit side **11.4.1-930b** of the lens element **11.4.1-930**, and protruding inward from the lens mount **11.4.1-316c**.

[2232] Referring again to FIGS. **11.4.1-3-11.4.1-4**, the lens mount **11.4.1-316c** and the removable lens assembly **11.4.1-120** are cooperatively configured for the removable lens assembly **11.4.1-120** to removably couple to the lens mount **11.4.1-316c** to be supported thereby. As shown, the lens mount **11.4.1-316c** and the removable lens assembly **11.4.1-120** include cooperative coupling features **11.4.1-316d**, **11.4.1-322**, respectively. The cooperative coupling features **11.4.1-316d**, **11.4.1-322** are magnetic and may be referred to as magnetic coupling features. The magnetic coupling features **11.4.1-316d**, **11.4.1-322** are arranged at corresponding positions of the lens mount **11.4.1-316c** and the lens frame **11.4.1-340** of the removable lens assembly **11.4.1-120** for the lens mount **11.4.1-316c** to support the removable lens assembly **11.4.1-120** in the one or more predetermined positions (i.e., vertically, laterally, and rotationally) at which the lens element **11.4.1-330** is properly aligned with the eye of the user.

[2233] For each pair of the corresponding magnetic coupling features **11.4.1-316d**, **11.4.1-322**, one may be a permanent magnet, while the other is an attractor element (e.g., an attractor plate made of a ferromagnetic material) or another permanent magnet of suitable orientation. For example, the magnetic coupling features **11.4.1-316d** of the lens mount **11.4.1-316c** may be attractor plates, while the magnetic coupling features **11.4.1-322** of the removable lens assembly **11.4.1-120** are permanent magnets.

[2234] The magnetic coupling features **11.4.1-316d**, **11.4.1-322** are provided in suitable size, number, and location for the lens mount **11.4.1-316c** to adequately support the removable lens assembly **11.4.1-120** during use, while still allowing the removable lens assembly **11.4.1-120** to be easily removed by the user. For example, as shown, the lens mount **11.4.1-316c** and the removable lens assembly **11.4.1-120** may include six pairs of the magnetic coupling features **11.4.1-316d**, **11.4.1-322**, more or less.

[2235] The removable lens assembly **11.4.1-120** and the lens mount **11.4.1-316c** may be configured for the removable lens assembly **11.4.1-120** to be mounted to the lens mount **11.4.1-316c** in only one predetermined position. For example, the lens element **11.4.1-330** may have an optical axis that is to be positioned off-center relative to the lens mount **11.4.1-316c**, thereby requiring the removable lens assembly **11.4.1-120** to be coupled to the lens mount **11.4.1-316c** in one predetermined position. If the removable lens assembly **11.4.1-120** were instead coupled the lens mount **11.4.1-316c** in another position rotated therefrom, the optical axis would be moved relative to the lens mount **11.4.1-316c** and, thereby, relative to the display **11.4.1-316a** and the eye of the user, resulting in reduced image quality perceived by the user.

[2236] The magnetic coupling features **11.4.1-316d**, **11.4.1-322** may be configured to prevent incorrect positioning of the removable lens assembly **11.4.1-120** relative to the lens mount **11.4.1-316c**. For example, as shown in FIG. **11.4.1-4** with the magnetic coupling features **11.4.1-322** illustrated in dashed lines in FIG. **11.4.1-4** (e.g., being embedded, hidden, or otherwise coupled to the lens frame **11.4.1-340**), the magnetic coupling features **11.4.1-316d**, **11.4.1-322** may be positioned and/or distributed asymmetrically to prevent alignment of corresponding pairs of the magnetic coupling features **11.4.1-316d**, **11.4.1-322** and, thereby, prevent coupling of the removable lens assembly **11.4.1-120** to the lens mount **11.4.1-316c** in another position (e.g., rotated by 180 degrees). Instead or additionally, the magnetic characteristics of the magnetic coupling features **11.4.1-316d**, **11.4.1-322** (e.g., orientation and whether being an attractor plate or a permanent magnetic) may prevent magnetic coupling of those magnetic coupling features **11.4.1-**

316d that do not correspond to each other in another position (e.g., rotated orientation). For example, if the removable lens assembly **11.4.1-120** were rotated by 180 degrees, the magnetic coupling features **11.4.1-316d**, **11.4.1-322** that are aligned but do not correspond to each other may not attract each other (e.g., two attractor plates) or may repel each other (e.g., permanent magnets with common poles positioned adjacent each other).

[2237] The lens mount **11.4.1-316c** and the removable lens assembly **11.4.1-120** may be further configured to mechanically prevent misalignment thereof. For example, the lens mount **11.4.1-316c** and the removable lens assembly **11.4.1-120** may further include mechanical engagement features **11.4.1-316e**, **11.4.1-324**, respectively. The mechanical engagement features **11.4.1-316e**, **11.4.1-324** mechanically engage each other to ensure proper positioning and/or alignment between the lens mount **11.4.1-316c** and the removable lens assembly **11.4.1-120**. The mechanical engagement features **11.4.1-316e**, **11.4.1-324** may further engage each other to further prevent relative movement of the removable lens assembly **11.4.1-120** to the lens mount **11.4.1-316c** (e.g., in vertical and left-to-right directions relative to the optical axis, while the magnetic coupling features **11.4.1-316d**, **11.4.1-322** prevent movement in along the optical axis).

[2238] The mechanical engagement features **11.4.1-316e**, **11.4.1-324** may, for example, be a recess and a protrusion, respectively, that is received by the recess, or vice versa. If the removable lens assembly **11.4.1-120** were arranged in different position relative to the lens mount **11.4.1-316c**, the protrusion would instead engage another surface of the lens mount **11.4.1-316c** to prevent mating of the removable lens assembly **11.4.1-120** thereto. Alternatively, the opposed axially-facing (e.g., mating) surfaces of the display module **11.4.1-116** (e.g., the lens mount **11.4.1-316c**) and the removable lens assembly **11.4.1-120** (e.g., of the lens frame **11.4.1-340**) may have three-dimensional contours that prevent misalignment of the removable lens assemble **11.4.1-120** relative to the lens mount **11.4.1-316c**.

[2239] Instead of having only one mounting position, the removable lens assembly **11.4.1-120** and the lens mount **11.4.1-316c** may be configured for the removable lens assembly **11.4.1-120** to be mounted to the lens mount **11.4.1-316c** in more than one position. For example, the lens frame **11.4.1-340** and the lens mount **11.4.1-316c**, the coupling features, and/or the mechanical alignment features may be cooperatively configured for the removable lens assembly **11.4.1-120** to couple to the lens mount **11.4.1-316c** in two predetermined positions (e.g., by being two-fold rotationally symmetric) or more (e.g., three, or four) and to not couple to the lens mount **11.4.1-316c** in other positions.

[2240] Referring to FIG. **11.4.1-10A-11.4.1-11B**, variations of the removable lens assembly **11.4.1-120** and the lens mount **11.4.1-316c** are configured for mechanically coupling the removable lens assembly **11.4.1-120** to the lens mount **11.4.1-316c**.

[2241] As shown in FIGS. **11.4.1-10A-11.4.1-10B**, a display module **11.4.1-1016** includes a lens mount **11.4.1-1016c** to which removable lens assembly **11.4.1-1020** is removably coupleable with interfitting structures (e.g., forming an interference fit). The removable lens assembly **11.4.1-1020** includes the lens element **11.4.1-330** and a lens frame **11.4.1-1040**, which may be coupled to each other and otherwise configured in the manners described previously (see, e.g., FIGS. **11.4.1-3A-11.4.1-9**). The lens frame **11.4.1-1040** includes a barrel **11.4.1-1042** having an annular protrusion **11.4.1-1044** (e.g., a lip or circumferential lip). The barrel **11.4.1-1042** extends axially rearward (e.g., of the lens element **11.4.1-330**), while the annular protrusion **11.4.1-1044** protrudes radially inward therefrom. The lens frame **11.4.1-1040** may further define an annular recess **11.4.1-1046** (e.g., a circumferential channel; not labeled) extending radially outward from the annular protrusion **11.4.1-1044** and positioned axially outward thereof (i.e., away from the display **11.4.1-316a**). The annular protrusion **11.4.1-1044** and the annular recess **11.4.1-1046** extend circumferentially around the axis of the lens element **11.4.1-330** entirely or substantially entirely (e.g., 80% or more).

[2242] The lens mount **11.4.1-1016c** includes a barrel **11.4.1-1016d** having an annular protrusion

11.4.1-1016e. The barrel **11.4.1-1016d** extends axially forward, while the annular protrusion **11.4.1-1016e** protrudes radially outward therefrom. The annular protrusion **11.4.1-1016e** further defines an annular recess **11.4.1-1016f** (e.g., a circumferential channel) positioned axially forward and extending radially inward thereof. The annular protrusion **11.4.1-1016e** and the annular recess **1016f** defined thereby extend circumferentially around the axis of the lens element **11.4.1-330** entirely or substantially entirely.

[2243] The barrel **11.4.1-1042** of the lens frame **11.4.1-1040** is configured to receive the barrel **11.4.1-1016d** of the lens mount **11.4.1-1016c** therein, so as to removably couple the removable lens assembly **11.4.1-1020** to the lens mount **11.4.1-1016c**. The annular protrusion **11.4.1-1044** of the lens frame **11.4.1-1040** is received by and retained in the annular recess **11.4.1-1016f** of the lens mount **11.4.1-1016c**, while the annular protrusion **11.4.1-1016e** of the lens mount **11.4.1-1016c** is received by and retained in the annular recess **11.4.1-1046** of the lens frame **11.4.1-1040**.

[2244] An inner periphery of the annular protrusion **11.4.1-1044** of the lens frame **11.4.1-1040** is smaller than an outer periphery of the annular protrusion **11.4.1-1016e** of the lens mount **11.4.1-1016c**, such that axial surfaces of the annular protrusions **11.4.1-1044**, **11.4.1-1016e** overlap each other radially and engage each other axially to prevent unintended decoupling of the removable lens assembly **11.4.1-1020** from the display module **11.4.1-1016**. The annular protrusion **11.4.1-1044**, as may be the barrel **11.4.1-1042**, of the lens frame **11.4.1-1040** is formed of a compliant material, such as rubber or other polymer, that allows the inner periphery of the annular protrusion **11.4.1-1044** to expand and stretch over the annular protrusion **11.4.1-1016e** of the lens mount **11.4.1-1016c**. The lens frame **11.4.1-1040** being formed of, or otherwise including, the compliant material forming the annular protrusion **11.4.1-1044** may also provide a soft material that may incidentally contact or otherwise engage the face of the user. Alternatively, the barrel **11.4.1-1042** and/or the annular protrusion **11.4.1-1044** may form a flexure. In a still further alternative, the barrel **11.4.1-1042** and the annular protrusion **11.4.1-1044** of the removable lens assembly **11.4.1-1020** are rigid, while the barrel **11.4.1-1016d** and/or the annular protrusion **11.4.1-1016e** of the display module **11.4.1-1016** are compliant or form a flexure.

[2245] While the lens frame **11.4.1-1040** and the lens mount **11.4.1-1016c** have been shown and described with the lens mount **11.4.1-1016c** (e.g., the barrel **11.4.1-1016d** and the annular protrusion **11.4.1-1016e**) being received within the lens frame **11.4.1-1040**, the lens frame **11.4.1-1040** and the lens mount **11.4.1-1016c** may be arranged in an opposite manner with the lens frame **11.4.1-1040** receiving the lens mount **11.4.1-1016c**.

[2246] As shown in FIGS. **11.4.1-11A-11.4.1-11B**, a display module **11.4.1-1116** includes a lens mount **11.4.1-1116c** to which a removable lens assembly **11.4.1-1120** is removably coupled with sprung mechanisms. The removable lens assembly **11.4.1-1120** includes retractable latches **11.4.1-1122** (e.g., latch members) that are normally sprung radially outward for receipt into receptacles **11.4.1-1116d** of the lens mount **11.4.1-1116c** corresponding thereto. For example, the retractable latches **11.4.1-1122** extend axially rearward from a rear axial surface of a lens frame **11.4.1-1140** and include inner ends **11.4.1-1124** (e.g., hooked or outwardly protruding ends) that extend radially outward therefrom (e.g., hooked ends). When the retractable latches **11.4.1-1122** are retracted (e.g., by a user pressing on exposed ends of the protrusions), the retractable latches **11.4.1-1122** are axially insertable into or removable from the receptacles **11.4.1-1116d** in the forward axial surface of the lens mount **11.4.1-1116c** corresponding thereto. The inner ends of the retractable latches **11.4.1-1122** are received behind lips **11.4.1-1116e** of the receptacles **11.4.1-1116d**. Springs (shown schematically; not labeled) bias the retractable latches **11.4.1-1122** radially outward upon release by the user, such that the inner ends of the retractable latches **11.4.1-1122** are arranged behind and engage the lips **11.4.1-1116e** of the receptacles **11.4.1-1116d** to retain the removable lens assembly **11.4.1-1120** on the lens mount **11.4.1-1116c**. While the retractable latches **11.4.1-1122** have been shown and described as being part of the removable lens assembly **11.4.1-1120**, retractable latches may instead be provided on the display module **11.4.1-1116** for releasably engaging the removable

lens assembly **11.4.1-1120** in a similar manner (e.g., engaging receptacles in an outer peripheral surface of the lens frame **11.4.1-1140**).

[2247] Referring to FIGS. **11.4.1-12A-11.4.1-12B**, in still further examples, variations of the removable lens assembly **11.4.1-120** may be coupled to the lens mount **11.4.1-1216c** in other manners. For example, a removable lens assembly **11.4.1-1220** and the lens mount **11.4.1-1216c** may include corresponding mating features **11.4.1-1222**, **11.4.1-1223** (e.g., tabs and detents or protrusions) that allow for coupling of the removable lens assembly **11.4.1-1220** to the lens mount **11.4.1-1216c** with one or more of an axial motion **11.4.1-1222a**, a vertical motion **11.4.1-1222b**, or a turning motion **11.4.1-1222c**. The axial motion **11.4.1-1222a**, the vertical motion **11.4.1-1222b**, or the turning motion **11.4.1-1222c** (e.g., a quarter turn) cause engagement and coupling of the removable lens assembly **11.4.1-1220** to the lens mount **11.4.1-1216c** via the corresponding mating features **11.4.1-1222**, **11.4.1-1223**.

[2248] Referring additionally to FIG. **11.4.1-12C**, a removable lens assembly **11.4.1-1220'** is coupled to a display unit **11.4.1-1216'** via a friction or interference fit. In particular, the removable lens assembly **11.4.1-1220** includes a lens frame **11.4.1-1240'** that is received by a lens mount **11.4.1-1216c'** of the display unit **11.4.1-1216'**, so as to be retained thereby via friction (e.g., with the lens frame **11.4.1-1240'** being compressed radially inward by the lens mount **11.4.1-1216c'**).

[2249] Referring additionally to FIG. **11.4.1-12D**, the removable lens assembly **11.4.1-1220'** is coupled to the display unit **11.4.1-1216'** via the retractable pins **11.4.1-1216d**. For example, the display unit **11.4.1-1216'** includes the lens mount **11.4.1-1216c'** that receives the lens frame **11.4.1-1240'** of the removable lens assembly **11.4.1-1220'** therein and the retractable pins **11.4.1-1216d**, which are selectively extended into corresponding recesses of the lens frame **11.4.1-1240'**. For example, the pins **11.4.1-1216d'** may be spring biased radially inward, so as to be received by the lens frame **11.4.1-1240'**.

[2250] Referring to FIGS. **11.4.1-13A-11.4.1-13F**, in a still further example, a removable lens assembly **11.4.1-1320** includes a mechanical coupling feature **11.4.1-1322** (e.g., a male component, or protrusion), which may be referred to as a lens mechanical coupling feature **11.4.1-1322**, while a display module **11.4.1-1310** includes a corresponding mechanical coupling feature **11.4.1-1312** (e.g., a female component, recess, or receptacle), which may be referred to as a display mechanical coupling feature **11.4.1-1312**. The lens mechanical coupling feature **11.4.1-1322** and the display mechanical coupling feature **11.4.1-1312** mechanically mate with each other to mechanically couple the removable lens assembly **11.4.1-1320** to the display module **11.4.1-1310**. For example, as discussed in further detail below, the lens mechanical coupling feature **11.4.1-1322** and the display mechanical coupling feature **11.4.1-1312** may prevent movement of the removable lens assembly **11.4.1-1320** relative to the display module **11.4.1-1310** in directions generally perpendicular to, generally parallel with, and/or about an optical axis of the removable lens assembly **11.4.1-1320**.

[2251] The removable lens assembly **11.4.1-1320** may further include one or more magnetic coupling features **11.4.1-1324**, which may be referred to as lens magnetic coupling features **11.4.1-1324**, while the display module **11.4.1-1310** includes one or more corresponding magnetic coupling features **11.4.1-1314**, which may be referred to as display magnetic coupling features **11.4.1-1314**. The lens magnetic coupling feature **11.4.1-1324** and the display magnetic coupling feature **11.4.1-1314** magnetically couple to each other to magnetically couple the removable lens assembly **11.4.1-1320** to the display module **11.4.1-1310**. For example, as discussed in further detail below, the lens magnetic coupling feature **11.4.1-1324** and the display magnetic coupling feature **11.4.1-1314** may prevent movement of the removable lens assembly **11.4.1-1320** relative to the display module **11.4.1-1310** in directions generally parallel with the optical axis of the removable lens assembly **11.4.1-1320**.

[2252] The lens mechanical coupling feature **11.4.1-1322** of the removable lens assembly **11.4.1-1320** may be arranged on a first side thereof (e.g., left, right, upper, lower, inner, or outer), while

the one or more lens magnetic coupling features **11.4.1-1324** may be arranged on a second side thereof, which may be spaced apart from and/or opposite the first side (e.g., right, left, lower, upper, outer, or inner, respectively, such as with the lens element **11.4.1-330** being positioned therebetween). The lens mechanical coupling feature **11.4.1-1322** may be formed by (e.g., be integrally-formed with) a lens frame **11.4.1-340**, while the one or more lens magnetic coupling features **11.4.1-1324** are coupled to the lens frame **11.4.1-340**. The lens magnetic coupling features **11.4.1-1324** may be or include one or more permanent magnets and/or an attractor material (e.g., ferromagnetic material or component).

[2253] The display module **11.4.1-1310** includes a lens mount **11.4.1-1311** having the lens mechanical coupling feature **11.4.1-1322** and the one or more lens magnetic coupling features **11.4.1-1324**. As discussed in further detail below, the display mechanical coupling feature **11.4.1-1312** of the lens mount **11.4.1-1311** is configured to receive the lens mechanical coupling feature **11.4.1-1322** therein. The one or more display magnetic coupling features **11.4.1-1314** of the lens mount **11.4.1-1311** magnetically couple to the lens magnetic coupling features **11.4.1-1324** (e.g., being a permanent magnet of opposite polarity or being an attractor material).

[2254] As referenced above, the lens mechanical coupling feature **11.4.1-1322** of the removable lens assembly **11.4.1-1320** and the display mechanical coupling feature **11.4.1-1312** of the display module **11.4.1-1310** are configured to mate with each other and, in particular, with the lens mechanical coupling feature **11.4.1-1322** being received by the display mechanical coupling feature **11.4.1-1312**. When received thereby, the lens mechanical coupling feature **11.4.1-1322** engages the display mechanical coupling feature **11.4.1-1312** to prevent relative movement therebetween. For example, the lens mechanical coupling feature **11.4.1-1322** engages the display mechanical coupling feature **11.4.1-1312** to prevent relative shearing movement between the removable lens assembly **11.4.1-1320** and the display module **11.4.1-1310**, such as upward movement, lateral movement (e.g., in a direction generally perpendicular to an optical axis of the removable lens assembly **11.4.1-1320**), and/or rotational movement of the removable lens assembly **11.4.1-1320** relative to the display module **11.4.1-1310** (e.g., generally about the optical axis of the removable lens assembly **11.4.1-1320**).

[2255] To prevent relative upward movement, an upwardly-facing surface of the lens mechanical coupling feature **11.4.1-1322** engages a downwardly facing surface of the display mechanical coupling feature **11.4.1-1312**. To prevent relative downward movement, a downwardly-facing surface of the lens mechanical coupling feature **11.4.1-1322** engages an upwardly-facing surface of the display mechanical coupling feature **11.4.1-1312**. For example, upper and lower ends of the lens mechanical coupling feature **11.4.1-1322** engaging respective upper and lower ends of the display mechanical coupling feature **11.4.1-1312** (see FIGS. **11.4.1-13A** and **11.4.1-13B**).

Alternatively, upward- and downward-facing surfaces at intermediate heights of the lens mechanical coupling feature **11.4.1-1322** and the display mechanical coupling feature **11.4.1-1312** engage each other to prevent upward and downward relative movement.

[2256] To prevent relative lateral outward movement (e.g., away from the nose of the user), a laterally outward-facing surface of the lens mechanical coupling feature **11.4.1-1322** engages a laterally inwardly-facing surface of the display mechanical coupling feature **11.4.1-1312** (e.g., a leftward facing surface and a rightward facing surface, respectively, for a left removable lens assembly **11.4.1-1320**). To prevent relative lateral inward movement (e.g., toward the nose of the user), a laterally inward-facing surface of the lens mechanical coupling feature **11.4.1-1322** engages a laterally outward-facing surface of the display mechanical coupling feature **11.4.1-1312** (e.g., a rightward facing surface and a leftward facing surface, respectively, for a left removable lens assembly **11.4.1-1320**), as is shown in FIG. **11.4.1-13F**. For example, as is illustrated, the lens mechanical coupling feature **11.4.1-1322** and the display mechanical coupling feature **11.4.1-1312** have curved shapes with outward and inward sides of the lens mechanical coupling feature **11.4.1-1322** (e.g., concave and convex surfaces) engaging respective outward and inward sides of the

display mechanical coupling feature **11.4.1-1312** (e.g., convex and concave surfaces, respectively). [2257] To prevent relative rotational movement, the lens mechanical coupling feature **11.4.1-1322**, in spaced apart locations on opposite sides thereof (e.g., two on each side), engages the display mechanical coupling feature **11.4.1-1312** on opposite sides thereof. For example, as a clockwise torque is applied to the removable lens assembly **11.4.1-1320**, upper inner surfaces and lower outer surfaces of the lens mechanical coupling feature **11.4.1-1322** and the display mechanical coupling feature **11.4.1-1312** engage each other to constrain (e.g., prevent) relative clockwise rotation therebetween. As a counterclockwise torque is applied to the removable lens assembly **11.4.1-1320**, upper outer surface and lower inner surfaces of the lens mechanical coupling feature **11.4.1-1322** and the display mechanical coupling feature **11.4.1-1312** engage each other to constrain (e.g., prevent) relative counterclockwise rotation therebetween.

[2258] The lens mechanical coupling feature **11.4.1-1322** and the display mechanical coupling feature **11.4.1-1312** may additionally be configured to prevent normal movement (e.g., in the general direction of the optical axis) away from the display module **11.4.1-1310**. For example, the display mechanical coupling feature **11.4.1-1312** may include a forward-facing surface (e.g., an undercut or toe-in feature) that engages a rearward facing surface of the lens mechanical coupling feature **11.4.1-1322**.

[2259] Referring to FIG. **11.4.1-13E**, to couple the removable lens assembly **11.4.1-1320** to the display module **11.4.1-1310**, the lens mechanical coupling feature **11.4.1-1322** is first inserted into the display mechanical coupling feature **11.4.1-1312**, and the removable lens assembly **11.4.1-1320** is rotated to bring the magnetic coupling features **11.4.1-1324**, **11.4.1-1314** closer to each other to magnetically couple, as discussed in further detail below.

[2260] As referenced above, the lens magnetic coupling features **11.4.1-1324** of the removable lens assembly **11.4.1-1320** and the display magnetic coupling features **11.4.1-1314** of the display module **11.4.1-1310** couple to each other magnetically. The lens magnetic coupling features **11.4.1-1324** and the display magnetic coupling features **11.4.1-1314** are arranged at corresponding positions and with corresponding magnetic properties (e.g., being opposite polarity, or one being a ferromagnetic material and the other a permanent magnetic). The lens magnetic coupling features **11.4.1-1324** and the display magnetic coupling features **11.4.1-1314** couple to each other magnetically with force therebetween applied generally in the direction of the optical axis. The lens magnetic coupling features **11.4.1-1324** and the display magnetic coupling features **11.4.1-1314** are also spaced apart from the pivot axis defined by the display mechanical coupling feature **11.4.1-1312**, such that the magnetic coupling for therebetween prevent rotation about the pivot axis defined the display mechanical coupling feature **11.4.1-1312**. For example, as shown, the lens magnetic coupling features **11.4.1-1324** and the display magnetic coupling features **11.4.1-1314** are positioned generally opposite the lens mechanical coupling feature **11.4.1-1322** and the display mechanical coupling feature **11.4.1-1312**, respectively.

[2261] Variations of the removable lens assembly **11.4.1-1320** and the display module **11.4.1-1310** are contemplated. For example, the lens mechanical coupling feature **11.4.1-1322** and the display mechanical coupling feature **11.4.1-1312** may have different shapes (e.g., being straight), may be provided in different numbers (e.g., two or more sets thereof, such as pins and apertures, at spaced apart locations to prevent movement as described above), and/or may be provided at different locations (e.g., at upper, lower, or inner sides thereof). Furthermore, the display mechanical coupling feature **11.4.1-1312** may be provided by a different part of the head-mounted display unit **11.4.1-110**, such as to the housing **11.4.1-112** (e.g., a curtain or other structure surrounding the display module **11.4.1-1310**). Further, instead of the magnetic coupling features **11.4.1-1324**, **11.4.1-1314**, the removable lens assembly **11.4.1-1320** and the display module **11.4.1-1310** may instead or additionally include mechanical coupling features (e.g., interfitting structures or latches as described above).

[2262] It should be noted that, for each of the manners for coupling described above with respect to

FIGS. **11.4.1-3A-3B** and **11.4.1-10A-11.4.1-13F**, the lens mount may be coupled to another portion of the head-mounted display unit **11.4.1-110**, such as the housing **11.4.1-112** (e.g., chassis), as opposed to the display module **11.4.1-116** or variations thereof.

[2263] Referring to FIGS. **11.4.1-14A-11.4.1-16**, the display system **11.4.1-100** and the removable lens assemblies **11.4.1-120** are cooperatively configured for the display system **11.4.1-100** to detect the removable lens assembly **11.4.1-120** coupled to the head-mounted display unit **11.4.1-110**. For example, the head-mounted display unit **11.4.1-110** includes one or more lens detection sensors **11.4.1-1418a** for detecting the one or more removable lens assemblies **11.4.1-120** coupled thereto. In detecting the removable lens assembly **11.4.1-120**, the display system **11.4.1-100** may determine whether or not any of the removable lens assemblies **11.4.1-120** is coupled to the display module **11.4.1-116** (e.g., using a proximity sensor, image recognition, and/or mechanical switch), whether the removable lens assembly **11.4.1-120** is properly coupled (e.g., using a proximity sensor, image recognition, or hall sensors with magnets coupled to the removable lens assembly **120**), and/or other information about the removable lens assembly **11.4.1-120** (e.g., corrective or not, serial or model number, associated user, and/or lens characteristics using other types of lens detection sensors **11.4.1-1418a** discussed below) to support other functionality. The display system **11.4.1-100** may provide an indication of whether or not the removable lens assembly **11.4.1-120** is coupled to the display module **11.4.1-116**, whether the removable lens assembly **11.4.1-120** is properly position relative to the display module **11.4.1-116**, and/or other indications (e.g., of the lens configuration and/or lens compatibility as discussed below).

[2264] Each removable lens assembly **11.4.1-120** may further include a lens tag **11.4.1-1428** that is in communication with or otherwise readable by the one or more lens detection sensors **11.4.1-1418a**. According to detection of the removable lens assembly **11.4.1-120**, the display system **11.4.1-100** may perform one or more subsequent operations, which may include changing output of the display module **11.4.1-116**, providing an indicator of the removable lens assembly **11.4.1-120**, and/or providing an indicator of compatibility of the removable lens assembly **11.4.1-120** with a user. The display system also includes a controller **11.4.1-1450** that controls operation of the head-mounted display unit **11.4.1-110**, for example, to detect the removable lens assemblies **11.4.1-120** coupled thereto and/or perform subsequent operations in response thereto. The controller **11.4.1-1450** may be wholly or partially physically incorporated into the head-mounted display unit **11.4.1-110** (as indicated by dash-lines) or may be wholly or partially provided physically separate therefrom (e.g., in an external computing device). An example hardware configuration for the controller **11.4.1-1450** is discussed below with reference to FIG. **11.4.1-14**.

[2265] In detecting the removable lens assembly **11.4.1-120**, the display system **11.4.1-100** may identify or determine information about the removable lens assembly **11.4.1-120**, which may be referred to as lens information. The lens information may include an identifier and/or technical characteristics of the removable lens assembly **11.4.1-120**. The identifier may be of the removable lens assembly **11.4.1-120** itself (e.g., a serial number or a model number; referred to as a lens identifier), a user associated with the lens (e.g., a user name or user ID; referred to as an associated user identifier), and/or other identifying information (e.g., manufacturer, date; referred to as manufacturing information). The technical characteristics may include refractive characteristics, such as sphere, cylinder, and/or axis parameters and/or an optical axis location of the removable lens assembly **11.4.1-120**, which may be referred to as lens characteristics, lens characteristic information, or prescription information. The lens information determined by the display system **11.4.1-100** may then be used by the display system **11.4.1-100** to perform the subsequent operations referenced above and described in further detail below.

[2266] The display system **11.4.1-100** may determine the lens information of the removable lens assembly **11.4.1-120** actively coupled (e.g., being coupled or currently coupled) to the head-mounted display unit **11.4.1-110** in various different manners and with different devices. The display system **11.4.1-100** may determine the lens information from the removable lens assembly

11.4.1-120, such as with the lens detection sensor **11.4.1-1418a** from the lens tag **11.4.1-1428**. In a first example, the lens tag **11.4.1-1428** is an electronic device that stores or otherwise electrically provides the lens information. For example, the lens tag **11.4.1-1428** each removable lens assembly **11.4.1-120** may be an electrically-erasable programmable read-only memory device (EEPROM), a radio frequency identification device (RFID), or an internal resistance device (e.g., an electrical resistance or resistor). The lens detection sensor **11.4.1-1418a** is a corresponding electrical device or system configured to receive and/or otherwise receive the lens information from the lens tag **11.4.1-1428**, such as by being in wireless or physical communication with the EEPROM, to read a signal of the RFID, and/or to measure or otherwise determine a resistance signature of the resistance device.

[2267] In a second example, the lens tag **11.4.1-1428** has a magnetic signature. The removable lens assembly **11.4.1-120** includes various magnets (e.g., the magnetic coupling features **11.4.1-322**) or other magnets, which may provide a magnetic signature by having varied strength and/or position and which are detected by the lens detection sensor **11.4.1-1418a**, which may include one or more hall sensors or other magnetic sensors that sense the magnetic field of the magnets. The combination of magnetic fields detected by the lens detection sensor **11.4.1-1418a** provide the lens information (e.g., a lens identifier of a particular magnetic signature). Hall sensors may instead or additionally be used to determine whether the removable lens assembly **11.4.1-120** is properly coupled.

[2268] In a third example, the lens tag **11.4.1-1428** is an optical marking that displays or otherwise communicates the lens information. For example, the lens tag **11.4.1-1428** may be an optical or infrared (IR) marking on the removable lens assembly **11.4.1-120**, such as a quick response code (QR Code), a bar code, or alphanumeric characters. The lens detection sensor **11.4.1-1418a** of the head-mounted display unit **11.4.1-110** is a corresponding optical reading device, such as a camera or a combined illuminator/sensor device (e.g., a QR Code scanner), which may be the eye camera **11.4.1-319** that also observes the eye as described previously. In one particular example, the lens tag **11.4.1-1428** is an infrared bar code.

[2269] The display system **11.4.1-100** may determine the lens information required for one or more further operations directly from the removable lens assembly **11.4.1-120** itself. Instead or additionally, the display system **11.4.1-100** may determine additional lens information from other sources for performing subsequent operations. For example, the display system **11.4.1-100** may receive only initial lens information (e.g., the lens identifier or the associated user identifier) from the removable lens assembly **11.4.1-120**, while other lens information (e.g., the associated user identifier and/or lens characteristics) are stored in association therewith by the display system **11.4.1-100** (e.g., by a storage of the controller **11.4.1-1450**). Thus, by determining initial lens information from the removable lens assembly **11.4.1-120**, the display system **11.4.1-100** may determine (e.g., retrieve) other lens information to support subsequent operations.

[2270] The lens information of the removable lens assembly **11.4.1-120** coupled to the display module **11.4.1-116** may be used to control various functionality of the display system **11.4.1-100**, which may include operating the display **11.4.1-316a**, providing an indication to the user of the removable lens assemblies **11.4.1-120** coupled thereto, and/or determining compatibility of the removable lens assemblies **11.4.1-120** with the user. Operation of the display **11.4.1-316a** may be performed according to the lens information of the removable lens assembly **11.4.1-120** coupled to the display module **11.4.1-116**, such that operation of the display **11.4.1-316a** may differ between one of the removable lens assemblies **11.4.1-120** having one set of lens characteristics and another of the removable lens assemblies **11.4.1-120** having another set of lens characteristics (e.g., different brightness and/or pixel output to account for different characteristics of the optical system formed by the primary lens **11.4.1-316b** and the lens elements **11.4.1-330** or different lens characteristics).

[2271] Instead or additionally, the lens information of the removable lens assembly **11.4.1-120**

coupled to the display module **11.4.1-116** may be used to provide an indication to the user or potential users of the removable lens assembly **11.4.1-120** coupled thereto, which may be referred to as a lens configuration indication. The lens configuration indication differs for different ones of the removable lens assemblies **11.4.1-120**, for example, those removable lens assemblies **11.4.1-120** associated with different users. A first lens configuration indication is provided for one of the removable lens assemblies **11.4.1-120** (or one pair of the removable lens assemblies **11.4.1-120** associated with each other), while a second, different lens configuration indication is provided another of the removable lens assemblies **11.4.1-120** (or another pair of the removable lens assemblies **11.4.1-120** associated with each other). The display indication (e.g., or output instruction) may be stored in association with the lens information (e.g., the lens identifier, associated user identifier, or lens characteristics), for example, by the display system **11.4.1-100** and retrieved upon determining the lens identifier. The configuration indication may be provided irrespective of the user (e.g., without detecting or identifying the user).

[2272] The lens configuration indication may be provided as, for example, an external light indicator, a display indicator, or an audible indicator. The external light indicator is provided with an external visual indicator **11.4.1-1418b** (e.g., a light-emitting diode) of the head-mounted display unit **11.4.1-110**, which is visible by potential users prior to wearing the head-mounted display unit **11.4.1-110** and may differ, for example, by color (e.g., red vs. blue to differentiate between first and second users). The display indicator is provided by the display **11.4.1-316a** and is viewable by the user or potential users through the removable lens assembly **11.4.1-120**. The display indicator may for example, be a color, symbol, or set of characters that differs for different users and is identifiable even for potential users having low acuity with the removable lens assembly **11.4.1-120** (e.g., for which the display **11.4.1-316a** may appear blurry). The audible indicator is provided by an audio output device **11.4.1-1418c** of or associated with the head-mounted display unit **11.4.1-110** (e.g., a speaker or headphones) and may be audible to potential users prior to or after wearing the head-mounted display unit **11.4.1-110**. The audible indicator may, for example, be spoken words (e.g., identifying the user associated with the removable lens assembly **11.4.1-120**) or tones that for different users associated with the removable lens assemblies **11.4.1-120**.

[2273] Referring to FIGS. **11.4.1-14A** and **11.4.1-14B**, a method **11.4.1-1400** is provided for indicating a lens configuration, which generally includes a first operation **11.4.1-1410** of identifying a removable lens assembly **11.4.1-120** coupled to a display module **11.4.1-116**, a second operation **11.4.1-1420** of retrieving a configuration indicator instruction, and a third operation **11.4.1-1430** of providing a configuration indication according to the instruction. The first operation **11.4.1-1410** of identifying the interchangeable lens assembly is performed with the lens detection sensor **11.4.1-1418a** and the controller **11.4.1-1450** determining lens information from the lens tag **11.4.1-1428** of the one or more removable lens assembly **11.4.1-120** coupled to the one or more display modules **11.4.1-116** of the head-mounted display unit **11.4.1-110**. The second operation **11.4.1-1420** of retrieving a configuration indication instruction is performed, for example, by the controller **11.4.1-1450** retrieving from a storage the instruction associated with the lens information. The third operation **11.4.1-1430** of providing the configuration indication is provided, for example, by the display module **11.4.1-116**, the external visual indicator **11.4.1-1418b**, and/or the audio output device **11.4.1-1418c**, as operated by the controller **11.4.1-1450**.

[2274] Referring to FIGS. **11.4.1-14A-11.4.1-16**, instead or additionally, the display system **11.4.1-100** may use the lens information of the removable lens assembly **11.4.1-120** coupled thereto to determine compatibility with the user and accordingly provide a compatibility indicator to the user (e.g., only of incompatibility is determined). The display system **11.4.1-100**, in addition to determining the lens information of the removable lens assembly **11.4.1-120** coupled to the head-mounted display unit **11.4.1-110**, identifies the current user, and compares user information with the lens information to determine compatibility.

[2275] The display system **11.4.1-100** may identify the user in various manners, for example, with

biometric sensing, receipt of user credentials, and/or authentication by another device. To identify the user via biometrics, the display system **11.4.1-100** may include a biometric sensor **11.4.1-1418d** to identify the user with biometrics (e.g., facial recognition, fingerprint recognition, voice recognition, iris recognition, and/or recognition of other biometric parameters, such as ear geometry, bone conduction or density characteristics, forehead features, or skin). The biometric sensor **11.4.1-1418d** is, for example, coupled physically coupled to the head-mounted display unit. For example, in the case of identifying the user with iris detection, the biometric sensor **11.4.1-1418d** may be the eye camera **11.4.1-319** described previously and which may be the same sensor for identifying the removable lens assembly **11.4.1-120**. To identify the user via user credentials, the display system **11.4.1-100** receives user credentials (e.g., a username and password) from an input device of or associated with the display system **11.4.1-100** (e.g., a microphone of the head-mounted display unit, or an external device **11.4.1-1460** in communication with the controller **11.4.1-1450** or the head-mounted display unit **11.4.1-110**). To identify the user by authentication from another device, the external device **11.4.1-1460** may authenticate the user (e.g., via biometrics or credentials) and communicate such authentication to the controller **11.4.1-1450**. The external device **11.4.1-1460** may be considered part of the display system **11.4.1-100** but may function independent thereof (e.g., a smartphone).

[2276] The display system **11.4.1-100** stores or otherwise receives user information. The user information may include a user identifier (e.g., a username or user number) stored in association with an associated lens identifier (e.g., a serial number or model number of one or more of the removable lens assemblies **11.4.1-120** associated with the user) and/or eye characteristic information (e.g., sphere, cylinder, and/or axis parameters; referred to herein as eye characteristics).

[2277] The display system **11.4.1-100** may receive the user information during an initializing process in which the user information is associated with the lens information. During the initializing process, the user information may be input by the user or received from another source (e.g., the external device **11.4.1-1460**). The lens information may be obtained from the removable lens assemblies **11.4.1-120** when coupled to the display module **11.4.1-116** as described above (e.g., electronically and/or optically), or be input by the user (e.g., inputting the serial or model number of the removable lens assembly **11.4.1-120**) or received from another source (e.g., the external device **11.4.1-1460**).

[2278] Referring to FIG. **11.4.1-15**, to determine compatibility, the display system **11.4.1-100** compares the lens information **11.4.1-1510** to the user information **11.4.1-1520**, for example, using a compatibility determiner **11.4.1-1500**. For example, or more of: (a) the lens information **11.4.1-1510** of the lens identifier **11.4.1-1512** is compared to the user information **11.4.1-1520** of the associated lens identifier **11.4.1-1522**, (b) the lens information **11.4.1-1510** of the associated user identifier **11.4.1-1514** is compared to the user information **11.4.1-1520** of the user identifier **11.4.1-1524**, or (c) the lens information **11.4.1-1510** of the lens characteristics **11.4.1-1516** is compared to the user information **11.4.1-1520** of the eye characteristics **11.4.1-1526**. The display system **11.4.1-100** determines the one or more removable lens assemblies **11.4.1-120** to be compatible if the lens information matches (e.g., equals) the user information, or incompatible if not matching.

[2279] If determined compatible, the compatibility indication may provide a positive indication (e.g., that the removable lens assembly **11.4.1-120** is compatible with the user) or be omitted (e.g., proceeding to normal operation). If determined incompatible, the compatibility indication is negative (e.g., indicating that the removable lens assembly **11.4.1-120** is not compatibility with the user). Such a negative indication may, for example, include an icon that graphically indicates incompatibility, textual instructions or information, and/or audible instructions or information.

[2280] Referring to FIG. **11.4.1-16**, a method **11.4.1-1600** is provided for operating the display system **11.4.1-100** to determine and indicate compatibility of the removable lens assembly **11.4.1-120** with the user. The method includes a first operation **11.4.1-1610** of identifying the one or more removable lens assemblies **11.4.1-120** coupled to the one or more display modules **11.4.1-116**, a

second operation **11.4.1-1620** of identifying the user, a third operation **11.4.1-1630** of comparing the lens information to the user information, and a fourth operation **11.4.1-1640** of indicating compatibility. The first operation **11.4.1-1610** of identifying the one or more removable lens assemblies **11.4.1-120** may be performed by the one or more lens detection sensors **11.4.1-1418a** in cooperation with the controller **11.4.1-1450**. The first operation **11.4.1-1610** may also include determining lens information. The second operation **11.4.1-1620** of identifying the user may be performed biometrically (e.g., cooperatively by the biometric sensor **11.4.1-1418d** in cooperation with controller **11.4.1-1450**), by receiving credentials from the user, and/or by user authentication with the external device **11.4.1-1460**. The second operation **11.4.1-1620** may also include determining user information. The third operation **11.4.1-1630** of determining lens compatibility is performed by comparing the lens information to the user information, for example, with the controller **11.4.1-1450** according to the compatibility determiner **11.4.1-1500**. The fourth operation **11.4.1-1640** of indicating compatibility is performed, for example, with the head-mounted display unit **11.4.1-110** as operated by the controller **11.4.1-1450**. If compatible, a positive indication is provided, which may be usual operation of the display system **11.4.1-100**. If incompatible, a negative indication is provided reflecting incompatibility (e.g., by the head-mounted display unit **11.4.1-110** as operated by the controller **11.4.1-1450**).

[2281] Referring to FIG. **11.4.1-17**, an example hardware configuration of the controller **11.4.1-1450** is shown. The controller **11.4.1-1450** is a computing device, which may generally include a processing device **11.4.1-1710** (e.g., a processor or CPU), a memory **11.4.1-1720** (e.g., a volatile short-term memory, such as a random-access memory module), a storage **11.4.1-1730** (e.g., a long-term storage device, such as a hard disk or solid state drive), a communications interface **11.4.1-1740** by which the controller sends and/or receives signals to and/or from other components (e.g., the display module **11.4.1-116** and various sensors), and a bus **11.4.1-1750** by which the other components of the controller **11.4.1-1450** communicate with each other. The controller **11.4.1-1450** may be physically connected to the head-mounted display unit **11.4.1-110** or otherwise be in communication therewith. The display system **11.4.1-100** may include additional controllers **11.4.1-1450** or components thereof. The controller **11.4.1-1450** may include software programming (e.g., stored by the storage **11.4.1-1730**) that includes instructions executable by the processing device **11.4.1-1710** to perform the methods and operations described herein.

11.4.2: Electronic Device System with Supplemental Lenses

[2282] Some users of head-mounted devices have visual defects such as myopia, hyperopia, astigmatism, or presbyopia. It can be challenging to ensure that an optical system in a head-mounted device operates satisfactorily for these users. If care is not taken, it may be difficult or impossible for a user with visual defects to focus properly on content that is being displayed.

[2283] A head-mounted device may have a display that displays content for a user. Head-mounted support structures in the device may support the device on the head of the user. A non-removable lens system may be supported by the head-mounted support structures and may be used to present content on the display to eye boxes. The user may view the content when the user's eyes are located in the eye boxes.

[2284] The head-mounted support structures may be configured to receive a removable supplemental lens system in alignment with the non-removable lens system. Magnetic coupling structures or other engagement structures may be used to removably couple the supplemental lens system to the head-mounted device.

[2285] The removable supplemental lens system may be used to adjust the lens characteristics of the non-removable lens system and thereby customize the head-mounted display to accommodate the user's vision. For example, the removable supplemental lens system may include supplemental left and right lenses that align with and supplemental corresponding left and right lenses in the non-removable lens system and that include astigmatic lens characteristics and other characteristics that allow the head-mounted device to be used by a user with astigmatism or other visual defects.

[2286] Removable supplemental lens systems associated with different users may be configured to operate with a shared head-mounted device. To ensure that the head-mounted device can customize its operation for each user, the removable supplemental lens system of each user may be provided with information that identifies the user associated with that removable supplemental lens system and other stored information. This information be stored in the removable supplemental lens system using bar codes or other optically readable patterns, programmable memory, or other data storage. A memory reader, gaze tracking system, or other sensor may be used in retrieving the stored information.

[2287] The information that is stored in the removable supplemental lens system may include information on the optical characteristics of the removable supplemental lens system, user information such as the user's eyeglass prescription, and/or other information. When the removable supplemental lens system is installed in the head-mounted device, control circuitry in the head-mounted device may retrieve the stored information and may take appropriate action. For example, the control circuitry may use lens positioners to adjust lens spacing and other operating parameters of the lens systems based on the user's interpupillary distance and other information on the user.

[2288] A head-mounted device may contain a display formed from one or more display panels (displays) for displaying visual content to a user. A lens system may be used to allow the user to focus on the display and view the visual content. To ensure that a wide range of users are able to clearly focus on the display and view visual content, the head-mounted device may receive removable supplemental lenses. The supplemental lenses may address the visual defects of users that are not otherwise addressed by the lens system. For example, a user with astigmatism may have supplemental lenses that correct for astigmatism. When this user desires to view content with the head-mounted device, the supplemental lenses may be installed within the head-mounted device to help correct for the user's astigmatism. With one illustrative arrangement, the supplemental lenses may be coupled to the head-mounted support structures using magnets or other removable fasteners that attach the supplemental lenses to non-removable lenses in the device.

[2289] If desired, removable supplemental lenses may be provided with the ability to store information that is subsequently retrieved and used by the head-mounted device. This information, which may sometimes be referred to as supplemental lens information or stored information may include lens power information and other information that is specific to the supplemental lenses and/or user information for a user associated with the lenses (e.g., a username, eyeglasses prescription information, etc.). Storage circuitry such as programmable read-only memory may be used to store the supplemental lens information and/or the supplemental lens information may be stored in other ways (e.g., using bar codes, text on the supplemental lenses, other patterned optically readable information, etc.). During operation, after a user has installed the supplemental lenses in a head-mounted device, the head-mounted device may retrieve and use the supplemental lens information. For example, information on the optical characteristics of the lens may be used by the head-mounted device to help correct for optical aberrations in the lenses (e.g., spherical aberration, chromatic aberration, pincushion distortion, barrel distortion, etc.), information on the user may be used to adjust the interpupillary distance of lenses or may be used to present the user with a login screen that has been prepopulated with the user's username, etc.

[2290] A schematic diagram of an illustrative system that may use supplemental head-mounted display lenses is shown in FIG. 11.4.2-1. As shown in FIG. 11.4.2-1, system 11.4.2-8 may include one or more electronic devices such as electronic device 11.4.2-10. The electronic devices of system 11.4.2-8 may include computers, cellular telephones, head-mounted devices, wristwatch devices, and other electronic devices. Configurations in which electronic device 11.4.2-10 is a head-mounted device are sometimes described herein as an example.

[2291] As shown in FIG. 11.4.2-1, electronic devices such as electronic device 11.4.2-10 may have control circuitry 11.4.2-12. Control circuitry 11.4.2-12 may include storage and processing circuitry for controlling the operation of device 11.4.2-10. Circuitry 11.4.2-12 may include storage such as

hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **11.4.2-12** may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, and other integrated circuits. Software code may be stored on storage in circuitry **11.4.2-12** and run on processing circuitry in circuitry **11.4.2-12** to implement control operations for device **11.4.2-10** (e.g., data gathering operations, operations involved in processing three-dimensional facial image data, operations involving the adjustment of components using control signals, etc.). Control circuitry **11.4.2-12** may include wired and wireless communications circuitry. For example, control circuitry **11.4.2-12** may include radio-frequency transceiver circuitry such as cellular telephone transceiver circuitry, wireless local area network (WiFi®) transceiver circuitry, millimeter wave transceiver circuitry, and/or other wireless communications circuitry.

[2292] During operation, the communications circuitry of the devices in system **11.4.2-8** (e.g., the communications circuitry of control circuitry **11.4.2-12** of device **11.4.2-10**), may be used to support communication between the electronic devices. For example, one electronic device may transmit video and/or audio data to another electronic device in system **11.4.2-8**. Electronic devices in system **11.4.2-8** may use wired and/or wireless communications circuitry to communicate through one or more communications networks (e.g., the internet, local area networks, etc.). The communications circuitry may be used to allow data to be received by device **11.4.2-10** from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, online computing equipment such as a remote server or other remote computing equipment, or other electrical equipment) and/or to provide data to external equipment.

[2293] Device **11.4.2-10** may include input-output devices **11.4.2-22**. Input-output devices **11.4.2-22** may be used to allow a user to provide device **11.4.2-10** with user input. Input-output devices **11.4.2-22** may also be used to gather information on the environment in which device **11.4.2-10** is operating. Output components in devices **11.4.2-22** may allow device **11.4.2-10** to provide a user with output and may be used to communicate with external electrical equipment.

[2294] As shown in FIG. **11.4.2-1**, input-output devices **11.4.2-22** may include one or more displays such as display(s) **11.4.2-14**. In some configurations, display **11.4.2-14** of device **11.4.2-10** includes left and right display panels in alignment with the user's left and right eyes, respectively. In other configurations, display **11.4.2-14** includes a single display panel that extends across both eyes.

[2295] Display **11.4.2-14** may be used to display images. The visual content that is displayed on display **11.4.2-14** may be viewed by a user of device **11.4.2-10**. Displays in device **11.4.2-10** such as display **11.4.2-14** may be organic light-emitting diode displays or other displays based on arrays of light-emitting diodes, liquid crystal displays, liquid-crystal-on-silicon displays, projectors or displays based on projecting light beams on a surface directly or indirectly through specialized optics (e.g., digital micromirror devices), electrophoretic displays, plasma displays, electrowetting displays, or any other suitable displays.

[2296] Display **11.4.2-14** may present computer-generated content such as virtual reality content and mixed reality content to a user. Virtual reality content may be displayed in the absence of real-world content. Mixed reality content, which may sometimes be referred to as augmented reality content, may include computer-generated images that are overlaid on real-world images. The real-world images may be captured by a camera (e.g., a forward-facing camera) and merged with overlaid computer-generated content or an optical coupling system may be used to allow computer-generated content to be overlaid on top of real-world images. As an example, a pair of mixed reality glasses or other augmented reality head-mounted display may include a display device that provides images to a user through a beam splitter, prism, holographic coupler, or other optical coupler. Configurations in which display **11.4.2-14** is used to display virtual reality content to a

user through lenses are described herein as an example.

[2297] Input-output circuitry **11.4.2-22** may include sensors **11.4.2-16**. Sensors **11.4.2-16** may include, for example, three-dimensional sensors (e.g., three-dimensional image sensors such as structured light sensors that emit beams of light and that use two-dimensional digital image sensors to gather image data for three-dimensional images from light spots that are produced when a target is illuminated by the beams of light, binocular three-dimensional image sensors that gather three-dimensional images using two or more cameras in a binocular imaging arrangement, three-dimensional LIDAR (light detection and ranging) sensors, three-dimensional radio-frequency sensors, or other sensors that gather three-dimensional image data), cameras (e.g., infrared and/or visible digital image sensors), gaze tracking sensors (e.g., a gaze tracking system based on an image sensor and, if desired, a light source that emits one or more beams of light that are tracked using the image sensor after reflecting from a user's eyes), touch sensors, buttons, capacitive proximity sensors, light-based (optical) proximity sensors, other proximity sensors, force sensors, sensors such as contact sensors based on switches, gas sensors, pressure sensors, moisture sensors, magnetic sensors, audio sensors (microphones), ambient light sensors, microphones for gathering voice commands and other audio input, sensors that are configured to gather information on motion, position, and/or orientation (e.g., accelerometers, gyroscopes, compasses, and/or inertial measurement units that include all of these sensors or a subset of one or two of these sensors), and/or other sensors.

[2298] User input and other information may be gathered using sensors and other input devices in input-output devices **11.4.2-22**. If desired, input-output devices **11.4.2-22** may include other devices **11.4.2-24** such as haptic output devices (e.g., vibrating components), light-emitting diodes and other light sources, speakers such as ear speakers for producing audio output, and other electrical components. Device **11.4.2-10** may include circuits for receiving wireless power, circuits for transmitting power wirelessly to other devices, batteries and other energy storage devices (e.g., capacitors), joysticks, buttons, and/or other components.

[2299] Electronic device **11.4.2-10** may have housing structures (e.g., housing walls, straps, etc.), as shown by illustrative support structures **11.4.2-26** of FIG. **11.4.2-1**. In configurations in which electronic device **11.4.2-10** is a head-mounted device (e.g., a pair of glasses, goggles, a helmet, a hat, etc.), support structures **11.4.2-26** may include head-mounted support structures (e.g., a helmet housing, head straps, temples in a pair of eyeglasses, goggle housing structures, and/or other head-mounted structures). The head-mounted support structures may be configured to be worn on a head of a user during operation of device **11.4.2-10** and may support display(s) **11.4.2-14**, sensors **11.4.2-16**, other components **11.4.2-24**, other input-output devices **11.4.2-22**, and control circuitry **11.4.2-12**.

[2300] FIG. **11.4.2-2** is a top view of electronic device **11.4.2-10** in an illustrative configuration in which electronic device **11.4.2-10** is a head-mounted device. As shown in FIG. **11.4.2-2**, electronic device **11.4.2-10** may include support structures (see, e.g., support structures **11.4.2-26** of FIG. **11.4.2-1**) that are used in housing the components of device **11.4.2-10** and mounting device **11.4.2-10** onto a user's head. These support structures may include, for example, structures that form housing walls and other structures for main unit **11.4.2-26-2** and straps or other supplemental support structures such as structures **11.4.2-26-1** that help to hold main unit **11.4.2-26-2** on a user's face so that the user's eyes are located within eye boxes **11.4.2-60**.

[2301] Display **11.4.2-14** may include left and right display panels (e.g., left and right pixel arrays, sometimes referred to as left and right displays) that are mounted respectively in left and right display modules **11.4.2-70** corresponding respectively to a user's left eye (and left eye box **11.4.2-60**) and right eye (and right eye box). Modules **11.4.2-70**, which may sometimes be referred to as lens support structures or lens housings, may be individually positioned relative to the housing wall structures of main unit **11.4.2-26-2** and relative to the user's eyes using respective left and right positioners **11.4.2-58** (e.g., stepper motors, piezoelectric actuators, motors, linear electromagnetic

actuators, and/or other electronic components for adjusting position). Positioners **11.4.2-58** may be controlled by control circuitry **11.4.2-12** during operation of device **11.4.2-10**. For example, positioners **11.4.2-58** may be used to adjust the spacing between modules **11.4.2-70** (and therefore the lens-to-lens spacing between the lenses of modules **11.4.2-70**) to match the interpupillary distance IPD of a user's eyes.

[2302] Device **11.4.2-10** may include a non-removable lens system (sometimes referred to as a permanent lens system or fixed lens system) to which the user of device **11.4.2-10** may attach removable supplemental lenses to accommodate the user's vision (astigmatism, etc.). When the removable supplemental lenses are absent, a user may use the non-removable lens system. Users with particular visual defects (e.g., astigmatism), may benefit from the attachment of the removable supplemental lenses to the non-removable lenses. When the supplemental lenses are coupled to device **11.4.2-10** in alignment with the non-removable lenses, the supplemental lenses and non-removable lenses operate together to form a combined lens system with combined optical properties determined by both the non-removable lenses and the supplemental lenses.

[2303] As shown in FIG. **11.4.2-2**, for example, each lens module **11.4.2-70** may include a non-removable lens **11.4.2-72**. Lenses **11.4.2-72** may be Fresnel lenses, single-element or multiple-element lenses formed from clear materials such as glass or plastic, mirror lenses, catadioptric lenses, and/or lenses of other types. The position of lenses **11.4.2-72** along the X axis may be adjusted so that lenses **11.4.2-72** are separated by a distance that matches the user's interpupillary distance (distance IPD). The position of lenses **11.4.2-72** along the Z axis may be adjusted to help a user focus on display **11.4.2-14**.

[2304] The amount of travel available to lenses **11.4.2-72** may be limited and/or lenses **11.4.2-72** may have attributes that make it difficult for lenses **11.4.2-72** alone to be used to view content on display **11.4.2-14** by all users. For example, a user may have astigmatism or may be strongly nearsighted or farsighted. In situations such as these, lenses **11.4.2-72** when used alone may not be able to correct for the user's vision. To ensure that the user's vision is fully corrected, supplemental lenses **11.4.2-76** may be coupled to lenses **11.4.2-72** to provide further adjustments. For example, if a user is astigmatic, supplemental lenses **11.4.2-76** may be configured to compensate for the astigmatism of the user. If there are multiple users associated with device **11.4.2-10**, each of the multiple users may potentially have a different respective set of supplemental lenses **11.4.2-76**. Lenses **11.4.2-76** may be joined together to form a supplemental lens system based on a single supplemental lens module or may be housed in separate lens modules that form a multi-lens-module supplemental lens system.

[2305] As shown in FIG. **11.4.2-2**, for example, supplemental lenses **11.4.2-76** may be mounted in supplemental lens modules (sometimes referred to as supplemental lens support structures or housings) such as supplemental lens modules **11.4.2-74**. Lens modules **11.4.2-74** may be removably coupled to support structures **11.4.2-26** by using coupling structures **11.4.2-78** in non-removable lens modules **11.4.2-70** and corresponding coupling structures that mate with coupling structures **11.4.2-78** such as coupling structures **11.4.2-80** in removable lens modules **11.4.2-74** to couple lens modules **11.4.2-74** to lens modules **11.4.2-70**. Coupling structures **11.4.2-78** and **11.4.2-80** may be formed from magnetic couplers (e.g., magnets and/or magnetic structures such as iron bars), clips, hook-and-loop fasteners, snaps, screws and other threaded fasteners, temporary adhesive, spring-loaded engagement structures, and/or other coupling mechanisms.

[2306] Device **11.4.2-10** may include a gaze detection system that is configured to monitor a user's eyes in eye boxes **11.4.2-60**. The gaze detection system may, as an example, use optical eye monitoring components to monitor the user's direction of gaze as the user is using device **11.4.2-10**. With one illustrative configuration, a gaze tracker such a gaze tracker **11.4.2-86** may be located in each lens module **11.4.2-70**. In this position, the gaze tracker **11.4.2-86** in each lens module **11.4.2-70** may monitor a user's eye in a respective eye box **11.4.2-60** through a respective lens **11.4.2-72** and a respective mating supplemental lens **11.4.2-76**. Gaze trackers **11.4.2-86** may be placed at

peripheral portions of lens modules **11.4.2-70** so that gaze trackers **11.4.2-86** do not block the user's view of display **11.4.2-14** through removable lenses **11.4.2-76** and non-removable lenses **11.4.2-72**. [2307] If desired, the removable lens system of FIG. **11.4.2-2** (e.g., one or both of lens modules **11.4.2-74** and/or a single removable lens holder that is configured to support both of removable lenses **11.4.2-76**) may be provided with electronic storage such as storage **11.4.2-84**. Storage **11.4.2-84** may be programmable read-only memory (e.g., electrically programmable read-only memory, memory formed from laser-programmed fuses, and/or other programmable memory), may be formed from a pattern of metal traces and/or resistive elements (e.g., to encode information in the form of a pattern of conductive paths, a pattern of resistors and/or their associated resistance values, and/or other circuitry that is configured to serve as storage that stores data in the removable lenses for electrical retrieval by device **11.4.2-10**). Storage **11.4.2-84**, which may be formed as part of the removable lens system (e.g., as part of one or more of modules **11.4.2-74** and associated removable lenses **11.4.2-76**), may be used in storing lens information, user information, and/or other information electrically. When the removable lens system is coupled to the non-removable lens system (e.g., one or both lens modules **11.4.2-70** such as storage **11.4.2-84**), electrically stored information reader **11.4.2-82** may retrieve the data stored in storage **11.4.2-84**. Information reader **11.4.2-82** may be, for example, an electrically programmed read-only memory reader, a memory reader that reads laser-programmed storage or other programmable memory, circuitry that electrically measures storage **11.4.2-84** (e.g., to detect a pattern of conductive traces and/or resistive elements that are being used to store data), and/or other reader circuitry.

[2308] Information such as lens information, user information, and other information may also be stored using non-electrical storage. This storage may include, for example, bar codes, text, patterns of dots, and/or other information that can be detected optically. Optical data reading operations may be performed using infrared and/or visible cameras or other equipment (e.g., bar code scanning equipment) that reads optically encoded information. As an example, optical data reading equipment that reads data on the removable lens by imaging the removable lens such as infrared digital image sensors in gaze tracking sensors **11.4.2-86** and/or visible-light digital image sensors may be used in reading bar codes, text, and/or other optically encoded information.

[2309] Gaze trackers **11.4.2-86** (sometimes referred to as gaze monitoring systems, gaze tracking systems, eye monitoring systems, etc.) may monitor the eyes of the user in eye boxes **11.4.2-60** during operation of device **11.4.2-10**. Gaze trackers **11.4.2-86** may include light sources and light-detecting components such as image sensors. Each gaze tracker **11.4.2-86** may, as an example, include a light source such as a light-emitting diode light source or laser light source that produces an array of light beams (e.g., infrared light beams that do not interfere with the vision of the user). Blanket (flood) illumination at infrared and/or visible wavelengths may also be provided. Each gaze tracker **11.4.2-86** may also include a respective infrared image sensor (and/or visible-light image sensor) that faces a user's eye in a respective eye box **11.4.2-60** and capture images of the user's eye through lenses **11.4.2-72** and lenses **11.4.2-76** while the user's eye is being illuminated with the light beams from the light source of that gaze tracker. During normal operation, control circuitry **11.4.2-20** may use gaze trackers **11.4.2-86** to track a user's gaze (e.g., so that images being displayed on displays **11.4.2-14** can be adjusted based on the direction of the user's gaze, etc.). When it is desired to read information in the removable lens system that has been optically stored (e.g., by encoding the information in the form of a bar code, text, etc.), one or more image sensors such as one or more infrared image sensors in one or more gaze trackers **11.4.2-86** may optically read the information from the supplemental lens system (e.g., from one or more lens modules **11.4.2-74**).

[2310] FIG. **11.4.2-3** is a front view of an illustrative removable lens module showing arrangements that may be used to store information in the lens module. As shown in FIG. **11.4.2-3**, lens module **11.4.2-74** may include lens support structures **11.4.2-74H**. Support structures **11.4.2-74H** may be formed from polymer, metal, and/or other materials and may surround or otherwise

support lens **11.4.2-76** to form lens module **11.4.2-74**. Coupling structures **11.4.2-80** may be supported by support structures **11.4.2-74H**, so that coupling structures **11.4.2-80** hold lens module **11.4.2-74** to lens module **11.4.2-70** and support structures **11.4.2-26** when coupling structures **11.4.2-80** are coupled to mating coupling structures **11.4.2-78** in lens module **11.4.2-70**.

[2311] Lens information, user information, and other information can be stored in removable lens module **11.4.2-74** using one or more approaches. With one illustrative arrangement, electrical storage **11.4.2-84** is included in lens module **11.4.2-74**. Data may be stored in electrical storage **11.4.2-84**. When removable lens module **11.4.2-74** is coupled to non-removable lens module **11.4.2-70**, reader **11.4.2-82** may couple to electrical storage **11.4.2-84** and may retrieve the stored data from electrical storage **11.4.2-84**.

[2312] With another illustrative arrangement, a bar code such as bar code **11.4.2-92** may be included in lens module **11.4.2-74**. Bar code **11.4.2-92** may, as an example, be an infrared bar code that is formed from an ink that is visibly transparent while being opaque to infrared light (e.g., infrared light at the wavelength emitted by an infrared light source in gaze tracker **11.4.2-86**) or other bar code structure that is transparent to visible light and opaque at infrared wavelengths. With this type of arrangement, bar code **11.4.2-92** will be transparent to the user and will not interfere with the user's viewing of content on display **11.4.2-14**, even though bar code **11.4.2-92** is formed on lens **11.4.2-76**. At the same time, gaze tracker **11.4.2-86** is pointed towards lens **11.4.2-76** so that gaze tracker **11.4.2-86** may capture an image of bar code **11.4.2-92** at infrared wavelengths.

[2313] Bar codes, text information, or other visible-light (and/or infrared light) markings may be formed from laser markings, patterned ink, patterned metal coatings, and/or other markings formed in regions such as region **11.4.2-90**. Visible-light-encoded information of this type (e.g., information stored optically) may be read by a visible light image sensor in lens module **11.4.2-70** or an image sensor elsewhere in device **11.4.2-10** (e.g., a visible light image sensor in sensors **11.4.2-16**). In some situations, visible-light-encoded information such as text, icons, or other optically readable patterns may contain information that is recognizable to a user (e.g., text that a user can read such as the user's name or user name, an icon that a user can recognize, etc.). User recognizable information may be used to label lens modules **11.4.2-74** so that lens modules that belong to different users are not mixed up. An image sensor in device **11.4.2-10** (e.g., an image sensor in gaze tracker **11.4.2-87** and/or other image sensors) may optically read text, icons, or other patterned structures on the removable lens system that are also recognizable to the user and/or the image sensor may read patterns such as bar codes and other patterns that do not contain user-recognizable text or icons.

[2314] In some situations, it may be desirable to encrypt the information that is being stored in removable lens module **11.4.2-74**. Stored information may be encrypted using a cryptographic key. Control circuitry **11.4.2-12** can decrypt the encrypted information using a corresponding cryptographic key suitable for decryption (which may be the same as the encrypting key or may be related to the encrypting key). Control circuitry **11.4.2-12** can obtain the key from the user (e.g., using input-output devices **11.4.2-22**). If desired, control circuitry **11.4.2-12** can obtain the key from a remote server (e.g., an online service). A user password, username, or other information may, if desired, be gathered from the user and/or storage in device **11.4.2-10** and may be used by control circuitry **11.4.2-12** in obtaining the key from the online service. In some arrangements, keys may be stored in control circuitry **11.4.2-12** for use in decrypting encrypted information stored in lens module **11.4.2-74** (e.g., when device **11.4.2-10** is only used by the members of a single household). These illustrative arrangements and/or other arrangements for encrypting and decrypting the information stored in removable lens module **11.4.2-74** may be used to enhance privacy for stored user information.

[2315] Illustrative steps involved in using system **11.4.2-8** are shown in FIG. **11.4.2-4**.

[2316] During the operations of block **11.4.2-100**, a user who desires to use device **11.4.2-10** but who has specific visual defects (astigmatism, etc.) may attach a removable lens system that is

associated with that user to device **11.4.2-10**. During attachment of the removable lens system, removable lens support structures such as lens modules **11.4.2-74** may be aligned with non-removable lens support structures such as lens modules **11.4.2-70**, so that lenses **11.4.2-76** are aligned with corresponding lenses **11.4.2-72**. Coupling structures **11.4.2-82** and **11.4.2-84** may be used to hold lens modules **11.4.2-74** and **11.4.2-72** together. The presence of lenses **11.4.2-76** in alignment with lenses **11.4.2-72** allows the optical properties of lenses **11.4.2-76** to modify the optical properties of lenses **11.4.2-72** (e.g., to increase or decrease lens power, to add lens asymmetry to compensate for user astigmatism, etc.). In this way, the optical system of device **11.4.2-10** may be tailored to the user's vision.

[2317] During the operations of block **11.4.2-102**, control circuitry **11.4.2-12** of head-mounted device **11.4.2-10** may read information that has been stored in the removable lens system (e.g., using gaze trackers **11.4.2-86**, a visible-light image sensor, a memory reader such as reader **11.4.2-82**, etc.). The information that is read may be stored using a bar code, text, other optically readable patterns (e.g., patterned ink, patterned metal, or other patterned structures), and/or using electrical data storage arrangements (see, e.g., bar code **11.4.2-92**, visible information **11.4.2-90**, and electrically stored information in storage **11.4.2-84** of FIG. **11.4.2-3**).

[2318] In general, any suitable data may be stored in the removable lens system. This information may include, for example, information regarding the user that is associated with the removable lens system such as the user's eyeglass prescription (e.g., the spherical component of the user's prescription to be handled by spherical lens power, the astigmatic component of the user's prescription to be handled by an aspherical lens power, and/or the interpupillary distance of the user's prescription), the date of the prescription, the expiration date of the prescription, the user's username and other user credentials, manufacturing information such as the date of manufacture of the removable lens system, the name of the manufacturer, a serial number, a lot number, and a model name or other model type information, lens materials for lenses **11.4.2-76**, lens powers (e.g., spherical lens power and/or aspherical lens power) and/or other optical characteristics of lenses **11.4.2-76** such as known aberrations that can be later corrected using compensating image warping operations performed by control circuitry **11.4.2-12** during video playback, and/or other information.

[2319] After reading this information from the removable lens system during the operations of block **11.4.2-102**, control circuitry **11.4.2-12** may, if desired, make adjustments to the components in device **11.4.2-10**. These adjustments may be made based on real-time measurements from gaze trackers **11.4.2-86** and/or based on the stored data obtained from the removable lens system. For example, control circuitry **11.4.2-12** may use positioners **11.4.2-58** to adjust lens-to-lens spacing based on real-time measurements of the user's interpupillary distance determined from the positions of the user's pupils measured with gaze trackers **11.4.2-86** and/or based on stored interpupillary distance information for the user that is obtained from removable lens modules **11.4.2-74**. Control circuitry **11.4.2-12** may also adjust the positions of the lenses in device **11.4.2-10** along the Z axis of FIG. **11.4.2-2** to place lenses **11.4.2-72** and associated supplemental lenses **11.4.2-76** in a satisfactory location to allow the user to view content on displays **11.4.2-14**. These focusing adjustments with lenses **11.4.2-72** and **11.4.2-76** may be made based on the user's prescription, the powers of lenses **11.4.2-76**, and other information obtained using the stored information in lenses **11.4.2-76**. In arrangements in which lenses **11.4.2-72** have adjustable power (e.g., when lenses **11.4.2-72** are tunable lenses), control circuitry **11.4.2-12** can adjust lenses **11.4.2-72** to implement the spherical component of the user's prescription. The asymmetric component of a user's prescription may be handled using the asymmetric portion of supplemental lenses **11.4.2-76** (as an example).

[2320] After making adjustments to device **11.4.2-10** based on information from the supplemental lens system formed from lenses **11.4.2-76**, a user may place device **11.4.2-10** on the user's head.

[2321] During the operations of block **11.4.2-104**, control circuitry **11.4.2-12** may make further

adjustments to the components in device **11.4.2-10**. For example, control circuitry **11.4.2-12** may use positioners **11.4.2-58** to adjust the lens-to-lens separation of left and right lenses **11.4.2-72** (and left and right lenses **11.4.2-76**) based on information from sensors in device **11.4.2-10** (e.g., based on a measured interpupillary distance gathered using gaze trackers **11.4.2-86**).

[2322] During the operations of block **11.4.2-104**, an iris scanner, fingerprint sensor, input-output devices that gather username and password information, or other components may be used in authenticating the user. The user may, for example, be directed to supply a username and password so that control circuitry **11.4.2-12** can establish a communications link with an online service that provides media content (block **11.4.2-108**). If the user is not authenticated by control circuitry **11.4.2-12** (e.g., if the user's username and password or other credentials (biometric information, etc.) are not authenticated, control circuitry **11.4.2-12** can, during the operations of block **11.4.2-110**, prevent the user from accessing the online service, selectively block content, and/or take other suitable action. If desired, the user may be allowed to override the authentication process at block **11.4.2-110** and/or additional authentication techniques may be used in scenarios in which the user fails authentication during the operations block **11.4.2-106**.

[2323] In some configurations of device **11.4.2-10**, lenses **11.4.2-76** may carry user information (e.g., user game preferences, user movie preferences, etc.). In these configurations, control circuitry **11.4.2-12** can refrain from decoding this information until the user has been authenticated during the operations of block **11.4.2-106** and an appropriate decryption key has been obtained (e.g., the user's password).

[2324] In some embodiments, user information is stored by control circuitry **11.4.2-12**. The user information may include information on a user's eye glasses prescription (e.g., a required optical prescription strength). The information stored in lenses **11.4.2-76** may include information on the optical prescription strength of lenses **11.4.2-76**. After determining the user of device **11.4.2-10** and retrieving the user's stored prescription from control circuitry **11.4.2-12**, control circuitry **11.4.2-12** may compare this retrieved user prescription information with the prescription of lenses **11.4.2-76** to determine if there is a satisfactory match. In the event that the prescription of lenses **11.4.2-76** does not match the user's prescription, control circuitry **11.4.2-12** may use display **11.4.2-14** and/or other output circuitry to issue a visible and/or audible alert for the user and may, if desired, decline to display content on display **11.4.2-14**.

[2325] In scenarios in which optical aberration information is included in the information stored in lenses **11.4.2-76** (e.g., information indicating that lenses **11.4.2-76** have a particular amount of barrel distortion, pincushion distortion, spherical aberration, chromatic aberration, etc.), control circuitry **11.4.2-12** may, during the playback operations of block **11.4.2-108**, apply geometric transformations or other video processing to the content being displayed on display **11.4.2-14** that compensates for this distortion. In this way, a user may view content that is free or nearly free of aberrations. If, as an example, lenses **11.4.2-76** exhibit pincushion distortion of a particular amount, information on this amount of pincushion distortion can be stored in lenses **11.4.2-76**. During playback operations, control circuitry **11.4.2-12** can apply compensating (barrel) image warping to the content for display **11.4.2-14** to ensure that the displayed content exhibits low distortion and aberrations, thereby enhancing the content viewing experience for the user. In general, control circuitry **11.4.2-12** may apply any image processing correction that helps enhance image quality and the user's viewing experience. For example, corrections can be applied to correct for chromatic aberration, spherical aberration, etc.

[2326] In accordance with an embodiment, a system is provided that includes a head-mounted support structure, a display coupled to the head-mounted support structure, control circuitry configured to supply content using the display, a non-removable lens coupled to the head-mounted support structure that is configured to present the content from the display in an eye box, and a removable supplemental lens removably coupled to head-mounted support structure between the non-removable lens and the eye box, the removable supplemental lens is configured to store

information that is retrieved by the control circuitry.

[2327] In accordance with another embodiment, the system includes a gaze tracking system having a light emitter and an infrared image sensor, the removable supplemental lens has a bar code that stores the information, the bar code is opaque at infrared wavelengths and transparent at visible wavelengths, and the image sensor is configured to read the bar code while the removable supplemental lens is removably coupled to the head-mounted support structure.

[2328] In accordance with another embodiment, the bar code includes infrared-opaque-and-visible-light-transparent ink on a clear lens element in the removable supplemental lens.

[2329] In accordance with another embodiment, the information includes information selected from the group consisting of an eyeglass prescription, an eyeglass prescription date of expiration, a username, a manufacturing date, a manufacturer, a serial number, a manufacturing lot number, a model name, a model type, and information on optical characteristics of the removable supplemental lens.

[2330] In accordance with another embodiment, the system includes a data reading device coupled to the head-mounted support structure, the control circuitry is configured to use the data reading device to obtain the information from the removable supplemental lens.

[2331] In accordance with another embodiment, the information includes an eyeglasses prescription.

[2332] In accordance with another embodiment, the removable supplemental lens includes memory and the data reading device is configured to obtain the information from the memory.

[2333] In accordance with another embodiment, the removable supplemental lens includes text and the data reading device is configured to obtain the information by optically reading the text.

[2334] In accordance with another embodiment, the removable supplemental lens includes circuitry configured to store data and the data reading device is configured to obtain the information from the circuitry.

[2335] In accordance with another embodiment, the removable supplemental lens includes programmable read only memory, the data reading device is configured to obtain the information from the programmable read only memory, and the information includes a lens characteristics of the removable supplemental lens.

[2336] In accordance with another embodiment, the lens characteristic includes a lens power associated with the removable supplemental lens.

[2337] In accordance with another embodiment, the lens characteristic includes an aspherical lens power associated with the removable supplemental lens.

[2338] In accordance with another embodiment, the lens characteristic includes an optical aberration associated with the removable supplemental lens.

[2339] In accordance with another embodiment, the control circuitry is configured to obtain the information when the removable supplemental lens is coupled to the head-mounted support structure.

[2340] In accordance with another embodiment, the information includes information selected from the group consisting of a username, a serial number, optical characteristics of the removable supplemental lens, and eyeglass prescription information.

[2341] In accordance with an embodiment, a head-mounted device, is provided that includes head-mounted support structures, a display configured to display content, left and right positioners, a non-removable lens system having left and right lenses that are positioned respectively by the left and right positioners, a removable lens system with removable supplemental left and right lenses that are removably coupled to the head-mounted support structures in alignment respectively with the right and left lenses, and control circuitry configured to obtain information stored in the removable lens system.

[2342] In accordance with another embodiment, the information includes an interpupillary distance for a user and the control circuitry is configured to use the left and right positioners to adjust a lens-

center-to-lens-center spacing associated with the left and right lenses to match the interpupillary distance.

[2343] In accordance with another embodiment, the head-mounted device includes an optically readable pattern on the removable lens system that stores the information.

[2344] In accordance with another embodiment, the head-mounted device includes an image sensor that optically reads the optically readable pattern to retrieve the information from the removable lens system.

[2345] In accordance with another embodiment, the image sensor includes an infrared image sensor in a gaze tracking system.

[2346] In accordance with another embodiment, the information includes a user's eyeglass prescription and the control circuitry is configured to adjust the left and right positioners based on the user's eyeglass prescription.

[2347] In accordance with an embodiment, an electronic device configured to operate with a removable supplemental lens system is provided that includes a display that displays content, control circuitry, a non-removable lens system configured to allow the display content to be viewed from eye boxes, a removable supplemental lens system, and head-mounted support structures that support the display and that support the non-removable lens system between the eye boxes and the display, the removable supplemental lens system is configured to be supported between the non-removable lenses and the eye boxes, the removable supplemental lens system is configured to store information, and the control circuitry is configured to obtain the information from the removable supplemental lens system when the removable supplemental lens system is between the non-removable lenses and the eye boxes.

[2348] In accordance with another embodiment, the electronic device includes lens positioners, the control circuitry is configured to use the lens positioners to adjust a lens-to-lens spacing associated with left and right lenses in the non-removable lens system based on the information obtained from the removable supplemental lens system.

[2349] In accordance with another embodiment, the information includes an optical prescription associated with the removable supplemental lens system and the control circuitry is configured to issue an alert to a user in response to determining that there is a mismatch between the user's prescription and the optical prescription associated with the removable supplemental lens system.

[2350] In accordance with another embodiment, the information includes distortion information specifying an optical distortion associated with the supplemental lens system and the control circuitry is configured to apply a compensating correction to the content being displayed on the display based on the distortion information.

[2351] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

11.4.3: RX Lenses

[2352] FIG. **11.4.3-1** illustrates an optical assembly **11.4.3-100** for use in an HMD device such as those described herein. The optical assembly **11.4.3-100** can be configured to project light toward the user's eyes when donning the HMD. In at least one example, the optical assembly **11.4.3-100** can include a removably attachable prescription ("Rx") lens **11.4.3-102**. The Rx lens **11.4.3-102** can be removably coupled to a rim **11.4.3-104**. The rim **11.4.3-104** can be fixed to an outer housing **11.4.3-106** of the optical assembly **11.4.3-106**. The Rx lens **11.4.3-102** can be swappable to accommodate the vision correction needs of various users of the HMD. The optical assembly **11.4.3-100** can also include a display, such as a display screen (not shown), configured to project light toward the user's eye(s) through the Rx lens **11.4.3-102**.

[2353] FIG. **11.4.3-2** illustrates the optical assembly **11.4.3-100** with the Rx lens removed to illustrate another lens **11.4.3-108** of the optical assembly. The fixed lens **11.4.3-108** shown in FIG. **11.4.3-2** can be disposed below/behind the Rx lens **11.4.3-102** shown in FIG. **11.4.3-1** (e.g., with

the Rx lens **11.4.3-102** disposed between the fixed lens **11.4.3-108** and the user's eye **11.4.3-108** during use) and substantially un-removably fixed to the optical assembly **11.4.3-100**, for example to the rim **11.4.3-104** of the optical assembly **11.4.3-100**.

[2354] FIG. **11.4.3-3** illustrates another example of the optical assembly with the Rx lens **11.4.3-102**, rim **11.4.3-104**, and housing **11.4.3-106** removed to illustrate an array of magnets **11.4.3-110** disposed peripherally to the lens **11.4.3-108** and the Rx lens **11.4.3-102**. The magnets **11.4.3-110** can be embedded in the rim or otherwise fixed in the optical assembly **11.4.3-100** as shown to provide magnetic securement of the Rx lens **11.4.3-102**, which can also include a complimentary array of magnets to removably attach the Rx lens **11.4.3-102** to the optical assembly **11.4.3-100** over the fixed lens **11.4.3-108**.

[2355] The magnet array **11.4.3-110** can be used for macro-alignment of the Rx lens **11.4.3-102** to the assembly **11.4.3-100** to ensure the Rx lens **11.4.3-102** is rotated correctly relative to the fixed lens **11.4.3-108** and the rest of the assembly **11.4.3-100**, including the rim **11.4.3-104** and the housing **11.4.3-106**.

[2356] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **11.4.3-1-11.4.3-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **11.4.3-1-11.4.3-3**.

[2357] FIG. **11.4.3-4** illustrates an example of an optical assembly **11.4.3-200** configured for receiving a removably attachable Rx lens. FIG. **11.4.3-4** shows the assembly **11.4.3-200** as assembled as well as an exploded view thereof. The two magnet arrays **11.4.3-210** and **11.4.3-212** are also shown. The assembly **11.4.3-200** can include a housing **11.4.3-206**, rim **11.4.3-204**, a first magnet array **11.4.3-210**, and a second magnet array **11.4.3-212**. In at least one example, the Rx lens (not shown) can be coupled with the rim **11.4.3-204** such that the Rx lens and the rim **11.4.3-204** are removably secured to the housing **11.4.3-206**. The second magnet array **11.4.3-212** can be embedded or otherwise secured with the rim **11.4.3-204** such that the second magnet array **11.4.3-212** magnetically couples to the first magnet array **11.4.3-210** to removably secure the Rx lens and the rim **11.4.3-204** to the housing **11.4.3-206**.

[2358] In at least one example, the magnet arrays **11.4.3-210**, **11.4.3-212**, either those fixed to the rim **11.4.3-204** or those fixed to the Rx lens, can be mapped when the lenses are installed by the user to ensure proper lens alignment with the system/assembly **11.4.3-200** in order to ensure the correct lens (Rx lens) is being installed. The Rx lens itself can be mapped (e.g., automatically mapped by imaging components such as rear-facing cameras, including gaze tracking cameras . . . etc.) upon user install to ensure the correct lens is being secured by the user.

[2359] FIG. **11.4.3-5** illustrates left and right magnet arrays **11.4.3-310a**, **11.4.3-310b**, respectively, which can be used for attaching left and right Rx lenses, respectively. In at least one example, the left and right arrays **11.4.3-310a-b** can include individual magnets with alternating polarities, where the polarities of the left magnet array **11.4.3-310a** are opposite relative to the corresponding magnets of the right magnet array **11.4.3-310b**. The polarities of the individual magnets of the left and right arrays **11.4.3-310a-b** can be flipped about a line of symmetry **11.4.3-314** between the user's eyes and/or between the left and right arrays **11.4.3-310a-b**.

[2360] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **11.4.3-4** and **11.4.3-5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the

devices, features, components, and parts shown in FIGS. **11.4.3-4** and **11.4.3-5**.

[2361] FIGS. **11.4.3-6** and **11.4.3-7** illustrate a perspective view and a side/cross-sectional view of an example of an Rx lens **11.4.3-402** configured to be removably attached to an optical assembly/module of an HMD device. Depending on the prescription of the Rx lens **11.4.3-402**, one or more proximal, perimeter edges of the lens **11.4.3-402** may be disposed proud of the housing or rim of the optical assembly. In at least one example, the proximal, peripheral edge **11.4.3-416** of the lens **11.4.3-402** can include a rounded and/or chamfered geometry to reduce sharp or abrupt edge features.

[2362] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **11.4.3-6** and **11.4.3-7** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **11.4.3-6** and **11.4.3-7**.

[2363] FIG. **11.4.3-8** illustrates an example of an Rx lens assembly including an Rx lens **11.4.3-502** secured to a rim **11.4.3-504** and a magnet array **11.4.3-512** embedded in the rim **11.4.3-504**. In at least one example, the magnets of the magnet array **11.4.3-512** can be molded into the rim **11.4.3-504**. In at least one example, the assembly includes a notched feature **11.4.3-520** where a protruding feature **11.4.3-524** of the lens **11.4.3-502** extends into a recess **11.4.3-522** of the rim **11.4.3-504** to form a mechanical fit between the lens **11.4.3-502** and the rim **11.4.3-504**. In at least one example, the notched feature **11.4.3-520** includes a protrusion of the rim **11.4.3-504** extending into a cavity of the lens **11.4.3-502**. In at least one example, the rim **11.4.3-504** is removably securable to the housing of an optical module via the magnets of the magnet array **11.4.3-512** and the lens **11.4.3-502** is removably securable to the rim **11.4.3-504** via the notched feature **502**. In addition, in at least one example, the optical assemblies described herein can include multiple notch assemblies similar to the notch assembly **11.4.3-520** shown in FIG. **8** between the lens **11.4.3-502** and the rim **11.4.3-504**, including one or more notch features securing the rim **11.4.3-504** to a housing or other component of the optical assembly, and/or a notch assembly securing the optical assembly to other component of the HMD. In at least one example, the magnets of the magnet array **11.4.3-512** can vary in size to achieve desired attraction forces where the magnets of the magnet array **11.4.3-512** include a shaved or chamfered corner to accommodate the lens edges adjacent the magnets.

[2364] In at least one example, the assembly can include one or more opaque films or inks applied to the proximal, peripheral edges **11.4.3-526**, **11.4.3-528** of the lens **11.4.3-502** and the rim **11.4.3-504**, respectively, to minimize edge glares and unsightly light features at the peripheral edge of the assembly. In at least one example, as shown in FIG. **11.4.3-8**, the proximal, peripheral edge **11.4.3-526** of the lens **11.4.3-502** can extend beyond the proximal, peripheral edge **11.4.3-528** of the rim **11.4.3-504**.

[2365] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **11.4.3-8** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **11.4.3-8**.

[2366] FIG. **11.4.3-9** illustrates another example of a removable Rx lens assembly including an Rx lens **11.4.3-602** secured to a rim **11.4.3-604** at a notch feature **11.4.3-620**. The assembly can also include one or more magnets of a magnet array **11.4.3-612** embedded within, e.g., molded with, the rim **11.4.3-604**. In the illustrated example of FIG. **11.4.3-9**, the proximal, peripheral edge **11.4.3-**

628 of the rim **11.4.3-604** extends beyond the proximal, peripheral edge **11.4.3-626** of the lens **11.4.3-602**.

[2367] FIG. **11.4.3-10** illustrates another example of a removable Rx lens assembly including an Rx lens **11.4.3-702** secured to a rim **11.4.3-704** at a notch feature **11.4.3-720**. The assembly can also include one or more magnets of a magnet array **11.4.3-712** embedded within, e.g., molded with, the rim **11.4.3-704**. In the illustrated example of FIG. **11.4.3-10**, the proximal, peripheral edge **11.4.3-728** of the rim **11.4.3-704** extends beyond the proximal, peripheral edge **11.4.3-726** of the lens **11.4.3-702**. In the illustrated example of FIG. **11.4.3-10**, a film or ink **11.4.3-718** can be applied to the edge **11.4.3-728** of the rim **11.4.3-702** to reduce glare and edge/lensing effects around a periphery of the assembly.

[2368] FIG. **11.4.3-11** illustrates another example of a removable Rx lens assembly including an Rx lens **11.4.3-802** secured to a rim **11.4.3-804** at a notch feature **11.4.3-820**. The assembly can also include one or more magnets of a magnet array **11.4.3-812** embedded within, e.g., molded with, the rim **11.4.3-804**. In the illustrated example of FIG. **11.4.3-11**, the proximal, peripheral edge **11.4.3-828** of the rim **11.4.3-804** can be substantially flush with the proximal, peripheral edge **11.4.3-826** of the lens **802**.

[2369] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **11.4.3-8-11.4.3-11** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in any other figures shown and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to any other figure shown and described herein, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **11.4.3-8-11.4.3-11**.

[2370] In at least one example of the present disclosure, the lenses of HMD devices disclosed herein, including the Rx lenses disclosed herein, can include multiple layers or lenses stacked together as layers. In at least one example, an HMD lens stack can include three or more lenses or layers of lenses stacked together to form a single lens. In at least one example, the lens stack can include four or more lenses. In at least one example, a four-lens stack can include a first lens, a second lens coupled to the first lens, a third lens coupled to the second with the second lens between the first lens and the third lens, and a fourth lens coupled to the third lens with the third lens between the fourth lens and the second lens. In such an example, the fourth lens can be an Rx lens customized to the prescription of the user. In one example, the lens stack can include a first, second, and third lens with the second lens disposed between the third lens and the first lens. In such an example, the third lens can include the Rx lens customized to the user's prescription. In such examples, e.g., the third lens of a three stack lens being the Rx lens or the fourth lens of a four-stack lens being Rx lens, the other lenses can be standard lenses of the HMD across other SKUs or HMD devices. The Rx lens of the stacks can be permanently fixed to the other lenses or lens layers of the stack or the Rx lenses can be removably coupled thereto to allow replacement.

[2371] In at least one example, any of the lenses, including Rx lenses, of HMD devices described herein, can include identification features recognizable by the HMD device, including via the various cameras, sensors, and controllers of the optical systems of the HMD, to recognize and verify the correct lens is attached for a given user. In one example, lenses described herein can include radio-frequency identification features detectable by the HMD. In one example, the lenses can include IR visible barcodes detectable by IR sensors/cameras of the HMD. In one example, the lenses can include QR codes or other codes or visual identification features which the HMD can detect and compare against server-based data or other data to verify the correct lenses for given users and/or for lens calibration purposes. In at least one example, the sensor system of the optical module of the HMD, including the cameras, sensors, controllers/processors, can be used to detect a distortion of light through an Rx lens and match the distortion to a certain prescription and/or user associated with the prescription to verify the correct lens is being used for the given user.

[2372] In at least one example of the present disclosure, any of the lenses described herein, including Rx lenses described herein, can be formed to accommodate a prescription based on a visual/focal distance provided by the display/lens system of the HMD to the user, which can be determined based on a traditional prescription of a user based on infinity and 60 cm standard prescription distances. In one example, if a user provides a standard prescription, for example a prescription associated with a visual distance of infinity and 60 cm, or other distances common in standard prescriptions, the Rx lenses of the HMD can be made based on that prescription to be a prescription for a visual distance of 1-meter. 1-meter is used as a non-limiting example of a focal distance of the optical display system of the HMD. Other distances can be used to match the effective focal distance provided by the HMD, if that effective focal distance displayed by the HMD is a distance other than 1-meter.

XII: Curtain

[2373] FIGS. **12.0-1** illustrates a view of an HMD **12.0-100** including a curtain assembly **12.0-102**. The curtain assembly **12.0-102** is described in more detail here in section XII.

12.1: Electronic Devices with Stretchable Fabric Covers

[2374] Electronic devices such as head-mounted electronic devices may include displays for presenting images to users. To accommodate variations in the interpupillary distances associated with different users, a head-mounted device may have left-eye and right-eye optical modules that move with respect to each other. Each optical module may include a display device for producing an image and an associated optical component such as a lens for providing the image to an associated eye box in which an eye of the user is located for viewing the image. The optical modules, which may sometimes be referred to as optical systems, display systems, lens systems, lens and display assemblies, etc., may each have a support structure such as a lens barrel that supports a respective display and lens.

[2375] Actuators may be used to position the lens barrels within the housing of a head-mounted device. To hide the actuators and other electrical components such as integrated circuits, batteries, sensors, etc. and to hide potentially unsightly internal housing structures from view, the rear of a head-mounted device that faces the user may be provided with a cosmetic covering. Openings in the cosmetic covering may receive the lens barrels of the optical modules. The cosmetic covering may be configured to accommodate movement in the positions of the optical modules for different interpupillary distances.

[2376] The covering may be a stretchable fabric cover having first strands that form a warp knit layer with diamond-shaped openings and second strands that zig-zag back and forth within the diamond-shaped openings. The first strands may be elastic and may provide stretch in the warp direction, while the second strands may be covered strands that provide stretch in the weft direction. The covered strands may have a drawn textured yarn covering that forms a fuzzy and opaque layer for hiding electrical components.

[2377] If desired, the cover may be a jacquard fabric with different zones of stretch, opacity, and/or other characteristics. For example, the cover may have a nose bridge region interposed between eye regions. The nose bridge region may have greater opacity and less stretch than the eye regions. Arrangements in which the cover is formed from three-dimensional fabric having cylindrical portions that move relative to one another in response to optical module movements may also be used.

[2378] An electronic device such as a head-mounted device may have a front face that faces away from a user's head and may have an opposing rear face that faces the user's head. Optical modules on the rear face may be used to provide images to a user's eyes. The positions of the optical modules may be adjusted to accommodate different user interpupillary distances. Internal device structures may be hidden from view by the user by covering the rear face of the device with a curtain. The curtain, which may sometimes be referred to as a cover, covering structure, rear housing cover, rear housing wall, rear housing structure, cosmetic covering, etc., may help block

potentially unsightly internal structures from view, while accommodating movement of the optical modules.

[2379] A top view of an illustrative head-mounted device with a curtain is shown in FIGS. **12.1-1**. As shown in FIGS. **12.1-1**, head-mounted devices such as electronic device **12.1-10** may have head-mounted support structures such as housing **12.1-12**. Housing **12.1-12** may include portions (e.g., support structures **12.1-12T**) to allow device **12.1-10** to be worn on a user's head. Support structures **12.1-12T** may be formed from fabric, polymer, metal, and/or other material. Support structures **12.1-12T** may form a strap or other head-mounted support structures that help support device **12.1-10** on a user's head. A main support structure (e.g., main housing portion **12.1-12M**) of housing **12.1-12** may support electronic components such as displays **12.1-14**. Main housing portion **12.1-12M** may include housing structures formed from metal, polymer, glass, ceramic, and/or other material. For example, housing portion **12.1-12M** may have housing walls on front face F and housing walls on adjacent top, bottom, left, and right side faces that are formed from rigid polymer or other rigid support structures and these rigid walls may optionally be covered with electrical components, fabric, leather, or other soft materials, etc. The walls of housing portion **12.1-12M** may enclose internal components **12.1-38** in interior region **12.1-34** of device **12.1-10** and may separate interior region **12.1-34** from the environment surrounding device **12.1-10** (exterior region **12.1-36**). Internal components **12.1-38** may include integrated circuits, actuators, batteries, sensors, and/or other circuits and structures for device **12.1-10**. Housing **12.1-12** may be configured to be worn on a head of a user and may form glasses, a hat, a helmet, goggles, and/or other head-mounted device. Configurations in which housing **12.1-12** forms goggles may sometimes be described herein as an example.

[2380] Front face F of housing **12.1-12** may face outwardly away from a user's head and face. Opposing rear face R of housing **12.1-12** may face the user. Portions of housing **12.1-12** (e.g., portions of main housing **12.1-12M**) on rear face R may form a cover such as curtain **12.1-12C**. In an illustrative configuration, curtain **12.1-12C** includes a fabric layer that separates interior region **12.1-34** from the exterior region to the rear of device **12.1-10**. Other structures may be used in forming curtain **12.1-12C**, if desired. The presence of curtain **12.1-12C** on rear face R may help hide internal housing structures, internal components **12.1-38**, and other structures in interior region **12.1-34** from view by a user.

[2381] Device **12.1-10** may have left and right optical modules **12.1-40** (sometimes referred to as optical assemblies). Each optical module may include a respective display **12.1-14**, lens **12.1-30**, and support structure **12.1-32**. Support structures **12.1-32**, which may sometimes be referred to as lens barrels or optical module support structures, may include hollow cylindrical structures with open ends or other supporting structures to house displays **12.1-14** and lenses **12.1-30**. Support structures **12.1-32** may, for example, include a left lens barrel that supports a left display **12.1-14** and left lens **12.1-30** and a right lens barrel that supports a right display **12.1-14** and right lens **12.1-30**. Displays **12.1-14** may include arrays of pixels or other display devices to produce images. Displays **12.1-14** may, for example, include organic light-emitting diode pixels formed on substrates with thin-film circuitry and/or formed on semiconductor substrates, pixels formed from crystalline semiconductor dies, liquid crystal display pixels, scanning display devices, and/or other display devices for producing images. Lenses **12.1-30** may include one or more lens elements for providing image light from displays **12.1-14** to respective eyes boxes **12.1-13**. Lenses may be implemented using refractive glass lens elements, using mirror lens structures (catadioptric lenses), using holographic lenses, and/or other lens systems. When a user's eyes are located in eye boxes **12.1-13**, displays (display panels) **12.1-14** operate together to form a display for device **12.1-10** (e.g., the images provided by respective left and right optical modules **12.1-40** may be viewed by the user's eyes in eye boxes **12.1-13** so that a stereoscopic image is created for the user). The left image from the left optical module fuses with the right image from a right optical module while the display is viewed by the user.

[2382] Not all users have the same interpupillary distance P. To provide device **12.1-10** with the ability to adjust the interpupillary spacing between modules **12.1-40** along lateral dimension X and thereby adjust the spacing P between eye boxes **12.1-13** to accommodate different user interpupillary distances, device **12.1-10** may be provided with actuators **12.1-42**. Actuators **12.1-42** can be manually controlled and/or computer-controlled actuators (e.g., computer-controlled motors) for moving support structures **12.1-32** relative to each other.

[2383] As shown in FIGS. **12.1-2**, curtain **12.1-12C** may cover rear face R while leaving lenses **12.1-30** of optical modules **12.1-40** uncovered (e.g., curtain **12.1-12C** may have openings that are aligned with and receive modules **12.1-40**). As modules **12.1-40** are moved relative to each other along dimension X to accommodate different interpupillary distances for different users, modules **12.1-40** move relative to fixed housing structures such as the walls of main portion **12.1-12M** and move relative to each other. To prevent undesired wrinkling and buckling of curtain **12.1-12C** as optical modules **12.1-40** are moved relative to rigid portions of housing **12.1-12M** and relative to each other, a fabric layer or other cover layer in curtain **12.1-12C** may be configured to slide, stretch, open/close, and/or otherwise adjust to accommodate optical module movement.

[2384] A schematic diagram of an illustrative electronic device such as a head-mounted device or other wearable device is shown in FIGS. **12.1-3**. Device **12.1-10** of FIGS. **12.1-3** may be operated as a stand-alone device and/or the resources of device **12.1-10** may be used to communicate with external electronic equipment. As an example, communications circuitry in device **12.1-10** may be used to transmit user input information, sensor information, and/or other information to external electronic devices (e.g., wirelessly or via wired connections). Each of these external devices may include components of the type shown by device **12.1-10** of FIGS. **12.1-3**.

[2385] As shown in FIGS. **12.1-3**, a head-mounted device such as device **12.1-10** may include control circuitry **12.1-20**. Control circuitry **12.1-20** may include storage and processing circuitry for supporting the operation of device **12.1-10**. The storage and processing circuitry may include storage such as nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **12.1-20** may be used to gather input from sensors and other input devices and may be used to control output devices. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors and other wireless communications circuits, power management units, audio chips, application specific integrated circuits, etc. During operation, control circuitry **12.1-20** may use display(s) **12.1-14** and other output devices in providing a user with visual output and other output.

[2386] To support communications between device **12.1-10** and external equipment, control circuitry **12.1-20** may communicate using communications circuitry **12.1-22**. Circuitry **12.1-22** may include antennas, radio-frequency transceiver circuitry, and other wireless communications circuitry and/or wired communications circuitry. Circuitry **12.1-22**, which may sometimes be referred to as control circuitry and/or control and communications circuitry, may support bidirectional wireless communications between device **12.1-10** and external equipment (e.g., a companion device such as a computer, cellular telephone, or other electronic device, an accessory such as a point device, computer stylus, or other input device, speakers or other output devices, etc.) over a wireless link. For example, circuitry **12.1-22** may include radio-frequency transceiver circuitry such as wireless local area network transceiver circuitry configured to support communications over a wireless local area network link, near-field communications transceiver circuitry configured to support communications over a near-field communications link, cellular telephone transceiver circuitry configured to support communications over a cellular telephone link, or transceiver circuitry configured to support communications over any other suitable wired or wireless communications link. Wireless communications may, for example, be supported over a Bluetooth® link, a WiFi® link, a wireless link operating at a frequency between 10 GHz and 400

GHz, a 60 GHz link, or other millimeter wave link, a cellular telephone link, or other wireless communications link. Device **12.1-10** may, if desired, include power circuits for transmitting and/or receiving wired and/or wireless power and may include batteries or other energy storage devices. For example, device **12.1-10** may include a coil and rectifier to receive wireless power that is provided to circuitry in device **12.1-10**.

[2387] Device **12.1-10** may include input-output devices such as devices **12.1-24**. Input-output devices **12.1-24** may be used in gathering user input, in gathering information on the environment surrounding the user, and/or in providing a user with output. Devices **12.1-24** may include one or more displays such as display(s) **12.1-14**. Display(s) **12.1-14** may include one or more display devices such as organic light-emitting diode display panels (panels with organic light-emitting diode pixels formed on polymer substrates or silicon substrates that contain pixel control circuitry), liquid crystal display panels, microelectromechanical systems displays (e.g., two-dimensional mirror arrays or scanning mirror display devices), display panels having pixel arrays formed from crystalline semiconductor light-emitting diode dies (sometimes referred to as microLEDs), and/or other display devices.

[2388] Sensors **12.1-16** in input-output devices **12.1-24** may include force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors such as a touch sensor that forms a button, trackpad, or other input device), and other sensors. If desired, sensors **12.1-16** may include optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, optical touch sensors, optical proximity sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, image sensors, fingerprint sensors, iris scanning sensors, retinal scanning sensors, and other biometric sensors, temperature sensors, sensors for measuring three-dimensional non-contact gestures (“air gestures”), pressure sensors, sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or inertial measurement units that contain some or all of these sensors), health sensors such as blood oxygen sensors, heart rate sensors, blood flow sensors, and/or other health sensors, radio-frequency sensors, depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices that capture three-dimensional images), optical sensors such as self-mixing sensors and light detection and ranging (LIDAR) sensors that gather time-of-flight measurements, humidity sensors, moisture sensors, gaze tracking sensors, electromyography sensors to sense muscle activation, facial sensors, and/or other sensors. In some arrangements, device **12.1-10** may use sensors **12.1-16** and/or other input-output devices to gather user input. For example, buttons may be used to gather button press input, touch sensors overlapping displays can be used for gathering user touch screen input, touch pads may be used in gathering touch input, microphones may be used for gathering audio input, accelerometers may be used in monitoring when a finger contacts an input surface and may therefore be used to gather finger press input, etc.

[2389] If desired, electronic device **12.1-10** may include additional components (see, e.g., other devices **12.1-18** in input-output devices **12.1-24**). The additional components may include haptic output devices, actuators for moving movable housing structures, audio output devices such as speakers, light-emitting diodes for status indicators, light sources such as light-emitting diodes that illuminate portions of a housing and/or display structure, other optical output devices, and/or other circuitry for gathering input and/or providing output. Device **12.1-10** may also include a battery or other energy storage device, connector ports for supporting wired communication with ancillary equipment and for receiving wired power, and other circuitry.

[2390] FIGS. **12.1-4** and **12.1-5** are top views of device **12.1-10** showing how the optical modules of device **12.1-10** move with respect to each other along lateral dimension X to accommodate different interpupillary distances P (the distance between a user's left and right eyes). In the example of FIGS. **12.1-4**, left optical module **12.1-40L** and right optical module **12.1-40R** have

been moved towards each other to accommodate a small interpupillary distance. In the example of FIGS. 12.1-5, left optical module 12.1-40L and right optical module 12.1-40R have been moved away from each other to accommodate a large interpupillary distance.

[2391] Curtain 12.1-12C has edge portions such as left portion 12.1-12C-L between left housing wall 12.1-12M-L and left optical module 12.1-40L and right portion 12.1-12C-R between right housing wall 12.1-12M-R and right optical module 12.1-40R. Middle portion 12.1-12C-M of curtain 12.1-12C extends between left optical module 12.1-40L and right optical module 12.1-40R. In the configuration of FIGS. 12.1-4, optical modules 12.1-40L and 12.1-40R are relatively close to each other, so middle portion 12.1-12C-M is relatively small and portions 12.1-12C-L and 12.1-12C-R are relatively large. In the configuration of FIGS. 12.1-5, optical modules 12.1-40L and 12.1-40R are relatively far from each other, so left portion 12.1-12C-L and right portion 12.1-12C-R are shorter along lateral dimension X and middle portion 12.1-12C-M has been enlarged relative to the configuration of FIGS. 12.1-4.

[2392] To help accommodate differences in size for curtain 12.1-12C (e.g., length changes for portions of curtain 12.1-12C along lateral dimension X), curtain 12.1-12C may include a cover layer such as cover layer 12.1-12CC formed from a stretchable material such as fabric. Cover layer 12.1-12CC may be supported by a rigid frame. The fabric of cover layer 12.1-12CC may be provided with a peripheral elastic band that helps allow the fabric to slide relative to the frame while being retained securely on the frame, thereby further helping curtain 12.1-12C to be dynamically adjusted without exhibiting undesired buckling and wrinkling. Cover layer 12.1-12CC may have left and right openings to receive respective left and right optical modules 12.1-40L and 12.1-40R.

[2393] Cover layer 12.1-12CC may be formed from fabric having the desired opacity (e.g., sufficiently opaque to hide electronic components in device 12.1-10), stretch (e.g., cover 12.1-12CC may be configured to stretch by at least 50%, at least 75%, at least 100%, or at least 150% of its length to allow dynamic expansion and contraction of cover 12.1-12CC as optical modules 12.1-40L and 12.1-40R are adjusted, as an example), modulus of elasticity (e.g., the elastic modulus of cover 12.1-12CC may be sufficiently low so that less electrical power is required for actuators 12.1-42 to stretch cover 12.1-12CC), durability (e.g., cover 12.1-12CC may be configured to maintain dimensional stability over long term use by avoiding wrinkling after being exposed to skin sebum and/or other chemicals, repeated stretching, etc.), cosmetic appeal, and/or other suitable characteristics.

[2394] FIGS. 12.1-6 is a front view of an illustrative cover layer for curtain 12.1-12C. As shown in FIGS. 12.1-6, cover layer 12.1-12CC may include fabric formed from fabric such as fabric 12.1-62. Fabric 12.1-62 may be warp knit fabric, weft knit fabric, or other knit fabric (e.g., to promote stretchiness), may be woven fabric, braided fabric, non-woven fabric, etc. If desired, a layer of stretchy plastic may be attached to fabric 12.1-62 and/or an elastomeric polymer layer may be used in place of some or all of fabric 12.1-62 in forming layer 12.1-12CC. Fabric 12.1-62 may be formed from interlaced (intertwined) strands of material such as polymer strands, strands of cotton or other natural material, synthetic material, and/or other materials. The strands in fabric 12.1-62 may include monofilament strands and/or multi-filament strands. In an illustrative configuration, some of the strands of fabric 12.1-62 may be selected to provide fabric 12.1-62 with strength, some of strands in fabric 12.1-62 may be formed from elastomeric material that enhances the ability of fabric 12.1-62 to stretch (and that has a lower elastic modulus than the strands that provide fabric 12.1-62 with strength). Examples of stretchable strand materials include elastomeric materials such as silicone, spandex, and thermoplastic polyurethane (TPU). Examples of strength-enhancing strand materials include polyester, nylon, etc. Other materials may be used in forming the strands in fabric 12.1-62, if desired.

[2395] Strands in fabric 12.1-62 may be dyed at the fabric level (e.g., after strands are knit, woven, or otherwise interlaced to form fabric 12.1-62), may be dyed at the strand or yarn level (e.g., after

material is made into monofilament or multifilament strands), and/or may be dyed at the polymerization level (e.g., before material is made into monofilament or multifilament strands). Dyeing yarn may cause some shrinkage and should therefore be taken into account when determining the appropriate amount of stretch for fabric **12.1-62**.

[2396] In the example of FIGS. **12.1-6**, fabric **12.1-62** includes warp knit strands **12.1-44** and zig-zag strands **12.1-46**. Warp knit strands **12.1-44** may be warp knit strands that form a pattern of openings such as diamond-shaped openings **12.1-64**. If desired, openings **12.1-64** may have other shapes. The use of diamond shapes for openings **12.1-64** is merely illustrative. Warp knit strands **12.1-44** may be formed from stretchable materials such as silicone, spandex, thermoplastic polyurethane (TPU), and or other stretchable materials, or warp strands **12.1-44** may be formed from strength-enhancing strand materials such as polyester, nylon, etc. Other materials may be used in forming strands **12.1-44**, if desired.

[2397] Zig-zag strands **12.1-46** may be interlaced with warp knit strands **12.1-44** and may zig-zag back and forth (e.g., back in forth in the weft or x-direction) multiple times within each diamond-shaped opening **12.1-64**. Zig-zag strands **12.1-46** (sometimes referred to as running strands, accordion strands, etc.) may be formed from stretchable materials such as silicone, spandex, thermoplastic polyurethane (TPU), and or other stretchable materials, or zig-zag strands **12.1-46** may be formed from strength-enhancing strand materials such as polyester, nylon, etc. Other materials may be used in forming strands **12.1-46**, if desired. Strands **12.1-46**, may, if desired, be covered strands that include a covering. The covering may be a different material than the core. For example, the core may be a stretchable material such as spandex and the covering may be a less-stretchable material such as polyester or polyethylene terephthalate. If desired, the covering may be relatively “fuzzy” (e.g., may have a frayed or otherwise not smooth texture) to increase opacity of cover **12.1-12CC**.

[2398] In some arrangements, warp knits strands **12.1-44** and zig-zag strands **12.1-46** are formed in the same layer (e.g., formed in single layer). In other arrangements, warp knit strands **12.1-44** may be formed in front of or behind zig-zag strands **12.1-46** (e.g., warp knit strands **12.1-44** may be formed in a first layer and zig-zag strands **12.1-46** may be formed in a second layer that overlaps the first layer). In one illustrative arrangement, zig-zag strands **12.1-46** are positioned on the side closer to the user's face and warp strands **12.1-44** are positioned on the side closer to the electrical components in device **12.1-10**. In configurations where zig-zag strands **12.1-46** include a covering (e.g., a covering of polyester, polyethylene terephthalate, and/or other suitable material), placing zig-zag strands **12.1-46** on the user-facing side of cover **12.1-12CC** may help protect the stretchable strands in fabric **12.1-62** (e.g., strands **12.1-44** and/or the core of strands **12.1-46**) from facial chemicals such as sebum that may otherwise cause wear to elastic strands in fabric **12.1-62**. If desired, warp knit strands **12.1-44** may be smooth strands and zig-zag strands **12.1-46** may be textured fuzzy strands.

[2399] The use of warp knit strands **12.1-44** and zig-zag strands **12.1-46** may provide cover **12.1-12CC** with different features. For example, warp knit strands **12.1-44** extend in the warp direction, thereby promoting shrinkage and increasing stretch in the warp direction. In contrast, zig-zag strands **12.1-46** may zig-zag in the weft direction, thereby promoting shrinkage and increasing stretch in the weft direction. For example, when fabric **12.1-62** is in an unstretched state, each diamond-shaped opening **12.1-64** has dimension D1 in the vertical (warp) direction (e.g., parallel to the y-axis of FIGS. **12.1-6**) and dimension D2 in the horizontal (weft) direction (e.g., parallel to the x-axis of FIGS. **12.1-6**). When fabric **12.1-62** is in a stretched state, as shown in FIGS. **12.1-7**, each diamond-shaped opening **12.1-64** has dimension D3 in the vertical (warp) direction (e.g., parallel to the y-axis of FIGS. **12.1-7**) and dimension D4 in the horizontal (weft) direction (e.g., parallel to the x-axis of FIGS. **12.1-7**). Dimension D3 may be larger than dimension D1, and dimension D4 may be larger than dimension D2. In other words, openings **12.1-64** may open and expand when cover **12.1-12CC** is stretched. Warp knit strands **12.1-44** stretch in the warp direction, thereby allowing

the opening and closing of openings **12.1-64**, whereas zig-zag strands **12.1-44** stretch in the weft direction, thereby providing fabric **12.1-62** with four-way stretch capabilities. The opacity provided by the fuzzy covering on zig-zag strands **12.1-46** may also serve to hide electrical components from view when openings **12.1-64** are expanded.

[2400] Another factor that may affect the stretchability of fabric **12.1-62** of cover **12.1-12CC** is heat setting. In arrangements where fabric **12.1-62** is heat set, fabric **12.1-62** may be stretched and heat set to remove wrinkles. This may, however, reduce some of the stretch in fabric **12.1-62**. If desired, fabric **12.1-62** of cover **12.1-12CC** may be produced without any heat setting (e.g., may only undergo washing, dyeing, and drying) to retain the desired stretchability of fabric **12.1-62**.

[2401] FIGS. **12.1-8** is a side view of an illustrative strand **12.1-44** that may be used to form warp knit strands **12.1-44** of fabric **12.1-62**. In the example of FIGS. **12.1-8**, strand **12.1-44** may be a single strand of material such as spandex, silicone, thermoplastic polyurethane (TPU), and/or other stretchable material, or strand **12.1-44** may be a single strand of strength-enhancing strand materials such as polyester, nylon, etc. In one illustrative configuration, strand **12.1-44** is a single strand of spandex having 20 denier, 40 denier, 30 denier, between 15 and 60 denier, between 20 and 50 denier, greater than 40 denier, less than 40 denier, etc. This is merely illustrative. If desired, strand **12.1-44** may be formed from other suitable materials and/or may have other denier values.

[2402] FIGS. **12.1-9** is a side view of an illustrative strand **12.1-46** that may be used to form zig-zag strands **12.1-46** of fabric **12.1-62**. In the example of FIGS. **12.1-9**, strand **12.1-46** may include a core such as core **12.1-48**. Core **12.1-48** may be an elastic core formed from one or more strands of material such as spandex, silicone, thermoplastic polyurethane (TPU), and/or other stretchable material, or core **12.1-48** may be one or more strands of strength-enhancing strand materials such as polyester, nylon, etc. In one illustrative configuration, core **12.1-48** is an elastic core formed from a single strand of spandex having 20 denier, 40 denier, 30 denier, between 15 and 60 denier, between 20 and 50 denier, greater than 40 denier, less than 40 denier, etc. The denier value of core **12.1-48** of zig-zag strand **12.1-46** may, for example, be less than the denier value of warp knit strands **12.1-44**. This is merely illustrative. If desired, strand **12.1-46** may be formed from other suitable materials and/or may have other denier values.

[2403] As shown in FIGS. **12.1-9**, core **12.1-48** of strand **12.1-46** may be covered with one or more covering strands **12.1-50**. Covering strands **12.1-50** may be formed from strength-enhancing strand materials such as polyester, nylon, etc., or covering strands **12.1-50** may be formed from elastic material such as spandex, silicone, thermoplastic polyurethane (TPU), and/or other stretchable material. Cover **12.1-50** may have any suitable denier value such as 30 denier, 20 denier, 40 denier, between 15 and 60 denier, between 20 and 50 denier, greater than 40 denier, less than 40 denier, etc. The denier value of cover **12.1-50** of zig-zag strand **12.1-46** may, for example, be greater than the denier value of core **12.1-48** of strands **12.1-46**. Covering strand **12.1-50** may be a single filament covering or may include multiple filaments such as **12.1-24** filaments, **12.1-30** filaments, **12.1-20** filaments, more than **12.1-20** filaments, less than **12.1-20** filaments, etc. Covering strand **12.1-50** may be textured, non-textured, bright, semi-dull, etc. In one illustrative configuration, covering strand **12.1-50** is a drawn textured yarn of polyester that is simultaneously twisted and drawn to provide a textured, fuzzy covering over core **12.1-48**. Because polyester is less sensitive to facial chemicals such as sebum than spandex, covering core **12.1-48** with a fuzzy yarn such as strand **12.1-50** ensures that most of the surface facing the user is polyester and thus more durable throughout continued use. The use of drawn textured yarn for covering **12.1-50** may also increase the opacity of cover **12.1-12CC** to help hide electronic components even when cover **12.1-12CC** and curtain **12.1-12C** are stretched and openings **12.1-64** expand.

[2404] If desired, covering strand **12.1-50** may be formed from a conductive material. Using conductive strands for covering **12.1-50** may provide a conductive layer on fabric **12.1-62** in which conductive covering strands **12.1-50** are in contact with one another across the surface of the fabric. In other configurations, strand **12.1-46** may be a polycarbonate or braided cord with an electrically

conductive core. Arrangements in which strand **12.1-46** includes a multi-component yarn with different polymers joined together within each filament may also be used. When exposed to heat, the different polymers may shrink to different degrees, thereby producing a smooth helical crimp. This may provide strands **12.1-46** with texture while still allowing strands **12.1-46** to maintain durable stretch and recovery.

[2405] If desired, cover **12.1-12CC** may have regions with different levels of stretch, opacity, and/or other characteristics. This type of arrangement is illustrated in FIGS. **12.1-10**. As shown in FIGS. **12.1-10**, cover **12.1-12CC** may have regions with different fabric characteristics such as eye regions **12.1-54** and nose region **12.1-56**. Eye regions **12.1-54** may surround left and right openings such as lens barrel openings **12.1-52** for receiving respective left and right lenses **12.1-30** of respective left and right optical modules **12.1-40**. Nose region **12.1-56** may be interposed between eye regions **12.1-54** and may be configured to overlap the nose bridge of the user when device **12.1-10** is mounted on the user's head.

[2406] If desired, fabric **12.1-62** of cover **12.1-12CC** may be a jacquard knit fabric that allows for different zones of fabric to have different features such as different fabric construction, different strand material(s), different stretch capabilities, different opacity levels, etc. For example, nose region **12.1-56** may be constructed to have a higher opacity and/or lower stretch capability than eye regions **12.1-54**. For example, to reduce stretch in nose region **12.1-56** relative to eye regions **12.1-54**, zig-zag strands **12.1-46** may be omitted from fabric **12.1-62** in region **12.1-56** and may only be present in eye regions **12.1-54**. Additionally or alternatively, strands in nose region **12.1-56** may be mostly or entirely formed from less-stretchable material such as polyester instead of spandex. More textured yarns may be present in nose region **12.1-56** than in eye regions **12.1-54** to increase opacity of nose region **12.1-56** relative to eye regions **12.1-54**, if desired.

[2407] FIGS. **12.1-11** is a perspective view of an illustrative example in which cover **12.1-12CC** is formed from three-dimensional fabric **12.1-62**. Three-dimensional fabric **12.1-62** of FIGS. **12.1-11** may have a fabric construction similar to a pair of pants, where fabric **12.1-62** forms left and right cylindrical portions around openings **12.1-52**. Instead of being mostly flat in the z-direction, fabric **12.1-62** may have a non-zero dimension P1 in the z-direction, may have a dimension P2 in the x-direction, with openings **12.1-52** having a diameter P3 and separated by a distance P4. With this type of configuration, movement of optical modules **12.1-40** may be accommodated by moving the "legs" of fabric **12.1-62** around openings **12.1-52** towards or away from one another (e.g., moving the left and right cylindrical portions of fabric **12.1-62** relative to one another), which involves less stretching of fabric **12.1-62** and instead involves moving an entire piece of fabric **12.1-62** in a given direction. The use of three-dimensional fabric **12.1-62** of FIGS. **12.1-11** is merely illustrative. If desired, cover **12.1-12CC** may be formed from a mostly flat fabric **12.1-62**.

[2408] In accordance with an embodiment, a fabric cover for a head-mounted device having first and second lenses is provided that includes first strands that form a warp knit layer having diamond-shaped openings, in which the warp knit layer is configured to stretch in a warp direction; second strands that are interlaced with the first strands and that zig-zag back and forth within the diamond-shaped openings, in which the second strands are configured to stretch in a weft direction; and first and second openings for receiving the respective first and second lenses.

[2409] In accordance with another embodiment, the first strands include elastic strands.

[2410] In accordance with another embodiment, the second strands include covered strands.

[2411] In accordance with another embodiment, the covered strands each include an elastic core and a covering strand.

[2412] In accordance with another embodiment, the covering strand includes a drawn textured yarn that increases an opacity within the diamond-shaped openings.

[2413] In accordance with another embodiment, the drawn textured yarn includes polyester.

[2414] In accordance with another embodiment, the elastic strands each have a first denier value and the elastic core has a second denier value that is less than the first denier value.

[2415] In accordance with another embodiment, at least some of the first strands and second strands are dyed at a polymerization level.

[2416] In accordance with another embodiment, at least some of the first strands and second strands are dyed at a fabric level.

[2417] In accordance with another embodiment, the fabric cover also includes a nose bridge region interposed between the first and second openings, in which the nose bridge region has higher opacity and lower stretch than other portions of the fabric cover.

[2418] In accordance with an embodiment, a head-mounted device is provided that includes a housing separating an interior region from an exterior region that surrounds the housing, first and second lenses in the housing that are configured to provide images respectively to first and second eye boxes, and a fabric cover configured to block the interior region from view, in which the fabric cover includes: first and second cover openings that are respectively aligned with the first and second lenses; smooth strands that stretch in a first direction; and textured strands that stretch in a second direction.

[2419] In accordance with another embodiment, the textured strands include an elastic core and a fuzzy covering.

[2420] In accordance with another embodiment, the smooth strands include warp knit strands.

[2421] In accordance with another embodiment, the warp knit strands have a higher denier value than the elastic core.

[2422] In accordance with another embodiment, the elastic core is less than 40 denier.

[2423] In accordance with another embodiment, the smooth strands form a pattern of openings and the textured strands zig-zig back and forth within the openings.

[2424] In accordance with another embodiment, the openings include diamond-shaped openings.

[2425] In accordance with another embodiment, the first direction is a warp direction and the second direction is a weft direction.

[2426] In accordance with an embodiment, a head-mounted device is provided that includes a housing, left and right lenses supported by the housing, in which the left lens is configured to provide a left image to a left eye box, in which the right lens is configured to provide a right image to a right eye box, and in which the left and right lenses are configured to move relative to each other, and a three-dimensional fabric cover that has a left cylindrical portion that is aligned with the left lens and a right cylindrical portion that is aligned with the right lens, in which the three-dimensional fabric cover is configured to block an interior region of the housing from view and in which the left and right cylindrical portions are configured to move relative to one another in response to movement of the right and left lenses to accommodate different interpupillary distances.

[2427] In accordance with another embodiment, the three-dimensional fabric cover includes a nose bridge portion coupled between the first and second cylindrical portions.

[2428] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

12.2: Curtain Assembly

[2429] FIGS. **12.2-1** illustrates a view of an example of an HMD **12.2-100** including a frame assembly **12.2-102** and first and second optical modules **12.2-104a**, **12.2-104b** configured to be coupled to the frame assembly **12.2-102**. In at least one example, the frame assembly **12.2-102** defines a rearward facing viewing opening through which the user donning the HMD **12.2-100** can view light projected from display screens of the optical module **12.2-104a-b**. The frame assembly **12.2-102** of the HMD **12.2-100** can define first and second apertures aligned with the first and second optical modules **12.2-104a-b** as shown in FIGS. **12.2-1** and described elsewhere herein with reference to other figures.

[2430] In at least one example, the optical modules **12.2-104a-b** can include optical components

and assemblies such as display screens, lenses, and so forth. The HMD **12.2-100** can also include a curtain assembly **12.2-106** coupled to the frame assembly **12.2-102** and extending across the viewing opening between the frame assembly **12.2-102** and the optical modules **12.2-104a-b**, for example between the frame assembly **12.2-102** and the lenses of the optical modules **12.2-104a-b**. The curtain assembly can include multiple components, parts, and layers, to manage airflow through the HMD **12.2-100** due to cooling systems having fans blowing air through an internal volume of the HMD **100**.

[2431] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **12.2-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **12.2-2-12.2-13** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **12.2-2-12.2-13** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **12.2-1**.

[2432] FIGS. **12.2-2** shows a rear, perspective view of an HMD **12.2-200** including a frame assembly **12.2-202** defining an external surface of the device, an internal volume **12.2-208**, and a rear-facing viewing opening **12.2-210**, which can also be referred to as an aperture or viewing aperture. The HMD **12.2-200** also includes a display module **12.2-204** including a lens **12.2-218** through which the user can view the light projected from a display screen of the optical module **12.2-204** disposed in the internal volume **12.2-208**. In at least one example, the HMD **12.2-200** also includes a curtain assembly **12.2-206** occluding the viewing opening **12.2-210** between the lens **12.2-218** and the frame assembly **12.2-202**.

[2433] In at least one example, the HMD **12.2-200** can include one or more motors mechanically coupled to the optical module **12.2-204** to alter a position of the optical module **12.2-204**, including the lens **12.2-218**, relative to the frame assembly **12.2-202**. Such motors are illustrated in other figures and described elsewhere herein. These motors can be used to adjust the optical module(s) **12.2-204** of the HMD **12.2-200** to match the interpupillary distance defined by the user's eyes. In at least one example, the curtain assembly **12.2-206** can be elastically deformable or stretchy to accommodate the different possible positions of the optical module **12.2-204**, including the lens **12.2-218**, such that the curtain assembly occludes the viewing aperture/opening **12.2-210** between the frame assembly **12.2-202** and the optical module(s) **12.2-204** or lens(es) **12.2-218** thereof regardless of the position or movement of the optical module **12.2-204** and lens **12.2-218**.

[2434] The HMD **12.2-200** shown in FIGS. **12.2-2** includes a first optical module **12.2-204** configured to align with a user's left eye but does not show a second optical module configured to align with a user's right eye so that the internal volume **12.2-208** and fan **12.2-212** can be seen. The HMD **12.2-200** can include one or more fans **12.2-212** in the internal volume. The HMD **12.2-200** can include a second fan, not seen, aligned with the optical module **12.2-204** shown. The frame assembly **12.2-202** can define one or more intake ports **12.2-214** and one or more exhaust ports **12.2-216** on an opposing side of the frame assembly **12.2-202** relative to the intake ports **12.2-214**. The fan **12.2-212** can be configured to draw air into the internal volume **12.2-208** through the intake ports **12.2-214** and exhaust the air from the internal volume **12.2-208** out through the exhaust ports **12.2-216**.

[2435] The fan **12.2-212** can be configured to draw air in and out of the internal volume **12.2-208** to cool various heat-generating components within the HMD **12.2-200**. Such components can include processors, display screens of the optical module(s) **12.2-204**, sensors, and so forth. In such an example, the curtain assembly **12.2-206** can be configured to prevent the flow of air from reaching the user's eyes when donning the HMD **12.2-200** such that the airflow through the HMD **12.2-200** does not dry the eyes out and cause discomfort.

[2436] Accordingly, in order serve the dual purpose of preventing airflow from reaching the user's eyes and maintaining an elastic, aesthetically pleasing barrier occluding the viewing

aperture/opening **12.2-210**, the curtain assembly **12.2-206** can include a first layer defining an external surface of the HMD **12.2-200** and a second layer defining the internal volume **12.2-208**, with the first layer being elastic and the second layer being non-elastic and/or air-impermeable. In one example, the second layer can be semi-elastic, or have some elasticity, but remain air-impermeable when stretched or deformed.

[2437] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **12.2-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **12.2-1** and **12.2-3-12.2-13** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **12.2-1** and **12.2-3-12.2-13** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **12.2-2**.

[2438] FIGS. **12.2-3** shows a rear, plan view of an example of an HMD **12.2-300** similar to the HMD **12.2-200** shown in FIGS. **12.2-2**, including a frame **12.2-302** defining an internal volume **12.2-308**, viewing opening **12.2-310**, intake port(s) **12.2-314**, and exhaust port(s) **12.2-316**. The HMD **12.2-300** can also include first and second fans **12.2-312a**, **12.2-312b** disposed in the internal volume **12.2-308** and drawing air through the internal volume **12.2-308** via the intake ports **12.2-314** and exhaust ports **12.2-316**. FIGS. **12.2-3** does not include optical modules in order to visualize the internal volume **12.2-308** and fans **12.2-312a-b** aligned with the apertures and optical module positions. The HMD **12.2-300** can also include a curtain assembly **12.2-306** similar to the curtain assembly **12.2-206** illustrated in FIGS. **12.2-2**, where the curtain assembly **12.2-306** occludes the viewing opening **12.2-310** between the frame **12.2-302** and the optical modules.

[2439] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **12.2-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **12.2-1-12.2-2** and **12.2-4-12.2-13** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **12.2-1-12.2-2** and **12.2-4-12.2-13** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **12.2-3**.

[2440] FIGS. **12.2-4** illustrates a side, cross-sectional view of an example of and HMD **12.2-400** donned by a user **12.2-401**. The HMD **12.2-400** can include a frame, also referred to as a housing **12.2-402**, defining an internal volume **12.2-408** and a fan assembly **12.2-412** disposed in the internal volume **12.2-408**. The housing **12.2-402** can define an intake port **12.2-414** and an exhaust port **12.2-416** and the fan assembly **12.2-412** can be configured to draw air **12.2-424** through the intake port **12.2-414** into the internal volume and push the air **12.2-424** out from the internal volume **12.2-408** through the exhaust port **12.2-416**.

[2441] In at least one example, the HMD **12.2-400** can include an optical module **12.2-404** directing light from a display screen toward the user's eye and a light seal **12.2-420** extending around a perimeter of the HMD **12.2-400** between the housing **12.2-402** and the user's face when the user **12.2-401** dons the HMD **12.2-400** as shown. In at least one example, the HMD **12.2-400** also includes a curtain assembly **12.2-406** extending between the housing **12.2-402** and the optical module **12.2-404**. The curtain assembly **12.2-406** can be configured to block the air **12.2-424** from entering a space **12.2-422** between the eyes/face of the user **12.2-401**, the space being defined by the user **12.2-401**, the light seal **12.2-420**, the optical module **12.2-404**, and the curtain assembly **12.2-406**.

[2442] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **12.2-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **12.2-1-12.2-3** and **12.2-5-12.2-13** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to

FIGS. 12.2-1-12.2-3 and 12.2-5-12.2-13 can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 12.2-4.

[2443] FIGS. 12.2-5 illustrates a perspective view of a curtain assembly 12.2-506 for use in HMD devices described herein. The curtain assembly 12.2-506 can define first and second apertures 12.2-526a and 12.2-526b. As noted above, examples of curtain assemblies can include multiple layers, including a first elastic layer and a second non-elastic, air-impermeable layer. For reference, FIGS. 12.2-3 shows the user facing side of a curtain assembly 12.2-306 including the first, elastic layer 12.2-334. The view of FIGS. 12.2-5 shows an internal side or surface of the curtain assembly 12.2-506 including a second, internal air-impermeable, layer 12.2-528 defining an internal volume of an HMD.

[2444] In at least one example, the second layer can extend between curtain trim rings 12.2-530a and 12.2-530b and a structural curtain backplate 12.2-532. The rings 12.2-530a-b and backplate 12.2-532 can be made of more rigid, structural material including plastics, metals, and so forth, to give structure and shape to the first and second layers of the curtain assembly 12.2-506 attached thereto.

[2445] In at least one example, the layers of the curtain assembly 12.2-506, including the second layer 12.2-528 shown in FIGS. 12.2-5, can be coupled around a first perimeter edge 12.2-542 thereof to the backplate 12.2-532 and to the rings 12.2-530a-b and/or one or more optical assemblies at a second and third inner edges 12.2-544 of the layers forming or defining the apertures 12.2-526a-b, respectively.

[2446] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 12.2-5 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. 12.2-1-12.2-4 and 12.2-6-12.2-13 and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. 12.2-1-12.2-4 and 12.2-6-12.2-13 can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 12.2-5.

[2447] FIGS. 12.2-6 shows a view of a curtain assembly 12.2-606, which can be similar to the assembled curtain assembly 12.2-506 shown in FIGS. 12.2-5. The view of FIGS. 12.2-6 illustrates a curtain assembly 12.2-606 including a structural backplate 12.2-632, trim rings 12.2-630, a second air-impermeable layer 12.2-628, and a first elastic layer 12.2-634. The curtain assembly 12.2-606 can also include first adhesive portions or layers 12.2-636 disposed between and coupling the second layer 12.2-628 to the backplate 12.2-632 and the trim rings 12.2-630 and second adhesive portions or layers 12.2-638 disposed between and coupling the second layer 12.2-628 and the first layer 12.2-634. The curtain assembly can also include additional stiffener components or assemblies 12.2-640 coupled to the first layer 12.2-634.

[2448] In at least one example, the first and second layers 12.2-628, 12.2-634 are coupled together and/or to the backplate 12.2-632 and/or stiffener components 12.2-640 around a first perimeter edge of the layers 12.2-628, 12.2-634. The layers 12.2-628, 12.2-634 can also be coupled together at second edges defining the apertures at the optical modules and/or trim rings 12.2-630. In at least one example, the layers 12.2-628, 12.2-634 are fixed in position relative to one another at these edges only and are free of affixation to one another between the edges. That is, the layers 12.2-628, 12.2-634 can extend between the first and second edges (i.e., perimeter and aperture edges) free of affixation to one another. In this way, the layers 12.2-628, 12.2-634 of the curtain assembly 12.2-606 are fixed relative to each other at the edges 12.2-542, 12.2-544 but free to move relative to one another between the edges 12.2-542, 12.2-544.

[2449] In at least one example, the first, elastic layer 12.2-634 includes a woven fabric material/textile. In at least one example, the first, elastic layer 634 includes an elastic polymer material. In at least one example, the first, elastic layer 12.2-634 includes natural woven fibers. In at least one example, the first, elastic layer 12.2-634 includes synthetic woven fibers. In at least one

example, the second, air-impermeable and/or non-elastic layer **12.2-628** includes plastic, or other non-elastic material configured to prevent air from passing through the curtain assembly **12.2-606**. [2450] In at least one example, the first layer **12.2-634** of the curtain assembly **12.2-606** extends between the lens and frame of an HMD, as described above, in tension. As such, as the lens is moved relative to the housing/frame of the HMD, when certain portions or areas of the first layer **12.2-634** are reduced in size or span a short distance, the portion or area of the layer **12.2-634** remains in tension to avoid wrinkling or folding during use. In at least one example, the second layer **628** of the curtain assembly **12.2-606** extends between the lens and frame of an HMD, as described above, in compression such that certain areas or portions of the second layer **12.2-628** can expand to cover a larger distance or area as the lens connected to the second edge **12.2-544** moves.

[2451] Along these lines, in at least one example, the first and second layers **12.2-634**, **12.2-628** occupy the same area between the lenses and the frame of an HMD, as described above, but have different surface areas. That is, in at least one example, the first layer **12.2-634** can include a first surface area and the second layer **12.2-628** can include a second surface greater than the first surface area while the first and second layers **12.2-634**, **12.2-628** extend across the same area defined between the frame/housing and the optical modules/lenses of an HMD, as shown in other figures. In this way, as the lenses are moved relative to the frame, the second layer **12.2-628**, which is non-elastic, can accommodate the movement without tearing or being damaged and the first layer **12.2-634** can accommodate the movement while staying taught.

[2452] In at least one example, the second layer **12.2-632** can include one more materials, including but not limited to, acrylic copolymers, silicone, TPU silicone blends, TPU, nitrile rubber, to achieve the impermeability as well as low crinkle noises as the optical modules are adjusted. In at least one example, the second layer **12.2-632** can be thermoformed to include dimples.

[2453] In at least one example, the curtain assembly **12.2-606** can prevent dust and chemical ingress into the HMD device and reliability performance can be enhanced due to the curtain assembly **12.2-606**.

[2454] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **12.2-6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **12.2-1-12.2-5** and **12.2-7-12.2-13** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **12.2-1-12.2-5** and **12.2-7-12.2-13** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **12.2-6**.

[2455] FIGS. **12.2-7** illustrates a plan view of an example of a curtain assembly **12.2-706** including first and second layers **12.2-734**, **12.2-728**, respectively, coupled to a frame or backplate **12.2-732** at an outer perimeter edge **12.2-742** and extending to inner edges **12.2-744a**, **12.2-744b** coupled to trim rings **12.2-730a**, **12.2-730b** and/or optical modules **12.2-704a**, **12.2-704b**, respectively. In at least one example, the first layer can extend to cover substantially all of an area between the backplate **12.2-732** and trim rings **12.2-744a-b** coupling the layers **12.2-734**, **12.2-728** to the optical modules **12.2-704a-b**. In the plan view of FIGS. **12.2-7**, the second layer **12.2-728** overlies the first layer **12.2-734** such that much of the first layer **12.2-734** is not visible. However, as noted above, the first layer **12.2-734** can occlude substantially all of the aperture/opening between the lenses of the optical modules **12.2-704a-b** and the frame or backplate **12.2-732**.

[2456] In at least one example, the second layer **12.2-728** can extend between the frame or backplate **12.2-732** and the lenses of the optical modules **12.2-704a-b** and/or the trim rings **12.2-730a-b** to define laterally positioned gaps **12.2-746a** and **12.2-746b** where the second layer **12.2-728** does not extend between the frame/backplate **12.2-732** and the lenses/optical modules **12.2-704a-b**. In at least one example, the second layer **12.2-728** covers between about 70% and about 90% of the area between the frame/backplate **12.2-732** and the optical modules **12.2-704a-b**/trim

rings **12.2-730a-b**. In some examples, the second layer **12.2-728** covers between about 75% and about 90% of the area, for examples between about 80% and about 85% of the area. The gaps **12.2-746a** and **12.2-746b** can also allow for some airflow through the curtain assembly **12.2-706** and to the user's eyes, which can keep the face and eye area of the user cool as well as provide de-fogging airflow for the screens of the displays and lenses of the HMD.

[2457] The curtain assembly **12.2-706** can define a middle portion **12.2-748** extending between the first lens or optical module **12.2-704a** and the second lens or optical module **12.2-704b**. The curtain assembly **12.2-706** can also define a first side portion **12.2-750** extending between the first lens or optical module **12.2-704a** and the frame/backplate **12.2-732** and a second side portion **12.2-752** extending between the second lens or optical module **12.2-704b** and the frame/backplate **12.2-732**. In such an example, the gaps **12.2-746a-b** can be disposed or positioned at the first and second side portions **12.2-750**, **12.2-752**, respectively. As such, as the lenses/optical modules **12.2-704a-b** are moved laterally back and forth in the direction indicated by arrows **12.2-754**, the absence of material of the second layer **12.2-728** at the gaps **12.2-746a-b** can reduce forces opposing the static positioning of the optical modules **12.2-704a-b** and/or opposing the forces of motors configured to move the optical modules **12.2-704a-b** relative to the frame/backplate **12.2-732**, as described above.

[2458] In at least one example, the curtain assembly **12.2-706** can bias the optical modules against the guide-rods to remove slack or backlash between the components of the IPD adjustment system.

[2459] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **12.2-7** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **12.2-1-12.2-6** and **12.2-8-12.2-13** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **12.2-1-12.2-6** and **12.2-8-12.2-13** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **12.2-7**.

[2460] FIGS. **12.2-8** illustrates a partial view of an example of a curtain assembly **12.2-806** including first and second layers **12.2-834**, **12.2-828** coupled to an optical module **12.2-804**. In particular, FIGS. **12.2-8** shows the first and second layers **12.2-834**, **12.2-828** coupled to the housing **12.2-856** of the optical module **12.2-804** coupled to a lens **12.2-818**. In at least one example, the layers **12.2-834**, **12.2-828** can be pressed or coupled between the housing **12.2-856** and a curtain trim ring **12.2-830** as shown. In at least one example, the layers **12.2-834**, **12.2-828** can be adhered to the ring **12.2-830** and the housing **12.2-856** of the optical module **12.2-804** via opposing adhesive portions **12.2-860a**, **12.2-860b**. In the illustrated example, the layers **12.2-834**, **12.2-828** can curve around the ring **12.2-830** and around between the ring **12.2-830** and housing **12.2-856** such that the ring **12.2-830** is hidden from view from the perspective of the user.

[2461] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **12.2-8** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **12.2-1-12.2-7** and **12.2-9-12.2-13** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **12.2-1-12.2-7** and **12.2-9-12.2-13** can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **12.2-8**.

[2462] FIGS. **12.2-9-12.2-13** shown partial, cross-sectional views of examples of curtain assemblies. FIGS. **12.2-9-12.2-13** illustrate various examples of how various parts of the curtain assemblies can be coupled together. For example, FIGS. **12.2-9** illustrates first and second layers **12.2-934**, **12.2-928** of a curtain assembly **12.2-906** can be coupled to the backplate **12.2-932** with various adhesive layers and glue portions. In one example, a first adhesive layer **12.2-960a** is disposed between and coupling a stiffener **12.2-940** and the first layer **12.2-934**. The first layer **12.2-934** can include an end abutting or adjacent a sidewall **12.2-933** of the backplate **12.2-932**. A

second adhesive layer **12.2-960b** can be disposed between and coupling the second layer **12.2-928** and the first layer **12.2-934**. A third adhesive layer **12.2-960c** can be disposed between and couple the second layer **12.2-928** and the backplate **12.2-932**. A glue portion **12.2-961** can be disposed between and couple the first layer **12.2-934** and the backplate **12.2-932** and extend between an end of the second layer **12.2-928** and the sidewall **12.2-933**.

[2463] FIGS. **12.2-10** illustrates an example of a curtain assembly **12.2-1006** including a first adhesive layer between and coupling a stiffener **12.2-1040** to the first layer **12.2-1034**, a second adhesive layer **12.2-1060b** between and coupling the first layer **12.2-1034** and the second layer **12.2-1028**, and a third adhesive layer **12.2-1060c** between and coupling the second layer **12.2-1028** and the backplate **12.2-1032**. In one example, a glue portion **12.2-1061** can couple the stiffener **12.2-1040**, first layer **12.2-1034**, and the second layer **12.2-1028**, as well as the various adhesive layers **12.2-1060a-c**, to the backplate **12.2-1032** at the ends of the various components and layers between the sidewall **12.2-1033** and the layers **12.2-1034**, **12.2-1028**. In at least one example, a space remains between the glue portion **12.2-1061** and a sidewall **12.2-1033** of the backplate **12.2-1032**.

[2464] FIGS. **12.2-11** illustrates an example of a curtain assembly **12.2-1106** including a first adhesive layer **12.2-1160a** between and coupling a stiffener **12.2-1140** and a first layer **12.2-1134**, a second adhesive layer **12.2-1160b** between and coupling the first layer **12.2-1134** and the second layer **12.2-1128**, and a third adhesive layer **12.2-1160c** between and coupling the second layer **12.2-1128** and the backplate **12.2-1132**. An end of the first layer **12.2-1134** can abut or be disposed adjacent to a sidewall **12.2-1133** of the backplate **12.2-1132** and a glue portion **12.2-1161** can be disposed between an end of the second layer **12.2-1128** and the sidewall **12.2-1133**. The glue portion **12.2-1161** can be disposed between and couple the first layer **12.2-1134** and the backplate **12.2-1132**. In at least one example, the backplate **12.2-1132** can define an aperture **12.2-1162** through which the glue portion **12.2-1162** can be dispensed and at least partially disposed, as shown.

[2465] FIGS. **12.2-12** illustrates an example of a curtain assembly **12.2-1206** including a first adhesive layer **12.2-1260a** disposed between and coupling a stiffener **12.2-1240** to a first layer **12.2-1234**, a second adhesive layer **12.2-1260b** disposed between and coupling the first layer **12.2-1234** and the second layer **12.2-1228**, and a third adhesive layer disposed between and coupling the second layer **12.2-1228** and the backplate **12.2-1232**. In the illustrates example of FIGS. **12.2-12**, the first layer **12.2-1234** can curve around and be coupled to the backplate **12.2-1232** via a fourth adhesive layer **12.2-1260d** such that a portion of the backplate is disposed between opposing portions of the first layer **1234** within the curved geometry thereof.

[2466] FIGS. **12.2-13** illustrates an example of a curtain assembly **12.2-1306** including a first adhesive layer between and coupling a stiffener **12.2-1340** to the first layer **12.2-1334**, a second adhesive layer **12.2-1360b** between and coupling the first layer **12.2-1334** and the second layer **12.2-1328**, and a third adhesive layer **12.2-1360c** between and coupling the second layer **12.2-1328** and the backplate **12.2-1332**. In one example, a glue portion **12.2-1361** can couple the stiffener **12.2-1340**, first layer **12.2-1334**, and the second layer **12.2-1328**, as well as the various adhesive layers **12.2-1360a-c**, to the backplate **12.2-1332** at the end of the various components and layers between the sidewall **12.2-1333** and the layers **12.2-1334**, **12.2-1328**. In at least one example, the glue portion **12.2-1361** can extend substantially all the way between the ends of the layers **12.2-1334**, **12.2-1328** and a sidewall **12.2-1333** of the backplate **12.2-1332**.

[2467] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **12.2-9-12.2-13** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in FIGS. **12.2-1-12.2-8** and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to FIGS. **12.2-1-12.2-8** can be included, either alone or in any combination, in the example of the

devices, features, components, and parts shown in FIGS. 12.2-9-12.2-13.

XIII: Light Seal

[2468] FIGS. 13.0-1 illustrates a view of an HMD 13.0-100 including a light seal 13.0-102. The light seal 13.0-102 is described in more detail here in section XIII.

[2469] FIGS. 13.0-2A illustrates a front perspective view of a device seal 13.0-202 according to one embodiment. FIGS. 13.0-2B illustrates a bottom-rear perspective view of the device seal 13.0-202. FIGS. 13.0-2C illustrates a rear view of the device seal 13.0-202. The device seal 13.0-202 illustrated in FIGS. 13.0-2A through FIGS. 13.0-2C can be substantially similar to, including some or all of the features of, the device seals or light seals described herein.

[2470] In some examples, the device seal 13.0-202 can be constructed and designed as depicted in FIGS. 13.0-2A through FIGS. 13.0-2C. The device seal 13.0-202 can include a facial interface 13.0-216 to contact a wearer's face. The facial interface 13.0-216 can include a compressible material, such as a foam, that is compliant and comfortable against the user's face.

[2471] The device seal 13.0-202 can include a cover 13.0-214 extending from the facial interface 13.0-216 and defining an aperture through which the user can view a display of the HMD. The cover 13.0-214 can help to block light from the outside environment. The device seal 13.0-202 can include a nose relief 13.0-215 positioned, sized, and shaped to accommodate for the user's nose. While other designs and configurations are envisioned, those depicted in FIGS. 13.0-2A through FIGS. 13.0-2C provide the user with exceptional user experience and engagement with the HMD.

13.1: Electronic Devices with Covering Structures

[2472] Electronic devices such as head-mounted devices are configured to be worn on a head of a user. A head-mounted device may have left and right optical systems for presenting images to a user's left and right eyes. Not all users have the same physical distance separating their eyes. To accommodate differences in interpupillary distance between different users, a head-mounted device may have a mechanism for adjusting the positions of the left and right optical systems.

[2473] An electronic device such as a head-mounted device may have a front face that faces away from a user's head and may have an opposing rear face that faces the user's head. Optical modules on the rear face may be used to provide images to a user's eyes. The positions of the optical modules may be adjusted to accommodate different user interpupillary distances. Internal device structures may be hidden from view by the user by covering the rear face of the device with a curtain. The curtain, which may sometimes be referred to as a cover, covering structure, rear housing cover, rear housing wall, rear housing structure, cosmetic covering, etc., may help block potentially unsightly internal structures from view, while accommodating movement of the optical modules.

[2474] A top view of an illustrative head-mounted device with a curtain is shown in FIGS. 13.1-1. As shown in FIGS. 13.1-1, head-mounted devices such as electronic device 13.1-10 may have head-mounted support structures such as housing 13.1-12. Housing 13.1-12 may include portions (e.g., support structures 13.1-12T) to allow device 13.1-10 to be worn on a user's head. Support structures 13.1-12T may be formed from fabric, polymer, metal, and/or other material. Support structures 13.1-12T may form a strap or other head-mounted support structures that help support device 13.1-10 on a user's head. A main support structure (e.g., main housing portion 13.1-12M) of housing 13.1-12 may support electronic components such as displays 13.1-14. Main housing portion 13.1-12M may include housing structures formed from metal, polymer, glass, ceramic, and/or other material. For example, housing portion 13.1-12M may have housing walls on front face F and housing walls on adjacent top, bottom, left, and right side faces that are formed from rigid polymer or other rigid support structures and these rigid walls may optionally be covered with electrical components, fabric, leather, or other soft materials, etc. The walls of housing portion 13.1-12M may enclose internal components 13.1-38 in interior region 13.1-34 of device 13.1-10 and may separate interior region 13.1-34 from the environment surrounding device 13.1-10 (exterior region 13.1-36). Internal components 13.1-38 may include integrated circuits, actuators,

batteries, sensors, and/or other circuits and structures for device **13.1-10**. Housing **13.1-12** may be configured to be worn on a head of a user and may form glasses, a hat, a helmet, goggles, and/or other head-mounted device. Configurations in which housing **13.1-12** forms goggles may sometimes be described herein as an example.

[2475] Front face F of housing **13.1-12** may face outwardly away from a user's head and face. Opposing rear face R of housing **13.1-12** may face the user. Portions of housing **13.1-12** (e.g., portions of main housing **13.1-12M**) on rear face R may form a cover such as curtain **13.1-12C**. In an illustrative configuration, curtain **13.1-12C** includes a fabric layer that separates interior region **13.1-34** from the exterior region to the rear of device **13.1-10**. Other structures may be used in forming curtain **13.1-12C**, if desired. The presence of curtain **13.1-12C** on rear face R may help hide internal housing structures, internal components **13.1-38**, and other structures in interior region **13.1-34** from view by a user.

[2476] Device **13.1-10** may have left and right optical modules **13.1-40**. Each optical module may include a respective display **13.1-14**, lens **13.1-30**, and support structure **13.1-32**. Support structures **13.1-32**, which may sometimes be referred to as lens barrels or optical module support structures, may include hollow cylindrical structures with open ends or other supporting structures to house displays **13.1-14** and lenses **13.1-30**. Support structures **13.1-32** may, for example, include a left lens barrel that supports a left display **13.1-14** and left lens **13.1-30** and a right lens barrel that supports a right display **13.1-14** and right lens **13.1-30**. Displays **13.1-14** may include arrays of pixels or other display devices to produce images. Displays **13.1-14** may, for example, include organic light-emitting diode pixels formed on substrates with thin-film circuitry and/or formed on semiconductor substrates, pixels formed from crystalline semiconductor dies, liquid crystal display pixels, scanning display devices, and/or other display devices for producing images. Lenses **13.1-30** may include one or more lens elements for providing image light from displays **13.1-14** to respective eyes boxes **13.1-13**. Lenses may be implemented using refractive glass lens elements, using mirror lens structures (catadioptric lenses), using holographic lenses, and/or other lens systems. When a user's eyes are located in eye boxes **13.1-13**, displays (display panels) **13.1-14** operate together to form a display for device **13.1-10** (e.g., the images provided by respective left and right optical modules **13.1-40** may be viewed by the user's eyes in eye boxes **13.1-13** so that a stereoscopic image is created for the user). The left image from the left optical module fuses with the right image from a right optical module while the display is viewed by the user.

[2477] Not all users have the same interpupillary distance IPD. To provide device **13.1-10** with the ability to adjust the interpupillary spacing between modules **13.1-40** along lateral dimension X and thereby adjust the spacing IPD between eye boxes **13.1-13** to accommodate different user interpupillary distances, device **13.1-10** may be provided with actuators **13.1-42**. Actuators **13.1-42** can be manually controlled and/or computer-controlled actuators (e.g., computer-controlled motors) for moving support structures **13.1-32** relative to each other.

[2478] As shown in FIGS. **13.1-2**, curtain **13.1-12C** may cover rear face F while leaving lenses **13.1-30** of optical modules **13.1-40** uncovered (e.g., curtain **13.1-12C** may have openings that are aligned with and receive modules **13.1-40**). As modules **13.1-40** are moved relative to each other along dimension X to accommodate different interpupillary distances for different users, modules **13.1-40** move relative to fixed housing structures such as the walls of main portion **13.1-12M** and move relative to each other. To prevent undesired wrinkling and buckling of curtain **13.1-12C** as optical modules **13.1-40** are moved relative to rigid portions of housing **13.1-12M** and relative to each other, a fabric layer or other cover layer in curtain **13.1-12C** may be configured to slide, stretch, open/close, and/or otherwise adjust to accommodate optical module movement.

[2479] A schematic diagram of an illustrative electronic device such as a head-mounted device or other wearable device is shown in FIGS. **13.1-3**. Device **13.1-10** of FIGS. **13.1-3** may be operated as a stand-alone device and/or the resources of device **13.1-10** may be used to communicate with external electronic equipment. As an example, communications circuitry in device **13.1-10** may be

used to transmit user input information, sensor information, and/or other information to external electronic devices (e.g., wirelessly or via wired connections). Each of these external devices may include components of the type shown by device **13.1-10** of FIGS. **13.1-3**.

[2480] As shown in FIGS. **13.1-3**, a head-mounted device such as device **13.1-10** may include control circuitry **13.1-20**. Control circuitry **13.1-20** may include storage and processing circuitry for supporting the operation of device **13.1-10**. The storage and processing circuitry may include storage such as nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **13.1-20** may be used to gather input from sensors and other input devices and may be used to control output devices. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors and other wireless communications circuits, power management units, audio chips, application specific integrated circuits, etc. During operation, control circuitry **13.1-20** may use display(s) **13.1-14** and other output devices in providing a user with visual output and other output.

[2481] To support communications between device **13.1-10** and external equipment, control circuitry **13.1-20** may communicate using communications circuitry **13.1-22**. Circuitry **13.1-22** may include antennas, radio-frequency transceiver circuitry, and other wireless communications circuitry and/or wired communications circuitry. Circuitry **13.1-22**, which may sometimes be referred to as control circuitry and/or control and communications circuitry, may support bidirectional wireless communications between device **13.1-10** and external equipment (e.g., a companion device such as a computer, cellular telephone, or other electronic device, an accessory such as a point device, computer stylus, or other input device, speakers or other output devices, etc.) over a wireless link. For example, circuitry **13.1-22** may include radio-frequency transceiver circuitry such as wireless local area network transceiver circuitry configured to support communications over a wireless local area network link, near-field communications transceiver circuitry configured to support communications over a near-field communications link, cellular telephone transceiver circuitry configured to support communications over a cellular telephone link, or transceiver circuitry configured to support communications over any other suitable wired or wireless communications link. Wireless communications may, for example, be supported over a Bluetooth® link, a WiFi® link, a wireless link operating at a frequency between 10 GHz and 400 GHz, a 60 GHz link, or other millimeter wave link, a cellular telephone link, or other wireless communications link. Device **13.1-10** may, if desired, include power circuits for transmitting and/or receiving wired and/or wireless power and may include batteries or other energy storage devices. For example, device **13.1-10** may include a coil and rectifier to receive wireless power that is provided to circuitry in device **13.1-10**.

[2482] Device **13.1-10** may include input-output devices such as devices **13.1-24**. Input-output devices **13.1-24** may be used in gathering user input, in gathering information on the environment surrounding the user, and/or in providing a user with output. Devices **13.1-24** may include one or more displays such as display(s) **13.1-14**. Display(s) **13.1-14** may include one or more display devices such as organic light-emitting diode display panels (panels with organic light-emitting diode pixels formed on polymer substrates or silicon substrates that contain pixel control circuitry), liquid crystal display panels, microelectromechanical systems displays (e.g., two-dimensional mirror arrays or scanning mirror display devices), display panels having pixel arrays formed from crystalline semiconductor light-emitting diode dies (sometimes referred to as microLEDs), and/or other display devices.

[2483] Sensors **13.1-16** in input-output devices **13.1-24** may include force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors such as a touch sensor that forms a button, trackpad, or other input device), and other sensors. If desired, sensors **13.1-16** may include

optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, optical touch sensors, optical proximity sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, image sensors, fingerprint sensors, iris scanning sensors, retinal scanning sensors, and other biometric sensors, temperature sensors, sensors for measuring three-dimensional non-contact gestures (“air gestures”), pressure sensors, sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or inertial measurement units that contain some or all of these sensors), health sensors such as blood oxygen sensors, heart rate sensors, blood flow sensors, and/or other health sensors, radio-frequency sensors, depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices that capture three-dimensional images), optical sensors such as self-mixing sensors and light detection and ranging (LIDAR) sensors that gather time-of-flight measurements, humidity sensors, moisture sensors, gaze tracking sensors, electromyography sensors to sense muscle activation, facial sensors, and/or other sensors. In some arrangements, device **13.1-10** may use sensors **13.1-16** and/or other input-output devices to gather user input. For example, buttons may be used to gather button press input, touch sensors overlapping displays can be used for gathering user touch screen input, touch pads may be used in gathering touch input, microphones may be used for gathering audio input, accelerometers may be used in monitoring when a finger contacts an input surface and may therefore be used to gather finger press input, etc.

[2484] If desired, electronic device **13.1-10** may include additional components (see, e.g., other devices **13.1-18** in input-output devices **13.1-24**). The additional components may include haptic output devices, actuators for moving movable housing structures, audio output devices such as speakers, light-emitting diodes for status indicators, light sources such as light-emitting diodes that illuminate portions of a housing and/or display structure, other optical output devices, and/or other circuitry for gathering input and/or providing output. Device **13.1-10** may also include a battery or other energy storage device, connector ports for supporting wired communication with ancillary equipment and for receiving wired power, and other circuitry.

[2485] FIGS. **13.1-4** and **13.1-5** are top views of device **13.1-10** showing how the optical modules of device **13.1-10** move with respect to each other along lateral dimension X to accommodate different interpupillary distances IPD (the distance between a user's left and right eyes). In the example of FIGS. **13.1-4**, left optical module **13.1-40L** and right optical module **13.1-40R** have been moved towards each other to accommodate a small interpupillary distance. In the example of FIGS. **13.1-5**, left optical module **13.1-40L** and right optical module **13.1-40R** have been moved away from each other to accommodate a large interpupillary distance.

[2486] Curtain **13.1-12C** has edge portions such as left portion **13.1-12C-L** between left housing wall **13.1-12M-L** and left optical module **13.1-40L** and right portion **13.1-12C-R** between right housing wall **13.1-12M-R** and right optical module **13.1-40R**. Middle portion **13.1-12C-M** of curtain **13.1-12C** extends between left optical module **13.1-40L** and right optical module **13.1-40R**. In the configuration of FIGS. **13.1-4**, optical modules **13.1-40L** and **13.1-40R** are relatively close to each other, so middle portion **13.1-12C-M** is relatively small and portions **13.1-12C-L** and **13.1-12C-R** are relatively large. In the configuration of FIGS. **13.1-5**, optical modules **13.1-40L** and **13.1-40R** are relatively far from each other, so left portion **13.1-12C-L** and right portion **13.1-12C-R** are shorter along lateral dimension X and middle portion **13.1-12C-M** has been enlarged relative to the configuration of FIGS. **13.1-4**.

[2487] To help accommodate differences in size for curtain **13.1-12C** (e.g., length changes for portions of curtain **13.1-12M** along lateral dimension X), curtain **13.1-12C** may include a cover layer formed from a stretchable material such as fabric. The cover layer may be supported by a rigid frame. The fabric may be provided with a peripheral elastic band that helps allow the fabric to slide relative to the frame while being retained securely on the frame, thereby further helping curtain **13.1-12M** to be dynamically adjusted without exhibiting undesired buckling and wrinkling.

[2488] FIGS. 13.1-6 is a side view of device 13.1-10 showing how device 13.1-10 may, if desired, include cooling features to help cool display 13.1-14 and other internal components 13.1-38. As shown in FIGS. 13.1-6, for example, device 13.1-10 may have a fan such as fan 13.1-50. Fan 13.1-50 may be mounted in housing 13.1-12 (e.g., a fan housing in housing 13.1-12, etc.) in a configuration that allows fan 13.1-50 to expel air from housing 13.1-12. To allow cool air to move past the face of the user while cooling electrical components in interior region 13.1-34 such as internal components 13.1-38 and display 13.1-14, curtain 13.1-12C may be permeable to air. Curtain 13.1-12C may, as an example, have one or more air-flow-promotion openings. In some examples, curtain 13.1-12C is formed from knit or woven fabric that has a sufficiently loose knit or weave to allow air to flow through the interstitial gaps between adjacent strands of material in the fabric (e.g., between warp and weft strands). Openings may also be formed by laser cutting and/or other opening formation techniques. In examples in which curtain 13.1-12C is permeable to air, air may flow into device 13.1-10 through curtain 13.1-12C, may flow past display 13.1-14 and other internal components 13.1-38, and may exit interior region 13.1-34 through fan 13.1-50. Air may flow in this way to cool device 13.1-10 while device 13.1-10 is being worn on a head of a user. If desired, housing 13.1-12 may have additional openings (e.g., slot shaped openings on an upper wall, side wall, and/or lower wall) that provide additional pathways for air flow.

[2489] To allow a cover layer in curtain 13.1-12C to slide back and forth during adjustments to the positions of modules 13.1-40, curtain 13.1-12C may be provided with a rigid frame that supports the cover layer without excessively restricting lateral motion of the cover layer. FIGS. 13.1-7 is an exploded perspective view of a left-hand portion of an illustrative curtain. As shown in FIGS. 13.1-7, curtain 13.1-12C may include frame 13.1-12CF and cover layer 13.1-12CC. Cover layer 13.1-12CC may have left and right openings such as lens barrel opening 13.1-52 to receive respective left and right optical modules 13.1-40. Frame 13.1-54 may have corresponding left and right openings such as illustrative opening 13.1-54. Opening 13.1-54 may be larger than opening 13.1-52 to accommodate lateral movement of module 13.1-40. When mounted in device 13.1-10, cover layer 13.1-12CC may be attached to frame 13.1-12CF so that cover layer 13.1-12CC can stretch and/or slide relative to frame 13.1-12CF.

[2490] If desired, cover layer 13.1-12CC may include multiple layers of material. As shown in the top view of FIGS. 13.1-8, for example, cover layer 13.1-12CC may, include outer layer 13.1-56 and inner layer 13.1-58. Outer layer 13.1-56 may be, for example, a fabric layer. Inner layer 13.1-58 may be, for example, a layer of fabric formed from polymer strands that have been thermally formed into an accordion shape. In this type of arrangement, outer layer 13.1-56 may be a fabric with a desired visual appearance, whereas inner layer 13.1-58 may be a fabric or other layer that provides cover layer 13.1-12CC with a desired opacity to help block internal device components from view. Layer 13.1-12CC may be attached to module 13.1-40 using attachment structures 13.1-60. Structures 13.1-60 may include retention rings, screws and other fasteners, clips, and other structures that mechanically attach layer 13.1-12CC to support structures 13.1-32 of module 13.1-40 and/or may include adhesive for attaching layer 13.1-12CC to module 13.1-40.

[2491] If desired, layer 13.1-12CC may be provided with a peripheral elastic band. This type of arrangement is shown in FIGS. 13.1-9. As shown in FIGS. 13.1-9, cover layer 13.1-12CC may have fabric 13.1-62 and peripheral elastic band 13.1-64. Fabric 13.1-62 may be warp knit fabric, weft knit fabric, or other knit fabric (e.g., to promote stretchiness), may be woven fabric, braided fabric, non-woven fabric, etc. If desired, a layer of stretchy plastic may be attached to fabric 13.1-62 and/or an elastomeric polymer layer may be used in place of some or all of fabric 13.1-62 in forming layer 13.1-12CC. Fabric 13.1-62 may be formed from interlaced (intertwined) strands of material such as polymer strands, strands of cotton or other natural material, synthetic material, and/or other materials. The strands in fabric 13.1-62 may include monofilament strands and/or multi-filament strands. In some examples, some of the strands of fabric 13.1-62 may be selected to provide fabric 13.1-62 with strength, whereas other strands in fabric 13.1-62 may be formed from

elastomeric material that enhances the ability of fabric **13.1-62** to stretch (and that has a lower elastic modulus than the strands that provide fabric **13.1-62** with strength). Examples of stretchable strand materials include elastomeric materials such as silicone and thermoplastic polyurethane (TPU). Examples of strength-enhancing strand materials include polyester, nylon, etc. Elastic band **13.1-64** may be formed from a strand of elastomeric material such as a strand of silicone or thermoplastic polyurethane or other stretchable material. Other materials may be used in forming the strands in fabric **13.1-62**, if desired.

[2492] Elastic band **13.1-64** may be attached to along the outer edge of fabric **13.1-62** by sewing, by knitting band **13.1-64** into a knit fabric, using adhesive, using crimped connections or other fasteners, and/or by otherwise attaching band **13.1-64** to the periphery of fabric **13.1-62**. If desired, band **13.1-64** may be formed from warp strands and weft strands in a woven fabric (see, e.g., band **13.1-64** of fabric **13.1-62** of FIGS. **13.1-10**).

[2493] The fabric or other material forming cover layer **13.1-12CC** may be stretchable. As shown in FIGS. **13.1-11**, for example, layer **13.1-12CC** may be configured to be stretched without damage from a first shape characterized by length **13.1-L1** along dimension X to a second larger shape characterized by length **13.1-L2** along dimension X. The amount of stretching ($(13.1-L2/13.1-L1)$) that layer **13.1-12CC** may accommodate may be, for example, at least 50%, at least 75%, at least 100%, or at least 150%.

[2494] When attaching cover layer **13.1-12CC** to frame **13.1-12CC**, band **13.1-64** may fit over the outside of frame **13.1-12CC**. Band **13.1-64** may then tug inwardly on the portions of cover layer **13.1-12CC** that overlap the edges of the frame. This, in turn, will help to tension the main portion of layer **13.1-12CC** outwardly (e.g., in lateral dimensions X and Y), thereby ensuring that cover layer **13.1-12CC** will remain taut. At the same time, there may be at least some allowed lateral slippage of layer **13.1-12CC** back and forth as needed to accommodate changes in the positions of modules **13.1-40**.

[2495] An example shape for frame **13.1-12CF** of curtain **13.1-12C** is shown in FIGS. **13.1-12**. As shown in FIGS. **13.1-12**, frame **13.1-12CF** may have left and right openings **13.1-54** to overlap the desired range of positions achievable by modules **13.1-40**. Frame **13.1-12CF** may have an outer ring-shaped portion **13.1-66** bridged in the portion of frame **13.1-12CF** that overlaps the user's nose by bridging middle portion **13.1-68**. Openings **13.1-54** may be rectangular, oval, teardrop shaped, circular, and/or may have other suitable shapes.

[2496] FIGS. **13.1-13** shows how elastic band **13.1-64** may help provide fabric **13.1-62** of cover **13.1-12CC** with the ability to slide laterally relative to frame to accommodate movement in optical modules **13.1-40** while helping to keep fabric **13.1-62** taut. Elastic band **13.1-64** is normally in a stretched state. As a result, band **13.1-64** attempts to contract and, in doing so, tends to pull fabric **13.1-62** around frame **13.1-12CF** in direction **13.1-74**. On the inner side of frame **13.1-12CF**, the tightening force from band **13.1-64** is therefore normally pulling fabric **13.1-62** in direction **13.1-76**, whereas on the opposing outer (rear-facing) side of frame **13.1-12CF**, the tightening force of band **13.1-64** tends to pull fabric **13.1-62** towards the periphery of frame **13.1-12CF** in direction **13.1-78**. The presence of band **13.1-64** therefore helps tighten fabric **13.1-62** and prevent wrinkles in cover layer **13.1-12CC**.

[2497] When it is desired to move optical module **13.1-40**, fabric **13.1-62** can slide back and forth over frame **13.1-12CF** as needed. Consider, as an example, a scenario in which module **13.1-40** is moved in direction **13.1-70**. This pulls fabric **13.1-62** on the outer side of frame **13.1-12CF** (e.g., frame portion **13.1-66**) in direction **13.1-80**. On the inner side of frame **13.1-12CF**, the edge of fabric **13.1-62** is pulled in direction **13.1-72**, causing band **13.1-64** to stretch and expand slightly (e.g., so that band **13.1-64** moves to position **13.1-64'**). Due to the pull on fabric **13.1-62** in direction **13.1-80**, fabric **13.1-62** slides around frame **13.1-12CF** and across the outer surface of frame **13.1-12CF** in direction **13.1-82**, thereby helping to accommodate movement of module **13.1-40** without wrinkling cover layer **13.1-12CC**.

[2498] To facilitate sliding movement of cover **13.1-12CC** around the edges of frame **13.1-12CF** in this way, at least the left and right edges of cover **13.1-12CC** (and adjacent portions along the upper and lower edges of cover **13.1-12CC**) may not be fixedly attached to frame **13.1-12CF** or housing **13.1-12**. An illustrative configuration for mounting curtain **13.1-12C** within housing portion **13.1-12M** of housing **13.1-12** of device **13.1-10** is shown in FIGS. **13.1-14**. In the example of FIGS. **13.1-14**, housing portion **13.1-12M** has a central support such as housing structure **13.1-84**. Curtain **13.1-12C** may be fixedly attached to structure **13.1-84** using attachment mechanism **13.1-86**. Mechanism **13.1-86** may include, for example, glue and/or mechanical structures that grip fabric **13.1-62** and/or other portions of curtain **13.1-12C** to hold curtain **13.1-12C** firmly in place within device **13.1-10**. Mechanism **13.1-86** may, if desired, include engagement structures (e.g., snap features) that allow curtain **13.1-12C** to be removed and replaced with another curtain. When mounted in device **13.1-10**, however, mechanism **13.1-86** will hold curtain **13.1-12C** securely. [2499] As shown in FIGS. **13.1-14**, device **13.1-10** may, if desired, have a ring-shaped opaque light seal such as light seal **13.1-90**. Light seal **13.1-90** may be configured to be removable (e.g. so that light seal **13.1-90** may be replaced when worn). Foam or other soft materials may be used in forming light seal **13.1-90**.

[2500] The attachment structures used in mechanism **13.1-86** of FIGS. **13.1-14** may or may not permit fabric **13.1-62** to slide freely with respect to frame **13.1-12CF**. To permit fabric **13.1-62** to slide freely with respect to frame **13.1-12CF** of curtain **13.1-12C** elsewhere in curtain **13.1-12C** (e.g., at the left and right edges of curtain **13.1-12C** and at other portions of curtain **13.1-12C** away from attachment mechanism **13.1-86**), curtain **13.1-12C** may float with respect to housing **13.1-12M** (except at attachment mechanism **13.1-86**). As an example, the opposing ends of curtain **13.1-12C** at the left and right edges of curtain **13.1-12C** may be separated from nearby portions of housing portion **13.1-12M** by air gaps **13.1-88**. This prevents the fabric of layer **13.1-12CC** from becoming caught between frame **13.1-12CF** and housing **13.1-12**. FIGS. **13.1-15** shows how attachment structures **13.1-86** may, if desired, be used to couple a bridging central portion of frame **13.1-12CF** such as portion **13.1-68** to housing structure **13.1-84** in the center of frame **13.1-12C** (e.g., along the upper and lower edges of curtain **13.1-12C**).

[2501] FIGS. **13.1-16** is a cross-sectional view of curtain **13.1-12C** showing how attachment mechanism **13.1-86** for attaching curtain **13.1-12** to housing structure **13.1-84** may include trim member **13.1-94**. Adhesive **13.1-92** may be used to attach fabric **13.1-62** and frame **13.1-12CF** to trim member **13.1-94** (e.g., in the vicinity of attachment mechanism **13.1-86** in the areas of curtain **13.1-12C** that overlap bridging portion **13.1-86** of frame **13.1-12CF**). Trim member **13.1-94** and housing structure **13.1-84** may have mating engagement structures. For example, trim member **13.1-94** may have a snap such as snap **13.1-96** that mates with a corresponding hook such as hook **13.1-98** on housing structure **13.1-84**. When it is desired to snap curtain **13.1-12C** in place, curtain **13.1-12C** may be pressed into housing **13.1-12M** in the $-Z$ direction. If desired, curtain **13.1-12C** may be removed by disengaging (e.g., unsnapping) the engagement structures (e.g., when it is desired to remove or replace curtain **13.1-12C**).

[2502] If desired, curtain **13.1-12C** may be used in equipment other than devices **13.1-10**. Consider, as an example, the arrangement of FIGS. **13.1-17**. As shown in FIGS. **13.1-17**, apparatus **13.1-100** may have a movable member such as movable member **13.1-102**. Apparatus **13.1-100** may be, for example, a joystick or automobile stick shift and movable member **13.1-102** may be a movable shaft. Movable member **13.1-102** may move in one or more directions **13.1-104**. Curtain **13.1-12C** may have a cover layer **13.1-12CC** formed from fabric **13.1-62** or other covering material and may have peripheral elastic band **13.1-64**. Cover layer **13.1-12CC** may be mounted over a frame such as frame **13.1-12CF**. This allows cover layer **13.1-12CC** to slip and/or stretch during movement of member **13.1-102** to help avoid wrinkling in curtain **13.1-12C**.

13.2: Device with a Removable Cushion

[2503] A head-mounted device may include a head-mounted support structure that allows the

device to be worn on the head of a user. The head-mounted device may have displays that are supported by the head-mounted support structure for presenting a user with visual content. The head-mounted device may also have sensors such as front-facing cameras and other sensors for gathering information on the environment surrounding the device.

[2504] The head-mounted device may include a removable cushion that is configured to be attached to the head-mounted support structure. The removable cushion may contact the user's face when the head-mounted device is worn by the user. There are many advantages to using a removable cushion of this type in the head-mounted device. Because the cushion contacts the user's face during operation, the cushion may tend to become dirty over time. If the cushion is not removable, it may be difficult to clean the cushion. However, the removable cushion is easy to remove from the head-mounted device to wash. The removable cushion is therefore easy to keep clean during repeated uses.

[2505] In addition to making the cushion easier to clean, the removability of the cushion makes the cushion easier to replace. If the cushion becomes worn down over time or damaged, the removable cushion may simply be replaced with a new removable cushion. This extends the lifetime of the head-mounted device.

[2506] The removable cushion may also be customizable. If the cushion is not removable, the properties of the cushion in the head-mounted device are fixed. With a removable cushion, an optimal removable cushion out of multiple options may be selected by a user. For example, the user may select a removable cushion with a desired color, softness, and dimensions. In this way, the user may select a custom removable cushion that is the most comfortable for them.

[2507] FIGS. **13.2-1** is a top view of a head-mounted electronic device **13.2-10**. As shown in FIGS. **13.2-1**, head-mounted device **13.2-10** may include housing **13.2-12**. Housing **13.2-12** is configured to be worn on a user's head and may sometimes be referred to as a head-mounted housing or head-mounted support structure. Housing **13.2-12** may have curved head-shaped surfaces, a nose-bridge portion such as portion NB that is configured to rest on a user's nose when device **13.2-10** is on a user's head, a strap **13.2-12T** for supporting device **13.2-10** on the user's head, and/or may have other features that allow device **13.2-10** to be worn by a user. Housing **13.2-12** may have walls or other structures that separate an interior region of device **13.2-10**, such as interior region **13.2-42**, from an exterior region surrounding device **13.2-10**, such as exterior region **13.2-44**. Electrical components **13.2-40** (e.g., integrated circuits, sensors, control circuitry, light-emitting diodes, lasers, and other light-emitting devices, other control circuits and input-output devices, etc.) may be mounted on printed circuits and/or other structures within device **13.2-10** (e.g., in interior region **13.2-42**).

[2508] To present a user with images for viewing from eye boxes such as eye box **13.2-34**, device **13.2-10** may include rear-facing displays in optical modules **13.2-16** (sometimes referred to as optical assemblies, etc.). There may be, for example, a left rear-facing display in left optical module **13.2-16L** for presenting an image through a left lens to a user's left eye in a left eye box and a right rear-facing display in right optical module **13.2-16R** for presenting an image through a right lens to a user's right eye in a right eye box.

[2509] The user's eyes are located in eye boxes **13.2-34** at rear side R of device **13.2-10** when inwardly facing surface **13.2-18** of housing **13.2-12** rests against the outer surface of the user's face. On rear side R, housing **13.2-12** may have cushioned structures (sometimes referred to as light seal structures) to enhance user comfort as surface **13.2-18** rests against the user's face.

[2510] Device **13.2-10** may have forward-facing components such as forward-facing cameras **13.2-20** on front side F that face outwardly away from the user. Cameras **13.2-20** may generally be oriented in the +Y direction of FIGS. **13.2-1**. If desired, the left-hand camera may face slightly to the left and the right-hand camera may face slightly to the right to enhance the overall coverage of cameras **13.2-20**. During operation, images captured by cameras **13.2-20** (sometimes referred to as pass-through video and/or pass-through images) and/or computer-generated content such as text,

graphics, etc. may be displayed for the user by displays in modules **13.2-16**.

[2511] FIGS. **13.2-2** is a cross-sectional top view of an optical module **13.2-16**. As shown in FIGS. **13.2-2**, optical module **13.2-16** may include display **13.2-14** and lens **13.2-22** mounted in optical module housing **13.2-24**. Display **13.2-14** presents images in eye box **13.2-34**.

[2512] FIGS. **13.2-3A** is a cross-sectional top view of a head-mounted device **13.2-10** with a removable cushion **13.2-60** in its unattached (removed) state. As shown in FIGS. **13.2-3A**, head-mounted support structure **13.2-12** may include a rigid front structure **13.2-52** (sometimes referred to as front portion, rigid structure, etc.) at the front side F of the head-mounted device **13.2-10**.

Head-mounted support structure **13.2-12** additionally includes support posts **13.2-54** (sometimes referred to as rigid support posts, rigid sidewall portions, side portions, sidewall portions, support structures, etc.). The support structures **13.2-54** extend from the front portion **13.2-52** towards the rear side R. The support structures **13.2-54** may couple the front portion **13.2-52** to a flexible rear portion **13.2-56** (sometimes referred to as flexible structure, flexible ring, flexible light seal structure, etc.) of the head-mounted support structure **13.2-12**. Interior **13.2-42** (with associated optical modules **13.2-16L/13.2-16R** and other electronic components) is formed in the volume between front structure **13.2-52** and rear structure **13.2-56**.

[2513] In some examples, a sidewall extending from front structure **13.2-52** in the -Y-direction may be formed continuously around the perimeter of the front structure **13.2-52**. In some examples, to minimize the weight of the head-mounted support structure **13.2-12**, discrete support posts **13.2-54** may be dispersed around the perimeter of the front structure **13.2-52**. The support posts are each coupled between the rigid front structure **13.2-52** and the flexible rear structure **13.2-56**. Air-filled gaps (or gaps filled with another desired light filler material) may separate adjacent support posts **13.2-54**.

[2514] Front structure **13.2-52** may be an opaque structure that extends across the entire front side of device **13.2-10**. Alternatively, front structure **13.2-52** may be a ring-shaped structure that extends in a ring around the front side of device **13.2-10**. Flexible rear structure **13.2-56** may be a ring-shaped structure that extends in a ring around some or all of the rear side of device **13.2-10**.

Flexible rear structure **13.2-56** may be ring-shaped (with a central opening) to allow the user to view optical modules **13.2-16** while device **13.2-10** is worn by the user.

[2515] Head-mounted support structure **13.2-12** may optionally be covered on one or more sides by a textile layer **13.2-64** (sometimes referred to as fabric layer, etc.). Textile layer **13.2-64** may be formed from an opaque or light-shielding material (e.g., black yarn) or may be formed from an underlying material coated with an opaque or light-shielding material (e.g., black dye or ink). The textile layer **13.2-64** may be formed by a woven fabric or a nonwoven fabric. As examples, the textile layer **13.2-64** may be formed from any suitable type of fabric such as knit fabric, woven fabric, braided fabric, etc.

[2516] In addition to head-mounted support structure **13.2-12**, the head-mounted device includes a removable cushion **13.2-60** (sometimes referred to as a removable cushion member, removable foam, removable foam member, cushion, foam, etc.). Removable cushion **13.2-60** includes a cushion member **13.2-58** (sometimes referred to as cushion, cushion structure, foam member, foam structure, foam, etc.) that is optionally covered by a textile layer **13.2-62**. Cushion member **13.2-58** may be formed from foam or another desired compressible material. Textile layer **13.2-62** may be formed from an opaque or light-shielding material (e.g., black yarn) or may be formed from an underlying material coated with an opaque or light-shielding material (e.g., black dye or ink). The textile layer **13.2-62** may be formed by a woven fabric or a nonwoven fabric. As examples, the textile layer **13.2-62** may be formed from any suitable type of fabric such as knit fabric, woven fabric, braided fabric, etc.

[2517] The removable cushion may be attached to flexible structure **13.2-56** of head-mounted support structure **13.2-12**. FIGS. **13.2-3B** is a cross-sectional top view of a head-mounted device **13.2-10** with a removable cushion **13.2-60** attached to an associated head-mounted support

structure **13.2-12**. As shown, when attached, the removable cushion **13.2-60** may be adjacent to flexible rear structure **13.2-56**. Flexible structure **13.2-56** and removable cushion **13.2-60** may combine to form light seal structures **13.2-66** for the head-mounted device **13.2-10**. Flexible structure **13.2-56** may therefore sometimes be referred to as a light seal structure, flexible light seal structure, non-removable light seal structure, etc. Removable cushion **13.2-60** may sometimes be referred to as a light seal structure, cushioning light seal structure, removable light seal structure, etc.

[2518] When head-mounted device **13.2-10** is worn on the head of a user, the light seal structures **13.2-66** may conform to the user's face. To provide an immersive experience to the viewer and preserve high contrast in the display viewed by the user during operation, it may be desirable to block ambient light from reaching the user's eyes. This ensures the only light viewed by the user is from the display in the head-mounted device **13.2-10**. Light seal structures **13.2-66** may be sufficiently flexible and/or compressible to conform to a user's face during operation, preventing stray light from entering the head-mounted device **13.2-10**.

[2519] Flexible light seal structure **13.2-56** may be formed from a flexible plastic structure. The flexible plastic structure may be sufficiently flexible to conform to the shape of a user's face when worn by the user. However, the flexible plastic structure may have sufficient rigidity to hold its shape while being worn (e.g., and does not bias towards or away from the user's face while being worn). Flexible light seal structure **13.2-56** may not be compressible (e.g., may be less compressible than cushion **13.2-60**). Removable cushion **13.2-60** may also be sufficiently flexible to conform to the shape of a user's face when worn by the user. Cushion **13.2-60** may be compressible. The compressible cushion may be pressed against the user's face to form a tight light seal while still remaining comfortable for the user.

[2520] There are many advantages to using a removable cushion as shown in FIGS. **13.2-3A** and **13.2-3B** in the head-mounted device **13.2-10** (as opposed to a permanently fixed cushion). Because the cushion **13.2-60** contacts the user's face during operation, the cushion **13.2-60** may tend to become dirty over time. If the cushion **13.2-60** is not removable, it may be difficult to clean the cushion **13.2-60**. However, the removable cushion **13.2-60** is easy to remove from the head-mounted device **13.2-10** to wash. The removable cushion **13.2-60** therefore is easy to keep clean during repeated uses. The removable cushion **13.2-60** may be formed using materials that will not be damaged by water to promote easy cleaning. The removable cushion **13.2-60** may be formed using materials that dry quickly so that the removable cushion **13.2-60** is ready for use quickly after cleaning.

[2521] In addition to making the cushion **13.2-60** easier to clean, the removability of the cushion **13.2-60** makes the cushion **13.2-60** easier to replace. If the cushion **13.2-60** becomes worn down over time or damaged, the removable cushion **13.2-60** may simply be replaced with a new removable cushion **13.2-60**. This extends the lifetime of the head-mounted device **13.2-10**.

[2522] The removable cushion **13.2-60** may also be customizable. If the cushion **13.2-60** is not removable, the properties of the cushion in the head-mounted device **13.2-10** are fixed. With a removable cushion **13.2-60**, an optimal removable cushion **13.2-60** out of multiple options may be selected by a user. For example, the user may select a removable cushion **13.2-60** with a desired color and/or dimensions. In this way, the user may select a custom removable cushion **13.2-60** that is the most comfortable for them.

[2523] There are numerous dimensions of the removable cushion **13.2-60** that may be customized for different users. One such dimension is the cushion's thickness **13.2-92** (e.g., the dimension parallel to the Y-direction) as shown in FIGS. **13.2-3B**. Some users may prefer a thicker cushion **13.2-60** for maximum comfort whereas other users may prefer a thinner cushion **13.2-60** for maximum comfort.

[2524] The cushion thickness **13.2-92** may also impact the eye relief of the optical system. The eye relief is defined as the distance **13.2-94** between an optical module and a respective eye box that

views that optical module. The thickness **13.2-92** of the removable cushion **13.2-60** therefore influences the eye relief distance **13.2-94**. A thicker removable cushion **13.2-60** will result in a larger associated eye relief distance whereas a thinner removable cushion **13.2-60** will result in a smaller associated eye relief distance.

[2525] In some examples, depending on the characteristics of a user's eye (e.g., a user's glasses prescription), a specific eye relief may be preferable. For example, a first user may not require glasses. For the first user, a first eye relief distance may be optimal. A second user may have a glasses prescription. The optical module may be customized for the user's glasses prescription (e.g., by adding removable lenses to the optical module). In this scenario, for the second user, a second eye relief distance that is different than the first eye relief distance may be optimal. In this situation, the first user may use a first removable cushion having a first thickness (that produces the first eye relief distance) whereas the second user may use a second removable cushion having a second thickness that is different than the first thickness (and that produces the second eye relief distance).

[2526] Thickness **13.2-92** may be any desired magnitude (e.g., between 1.5 millimeters and 2.5 millimeters, between 2.5 millimeters and 3.5 millimeters, between 3.5 millimeters and 4.5 millimeters, between 1 millimeter and 5 millimeters, between 1 millimeter and 10 millimeters, etc.).

[2527] The removable cushion selected by a user may therefore have a thickness **13.2-92** that is optimized for their comfort and/or optimizes eye relief distance for the system.

[2528] Instead of (or in addition to) customizing the thickness (or other dimensions) of the removable cushion, a user may customize the material of the removable cushion. For example, some users may prefer a softer foam in the removable cushion whereas other users may prefer a harder foam in the removable cushion. Different options for the removable cushions may therefore have the same dimensions but be formed from different materials.

[2529] FIGS. **13.2-4** is a perspective view of head-mounted support structure **13.2-12** (without the removable cushion **13.2-60**). As shown, front structure **13.2-52** covers the entire front side of the head-mounted device **13.2-10** (e.g., front structure **13.2-52** does not have a central opening). In contrast, flexible rear structure **13.2-56** is ring-shaped with a central opening (e.g., the flexible rear structure **13.2-56** defines a central opening). The user may view optical modules **13.2-16** through the central opening while using the head-mounted device. As shown in FIGS. **13.2-4**, support posts **13.2-54** connect the flexible rear structure **13.2-56** to rigid front structure **13.2-52**. Interior volume **13.2-42** (with associated optical modules **13.2-16** and other electronic components) is formed between front structure **13.2-52** and rear structure **13.2-56**. Any desired number of support structures may be included in the device. The support structures may be distributed around the periphery of the device to ensure that the central opening with optical modules **13.2-16** is not blocked.

[2530] Each support structure may optionally be attached to the flexible structure **13.2-56** using a pivoting portion with a spherical joint. The spherical joint may allow for the flexible structure **13.2-56** to be pressed against the user's face at any desired angle, allowing the flexible structure **13.2-56** to conform to the user's face as much as possible (promoting a tight light seal).

[2531] It should be noted that head-mounted support structure **13.2-12** in FIGS. **13.2-4** may also be covered by textile layer **13.2-64**, similar to as shown and discussed in connection with FIGS. **13.2-3A**. The textile layer is omitted from FIGS. **13.2-4** so as to not obfuscate the drawing.

[2532] FIGS. **13.2-5A** is a rear view of flexible structure **13.2-56**. As shown, flexible structure **13.2-56** may have a ring-shape that extends partially or completely in a ring around a central opening **13.2-70**. The flexible structure **13.2-56** may have a gap along the lower edge to accommodate the nose of the user during operation. If desired, an optional nasal-region-mounted structure **13.2-72** that is configured to rest on the nose of a user may bridge the gap between the first and second ends of the flexible structure. Structure **13.2-72** may conform to the facial topology of the user around the nasal region and block light from entering the head-mounted device during

operation. Alternatively, nasal-region-mounted structure **13.2-72** may be omitted and flexible structure **13.2-56** may form a complete ring around the central opening such that a portion of the flexible structure **13.2-56** rests on the nose of the user during operation of the head-mounted device **13.2-10**.

[2533] The flexible structure **13.2-56** be attached to support posts **13.2-54** at various points around the flexible structure. FIGS. **13.2-5A** shows an example where four support posts **13.2-54** are coupled to the flexible structure. As previously mentioned, each support post may be coupled to the flexible structure with a spherical joint to accommodate the different curvatures of different users' faces.

[2534] FIGS. **13.2-5B** is a rear view of removable cushion **13.2-60**. As shown, removable cushion **13.2-60** may have a ring-shape that extends partially or completely in a ring around a central opening **13.2-74**. The cushion **13.2-60** may have a gap along the lower edge to accommodate the nose of the user during operation. If desired, an optional nasal-region-mounted structure **13.2-76** that is configured to rest on the nose of a user may bridge the gap between the first and second ends of the removable cushion. Structure **13.2-76** may conform to the facial topology of the user around the nasal region and block light from entering the head-mounted device during operation. Alternatively, nasal-region-mounted structure **13.2-76** may be omitted and cushion **13.2-60** may form a complete ring around the central opening such that a portion of the cushion **13.2-60** rests on the nose of the user during operation of the head-mounted device.

[2535] Although it is generally desirable for cushion **13.2-60** to be compressible (for user comfort and a tight light seal), cushion **13.2-60** may include one or more high-rigidity portions **13.2-60-1**. The high-rigidity portions **13.2-60-1** may provide sufficient structural integrity to prevent cushion **13.2-60** from being totally compressed. This may, for example, prevent a user's face from striking internal electronic components of the head-mounted device and/or flexible structure **13.2-56** during an impact event. High-rigidity portions **13.2-60-1** are distributed across cushion **13.2-60**. Low-rigidity portions **13.2-60-2** are interposed in the remaining areas of the cushion (e.g., between each adjacent pair of high-rigidity portions **13.2-60-1**).

[2536] The high-rigidity portions **13.2-60-1** have a greater rigidity and a lower compressibility than low-rigidity portions **13.2-60-2**. The high-rigidity portions **13.2-60-1** may be formed from a different material than low-rigidity portions **13.2-60-2**. In other words, cushion structure **13.2-58** (see FIGS. **13.2-3A** and **13.2-3B**) includes high-rigidity portions **13.2-60-1** formed from a first material and low-rigidity portions **13.2-60-2** formed from a second material that is different than the first material. As another example, the entire cushion structure **13.2-58** may be formed from the same base material. However, high-rigidity portions **13.2-60-1** may include an additive (e.g., a second material) that increases rigidity relative to low-rigidity portions **13.2-60-2**.

[2537] As shown in FIGS. **13.2-5B**, the cushion **13.2-60** may have approximately the same footprint as flexible rear structure **13.2-56** from FIGS. **13.2-5A** (when viewed from the rear). Additionally, high-rigidity portions **13.2-60-1** may be positioned to overlap support posts **13.2-54** (in the Y-direction) when the removable cushion **13.2-60** is attached to flexible rear structure **13.2-56**. In other words, each high-rigidity portion **13.2-60-1** is aligned (in the Y-direction) with a respective portion of flexible structure **13.2-56** that is attached to a support post. Each high-rigidity portion **13.2-60-1** is therefore aligned (in the Y-direction) with a respective support post. This type of arrangement may ensure that the high-rigidity portions **13.2-60-1** provide the target structural integrity for cushion **13.2-60** while minimizing detectability to the user. The presence of support posts **13.2-54** make flexible structure **13.2-56** more rigid in these locations. Therefore, positioning high-rigidity portions **13.2-60-1** in the same location consolidates the high-rigidity portions.

[2538] FIGS. **13.2-5B** shows another dimension **13.2-96** of the removable cushion that may be customized. Dimension **13.2-96** may be referred to as the height (e.g., in the Z-direction) of the cushion along an upper edge of the cushion. The upper edge of the cushion may be adjacent to the user's forehead during use and may therefore sometimes be referred to as a forehead portion of the

cushion. Some users may prefer a larger upper-edge height **13.2-96** than others for maximum comfort. A user may therefore select a removable cushion **13.2-60** that has an optimal upper-edge height **13.2-96** for their needs. Height **13.2-96** may be greater than 3 millimeters, greater than 5 millimeters, greater than 10 millimeters, greater than 15 millimeters, greater than 20 millimeters, greater than 30 millimeters, greater than 50 millimeters, less than 3 millimeters, less than 5 millimeters, less than 10 millimeters, less than 15 millimeters, less than 20 millimeters, less than 30 millimeters, less than 50 millimeters, between 5 millimeters and 30 millimeters, etc.

[2539] Dimension **13.2-98** (as shown in FIGS. **13.2-5B**) may also be customized. Dimension **13.2-98** may be referred to as the total height (e.g., in the Z-direction) of the cushion. The total height may influence factors such as the separation between the upper edge of the cushion and a user's eyebrow. Some users may prefer greater separation between the upper edge of the cushion and the user's eyebrow than others for maximum comfort. A user may therefore select a removable cushion **13.2-60** that has an optimal height **13.2-98** for their needs. Height **13.2-98** may be greater than 5 centimeters, greater than 8 centimeters, greater than 10 centimeters, greater than 15 centimeters, greater than 20 centimeters, less than 8 centimeters, less than 10 centimeters, less than 15 centimeters, less than 20 centimeters, between 5 centimeters and 15 centimeters, etc.

[2540] Head-mounted support structure **13.2-12** and/or removable cushion **13.2-60** may include attachment structures that are used to secure removable cushion **13.2-60** to the head-mounted support structure. FIGS. **13.2-6A** is a rear view of flexible rear structure **13.2-56** (of head-mounted support structure **13.2-12**) showing how the flexible rear structure may include one or more primary attachment structures **13.2-82P** and, optionally, one or more auxiliary attachment structures **13.2-82A**. FIGS. **13.2-6B** is a rear view of removable cushion **13.2-60** showing how the removable cushion may include one or more primary attachment structures **13.2-84P** and, optionally, one or more auxiliary attachment structures **13.2-84A**.

[2541] Each primary attachment structure **13.2-82P** in flexible rear structure **13.2-56** may be configured to attach to a corresponding primary attachment structure **13.2-84P** in removable cushion **13.2-60**. Similarly, each auxiliary attachment structure **13.2-82A** in flexible rear structure **13.2-56** may be configured to attach to a corresponding auxiliary attachment structure **13.2-84A** in removable cushion **13.2-60**. As shown in FIGS. **13.2-6A**, each primary attachment structure **13.2-82P** in flexible rear structure **13.2-56** may align in the Y-direction with a corresponding primary attachment structure **13.2-84P** in removable cushion **13.2-60** and each auxiliary attachment structure **13.2-82A** in flexible rear structure **13.2-56** may align in the Y-direction with a corresponding auxiliary attachment structure **13.2-84A** in removable cushion **13.2-60**.

[2542] A wide range of attachment structures may be used for attachment structures **13.2-82P**, **13.2-84P**, **13.2-82A**, and **13.2-84A**. The attachment structures may include protrusions, recesses, grooves, posts, magnets, flexible fabric, hooks, loops, snaps, buttons, suction cups, a draw string, a zipper, adhesive (e.g., tape), a flexible band, or any other desired type of attachment structure.

[2543] As some specific examples of possible attachment structures, primary attachment structures **13.2-82P** on flexible structure **13.2-56** may be protrusions that are configured to mate (interlock) with corresponding recesses (**13.2-84P**) on cushion **13.2-60**. Primary attachment structures **13.2-82P** on flexible structure **13.2-56** may be magnets (e.g., permanent magnets) that are configured to magnetically couple with corresponding magnets (**13.2-84P**) on cushion **13.2-60**. Primary attachment structures **13.2-82P** and **13.2-84P** may form hook-and-loop fasteners (e.g., with hooks on structures **13.2-56** and loops on cushion **13.2-60** or vice versa). Primary attachment structures **13.2-82P** and **13.2-84P** may form snap fasteners. The first halves of the snaps may be formed on structures **13.2-56** and may have recesses and the second halves of the snaps may be formed on cushion **13.2-60** and may have protrusions with grooves that snap into place when pressed into the recesses (or vice versa). As yet another example, primary attachment structures **13.2-82P** on flexible structure **13.2-56** may be grooves that are configured to mate (interlock) with corresponding posts (**13.2-84P**) on cushion **13.2-60**. As yet another example, primary attachment

structures **13.2-82P** on flexible structure **13.2-56** may be posts. Cushion **13.2-60** may have a layer (ring) of flexible fabric that is configured to be stretched around the posts. As yet another example, rear flexible structure **13.2-56** may include an attached fabric channel. The removable cushion is then placed in the fabric channel during operation.

[2544] The specific example of a locations for primary attachment structures **13.2-82P** and **13.2-84P** in FIGS. **13.2-6A** and **13.2-6B** are merely illustrative. In general, primary attachment structures may be included at any desired locations.

[2545] The primary attachment structures may be the primary mechanism by which the removable cushion is secured onto the head-mounted support structure. However, in order to provide additional attachment strength and/or to ensure a tight light seal, auxiliary attachment structures **13.2-82A** and **13.2-84A** may also be included. As shown in FIGS. **13.2-6B**, auxiliary attachment structures **13.2-82A** and **13.2-84A** may be included along the left and right edges of the rear flexible structure and removable cushion. The rear flexible structure and removable cushion may commonly have high degrees of curvature in these areas when the head-mounted device is in operation (due to the curvature of the user's face). These high degrees of curvature may make the cushion and rear flexible structure susceptible to light leakage (e.g., through a gap between the cushion and rear flexible structure when the cushion and rear flexible structure are bent to conform to the user's face). The auxiliary attachment structures **13.2-82A** and **13.2-84A** in these high-risk areas may ensure a tight seal around the entire perimeter of the removable cushion and the rear flexible structure. The specific example of a locations for auxiliary attachment structures **13.2-82A** and **13.2-84A** in FIGS. **13.2-6A** and **13.2-6B** are merely illustrative. In general, auxiliary attachment structures may be included at any desired locations (e.g., adjacent/between any desired primary attachment structures).

[2546] Auxiliary attachment structures **13.2-82A** and **13.2-84A** may form hook-and-loop fasteners (e.g., with hooks on structures **13.2-56** and loops on cushion **13.2-60** or vice versa). Auxiliary attachment structures **13.2-82A** and **13.2-84A** may alternatively form snap fasteners. The first halves of the snaps may be formed on structures **13.2-56** and may have recesses and the second halves of the snaps may be formed on cushion **13.2-60** and may have protrusions with grooves that snap into place when pressed into the recesses (or vice versa). In general, any of the aforementioned attachment structures may be used for the auxiliary attachment structures.

[2547] If desired, multiple types of attachment structures may be used at each attachment point to ensure a secure attachment between the removable cushion **13.2-60** and the head-mounted support structure **13.2-12**. FIGS. **13.2-7** is a cross-sectional top view of an illustrative head-mounted device showing how protrusions, recesses, and magnets may be used to secure the removable cushion **13.2-60** to head-mounted support structure **13.2-12**. As shown in FIG. **13.2-7**, protrusions **13.2-82P-2** extend from flexible rear structure **13.2-56** in the -Y-direction (e.g., towards the removable cushion **13.2-60**). Removable cushion **13.2-60** has corresponding recesses **13.2-84P-2**. Each recess **13.2-84P-2** may mate with a corresponding protrusion **13.2-82P-2**.

[2548] In addition to the protrusions and recesses, magnets are also used as attachment structures for the head-mounted device. As shown in FIGS. **13.2-7**, magnets **13.2-82P-1** may be embedded in protrusions **13.2-82P-2**. A single magnet may be embedded in each protrusion or multiple magnets may be embedded in each protrusion. A corresponding magnet **13.2-84P-1** may be formed in cushion structure **13.2-58** adjacent to each respective recess **13.2-84P-2**. When removable structure **13.2-60** is attached to the head-mounted support structure, protrusions **13.2-82P-2** extend into and mate with recesses **13.2-84P-2**. When the protrusions **13.2-82P-2** and recesses **13.2-84P-2** are mated, magnets **13.2-82P-1** will also be magnetically coupled to magnets **13.2-84P-1**. This provides additional security for the removable cushion attachment and prevents gravity from causing the removable cushion to fall off of the head-mounted support structure.

[2549] The example of FIGS. **13.2-7** is merely illustrative. In addition to the primary attachment structures of protrusions, recesses, and magnets, auxiliary attachment structures such as snaps or

hook-and-loop fasteners may also be included to secure the removable cushion to head-mounted support structure **13.2-12** and ensure a tight light seal.

[2550] In some examples, protrusions **13.2-82P-2** may be formed integrally with rear flexible structure **13.2-56** or may be formed from a separate material that is attached to rear flexible structure **13.2-56**. Alternatively, in the arrangement of FIGS. **13.2-7**, the protrusions **13.2-82P-2** are formed integrally with support posts **13.2-54**. In other words, portions of support posts **13.2-54** extend through openings in rear flexible structure **13.2-56** to form attachment structures **13.2-82P-2**. The support posts **13.2-54** may also extend through openings in textile layer **13.2-64** (which covers head-mounted support structure **13.2-12**). In this way, the support post protrusions **13.2-82P-2** serve as a visual indicator to a user that the head-mounted support structure is not yet attached to a removable cushion. The support post protrusions **13.2-82P-2** also provide recognizable alignment markers that allow the user to easily align and attach the removable cushion to the head-mounted support structure.

[2551] FIGS. **13.2-8** is a rear view of an illustrative removable cushion **13.2-60**. As shown in FIGS. **13.2-8**, the removable cushion may have one or more incorporated hinge structures **13.2-86**. The hinge structures **13.2-86** may be formed by a highly flexible portion of the cushion member. Alternatively, the foldable portions **13.2-86** may be formed by a post that is formed separately from the cushion member. Hinge structures **13.2-86** allow for rotation of portions of the cushion member relative to other portions of the cushion member. Each hinge structure **13.2-86** may allow for rotation of the cushion member around a respective bend axis **13.2-88**. The presence of hinge structures **13.2-86** may therefore allow for the cushion member to be folded to have a smaller footprint than in the unfolded state, increasing the portability of the cushion member.

[2552] In one possible arrangement, each hinge structure **13.2-86** may be formed by a post that allows rotation of adjacent cushion portions relative to each other. The post may optionally double as an attachment structure for attaching the removable cushion to the head-mounted support structure. For example, the head-mounted support structure may have a groove and the post (which defines a hinge structure for the cushion) slides into the groove to attach the removable cushion to the head-mounted support structure.

[2553] The example of two hinge structures being included in the removable cushion is merely illustrative. In general, any desired number of hinge structures may be included in the removable cushion.

[2554] FIGS. **13.2-9** is a schematic diagram of a system **13.2-200** that includes a head-mounted support structure **13.2-12** (with optical modules **13.2-16L/13.2-16R** and other associated electronics at its interior) and numerous removable cushions **13.2-60**. As previously shown and discussed, one of the removable cushions may be attached to the head-mounted support structure to form a head-mounted device **13.2-10** with an attached cushion.

[2555] System **13.2-200** may include any desired number of removable cushions **13.2-60-1** through **13.2-60-N** having different properties. As previously discussed, the varying properties of the removable cushions may include color, thickness, upper-edge height, total height, any other desired dimension, the softness of the material, or any other desired characteristic.

[2556] As specific examples, a user may select a removable cushion having a desired color out of many color options. A user may select a removable cushion having a thickness that optimizes the optical properties (e.g., eye relief) of the head-mounted device. A user may select a removable cushion having a desired upper-edge height to maximize their comfort when wearing the head-mounted device.

[2557] System **13.2-200** may include any desired number removable cushions, each with a unique set of properties. As one example, a user may purchase a head-mounted device and receive the head-mounted support structure and multiple removable cushions having various sizes (e.g., thicknesses and/or total heights). The multiple removable cushions may include a small removable cushion, a medium removable cushion that is larger in at least one dimension than the small

removable cushion, and a large removable cushion that is larger than the medium removable cushion in at least one dimension. The user may attach their preferred removable cushion to the head-mounted support structure before using the head-mounted device.

[2558] As another example, a user may select their preferred removable cushion (e.g., preferred color, size, etc.) when purchasing a head-mounted device. The user receives their preferred removable cushion in addition to the head-mounted support structure and may attach their preferred removable cushion to the head-mounted support structure before using the head-mounted device. The removable cushion may then repeatedly be removed and washed during the lifetime of the head-mounted device.

[2559] In accordance with some examples, a head-mounted device is provided that includes a head-mounted support structure including a rigid structure, a flexible structure, and support structures that couple the flexible structure to the rigid structure, the flexible structure at least partially defines a central opening, left and right optical assemblies configured to display images, the left and right optical assemblies are coupled to the head-mounted support structure, and a removable cushion that is configured to be selectively attached to the flexible structure, the removable cushion at least partially surrounds the central opening.

[2560] In accordance with some examples, the left and right optical assemblies are viewable through the central opening, the head-mounted support structure is covered by a first textile layer, the removable cushion is covered by a second textile layer, the removable cushion has first portions having a first rigidity and second portions having a second rigidity, the removable cushion has recesses and first magnets, at least one of the first magnets is adjacent to each recess, and the head-mounted support structure includes protrusions at the flexible structure that extend away from the rigid structure, each protrusion is configured to mate with a respective recess of the removable cushion when the removable cushion is attached to the flexible structure, and second magnets, at least one of the second magnets is positioned in each protrusion and the first magnets are configured to magnetically couple to the second magnets when the removable cushion is attached to the flexible structure.

[2561] In some examples, the left and right optical assemblies are viewable through the central opening. In some examples the head-mounted support structure is covered by a textile layer. In some examples, the support structures are configured to protrude past the flexible structure and through openings in the textile layer. In some examples, the removable cushion has recesses that are configured to receive the protruding portions of the support structures when the removable cushion is attached to the flexible structure. In some examples, first magnets are formed in the removable cushion adjacent to the recesses, second magnets are formed in protruding portions of the support structures, and the first magnets are configured to magnetically couple to the second magnets when the removable cushion is attached to the flexible structure.

[2562] In accordance with some examples, the flexible structure includes a first auxiliary attachment structure between first and second protruding portions, the removable cushion includes a second auxiliary attachment structure between first and second recesses, and the first auxiliary attachment structure is configured to couple to the second auxiliary attachment structure when the removable cushion is attached to the flexible structure. In some examples, the first and second auxiliary attachment structures include a hook-and-loop fastener. In some examples, the removable cushion includes a cushion that is covered by a textile layer.

[2563] In accordance with some examples, first magnets are formed in the removable cushion, second magnets are formed in the head-mounted support structure, and the first magnets are configured to magnetically couple to the second magnets when the removable cushion is attached to the flexible structure. In some examples, the removable cushion has first portions having a first rigidity and second portions having a second rigidity that is lower than the first rigidity. In some examples, each one of the first portions is aligned with a respective support structure. In some examples, the removable cushion includes a hinge structure that allows a first portion of the

removable cushion to rotate relative to a second portion of the removable cushion.

[2564] In accordance with some examples, a head-mounted device is provided that includes a head-mounted support structure including a flexible portion that is configured to conform to a face when the head-mounted support structure is worn, a rigid portion that is separated from the flexible portion by a gap, side portions that bridge the gap between the rigid portion and the flexible portion, and protrusions at the flexible portion that extend away from the rigid portion, and a first set of magnets, each one of the first set of magnets is positioned in a respective one of the protrusions, and a removable cushion that is configured to be selectively attached to the flexible portion, the removable cushion is configured to conform to and directly contact the face when the head-mounted support structure is worn while the removable cushion is attached and the removable cushion includes recesses that are configured to mate with the protrusions of the head-mounted support structure, and a second set of magnets that is configured to magnetically couple to the first set of magnets. In some examples, the head-mounted support structure is covered by a textile layer and the protrusions extend through openings in the textile layer. In some examples, the protrusions are formed by the side portions. In some examples, the head-mounted device includes left and right optical assemblies configured to display images, the left and right optical assemblies are formed in an interior volume of the head-mounted support structure.

[2565] In accordance with some examples, a system is provided that includes a head-mounted support structure, left and right optical assemblies configured to display images, the left and right optical assemblies are coupled to the head-mounted support structure, and a plurality of removable cushions, each one of the removable cushions is configured to be selectively attached to the head-mounted support structure and each one of the plurality of removable cushions has a property that is unique relative to that property in the remaining removable cushions. In some examples, the property includes a property selected from the group consisting of a color, a thickness, a cushion material, a total height, and an upper-edge height.

13.3: Electronic Devices with Light-Blocking Fabrics

[2566] An electronic device, such as a head-mounted device, may have a front face that faces away from a user's head and may have an opposing rear face that faces the user's head. The head-mounted device may include a main housing portion with displays that display images and optical modules through which the images are viewable from eye boxes. A face frame that is coupled to the main housing portion may form a light seal around the eye boxes. The light seal may include one or more fabric layers such as knit fabric layers. The light seal may include inner and outer fabric layers, a face portion that rests against the user's face, and a nose bridge portion that accommodates the user's nose. The outer fabric layer may be a seamless tube of knit fabric that forms an outermost layer of the light seal. The inner fabric layer may be a light-blocking fabric that lines the inner surface of the seamless tube of knit fabric. The knit fabric may have a modified bird's eye pattern that incorporates knit stitches and tuck stitches. The tuck stitches may provide the fabric with additional lengthwise stretch so that the light seal fabric does not buckle when the head-mounted device is placed on the user's head. The tuck stitches also allow for strands of different colors to be visible on the outer surface of the light seal.

[2567] The light-blocking fabric may include a dark-colored weft knit layer facing the eye box, a light-colored weft knit layer facing the seamless tube of knit fabric, and a middle layer joining the light-colored weft knit layer and the dark-colored weft knit layer. The light-colored weft knit layer may match a color of the seamless tube of knit fabric. The dark-colored weft knit layer may ensure sufficient opacity without being visible through the seamless tube of knit fabric.

[2568] A top view of an illustrative head-mounted device that may include a fabric light seal is shown in FIGS. 13.3-1. As shown in FIGS. 13.3-1, head-mounted devices such as electronic device 13.3-10 may have head-mounted support structures such as housing 13.3-12. Housing 13.3-12 may include portions (e.g., support structures 13.3-12T) to allow device 13.3-10 to be worn on a user's head. Support structures 13.3-12T (sometimes referred to as temple housing structures or temple

housing portions) may be formed from fabric, polymer, metal, and/or other material. Support structures **13.3-12T** may form a strap or other head-mounted support structures that help support device **13.3-10** on a user's head. Some or all of temple housing portions **13.3-12T** may overlap a user's temples when device **13.3-10** is worn on the user's head. A main support structure (e.g., main housing portion **13.3-12M**) of housing **13.3-12** may support electronic components such as displays **13.3-14**. Main housing portion **13.3-12M** may include housing structures formed from metal, polymer, glass, ceramic, and/or other material. For example, housing portion **13.3-12M** may have housing walls on front face **F** and housing walls on adjacent top, bottom, left, and right side faces that are formed from rigid polymer or other rigid support structures and these rigid walls may optionally be covered with electrical components, fabric, leather, or other soft materials, etc. The walls of housing portion **13.3-12M** may enclose internal components **13.3-38** in interior region **13.3-34** of device **13.3-10** and may separate interior region **13.3-34** from the environment surrounding device **13.3-10** (exterior region **13.3-36**). Internal components **13.3-38** may include integrated circuits, actuators, batteries, sensors, and/or other circuits and structures for device **13.3-10**. Housing **13.3-12** may be configured to be worn on a head of a user and may form glasses, a hat, a helmet, goggles, and/or other head-mounted device. Configurations in which housing **13.3-12** forms goggles may sometimes be described herein as an example.

[2569] Front face **F** of housing **13.3-12** may face outwardly away from a user's head and face. Opposing rear face **R** of housing **13.3-12** may face the user. Portions of housing **13.3-12** (e.g., portions of main housing **13.3-12M**) on rear face **R** may form a cover such as curtain **13.3-12C**. In some examples, curtain **13.3-12C** includes a fabric layer that separates interior region **13.3-34** from the exterior region to the rear of device **13.3-10**. Other structures may be used in forming curtain **13.3-12C**, if desired. The presence of curtain **13.3-12C** on rear face **R** may help hide internal housing structures, internal components **13.3-38**, and other structures in interior region **13.3-34** from view by a user.

[2570] Device **13.3-10** may have left and right optical modules **13.3-40**. Each optical module may include a respective display **13.3-14**, lens **13.3-30**, and support structure **13.3-32**. Support structures **13.3-32**, which may sometimes be referred to as lens barrels or optical module support structures, may include hollow cylindrical structures with open ends or other supporting structures to house displays **13.3-14** and lenses **13.3-30**. Support structures **13.3-32** may, for example, include a left lens barrel that supports a left display **13.3-14** and left lens **13.3-30** and a right lens barrel that supports a right display **13.3-14** and right lens **13.3-30**. Displays **13.3-14** may include arrays of pixels or other display devices to produce images. Displays **13.3-14** may, for example, include organic light-emitting diode pixels formed on substrates with thin-film circuitry and/or formed on semiconductor substrates, pixels formed from crystalline semiconductor dies, liquid crystal display pixels, scanning display devices, and/or other display devices for producing images. Lenses **13.3-30** may include one or more lens elements for providing image light from displays **13.3-14** to respective eyes boxes **13.3-13**. Lenses may be implemented using refractive glass lens elements, using mirror lens structures (catadioptric lenses), using holographic lenses, and/or other lens systems. When a user's eyes are located in eye boxes **13.3-13**, displays (display panels) **13.3-14** operate together to form a display for device **13.3-10** (e.g., the images provided by respective left and right optical modules **13.3-40** may be viewed by the user's eyes in eye boxes **13.3-13** so that a stereoscopic image is created for the user). The left image from the left optical module fuses with the right image from a right optical module while the display is viewed by the user.

[2571] Not all users have the same interpupillary distance **P**. To provide device **13.3-10** with the ability to adjust the interpupillary spacing between modules **13.3-40** along lateral dimension **X** and thereby adjust the spacing **P** between eye boxes **13.3-13** to accommodate different user interpupillary distances, device **13.3-10** may be provided with one or more actuators **13.3-42**. Actuators **13.3-42** can be manually controlled and/or computer-controlled actuators (e.g., computer-controlled motors) for moving support structures **13.3-32** relative to each other.

[2572] As shown in FIGS. **13.3-2**, curtain **13.3-12C** may cover rear face F while leaving lenses **13.3-30** of optical modules **13.3-40** uncovered (e.g., curtain **13.3-12C** may have openings that are aligned with and receive modules **13.3-40**). As modules **13.3-40** are moved relative to each other along dimension X to accommodate different interpupillary distances for different users, modules **13.3-40** move relative to fixed housing structures such as the walls of main portion **13.3-12M** and move relative to each other. To prevent undesired wrinkling and buckling of curtain **13.3-12C** as optical modules **13.3-40** are moved relative to rigid portions of housing **13.3-12M** and relative to each other, a fabric layer or other cover layer in curtain **13.3-12C** may be configured to slide, stretch, open/close, and/or otherwise adjust to accommodate optical module movement.

[2573] A schematic diagram of an illustrative electronic device such as a head-mounted device or other wearable device is shown in FIGS. **13.3-3**. Device **13.3-10** of FIGS. **13.3-3** may be operated as a stand-alone device and/or the resources of device **13.3-10** may be used to communicate with external electronic equipment. As an example, communications circuitry in device **13.3-10** may be used to transmit user input information, sensor information, and/or other information to external electronic devices (e.g., wirelessly or via wired connections). Each of these external devices may include components of the type shown by device **13.3-10** of FIGS. **13.3-3**.

[2574] As shown in FIGS. **13.3-3**, a head-mounted device such as device **13.3-10** may include control circuitry **13.3-20**. Control circuitry **13.3-20** may include storage and processing circuitry for supporting the operation of device **13.3-10**. The storage and processing circuitry may include storage such as nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **13.3-20** may be used to gather input from sensors and other input devices and may be used to control output devices. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors and other wireless communications circuits, power management units, audio chips, application specific integrated circuits, etc. During operation, control circuitry **13.3-20** may use display(s) **13.3-14** and other output devices in providing a user with visual output and other output.

[2575] To support communications between device **13.3-10** and external equipment, control circuitry **13.3-20** may communicate using communications circuitry **13.3-22**. Circuitry **13.3-22** may include antennas, radio-frequency transceiver circuitry, and other wireless communications circuitry and/or wired communications circuitry. Circuitry **13.3-22**, which may sometimes be referred to as control circuitry and/or control and communications circuitry, may support bidirectional wireless communications between device **13.3-10** and external equipment (e.g., a companion device such as a computer, cellular telephone, or other electronic device, an accessory such as a point device, computer stylus, or other input device, speakers or other output devices, etc.) over a wireless link. For example, circuitry **13.3-22** may include radio-frequency transceiver circuitry such as wireless local area network transceiver circuitry configured to support communications over a wireless local area network link, near-field communications transceiver circuitry configured to support communications over a near-field communications link, cellular telephone transceiver circuitry configured to support communications over a cellular telephone link, or transceiver circuitry configured to support communications over any other suitable wired or wireless communications link. Wireless communications may, for example, be supported over a Bluetooth® link, a WiFi® link, a wireless link operating at a frequency between 10 GHz and 400 GHz, a 60 GHz link, or other millimeter wave link, a cellular telephone link, or other wireless communications link. Device **13.3-10** may, if desired, include power circuits for transmitting and/or receiving wired and/or wireless power and may include batteries or other energy storage devices. For example, device **13.3-10** may include a coil and rectifier to receive wireless power that is provided to circuitry in device **13.3-10**.

[2576] Device **13.3-10** may include input-output devices such as devices **13.3-24**. Input-output

devices **13.3-24** may be used in gathering user input, in gathering information on the environment surrounding the user, and/or in providing a user with output. Devices **13.3-24** may include one or more displays such as display(s) **13.3-14**. Display(s) **13.3-14** may include one or more display devices such as organic light-emitting diode display panels (panels with organic light-emitting diode pixels formed on polymer substrates or silicon substrates that contain pixel control circuitry), liquid crystal display panels, microelectromechanical systems displays (e.g., two-dimensional mirror arrays or scanning mirror display devices), display panels having pixel arrays formed from crystalline semiconductor light-emitting diode dies (sometimes referred to as microLEDs), and/or other display devices.

[2577] Sensors **13.3-16** in input-output devices **13.3-24** may include force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors such as a touch sensor that forms a button, trackpad, or other input device), and other sensors. If desired, sensors **13.3-16** may include optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, optical touch sensors, optical proximity sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, image sensors, fingerprint sensors, iris scanning sensors, retinal scanning sensors, and other biometric sensors, temperature sensors, sensors for measuring three-dimensional non-contact gestures (“air gestures”), pressure sensors, sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or inertial measurement units that contain some or all of these sensors), health sensors such as blood oxygen sensors, heart rate sensors, blood flow sensors, and/or other health sensors, radio-frequency sensors, depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices that capture three-dimensional images), optical sensors such as self-mixing sensors and light detection and ranging (LIDAR) sensors that gather time-of-flight measurements, humidity sensors, moisture sensors, gaze tracking sensors, electromyography sensors to sense muscle activation, facial sensors, and/or other sensors. In some arrangements, device **13.3-10** may use sensors **13.3-16** and/or other input-output devices to gather user input. For example, buttons may be used to gather button press input, touch sensors overlapping displays can be used for gathering user touch screen input, touch pads may be used in gathering touch input, microphones may be used for gathering audio input, accelerometers may be used in monitoring when a finger contacts an input surface and may therefore be used to gather finger press input, etc.

[2578] If desired, electronic device **13.3-10** may include additional components (see, e.g., other devices **13.3-18** in input-output devices **13.3-24**). The additional components may include haptic output devices, actuators for moving movable housing structures, audio output devices such as speakers, light-emitting diodes for status indicators, light sources such as light-emitting diodes that illuminate portions of a housing and/or display structure, other optical output devices, and/or other circuitry for gathering input and/or providing output. Device **13.3-10** may also include a battery or other energy storage device, connector ports for supporting wired communication with ancillary equipment and for receiving wired power, and other circuitry.

[2579] FIGS. **13.3-4** is a perspective view of device **13.3-10** showing how a face frame may form a light seal around eye boxes **13.3-13** (FIGS. **13.3-1**) to help prevent outside light from leaking into the viewing area of head-mounted device **13.3-10**. As shown in FIGS. **13.3-4**, device **13.3-10** may include main housing portion **13.3-12M** which is configured to be mounted on a user's head. To help block outside light (e.g., ambient light in the user's environment that is not emitted by displays **13.3-14** of device **13.3-10**) from entering the viewing area of head-mounted device **13.3-10** where eye boxes **13.3-13** are located, a light seal such as light seal **13.3-52** may be formed between main housing portion **13.3-12M** and the user's face. For example, light seal **13.3-52** may extend between main housing portion **13.3-12M** and temple housing portions **13.3-12T** (FIGS. **13.3-1**) to help prevent light from entering any gaps between device **13.3-10** and the user's face.

[2580] Light seal **13.3-52** (sometimes referred to as face frame **13.3-52**) may include one or more rigid structures such as a rigid internal frame member or other stiff structure and one or more flexible materials such as fabric, foam, polymer, or other suitable materials. For example, light seal **13.3-52** may include a ring-shaped or horseshoe-shaped frame that surrounds eye boxes **13.3-13** (FIGS. **13.3-1**) and that is covered by one or more layers of fabric. As shown in FIGS. **13.3-4**, light seal **13.3-52** may include one or more different fabric layers such as outer fabric layer **13.3-72**, inner fabric layer **13.3-58**, nose bridge fabric **13.3-60**, and face fabric **13.3-56**. Face fabric **13.3-56** may rest against the user's face when device **13.3-10** is worn on the user's head. Face fabric **13.3-56** may include one or more layers of foam covered in one or more layers of fabric (e.g., a warp knit fabric, a weft knit fabric, a spacer fabric, a woven fabric, and/or any other suitable fabric). Nose bridge fabric **13.3-60** may be formed from a stretchable textile to accommodate different nose shapes.

[2581] Outer fabric layer **13.3-72** may be a seamless tube of fabric that loops around the optical axes A of lenses **13.3-30** of optical modules **13.3-40**. The optical axis A of each lens **13.3-30** extends parallel to the Y-direction of FIGS. **13.3-4**. The lengthwise direction L of outer fabric layer **13.3-72** may extend parallel to the optical axes A of lens **13.3-30** (and thus parallel to the Y-axis of FIGS. **13.3-4**), whereas the widthwise direction W of outer fabric layer **13.3-72** may extend perpendicular to the optical axes A of lens **13.3-30** of optical modules **13.3-40**. In a weft knit fabric, the rows of loops (referred to as courses) extend in the width direction W, while the columns of loops (referred to as wales) extend in the length direction L.

[2582] Outer fabric layer **13.3-72** may form an outermost surface of device **13.3-10**, if desired. Outer layer **13.3-72** may be formed from fabric such as knit fabric (e.g., warp knit fabric, weft knit fabric, etc.), woven fabric, spacer fabric (e.g., inner and outer knit layers separated by a gap and joined by a spacer layer such as a monofilament strand), braided fabric, and/or any other suitable fabric. In some examples, outer fabric layer **13.3-72** is a stretchable, seamless tube of weft knit fabric having a bird's eye pattern or other suitable two-color pattern (as an example). Arrangements in which outer layer **13.3-72** is formed from non-fabric materials such as polymer, silicone, or elastomer may also be used. Arrangements in which outer fabric layer **13.4-72** of light seal **13.4-52** is formed from a stretchable fabric are sometimes described herein as an example.

[2583] If desired, outer fabric layer **13.3-72** may be formed from weft knit fabric with a bird's eye pattern that uses two types of strands such as strands with a first property (e.g., a first color, material, fuzziness, denier, elasticity, etc.) and strands with a second property (e.g., a second color, material, fuzziness, denier, elasticity, etc.). The two types of strands may be visible on the outer surface of outer fabric layer **13.3-72** to provide the outer surface of outer fabric layer **13.3-72** with color variation, material variation, texture variation, etc. For example, the color variation on the outer surface of outer fabric layer **13.3-72** may form a checker pattern, a stripe pattern, a diamond pattern, a grid pattern, a dot pattern, or any other suitable pattern alternating between a first color and a second color (e.g., black and white, white and grey, grey and black, etc.).

[2584] If care is not taken, outer fabric layer **13.3-72** may not have sufficient stretch in the length direction L, which in turn may cause outer fabric layer **13.3-72** to buckle (e.g., wrinkle) when face seal **13.3-52** is pressed against the user's face. To avoid buckling in outer fabric layer **13.3-72**, outer fabric layer **13.3-72** may incorporate one or more tuck stitches. The tuck stitches may be used in place of miss stitches (sometimes referred to as float stitches) to add stretch in length direction L. Tuck stitches allow outer fabric layer **13.3-72** to have a bird's eye pattern (to allow for color variation across the outer surface of outer fabric layer **13.3-72**) while providing additional stretch in lengthwise direction L of outer fabric layer **13.3-72**.

[2585] Inner fabric layer **13.3-58** may be a light-blocking fabric that lines the interior surface of outer fabric layer **13.3-72** of light seal **13.3-52**. Inner fabric layer **13.3-58** may be formed from one or more opaque (e.g., black) fabric layers. Inner fabric layer **13.3-58** may include one or more layers of knit fabric, warp knit fabric, weft knit fabric, woven fabric, spacer fabric, braided fabric,

and/or any other suitable type of fabric. It may be desirable to use dark-colored fabric for inner fabric layer **13.3-58** to help keep the viewing area around eye boxes **13.3-13** sufficiently dark while the user is viewing images on displays **13.3-14** of device **13.3-10**. If care is not taken, however, the dark color of inner fabric layer **13.3-58** may be visible through outer fabric layer **13.3-72**. If outer fabric layer **13.3-72** is a light color such as white or gray and inner fabric layer **13.3-58** is a dark color such as black, for example, the black color of inner fabric layer **13.3-58** may be visible through openings or gaps between strands in outer fabric layer **13.3-72**, which may be visually distracting. If inner fabric layer **13.3-58** is instead made from a light-colored fabric such as white or gray, inner fabric layer **13.3-58** may not be as visible through outer fabric layer **13.3-72**, but the opacity of inner fabric layer **13.3-58** may be reduced, which can lead to an unsatisfactory viewing experience if care is not taken.

[2586] To prevent inner fabric layer **13.3-58** from being overly visible through outer fabric layer **13.3-72** without compromising opacity, inner fabric layer **13.3-58** may include one or more dark-colored inner fabric layers on the viewing side (e.g., facing eye boxes **13.3-13** and lenses **13.3-30** of optical modules **13.3-40**) and one or more light-colored outer fabric layers on the non-viewing side (e.g., facing outer fabric layer **13.3-72**). The dark-colored inner fabric layer of inner fabric layer **13.3-58** may be black, dark gray, or other suitable dark color, whereas the light-colored outer fabric layer of inner fabric layer **13.3-58** may be white, gray, light gray, cream, off-white, or other suitable light color. The dark inner layer and light outer layer of inner fabric layer **13.3-58** may be weft knit layers that are coupled by a middle spacer layer, if desired. For example, a spacer layer formed from a multifilament or monofilament strand may be used to join the dark inner layer and light outer layer of inner fabric layer **13.3-58**. The spacer layer may be a drawn textured yarn with a low denier value such as 20D or other suitable denier value to ensure that the spacer layer has sufficient density to block outside light from entering the viewing area. If desired, strands that make up inner fabric layer **13.3-58** may be extruded, solution-dyed strands with high texture and high stretch, thereby increasing the shrinkage and opacity of inner fabric layer **13.3-58**.

[2587] A knitting machine or other equipment may be used in forming fabric for device **13.3-10** such as inner fabric layer **13.3-58**. FIGS. **13.3-5A** is a schematic diagram of an illustrative knitting system. As shown in FIGS. **13.3-5A**, strand source **13.3-66** in knitting system **13.3-64** may be used in supplying strands **13.3-68** to guide and needle structures **13.3-70**. Structures **13.3-70** may include strand guide structures (e.g., a system of movable guide bars with eyelets that guide strands **13.3-68**) and needle systems (e.g., needle guide systems that guide sets of individually adjustable needles so that the needles may interact with the strands dispensed by the guide bars). During operations, a controller may control electrically adjustable positioners in system **13.3-64** to manipulate the positions of guide bars and needles in system **13.3-64** and thereby knit strands **13.3-68** into fabric **13.3-58**. Take down **13.3-74** (e.g., a pair of mating rollers or other equipment forming a takedown system) may be used to gather fabric **13.3-58** that is produced during knitting.

[2588] A knitting machine or other equipment may be used in forming fabric for device **13.4-10** such as outer fabric layer **13.4-72**. FIGS. **13.3-5B** is a schematic diagram of an illustrative knitting system. As shown in FIGS. **13.3-5B**, strand source **13.4-66** in knitting system **13.4-64** may be used in supplying strands **13.4-68** to guide and needle structures **13.4-70**. Structures **13.4-70** may include strand guide structures (e.g., a system of movable guide bars with eyelets that guide strands **13.4-68**) and needle systems (e.g., needle guide systems that guide sets of individually adjustable needles so that the needles may interact with the strands dispensed by the guide bars). During operations, a controller may control electrically adjustable positioners in system **13.4-64** to manipulate the positions of guide bars and needles in system **13.4-64** and thereby knit strands **13.4-68** into fabric **13.4-72**. Take down **13.4-74** (e.g., a pair of mating rollers or other equipment forming a takedown system) may be used to gather fabric **13.4-72** that is produced during knitting.

[2589] FIGS. **13.3-6** is a diagram of a layer of a weft knit fabric that may be included in inner fabric layer **13.3-58**. Knit fabric such as fabric **13.3-58** may be made up of courses **13.3-78** (e.g.,

rows of loops formed by strands **13.3-68**) and wales **13.3-76** (e.g., columns of loops formed by strands **13.3-68**). In a weft knit fabric of the type shown in FIGS. **13.3-6** (sometimes referred to as a flat knit fabric), strands **13.3-68** form loops that extend horizontally across the fabric. An illustrative strand **13.3-68'** among strands **13.3-68** has been highlighted to show the horizontal path taken by each strand **13.3-68** in fabric **13.3-72**. In contrast, a warp knit fabric includes wales **13.3-76** formed from strands **13.3-68** that follow zig-zag paths vertically down the fabric.

[2590] A cross-sectional side view of a portion of light seal **13.3-52** of device **13.3-10** is illustrated in FIGS. **13.3-7**. As shown in FIGS. **13.3-7**, light seal **13.3-52** may include inner fabric layer **13.3-58** and outer fabric layer **13.3-72**. Outer fabric layer **13.3-72** may form an outermost surface of light seal **13.3-52** and may be viewed in direction **13.3-82** by an outside viewer such as viewer **13.3-80**. Inner fabric layer **13.3-58** may line the inner surface of outer fabric layer **13.3-72** and may face towards eye boxes such as eye box **13.3-13**. When device **13.3-10** is placed on a user's head, the user's eyes may be located in eye boxes such as eye box **13.3-13**.

[2591] Inner fabric layer **13.3-58** may have inner and outer fabric layers such as light-colored outer fabric layer **13.3-58L** and dark-colored inner fabric layer **13.3-58D** coupled by a spacer layer such as spacer layer **13.3-58M**. Light-colored outer fabric layer **13.3-58L** may be white, off-white, cream, gray, light gray, or other suitable light color. Light-colored outer fabric layer **13.3-58L** may, for example, match a color of outer fabric layer **13.3-72**. Dark-colored inner fabric layer **13.3-58D** may be black, dark gray, dark blue, or other suitable dark color. Dark-colored inner fabric layer **13.3-58D** and light-colored outer fabric layer **13.3-58L** may be weft knit layers (e.g., of the type shown in FIGS. **13.3-6**), and may be formed from strands **13.3-68** (e.g., drawn textured yarns) having a denier value of 40D (or two-ply 20D strands) or other suitable denier value.

[2592] Spacer layer **13.3-58M** (sometimes referred to as middle layer, etc.) may be formed from one or more multifilament strands **13.3-68** with a low denier value such as 20D, 25D, or other suitable denier value. Spacer layer **13.3-58M** may be formed from black strands, white strands, gray strands, or strands of any other suitable color. By selecting strands **13.3-68** for spacer layer **13.3-58M** that have half the denier value of strands **13.3-68** of light-colored inner fabric layer **13.3-58L** and dark-colored fabric layer **13.3-58D**, fabric **13.3-58** may appear truly white (or other suitable light color) when viewed in direction **13.3-82**) and may appear truly black (or other suitable dark color) when viewed from eye box **13.3-13**), regardless of the color of spacer layer **13.3-58M**.

[2593] Strands in light-colored inner fabric layer **13.3-58L**, dark-colored outer fabric layer **13.3-58D**, and spacer layer **13.3-58M** may be formed from materials such as nylon, polyester, elastic material such as spandex, or other suitable materials. In general, nylon is bulkier than polyester due to less intermingling points and intermingling intensity among individual filaments. If polyester strands are used, the polyester may be modified to reduce intermingling between filaments such that the filaments in each polyester strand spread out more and thus block more light.

[2594] Inner fabric layer **13.3-58** may be formed using a circular knitting machine. The circular knitting machine may have a high gauge such as 80 gauge or other suitable gauge to ensure a higher number of strands per inch in the fabric. The higher number of strands per inch in fabric **13.3-58**, the higher the opacity will be. For example, fabric **13.3-58** may have more than 80 strands per inch, more than 90 strands per inch, more than 100 strands per inch, less than 100 strands per inch, or other suitable density of strands **13.3-68**. Additionally, fabric **13.3-58** may include stretchable strands (e.g., strands formed from spandex or other elastic material) to add shrinkage to fabric **13.3-58** and further increase the opacity of light seal **13.3-52**. Fabric **13.3-58** may be heat set at a low heat setting (or not heat set at all) to help maintain opacity and stretch.

[2595] FIGS. **13.3-8** is a perspective view of inner fabric layer **13.3-58**. As shown in FIGS. **13.3-8**, inner fabric layer **13.3-58** may include dark-colored inner fabric layer **13.3-58D** facing eye box **13.3-13** and light-colored outer fabric layer **13.3-58L** facing outside viewers such as viewer **13.3-80** viewing light seal **13.3-52** in direction **13.3-82**. Spacer layer **13.3-58M** may include one or more

multifilament strands such as multifilament strand **13.3-68M** coupled between dark-colored inner fabric layer **13.3-58D** and light-colored outer fabric layer **13.3-58L**. Spacer layer **13.3-58M** may be a low denier yarn (e.g., 20D, 25D, or other suitable denier value) to increase fabric density and minimize the gap **G** between dark-colored inner fabric layer **13.3-58D** and light-colored outer fabric layer **13.3-58L**.

[2596] Strands **13.3-68D** of dark-colored inner fabric layer **13.3-58D** and strands **13.3-68L** of light-colored outer fabric layer **13.3-58L** may have twice the denier value of spacer strand **13.3-68M** (e.g., strands **13.3-68M** may have a denier value of 20D, whereas strands **13.3-68D** and **13.3-68L** may have a denier value of 40D or may be two-ply strands with each ply having a denier value of 20D) such that fabric **13.3-58** appears truly white (or other light color) when viewed in direction **13.3-82** by viewer **13.3-80** and truly black (or other dark color) when viewed from eye box **13.3-13**. In general, strands **13.3-68L**, **13.3-68D**, and **13.3-68M** may be formed from any suitable material, such as nylon, polyester, and spandex or other elastic material, etc. If desired, strands **13.3-68D** and **13.3-68L** may be two-ply strands that each include an elastic strand having a denier value of 20D and a polyester strand having a denier value of 20D.

[2597] FIGS. **13.3-9** is a cross-sectional side view of light seal **13.3-52** in an example in which outer fabric layer **13.3-72** has been omitted. With this type of arrangement, light-colored outer fabric layer **13.3-58L** may form the outermost surface of light seal **13.3-52** and may be viewed directly by viewer **13.3-80** in direction **13.3-82**. Due to the presence of middle layer **13.3-58M**, dark-colored fabric **13.3-58D** may not be viewable through light-colored outer fabric layer **13.3-58L**. In other words, fabric layer **13.3-58** may appear truly white (or other light color) when viewed in direction **13.3-82** and truly black (or other dark color) when viewed from eye box **13.3-13**.

13.4: Electronic Devices with Stretchable Fabrics

[2598] FIGS. **13.4-6**, **13.4-7**, and **13.4-8** show illustrative types of stitches that may be incorporated into fabric **13.4-72** to adjust the amount of stretch in the fabric.

[2599] In the example of FIGS. **13.4-6**, fabric **13.4-72** is a weft knit fabric made up of courses **13.4-78** (e.g., rows of loops formed by strands **13.4-68**) and wales **13.4-76** (e.g., columns of loops formed by strands **13.4-68**). In a weft knit fabric of the type shown in FIGS. **13.4-6** (sometimes referred to as a flat knit fabric), strands **13.4-68** form loops that extend horizontally across the fabric. An illustrative strand **13.4-68'** among strands **13.4-68** has been highlighted to show the horizontal path taken by each strand **13.4-68** in fabric **13.4-72**. In contrast, a warp knit fabric includes wales **13.4-76** formed from strands **13.4-68** that follow zig-zag paths vertically down the fabric.

[2600] The example of FIGS. **13.4-6** shows courses (rows) **13.4-78** that are made up entirely of knit stitches. This is sometimes referred to as a plain jersey or single jersey fabric. If desired, color variations may be incorporated into fabric **13.4-72** by incorporating miss stitches and/or float stitches. A miss stitch is illustrated in FIGS. **13.4-7**.

[2601] As shown in FIGS. **13.4-7**, fabric **13.4-72** may include courses **13.4-78** such as first course **13.4-78-1**, second course **13.4-78-2**, third course **13.4-78-3**, and fourth course **13.4-78-4**. First course **13.4-78-1**, third course **13.4-78-3**, and fourth course **13.4-78-4** each include a series of knit stitches **13.4-80**. The legs of each knit stitch loop are connected to the head of the previous loop in the previous course. Second course **13.4-78-2** incorporates miss stitch **13.4-84**. To create a miss stitch such as miss stitch **13.4-84** during knitting, a given strand is not collected by a needle, which causes the strand to float behind the needle while remaining connected to the loops on either side of miss stitch **13.4-84**. If the strands of courses **13.4-78** include different colors, miss stitch **13.4-84** will create a dual color pattern on each side of the fabric.

[2602] If desired, outer fabric layer **13.4-72** may have a bird's eye pattern that incorporates alternating knit stitches **13.4-80** and miss stitches **13.4-84**. For example, outer fabric layer **13.4-72** may have a four row repeat pattern. In the first and second rows, the odd-numbered wales **13.4-76** may be knit stitches **13.4-80**, while the even-numbered wales **13.4-76** may be tuck stitches **13.4-84**.

In the third and fourth rows, the odd-numbered wales **13.4-76** may be tuck stitches **13.4-84**, while the even-numbered wales **13.4-76** may be knit stitches **13.4-80**. This pattern may repeat to create a two-color bird's eye pattern on the outer surface of outer fabric layer **13.4-72**.

[2603] As shown in FIGS. **13.4-7**, miss stitches **13.4-84** extend in the width direction W of fabric **13.4-72** and float horizontally across the fabric. If additional stretch is desired in lengthwise direction L of fabric **13.4-72**, some or all of the miss stitches **13.4-84** may be replaced by tuck stitches. This type of arrangement is illustrated in FIGS. **13.4-8**.

[2604] As shown in FIGS. **13.4-8**, fabric **13.4-72** may include courses **13.4-78** such as first course **13.4-78-1**, second course **13.4-78-2**, and third course **13.4-78-3**. First course **13.4-78-1**, second course **13.4-78-2**, and third course **13.4-78-3** may each include alternating knit stitches **13.4-80** and tuck stitches **13.4-82**. Tuck stitches **13.4-82** may be created by placing two strands **13.4-68** on a single needle, thereby tucking the extra strand **13.4-68** behind the first strand **13.4-68**. The legs of tuck stitch **13.4-82** are not connected to the head of the previous loop. For example, as shown in FIGS. **13.4-8**, the legs of each tuck stitch **13.4-82** in course **13.4-78-3** are not connected to head **13.4-84** of the previous loop in course **13.4-78-2**.

[2605] If desired, outer fabric layer **13.4-72** may have a bird's eye pattern that incorporates alternating knit stitches **13.4-80** and tuck stitches **13.4-82**. For example, outer fabric layer **13.4-72** may have a four-row-repeat-pattern of the type shown in FIGS. **13.4-9**.

[2606] As shown in FIGS. **13.4-9**, the odd numbered wales of the first course may be tuck stitches **13.4-82** (represented as a "V" in FIGS. **13.4-9**), whereas the even numbered wales of the first course may be knit stitches **13.4-80** (represented as an "O" in FIGS. **13.4-9**). The second course may be all knit stitches **13.4-80**. The odd numbered wales of the third course may be knit stitches **13.4-80**, whereas the even numbered wales of the second course may be tuck stitches **13.4-82**. The fourth course may be all knit stitches **13.4-80**. This pattern may repeat to create a two-color modified bird's eye pattern on the outer surface of outer fabric layer **13.4-72**.

[2607] The incorporation of tuck stitches **13.4-82** provides additional stretch in the lengthwise direction L because tuck stitches **13.4-82** have strand segments that extend in the L direction to form the tuck stitch. The longer stitch length in length direction L from incorporating one or more tuck stitches **13.4-82** allows the seamless tube of fabric **13.4-72** to stretch and shrink in direction L without buckling or wrinkling when device **13.4-10** is placed on the user's head and light seal **13.4-52** is pressed against the user's face. If desired, strands **13.4-68** of fabric **13.4-72** may include strands of first and second colors to create a two-color pattern on the outside of light seal **13.4-52**. This is merely illustrative, however. If desired, more than two colors may be used or strands **13.4-68** may have other distinct properties (e.g., different deniers, different amounts of fuzziness, different textures, different diameters, different materials, etc.).

[2608] If desired, fabric **13.4-72** may be placed in boiling water after fabric **13.4-72** is formed. The post-processing step of boiling fabric **13.4-72** may cause additional shrinkage in the length direction L of fabric **13.4-72**, which in turn reduces the tendency of light seal **13.4-52** to buckle when pressed against the user's face. The boiling step may, for example, result in more shrinkage in length direction L of fabric **13.4-72** than in width direction W of fabric **13.4-72**. Additionally, the increased stretch in the length direction L of modified bird's eye fabric **13.4-72** results in greater frame mobility and deflection (e.g., light seal **13.4-52** may be more mobile and deflect more easily when covered by fabric **13.4-72** having increased lengthwise stretch), which can significantly increase user comfort.

[2609] The pattern of FIGS. **13.4-8** and **13.4-9** is merely illustrative. If desired, fabric **13.4-72** may include both miss stitches **13.4-84** and tuck stitches **13.4-82**, depending on the amount of stretch desired in length direction L. For example, if more or maximum stretch in lengthwise direction L is desired, fabric **13.4-72** may be free of miss stitches **13.4-84** and may follow the four-row-repeat-pattern of FIGS. **13.4-9**.

13.5: Contactless Sensors for a Head-Mountable Device

[2610] The following disclosure relates to a facial interface in a head-mountable device. More particularly, the present embodiments relate to a facial interface that contains sensorally transparent materials for head-mountable devices used for AR/VR experiences. These facial interfaces can enable sensors to interact with a user through sensorally transparent materials. As used herein, the term “sensorally transparent materials” include materials that allow for the transfer of sensor signals therethrough.

[2611] In one example, the head-mountable device of the present disclosure includes a display and a light seal portion (hereafter “light seal”). Light seals enable a user to experience a light shielded environment, where outside ambient light and possibly other environmental items, are blocked from the user field of view. The shielded environment allows for better user interaction and a more immersive experience. The light seal, as a facial interface, can be customized to a user's facial profile such that the light seal physically interacts with the user's face to fit snugly on or around the forehead, eyes, nose, and other features or bones, such as the maxilla regions, that can vary user to user. Additionally, a light seal can include components connecting the display to the facial interface, such as a webbing, housing or a frame positioned between the display and facial interface.

[2612] Conventional light seals of conventional head-mountable devices are passive and do not include a facial interface with sensorally transparent materials. Indeed, passive light seals create a light shielded environment, but do not include active component integration to enable contactless readings from sensors embedded in a facial interface. Therefore, conventional light seals do not provide contactless readings of a user via sensorally transparent materials and sensors.

[2613] By contrast, a light seal of the present disclosure includes a facial interface with sensorally transparent materials for active component integration. A light seal with active components has advantages over a traditional passive light seal. A light seal with sensorally transparent materials can include active components that can monitor user responses without direct contact, leading to improved user comfort while wearing the head-mountable device. Sensors configured in this contactless manner can also avoid biological ingress from user skin (e.g., lotion, makeup, sunscreen, etc.). A head-mountable device that monitors such user responses can also create a highly customized user experience (unlike the sensors of conventional head-mountable devices that are “blind” to the user experience).

[2614] Sensors can be important for creating a customized user experience. An active light seal can contain sensors to measure a user's response or engagement via indicators, such as core body temperature, sweat, heart rate, electrical signals from the heart (e.g., ECG, EKG, EXG, etc.), brain activity (e.g., EEG signals, frontal lobe activity), etc. Additionally, the sensor data can be used as feedback data, for example, to monitor user fatigue or obtain activity-specific metrics.

[2615] Sensors of the present disclosure can be implemented on or within a facial interface in myriad different ways. For example, a sensor can be oriented towards a user and positioned on a facial interface surface opposite of the surface that contacts a user. In another example, a sensor can be oriented towards a user and embedded inside the facial interface. In these or other examples, the sensor can include a field of view that projects towards a user and through at least a portion of the facial interface. Thus, such portions of the facial interface can be sensorally transparent to allow the sensor to obtain a sensor reading through at least a portion of the facial interface.

[2616] Sensors can also be implemented in different ways for different facial interfaces. That is, the head-mountable device of the present disclosure can implement a facial interface with a base layer and an interchangeable layer. The interchangeable layer can be exchanged or swapped out for a different interchangeable layer. In some examples, the different interchangeable layer can correspond to a different user activity, such as a yoga activity versus a movie-watching activity. In certain implementations, the yoga interchangeable layer can include a different sensor arrangement than the movie-watching interchangeable layer (e.g., for obtaining different, activity-specific metrics).

[2617] These and other embodiments are discussed below with reference to FIGS. **13.5-1-13.5-7B**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[2618] FIGS. **13.5-1** illustrates a top view profile of a head-mountable device **13.5-100** worn on a user head. The head-mountable device **13.5-100** can include a display **13.5-102** (e.g., one or more optical lenses or display screens in front of the eyes of the user). The display **13.5-102** can include a display for presenting an augmented reality visualization, a virtual reality visualization, or other suitable visualization.

[2619] The head-mountable device **13.5-100** also includes a facial interface **13.5-103** and a sensor **13.5-108** positioned (e.g., attached to or embedded within) on the facial interface **13.5-103**. As used herein, the terms “facial interface” or “engagement interface” refer to a portion of the head mountable device **13.5-100** that engages a user face via direct contact. In particular, a facial interface includes portions of the head-mountable device **13.5-100** that conform to (e.g., compress against) regions of a user face. To illustrate, a facial interface can include a pliant (or semi-pliant) facetrack that spans the forehead, wraps around the eyes, contacts the zygoma and maxilla regions of the face, and bridges the nose. In addition, a facial interface can include various components forming a structure, webbing, cover, fabric, or frame of a head-mountable device disposed between the display **13.5-102** and the user skin. In particular implementations, a facial interface can include a seal (e.g., a light seal, environment seal, dust seal, air seal, etc.). It will be appreciated that the term “seal” can include partial seals or inhibitors, in addition to complete seals (e.g., a partial light seal where some ambient light is blocked and a complete light seal where all ambient light is blocked when the head-mountable device is donned).

[2620] In addition, the term “sensor” refers to one or more different sensing devices, such as a camera or imaging device, temperature device, oxygen device, movement device, brain activity device, sweat gland activity device, breathing activity device, muscle contraction device, etc. Some particular examples of sensors include an electrooculography sensor, electrocardiography sensor, EKG sensor, heart rate variability sensor, blood volume pulse sensor, SpO2 sensor, compact pressure sensor, electromyography sensor, core-body temperature sensor, galvanic skin sensor, accelerometer, gyroscope, magnetometer, inclinometer, barometer, infrared sensor, global positioning system sensor, etc.

[2621] In one example, the head mountable device **13.5-100** includes a sensor controller **13.5-104**. The sensor controller **13.5-104** can include a processor (e.g., a system on chip, integrated circuit, driver, microcontroller, application processor, crossover processor, etc.). Further the sensor controller **13.5-104** can include one or more memory devices (e.g., individual nonvolatile memory, processor-embedded nonvolatile memory, random access memory, memory integrated circuits, DRAM chips, stacked memory modules, storage devices, memory partitions, etc.). In certain implementations, the sensor controller **13.5-104** is positioned within one or both arms **13.5-105**, **13.5-106** of the head-mountable device **13.5-100** (e.g., for integration with an HMD processor/memory component). In alternative implementations, the sensor controller **13.5-104** is physically integrated within the sensors **13.5-108** themselves.

[2622] The sensor controller **13.5-104** can perform myriad different functions. For example, the memory device can store computer-executable instructions that, when executed by the processor, cause the sensor controller **13.5-104** to receive sensor data from the sensors **13.5-108** and transmit

a signal based on the sensor data. For instance, the sensor controller **13.5-104** can transmit a sensor signal to the display **13.5-102**. In response to the sensor signal, the display **13.5-102** can power off, present a digital notification (e.g., user-generated notification, push notification, context-generated notification, system-generated notification, smart notification, etc.), or render at least one of an avatar or an avatar emotional response. As used herein, the term “avatar” is a visual representation of a person for use in digital context, such as with the head mountable device **13.5-100**. An avatar can include animated characters, animals, objects, emojis, etc. that can depict human emotion (e.g., as detected via the sensors **13.5-108** of the head-mountable device **13.5-100**). The depiction of human emotion through an avatar constitutes an avatar emotional response.

[2623] Additionally shown in FIGS. **13.5-1**, the head-mountable device **13.5-100** includes one or more arms **13.5-105**, **13.5-106**. The arms **13.5-105**, **13.5-106** are connected to the display **13.5-102** and extend distally toward the rear of the head. The arms **13.5-105**, **13.5-106** are configured to secure the display **13.5-102** in a position relative to the head (e.g., such that the display **13.5-102** is maintained in front of a user's eyes). For example, the arms **13.5-105**, **13.5-106** extend over the user's ears **13.5-107**. In certain examples, the arms **13.5-105**, **13.5-106** rest on the user's ears **13.5-107** to secure the head-mountable device **13.5-100** via friction between the arms **13.5-105**, **13.5-106** and the head. Additionally, or alternatively, the arms **13.5-105**, **13.5-106** can rest against the head. For example, the arms **13.5-105**, **13.5-106** can apply opposing pressures to the sides of the head to secure the head-mountable device **13.5-100** to the head. Optionally, the arms **13.5-105**, **13.5-106** can be connected to each other via a strap (shown in dashed lines) that can compress the head-mountable device **13.5-100** against the head).

[2624] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.5-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.5-1**.

[2625] FIGS. **13.5-2A-13.5-2B** respectively illustrate a side view and front view profiles of an example of the head mountable device **13.5-100**. As discussed above, the head mountable device **13.5-100** includes the display **13.5-102**, the facial interface **13.5-103**, the sensor controller **13.5-104**, and the sensors **13.5-108**. Indeed, at least one sensor **13.5-108** is positioned on or within the facial interface **13.5-103**. Additionally, the facial interface **13.5-103** can wrap around eyes **13.5-201** and bridge a nose **13.5-202** of a user. The head-mountable device **13.5-100** can also include connections **13.5-206** that movably constrain the display **13.5-102** and the facial interface **13.5-103** (e.g., at forehead and cheek regions of a user face). Examples of the connections **13.5-206** include a pivot connection, spring connection, etc.

[2626] The sensors **13.5-108** can be positioned in myriad different configurations. In one example, at least one of the sensors **13.5-108** is positioned in a pliant region **13.5-212** of the facial interface **13.5-103** between the connections **13.5-206**. A “pliant region” refers to the portion(s) of the facial interface **13.5-103** disposed between the connections **13.5-206**, where the facial interface **13.5-103** is more flexible and conformable. In certain implementations, one or more of the sensors **13.5-108** is positioned in a middle portion of the pliant region **13.5-212** (approximately equidistant from the connections **13.5-206**). By positioning one or more of the sensors **13.5-108** within the pliant region **13.5-212**, pressure points felt by the user can be mitigated. Additionally, or alternatively, the sensors **13.5-108** can be positioned in certain configurations, depending on the desired location (on the user) to be sensed (e.g., a forehead region, an eye region, a nasal region, etc.).

[2627] Further, the term “forehead region” refers to the anatomical area of a human head between the eyes and scalp. Additionally, the term “nasal region” refers to the anatomical area of a human nose.

[2628] Also shown in FIGS. **13.5-2A-13.5-2B**, the head-mountable device **13.5-100** can include a power source **13.5-203**. In some examples, the power source **13.5-203** can include one or more electrochemical cells with connections for powering electrical devices. For example, in some examples, the power source **13.5-203** includes a lithium-ion battery, alkaline battery, carbon zinc battery, lead-acid battery, nickel-cadmium battery, nickel-metal hydride battery, etc. It will therefore be appreciated that the power source **13.5-203** can be dispensable or rechargeable, as may be desired. In certain implementations, the power source **13.5-203** is connected to the sensor controller **13.5-104** via one or more electrical connections. In some examples, though not required, the power source **13.5-203** is mounted to the sensor controller **13.5-104**.

[2629] The head-mountable device **13.5-100** can also include an interface **13.5-210**, which can be electromechanical or wireless. The interface **13.5-210** can communicatively couple the sensors **13.5-108** to at least one of the power source **13.5-203**, the sensor controller **13.5-104**, or an HMD processor/memory component (not shown).

[2630] In some examples, the sensor **13.5-108** can connect to the sensor controller **13.5-104** (or the HMD processor/memory component, not shown) via certain wireless communication protocols, such as via a wireless local area network protocol, wireless area network protocol, wireless personal area network protocol, wide area protocol, etc. Some particular examples of wireless communication via such protocols include a Wi-Fi based communication, mesh network communication, Bluetooth® communication, near-field communication, low-energy communication, Zigbee communication, Z-wave communication, and 6LoWPAN communication. In a particular implementation, the sensor **13.5-108** is communicatively coupled to the sensor controller **13.5-104** (or the HMD processor/memory component, not shown) via a wireless 60 GHz frequency.

[2631] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.5-2A-13.5-2B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.5-2A-13.5-2B**.

[2632] As discussed above, sensors can be disposed on or within the facial interface of the present disclosure. In accordance with one or more such examples, FIGS. **13.5-3** illustrates one of the sensors **13.5-108** positioned on a surface of the facial interface **13.5-103**. Specifically, as shown in FIGS. **13.5-3**, the sensor **13.5-108** is positioned on a first surface **13.5-318** oriented towards a user side **13.5-324** (and away from a display side **13.5-326**). Additionally, the sensor **13.5-108** includes a field of view **13.5-320** that crosses through an interface material **13.5-301** from the first surface **13.5-318** to a second surface **13.5-322** opposing the first surface **13.5-318**. Thus, the portion of the interface material **13.5-301** corresponding to the field of view **13.5-320** between the first surface **13.5-318** and the second surface **13.5-322** can be sensorally transparent-constituting a sensorally transparent window through the interface material **13.5-301**.

[2633] The terms “sensorally transparent” refers to a type of material penetrable by a sensor measurement signal without substantial loss to the quality or accuracy of the sensor measurement signal (where “substantial” means greater than about 5%, about 10%, about 25%, about 50%, or greater than 50% discrepancy from a ground truth signal). For example, a sensorally transparent material can allow a heart rate sensor to accurately detect electrical, magnetic, or audio heart data indicative of a heart palpation, heartbeat, heart rhythm, etc. despite the sensorally transparent material being disposed between the heart rate sensor and the user. The sensor measurement signal is therefore a wireless signal to and/or from a sensor, where the wireless signal comprises wavelike properties (e.g., frequency, amplitude, etc.) that allow the wireless signal to propagate through the sensorally transparent material.

[2634] Relatedly, the term “sensorally transparent window” refers to the portion of the interface material **13.5-301** that is sensorally transparent. In some examples, the sensorally transparent window includes an entirety of the interface material **13.5-301**. In other examples, the sensorally transparent window includes at least a portion of the interface material **13.5-301** for the field of view **13.5-320**. In these or other examples, the sensorally transparent window can be sized and shaped according to the field of view **13.5-320**.

[2635] The interface material **13.5-301** composes or defines (at least in part) the facial interface **13.5-103**. The interface material **13.5-301** can include the first surface **13.5-318** and the second surface **13.5-322** opposing the first surface **13.5-318**, as illustrated in at least FIGS. **13.5-3**. In at least some examples, the second surface **13.5-322** is configured to contact user skin.

[2636] Additionally, the interface material **13.5-301** can include at least one of foam, gel, or fabric. The interface material **13.5-301** can likewise include a combination of foam (e.g., polyurethane foam cushion, cotton foam), gel (e.g., silicone, polyurethane, etc.), or fabric (e.g., cotton, leather, leatherette, etc.). For example, the interface material **13.5-301** can include multiple different layers (e.g., an outer leatherette layer forming the second surface **13.5-322** and a foam layer underneath forming the first surface **13.5-318**). The combination described is merely exemplary and other embodiments, materials, configurations and/or combinations are contemplated herein.

[2637] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.5-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.5-3**.

[2638] Other sensor arrangements are also within the scope of the present disclosure. In accordance with one or more examples, FIGS. **13.5-4** illustrates the facial interface **13.5-103** which can include sensorally transparent windows through which sensor measurement signals to or from the sensors **13.5-108a**, **13.5-108b** can pass. The sensors **13.5-108a**, **13.5-108b** can detect physiological or biological changes of the user's body through corresponding fields of view **13.5-320a**, **13.5-320b**. For example, a head-mountable device **13.5-100** can detect changes to a user body temperature or a heat profile via an infrared sensor. Other biological sensors can be added or substituted, as may be desired.

[2639] In particular, FIGS. **13.5-4** shows the sensor **13.5-108a** in a same or similar position as described above in relation to FIGS. **13.5-3**. The sensor **13.5-108a** is positioned as a contactless sensor on the first surface **13.5-318** (i.e., on the display side **13.5-326**). In addition, FIGS. **13.5-4** shows the facial interface **13.5-103** includes the sensor **13.5-108b** disposed between the first surface **13.5-318** and the second surface **13.5-322**. That is, the sensor **13.5-320b** is embedded within the facial interface **13.5-103** such that the sensor **13.5-320b** is invisible and inaccessible from the second surface **13.5-322** (i.e., the skin-facing surface). By being offset from the second surface **13.5-322**, the sensor **13.5-108b** is also a contactless sensor.

[2640] In one or more examples, the sensors **13.5-108a**, **13.5-108b** are the same type of sensors (albeit positioned differently). In other examples, the sensors **13.5-108a**, **13.5-108b** are different types of sensors. Similarly, the sensors **13.5-108a**, **13.5-108b** can have the same field of view or a different field of view, as may be desired. Alternative embodiments can also include the same or different sensors with alternative fields of view. For instance, the fields of view **13.5-320a**, **13.5-320b** can be oriented or angled towards a particular location along the second surface **13.5-322** (e.g., for measuring a particular location on the user). Additionally, or alternatively, the fields of view **13.5-320a**, **13.5-320b** can intersect, overlap, and/or include mutually exclusive measurement regions.

[2641] One of ordinary skill in the art will appreciate that the sensor depth for the sensor **13.5-320b**

can vary any distance from the first surface **13.5-318** to the second surface **13.5-322**. An example of sensor depth variation is shown where the sensor **13.5-108a** is on the first surface **13.5-318** of the interface material **13.5-301**, which increases the sensor field of view **13.5-320a**. Similarly, sensor **13.5-108b** is a distance from the first surface **13.5-318**, being disposed within the interface material **13.5-301**, and therefore closer to the second surface **13.5-322** than the sensor **13.5-108a**. In some instances, this closer positioning of the sensor **13.5-108b** to the second surface **13.5-322** can correspondingly reduce the field of view **13.5-320b**. This is only one example variation of sensor depth, as a multitude of sensors can be disposed on the first surface **13.5-318** of the interface material **13.5-301**, or between the first surface **13.5-318** and the second surface **13.5-322**, within the interface material **13.5-301**. In one example, the second surface **13.5-322** abuts a forehead region or a nasal region of a user head when the head-mountable device **13.5-100** is donned.

[2642] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.5-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.5-4**.

[2643] FIGS. **13.5-5** illustrates yet another example of the facial interface **13.5-103**, which can include multiple sensors embedded within the interface material **13.5-301**. FIGS. **13.5-5** also illustrates none of the sensors **13.5-108a**, **13.5-108b** are positioned on the first surface **13.5-318**. In this example, the sensors **13.5-108a**, **13.5-108b** can be wirelessly coupled to a power source and/or an HMD memory/processor component. Indeed, the sensors **13.5-108a**, **13.5-108b** can be powered by an inductive coil running through the head-mountable device **13.5-100** (e.g., adjacent to the first surface **13.5-318**). Similarly, the sensors **13.5-108a**, **13.5-108b** can be communicatively coupled to an HMD memory/processor component via a wireless communication protocol (e.g., for sending sensor data/sensor signals or receiving sensor feedback).

[2644] As further shown in FIGS. **13.5-5**, sensor **13.5-108a** and sensor **13.5-108b** are disposed within the interface material **13.5-301**. Sensors **13.5-108a**, **13.5-108b** can be positionally modified to different locations (e.g., laterally, or depth-wise) than illustrated in FIGS. **13.5-5**. Similarly, the sensor **13.5-108a** can vary in type and configuration, transmission power, shape, and size from the sensor **13.5-108b** (as indicated above). The field of view **13.5-320a** of the sensor **13.5-108a** can vary from the field of view **13.5-320b** of **13.5-108b**. In another example, the fields of view **13.5-320a**, **13.5-320b** can be the same configuration (also indicated above).

[2645] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.5-5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.5-5**.

[2646] FIGS. **13.5-6A** and **13.5-6B** illustrate another example of the facial interface **13.5-103** comprising multiple layers of interface material. In a particular example shown in FIGS. **13.5-6A-13.5-6B**, the interface material includes a base layer **13.5-624** and an interchangeable layer **13.5-626**. The base layer **13.5-624** is affixed to the head-mountable device **13.5-100**—thereby forming a permanent portion of the facial interface **13.5-103**. By contrast, the interchangeable layer **13.5-626** is removably attached to the base layer **13.5-624** (e.g., via fasteners **13.5-628a**, **13.5-628b**) such that the interchangeable layer **13.5-626** can be swapped out for a different interchangeable layer (e.g., that supports or is designated for a different user activity).

[2647] The fasteners **13.5-628a**, **13.5-628b** can include myriad different fasteners. For example, the

fasteners **13.5-628a**, **13.5-628b** can include brooches, buttons, buckles, clasps, eyelets, fabric ties, frog closures, grommets, hook and eyes, laces, loop fasteners, pins, poppers, press studs, snap fasteners, toggles, hook and loop Velcro® tape, zippers, etc. Additionally, rather than fasteners, temporary adhesives (e.g., tape, glue, tack, etc.) can be implement. It will be appreciated that more than two fasteners can be utilized. Additionally, only a single fastener may be implanted in certain instances.

[2648] In one example, FIGS. **13.5-6A** shows a configuration where the interchangeable layer **13.5-626** of the facial interface **13.5-103** can be removably attached to the base layer **13.5-624** via the fasteners **13.5-628a**, **13.5-628b**. The interchangeable layer **13.5-626** can include sensor **13.5-108a** with a field of view **13.5-320a**. The sensor **13.5-108a** can be a wireless biometric sensor, which can include a temperature sensor, a respiration sensor, a heart activity sensor, or a brain activity sensor. Additionally or alternatively, the sensor **13.5-108a** can include a sensor indicative of certain biological responses (e.g., a stress response). The sensor **13.5-108a** can be disposed such that the sensor field of view **13.5-320a**, when oriented toward a region of the face, can detect characteristics unique to the user. In a particular example, the sensor **13.5-108a** can include an infrared sensor capable of detecting user characteristics, such as body temperature or changes to body temperature.

[2649] In another example, FIGS. **13.5-6B** shows a configuration where an additional interchangeable layer **13.5-630** of the facial interface **13.5-103** can be removably attached to the base layer **13.5-624**. The additional interchangeable layer **13.5-630** can be removably attached to the base layer **13.5-624** via the fasteners **13.5-628a**, **13.5-628b**, as similarly described above. Different from the interchangeable layer **13.5-626**, the additional interchangeable layer **13.5-630** can include an additional sensor **13.5-108b** with an additional field of view **13.5-320b**. Here, the sensor **13.5-108b** and corresponding field of view differs from the sensor **13.5-108a**. In this manner, facial interface **13.5-103** can be compatible with myriad different interchangeable layers as may be desired for different user activities (or different types of users, such as kids or adults).

[2650] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.5-6A-13.5-6B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.5-6A-13.5-6B**.

[2651] FIGS. **13.5-7A** illustrates a perspective view of an electronic device **13.5-700** including a wearable display **13.5-702**, an engagement interface **13.5-726**, and a contactless sensor **13.5-714** coupled to the engagement interface **13.5-726**. In one or more examples, the electronic device **13.5-700** is the same as or similar to the head-mountable device **13.5-100** described above. For example, the engagement interface **13.5-726** can be the same as or similar to the facial interface **13.5-103** described above. Similarly, the wearable display **13.5-702** can be the same as or similar to the display **13.5-102** described above.

[2652] The engagement interface **13.5-726** is adjustable for different size, shapes, and contours of facial features. For example, the engagement interface **13.5-726** can flexibly conform to a user face via connections **13.5-628a-13.5-628d** (e.g., that are the same as or similar to the connections **13.5-206** described above). To illustrate, the connections **13.5-628a-13.5-628d** each include pivot connections. However, the connections **13.5-628a-13.5-628d** can be adjusted to include different types of connections (e.g., foam hard stops, leaf springs, compliant mechanisms, etc.).

[2653] In another example, the engagement interface **13.5-726** includes an interfacing material **13.5-727**, wherein the contactless sensor **13.5-714a** can be invisible and inaccessible through a skin-facing surface of the interfacing material **13.5-727**. The contactless sensor **13.5-714a** can be oriented toward the first facial region **13.5-725a**, such a user's forehead, when the device is worn.

[2654] In another example, the electronic device includes additional contactless sensors **13.5-714b**,

13.5-714c coupled to the engagement interface **13.5-726**. The additional contactless sensors **13.5-714b**, **13.5-714c** can be mounted on (or within) an engagement interface **13.5-732** (e.g., a nose piece). The additional contactless sensors **13.5-714b**, **13.5-714c** can be oriented toward a second facial region **13.5-730**, such as a user's nose, different from the first facial region **13.5-725a** when the device is worn. In this manner, the contactless sensors **13.5-714a-13.5-714c** can be oriented away from the wearable display **13.5-702** (and instead oriented towards a user head or skin, not shown).

[2655] FIGS. **13.5-7B** shows a perspective exploded view of the electronic device **13.5-700** with sensors **13.5-714a**, **13.5-714b**, and **13.5-714c**. In one example, the sensors can be removably attached to the engagement interfaces **13.5-726**, **13.5-732** (e.g., for swapping out with different sensors). In other examples, the sensors are permanently affixed to the engagement interfaces **13.5-726**, **13.5-732**. Additionally, the sensors **13.5-714a-13.5-714c** can be disposed on the engagement interfaces **13.5-726**, **13.5-732** such that the sensors are positioned over corresponding sensorally transparent windows **13.5-734a-13.5-734c**. The sensorally transparent windows **13.5-734a**, **13.5-734b**, and **13.5-734c** allow sensor measurement signals to pass to and from the sensors **13.5-714a-13.5-714c**.

[2656] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.5-7A-13.5-7B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.5-7A-13.5-7B**.

13.6: Integrated Health Sensors

[2657] The following disclosure relates to a facial interface of a head-mountable device. More particularly, the present embodiments relate to a facial interface of a head-mountable device including health sensors used for acquiring biometric information of the user, and performing actions in response to the detected biometric information. These facial interfaces can enable health sensors to interact with a user to collect biometric information. In some examples, a processor causes a component of the HMD to perform an action in response to the biometric information collected by the sensor. As used herein “an action” or “perform an action” can refer to any electrical or mechanical act that causes a change in one or more components of the HMD. For example, the action can include displaying a notification for the user, or activating or deactivating one or more components. In some examples, a sensor can collect biometric information of a user who is exercising. The biometric information can be related to temperature, respiration, stress response, heart activity, or brain activity or any other pertinent data. Further, in response to the collected data, the system can perform an action (e.g., notify the user of the collected data).

[2658] Head-mountable devices equipped with sensors are utilized for various purposes that detect limited user feedback, such as movement or positioning feedback, providing limited information in response to the user. For example, a user may have an increase in heart rate or brain activity while performing an action or movement. Conventional head-mountable devices are not suitably equipped to capture all the necessary biometric information generated during use.

[2659] In contrast, the head-mountable devices of the present disclosure include a facial interface that can be integrated or incorporated with sensors configured to collect user data, such as biometric information, including heart rate, respiration, brain activity, etc. By capturing the user data, the facial interface sensors can provide improved and increased feedback to the user. A head-mountable device with sensors monitoring user biometric or health feedback can create highly customized user experiences, unlike the sensors of conventional head-mountable devices that are unable to consider, react to, or further enhance the user experience.

[2660] Sensors positioned on the facial interface are important to create a customized user

experience. The head-mountable device of the present disclosure can contain sensors to measure a user's response or engagement via indicators, such as heart rate, electrical signals from the heart (e.g., ECG, EKG, EXG, etc.), brain activity (e.g., EEG signals, frontal lobe activity), core body temperature, etc. Additionally, the sensor data can be used as feedback data, for example, to monitor user fatigue, facial expressions, or obtain activity-specific metrics.

[2661] The head-mountable device can have different sensors implemented in different ways. For example, the head-mountable device of the present disclosure can implement sensors removably attached to the facial interface. In some examples, the removably attached sensors correspond to a different user activity, such as exercise, learning activities, health, clinical settings, etc. In certain implementations, the exercise sensors can include a different sensor, such as a heart rate monitor sensor, whereas the learning activity sensor may include a brain activity sensor. The sensors can be implemented in different applications and arrangement/combinations for obtaining different, activity biometric readings.

[2662] These and other embodiments are discussed below with reference to FIGS. **13.6-1-13.6-6**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[2663] FIGS. **13.6-1** illustrates a block diagram of a head-mountable device **13.6-100** including a frame **13.6-116**, a display **13.6-108**, a facial interface **13.6-112**, and a support or retention band **13.6-123**. The display **13.6-108** can include one or more optical lenses or display screens in front of the eyes of a user. The display **13.6-108** can include a display or display unit for presenting an augmented reality visualization, a virtual reality visualization, or other suitable visualization to a user. Additionally, the display **13.6-108** can be positioned in or on the frame **13.6-116**. Similarly, a facial interface **13.6-112** can be connected to the frame **13.6-116**. In some examples, the frame **13.6-116** can be a housing for the display **13.6-108**. Alternatively, a separate housing can form an exterior portion of the head-mountable device **13.6-100**.

[2664] The facial interface **13.6-112** can include one or more sensors **13.6-124**, support attachments **13.6-132**, display attachments **13.6-128**, and feedback or output modules **13.6-136**. The sensors **13.6-124** can be removably attached to the facial interface **13.6-112**. As used herein, the terms “facial interface”, “engagement interface”, or “light seal” refer to a portion of the head mountable device **13.6-100** that engages (i.e., contacts or conforms to) a user's face. In particular, the facial interface **13.6-112** can include portions of a head-mountable device that conform or press against regions of a user's face. The facial interface **13.6-112** can be positioned between the display **13.6-108** and the user's face. In some examples, the facial interface **13.6-112** can include a pliant (or semi-pliant) facetrack or lumen that spans the forehead, wraps around the eyes, contacts other regions of the face (etc., zygoma and maxilla regions), and bridges the nose.

[2665] In addition, the facial interface **13.6-112** can include various components forming a frame, structure, or webbing of a head mountable device disposed between the display **13.6-108** and the user's skin. In particular implementations, the facial interface **13.6-112** can include a seal (e.g., a light seal, environment seal, dust seal, air seal, etc.). It will be appreciated that the term “seal” can include partial seals or inhibitors, in addition to complete seals (e.g., a partial light seal where some ambient light is blocked and a complete light seal where all ambient light is blocked when the head-mountable device **13.6-100** is donned). The facial interface **13.6-112** can be removably attached to the frame **13.6-116** and in electrical communication with the display **13.6-108**.

[2666] The sensor **13.6-124** of the facial interface **13.6-112** can collect biometric information, such as the user's vital sign (including body temperature, pulse data, respiration data, and blood pressure). The sensor **13.6-124** can generate a signal based on the collected user information and transmit the signal to a processor that can cause an output **13.6-136** to perform an action in response to the signal (i.e., in response to the biometric information collected by the sensor **13.6-124**). For example, a user may perform a rigorous activity, such as lifting weights or working out while wearing the HMD **13.6-100**. During such activity, the user's heart rate or other vital signs may be elevated or changed and is detectable by the sensor **13.6-124**. The sensor **13.6-124** can generate one or more signals based on the received input. The sensor **13.6-124** can transmit the signal to one or more components of the HMD **13.6-100** (e.g., to the display **13.6-108**, to the output **13.6-136**). The display **13.6-108**, being in electrical communication with the facial interface **13.6-112**, can receive electrical communication and provide feedback to the user related to their biometric readings (e.g., visual feedback, audio feedback, haptic feedback, etc.). Feedback can include determining when the user needs to take a break, or when the difficulty of an activity needs to be lowered or raised, the output can include scheduling various activities for the user or recommending adjustment of the HMD.

[2667] As used herein, the term “sensor” refers to one or more different sensing devices, such as a camera or imaging device, temperature device, oxygen device, movement device, brain activity device, sweat gland activity device, breathing activity device, muscle contraction device, etc. As used herein, the term “biometric,” “biometric information,” or “biometric feature” can refer to vital signs, including heart rate, pulse, respiration rate, respiration amplitude, or any other health related data. The term “biometric,” “biometric information,” or “biometric feature” can also refer to biological measurements or physical characteristics, such as facial expressions or emoting. In some examples, the sensor can detect or sense biometric features including features of the autonomic nervous system. Some particular examples of sensors include an electrooculography (EOG) sensor, electrocardiogram (EKG) sensor, photoplethysmography (PPG) sensor, heart rate sensor, heart rate variability sensor, blood volume pulse sensor, oxygen saturation (SpO₂) sensor, compact pressure sensor, electromyography (EMG) sensor, core-body temperature sensor, galvanic skin response (GSR) sensor, functional near-infrared spectroscopy (FNIR) sensor, non-contact passive infrared (IR) sensor, accelerometer, gyroscope, magnetometer, inclinometer, barometer, infrared sensor, global positioning system sensor, etc. Additional sensor examples can include, contact microphones (e.g., pressure-based MEMS), bioelectrical activity sensors, UV exposure sensors, or particle sensors. In some examples, certain sensors can be used to assess stress and emotion. The HMD can then provide feedback or output related to the detected stress and emotion. In some examples, the sensors can operate through coin cell battery or Bluetooth connectivity. In some examples, the sensors are powered by a primary battery of the HMD.

[2668] The sensors described herein can allow for observations of the autonomic nervous system (ANS), to observe relaxation and stress indicators, mental health, medical treatments, etc. Using the sensors, physicians and care takers could have live feedback of biometrics. Use cases can include fitness settings, user content, workplace, telepresence, clinical, education, training, pain, therapy, etc. In some examples, the sensors can be used to capture facial expressions. This is particularly relevant given the user's face is covered by HMD. For example, the HMD could use MEMS or motion tracking sensors to detect facial expressions

[2669] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.6-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.6-1**.

[2670] FIGS. **13.6-2** illustrates a head-mountable device **13.6-200**. The head-mountable device **13.6-200** can be substantially similar to, including some or all of the features of, the head-mountable devices described herein, such as head-mountable device **13.6-100**. In some examples, the head-mountable device **13.6-200** includes a controller **13.6-213** (e.g., sensor controller). While the controller **13.6-213** is illustrated as being positioned in the support **13.6-223**, the position of the controller **13.6-213** is not limited to the support **13.6-223**, but can be positioned in/on the facial interface **13.6-212** or in the HMD display housing. The controller **13.6-213** can include a processor and a memory device storing computer-executable instructions that, when executed by the processor, cause the controller to receive biometric data from one or more sensors **13.6-224**, transmit a signal based on the sensor data, and generate a signal to cause a component to perform an action in response to the signal.

[2671] The controller **13.6-213** can include one or more processors (e.g., a system on chip, integrated circuit, driver, microcontroller, application processor, crossover processor, etc.). Further, the controller **13.6-213** can include one or more memory devices (e.g., individual nonvolatile memory, processor-embedded nonvolatile memory, random access memory, memory integrated circuits, DRAM chips, stacked memory modules, storage devices, memory partitions, etc.). In certain implementations, the controller **13.6-213** is communicatively coupled to a power source (e.g., a battery).

[2672] In some examples, the controller **13.6-213** stores sensor data received from the sensor(s) **13.6-224** in the memory. The controller **13.6-213** can receive and/or transmit signals based on sensor data. For example, as will be described below, the controller **13.6-213**, by way of the processor and memory, can transmit a signal to the display **13.6-208** based on the sensor data (e.g., causing the display **13.6-208** or the head-mountable device **13.6-200** to perform an action, such as present a certain message, power off, react to biometric feedback, etc.

[2673] The controller **13.6-213** can perform any number of different functions. For example, the memory device can store computer-executable instructions that, when executed by the processor, cause the controller to receive sensor data from the sensors **13.6-224** and transmit a signal based on the sensor data. For instance, the controller **13.6-213** can transmit a sensor signal to a display **13.6-208**. In response to the controller **13.6-213** the display **13.6-208** can perform a wide variety of actions, including power off or power on, react to a user generated facial expression, present a digital notification (e.g., user-generated notification, push notification, context-generated notification, system-generated notification, smart notification, etc.). In some examples, the memory device storing computer-executable instructions that, when executed by the processor, cause the controller **13.6-213** to receive biometric data from the sensor(s) **13.6-224**, transmit a signal based on the sensor data, and perform an action in response to the signal.

[2674] As illustrated, the head-mountable device (e.g., wearable electronic device) **13.6-200** can include the display **13.6-208**, the facial interface **13.6-212**, and the sensor(s) **13.6-224** coupled to the engagement interface **13.6-212**. The sensors **13.6-224** can be embedded, encapsulated, deposited, adhered, or otherwise attached to the facial interface **13.6-212**. The sensors **13.6-224** can be non-contact sensors or can directly contact or touch the user's head. The sensor(s) **13.6-224** can be configured to detect a biometric feature of the user **13.6-220**. The sensor(s) **13.6-224**, upon detecting a signal from the biometric feature, can transmit the signal to a component of the head-mountable device **13.6-200** causing the head-mountable device **13.6-200** to perform an action. For example, the head-mountable device **13.6-200** can change shape (i.e., tighten or loosen), move, vibrate, rotate, recalibrate, or reposition in response to the sensor signal of the biometric feature.

[2675] The head-mountable device **13.6-200** can include a support, headband, or retention band **13.6-223** connected to the display **13.6-208** and/or frame **13.6-216**. The retention band **13.6-223** is configured to secure the display **13.6-108** and/or frame **13.6-216** relative to the user's head **13.6-220** (e.g., such that the display **13.6-208** is maintained in front of a user's eyes). The retention band **13.6-223** can be constructed from elastic material, inelastic material, or a combination of elastic

and inelastic material. The retention band **13.6-223** is adjustable such that the retention band **13.6-223** conforms to the various shapes and sizes of a user's head **13.6-220**. In some examples, the retention band **13.6-223** secures the head-mountable device **13.6-200** via friction between the user's head **13.6-220** and the retention band **13.6-223**. In some examples, the retention band **13.6-223** elastically secures the head-mountable device **13.6-200** to the user's head **13.6-220**. In some examples, the retention band **13.6-223** is coupled to a ratchet system or mechanism securing the head-mountable device **13.6-200** to the user's head **13.6-220**. In some examples, the retention band **13.6-223** is disposed above or on an ear **13.6-221** of the user **13.6-220**, supporting the head-mountable device **13.6-200**.

[2676] In some examples, the housing or frame **13.6-216** of the head-mountable display **13.6-200** is connected via a connector **13.6-215** to the facial interface **13.6-212**, the retention band **13.6-223** being connected to the frame **13.6-216** and/or the facial interface **13.6-212**, securing the head-mountable device **13.6-200** to the user's head **13.6-220** above or over the ears **13.6-221** of the user.

[2677] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.6-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.6-2**.

[2678] FIGS. **13.6-3** shows a rear view of a head-mountable device **13.6-300**. The head-mountable device **13.6-300** can be substantially similar to, including some or all of the features of, the head-mountable devices described herein, such as head-mountable device **13.6-100** and **13.6-200**. The head-mountable device **13.6-300** can include a facial interface (e.g., a facial engagement feature) **13.6-312** and a retention band **13.6-323**. The head-mountable device **13.6-300** further includes connectors **13.6-332a**, **13.6-332b** (collectively referred to as connector(s) **13.6-332**) connecting the retention band **13.6-323** and the facial engagement interface **13.6-312**. In some examples, the facial interface **13.6-312** is adjustable, accommodating to a user's head in a comfortable and secure way that shields the user from ambient light.

[2679] The connector **13.6-332** connecting the retention band **13.6-323** to the facial engagement interface **13.6-312** includes a first connector side **13.6-332a** attached to the retention band **13.6-323** and a second connector side **13.6-332b** attached to the facial engagement feature **13.6-312**. In addition to forming a mechanical attachment between the retention band **13.6-323** and the facial interface **13.6-312**, the connector **13.6-332** can include various types of electrical connections, such as a pogo-pin connection, or other connection types to electrically connect the retention band **13.6-323** with the facial interface **13.6-312**. When mated, the first connector side **13.6-332a** and the second connector side **13.6-332b** mate (e.g., mechanically mate, magnetically mate via a magnet, electrically mate, etc.) such that the connectors **13.6-332** are in mechanical contact and electrical contact. When the first connector side **13.6-332a** and the second connector side **13.6-332b** are connected in this manner, electrical signals can be relayed through the from the retention band **13.6-323** to the facial engagement interface **13.6-312**, and vice versa. That is, when connected, the facial engagement interface **13.6-312** can provide data and/or power to the retention band **13.6-323**, for example to a sensor or electrical component **13.6-324c** of the retention band. Likewise, when connected via connector **13.6-332**, the retention band **13.6-323** can provide data and/or power to the facial interface **13.6-312**, for example, to the sensors **13.6-324a** and/or **13.6-324b**. In some examples, the connectors **13.6-332a**, **13.6-332b** can magnetically mate or couple one to the other, transmitting information via a wireless communication connection and receiving power from a wireless power source (e.g., an inductive charger).

[2680] The head-mountable device **13.6-300** includes one or more sensor(s) **13.6-324a**, **13.6-324b**, **13.6-324c** (collectively “sensors **13.6-324**”). In some examples, the sensors **13.6-324** can be in

contact (indirect or direct) with a user. For example, the sensors be positioned to receive input from a user's forehead, cheek, nose, temple region, back of the head, or at any location on the user's head/face. In some examples, the sensors **13.6-324** can be in indirect contact with a user. For example, the sensors **13.6-324** may detect a user's biometric features via infrared, IR, optics, imaging, or other means of detecting biometric features in an indirect or contactless way. The sensors can be a biometric or health sensor, such as any of those described above.

[2681] In some examples, the sensors **13.6-324** include a first sensor **13.6-324a** oriented towards a first facial region (e.g., the forehead) and coupled to the engagement interface **13.6-312**, a second sensor **13.6-324b** oriented towards a second facial region (e.g., a nasal region) and coupled to the engagement interface **13.6-312**, and a third sensor or sensor set **13.6-324c** coupled to the retention band **13.6-323** and configured to contact a side or back of the user's head. When wearing the head-mountable device **13.6-300** the first sensor **13.6-324a** orients towards the first facial region is different in orientation and location from the second sensor **13.6-324** which orients toward the second facial region. It will be appreciated by one skilled in the art that other configurations are contemplated herein, and the above example is for illustrative purposes only.

[2682] The sensors **13.6-324** can be located at various locations of the head-mountable device **13.6-300** and can differ in function or sensing capacity depending on where the sensor is located. For example, the sensors **13.6-324b** are positioned on the engagement interface **13.6-312** proximate a nose of the user and may sense a user's respiratory biometrics providing a user with feedback indicating oxygen saturation (e.g., SPO.sub.2). In some examples, the sensor **13.6-324a** is positioned on the engagement interface **13.6-312** contacting a forehead of a user. In some examples, the sensor **13.6-324a** detects pulse, facial expression, brain activity or stress response while performing an activity. In some examples, the sensor **13.6-324c** is positioned on the retention band **13.6-323**, adjacent the temple(s) of a user or at another location where the retention band **13.6-323** is contacting the user. In some examples, a sensor fusion exists in which one or more of the sensors **13.6-324** coordinate and communicate with each other and/or with other sensors on the HMD **13.6-300** to collectively gather user data.

[2683] In some examples, forehead sensors, such as sensor **13.6-324a**, can be used for brain imaging and observations of the central nervous system. While a single forehead sensor **13.6-324a** is shown in FIGS. **13.6-3B**, it will be understood that multiple sensors or a sensor array can be located on the facial interface **13.6-312** to contact, or be proximate to, the user's head. In some examples, the sensors can be configured to perform EEG (electroencephalography) or functional near-infrared spectroscopy (fNIR).

[2684] In some examples, the position of the forehead sensors **13.6-324a** can be tuned or tailored to sense brain areas related to language, learning, memory, comprehension, sleep, stress, pain, attention, fear, discomfort, etc. In some examples, the facial interface **13.6-312** can be integrated with a sensor array that includes one or more transmitters that emit a signal. The transmitter(s) can be positioned a certain distance away from one or more detectors that sense the emitted signal. Based on the received signals by the detectors, a processor can infer or determine certain brain activity. In some examples, the sensors **13.6-324a** can form a brain-computer interface (BCI), such as a non-invasive neural interface. In some examples, the integrated sensor array can be used to detect the parieto-frontal network of the brain.

[2685] In some examples, the sensors **13.6-324a**, **13.6-324c** can be used to perform EEG detections. The sensors **13.6-324a**, **13.6-324c** can measure the electrical activity in the cerebral cortex (the outer layer of the brain). The sensors **13.6-324a**, **13.6-324c** can include electrodes that are placed on a user's head, then the electrodes can non-invasively detect brainwaves from the subject. The EEG sensors **13.6-324a**, **13.6-324c** can record up to several thousands of snapshots of the electrical activity generated in the brain every second. The recorded brainwaves can be sent to amplifiers, then to a local processor, a remote electronic device, or the cloud to process the data.

[2686] In some examples, the sensors **13.6-324a**, **13.6-324c** can be used to perform functional near-

infrared spectroscopy (FNIR). The sensors **13.6-324a**, **13.6-324c** can use low levels of non-ionizing light to record changes in cerebral blood flow in the brain through optical sensors placed on the surface of the head.

[2687] The sensors described herein can perform both passive and active EEG. The passive electrode signals can be sent to a remote processor to analyze. Amplification of the active EEG signals can be performed locally at the site to eliminate coexistence and interference that may be present with passive signal.

[2688] The HMD can include an output or feedback module that provides feedback based on the detections from the sensors. The feedback from the system can include displaying visualizations to the user. For example, the feedback can include breathing visualizations, visualization that relate to the user's attention, promptings or suggestions to the user to take a break or change an activity, or make other recommendations or notifications to the user.

[2689] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.6-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.6-3**.

[2690] FIGS. **13.6-4** shows a side or cross-sectional view of portion of a facial interface **13.6-412** wherein a sensor(s) **13.6-424** is embedded in the facial interface **13.6-412**. The facial interface **13.6-412** includes a first side or surface **13.6-407** facing the display side **13.6-429** (i.e., toward the HMD display) and a second side or surface **13.6-409** facing toward a user side **13.6-427**. In some examples, the first surface **13.6-407** attaches to a frame of the HMD. In some examples, the second surface **13.6-409** directly contacts or touches the user's head when the HMD is donned or being worn by the user.

[2691] The facial interface **13.6-412** can include one or more sensors **13.6-424a**, **13.6-424b**, **13.6-424c** (collectively "sensors **13.6-424**"). The sensor **13.6-424a** can be positioned on the first surface **13.6-407** of the facial interface **13.6-412**. In some examples, the first surface **13.6-409** can be an interior and the second surface **13.6-409** can be an exterior of the facial interface **13.6-412**. Thus, the sensor **13.6-424a** can be positioned on an interior of the facial interface **13.6-412**. In some examples, the facial interface **13.6-412** can include a sensor **13.6-424b** is embedded, encapsulated, or otherwise surrounded by the facial interface **13.6-412**. The facial interface **13.6-412** can include, a sensor **13.6-424c** that positioned such that a portion of the sensor **13.6-424c** is exposed through the second surface **13.6-409**. In other words, the sensor **13.6-424c** can at least partially define the second surface **13.6-409** and can directly contact or touch the user's skin. In some examples, the sensor **13.6-424c** can be partially surrounded or embedded in the facial interface **13.6-412**. In some examples, the sensor **13.6-424c** is external to the facial interface **13.6-412**, positioned on the exterior **13.6-409** of the facial interface **13.6-412**.

[2692] The sensors **13.6-424a** and **13.6-424b** can have corresponding sensor areas **13.6-425a**, **13.6-425b**, respectively. The sensor areas **13.6-425a**, **13.6-425b** can represent a field of view or cone of influence that is transparent to a signal emitted by the sensors **13.6-424a**, **13.6-424b**. In some examples, the field of view of the sensors **13.6-424** can be approximately 50 degrees. The sensors **13.6-424a**, **13.6-424b** can detect physiological, biological, and/or biometric changes of the user's body through corresponding sensor areas **13.6-425a**, **13.6-425b**. For example, the sensors **13.6-424a**, **13.6-424b** can detect changes to a user through the material of the facial interface **13.6-412**. Other sensors may be added or substituted, as may be desired. In some examples, the sensor areas **13.6-425a**, **13.6-425b** can comprises a different material than the rest of the facial interface **13.6-412**. For example, the sensor areas **13.6-425a**, **13.6-425b** can be transparent to certain signals from the sensors **13.6-424a**, **13.6-424b** or from the user, while the remainder of the facial interface **13.6-**

412 is not transparent or transmissive to such signals.

[2693] Sensor **13.6-424b** is a distance from the first surface **13.6-407**, disposed with in the facial interface **13.6-412**, and therefore closer to the second surface **13.6-409** than the sensor **13.6-424a**. In some instances, this closer positioning of the sensor **13.6-424b** to the second surface **13.6-409** can correspondingly reduce the field of view **13.6-425b**. In some examples, the sensor **13.6-424c** may be flush with the second surface **13.6-409** and/or directly contacting a user's face and/or skin. This is only one example of sensor depth variation, as a plurality of sensors can be disposed on the first surface **13.6-407** of the facial interface **13.6-412**, or between the first surface **13.6-407** and the second surface **13.6-409** of the facial interface **13.6-412**. In some examples, the second surface **13.6-409** (e.g., user side **13.6-427**) abuts a forehead region or nasal region of a user head when the head-mountable device **13.6-100** is donned.

[2694] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.6-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.6-4**.

[2695] FIGS. **13.6-5** shows a front perspective view of a facial interface **13.6-512**. The facial interface **13.6-512** can be substantially similar to, including some or all of the features of, the facial interface described herein, such as facial interfaces **13.6-112**, **13.6-212**, **13.6-312**, and **13.6-412**. In some examples, the facial interface **13.6-512** can include a framed or clamshell design. In some examples, various components and/or circuitry can be discretely positioned within the volume of the clamshell design. The facial interface **13.6-512** can include a first side **13.6-540a** and a second side **13.6-540b**. The first side **13.6-540a** can be proximate the user when donned. In some examples, the first side **13.6-540a** can contact the user's face. The second side **13.6-540b** can be positioned between the first side **13.6-540a** and a display of the HMD. In some examples, the second side **13.6-540b** is attached to the display housing.

[2696] The first side **13.6-540a** can include a connector **13.6-532** configured to attach the facial interface **13.6-512** to a component, such as a retention band. The connector **13.6-532** can be mechanical and/or electrical. In some examples, the connector **13.6-532** is electrically connected to a sensor **13.6-524c**. The first side **13.6-540a** can be configured to contact a face of a user. The first side **13.6-540a** can include sensors **13.6-524a**, **13.6-524b**, **13.6-524c**, **13.6-524d** that can be configured to detect a biometric feature(s) of the user. In some examples, the facial interface includes a nasal support **13.6-539**. The nasal support **13.6-539** can be configured to contact the user's nose. The nasal support **13.6-539** can include at least one sensor **13.6-524b**.

[2697] In some examples, the facial interface **13.6-512** can be interchangeable with a different facial interface having different sensors or structure. For example, a user may exercise with the head-mountable device **13.6-100** using a first facial interface with specific sensors for exercising. The same user then may game or perform a leisure activity using the head-mountable device **13.6-100** fitted with a different facial interface with different sensors or structure for those activities. This is only one example and one of ordinary skill in the art will appreciate other examples are contemplated herein.

[2698] In some examples, the sensors **13.6-524** are removably attached to the facial interface **13.6-512**. For example, a user may wish to replace or add certain sensors to the facial interface **13.6-512**. A user can selectively attach or remove a sensor **13.6-524** and replace it with a different sensor depending on the activity of the user or purpose of the sensor. In this way, the sensors can accommodate the needs or demands of a user for a specific activity requiring specific sensors without needing to replace the entire HMD or facial interface **13.6-512**.

[2699] In some examples, the facial interface **13.6-512** includes a connector **13.6-536**. The

connector **13.6-536** can be electrically connected one or more of the sensors **13.6-524** via a wired or wireless connection **13.6-543**. In some examples, the facial interface **13.6-512** is in electrical communication with a display via the wired connection link **13.6-543** and/or the connector **13.6-536**, such as, a braided electrical cable, a pogo connection, or other type of wired connection. In some examples, the facial interface **13.6-512** can be in electrical communication with the display **13.6-208** via a wireless connection (e.g., low-power Bluetooth BLE connection). In some examples, data or signals from the sensors **13.6-524** are transmitted through the link **13.6-543** to the connector **13.6-536**, where they are further transmitted to the HMD itself to be processed. Thus, in some examples, the facial interface **13.6-512** serves as an interconnect between the retention band and the display unit. In some examples, the retention band can mate directly into the display unit, bypassing the facial interface **13.6-612**. In some examples, the facial interface **13.6-512** can include an Orion style connection (e.g., 3 contact pads and 3 contact pins).

[2700] In some examples, the facial interface **13.6-512** includes a bridging feature **13.6-541** structurally connecting the first side **13.6-540a** and the second side **13.6-540b**. In some examples, the bridging features **13.6-541** can provide a conduit for wires or electronics, such as the link **13.6-543**. In some examples, the bridging feature **13.6-541** is flexible permitting the facial interface first side **13.6-540a** to move, pivot, or change shape relative to the facial interface second side **13.6-540b**. In some examples, the bridging feature **13.6-541** is rigid maintaining the facial interface first side **13.6-540a** unmoved relative to the facial interface second side **13.6-540b**.

[2701] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.6-5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.6-5**.

[2702] FIGS. **13.6-6** illustrates a perspective view of select components of a head-mountable device **13.6-600**. The HMD **13.6-600** can be substantially similar to, including some or all of the features of, the HMDs described herein. The HMD **13.6-600** can include chassis or a frame **13.6-609** that can at least partially house a display, and a facial interface **13.6-612** that can contact a user's face. In some examples, the HMD **13.6-600** changes shape in response to a signal from one or more sensor(s) **13.6-624a**, **13.6-624b**, **13.6-624c** (collectively "sensors **13.6-624**"). The frame **13.6-609** can be connected via a connector(s) **13.6-615** to the facial interface **13.6-612**. In some examples, the connectors **13.6-615** are electrical and/or mechanical connectors configured to establish electrical and/or mechanical connections between the frame **13.6-609** and the facial interface **13.6-612**. In some examples, the connectors **13.6-615** are articulable or movable, such that the connectors **13.6-615** can move to change a shape of the HMD **13.6-600**. In some examples, the HMD **13.6-600** can include strain gauges or point load pressure sensors that can be used on posts **13.6-615** to act as pressure sensors to extrapolate retention band tension. The system can then guide the user to adjust for proper fitting of the HMD (e.g., prompt to tighten or loosen or change an angle of the band). In some examples, the display can illustrate for the user how to adjust band up/down based on the detected tension. In some examples, semiconductor(s) can be embedded in the facial interface (e.g., in the foam of the facial interface).

[2703] In some examples, the facial interface **13.6-612** can be integrated with one or more feedback modules or outputs **13.6-650**, **13.6-653**. The sensors **13.6-624** can receive biometric input from a user. Based on the input detected by the sensors **13.6-624**, the outputs **13.6-650**, **13.6-653** can perform some action. The outputs **13.6-650**, **13.6-653** can include LEDs, haptics, speakers, motors, displays, or any other feedback devices that creates an output in response to the signals of biometric information from the sensors **13.6-624**. For example, one or more of the sensors **13.6-624** can detect when the HMD is donned, for example through capacitive or pressure sensors. The

sensors **13.6-624** can then generate a signal to be analyzed by a processor.

[2704] The processor can then cause a first feedback output **13.6-650**, such as an LED to emit a certain color of light indicating that the head-mountable device **13.6-600** is donned. In some examples, the second feedback output **13.6-653** can be a feedback output such as a haptic feedback output alerting a user of a notification (e.g., message notification, text notification, email notification, etc.) In some examples, the output **13.6-653** can be a display visible by the user and which can extend the peripheral view of the user. It should be appreciated by one of normal skill in the art that the above example is one example and other examples of feedback outputs are contemplated herein.

[2705] In some examples, the chassis or frame **13.6-609** can at least partially define an exterior surface of the HMD **13.6-600**. In some examples, health sensors **13.6-616** can be mounted, attached, or otherwise integrated into/onto the chassis **13.6-609**. While two health sensors **13.6-616** are illustrated, it will be understood that one, two, or more than two sensors can be used and are contemplated by the present disclosure. The health sensors **13.6-616** can be substantially similar to, including some or all of the features of, the sensor described herein (e.g., sensors **13.6-624**). In some examples, the health sensors **13.6-616** can be non-contact sensors (i.e., the health sensors **13.6-616** can collect the intended data without needing to be in direct physical contact with the user. The health sensors **13.6-616** can be positioned on a bottom surface of the chassis **13.6-609**. The health sensors **616** can at least partially define an exterior surface of the HMD **13.6-600**.

[2706] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.6-6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.6-6**.

13.7: Health Sensing Retention Band

[2707] According to some aspects of the present disclosure, a head-mountable device can include a housing, a display positioned in the housing, a processor positioned in the housing, and a retention band connected to the housing, the retention band including a sensor to monitor a brain activity of a user.

[2708] In some examples, the retention band includes a sensor array positionable adjacent a back of a head of the user when the head-mountable device is worn by the user. The sensor array can detect brain activity of the user. The sensor can transmit a signal to the processor, the signal based on the biometric information. The processor can analyze the signal and cause the head-mountable device to perform an action in response to the analysis of the signal. The action can be at least one of providing a visual feedback, providing an audio feedback, or providing a haptic feedback.

[2709] In some examples, the retention band is removably attached to the housing. The sensor can be removably attached to the retention band. The retention band can be in electrical communication with the display. The retention band can be pivotably attached to the housing.

[2710] According to some aspects, a headband for a head-mountable device can include a first end to attach to the head-mountable device, a second end to attach to the head-mountable device, and a sensor positioned between the first end and the second end. The sensor can detect brain activity of a user, and generate a signal based on the detected brain activity.

[2711] In some examples, the sensor can be embedded in the headband. The headband can include an area that is transparent to the signal emitted by the sensor. The headband can be articulable relative to the head-mountable device. The headband can include an expandable surface area. The signal can include a display command signal.

[2712] According to some aspects, a wearable electronic device can include a display, a retention band attached to the display, the retention band having a first orientation and a second orientation,

and a sensor connected to the retention band. The sensor can detect a brain activity of a user and produce a signal based on the brain activity. A processor can perform a first analysis of the signal in response to the retention band being in the first orientation, and can perform a second analysis of the signal in response to the retention band being in the second orientation.

[2713] In some examples, the wearable electronic device can perform an action in response to the signal. The wearable electronic device can include a head-mountable device. The retention band can be adjustable. The sensor can perform at least one of functional near-infrared spectroscopy or electroencephalography.

[2714] In some examples, the retention band is movable between a first position and a second position. The sensor can be oriented toward a first brain region in the first position, and a second brain region in the second position when the wearable electronic device is worn by the user. The retention band can produce at least one of visual, audio, or haptic feedback in response to the signal from the sensor. The sensor can be a first sensor. The wearable device can further include a second sensor connected to the retention band. The second sensor can collect vital signs.

[2715] The present disclosure includes details of a retention band for a head-mountable device that is integrated with health sensors. The integrated retention band can enable both optical and electrical neural imaging of the user's head. In some examples, the sensor-integrated headband can be used for brain imaging. For example, the sensors can be configured to perform functional near-infrared spectroscopy (FNIR). In some examples, the sensors can be used to perform EEG (electroencephalography).

[2716] The retention band can be adjustable or articulable to position the sensors at desired locations on the user's head. By having the ability to adjust the retention band, the sensors can observe specific areas of the brain. For example, the position of the sensors can be tuned or tailored to look at brain areas related to language, learning, memory, comprehension, sleep, stress, pain, attention, fear, discomfort, etc.

[2717] In some examples, the retention band can be integrated with a sensor array that includes one or more transmitters that emit a signal. The transmitter(s) can be positioned a certain distance away from one or more detectors that sense the emitted signal. Based on the received signals by the detectors, a processor can infer or determine certain brain activity. In some examples, the integrated sensors on the retention band form a brain-computer interface (BCI), such as a non-invasive neural interface. In some examples, the integrated sensor array can be used to detect the parieto-frontal network of the brain.

[2718] The HMD can include an output or feedback module that provides feedback based on the detections from the sensors. The feedback from the system can include displaying visualizations to the user. For example, the feedback can include breathing visualizations, visualization that relate to the user's attention, promptings or suggestions to the user to take a break or change an activity, or make other recommendations or notifications to the user.

[2719] These and other embodiments are discussed below with reference to FIGS. **13.7-1-13.7-6**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[2720] FIGS. **13.7-1** shows a schematic block diagram of a head-mountable device **13.7-100** including a display **13.7-104**, a retention band **13.7-108**, a memory **13.7-106**, and a processor **13.7-120**. The display **13.7-104** can include one or more optical lenses or display screens in front of the

eyes of a user. The display **13.7-104** can present an augmented reality visualization, a virtual reality visualization, or other suitable visualization to a user. Additionally, the display **13.7-104** can be positioned in or on a housing or frame that supports the display **13.7-104** and additional electronic components. The display **13.7-104** in combination with the frame or the housing and other electronic components supported by the housing can be referred to as the display unit **13.7-104** and the display **13.7-104**. The retention band **13.7-108** can be connected to the frame or housing of the display **13.7-104**.

[2721] In some examples, the retention band **13.7-108** can include a support, strap, headband, belt, arms, or any other attachment mechanism capable of supporting the HMD **13.7-100** on the user's head. In some examples, the retention band **13.7-108** can include a pliant (or semi-pliant) material or fabric that wraps partially around the user's head.

[2722] The retention band **13.7-108** can be made from a flexible material and can securely and snugly fit around a head of the user. The retention band **13.7-108** can be made from a woven fabric, leather, polymer, or any other material compatible to be integrated with the sensors **13.7-112** or having electromagnetic transmissive properties. In some examples, the retention band **13.7-108** can be made from silicone or thermoplastic polyurethane (TPU). In some examples, the retention band **13.7-108** can be made from compression molded materials, such as rubber. The sensors **13.7-112** can be integrated into the compression molded materials. In some examples, the retention band **13.7-108** is semi-rigid. For example, the retention band **13.7-108** can be rigid where the sensors **13.7-112** are located. The sensors **13.7-112** can be flexible or deformable relative to each other such that the sensor array can move, flex, or stretch with the retention band **13.7-108**. The retention band **13.7-108** can include rigid sections where electronic components are located.

[2723] The retention band **13.7-108** can include two ends, straps or sidebands **13.7-108a** that connect to the display **13.7-104**. The retention band **13.7-108** can include a middle section **13.7-108b** that is positioned between the ends **13.7-108a** of the retention band **13.7-108**. The middle section **13.7-108b** can be configured to contact the back of the user's head, while the sidebands **13.7-108a** can be considered to contact the sides of the user's head. The retention band **13.7-108** can include pivots **13.7-115**. The pivots **13.7-115** can be located between the display **13.7-104** and the sidebands **13.7-108a** such that the entire retention band **13.7-108** can pivot or rotate relative to the display **13.7-104**. In some examples, the pivots **13.7-115** can be positioned between the sidebands **13.7-108a** and the middle section **13.7-108b** to enable the middle section **13.7-108b** to pivot or rotate relative to the sidebands **13.7-108b** and display **13.7-104**. Further detail of the pivots will be provided below with regard to FIGS. **13.7-4A-13.7-4C**.

[2724] The retention band **13.7-108** can include one or more sensors **13.7-112**. It will be understood that throughout the disclosure, reference to “sensors” can refer to one or more sensors. The phrase “sensors” allows for, but does not necessarily require, multiple sensors. In some examples, the sensors **13.7-112** can be removably attached to the retention band **13.7-108**. The retention band **13.7-108** can be removably attached to the display **13.7-104** and in electrical communication with the display **13.7-104**. In other words, the sensors **13.7-112** can be in electrical communication with the display **13.7-104**.

[2725] The sensor **13.7-112** of the retention band **13.7-108** can collect biometric information, such as brain activity and vital signs (including body temperature, pulse data, respiration data, and blood pressure). The sensor **13.7-112** can generate a signal based on the collected user information and transmit the signal to the processor **13.7-120** to be analyzed. In some examples, the memory **13.7-106** can include programmed instructions responsive to the signals from the sensors **13.7-112**. In some examples, the instructions can cause an output or feedback module to perform an action in response to the signal (i.e., in response to the biometric information collected by the sensor **13.7-112**). In some examples, the retention band **13.7-108** itself provides feedback (e.g., through LEDs, tactile or haptic feedback, changing its shape, color, etc.) based on the collected biometric information.

[2726] For example, a user may be in a meditation or mental awareness session that is facilitated by the HMD **13.7-100**. During the activity, the user's brain waves or other biometric may change in a way that is detectable by the sensors **13.7-112**. The sensors **13.7-112** can generate one or more signals based on the received input. The sensors **13.7-112** can transmit the signals to the processor **13.7-120**, which may cause an action by one or more components of the HMD **13.7-100** (e.g., to the display **13.7-104** to provide a visual feedback to the user). The display **13.7-104**, being in electrical communication with the retention band **13.7-108**, can receive electrical communication and provide feedback to the user related to their biometric readings (e.g., visual feedback, audio feedback, haptic feedback, etc.). Feedback can include determining when the user needs to take a break, or when the difficulty of an activity needs to be lowered or raised, the output can include scheduling various activities for the user or recommending adjustment of the HMD **13.7-100**.

[2727] The sensors **13.7-112** can include a variety of different sensing devices. The sensors **13.7-112** can include, but are not limited to, brain activity sensors, a camera or imaging device, temperature device, oxygen device, movement device, brain activity device, sweat gland activity device, breathing activity device, muscle contraction device, etc. In some examples, the sensor can detect or sense biometric features including features of the autonomic nervous system.

[2728] Some particular examples of sensors include an electrooculography (EOG) sensor, electrocardiogram (EKG) sensor, EEG (electroencephalography), photoplethysmography (PPG) sensor, heart rate sensor, heart rate variability sensor, blood volume pulse sensor, oxygen saturation (SpO2) sensor, compact pressure sensor, electromyography (EMG) sensor, core-body temperature sensor, galvanic skin response (GSR) sensor, functional near-infrared spectroscopy (fNIR) sensor, functional magnetic resonance imaging (fMRI) sensor, non-contact passive infrared (IR) sensor, accelerometer, gyroscope, magnetometer, inclinometer, barometer, infrared sensor, global positioning system sensor, etc.

[2729] In some examples, the sensors **13.7-112** can include, contact microphones (e.g., pressure-based MEMS), bioelectrical activity sensors, UV exposure sensors, or particle sensors. In some examples, certain sensors can be used to assess stress and emotion. In some examples, the sensors can operate through coin cell battery or Bluetooth connectivity. In some examples, the sensors are powered by a primary battery of the HMD.

[2730] The sensors **13.7-112** described herein can allow for observations of the autonomic nervous system (ANS), to observe relaxation and stress indicators, mental health, medical treatments, etc. Using the sensors, physicians and care takers could have live feedback of biometrics. Use cases can include fitness settings, user content, workplace, telepresence, clinical, education, training, pain, therapy, etc. In some examples, the sensors can be used to capture facial expressions. This is particularly relevant given the user's face is covered by HMD. For example, the HMD could use MEMS or motion tracking sensors to detect facial expressions.

[2731] In some examples, the sensors **13.7-112** can be used to perform EEG detections. The sensors **13.7-112** can measure the electrical activity in the cerebral cortex (the outer layer of the brain). The sensors **13.7-112** can include electrodes that are placed on a participant's head, then the electrodes can non-invasively detect brainwaves from the subject. The EEG sensors **13.7-112** can record up to several thousands of snapshots of the electrical activity generated in the brain every second. The recorded brainwaves can be sent to amplifiers, then to the processor **13.7-120**, a remote electronic device, or the cloud to process the data.

[2732] In some examples, the sensors **13.7-112** can be used to perform functional near-infrared spectroscopy (fNIR). The sensors **13.7-112** can use low levels of non-ionizing light to record changes in cerebral blood flow in the brain through the optical sensors **13.7-112** placed on the surface of the scalp. The signals can be recorded via flexible fiber optic cables.

[2733] The HMD **13.7-100** can be a wearable device can include electronic components that are communicatively coupled to each other and to the sensors **13.7-112** via a wired or wireless communications link. The communications link can be a physical connection, such as an electrical

wire, or can be a wireless connection, such as Bluetooth, Wi-Fi, proximity sensors, etc. In some examples, the HMD **13.7-100** can be communicatively coupled to a companion electronic device, such as a remote, or a personal computing device such as a smart phone, a smart watch, a laptop, a tablet, an HMD, or any other form of electronic device. As described in further detail below, the signals from the sensors **13.7-112** can influence the HMD **13.7-100**. For example, the sensors **13.7-112** can influence the visual information, content, style, frequency, and operation of the content provided by the sensors **13.7-212**.

[2734] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.7-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.7-1**.

[2735] FIGS. **13.7-2** shows a top view of a head-mountable device (HMD) **13.7-200**. The HMD **13.7-200** can be substantially similar to, and can include some or all of the features of, the head-mountable devices described herein, such as HMD **13.7-100**. The HMD **13.7-200** can include a display unit **13.7-204** and a retention band **13.7-208**. The display unit **13.7-204** can include electronic components, such as a display, processor, memory, controller, battery, and other circuitry and electronics. The display unit **13.7-204** can include a frame or housing structurally supporting the electronic components. In some examples, the display **13.7-204** includes an opaque, translucent, transparent, or semi-transparent screen, including any number lenses, for presenting visual data.

[2736] The frame can at least partially border one or more edges of the display. In some examples, the frame can be configured to contact a user's head or face. In some examples, the frame blocks outside light and limits the peripheral view of the user. Various components of the display unit **13.7-204** can be housed within the frame. For example, the hardware and electronics which allow functionality of the HMD can be housed within the frame. In some examples, the frame can include attachment mounts for connecting the display unit **13.7-204** to the retention band **13.7-208**. For example, the frame can include mechanical attachments, magnetic attachments, or any other suitable connector to attach the retention band **13.7-208** to the display unit **13.7-204**.

[2737] The HMD **13.7-200** can be worn on the user's head such that the display unit **13.7-204** is positioned over the user's face and disposed over one or both of the user's eyes. In some examples, the retention band **13.7-208** can be positioned against (i.e., contacting or pressing against) the side and/or back of a user's head when worn. In some examples, the retention band **13.7-208** can be at least partially positioned above the user's ear or ears. In some examples, the retention band **13.7-208** can be positioned adjacent to the user's ear or ears. The display unit **13.7-204** and the retention band **13.7-208** can form a loop defining an aperture **13.7-205** for the user's head. It should be understood, however, that this configuration is just one example of how the components of an HMD **13.7-200** can be arranged, and that in some examples, a different number of connector straps and/or retention bands can be included.

[2738] In some examples, the retention band **13.7-208** includes a first sensor array **13.7-212a** including one or more sensors. The sensor array **13.7-212a** can be positioned on the back of the retention band **13.7-208**, located to contact or be proximate the back of the user's head when the HMD **13.7-200** is donned by the user. In some examples, the retention band **13.7-208** includes a second sensor array **13.7-212b** including one or more sensors. The second sensor array **13.7-212b** can be positioned on a side strap or arm of the retention band **13.7-208**, located to contact or be proximate the side of the user's head when the HMD **13.7-200** is donned by the user. In some examples, the first sensor array **13.7-212a** and the second sensor array **13.7-212b** are separate and distinct, both in form and functionality. The first sensor array **13.7-212a** can include different

sensors than the second sensor array **13.7-212b**.

[2739] In some examples, the first sensor array **13.7-212a** can translate or move along the retention band **13.7-208**. For examples, the first sensor array **13.7-212a** can slide from a first position on the retention band **13.7-208** (e.g., the back of the band) to a second position on the retention band **13.7-208** (e.g., the side strap). Translation of the sensor array **13.7-212a** can be manually or automatically actuated. In the manner, the sensor array **13.7-212a** can move to monitor different regions of the user's head.

[2740] In some examples, the first sensor array **13.7-212a** and the second sensor array **13.7-212b** (collectively referred to as “sensors **13.7-212**”) include the same sensors and are considered part of a single array of sensors. The sensors **13.7-212** can be positioned on an interior surface **13.7-207** of the retention band **13.7-208**. In some examples, the sensors **13.7-212** are positioned in the retention band **13.7-208** (i.e., between the interior surface **13.7-207** and the exterior surface **13.7-209**). In some examples, the sensors **13.7-212** can span substantially across an entirety of a length of the retention band **13.7-208**. As will be discussed in greater detail herein, the sensors **13.7-212** can be integrated with the retention band **13.7-208** in a variety of ways.

[2741] In some examples, the sensors **13.7-212** can be used to perform EEG detections. The sensors **13.7-212** can measure the electrical activity in the cerebral cortex (the outer layer of the brain). The sensors **13.7-212** can include electrodes that are placed on a participant's head, then the electrodes can non-invasively detect brainwaves from the subject. The EEG sensors **13.7-212** can record up to several thousands of snapshots of the electrical activity generated in the brain every second. The recorded brainwaves can be sent to amplifiers, then to the HMD computer, a remote electronic device, or the cloud to process the data.

[2742] In some examples, the sensors **13.7-212** can be used to perform functional near-infrared spectroscopy (FNIR). The sensors **13.7-212** can use low levels of non-ionizing light to record changes in cerebral blood flow in the brain through the optical sensors **13.7-212** placed on the surface of the scalp. The signals can be recorded via flexible fiber optic cables.

[2743] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.7-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.7-2**. Further details regarding the integration of sensors into an HMD retention band are described below with reference to FIGS. **13.7-3**.

[2744] FIGS. **13.7-3** shows a cross-sectional side view of a head-mountable device (HMD) **13.7-300**. The HMD **13.7-300** can be substantially similar to, and can include some or all of the features of, the head-mountable devices described herein, such as HMDs **13.7-100** and **13.7-200**. In some examples, sensor(s) **13.7-312** positioned in/on the retention band **13.7-308** can be operatively coupled to one or more electronic components in the display unit **13.7-304**. For example, the sensors **13.7-312** can be electrically coupled a power source, such as a battery **13.7-335** that is disposed within or on the housing of the display unit **13.7-304** via an electrical connection **13.7-311**. The sensor **13.7-312** can be communicatively coupled with a processor **13.7-320** or other controller that is housed within the head-mountable device **13.7-300**. The battery **13.7-335** and/or the processor **13.7-320** can be located with the housing of the display unit **13.7-304**, located within the retention band **13.7-108**, disposed exterior to the housing of the display unit **13.7-304** and the retention band **13.7-108**, or combinations thereof. The sensor **13.7-312** can generate and transmit signals reflective of the biometrics of the user to the processor **13.7-320**. The processor **13.7-320** can then cause commands or instructions to be issued.

[2745] Although the sensors **13.7-312** are shown as being connected to the battery **13.7-335** and the processor **13.7-320** through a wired connection **13.7-311**, in some examples, the sensor **13.7-312**

can wirelessly receive data and/or power from the battery **13.7-335** and/or processor **13.7-320** by any desired method or technology. In some examples, the sensors **13.7-312** can be communicatively coupled to an external device that is not located on the HMD **13.7-300**.

[2746] Further, although the components of the wearable electronic device **13.7-300** are shown as being connected to one another at certain locations, it should be understood that any of the components of the HMD **13.7-300** can be electrically and/or mechanically connected to one or more of any of the other components of the HMD **13.7-300**, in any manner and location, as desired.

[2747] In some examples, the retention band **13.7-308** can include other operational or functional components in addition to the sensors **13.7-312**. For example, the retention band **13.7-108** can include a battery **13.7-335** that can be in electrical communication with the display unit **13.7-304** and/or the sensors **13.7-312**.

[2748] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.7-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.7-3**.

[2749] FIGS. **13.7-4A** shows a rear perspective view of a retention band **13.7-408**. The retention band **13.7-408** can be substantially similar to, and can include some or all of the features of, the retention bands described herein, such as retention bands **13.7-108**, **13.7-208**, and **13.7-308**. While FIGS. **13.7-4A** illustrates the sensors **13.7-412** being visible on an exterior of the retention band **13.7-408**, it will be understood that in some examples, the sensors **13.7-412** are concealed or hidden behind a cover or layer of the retention band **13.7-408**. The sensors **13.7-412** can be configured to directly contact or touch the user's head. In some examples, the sensors **13.7-412** are non-contact sensors. In some examples, the retention band **13.7-408** is made up of multiple components. For example, the retention band **13.7-408** can include sidebands or arms **13.7-408a** and a middle section or back **13.7-408b** (collectively retention band **13.7-408**). The back **13.7-408b** can be connected at each end to the arms **13.7-408a** via pivots **13.7-415**.

[2750] In some examples, the sensors **13.7-412** can be interconnected (i.e., electrically connected) via wires **13.7-411**. The wires **13.7-411** can further define a communications link between the display unit and the retention band **13.7-408**. The sensors **13.7-412** can be disposed substantially across the entirety of the back **13.7-408b** of the retention band **13.7-408**. In some examples, the sensors **13.7-412** can be intentionally positioned to align with specific portions of the user's head when donned. As shown in greater detail with reference to FIGS. **13.7-4B** and **13.7-4C**, the retention band **13.7-408** can be articulable. By adjusting, articulating, or moving the position of the retention band **13.7-408** relative to the HMD, the sensors **13.7-412** can be positioned at various regions of the user's head.

[2751] In some examples, the sensors **13.7-412** can include a grounding sensor. For example, one or more of the sensors **13.7-412** can be a grounding sensor that is used as a reference sensor. Detections from the sensors **13.7-412** can be compared against the grounding reference sensor to analyze the detections. In some examples, the sensors **13.7-412** can include rounded edges and/or elastomeric or soft material that is transparent to the signal produced by the sensors **13.7-412**. The rounded edges and soft material can help to reduce discomfort when the sensors **13.7-412** are contacting the user's head. In some examples, the geometry of the sensors **13.7-412** can be specifically designed to assist in parting the user's hair to ensure efficient contact against the user's head. In some examples, the sensors move (e.g., vibrate, rotate or oscillate) to aid in proper contact against the user's skull.

[2752] In some examples, the retention band **13.7-408** is a specialized band that can be interchanged with other headbands that are customized for different uses. For example, there can be

individual custom retention bands for education, sports, industrial applications, health, learning application, training application, etc. Each different retention band can be integrated with sensors specific for its needs.

[2753] In some examples, the retention band **13.7-408** can be detachable or removable from the HMD. In other words, the HMD can include a standard headband that is replaced or modified with the retention band **13.7-408**. In some examples, the retention band **13.7-408** can be an add-on component that is added to the HMD when desired or needed. The retention band **13.7-408** can include electrical/mechanical attachment mechanisms to attach to the HMD. For example, the retention band **13.7-408** can include attachment mechanisms **13.7-418** that can electrically, mechanically, and or magnetically connect the retention band **13.7-408** to an HMD. In some examples, the retention band wirelessly connects to the HMD to transmit data. The retention band **13.7-408** can include its own power source and processing capabilities.

[2754] In some examples, the retention band **13.7-408** can be expandable to reveal additional sensors **13.7-412**. For example, the retention band **13.7-408** can fold or overlap itself to reduce a surface area of the retention band **13.7-408**. Then when desired, the retention band **13.7-408** can be stretched, unfolded, or expanded, revealing additional sensors that were previously covered.

[2755] FIGS. **13.7-4B** shows a side view of the retention band **13.7-408** rotated upward. As illustrated, the back **13.7-408b** of the retention band **13.7-408** can rotate relative to the side arms **13.7-408a** via the pivots **13.7-415**. In some examples, the entire retention band **13.7-408** can rotate relative to a housing or frame of the HMD (i.e., the pivots **13.7-415** can be located on the HMD housing). Movement of the retention band **13.7-408** can be manual or automated. In some examples, the user is prompted, guided, or instructed to move the retention band **13.7-408** into a certain position or configuration. For example, the system may prompt the user to adjust the retention band **13.7-408** to optimize the ability of the sensors **13.7-412** to detect the desired biometric data. In this configuration the sensors **13.7-412** would be positioned at or near the top of the user's head.

[2756] In some examples, the HMD include multiple retention bands **13.7-408**. For example, the HMD can include a back-of-the-head band that wrap around the back of the user's head, and an over-the-head band that wraps over the top of the user's head. Each of the bands can include the sensors described herein. In some examples, the retention band **13.7-408** can be a bifurcated strap with sensors on each bifurcated section to increase the coverage area of the sensors over the head.

[2757] FIGS. **13.7-4C** shows a side view of the retention band **13.7-408** rotated downward. As illustrated, the back **13.7-408b** of the retention band **13.7-408** can rotate relative to the side arms **13.7-408a** via the pivots **13.7-415**. In some examples, the entire retention band **13.7-408** can rotate relative to a housing or frame of the HMD (i.e., the side straps **13.7-408a** also rotate and the pivots **13.7-415** are located on the HMD housing). Movement of the retention band **13.7-408** can be manual or automated. In some examples, the user is prompted, guided, or instructed to move the retention band **13.7-408** into a certain position or configuration. For example, the system may prompt the user to adjust the retention band **13.7-408** downward to optimize the ability of the sensors **13.7-412** to detect the desired biometric data. In this configuration the sensors **13.7-412** would be positioned at or near the bottom of the user's head, near the top of the neck. In this configuration, the sensors **13.7-412** can be used to detect information about neck muscles. In some examples, the location or position of the retention band **13.7-408** can determine what type of analysis is performed by the processor. For example, the processor can interpret a signal differently based on the detected position of the retention band **13.7-408** relative to the HMD and/or user when the signal was acquired. In some examples, the signal analysis changes based on the region of the brain being monitored by the sensors **13.7-412**. In some examples, operation of the sensors themselves changes based on the position of the retention band **13.7-408**. For example, different sensors can be used when monitoring different areas of the brain, or different sensor emission/detections can be used based on the location of the sensors relative to the user. According

to one example, a position detecting IMU, sensor, encoder, or the like can be positioned in the retention band **13.7-408** to detect the relative position of the band and generate a signal indicative of the position. The processor can then use that signal to determine the type of analysis to be performed on the data collected from the sensors **13.7-412**.

[2758] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.7-4A-13.7-4C** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.7-4A-13.7-4C**.

[2759] FIGS. **13.7-5** illustrates a perspective exploded view of an HMD **13.7-500**. The HMD **13.7-500** can be substantially similar to, and can include some or all of the features of, any of the HMDs described herein, such as HMD **13.7-100**, **13.7-200**, **13.7-300**, and **13.7-500**. The HMD **13.7-500** can include a display unit **13.7-504** and a retention band **13.7-508**. The retention band **13.7-508** can include an inner portion **13.7-507** that contacts a user's head when worn, and an outer portion **13.7-509** at least partially defining an exterior surface of the wearable device **13.7-500**. The outer portion **13.7-509** can be integrally formed with the inner portion **13.7-507**. In other words, the inner portion **13.7-507** and the outer portion **13.7-509** can be made from a single piece of material (shown separated here for simplicity). In some examples, the outer portion **13.7-509** and the inner portion **13.7-507** can be coupled together, for example using adhesive, thread, or other coupling techniques.

[2760] In some examples, the HMD **13.7-500** can include a first sensor array **13.7-512a** and a second sensor array **13.7-512b** (collectively referred to as “sensors **13.7-512**”). In some examples, the inner portion **13.7-507** can include a material that is transmissive to the sensors **13.7-512**. The sensors **13.7-512** can be positioned between the inner portion **13.7-507** and the outer portion **13.7-509**, such that the sensors **13.7-512** are positioned within the retention band **13.7-508**. In some examples, the inner portion **13.7-507** of the retention band **13.7-508** can include a perforated section adjacent the sensors **13.7-512**. The perforated section can include holes, gaps, or apertures formed in the material of the inner portion **13.7-507**.

[2761] The holes in the inner portion **13.7-507** can correspond to locations of the sensors **13.7-512** and can allow signals from the sensors, or the sensors themselves to pass through the holes. In other words, the sensors **13.7-512** can be exposed and accessible at the inner portion **13.7-507**, for example, to contact the user's head. In some examples, the sensors **13.7-512** are embedded or sandwiched between the inner portion **13.7-507** and the outer portion **13.7-509** of the retention band **13.7-508**. In some examples, the sensors **13.7-512** can be position adjacent a sensorally transparent or transmissive window **13.7-530** that allows for the transmission and reception of signals between the user's head and the sensors **13.7-512**. The sensors **13.7-512** can be electrically coupled to an electronic component **13.7-520** in the display unit **13.7-504**, such as a processor or battery, via one or more electrical connections **13.7-511**.

[2762] In some examples, the sensor arrays **13.7-512** can flexible and capable of bending or curving to according to a shape of the retention band **13.7-508** on the user's head. In some examples, the sensors **13.7-512** include flex cables and/or optical fibers or fiber optic cables that are woven into the fabric of the retention band **13.7-508**. In some examples, the retention band **13.7-508** can include conductive fibers woven into the fabric of the retention band **13.7-508**.

[2763] The retention band **13.7-508** can be made from a flexible material and can securely and snugly fit around a head of the user. The retention band **13.7-508** can be made from a woven fabric, leather, polymer, or any other material compatible with micro-perforations or having electromagnetic transmissive properties. In some examples, the retention band **13.7-508** can be made from silicone or thermoplastic polyurethane (TPU). In some examples, the retention band

13.7-508 can be made from compression molded materials, such as rubber. The sensors **13.7-512** can be integrated into the compression molded materials. In some examples, the retention band **13.7-508** is semi-rigid. For example, the retention band **13.7-508** can be rigid where the sensors **13.7-512** are located.

[2764] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.7-5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.7-5**.

[2765] FIGS. **13.7-6** shows a side or cross-sectional view of portion of a retention band **13.7-608**. The retention band **13.7-608** can be substantially similar to, and can include some or all of the features of, the retention bands described herein, such as retention band **13.7-108**, **13.7-208**, **13.7-308**, **13.7-408**, and **13.7-508**. The retention band **13.7-608** can include one or more sensors **13.7-612a**, **13.7-612b**, and **13.7-612c** (collectively referred to as “sensors **13.7-612**”). The sensors **13.7-612** can be partially or entirely embedded or encapsulated in the retention band **13.7-608**. The retention band **13.7-608** can include a first, interior surface **13.7-607** facing the user's head when donned, and a second, exterior surface **13.7-609** facing away from the user and defining an exterior of the retention band **13.7-608**. In some examples, the first surface **13.7-607** directly contacts or touches the user's head when the HMD is donned or being worn by the user.

[2766] The sensor **13.7-612a** can be positioned on the exterior surface **13.7-609** of the retention band **13.7-608**. In some examples, the first surface **13.7-607** can define an interior and the second surface **13.7-609** can be an exterior of the retention band **13.7-608**. Thus, the sensor **13.7-612a** can be positioned on an exterior of the retention band **13.7-608**. In some examples, the retention band **13.7-608** can include a sensor **13.7-612b** that is embedded, encapsulated, or otherwise surrounded by the retention band **13.7-608**. The retention band **13.7-608** can include, a sensor **13.7-612c** that is positioned such that a portion of the sensor **13.7-612c** is exposed through the inner surface **13.7-607**. In other words, the sensor **13.7-612c** can at least partially define an interior of the retention band **13.7-608** and can directly contact or touch the user's head. In some examples, the sensor **13.7-612c** can be partially surrounded or embedded in the retention band **13.7-608**. In some examples, the sensor **13.7-612c** is externally attached to the retention band **13.7-608**, positioned on the interior of the retention band **13.7-608**.

[2767] The sensors **13.7-612a** and **13.7-612b** can have corresponding sensor areas **13.7-625a**, **13.7-625b**, respectively. The sensor areas **13.7-625a**, **13.7-625b** can represent a field of view or cone of influence that is transparent to a signal emitted by the sensors **13.7-612a**, **13.7-612b**. In some examples, the field of view of the sensors **13.7-612** can be approximately 50 degrees. The sensors **13.7-612a**, **13.7-612b** can detect physiological, biological, and/or biometric changes of the user's body through corresponding sensor areas **13.7-625a**, **13.7-625b**. For example, the sensors **13.7-612a**, **13.7-612b** can detect changes to a user through the material of the retention band **13.7-608**. Other sensors may be added or substituted, as may be desired. In some examples, the sensor areas **13.7-625a**, **13.7-625b** can include a different material than the rest of the retention band **13.7-608**. For example, the sensor areas **13.7-625a**, **13.7-625b** can be transparent to certain signals from the sensors **13.7-612a**, **13.7-612b** or from the user, while the remainder of the retention band **13.7-608** is not transparent or transmissive to such signals.

[2768] In some instances, the closer positioning of sensor **13.7-612b** to the first surface **13.7-607** than the sensor **13.7-612a** can correspondingly reduce the field of view **13.7-625b**. In some examples, the sensor **13.7-612c** may be flush with the inner surface **13.7-607** and/or directly contacting a user's head. This is only one example of sensor depth variation, as a plurality of sensors can be disposed on the first surface **13.7-607** of the retention band **13.7-608**, or between the

first surface **13.7-607** and the second surface **13.7-609** of the retention band **13.7-608**.
[2769] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.7-6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.7-6**.

13.8: Conductive Fabric Architecture

[2770] In some examples, a head mountable device includes a display, a housing at least partially surrounding the display, a facial interface attached to the housing, and a cover positioned between the housing and the facial interface, the cover including a conductive fabric.

[2771] In some examples of the head-mountable device, the facial interface directly contacts a face of a user, the cover blocks ambient light, the conductive fabric elastically deforms in response to a movement of the cover, and an electrical property of the conductive fabric detectably changes in response to the movement.

[2772] In additional examples, the conductive fabric is elastic. Additionally, the conductive fabric can be interwoven with the cover. During use, the geometry of the conductive fabric can change in response to a user input. In other examples, the conductive fabric is configured to detect a change in a distance between the housing and the facial interface. In yet other examples, the conductive fabric forms a serpentine pattern. In some instances, the conductive fabric includes an array of parallel conductive threads arranged perpendicular to a direction of deformation of the cover, and a distance between two of the parallel conductive threads changes in response to a facial movement of a user. The conductive fabric can include a user input member.

[2773] In some examples, a wearable electronic device can include a display, a frame at least partially surrounding the display, and a light-blocking material attached to the frame, the light-blocking material including a conductive fabric.

[2774] In some examples the conductive fabric elastically deforms in response to a facial movement of a user. Additionally, a capacitance of the conductive fabric can detectably change in response to a deformation of the conductive fabric. In some examples, the conductive fabric includes or acts as a switch. In other examples, the conductive fabric electrically connects a first electronic component and a second electronic component. In other examples the conductive fabric directly contacts a user's skin. In some instances the conductive fabric includes a touch-sensitive input member.

[2775] In other embodiments a facial interface for a head-mountable device can include a facial contact, a conductive component incorporated into the facial contact, the conductive component configured to generate a signal based on a change in a capacitance of the conductive component, and a processor connected to the conductive component.

[2776] In some examples the facial contact is positioned adjacent a nasal region when the head-mountable device is worn by a user. In other examples, the conductive component is configured to detect a facial expression.

[2777] The following embodiments relate to a light seal of a head-mountable device including conductive fabric. As used herein “conductive fabric” can refer to the material or fabric into/onto which conductive elements are incorporated, or can also refer to the conductive elements themselves, separate from the material or fabric with which they are integrated. The conductive fabric can act as sensors, input members, and electrical interconnects to enable sensors and other electronic components to interact with one another. An input member can be any button, switch, capacitive surface, dial, slider, or other physically activated input system that can be used to provide an indication to a processor of a user's intent.

[2778] Conventional head-mountable devices equipped with sensors are utilized for various

purposes that detect limited user feedback, such as movement or positioning feedback, providing limited information in response to the user. These conventional sensors and electrical interconnects are often located in non-ideal and burdensome locations, taking up much needed space and unable to seamlessly integrated with the existing HMD architecture.

[2779] By contrast, the head-mountable device of the present disclosure includes a light seal with conduct fabric that collects user information, captures changes in user motion, such as facial movements, and can receive user input, such as touch input. The conductive fabric can accomplish this while being unobtrusively integrated with the light seal. A head-mountable device with such conductive fabrics accomplishes increased functionality without adding burdensome components or structures.

[2780] There are a number of different uses for “smart” conductive fabric described herein. For example, the disclosed conductive fabrics can be used for: detecting respiration patterns and amplitudes, receiving user input (such as touch/force inputs and playback controls), detecting facial expressions, detecting tension or force on a user's face from the facial interface and/or headband support, raising or lowering the temperature of the light seal, moisture detection, and low power “on-head” detection of when the HMD is being handled or worn by the user.

[2781] These and other embodiments are discussed below with reference to FIGS. **13.8-1A-13.8-7**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[2782] FIGS. **13.8-1A** illustrates a block diagram of a head-mountable device **13.8-100** including a frame **13.8-104**, a display **13.8-105**, a support **13.8-109**, and a light seal **13.8-112**. The display **13.8-105** can include one or more optical lenses or display screens in front of the eyes of a user. The display **13.8-105** can include a display for presenting an augmented reality visualization, a virtual reality visualization, or other suitable visualization. Additionally, the display **13.8-105** can be positioned at least partially in or on the frame **13.8-104**. Similarly, the light seal **13.8-112** can be connected to the frame **13.8-104**. In some examples, the light seal **13.8-112** includes the frame **13.8-104** (i.e., the frame **13.8-104** is part of the light seal **13.8-112**).

[2783] The light seal **13.8-112** can include electrical components (e.g., sensors **13.8-120**), a cover **13.8-113**, a facial interface **13.8-108**, and conductive fabric **13.8-116**. The frame **13.8-104** can be a housing of the display **13.8-105**. Further, the frame **13.8-104** can also be considered to be a part of or separate from the light seal **13.8-112**. As used herein, the term “light seal” can refer to a portion of the head mountable device **13.8-100** that engages or shields a user's face. In particular, the light seal **13.8-112** includes portions (e.g., the facial interface) that conform to, contact, or press against regions of the user's face.

[2784] The light seal **13.8-112** can include a pliant (or semi-pliant) facetrack or face engagement component that spans the forehead, wraps around the eyes, contacts other regions of the face (e.g., zygoma and maxilla regions), and bridges the nose. In addition, the light seal **13.8-112** can include various components forming a frame, structure, or webbing of a head-mountable device disposed between the display **13.8-105** and the user's skin, such as the cover **13.8-113**. The cover **13.8-113** can include a seal, environment seal, dust seal, air seal, etc. that is positioned between the gap between the display **13.8-105** and the user's face. The cover **13.8-113** can be a woven fabric that is non-rigid or deformable. The cover **13.8-113** can be elastically deformable. In some examples, the cover **13.8-113** can be a plastic, rubber, or polymer material. In some examples, the cover **13.8-113**

can be rigid.

[2785] The cover **13.8-113** can form an eye-box through which the user can view the display **13.8-105**. It will be appreciated that the term “seal” can include partial seals or inhibitors, in addition to complete seals (e.g., a partial light seal where some ambient light is blocked and a complete light seal where all ambient light is blocked when the head-mountable device **13.8-100** is donned).

[2786] The light seal **13.8-112** can be removably attached to the frame **13.8-104** and in electrical communication with the display **13.8-105**. The light seal **13.8-112** can include an electrical component **13.8-120**, such as a sensor **13.8-120**. The sensor **13.8-120** can collect user or environmental data, such as biometric information. The sensor **13.8-120** can transmit signals to the HMD, and more particularly to the display **13.8-105**. The sensor **13.8-120** can transmit signals to an output configured to perform an action in response to the information collected by the sensor **13.8-120**.

[2787] The light seal **13.8-112** can include conductive fabric **13.8-116**. In some examples, the conductive fabric **13.8-116** can include highly elastic copper threads that can compress and stretch while still maintaining electrical connectivity. The conductive fabric **13.8-116** can be elastically deformable (i.e., capable of temporary change in length, volume, or shape). In some examples, the conductive fabric **13.8-116** can include electrically conductive carbon fibers. In some examples, the conductive fabric **13.8-116** can collect biometric information of the user. More specifically, the biometric information of the user can manifest itself in movement of the user's face (e.g., nasal movements indicative of respiration or breathing patterns). These movement can cause a change in the conductive fabric (i.e., stretching or compressing). The degree or amount to which the conductive fabric is moved by the user's facial movements can be detectable by the conductive fabric and can be used to generate signals that can be analyzed by a processor of the HMD **13.8-100**. As described in greater detail below, the conductive fabric can be a thread, line, wire, plate, or any other structure that is capable of conducting electricity. The conductive fabric **13.8-116** can be integrated with the light seal **13.8-112**. For example, the conductive fabric **13.8-116** can be embedded, interwoven, or encapsulated with the cover **13.8-113** or facial interface **13.8-108**.

[2788] The display **13.8-105**, being in electrical communication with the light seal **13.8-112**, and more specifically with the conductive fabric **13.8-116**, can receive electrical communication and provide feedback to the user related to the readings of the conductive fabric **13.8-116** (e.g. visual feedback, audio feedback, haptic feedback, etc.).

[2789] As used herein, the term “sensor” refers to one or more different sensing devices, such as a camera or imaging device, temperature device, oxygen device, movement device, brain activity device, sweat gland activity device, breathing activity device, muscle contraction device, etc. In some examples, the sensor can sense biometric features including features of the autonomic nervous system. Some particular examples of sensors include an electrooculography sensor, electrocardiogram sensor, EKG sensor, heart rate variability sensor, blood volume pulse sensor, SpO2 sensor, compact pressure sensor, electromyography sensor, core-body temperature sensor, galvanic skin sensor, accelerometer, gyroscope, magnetometer, inclinometer, barometer, infrared sensor, global positioning system sensor, etc. Additional sensor examples can include, contact microphones (e.g., piezo based MEMS), bioelectrical activity sensors, UV exposure sensors, or particle sensors.

[2790] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.8-1A** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.8-1A**.

[2791] FIGS. **13.8-1B** shows a top partial view of a head-mountable device **13.8-100**. The head-mountable device **13.8-100** of FIGS. **13.8-1** can be substantially similar to, including some or all of

the features of, the head-mountable device **13.8-100** described in FIGS. **13.8-1A**. The HMD **13.8-100** can include a display (also referred to as a display unit, or housing) **13.8-105** and a retention band **13.8-109**. The display **13.8-105** can include any number of internal electronic components **13.8-107**. The HMD **13.8-100** can include a frame **13.8-104** (which can also be referred to as a housing) attached to the display **13.8-105**. In some examples, the display **13.8-105** includes an opaque, translucent, transparent, or semi-transparent screen, including any number lenses, for presenting visual data. The frame **13.8-104** can at least partially border one or more edges of the display **13.8-105**. The frame **13.8-104** can be attached to a cover **13.8-128** at one end of the cover **13.8-128**. At an opposite end, the cover **13.8-128** can form, or be attached to, a facial interface **13.8-108**. In some examples, the frame **13.8-104**, cover **13.8-113**, and facial interface **13.8-108** can together form the light seal **13.8-112**. It will be understood, however, that the light seal **13.8-112** can include fewer or more components than those listed or shown.

[2792] The HMD **13.8-100** can be worn on a user's head **13.8-102** such that the display **13.8-105** is positioned over the user's face and disposed over one or both of the user's eyes. The display **13.8-105** can be connected to the retention band **13.8-109** and/or the light seal **13.8-112**. In some examples, the retention band **13.8-109** can be positioned against the side of a user's head **13.8-102** and in contact therewith. In some examples, the retention band **13.8-109** can be at least partially positioned above the user's ear or ears. In some examples, the retention band **13.8-109** can be positioned adjacent to the user's ear or ears. The retention band **13.8-109** can extend around the user's head **13.8-102**. In this way, the display **13.8-105** and the retention band **13.8-109** can form a loop that can retain the wearable electronic device **13.8-100** on the user's head **13.8-102**. It should be understood, however, that this configuration is just one example of how the components of a modular wearable electronic device **13.8-100** can be arranged, and that in some examples, a different number of connector straps and/or retention bands can be included. Although the particular component **13.8-100** can be referred to as an HMD, it should be understood that the terms wearable device, wearable electronic device, HMD, HMD device, and/or HMD system can be used to refer to any wearable device, including smart glasses.

[2793] In some examples, the frame **13.8-104** is attached to a facial interface **13.8-108**. The facial interface **13.8-108** can contact a user's head and/or face. In some examples, the cover **13.8-113** can be a light blocking component that extends between the frame **13.8-104** and the facial interface **13.8-108**. The light blocking component can cover or surround a perimeter of the frame **13.8-104** and/or the facial interface **13.8-108**.

[2794] The cover **13.8-113** can be a cloth, fabric, woven material, plastic, rubber, or any other suitable opaque or semi-opaque material. In some examples, the cover **13.8-113** is flexible, having the ability to repeatedly stretch, compress, and deform. The cover **13.8-113** can be elastically or inelastically deformable. The facial interface **13.8-108** in combination with the cover **13.8-113** can block outside light and limits the peripheral view of the user **13.8-102**. In some examples, the cover **13.8-113** and the facial interface **13.8-108** is the same or a unitary component. As will be discussed in greater detail below, the light seal **13.8-112** can include a conductive fabric that is able to serve multiple functions and provide added benefits and functionality to the HMD **13.8-100**.

[2795] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.8-1B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.8-1B**.

[2796] FIGS. **13.8-2** shows a bottom perspective view of select components of a light seal **13.8-212**. The light seal **13.8-212** can be substantially similar to, including some or all of the features of, the light seals described herein, such as light seal **13.8-112**. The light seal **13.8-212** can be

implemented on an HMD, such as HMD **13.8-100**.

[2797] The light seal **13.8-212** can be integrated with conductive fabrics **13.8-216a**, **13.8-216b**, **13.8-216c**, **13.8-216d**, and **13.8-216e** (collectively referred to as conductive fabric **13.8-216**). The conductive fabric **13.8-216** or thread can be positioned at various locations on the light seal **13.8-212**. For example, the conductive fabric **13.8-216** can be positioned on or in the cover (e.g., embedded, encapsulated, or interwoven into the cover)

[2798] In some examples, the conductive fabric **13.8-216** can be positioned on the facial interface **13.8-208** of the light seal **13.8-212**. In one example, the conductive fabric **13.8-216** can be in direct contact with a user, for example, touching a user's forehead, cheek, nose, temple region, back of the head, or at any location where the HMD contacts the user.

[2799] The light seal **13.8-212** can include a nasal region **13.8-218** that is configured to contact a user's nose. The nasal region **13.8-218** can be considered part of the facial interface **13.8-208** and/or the cover **13.8-213**, or can be its own separate section of the light seal **13.8-212**. The nasal region **13.8-218** can include two sections corresponding to each side of a user's nose. As illustrated in FIGS. **13.8-2**, conductive threads **13.8-216a** can be positioned in the nasal region **13.8-218**. In some examples, the conductive thread **13.8-216a** is a single continuous wire that spans the nasal region **13.8-218** (i.e., both sides of the nose and across the bridge of the nose). In some examples, each side of the nasal region **13.8-218** of the light seal **13.8-212** can include separate or distinct conductive threads **13.8-216a**.

[2800] The conductive thread **13.8-216a** can be intentionally located to detect movements of the user's nose. For example, when a user breathes, the nose can move a certain amount based on the intensity or amplitude of the breath. This movement can in turn cause the nasal region **13.8-218** of the light seal **13.8-212** to change shape or move (e.g., compress or stretch). The conductive thread **13.8-216a** can be ideally positioned such that as the nasal region **13.8-218** moves in response to nasal movements. The conductive threads **13.8-216** are likewise stretched or compressed by nasal movements.

[2801] The movement of the conductive thread **13.8-216a** can result in a detectable change in the conductivity and/or capacitance of the conductive threads **13.8-216**, which can be used to infer respiration or facial expression data. As used herein “detectable” can refer to a change in a signal generated by the conductive fabric that is able to be recognized and analyzed by a processor. In some examples, the conductive threads **13.8-216** can directly contact the user's nose and can be impacted by the capacitance of the user's nose. In some examples, the conductive threads **13.8-216a** are embedded, encapsulated, or interwoven into the material of the nasal region (i.e., the fabric of the cover **13.8-213** and/or the facial interface **13.8-208**) such that the conductive threads **13.8-216a** do not directly contact the user's skin, but still respond to movements of the user's face relative to the light seal **13.8-212**.

[2802] The light seal **13.8-212** can include a forehead region that is configured to contact a user's forehead when the HMD is donned by the user. The forehead region can be impacted (e.g., moved by) movements of the user's forehead and/or eyebrows. As illustrated in FIGS. **13.8-2**, conductive threads **13.8-216d** can be positioned in the forehead region of the cover **13.8-213**. In some examples, the conductive thread **13.8-216d** is a single continuous wire that spans a substantial portion of the forehead region. In some examples, the forehead region of the cover **13.8-213** includes multiple conductive threads **13.8-216d** (e.g., one over each eyebrow).

[2803] The conductive thread **13.8-216d** can be intentionally located to be able to detect movements of the user's forehead and/or eyebrows. For example, a user's facial expression can be inferred from the position and movements of the forehead. This movement can in turn result in movements of the light seal **13.8-212**. The conductive thread **13.8-216d** can be ideally positioned such that as the forehead moves, the conductive thread **13.8-216d** are likewise stretched or compressed. This movement of the conductive thread **13.8-216d** can result in a detectable change in an electrical property (e.g., conductivity and/or capacitance) of the conductive thread **13.8-216d**,

which can be used to infer biometric information, such as facial expressions. In some examples, the conductive threads **13.8-216d** can directly contact the user's skin and can be excited by the capacitance of the user's skin. In some examples, the conductive threads **13.8-216d** are embedded, encapsulated, or interwoven into the material of the nasal region (i.e., the fabric of the cover **13.8-213** and/or the facial interface **13.8-208**) such that the conductive threads **13.8-216d** do not directly contact the user's skin, but still respond to movements of the user's face relative to the light seal **13.8-212**.

[2804] The light seal **13.8-212** can include a facial interface **13.8-208** that is configured to directly contact and conform against the user's face when the HMD is donned by the user. The facial interface **13.8-208** can be a distinct components from the cover **13.8-213** or can be a single unitary piece of the cover **13.8-213**. For example, the facial interface **13.8-208** can include foam that is attached to the cover **13.8-213**. The facial interface can be impacted (e.g., moved by) movements of the user's face. As illustrated in FIGS. **13.8-2**, conductive threads **13.8-216e** can be positioned in/on the facial interface **13.8-208**.

[2805] The conductive thread **13.8-216e** can be intentionally located to be able to detect movements of the user's face. Movement of the user's face can transfer to movement of the facial interface **13.8-208** and movements of the conductive thread **13.8-216e**, resulting in a detectable change in an electrical property (e.g., conductivity and/or capacitance) of the conductive thread **13.8-216e**. In some examples, the conductive threads **13.8-216e** can directly contact the user's skin and can be excited by the capacitance of the user's skin. In some examples, the conductive threads **13.8-216e** are embedded, encapsulated, or interwoven into the material of the facial interface such that the conductive threads **13.8-216e** do not directly contact the user's skin, but still respond to movements of the user's face relative to the light seal **13.8-212**. In some examples, the conductive fabric **13.8-216e** can be interwoven into the facial interface **13.8-208** while still being exposed and capable of contacting the user's face.

[2806] The light seal **13.8-212** can include one or more electronic components **13.8-220**. The electronic component **13.8-220** can be any form of module, sensor, unit, input, or output that performs a function of the light seal **13.8-212** and/or the HMD. For example, the component **13.8-220** can be a battery, processor, haptic actuator, LED, display, speaker, biometric sensor, etc.

[2807] The component **13.8-220** can be positioned at various locations on the light seal **13.8-212**. The component **13.8-220** can be positioned on or in the cover (e.g., embedded, encapsulated, or interwoven into the cover). In some examples, the component **13.8-220** can be positioned on the facial interface **13.8-208** or the frame of the light seal **13.8-212**. In one example, the component **13.8-220** can be in direct contact with a user, for example, touching a user's forehead, cheek, nose, temple region, back of the head, or at any location where light seal **13.8-212** contacts the user's head.

[2808] In some examples, a conductive fabric **13.8-216c** can be electrically connected to the component **13.8-220**. The conductive fabric **13.8-216c** can be an interconnect to electrically connect the components with another component of the HMD. In some examples, the component **13.8-220** is a processor configured to analyze the signals transmitted/generated by the conductive thread **13.8-216c**.

[2809] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.8-2** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.8-2**.

[2810] FIGS. **13.8-3** shows a top view of a head-mountable device (HMD) **13.8-300**. The HMD **13.8-300** can be substantially similar to, including some or all of the features of, the devices

described herein, such as HMD **13.8-100** and light seal **13.8-212**. In some examples, the HMD **13.8-300** can include an electronic module **13.8-307**. The electronic module **13.8-307** can be positioned on or housed in the display unit **13.8-305**. The electronic module **13.8-307** can be a battery, a processor, a display, a camera, or any other electronic component. In some examples the electronic module **13.8-307** can be attached to the frame **13.8-304**. The electronic module **13.8-307** can be in electrical communication with an electronic component **13.8-320** attached to the light seal **13.8-312**.

[2811] In some examples, the electronic component **13.8-307** and the electronic module **13.8-320** can be electrically connected by a conductive fabric **13.8-316a** that runs through the cover **13.8-313** of the light seal **13.8-312**. In some examples, the conductive fabric **13.8-316a** runs through or across the headband or support **13.8-309**. In some examples, the conductive fabric **13.8-316a** runs through or across the frame **13.8-304**. In this manner, the conductive fabric **13.8-316a** serves as an interconnect (i.e., electrical connection) between the electronic component **13.8-320** on the light seal **13.8-312** and the electronic module **13.8-307**. In some examples, the detections of the conductive fabrics **13.8-216** can be combined with other sensors on-board or remote from the HMD **13.8-300** to gather data in a sensor fusion manner. For example, the conductive threads in combination with user facing cameras can be used for facial expression detection.

[2812] In some examples, the HMD **13.8-300** can include a conductive fabric **13.8-316b** incorporated onto and/or into the support **13.8-309** of the HMD **13.8-300**. The support **13.8-309** can be a headband having flexible sections. The conductive fabric **13.8-316b** can be configured to stretch or compress in response to the headband **13.8-309** stretching or compressing. In some examples, the conductive fabric **13.8-316b** can generate signals indicative of the tightness or position of the headband. In some examples, a user can be notified based on the signals generated by the conductive fabric **13.8-316b**. The headband **13.8-309** can be automatically adjusted based on the signals generated by the conductive fabric **13.8-316b**.

[2813] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.8-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.8-3**.

[2814] FIGS. **13.8-4A** shows a conductive fabric **13.8-416** in a neutral or rest state. In some examples, the conductive fabric **13.8-416** is positioned in a serpentine pattern (curving, undulating, or winding in a repeated pattern). As illustrated in FIGS. **13.8-4A**, in a rest or neutral state (i.e., when no external forces are acting on the conductive fabric **13.8-416**, the nodes or peaks of the conductive fabric **13.8-416** can be a distance $d_{sub.1}$ from each other. The distance $d_{sub.1}$ can be visualized as the wavelength or distance between parallel sections of the conductive fabric **13.8-416**.

[2815] By monitoring the electrical current passing through the conductive fabric, a processor can determine the amount and direction of strain the fabric is experiencing. In some examples, the conductive fabric **13.8-416** arranged in a serpentine pattern can be excited using a 100-140 kHz sine or square wave to generate a capacitive field. The system can then monitor for changes in the alternating current (AC) component over time (e.g., to determine respiration data). In some examples, a frequency sweep is performed to acquire a frequency/phase dependent signal.

[2816] In some examples, the geometry or shape of conductive threads **13.8-416** can be tuned depending on what is being monitored and the required sensitivity levels. In some examples, the system primarily monitors the change in capacitance of the conductive threads **13.8-416**.

[2817] FIGS. **13.8-4B** shows the conductive fabric **13.8-416** in a compressed state. The compressed state may be the result of the light seal (such as the cover and/or facial interface) being acted upon

by the user. In the compressed state, the distance separating the peaks or the parallel sections of the conductive fabric is a distance $d_{sub.2}$. The distance $d_{sub.2}$ can be less than the distance $d_{sub.1}$. As a result, the conductive properties (e.g., capacitance) of the conductive fabric can change in a detectable manner. Thus, it can be determined that the conductive fabric **13.8-416** is compressed and even the degree to which the conductive fabric **13.8-416** is compressed. From this, biometric information, facial expression, breathing patterns, user input, and other user information can be determined.

[2818] FIGS. **13.8-4C** shows the conductive fabric **13.8-416** in a stretched state. The stretched state may be the result of the light seal (such as the cover and/or facial interface) being acted upon by the user. In the stretched state, the distance separating the peaks or the parallel sections of the conductive fabric is a distance $d_{sub.3}$. The distance $d_{sub.3}$ can be greater than the rest distance $d_{sub.1}$. As a result, the conductive properties of the conductive fabric change in a detectable manner as the area and geometry of the conductive fabric **13.8-416** is modified. Thus, it can be determined that the conductive fabric **13.8-416** is stretched and even the degree to which the conductive fabric **13.8-416** is stretched. From this, biometric information, facial expression, breathing patterns, user input, and other user information can be determined.

[2819] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.8-4A-13.8-4C** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.8-4A-13.8-4C**.

[2820] FIGS. **13.8-5A** shows a conductive component **13.8-516a** on an exterior **13.8-517** of a cover **13.8-513**. The conductive component **13.8-516a** can be substantially similar to, including some or all of the features of, the conductive fabrics described herein, such as conductive fabric **13.8-216**, **13.8-316**, and **13.8-416**. As described herein, a light seal can include a cover **13.8-513** that extends between the HMD and the user's face. The cover **13.8-513** can define a hollow, aperture, or opening that forms an internal volume or eye-box. The cover **13.8-513** can therefore include an internal or interior surface **13.8-515** that defines the internal volume, and an exterior surface **13.8-517** that defines an exterior of the light seal.

[2821] As illustrated in FIGS. **13.8-5A**, the conductive component **13.8-516a**, can be positioned on the exterior surface **13.8-517**. In some examples, the conductive component **13.8-516a** at least partially defines the exterior of the light seal. In some examples, the conductive component **13.8-516a** is accessible to the user from the exterior of the HMD. For example, the conductive component **13.8-516a** can be responsive to a user's touch or deformation of the conductive component **13.8-516a**. In this manner, a user who is wearing the HMD can control or provide input by touch (e.g., tapping or sliding a finger) or physically manipulation (e.g., deforming) the conductive component **13.8-516a** with their hand. The conductive component **13.8-516a** can be attached to the cover **13.8-513** using any suitable methods, including but not limited to adhesion, molding, mechanical fasteners, friction, weaving, magnets, etc.

[2822] FIGS. **13.8-5B** shows a conductive component **13.8-516b** interwoven into the cover **13.8-513**. The conductive component **13.8-516b** can be substantially similar to, including some or all of the features of, the conductive fabrics described herein, such as conductive fabric **13.8-216**, **13.8-316**, **13.8-416**, and **13.8-516a**. As described herein, a light seal can include a cover **13.8-513** that extends between the HMD and the user's face. The cover **13.8-513** can be made from a fabric or material having a thickness within which the conductive component **13.8-516b** can be deposited.

[2823] As illustrated in FIGS. **13.8-5B**, the conductive component **13.8-516b**, can be positioned in (e.g., surrounded, encapsulated, embedded, interwoven, etc.) the cover **13.8-513**. In some examples, the conductive component **13.8-516b** at least partially defines the exterior of the light

seal. In some examples, at least a portion of the conductive component **13.8-516b** is accessible to the user from the exterior of the HMD, even while being woven into the cover **13.8-513**. For example, the conductive component **13.8-516b** can be responsive to a user's touch or deformation of the conductive component **13.8-516b**. In this manner, a user who is wearing the HMD can control or provide input by touch or physically manipulation the conductive component **13.8-516b** (e.g., with their hand). The conductive component **13.8-516b** can be attached to the cover **13.8-513** using any suitable methods, including but not limited to adhesion, molding, mechanical fasteners, friction, weaving, magnets, etc. Further, it will be understood that as the cover **13.8-513** is deformed (e.g., by movements of the user's face) the embedded or interwoven conductive component **13.8-516b** can likewise be deformed. This deformation change the conductive properties of the conductive component **13.8-516b** is a detectable manner, which can then cause the HMD to perform an action.

[2824] FIGS. **13.8-5C** shows a conductive component **13.8-516c** on an interior **13.8-515** of the cover **13.8-513**. The conductive component **13.8-516c** can be substantially similar to, including some or all of the features of, the conductive fabrics described herein, such as conductive fabric **13.8-216**, **13.8-316**, **13.8-416**, **13.8-516a**, and **13.8-516b**. As described herein, a light seal can include a cover **13.8-513** that extends between the HMD and the user's face. The cover **13.8-513** can define a hollow, aperture, or opening that forms an internal volume or eye-box. The cover **13.8-513** can therefore include an internal or interior surface **13.8-515** that defines the internal volume, and an exterior surface **13.8-517** that defines an exterior of the light seal.

[2825] As illustrated in FIGS. **13.8-5C**, the conductive component **13.8-516c**, can be positioned on the interior surface **13.8-515**. In some examples, the conductive component **13.8-516c** at least partially defines the interior surface of the light seal. In some examples, the conductive component **13.8-516c** is accessible to the face of the user from the interior of the light seal. For example, the conductive component **13.8-516c** can be responsive to a user's facial touch or movements. The conductive component **13.8-516c** can be attached to the cover **13.8-513** using any suitable methods, including but not limited to adhesion, molding, mechanical fasteners, friction, weaving, magnets, etc.

[2826] FIGS. **13.8-5D** shows free-floating conductive components. In some examples, the light seal includes a continuous conductive plate **13.8-519** free-floating (i.e., not anchored) in/on the cover **13.8-513** in a position that cover, overlays, overlaps, or is adjacent to conductors **13.8-521** deposited in the cover **13.8-513**. In this manner, the deposited conductors **13.8-521** can freely move and still provide signals representative of the deformations of the cover **13.8-513**, even if not galvanically connected to the plate **13.8-519**. It will be understood that a light seal can include multiple conductive components, conductive fabrics, and plates. For example, a cover **13.8-513** can include one or more of the conductive component arrangements described with reference to FIGS. **13.8-5A-13.8-5D**. In some examples, the conductive components includes metal material that is deposited (e.g., using laser direct structuring (LDS) methods) onto fabric of the cover and/or facial interface. In some examples, the conductive components could be a combination of rigid and flexible conductive materials.

[2827] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.8-5A-13.8-5D** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.8-5A-13.8-5D**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.8-5A-13.8-5D**.

[2828] FIGS. **13.8-6** shows a side perspective view of a light seal **13.8-612**. The light seal **13.8-612** can be substantially similar to, including some or all of the features of, any of the light seals

described herein, such as light seal **13.8-112**, **13.8-212**, and **13.8-312**. The light seal **13.8-612** can include a frame **13.8-604** configured to be attached to a display unit, a facial interface **13.8-608** configured to contact and conform to a user's face, and a cover **13.8-613** extending between the frame **13.8-604** and the facial interface **13.8-608** and establishing a light-blocking component or shielding.

[2829] The cover **13.8-613** can include a first conductive fabric **13.8-616a** that is connected to the cover **13.8-613** in a fashion described herein. The first conductive fabric **13.8-616a** can include a wire or thread arranged in a generally serpentine pattern or in a zig-zag pattern. In some examples, the first conductive fabric **13.8-616a** can include a geometry that is responsive to (i.e., prone to move) force in the vertical direction (as oriented in FIGS. **13.8-6**).

[2830] The configuration of the conductive fabric **13.8-616a** can be ideal for receiving user input. For example, the user can squeeze or pinch the cover **13.8-613** along the vertical axis to provide input to the system. In some examples, the conductive fabric **13.8-616a** can be configured to detect an inward force, orthogonal to the surface plane of the side of the cover **13.8-613**. For example, the conductive fabric **13.8-616a** can detect the user pushing the cover inward with their finger (e.g., toward the user's temple).

[2831] In some examples, the cover **13.8-613** can include a second conductive fabric **13.8-616b** that is connected to the cover **13.8-613** in a fashion described herein. The second conductive fabric **13.8-616b** can be oriented substantially perpendicular to the first conductive fabric **13.8-616a**. In other words, a longitudinal axis of the first conductive fabric **13.8-616a** can be perpendicular to the longitudinal axis of the second conductive fabric **13.8-616b**. It will be understood that the light seal **13.8-612** can include one of or both the first conductive fabric **13.8-616a** and the second conductive fabric **13.8-616b**. The conductive fabric **13.8-616b** can include a wire or thread arranged in a generally serpentine pattern or in a zig-zag pattern. In some examples, the conductive fabric **13.8-616b** can include a geometry that is responsive to a force in the horizontal direction (as oriented in FIGS. **13.8-6**).

[2832] The configuration of the conductive fabric **13.8-616b** can be ideal for receiving user input. For example, the user can squeeze or pinch the cover **13.8-613** along the horizontal axis to provide input to the system. In some examples, the conductive fabric **13.8-616b** is sensitive to changes in the distance between the frame **13.8-604** and the facial interface **13.8-608**. A user can provide input by pressing the display unit or frame **13.8-604** toward the facial interface **13.8-608**. In some examples, the conductive fabric **13.8-616b** can be configured to detect an inward force, orthogonal to the surface plane of the side of the cover **13.8-613**. For example, the conductive fabric **13.8-616b** can detect the user pushing the cover inward with their finger (e.g., toward the user's temple). In some examples, the light seal **13.8-612** can include multiple serpentine patterned conductive threads that run parallel to one another.

[2833] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.8-6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.8-6**.

[2834] FIGS. **13.8-7** shows a bottom perspective view of a light seal **13.8-712**. The light seal **13.8-712** can be substantially similar to, including some or all of the features of, any of the light seals described herein, such as light seal **13.8-112**, **13.8-212**, **13.8-312**, and **13.8-612**. The light seal **13.8-712** can include a cover **13.8-713** and a first and second nasal sections **13.8-718a** and **13.8-718b**, respectively (collectively referred to as nasal sections **13.8-718**). In some examples, the light seal can include an array of conductive threads **13.8-716a** and **13.8-716b** (collectively referred to as the array of conductive threads **13.8-716**). The array of conductive thread **13.8-716** can be arranged

parallel to one another such that changes in a distance separating the parallel threads is detectable and can be used to infer nasal movements and respiration data, such as instantaneous estimation of breath amplitude. The parallel conductive threads **13.8-716** can be arranged perpendicular to the direction of travel or deformation of the nasal regions **13.8-718**.

[2835] For example, it can be seen in FIGS. **13.8-7** that the distances separating the first array of conductive threads **13.8-716a** is less than the distances between the threads of the second array of conductive threads **13.8-716b**. It can therefore be determined (e.g., by a processor) that the first nasal section **13.8-718a** is being compressed and/or the second nasal section **13.8-718b** is being stretched.

[2836] It will be understood that location of the array of conductive threads **13.8-716** is not limited to the nasal sections **13.8-718**. For example, any of the conductive fabrics described herein can instead be arranged as a parallel array of conductive threads.

[2837] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.8-7** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.8-7**.

13.9: Facial Interface Having Integrated Health Sensors

[2838] The following disclosure relates to a facial interface of a head-mountable device. More particularly, the present embodiments relate to a facial interface of a head-mountable device including a pressure sensor assembly used for acquiring biometric information. It will be understood the head-mountable device (HMD) in the present disclosure can apply to AR/VR head sets and also to smart computer glasses. In some examples, a processor causes a component of the HMD to perform an action in response to the biometric information collected by the sensor(s). As used herein “an action” or “perform an action” can refer to any electrical or mechanical act causing a change in one or more of the components of the HMD. For example, an action can include displaying a notification for the user and/or activating or deactivating one or more components.

[2839] In some examples, a sensor can collect biometric information from the nose and/or forehead of the user wearing the device, including when they are performing an activity, like exercising. The biometric information can be related to pulse, respiration, facial expression, heart rate, blood pressure, changes in the cartilage, chewing, or any other pertinent data. In some examples, a combination of sensors (sensor fusion) can be used to determine the biometric information. For example, a facial expression can be determined using both the pressure-based MEMS sensor described herein as well as an internal facing camera of the HMD. Further, in response to the collected data, the system can perform an action, for example, notifying the user of the collected data.

[2840] Head-mountable devices equipped with sensors are utilized for various purposes that detect user feedback, such as positioning or movement feedback, providing limited information in response to a user. For example, a user may have an increase in respiration or heart rate while performing a movement or action. Conventional head-mountable devices are not appropriately equipped to capture all the desired biometric information.

[2841] In contrast, the head-mountable devices of the present disclosure include a facial interface that can be incorporated or integrated with sensor(s), such as a pressure sensor assembly, configured to collect user data, such as biometric information, including pulse, respiration, facial expressions, heart rate, blood pressure, etc. By capturing the user data, the pressure sensor assembly of the facial interface can provide increased and improved feedback to the user. A head-mountable device with sensors monitoring user biometrics or health feedback creates a highly customized user experience, unlike the sensors of a conventional head-mountable device, which are

unable to consider the user experience to such a degree.

[2842] Pressure sensor assemblies positioned on the facial interface can be used to create a customized user experience. The head-mountable device of the present disclosure can include pressure sensors to measure a user's response or engagement via indicators, such as pulse, respiration, facial expressions, heart rate, blood pressure, etc. Additionally, the sensor data can be used to monitor user fatigue, facial expressions, change in a user state, or obtain activity-specific metrics.

[2843] The head-mountable device can include sensors that are integrated on/in the facial interface in a variety of different ways. For example, the head-mountable device of the present disclosure can implement a first facial interface with a first sensor assembly being interchangeable, along with a second facial interface with a second sensor assembly. In some examples, the removeably attached sensor assembly can correspond to a different user activity, such as exercise, health, clinical settings, learnings activities, etc.

[2844] These and other embodiments are discussed below with reference to FIGS. **13.9-1** through **13.9-6**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[2845] FIGS. **13.9-1** illustrates a block diagram of a head-mountable device **13.9-100** including a frame or housing **13.9-116**, a display **13.9-108**, a facial interface **13.9-112**, and a support **13.9-123**. The display **13.9-108** can include one or more optical lenses or display screens in front of the eyes of a user. The display **13.9-108** can include a screen or display unit for presenting an augmented reality visualization, a virtual reality visualization, or other suitable visualization to a user.

Additionally, the display **13.9-108** can be positioned in or on the housing **13.9-116**. Similarly, a facial interface **13.9-112** can be connected to the housing **13.9-116**. In some examples, the housing **13.9-116** can support or house a variety of electronic components, including the display **13.9-108**.

[2846] The facial interface **13.9-112** can include one or more sensors (e.g., pressure sensors) **13.9-124**, support attachments **13.9-132**, display attachments **13.9-128**, and feedback or output modules **13.9-136**. The sensors **13.9-124** can be removably attached to the facial interface **13.9-112**. As used herein, the terms “facial interface,” “engagement interface,” or “light seal” refer to a portion of the head mountable device **13.9-100** that engages (i.e., contacts or conforms to) a user's face.

[2847] In particular, the facial interface **13.9-112** can include portions of a head-mountable device that conform to, or press against, regions of a user's face. The facial interface **13.9-112** can be positioned between the display **13.9-108** or frame **13.9-116** and the user's face. In some examples, the facial interface **13.9-112** can include a pliant (or semi-pliant) facetrack or lumen that spans the forehead, wraps around the eyes, contacts other regions of the face (e.g., zygoma and maxilla regions), and bridges the nose.

[2848] In addition, the facial interface **13.9-112** can include various components forming a frame, structure, housing, or webbing of the head mountable device **13.9-100** disposed between the display **13.9-108** and the user's skin. In some examples, the facial interface **13.9-112** can include a seal (e.g., a light seal, environment seal, dust seal, air seal, etc.). As used herein, the term “seal” can include partial seals or inhibitors, in addition to complete seals (e.g., a partial light seal where some ambient light is blocked and a complete light seal where all ambient light is blocked when the head-mountable device **13.9-100** is donned). The facial interface **13.9-112** can be removably attached to the housing **13.9-116** and can be in electrical communication with the display **13.9-108**.

[2849] The sensor(s) **13.9-124** of the facial interface **13.9-112** can collect biometric information, such as the user's vital sign (including pulse data, respiration data, and blood pressure). The pressure sensor assembly **13.9-124** can generate a signal based on the collected user information and transmit the signal to a processor that can cause an output **13.9-136** to perform an action in response to the signal (i.e., in response to the biometric information collected). For example, a user can perform a rigorous activity, such as lifting weights or working out while wearing the HMD **13.9-100**. During such activity, the user's heart rate or other vital signs can elevate or change, being detectable by the sensor **13.9-124**.

[2850] The sensor **13.9-124** can generate one or more signals based on the received input. The sensor **13.9-124** can transmit the signal to one or more components of the HMD **13.9-100** (e.g., to the display **13.9-108** and/or the output **13.9-136**). The display **13.9-108**, being in electrical communication with the facial interface **13.9-112**, can receive electrical communication and provide feedback to the user related to their biometric readings (e.g., visual feedback, audio feedback, haptic feedback, etc.). Feedback can include determining when a user needs to take a break, or when the difficulty of an activity needs to be lowered or raised, the output **13.9-136** can include scheduling various activities for the user or recommending adjustment of the HMD **13.9-100**.

[2851] As used herein, the term “sensor,” “sensors,” “pressure sensor assembly,” or “sensor assembly” can refer to one or more sensing devices, such as a MEMS pressure sensor. In some examples, the HMD **13.9-100** includes a pressure sensing assembly having a deformable pad that deflects in response to a movement of a user's face, creating an increase or decrease in internal pressure of a pressure channel (e.g., a fluid volume connecting a deformable pad and a pressure sensors in fluid communication with the deformable pad) that changes in response to the deflection. The change in internal pressure caused by a movement in the user's face can be detected by a pressure sensor connected fluidly connected to the channel and attached to the deformable pad.

[2852] In some examples, the sensor is capable of detecting a deflection of the deformable pad as small as 10 microns. In some examples, the sensor is capable of detecting a deflection of the deformable pad that is less than as 10 microns. In some examples, the sensor is capable of detecting a deflection of the deformable pad as small as 20 microns. In some examples, the sensor is capable of detecting a deflection of the deformable pad as small as 50 microns. In some examples, the sensor is capable of detecting a deflection of the deformable pad as small as 100 microns. In some examples, the sensor is capable of detecting a deflection of the deformable pad greater than 100 microns. In some examples, the sensor is capable of detecting a deflection of the deformable pad that is approximately 100 nanometers or less.

[2853] The term “pressure sensor” or “sensor” used within the context of this application can refer to a sensor capable of sensing pressure, such as a compact based microphone (e.g., a MEMS sensor) with a sensitivity of about 2.5 millibar. In some examples, the pressure-based MEMS sensor can have a sensitivity of about 250 Pascals. In some examples, the pressure-based MEMS sensor can include an electrical circuit, such as a Winston bridge, to detect nano-metric deflections of the deformable pad. A pressure sensor or a sensor used for this application can be sensitive enough to detect a pulse, respiration, or other pressure related feedback related to a user.

[2854] In some instances, the change in pressure detected in a pressure sensor or sensor can generate an electrical signal capable of being interpreted by a controller/processor as a facial expression. In some examples, the pressure assembly is tuned to capture a specific pressure range applicable to a certain activity requiring other sensitivity values including approximately 1 millibar, 1.5 millibar, 3 millibar, or 4.5 millibar. When the pressure sensor is paired with another pressure sensor, a rate can be captured, such as blood volume, pulse rate, or other rates applicable and interpretable from the cardiovascular system.

[2855] In some examples, certain sensors can be used to assess stress and emotion through changes in pressures via direct contact with a user's face. The HMD can then provide feedback or output

related to the detected stress and emotion. In some examples, the sensors can operate through coin cell battery or Bluetooth connectivity. In some examples, the sensors are powered by a primary battery of the HMD.

[2856] The sensors described herein can allow for observations of the cardiovascular system and respiration, to observe relaxation and stress indicators, mental health, medical treatments, etc. Using the disclosed sensors integrated on the facial interface, physicians and care takers could have live feedback of biometrics. Use cases can include fitness settings, user content, workplace, telepresence, clinical, education, training, pain, therapy, etc. In some examples, the sensors can be used to capture facial expressions. This is particularly relevant given the user's face is covered by the HMD when worn. For example, the HMD could include a micro electro-mechanical system (MEMS) to detect changes in blood vessel diameter, blood pressure, pulse, heart rate, respiration, and detecting facial expressions.

[2857] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.9-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.9-1**.

[2858] FIGS. **13.9-2A** illustrates a head-mountable device **13.9-200**. The head-mountable device **13.9-200** can be substantially similar to, including some or all of the features of, the head-mountable devices described herein, such as head-mountable device **13.9-100**. In some examples, the head-mountable device **13.9-200** includes a controller **13.9-213** (e.g., a sensor controller). While the controller **13.9-213** is illustrated as being positioned in a support (e.g., a support arm) **13.9-223**, the position of the controller **13.9-213** is not limited to the support **13.9-223**, and the controller **13.9-213** can be positioned in/on the facial interface **13.9-212** or in the HMD display housing.

[2859] The controller **13.9-213** can include a processor and a memory device storing computer-executable instructions that, when executed by the processor, cause the controller to receive biometric data from one or more pressure sensor assemblies **13.9-224**, transmit a signal based on the sensor data, and generate a signal to cause a component to perform an action in response to the received signal.

[2860] The controller **13.9-213** can include one or more processors (e.g., a system on chip, integrated circuit, driver, microcontroller, application processor, crossover processor, etc.). Further, the controller **13.9-213** can include one or more memory devices (e.g., individual nonvolatile memory, processor-embedded nonvolatile memory, random access memory, memory integrated circuits, DRAM chips, stacked memory modules, storage devices, memory partitions, etc.). In certain implementations, the controller **13.9-213** is communicatively coupled to a power source (e.g., a battery).

[2861] In some examples, the controller **13.9-213** stores sensor data received from the sensor assembly **13.9-224** in the memory. The controller **13.9-213** can receive and/or transmit signals based on sensor data. For example, as will be described below, the controller **13.9-213**, by way of the processor and memory, can transmit a signal to the display **13.9-208** based on the sensor data from the pressure sensor (e.g., causing the display **13.9-208** or the head-mountable device **13.9-200** to perform an action, such as present a certain message, power off, react to biometric feedback, etc.).

[2862] The controller **13.9-213** can perform any number of different functions. For example, the memory device can store computer-executable instructions that, when executed by the processor, cause the controller to receive sensor data from the sensor assemblies **13.9-224** and transmit a signal based on the sensor data. For instance, the controller **13.9-213** can transmit a sensor signal to

a display **13.9-208**.

[2863] In response to the controller **13.9-213** the display **13.9-208** can perform a wide variety of actions, including power off or power on, react to a user generated facial expression, present a digital notification (e.g., a user-generated notification, a push notification, a context-generated notification, a system-generated notification, a smart notification, etc.). In some examples, the memory device can store computer-executable instructions that, when executed by the processor, cause the controller **13.9-213** to receive biometric data from the sensor assemblies **13.9-224**, transmit a signal based on the sensor data, and perform an action in response to the signal.

[2864] As illustrated, the head-mountable device (e.g., a wearable electronic device) **13.9-200** can include the display **13.9-208**, the facial interface **13.9-212**, and the pressure sensor(s) **13.9-224** coupled to the facial interface **13.9-212**. The pressure sensor assembly **13.9-224** can be embedded, encapsulated, deposited, adhered, or otherwise attached to the facial interface **13.9-212**. A portion of the pressure sensor assembly **13.9-224**, such as a deformable pad configured to contact a user's face, can deflect in response to a movement of a user's face.

[2865] The pressure sensor assembly **13.9-224** can be configured to detect a biometric feature of the user **13.9-220**. The pressure sensor assembly **13.9-224**, upon detecting a signal from the biometric feature, can transmit the signal to a component of the head-mountable device **13.9-200** causing the head-mountable device **13.9-200** to perform an action. For example, the head-mountable device **13.9-200** can change shape (i.e., tighten or loosen), move, vibrate, rotate, recalibrate, or reposition in response to the sensor signal of the biometric feature.

[2866] The head-mountable device **13.9-200** can include a headband, a retention band, a strap, or a support **13.9-223** connected to the display **13.9-208** and/or the frame **13.9-216**. The support **13.9-223** can secure the display **13.9-108** and/or housing **13.9-216** relative to the user's head **13.9-220** (e.g., such that the display **13.9-208** is maintained in front of a user's eyes). The support **13.9-223** can be constructed from elastic material, inelastic material, or a combination of elastic and inelastic material.

[2867] The support **13.9-223** can be adjustable such that the support **13.9-223** conforms to the various shapes and sizes of a user's head **13.9-220**. In some examples, the support **13.9-223** secures the head-mountable device **13.9-200** via friction between the user's head **13.9-220** and the retention band **13.9-223**. In some examples, the support **13.9-223** elastically secures the head-mountable device **13.9-200** to the user's head **13.9-220**. In some examples, the support **13.9-223** is coupled to a ratchet system or mechanism securing the head-mountable device **13.9-200** to the user's head **13.9-220**. In some examples, the support **13.9-223** is disposed above or on an ear **13.9-221** of the user's head **13.9-220**, supporting the head-mountable device **13.9-200**.

[2868] In some examples, the housing or frame **13.9-216** of the head-mountable display **13.9-200** is connected via a connector **13.9-215** to the facial interface **13.9-212**, the retention band **13.9-223** being connected to the frame **13.9-216** and/or the facial interface **13.9-212**, securing the head-mountable device **13.9-200** to the user's head **13.9-220** above or over the ears **13.9-221** of the user.

[2869] FIGS. **13.9-2B** shows an example facial interface **13.9-212** including one or more deformable pads **13.9-225a** and **13.9-225b** (collectively referred to as “deformable pads **13.9-225**”) disposed on the facial interface **13.9-212**. The deformable pads **13.9-225** can include thin membranes or sheets. The deformable pads **13.9-225** can be inflatable membranes or pillows. While multiple deformable pads **13.9-225** are shown, it will be understood that one deformable pad **13.9-225** and sensor **13.9-224** can be used to achieve the same or similar results. The deformable pads **13.9-225** can be in fluid communication with one or more pressure sensors **13.9-224a** and **13.9-224b** (collectively referred to as “pressure sensors **13.9-224**”).

[2870] It will be noted that because of the use of deformable pad **13.9-225**, the pressure sensors **13.9-225** do not need to be in direct contact with the user's skin. This allows the sensors **13.9-225** to be more securely housed within the facial interface **13.9-212** where motion on the user's skin can be transferred through various materials of the facial interface **13.9-212** to the deformable pad **13.9-**

225. The deformable pads **13.9-225** can be located at any location of the facial interface **13.9-212** that enables them to contact a face of a user. In some examples, the deformable pads **13.9-225a** are located at/on a nose of a user when the head-mountable device is worn. The deformable pads **13.9-225a** can be in fluid communication with the pressure sensors **13.9-224a**. The pressure sensors **13.9-224a** can detect changes in pressure caused by a movement in the deformable pads **13.9-225a**. [2871] The movement in the deformable pads **13.9-225a** can be caused by a movement of the user's face. For example, a user may breathe through his/her nose, deflecting the deformable pads **13.9-225a**, and thereby a change in internal pressure of the pressure sensor assembly. The pressure sensors **13.9-224** can detect the change in pressure and transmit a signal based on the change to the head-mountable device, such as the head-mountable device **13.9-200** shown in FIGS. **13.9-2A**, via an electrical signal. In some examples the change in pressure results in a piezoelectric reaction, a change in resistance, or a diaphragm deflection to generate the signal.

[2872] Similarly, the deformable pads **13.9-225b** can be located near a forehead region of a user's face when the facial interface **13.9-212** is donned. The deformable pads **13.9-225b** can be in fluid communication with the pressure sensors **13.9-224b**. The pressure sensors **13.9-224b** can detect changes in pressure caused by a movement in the deformable pads **13.9-225b** in response to a movement of a user's face. For example, a user may perform an activity, such as making a facial expression or exercising, increasing the user's blood pressure, and thereby causing the detectable forehead vasculature to enlarge.

[2873] As the forehead vasculature enlarges, deformable pads **13.9-225b** deflect, causing a change in internal pressure to the pressure assembly. The pressure sensors **13.9-224b** detects the change in pressure and communicates the pressure change by generating an electrical signal in response to the pressure change, which signal can then be transmitted to the head-mountable device. Likewise, a change in the user's facial expression can cause a deflection of the deformable pads **13.9-225b** which can be detected by the pressure sensors **13.9-224b**. The pressure sensors **13.9-224b** then generate an electrical signal in response to the detected deflection, which signal can then be analyzed by a processor to infer a characteristic of the user's facial expression.

[2874] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.9-2A** and **13.9-2B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.9-2A** and **13.9-2B**.

[2875] FIGS. **13.9-3** shows a rear perspective view of a head-mountable device **13.9-300**. The head-mountable device **13.9-300** can be substantially similar to, and can include some or all of the features of, the head-mountable devices described herein, such as head-mountable device **13.9-100** and **13.9-200**. The head-mountable device **13.9-300** shown in FIGS. **13.9-3** includes a support **13.9-323**, a facial interface **13.9-312**, overmolded portions **13.9-313a** and **13.9-313b** (collectively referred to as "overmolded portions **13.9-313**") connected to the facial interface **13.9-312**, deformable pads **13.9-325**, channels **13.9-326**, and sensors **13.9-324**.

[2876] The channel **13.9-326** allows for fluid communication between the deformable pad **13.9-325** and the sensor **13.9-324**. As described in further detail with regards to FIGS. **13.9-4A** and **13.9-4B**, the channel **13.9-326** can define an internal volume. In some examples, the channel **13.9-326** defines a support arm for the deformable pad **13.9-325**. The channel **13.9-326** can be adjustable such that the position of the deformable pad **13.9-325** relative to the user's nose can be modified for varied applications or users. For example, one user may have specific nose shape, and another user may have a substantially different nose shape. The channel **13.9-326** can be adjusted, such that the nose shape can be accommodated in a comfortable and precise manner.

[2877] In some examples, the deformable pads **13.9-325** can be adjustable. For example, the

deformable pads **13.9-325** can move or pivot relative to the facial interface **13.9-312**. The deformable pads **13.9-325** can change their pitch, yaw, or roll in order to establish a better fit and comfort on the user's face. In some examples, the stiffness or rigidity of the deformable pads **13.9-325** can be adjusted or tuned based on needs and preferences. The stiffness of the deformable pads **13.9-325** can be adjusted based on the pressure in the channels **13.9-326**. In some examples, the deformable pads **13.9-325** are automatically adjusted based on the user or application. Further details regarding deformable pads are provided below with reference to FIGS. **13.9-4A** and **13.9-4B**.

[2878] In some examples, a user may perform a rigorous activity, requiring more rigid contact of the deformable pad **13.9-325**. The deformable pad **13.9-325** can be replaced with a more rigid deformable pad. In some examples, the deformable pad **13.9-325** can be replaced with a less rigid deformable pad to accommodate other certain activities, such as test taking or learning activities.

[2879] The overmolded portion **13.9-313** can include portions contacting the user's face. For example, overmolded portion **13.9-313b** can contact the maxilla and/or zygoma facial regions, creating a comfortable conforming fit to a user's face. Similarly, the overmolded portion **13.9-313a** can contact a user's forehead conforming to each user's forehead unique attributes, providing a comfortable customized fit for an individual user. In some examples, the overmolded portion **13.9-313** can be considered part of the facial interface **13.9-312**. The overmolded portions **13.9-313** can secure the sensor assembly to the facial interface **13.9-312**, creating an esthetically pleasing head-mountable display. In some examples, the overmolded portions **13.9-313** can create or define the channels **13.9-326**. In some examples, the overmolded portions **13.9-313** can form a housing to house or conceal the sensors **13.9-324** and/or channels **13.9-326**.

[2880] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.9-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.9-3**.

[2881] FIGS. **13.9-4A** shows a perspective view of a portion of a facial interface **13.9-412**. The facial interface **13.9-412** can be substantially similar to, and can include some or all of the features of, the facial interfaces described herein, such as facial interfaces **13.9-112**, **13.9-212**, and **13.9-312**. The facial interface **13.9-412** can include an arm or channel **13.9-426** that defines an internal volume **13.9-428**. The internal volume **13.9-428** can establish a fluid communication between a deformable pad **13.9-425** and a sensor **13.9-424**. In some examples, the internal volume includes air or some other gas. In some examples, the internal volume **13.9-428** include a liquid or pneumatic fluid. The sensor **13.9-424** can be a pressure-based MEMS sensor **13.9-424**. The sensor **13.9-424** can be disposed within the arm **13.9-426** of the facial interface **13.9-412**. In some examples, the sensor **13.9-424** is positioned on an exterior of the arm **13.9-426**.

[2882] The deformable pad **13.9-425** can include an inflatable member that can flex or deform in response to an external force. The deformable pad **13.9-425** can be removably attached to the arm **13.9-426** via a connector **13.9-433**. The connector **13.9-433** can any suitable mechanical or magnetic attachment mechanisms. In some examples, the connector **13.9-433** can include a barbed configuration capable of removably securing the deformable pad **13.9-425** to the arm **13.9-426**. The deformable pad **13.9-425** can include a female connector **13.9-435** to receive a male connector **13.9-433** on the arm **13.9-426**. In some examples, the connector **13.9-433** can be threaded such that the deformable pad **13.9-425** is threaded onto the connector **13.9-433**.

[2883] The term "barbed" used within the context of this application can define one end of a component connecting to another end of a different component. For example, the deformable pad can connect to the facial interface **13.9-412** via a barbed connection. The connection can be

physical and contain egressed portions on one component and ingress portions on another components. These egressed portions and ingress portions can form a threaded fit, snap fit, compressive fit, or other fit that can produce fluid communication between the deformable pad **13.9-425** and the sensor **13.9-424**.

[2884] As shown in FIGS. **13.9-4B**, in some examples, the connector **13.9-433** can establish a substantially air-tight seal between the deformable pad **13.9-425** and the internal volume **13.9-428** of the channel. In some examples, the gasket, O-ring, or other type of seal is positioned around the connector **13.9-433** to improve the seal. In some examples, a known and predetermined amount of air is permitted to escape the internal volume **13.9-428** as the deformable pad **13.9-425** moves. By controlling the amount of escaped air, the pressure sensor is still able to deduce facial movement.

[2885] The deformable pad **13.9-425** can include an elastomeric material. The deformable pad **13.9-425** can include a soft material that is comfortable on the user's skin. The deformable pad **13.9-425** can include, but is not limited to, any of the following: Natural Rubber, Polyisoprene, Butyl Rubber (IIR, Isobutene-isoprene), Chloroprene (CR, Neoprene®), Ethylene Propylene Diene (EPDM), Fluorocarbon (FMK, Viton®), Fluorosilicone (FSI), Nitrile Butadiene (NBR), Saturated Nitrile (HNBR), Silicone Rubber (SI, Gum and Liquid), Styrene Butadiene (SBR), and Urethane (PU, Polyurethane).

[2886] In some examples, the deformable pad can include a compressible elastomer such as a silicon material that is durable and not likely to wear down. The deformable pad can be soft and resilient to moisture, such as sweat. In some examples, the deformable pad is overmolded or can be a stand-alone piece that is attached to the facial interface.

[2887] Further, the deformable pad **13.9-425** can be tuned for specific needs or users. For example, one deformable pad can be constructed of one elastomeric material for one application or user, and another deformable pad can include elastomeric material designed for another application or user. In some examples, the deformable pad **13.9-425** can be shaped or manufactured to increase or decrease sensitivity of the deformable pad **13.9-425** for a certain activity. In some examples, the deformable pad **13.9-425** can be constructed of a combination of materials, such as two or more types of elastomeric materials.

[2888] An example operation of the sensor module can occur when a user, wearing the HMD, has facial movement. A facial movement can include movement in the nose, forehead, cheeks, or any other portion of the face/head. Facial movements can be indicative of respiration, pulse, and facial expressions. Because the deformable pad **13.9-425** is in close proximity or touching the user, the facial movement can push or deflect the deformable pad **13.9-425** a certain amount. The deflection of the deformable pad **13.9-425** can cause a temporary increase in the pressure within the channel **13.9-428**. The change in pressure can be detected by the sensor **13.9-424**, which transmits a signal, based on the pressure fluctuation to be processed (e.g., by a processor on board the HMD). In some examples, the sensor **13.9-424** can be integrated into other components of the HMD (e.g., integrated onto the headband of the HMD).

[2889] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.9-4A** and **13.9-4B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.9-4A** and **13.9-4B**.

[2890] FIGS. **13.9-5A** shows a rear view of a facial interface **13.9-512**. The facial interface **13.9-512** can be substantially similar to, and can include some or all of the features of, the facial interfaces described herein, such as facial interfaces **13.9-112**, **13.9-212**, **13.9-312**, and **13.9-412**. In some examples, the facial interface **13.9-512** can include a pressure sensor located in a forehead-contacting region of the facial interface **13.9-512**. The pressure sensor can include a deformable

pad (e.g., inflatable member) **13.9-525**. In some examples, the deformable pad is overmolded to the facial interface **13.9-512**, sealing the deformable pad **13.9-525** and creating a pressure chamber wherein facial movements of a user cause the deformable pad **13.9-525** to deflect or deform. The deflection of the deformable pad **13.9-525** creates a change in pressure within the pressure chamber. The change in pressure can be detectable to the pressure sensor which can then generate a signal to be analyzed or interpreted by a processor/controller.

[2891] As illustrated in FIGS. **13.9-5A**, the deformable pad **13.9-525** can include an elongated shape. The elongated shape of the deformable pad **13.9-525** can span across a substantial portion of the user's forehead when the user is wearing the HMD. The elongated horizontal shape of the deformable pad **13.9-525** can be predetermined to cover a maximum amount of veins/vessels in the forehead. In some examples, the deformable pad **13.9-525** can be centered on the facial interface **13.9-512**, and therefore positioned to contact a center of the user's forehead. However, other positions are also possible. In some examples, the deformable pad **13.9-525** is off-center from the center or middle of the facial interface **13.9-512**. In some examples, the facial interface **13.9-512** wraps partially around the user's head, such that the deformable pad **13.9-525** can be positioned proximate a temple of the user when the HMD is donned.

[2892] FIGS. **13.9-5B** shows a rear view of the facial interface **13.9-512** having a plurality of deformable pads **13.9-525a**, **13.9-525b**, **13.9-525c**, **13.9-525d** (collectively referred to as "deformable pads **13.9-525**"). The deformable pads **13.9-525** can each deflect to cause a change in pressure. In some examples, the facial interface **13.9-512** includes a plurality of pressure chambers. The pressure chambers can be isolated and distinct or can be in fluid communication with each other. The pressure within the channel(s) can be equalized. In some examples, each deformable pad **13.9-525** can be fluidically connected to a single sensor.

[2893] In some examples, the system includes multiple sensors, each of which is fluidically connected to one or more of the deformable pads **13.9-525**. The increased number of deformable pads **13.9-525** can increase accuracy and sensitivity for a head-mountable device used in combination with the facial interface **13.9-512**. In some examples, a combination of the deformable pads **13.9-525** can be used to detect or track motions along the user's forehead. For example, by using at least pad **13.9-525a** and pad **13.9-525b**, a vertical signal or input can be monitored or tracked across the user's forehead. Likewise, using at least pad **13.9-525c** and pad **13.9-525d**, a horizontal signal or input can be tracked as it traverses horizontally across a user's forehead.

[2894] In some examples, the combination of pads **13.9-525** can detect a user's pulse transit time (e.g., the time it takes a pulse wave to travel between two pulse sites). For example, deformable pad **13.9-525a** and deformable pad **13.9-525b** can be placed over the supratrochlear artery (e.g. the arterial supply to the forehead and the anterior part of the scalp) measuring the pulse transit time from deformable pad **13.9-525a** to deformable pad **13.9-525b**. Similarly, any artery or combination thereof can be used to measure arterial functions, such as pulse transit times in a vertical, horizontal, or combinatory manner.

[2895] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.9-5A** and **13.9-5B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures, can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.9-5A** and **13.9-5B**.

[2896] FIGS. **13.9-6** illustrates a side cross-sectional view of a portion of a sensor module of a facial interface **13.9-612**. The facial interface **13.9-612** can be substantially similar to, and can include some or all of the features of, the facial interfaces described herein, such as facial interface **13.9-112**, **13.9-212**, **13.9-312**, **13.9-412**, and **13.9-512**. The sensor module of FIG. **13.9-6** includes a deformable pad **13.9-625**, a contact portion **13.9-627** configured to contact a user's head, and a

sensor **13.9-624**.

[2897] A channel **13.9-628** can define an internal volume that places the deformable pad **13.9-625** and the sensor **13.9-624** in fluid communication. In some examples, the deformable pad **13.9-625** can deform in response to facial movements. The deformable pad **13.9-625** can be constructed of a compressible elastomer. The facial interface **13.9-612** can include an overmolded portion **13.9-627**. The overmolded portion **13.9-627** can contact a user's face and can provide a comfortable point of contact to the user's forehead, nose, around the user's eyes, and other contact regions the overmolded portion contacts the user's face.

[2898] In some examples, the overmolded portion **13.9-627** is formed of a soft elastomer that is resilient to sweat/perspiration from a user. As illustrated, a pressure-based MEMS sensor **13.9-624** can be embedded or housed within the facial interface **13.9-612**. The MEMS sensor can have dimension of approximately 2 mm by 2 mm. FIGS. **13.9-6** illustrates the deformable pad **13.9-625** being proud from (extending or protruding from) the facial interface **13.9-612**.

[2899] However, in some examples, the deformable pad **13.9-625** can be flush with the facial interface **13.9-612**, specifically flush with the section of the facial interface **13.9-612** that abuts or contacts the forehead of the user.

[2900] An example operation of the sensor module can occur when a user, wearing the HMD, has facial movement. A facial movement can include movement in the nose, forehead, cheeks, or any other portion of the face/head. Facial movements can be indicative of respiration, pulse, and facial expressions. Because the deformable pad **13.9-625** is in close proximity or touching the user, the facial movement can push or deflect the deformable pad **13.9-625** a certain amount.

[2901] The deflection of the deformable pad **13.9-625** causes a temporary increase in the pressure within the channel **13.9-628**. The change in pressure can be detected by the sensor **13.9-624**, which transmits a signal, based on the pressure fluctuation to be processed (e.g., by a processor on board the HMD). In some examples, the sensor **13.9-624** can be integrated into other components of the HMD (e.g., integrated onto the headband of the HMD).

[2902] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.9-6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.9-6**.

13.10: Touch Sensitive Input Surface

[2903] The following disclosure relates to a head-mountable device. More particularly, the present embodiments relate to a light seal of a head-mountable device including manual touch sensitive input surfaces for a user donning the head-mountable device to provide commands to the head-mountable device. As used herein the term “conductive fabric” can refer to the material or fabric into/onto which conductive elements are incorporated, or can refer to the conductive elements themselves, separate from the material or fabric with which they are integrated. The conductive fabric can include a plurality of conductive fibers. The conductive fabric can act as sensors, input members, and electrical interconnects to enable sensors and other electronic components to interact with one another. A manual input sensor can be a physically activated input system that can be used to provide an indication to a processor of a user's intent.

[2904] The head-mountable device of the present disclosure includes a manual input sensor that utilizes a touch sensitive surface that receives user input, such as touch input. The touch sensitive surface can accomplish this while being unobtrusively integrated with the light seal. A head-mountable device with such touch sensitive surfaces adds another way for a user to provide input or commands to a head-mountable device and adding functionality without adding burdensome components or structures.

[2905] These and other embodiments are discussed below with reference to FIGS. **13.10-1A** through **13.10-8B**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature including at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[2906] FIGS. **13.10-1A** illustrates a block diagram of a head-mountable device (“HMD”) **13.10-100** including a frame **13.10-104**, a display **13.10-105**, a support (e.g., a retention band **13.10-109**), and a light seal or cover **13.10-112**. The display **13.10-105** can include one or more optical lenses or display screens in front of the eyes of a user. The display **13.10-105** can include a display for presenting an augmented reality visualization, a virtual reality visualization, or other suitable visualization. Additionally, the frame **13.10-104** can at least partially surround the display **13.10-105**. Similarly, the light seal **13.10-112** can be connected to the frame **13.10-104**. In some examples, the light seal **13.10-112** includes the frame **13.10-104** (i.e., the frame **13.10-104** is part of the light seal **13.10-112**).

[2907] The light seal **13.10-112** can include electrical components (e.g., sensors **13.10-120**), a cover **13.10-113**, a facial interface **13.10-108**, and a conductive fabric **13.10-116**. The frame **13.10-104** can be a housing of the display **13.10-105**. Further, the frame **13.10-104** can also be considered to be a part of or separate from the light seal **13.10-112**. As used herein, the term “light seal” can refer to a portion of the HMD **13.10-100** that engages or shields a user's face. In particular, the light seal **13.10-112** includes portions (e.g., the facial interface) that conform to, contact, or press against regions of the user's face.

[2908] The light seal **13.10-112** includes a body with an inner surface and an outer surface. The outer surface is exposed to the external environment and the inner surface is disposed within the space formed by the HMD **13.10-100** and the user's face. The light seal can include a pliant (or semi-pliant) facetrack or face engagement component that spans the forehead, wraps around the eyes, contacts other regions of the face (e.g., zygoma and maxilla regions), and bridges the nose. In addition, the light seal **13.10-112** can include various components forming a frame, structure, or webbing of a head-mountable device disposed between the display **13.10-105** and the user's skin, such as the cover **13.10-113**. The cover **13.10-113** can include a seal, environment seal, dust seal, air seal, etc. that is positioned between the gap between the display **13.10-105** and the user's face. The cover **13.10-113** can be a woven fabric that is non-rigid or deformable. The cover **13.10-113** can be elastically deformable. In some examples, the cover **13.10-113** can be a plastic, rubber, or polymer material. In some examples, the cover **13.10-113** can be rigid.

[2909] The cover **13.10-113** can form an eye-box through which the user can view the display **13.10-105**. It will be appreciated that the term “seal” can include partial seals or inhibitors, in addition to complete seals (e.g., a partial light seal where some ambient light is blocked and a complete light seal where all ambient light is blocked when the HMD **13.10-100** is donned).

[2910] The light seal **13.10-112** can have a removable fastener that enables the light seal to be removably attachable to the frame **13.10-104** and in electrical communication with the display **13.10-105**. The light seal **13.10-112** can include an electrical component, such as a sensor **13.10-120**. The sensor **13.10-120** can collect user or environmental data, such as biometric information. The sensor **13.10-120** may receive input from the user, such as the user physically touching the sensor to input or command to the HMD **13.10-100** as an indication of the user's intent. The sensor **13.10-120** can transmit signals to the HMD **13.10-100**, and more particularly to the display **13.10-**

105. The sensor **13.10-120** can transmit signals to an output configured to perform an action in response to the information collected by the sensor **13.10-120**.

[2911] The light seal **13.10-112** can include conductive fabric **13.10-116**. In some examples, the conductive fabric **13.10-116** can include highly elastic copper threads that can compress and stretch while still maintaining electrical connectivity. The conductive fabric **13.10-116** can be elastically deformable (i.e., capable of temporary change in length, volume, or shape). In some examples, the conductive fabric **13.10-116** can include electrically conductive carbon fibers. In some examples, the conductive fabric **13.10-116** can collect input from the user by the user engaging the conductive fabric through touch, such as with a finger or fingers. These movements can cause a change in the conductive fabric **13.10-116** (i.e., stretching or compressing). The degree or amount to which the conductive fabric **13.10-116** is moved by the user's finger can be detectable by the conductive fabric **13.10-116** and can be used to generate signals that can be analyzed by a processor of the HMD **13.10-100**. As described in greater detail below, the conductive fabric **13.10-116** can be a thread, line, wire, plate, or any other structure that is capable of conducting electricity. The conductive fabric **13.10-116** can be integrated with the light seal **13.10-112**. For example, the conductive fabric **13.10-116** can be embedded, interwoven, or encapsulated with the cover **13.10-113** or facial interface **13.10-108**.

[2912] The display **13.10-105**, being in electrical communication with the light seal **13.10-112**, and more specifically with the conductive fabric **13.10-116**, can receive electrical communication and provide feedback to the user related to the readings of the conductive fabric **13.10-116** (e.g., visual feedback, audio feedback, haptic feedback, etc.).

[2913] As used herein, the term “sensor” refers to one or more different sensing devices, such as a camera or imaging device, strain gauge, capacitance sensors, proximity sensors, touch sensitive sensors, accelerometers, and the like. Additional sensors can include a temperature sensor, oxygen sensor, movement sensor, brain activity sensor, sweat gland activity sensor, breathing activity sensor, muscle contraction sensor, etc. In some examples, the sensor can sense biometric features including features of the autonomic nervous system. Some examples of sensors include an electrooculography sensor, electrocardiogram sensor, EKG sensor, heart rate variability sensor, blood volume pulse sensor, SpO2 sensor, compact pressure sensor, electromyography sensor, core-body temperature sensor, galvanic skin sensor, accelerometer, gyroscope, magnetometer, inclinometer, barometer, infrared sensor, global positioning system sensor, etc. Additional sensor examples can include, contact microphones (e.g., press-based MEMS), bioelectrical activity sensors, UV exposure sensors, or particle sensors.

[2914] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.10-1A** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.10-1A**.

[2915] FIGS. **13.10-1B** shows a top partial view of an HMD **13.10-100**. The HMD **13.10-100** of FIGS. **13.10-1B** can be substantially similar to, including some or all of the features of, the HMD **13.10-100** described in FIGS. **13.10-1A**. The HMD **13.10-100** can include a display (also referred to as a display unit) **13.10-105**. The display **13.10-105** can include any number of internal electronic components **13.10-107**. The HMD **13.10-100** can include a frame **13.10-104** (which can also be referred to as a housing) attached to the display **13.10-105**. In some examples, the display **13.10-105** includes an opaque, translucent, transparent, or semi-transparent screen, including any number of lenses, for presenting visual data. The frame **13.10-104** can at least partially border one or more edges of the display **13.10-105**. The frame **13.10-104** can be attached to a cover **13.10-128** at one end of the cover **13.10-128**. At an opposite end, the cover **13.10-128** can form, or be attached

to, a facial interface **13.10-108**. In some examples, the frame **13.10-104**, cover **13.10-113**, and facial interface **13.10-108** can together form the light seal **13.10-112**. It will be understood, however, that the light seal **13.10-112** can include fewer or more components than those listed or shown.

[2916] The HMD **13.10-100** can be worn on a user's head **13.10-20** such that the display **13.10-105** is positioned over the user's face and disposed over one or both of the user's eyes. The HMD **13.10-100** can further include a retention band **13.10-109**. The retention band **13.10-109** can secure the HMD **13.10-100** to the user's head **13.10-20** during use so that the user can enjoy the experience provided by the HMD **13.10-100**. The display **13.10-105** can be connected to the retention band **13.10-109** and/or the light seal **13.10-112**. In some examples, the retention band **13.10-109** can be positioned against the side of a user's head **13.10-20** and in contact therewith. In some examples, the retention band **13.10-109** can be at least partially positioned above the user's ear or ears. In some examples, the retention band **13.10-109** can be positioned adjacent to the user's ear or ears. The retention band **13.10-109** can extend around the user's head **13.10-20**. In this way, the display **13.10-105** and the retention band **13.10-109** can form a loop that can retain the HMD **13.10-100** on the user's head **13.10-20**. It should be understood, however, that this configuration is just one example of how the components of the HMD **13.10-100** can be arranged, and that in some examples, a different number of connector straps and/or retention bands can be included. Although the HMD **13.10-100** is referred to as an HMD, it should be understood that the terms wearable device, wearable electronic device, head-mountable device, HMD, HMD device, and/or HMD system can be used to refer to any wearable device, including smart glasses.

[2917] In some examples, the frame **13.10-104** is attached to a facial interface **13.10-108**. The facial interface **13.10-108** can contact a user's head **13.10-20** and/or face. In some examples, the cover **13.10-113** can be a light blocking component that extends between the frame **13.10-104** and the facial interface **13.10-108**. The light blocking component can cover or surround a perimeter of the frame **13.10-104** and/or the facial interface **13.10-108**.

[2918] The cover **13.10-113** can be a cloth, fabric, woven material, plastic, rubber, or any other suitable opaque or semi-opaque material. In some examples, the cover **13.10-113** is flexible, having the ability to repeatedly stretch, compress, and deform. The cover **13.10-113** can be elastically or in-elastically deformable. The facial interface **13.10-108** in combination with the cover **13.10-113** can block outside light and limit the peripheral view of the user. In some examples, the cover **13.10-113** and the facial interface **13.10-108** is the same or a unitary component. As will be discussed in greater detail below, the light seal **13.10-112** can include the conductive fabric **13.10-116** that is able to serve multiple functions and provide added benefits and functionality to the HMD **13.10-100**.

[2919] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.10-1B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.10-1B**.

[2920] FIGS. **13.10-2** shows a bottom perspective view of select components of a light seal **13.10-212**. The light seal **13.10-212** can be substantially similar to, including some or all of the features of, the light seals described herein, such as light seal **13.10-112**. The light seal **13.10-212** can be implemented on an HMD, such as HMD **13.10-100**.

[2921] The light seal **13.10-212** can be integrated with conductive fabrics **13.10-216a**, **13.10-216b** (collectively referred to as conductive fabric **13.10-216**). The conductive fabric **13.10-216** or thread can be positioned at various locations on the light seal **13.10-212**. For example, the conductive fabric **13.10-216** can be positioned on or in the cover **13.10-213** (e.g., embedded, encapsulated, or

interwoven into the cover).

[2922] In some examples, the conductive fabric **13.10-216** can be positioned on the facial interface **13.10-208** of the light seal **13.10-212**. In one example, the conductive fabric **13.10-216** can be in direct contact with a user, for example, touching a user's forehead, cheek, nose, temple region, back of the head, or at any location where the HMD contacts the user. In some examples, the conductive fabrics **13.10-216** is disposed on an outer surface of the light seal **13.10-212**.

[2923] In the illustrated example of FIGS. **13.10-2**, conductive fabrics **13.10-216a** are disposed on a left side of the light seal and conductive fabrics **13.10-216b** are disposed on a right side of the light seal **13.10-212**. The conductive fabrics **13.10-216a** can be disposed in a specific touch sensitive surface **13.10-218a** and conductive fabrics **13.10-216b** can be disposed in a touch sensitive surface **13.10-218b**. The touch sensitive surfaces **13.10-218a**, **13.10-218b** can be manual input sensors to receive input from the user by manually or physically touching the touch sensitive surfaces **13.10-218a**, **13.10-218b** to provide input and/or commands to the HMD **13.10-100**. However, the conductive fabric **13.10-216** can be located in a variety of different locations of the light seal **13.10-212**.

[2924] FIGS. **13.10-3A** shows a top view of an HMD **13.10-300**. The HMD **13.10-300** can be substantially similar to, including some or all of the features of, the devices described herein, such as the HMD **13.10-100** and the light seal **13.10-212**. In some examples, the HMD **13.10-300** can include an electronic component (e.g., a sensor **13.10-320**). The electronic component can be positioned on or housed in the display unit **13.10-305**. The electronic component can be a battery, a processor, a display, a camera, or any other electronic component. In some examples, the electronic module can be attached to the frame **13.10-304**.

[2925] In some examples, the electronic component (e.g., the sensor **13.10-320**) can be electrically connected by a conductive fabric **13.10-316** that runs through the cover **13.10-313** of the light seal **13.10-312**. In some examples, the conductive fabric **13.10-316** runs through or across a retention band **13.10-309**. In some examples, the conductive fabric **13.10-316** runs through or across the frame **13.10-304**. In this manner, the conductive fabric **13.10-316** serves as an interconnect (i.e., an electrical connection) between the sensor **13.10-320** on the light seal **13.10-312**. In some examples, the detections of the conductive fabrics **13.10-316** can be combined with other sensors on-board or remote from the HMD **13.10-300** to gather data in a sensor fusion manner. In the illustrated example, the light seal **13.10-312** includes a conductive fabric **13.10-316a** disposed on a left side of the light seal **13.10-312** and a conductive fabric **13.10-316b** disposed on a right side of the light seal **13.10-312**. The conductive fabric **13.10-316a** can be disposed in a specific touch sensitive surface **13.10-318a** and the conductive fabric **13.10-316b** can be disposed in a specific touch surface **13.10-318b**. These specific touch sensitive surfaces **13.10-318a**, **13.10-318b** are configured to allow a user to provide input to the HMD **13.10-100** by touching the specific touch sensitive surfaces **13.10-318a**, **13.10-318b**.

[2926] In the illustrated example, the light seal **13.10-312** includes two touch sensitive surfaces **13.10-318a**, **13.10-318b**, however, the present disclosure is not so limited and can include more or less than two touch sensitive surfaces. In some examples, the entire light seal **13.10-312** can be a touch sensitive surface that allows the user to input information or commands to the HMD **13.10-300** from any location on the light seal **13.10-312**. In some examples, there can be a distinct number of touch sensitive surfaces to enable the user to input information or commands into the HMD **13.10-300**. In some examples, each distinct touch sensitive surface can be directed to a specific type of input or functionality. For example, one touch sensitive surface can be for volume control, another touch sensitive input can be for scrolling, and the like. The HMD **13.10-300** can include an indicator that directs the user to the location of the distinct touch sensitive surfaces on the outer surface of the light seal **13.10-312**.

[2927] In some examples, the HMD **13.10-300** can include conductive fabrics **13.10-316c**, **13.10-316d** incorporated onto and/or into the retention band **13.10-309** of the HMD **13.10-300**. The

retention band **13.10-309** can be a headband having flexible sections. The conductive fabrics **13.10-316c**, **13.10-316d** can be configured to stretch or compress in response to the retention band **13.10-309** stretching or compressing. In some examples, the conductive fabrics **13.10-316c**, **13.10-316d** can generate signals indicative of the tightness or position of the retention band **13.10-309**. In some examples, a user can be notified based on the signals generated by the conductive fabrics **13.10-316c**, **13.10-316d**. The retention band **13.10-309** can be automatically adjusted based on the signals generated by the conductive fabrics **13.10-316c**, **13.10-316d**. In the illustrated example, the retention band **13.10-309** include the conductive fabric **13.10-316c** disposed on a left side of the retention band **13.10-309** and the conductive fabric **13.10-316d** disposed on a right side of the retention band **13.10-309** and can be connected to electrical components **13.10-320** on the frame **13.10-304** of the HMD **13.10-300**.

[2928] FIGS. **13.10-3B** illustrates a top view of a user interacting with the touch sensitive surface **13.10-318b**. For ease of illustration, the touch sensitive surface **13.10-318b** depicts a single conductive fiber but can includes a plurality of conductive fibers. In some examples, the conductive fabric **13.10-316b** is woven into a strain gauge pattern and the conductive fabric **13.10-316b** is electrically connected to an electrical component (e.g., a sensor **13.10-320**) of the HMD **13.10-300**. The user's engagement with the touch sensitive surface **13.10-318b** can be interpreted as input to provide commands to the HMD **13.10-300**.

[2929] In some examples, the conductive fabric **13.10-316b** is physically connected to the sensor **13.10-320**. The sensor **13.10-320** can be a strain gauge and the conductive fabric **13.10-316b** is physically coupled to the strain gauge **13.10-320**. The strain gauge **13.10-320** can measure the displacement of the conductive fabric **13.10-316b** and interpret or transform the displacement of the conductive fabric **13.10-316b** into input for the HMD **13.10-300**.

[2930] In some examples, the touch sensitive surface **13.10-318b** includes yarn that is physically connected to the strain gauge **13.10-320**. All the yarn disposed within a perimeter of the touch sensitive surface **13.10-318b** can be physically connected to the strain gauge **13.10-320**. The touch sensitive surface **13.10-318b** leverages the elastic nature of the yarn. In other words, as the user presses or touches the yarn of the touch sensitive surface **13.10-318b**, the strain is transferred to the strain gauge **13.10-320** that is disposed away from the touch sensitive surface **13.10-318b** and the strain gauge can measure the displacement of the yarn and interpreted the displacement as input for the HMD **13.10-300**. In some examples, the touch sensitive surface **13.10-318** can include both conductive fabric **13.10-316b** and yarn.

[2931] FIGS. **13.10-3B** illustrates the user pushing or poking the specific touch sensitive surface **13.10-318b** with a finger **13.10-10**, which displaces the conductive fabric **13.10-316b** and transfers the displacement to the strain gauge **13.10-320**, which interprets the displacement as a command or input to the HMD **13.10-300**. However, the user can use a number of different gestures with their finger **13.10-10** to provide input to the HMD **13.10-300**. As discuss above, the user can press (e.g., poke) against the touch sensitive surface **13.10-318b**. The user can also slide their finger **13.10-10** along the touch sensitive surface **13.10-318b**. The user can stretch the touch sensitive surface **13.10-318b** by using two fingers to stretch the conductive fabric **13.10-316b** between the two fingers or push the conductive fabric **13.10-316b** closer together. This gesture can be interpreted as zooming in or zooming out. The user can also pinch the conductive fabric **13.10-316**. This gesture can be interpreted as taking a picture. In some examples, the number of fingers used in the gesture can indicate a different type of input. For example, two fingers swiping can be interpreted by the system as a different input relative to a single finger swiping, or three fingers swiping.

[2932] The strain gauge **13.10-320** can be incorporated into the frame **13.10-304** of the HMD **13.10-300**. In some examples, the strain gauge **13.10-320** can be disposed on the frame **13.10-304** of the HMD **13.10-300**. In some examples, the strain gauge **13.10-320** can be disposed within the frame **13.10-304**. For example, the frame **13.10-304** can includes a slot in which the strain gauge **13.10-320** is disposed and the yarn and/or conductive fabric **13.10-316b** can enter to physically

connect to the strain gauge **13.10-320**. Accordingly, the strain gauge **13.10-320** is shielded from environmental conditions, such as sweat, fluid, debris, and the like.

[2933] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.10-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.10-3**.

[2934] FIGS. **13.10-4** illustrates a touch sensitive surface **13.10-418** on a light seal **13.10-412** of an HMD **13.10-400**. The HMD **13.10-400** can be substantially similar to, including some or all of the features of, the devices described herein, such as the HMDs **13.10-100**, **13.10-300** and the light seal **13.10-212**. The touch sensitive surface **13.10-418** can include a distinct perimeter **13.10-419**. In the illustrated example, the perimeter **13.10-419** of the touch sensitive surface **13.10-418** is a circle, however, the perimeter **13.10-419** can include a plurality of different shapes. For example, the perimeter **13.10-419** can be semi-circular, ovoid, rectangular, trapezoidal, polygonal, triangular, and a variety of different shapes. As discuss above, the touch sensitive surface **13.10-418** can include a conductive fabric with conductive fibers (not shown) and/or yarn that can be connected to a strain gauge.

[2935] The touch sensitive surface **13.10-418** can include an indicator to alert or guide the user to the touch sensitive surface **13.10-418**. For example, when the user is donning the HMD **13.10-400**, the user cannot see the outer surface of the light seal **13.10-412** or the touch sensitive surface **13.10-418** on the light seal **13.10-412** as the HMD **13.10-400** is intentionally blocking their view outside of light seal **13.10-412** and HMD **13.10-400**. Therefore, when the user is donning the HMD **13.10-400**, the user attempts to engage with the touch sensitive surface **13.10-418** blindly. The indicator can provide feedback to help to user locate the touch sensitive surface **13.10-418** in a variety of different ways so that the user can input commands to the HMD **13.10-400** using the touch sensitive surface **13.10-418**.

[2936] In the illustrated example of FIGS. **13.10-4**, the indicator can use tactile feedback to guide the finger **13.10-10** of the user to the touch sensitive surface **13.10-418**. For example, a plurality of projections **13.10-417** or bumps can project outward from the surface of a light seal **13.10-412** along the perimeter **13.10-419** of the touch sensitive surface. Therefore, when the user slides their finger **13.10-10** over the projections **13.10-417**, the user's feels a tactile sensation on the tip of their finger **13.10-10**. These tactile sensations can orient and alert the user to the perimeter of the touch sensitive surface **13.10-418**. In other words, as the finger **13.10-10** of the user approaches the touch sensitive surface **13.10-418** the projections **13.10-417** provide tactile feedback to the user to know that they have arrived at the touch sensitive surface **13.10-418**. The projections **13.10-417** can be spaced along the perimeter **13.10-419** of the touch sensitive surface **13.10-418**. The space between adjacent projections **13.10-417** may be less than the width a fingertip, so that user can feel it. In some examples, the projection **13.10-417** may be a continuous projection that extends along the entire perimeter **13.10-419** of the touch sensitive surface **13.10-418**. In this manner, the user knows when they are approaching the perimeter **13.10-419** of the touch sensitive surface **13.10-418**, either from the outside of the touch sensitive surface **13.10-418** or from the inside of the touch sensitive surface **13.10-418**.

[2937] The projections **13.10-417** can have a variety of different shapes. In the illustrated example, the projections **13.10-417** have a circular shape. However, the scope of the present disclosure incorporates a variety of different shapes for the projections, such a rectangular, triangular, polygonal, and the like.

[2938] In some examples, the projections **13.10-417** can be positioned in patterns to help orient the user. For example, the bottom portion of the perimeter **13.10-419** can have a different pattern than

the rest of the perimeter **13.10-419**. For example, the bottom portion of the perimeter can have a cluster of three projections **13.10-419** that are closer to each other to indicate to the user that they are approaching the bottom portion of the perimeter **13.10-419** of the touch sensitive surface **13.10-418**. Similarly, a top portion of the perimeter **13.10-419** can have a similar pattern as the bottom portion of a different pattern. The same goes for a left portion of the perimeter **13.10-419** and a right portion of the perimeter **13.10-419**.

[2939] In some examples, the tactile feedback can be dynamic. In other words, in some examples the projections **13.10-417** can be felt, and in other examples the projections **13.10-417** cannot be felt. For example, in situations where the touch sensitive surface **13.10-418** is turned on and active, the projections **13.10-417** can be felt and in situations where the touch sensitive surface **13.10-418** is turned off and not active the projections **13.10-417** cannot be felt. In some examples, the projections **13.10-417** can be part of an inflatable bladder that inflates to make the projections **13.10-417** project outward from the outer surface of the light seal **13.10-412** and deflate to make the projections **13.10-417** retract back into the light seal **13.10-412**. In some other examples, the projections **13.10-417** can be selectively activated by a solenoid or other electrically actuated system that extends the projections **13.10-417** outward from the outer surface of the light seal **13.10-412** when the touch sensitive surface **13.10-418** is active, and retracts the projections **13.10-417** when the touch sensitive surface **13.10-418** is not active. As discussed below in more detail, the touch sensitive surface **13.10-418** can become active when a position sensor detects a near touch. When a near touch is detected, the touch sensitive surface **13.10-418** can be turned on and the projections **13.10-417** can be activated and extended.

[2940] In other examples, the indicator can include visual feedback that alerts or guides the finger **13.10-10** of the user to the touch sensitive surface **13.10-418**. Visual feedback can be displayed on a display **13.10-405** of the HMD **13.10-400** to the user. The HMD **13.10-400** can include a sensor, such as a camera, that is disposed on the frame **13.10-404** of the HMD **13.10-400**. The sensor can capture images or video of the touch sensitive surface **13.10-418** on the outer surface of the light seal **13.10-412**. Accordingly, the display **13.10-405** can display the images or video from the sensor directly to the user on the display **13.10-405** to show the user where their finger **13.10-10** is in relation to the touch sensitive surface **13.10-418** to help the user find and utilize the touch sensitive surface **13.10-418**.

[2941] In another example, the display **13.10-405** can merely display animations or visual cues to the user to orient the finger **13.10-10** of the user toward the touch sensitive surface **13.10-418**. For example, the display **13.10-405** can display to the user an animation, rather than a real-time video or image of how close the finger **13.10-10** of the user is relative to the touch sensitive surface **13.10-418**. The animations or visual cues may be display in a corner on a peripheral of the display **13.10-405** so that the animations or visual cues are not overlap over the central area of the display **13.10-405**.

[2942] In another example, an inner surface of the light seal **13.10-412** can include a plurality of light emitting diodes. The light emitting diodes can be used to provide feedback to the user to guide the user to the touch sensitive surface **13.10-418**. In some examples, the light emitting diodes can outline the touch sensitive surface **13.10-418** on the inner surface of the light seal **13.10-412**. When the light emitting diodes light up, the user can see the relative location of the touch sensitive surface **13.10-418**, albeit, from the inside of the light seal **13.10-412**. In some examples, the light emitting diodes on the inner surface of the light seal **13.10-412** may light up to show the user the current location of the finger **13.10-10** of the user from the inside of the light seal so that user can see the approximate location of their finger **13.10-10** relative to the touch sensitive surface **13.10-418**.

[2943] In another example, the color of the light emitting diodes can change color based on the location of the finger **13.10-10** of the user relative to the location of the touch sensitive surface **13.10-418**. For example, the light emitting diode can transition from a red color to a yellow color to

a green color based on the location of the finger **13.10-10**. The light emitting diode can be red when the finger **13.10-10** is not near the touch sensitive surface **13.10-418**, transition to a yellow color as the finger **13.10-10** get closer to the perimeter **13.10-419** of the touch sensitive surface **13.10-418** and turn green when the finger **13.10-10** engages the touch sensitive surface **13.10-418**.

[2944] In some examples, the indicator can include haptic feedback. The light seal **13.10-412** can include a haptic engine **13.10-434** that is electrically connected to processor **13.10-430** via a wire **13.10-432**. The haptic engine is configured to provide haptic feedback (e.g., vibration) to the user when their finger **13.10-10** approaches the perimeter **13.10-419** of the touch sensitive surface **13.10-418**. The vibration of the haptic engine **13.10-434** can orient and alert the user to the perimeter **13.10-419** of the touch sensitive surface **13.10-418**. In other words, as the finger **13.10-10** of the user approaches the touch sensitive surface **13.10-418** the haptic engine **13.10-434** vibrates to provide feedback to the user to know that they have arrived at the touch sensitive surface **13.10-418**. In this manner, the user knows when they are approaching the perimeter **13.10-419** of the touch sensitive surface **13.10-418**, either from the outside of the touch sensitive surface **13.10-418** or from the inside of the touch sensitive surface **13.10-418**.

[2945] The processor **13.10-430** can also perform a variety of different functions, including communicating with each of the feedback mechanisms discussed above. The processor **13.10-430** can function as a triggering device that can turn on and turn off the touch sensitive surface. In an attempt to conserve power, the touch sensitive surface **13.10-418** can be turned on for a user to provide input to the HMD **13.10-400** and turned off when the touch sensitive surface **13.10-418** is not in use. The processor **13.10-430** can be in communication with a number of different sensors.

[2946] The processor **13.10-430** can be in communication with a position sensor that detects a near touch, which is when the finger **13.10-10** or hand of the user is within a predetermined distance of the touch sensitive surface **13.10-418**. For example, the processor **13.10-430** can be in communication with a plurality of proximity sensors that are disposed around the perimeter **13.10-419** of touch sensitive surface **13.10-418** or disposed within the touch sensitive surface **13.10-418**. The proximity sensor can detect a near touch. Accordingly, when the proximity sensor senses a near touch, the processor **13.10-430** triggers the touch sensitive surface **13.10-418** to turn on so that the user can engage with the touch sensitive surface **13.10-418**. In other words, the processor **13.10-430** can turn on the touch sensitive surface **13.10-418** when the sensor detects a near touch.

[2947] In another example, the processor **13.10-430** can be in communication with a camera that captures images and/or video of the touch sensitive surface **13.10-418**. The camera monitors the touch sensitive surface **13.10-418** on the outer surface of the light seal **13.10-412** and captures images and/or video of the touch sensitive surface **13.10-418**. When the camera detects a near touch, the processor **13.10-430** determines that the user is about to touch the touch sensitive surface **13.10-418** and triggers the touch sensitive surface **13.10-418** to turn on so that the user can engage with the touch sensitive surface **13.10-418**.

[2948] The HMD **13.10-400** can use a number of different position sensors, such as proximity sensors, cameras, capacitance sensors, or any other type of sensor to detect a near touch (e.g., the approach of the user's finger, hand, etc. to trigger turning on the touch sensitive surface **13.10-418**). These systems can be used to avoid having environmental factors, such as fluid, water, humidity, sweat, accidentally engaging with the touch sensitive surface **13.10-418** and triggering input to the HMD **13.10-400** when the user was not intending to.

[2949] FIGS. **13.10-5** through **13.10-8B** illustrate additional ways in which the user can provide input for the HMD in a variety of different manners. FIGS. **13.10-5** illustrates an HMD **13.10-500** with a frame **13.10-504**. The frame **13.10-504** of the HMD **13.10-500** is coupled to the light seal **13.10-512**. The light seal **13.10-512** includes a touch sensitive surface **13.10-518** that is disposed on an outer surface of a light seal **13.10-512**. In the illustrated example, the touch sensitive surface **13.10-518** can include a plurality of conductive fabrics **13.10-516** organized (e.g., weaved) in a specific pattern. In the illustrated example, the conduct fabric **13.10-516** is organized in a matrix

pattern. When the conductive fabric **13.10-516** is organized in this pattern, the conductive fabric **13.10-516** can track the movement of the finger **13.10-10** along the touch sensitive surface **13.10-518**. The conductive fabrics **13.10-516** can be spaced apart a predetermined distance such that environmental factors can not adversely affect input to the touch sensitive surface **13.10-518**.

[2950] FIGS. **13.10-6** illustrates an HMD **13.10-600** with a frame **13.10-604**. The frame **13.10-604** of the HMD **13.10-600** is connected to the light seal **13.10-612**. The light seal **13.10-612** includes a touch sensitive surface **13.10-618** that is disposed on an outer surface of a light seal **13.10-612**. In the illustrated example, the touch sensitive surface **13.10-618** includes a plurality of planes **13.10-616** that are woven into the light seal **13.10-612** and are spaced vertically from each other. The planes **13.10-616** can get a capacitance reading to track the movement of the finger **13.10-10** of the user. In other words, the finger **13.10-10** can do a swiping motion or gesture in a vertical direction and the planes **13.10-616** can track the movement. This gesture can be used for volume control or scrolling. In some examples, the planes **13.10-616** can be spaced horizontally from each other and the user can do a swiping motion or gesture in a horizontal direction.

[2951] FIGS. **13.10-7** illustrates an HMD **13.10-700** with a frame **13.10-704**. The frame **13.10-704** can include a sensor **13.10-720**. The sensor **13.10-720** can be an accelerometer that can detect vibrations in the frame **13.10-704**. Accordingly, the user can tap the frame **13.10-704** of the HMD **13.10-700** with their finger **13.10-10**. Tapping on the frame **13.10-704** can be interpreted as input from the user to the HMD **13.10-700**.

[2952] FIGS. **13.10-8A** and **13.10-8B** illustrate an HMD **13.10-800** with a frame **13.10-804**. The frame **13.10-804** can include a sensor **13.10-820**. The sensor **13.10-820** can be a strain gauge that can detect mechanical deflection of the frame **13.10-804**. FIGS. **13.10-8A** illustrates the frame **13.10-804** in a natural or unconstrained state. FIGS. **13.10-8B** illustrates the frame **13.10-804** in a constrained state in which the user is squeezing a top and a bottom of the frame **13.10-804** toward each other, thereby mechanically deflecting the frame **13.10-804**. The deflection of the frame **13.10-804** can be interpreted as input to the HMD **13.10-800**. For example, squeezing the HMD **13.10-800** as illustrated can be interpreted as a command to take a picture of the display of the HMD **13.10-800**.

13.11: Face Engaging Structures

[2953] The following disclosure relates to wearable electronic devices (e.g., head-mountable device). More particularly, the present embodiments relate to connections between a display and a facial interface of a head-mountable device. Additionally, the present embodiments relate to lightweight variations of a facial interface.

[2954] When donning a head-mountable device or during use, the position and weight of the head-mountable device can affect the quality of user experience. Indeed, conventional head-mountable devices can be heavy due to the mass of materials (or combination of materials) used in manufacture. The additional weight can apply excess pressure to a user's face causing discomfort and pressure points at certain regions of the face. This user discomfort can degrade the overall AR/VR experience.

[2955] Exacerbating the weight factor, facial regions that contact a head-mountable device can have unique facial characteristics, such as variations of cranial width or length or variations in facial bones (e.g., frontal bone, zygoma, maxilla, etc.). Thus, these areas of high facial variability can undergo acute pressure or lead to faster user exhaustion.

[2956] Connections of the present disclosure address these and/or other issues of conventional head-mountable devices by providing dynamic adjustment. Indeed, connections of the present disclosure can dynamically adjust to the facial variability of users to provide a comfortable user experience. Additionally, connections (and associated connector positioning) of the present disclosure can better distribute applied loads across the user face for improved comfort compared to conventional head-mountable device. These connections can range from mechanical-based connectors providing more mechanical ranges of motion to organic connectors providing more

organic ranges of motion.

[2957] To illustrate, the connectors may be positioned at various locations (e.g., maxilla, zygoma, frontal bone, etc.). The connectors can include various types of joints (e.g., pivot joint, soft joint, flexure joint, spring joint, etc.). The connectors can, in combination, provide a comfortable pressure distribution.

[2958] Additionally, or alternatively to connectors, a head-mountable device of the present disclosure includes facial interface with a flexible (or flexure) portion. The flexure portion of the facial interface can include an elastomer material capable of recovering to an original shape after being flexed or stretched. In certain instances, the flexure region includes a particular pattern (e.g., a serpentine pattern) that can naturally flex, move, and conform to a user face.

[2959] Furthermore, embodiments described below provide examples of adjustment mechanisms, connectors, and facial interfaces of head-mountable devices that are lighter and more comfortable than conventional head-mountable devices. To illustrate, a head-mountable device can include a display frame with relief cutouts (e.g., lozenge-shaped thru-holes, cored-out portions, dimples, etc.). The relief cutouts can reduce the weight of the head-mountable device while maintaining structural integrity of the head-mountable device.

[2960] In another example, the head-mountable device includes a stiffener member that allows for a reduced amount of implemented material (and hence, weight savings). The stiffener member can be positioned between the display frame and the facial interface. The stiffener member can also be implemented as an additional shot of material in manufacturing. Additionally, or alternatively, the stiffener member can be overmolded to strengthen certain areas (e.g., at connection sites).

[2961] These and other embodiments are discussed below with reference to FIGS. **13.11-1-13.11-13**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature including at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[2962] FIGS. **13.11-1** illustrates a top view of a head-mountable device **100** worn on a user head **13.11-101**. The head mountable device can include a display **13.11-102** (e.g., one or more optical lenses or display screens in front of the eyes of the user). The display **13.11-102** can include a display for presenting a virtual reality visualization, an augmented reality visualization, or other suitable visualization.

[2963] The head-mountable device **13.11-100** also includes a facial interface **13.11-104**. As used herein, the term “facial interface” refers to a portion of the head-mountable device **13.11-100** that engages a user face via direct contact. In particular, a facial interface includes portions of the head-mountable device **13.11-100** that conform to (e.g., compress against) regions of the user face. For example, a facial interface may include a pliant (or semi-pliant) face track that spans the forehead, wraps around the eyes, contacts the zygoma and maxilla regions of the face, and bridges the nose. Furthermore, a facial interface can include various components forming a structure, webbing, cover, fabric, or frame of a head-mountable device disposed between the display **13.11-102** and the user skin. In particular implementations, a facial interface can include a seal (e.g., a light seal, environment seal, dust seal, air seal, etc.). It will be appreciated that the term “seal” can include partial seals or inhibitors, in addition to complete seals (e.g., a partial light seal where some ambient light is blocked and a complete light seal where all ambient light is blocked when the head-mountable device is donned).

[2964] In addition, the head-mountable device **13.11-100** includes connector(s) **13.11-106**. As used

herein, the terms “connector” or “joint” refer to a joining between the display **13.11-102** and facial interface **13.11-104**. In some examples, a connector allows the facial interface **13.11-104** to translate or rotate relative to the display **13.11-102** via the connector. In other examples, the connector allows the facial interface **13.11-104** to both translate and rotate relative to the facial interface **13.11-104**. For example, the connector(s) **13.11-106**, when acted on can translate the facial interface **13.11-104** toward or away from the display **13.11-102** (e.g., in a linear fashion). In another example, the connection(s) **13.11-106** when acted on can rotate the facial interface **13.11-104** up or down relative to the display **13.11-102** (e.g., in an angular fashion). In particular implementations, the connector(s) **13.11-106** moveably constrain the facial interface **13.11-104** to the display **13.11-102** at a forehead location, a zygoma location, or a maxilla location.

[2965] In this manner, the connection(s) **13.11-106** can movably constrain the display **13.11-102** relative to the facial interface **13.11-104**. As used herein, the term “movably constrain” refers to the type of connection that can dynamically move (e.g., translate or rotate), yet retain control over a particular element's movement or position. For example, to “movably constrain” means the connection(s) **13.11-106** can bound movement of the display **13.11-102** within two degrees of freedom (e.g., along a horizontal plane and along an additional plane non-planar with the horizontal plane) relative to the facial interface **13.11-104**.

[2966] As used herein, the term “forehead region” refers to an area of a human face between the eyes and the scalp of a human. Additionally, the term “maxilla region” refers to an area of a human face corresponding to the zygomatic bone structure of a human. Similarly, the term “maxilla region” refers to an area of a human face corresponding to the maxilla bone structure of a human. Further, the term “temple region” refers to an area of a human face between a respective eye and ear on a particular side of a face (e.g., between cheek bones and a forehead region). It will be appreciated that the foregoing regions can correspond to particular structure of the head-mountable device **13.11-100**. However, such structure of the head-mountable device **13.11-100** is not dependent on a face or a user.

[2967] The connector(s) **13.11-106** allow the facial interface **13.11-104** to conform freely to a wide range of facial topographies, thereby allowing the facial interface **13.11-104** to pivot and flex relative to the display **13.11-102**. The connector(s) **13.11-106** can also distribute loads (e.g., forces exerted from different facial topographies or compression from the strap **13.11-103**) evenly on a user's face. The connector(s) **13.11-106** can include one or more joints (e.g., pivot joint, soft joint, flexure joint, spring joint, etc.) that allow (or actively provide) translation or rotation of the facial interface **13.11-104** relative to the display **13.11-102**.

[2968] Furthermore, locations of the connector(s) **13.11-106** can be modified or tuned, as can the number of connections. For example, the location and or number of the connector(s) **13.11-106** can correlate to an amount of force or pressure exerted on the user at any one datum (e.g., forehead region, maxilla region, zygoma region, etc.) In other instances, the location and/or the number of connector(s) **13.11-106** can correspond to rigidity (or rigidity variances) between the connector(s) **13.11-106**.

[2969] Additionally shown in FIGS. **13.11-1**, the head-mountable device **13.11-100** includes one or more arms **13.11-108**, **13.11-110**. The arms **13.11-108**, **13.11-110** are connected to the display **13.11-102** and extend distally toward the rear of the head. The arms **13.11-108**, **13.11-110** are configured to secure the display in a position relative to the user head **13.11-101** (e.g., such that the display **13.11-102** is maintained in front of a user's eyes). For example, the arms **13.11-108**, **13.11-110** extend over the user's ears **13.11-112**. In certain examples, the arms **13.11-108**, **13.11-110** rest on the user's ears **13.11-112** to secure the head-mountable device **13.11-100** via friction between the arms **13.11-108**, **13.11-110** and the user head **13.11-101**. For example, the arms **13.11-108**, **13.11-110** can apply opposing pressures to the sides of the user head **13.11-101** to secure the head-mountable device **13.11-100** to the user head **13.11-101**. Optionally, the arms **13.11-108**, **13.11-110** can be connected to each other via a strap **13.11-103** (shown in the dashed lines) that can compress

the head-mountable device **13.11-100** against the user head **13.11-101**.

[2970] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-1**.

[2971] FIGS. **13.11-2A-13.11-2B** respectively illustrate side and front view profiles of an example of the head-mountable device **13.11-100**. As discussed above, the head-mountable device **13.11-100** includes the display **13.11-102**, the facial interface **13.11-104**, and the connector(s) **13.11-106**. In particular, as shown in FIGS. **13.11-2A-13.11-2B**, the facial interface **13.11-104** can indeed wrap around the eyes **13.11-201**, bridge the nose **13.11-202**, span the forehead **13.11-203**, and contact a maxilla facial region **13.11-204** and a zygoma facial region **13.11-205**.

[2972] Additionally shown in FIGS. **13.11-2B** are the locations of the connector(s) **13.11-106**. In particular examples, the connector(s) **13.11-106** are located at the forehead **13.11-203**, the maxilla facial region **13.11-204**, and the zygoma facial region **13.11-205**. Other locations of the connector(s) **13.11-106** are herein contemplated. However, the connector(s) **13.11-106** in at least these positions can provide a dynamic, yet stable connection between the display **13.11-102** and the facial interface **13.11-104**.

[2973] It will be appreciated that the connector(s) **13.11-106** at the different locations can be the same, or in certain cases, different. For instance, the connector(s) **13.11-106** can include a first joint (e.g., socket joint, soft joint, molded hinge joint, butterfly flexure joint, cam pivot joint, cross axis pivot joint, etc.) positioned at the forehead **13.11-203**. In addition, the connector(s) **13.11-106** can include a second joint, different from the first joint, such as a pivot joint, elastomer spring joint, soft joint, single ball joint, etc. Indeed, different arrangements and types of the connector(s) **13.11-106** can be implemented to provide a particular force profile, amount of rigidity, etc.

[2974] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-2A-13.11-2B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-2A-13.11-2B**.

[2975] As previously discussed, head-mountable device **13.11-100** can include connections at various locations. The forehead location of the connector(s) will now be discussed below.

[2976] FIGS. **13.11-3A** illustrates an example of the connection(s) **13.11-106** located at the forehead **13.11-203**. The connector(s) **13.11-106** can translate and/or rotate the facial interface **13.11-104** relative to the display **13.11-102** via the connector(s) **13.11-106**. The connector(s) **13.11-106** can vary in mobility (e.g., ranging from mechanical to organic), as will be discussed below. Specifically, FIGS. **13.11-3B-13.11-3E** illustrate a few examples of the connector(s) **13.11-106**, namely connectors **13.11-300-13.11-306**, respectively.

[2977] FIGS. **13.11-3B** illustrates the connector **13.11-300**. The connector **13.11-300** includes a single degree of freedom. In particular, the connector **13.11-300** can include a pivot connection that rotatably connects the display **13.11-102** to the facial interface **13.11-104**. For instance, the facial interface **13.11-104** can laterally pivot relative to the display **13.11-102**, where the display **13.11-102** includes a pivot point. In these or other examples, the connector **13.11-300** limits the amount of motion (e.g., to certain angular thresholds relative to the pivot point).

[2978] FIGS. **13.11-3C** illustrates the connector **13.11-302**. The connector **13.11-302** includes one degree of freedom plus translation. In certain implementations, the connector **13.11-302** includes a

cam pivot. The connector **13.11-302** can rotationally pivot similar to the connector **13.11-300**. In addition, the connector **13.11-302** can slide laterally such that the pivot point can also move. In this manner, the connector **13.11-302** can slidably and rotationally connect the display **13.11-102** to the facial interface **13.11-104**. With more free range of motion, the connector **13.11-302** allows more organic motion than the connector **13.11-300**.

[2979] FIGS. **13.11-3D** illustrates the connector **13.11-304**. The connector **13.11-304** includes a cross-axis pivot joint connector (e.g., butterfly flexure). The connector **13.11-304** therefore includes a cross-axis design allowing each contact point on the facial interface **13.11-104** to move independently in a plane (e.g., x-y plane). The thickness of the cross-axis pivot joint can vary in flexure dependent on the thickness of a cross-axis member. For example, increasing the thickness (e.g., cross-sectional area) or decreasing the length of the cross-axis member(s) increases the flexural rigidity of the cross-axis pivot joint. Inversely, decreasing the thickness or increasing the length of the cross-axis member(s) decreases the flexural rigidity of the cross-axis pivot joint. For instance, a cross-axis member with a larger cross-sectional area while maintaining the length of the cross-axis member(s) increases the amount of compression force required to deflect the cross-axis member(s). In addition, the flexural rigidity can be altered by varying the material type or durometer.

[2980] FIGS. **13.11-3E** illustrates the connector **13.11-306**. The connector **13.11-306** includes a flexure joint connector (e.g., a compliant mechanism capable of storing strain energy in one or more deformable/elastic members). The connector **13.11-306** can include a symmetrical design. In addition, the connector **13.11-306** can include a predetermined amount of stiffness in one or more directions, and a predetermined amount of flexibility in one or more different directions. The topology or architecture of the flexure joint can coincide with desired rigidity/flexibility properties in particular directions for providing mobility and/or support of the facial interface **13.11-104** relative to the display **13.11-102**.

[2981] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-3A-13.11-3E** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-3A-13.11-3E**.

[2982] FIGS. **13.11-4A** illustrates the connector(s) **13.11-106** located at the zygoma facial region **13.11-205**. The connector(s) **13.11-106** can moveably constrain the facial interface **13.11-104** to the display **13.11-102** in the same or similar ways as discussed above. The connector(s) **13.11-106** can include a variety of different connectors. Examples of the connector(s) **13.11-106**, namely connectors **13.11-400-13.11-414**, are respectively discussed below in relation to FIGS. **13.11-4B-13.11-4E**.

[2983] FIGS. **13.11-4B** illustrates the connector **13.11-400**. The connector **13.11-400** can include a pivot joint (e.g., a single ball joint). The connector **13.11-400** includes a male portion and a female portion. The male portion is positioned on the display **13.11-102**, and the female portion is positioned on the facial interface **13.11-104**. When the male and female portions of the connector **13.11-400** are joined, the facial interface **13.11-104** can move relative to the display **13.11-102**. In particular, the connector **13.11-400** can allow for spherical motion between the facial interface **13.11-104** and the display **13.11-102** of about 5 degrees, about 10 degrees, about 25 degrees, about 50 degrees, about 70 degrees, or more.

[2984] FIGS. **13.11-4C** illustrates the connector **13.11-402**. The connector **13.11-402** can include a spring joint connector (e.g., metal spring, elastomer spring, plastic spring, composite spring, etc.) located at the zygomatic location. The connector **13.11-402** can translate and rotate (e.g., to accommodate cheek or eye movement, or a user's particular facial features). The connector **13.11-**

402 can be rigidly attached to the display **13.11-102**, while being rotatably and translatably connected to the facial interface **13.11-104**.

[2985] FIGS. **13.11-4D-13.11-4E** illustrate the connectors **13.11-406-13.11-408**, respectively. The connectors **13.11-406-13.11-408** include soft joints with compressible portions **13.11-407**. The term “soft joint” references a joint that can compress and decompress. For example, a soft joint includes datums or hardstops, between which the soft joint can compress. The connector **13.11-406** includes two compressible portions **13.11-407**, while the connector **13.11-408** includes a single compressible portion. The compressible portions **13.11-407** can be disposed between rigid portions of the connectors **13.11-406, 13.11-408**. In addition, the positioning of the compressible portions **13.11-407** within the connectors **13.11-406, 13.11-408** can provide corresponding mobility in a predetermined direction. For instance, the connector **13.11-406** with two of the compressible portions **13.11-407** can include mobility (e.g., compressibility) in a first direction and mobility in a second direction different than the first direction. The connector **13.11-408** includes mobility in at least the first direction by virtue of a single compressible portion.

[2986] In one example, the compressible portions **13.11-407** include a foam material. The foam material can be adhered (e.g., glued, pasted, melted, etc.) in between portions of the connectors **13.11-406, 13.11-408**. Additionally, or alternatively, the compressible portions **13.11-407** can include materials with properties that impart flexibility, softness, compressibility, deformability, etc. Examples of such material can include silicone, polymers, elastomers, hydrogels, etc. In addition, the compressible portions **13.11-407** can include molded material, printed material, cast material, etc.

[2987] FIGS. **13.11-4F-13.11-4G** respectively illustrate cross-sectional views of the connectors **13.11-410, 13.11-412**. The connector **13.11-410** includes a socket connector. The connector **13.11-412** includes a mini-socket connector. The connectors **13.11-410, 13.11-412** are similar to the connector **13.11-400**—each including a male and female portion. The female portion (i.e., socket) corresponds to the facial interface **13.11-104**, and the male portion (i.e., pin and ball) corresponds to the display **13.11-102**. In particular, however, the connectors **13.11-410, 13.11-412** include a pivot center located at a specific place. For instance, the connector **13.11-410** includes a pivot center **13.11-411** located on the surface of the facial interface **13.11-104**. By contrast, the connector **13.11-412** includes a pivot center **13.11-413** that is positionally offset from the surface of the facial interface **13.11-104** (e.g., by about 0.5 mm to about 5 mm, or about 1.4 mm). In at least some examples, these variations of a pivot center can provide different rotational mobility.

[2988] FIGS. **13.11-4H** illustrates the connector **13.11-414**. The connector **13.11-414** is the same as the connector **13.11-306** discussed above in relation to FIGS. **13.11-3E**. Indeed, connector **13.11-414** can include a flexure joint providing compliant-mechanism based mobility between the facial interface **13.11-104** and the display **13.11-102**.

[2989] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-4A-13.11-4H** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-4A-13.11-4H**.

[2990] As briefly indicated above, examples of a head-mountable device disclosed herein include a facial interface with one or more flexure portions that conform to myriad facial profiles and contours, thereby evenly distributing pressure and providing a comfortable user experience. These flexure portions can, via connectors described above, allow the facial interface **13.11-104** to more easily translate or rotate relative to the display **13.11-102**. In accordance with one or more such examples, FIGS. **13.11-5A-13.11-5G** illustrate example facial interfaces with flexure portions.

[2991] FIGS. **13.11-5A** illustrates a perspective view of a head-mountable device **13.11-500**

including the display **13.11-102**, a facial interface **13.11-502**, and the connector(s) **13.11-106**. The facial interface **13.11-502** includes a flexure portion **13.11-504**.

[2992] As used herein, the term “flexure portion” refers to a section of a facial interface that is independently movable from other sections of the facial interface. In particular examples, a flexure portion is movable, bendable, flexible, deformable, translatable, rotatable (i.e., twistable), etc. relative to one or more other sections of the facial interface **13.11-104**. A flexure portion can include various patterns, relief cutouts, slits, holes, etc. defined by the facial interface **13.11-104** to impart the desired flexibility. In certain implementations, a flexure portion includes a serpentine pattern (e.g., winds, turns, curvature, folds, etc.). Additionally, a flexure portion can include a same material as other non-flexure (e.g., solid) sections of the facial interface **13.11-104**. In other examples, the flexure portion includes a different material than other non-flexure sections of the facial interface **13.11-104**.

[2993] As will be described below in relation to FIGS. **13.11-5B-13.11-5G**, the flexure portion **13.11-504** can be positioned at various locations (e.g., top corner region, bottom corner region, forehead region, temple region, etc.). In particular, the various flexure portions will be described positionally in relation to connectors **13.11-106a-13.11-106f**. The connectors **13.11-106a**, **13.11-106b** correspond to forehead connectors for engaging a forehead of a face. The connectors **13.11-106c**, **13.11-106d** correspond to zygoma connectors for engaging a zygoma region of a face. The connectors **13.11-106e**, **13.11-106f** correspond to maxilla connectors for engaging a maxilla region of a face.

[2994] FIGS. **13.11-5B** illustrates a facial interface **13.11-506**. The facial interface **13.11-506** is devoid of a flexure portion. That is, each section between the connectors **13.11-106a-106f** includes a solid section.

[2995] By contrast, FIGS. **13.11-5C** illustrates a facial interface **13.11-508** with flexure portions **13.11-510a-13.11-510b** positioned a temple region of a face. Indeed, the flexure portions **13.11-510a-13.11-510b** are respectively positioned in between the forehead connectors (i.e., the connectors **13.11-106a**, **13.11-106b**) and the zygoma connectors (i.e., the connectors **13.11-106c**, **13.11-106d**). The flexure portions **13.11-510a-13.11-510b** can span an entire section of the facial interface **13.11-508** between the forehead connectors and the zygoma connectors. In other implementations, the flexure portions **13.11-510a-13.11-510b** can span a partial section of the facial interface **13.11-508** between the forehead connectors and the zygoma connectors. For instance, the flexure portions **13.11-510a-13.11-510b** may stop at a connector support area that encompasses a threshold distance around one or more of the connectors **13.11-106a-106d**, thereby leaving a more supportive region underneath and immediately adjacent to the connectors **13.11-106a-13.11-106d**.

[2996] In FIGS. **13.11-5C**, solid sections—devoid of flexure portions—of the facial interface **13.11-508** also exist. In particular, the forehead region between the connectors **13.11-106a**, **13.11-106b** includes a solid section. Similarly, the bottom corner region between the zygoma connectors and the maxilla connectors are solid sections.

[2997] FIGS. **13.11-5D** illustrates a facial interface **13.11-512** with flexure portions **13.11-514a**, **13.11-514b** positioned at the temple region, and flexure portions **13.11-514c**, **13.11-514d** positioned at a bottom corner region. In particular, the flexure portions **13.11-514a**, **13.11-514b** are similarly positioned between the forehead connectors (i.e., the connectors **13.11-106a**, **13.11-106b**) and the zygoma connectors (i.e., the connectors **13.11-106c**, **13.11-106d**). In addition, the flexure portions **13.11-514c**, **13.11-514d** are respectively positioned between the zygoma connectors and the maxilla connectors (i.e., the connectors **13.11-106e**, **13.11-106f**). FIG. **13.11-5D** also shows a solid section in the forehead region between the connectors **13.11-106a**, **13.11-106b**.

[2998] FIGS. **13.11-5E** illustrates a facial interface **13.11-516** with flexure portions **13.11-518a-13.11-518e**. The flexure portions **13.11-518a**, **13.11-518c** are respectively positioned at temple regions of a face between the forehead connectors (i.e., the connectors **13.11-106a**, **13.11-106b**)

and the zygoma connectors (i.e., the connectors **13.11-106c**, **13.11-106d**). The flexure portion **13.11-518b** is positioned at a forehead region of a face (i.e., between the connectors **13.11-106a**, **13.11-106b**). The flexure portions **13.11-518d**, **13.11-518e** are positioned at a bottom corner region between the zygoma connectors and the maxilla connectors (i.e., the connectors **13.11-106e**, **13.11-106f**). Further shown in FIGS. **13.11-5E**, the only portions of the facial interface **13.11-516** devoid of a flexure portion includes the regions at the joints or connectors themselves (and a surrounding radius).

[2999] FIGS. **13.11-5F** illustrates a facial interface **13.11-520** with flexure portions positioned at the forehead region and a bottom temple region (as a temple flare). In particular, the facial interface **13.11-520** includes the flexure portion **13.11-522a** positioned at a forehead region of a face between the connectors **13.11-106a**, **13.11-106b**. In addition, the facial interface **13.11-520** includes flexure portions **13.11-522b**, **13.11-522c** respectively positioned between the forehead connectors and the zygoma connectors, and adjacent to the zygoma connectors (i.e., connectors **13.11-106c**, **13.11-106d**). In FIGS. **13.11-5F**, solid regions devoid of flexure portions also include portions of the forehead region, portions of the temple region, and all of the bottom corner region between the zygoma and maxilla connectors.

[3000] FIGS. **13.11-5G** illustrates a facial interface **13.11-524** with flexure portions **13.11-526a-13.11-526e**. The flexure portions **13.11-526a**, **13.11-526e** are respectively positioned between the forehead connectors and the zygoma connectors, and adjacent to the zygoma connectors (i.e., connectors **13.11-106c**, **13.11-106d**). The flexure portions **13.11-526b**, **13.11-526d** are respectively positioned between the forehead connectors and the zygoma connectors, and adjacent to the forehead connectors (i.e., connectors **13.11-106a**, **13.11-106b**). In addition, the flexure portion **13.11-526c** is positioned at a forehead region of a face between the connectors **13.11-106a**, **13.11-106b**. In FIGS. **13.11-5G**, solid regions devoid of flexure portions include portions of the temple region and all of the bottom corner region between the zygoma and maxilla connectors.

[3001] It will be appreciated that the foregoing examples of flexure portions can be implemented to provide varying degrees (and places) of flexibility and conformability. In this manner, the head-mountable device of the present disclosure can provide myriad different forms of flexibility as may be desired.

[3002] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-5A-13.11-5G** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-5A-13.11-5G**.

[3003] The head-mountable device of the present disclosure can be manufactured in myriad different ways. Some methods or manufacturing processes can provide various advantages (e.g., decreased material consumption, weight savings, manufacturing friendliness, etc.). The following description in relation to FIGS. **13.11-6A-12** describes manufacturing and/or associated apparatus aspects.

[3004] FIGS. **13.11-6A-13.11-6B** illustrate an example assembly **13.11-600** of a head-mountable device implementing a double shot of material in accordance with one or more examples of the present disclosure. As shown, the assembly **13.11-600** includes the connectors **13.11-106a-13.11-106f** described above. In addition, the assembly **13.11-600** includes a facial interface **13.11-602** having flexure portions **13.11-604**. The flexure portions **13.11-604** may be the same as, or similar to the flexure portions **13.11-510a**, **13.11-510b** described above in relation to FIGS. **13.11-5C**.

[3005] In particular implementations, the assembly **13.11-600** in FIGS. **13.11-6A** includes only a first shot of material. In some examples, the first shot of material includes an injection molded material. For instance, the first shot of material includes a more rigid material, such as at least one

of Pebax, TPU, TPE, PP, ABS, etc. In other examples, the first shot of material is less rigid (and more flexible).

[3006] Additionally shown in FIGS. **13.11-6A**, the assembly **13.11-600** can include tooling inserts **13.11-606**. The tooling inserts **13.11-606** can be positioned at locations configured to receive a second shot of material. The tooling inserts **13.11-606** can be positioned at various locations along the facial interface **13.11-602**. In particular implementations, however, the tooling inserts **13.11-606** are positioned at the forehead region between the connectors **13.11-106a**, **13.11-106b**, and at temple regions between the forehead connectors and the zygoma connectors (i.e., between the connectors **13.11-106a**, **13.11-106b** and the connectors **13.11-106c**, **13.11-106d**).

[3007] In FIGS. **13.11-6B**, the assembly **13.11-600** includes a second shot of material applied to the tooling inserts **13.11-606**. In one or more examples, the second shot of material includes an overmolded, flexible material. For instance, the second shot of material includes an elastomer material (e.g., Neoprene, EVA, PVC, Acrylonitrile Butadiene Rubber, Ethylene Propylene, Fluorelastomer, Silicone, etc.). Thus, in some examples, the first shot of material and the second shot of material can be different. In these or other examples, the assembly **13.11-600** implementing double shots of material can provide various advantages, such as strategically positioned support, improved flexibility, etc.

[3008] In alternative examples, more than two shots of material can be implemented. Further, in some alternative examples, only a single shot of material is implemented. For instance, the assembly **13.11-600** can include a single, lightweight material (e.g., all aluminum).

[3009] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-6A-13.11-6B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-6A-13.11-6B**.

[3010] Other variations of combined materials are also herein contemplated. For example, some head-mountable devices of the present disclosure include a display frame of the display **13.11-102**, where the display frame includes a combination of frame shells forming a box-type frame enclosure. In accordance with one or more such examples, FIGS. **13.11-7A-13.11-7B** illustrate an example display **13.11-700** of a head-mountable device. In particular, FIGS. **13.11-7A-13.11-7B** respectively illustrate assembled and exploded views of the display **13.11-700**.

[3011] As shown, the display **13.11-700** includes a first frame shell **13.11-702** and a second frame shell **13.11-704**. The first frame shell **13.11-702** includes a pocket **13.11-706** defined by the surface of the first frame shell **13.11-702**. The pocket **13.11-706** can include one or more recesses, cutouts, etc. In certain implementations, the pocket **13.11-706** is defined by tooling surfaces (e.g., for injection molding, casting, etc.). The pocket **13.11-706** can additionally include webbing, such as a mold webbing, (indicated in dashed lines) that sections off or separates different portions of the pocket **13.11-706**.

[3012] In some examples, the first frame shell **13.11-702** and the second frame shell **13.11-704** include different materials (albeit not required). For instance, the first frame shell **13.11-702** can include a plastic material, and the second frame shell **13.11-704** can include a metal material. In this manner, the second frame shell **13.11-704** can be combined with the first frame shell **13.11-702** to increase a stiffness or rigidity across the display **13.11-700**. Indeed, the second frame shell **13.11-704** as a stiffener member can provide structural support, thereby stabilizing the display frame and increasing resistance to stresses (e.g., torsional stress, compressive stress, tension stress, etc.).

[3013] The first frame shell **13.11-702** and the second frame shell **13.11-704** can be formed or assembled in a variety of different ways. In some examples, the first frame shell **13.11-702** and the

second frame shell **13.11-704** can be manufactured using molding techniques (e.g., injection molding, casting, etc.). In certain implementations, the first frame shell **13.11-702** includes a first-shot mold, and the second frame shell **13.11-704** includes a second-shot mold.

[3014] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-7A-13.11-7B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-7A-13.11-7B**.

[3015] As discussed above, displays (specifically display frames) can include a combination of materials. Additionally, or alternatively, display frames have one or more a relief cutouts. In accordance with one or more such examples, FIGS. **13.11-8A-13.11-8B** illustrate an example display **13.11-800**.

[3016] As shown, the display **13.11-800** includes relief cutouts **13.11-802** defined by the surface of the display frame. As used herein, the term “relief cutouts” refers to thru-holes, dimples, slits, slots, core-outs, recesses, etc. In addition, a relief cutout provides weight advantages over a conventional head-mountable device while maintaining the structural integrity (e.g., deflection integrity, stress integrity, etc.) of the display frame.

[3017] Here, the relief cutouts **13.11-802** include thru-holes in the display frame. The thru-holes can be sized and shaped in myriad different ways. In some examples, the thru-holes are rectangular, square, triangular, circular, etc. In other examples, the thru-holes are lozenge-shaped (e.g., cylindrical, pill-shaped) thru-holes. The relief cutouts **13.11-802** also include a spacing that can be optimized or tuned (as may be desired). Indeed, the relief cutouts **13.11-802** can be spaced apart to satisfy (e.g., meet or exceed) a threshold force or stress/strain profile of the display **13.11-800**.

Additionally, or alternatively, the relief cutouts **13.11-802** can be spaced apart to provide a particular amount of flexibility or rigidity. In certain implementations, the relief cutouts **13.11-802** are omitted from certain areas (e.g., at connector support areas corresponding to the connector(s) **13.11-106**). Thus, some connector support areas are devoid of the relief cutouts **13.11-802**.

[3018] Additionally, as shown in FIGS. **13.11-8B**, the display **13.11-800** includes a support structure **13.11-804** and an overmolding **13.11-806**. The support structure **13.11-804** can be rigid (e.g., a metal material). By contrast, the overmolding **13.11-806** can include a plastic material, elastomer material, etc. The support structure **13.11-804** can be positioned at predetermined areas of the display **13.11-800**, for instance, at areas corresponding to the connector(s) **13.11-106** (e.g., for increased rigidity and/or structural integrity per finite element analysis). To illustrate, the support structure **13.11-804** is positioned directly underneath the connector(s) **13.11-106** and extends a distance past at least one relief cutout. In certain implementations, the support structure **13.11-804** is positioned all along the display **13.11-800**.

[3019] In particular implementations, the support structure **13.11-804** is positioned at an offset distance **13.11-808** from the overmolding **13.11-806**. The offset distance **13.11-808** can range from about 0.2 mm to about 4 mm. In certain examples, the offset distance **13.11-808** is about 0.75 mm.

[3020] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-8A-13.11-8B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-8A-13.11-8B**.

[3021] The head-mountable devices, and in particular the frames of the displays of the present disclosure, can be formed of various materials and with different geometric variations. These

example display frames can provide various advantages, including decreased weight. In accordance with one or more such examples, FIGS. **13.11-9A-13.11-9B** respectively show example head-mountable devices **13.11-900a**, **13.11-900b** with example displays **13.11-902**, **13.11-904**.

[3022] In FIGS. **13.11-9A**, the head-mountable device **13.11-900a** includes the display **13.11-902** connected to the facial interface **13.11-104** via the connector(s) **13.11-106**. The display **13.11-902** in this example includes a frame including a single, lightweight material. For instance, the display **13.11-902** includes an all-metal frame. In certain implementations, the display **13.11-902** includes stainless steel (e.g., SUS 17-4PH). In other implementations, the display **13.11-902** includes aluminum (e.g., AL 6061). In alternative embodiments, the display **13.11-902** includes at least one of a composite material (e.g., carbon fiber), polycarbonate or thermoplastic material (e.g., Lexan), or other suitable material with a higher strength-to-weight ratio (e.g., titanium, magnesium, graphene, ceramics, etc.).

[3023] In FIGS. **13.11-9B**, the head-mountable device **13.11-900b** similarly includes the display **13.11-904** connected to the facial interface **13.11-104** via the connector(s) **13.11-106**. The display **13.11-904** can include one or more materials as just described for the display **13.11-902**. In particular, however, the display **13.11-904** includes a frame including cored-out regions **13.11-906** in the forehead and temple areas of a face. The cored-out regions **13.11-906** can reduce the amount of material in the display **13.11-904**, thereby reducing the weight of the display **13.11-904**. In certain implementations, the cored-out regions **13.11-906** correspond to areas of the display **13.11-904** between the connector(s) **13.11-106**. However, it will be appreciated that the cored-out regions **13.11-906** can be positioned in additional or alternative locations along the frame of the display **13.11-904**, as may be desired.

[3024] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-9A-13.11-9B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-9A-13.11-9B**.

[3025] The present disclosure also contemplates stiffener members disposed between the facial interface **13.11-104** and the display **13.11-102**. In accordance with one or more such examples, FIGS. **13.11-10A-13.11-10B** illustrate head-mountable devices **13.11-1000a**, **13.11-1000b**. In particular, FIGS. **13.11-10A** shows the head-mountable device **13.11-1000a** includes a display **13.11-1002a** (i.e., a display frame) connected to the facial interface **13.11-104** via the connector(s) **13.11-106**, as similarly described above. In addition, the head-mountable device **13.11-1000a** includes a stiffener member **13.11-1004a** disposed between the display **13.11-1002a** and the facial interface **13.11-104**. The stiffener member **13.11-1004a** can include an additional frame, skeleton, hoop, or support structure. The stiffener member **13.11-1004a** can extend partially along the display **13.11-1002a** (e.g., until the zygoma connector), or in certain cases all along the display **13.11-1002a** (e.g., until the maxilla connector).

[3026] In particular implementations, a gap exists between the stiffener member **13.11-1004a** and the display **13.11-1002a**. In these cases, the connectors **13.11-106** can bridge the gap between the stiffener member **13.11-1004a** and the display **13.11-1002a**.

[3027] The display **13.11-1002a** further includes cored-out regions **13.11-1006**. The cored-out regions **13.11-1006** can provide weight savings, as may be desired.

[3028] Likewise, FIGS. **13.11-10B** shows the head-mountable device **13.11-1000b** includes a display **13.11-1002b** connected to the facial interface **13.11-104** via the connector(s) **13.11-106**, as similarly described above. In addition, the head-mountable device **13.11-1000b** includes a stiffener member **13.11-1004b** disposed between the display **13.11-1002b** and the facial interface **13.11-104**. The stiffener member **13.11-1004b** can include an additional frame, skeleton, hoop, or support

structure. The stiffener member **13.11-1004b** can extend partially along the display **13.11-1002b** (e.g., until the zygoma connector), or in certain cases all along the display **13.11-1002a** (e.g., until the maxilla connector). The display **13.11-1002b** further includes cored-out regions **13.11-1010**, similar to the cored-out regions **13.11-1006** described above.

[3029] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-10A-13.11-10B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-10A-13.11-10B**.

[3030] As mentioned above, the head-mountable devices of the present disclosure can include myriad different configurations of cored-out portions. In accordance with one or more such examples, FIGS. **13.11-11A-13.11-11C** illustrate display frames **13.11-1100a-13.11-1100c** with respective cored-out portions (e.g., thru-holes, partial holes, recesses, etc.). In particular, FIGS. **13.11-11A** shows the display frame **13.11-1100a** includes cored-out portions **13.11-1102a-13.11-1102d** at the forehead and temple regions of a face. FIGS. **13.11-11B** shows the display frame **13.11-1100b** including cored-out portions **13.11-1102a-13.11-1102b** (thus only including cored-out regions at the forehead). In addition, FIGS. **13.11-11C** shows the display frame **13.11-1100c** including cored-out portions **13.11-1104**. The cored-out portions **13.11-1104** are defined by one or more backside surfaces oriented towards a facial interface (not shown). Thus, the cored-out portions **13.11-1104** are perpendicular to the cored-out portions shown in FIGS. **13.11-11A-13.11-11B**.

[3031] It will be appreciated that the cored-out portions of the display frames **13.11-1100a-13.11-1100c** can be located in various positions and in various configurations or patterns (e.g., to decrease the overall weight of the display frame). The cored-out portions can also be configured to maintain a structural stability of the display frame.

[3032] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-11A-13.11-11C** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-11A-13.11-11C**.

[3033] FIGS. **13.11-12** illustrates a display frame **13.11-1200** including an array of holes **13.11-1202** in accordance with one or more examples of the present disclosure. The array of holes **13.11-1202** can reduce an amount of material (and therefore, weight) of the display frame **13.11-1200**. The array of holes **13.11-1202** can include a hole density such that at least (or at most) a certain number of holes is found within a predetermined area of the display frame **13.11-1200**. As an alternative to the array of holes **13.11-1202**, dimples or non-thru-holes can similarly be implemented in the display frame **13.11-1200**.

[3034] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-12** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-12**.

[3035] FIGS. **13.11-13** shows a display frame **13.11-1300** in accordance with one or more examples of the present disclosure. As illustrated, the display frame **13.11-1300** includes a single-

shot plastic portion **13.11-1302**. A variety of different plastics can be implemented. Moreover, the single-shot plastic portion **13.11-1302** can assume any number of different shapes and forms (albeit schematically shown as a block with various straight and curved edges).

[3036] In addition, the display frame **13.11-1300** can include metal stiffeners **13.11-1304**.

Examples of the metal stiffeners **13.11-1304** include sheets or pieces of metal (whether molded, shaped, cast, etc.). Further, the metal stiffeners **13.11-1304** can be arranged in different ways. For example, the metal stiffeners **13.11-1304** can be arranged in an array that structurally couples or connects them to each other (e.g., via attachment mechanisms **13.11-1308** and fasteners **13.11-1310**). For instance, the metal stiffeners **13.11-1304** can be arranged partially around a perimeter of the single-shot plastic portion **13.11-1302**. In other examples, the metal stiffeners can be arranged entirely around a perimeter of the single-shot plastic portion **13.11-1302**.

[3037] The attachment mechanisms **13.11-1308** can include various pins, clasps, hinges, cams, swivels, balls, knuckles, springs, etc. Likewise, the fasteners **13.11-1310** can include screws, bolts, pins, press-fits, welds, bonds, etc.

[3038] In some examples, the display frame **13.11-1300** includes adhesives (not shown) to structurally couple the metal stiffeners **13.11-1304** (and associated attachment mechanisms **13.11-1308**) and the single-shot plastic portion **13.11-1302**. Examples of adhesives include glue, epoxy, thermal bonds, tape strips, etc. In alternative examples, the metal stiffeners **13.11-1304** and the single-shot plastic portion **13.11-1302** include no adhesives in between these components. For example, the metal stiffeners **13.11-1304** and the single-shot plastic portion **13.11-1302** can be sandwiched together via applied pressure from one or more additional frame layers, fabric, etc.

[3039] In some examples, the display frame **13.11-1300** can include additional layers (not shown). For instance, the display frame **13.11-1300** can include a final plastic “cover” to provide cosmetic and/or additional stiffness properties to the display frame **13.11-1300**. The additional cover can be attached to the metal stiffeners **13.11-1304** in a same or similar fashion as just described between the metal stiffeners **13.11-1304** and the single-shot plastic portion **13.11-1302**.

[3040] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-13** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-13**.

[3041] FIGS. **13.11-14A** illustrates a top view of a frame **13.11-1410** of a device seal according to one embodiment. The frame **13.11-1410** can be substantially similar to, including some or all of the features of, the frames described herein.

[3042] In some examples, the frame **13.11-1410** can include one or more stiffeners. For example, the frame **13.11-1410** can include a top stiffener **13.11-1421** that is positioned on a top forehead region of the frame **13.11-1410**. The top stiffener **13.11-1421** can be curved or angled to match a curve of the frame **13.11-1410**. In some examples, the curve of the top of the frame **13.11-1410** and the top stiffener **13.11-1421** are designed to match the curvature of a user's forehead. In some examples, the curvature of the top of the frame **13.11-1410** and the top stiffener **13.11-1421** are designed to allow for a gap between the user's forehead and the frame **13.11-1410**. The top stiffener **13.11-1421** can aid in maintaining the gap between the frame **13.11-1410** and the user's forehead. Thus, the top stiffener **13.11-1421** can provide structural support to the frame **13.11-1410** and can be used to modify the stiffness and/or structural integrity of the frame. In some examples, there can be multiple top stiffeners in the forehead region of the frame **13.11-1410**, and each of the multiple top stiffeners can have different profiles, geometries, and/or material properties that effect the properties imparted to the frame **13.11-1410**.

[3043] The top stiffener **13.11-1421** can include or can be made from any suitable stiff or rigid

material capable of maintaining its shape under the amounts of stress likely to be encountered from typical use of the HMD, thereby increasing comfort and predictability to the user. In some examples, the top stiffener **13.11-1421** can undergo elastic deformation. The top stiffener **13.11-1421** can include sheet metal, aluminum, steel, carbon fiber, plastic, combinations thereof, or any other suitable material capable of stiffening the frame **13.11-1410**.

[3044] The top stiffener **13.11-1421** can be secured to the frame **13.11-1410** using any suitable attachment methods. In some examples, the top stiffener **13.11-1421** is fastened to the frame **13.11-1410** using fastening screws **13.11-1423**. In some examples, the fastening screws **13.11-1423** can attached ends of the top stiffener **13.11-1421** to a metallic anchor **13.11-1425** of the frame **13.11-1410**, which in turn is connected to posts **13.11-1427** that engage the facial interface of the device seal. In some examples, the top stiffener **13.11-1421** can be laser welded to the frame **13.11-1410**, such as via the metallic anchor **13.11-1425** of the frame **13.11-1410**. In some examples, the top stiffener **13.11-1421** can be adhered to the frame **13.11-1421** using any suitable adhesive.

[3045] In some examples, the frame **13.11-1410** can include one or more side stiffeners **13.11-1429** that are positioned on along a side region of the frame **13.11-1410**. The side stiffeners **13.11-1429** can be curved or angled to match a curvature of the frame **13.11-1410**. In some examples, the curve of the side of the frame **13.11-1410** and the side stiffeners **13.11-1429** are designed to match the curvature of a user's brow, temple, and/or cheeks. In some examples, the curvature of the side of the frame **13.11-1410** and the side stiffener **13.11-1429** are designed to allow for a gap between the user's face and the frame **13.11-1410**. The side stiffeners **13.11-1429** can aid in maintaining the gap between the frame **13.11-1410** and the user's face. Thus, the side stiffeners **13.11-1429** can provide structural support and added comfort to the frame **13.11-1410**.

[3046] The side stiffeners **13.11-1429** can include or be made from any suitable stiff or rigid material. In some examples, the side stiffeners **13.11-1429** can experience elastic deformation during use. For example, the side stiffeners **13.11-1429** can include sheet metal, aluminum, steel, carbon fiber, polymer, or any other suitable material capable of stiffening the frame **13.11-1410**. The side stiffener **13.11-1429** can be made from the same material or a different material than the top stiffener **13.11-1421**.

[3047] The side stiffeners **13.11-1429** can be secured to the frame **13.11-1410** using any suitable attachment methods or systems. In some examples, the side stiffeners **13.11-1429** are fastened to the frame **13.11-1410** using fastening screws **13.11-1423**. In some examples, the fastening screws **13.11-1423** can attached ends of the side stiffeners **13.11-1429** to a metallic anchor **13.11-1425** of the frame **13.11-1410**, which in turn is connected to posts **13.11-1427** that engage the facial interface of the device seal. In some examples, the side stiffeners **13.11-1429** can be laser welded to the frame **13.11-1410**, such as the metallic anchor **13.11-1425** of the frame **13.11-1410**. In some examples, the side stiffeners **13.11-1429** can be adhered to the frame **13.11-1421** using any suitable adhesive.

[3048] FIGS. **13.11-14B** illustrates a cross-sectional view of the frame **13.11-1410** of a device seal according to one embodiment. As illustrated, the stiffener **13.11-1421** can be positioned within a housing of the frame **13.11-1410**. For instance, the frame **13.11-1410** can include an upper portion **13.11-1411** and a lower portion **13.11-1412** that engage to form an internal volume. The upper portion **13.11-1411** and the lower portion **13.11-1412** can be plastic, metal, or any other suitable material for the frame **13.11-1410**. The stiffener **13.11-1421** can be positioned directly on the lower portion **13.11-1412** of the frame **13.11-1410** to directly strengthen and stabilize the frame **13.11-1410**. In some examples, the stiffener **13.11-1421** is secured in place by adhesive **13.11-1433**. The adhesive **13.11-1433** can be located at a number of locations between the stiffener **13.11-1421** and the lower portion **13.11-1412**. Adhesive can also be placed between the stiffener **13.11-1421** and the upper cover **13.11-1411**. In this way, forces experienced by the lower portion **13.11-1412** and the upper cover **13.11-1411** can be passed through the adhesive **13.11-1433** to be carried by the stiffener **13.11-1421**. In some examples, the stiffener **13.11-1421** is secured in place by being

sandwiched or pressed between the upper cover **13.11-1411** and the lower portion **13.11-1412** of the frame **13.11-1410**.

[3049] In some examples, the stiffener **13.11-1421** can include an angled edge. For example, the edge of the stiffener **13.11-1421** closest to the user's face can be angle so as to run substantially perpendicular or flat to the user's face. The angled edge can prevent a hard or sharp edge of the stiffener from being directed toward the user to enhance prolonged user comfort. The angled edge of the stiffener **13.11-1421** can also assist in securing the stiffener **13.11-1421** in place and prevent movement within the internal volume of the frame **13.11-1410**.

[3050] FIGS. **13.11-14C** illustrates a bottom view of the frame **13.11-1410** of a device seal according to one exemplary embodiment. The frame **13.11-1410** can include one or more stiffeners along the sides of the frame **13.11-1410**. For example, the frame **13.11-1410** can include two stiffeners **13.11-1429** along each side of the frame **13.11-1410**. In some examples, the frame **13.11-1410** can include a nose stiffener **13.11-1431** positioned along the nose region **13.11-1415**.

[3051] The nose stiffener **13.11-1431** can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as top stiffener **13.11-1421** and side stiffeners **13.11-1429**. In some examples, there can be multiple nose stiffeners in the nose region **13.11-1415** of the frame **13.11-1410**

[3052] The nose stiffener **13.11-1431** can be curved or angled to match a curve of the nose region **13.11-1415**. In some examples, the curve of the nose region **13.11-1415** and the nose stiffener **13.11-1431** is designed to match the curvature of a user's nose. In some examples, the curvature of the nose region **13.11-1415** and the nose stiffener **13.11-1431** is designed to allow for a gap between the user's nose and the frame **13.11-1410**. The nose stiffener **13.11-1431** can aid in maintaining the gap between the frame **13.11-1410** and the user's nose. Thus, the nose stiffener **13.11-1431** can provide structural support to the frame **13.11-1410**. The nose stiffener **13.11-1431** can be made from the same or different material than the top stiffener **13.11-1421** or the side stiffeners **13.11-1429**. The nose stiffener **13.11-1431** can be attached to the nose portion **13.11-1415** of the frame **13.11-1410** by at least one of the following: laser welding, fastening screws, adhesive, or any other suitable coupling method.

[3053] FIGS. **13.11-14D** illustrates a top view of the frame **13.11-1410** including stiffeners **13.11-1421**, **13.11-1429** to improve the structural stability and rigidity of the frame **13.11-1410**. In some examples, adhesive **13.11-1433** can be placed on a top surface of the stiffeners **13.11-1421**, **13.11-1429** to secure the stiffeners to the top cover of the frame (shown in FIG. **13.11-14B**).

[3054] In some examples, the frame **13.11-1410** can include a hard stop component **13.11-1435**. The hard stop component **13.11-1435** can be an adhesive or any other suitable coupler to fixedly secure the ends of the stiffeners, and the anchor **13.11-1425** in place on the frame **13.11-1410**. Thus, when a force is applied to the post **13.11-1427**, the hard stop component **13.11-1435** can prevent the anchor **13.11-1425** from moving relative to the frame **13.11-1410** and stiffeners **13.11-1421**, **13.11-1429**. Thus, the hard stop component **13.11-1435** acts as a deflection limiter. In the case that the hard stop component **13.11-1435** is an adhesive, it can be the same or a different adhesive than the adhesives **13.11-1433** applied on the stiffeners.

[3055] As mentioned above, many different types of connectors between a facial interface and a display frame are herein contemplated. The following description for FIGS. **13.11-14-13.11-19** corresponds to an example connector that includes a floating connection between a facial interface and a display frame. As a floating connection, the connector can be attached to one of the display frame or the facial interface, but not both, as will be described in more detail below.

[3056] FIGS. **13.11-14** illustrates a perspective view of an example connector **13.11-1400** in accordance with one or more examples of the present disclosure. The connector **13.11-1400** can include a floating connection. The connector **13.11-1400** can therefore include a bumper, a compliant stop, or a contact mechanism positioned between a display frame **13.11-1402** and a facial interface **13.11-1404**. The display frame **13.11-1402** and the facial interface **13.11-1404** are,

respectively, the same as or similar to the display frames and facial interfaces discussed above. However, as shown specifically in FIGS. **13.11-14**, the connector **13.11-1400** is only attached to the display frame **13.11-1402** (i.e., not also attached to the facial interface **13.11-1404**). This constitutes a floating connection where the connector **13.11-1400** can contact or abut the facial interface **13.11-1404**, but can also move relative to the facial interface **13.11-1404**. For example, the connector **13.11-1400** can translate or rotate along a surface of the facial interface **13.11-1404**. In certain examples, rounded edges or smooth surfaces of the connector **13.11-1400** can facilitate such movement or range of motion. As another example, the connector **13.11-1400** can compress against (or have more surface area contact with) the facial interface **13.11-1404**, such as when donning the head-mountable device. In yet another example, the connector **13.11-1400** can retract or bias away from the facial interface **13.11-1404**. Accordingly, the connector **13.11-1400** can provide desired flexibility and compliance (e.g., for improved comfort when wearing or adjusting the head-mountable device), yet also provide desired rigidity, support, and load-bearing functionality.

[3057] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-14** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-14**.

[3058] FIGS. **13.11-15A** illustrates a side view of an example connector positioned between a display frame and a facial interface in accordance with one or more examples of the present disclosure. As shown, the connector **13.11-1400** can include one or more posts. In particular implementations, the connector **13.11-1400** includes a post **13.11-1502a**. In other implementations, the connector **13.11-1400** includes a post **13.11-1502b**.

[3059] In these or other examples, a post can include an element with rigid and/or pliant portions. In some examples, a post can include a soft outer portion and a rigid inner portion (e.g., a sheath or sheet metal overmolded by a softer, cushioning material). A post can include a contact mechanism for contacting the facial interface **13.11-1404**. Additionally or alternatively, a post can include an arm, a bumper, a mechanical stop, a damper, a shock absorber, a spring, an actuator, a crumple element, a mechanical fuse, a break-away element, etc. In at least some examples, a post can include surfaces, joints, etc. that include a radius (or curvature) allowing movement, deflection, and/or energy absorption.

[3060] As shown in FIGS. **13.11-15A**, the post **13.11-1502a** is longer than the post **13.11-1502b**. It will be appreciated that the length of the posts **13.11-1502a**, **13.11-1502b** can determine the gap or distance between the connector **13.11-1400** and the facial interface **13.11-1404**. For example, the post **13.11-1502a** is positioned a distance **13.11-1506** from a contact surface **13.11-1508** of the facial interface **13.11-1404**. Similarly, the post **13.11-1502b** is positioned a distance **13.11-1504** from the contact surface **13.11-1508** of the facial interface **13.11-1404**.

[3061] In addition to length of the posts **13.11-1502a**, **13.11-1502b**, an angle of the posts **13.11-1502a**, **13.11-1502b** relative to the display frame **13.11-1402** can also affect the distances **13.11-1504**, **13.11-1506** to the facial interface **13.11-1404**. The post **13.11-1502a** is positioned at an angle **13.11-1510** relative to the display frame **13.11-1402**. The post **13.11-1502b** is positioned at an angle **13.11-1512** relative to the display frame **13.11-1402**. These angles may be preset, adjusted, or actuated. In particular examples, the angles **13.11-1510**, **13.11-1512** can be dynamically modified (e.g., decreased) in response to the post contacting the contact surface **13.11-1508** and flexing upward. The angles **13.11-1510**, **13.11-1512** can also correspond to the angle of incidence or angle of contact with the contact surface **13.11-1508**. Likewise, the angles **13.11-1510**, **13.11-1512** can affect the point of contact with the contact surface **13.11-1508**. The point of contact and angle of incidence (as well as the distances **13.11-1504**, **13.11-1506**) can include tunable factors for tuning

comfort, fit, adjustability, load responsiveness, etc. as may be desired.

[3062] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-15A** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-15A**.

[3063] With respect to the facial interface **13.11-1404**, it will be appreciated that the facial interface **13.11-1404** can include many different variations. In some examples, the facial interface **13.11-1404** can include thicker regions and/or thinner regions at certain locations along the facial interface **13.11-1404**, can be fixed or removable, and can be removably attached by any number of systems such as hook and loop, magnets, fasteners, and the like. In accordance with one or more such examples, FIGS. **13.11-15B** illustrates a cross-sectional view of a facial interface **15-1502**. FIG. **15C** illustrates a cross-sectional view of a forehead region **13.11-15-1503** of the facial interface **13.11-15-1502**, while FIG. **15D** illustrates a cross-sectional view of a cheek region **13.11-15-1505** of the facial interface **13.11-15-1502**. The facial interface **13.11-15-1502** can be substantially similar to, including some or all of the features of, the facial interfaces described herein.

[3064] In some examples, the facial interface **13.11-15-1502** includes a compressible portion that can include a central region **13.11-15-1509** and an extended region **13.11-15-1507**. In some examples, the central region **13.11-15-1509** and the extended region **13.11-15-1507** can be a single unitary component (e.g., a foam piece). In some examples, the central region **13.11-15-1509** can be a separate and distinct material from the extended region **13.11-15-1507** (e.g., two different types of foam pieces). The central region **13.11-15-1509** can be shaped, sized, and positioned to correspond to the base of the frame. In other words, the central region **13.11-15-1509** can reside directly adjacent the base plastic. In contrast, the extended region **13.11-15-1507** can extend beyond a footprint of the base, such that the extended region **13.11-15-1507** overhangs the base.

[3065] In some examples, the extended region **13.11-15-1507** can run along an entirety of the facial interface **13.11-15-1502**. The compressible portion (or combination of the extended region **13.11-15-1507** and the central region **13.11-15-1509**) can have a varying cross-section or be asymmetric relative to the base. For example, the extended region **13.11-15-1507** can taper as it travels from the forehead region **13.11-15-1503** to the cheek region **13.11-15-1505**. In other words, the extended region **13.11-15-1507** can decrease in size as it approaches the cheek region **13.11-15-1505**, so that it is wider at the forehead, and less wide at the cheeks. For example, at the forehead region **13.11-15-1503** shown in FIGS. **13.11-15C**, the extended region **13.11-15-1507** can overhang a base by approximately equal to or greater than 3.4 mm. In contrast, at the cheek region **13.11-15-1505** shown in FIGS. **13.11-15C**, the extended region **13.11-15-1507** can overhang a base by approximately equal to or less than 3.4 mm. Accordingly, the extended region **13.11-15-1507** can extend beyond or overhang the central region **13.11-15-1509** and the base plastic both above and below, as oriented in the figures. The varying asymmetric cross-section of the extended region **13.11-15-1507** can be intentionally varied to correspond to user facial features, thereby increasing comfort and fit of the head-mountable device.

[3066] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-15B-13.11-15D** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-15B-13.11-15D**.

[3067] As mentioned above, a connector of the present disclosure can include multiple portions with differing material properties. In accordance with one or more such examples, FIGS. **13.11-16** illustrates a cross-sectional view of the connector **13.11-1400** including a connector frame **13.11-1600** for a post **13.11-1602**. The cross-sectional view of FIGS. **13.11-16** is taken along the cross-sectional slice indicated in FIGS. **13.11-15A**.

[3068] As shown in FIGS. **13.11-16**, the connector frame **13.11-1600** can include a rigid or semi-rigid core extending at least partially through the post **13.11-1602**. The connector frame **13.11-1600** can provide a desired stiffness or shape to the post **13.11-1602**. For example, as shown in FIGS. **13.11-16**, the connector frame **13.11-1600** includes a bend adjoining a portion attached to the display frame **1402** and the post **13.11-1602** extending away from the display frame **13.11-1402**. In certain examples, the post **13.11-1602** is overmolded onto, adhered to, or fastened to the connector frame **13.11-1600**. In other examples, the post **13.11-1602** is formed integral with the connector frame **13.11-1600**. In at least one example, the connector frame **13.11-1600** includes sheet metal and the surrounding portion of the post **13.11-1602** can include an elastomer material, a rubber, a silicone, a composite, a plastic, or any combination thereof.

[3069] As will be shown more below in relation to FIGS. **13.11-17**, the connector frame **13.11-1600** can include one or more elements for attaching the connector **13.11-1400** to the display frame **13.11-1402**.

[3070] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-16** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-16**.

[3071] FIGS. **13.11-17** illustrates a top view of an example connector in accordance with one or more examples of the present disclosure. As shown, the connector **13.11-1400** can include the connector frame **13.11-1600** discussed above. In particular, the connector frame **1600** can include one or more through holes **13.11-1700**. Via the through-holes **13.11-1700**, the connector frame **13.11-1600** can be fastened to the display frame **13.11-1402** (having corresponding holes that are align-able with the connector frame **13.11-1600**). In this way, the connector frame **13.11-1600** can be secured to the display frame **13.11-1402**.

[3072] In some examples, the connector **13.11-1400** can be attached to a stiffener **13.11-1604**. The stiffener **13.11-1604** can provide certain material property advantages to the connector **13.11-1400**. For example, the stiffener **13.11-1604** can be used to tune the spring force, biasing strength, flexibility, etc. of the connector frame **13.11-1600**. Additionally or alternatively, the spring force, biasing strength, flexibility, etc. for the connector **13.11-1400** as a whole can be tuned via pre-biasing the connector frame **13.11-1600** according to a predetermined force-displacement curve. In other examples, the spring force, biasing strength, flexibility, etc. for the connector **13.11-1400** can be tuned via an overmolded material for the post **13.11-1602** or by adding a spring member, a torsion bar, or an actuator to the connector **13.11-1400**.

[3073] Additionally or alternatively, the stiffener **13.11-1604** can provide a datum surface (e.g., a flat mating surface) for the connector frame **1600**. In certain examples, the fixture biasing can be implemented to properly position, orient, and/or attach the connector frame **13.11-1600** to the stiffener **13.11-1604**. Additionally or alternatively to the stiffener being used as a datum, an adhesive can be used to flow into a desired position and solidify into a datum-like element relative to the connector frame **13.11-1600**.

[3074] Additionally, in particular examples, the connector frame **13.11-1600** (and the connector **13.11-1400** as a whole) can be removed from the display frame **13.11-1402**. For example, fasteners (not shown) can be removed from the through-holes **13.11-1700** to allow removal of the connector

13.11-1400 from the display frame **13.11-1402**. In some examples, the fasteners and the through-holes **13.11-1700** can be user-accessible and/or manufacturer-accessible for swapping out the connector **13.11-1400** for a connector with a different size post and/or a post with a different angle relative to the display frame (e.g., a shorter or longer post as shown in FIGS. **13.11-15A**).

Accordingly, in some examples, the connector **13.11-1400** is modular.

[3075] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-17** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-17**. For example, a connector **13.11-1606** shown in the background of FIGS. **13.11-16** can include one or more other connectors described above in the present disclosure.

[3076] It will be appreciated that one or more connectors of the present disclosure can be modified to exhibit certain material properties (e.g., increased strength, resilience, flexibility, etc.). In accordance with one or more such examples, FIGS. **13.11-18** illustrates a side perspective view of a base of a connector where it is attached to a display frame. In particular, FIGS. **13.11-18** illustrates a gusset **13.11-1800** formed on the surface of the connector frame **13.11-1600**. The gusset **13.11-1800** can include a feature on the connector **13.11-1400** that helps to reinforce the connector attachment to the display frame **13.11-1402** (e.g., via the stiffener **13.11-1604**). In some examples, the gusset **13.11-1800** includes a plate adhered, bonded, welded, or molded onto the connector frame **13.11-1600**. In certain examples, the gusset **1800** includes a reinforced joint. In at least one example, the gusset **13.11-1800** is shaped to distribute loads from concentrating around fasteners **13.11-1802** on the connector frame **13.11-1600**.

[3077] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-18** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-18**.

[3078] FIGS. **13.11-19** illustrates another cross-sectional view of the connector **13.11-1400** in accordance with one or more examples of the present disclosure. As shown in FIGS. **13.11-19**, the connector frame **13.11-1600** can be discontinuous at certain areas of the connector frame **1600**. For example, the connector frame **13.11-1600** can include one or more reliefs **13.11-1900** (e.g., cutouts for added flexibility, decreased weight, or less material consumption). In these or other examples, the post **13.11-1602** can include corresponding portions positioned inside the relief **13.11-1900** that are solid plastic, elastomer, or other material.

[3079] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-19** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.11-19**. The following description discusses example adhesives in a head-mountable device. Indeed, a head-mountable device of the present disclosure can include a variety of different adhesives that perform various functions and/or impart certain properties.

[3080] In particular, FIGS. **13.11-20-13.11-21** respectively show perspective and top views of

example adhesives in an example head-mountable device. The term adhesives can refer to one or more types of glue, epoxy adhesives, polyurethane adhesives, polyimide adhesives, paste adhesives, liquid adhesives, film adhesives, pellet adhesives, strip adhesives, hot melt, thermosetting adhesives, pressure sensitive adhesives, contact adhesives, structural adhesives, non-structural adhesives, semi-structure adhesives, tape, bonds, welds, etc.

[3081] Specifically, as shown, the display frame **13.11-1402** is depicted with a cover or housing portion removed for illustration purposes (e.g., to expose various adhesives that can be included within the display frame **13.11-1402**). In some examples, the display frame **13.11-1402** includes a connector **13.11-2000**, as described above. In at least certain implementations, the connector **13.11-2000** can include a different material than the stiffener **13.11-1604**. For instance, in one example, the connector **13.11-2000** can include a **13.11-316** stainless steel, and the stiffener **13.11-1604** can include a 17-4 steel. In such an example, laser welding the dissimilar materials can cause oxidation, such as oxidation of the chromium in the **13.11-316** stainless steel. In specific examples, adhesives **13.11-2002** can help prevent (or at least reduce an amount of) oxidation on one or both of the connector **2000** or the stiffener **13.11-1604**. Thus, the adhesives **13.11-2002** can be positioned on the surface of the stiffener **13.11-1604** at an end portion adjacent to the connector **13.11-2000**. In at least one example, the adhesives **13.11-2002** are positioned at least partially around fasteners attaching the stiffener **13.11-1604** to the display frame **13.11-1402**. In these or other examples, the adhesives **13.11-2002** can interact with the surface of the stiffener **13.11-1604** and/or the connector **13.11-2000** (e.g., when under thermal loads from laser welding applications) to inhibit oxidation. Other cosmetic advantages can also be imparted by the adhesives **13.11-2002**, as desired.

[3082] The display frame **13.11-1402** can also include adhesives **13.11-2004**. The adhesives **13.11-2004** are strips of adhesive, which can be positioned longitudinally along a top surface of the stiffener **13.11-1604**. In particular, the adhesives **13.11-2004** can couple the outer portion or shell (removed for interior illustration purposes) of the display frame to the stiffener **13.11-1604**. This coupling can impart increased stiffness to the display frame **13.11-1402** as a whole (and in particular, impart more stiffness to the less rigid or stiff outer shell)—by leveraging stiffness of the stiffener **13.11-1604**.

[3083] In addition, the display frame **13.11-1402** can include adhesives **13.11-2006**. The adhesives **13.11-2006** can include adhesives positioned underneath the stiffener **13.11-1604**. The adhesives **13.11-2006** can specifically adhere the stiffener **13.11-1604** to the bottom housing or shell of the display frame **13.11-1402**.

[3084] In some examples, the display frame **13.11-1402** can include adhesives **13.11-2008**. The adhesives **13.11-2008** can include a discontinuous glue strip between top and bottom shells of the display frame **13.11-1402**. In at least one example, the gaps between the adhesives **13.11-2008** can impart thermal advantages. For instance, the adhesives **13.11-2008** can reduce or eliminate deformation in the display frame **13.11-1402** caused by the different coefficients of thermal expansion in response to higher heat and/or humid environments.

[3085] In one or more examples, the display frame **13.11-1402** can include an adhesive **13.11-2010**. The adhesive **13.11-2010** can include a gap filler or hardstop glue to limit travel of the connector **13.11-2000**. For example, the adhesive **13.11-2010** can bound the travel of the connector **2000** towards the optical system such that the adhesive **13.11-2010** serves as a mechanical stop. In these or other examples, the adhesive **13.11-2010** can include an adhesive with higher compressive stiffness compared to other adhesives. For example, the adhesive **13.11-2010** can be load bearing (e.g., to at least partially transfer an applied load to the connector **13.11-2000**).

[3086] Additionally or alternatively to the foregoing adhesives, it will be appreciated that other examples are also contemplated. For instance, in some implementations, the display frame **13.11-1402** can include near net molding (which can obviate one or more adhesives discussed above).

[3087] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.11-20-13.11-21** can be included, either alone or in any

combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 13.11-20-13.11-21. The following description discusses example adhesives in a head-mountable device. Indeed, a head-mountable device of the present disclosure can include a variety of different adhesives that perform various functions and/or impart certain properties.

[3088] It will be noted that in addition to forming the frame in the manner described herein, it is also contemplated to form the frame and its structural components using machined configurations, carbon fiber configurations, 3D modeled configurations, fixed molded configurations, etc.

13.12: Face Engaging Structure

[3089] The following disclosure relates to connectors of a head-mountable device used for AR/VR experiences. More particularly, the present embodiments relate to connectors positioned at a maxilla region (and nearby zygoma regions) of a face when the head-mountable device is donned. These connectors can dynamically adjust to different heads, as well as dynamically adjust in real time in response to force events. The connectors can also include a variety of different types of connections, as will be described below.

[3090] In one example, a head-mountable device includes a connector between a display and a facial interface. In certain implementations, the connector between the display and the facial interface has a stop (e.g., a hard stop). The connector allows for free movement (or multiple degrees of freedom) of the facial interface while still grossly constraining the facial interface. Because the facial interface can freely react to the user's face, pressure can be more evenly distributed to the user's face. The improved load distribution across a user's face can create a pleasant and enjoyable AR/VR experience for the user.

[3091] Conventional head-mountable devices do not include connections with a stop. By contrast, the head-mountable device of the present disclosure includes connections with a stop. A stop can stabilize (and in some cases protect from undue force distribution to) the HMD and internal components. A stop can also improve HMD robustness by preventing damage to sensitive components and efficiently transferring a force load through the stop (e.g., to a force-dissipation member, force-spreader member, or predetermined facial feature), thereby improving user comfort.

[3092] In one example, a head-mountable device of the present disclosure includes a display and a facial interface with a connector positioned at the maxilla facial region. The connector can include two degrees of freedom and can be a pin-and-bowl connection.

[3093] In another example, the connector can include a foam stop adhered to the facial interface. In certain implementations, the foam stop is also coupled to the display. For instance, the foam stop can be preloaded to remain positioned in a "home" (e.g., non-deflected or deflected) state until the head-mountable device is donned.

[3094] In yet another example, the connector can include a slidable post. The slidable post can slide relative to the facial interface or the display, as desired.

[3095] Accordingly, the apparatuses and systems described herein can provide facial feature adjustments and connections that can increase user customization, reduce facial pressure, and improve distribution of force loads for a head-mountable device, thereby increasing comfort.

[3096] These and other embodiments are discussed below with reference to FIGS. 13.12-1-13.12-8B. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature including at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second

option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[3097] FIGS. **13.12-1** illustrates a top view profile of a head-mountable device **13.12-100** worn on a user head **13.12-101**. The head-mountable device **13.12-100** can include a display **13.12-102** (e.g., one or more optical lenses or display screens in front of the eyes of the user). The display **13.12-102** can include a display for presenting augmented reality visualizations, a virtual reality visualization, or other suitable visualizations.

[3098] The head-mountable device **13.12-100** also includes a facial interface **13.12-104**. As used herein, the term “facial interface” refers to a portion of the head-mountable device **13.12-100** that engages with a user face via direct contact. In particular, a facial engagement portion includes portions of the head-mountable device **13.12-100** that conform to (e.g., press against) regions of a user face. To illustrate a facial interface can include a pliant (or semi-pliant) face track that spans the forehead, wraps around the eyes, contacts the zygoma and maxilla regions of the face, and bridges the nose. In addition, a facial interface can include various components forming a structure, webbing, cover, fabric, or frame of a head-mountable device disposed between the display **13.12-102** and the user skin. In particular implementations, a facial interface can include a seal (e.g., a light seal, environment seal, dust seal, air seal, etc.). It will be appreciated that the term “seal” can include partial seals or inhibitors, in addition to complete seals (e.g., a partial light seal where some ambient light is blocked and a complete light seal where all ambient light is blocked when the head-mountable device is donned).

[3099] The head-mountable device **13.12-100** further includes connector(s) **13.12-103**. As used herein, the term “connector” refers to a connection or touch point between the display **13.12-102** and the facial interface **13.12-104**. For instance, a connector can movably constrain the facial interface **13.12-104** to the display **13.12-102**. In certain implementations, a connector includes a joint that physically connects the display **13.12-102** and the facial interface **13.12-104**. In these or other implementations, a connector includes a floating joint or mechanical stop where the facial interface **13.12-104** and the display **13.12-102** can contact each other (e.g., at rest, after certain amounts of deflection, etc.).

[3100] To illustrate, the connector(s) **13.12-103** can allow the facial interface **13.12-104** to translate (e.g., left-right, up-down,) in one plane. Additionally or alternatively, the connector(s) **13.12-103** can allow the facial interface **13.12-104** to translate (e.g., depth-wise) in another plane relative to the display **13.12-102**. For example, the connector(s) **13.12-103** allow the facial interface **13.12-104** to move between a first state (e.g., a home state when the head-mountable device **13.12-100** is not donned by a user) to a second state (e.g., when the head-mountable device **13.12-100** is donned by a user). The connector(s) **13.12-103** help enable the facial interface **13.12-104** to flexibly accommodate the facial profile of the user by deflecting according to the user's facial features. When the head-mountable device **13.12-100** is removed from the user's head **13.12-101**, the connector(s) **13.12-103** can return to the first state-thereby causing the facial interface **13.12-104** to correspondingly move.

[3101] The connector(s) **13.12-103** can include one or more components (e.g., pins, bowls, slidable posts, foam, mechanical stops, dampeners, spring connections, etc.) that can allow (or actively provide) translation in one or more planes (e.g., at a maxilla region of a face). The connector(s) **13.12-103** can include proud engagement portions (e.g., pin, slidable post) and recessed engagement portions (e.g., receptacle, cup, slidable post-track). The proud engagement portions and recessed engagement portions can mate, forming a floating connection (e.g., a receptacle, touch point, mechanical stop, etc. constraining a post from moving in certain directions or beyond certain distances or depths).

[3102] Additionally, as shown in FIGS. **13.12-1**, the head-mountable device **13.12-100** can include one or more arms **13.12-108**, **13.12-110**. The arm **13.12-108**, **13.12-110** are connected to the

display **13.12-102** and extend distally toward the rear of the head. The arms **13.12-108**, **13.12-110** are configured to secure the display **13.12-102** in a position relative to the user head **13.12-101** (e.g., such that the display is maintained in front of a user's eyes). For example, the arms **13.12-108**, **13.12-110** extend over the user's ears **13.12-106**. In certain examples, the arms **13.12-108**, **13.12-110** rest on the user's ears **13.12-106** to secure the head-mountable device **13.12-100** via friction between the arms **13.12-108**, **13.12-110** and the user head **13.12-101**. Optionally, the arms **13.12-108**, **13.12-110** can be connected to each other via a strap **13.12-112** (shown in the dashed lines) that can compress the head-mountable device **13.12-100** against the user head **13.12-101**.

[3103] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-1**.

[3104] FIGS. **13.12-2A-13.12-2B** illustrate side and front view profiles, respectively, of an example of the head-mountable device **13.12-100**. As discussed above, the head-mountable device **13.12-100** includes the display **13.12-102**, the facial interface **13.12-104**, and the connector(s) **13.12-103**. In particular, as shown in FIGS. **13.12-2A-13.12-2B**, the facial interface **13.12-104** can indeed wrap around the eyes **13.12-206**, bridge the nose **13.12-208**, span a forehead region **13.12-203**, contact a zygoma facial region **13.12-202**, and contact a maxilla facial region **13.12-204**.

[3105] In addition, the connector(s) **13.12-103** can be positioned along the maxilla facial region **13.12-204** and connect directly (e.g., having a direct connection) or indirectly (e.g., having a floating connection) to the facial interface **13.12-104** and the display **13.12-102**. The connector(s) **13.12-103** can similarly be positioned along the zygoma facial region **13.12-202** and/or the forehead region **13.12-203**. In at least some examples, the connector(s) **13.12-103** are spatially configured to provide a particular force profile when donned and/or during a force event applying a force load to the display **13.12-102**.

[3106] Further below, this disclosure describes with more particularity how the connector(s) **13.12-103** can be constructed. For example, the connector(s) **13.12-103** can include pivot connectors, floating connectors, foam connectors, stop connectors, or other connection types (e.g., sliding connectors, rigid connectors), including combinations thereof (such as a hybrid elastomer and mechanical slider).

[3107] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-2A-13.12-2B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-2A-13.12-2B**.

[3108] The connector(s) **13.12-103** can provide a stable and flexible interface between the facial interface **13.12-104** and the display **13.12-102**. Additionally or alternatively, the connector(s) **13.12-103** include an amount of travel (e.g., an amount of possible displacement). The connector(s) **13.12-103** can also provide a datum or a hard stop limiting the travel to a particular separation distance (e.g., a maximum displacement) of the display **13.12-102** relative to the facial interface **13.12-104**. These connector(s) **13.12-103** can also increase movement of the facial interface **13.12-104** in the maxilla facial region **13.12-204** in such a way that the facial interface **13.12-104** freely interacts (e.g., conforms) with the user's face without applying excessive pressure back on to the user's face.

[3109] In accordance with one or more such examples of the present disclosure, FIGS. **13.12-3**

illustrates an example of the head-mountable device **13.12-100** including the display **13.12-102**, the facial interface **13.12-104**, and the connector(s) **13.12-103** (namely, connectors **13.12-303a-13.12-303f**). The connector(s) **13.12-103** (albeit positionable in different places) can specifically be positioned at the maxilla facial region **13.12-204** to distribute pressure exerted on a user's face. For example, elastic forces exerted by the strap **13.12-112** (not shown) when donning the head-mountable device **13.12-100** can be distributed from the display **13.12-102** to the facial interface **13.12-104** in a comfortable manner via the connectors **13.12-303a-13.12-303f**. In particular, the connectors **13.12-103a-13.12-103b** positioned at the maxilla facial region **13.12-204** can conform (e.g., flex, pivot, rotate, etc.) to a user's face evenly distributing the weight of the head-mountable device **13.12-100** across the maxilla facial region **13.12-204**, thereby helping to provide increased comfort to a user.

[3110] As shown in FIGS. **13.12-3**, the connectors **13.12-303a-13.12-303b** at the maxilla facial region **13.12-204** include a pivot connection. A pivot connection can include one or more different connections that allow pivot action relative to an anchored portion. Thus, a pivot connection can include a ball-and-socket joint, hinged joint, condyloid joint, atlantoaxial joint, radioulnar joint, serial manipulators, etc.

[3111] In certain implementations, the anchored portion of the pivot connection for the connectors **13.12-303a-13.12-303b** is positioned on the display **13.12-102**. Thus, the facial interface **13.12-104** can move or pivot relative to the display **13.12-102** (e.g., for a comfortable, conforming fit to the user face). In other implementations, however, the anchored portion of the pivot connection for the connectors **13.12-303a-13.12-303b** is positioned on the facial interface **13.12-104**.

[3112] The pivot connection for the connectors **13.12-303a-13.12-303b** can include travel defined by the amount of possible deflection or linear distance movement. In some cases, the pivot connection includes a range of travel between about 2 mm to about 20 mm. In particular implementations, the pivot connection includes about 6 mm of travel. It will be appreciated that the pivot connection also includes a stop (e.g., a portion of the socket joint) bounding the travel.

[3113] In at least some examples, the head-mountable device **13.12-100** implements additional connectors to change a pressure profile to reduce or avoid "hot-spot loading" lending to acute pressure against a user's face. For example, the head-mountable device **13.12-100** includes the connectors **13.12-303e-13.12-303f** positioned at the zygoma facial region **13.12-202** to at least partially offload an applied force distributed through the connectors **13.12-303a-13.12-303b** at the maxilla facial region **13.12-204**. In certain implementations, the connectors **13.12-303e-13.12-303f** include the same type of connection as the connectors **13.12-303a-13.12-303b**. In other implementations, the connectors **13.12-303e-13.12-303f** include a different type of connection than the connectors **13.12-303a-13.12-303b**. For instance, the connectors **13.12-303e-13.12-303f** can include a floating foam joint or a leaf spring joint, and the connectors **13.12-303a-13.12-303b** can include a pivot connection.

[3114] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-3**.

[3115] As mentioned above, the head-mountable device of the present disclosure can implement a pin-and-bowl connection between the display **13.12-102** and the facial interface **13.12-104**. In accordance with one or more such examples, FIGS. **13.12-4A** illustrates an example of the head-mountable device **13.12-100** including the display **13.12-102**, the facial interface **13.12-104**, and the connector(s) **13.12-103** (namely connectors **13.12-402**). FIGS. **13.12-4B-13.12-4C** show a front profile view and side profile view, respectively, of the connectors **13.12-402** located at the maxilla

facial region **13.12-204**.

[3116] As shown, the connectors **13.12-402** include a pin-and-bowl connection. In particular, the connectors **13.12-402** include a bowl **13.12-404** and a pin **13.12-405**. The bowl **13.12-404** includes a receptacle defined within the display **13.12-102** that is configured to receive the pin **13.12-405** affixed to the facial interface **13.12-104**. In particular implementations, the bowl **13.12-404** is sized and shaped to receive a particular portion of the pin **13.12-405** (e.g., an entirety of the pin **13.12-405**, an end portion of the pin **13.12-405**, or a predetermined depth of the pin **13.12-405**).

[3117] In these or other examples, the bowl **13.12-404** includes a stop **13.12-410** (e.g., an axial dead stop) coupled to the display **13.12-102**. The stop **13.12-410** can control the penetration depth of the pin **13.12-405** into the bowl **13.12-404**. Specifically, the stop **13.12-410** allows a maximum displacement or penetration of the pin **13.12-405** in an axial direction **13.12-412**. When the pin **13.12-405** abuts the stop **13.12-410**, the pin **13.12-405** is stopped out and cannot proceed further in the axial direction **13.12-412**—at which point the facial interface **13.12-104** and the display **13.12-102** are positioned at a predetermined distance from each other (e.g., a minimum separation distance).

[3118] The stop **13.12-410** can also control other movement of the pin **13.12-405** within the bowl **13.12-404**. In particular, the stop **13.12-410** can limit or define translational movement of the pin **13.12-405** in directions perpendicular to the axial direction **13.12-412**.

[3119] In particular implementations, the connectors **13.12-402** include two degrees of freedom. As illustrated in FIGS. **13.12-4B**, a first degree of freedom includes a translational degree of freedom in a first plane (e.g., defined by a first axis **13.12-406** and a second axis **13.12-408**). A second degree of freedom includes depth-based motion in the axial direction **13.12-412**, as illustrated in FIGS. **13.12-4C**. The combination of the first degree of freedom and the second degree of freedom allow the facial interface **13.12-104** to conform to a user's face while also providing rigidity and stability where appropriate.

[3120] According to some examples, the connectors **13.12-402** can move along the first degree of freedom (i.e., along the first axis **13.12-406**) when the pin **13.12-405** is positioned at a variety of different depths along the axial direction **13.12-412**. For example, the connectors **13.12-402** can move along the first axis **13.12-406** even when the pin **13.12-405** is fully depressed (i.e., when the pin **13.12-405** is stopped out). In this manner, the connectors **13.12-402** can move in response to various forces, such as at least one of an applied load or a shear force. Nonetheless, such movement of the connectors **13.12-402** can be bounded by virtue of the pin **13.12-405** being retained within the bowl **13.12-404**.

[3121] The pin **13.12-405** can include a variety of components and/or materials. In one example, the pin **13.12-405** includes a contact portion **13.12-418**. The contact portion **13.12-418** can be firm and rigid. In other examples, the contact portion **13.12-418** is soft or compressive. In some examples, the contact portion **13.12-418** includes a material such as a foam, an elastomer, a gel, a silicon, a metal, or a thermoplastic.

[3122] The pin **13.12-405** can also include a variety of different shapes and sizes. In some examples, the pin **13.12-405** is cone-shaped. In some instances, a cone shape can allow more freedom of motion at greater distances of separation between the display **13.12-102** and the facial interface **13.12-104**. That is, as the pin **13.12-405** compresses further into the bowl **13.12-404**, the cone-shape of the pin **13.12-405** fills more and more volume of the bowl **13.12-404**, leaving little or no motion for the pin **13.12-405**. Other shapes, however, are herein contemplated. For instance, the pin **13.12-405** can be shaped to resemble a square, rectangle, cylinder, oval, triangle, or other suitable shape.

[3123] Further, the connectors **13.12-402** can include one or more mechanisms for retaining the pin **13.12-405** within the bowl **13.12-404** (not shown). For example, a tip of the pin **13.12-405** can be sized greater than the opening of the bowl **13.12-404**. According to another example, the pin **13.12-405** and/or the bowl **13.12-404** can be sized and shaped such that the pin **13.12-405** cannot escape

the bowl **13.12-404**. For example, the pin **13.12-405** and/or the bowl **13.12-404** can be sized greater than the amount of pin displacement between positional states. In yet another example, the bowl **13.12-404** can include a socket-like structure that retains the pin **13.12-405**.

[3124] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-4A-13.12-4C** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-4A-13.12-4C**.

[3125] In one example, the connectors of the present disclosure are preloaded (e.g., via springs, elastic materials, etc.) to return to a certain position or state after being moved. In accordance with one or more such examples, FIGS. **13.12-5A-13.12-5B** respectively illustrate states (or positions) of the connectors **13.12-402** described above. In particular, FIGS. **13.12-5A** illustrates a first state **13.12-500** of the connectors **13.12-402** where the pin **13.12-405** is stopped out (i.e., bottomed out) within the bowl **13.12-404**. When in the first state **13.12-500**, the display **13.12-102** and the facial interface **13.12-104** are positioned at a predetermined separation distance (e.g., a minimum separation distance). In the first state **13.12-500**, further travel of the pin **13.12-405** into the bowl **13.12-404** is disallowed.

[3126] In some examples, the first state **13.12-500** includes a home state. The home state refers to a position that the connectors **13.12-402** rebound to after being deflected. In particular, the home state is the unperturbed position of the connectors **13.12-402**.

[3127] In other examples, however, the first state **13.12-500** includes a deflected state. The deflected state refers to a position that the connectors **13.12-402** can achieve in response to deflection (e.g., when donning the head-mountable device or during a force event).

[3128] FIGS. **13.12-5B** illustrates a second state **13.12-502** of the connectors **13.12-402** where the pin **13.12-405** is only partially disposed within the bowl **13.12-404**. In certain cases, the pin **13.12-405** is complete withdrawn from the bowl **13.12-404**. When in the second state **13.12-502**, the display **13.12-102** and the facial interface **13.12-104** are movable relative to each other (either closer to or farther away). In the second state **13.12-502**, further travel of the pin **13.12-405** into the bowl **13.12-404** is allowed.

[3129] Similar to the first state **13.12-500**, the second state **13.12-502** can include a home state. In other implementations, the second state **13.12-502** includes a deflected state.

[3130] In at least some examples, fabric **13.12-504** shown in FIGS. **13.12-5A-13.12-5B** can be stretched or tensioned to accommodate the first state **13.12-500** and the second state **13.12-502** just described. The fabric **13.12-504** can include an adjustable external shell that flexes with movement of the display **13.12-102** and the facial interface **13.12-104**. Thus, as the connectors **13.12-402** move to and from the first state **13.12-500** and the second state **13.12-502**, the fabric **13.12-504** can correspondingly adjust (e.g., stretch, tighten, loosen, wrinkle, fold, etc.).

[3131] According to some examples, the fabric **13.12-504** shown in FIGS. **13.12-5A-13.12-5B** can be implemented to bias the connectors of the present disclosure. For example, the fabric **13.12-504** can be stretched or tensioned to bias the connectors **13.12-402** in a particular home state or a deflected state. In at least some examples, the induced bias of the connectors **13.12-402** (provided by the tension from the fabric **13.12-504**) can facilitate a broad range of connector architectures, ranges of motion, etc.

[3132] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-5A-13.12-5B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other

figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-5A-13.12-5B**.

[3133] As mentioned above, the connectors of the present disclosure can include a variety of different connectors, including foam connectors positioned at a maxilla facial region. In accordance with one or more such examples, FIGS. **13.12-6A** illustrates another example of the head-mountable device **13.12-100** including the display **13.12-102** and the facial interface **13.12-104**, and the connector(s) **13.12-103** (namely connectors **13.12-602**).

[3134] In these or other examples, the connectors **13.12-602** include foam connectors. Specifically, the connectors **13.12-602** include a foam portion **13.12-604** affixed to the facial interface **13.12-104** and a base **13.12-606** affixed to the display **13.12-102**. The base **13.12-606** is sized and shaped to receive the foam portion **13.12-604**. In particular implementations, the foam portion **13.12-604** is permanently adhered to the base **13.12-606** via an attachment **13.12-608**. The attachment **13.12-608** can include glue, bond, or other type of adhesion. Additionally or alternatively, the attachment **13.12-608** includes a fastener, such as a bolt, screw, etc.

[3135] In at least one example, the connectors **13.12-602** include a compressible portion **13.12-610**, as illustrated in **13.12-6B**. The compressible portion **13.12-610** can also include a foam or other compressible material. In certain implementations, the compressible portion **13.12-610** may include a predefined thickness (e.g., about 2 mm, about 3 mm, about 10 mm, etc.) that dictates a corresponding amount of displacement or compression.

[3136] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-6A-13.12-6B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-6A-13.12-6B**.

[3137] As similarly described above, the connectors of the present disclosure can move between states or positions. Foam connections can likewise move between states or positions. In accordance with one or more such examples, FIGS. **13.12-7A-13.12-7B** illustrate positional states of the connectors **13.12-602**.

[3138] A first state **13.12-700** of the connector **13.12-602** is shown in FIGS. **13.12-7A**. In the first state **13.12-700**, the compressible portion **13.12-610** is uncompressed and has a first thickness. The second state **13.12-702** of the connector **13.12-602** is shown in FIGS. **13.12-7B**. Specifically, in the second state **13.12-702**, the compressible portion **13.12-610** is compressed (e.g., reduced in thickness from the first thickness) and has a second thickness less than the first thickness. The compressible portion **13.12-610** in the second state **13.12-702** need not fully compress. Indeed, the compressible portion **13.12-610** can compress partially or fully, as may be the case (e.g., when donning the head-mountable device or during a force event).

[3139] It will be appreciated that the first state **13.12-700** and the second state **13.12-702** can be, respectively, home states and deflected states. However, in certain implementations, the first state **13.12-700** and the second state **13.12-702** can oppositely configured (i.e., deflected states and home states, respectively).

[3140] In at least some examples, the fabric **13.12-504** described above can correspondingly move and flex to accommodate the first state **13.12-700** and the second state **13.12-702** for the connectors **13.12-602**. Alternatively, the fabric **13.12-504** can be attached to non-deflecting portions (e.g., exclusive of the compressible portion **13.12-610**) such that the fabric **13.12-504** need not flex or move.

[3141] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-7A-13.12-7B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the

other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-7A-13.12-7B**.

[3142] Also discussed above, the connectors of the present disclosure can include sliding posts that slide relative to the display **13.12-102** or the facial interface **13.12-104**. These sliding connectors can provide improved ranges of motion (e.g., lateral motion of the maxilla region) while still maintaining comfortable stability and rigidity. In accordance with one or more such examples, FIGS. **13.12-8A-13.12-8B** illustrate connectors **13.12-802a-13.12-802b** positioned at the maxilla region of the facial interface **13.12-104**.

[3143] As shown in FIGS. **13.12-8A**, the connector **13.12-802a** is affixed to the facial interface **13.12-104** and slidably engaged with the display **13.12-102**. In particular, the connector **13.12-802a** includes a slidable post with a proud engagement portion **13.12-801**. The proud engagement portion **13.12-801** interfaces with a recessed engagement portion **13.12-804** positioned within the display **13.12-102**. The recessed engagement portion **13.12-804** can include a slidable-post track (e.g., slide mechanism, track, guide, groove, recess, or raceway) defined by one or more surfaces of the display **13.12-102**. The recessed engagement portion **13.12-804** can guide a range of motion for the connector **13.12-802a** along a direction **13.12-808**. In this manner, the connector **13.12-802a** allows the facial interface **13.12-104** to comfortably conform to or move with the user's face in a stabilized fashion.

[3144] Alternatively, as shown in FIGS. **13.12-8B**, the connector **13.12-802b** is affixed to the display **13.12-102** and slidably engaged with the facial interface **13.12-104**. The connector **13.12-802b** is the same as the connector **13.12-802a**, except configured in a different positional relationship. In particular, the connector **13.12-802b** includes a slidable post with a proud engagement portion **13.12-801**. The proud engagement portion **13.12-801** interfaces with a recessed engagement portion **13.12-804** positioned within the facial interface **13.12-104**. The proud engagement portion **13.12-801** can move along the direction **13.12-808** within the facial interface **13.12-104**.

[3145] According to some examples, the mechanical slider examples illustrated in FIGS. **13.12-8A-13.12-8B** can be implemented utilizing a variety of different materials. For example, at least one of the proud engagement portion **13.12-801** or the recessed engagement portion **13.12-804** can be an elastomer material, metal material, plastic material, composite materials, and the like.

[3146] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-8A-13.12-8B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-8A-13.12-8B**.

[3147] Myriad other types of connectors are herein contemplated. For instance, the connector(s) **13.12-106** of the present disclosure can include a piston-based mechanism. In accordance with one or more such examples, FIGS. **13.12-9A-13.12-9F** illustrate a head-mountable device **13.12-900** with a maxilla connector **13.12-902** and a zygoma connector **13.12-904**. The maxilla connector **13.12-902** and the zygoma connector **13.12-904** movably join the facial interface **13.12-104** and the display **13.12-102** as similarly described above.

[3148] As shown, however, the maxilla connector **13.12-902** includes a piston connection, and the zygoma connector **13.12-904** includes a pivot connection. The maxilla connector **13.12-902** can move in and out in response to application and release of a compressive force at the maxilla region of a face. As the maxilla connector **13.12-902** moves, the zygoma connector **13.12-904** correspondingly moves (e.g., pivots).

[3149] FIGS. **13.12-9B** shows that the maxilla connector **13.12-902** includes a cap **13.12-906**, a plunger **13.12-908**, a compression spring **13.12-910**, and a housing **13.12-912**. The cap **13.12-906** maintains the components of the maxilla connector **13.12-902** in place. The plunger **13.12-908** moves in and out of the housing **13.12-912**, with resistance provided by the compression spring **13.12-910**. The plunger **13.12-908** is also a stop, where compression of the plunger **13.12-908** is mechanically stopped due to the body of the plunger **13.12-908** abutting a bottom of the housing **13.12-912**.

[3150] FIGS. **13.12-9C** illustrates the maxilla connector **13.12-902** in a nominal (e.g., uncompressed) position. FIGS. **13.12-9D** shows the maxilla connector **13.12-902** in a compressed position.

[3151] FIGS. **13.12-9E-13.12-9F** show a positional relationship of the maxilla connector **13.12-902** relative to the zygoma connector **13.12-904** at depth positions **13.12-914-918** of the maxilla connector **13.12-902**. The maxilla connector **13.12-902** is positioned in a perpendicular fashion relative to the zygoma connector **13.12-904**. In particular, the maxilla connector **13.12-902** is positioned perpendicular to the zygoma connector **13.12-904** at a median (i.e., middle) position, namely the depth position **13.12-916**. By aligning the piston direction of the maxilla connector **13.12-902** to be perpendicular to the depth position **13.12-916**, an amount of horizontal displacement can be minimized or reduced. In turn, the risk of binding the maxilla connector **13.12-902** due to horizontal displacement is also minimized or reduced.

[3152] Specifically, a displacement distance **13.12-920** in FIGS. **13.12-9F** corresponds to the piston alignment with the depth position **13.12-916**. In addition, a displacement distance **13.12-922** corresponds to an alternative piston alignment (i.e., with the depth position **13.12-914**).

Comparatively, the displacement distance **13.12-922** is far greater than the displacement distance **13.12-920**. In certain implementations, the displacement distance **13.12-920** is (for example) about 0.2 mm, and the displacement distance **13.12-922** is (for example) about 0.8 mm (e.g., for a 0 to 15 degree rotation of the zygoma connector **13.12-904** and a radial distance of about 24 mm). Myriad other positional relationships of the maxilla connector **13.12-902** relative to the zygoma connector **13.12-904** are herein contemplated.

[3153] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-9A-13.12-9F** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-9A-13.12-9F**.

[3154] FIGS. **13.12-10** illustrates a cutaway view of an example connector **13.12-1000** in accordance with one or more examples of the present disclosure. As shown, the connector **13.12-1000** includes a spring element **13.12-1002**. The spring element **13.12-1002** can bias internal portions of the connector **13.12-1000** (e.g., to facilitate more or less travel or compression of the connector **13.12-1000**).

[3155] Additionally or alternatively to the spring element **13.12-1002** (or other suitable internal biasing components), the connector **13.12-1000** can include external biasing component(s). For example, a textile **13.12-1004** forming an exterior surface of the connector **13.12-1000** can also bias the connector **13.12-1000**. Indeed, the textile **13.12-1004** can be stretched or otherwise positioned over the connector **13.12-1000** to bias internal components of the connector **13.12-1000**. In these or other examples, the textile **13.12-1004** can include silicone, stretchy fabric (e.g., spandex, polyester, etc.), or other suitable materials.

[3156] It will be appreciated that the connector **13.12-1000** can be implemented in a variety of ways. In some examples, the connector **13.12-1000** includes a floating connection that does not integrally secure the display **13.12-102** and the facial interface **13.12-104**. Rather, the connector

13.12-1000 can make contact with (and can compress against) the display **13.12-102** or the facial interface **13.12-104**.

[3157] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-10** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-10**.

[3158] As discussed above, the connectors of the present disclosure can include a variety of different materials. In accordance with one or more examples of the present disclosure, FIGS. **13.12-11** illustrates an example connector **13.12-1100** with an elastomer portion **13.12-1102** adjoining the display **13.12-102** and the facial interface **13.12-104**. In some examples, the elastomer portion **13.12-1102** can include a thickness of a predetermined amount (e.g., based on a force-displacement profile). Further, in some examples, the elastomer portion **13.12-1102** can vary in thickness (e.g., in a tapered fashion). In these or other examples, the elastomer portion **13.12-1102** can include various different types of elastomers, such as cis-polyisoprene (natural rubber), cis-polybutadiene (butadiene rubber), styrene-butadiene rubber, and ethylene-propylene monomer. Additionally or alternatively, the elastomer portion **13.12-1102** can include a hybrid elastomer material (e.g., part elastomer and part non-elastomer).

[3159] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-11** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-11**.

[3160] FIGS. **13.12-12** illustrates another example connector **13.12-1200** in accordance with one or more examples of the present disclosure. As shown, the connector **13.12-1200** includes a ball joint **13.12-1202** and a leaf spring **13.12-1204**. In this manner, the connector **13.12-1200** can allow at least some pivoting or twisting about the ball joint **13.12-1202**. In addition, the connector **13.12-1200** can allow tensioned (and tunable) depth adjustment via the leaf spring **13.12-1204**. Indeed, the leaf spring **13.12-1204** can include a variety of different lengths, thicknesses, etc. to provide the desired stiffness (or flexibility) of connection between the display **13.12-102** and the facial interface **13.12-104**. As with other connectors, the leaf spring **13.12-1204** can also be formed of various materials, such as metal, plastic, or composite material.

[3161] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.12-12**, can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.12-12**.

13.13: Adjustment Mechanism

[3162] The following disclosure relates to adjustment mechanisms of a head-mountable device (e.g., for AR/VR experiences). More particularly, the present embodiments relate to adjustment mechanisms of a head-mountable device for altering angle and/or distance of a head-mountable device relative to the user head (or eyes). These adjustment mechanisms include depth (e.g., translation) adjustments and/or angular (e.g., rotation) adjustments.

[3163] In one example, a head-mountable device includes a connection between a display and a

facial interface. The connection allows the display to be translatable or rotatable relative to the facial interface. The translatable or rotatable movements allow the head-mountable device to achieve positions and combinations of positions that otherwise could not be achieved.

[3164] Conventional head-mountable devices do not include connections between the display and facial interfaces that allow translatable or rotatable movements. Therefore, conventional head-mountable devices do not allow the movement required to adjust depth for eye relief or angular adjustments to optimize viewing. Thus, conventional head-mountable devices cannot accommodate for different facial profiles, viewing preferences, eyesight capacities, etc.

[3165] By contrast, a head-mountable device of the present disclosure provides connections allowing translatable or rotatable motion relative to a facial interface. Such a head-mountable device has many advantages over a conventional head-mountable device. A head-mountable device with connections allowing movement of a display relative to the user's face allows a user adjust the depth and angle of the display screen as needed to accommodate their personal preferences or requirements. Additionally, a head mountable display with connections allows motion of the display relative to different users' faces, thereby improving a shareability of head-mountable device over conventional HMDs.

[3166] As disclosed herein, controlling the movement (e.g., linear translation or angular tilt) of a head-mounted display entails user friendly and simple actuation controls (e.g., lever, button, dial, rocker, slider, toggle, etc.) that are intuitive and easy to manipulate. Several examples of these devices are discussed below.

[3167] In one example, a head-mountable device includes a display, a strap connected to the display (e.g., the display frame), a facial interface, which is customizable to a user's facial profile, and a connection. The connection can be between the display and the facial interface such that the display is translatable and/or rotatable relative to the facial interface via the connection.

[3168] In another example, a head-mountable device includes a display, a strap connected to the display (e.g., the display frame), a facial interface, which is customizable to a user's facial profile, and an actuator. The actuator can moveably constrain the display relative to the facial interfaces.

[3169] In yet another example, a head-mountable device includes a display, a facial interface, which is customizable to a user's facial profile, and an angular and linear adjustment connection between the display and facial interface.

[3170] Accordingly, the apparatus and systems described herein provide angular and linear adjustments for a display of a head-mountable device in a user-friendly way, creating an enhanced more customizable user experience.

[3171] These and other embodiments are discussed below with reference to FIGS. **13.13-1-13.13-26B**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature including at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[3172] FIGS. **13.13-1** illustrates a top view profile of a head-mountable device **13.13-100** worn on a user head **13.13-101**. While the present systems and methods are described in the context of a head-mountable device **13.13-100**, the systems and methods can be used with any wearable apparatus, or any apparatus or system that can be physically attached to a user's body, but are particularly relevant to an electronic device worn on a user's head. The head-mountable device **13.13-100** can include a display **13.13-102** (e.g., one or more optical lenses or display screens in front of the eyes of the user). The display **13.13-102** can include a display for presenting

augmented reality visualizations, a virtual reality visualization, or other suitable visualization. The display **13.13-102** can also include a display frame **13.13-105** that houses optical components of the display **13.13-102**. It will therefore be appreciated that when referencing the display **13.3-102** herein, the display frame **13.3-105** can be implemented as part of specific aspects or functionality for the display **13.3-102**. For example, connection(s) **13.3-106** (described further below) can connect the facial interface **13.3-104** to at least one of the display **13.3-102** (e.g., optical elements of the display **13.3-102**) or structural elements of the display **13.3-102** (e.g., the display frame **13.3-105**).

[3173] The head-mountable device **13.13-100** also includes a facial interface **13.13-104** and a connection(s) **13.13-106** between the display **13.13-102** (e.g., the display frame **13.13-105**) and the facial interface **13.13-104**. As used herein, the term “facial interface” refers to a portion of the head-mountable device **13.13-100** that engages a user face via direct contact. In particular, a facial interface includes portions of the head-mountable device **13.13-100** that conform to (e.g., compress against) regions of a user face. To illustrate, a facial interface can include a pliant (or semi-pliant) face track that spans the forehead, wraps around the eyes, contacts the zygoma and maxilla regions of the face, and bridges the nose. In addition, a facial interface can include various components forming a structure, webbing, cover, fabric, or frame of a head-mountable device disposed between the display **13.13-102** and the user skin. In particular implementations, a facial interface can include a seal (e.g., a light seal, environment seal, dust seal, air seal, etc.). It will be appreciated that the term “seal” can include partial seals or inhibitors, in addition to complete seals (e.g., a partial light seal where some ambient light is blocked and a complete light seal where all ambient light is blocked when the head-mountable device is donned).

[3174] In addition, the term “connection” refers to a joining between the display **13.13-102** (e.g., the display frame **13.13-105**) and the facial interface **13.13-104**. In some examples, a connection allows the display **13.13-102** to translate or rotate relative to the facial interface **13.13-104**. In other examples, a connection allows the display **13.13-102** to both translate and rotate relative to the facial interface **13.13-104**. For instance, the connection(s) **13.13-106**, when adjusted, can slide (e.g., translate) the display **13.13-102** toward or away from the facial interface **13.13-104** (e.g., in a linear fashion). In another example, the connection(s) **13.13-106**, when adjusted, can rotate the display **13.13-102** up or down relative to the facial interface **13.13-104** (e.g., in an angular fashion). Thus a connection can slidably join or rotatably join the display **13.13-102** and the facial interface **13.13-104**.

[3175] In this manner, the connection(s) **13.13-106** can movably constrain the display **13.13-102** relative to the facial interface **13.13-104**. As used herein, the term “movably constrain” or “movably constraining” refers to the type of connection that can dynamically move (e.g., translate or rotate), yet retain control over a particular element's movement or position. For example, to “movably constrain” means the connection(s) **13.13-106** can bound movement of the display **13.13-102** within two degrees of freedom (e.g., along a horizontal plane and along an additional plane non-planar with the horizontal plane) relative to the facial interface **13.13-104**.

[3176] The connection(s) **13.13-106** can include one or more components (e.g., actuators, sliders, gears, levers, mechanical advantage devices, posts, mechanical stops, dampeners, etc.) that can allow (or actively provide) translation or rotation of the display **13.13-102** relative to the facial interface **13.13-104**. The connection(s) **13.13-106** can have an adjustable resolution that is a continuous resolution or a non-continuous resolution, and capable of transitioning between states (e.g., between a first display position and a second display position). The connection(s) **13.13-106** can include a manual actuator control or an automated actuator control that is operable from a single interface location (or multiple interface locations).

[3177] In certain implementations, the connection(s) **13.13-106** are positioned between inner and outer surfaces of the head-mountable device **13.13-100**. As used herein, an “outer surface” refers to an exterior surface of the head-mountable device **13.13-100** that outwardly faces the ambient

environment. By contrast, as used herein, an “inner surface” refers to an exterior surface of the head-mountable device **13.13-100** that is oriented to face towards (or contact) a human face or skin. [3178] Furthermore, the connection **13.13-106** can include a variety of different adjustment mechanisms, members, or movement mechanisms or movable connectors (e.g., bearings/bushings, dovetail slides, scissor mechanisms, sarrus linkages, gears, movable arms, etc.), cables (e.g., tensioned cables), actuators (e.g., mechanical, electromechanical, hydraulic, pneumatic, piezoelectric, etc.), and locking mechanisms (toggles, latches, ratchets, clamps, friction brakes, toothed brakes, hydraulic brakes, pneumatic bladders, non-back-drivable mechanisms, etc.), and other components. Locations of the connection **13.13-106** can be modified or tuned, as can the number of connections. For instance, the location and/or the number of the connection(s) **13.13-106** can correlate to an amount of force or pressure exerted on the user at any one datum (e.g., forehead region, maxilla region, zygoma region, etc.). In other instances, the location and/or the number of the connection(s) **13.13-106** can correspond to rigidity (or rigidity variances) between the connection(s) **13.13-106**.

[3179] As used herein, the term “forehead region” refers to an area of a human face between the eyes and the scalp of a human. Additionally, the term “maxilla region” refers to an area of a human face corresponding to the zygomatic bone structure of a human. Similarly, the term “maxilla region” refers to an area of a human face corresponding to the maxilla bone structure of a human. [3180] Additionally shown in FIGS. **13.13-1**, the head-mountable device **13.13-100** includes one or more arms **13.13-108**, **13.13-110**. The arms **13.13-108**, **13.13-110** are connected to the display **13.13-102** and extend distally toward the rear of the head. The arms **13.13-108**, **13.13-110** are configured to secure the display in a position relative to the user head **13.13-101** (e.g., such that the display **13.13-102** is maintained in front of a user's eyes). For example, the arms **13.13-108**, **13.13-110** extend over the user's ears **13.13-112**. In certain examples, the arms **13.13-108**, **13.13-110** rest on the user's ears **13.13-112** to secure the head-mountable device **13.13-100** via friction between the arms **13.13-108**, **13.13-110** and the user head **13.13-101**. For example, the arms **13.13-108**, **13.13-110** can apply opposing pressures to the sides of the user head **13.13-101** to secure the head-mountable device **13.13-100** to the user head **13.13-101**. Optionally, the arms **13.13-108**, **13.13-110** can be connected to each other via a strap **13.13-103** (shown in the dashed lines) that can compress the head-mountable device **13.13-100** against the user head **13.13-101**. In particular examples, the strap **13.13-103** is connected to at least the frame of the display **13.13-102** or the facial interface **13.13-104**.

[3181] In specific examples, the arms **13.13-108**, **13.13-110** are connected to the display **13.13-102** (by way of the display frame **13.13-105**) via a pivot joint **13.13-114**. The pivot joint **13.13-114** can be external or internal with respect to the arms **13.13-108**, **13.13-110** and/or the display frame **13.13-105**. In these or other examples, the pivot joint **13.13-114** can allow the display frame **13.13-105** to rotate about the pivot joint **13.13-114** independent of the arms **13.13-108**, **13.13-110**. To do so, the pivot joint **13.13-114** can include a detent hinge, a friction hinge, a passive hinge, a full lock-out hinge, etc.

[3182] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-1**.

[3183] FIGS. **13.13-2A-13.13-2B** respectively illustrate side and front view profiles of an example of the head-mountable device **13.13-100**. As discussed above, the head-mountable device **13.13-100** includes the display **13.13-102**, the facial interface **13.13-104**, and the connection(s) **13.13-106**. In particular, as shown in FIGS. **13.13-2A-13.13-2B**, the facial interface **13.13-104** can indeed

wrap around the eyes **13.13-201**, bridge the nose **13.13-202**, span the forehead **13.13-203**, and contact the zygoma and maxilla regions **13.13-205** of the face.

[3184] Additionally shown in FIGS. **13.13-2A-13.13-2B**, the head-mountable device **13.13-100** includes sensors **13.13-204** which can be attached to (or embedded within) the facial interface **13.13-104**. As used herein, the term “sensor” refers to one or more different sensing devices, such as a camera or imaging device, temperature device, oxygen device, movement device, brain activity device, sweat gland activity device, breathing activity device, muscle contraction device, etc. some particular examples of sensors include an electrooculography sensor, electrocardiogram sensor, EKG sensor, heart rate variability sensor, blood volume pulse sensor, SpO2 sensor, compact pressure sensor, electromyography sensor, core-body temperature sensor, galvanic skin sensor, accelerometer, gyroscope, magnetometer, inclinometer, barometer, infrared sensor, global positioning system sensor, etc.

[3185] Further below, this disclosure describes with more particularity how the connection(s) **13.13-106** as adjustment mechanisms can be positioned within the head-mountable device **13.13-100**. In addition, subsequent portions of this disclosure provide additional detail with respect to user interactions for controlling or actuating adjustment mechanisms.

[3186] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-2A-13.13-2B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-2A-13.13-2B**.

[3187] The location and positioning of the connection(s) **13.13-106** can provide a stable yet flexible and adjustable interface between the facial interface **13.13-104** and the display **13.13-102**. In accordance with one or more examples of the present disclosure, FIGS. **13.13-3A-13.13-3D** illustrate a plurality of positions where the connection(s) **13.13-106** can be located between the display **13.13-102** and the facial interface **13.13-104**. Although FIGS. **13.13-3A-13.13-3D** illustrate particular locations, it will be appreciated that the locations of the connection(s) **13.13-106** can be modified within the scope of the present disclosure.

[3188] In one example, FIGS. **13.13-3A**, shows the connection(s) **13.13-106** includes a single adjustment mechanism located at the forehead **13.13-203** of a user. The connection(s) **13.13-106** can be configured to provide a linear or angular connection between the display **13.13-102** and the facial interface **13.13-104**. In particular implementations, the connection(s) **13.13-106** provides only angular (tilt) adjustment by pivoting the display **13.13-102** relative to the facial interface **13.13-104**.

[3189] In at least some instances, a single-point system as shown in FIGS. **13.13-3A** can provide the simplest and most direct means of adjustment. For example, the connection(s) **13.13-106** in FIGS. **13.13-3A** can avoid potential angular misalignment or racking issues of multi-point systems. As another example, the connection(s) **13.13-106** in FIGS. **13.13-3A** can facilitate single-handed user adjustment of the head-mountable device **13.13-100**.

[3190] FIGS. **13.13-3B** shows another example where the connection(s) **13.13-106** include two adjustment mechanisms oppositely positioned between the facial interface **13.13-104** and the display **13.13-102**. Specifically, the connection(s) **13.13-106** in FIGS. **13.13-3B** include a first adjustment mechanism on a left side and a second adjustment mechanism on a right side (e.g., near regions of the user eye). Similarly, in another example, the connection(s) **13.13-106** may be located near the zygoma or maxilla regions **13.13-205** of the face. The two connections can provide at least two points of angular or linear adjustment of the display **13.13-102** relative to the facial interface **13.13-104**. For example, both of the connection(s) **13.13-106** in FIGS. **13.13-3B** can be adjusted linearly together to provide a depth adjustment, or adjusted angularly together to provide a vertical

(up/down) tilt adjustment. In another example, one of the connections **13.13-106** can be extended outward and the other of the connection(s) **13.13-106** can be pushed inward (so as to provide a side-to-side or horizontal tilt adjustment).

[3191] FIGS. **13.13-3C** shows the connection(s) **13.13-106** include three adjustment mechanisms—one near the forehead **13.13-203** and two at bottom corner areas at the zygoma or maxilla regions. Additionally, FIGS. **13.13-3D** shows the connection(s) **13.13-106** include four connection(s) **13.13-106**—two at the top corners and two at the bottom corners. In at least some examples, multi-point systems like those shown in FIGS. **13.13-3B-13.13-3D** can provide improved load distribution and/or rigidity. Multi-point systems like those shown in FIGS. **13.13-3B-13.13-3D** can also satisfy larger load requirements, such as increased loads during a sudden impact event to the head-mountable device **13.13-100**. In the case of multi-point systems, synchronization mechanisms can be added to ensure that each of the connection(s) **13.13-106** move in lockstep.

[3192] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-3A-13.13-3D** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-3A-13.13-3D**. Moreover, although not shown, it will be appreciated that more than four of the connection(s) **13.13-106** are also contemplated, and the locations of any one the connection(s) **13.13-106** described can vary in position and location relative to the figures in which they were described.

[3193] As mentioned above, the connection(s) **13.13-106** can include adjustment mechanisms that allow (or provide) movement of the display **13.13-102** relative to the facial interface **13.13-104**. This movement can include motion in a single degree of freedom or multiple degrees of freedom. In accordance with one or more such examples, FIGS. **13.13-4A-13.13-4C** illustrate an example of depth adjustment via depth adjustment mechanisms **13.13-402** (e.g., linear adjustment connections). For example, FIGS. **13.13-4A** shows the display **13.13-102** positioned relative to the facial interface **13.13-104** in a home state. In the home state, depth adjustment mechanisms **13.13-402** can be positioned at a default position, a customized position, etc. in which the display **13.13-102** is a predetermined distance (i.e., eye relief distance) away from the facial interface **13.13-104**. In other instances, the home state can include a current positioning of the display **13.13-102** relative to the display **13.13-102**.

[3194] FIGS. **13.13-4B** shows the depth adjustment mechanisms **13.13-402** in an extended state. In the extended state, the depth adjustment mechanisms **13.13-402** are extended or otherwise altered from the home state (or another state) to increase a relative distance between the display **13.13-102** and the facial interface **13.13-104**.

[3195] FIGS. **13.13-4C** shows the depth adjustment mechanisms **13.13-402** in a retracted state. In the retracted state, the depth adjustment mechanisms **13.13-402** are retracted or otherwise altered from the home state (or another state) to decrease a relative distance between the display **13.13-102** and the facial interface **13.13-104**.

[3196] In some examples, the various states shown in FIGS. **13.13-4A-13.13-4C** can include discrete or predefined positions of the display **13.13-102** relative to the facial interface **13.13-104**—where the predefined positions are differentiated by set (or customizable) step-sizes. For instance, the depth adjustment mechanisms **13.13-402** can be utilized to vary the position of the display **13.13-102** relative to the facial interface **13.13-104** in positional steps from 0 mm eye relief distance in a fully retracted state, to 6 mm eye relief distance in a fully extended state. In other examples, however, the various states shown in FIGS. **13.13-4A-13.13-4C** can include non-discrete positions of the display **13.13-102** relative to the facial interface **13.13-104**—where the positions are configurable via continuous adjustment of the depth adjustment mechanisms **13.13-402**, due to

a continuous interface that is gradually adjusting over a substantially smooth surface, providing a continuous adjustment resolution.

[3197] Adjustment of the depth adjustment mechanisms **13.13-402** can be performed in different ways. In some examples, user interactions to the depth adjustment mechanisms **13.13-402** (e.g., via one or more user controls) can actuate or move the depth adjustment mechanisms **13.13-402**. Additionally or alternatively, the depth adjustment mechanisms **13.13-402** can actuate in response to one or more sensors of the head-mountable device **13.13-100** detecting a voice command or gesture (e.g., eye gesture, hand gesture, touch, etc.). Still further, in some examples, the head-mountable device **13.13-100** can automatically adjust the depth adjustment mechanisms **13.13-402** (e.g., based on one or more sensors of the head-mountable device **13.13-100** detecting a user field of view indicating adjustment is needed). Thus, in certain implementations, the head-mountable device **13.13-100** includes a motor to auto-drive certain actuation. In this example, the motor can mechanically interface with the connections or adjustment mechanisms to automate or controllably drive the adjustment mechanisms.

[3198] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-4A-13.13-4C** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-4A-13.13-4C**. Moreover, although not shown, it will be appreciated that more than four of the connection(s) **13.13-106** are also contemplated, and the locations of any one the connection(s) **13.13-106** described can vary in position and location relative to the figures in which they were described.

[3199] The connection(s) **13.13-106** can also allow (or provide) angular or rotational adjustment of the display **13.13-102** relative to the facial interface **13.13-104**. In accordance with one or more such examples, FIGS. **13.13-5A-13.13-5C** illustrate an example of tilt adjustment via angular adjustment mechanisms **13.13-502** (e.g., angular adjustment connections). Via adjustment of the angular adjustment mechanisms **13.13-502**, the display **13.13-102** can pivot (e.g., rotate) the display **13.13-102** up or down by an angular displacement relative to the facial interface **13.13-104**, as illustrated in FIGS. **13.13-5A-C**.

[3200] In particular, FIGS. **13.13-5A** shows the angular adjustment mechanisms **13.13-502** in a home state (e.g., with no angular tilt). It will be appreciated, however, that alternative home states with angular tilt are herein contemplated.

[3201] FIGS. **13.13-5B** shows the angular adjustment mechanisms **13.13-502** in a negative angular state (e.g., downward angular tilt) relative to the home state. By contrast, FIGS. **13.13-5C** shows the angular adjustment mechanisms **13.13-502** in a positive angular state (e.g., upward angular tilt) relative to the home state.

[3202] As with the depth adjustment mechanisms **13.13-402** described above, the angular adjustment mechanisms **13.13-502** can also be adjusted in myriad different ways. For example, the angular adjustment mechanisms **13.13-502** can be adjusted manually (e.g., via user controls, voice commands, or user gestures). As another example, the head-mountable device **13.13-100** can automatically adjust the angular adjustment mechanisms **13.13-502** based on a detected user field of view.

[3203] In certain implementations, the angular state can have angular range of negative 7.5 degrees in a fully extended negative angular tilt state, to positive 7.5 degrees in a fully extended positive angular tilt state. The angular state can be discrete (e.g., with limited positional states) or continuous (e.g., unlimited positional states).

[3204] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-5A-13.13-5C** can be included, either alone or in any

combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-5A-13.13-5C**.

[3205] In some cases, the connection(s) **13.13-106** can be statically adjusted, customized, or set. For instance, the connection(s) **13.13-106** may be adjusted upon initially donning the head-mountable device **13.13-100**, and thereafter the connection(s) **13.13-106** is set without the ability to externally modify the connection(s) **13.13-106**. That is, the connection(s) **13.13-106** nor a corresponding actuator control may be externally accessible when the apparatus is donned (e.g., without some disassembly of the head-mountable device **13.13-100**). In accordance with one or more such examples, FIGS. **13.13-6A-13.13-6B** illustrate an example head-mountable device **13.13-600** with adjustment connections **13.13-602** between the display **13.13-102** and the facial interface **13.13-104**. In particular, FIGS. **13.13-6A** shows the head-mountable device assembly, while FIGS. **13.13-6B** provides a zoomed-in view of one of the adjustment connections **13.13-602**.

[3206] The adjustment connections **13.13-602** can connectively join the display **13.13-102** at a distance from the facial interface **13.13-104** (as similarly described above for the connection(s) **13.13-106**). The adjustment connections **13.13-602** can be positioned in myriad different locations. As shown, however, the adjustment connections **13.13-602** are positioned in upper corners of the forehead region and at the zygoma regions.

[3207] Additionally, the adjustment connections **13.13-602** as shown include an adjustable pillar **13.13-604**. The adjustable pillar **13.13-604** is configured to slide into and out of an adjustment connection. Specifically, the adjustable pillar **13.13-604** moves relative to a base **13.13-603** of the adjustment connection. The adjustable pillar **13.13-604** is connected to the facial interface **13.13-104**, and the base **13.13-603** is connected to the display **13.13-102**. Alternatively, opposite configurations are herein contemplated (e.g., the adjustable pillar **13.13-604** connected to the display **13.13-102** and the base **13.13-603** connected to the facial interface **13.13-104**). Thus, adjusting the adjustable pillar **13.13-604** (in or out of the base **13.13-603**) causes a corresponding change in distance between the display **13.13-102** and the facial interface **13.13-104**. The adjustable pillar **13.13-604** further includes holes **13.13-606** configured to receive a set screw (or other fastener) by which the adjustable pillar **13.13-604** can be positionally affixed upon setting to the desired position.

[3208] The adjustable pillar **13.13-604** can include a variety of different specifications (e.g., according to desired load distribution). For example, in some instances, the adjustable pillars **13.13-604** are constructed to withstand the compressive force of the strap **13.13-103** (not shown) pulling the display **13.13-102** towards the facial interface **13.13-104**. In certain implementations, the adjustment connections **13.13-602** can distribute up to a 6 kN applied load (e.g., from the display **13.13-102** along the facial interface **13.13-104**).

[3209] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-6A-13.13-6B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-6A-13.13-6B**.

[3210] In contrast to static adjustment, the following description for FIGS. **13.13-7A-27** discusses various examples of dynamic adjustment (e.g., for different users, head sizes, head shapes, etc.)—where “dynamic” refers to on-the-fly or real time adjustment as may be desired for a given user at a given time. Moreover, as discussed above, the head-mountable device of the present disclosure can include single-point systems with a singular adjustment mechanism. In accordance with one or

more examples of the present disclosure, FIGS. **13.13-7A-13.13-7B** show an example head-mountable device **13.13-700** including the display **13.13-102** and the facial interface **13.13-104**. In particular, the head-mountable device **13.13-700** includes an angular adjustment connection **13.13-702** that movably constrains the display **13.13-102** relative to the facial interface **13.13-104**. [3211] The angular adjustment connection **13.13-702** specifically includes a linkage **13.13-704** attached to the display **13.13-102**. The linkage **13.13-704** includes notches **13.13-706** configured to receive a locking element **13.13-708**, where the locking element **13.13-708** is connected to the facial interface **13.13-104**. The notches **13.13-706** correspond to positions at which the locking element **13.13-708** can engage the linkage **13.13-704** for maintaining a position of the linkage **13.13-704**.

[3212] The display **13.13-102** can be moved (e.g., rotated) by sliding the linkage **13.13-704** relative to the locking element **13.13-708**. For example, the linkage **13.13-704** allows the display **13.13-102** to pivot relative to the facial interface **13.13-104** when the linkage **13.13-704** extends or the linkage **13.13-704** retracts relative to the locking element **13.13-708**.

[3213] In certain implementations, the linkage **13.13-704** automatically moves in response to direct manipulation of the display **13.13-102**. For example, the linkage **13.13-704** can extend or retract relative to the locking element **13.13-708** in response to user manipulation (e.g., a user hand tilting) of the display **13.13-102**.

[3214] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-7A-13.13-7B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-7A-13.13-7B**.

[3215] The head-mountable device of the present disclosure can also include a multi-point system where multiple adjustment mechanisms in combination can allow (or provide) multiple degrees of freedom (e.g., multiple directions of display mobility). In accordance with one or more such examples, FIGS. **13.13-8** illustrates the display **13.13-102** and facial interface **13.13-104** connected by adjustable connections **13.13-802a-13.13-802d**. The adjustable connections **13.13-802a-13.13-802d** are disposed between the display **13.13-102** and the facial interface **13.13-104**. Each of the adjustable connections **13.13-802** can be individually adjusted such that the display and be displaced linearly and/or angularly.

[3216] For example, the adjustable connections **13.13-802a, 13.13-802b** can be adjusted so the top of the display is a first distance d_1 from the facial interface. The adjustable connections **13.13-802c, 13.13-802d** can be adjusted differently than the adjustable connections **13.13-802a, 13.13-802b** so that the bottom of the display is a second distance d_2 from the facial interface **13.13-104**. In downward tilting cases, the second distance d_2 can be less than the first distance d_1 . By contrast, for upward tilting cases, the second distance d_2 can be more than the first distance d_1 . In this way, the adjustable connections **13.13-802a-13.13-802d** make the angular adjustment connection between the facial interface **13.13-104** and the display **13.13-102**, and the adjustable connections **13.13-802** make the linear adjustment connection between the facial interface **13.13-104** and the display **13.13-102**.

[3217] As another example, the adjustable connections **13.13-802a-13.13-802d** can be adjusted in other ways. For instance, the adjustable connections **13.13-802a-13.13-802d** can be adjusted in a same or similar fashion (e.g., all lengthened or retracted by a same distance) to linearly translate the display **13.13-102** towards or away from the facial interface **13.13-104**.

[3218] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-8** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures

described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-8**.

[3219] Where FIGS. **13.13-6-8** depict example adjustment mechanisms without actuator controls (moving instead via direct user manipulation or setting of the head-mountable device), the following figures depict adjustment mechanisms with dynamic actuator controls. Indeed, as mentioned above, the connections (i.e., adjustment mechanisms or actuators) of the present disclosure can include adjustment controls configured for myriad different modes of user input and in various locations. In accordance with one or more such embodiments, FIGS. **13.13-9A-13.13-24B** show example adjustment mechanism variations of the head-mountable device **13.13-100**. In particular, the head-mountable device **13.13-100** includes the display **13.13-102**, the facial interface **13.13-104**, and the strap **13.13-103** (each described above).

[3220] In addition, FIGS. **13.13-9A-13.13-24B** respectively depict front and side-view profiles of head-mountable devices **13.13-900-13.13-2400** with corresponding connections **13.13-902-13.13-2402** (which are the same as or similar to the connection(s) **13.13-106** also described above). The connections **13.13-902-13.13-2402** movably constrain the display **13.13-102** relative to the facial interface **13.13-104**. The connections **13.13-902-13.13-2402** can therefore include myriad different forms of connections (e.g., mechanical connections) as indicated previously. Further, the connections **13.13-902-13.13-2402** can include myriad different forms of control mechanisms (e.g., actuator controls) that actuate the connections **13.13-902-13.13-2402** in response to user input.

[3221] In many examples, the head-mountable devices **13.13-900-13.13-2400** include actuator controls such as levers (e.g., side peel lever, vertical slide lever, etc.), buttons (push-button, side push button, top push button, automatic release button, etc.), dials (vertical scroll wheel dial, horizontal scroll wheel dial, top scroll wheel dial, etc.), rockers, sliders, or toggles. The actuator controls can be combined with multiple control mechanisms. For example, a sliding actuator controller can include a lever which can be user to lock a head-mountable device in position at the desired viewing state. In another example, a combined actuator control may include a rocker combined with a scroll wheel, such that the scroll wheel, when engaged by the rocker rotates, causing a change in distance between the display **13.13-102** and the facial interface **13.13-104**. It can be appreciated that a number of variations and examples are contemplated herein and the examples provided below serve to illustrate a few of the possible configurations.

[3222] FIGS. **13.13-9A-13.13-9B** illustrate a head-mountable device **13.13-900** with a connection **13.13-902**. The connection **13.13-902** includes an actuator control **13.13-904**. The actuator control **13.13-904** is positioned on a side portion of the display **13.13-102** or the facial interface **13.13-104**. The actuator control **13.13-904** includes a side push button **13.13-906**. In response to actuation of the side push button **13.13-906** (e.g., an inward push, press, tap), the connection **13.13-902** can translate the display **13.13-102** to or from a first state **13.13-908**, a second state **13.13-910**, or a third state **13.13-912**. Although FIGS. **13.13-9A-13.13-9B** illustrate discrete states or positions (i.e., the first state **13.13-908**, the second state **13.13-910**, and the third state **13.13-912**), it will be appreciated that head-mountable device **13.13-900** can include more or fewer positional states. In certain implementations, the head-mountable device **13.13-900** includes a continuous resolution of positional states (as opposed to discrete, predetermined positional states).

[3223] In one or more examples, the side push button **13.13-906** can be actuated in various ways. For instance, the side push button **13.13-906** can include a push-to-release button, a push-and-hold button, slide-push buttons, top-push buttons, etc.). The buttons can be located in various locations (e.g., top portion, side portion, bottom portion, etc.) of the head-mountable device **13.13-900**, either on the display **13.13-102** or on the facial interface **13.13-104**.

[3224] FIGS. **13.13-10A-13.13-10B** illustrate a head-mountable device **13.13-1000** with a connection **13.13-1002**. The connection **13.13-1002** includes an actuator control **13.13-1004**. The

actuator control **13.13-1004** is similar to the actuator control **13.13-904**. In particular, the actuator control **13.13-1004** includes a top push button positioned at a top portion of the display **13.13-102** or the facial interface **13.13-104** (e.g., at one or both top corners of the head-mountable device **13.13-1000**). When the actuator control **13.13-1004** is pressed, the connection **13.13-1002** can translate the display **13.13-102** to or from the first state **13.13-908**, the second state **13.13-910**, or the third state **13.13-912**, as described above.

[3225] In another example, the head-mountable device **13.13-100** includes actuation components that have actuator controls that slide or twist (e.g., by levers, buttons, rotatable components, etc.) causing the display **13.13-102** to move relative to the facial interface **13.13-104**. FIGS. **13.13-11A-13.13-14B** illustrate such examples.

[3226] In particular, FIGS. **13.13-11A** and **13.13-11B** illustrate a head-mountable device **13.13-1100** with a connection **13.13-1102**. The connection **13.13-1102** includes an actuator control **13.13-1104**. The actuator control **13.13-1104** is positioned on a side portion of the display **13.13-102** or the facial interface **13.13-104**. The actuator control **13.13-1104** includes a side peel lever. The side peel lever of the actuator control **13.13-1104**, when pulled downward, releases a lock (e.g., a cam lock) allowing the connection **13.13-1102** to move the display **13.13-102** relative to the facial interface **13.13-104**. Additionally, the actuator control **13.13-1104**, when pushed upward into a locked position, causes the connection **13.13-1102** to lock the display **13.13-102** into position relative to the facial interface **13.13-104**.

[3227] FIGS. **13.13-12A** and **13.13-12B** illustrate a head-mountable device **13.13-1200** with a connection **13.13-1202**. The connection **13.13-1202** includes an actuator control **13.13-1204**. The actuator control **13.13-1204** is also positioned on a side portion of the display **13.13-102** or the facial interface **13.13-104**. The actuator control **13.13-1204** includes a twist sleeve lever. Thus, the twist sleeve lever of the actuator control **13.13-1204** can rotate (e.g., outward in a clockwise direction), allowing the connection **13.13-1202** to move the display **13.13-102** relative to the facial interface **13.13-104**. Likewise, the actuator control **13.13-1204** can be rotated in an opposite direction (e.g., inward in a counterclockwise direction), allowing the connection **13.13-1202** to positionally fix the display **13.13-102** relative to the facial interface **13.13-104**.

[3228] FIGS. **13.13-13A** and **13.13-13B** illustrate a head-mountable device **13.13-1300** with a connection **13.13-1302**. The connection **13.13-1302** includes an actuator control **13.13-1304**. The actuator control **13.13-1304** is similarly positioned on a side portion of the display **13.13-102** or the facial interface **13.13-104**. The actuator control **13.13-1304** includes a finger pinch slider. The finger pinch slider of the actuator control **13.13-1304** can slide frontwards and backwards allowing the connection **13.13-1302** to positionally move (or lock) the display **13.13-102** relative to the facial interface **13.13-104**.

[3229] FIGS. **13.13-14A** and **14B** illustrate a head-mountable device **13.13-1400** with a connection **13.13-1402**. The connection **13.13-1402** includes an actuator control **13.13-1404**. The actuator control **13.13-1404** is positioned on a side portion of the display **13.13-102** or the facial interface **13.13-104**. The actuator control **13.13-1404** includes a sleeve slide. The sleeve slide of the actuator control **13.13-1404** can slide along the actuator control **13.13-904** allowing the connection **13.13-1402** to move the display **13.13-102** relative to the facial interface **13.13-104**.

[3230] FIGS. **13.13-15A** and **13.13-15B** illustrate a head-mountable device **13.13-1500** with a connection **13.13-1502**. The connection **13.13-1502** includes an actuator control **13.13-1504**. The actuator control **13.13-1504** is positioned along an outer periphery of the head-mountable device **13.13-1500** (e.g., from top to bottom corners at a side portion of the display **13.13-102** or the facial interface **13.13-104**). The actuator control **13.13-1504** includes a vertical slide that moves approximately up and down along the outer periphery of the head-mountable device **13.13-1500**. The vertical slide of the actuator control **13.13-1504** can be toggled between positions (e.g., the first state **13.13-908**, the second state **13.13-910**, and the third state **13.13-912**). Additional detail of this embodiment is discussed further below in relation to FIGS. **13.13-25A-13.13-25D**.

[3231] FIGS. 13.13-16A and 13.13-16B illustrate a head-mountable device 13.13-1500 with a connection 13.13-1602. The connection 13.13-1602 includes an actuator control 13.13-1604. The actuator control 13.13-1604 is positioned on a side portion of the display 13.13-102 or the facial interface 13.13-104. The actuator control 13.13-1604 includes a side temple squeeze button (e.g., spring button) that, when squeezed, allows the connection 13.13-1602 to contemporaneously move the display 13.13-102 relative to the facial interface 13.13-104. The side temple squeeze button of the actuator control 13.13-1604 can automatically release when one of the first state 13.13-908, the second state 13.13-910, or the third state 13.13-912 is reached. The side temple squeeze button of the actuator control 13.13-1604 can have the button oriented upward (or in alternative cases, downward) relative to the head-mountable device 13.13-1600.

[3232] FIGS. 13.13-17A and 13.13-17B illustrate a head-mountable device 13.13-1700 with a connection 13.13-1702. The connection 13.13-1702 includes an actuator control 13.13-1704. The actuator control 13.13-1704 is positioned on a side portion of the display 13.13-102 or the facial interface 13.13-104. The actuator control 13.13-1704 includes a scroll wheel rotatable about an axis 13.13-1706 proceeding inward and outward of the head-mountable device 13.13-1700. In some examples, the scroll wheel of the actuator control 13.13-1704 rotates in response to tangential forces (e.g., a finger swipe from front to back or up and down). In other examples, the scroll wheel of the actuator control 13.13-1704 rotates in response to twisting fingers. The scroll wheel of the actuator control 13.13-1704, when rotated about the axis 13.13-1706, can cause the connection 13.13-1702 to correspondingly adjust the distance between the display 13.13-102 and the facial interface 13.13-104.

[3233] FIGS. 13.13-18A and 13.13-18B illustrate a head-mountable device 13.13-1800 with a connection 13.13-1802. The connection 13.13-1802 includes an actuator control 13.13-1804. The actuator control 13.13-1804 is positioned on a side portion of the display 13.13-102 or the facial interface 13.13-104. The actuator control 13.13-1804 includes a scroll wheel rotatable about an axis 13.13-1806. The actuator control 13.13-1804, like the actuator control 13.13-1704, can respond to tangential forces (such as a finger swipe up or down). The scroll wheel of the actuator control 13.13-1804, when rotated about the axis 13.13-1806, can cause the connection 13.13-1802 to correspondingly adjust the distance between the display 13.13-102 relative to the facial interface 13.13-104.

[3234] FIGS. 13.13-19A and 13.13-19B illustrate a head-mountable device 13.13-1900 with a connection 13.13-1902. The connection 13.13-1902 includes an actuator control 13.13-1904. The actuator control 13.13-1904 is positioned on a side portion of the display 13.13-102 or the facial interface 13.13-104. The actuator control 13.13-1904 is similar to the actuator control 13.13-1804. In particular, the actuator control 13.13-1904 includes a scroll wheel rotatable about an axis 13.13-1906. The scroll wheel of the actuator control 13.13-1904 can therefore respond to tangential forces (e.g., a finger swipe front to back, and vice-versa). In response to this user interaction, the connection 13.13-1902 can move the display 13.13-102 relative to the facial interface 13.13-104.

[3235] FIGS. 13.13-20A and 13.13-13.13-20B illustrate a head-mountable device 13.13-2000 with a connection 13.13-2002. The connection 13.13-2002 includes an actuator control 13.13-2004. The actuator control 13.13-2004 includes a slider (e.g., toggle) positioned on the top region of the display 13.13-102 or the facial interface 13.13-104 (e.g., along a forehead region). The actuator control 13.13-904 moves laterally (e.g., inward and outward) to correspondingly cause the connection 13.13-2002 to adjust the distance between the display 13.13-102 relative to the facial interface 13.13-104 (e.g., frontwards or backwards).

[3236] FIGS. 13.13-21A and 13.13-13.13-21B illustrate a head-mountable device 13.13-2100 with a connection 13.13-2102. The connection 13.13-2102 includes an actuator control 13.13-2104. The actuator control 13.13-2104 includes a rocker positioned on the top region of the display 13.13-102 or the facial interface 13.13-104. The actuator control 13.13-2104 moves up or down about a pivot 13.13-2106. In response to up/down actuation of the actuator control 13.13-2104, the connection

13.13-2102 adjusts the distance between the display **13.13-102** relative to the facial interface **13.13-104**.

[3237] FIGS. **13.13-22A-13.13-24B** illustrate an actuator control that includes a scroll wheel disposed on top of a head-mountable device. The scroll wheel can be rotated around different axes to adjust the distance between the display **13.13-102** relative to the facial interface **13.13-104**. Further detail is provided below.

[3238] In particular, FIGS. **13.13-22A** and **13.13-13.13-22B** illustrate a head-mountable device **13.13-2200** with a connection **13.13-2202**. The connection **13.13-2202** includes the actuator control **13.13-1804** described above. Here, however, the actuator control **13.13-1804** is positioned on a top portion of the display **13.13-102** or the facial interface **13.13-104**. Indeed, the actuator control **13.13-1804** includes a scroll wheel rotatable about an axis **13.13-2204**. The scroll wheel of the actuator control **13.13-1804**, when rotated up or down about the axis **13.13-2204**, causes the connection **13.13-2202** to adjust the distance between the display **13.13-102** relative to the facial interface **13.13-104**.

[3239] FIGS. **13.13-23A** and **13.13-13.13-23B** illustrate a head-mountable device **13.13-2300** with a connection **13.13-2302**. The connection **13.13-2302** also includes the actuator control **13.13-1804** described above. Differently though, the actuator control **13.13-1804** is positioned on a top portion of the display **13.13-102** or the facial interface **13.13-104**. The actuator control **13.13-1804** also includes a scroll wheel rotatable about an axis **13.13-2304**. The scroll wheel of the actuator control **13.13-1804**, when rotated up or down about the axis **13.13-2304**, causes the connection **13.13-2302** to adjust the distance between the display **13.13-102** relative to the facial interface **13.13-104**.

[3240] FIGS. **13.13-24A** and **13.13-13.13-24B** illustrate a head-mountable device **13.13-2400** with a connection **13.13-2402**. The connection **13.13-2402** includes the actuator control **13.13-1704** described above, albeit differently located in FIGS. **13.13-24A-13.13-24B**. In particular, the actuator control **13.13-1704** is positioned on a top portion of the display **13.13-102** or the facial interface **13.13-104**. The actuator control **13.13-1704** includes a scroll wheel rotatable about an axis **13.13-2404**. The scroll wheel of the actuator control **13.13-1704**, when rotated about the axis **13.13-2404**, causes the connection **13.13-2402** to adjust the distance between the display **13.13-102** relative to the facial interface **13.13-104**.

[3241] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-9A-13.13-24B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-9A-13.13-24B**.

[3242] As discussed above, a head-mountable device of the present disclosure can include connections (e.g., adjustment mechanisms) that can toggle from position to position via a sliding actuation. In accordance with one or more such examples, FIGS. **13.13-25A** shows a perspective view of the head-mountable device **13.13-1500** shown in FIGS. **13.13-15A** and **13.13-15B**. The head-mountable device **13.13-1500** includes connections **13.13-2502a**, **13.13-2502b** having an actuator control **13.13-1504**. The actuator control **13.13-1504** includes a handle or lever that juts out from the head-mountable device **13.13-1500** for convenient user manipulation of the connections **13.13-2502a**, **13.13-2502b** (e.g., a sliding motion along the periphery of the head-mountable device **13.13-1500**).

[3243] As shown, user interaction with the actuator control **13.13-1504** can move the connections **13.13-2502a**, **13.13-2502b** in a manner that toggles the connections **13.13-2502a**, **13.13-2502b** between three discrete states (e.g., positions) including the first state **13.13-908**, the second state **13.13-910**, and the third state **13.13-912**. The connections **13.13-2502a**, **13.13-2502b** are independently movable. In other implementations, however, the connections **13.13-2502a**, **13.13-**

2502b are movably connected to each other. It will be appreciated that the connections **13.13-2502a**, **13.13-2502b** can move within the head-mountable device **13.13-1500** in myriad different ways. In particular examples, the connections **13.13-2502a**, **13.13-2502b** follow a track having a constant radius.

[3244] As the connections **13.13-2502a**, **13.13-2502b** move, the display **13.13-102** correspondingly moves relative to the facial interface **13.13-104**. The particular movement of the connections **13.13-2502a**, **13.13-2502b** causing the head-mountable device adjustment is described below in relation to FIGS. **13.13-25B-13.13-25D** (depicting the connection **13.13-2502a** in greater detail). In particular, FIGS. **13.13-25B** illustrates the connection **13.13-2502a** including a stepped adjustment portion **13.13-2503** in a first state. In the first state, the facial interface **13.13-104** and the display **13.13-102** are positioned a distance **13.13-2504a** apart (e.g., about zero mm).

[3245] By contrast, FIGS. **13.13-25C** illustrates the connection **13.13-2502a** moving left as indicated in the provided arrow. In doing so, the stepped adjustment portion **13.13-2503** correspondingly moves in a stepwise fashion, thereby creating increased separation between the display **13.13-102** and the facial interface **13.13-104** of a distance **13.13-2504b** (e.g., about 3 mm) to a second positional state. Still further, FIGS. **13.13-1-13.13-25D** illustrates the connection **13.13-2502a** moving further left as indicated in the provided arrow. With this additional movement, the stepped adjustment portion **13.13-2503** moves another step to create more separation between the display **13.13-102** and the facial interface **13.13-104** of a distance **13.13-2504c** (e.g., about 6 mm) to a third positional state. Thus, the geometry or structure of the stepped adjustment portion **13.13-2503** can provide a desired stepwise resolution between positional states by causing the display **13.13-102** to move away from (or towards) the facial interface **13.13-104**.

[3246] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-25A-13.13-25D** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-25A-13.13-25D**.

[3247] As briefly mentioned above, multiple connections of a head-mountable device can be linked together as a single connection. The links between connections of a head-mountable device form a timing mechanism that causes the connections to move together uniformly. In accordance with one or more such examples, FIGS. **13.13-26-27** illustrate respective perspective views of assemblies **13.13-2600-2700** for head-mountable devices.

[3248] In particular, FIGS. **13.13-26** depicts the assembly **13.13-2600** for a head-mountable device with a connection **13.13-2602** that is movably adjustable via an actuator control **13.13-2604**. The actuator control **13.13-2604** is positioned along a forehead region, but oriented downward (albeit myriad other orientations are herein contemplated). In response to actuation of the actuator control **13.13-2604**, stepped adjustment portions **13.13-2606** each move in unison in a stepwise fashion (as similarly described above for the stepped adjustment portion **13.13-2503** in relation to FIGS. **13.13-25B-13.13-25D**). Linkage arms **13.13-2608** movably connect portions of the connection **13.13-2602** to provide such a timing mechanism that transfers motion in a coordinated manner. While FIGS. **13.13-26** illustrates stepped adjustment portions **13.13-2606**, the adjustment portions can have additional steps or even a gradual incline surface that provides a finer adjustment capability.

[3249] In contrast to stepped adjustment portions, FIGS. **13.13-27** depicts the assembly **13.13-2700** for a head-mountable device with a connection **13.13-2702** with toggle links. In particular, the connection **13.13-2702** is movably adjustable via the actuator control **13.13-2604**. In response to actuation of the actuator control **13.13-2604**, toggle links **13.13-2704** toggle between stowed and locked positions. When the toggle links **13.13-2704** move in unison, the display **13.13-102**

correspondingly moves relative to the facial interface **13.13-104**. Moreover, in this configuration, the connection **13.13-2702** moves in binary fashion between a first positional state and a second positional state. The linkage arms **13.13-2608** in FIGS. **13.13-27** also movably connect portions of the connection **13.13-2702** to provide such a timing mechanism that transfers motion in a coordinated manner.

[3250] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-26-27** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-26-27**. The following figure descriptions for FIGS. **13.13-28-13.13-33** relate to example head-mountable devices, wearable apparatuses, or wearable electronic devices that can implement two degrees of freedom—namely angular tilt and linear translation toward and away from the user's face. In these examples, a linear slide can be implemented with certain structure to control actuation of the linear slide (whether in response to user adjustment or an applied force).

[3251] FIGS. **13.13-28-13.13-30** illustrate top, front, and side views, respectively, of a head-mountable device **13.13-2800** in accordance with one or more examples of the present disclosure. As shown, the head-mountable device **13.13-2800** can include a display frame **13.13-2802**, a facial interface **13.13-2804**, and connections **13.13-2806a-13.13-2806d**. The display frame **13.13-2802** and associated display can be the same as, or similar to, the display frame **13.13-105** (and associated display **13.13-102**) discussed above. The facial interface **13.13-2804** can be the same as, or similar to, the facial interface **13.13-104** discussed above. Likewise, the connections **13.13-2806a-13.13-2806d** can be the same as or similar to the connection(s) **13.13-106** discussed above.

[3252] In particular, however, the head-mountable device **13.13-2800** can include actuators **13.13-2808, 13.13-2810**. The actuators **13.13-2808, 13.13-2810** can include a variety of different actuators. In particular examples, the actuators **13.13-2808, 13.13-2810** include linear actuators of a linear adjustment system (e.g., that facilitate translational movement of the facial interface **13.13-2804** toward or away from the display frame **13.13-2802**). For instance, the actuators **13.13-2808, 13.13-2810** can include linear slides and/or linear springs that can translate in and out to vary a distance (e.g., a depth distance, eye relief, etc.) between the facial interface **13.13-2804** and the display frame **13.13-2802**. In particular, the actuators **13.13-2808** correspond to forehead actuators or top actuators, and the actuators **13.13-2810** correspond to zygoma actuators or side actuators. It will be appreciated that additional or alternative configurations of the actuators **13.13-2808, 13.13-2810** are herein contemplated. For instance, actuators can be positioned in the cheek (maxilla region) along a bottom portion of the head-mountable device **13.13-2800**. Additional details regarding the actuators **13.13-2808, 13.13-2810** are described below in relation to FIGS. **13.13-31-13.13-33**.

[3253] In some examples, the actuators **13.13-2808, 13.13-2810** are controlled or manipulated via at least one of actuator controls **13.13-2812a, 13.13-2812b**. In these or other examples, the actuator controls **13.13-2812a, 13.13-2812b** can include a variety of different types of controls, such as a lever, button, dial, rocker, slider, or toggle (as described above). The actuator controls **13.13-2812a, 13.13-2812b** can be engaged via manual manipulation and/or via hands-free or automated methods (also mentioned above). The actuator controls **13.13-2812a, 13.13-2812b** can be positioned in various locations on or within the head-mountable device **13.13-2800**. In particular examples, the actuator controls **13.13-2812a, 13.13-2812b** can be positioned adjacent to a maxilla region of a human face when the head-mountable device **13.13-2800** is donned. Other example positions for the actuator controls **13.13-2812a, 13.13-2812b** can include the zygoma region, forehead region, on the arms (not shown), or within an interior portion of the optical space adjacent the display.

[3254] In these or other examples, the actuator controls **13.13-2812a**, **13.13-2812b** can control or manipulate the actuators in a variety of ways. In some examples, the actuator controls **13.13-2812a**, **13.13-2812b** are connected to flexible drive shafts **13.13-2814**, **13.13-2816**. In particular examples, the flexible drive shafts **13.13-2814**, **13.13-2816** can transfer energy or movement from engagement (e.g., depression) of at least one of the actuator controls **13.13-2812a**, **13.13-2812b** to the actuators **13.13-2808**, **13.13-2810** (or associated locks, as described below). Examples of the flexible drive shafts **13.13-2814**, **13.13-2816** can include cables, tensioned wires, rods, dowels, etc. In some examples, the flexible drive shafts **13.13-2814**, **13.13-2816** are positioned inside the display frame **13.13-2802** (hidden from exterior viewing). In other examples, the flexible drive shafts **13.13-2814**, **13.13-2816** are positioned externally (e.g., on an outside surface of the display frame **13.13-2802**).

[3255] In at least some examples, the flexible drive shafts **13.13-2814**, **13.13-2816** can provide synchronized movement or parallel motion for each of the actuators **13.13-2808**, **13.13-2810**. For example, the flexible drive shafts **13.13-2814**, **13.13-2816** can be connected to each other via a linkage **13.13-2818**. Examples of the linkage **13.13-2818** include a gear system, a pulley system, a pivot connection, etc.

[3256] The flexible drive shafts **13.13-2814**, **13.13-2816** can be motion-synchronized via the linkage **13.13-2818** to mitigate (or prevent) uneven translation between the actuators **13.13-2808**, **13.13-2810**. Further, in some examples, the linkage **13.13-2818** can allow a single input to one of the actuators **13.13-2808**, **13.13-2810** to control both of the flexible drive shafts **13.13-2814**, **13.13-2816**. That is, in certain implementations, engagement of both of the actuator controls **13.13-2812a**, **13.13-2812b** is not required to actuate all of the actuators **13.13-2808**, **13.13-2810** (and/or their corresponding locks, described below).

[3257] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-28-13.13-30** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-28-13.13-30**. The following figure descriptions in relation to FIGS. **13.13-31-13.13-33** relate to specific examples of the actuators **13.13-2808**, **13.13-2810** discussed above. Indeed, the actuators **13.13-2808**, **13.13-2810** can include a linear adjustment connection between the display and the facial interface, where the linear adjustment connection includes a linear slide actuator with a locking mechanism.

[3258] FIGS. **13.13-31-13.13-33** respectively illustrate a perspective view of lock-slider disengagement, a front view of lock-slider disengagement, and a front view of lock-slider engagement of a portion of a linear adjustment connection **13.13-3100** in accordance with one or more examples of the present disclosure. In these or other examples, the linear adjustment connection **13.13-3100** includes an actuator **13.13-3102** (i.e., a linear slide), an actuator lock **13.13-3104** (otherwise referred to as a lock), and an actuator control (e.g., the actuator controls **13.13-2808**, **13.13-2810** not shown). As will be discussed below, in response to an actuator control, the actuator lock **13.13-3104** can engage the actuator **13.13-3102** to hold the actuator **13.13-3102** in position, or else to release the actuator **13.13-3102** for desired adjustability.

[3259] As shown, the actuator **13.13-3102** can be designed to translate in a slide direction **13.13-3116** (e.g., to move the facial interface **13.13-2804** away from the display frame **13.13-2802**). The actuator **13.13-3102** can also freely translate in a direction opposite the slide direction **13.13-3116** (e.g., to move the facial interface **13.13-2804** toward the display frame **13.13-2802**) when the actuator lock **13.13-3104** is disengaged from the actuator **13.13-3102**. To perform this function, the actuator **13.13-3102** can include certain structures. For example, the actuator **13.13-3102** can include teeth **13.13-3106** positioned on a top surface **13.13-3107**. The teeth **13.13-3106** can extend

upward from the top surface **13.13-3107** for a tooth height **13.13-3110**. In some examples, the tooth height **13.13-3110** can be tuned based on various design factors. For example, the tooth height **13.13-3110** can affect a stack thickness of one or more portions of the display frame **13.13-2802**. As another example, the tooth height **13.13-3110** can affect an ability of the actuator lock **13.13-3104** to deflect upwards in a direction **13.13-3122** (e.g., without damage to the actuator lock **13.13-3104**). Thus, in some examples, the tooth height **13.13-3110** can be minimized or reduced.

[3260] As an additional example, the teeth **13.13-3106** can include a tooth pitch (e.g., an amount of teeth per inch defining a distance **13.13-3112** between teeth). Like the tooth height **13.13-3110**, the tooth pitch can also be tuned based on various design factors. For example, a finer tooth pitch (e.g., a smaller distance **13.13-3112** or more teeth per inch) can be utilized as the tooth height **13.13-3110** is decreased. Additionally or alternatively, tooth pitch and/or the tooth height **13.13-3110** can be tuned for a desired amount of force at which back-drivability can occur (or to entirely prevent back-drivability according to some examples). In these or other examples, the term “back-drive” or “back-drag” refers to translation of the actuator **13.13-3102** in a direction opposite the slide direction **13.13-3116** (e.g., when the facial interface **13.13-2804** and the display frame **13.13-2802** move toward each other in response to an applied force from one of the display frame **13.13-2802** or the facial interface **13.13-2804**).

[3261] In more detail, the teeth **13.13-3106** can include leading tips **13.13-3114**. The leading tips **13.13-3114** can include a lead-in structure where teeth **13.13-3108** (and specifically leading tips **13.13-3118**) of the actuator lock **13.13-3104** can begin to engage. The leading tips **13.13-3114**, **13.13-3118** can include tapered portions or angled portions of the teeth **13.13-3106**, **13.13-3108** that serve as ramps (e.g., to avoid dead-band or teeth misalignment/blunt collisions where the actuator lock **13.13-3104** is prevented from engaging the actuator **13.13-3102**). That is, when transitioning from slider-lock disengagement (shown in FIGS. **13.13-32**) to slider-lock engagement (shown in FIGS. **13.13-33**), the leading tips **13.13-3114**, **13.13-3118** allow the teeth **13.13-3108** of the actuator lock to smoothly slide into and seat between the teeth **13.13-3106** of the actuator **13.13-3102**.

[3262] With respect to FIGS. **13.13-32** specifically, the actuator lock **13.13-3104** can translate in a lock translation direction **13.13-3120** (e.g., perpendicular to the slide direction **13.13-3116**). In particular, the actuator lock **13.13-3104** can translate in the lock translation direction **13.13-3120** in response to at least one of the actuator controls **13.13-2812a**, **13.13-2812b** being engaged (e.g., depressed) and the flexible drive shafts **13.13-2814**, **13.13-2816** correspondingly engaging the actuator lock **13.13-3104**. Thereafter, the actuator **13.13-3102** can be adjusted to translate the facial interface **13.13-2804** toward or away from the display frame **13.13-2802** (or associated display). Additionally, upon release of the actuator controls **13.13-2812a**, **13.13-2812b**, the actuator lock **13.13-3104** can move to an engaged position (shown in FIGS. **13.13-33**) in which the teeth **13.13-3106** and the teeth **13.13-3108** are engaged or otherwise positioned adjacent each other. In this position, the actuator **13.13-3102** is positionally locked in place to inhibit translation of the facial interface **13.13-2804** toward or away from the display frame **13.13-2802** (or associated display). As further shown in FIGS. **13.13-33**, the leading tips **13.13-3114** can extend beyond the teeth **13.13-3108**, and the leading tips **13.13-3118** can extend beyond the teeth **13.13-3106**.

[3263] When the teeth **13.13-3106**, **13.13-3108** are engaged (as shown in FIGS. **13.13-33**), the actuator lock **13.13-3104** can be deflected upward in a lock deflection direction **13.13-3122**. The lock deflection direction **13.13-3122** can be perpendicular to both the slide direction **13.13-3116** and the lock translation direction **13.13-3120**. In these or other examples, the actuator lock **13.13-3104** can deflect in the lock deflection direction **13.13-3122** during a back-driving or back-dragging event in which the actuator **13.13-3102** moves opposite the slide direction **13.13-3116**. To facilitate this back-driving or back-dragging, the actuator lock **13.13-3104** can act as a cantilever spring to deflect in the lock deflection direction **13.13-3122** in response to an applied force at the facial interface **13.13-2804** or the display frame **13.13-2802**. In some examples, actuator lock

13.13-3104 can be tuned to deflect in response to a predetermined amount of force (e.g., about 5 Newtons to about 500 Newtons per actuator, about 10 Newtons to about 100 Newtons per actuator, or about 20 Newtons to about 40 Newtons per actuator). In these or other examples, various design factors for the actuator lock **13.13-3104**, such as tooth angles or bending stiffness, can be adjusted to provide the desired force tuning.

[3264] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-31-13.13-33** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-31-13.13-33**. The following description provides more detail regarding an example implementation of multiple linear adjustment connections connected to a flexible driveshaft (or multiple flexible drive shafts).

[3265] FIGS. **13.13-34** illustrates a perspective view of a portion of a head-mountable device **13.13-3400** in accordance with one or more examples of the present disclosure. As shown, the head-mountable device **13.13-3400** includes multiple actuators (e.g., multiple linear adjustment connections **13.13-3100a-13.13-3100d**) discussed above with respect to the linear adjustment connection **13.13-3100**. As also mentioned, each linear adjustment connection **13.13-3100a-13.13-3100d** can be actuated via the flexible drive shafts **13.13-2814**, **13.13-2816**. The flexible drive shafts **13.13-2814**, **13.13-2816** can be motion-synchronized via the linkage **13.13-2818** to mitigate (or prevent) uneven translation between each linear adjustment connection **13.13-3100a-13.13-3100d**. Further, in some examples, the linkage **13.13-2818** can allow a single input (e.g., a single button press) to actuate each of the linear adjustment connections **13.13-3100a-13.13-3100d**. The following description of FIGS. **13.13-34** gives an example implementation in which this can be performed.

[3266] As shown in FIGS. **13.13-34**, the head-mountable device **13.13-3400** can include actuator controls **13.13-3401a**, **13.13-3401b** (which can be the same as or similar to the actuator controls **13.13-2812a**, **13.13-2812b** discussed above). In response to a force **13.13-3402a** applied to the actuator control **13.13-3401a**, the lock **13.13-3104a** can correspondingly deflect, translate, or otherwise move to release the slide **13.13-3102a** of the linear adjustment connection **13.13-3100a**. In addition, the flexible drive shaft **13.13-2816** can be attached to a joint **13.13-3404** (e.g., a ball joint) connected to the lock **13.13-3104a**. As the lock **13.13-3104a** deflects, translates, or otherwise moves in response to the applied force **13.13-3402a**, the flexible drive shaft **13.13-2816** can be tensioned or pulled in a direction **13.13-3406**.

[3267] The flexible drive shaft **13.13-2816** can be attached to a gear anchor **13.13-3410** such that, upon pulling of the flexible drive shaft **13.13-2816** in the direction **13.13-3406**, a first gear **13.13-3409a** connected to the gear anchor **13.13-3410** rotates in a direction **13.13-3412**. In turn, the lock **13.13-3104b** (which is movably attached to the first gear **13.13-3409a**) is pulled back away from the slide **13.13-3102b** of the linear adjustment connection **13.13-3100b** as the first gear **13.13-3409a** rotates in the direction **13.13-3412**.

[3268] Furthermore, rotation of the first gear **13.13-3409a** in the direction **13.13-3412** can simultaneously induce rotation of the second gear **13.13-3409b** in a direction **13.13-3414**. The rotation of the second gear **13.13-3409b** in the direction **13.13-3414** can cause the lock **13.13-3104c** (which is movably attached to the second gear **13.13-3409b**) to pull back away from the slide **13.13-3102c** of the linear adjustment connection **13.13-3100c**.

[3269] In addition, rotation of the second gear **13.13-3409b** in the direction **13.13-3414** can cause a gear anchor **13.13-3416** connected to the second gear **13.13-3409b** to push the flexible drive shaft **13.13-2814**. Specifically, the flexible drive shaft **13.13-2814** can be connected to the gear anchor **13.13-3416** such that, upon rotation of the second gear **13.13-3409b** in the direction **13.13-3414**,

the flexible drive shaft **13.13-2814** is pushed in the direction **13.13-3408**. Pushing of the flexible drive shaft **13.13-2814** can cause the lock **13.13-3104d** to release from the slide **13.13-3102d** of the linear adjustment connection **13.13-3100d**. For instance, the flexible drive shaft **13.13-2814** can be connected to the lock **13.13-3104d** in a same or similar manner as the flexible drive shaft **13.13-2816** is attached to the lock **13.13-3104a** to impart slide-lock disengagement.

[3270] Additionally or alternatively, it will be appreciated that a force **13.13-3402b** can be applied to the actuator control **13.13-3401b**, which is in mechanical communication with the lock **13.13-3104d** (in a same or similar manner as the actuator control **13.13-3401a** is mechanically coupled to the lock **13.13-3104a**). Thus, dual inputs via both of the actuator controls **13.13-3401a**, **13.13-3401b** can be applied, but are not required, for synchronized actuation of each of the linear adjustment connections **13.13-3100a-13.13-3100d**. Alternatively, the applied force **13.13-3402b** alone (i.e., without the applied force **13.13-3402a**) can be used to actuate each of the linear adjustment connections **13.13-3100a-13.13-3100d**.

[3271] Further, in some examples, additional or alternative modes of actuating the linear adjustment connections **13.13-3100a-13.13-3100d** can be implemented. For example, solenoids or small motors can be utilized to actuate linear adjustment connections **13.13-3100a-13.13-3100d** (with or without the flexible drive shafts **13.13-2814**, **13.13-2816**). Similarly, in some examples, the actuator controls **13.13-3401a**, **13.13-3401b** can include alternative types of actuator controls. For example, the actuator controls **13.13-3401a**, **13.13-3401b** can be arch sliders that move in an arc path (e.g., along a display frame not shown of the head-mountable display **13.13-3400**). The arch sliders can be connected to the flexible drive shafts **13.13-2814**, **13.13-2816** as discussed above to actuate the linear adjustment connections **13.13-3100a-13.13-3100d**.

[3272] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.13-34** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.13-34**.

13.14: Nosepiece

[3273] Head-mounted devices include head-mounted support structures that allow the devices to be worn on the heads of users. The head-mounted support structures may include device housings for housing components such as displays that are used for presenting a user with visual content. Head-mounted devices may also include a light-shielding nosepiece that rests on the nose of the user. The light-shielding nosepiece may include an elastomeric layer with perforations and fabric that covers the elastomeric layer. The elastomeric layer with the perforations may allow the light-shielding nosepiece to conform to the user's nose while the device is worn, while maintaining enough rigidity to be wrapped by a low force, high stretch textile or other low force, high stretch material.

Additional structures, such as rigid structures or semi-rigid structures, may be included in the light-shielding nosepiece to provide additional support for the device while it is worn.

[3274] A schematic diagram of an illustrative system having an electronic device with a light-shielding nosepiece is shown in FIGS. **13.14-1**. As shown in FIGS. **13.14-1**, system **13.14-8** may include one or more electronic devices such as electronic device **13.14-10**. The electronic devices of system **13.14-8** may include computers, cellular telephones, head-mounted devices, wristwatch devices, and other electronic devices. Configurations in which electronic device **13.14-10** is a head-mounted device are sometimes described herein as an example.

[3275] As shown in FIGS. **13.14-1**, electronic devices such as electronic device **13.14-10** may have control circuitry **13.14-12**. Control circuitry **13.14-12** may include storage and processing circuitry for controlling the operation of device **13.14-10**. Circuitry **13.14-12** may include storage such as hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory

configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc.

[3276] Processing circuitry in control circuitry **13.14-12** may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, and other integrated circuits. Software code may be stored on storage in circuitry **13.14-12** and run on processing circuitry in circuitry **13.14-12** to implement control operations for device **13.14-10** (e.g., data gathering operations, operations involved in processing three-dimensional facial image data, operations involving the adjustment of components using control signals, etc.). Control circuitry **13.14-12** may include wired and wireless communications circuitry. For example, control circuitry **13.14-12** may include radio-frequency transceiver circuitry such as cellular telephone transceiver circuitry, wireless local area network (WiFi®) transceiver circuitry, millimeter wave transceiver circuitry, and/or other wireless communications circuitry.

[3277] During operation, the communications circuitry of the devices in system **13.14-8** (e.g., the communications circuitry of control circuitry **13.14-12** of device **13.14-10**), may be used to support communication between the electronic devices. For example, one electronic device may transmit video and/or audio data to another electronic device in system **13.14-8**. Electronic devices in system **13.14-8** may use wired and/or wireless communications circuitry to communicate through one or more communications networks (e.g., the internet, local area networks, etc.). The communications circuitry may be used to allow data to be received by device **13.14-10** from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, online computing equipment such as a remote server or other remote computing equipment, or other electrical equipment) and/or to provide data to external equipment.

[3278] Device **13.14-10** may include input-output devices **13.14-22**. Input-output devices **13.14-22** may be used to allow a user to provide device **13.14-10** with user input. Input-output devices **13.14-22** may also be used to gather information on the environment in which device IO is operating. Output components in devices **13.14-22** may allow device **13.14-10** to provide a user with output and may be used to communicate with external electrical equipment.

[3279] As shown in FIG. 1, input-output devices **13.14-22** may include one or more displays such as display **13.14-14**. In some configurations, display **13.14-14** of device **13.14-10** includes left and right display panels (sometimes referred to as left and right portions of display **13.14-14** and/or left and right displays) that are in alignment with the user's left and right eyes and are viewable through left and right lens assemblies, respectively. In other configurations, display **13.14-14** includes a single display panel that extends across both eyes.

[3280] Display **13.14-14** may be used to display images. The visual content that is displayed on display **13.14-14** may be viewed by a user of device **13.14-10**. Displays in device **13.14-10** such as display **13.14-14** may be organic light-emitting diode displays or other displays based on arrays of light-emitting diodes, liquid crystal displays, liquid-crystal-on-silicon displays, projectors or displays based on projecting light beams on a surface directly or indirectly through specialized optics (e.g., digital micromirror devices), electrophoretic displays, plasma displays, electrowetting displays, microLED displays, or any other suitable displays.

[3281] Display **13.14-14** may present computer-generated content such as virtual reality content and mixed reality content to a user. Virtual reality content may be displayed in the absence of real-world content. Mixed reality content, which may sometimes be referred to as augmented reality content, may include computer-generated images that are overlaid on real-world images. The real-world images may be captured by a camera (e.g., a forward-facing camera) and merged with overlaid computer-generated content or an optical coupling system may be used to allow computer-generated content to be overlaid on top of real-world images. As an example, a pair of mixed reality glasses or other augmented reality head-mounted display may include a display device that provides images to a user through a beam splitter, prism, holographic coupler, or other optical

coupler. Configurations in which display **13.14-14** is used to display virtual reality content to a user through lenses are described herein as an example.

[3282] Input-output devices **13.14-22** may include sensors **13.14-16**. Sensors **13.14-16** may include, for example, three-dimensional sensors (e.g., three-dimensional image sensors such as structured light sensors that emit beams of light and that use two-dimensional digital image sensors to gather image data for three-dimensional images from light spots that are produced when a target is illuminated by the beams of light, binocular three-dimensional image sensors that gather three-dimensional images using two or more cameras in a binocular imaging arrangement, three-dimensional LIDAR (light detection and ranging) sensors, three-dimensional radio-frequency sensors, or other sensors that gather three-dimensional image data), cameras (e.g., infrared and/or visible digital image sensors), gaze tracking sensors (e.g., a gaze tracking system based on an image sensor and, if desired, a light source that emits one or more beams of light that are tracked using the image sensor after reflecting from a user's eyes), touch sensors, buttons, force sensors, sensors such as contact sensors based on switches, gas sensors, pressure sensors, moisture sensors, magnetic sensors, audio sensors (microphones), ambient light sensors, microphones for gathering voice commands and other audio input, sensors that are configured to gather information on motion, position, and/or orientation (e.g., accelerometers, gyroscopes, compasses, and/or inertial measurement units that include all of these sensors or a subset of one or two of these sensors), fingerprint sensors and other biometric sensors, optical position sensors (optical encoders), and/or other position sensors such as linear position sensors, and/or other sensors.

[3283] User input and other information may be gathered using sensors and other input devices in input-output devices **13.14-22**. If desired, input-output devices **13.14-22** may include other devices **13.14-24** such as haptic output devices (e.g., vibrating components), light-emitting diodes and other light sources, speakers such as ear speakers for producing audio output, and other electrical components. Device **13.14-10** may include circuits for receiving wireless power, circuits for transmitting power wirelessly to other devices, batteries and other energy storage devices (e.g., capacitors), joysticks, buttons, and/or other components.

[3284] Electronic device **13.14-10** may have housing structures (e.g., housing walls, straps, etc.), as shown by illustrative support structures **13.14-26** of FIGS. **13.14-1**. In configurations in which electronic device IO is a head-mounted device (e.g., a pair of glasses, goggles, a helmet, a hat, a headband, etc.), support structures **13.14-26** may include head-mounted support structures (e.g., a helmet housing, head straps, temples in a pair of eyeglasses, goggle housing structures, and/or other head-mounted structures). The head-mounted support structures may be configured to be worn on a head of a user during operation of device IO and may support display(s) **13.14-14**, sensors **13.14-16**, other components **13.14-24**, other input-output devices **13.14-22**, and control circuitry **13.14-12**.

[3285] In some embodiments, support structures **13.14-26** may include a light-shielding nosepiece. The light-shielding nosepiece may be attached to support structures **13.14-26**, such as a main housing portion of electronic device **13.14-10**, and may rest on the user's nose while device **13.14-10** is worn. The light-shielding nosepiece may be flexible, to allow the nosepiece to conform to the user's nose, while retaining enough rigidity to support device **13.14-10** on the user's face while it is being worn (i.e., to maintain its shape on the user's nose while the device is worn) and to be wrapped by a fabric or other material. If desired, the light-shielding nosepiece may also include stiffeners or other components that help maintain the nosepiece on the user's nose to prevent light from reaching the user's eyes. An example of an illustrative electronic device having a nosepiece is shown in FIGS. **13.14-2**.

[3286] As shown in FIGS. **13.14-2**, head-mounted device **13.14-10** may include support structures **13.14-26**, which may include a head-mounted housing (sometimes referred to as a main housing, main housing unit, head-mounted support structure, etc.). The housing may have walls or other structures that separate an interior housing region from an exterior region surrounding the housing.

For example, the housing may have walls formed from polymer, glass, metal, and/or other materials. Electrical and optical components may be mounted in the housing. These components may include components such as integrated circuits, sensors, control circuitry, input-output devices, etc.

[3287] To present a user with images for viewing from eye boxes (e.g., eye boxes in which the user's eyes are located when device **13.14-10** is being worn on the user's head), device **13.14-10** may include displays and lenses. These components may be mounted in optical modules or other supporting structure in the housing to form respective left and right optical systems. There may be, for example, a left display for presenting an image through a left lens to a user's left eye in a left eye box and a right display for presenting an image through a right lens to a user's right eye in a right eye box.

[3288] If desired, the housing may have forward-facing components such as cameras and other sensors on a front side for gathering sensor measurements and other input and may have a soft cushion on an opposing rear side of the housing. The rear side of the housing may have openings that allow the user to view images (image light **13.14-32**) from the left and right optical systems (e.g., when the rear side of the housing is resting on the user's head).

[3289] If desired, device **13.14-10** may have an adjustable strap or headband, and if desired, may have other structures (e.g., an over-the-head strap) to help hold the housing on the user's head.

[3290] As shown in FIGS. **13.14-2**, when worn by the user, device **13.14-10** may include a nosepiece, such as nosepiece **13.14-28** that rests on the nasal region the user's head (e.g., on the user's nose). In particular, nosepiece **13.14-28** (sometimes referred to as a light-shielding structure or a light-shielding nosepiece herein) may serve as an extension of the housing that rests on the user's nose and bridges between the opposing cheeks of the user. If desired, nosepiece **13.14-28** may be attached to the housing and/or may include one or more members formed from a portion of the housing. In some embodiments, nosepiece **13.14-28** may be a portion of a light seal or attached to a light seal that extends around some or all of a periphery of the housing (e.g., the light seal is attached to the housing around the periphery). When device **13.14-10** is in use, the light seal may compress against the user's face and prevent interference from ambient light. In general, however, nosepiece **13.14-28** may be attached to the housing in any desired manner.

[3291] Nosepiece **13.14-28** may be configured as a light-shielding structure and may therefore be sometimes referred to as light-shielding structure **13.14-28** or light-shielding nosepiece **13.14-28**. As an example, it may be desirable to enhance the viewing experience of the user by blocking external environmental light from entering the interior of device **13.14-10** (e.g., from entering the eye boxes) when device **13.14-10** is worn by the user. Nosepiece **13.14-28** may conform to the facial topology of the user around the user's nose and block light from entering the eye boxes. In some illustrative configurations, nosepiece **13.14-28** may be adjustable to conform to varying facial topologies of different users (e.g., portions of nosepiece **13.14-28** may deform differently based on the nose shapes of the users).

[3292] Nosepiece **13.14-28** may be mounted to a housing portion of electronic device **13.14-10**, such as head-mounted support structures **13.14-26**, at mounting points **13.14-30**. The housing may include a housing frame that runs along the periphery of device **13.14-10**. If desired, the housing frame may be overlapped by a cushion member on the rear side of the housing facing the user. As an example, the cushion member may include foam structures or other soft compressible structures affixed to the housing frame. A fabric may overlap and extend over the housing frame and/or the cushion member on the rear side of the housing. If desired, the fabric may enclose only the cushion member, and the fabric-enclosed cushion member may be removably coupled to the housing frame.

[3293] Mounting points **13.14-30** may be located at a bottom portion of the housing frame (e.g., a bottom portion of support structures **13.14-26**). As examples, mounting points **13.14-30** may include coupling mechanisms such as magnets, adhesive, hinges, or any other suitable coupling mechanisms.

[3294] In the example of FIGS. **13.14-2**, nosepiece **13.14-28** is attached to support structures **13.14-26** such that nosepiece **13.14-28** extends into a portion of support structures **13.14-26**. This is merely illustrative. If desired, the housing frame, a cushion member, and/or fabric may split into multiple portions to surround nosepiece **13.14-28**. As an example, a fabric-enclosed cushion member may run in front of nosepiece **13.14-28**, and a housing frame may run behind nosepiece **13.14-28**. In general, the housing portions, cushion member, and/or fabric may define an opening in which nosepiece **13.14-28** is disposed.

[3295] In some illustrative examples, nosepiece **13.14-28** may be removably coupled (via magnetics) to the housing frame or other portions of the housing. In some illustrative examples, a portion of nosepiece **13.14-28** may form an integral portion of the housing frame and/or may not be removable from the housing.

[3296] Support structures **13.14-26** (e.g., housing frames) and nosepiece **13.14-28** (along with other desired structures) may define the periphery of the eye boxes of device **13.14-10** at which the user's eyes are located. Components, such as displays, lenses, sensors, etc., may overlap and/or be located within the eye boxes of device **13.14-10**, and may be enclosed by and/or mounted to support structures **13.14-26** and/or nosepiece **13.14-28**. As illustratively shown in FIGS. **13.14-2**, display **13.14-14** emit image light **13.14-32** through a lens to an eye box. Nosepiece **13.14-28** may be configured to block environmental light from an exterior of device **13.14-10** from entering the eye box and interfering with image light **13.14-24**. An example of a nosepiece that may be used in device **13.14-10** is shown in FIGS. **13.14-3**.

[3297] As shown in FIGS. **13.14-3**, a nosepiece, such as nosepiece **13.14-28**, may include elastomer **13.14-34** (also referred to as elastomeric layer **13.14-34** herein). Elastomer **13.14-34** may be any desired elastomeric material, such as thermoplastic polyurethane (TPU), nitrile butadiene rubber (NBR), or silicone (such as a low-durometer silicone). Elastomer **13.14-34** may have a thickness of greater than 0.1 mm, greater than 0.25 mm, greater than 0.5 mm, about 1 mm, less than 2 mm, less than 5 mm, etc.

[3298] In some embodiments, elastomer **13.14-34** may include perforations **13.14-36**. Perforations **13.14-36** may allow elastomer **13.14-34** to bend to accommodate a user's nose (i.e., in the horizontal left and right directions of FIGS. **13.14-3**), while maintaining sufficient rigidity (i.e., in the vertical up and down directions of FIGS. **13.14-3**) to support device IO on the user's nose. The rigidity of elastomer **13.14-34** may also allow elastomer **13.14-34** to be wrapped by fabric **13.14-38** or other low force, high stretch material.

[3299] Perforations **13.14-36** may be formed in any desired pattern. In the example of FIGS. **13.14-3**, perforations **13.14-36** are formed as an array of perforations across the entire surface of elastomer **13.14-34**. Perforations **13.14-36** may be formed in a brick pattern, as in FIGS. **13.14-3**, or may be formed in any other desired pattern. Additionally, any desired number of perforations **13.14-36** may be formed in elastomer **13.14-34**, such as at least one perforation, at least **13.14-5** perforations, or any other desired number of perforations. Perforations **13.14-36** may extend entirely through elastomer **13.14-34**, or may extend partially through elastomer **13.14-34**.

[3300] Elastomer **13.14-34** may have a shape that generally conforms to a user's nose, such as a chevron shape, as shown in FIGS. **13.14-3**, or any other desired shape, such as a rounded shape, or a rectangular shape. Elastomer **13.14-34** may be covered by fabric **13.14-38**. In particular, fabric **13.14-38** may overlap and/or encapsulate elastomer **13.14-34** in region **13.14-40**. Fabric region **13.14-40** may have the same shape as elastomer **13.14-34**, as shown in FIGS. **13.14-3**, or may have a different shape than elastomer **13.14-34**. Fabric members **13.14-40** and/or **13.14-38** may serve as a light-shielding member or layer for light-shielding structure **13.14-28**. In particular, illustrative configurations in which fabric **13.14-40/38** is a fabric cover (sometimes referred to as a fabric cover layer or a cover layer) are described herein as an illustrative example. To serve light-shielding functions, the fabric cover may be formed from an opaque or light-shielding material (e.g., black yam) or may be formed from an underlying material coated with an opaque or light-

shielding material (e.g., black dye or ink). As examples, the fabric cover may be formed any suitable type of fabric such as knit fabric, woven fabric, braided fabric, etc.

[3301] Fabric **13.14-40** may be tented over elastomer **13.14-34**. Tenting of the fabric cover over an underlying structure may be achieved by the underlying structure contacting or otherwise supporting the fabric cover at one or more points or areas of support as the fabric cover extends over one or more sides of the underlying structure. The tenting of the fabric cover over the underlying structure may cause the fabric cover to follow the general outline of the underlying structure, especially around the areas of support. If desired, differences in the outlines of the fabric cover and of the underlying structure may exist, especially in some regions away from the areas of support, thereby causing some portions of the fabric cover to be suspended in air and therefore readily deflectable. As an example, the fabric cover may be deflectable to the boundary of the underlying structure (or even beyond the boundary of the underlying structure if the boundary is defined by a flexible or deformable member).

[3302] The rigidity of elastomer **13.14-34**, which is preserved in the vertical direction by perforations **13.14-36**, may allow elastomer **13.14-34** to be wrapped by fabric **13.14-40** (or other low force, high stretch textile, or other low force, high stretch material) without deforming. In other words, elastomer **13.14-34** may be rigid enough to maintain its shape while fabric **13.14-40** is applied to/wrapped around elastomer **13.14-34** (as well as to maintain its shape when the device is worn by a user), but flexible enough that it can conform to a user's nose.

[3303] In such a way, the fabric cover may have a three-dimensional shape (based on an outline of the underlying structure) that includes portions (e.g., directly supported by the underlying structure) that are more defined and portions (e.g., not directly supported by the underlying structure, suspended in air, etc.) that are less defined, and more flexible or yielding. These less-defined portions (e.g., a yielding fabric surface) may help form flexible boundaries such as those for an opening configured to receive a user's nose.

[3304] As a particular illustrative example, the underlying structure may have surfaces that define an opening for accommodating a user's nose. The surfaces may be surrounded by peripheral edges. The fabric cover may be tented over the underlying structure such that the fabric cover is directly supported by the underlying structure along one or more of the peripheral edges of the underlying structure and may be suspended in air around the opening, thereby providing a fabric surface that is deflectable by the user's nose. This may help with improving user comfort as well as providing a more conformal fit when the light shielding structure rests on the user's nose.

[3305] Regardless of the shape of fabric region **13.14-40**, fabric region **13.14-40** (and/or elastomer **13.14-34**) may be bonded to support structures **13.14-26** (such as a housing frame) using adhesive **13.14-42**. However, the use of adhesive **13.14-42** is merely illustrative. Fabric **13.14-40** and/or elastomer **13.14-34** may be formed integrally with support structures **13.14-26**, or may be attached to support structures **13.14-26** using any desired attachment mechanism.

[3306] Although nosepiece **13.14-28** is shown as including both elastomer **13.14-34** and fabric **13.14-40**, this is merely illustrative. If desired, nosepiece **13.14-28** may include elastomer **13.14-34** without an overlapping fabric layer. In this case, another layer, such as a polymer or rubber, may overlap elastomer **13.14-34**, or elastomer **13.14-34** may directly contact a user's nose as they wear device **13.14-10**. Alternatively, nosepiece **13.14-28** may include fabric **13.14-40** without elastomer **13.14-34**, if desired. In this case, fabric **13.14-40** may be a flat knit fabric to provide sufficient stretching over a user's nose, while providing enough support for device **13.14-10**.

[3307] Although fabric **13.14-38/40** is described as fabric, this is merely illustrative. In general, fabric **13.14-38/40** may be any low force, high stretch material, such as a low force, high stretch textile.

[3308] In some embodiments, although elastomer **13.14-34** may have rigidity to support device **13.14-10** on a user's nose (i.e., in the vertical direction of FIGS. **13.14-3**) and to be wrapped by fabric, it may be desirable to add additional rigid or semi-rigid structures to a nosepiece. An

illustrative example of a nosepiece having a rigid structure is shown in FIGS. **13.14-4**.

[3309] As shown in FIGS. **13.14-4**, nosepiece **13.14-28** may include fabric **13.14-40** (which may cover an elastomeric layer, such as elastomer **13.14-34**) and adhesive **13.14-42**, which in turn may attach fabric **13.14-40** to structural frame **13.14-43**. Structural frame **13.14-43** may be a plastic frame, as an example. Structural frame **13.14-43** may provide additional rigidity to ensure that nosepiece **13.14-28** retains its shape and support when worn by a user.

[3310] Although FIGS. **13.14-4** shows the use of adhesive **13.14-42**, this is merely illustrative. Adhesive **13.14-42** may be omitted, if desired. In some examples, elastomer **13.14-34** may be co-molded to structural frame **13.14-43**. Alternatively, elastomer **13.14-34** may be fit into a recessed portion of structural frame **13.14-43** or otherwise attached to structural frame **13.14-43**.

[3311] Additionally, although FIGS. **13.14-4** shows structural frame **13.14-43** on three sides of fabric **13.14-40**/elastomer **13.14-34**, this is merely illustrative. Structural frame **13.14-43** may be attached to one side (e.g., the bottom side) of fabric **13.14-40**/elastomer **13.14-34** or any other desired number of sides. Regardless of the attachment of structural frame **13.14-43** to elastomer **13.14-34** and/or fabric **13.14-40**, fabric **13.14-40** may extend over all or some of both frame **13.14-43** and/or elastomer **13.14-34**. An example of a stack up of nosepiece **13.14-28** is shown in FIGS. **13.14-5**.

[3312] As shown in FIGS. **13.14-5**, nosepiece **13.14-28** may include fabric **13.14-40**, which is shown as having fabric portions **13.14-40-1** and **13.14-40-2**, on either side of elastomer **13.14-34**. If desired, fabric **13.14-40** may be bonded to elastomer **13.14-34** at a periphery of elastomer **13.14-34**, which is shown in FIGS. **13.14-5** as points **13.14-44**. For example, fabric **13.14-40** may be bonded to elastomer **13.14-34** around a periphery (such as an entire periphery or a portion of the periphery) of the elastomer layer. Fabric **13.14-40** may be bonded to elastomer **13.14-34** using an adhesive or other desired mounting mechanism. In some cases, it may be desirable to allow elastomer **13.14-34** to move to a greater extent, and therefore leave elastomer **13.14-34** un-bonded from fabric **13.14-40**. For example, elastomer **13.14-34** may be fully surrounded fabric **13.14-40** and float within the fabric. In some embodiments portions of fabric **13.14-40** may be bonded directly to each other, rather than to elastomer **13.14-34**. In this way, elastomer **13.14-34** may float within fabric **13.14-40**, which may allow **13.14-34** to move more freely.

[3313] Because nosepiece **13.14-28** is designed to block light from reaching the eye boxes of a user wearing device **13.14-10**, it may be desirable to ensure a tight fit between nosepiece **13.14-28** and the user's nose and a similarly tight fit between nosepiece **13.14-28** and device **13.14-10**. An example of an extension piece that may allow for nosepiece **13.14-28** to fit tightly with device **13.14-10** is shown in FIGS. **13.14-6**.

[3314] As shown in FIGS. **13.14-6**, extension piece **13.14-46** may be added below nosepiece **13.14-28**. In particular, if a user's nose bridge protrudes only a small amount, nosepiece **13.14-28** may not reach the user's nose. Therefore, extension piece **13.14-46** may be attached between support structures of device **13.14-10**, such as support structures **13.14-26**, and nosepiece **13.14-28**. Extension piece **13.14-28** may be formed from fabric, foam, plastic, and/or any other desired material. In some cases, extension piece **13.14-28** may allow nosepiece **13.14-28** to align with other light-blocking structures in device **13.14-10**, such as a foam member that fits against the user's face. In this way, light may be prevented from reaching the user's eye boxes and interfering with the user's visibility of a display in device **13.14-10**.

[3315] In addition or as an alternative to extension piece **13.14-46**, it may be desirable to ensure a close fit to a user's nose to prevent light from entering the user's eye boxes. Examples of nosepieces that have additional structures to improve the fit to a user's nose are shown in FIGS. **13.14-7** and **13.14-8**.

[3316] As shown in FIGS. **13.14-7**, service loop **13.14-49** may be included within nosepiece **13.14-28**. In particular, service loop **13.14-49** (and other portions of nosepiece **13.14-28**, if desired) may be attached to supports **13.14-50-1** and **13.14-50-2**. Service loop **13.14-49** may be tightened and

loosened as desired to conform nosepiece **13.14-28** to the nose of a user. For example, as shown in FIGS. **13.14-7**, service loop **13.14-49** may be tightened to move nosepiece **13.14-28** to position **13.14-28'** by moving nosepiece **13.14-28** downward in direction **13.14-56**, right in direction **13.14-52**, and left in direction **13.14-54** toward to the nose of the user. In this way, service loop **13.14-49** may allow nosepiece **13.14-28** to fit more securely to the user's nose, preventing light from entering the user's eye boxes through gaps between nosepiece **13.14-28** and the nose.

[3317] Alternatively or additionally, nosepiece **13.14-28** may include foam **13.14-48**. Foam **13.14-48** may fill a gap between nosepiece **13.14-28** and the user's nose, and may be compressible to allow for a secure fit between nosepiece **13.14-28** and the nose. Foam **13.14-48** may directly contact the nose of the user (e.g., may be on a surface of a fabric layer, such as fabric **13.14-40**), as shown in FIGS. **13.14-7**, may be embedded within nosepiece **13.14-28** (e.g., covered by fabric, such as fabric **13.14-40**), or may otherwise be attached to nosepiece **13.14-28**.

[3318] Instead of service loop **13.14-49**, a deformable stiffener, such as deformable stiffener **13.14-58**, may be incorporated into nosepiece **13.14-28**. As shown in FIGS. **13.14-8**, deformable stiffener **13.14-58** may allow for nosepiece **13.14-28** to be adjusted in directions **13.14-52**, **13.14-54**, and **13.14-56** toward the nose of the user to conform nosepiece **13.14-28** to the user's nose at position **13.14-28'**. Deformable stiffener **13.14-58** may be formed within nosepiece **13.14-28** (i.e., may be covered by fabric), as shown in FIGS. **13.14-8**, or may be formed on a surface of the fabric (e.g., an inner or outer surface of fabric **13.14-40**). Although not shown in FIGS. **13.14-8**, a gap-filling foam, such as foam **13.14-48** of FIGS. **13.14-7**, may be incorporated in or on nosepiece **13.14-28** in addition to deformable stiffener **13.14-58**.

[3319] In some embodiments, to ensure that nosepiece **13.14-28** is tightly sealed to the nose of a user, internal components of device **13.14-10**, such as fans, may be used to move air toward nosepiece **13.14-28**, thereby sealing nosepiece **13.14-28** around the user's nose.

[3320] In addition to improving the fit of nosepiece **13.14-28** to prevent light from entering the eye boxes of the user, it may also be desirable to incorporate layers into nosepiece **13.14-28** that improve the comfort for the user. Examples of various modifications that may be made to nosepiece **13.14-28** to improve user comfort are shown in FIGS. **13.14-9A-13.14-9G**.

[3321] As shown in FIGS. **13.14-9A**, a nosepiece, which includes elastomer **13.14-34** and fabric **13.14-40** (such as nosepiece **13.14-28**) may have a rolled edge and may approach the user's nose at a shallow angle of contact to conform to a curved portion of nose **13.14-60**. The rolled edge may prevent elastomer **13.14-34** from digging into the user's skin on nose **13.14-60**, while the shallower angle of contact may allow the nosepiece to move up the user's nose rather than hitting the user's nose directly. Therefore, by having a rolled edge and/or a shallower angle of contact, nosepiece **13.14-28** may improve comfort for a user of device **13.14-10**.

[3322] Alternatively or additionally, a nosepiece may include foam, such as foam **13.14-62**, between some or all of elastomer **13.14-34** and fabric **13.14-40**, as shown in FIGS. **13.14-9B**. In particular, foam **13.14-62** may be at an edge portion of the nosepiece that contacts the user's nose, and therefore prevent elastomer **13.14-34** from hurting the user's nose or reducing discomfort as the nosepiece pushes against the nose.

[3323] In some examples, it may be desirable to have a series of openings in elastomer **13.14-34** that are either unfilled or filled with different material, such as foam. For example, in FIGS. **13.14-9C**, openings **13.14-64** may be provided in a portion of the nosepiece that will contact the user's nose. Openings **13.14-64** may be through openings, partial openings, may be filled with other material, such as foam, or may otherwise be more flexible than the surrounding regions of elastomer **13.14-34**. As the nosepiece pushes against the user's nose, openings **13.14-64** may crumple, thereby preventing or reducing discomfort to the user.

[3324] If desired, a portion of fabric may be extended from the portion that contacts the user's nose to provide an additional buffer between the nose and elastomer **13.14-34**. As shown in FIGS. **13.14-9D**, fabric portion **13.14-66** may extend from fabric **13.14-40**. When worn by a user, fabric

portion **13.14-66** may contact the nose of the user first, and therefore increase the amount of fabric between the nose and elastomer **13.14-34**, which may improve user comfort. In some examples, fabric portion **13.14-66** may be a hemmed edge of fabric **13.14-40**.

[3325] If desired, at least some portions of elastomer **13.14-34** that would otherwise contact the user's nose may be cut and replaced by more flexible material, such as foam. As shown in FIGS. **13.14-9E**, foam portions **13.14-68** may replace portions of elastomer **13.14-34** that would otherwise contact the user's nose. Although fabric **13.14-40** does not cover foam portions **13.14-68** in FIGS. **13.14-9E**, fabric **13.14-40** may overlap foam portions **13.14-68**, if desired.

[3326] In addition to, or instead of, the modifications of FIGS. **13.14-9A-13.14-9E**, in which an edge portion of a nosepiece is modified to prevent the elastomer from providing discomfort to a user's nose, a bottom portion of the nosepiece may be modified. As shown in FIGS. **13.14-9F**, segmented material **13.14-69** may be added to a bottom surface of nosepiece **13.14-28**. For example, segmented material **13.14-69** may contact the top of the user's nose (i.e., the nose bridge), and prevent nosepiece **13.14-28** from directly contacting the nose and causing discomfort. Material **13.14-69** may be foam or may be elastomer flaps.

[3327] Although FIGS. **13.14-9F** shows material **13.14-69** as being segmented and non-overlapping, material **13.14-69** may be segmented and overlapping, or may be one connected piece of material, if desired. For example, in FIGS. **13.14-9G**, foam **13.14-73** may be provided between nosepiece **13.14-28** and the user's nose while device **13.14-10** is worn. If desired, optional deformable stiffener **13.14-71** may be provided on the bottom surface of nosepiece **13.14-28** to improve the fit of nosepiece **13.14-28** on the user's nose. If stiffener **13.14-71** is included, foam **13.14-73** may also prevent stiffener **13.14-71** from directly contacting the user's nose and providing discomfort.

[3328] All of the examples in FIGS. **13.14-9A-13.14-9F** are merely illustrative examples of improving user comfort while ensuring that a nosepiece fits tightly to the user's nose. The examples are not limiting and may be implemented individually or together in any combination. Regardless of whether any structures in FIGS. **13.14-9A-13.14-9F** are used, it may be desirable to strengthen a portion of nosepiece **13.14-28**. Although the use of a structural frame was described in connection with FIGS. **13.14-4**, such a frame may add unnecessary bulk or add too much rigidity to nosepiece **13.14-28** in some cases. Therefore, a semi-rigid member may be used. An example of using a semi-rigid member on a nosepiece is shown in FIGS. **13.14-10**.

[3329] As shown in FIGS. **13.14-10**, a nosepiece, such as nosepiece **13.14-28**, may include portions **13.14-70** and **13.14-72**. Portion **13.14-70** may include an elastomer, such as elastomer **13.14-34**, and/or a fabric, such as fabric portion **13.14-72** may be coupled to portion **13.14-70** and may include a semi-rigid member. In particular, the semi-rigid member may be formed on a bottom portion of nosepiece **13.14-28** (i.e., toward the bottom of the user's nose when device **13.14-10** is worn), and may provide additional rigidity to nosepiece **13.14-28**. The semi-rigid members may be formed from any desired material, such as rubber or plastic. In this way, nosepiece **13.14-28** may remain sufficiently flexible to conform to the shape of a user's nose, while retaining its shape when after it has been conformed to the shape of the nose and the device is being worn by the user.

[3330] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

13.15: Removable Facial Interface

[3331] According to some aspects of the present disclosure, a head-mountable device (HMD) can include a display; a facial interface frame at least partially surrounding the display; a removable facial interface attached to the facial interface frame; a first attachment mechanism attached to one of the facial interface frame or the removable facial interface; and a second attachment mechanism attached to the other of the facial interface frame or the removable facial interface. In some examples, the removable facial interface can be attached to the facial interface frame by the first

attachment mechanism and the second attachment mechanism.

[3332] In some examples, the first attachment mechanism and the second attachment mechanism can include magnets. In some examples, the first attachment mechanism can include a spherical magnetic receptacle. The second attachment mechanism can include a spherical magnet. In some examples, the first attachment mechanism can include a capsule-dished magnetic receptacle. The second attachment mechanism can include a spherical magnet. In some examples, the first attachment mechanism can include a magnetic post. The second attachment mechanism can include a magnetic protrusion. The first attachment mechanism can be configured to slidably engage and disengage the second attachment mechanism.

[3333] In some examples, the first attachment mechanism and the second attachment mechanism can include hook-and-loop fasteners. In some examples, the first attachment mechanism can include a post and a flange on the post. The second attachment mechanism can include a spring-snap feature. The spring-snap feature can include an opening; a detent adjacent the opening; and a spring attached to the detent.

[3334] In some examples, the first attachment mechanism can include a receptacle and a first magnet in the receptacle. The second attachment mechanism can include a protrusion and a second magnet in the protrusion. The protrusion can have a first shape complementary to a second shape of the receptacle.

[3335] In some examples, the first attachment mechanism and the second attachment mechanism can include interlocking elastic fasteners. In some examples, the first attachment mechanism can include a suction cup. The second attachment mechanism can include a curved surface shaped to interface with the suction cup.

[3336] In some examples, the removable facial interface can be C-shaped. The facial interface frame can include a C-shaped base complementary to the removable facial interface.

[3337] In some examples, the removable facial interface can include a fabric material encircling one of the first attachment mechanism or the second attachment mechanism. The facial interface frame can include a compressible material encircling the other of the first attachment mechanism or the second attachment mechanism. The first attachment mechanism can be directly attached to the second attachment mechanism.

[3338] According to some aspects, a wearable electronic device can include a display; a facial interface frame physically coupled to the display; and a removable facial interface removably attached to the connector of the facial interface frame by a first attachment mechanism. The facial interface frame may include a frame partially surrounding the display; a connector attached to the frame; and a base attached to the connector. The connector may extend through the base.

[3339] In some examples, the first attachment mechanism can include a magnetic attachment mechanism. The magnetic attachment mechanism can include a post on one of the connector or the removable facial interface, and a receptacle on the other of connector or the removable facial interface.

[3340] In some examples, the first attachment mechanism can include a hook-and-loop attachment mechanism. The hook-and-loop attachment mechanism can include a post on one of the connector or the removable facial interface and a receptacle on the other of connector or the removable facial interface.

[3341] In some examples, the wearable electronic device can further include a compressible member attached to the base. The compressible member can encircle the connector and the connector can be exposed through the compressible member.

[3342] According to some aspects, a light seal for a head-mountable device can include a facial interface frame. The light seal can include a base; a compressible portion; and a magnet attached to the base.

[3343] In some examples, the light seal can further include a hook-and-loop fastener attached to the base. In some examples, the light seal can further include an interlocking fastener attached to the

base. In some examples, the light seal can further include a receptacle attached to the base.

[3344] The following disclosure relates to wearable electronic devices (e.g., head-mountable devices (HMDs)), including those that include a removable facial interface. Prolonged use of a head-mountable device can cause components of the head-mountable device to wear and to become dirty. In particular, components of the head-mountable device that contact a user's face are subject to wear and becoming dirty due to the contact with the user's face. It can be desirable to clean the portions of the head-mountable device that contact the user's face in order to remove oils, sweat, and the like. However, the head-mountable device can include sensitive components, small components, and oddly shaped components that are difficult to clean in-situ. Further, it can be desirable to replace portions of the head-mountable device that contact the user's face when those portions become worn out, in order to obtain a more comfortable fit for specific users, based on other user preferences, or the like.

[3345] A head-mountable device of the present disclosure includes a removable facial interface. As will be discussed in detail below, the removable facial interface can be attached to a facial interface frame through various attachment mechanisms, such as magnets, interlocking features, sliding features, hook-and-loop features (e.g., Velcro), spring snaps, suction features, bi-stable features, stretch features, re-usable adhesives, mating posts, combinations thereof, or the like. The removable facial interface can be easily removed from the facial interface frame, and can be re-installed on the facial interface frame in order to allow for the removable facial interface to be cleaned, replaced, or the like. Allowing the removable facial interface to be removed and cleaned increases the longevity of the removable facial interface. Replacing the removable facial interface improves the longevity of the head-mountable device, reduces user costs, provides users with choices of removable facial interfaces based on fit and personal preferences, and provides additional benefits.

[3346] These and other examples are discussed below with reference to FIGS. **13.15-1A** through **13.15-15C**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature including at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[3347] FIGS. **13.15-1A** illustrates a block diagram of a head-mountable device (HMD) **13.15-100** including a display **13.15-106**, a frame **13.15-104**, a device seal **13.15-116**, and a retention band **13.15-118**. The display **13.15-106** can include one or more optical lenses or display screens that are configured to be positioned in front of the eyes of a user. The display **13.15-106** can be configured to present an augmented reality visualization, a virtual reality visualization, or another suitable visualization to the user. The display **13.15-106** can be positioned at least partially in or on the frame **13.15-104**. The frame **13.15-104** can be a housing of the display **13.15-106**. The device seal **13.15-116** can be physically coupled to the frame **13.15-104**. The retention band **13.15-118** can be physically coupled to the device seal **13.15-116** and/or the frame **13.15-104** and can be used to secure the HMD **13.15-100** on the head of the user. In some examples, the device seal **13.15-116** includes the frame **13.15-104** (e.g., the frame **13.15-104** is part of the device seal **13.15-116**).

[3348] The device seal **13.15-116** includes a facial interface frame **13.15-112**, a removable facial interface **13.15-110**, a cover **13.15-114**, and electrical components (e.g., sensors **13.15-120**). The device seal **13.15-116** can also be referred to as a light seal. In some examples, the device seal **13.15-116** can refer to a portion of the HMD **13.15-100** that engages or shields a user's face. The device seal **13.15-116** can include portions of the HMD **13.15-100** that conform to, contact, or

press against regions of the user's face (e.g., the facial interface frame **13.15-112** and the removable facial interface **13.15-110**).

[3349] The removable facial interface **13.15-110** refers to a portion of the HMD **13.15-100** that directly contacts a user's face. The facial interface frame **13.15-112** refers to a portion of the HMD **13.15-100** to which the removable facial interface **13.15-110** is attached, and which is physically coupled between the removable facial interface **13.15-110** and the frame **13.15-104**. The removable facial interface **13.15-110** and/or the facial interface frame **13.15-112** can also be referred to as a face track. The removable facial interface **13.15-110** and the facial interface frame **13.15-112** can conform to (e.g., compress against and assume the shape of) regions of the user's face. In some examples, the removable facial interface **13.15-110** and the facial interface frame **13.15-112** include pliant (or semi-pliant) materials that span the forehead, wrap partially around the eyes, and contact the zygoma and maxilla regions of the user's face.

[3350] In some examples, the facial interface frame **13.15-112** can be formed of a relatively stiff or rigid material, while the removable facial interface **13.15-110** can be formed of a relatively soft, deformable, elastic material. In some examples, the facial interface frame **13.15-112** can be formed from plastics, metals, polymers, combinations thereof, or the like. The removable facial interface **13.15-110** can be formed from plastics, metals, polymers, fabrics, foams, rubbers, silicone, elastomers, hydrogels, combinations thereof, or the like. In some examples, the removable facial interface **13.15-110** can include a stiffener, which can be formed from metals, plastics, polymers, combinations thereof, or the like. The removable facial interface **13.15-110** can include a pliable material surrounding the stiffener, which can be formed from rubbers, foams, polymers, silicone, elastomers, hydrogen gels, combinations thereof, or the like. The pliable material can be formed around the stiffener through overmolding or the like. The removable facial interface **13.15-110** can further include a fabric material surrounding the pliable material. The fabric material can be wrapped around the pliable material. The stiffener can provide the removable facial interface **13.15-110** with a desired degree of stiffness. The pliable material can be pliant or semi-pliant and can be provided to increase user comfort. The fabric material can be configured to contact a user's skin and can also be provided to increase user comfort.

[3351] The removable facial interface **13.15-110** can be attached to the facial interface frame **13.15-112** through various attachment mechanisms. For example, the attachment mechanisms can include magnets, interlocking features, sliding features, hook-and-loop features (e.g., Velcro), spring snaps, suction features, bi-stable features, stretch features, re-usable adhesives, mating posts, combinations thereof, or the like. The attachment mechanisms can be provided on both the removable facial interface **13.15-110** and the facial interface frame **13.15-112**. For example, a male attachment mechanism can be provided on the removable facial interface **13.15-110** and a female attachment mechanism can be provided on the facial interface frame **13.15-112**, or vice versa. The attachment mechanisms can be self-aligning or can entail alignment by a user. The attachment mechanisms can be configured such that the removable facial interface **13.15-110** is removed from, and attached to, the facial interface frame **13.15-112** in one-handed operations. Each of the respective attachment mechanisms can datum the removable facial interface **13.15-110** to the facial interface frame **13.15-112**, or allow the removable facial interface **13.15-110** to float or move relative to the facial interface frame **13.15-112**. The respective attachment mechanisms can have variable shapes and/or sizes around the perimeters of the removable facial interface **13.15-110** and the facial interface frame **13.15-112**, which can improve the ability to interchange various removable facial interfaces **13.15-110** and facial interface frames **13.15-112**. More specifically, the variable shapes and/or sizes of the attachment mechanisms around the perimeters of the removable facial interface **13.15-110** and the facial interface frame **13.15-112** can allow for different shaped removable facial interfaces **13.15-110** and facial interface frames **13.15-112** to be connected together.

[3352] Providing the removable facial interface **13.15-110** and the facial interface frame **13.15-112**

allows for the removable facial interface **13.15-110** to be easily removed and cleaned. The removable facial interface **13.15-110** can be replaced based on wear, user preferences, or the like. For example, specific removable facial interfaces **13.15-110** can be provided based on different facial structures of various users. Removable facial interfaces **13.15-110** can be provided based on different intended user uses, such as a specific removable facial interface **13.15-110** designed for sport use. Providing the removable facial interface **13.15-110** allows for a portion of the HMD **13.15-100** that is subject to the greatest amount of wear to be cheaply and easily replaced.

[3353] The cover **13.15-114** can include a seal, such as an environment seal, a dust seal, an air seal, a light seal, or the like. The cover **13.15-114** can be positioned in a gap between the display **13.15-106** and the user's face. The cover **13.15-114** can form an eye-box through which the user can view the display **13.15-106**. It will be appreciated that the term “seal” can include partial seals or inhibitors, in addition to complete seals. For example, a seal can be a partial light seal, where some ambient light is blocked. In some examples, a seal can be a complete light seal where all ambient light is blocked when the HMD **13.15-100**.

[3354] The cover **13.15-114** can be a woven fabric that is non-rigid or deformable. The cover **13.15-114** can be elastically deformable. In some examples, the cover **13.15-114** can be formed from a plastic, a rubber, a polymer material, or the like. The cover **13.15-114** can be a cosmetic textile material with stretch and can feature an open mesh pattern. In some examples, the cover **13.15-114** can be rigid. Providing the removable facial interface **13.15-110** separate from the cover **13.15-114** allows for the cover **13.15-114** and a material covering the removable facial interface **13.15-110** to be formed from different materials, such as the cover **13.15-114** being formed from a cosmetic material and the removable facial interface **13.15-110** being formed from a comfortable material. In some examples, the cover **13.15-114** and the covering of the removable facial interface **13.15-110** can be formed from the same materials.

[3355] The device seal **13.15-116** can be removably attached to the frame **13.15-104**. The device seal **13.15-116** can be electrically coupled to the display **13.15-106**. The device seal **13.15-116** can include various electrical components, such as the sensors **13.15-120**. The sensors **13.15-120** can include various sensors, such as sensors that collect user data, or environmental data. In some examples, the sensors **13.15-120** can collect biometric information. The sensors **13.15-120** can transmit signals to other components of the HMD **13.15-100**, such as the display **13.15-106**. The sensors **13.15-120** can transmit signals to various outputs, such as an output configured to perform an action in response to information collected by the sensors **13.15-120**.

[3356] FIGS. **13.15-1B** illustrates a top view of an HMD **13.15-100**. The HMD **13.15-100** of FIGS. **13.15-1B** can be substantially similar to, including some or all of the features of, the HMD **13.15-100** described with respect to FIGS. **13.15-1A**. The HMD **13.15-100** includes a display (also referred to as a display unit) **13.15-106** and a retention band **13.15-118**. In some examples, the display **13.15-106** includes an opaque, translucent, transparent, or semi-transparent screen, including any number lenses, for presenting visual data to a user. The display **13.15-106** can include any number of internal electronic components **13.15-108**. The HMD **13.15-100** can be mounted on a user's head **13.15-102** using the retention band **13.15-118**.

[3357] The HMD **13.15-100** further includes a frame **13.15-104** (also referred to as a housing), a facial interface frame **13.15-112**, a removable facial interface **13.15-110**, and a cover **13.15-114**. The frame **13.15-104** can be physically coupled to the display **13.15-106**. The frame **13.15-104** can at least partially border one or more edges of the display **13.15-106**. One end of the cover **13.15-114** can be attached to the frame **13.15-104** and an opposite end of the cover **13.15-114** can be attached to the facial interface frame **13.15-112**. The facial interface frame **13.15-112** and the removable facial interface **13.15-110** provide an interface between a user's head **13.15-102** and the frame **13.15-104**. The combination of the frame **13.15-104**, the facial interface frame **13.15-112**, the removable facial interface **13.15-110**, and the cover **13.15-114** can form a device seal **13.15-116**. It will be understood, however, that the device seal **13.15-116** can include fewer or additional

components from those listed or shown.

[3358] The HMD **13.15-100** can be worn on the user's head **13.15-102** such that the display **13.15-106** is positioned on the user's face and disposed in front of one or both of the user's eyes. The display **13.15-106** can be physically coupled to the retention band **13.15-118** and/or the device seal **13.15-116**. In some examples, the retention band **13.15-118** can be positioned against sides of the user's head **13.15-102** and in contact therewith. In some examples, the retention band **13.15-118** can be at least partially positioned above the user's ear or ears. In some examples, the retention band **13.15-118** can be positioned adjacent to the user's ear or ears. The retention band **13.15-118** can extend around the user's head **13.15-102**. In this way, the display **13.15-106** and the retention band **13.15-118** can form a loop configured to retain the HMD **13.15-100** on the user's head **13.15-102**. It should be understood, however, that this configuration is just one example of how the components of the HMD **13.15-100** can be arranged. In some examples, a different number of connector straps and/or retention bands can be included. Although the HMD **13.15-100** is referred to as an HMD, it should be understood that the terms wearable device, wearable electronic device, HMD device, and/or HMD system can be used to refer to any wearable device, including smart glasses.

[3359] In some examples, the frame **13.15-104** is physically coupled to the facial interface frame **13.15-112**. The removable facial interface **13.15-110** is removably attached to the facial interface frame **13.15-112**. The removable facial interface **13.15-110** can contact the user's head **13.15-102**, such as a user's face. In some examples, the cover **13.15-114** can be a light blocking component that extends between the frame **13.15-104** and the removable facial interface **13.15-110**, such as along portions of the facial interface frame **13.15-112**. The cover **13.15-114** can cover or surround a perimeter of the frame **13.15-104** and/or the facial interface frame **13.15-112**.

[3360] The cover **13.15-114** can be formed from a cloth, fabric, woven material, plastic, rubber, or any other suitable opaque or semi-opaque material. In some examples, the cover **13.15-114** is flexible, having the ability to repeatedly stretch, compress, and deform. The cover **13.15-114** can be elastically or in-elastically deformable. The facial interface frame **13.15-112**, the cover **13.15-114**, and the removable facial interface **13.15-110** can be configured to block outside light and limit the peripheral view of the user. In some examples, the cover **13.15-114** and the facial interface frame **13.15-112** are part of the same or a unitary component.

[3361] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-1A** and **13.15-1B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-1A** and **13.15-1B**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-1A** and **13.15-1B**.

[3362] FIGS. **13.15-2A** illustrates a perspective view of a device seal **13.15-200**. The device seal **13.15-200** can be substantially similar to, including some or all of the features of, the device seals described herein, such as the device seal **13.15-116**. The device seal **13.15-200** can be implemented on an HMD, such as the HMD **13.15-100** described in reference to FIGS. **13.15-1A** and **13.15-1B**. The device seal **13.15-200** can include a removable facial interface **13.15-202**, a facial interface frame **13.15-204**, and a cover **13.15-206**. The removable facial interface **13.15-202**, the facial interface frame **13.15-204**, and the cover **13.15-206** can be substantially similar to, including some or all of the features of, the removable facial interface **13.15-110**, the facial interface frame **13.15-112**, and the cover **13.15-114**, respectively. The cover **13.15-206** can be included in the device seal **13.15-200** to provide a seal between the device seal **13.15-200** and the outer environment.

[3363] The facial interface frame **13.15-204** can include a frame **13.15-210**, connectors **13.15-212**, a base **13.15-214**, and a compressible portion **13.15-216**. The frame **13.15-210** can be physically

coupled to a display of the HMD, and the frame **13.15-210** can be configured provide an interface between the display and the facial interface frame **13.15-204**. The removable facial interface **13.15-202** can be removably attached to the base **13.15-214**, and the base **13.15-214** can be configured provide an interface between the removable facial interface **13.15-202** and the facial interface frame **13.15-204**. The connectors **13.15-212** can be configured to provide a desired distance between the display and a user's eyes when the HMD is donned by the user. The connectors **13.15-212** and/or the base **13.15-214** can include attachment mechanisms that are used to attach the removable facial interface **13.15-202** to the facial interface frame **13.15-204**.

[3364] The compressible portion **13.15-216** can be included in the facial interface frame **13.15-204** to fill a gap between the base **13.15-214** and the removable facial interface **13.15-202**, and to provide improved user comfort. The compressible portion **13.15-216** can also be referred to as a compressible member or a compressible material. In some examples, the connectors **13.15-212** (e.g., an attachment mechanism) can extend through the compressible portion **13.15-216**. The compressible portion **13.15-216** can encircle portions of the connectors **13.15-212** that extend through the compressible portion **13.15-216**. The connectors **13.15-212** can also extend through the base **13.15-214**, and attachment mechanisms provided on the connectors **13.15-212** can be exposed through both the compressible portion **13.15-216** and the base **13.15-214** for access and connection. In some examples, the compressible portion **13.15-216** can include a foam material. In some examples, the compressible portion **13.15-216** can include materials with properties that impart flexibility, softness, compressibility, deformability, and the like. Examples of materials that can be used for the compressible portion **13.15-216** include silicone, polymers, elastomers, hydrogels, combinations thereof, or the like. The compressible portion **13.15-216** can be formed by molding, printing, casting, or the like. In some examples, attachment mechanisms, such as perimeter attachments, can be included on the compressible portion **13.15-216**. The compressible portion **13.15-216** provides a soft, flexible, deformable interface between the removable facial interface **13.15-202** and rigid components of the removable facial interface **13.15-202**, such as the frame **13.15-210**, the connectors **13.15-212**, and the base **13.15-214**. This interface provides for improved user comfort when the HMD is donned.

[3365] The removable facial interface **13.15-202** is a removable component that is configured to contact a user's face. The removable facial interface **13.15-202** can be formed of conformable and comfortable materials. The removable facial interface **13.15-202** is configured to be easily removed from and attached to the facial interface frame **13.15-204**. This allows for the removable facial interface **13.15-202** to be removed, cleaned, replaced, and the like. In some examples, the attachment mechanisms used to attach the removable facial interface **13.15-202** to the facial interface frame **13.15-204** can be simple mechanisms that allow the removable facial interface **13.15-202** to be removed from and attached to the facial interface frame **13.15-204** in a one-handed manner. Alignment features can be provided on the removable facial interface **13.15-202** and/or the facial interface frame **13.15-204** to aid in the alignment of the removable facial interface **13.15-202** with respect to the facial interface frame **13.15-204**. Providing the removable facial interface **13.15-202** as a removable component allows the removable facial interface **13.15-202** to be easily cleaned, replaced as a result of wear, replaced due to user preferences, and the like. This reduces maintenance costs for HMDs including the removable facial interface **13.15-202**.

[3366] The removable facial interface **13.15-202** can be formed of a relatively pliant material, while the components of the facial interface frame **13.15-204** are formed of relatively rigid materials. This allows the removable facial interface **13.15-202** to conform to the shape of the facial interface frame **13.15-204** and the shape of a user's face or head when the user dons the HMD. As illustrated in FIGS. **13.15-2A**, the base **13.15-214**, the compressible portion **13.15-216**, and the removable facial interface **13.15-202** can be C-shaped and can have shapes corresponding to one another. In other words, the removable facial interface **13.15-202** is C-shaped, and the base **13.15-214** and the compressible portion **13.15-216** are C-shaped and complementary to the C-

shaped removable facial interface **13.15-202**. This provides a visual cue to users when the removable facial interface **13.15-202** is removed from the base **13.15-214** that the removable facial interface **13.15-202** is intended to be attached to the base **13.15-214**. In some examples, attachment mechanisms on the removable facial interface **13.15-202** and the facial interface frame **13.15-204** can be visible when the removable facial interface **13.15-202** is removed from the facial interface frame **13.15-204**, which provides further visual cues as to how the removable facial interface **13.15-202** is intended to be attached to the facial interface frame **13.15-204**. Moreover, upon seeing the attachment mechanisms on the facial interface frame **13.15-204**, a user can realize that something is missing from the facial interface frame **13.15-204** when the removable facial interface **13.15-202** is removed from the facial interface frame **13.15-204**.

[3367] FIGS. **13.15-2B** illustrates a perspective view of a facial interface frame **13.15-204**. The facial interface frame **13.15-204** can include a frame **13.15-210**, connectors **13.15-212**, and a base **13.15-214**. The frame **13.15-210**, the connectors **13.15-212**, and the base **13.15-214** of the facial interface frame **13.15-204** can be formed of relatively rigid materials, such as plastics, metals, polymers, combinations thereof, or the like. In some examples, the base **13.15-214** can be formed of a relatively flexible material, such that the base **13.15-214** is allowed to flex when an HMD including the facial interface frame **13.15-204** is donned by a user. In some examples, the frame **13.15-210**, the connectors **13.15-212**, and/or the base **13.15-214** can include rigid materials that can be coated in relatively soft materials, such as polymers, fabrics, foams, rubbers, silicone, elastomers, hydrogels, combinations thereof, or the like to increase user comfort.

[3368] The frame **13.15-210** can provide an interface between a display of the HMD and the facial interface frame **13.15-204**. For example, the frame **13.15-210** can be physically coupled to the display of the HMD and can hold the display in a desired position. The connectors **13.15-212** can control the spacing between the display and a user's face. For example, depending on a length of the connectors **13.15-212**, the display can be disposed closer to or further from a user's face. The base **13.15-214** and/or the connectors **13.15-212** can provide an interface between the facial interface frame **13.15-204** and a removable facial interface. For example, attachment mechanisms can be included on the base **13.15-214** and/or the connectors **13.15-212** and can be used to removably attach the removable facial interface to the base **13.15-214** and/or the connectors **13.15-212**.

[3369] In some examples, post attachments can be provided on the connectors **13.15-212** and perimeter attachments can be provided on the base **13.15-214**. As will be discussed in detail below, the post attachments and corresponding attachment mechanisms on the removable facial interface can be configured to remain fixed relative to one another or have a limited amount of travel when the removable facial interface is mounted to the facial interface frame **13.15-204**. The perimeter attachments and corresponding attachment mechanisms on the removable facial interface can be configured to slide relative to one another when the removable facial interface is mounted to the facial interface frame **13.15-204**. Including both the post attachments and the perimeter attachments allows the removable facial interface to be securely mounted on the facial interface frame **13.15-204**, while allowing the removable facial interface to conform comfortably to a user's face.

[3370] FIGS. **13.15-2C** illustrates a perspective view of portions of a removable facial interface **13.15-202** and a facial interface frame **13.15-204**. The removable facial interface **13.15-202** can include a body portion **13.15-220**, a side mount **13.15-222**, a top mount **13.15-224**, and a mounting area **13.15-226**. The body portion **13.15-220** can be formed of materials such as plastics, metals, polymers, fabrics, foams, rubbers, silicone, elastomers, hydrogels, combinations thereof, or the like. The facial interface frame **13.15-204** can include a frame **13.15-210**, connectors **13.15-212**, and a base **13.15-214**.

[3371] The side mount **13.15-222**, the top mount **13.15-224**, and the mounting area **13.15-226** are examples of attachment mechanisms that can be included in the removable facial interface **13.15-202**. The side mount **13.15-222**, the top mount **13.15-224**, and the mounting area **13.15-226** can be

used to removably attach the removable facial interface **13.15-202** to the facial interface frame **13.15-204**. In some examples, post attachments can be provided on the connectors **13.15-212** corresponding to the side mount **13.15-222** and the top mount **13.15-224**. Perimeter attachments can be provided on the base **13.15-214** corresponding to the mounting area **13.15-226**. Specifically, when the removable facial interface **13.15-202** is aligned with and mounted to the facial interface frame **13.15-204**, the side mount **13.15-222** and the top mount **13.15-224** are aligned with corresponding post attachments on the connectors **13.15-212** and the mounting area **13.15-226** is aligned with corresponding perimeter attachments on the base **13.15-214**.

[3372] The side mount **13.15-222**, the top mount **13.15-224**, and corresponding post attachments of the connectors **13.15-212** can include attachment mechanisms that are configured to remain fixed relative to one another, or to have a relatively small travel relative to one another (e.g., the attachment mechanisms can float relative to one another). In some examples, the side mount **13.15-222** and a corresponding post attachment can be configured to remain fixed relative to one another and the top mount **13.15-224** and a corresponding post attachment can be configured to have some travel relative to one another. In some examples, the side mount **13.15-222**, the top mount **13.15-224**, and corresponding post attachments can include magnets, interlocking features, sliding features, hook-and-loop features (e.g., Velcro), spring snaps, suction features, bi-stable features, stretch features, re-usable adhesives, mating posts, combinations thereof, or the like. Posts and corresponding holes or other self-aligning features can be included in the side mount **13.15-222**, the top mount **13.15-224**, and the post attachments in order to align the removable facial interface **13.15-202** with the facial interface frame **13.15-204**. Male features can be provided in the side mount **13.15-222** and the top mount **13.15-224** and corresponding female features can be provided in the post attachments of the facial interface frame **13.15-204**, or vice versa. By using the prescribed attachment mechanisms and fixing the positions of the side mount **13.15-222** and the top mount **13.15-224** relative to the post attachments, the removable facial interface **13.15-202** can be aligned with the facial interface frame **13.15-204**. Moreover, the relative positions of the removable facial interface **13.15-202** and the facial interface frame **13.15-204** can be fixed to one another during use of an HMD by a user. This prevents misalignment of the removable facial interface **13.15-202** and the facial interface frame **13.15-204** from negatively impacting a user's experience when using the HMD.

[3373] The mounting area **13.15-226** and the corresponding perimeter attachments of the facial interface frame **13.15-204** can include attachment mechanisms that are configured to slide relative to one another. For example, the mounting area **13.15-226** and the corresponding perimeter attachments can include longitudinal interlocking features, rows of interlocking features, rows of magnets, lengths of flexible magnets, lengths of hook-and-loop fasteners, combinations thereof, or the like. Male features can be provided in the mounting area **13.15-226** and corresponding female features can be provided in the perimeter attachments of the facial interface frame **13.15-204**, or vice versa. In some examples, the mounting area **13.15-226** and the corresponding perimeter attachments are disposed in curved or corner regions of the removable facial interface **13.15-202** and the facial interface frame **13.15-204**, respectively. The mounting area **13.15-226** and the corresponding perimeter attachments can be disposed in areas of the removable facial interface **13.15-202** and the facial interface frame **13.15-204** that move or bend when the removable facial interface **13.15-202** is mounted on the facial interface frame **13.15-204** and when an HMD including the removable facial interface **13.15-202** and the facial interface frame **13.15-204** is donned by a user. This causes an arc length of the mounting area **13.15-226** and the corresponding perimeter attachments to change. By using the prescribed attachment mechanisms, the mounting area **13.15-226** and the corresponding perimeter attachments are allowed to slide relative to one another, and the removable facial interface **13.15-202** is securely fastened to the facial interface frame **13.15-204** even when the removable facial interface **13.15-202** moves relative to the facial interface frame **13.15-204**.

[3374] In the example of FIGS. **13.15-2C**, a single top mount **13.15-224** and a single side mount **13.15-222** are illustrated. The top mount **13.15-224** can be configured to be disposed adjacent a user's forehead when the HMD is donned and the side mount **13.15-222** can be configured to be disposed adjacent the user's cheek bones when the HMD is donned. However, any number of top mounts **13.15-224** and side mounts **13.15-222** can be included in the removable facial interface **13.15-202**, and the top mounts **13.15-224** and the side mounts **13.15-222** can be configured to be disposed adjacent any portions of the user's face or head when the HMD is donned. Moreover, although the mounting area **13.15-226** is illustrated as being in a corner area of the removable facial interface **13.15-202** between the top mount **13.15-224** and the side mount **13.15-222**, the mounting area **13.15-226** can be disposed in any area along the perimeter of the removable facial interface **13.15-202** and any number or length of mounting areas **13.15-226** can be included.

[3375] FIGS. **13.15-2D** illustrates a cross-sectional view of a facial interface shim **13.15-219**. The facial interface shim **13.15-219** can be used in conjunction with connectors of an HMD in order to control the spacing between a display of the HMD and a user's face. In some examples, the facial interface shim **13.15-219** can be disposed between a removable facial interface and a connector of a facial interface frame. For example, the facial interface shim **13.15-219** can be disposed in a mount opening **13.15-218** of the removable facial interface to which the connector of the facial interface frame is attached. In some examples, the facial interface shim **13.15-219** can be disposed between a base of a facial interface frame and a connector of the facial interface frame. For example, the connector can pass through the mount opening **13.15-218**, and the facial interface shim **13.15-219** can be disposed between the connector and the base. The facial interface shim **13.15-219** can have a thickness in a range from about 2 mm to about 4 mm. The facial interface shim **13.15-219** allows for distances between a display of an HMD and a user's face to be altered while standard length connectors are used, which decreases manufacturing costs.

[3376] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-2A** through **13.15-2D** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-2A** through **13.15-2D**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-2A** through **13.15-2D**.

[3377] FIGS. **13.15-3A** and **13.15-3B** illustrate a perspective view of a device seal **13.15-200** and a plan view of a removable facial interface **13.15-202**, respectively. The device seal **13.15-200** includes the removable facial interface **13.15-202**, a facial interface frame **13.15-204**, and a cover **13.15-206**. The device seal **13.15-200** can be substantially similar to, including some or all of the features of, the device seals described herein, such as the device seals **13.15-116** and **13.15-200**. The device seal **13.15-200** can be implemented on an HMD, such as the HMD **13.15-100** described in reference to FIGS. **13.15-1A** and **13.15-1B**.

[3378] FIGS. **13.15-3A** and **13.15-3B** illustrate an example in which the attachment mechanisms used to attach the removable facial interface **13.15-202** to the facial interface frame **13.15-204** include magnets. In some examples, six connectors **13.15-212** can be provided between a base **13.15-214** and a frame **13.15-210** of the facial interface frame **13.15-204**. Four circular receptacles **13.15-250** can be provided corresponding to four of the connectors **13.15-212**. The connectors **13.15-212** can extend through the base **13.15-214**, such that the attachment mechanisms provided on the connectors **13.15-212** are exposed through the base **13.15-214**. The circular receptacles **13.15-250** can be disposed along portions of the facial interface frame **13.15-204** configured to be disposed adjacent to a user's cheek bones when an HMD including the device seal **13.15-200** is donned. A spherically-dished magnet **13.15-252** can be disposed in each of the circular receptacles **13.15-250**. Each of the circular receptacles **13.15-250** in combination with a corresponding

spherically-dished magnet **13.15-252** may be referred to as a spherical magnetic receptacle. Two stadium or lozenge-shaped receptacles **13.15-254** can be provided corresponding to two of the connectors **13.15-212**. The stadium or lozenge-shaped receptacles **13.15-254** can be disposed along portions of the facial interface frame **13.15-204** configured to be disposed adjacent to a user's forehead when the HMD is donned. A capsule-dished magnet **13.15-256** can be disposed in each of the stadium or lozenge-shaped receptacles **13.15-254**. Each of the stadium or lozenge-shaped receptacles **13.15-254** in combination with a corresponding capsule-dished magnet **13.15-256** may be referred to as a capsule-dished magnetic receptacle.

[3379] In the example of FIGS. **13.15-3B**, four circular receptacles **13.15-260** can be provided in the replaceable facial interface **13.15-202**. The circular receptacles **13.15-260** can be disposed along portions of the replaceable facial interface **13.15-202** configured to be disposed adjacent to a user's cheek bones when the HMD is donned. A spherical magnet **13.15-262** can be disposed in each of the circular receptacles **13.15-260**. Two stadium or lozenge-shaped receptacles **13.15-264** can be provided in the replaceable facial interface **13.15-202**. The stadium or lozenge-shaped receptacles **13.15-264** can be disposed along portions of the replaceable facial interface **13.15-202** configured to be disposed adjacent to a user's forehead when the HMD is donned. A spherical magnet **13.15-266** can be disposed in each of the stadium or lozenge-shaped receptacles **13.15-264**. Although the receptacles **13.15-250**, **13.15-254**, **13.15-260**, and **13.15-264** and the magnets **13.15-252**, **13.15-256**, **13.15-262**, and **13.15-266** are described as being various rounded shapes, any suitable shapes can be used.

[3380] The spherical magnets **13.15-262** can be configured to interface with the spherically-dished magnets **13.15-252**, and the spherical magnets **13.15-266** can be configured to interface with the capsule-dished magnets **13.15-256**. The spherical magnets **13.15-262** and the spherically-dished magnets **13.15-252** can have corresponding shapes, which provides for a small amount of relative travel between the spherical magnets **13.15-262** and the spherically-dished magnets **13.15-252** in the x direction and the y direction illustrated in FIGS. **13.15-3A** and **13.15-3B**. The spherical magnets **13.15-266** and the capsule-dished magnets **13.15-256** can have corresponding shapes, which provides for a small amount of relative travel between the spherical magnets **13.15-266** and the capsule-dished magnets **13.15-256** in the y direction illustrated in FIGS. **13.15-3A** and **13.15-3B**. The capsule-dished magnets **13.15-256** have a longitudinal axis extending in the x direction of FIGS. **13.15-3A**. This provides for a greater amount of relative travel between the spherical magnets **13.15-266** and the capsule-dished magnets **13.15-256** in the x direction illustrated in FIGS. **13.15-3A** and **13.15-3B**. The distances and directions that each of the magnets **13.15-252**, **13.15-256**, **13.15-262**, and **13.15-266** are allowed to travel relative to one another can be controlled by appropriately selecting the sizes and shapes of each of the magnets **13.15-252**, **13.15-256**, **13.15-262**, and **13.15-266**.

[3381] In the example illustrated in FIGS. **13.15-3A** and **13.15-3B**, the receptacles **13.15-250** and **13.15-254** and the magnets **13.15-252** and **13.15-256** in the facial interface frame **13.15-204** and the receptacles **13.15-260** and **13.15-264** and the magnets **13.15-262** and **13.15-266** in the removable facial interface **13.15-202** are visible to a user. In addition, the base **13.15-214** of the facial interface frame **13.15-204** and the removable facial interface **13.15-202** are C-shaped and have shapes corresponding to one another. This provides visual cues to users that the removable facial interface **13.15-202** is meant to mount to the facial interface frame **13.15-204**. Moreover, if the removable facial interface **13.15-202** is not attached to the facial interface frame **13.15-204**, the facial interface frame **13.15-204** looks as though a piece is missing. This encourages a user to install the removable facial interface **13.15-202** onto the facial interface frame **13.15-204** so that the HMD can be comfortably donned. In some examples, any of the receptacles **13.15-250**, **13.15-254**, **13.15-260**, and **13.15-264** and/or the magnets **13.15-252**, **13.15-256**, **13.15-262**, and **13.15-266** can be covered or otherwise hidden from view.

[3382] The magnets **13.15-252**, **13.15-262** can datum relative to one another (e.g., maintain

relatively fixed positions relative to one another), and the magnets **13.15-256**, **13.15-266** can float relative to one another (e.g., move relative to one another). Although the magnets **13.15-252**, **13.15-256**, **13.15-262**, and **13.15-266** are illustrated in particular positions in the example of FIGS. **13.15-3A** and **13.15-3B**, any of the magnets **13.15-252**, **13.15-256**, **13.15-262**, and **13.15-266** can be positioned in any desired positions relative to the removable facial interface **13.15-202** and the facial interface frame **13.15-204**, depending on areas of the removable facial interface **13.15-202** and the facial interface frame **13.15-204** that are desired to remain fixed or float. Moreover, the magnets **13.15-252**, **13.15-256**, **13.15-262**, and **13.15-266** and the receptacles **13.15-250**, **13.15-254**, **13.15-260**, and **13.15-264** can have varying shapes and sizes, which can improve the ability to interchange various removable facial interfaces **13.15-202** and facial interface frames **13.15-204**, and can aid in the alignment of the removable facial interface **13.15-202** relative to the facial interface frame **13.15-204**. In some examples, the variable shapes and/or sizes of the magnets **13.15-252**, **13.15-256**, **13.15-262**, and **13.15-266** and the receptacles **13.15-250**, **13.15-254**, **13.15-260**, and **13.15-264** around the perimeters of the removable facial interface **13.15-202** and the facial interface frame **13.15-204** can allow for different shaped removable facial interfaces **13.15-202** and facial interface frames **13.15-204** to be connected together.

[3383] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-3A** and **13.15-3B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-3A** and **13.15-3B**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-3A** and **13.15-3B**.

[3384] FIGS. **13.15-4** illustrates a cross-sectional view of a magnetic attachment mechanism **13.15-300**, which is an example of a post mount. The magnetic attachment mechanism **13.15-300** includes a connector **13.15-301**, which can be magnetically attached to a base **13.15-310**. The connector **13.15-301** can be substantially similar to, including some or all of the features of, the connectors described herein, such as the connectors **13.15-212**. The base **13.15-310** can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces **13.15-202**. The connector **13.15-301** can be physically coupled to a receptacle **13.15-302**, which can be physically coupled to a magnet **13.15-304**. The base **13.15-310** can be physically coupled to a protrusion **13.15-312**, which can surround or otherwise house a magnet **13.15-316**.

[3385] As illustrated in FIGS. **13.15-4**, the protrusion **13.15-312** and the receptacle **13.15-302** can have shapes that are configured to correspond to one another. For example, the receptacle **13.15-302** can include a squared opening, and the protrusion **13.15-312** can have a squared shape configured to fit into the squared opening of the receptacle **13.15-302**. The protrusion **13.15-312** can be an alignment post. Providing the protrusion **13.15-312** and the receptacle **13.15-302** with corresponding shapes aids in self-aligning a removable facial interface with a facial interface frame when the removable facial interface is attached to the facial interface frame. The magnet **13.15-304** and the magnet **13.15-316** are then attracted to one another and retain the protrusion **13.15-312** in the receptacle **13.15-302**.

[3386] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-4**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-4**.

[3387] FIGS. **13.15-5A** illustrates a cross-sectional view of an interlocking attachment mechanism **13.15-400**, which is an example of a post mount. The interlocking attachment mechanism **13.15-400** can be referred to as an interlocking fastener. The interlocking attachment mechanism **13.15-400** includes a connector **13.15-401**, to which a base **13.15-410** can be attached. The connector **13.15-401** can be substantially similar to, including some or all of the features of, the connectors described herein, such as the connectors **13.15-212**. The base **13.15-410** can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces **13.15-202**. The connector **13.15-401** can be physically coupled to a receptacle **13.15-402**, which can include one or more openings **13.15-404**. In the cross-sectional view illustrated in FIGS. **13.15-5A**, the receptacle **13.15-402** includes three openings **13.15-404**; however, the receptacle **13.15-402** of

[3388] FIGS. **13.15-5A** can include two openings **13.15-404** that are circular- or annular-shaped in a bottom-up view. Any number of openings **13.15-404** can be included in the receptacle **13.15-402**. The base **13.15-410** can be physically coupled to protrusions **13.15-412**. The protrusions **13.15-412** can have shapes corresponding to the openings **13.15-404**, such as having circular- or annular-shapes in a top-down view. In some examples, the openings **13.15-404** and the protrusions **13.15-412** can have triangular shapes, rectangular shapes, or any other suitable shapes. Providing the openings **13.15-404** and the protrusions **13.15-412** with corresponding shapes aids in self-aligning the base **13.15-410** with the connector **13.15-401**.

[3389] The protrusions **13.15-412** can be configured to interface with the openings **13.15-404** in the receptacle **13.15-402** to attach the base **13.15-410** to the connector **13.15-401** when the base **13.15-410** and the connector **13.15-401** are pressed together. The protrusions **13.15-412** and the receptacle **13.15-402** can be formed of elastic, flexible materials with high coefficients of friction. In examples in which the protrusions **13.15-412** and the receptacle **13.15-402** are formed from elastic materials, the interlocking attachment mechanism **13.15-400** may be referred to as an interlocking elastic fastener. Forming the protrusions **13.15-412** and the receptacle **13.15-402** from flexible materials allows the protrusions **13.15-412** to be inserted into the openings **13.15-404**. Forming the protrusions **13.15-412** and the receptacle **13.15-402** from elastic materials having high coefficients of friction helps to retain the protrusions **13.15-412** in the openings **13.15-404** once the protrusions **13.15-412** are inserted into the openings **13.15-404**. In some examples, the protrusions **13.15-412** and the receptacle **13.15-402** can be formed of polymers, elastomers, plastics, combinations thereof, or the like. Providing the protrusions **13.15-412** and the openings **13.15-404** in the receptacle **13.15-402** with corresponding shapes aids in self-aligning a removable facial interface with a facial interface frame when the removable facial interface is attached to the facial interface frame.

[3390] FIGS. **13.15-5B** illustrates a cross-sectional view of an interlocking attachment mechanism **13.15-420**, which is an example of a perimeter mount. The interlocking attachment mechanism **13.15-420** can be referred to as an interlocking fastener. The interlocking attachment mechanism **13.15-420** includes a base **13.15-421**, which can be physically attached to a base **13.15-430**. The base **13.15-421** can be substantially similar to, including some or all of the features of, the bases of the facial interface frames described herein, such as the base **13.15-214**. The base **13.15-430** can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces **13.15-202**. The base **13.15-421** can be physically coupled to an outer layer **13.15-424**. The base **13.15-421** can include one or more protrusions **13.15-422** and the outer layer **13.15-424** can include one or more protrusions **13.15-426**. The base **13.15-430** can be physically coupled to an outer layer **13.15-434**. The base **13.15-430** can include one or more protrusions **13.15-436** and the outer layer **13.15-434** can include one or more protrusions **13.15-436**. Any number of protrusions can be included in the base **13.15-421**, the outer layer **13.15-424**, the base **13.15-430**, and the outer layer **13.15-434**.

[3391] The protrusions **13.15-432** and **13.15-436** of the base **13.15-430** can have shapes

corresponding to the protrusions **13.15-422** and **13.15-426** of the base **13.15-421**. For example, the protrusion **13.15-422** can have a shape corresponding to an opening between the protrusions **13.15-432**; the protrusions **13.15-432** can have shapes corresponding to openings between the protrusion **13.15-422** and the protrusions **13.15-426**; and the protrusions **13.15-426** can have shapes corresponding to openings between the protrusions **13.15-432** and the protrusions **13.15-436**. The protrusions **13.15-422**, **13.15-426**, **13.15-432**, and **13.15-436** can have longitudinal shapes with open ends and longitudinal axes extending in a direction into and out of the page in the view of FIGS. **13.15-5B**. This allows for the base **13.15-430** to slide relative to the base **13.15-421**. This can aid a user in attaching a removable facial interface of an HMD to a facial interface frame. Further, this allows for the removable facial interface to conform to the user's face when the user dons the HMD.

[3392] The protrusions **13.15-432** and **13.15-436** of the base **13.15-430** can be configured to interface with the protrusions **13.15-422** and **13.15-426** of the base **13.15-421** to attach the base **13.15-430** to the base **13.15-421** when the base **13.15-430** and the base **13.15-421** are pressed together. The protrusions **13.15-422**, **13.15-426**, **13.15-432**, and **13.15-436**, the outer layers **13.15-424** and **13.15-434**, and the bases **13.15-421** and **13.15-430** can be formed of elastic, flexible materials with high coefficients of friction. In examples in which the protrusions **13.15-422**, **13.15-426**, **13.15-432**, and **13.15-436** are formed from elastic materials, the interlocking attachment mechanism **13.15-420** may be referred to as an interlocking elastic fastener. The outer layers **13.15-424** and **13.15-434** and the bases **13.15-421** and **13.15-430** can be formed of any combinations of materials, such as the base **13.15-421** and the outer layer **13.15-424** being formed of the same or different materials and the base **13.15-430** and the outer layer **13.15-434** being formed of the same or different materials. Forming the protrusions **13.15-422**, **13.15-426**, **13.15-432**, and **13.15-436** from flexible materials allows the protrusions **13.15-422**, **13.15-426**, **13.15-432**, and **13.15-436** to be inserted into corresponding openings. Forming the protrusions **13.15-422**, **13.15-426**, **13.15-432**, and **13.15-436** from elastic materials having high coefficients of friction helps to retain the protrusions **13.15-422**, **13.15-426**, **13.15-432**, and **13.15-436** in the corresponding openings once the protrusions **13.15-422**, **13.15-426**, **13.15-432**, and **13.15-436** are inserted into the corresponding openings. In some examples, the protrusions **13.15-422**, **13.15-426**, **13.15-432**, and **13.15-436**, the outer layers **13.15-424** and **13.15-434**, and the bases **13.15-421** and **13.15-430** can be formed of polymers, elastomers, plastics, combinations thereof, or the like.

[3393] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-5A** and **13.15-5B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-5A** and **13.15-5B**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-5A** and **13.15-5B**.

[3394] FIGS. **13.15-6** illustrates a cross-sectional view of a magnetic slide attachment mechanism **13.15-500**, which is an example of a post mount. The magnetic slide attachment mechanism **13.15-500** includes a connector **13.15-501**, to which a base **13.15-510** can be attached. The connector **13.15-501** can be substantially similar to, including some or all of the features of, the connectors described herein, such as the connectors **13.15-212**. The base **13.15-510** can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces **13.15-202**.

[3395] The connector **13.15-501** can be physically coupled to a magnetic post **13.15-504** through a fastener **13.15-502**. The magnetic post **13.15-504** can include a flange **13.15-506**. The base **13.15-510** can be physically coupled to a magnetic protrusion **13.15-512**. The magnetic post **13.15-504** is magnetically attracted to the magnetic protrusion **13.15-512**, which helps to align the connector

13.15-501 with the base **13.15-510** and helps to secure the flange **13.15-506** of the magnetic post **13.15-504** under a lip **13.15-514** of the magnetic protrusion **13.15-512**. A magnet **13.15-516** of the magnetic protrusion **13.15-512** can be disposed under the lip **13.15-514** of the magnetic protrusion **13.15-512** to pull the magnetic post **13.15-504** into an opening **13.15-518** of the magnetic protrusion **13.15-512** under the lip **13.15-514** of the magnetic protrusion **13.15-512**. In the view of FIGS. **13.15-6**, the magnetic post **13.15-504** can move from left to right as the connector **13.15-501** is attached to the base **13.15-510**. In other words, the magnetic post **13.15-504** can be slidably engaged with the magnetic protrusion **13.15-512** by sliding the flange **13.15-506** under the lip **13.15-514**. The connector **13.15-501** can be removed from the base **13.15-510** by moving the magnetic post **13.15-504** from right to left, then moving the magnetic post **13.15-504** in an upwards direction once the flange **13.15-506** of the magnetic post **13.15-504** clears the lip **13.15-514** of the protrusion **13.15-512**. In other words, the magnetic post **13.15-504** can be slidably disengaged with the magnetic protrusion **13.15-512** by sliding the flange **13.15-506** out from under the lip **13.15-514**.

[3396] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-6**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-6**.

[3397] FIGS. **13.15-7** illustrates a cross-sectional view of a hook-and-loop attachment mechanism **13.15-600**, which is an example of a post mount. The hook-and-loop attachment mechanism **13.15-600** includes a connector **13.15-601**, to which a base **13.15-610** can be attached. The connector **13.15-601** can be substantially similar to, including some or all of the features of, the connectors described herein, such as the connectors **13.15-212**. The base **13.15-610** can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces **13.15-202**.

[3398] The connector **13.15-601** can be physically coupled to a hook-and-loop feature **13.15-604** through a fastener **13.15-602**. The hook-and-loop feature **13.15-604** can include a post **13.15-606**. The base **13.15-610** can be physically coupled to a hook-and-loop feature **13.15-612**. The hook-and-loop feature **13.15-604** can include an opening **13.15-614** configured to receive the post **13.15-606**. In the example of FIGS. **13.15-7**, the post **13.15-606** and the opening **13.15-614** can have corresponding frustum shapes, however any suitable shapes can be used. The opening **13.15-614** and the post **13.15-606** can be included to aid in self-aligning the connector **13.15-601** and the base **13.15-610**; however, in some examples, the opening **13.15-614** and the post **13.15-606** can be omitted.

[3399] One of the hook-and-loop feature **13.15-604** or the hook-and-loop feature **13.15-612** can include a plurality of hooks, and the other of the hook-and-loop feature **13.15-604** or the hook-and-loop feature **13.15-612** can include a plurality of loops. The hooks of the hook-and-loop feature **13.15-604** or the hook-and-loop feature **13.15-612** are retained in the corresponding loops of the hook-and-loop feature **13.15-604** or the hook-and-loop feature **13.15-612** in order to attach the base **13.15-610** to the connector **13.15-601**. Velcro is an example of a hook-and-loop attachment mechanism that can be used for the hook-and-loop feature **13.15-604** and the hook-and-loop feature **13.15-612**. Fine hook and loop features can be provided in the hook-and-loop feature **13.15-604** and/or the hook-and-loop feature **13.15-612** in order to lessen a tearing sound when the connector **13.15-601** and the base **13.15-610** are separated from one another.

[3400] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-7** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures,

including in any of FIGS. 13.15-7, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 13.15-7.

[3401] FIGS. 13.15-8 illustrates a cross-sectional view of a magnetic attachment mechanism 13.15-700, which is an example of a post mount. The magnetic attachment mechanism 13.15-700 includes a connector 13.15-701, which can be magnetically attached to a base 13.15-710. The connector 13.15-701 can be substantially similar to, including some or all of the features of, the connectors described herein, such as the connectors 13.15-212. The base 13.15-710 can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces 13.15-202.

[3402] The connector 13.15-701 can be physically coupled to a magnet 13.15-706 through a fastener 13.15-702. The connector 13.15-701 can be further physically coupled to a post 13.15-704, which houses the magnet 13.15-706. In some examples, the post 13.15-704 can be referred to as a protrusion, and the magnet 13.15-706 can be disposed in the protrusion. The base 13.15-710 can be physically coupled to a magnet 13.15-714, and a receptacle 13.15-712 can be physically coupled to the base 13.15-710 and/or the magnet 13.15-714. In some examples, the magnet 13.15-714 can be disposed in the receptacle 13.15-712. As illustrated in FIGS. 13.15-8, the receptacle 13.15-712 can include an opening configured to receive the post 13.15-704. In the example of FIGS. 13.15-8, the post 13.15-704 and the opening of the receptacle 13.15-712 can have corresponding frustum shapes, however any suitable shapes can be used. In some examples, the post 13.15-704 (e.g., the protrusion) and the receptacle 13.15-712 have shapes that are complementary to one another, such that they can nest, stack, or be received in one-another. The opening of the receptacle 13.15-712 and the post 13.15-704 can be included to aid in self-aligning the connector 13.15-701 and the base 13.15-710; however, in some examples, the opening of the receptacle 13.15-712 and the post 13.15-704 can be omitted. The magnet 13.15-706 and 13.15-714 are magnetically attracted to one another and retain the post 13.15-704 of the connector 13.15-701 within the opening of the receptacle 13.15-712 of the base 13.15-710.

[3403] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 13.15-8 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. 13.15-8, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 13.15-8.

[3404] FIGS. 13.15-9 illustrates a cross-sectional view of a spring-snap attachment mechanism 13.15-800, which is an example of a post mount. The spring-snap attachment mechanism 13.15-800 includes a connector 13.15-801, which can be physically attached to a base 13.15-810. The connector 13.15-801 can be substantially similar to, including some or all of the features of, the connectors described herein, such as the connectors 13.15-212. The base 13.15-810 can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces 13.15-202.

[3405] The connector 13.15-801 is physically coupled to a spring-snap feature 13.15-804 through a fastener 13.15-802. The spring-snap feature 13.15-804 includes detents 13.15-806 that are attached to the spring-snap feature 13.15-804 through springs in the spring-snap feature 13.15-804. The base 13.15-810 is physically coupled to a post 13.15-812, which includes flange 13.15-814. When the base 13.15-810 is attached to the connector 13.15-801, the flange 13.15-814 of the post 13.15-812 push the detents 13.15-806 outwards relative to the post 13.15-812 and into openings in the spring-snap feature 13.15-804. Once the flange 13.15-814 pass the detents 13.15-806, the springs of the spring-snap feature 13.15-804 push the detents 13.15-806 against the post 13.15-812 to retain the

flange **13.15-814** of the post **13.15-812** within an opening of the spring-snap feature **13.15-804**. In other words, the spring force of the spring-snap feature **13.15-804** pressing the detents **13.15-806** against the post **13.15-812** adjacent the flange **13.15-814** retains the post **13.15-812** within the spring-snap feature **13.15-804**, attaching the base **13.15-810** to the connector **13.15-801**. When the base **13.15-810** is removed from the connector **13.15-801**, this same process occurs in reverse. In the example of FIGS. **13.15-9**, the post **13.15-812** and the opening of the spring-snap feature **13.15-804** that receives the post can have corresponding cylindrical shapes, however any suitable shapes can be used. The opening of the spring-snap feature **13.15-804** and the post **13.15-812** aid in self-aligning the connector **13.15-801** and the base **13.15-810**.

[3406] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-9** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-9**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-9**.

[3407] FIGS. **13.15-10** illustrates a cross-sectional view of an interlocking attachment mechanism **13.15-900**, which is an example of a post mount or a perimeter mount. The interlocking attachment mechanism **13.15-900** can be referred to as an interlocking fastener. The interlocking attachment mechanism **13.15-900** includes a base **13.15-910**, which can be interlocked with a base **13.15-901**. The base **13.15-901** or **13.15-910** can be substantially similar to, including some or all of the features of, the bases of the facial interface frames described herein, such as the base **13.15-214**. The other of the base **13.15-901** or **13.15-910** can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces **13.15-202**. The base **13.15-901** can include a plurality of protrusions **13.15-902**, such as two protrusions **13.15-902** in the example of FIGS. **13.15-10**. The base **13.15-910** can include a protrusion **13.15-912**. In some examples, any number of protrusions **13.15-912** and **13.15-902** can be included in the base **13.15-910** and the base **13.15-901**.

[3408] As illustrated in FIGS. **13.15-10**, the protrusion **13.15-912** can be configured to interlock with the protrusions **13.15-902**. The protrusion **13.15-912** can be mushroom-shaped and can have a flange at an end of a post. The protrusions **13.15-902** can define an opening and include lips adjacent to the protrusion **13.15-912** when the protrusion **13.15-912** is inserted into the opening defined by the protrusions **13.15-902**. The lips of the protrusions **13.15-902** and the flange of the protrusion **13.15-912** can secure the protrusion **13.15-912** within the opening defined between the protrusions **13.15-902**.

[3409] In some examples, the protrusions **13.15-902** and **13.15-912** can have longitudinal shapes with open ends and longitudinal axes extending in a direction into and out of the page in the view of FIGS. **13.15-10**. This allows for the base **13.15-901** to slide relative to the base **13.15-910**. This can aid a user in attaching a removable facial interface of an HMD to a facial interface frame. Further, this allows for the removable facial interface to conform to the user's face when the user dons the HMD. In some examples, the protrusion **13.15-912** can have a circular shape in a bottom-up view and the protrusions **13.15-902** can have an annular shape in a top-down view. In some examples, the protrusions **13.15-902** and **13.15-912** can have triangular, rectangular, or other shapes in bottom-up and top-down views, respectively. Providing the protrusions **13.15-902** and **13.15-912** with corresponding shapes aids in self-aligning the base **13.15-901** with the base **13.15-910**.

[3410] The protrusions **13.15-902** and **13.15-912** can be formed of elastic, flexible materials with high coefficients of friction. In examples in which the protrusions **13.15-902** and **13.15-912** are formed from elastic materials, the interlocking attachment mechanism **13.15-900** may be referred to as an interlocking elastic fastener. The protrusions **13.15-902** and **13.15-912** can be formed from

the same or different materials. Forming the protrusions **13.15-902** and **13.15-912** from flexible materials allows the protrusion **13.15-912** to be inserted into the corresponding opening between the protrusions **13.15-902**. Forming the protrusions **13.15-902** and **13.15-912** from elastic materials having high coefficients of friction helps to retain the protrusion **13.15-912** in the corresponding opening between the protrusions **13.15-902**. In some examples, the protrusions **13.15-902** and **13.15-912** can be formed of polymers, elastomers, plastics, combinations thereof, or the like.

[3411] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-10** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-10**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-10**.

[3412] FIGS. **13.15-11** illustrates a cross-sectional view of a suction attachment mechanism **13.15-1000**, which is an example of a post mount. The suction attachment mechanism **13.15-1000** includes a connector **13.15-1010**, which can be attached to a base **13.15-1001** through a suction force. The connector **13.15-1010** can be substantially similar to, including some or all of the features of, the connectors described herein, such as the connectors **13.15-212**. The base **13.15-1001** can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces **13.15-202**.

[3413] The base **13.15-1001** can include a curved opening **13.15-1002** formed at a top surface thereof. In some examples, the curved opening **13.15-1002** may be referred to as a curved surface. A suction cup **13.15-1012** can be physically coupled to the connector **13.15-1010**. The base **13.15-1001** can be attached to the connector **13.15-1010** by pressing the suction cup **13.15-1012** into the curved opening with sufficient force. The connector **13.15-1010** can be removed from the base **13.15-1001** by pulling the connector **13.15-1010** away from the base **13.15-1001** with sufficient force. Removing the connector **13.15-1010** from the base **13.15-1001** can be aided by tilting the connector **13.15-1010** relative to the base **13.15-1001**. The curved opening **13.15-1002** of the base **13.15-1001** can have a smooth hard surface, which can be surrounded by softer surfaces, such as fabrics or the like. This aids in self-aligning the base **13.15-1001** and the connector **13.15-1010**, as the suction cup **13.15-1012** cannot easily be mounted to portions of the base **13.15-1001** surrounding the curved opening **13.15-1002**.

[3414] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-11** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-11**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-11**.

[3415] FIGS. **13.15-12** illustrates a cross-sectional view of a bi-stable attachment mechanism **13.15-1100**, which is an example of a post mount. The bi-stable attachment mechanism **13.15-1100** includes a connector **13.15-1101**, which can be attached to a base **13.15-1110** through a suction force. The connector **13.15-1101** can be substantially similar to, including some or all of the features of, the connectors described herein, such as the connectors **13.15-212**. The base **13.15-1110** can be substantially similar to, including some or all of the features of, the stiffeners described herein, such as the stiffeners of the removable facial interfaces **13.15-202**.

[3416] The connector **13.15-1101** can include a curved opening **13.15-1102** formed at a bottom surface thereof. A bi-stable element **13.15-1114** can be physically coupled to the base **13.15-1110** through a joint **13.15-1112**. The bi-stable element **13.15-1114** can be stable in two positions, such as a first position **13.15-1114A** and a second position **13.15-1114C**. The third position **13.15-1114B**

is an intermediate position between the first position **13.15-1114A** and the second position **13.15-1114C**. The bi-stable element **13.15-1114** can flip between the first position **13.15-1114A** and the second position **13.15-1114C**. For example, when the connector **13.15-1101** is pressed into the base **13.15-1110**, the bi-stable element **13.15-1114** can move from the first position **13.15-1114A** to the second position **13.15-1114C**. When the connector **13.15-1101** is pulled from the base **13.15-1110**, the bi-stable element **13.15-1114** can move from the second position **13.15-1114C** to the first position **13.15-1114A**. The connector **13.15-1101** can be attached to the base **13.15-1110** through a suction force similar to the suction attachment mechanism **13.15-1100**. Removing the connector **13.15-1101** from the base **13.15-1110** can be aided by tilting the connector **13.15-1101** relative to the base **13.15-1110**. The curved opening **13.15-1102** of the connector **13.15-1101** can have a smooth hard surface, which can be surrounded by softer surfaces, such as fabrics or the like. This aids in self-aligning the base **13.15-1110** and the connector **13.15-1101**, as the bi-stable element **13.15-1114** cannot easily be mounted to portions of the connector **13.15-1101** surrounding the curved opening **13.15-1102**.

[3417] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-12** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-12**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-12**.

[3418] FIGS. **13.15-13A** and **13.15-13B** are plan views of removable facial interfaces **13.15-1200A** and **13.15-1200B**, respectively. The removable facial interfaces **13.15-1200A** and **13.15-1200B** include a base **13.15-1204A** and a base **13.15-1204B**, respectively. A pliant material **13.15-1202** can be formed surrounding the respective bases **13.15-1204A** and **13.15-1204B**. Top mounts **13.15-1214**, side mounts **13.15-1210**, and bottom mounts **13.15-1208** can be physically coupled to the respective bases **13.15-1204A** and **13.15-1204B** through the pliant material **13.15-1202**. Any of the above-described attachment mechanisms, such as the post attachments, can be provided on the top mounts **13.15-1214**, the side mounts **13.15-1210**, and the bottom mounts **13.15-1208**. Additional attachment mechanisms, such as the perimeter attachments, can be provided on the removable facial interfaces **13.15-1200A** and **13.15-1200B**, such as along the bases **13.15-1204A** and **13.15-1204B** between the top mounts **13.15-1214**, the side mounts **13.15-1210**, and the bottom mounts **13.15-1208**. FIGS. **13.15-13A** and **13.15-13B** illustrate magnets **13.15-1212** and **13.15-1206** physically coupled to the top mounts **13.15-1214** and the bottom mounts **13.15-1208**, respectively. Although **13.15-6** mounts are illustrated as being provided for the removable facial interfaces **13.15-1200A** and **13.15-1200B**; any number of mounts can be provided.

[3419] In the example of FIGS. **13.15-13A**, the base **13.15-1204A** includes thick portions and thin portions. The thick portions can be included to provide stiffness to the removable facial interface **13.15-1200A**, while the thin portions can be included to provide flexibility to the removable facial interface **13.15-1200A**. In the example of FIGS. **13.15-13B**, the base **13.15-1204B** includes slits formed along the perimeter of the base **13.15-1204B**. The slits portions can be included to provide flexibility to the removable facial interface **13.15-1200B**. Any suitable bases with any suitable configurations can be included to provide desired flexibility and stiffness to the removable facial interfaces.

[3420] The bases **13.15-1204A** and **13.15-1204B** can be formed from metals, plastics, polymers, combinations thereof, or the like. The pliable material **13.15-1202** can be formed from rubbers, foams, polymers, silicone, elastomers, hydrogen gels, combinations thereof, or the like. The removable facial interfaces **13.15-1200A** and **13.15-1200B** further include a fabric material surrounding the pliable material **13.15-1202**, the top mounts **13.15-1214**, the side mounts **13.15-1210**, and the bottom mounts **13.15-1208**. The fabric material may encircle the top mounts **13.15-**

1214, the side mounts **13.15-1210**, and the bottom mounts **13.15-1208**, and the top mounts **13.15-1214**, the side mounts **13.15-1210**, and the bottom mounts **13.15-1208** may be exposed through the fabric material. The bases **13.15-1204A** and **13.15-1204B** can provide the removable facial interfaces **13.15-1200A** and **13.15-1200B** with a desired degree of stiffness. The pliable material **13.15-1202** can be pliant or semi-pliant and can be provided to increase user comfort. The fabric material can be configured to contact a user's skin and can also be provided to increase user comfort.

[3421] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **13.15-13A** and **13.15-13B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures, including in any of FIGS. **13.15-13A** and **13.15-13B**, described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **13.15-13A** and **13.15-13B**.

[3422] In some examples, the device seal can include an asymmetric compressible portion that contacts a user's face. The asymmetry in the design of the compressible portion improves comfort and reduces contact between the user's face and hard edges of the device seal. While the asymmetric compressible portion is described herein as part of a removable device seal, the asymmetric compressible portion can similarly be a part of a fixed or non-removable facial interface to be used with an HMD.

[3423] FIGS. **13.15-14** illustrates a cross-sectional view of a device seal **13.15-1400**. The device seal **13.15-1400** can be substantially similar to, including some or all of the features of, the device seals described herein. The device seal **13.15-1400** can include a facial interface **13.15-1402**, and a cover **13.15-1406**. The cover **13.15-1406** can be included in the device seal **13.15-1400** to provide a seal between the device seal **13.15-1400** and the outer environment.

[3424] The device seal **13.15-1400** can include a frame **13.15-1410**, a base **13.15-1414**, and a compressible portion **13.15-1416**. The frame **13.15-1410** can be physically coupled to a display portion of the HMD, and the frame **13.15-1410** can be configured provide an interface between the display portion and the facial interface **13.15-1402**. The facial interface **13.15-1402** can be removably attached to the base **13.15-1414**, or non-removably fixed to the base.

[3425] The compressible portion **13.15-1416** can fill a gap between the base **13.15-1414** and the facial interface **13.15-1402**, and can provide improved user comfort relative to traditional interfaces. The compressible portion **13.15-1416** can also be referred to as a compressible member or a compressible material. In some examples, the compressible portion **13.15-1416** can include a foam material. In some examples, the compressible portion **13.15-1416** can include materials with properties that impart flexibility, softness, compressibility, deformability, and the like. Examples of materials that can be used for the compressible portion **13.15-1416** include, but are in no way limited to, silicone, polymers, elastomers, hydrogels, combinations thereof, or the like. The compressible portion **13.15-1416** can be formed by molding, printing, casting, or the like. In some examples, attachment mechanisms, such as perimeter attachments, can be included on the compressible portion **13.15-1416**. The compressible portion **13.15-1416** provides a soft, flexible, deformable interface between the removable facial interface **202** and rigid components of the removable facial interface **13.15-1402**, such as the frame **13.15-1410**, the connectors, and the base **13.15-1414**. This interface can provide for improved user comfort when the HMD is donned.

[3426] In some examples, the compressible portion **13.15-1416** extends beyond the edges of the base **13.15-1414**. In other words, the compressible portion **13.15-1416** can be wider than the base **13.15-1414**, such that the compressible portion **13.15-1416** overhangs the base **13.15-1414**. The compressible portion **13.15-1416** can overhang, extend from, or be proud of the cover **13.15-1406**. For example, the compressible portion **13.15-1416** can extend a distance d above the edge of the

base **13.15-1414**. In some examples, the distance “d” can be approximately 3.4 mm. However, as discussed in greater detail below, the width of the compressible portion **13.15-1416** can vary based on the position of the compressible portion **13.15-1416** relative to the user's face when donned. In some examples, the lower part of the compressible portion **13.15-1416** is substantially even or flush with the lower edge of the base **13.15-1416**, while the upper part of the compressible portion **13.15-1416** is proud of or extended beyond the upper edge of the base **13.15-1414**.

[3427] The increased width of the compressible portion **13.15-1416** advantageously increases comfort and wearability of the HMD. As the HMD is worn, the compressible portion **13.15-1416** can bend or fold around the upper edge of the base **13.15-1414**. The base **13.15-1414** can, in some examples, be made from a rigid material, such as plastic. Thus, the overhanging compressible portion **13.15-1416** can shield the user's face from the edges of the base **13.15-1414** during use. The overhanging compressible portion **13.15-1416** can also, in some examples, become more compliant over time, improving comfort.

[3428] FIGS. **13.15-15A** illustrates a cross-sectional view of a facial interface **13.15-1502**. FIGS. **13.15-15B** illustrates a cross-sectional view of a forehead region **13.15-1503** of the facial interface **13.15-1502**. FIGS. **13.15-15C** illustrates a cross-sectional view of a cheek region **13.15-1505** of the facial interface **13.15-1502**. The facial interface **13.15-15** can be substantially similar to, including some or all of the features of, the facial interfaces described herein.

[3429] In some examples, the compressible portion **13.15-1516** can include a central region **13.15-1509** and an extended region **13.15-1507**. In some examples, the central region **13.15-1509** and the extended region **13.15-1507** can be a single unitary component (e.g., a foam piece). In some examples, the central region **13.15-1509** can be a separate and distinct material from the extended region **13.15-1507** (e.g., two different types of foam pieces). The central region **13.15-1509** can be shaped, sized, and positioned to correspond to the base of the frame. In other words, the central region **13.15-1509** can reside directly adjacent the base plastic. In contrast, the extended region **13.15-1507** can extend beyond a footprint of the base, such that the extended region **13.15-1507** overhangs the base.

[3430] In some examples, the extended region **13.15-1507** can run along an entirety of the facial interface **13.15-1502**. The compressible portion **13.15-1516** can have a varying cross-section or be asymmetric relative to the base. The extended region **13.15-1507** can taper as it travels from the forehead region **13.15-1503** to the cheek region **13.15-1505**. In other words, the extended region **13.15-1507** can decrease in size as it approaches the cheek region **13.15-1505**, so that it is wider at the forehead, and less wide at the cheeks. For example, at the forehead region **13.15-1503**, the extended region **13.15-1507** can overhang a base by approximately equal to or greater than 3.4 mm, while at the forehead region **13.15-1505**, the extended region **13.15-1507** can overhang a base by approximately equal to or less than 3.4 mm. As shown in FIGS. **13.15-15B**. As shown, the compressible portion **13.15-1516** can extend beyond or overhang the central region **13.15-1509** and the base plastic both above and below, as oriented in the figures. The varying asymmetric cross-section of the extended region **13.15-1507** can be intentionally varied to correspond to user facial features, thereby increasing comfort and fit of the device seal **13.15-1400** during use.

13.16: Electronic Devices with Light Blocking Structures

[3431] Head-mounted devices include head-mounted support structures that allow the devices to be worn on the heads of users. The head-mounted support structures may include device housings for housing components such as displays that are used for presenting a user with visual content. Head-mounted devices may also include a light-shielding nosepiece that rests on the nose of the user. The light-shielding nosepiece may include an elastomeric layer with perforations and fabric that covers the elastomeric layer. The elastomeric layer with the perforations may allow the light-shielding nosepiece to conform to the user's nose while the device is worn, while maintaining enough rigidity to be wrapped by a low force, high stretch textile or other low force, high stretch material.

Additional structures, such as rigid structures or semi-rigid structures, may be included in the light-

shielding nosepiece to provide additional support for the device while it is worn.

[3432] A schematic diagram of an illustrative system having an electronic device with a light-shielding nosepiece is shown in FIGS. **13.16-1**. As shown in FIGS. **13.16-1**, system **13.16-8** may include one or more electronic devices such as electronic device **13.16-10**. The electronic devices of system **13.16-8** may include computers, cellular telephones, head-mounted devices, wristwatch devices, and other electronic devices. Configurations in which electronic device **13.16-10** is a head-mounted device are sometimes described herein as an example.

[3433] As shown in FIGS. **13.16-1**, electronic devices such as electronic device **13.16-10** may have control circuitry **13.16-12**. Control circuitry **13.16-12** may include storage and processing circuitry for controlling the operation of device **13.16-10**. Circuitry **13.16-12** may include storage such as hard disk drive storage, nonvolatile memory (e.g., electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **13.16-12** may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio chips, graphics processing units, application specific integrated circuits, and other integrated circuits. Software code may be stored on storage in circuitry **13.16-12** and run on processing circuitry in circuitry **13.16-12** to implement control operations for device **13.16-10** (e.g., data gathering operations, operations involved in processing three-dimensional facial image data, operations involving the adjustment of components using control signals, etc.). Control circuitry **13.16-12** may include wired and wireless communications circuitry. For example, control circuitry **13.16-12** may include radio-frequency transceiver circuitry such as cellular telephone transceiver circuitry, wireless local area network (WiFi®) transceiver circuitry, millimeter wave transceiver circuitry, and/or other wireless communications circuitry.

[3434] During operation, the communications circuitry of the devices in system **13.16-8** (e.g., the communications circuitry of control circuitry **13.16-12** of device **13.16-10**), may be used to support communication between the electronic devices. For example, one electronic device may transmit video and/or audio data to another electronic device in system **13.16-8**. Electronic devices in system **13.16-8** may use wired and/or wireless communications circuitry to communicate through one or more communications networks (e.g., the internet, local area networks, etc.). The communications circuitry may be used to allow data to be received by device **13.16-10** from external equipment (e.g., a tethered computer, a portable device such as a handheld device or laptop computer, online computing equipment such as a remote server or other remote computing equipment, or other electrical equipment) and/or to provide data to external equipment.

[3435] Device **13.16-10** may include input-output devices **13.16-22**. Input-output devices **13.16-22** may be used to allow a user to provide device **13.16-10** with user input. Input-output devices **13.16-22** may also be used to gather information on the environment in which device **13.16-10** is operating. Output components in devices **13.16-22** may allow device **13.16-10** to provide a user with output and may be used to communicate with external electrical equipment.

[3436] As shown in FIGS. **13.16-1**, input-output devices **13.16-22** may include one or more displays such as display **13.16-14**. In some configurations, display **13.16-14** of device **13.16-10** includes left and right display panels (sometimes referred to as left and right portions of display **13.16-14** and/or left and right displays) that are in alignment with the user's left and right eyes and are viewable through left and right lens assemblies, respectively. In other configurations, display **13.16-14** includes a single display panel that extends across both eyes.

[3437] Display **13.16-14** may be used to display images. The visual content that is displayed on display **13.16-14** may be viewed by a user of device **13.16-10**. Displays in device **13.16-10** such as display **13.16-14** may be organic light-emitting diode displays or other displays based on arrays of light-emitting diodes, liquid crystal displays, liquid-crystal-on-silicon displays, projectors or displays based on projecting light beams on a surface directly or indirectly through specialized optics (e.g., digital micromirror devices), electrophoretic displays, plasma displays, electrowetting

displays, microLED displays, or any other suitable displays.

[3438] Display **13.16-14** may present computer-generated content such as virtual reality content and mixed reality content to a user. Virtual reality content may be displayed in the absence of real-world content. Mixed reality content, which may sometimes be referred to as augmented reality content, may include computer-generated images that are overlaid on real-world images. The real-world images may be captured by a camera (e.g., a forward-facing camera) and merged with overlaid computer-generated content or an optical coupling system may be used to allow computer-generated content to be overlaid on top of real-world images. As an example, a pair of mixed reality glasses or other augmented reality head-mounted display may include a display device that provides images to a user through a beam splitter, prism, holographic coupler, or other optical coupler. Configurations in which display **13.16-14** is used to display virtual reality content to a user through lenses are described herein as an example.

[3439] Input-output devices **13.16-22** may include sensors **13.16-16**. Sensors **13.16-16** may include, for example, three-dimensional sensors (e.g., three-dimensional image sensors such as structured light sensors that emit beams of light and that use two-dimensional digital image sensors to gather image data for three-dimensional images from light spots that are produced when a target is illuminated by the beams of light, binocular three-dimensional image sensors that gather three-dimensional images using two or more cameras in a binocular imaging arrangement, three-dimensional lidar (light detection and ranging) sensors, three-dimensional radio-frequency sensors, or other sensors that gather three-dimensional image data), cameras (e.g., infrared and/or visible digital image sensors), gaze tracking sensors (e.g., a gaze tracking system based on an image sensor and, if desired, a light source that emits one or more beams of light that are tracked using the image sensor after reflecting from a user's eyes), touch sensors, buttons, force sensors, sensors such as contact sensors based on switches, gas sensors, pressure sensors, moisture sensors, magnetic sensors, audio sensors (microphones), ambient light sensors, microphones for gathering voice commands and other audio input, sensors that are configured to gather information on motion, position, and/or orientation (e.g., accelerometers, gyroscopes, compasses, and/or inertial measurement units that include all of these sensors or a subset of one or two of these sensors), fingerprint sensors and other biometric sensors, optical position sensors (optical encoders), and/or other position sensors such as linear position sensors, and/or other sensors.

[3440] User input and other information may be gathered using sensors and other input devices in input-output devices **13.16-22**. If desired, input-output devices **13.16-22** may include other devices **13.16-24** such as haptic output devices (e.g., vibrating components), light-emitting diodes and other light sources, speakers such as ear speakers for producing audio output, and other electrical components. Device **13.16-10** may include circuits for receiving wireless power, circuits for transmitting power wirelessly to other devices, batteries and other energy storage devices (e.g., capacitors), joysticks, buttons, and/or other components.

[3441] Electronic device **13.16-10** may have housing structures (e.g., housing walls, straps, etc.), as shown by illustrative support structures **13.16-26** of FIGS. **13.16-1**. In configurations in which electronic device **13.16-10** is a head-mounted device (e.g., a pair of glasses, goggles, a helmet, a hat, a headband, etc.), support structures **13.16-26** may include head-mounted support structures (e.g., a helmet housing, head straps, temples in a pair of eyeglasses, goggle housing structures, and/or other head-mounted structures). The head-mounted support structures may be configured to be worn on a head of a user during operation of device **13.16-10** and may support display(s) **13.16-14**, sensors **13.16-16**, other components **13.16-24**, other input-output devices **13.16-22**, and control circuitry **13.16-12**.

[3442] In some embodiments, support structures **13.16-26** may include a light-shielding nosepiece. The light-shielding nosepiece may be attached to support structures **13.16-26**, such as a main housing portion of electronic device **13.16-10**, and may rest on the user's nose while device **13.16-10** is worn. The light-shielding nosepiece may be flexible, to allow the nosepiece to conform to the

user's nose, while retaining enough rigidity to support device **13.16-10** on the user's face while it is being worn (i.e., to maintain its shape on the user's nose while the device is worn) and to be wrapped by a fabric or other material. If desired, the light-shielding nosepiece may also include stiffeners or other components that help maintain the nosepiece on the user's nose to prevent light from reaching the user's eyes. An example of an illustrative electronic device having a nosepiece is shown in FIGS. **13.16-2**.

[3443] As shown in FIGS. **13.16-2**, head-mounted device **13.16-10** may include support structures **13.16-26**, which may include a head-mounted housing (sometimes referred to as a main housing, main housing unit, head-mounted support structure, etc.). The housing may have walls or other structures that separate an interior housing region from an exterior region surrounding the housing. For example, the housing may have walls formed from polymer, glass, metal, and/or other materials. Electrical and optical components may be mounted in the housing. These components may include components such as integrated circuits, sensors, control circuitry, input-output devices, etc.

[3444] To present a user with images for viewing from eye boxes (e.g., eye boxes in which the user's eyes are located when device **13.16-10** is being worn on the user's head), device **13.16-10** may include displays and lenses. These components may be mounted in optical modules or other supporting structure in the housing to form respective left and right optical systems. There may be, for example, a left display for presenting an image through a left lens to a user's left eye in a left eye box and a right display for presenting an image through a right lens to a user's right eye in a right eye box.

[3445] If desired, the housing may have forward-facing components such as cameras and other sensors on a front side for gathering sensor measurements and other input and may have a soft cushion on an opposing rear side of the housing. The rear side of the housing may have openings that allow the user to view images (image light **13.16-32**) from the left and right optical systems (e.g., when the rear side of the housing is resting on the user's head).

[3446] If desired, device **13.16-10** may have an adjustable strap or headband, and if desired, may have other structures (e.g., an over-the-head strap) to help hold the housing on the user's head.

[3447] As shown in FIGS. **13.16-2**, when worn by the user, device **13.16-10** may include a nosepiece, such as nosepiece **13.16-28** that rests on the nasal region the user's head (e.g., on the user's nose). In particular, nosepiece **13.16-28** (sometimes referred to as a light-shielding structure or a light-shielding nosepiece herein) may serve as an extension of the housing that rests on the user's nose and bridges between the opposing cheeks of the user. If desired, nosepiece **13.16-28** may be attached to the housing and/or may include one or more members formed from a portion of the housing. In some embodiments, nosepiece **13.16-28** may be a portion of a light seal or attached to a light seal that extends around some or all of a periphery of the housing (e.g., the light seal is attached to the housing around the periphery). When device **13.16-10** is in use, the light seal may compress against the user's face and prevent interference from ambient light. In general, however, nosepiece **13.16-28** may be attached to the housing in any desired manner.

[3448] Nosepiece **13.16-28** may be configured as a light-shielding structure and may therefore be sometimes referred to as light-shielding structure **13.16-28** or light-shielding nosepiece **13.16-28**. As an example, it may be desirable to enhance the viewing experience of the user by blocking external environmental light from entering the interior of device **13.16-10** (e.g., from entering the eye boxes) when device **13.16-10** is worn by the user. Nosepiece **13.16-28** may conform to the facial topology of the user around the user's nose and block light from entering the eye boxes. In some illustrative configurations, nosepiece **13.16-28** may be adjustable to conform to varying facial topologies of different users (e.g., portions of nosepiece **13.16-28** may deform differently based on the nose shapes of the users).

[3449] Nosepiece **13.16-28** may be mounted to a housing portion of electronic device **13.16-10**, such as head-mounted support structures **13.16-26**, at mounting points **13.16-30**. The housing may

include a housing frame that runs along the periphery of device **13.16-10**. If desired, the housing frame may be overlapped by a cushion member on the rear side of the housing facing the user. As an example, the cushion member may include foam structures or other soft compressible structures affixed to the housing frame. A fabric may overlap and extend over the housing frame and/or the cushion member on the rear side of the housing. If desired, the fabric may enclose only the cushion member, and the fabric-enclosed cushion member may be removably coupled to the housing frame. [3450] Mounting points **13.16-30** may be located at a bottom portion of the housing frame (e.g., a bottom portion of support structures **13.16-26**). As examples, mounting points **13.16-30** may include coupling mechanisms such as magnets, adhesive, hinges, or any other suitable coupling mechanisms.

[3451] In the example of FIGS. **13.16-2**, nosepiece **13.16-28** is attached to support structures **13.16-26** such that nosepiece **13.16-28** extends into a portion of support structures **13.16-26**. This is merely illustrative. If desired, the housing frame, a cushion member, and/or fabric may split into multiple portions to surround nosepiece **13.16-28**. As an example, a fabric-enclosed cushion member may run in front of nosepiece **13.16-28**, and a housing frame may run behind nosepiece **13.16-28**. In general, the housing portions, cushion member, and/or fabric may define an opening in which nosepiece **13.16-28** is disposed.

[3452] In some illustrative examples, nosepiece **13.16-28** may be removably coupled (via magnetics) to the housing frame or other portions of the housing. In some illustrative examples, a portion of nosepiece **13.16-28** may form an integral portion of the housing frame and/or may not be removable from the housing.

[3453] Support structures **13.16-26** (e.g., housing frames) and nosepiece **13.16-28** (along with other desired structures) may define the periphery of the eye boxes of device **13.16-10** at which the user's eyes are located. Components, such as displays, lenses, sensors, etc., may overlap and/or be located within the eye boxes of device **13.16-10**, and may be enclosed by and/or mounted to support structures **13.16-26** and/or nosepiece **13.16-28**. As illustratively shown in FIGS. **13.16-2**, display **13.16-14** emit image light **13.16-32** through a lens to an eye box. Nosepiece **13.16-28** may be configured to block environmental light from an exterior of device **13.16-10** from entering the eye box and interfering with image light **13.16-24**. An example of a nosepiece that may be used in device **13.16-10** is shown in FIGS. **13.16-3**.

[3454] As shown in FIGS. **13.16-3**, a nosepiece, such as nosepiece **13.16-28**, may include elastomer **13.16-34** (also referred to as elastomeric layer **13.16-34** herein). Elastomer **13.16-34** may be any desired elastomeric material, such as thermoplastic polyurethane (TPU), nitrile butadiene rubber (NBR), or silicone (such as a low-durometer silicone). Elastomer **13.16-34** may have a thickness of greater than 0.1 mm, greater than 0.25 mm, greater than 0.5 mm, about 1 mm, less than 2 mm, less than 5 mm, etc.

[3455] In some embodiments, elastomer **13.16-34** may include perforations **13.16-36**. Perforations **13.16-36** may allow elastomer **13.16-34** to bend to accommodate a user's nose (i.e., in the horizontal left and right directions of FIGS. **13.16-3**), while maintaining sufficient rigidity (i.e., in the vertical up and down directions of FIGS. **13.16-3**) to support device **13.16-10** on the user's nose. The rigidity of elastomer **13.16-34** may also allow elastomer **13.16-34** to be wrapped by fabric **13.16-38** or other low force, high stretch material.

[3456] Perforations **13.16-36** may be formed in any desired pattern. In the example of FIGS. **13.16-3**, perforations **13.16-36** are slits formed as an array of perforations across the entire surface of elastomer **13.16-34**. Perforations **13.16-36** may be formed in a brick pattern, as in FIGS. **13.16-3**, or may be formed in any other desired pattern. Additionally, any desired number of perforations **13.16-36** may be formed in elastomer **13.16-34**, such as at least 1 perforation, at least 5 perforations, or any other desired number of perforations. Perforations **13.16-36** may extend entirely through elastomer **13.16-34**, or may extend partially through elastomer **13.16-34**.

[3457] The use of slits formed in a brick pattern, as shown in FIGS. **13.16-3**, are merely illustrative.

Other illustrative examples of perforations in elastomer **13.16-34** are shown in FIGS. **13.16-4A** and **13.16-4B**.

[3458] As shown in FIGS. **13.16-4A**, elastomer **13.16-34** may include three-part openings **13.16-37**. Three-part openings **13.16-37** may be arranged in a brick pattern, similar to the slits of FIGS. **13.16-3**. Alternatively, three-part openings **13.16-37** may be formed in any other suitable pattern.

[3459] Three-part openings **13.16-37** may allow elastomer **13.16-34** to stretch/deform in two or more axes. As a result, elastomer **13.16-34** (and therefore nosepiece **13.16-28**) may contour to a user's nose more accurately (e.g., elastomer **13.16-34** may be able to deform to a user's nose and therefore adapt to the shape of the user's nose). In this way, elastomer **13.16-34** may seal to the user's nose and prevent light from interfering with displays in the head-mounted device.

[3460] Different types of perforations may be used in a single elastomeric member. For example, as shown in FIGS. **13.16-4B**, slits **13.16-36** may be formed on a top half of elastomer **13.16-34**, while three-part openings **13.16-37** may be formed on a bottom half of elastomer **13.16-34**. In some embodiments, using three-part openings **13.16-37** at the bottom portion of elastomer **13.16-34** may allow elastomer **13.16-34** to stretch more (e.g., along two or more axes) at the user's nose.

However, the example of an upper half of elastomer **13.16-34** having slits **13.16-36** and the bottom half having three-part openings **13.16-37** is merely illustrative. In general, slits **13.16-36** and three-part openings (or other suitable perforations) may be formed anywhere on elastomer **13.16-34**, and may fill any suitable portion of elastomer **13.16-34** to allow elastomer **13.16-34** to conform to a user's nose.

[3461] Elastomer **13.16-34** may have a shape that generally conforms to a user's nose, such as a chevron shape, as shown in FIGS. **13.16-3**, or any other desired shape, such as a rounded shape, or a rectangular shape.

[3462] As shown in FIGS. **13.16-3**, elastomer **13.16-34** may be covered by fabric **13.16-38**. In particular, fabric **13.16-38** may overlap and/or encapsulate elastomer **13.16-34** in region **13.16-40**. Fabric region **13.16-40** may have the same shape as elastomer **13.16-34**, as shown in FIGS. **13.16-3**, or may have a different shape than elastomer **13.16-34**. Fabric members **13.16-40** and/or **13.16-38** may serve as a light-shielding member or layer for light-shielding structure **13.16-28**. In particular, illustrative configurations in which fabric **13.16-40/13.16-38** is a fabric cover (sometimes referred to as a fabric cover layer or a cover layer) are described herein as an illustrative example. To serve light-shielding functions, the fabric cover may be formed from an opaque or light-shielding material (e.g., black yarn) or may be formed from an underlying material coated with an opaque or light-shielding material (e.g., black dye or ink). As examples, the fabric cover may be formed any suitable type of fabric such as knit fabric, woven fabric, braided fabric, etc.

[3463] Fabric **13.16-40** may be tented over elastomer **13.16-34**. Tenting of the fabric cover over an underlying structure may be achieved by the underlying structure contacting or otherwise supporting the fabric cover at one or more points or areas of support as the fabric cover extends over one or more sides of the underlying structure. The tenting of the fabric cover over the underlying structure may cause the fabric cover to follow the general outline of the underlying structure, especially around the areas of support. If desired, differences in the outlines of the fabric cover and of the underlying structure may exist, especially in some regions away from the areas of support, thereby causing some portions of the fabric cover to be suspended in air and therefore readily deflectable. As an example, the fabric cover may be deflectable to the boundary of the underlying structure (or even beyond the boundary of the underlying structure if the boundary is defined by a flexible or deformable member).

[3464] The rigidity of elastomer **13.16-34**, which is preserved in the vertical direction by perforations **13.16-36**, may allow elastomer **13.16-34** to be wrapped by fabric **13.16-40** (or other low force, high stretch textile, or other low force, high stretch material) without deforming. In other words, elastomer **13.16-34** may be rigid enough to maintain its shape while fabric **13.16-40** is

applied to/wrapped around elastomer **13.16-34** (as well as to maintain its shape when the device is worn by a user), but flexible enough that it can conform to a user's nose.

[3465] In such a way, the fabric cover may have a three-dimensional shape (based on an outline of the underlying structure) that includes portions (e.g., directly supported by the underlying structure) that are more defined and portions (e.g., not directly supported by the underlying structure, suspended in air, etc.) that are less defined, and more flexible or yielding. These less-defined portions (e.g., a yielding fabric surface) may help form flexible boundaries such as those for an opening configured to receive a user's nose.

[3466] As a particular illustrative example, the underlying structure may have surfaces that define an opening for accommodating a user's nose. The surfaces may be surrounded by peripheral edges. The fabric cover may be tented over the underlying structure such that the fabric cover is directly supported by the underlying structure along one or more of the peripheral edges of the underlying structure and may be suspended in air around the opening, thereby providing a fabric surface that is deflectable by the user's nose. This may help with improving user comfort as well as providing a more conformal fit when the light shielding structure rests on the user's nose.

[3467] Regardless of the shape of fabric region **13.16-40**, fabric region **13.16-40** (and/or elastomer **13.16-34**) may be bonded to support structures **13.16-26** (such as a housing frame) using adhesive **13.16-42**. However, the use of adhesive **13.16-42** is merely illustrative. Fabric **13.16-40** and/or elastomer **13.16-34** may be formed integrally with support structures **13.16-26**, or may be attached to support structures **13.16-26** using any desired attachment mechanism.

[3468] Although nosepiece **13.16-28** is shown as including both elastomer **13.16-34** and fabric **13.16-40**, this is merely illustrative. If desired, nosepiece **13.16-28** may include elastomer **13.16-34** without an overlapping fabric layer. In this case, another layer, such as a polymer or rubber, may overlap elastomer **13.16-34**, or elastomer **13.16-34** may directly contact a user's nose as they wear device **13.16-10**. Alternatively, nosepiece **13.16-28** may include fabric **13.16-40** without elastomer **13.16-34**, if desired. In this case, fabric **13.16-40** may be a flat knit fabric to provide sufficient stretching over a user's nose, while providing enough support for device **13.16-10**.

[3469] Although fabric **13.16-38/13.16-40** is described as fabric, this is merely illustrative. In general, fabric **13.16-38/13.16-40** may be any low force, high stretch material, such as a low force, high stretch textile.

[3470] In some embodiments, although elastomer **13.16-34** may have rigidity to support device **13.16-10** on a user's nose (i.e., in the vertical direction of FIGS. **13.16-3**) and to be wrapped by fabric, it may be desirable to add additional rigid or semi-rigid structures to a nosepiece. An illustrative example of a nosepiece having a rigid structure is shown in FIGS. **13.16-5**.

[3471] As shown in FIGS. **13.16-5**, nosepiece **13.16-28** may include fabric **13.16-40** (which may cover an elastomeric layer, such as elastomer **13.16-34**) and adhesive **13.16-42**, which in turn may attach fabric **13.16-40** to structural frame **13.16-43**. Structural frame **13.16-43** may be a plastic frame, as an example. Structural frame **13.16-43** may provide additional rigidity to ensure that nosepiece **13.16-28** retains its shape and support when worn by a user.

[3472] Although FIGS. **13.16-5** shows the use of adhesive **13.16-42**, this is merely illustrative. Adhesive **13.16-42** may be omitted, if desired. In some examples, elastomer **13.16-34** may be co-molded to structural frame **13.16-43**. Alternatively, elastomer **13.16-34** may be fit into a recessed portion of structural frame **13.16-43** or otherwise attached to structural frame **13.16-43**.

[3473] Additionally, although FIGS. **13.16-5** shows structural frame **13.16-43** on three sides of fabric **13.16-40**/elastomer **13.16-34**, this is merely illustrative. Structural frame **13.16-43** may be attached to one side (e.g., the bottom side) of fabric **13.16-40**/elastomer **13.16-34** or any other desired number of sides.

[3474] Regardless of the attachment of structural frame **13.16-43** to elastomer **13.16-34** and/or fabric **13.16-40**, fabric **13.16-40** may extend over all or some of both frame **13.16-43** and/or elastomer **13.16-34**. An example of a stack up of nosepiece **13.16-28** is shown in FIGS. **13.16-6**.

[3475] As shown in FIGS. **13.16-6**, nosepiece **13.16-28** may include fabric **13.16-40**, which is shown as having fabric portions **13.16-40-1** and **13.16-40-2**, on either side of elastomer **13.16-34**. If desired, fabric **13.16-40** may be bonded to elastomer **13.16-34** at a periphery of elastomer **13.16-34**, which is shown in FIGS. **13.16-6** as points **13.16-44**. For example, fabric **13.16-40** may be bonded to elastomer **13.16-34** around a periphery (such as an entire periphery or a portion of the periphery) of the elastomer layer. Fabric **13.16-40** may be bonded to elastomer **13.16-34** using an adhesive or other desired mounting mechanism.

[3476] In some cases, it may be desirable to allow elastomer **13.16-34** to move to a greater extent, and therefore leave elastomer **13.16-34** un-bonded from fabric **13.16-40**. For example, elastomer **13.16-34** may be fully surrounded fabric **13.16-40** and float within the fabric. In some embodiments portions of fabric **13.16-40** may be bonded directly to each other, rather than to elastomer **13.16-34**. In this way, elastomer **13.16-34** may float within fabric **13.16-40**, which may allow **13.16-34** to move more freely.

[3477] Because nosepiece **13.16-28** is designed to block light from reaching the eye boxes of a user wearing device **13.16-10**, it may be desirable to ensure a tight fit between nosepiece **13.16-28** and the user's nose and a similarly tight fit between nosepiece **13.16-28** and device **13.16-10**. An example of an extension piece that may allow for nosepiece **13.16-28** to fit tightly with device **13.16-10** is shown in FIGS. **13.16-7**.

[3478] As shown in FIGS. **13.16-7**, extension piece **13.16-46** may be added below nosepiece **13.16-28**. In particular, if a user's nose bridge protrudes only a small amount, nosepiece **13.16-28** may not reach the user's nose. Therefore, extension piece **13.16-46** may be attached between support structures of device **13.16-10**, such as support structures **13.16-26**, and nosepiece **13.16-28**. Extension piece **13.16-28** may be formed from fabric, foam, plastic, and/or any other desired material. In some cases, extension piece **13.16-28** may allow nosepiece **13.16-28** to align with other light-blocking structures in device **13.16-10**, such as a foam member that fits against the user's face. In this way, light may be prevented from reaching the user's eye boxes and interfering with the user's visibility of a display in device **13.16-10**.

[3479] In addition or as an alternative to extension piece **13.16-46**, it may be desirable to ensure a close fit to a user's nose to prevent light from entering the user's eye boxes. Examples of nosepieces that have additional structures to improve the fit to a user's nose are shown in FIGS. **13.16-8** and **13.16-9**.

[3480] As shown in FIGS. **13.16-8**, service loop **13.16-49** may be included within nosepiece **13.16-28**. In particular, service loop **13.16-49** (and other portions of nosepiece **13.16-28**, if desired) may be attached to supports **13.16-50-1** and **13.16-50-2**. Service loop **13.16-49** may be tightened and loosened as desired to conform nosepiece **13.16-28** to the nose of a user. For example, as shown in FIGS. **13.16-7**, service loop **13.16-49** may be tightened to move nosepiece **13.16-28** to position **13.16-28'** by moving nosepiece **13.16-28** downward in direction **13.16-56**, right in direction **13.16-52**, and left in direction **13.16-54** toward to the nose of the user. In this way, service loop **13.16-49** may allow nosepiece **13.16-28** to fit more securely to the user's nose, preventing light from entering the user's eye boxes through gaps between nosepiece **13.16-28** and the nose.

[3481] Alternatively or additionally, nosepiece **13.16-28** may include foam **13.16-48**. Foam **13.16-48** may fill a gap between nosepiece **13.16-28** and the user's nose, and may be compressible to allow for a secure fit between nosepiece **13.16-28** and the nose. Foam **13.16-48** may directly contact the nose of the user (e.g., may be on a surface of a fabric layer, such as fabric **13.16-40**), as shown in FIGS. **13.16-8**, may be embedded within nosepiece **13.16-28** (e.g., covered by fabric, such as fabric **13.16-40**), or may otherwise be attached to nosepiece **13.16-28**.

[3482] Instead of service loop **13.16-49**, a deformable stiffener, such as deformable stiffener **13.16-58**, may be incorporated into nosepiece **13.16-28**. As shown in FIGS. **13.16-9**, deformable stiffener **13.16-58** may allow for nosepiece **13.16-28** to be adjusted in directions **13.16-52**, **13.16-54**, and **13.16-56** toward the nose of the user to conform nosepiece **13.16-28** to the user's nose at position

13.16-28'. Deformable stiffener **13.16-58** may be formed within nosepiece **13.16-28** (i.e., may be covered by fabric), as shown in FIGS. **13.16-9**, or may be formed on a surface of the fabric (e.g., an inner or outer surface of fabric **13.16-40**). Although not shown in FIGS. **13.16-9**, a gap-filling foam, such as foam **13.16-48** of FIGS. **13.16-8**, may be incorporated in or on nosepiece **13.16-28** in addition to deformable stiffener **13.16-58**.

[3483] In some embodiments, to ensure that nosepiece **13.16-28** is tightly sealed to the nose of a user, internal components of device **13.16-10**, such as fans, may be used to move air toward nosepiece **13.16-28**, thereby sealing nosepiece **13.16-28** around the user's nose.

[3484] In addition to improving the fit of nosepiece **13.16-28** to prevent light from entering the eye boxes of the user, it may also be desirable to incorporate layers into nosepiece **13.16-28** that improve the comfort for the user. Examples of various modifications that may be made to nosepiece **13.16-28** to improve user comfort are shown in FIGS. **13.16-10A-13.16-10G**.

[3485] As shown in FIGS. **13.16-10A**, a nosepiece, which includes elastomer **13.16-34** and fabric **13.16-40** (such as nosepiece **13.16-28**) may have a rolled edge and may approach the user's nose at a shallow angle of contact to conform to a curved portion of nose **13.16-60**. The rolled edge may prevent elastomer **13.16-34** from digging into the user's skin on nose **13.16-60**, while the shallower angle of contact may allow the nosepiece to move up the user's nose rather than hitting the user's nose directly. Therefore, by having a rolled edge and/or a shallower angle of contact, nosepiece **13.16-28** may improve comfort for a user of device **13.16-10**.

[3486] Alternatively or additionally, a nosepiece may include foam, such as foam **13.16-62**, between some or all of elastomer **13.16-34** and fabric **13.16-40**, as shown in FIGS. **13.16-10B**. In particular, foam **13.16-62** may be at an edge portion of the nosepiece that contacts the user's nose, and therefore prevent elastomer **13.16-34** from hurting the user's nose or reducing discomfort as the nosepiece pushes against the nose.

[3487] In some examples, it may be desirable to have a series of openings in elastomer **13.16-34** that are either unfilled or filled with different material, such as foam. For example, in FIGS. **13.16-10C**, openings **13.16-64** may be provided in a portion of the nosepiece that will contact the user's nose. Openings **13.16-64** may be through openings, partial openings, may be filled with other material, such as foam, or may otherwise be more flexible than the surrounding regions of elastomer **13.16-34**. As the nosepiece pushes against the user's nose, openings **13.16-64** may crumple, thereby preventing or reducing discomfort to the user.

[3488] If desired, a portion of fabric may be extended from the portion that contacts the user's nose to provide an additional buffer between the nose and elastomer **13.16-34**. As shown in FIGS. **13.16-10D**, fabric portion **13.16-66** may extend from fabric **13.16-40**. When worn by a user, fabric portion **13.16-66** may contact the nose of the user first, and therefore increase the amount of fabric between the nose and elastomer **13.16-34**, which may improve user comfort. In some examples, fabric portion **13.16-66** may be a hemmed edge of fabric **13.16-40**.

[3489] If desired, at least some portions of elastomer **13.16-34** that would otherwise contact the user's nose may be cut and replaced by more flexible material, such as foam. As shown in FIGS. **13.16-10E**, foam portions **13.16-68** may replace portions of elastomer **13.16-34** that would otherwise contact the user's nose. Although fabric **13.16-40** does not cover foam portions **13.16-68** in FIGS. **13.16-10E**, fabric **13.16-40** may overlap foam portions **13.16-68**, if desired.

[3490] In addition to, or instead of, the modifications of FIGS. **13.16-10A-13.16-10E**, in which an edge portion of a nosepiece is modified to prevent the elastomer from providing discomfort to a user's nose, a bottom portion of the nosepiece may be modified. As shown in FIGS. **13.16-10F**, segmented material **13.16-69** may be added to a bottom surface of nosepiece **13.16-28**. For example, segmented material **13.16-69** may contact the top of the user's nose (i.e., the nose bridge), and prevent nosepiece **13.16-28** from directly contacting the nose and causing discomfort. Material **13.16-69** may be foam or may be elastomer flaps.

[3491] Although FIGS. **13.16-10F** shows material **13.16-69** as being segmented and non-

overlapping, material **13.16-69** may be segmented and overlapping, or may be one connected piece of material, if desired. For example, in FIGS. **13.16-10G**, foam **13.16-73** may be provided between nosepiece **13.16-28** and the user's nose while device **13.16-10** is worn. If desired, optional deformable stiffener **13.16-71** may be provided on the bottom surface of nosepiece **13.16-28** to improve the fit of nosepiece **13.16-28** on the user's nose. If stiffener **13.16-71** is included, foam **13.16-73** may also prevent stiffener **13.16-71** from directly contacting the user's nose and providing discomfort.

[3492] All of the examples in FIGS. **13.16-10A-13.16-10G** are merely illustrative examples of improving user comfort while ensuring that a nosepiece fits tightly to the user's nose. The examples are not limiting and may be implemented individually or together in any combination. Regardless of whether any structures in FIGS. **13.16-10A-13.16-10G** are used, it may be desirable to strengthen a portion of nosepiece **13.16-28**. Although the use of a structural frame was described in connection with FIGS. **13.16-5**, such a frame may add unnecessary bulk or add too much rigidity to nosepiece **13.16-28** in some cases. Therefore, a semi-rigid member may be used. An example of using a semi-rigid member on a nosepiece is shown in FIGS. **13.16-11**.

[3493] As shown in FIGS. **13.16-11**, a nosepiece, such as nosepiece **13.16-28**, may include portions **13.16-70** and **13.16-72**. Portion **13.16-70** may include an elastomer, such as elastomer **13.16-34**, and/or a fabric, such as fabric **13.16-40**. Portion **13.16-72** may be coupled to portion **13.16-70** and may include a semi-rigid member. In particular, the semi-rigid member may be formed on a bottom portion of nosepiece **13.16-28** (i.e., toward the bottom of the user's nose when device **13.16-10** is worn), and may provide additional rigidity to nosepiece **13.16-28**. The semi-rigid members may be formed from any desired material, such as rubber or plastic. In this way, nosepiece **13.16-28** may remain sufficiently flexible to conform to the shape of a user's nose, while retaining its shape when after it has been conformed to the shape of the nose and the device is being worn by the user.

[3494] Although nosepieces, such as nosepiece **13.16-28**, have been described as including an elastomeric member, such as elastomer **13.16-34**, this is merely illustrative. In some embodiments, elastomer **13.16-34** may be omitted. For example, in the example of FIG. **13.16-3**, elastomer **13.16-34** may be omitted or may be replaced with thick fabric, which may be surrounded by thinner, more stretchy fabric. In the illustrative example of FIGS. **13.16-12**, nosepiece **13.16-28** may include fabric **13.16-73** that may be formed from fabric layers **13.16-74** and **13.16-76**. Fabric layers **13.16-74** and **13.16-76** may be bonded at points **13.16-78**, such as with laser welding, ultrasonic welding, or adhesive. The use of two (or more) fabric layers **13.16-74** and **13.16-76** may make fabric **13.16-73** thicker, thereby providing more support in the region of a user's nose. In general, however, fabric **13.16-73** may include any suitable number of fabric layers.

[3495] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

XIV: Powerstraps and Securement Band

[3496] FIGS. **14.0-1** illustrates a view of an HMD **14.0-100** including power straps **14.0-102** and a securement band **14.0-104**. The power straps **14.0-102** and a securement band **14.0-104** are described in more detail here in section XIV.

14.1: Electrical Connectors

[3497] Portable electronic devices, such as smart phones, laptops, tablet computing devices, smart watches, head-mountable devices (also referred to as head-mountable displays (HMDs)), and headphones, have become commonplace for persons undertaking daily activities (travel, communication, education, entertainment, employment, and the like). Indeed, portable electronic devices can provide assistance in completing daily tasks and errands, such as, watching an instructional video or monitoring progress during and after an exercise routine. However, some electronic devices incorporate a temporary or permanent cabled connection to operate (e.g., charging the device, providing electrical power to an electronic component, interconnecting a

peripheral input or output device, and the like).

[3498] Regarding HMDs, an electronic device can include one or more cabled connections, each of which can have unique and specific requirements. As an example, a temporary or permanent cabled connection can be coupled to a display portion of the HMD. The HMD can include one or more battery packs or electrical power sources that demand regular charging to operate the HMD for an extended period of time. However, a cabled connection to the display portion can inhibit (e.g., snag on a user or objects around the user while in use) or otherwise limit a user from operating the HMD while charging.

[3499] In some examples of the present disclosure, a display portion of the HMD can receive electrical power and/or control signals through a support electrically coupled to the display portion. The support can include a plug connector which is connected (structurally and electrically) within a receptacle connector of the display portion. While connected, the support can define an electrical path that enables electrical power and/or control signals to pass between the display portion and another electronic device (e.g., an electrical power source, such as, a battery pack) electrically connected to the support. The support can include a receptacle connector configured to electrically connect an electronic device to the support.

[3500] One aspect of the present disclosure relates to an electronic device, (e.g., an HMD) including an enclosure and a receptacle connector coupled or joined to the enclosure. The receptacle connector can include a housing defining an aperture and an internal volume. In some examples, the housing is configured to receive a plug connector of a support within the internal volume. The receptacle connector can include one or more electrical contacts at least partially disposed within the internal volume and contacting one or more correlating electrical contacts on the plug connector. In some examples, the plug connector and the receptacle connector can define respective convex surfaces and respective concave surfaces such that a cross-sectional shape of the plug connector and/or the receptacle connector has a curved longitudinal axis. The electrical contacts of the receptacle connector can be coupled or connected to the housing at angles normal to the concave and convex surfaces of the plug connector, which provides for improved electrical contact between the electrical contacts of the receptacle connector and the plug connector. In some examples, the receptacle connector can include a spring-driven fastener configured to retain the plug connector within the receptacle connector and configured to release the plug connector in response to a tool actuation. Bumpers can be provided in the housing to supply an ejection force to the plug connector, and seals can be provided adjacent an opening of the cavity to provide sealing and/or an ejection force to the plug connector.

[3501] In some examples, abruptly or accidentally removing electrical power and/or control signals from an electronic device can damage the components of the electronic device and/or data stored on the electronic device. As such, cabled connections that do not unwantedly or accidentally disconnect can be desirable. In some examples, a cable assembly that reliably interlocks with the electronic device to limit unwanted or accidental extraction of the cable assembly is provided. In some examples, a connection that requires more force to remove the connection than to establish the connection (e.g., disconnecting a cable assembly from a port versus inserting the cable assembly into the port) is provided. In some examples, a cable assembly and a receptacle connector of an electronic device can be configured to prevent damage to the electronic device and loss of data, even in cases in which the cable assembly is accidentally or undesirably extracted from the receptacle connector.

[3502] One aspect of the present disclosure relates to an electronic device including an enclosure and a receptacle connector attached to the enclosure. The receptacle connector can form a recess that interlocks or otherwise engages with a plug connector of a cable assembly. In some examples, the plug connector can include a boot, a central member coupled to the boot, and one or more protrusions extending laterally from the central member. The one or more protrusions can extend laterally into a channel, or respective channels, formed within the recess of the receptacle

connector.

[3503] In some examples, the receptacle connector can include one more detents disposed within the recess. Each of the one or more detents can interlock with a respective protrusion of the central member. The detents can be biased to extend into the recess by one or more leaf springs, canted coil springs, elastic foam, a combination thereof, or another biasing element. Each of the detents can form an angled surface which interfaces with a respective protrusion to retain the plug connector within the receptacle connector. At least one of the spring force of the biasing element and/or the angle of the angled surface can correlate to a force required to extract the plug connector from the receptacle connector. In some examples, each respective angle of the angled surfaces can vary so the plug connector can be more easily extracted by lifting one side of the plug connector (e.g., a non-cable-side) than another side of the plug connector (e.g., a cable-side of the plug connector).

[3504] In some examples, the central member can be rotated to interlock (or release) the plug connector and the receptacle connector. In other words, the plug connector can be rotatably received within the receptacle connector. Alternatively, a push-button, a latch, a slide-button, or another actuating mechanism can be used to interlock (or release) the plug connector and the receptacle connector. For example, an actuating mechanism can be incorporated into the boot of the plug connector. Additionally, or alternatively, one or more magnets can be disposed within the plug connector and/or the receptacle connector to orient the plug connector relative to the receptacle connector and/or retain the plug connector to the receptacle connector.

[3505] One aspect of the present disclosure relates to an electronic device including a housing that defines an exterior surface of the electronic device. The housing can form an aperture within the exterior surface. An electronic component can be disposed within the housing and can include a receptacle connector coupled to the electronic component. The receptacle connector can be fastened to a trim ring, which can be fastened to the housing. The housing, the trim ring, and the receptacle connector can form an electrical port. The electronic device can further include a first engagement feature and a cable having a plug connector. The plug connector can include a boot, a plug, and a second engagement feature. The boot is receivable within the aperture formed within the exterior surface of the housing and can be receivable within the trim ring. The plug is receivable within the receptacle connector to electrically couple the electronic device with the cable. The second engagement feature can interconnect or otherwise engage the first engagement feature to retain the cable to the electronic device.

[3506] In some examples, the first and second engagement features can be one or more: biasing elements, latches, arms, protrusions, recesses, channels, magnets, canted coiled springs, rotating members, sliding members, another type of engagement feature, or a combination thereof. In some examples, the first engagement feature can be included in the trim ring, the second engagement feature can be included in the boot, and the first and second engagement features can latch, interlock, or otherwise secure the boot in the trim ring such that the cable is interconnected with the electronic device. In some examples, the first engagement feature and the second engagement feature can involve a lesser force to connect the cable to the electronic device (e.g., insert the plug connector into the electrical port) and a greater force to disconnect the cable from the electronic device (e.g., remove the plug connector from the electrical port). In other words, a force required to withdraw the plug connector from the electrical port can be sufficiently large to prevent or limit unwanted or accidental removal of the cable from the electronic device. In some examples, the first engagement feature and the second engagement feature can be spring-driven features that are actuated by a tool in order to release the plug connector from the electrical port. In other words, a tool actuation to withdraw the plug connector from the electrical port can prevent or limit unwanted or accidental removal of the cable from the electronic device.

[3507] While the electronic devices are described herein as HMDs, the electronic devices can include a smart phone, a laptop, a tablet computing device, a smart watch, headphones, or any other

electronic device.

[3508] These and other examples are discussed below with reference to FIGS. **14.1-1A** through **14.1-35B**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[3509] FIGS. **14.1-1A** and **14.1-1B** show a first electronic device **14.1-100**, a cable assembly **14.1-140**, and a second electronic device **14.1-130**. In some examples, the first electronic device **14.1-100** can be a head-mountable display (HMD) including a display portion **14.1-102** and one or more supports **14.1-120**. While the first electronic device **14.1-100** is illustrated as an HMD, the first electronic device **14.1-100** can be a tablet computing device, smart phone, smart watch, or any other electronic device in other examples. The display portion **14.1-102** can output visual content viewable by a user of the electronic device **14.1-100**. For example, the display portion **14.1-102** can include a light-emitting diode (LED) display, an organic light-emitting diode (OLED) display, a liquid-crystal display (LCD) display, a micro-LED display, or the like. In some examples, the display portion **14.1-102** can be any form of display now known in the art, or as can be developed in the future.

[3510] The supports **14.1-120** can retain the electronic device **14.1-100** relative to a user's head **14.1-110**. In some examples, the first electronic device **14.1-100** can include a first support **14.1-120** disposed on a first side of the user's head **14.1-110** and illustrated in FIGS. **14.1-1A** and **14.1-1B**, and a second support **14.1-120** (not separately illustrated) coupled to the display portion **14.1-102** and disposed on a second side of the user's head **14.1-110** opposite the side of the user's head **14.1-110** illustrated in FIGS. **14.1-1A** and **14.1-1B**. In some examples, a single support **14.1-120** can be included, and can be a band **14.1-123** or can include a band portion coupled to the display portion **14.1-102**. The support **14.1-120** can be configured to wrap around or otherwise encircle a portion of the user's head **14.1-110** and can be coupled to the display portion **14.1-102** at two or more locations. In some examples in which the support **14.1-120** includes a band **14.1-123** or band portion, the support **14.1-120** can be made of an elastomer material which can flex or stretch and thereafter return to an initial state. In other examples, the supports **14.1-120** can be separate structural straps including electronics, processors, speakers, plugs, and/or other functional elements. These structural straps can then be connected to a band **14.1-123** that flexibly engages a user's head to impart a retention force on the system, in connection with facial engagement points. As used herein, the supports will be described as separate structural straps including wires, electronics, speakers, and/or other functional elements.

[3511] The supports **14.1-120** can include plug connectors **14.1-122**, which can be coupled to receptacle connectors **14.1-104** of the display portion **14.1-120**. The receptacle connectors **14.1-104** can include recesses that are configured to receive portions of the supports **14.1-120**, such as the plug connectors **14.1-122**. As illustrated in FIGS. **14.1-1A**, the plug connectors **14.1-122** can be disposed at proximal ends of the supports **14.1-120**. The plug connectors **14.1-122** and the receptacle connectors **14.1-104** can each include one or more electrical contacts (not separately illustrated) that electrically couple the plug connectors **14.1-122** and the supports **14.1-120** to the display portion **14.1-102**. In other words, as shown in FIGS. **14.1-1B**, the supports **14.1-120** can be electrically coupled to the display portion **14.1-102** through the plug connectors **14.1-122** and the receptacle connectors **14.1-104** when the plug connectors **14.1-122** are coupled to the receptacle connectors **14.1-104** (e.g., the plug connectors **14.1-122** are at least partially disposed within

recesses of the receptacle connectors **14.1-104**). Examples of the plug connectors **14.1-122** and the receptacle connectors **14.1-104** will be discussed in greater detail below with reference to FIGS. **14.1-2** through **14.1-15**. In some examples, the supports **14.1-120** can be welded, adhered, fastened, crimped, clipped, mechanically engaged with, or otherwise retained by the display portion **14.1-102**.

[3512] The supports **14.1-120** can each include an enclosure **14.1-121** (also referred to as a housing) formed from a polymer, metal, ceramic, or combinations thereof. In some examples, the enclosure **14.1-121** can form a channel or cavity extending between a receptacle connector **14.1-124** of each support **14.1-120** and the plug connector **14.1-122**. The supports **14.1-120** can be electrically coupled to the plug connector **14.1-122** (and the display portion **14.1-102** through the plug connector **14.1-122** and the receptacle connector **14.1-104**) such that electrical signals and/or electrical power received at the receptacle connector **14.1-124** can be provided to the plug connector **14.1-122** or other electronic components of the first electronic device **14.1-100**. For example, one or more electronic components (e.g., printed circuit boards, processors, electrical wires, digital logic circuitry, digital processing circuitry, or the like) can be positioned within the cavity formed within the enclosure **14.1-121** and extend between the receptacle connector **14.1-124** and the plug connector **14.1-122** to form an electrical path between the receptacle connector **14.1-124** and the plug connector **14.1-122**.

[3513] Each of the supports **14.1-120** can include a receptacle connector **14.1-124** disposed on or within an enclosure **14.1-121**. The receptacle connector **14.1-124** can form a recess **14.1-126** that can receive a portion of the cable assembly **14.1-140**. The cable assembly **14.1-140** can include a plug connector **14.1-142** at a distal end of the cable assembly **14.1-140**. The plug connector **14.1-142** can include a boot **14.1-143** and a central member **14.1-145** at least partially disposed within the boot **14.1-143**. The central member **14.1-145** can include one or more electrical contacts (not separately illustrated) that electrically couple the plug connector **14.1-142** to the first electronic device **14.1-100**. In other words, as shown in FIGS. **14.1-1B**, the first electronic device **14.1-100** can be electrically coupled to the second electronic device **14.1-130** through the cable assembly **14.1-140** when the plug connector **14.1-142** is at least coupled to the receptacle connector **14.1-124** (e.g., the central member **14.1-145** is at least partially disposed within the recess **14.1-126**). Examples of the receptacle connector **14.1-124** and the plug connector **14.1-142** will be discussed in greater detail below with reference to FIGS. **14.1-16A** through **14.1-25B**.

[3514] As illustrated in FIGS. **14.1-1A** and **14.1-1B**, each receptacle connector **14.1-124** can be positioned on or within the enclosure **14.1-121** of a respective support **14.1-120**. In some examples, the receptacle connector **14.1-124** can be positioned on the enclosure **14.1-121** and disposed a distance from the display portion **14.1-102**. The plug connector **14.1-142** can be operably coupled to the enclosure **14.1-121** (e.g. . . . , through the receptacle connector **14.1-124** of the enclosure **14.1-121**) rather than directly coupling the plug connector **14.1-142** to the display portion **14.1-102**. The receptacle connector **14.1-124** can be disposed on or within the enclosure **14.1-121** such that the display portion **14.1-102** and the receptacle connector **14.1-124** are disposed on opposing sides of a user's ear **14.1-112**. In some examples, the display portion **14.1-102** can be coupled to a proximal end of the support **14.1-120** and the receptacle connector **14.1-124** can be positioned on or within a distal end of the support **14.1-120** opposite the display portion **14.1-102**. In some examples, the receptacle connector **14.1-124** can be disposed on or within the enclosure **14.1-121** between the proximal end and the distal end of the support **14.1-120** such that the display portion **14.1-102** and the user's ear **14.1-112** are disposed on opposing sides of the receptacle connector **14.1-124** (as shown in FIGS. **14.1-1A** and **14.1-1B**). In some examples, the receptacle connector **14.1-124** can be disposed within the enclosure **14.1-121** at a location that is closer to the distal end than the proximal end. In some examples, the receptacle connector **14.1-124** can be disposed within the enclosure **14.1-121** at a location that is closer to the proximal end than the distal end. In some examples, the receptacle connector **14.1-124** can be disposed behind the user's ear **14.1-112** such

that the user's ear **14.1-112** is between the display portion **14.1-102** and the receptacle connector **14.1-124**.

[3515] Some HMDs utilize a re-chargeable power source (e.g., a battery) affixed to the HMD to provide electrical power to the electronic components (e.g., processors, displays, speakers, and the like). The size or capacity of the re-chargeable power source can be limited by the desired size, shape, and weight of the HMD. After the re-chargeable power source has been substantially depleted of electrical power, the user can be required to discontinue use of the HMD to allow re-charging of the electrical power source.

[3516] In some aspects of the present disclosure, rather than solely relying on a power source disposed within the HMD (e.g., the first electronic device **14.1-100**), at least one power source (e.g., the second electronic device **14.1-130**) can be additionally, or alternatively, electrically coupled to the HMD by a cabled connection (e.g., the cable assembly **14.1-140**), which is electrically coupled to the enclosure **14.1-121** of the support **14.1-120** via the receptacle connector **14.1-124**. Electrically coupling the power source (e.g., the second electronic device **14.1-130**) to the receptacle connector **14.1-124** of the support **14.1-120** can be advantageous. For example, electrical power and/or electrical signals can be provided to the first electronic device **14.1-100**, while the second electronic device **14.1-130** is disposed within a case, pocket, pouch, or otherwise retained by the user. Relocating the electrical power source away from the first electronic device **14.1-100** can accommodate a larger electrical power source than can be directly disposed on the first electronic device **14.1-100**, which can provide for extended use of the first electronic device **14.1-100**. Positioning the receptacle connector **14.1-124** between the proximal end and distal end of the support **14.1-120** can be beneficial in at least partially limiting user contact with the cable assembly **14.1-140** by positioning the cable assembly **14.1-140** at the side of the user rather than dangling or hanging the cable assembly **14.1-140** in front of the user near the first electronic device **14.1-100**.

[3517] Electrically coupling a power source (e.g., the second electronic device **14.1-130**) to the receptacle connector **14.1-124** of the support **14.1-120**, as opposed to disposing the power source directly on the display portion **14.1-102** can also enable a reduction in the weight and/or a size of the display portion **14.1-102**. A reduction in the weight and/or size of the display portion **14.1-102** can render the first electronic device **14.1-100** more comfortable during use, more convenient to transport, and more convenient to store.

[3518] While the receptacle connectors **14.1-124** and the plug connectors **14.1-142** are shown in FIGS. **14.1-1A** and **14.1-1B** as having cubic shape or square cross-sections, the receptacle connectors **14.1-124** and the plug connectors **14.1-142** can define other shapes and cross-sections in some examples. For example, the cross-sections of the receptacle connectors **14.1-124** and the plug connectors **14.1-142** can be circular or rounded to enable rotation of the plug connectors **14.1-142** relative to the receptacle connectors **14.1-124**. In other words, the plug connectors **14.1-142** can be rotatably received within the receptacle connectors **14.1-124**.

[3519] In some examples, the second electronic device **14.1-130** can provide electrical power and/or electrical signals to the first electronic device **14.1-100** through the cable assembly **14.1-140**. For example, the second electronic device **14.1-130** can be an external electrical power source, such as, a lithium battery pack or other device capable of supplying electrical power to the first electronic device **14.1-100**. The cable assembly **14.1-140** can include a plug connector **14.1-144**, which can be coupled to a receptacle connector **14.1-132** of the second electronic device **14.1-130**.

[3520] The plug connector **14.1-144** can include a boot **14.1-146** and a plug **14.1-147** at least partially disposed within the boot **14.1-146**. The receptacle connector **14.1-132** of the second electronic device **14.1-130** can include a recess configured to receive portions of the plug connector **14.1-144**. The plug **14.1-147** can include one or more electrical contacts (not separately illustrated) that electrically couple the plug connector **14.1-144** to the receptacle connector **14.1-132** of the second electronic device **14.1-130**. In other words, as shown in FIGS. **14.1-1B**, the second first

electronic device **14.1-130** can be electrically coupled to the cable assembly **14.1-140** through the plug connector **14.1-144** and the receptacle connector **14.1-132**. The second electronic device **14.1-130** can be electrically coupled to the first electronic device **14.1-100** through the cable assembly **14.1-140** and the supports **14.1-120**. Examples of the receptacle connector **14.1-132** and the plug connector **14.1-144** will be discussed in greater detail below with reference to FIGS. **14.1-26A** through **14.1-35B**.

[3521] As will be discussed in detail below, the various plug connectors and receptacle connectors included in an HMD (e.g., coupling the first electronic device **14.1-100** and the second electronic device **14.1-130**) can be fastened to one another through various fastening mechanisms. In some examples, the fastening mechanisms can require a relatively small force to mate a plug connector and a receptacle connector and can require a relatively large force to un-mate the plug connector and the receptacle connector. In some examples, a tool actuation can be required to un-mate a plug connector and a receptacle connector once the plug connector and the receptacle connector are mated. In some examples, various seals can be provided on plug connectors and/or receptacle connectors to provide sealing between the plug connectors and receptacle connectors. The seals can provide forces between the plug connectors and receptacle connectors, such as ejection forces to aid in the removal of plug connectors from receptacle connectors.

[3522] FIGS. **14.1-2** through **14.1-15** illustrate various examples of receptacle connectors and plug connectors that can be used as the receptacle connectors **14.1-104** of the display portion **14.1-102** and the plug connectors **14.1-122** of the supports **14.1-120**.

[3523] FIGS. **14.1-2** shows a first electronic device **14.1-200** including a display portion **14.1-202**, supports **14.1-210**, and a cable assembly **14.1-220**. In some examples, the first electronic device **14.1-200** can be a head-mountable device (HMD). While the first electronic device **14.1-200** is illustrated as an HMD, the first electronic device **14.1-200** can be a tablet computing device, a smart phone, a smart watch, headphones, or any other electronic device. The display portion **14.1-202** can output visual content viewable by a user of the first electronic device **14.1-200**. The first electronic device **14.1-200**, the display portion **14.1-202**, the supports **14.1-210**, and the cable assembly **14.1-220** can be similar to or the same as the first electronic device **14.1-100**, the display portion **14.1-102**, the supports **14.1-120**, and the cable assembly **14.1-140**, respectively, discussed above with respect to FIGS. **14.1-1A** and **14.1-1B**.

[3524] The supports **14.1-210** can retain the first electronic device **14.1-200** relative to a user's head. As illustrated in FIGS. **14.1-2**, the first electronic device **14.1-200** can include two supports **14.1-210** coupled to the display portion **14.1-202** and configured to be positioned on either side of the user's head. The supports **14.1-210** can be connected to the display portion **14.1-202** at two or more locations.

[3525] The supports **14.1-210** can each include a housing or an enclosure **14.1-212** formed from a polymer, metal, ceramic, combination thereof, or the like. In some examples, the enclosure **14.1-212** can form a channel or cavity extending between a receptacle connector **14.1-222** of the support **14.1-210** and a plug connector **14.1-214** of the support **14.1-210**. The support **14.1-210** can be electrically coupled to the display portion **14.1-202** such that electrical signals and/or electrical power received at the receptacle connector **14.1-222** can be provided to the display portion **14.1-202** through the plug connector **14.1-214**. For example, one or more electronic components (e.g., printed circuit boards, processors, electrical wires, digital logic circuitry, digital processing circuitry, or the like) can be positioned within the cavity formed within the enclosure **14.1-212** and extend between the receptacle connector **14.1-222** and the plug connector **14.1-214** to form an electrical path between the receptacle connector **14.1-222** and the plug connector **14.1-214**.

[3526] In some examples, the support **14.1-210** can be coupled to the display portion **14.1-202**. For example, the support **14.1-210** can be welded, adhered, fastened, crimped, clipped, or otherwise retained within a receptacle connector **14.1-206** of the display portion **14.1-202**. In some examples, a proximal end of the support **14.1-210** can be electrically conductive or include the plug connector

14.1-214 that enables electrical power and/or electrical signals received by the receptacle connector **14.1-222** of the support **14.1-210** to be transferred to the display portion **14.1-202** through the plug connector **14.1-214** and the receptacle connector **14.1-206**.

[3527] As illustrated in FIGS. **14.1-2**, each of the supports **14.1-210** can include a plug connector **14.1-214** disposed at a proximal end of the support **14.1-210**. The plug connector **14.1-214** can be coupled (e.g., electrically and structurally) to the display portion **14.1-202** by disposing the proximal end of the support **14.1-210** into a receptacle connector **14.1-206** coupled to a housing **14.1-204** of the display portion **14.1-202**. The receptacle connector **14.1-206** can be positioned on or within the housing **14.1-204** of the display portion **14.1-202**. In some examples, the receptacle connector **14.1-206** of the display portion **14.1-202** can be in electrical communication with the receptacle connector **14.1-222** of the support **14.1-210**. That is, the cable assembly **14.1-220** can operably couple to the receptacle connector **14.1-222** of the support **14.1-210** rather than directly coupling to the display portion **14.1-202** to electrical power and/or electrical signals can be transferred between the receptacle connector **14.1-222** of the support **14.1-210** and the receptacle connector **14.1-206** of the display portion **14.1-202**.

[3528] The plug connector **14.1-214** can include one or more electrical contacts **14.1-216A**, **14.1-216B**, **14.1-216C** which electrically couple to respective electrical contacts **14.1-208A**, **14.1-208B**, and **14.1-208C** disposed within the receptacle connector **14.1-206** while the plug connector **14.1-214** is disposed within the receptacle connector **14.1-206**. The one or more electrical contacts **14.1-216A**, **14.1-216B**, **14.1-216C** and the correlating one or more electrical contacts **14.1-208A**, **14.1-208B**, **14.1-208C** can provide electrical signals, electrical power, a grounding path, another electrical communication, or a combination thereof between the support **14.1-210** and the display portion **14.1-202**.

[3529] Electrically coupling a power source (e.g., the second electronic device **104**) to the receptacle connector **14.1-222** within the support **14.1-210**, as opposed to disposing the power source directly on the display portion **14.1-202** can also enable a reduction in the weight and/or a size of the display portion **14.1-202**. A reduction in the weight and/or size of the display portion **14.1-202** can render the first electronic device **14.1-200** more comfortable during use, more convenient to transport, and more convenient to store. While the receptacle connector **14.1-206** and the plug connector **14.1-214** are shown in FIGS. **14.1-2** as having a cubic shape or square cross-section, the receptacle connector **14.1-206** and the plug connector **14.1-214** can define other shapes and cross-sections in other examples. For example, the cross-section of the receptacle connector **14.1-206** and the plug connector **14.1-214** can be elongated and/or curved such that the support **14.1-210** can only be inserted into the receptacle connector **14.1-206** of the display portion **14.1-202** in a single orientation.

[3530] FIGS. **14.1-3A** and **14.1-3B** show perspective views of a receptacle connector **14.1-300** and a plug connector **14.1-320**. The receptacle connector **14.1-300** can be used as the receptacle connector **14.1-104** or the receptacle connector **14.1-206** and the plug connector **14.1-320** can be used as the plug connector **14.1-122** or the plug connector **14.1-214** of the first electronic devices **14.1-100/14.1-200** of FIGS. **14.1-1A** through **14.1-2**. The receptacle connector **14.1-300** can include a receptacle housing **14.1-304**, one or more electrical contacts (e.g., electrical contacts **14.1-308A**, **14.1-308B**, and **14.1-308C**), a cover member **14.1-302**, and a printed circuit board (PCB) **14.1-316**.

[3531] In some examples, the receptacle housing **14.1-304** can be manufactured from an electrically insulating material, such as, a polymer or ceramic and can be molded, machined, cast, stamped, or a combination thereof. The receptacle housing **14.1-304** can one include or more sidewalls defining a recess **14.1-310**. The recess **14.1-310** can be configured to accept and retain the plug connector **14.1-320**. For example, at least a portion of the plug connector **14.1-320** can be inserted into and received within the recess **14.1-310**. The recess **14.1-310** can define a cross-sectional shape having a longitudinal axis that curves or bends. For example, the recess **14.1-310**

can have a cross-sectional shape that substantially conforms to a cross-sectional shape of a plug connector (e.g., the plug connector **14.1-320**). In other words, the sidewalls of the receptacle housing **14.1-304** can define a convex surface and a concave surface that correlate in size, shape, and contour with convex and concave surfaces of the plug connector **14.1-320**.

[3532] The one or more electrical contacts (e.g., the electrical contacts **14.1-308A**, **14.1-308B**, and **14.1-308C**) can extend from a sidewall of the receptacle connector **14.1-300** into the recess **14.1-310**. The electrical contacts can provide electrical signals, electrical power, a grounding path, another electrical communication, or a combination thereof to the PCB **14.1-316**. In some examples, a first set of electrical contacts, including the electrical contacts **14.1-308A**, **14.1-308B**, and **14.1-308C**, can be disposed on a first side of the recess **14.1-310** and a second set of electrical contacts (not separately illustrated) can be disposed on a second, opposite, side of the recess **14.1-310**. As will be discussed in detail below, the electrical contacts can be attached to the receptacle housing **14.1-304** at angles normal to a surface of the plug connector **14.1-320**, which the electrical contacts are configured to contact. This provides improved contact between the electrical contacts of the receptacle connector **14.1-300** and electrical contacts of the plug connector **14.1-320**.

[3533] In FIGS. **14.1-3B**, the plug connector **14.1-320** can define a top surface and a bottom surface. The top surface can be a convex external surface or otherwise form a radius of curvature. The bottom surface can be a concave external surface extending parallel or substantially parallel to the top surface. One or more electrical contacts (e.g., electrical contacts **14.1-324A**, **14.1-324B**, and **14.1-324C**) can be disposed on the top surface and/or the bottom surface of the plug connector **14.1-320**. In some examples, the electrical contacts can be provided on the bottom surface of the plug connector **14.1-320**, and the top surface of the plug connector **14.1-320** can be free from any electrical contacts; or the electrical contacts can be provided on the top surface of the plug connector **14.1-320**, and the bottom surface of the plug connector **14.1-320** can be free from any electrical contacts.

[3534] In some examples, the plug connector **14.1-320** can include one or more guide channels **14.1-328** and one or more retention channels **14.1-326**. The guide channels **14.1-328** and/or the retention channels **14.1-326** can be formed within the top surface, the bottom surface, or other surfaces (e.g., side surfaces) of the plug connector **14.1-320**. The guide channels **14.1-328** can receive guide rails (such as guide rails **14.1-313**, illustrated in FIGS. **14.1-5A**) of a receptacle connector of a display portion to limit or inhibit movement between a support including the plug connector **14.1-320** and the display portion. The guide channels **14.1-328** can also be beneficial to enable insertion of the plug connector **14.1-320** into a receptacle connector in a limited combination of configurations. For example, a guide rail will prevent insertion of the plug connector **14.1-320** within a receptacle connector unless the plug connector **14.1-320** has a correlating guide channel **14.1-328**.

[3535] The retention channels **14.1-326** can receive latches **14.1-312** of the receptacle connector **14.1-300** in order to retain the plug connector **14.1-320** within the recess **14.1-310** of the receptacle connector **14.1-300**. The retention channels **14.1-326** and the latches **14.1-312** can prevent undesired removal of the plug connector **14.1-320** from the recess **14.1-310**. In some examples, the retention channels **14.1-326** and the latches **14.1-312** can retain the plug connector **14.1-320** in the recess **14.1-310** of the receptacle connector **14.1-300** in a position in which electrical contacts of the plug connector **14.1-320** are aligned with electrical contacts of the receptacle connector **14.1-300**.

[3536] As will be discussed in detail below, the receptacle connector **14.1-314** can include a push block **14.1-314** operably coupled to the latches **14.1-312**. The push block **14.1-314** can be actuated by a tool and actuating the push block **14.1-314** can rotate the latches **14.1-312** out of the recess **14.1-310** such that the plug connector **14.1-320** can be removed from the recess **14.1-310**. The latches **14.1-312** can be configured such that a large force is required to pull the plug connector **14.1-320** from the recess **14.1-310**, but a small force is required to pull the plug connector **14.1-320**

from the recess **14.1-310** when the push block **14.1-314** is actuated and the latches **14.1-312** release the plug connector **14.1-320**. This allows for the plug connector **14.1-320** to be removed when needed but prevents the plug connector **14.1-320** from being accidentally or unintentionally removed from the receptacle connector **14.1-300**.

[3537] The guide channels **14.1-328** and the retention channels **14.1-326** are illustrated with partial cylindrical shapes. However, any suitable shapes can be used for the guide channels **14.1-328** and the retention channels **14.1-326**, and the guide channels **14.1-328** and the retention channels **14.1-326** can have square, rectangular, semi-circular, triangular, U-shaped, V-shaped, other geometric-shaped, combinations thereof, or the like cross-sectional shapes. In some examples, the cross-sectional shape of the guide channels **14.1-328** and the retention channels **14.1-326** can vary along a length of the channels. For example, the guide channels **14.1-328** and the retention channels **14.1-326** can form rectangular cross-sectional shapes that taper to square cross-section shapes along the length of the respective channels.

[3538] The receptacle connector **14.1-300** can include a face seal **14.1-306** adjacent the recess **14.1-310**. The face seal **14.1-306** can encircle at least a portion of the recess **14.1-310**, such as a distal portion of the recess **14.1-310**. The face seal **14.1-306** can act as a seal which prevents or limits ingress of contaminants (e.g., liquid, dust, lint, debris, or the like) into the receptacle connector **14.1-300** through the recess **14.1-310**, such as when the plug connector **14.1-320** is inserted into the recess **14.1-310** of the receptacle connector **14.1-300**. In some examples, the face seal **14.1-306** can be configured to provide an ejection force to the plug connector **14.1-320**. This can aid a user in removing the plug connector **14.1-320** from the receptacle connector **14.1-300**, such as when the push block **14.1-314** is actuated.

[3539] The cover member **14.1-302** can include an upper portion and a lower portion. The upper portion and the lower portion can be machined, molded, stamped, extruded, or otherwise manufactured from one or more materials, such as a metal, a ceramic, or a polymer. In some examples, the lower portion can at least partially form the holes, which can be used to mount the receptacle connector to a housing or enclosure of a display portion. Each of the upper portion and the lower portion can provide a support structure for the receptacle connector **14.1-300**. The upper portion and the lower portion can shield the receptacle connector **14.1-300** from propagating errant electromagnetic waves out of the receptacle connector **14.1-300**.

[3540] The PCB **14.1-316** can be electrically coupled to the electrical contacts of the receptacle connector **14.1-300** (e.g., the electrical contacts **14.1-308A**, **14.1-308B**, and **14.1-308C**). For example, the PCB **14.1-316** can include one or more electrical traces which carry electrical signals and/or electrical power from the electrical contacts to an electronic component (e.g., a processor, electrical wires, digital logic circuitry, digital processing circuitry, or the like) disposed on the PCB **14.1-316**. The PCB **14.1-316** and/or the one or more electronic components electrically coupled to the PCB **14.1-316** can be electrically coupled to one or more electrical wires (not shown) to form at least a portion of an electrical path for electrical signals, electrical power, an electrical ground, another electrical communication, or a combination thereof to transfer between the receptacle connector **14.1-300** and a display or other electronic components within the display portion (e.g., the display portion **14.1-102/14.1-202**). Similarly, the PCB **14.1-316** can form at least a portion of an electrical path for electrical signals, electrical power, an electrical ground, another electrical communication, or a combination thereof to transfer between the receptacle connector **14.1-300** and the plug connector **14.1-320**.

[3541] FIGS. **14.1-4** shows an exploded view of the receptacle connector **14.1-300** including the receptacle housing **14.1-304**. The receptacle housing **14.1-304** can form slits or slots within the sidewalls of the receptacle housing **14.1-304**, which enable a first set of electrical contacts (e.g., the electrical contacts **14.1-308A**, **14.1-308B**, and **14.1-308C**) to extend through the sidewalls and into the recess **14.1-310**. In some examples, the receptacle housing **14.1-304** can further include slits or slots within the sidewalls, which enable a second set of electrical contacts (e.g., electrical contacts

14.1-309A, **14.1-309B**, and **14.1-309C**) to extend through the sidewalls of the receptacle housing **14.1-304** and into the recess **14.1-310**. The second set of electrical contacts can extend into the recess **14.1-310** through a different sidewall of the receptacle housing **14.1-304** from the first set of electrical contacts. In some examples, the plug connector **14.1-320** can be devoid of electrical contacts that make contact with both the first set of electrical contacts and the second set of electrical contacts. In other words, one example of a support can include electrical contacts that only utilize one of the sets of electrical contacts shown in FIGS. **14.1-4**. Another example of a support can include electrical contacts that utilize both sets of electrical contacts shown in FIGS. **14.1-4**. Thus, a single configuration of the receptacle connector **14.1-300** having two sets of electrical contacts can be operable with multiple configurations of supports (e.g., supports having varied electrical contact configurations).

[3542] As shown in FIGS. **14.1-4**, in some examples, each of the electrical contacts of the first set of electrical contacts can be provided with a corresponding electrical contact of the second set of electrical contacts. Each set of corresponding electrical contacts can be disposed in and coupled to the receptacle housing **14.1-304** at an oblique angle to a longitudinal axis of the recess **14.1-310** (e.g., a lateral axis of the receptacle housing **14.1-304**). Each set of electrical contacts can be coupled to the receptacle housing **14.1-304** at an angle normal to a surface of the plug connector **14.1-310** that the electrical contacts are configured to contact. This improves the electrical contact between the electrical contacts of the receptacle connector **14.1-300** and the plug connector **14.1-320**. Each set of electrical contacts can also be disposed in and coupled to the receptacle housing **14.1-304** at an angle oblique to other sets of the electrical contacts, such as adjacent sets of electrical contacts.

[3543] The receptacle connector **14.1-300** can include a fastening mechanism for retaining the plug connector **14.1-320** in the recess **14.1-310**. In the example illustrated in FIGS. **14.1-4**, the fastening mechanism includes a push block **14.1-314**, a spring **14.1-318**, a guide rail **14.1-319**, and latches **14.1-312**. The latches **14.1-312** extend through openings in the receptacle housing **14.1-304** into the recess **14.1-310**. The latches **14.1-312** are pivotally coupled to the receptacle housing **14.1-304**. The latches **14.1-312** can be in an engaged configuration when the latches **14.1-312** are pivoted into the recess **14.1-310**, and in an un-engaged configuration when the latches are pivoted out of the recess **14.1-310**.

[3544] The push block **14.1-314** is coupled to the latches **14.1-312**, the spring **14.1-318**, and the guide rail **14.1-319**. The push block **14.1-314** is configured to slide along the guide rail **14.1-319** and rotate the latches **14.1-312** between the engaged configuration and the un-engaged configuration. The spring **14.1-318** is configured to retain the push block **14.1-314** in an initial position in which the latches **14.1-312** are in the engaged configuration. A user can use a tool or the like to apply force to the push block **14.1-314** and push the push block **14.1-314** along the guide rail **14.1-319**. This rotates the latches **14.1-312** to the un-engaged configuration. Once the latches are in the un-engaged configuration, a plug connector **14.1-320** can be released from the recess **14.1-310** and removed from the receptacle connector **14.1-300**. The spring **14.1-318** resists force applied to the push block **14.1-314** and movement of the push block **14.1-314** and returns the push block **14.1-314** to the initial position with the latches **14.1-312** in the engaged configuration once the force is no longer applied to the push block **14.1-314**. As such, the latches **14.1-312**, the push block **14.1-314**, the spring **14.1-318**, and the guide rail **14.1-319** are used to retain the plug connector **14.1-320** in the recess **14.1-310** of the receptacle connector **14.1-300** and release the plug connector **14.1-320** from the receptacle connector **14.1-300**.

[3545] The face seal **14.1-306** can be coupled to a front surface of the receptacle housing **14.1-304** and can encircle the recess **14.1-310**. The face seal **14.1-306** can be molded, adhered, welded, fastened, or otherwise secured to a surface of the receptacle housing **14.1-304**. In some examples, the face seal **14.1-306** can include an elastic material that can be stretched around a lip of the receptacle housing **14.1-304** and can be secured to the receptacle housing **14.1-304** by interference

with the receptacle housing **14.1-304**. Bumpers **14.1-307** can be coupled to the receptacle housing **14.1-304** at a back surface of the recess **14.1-310**. The bumpers **14.1-307** can be molded, adhered, welded, fastened, or otherwise secured to a surface of the receptacle housing **14.1-304**. The bumpers **14.1-307** and the face seal **14.1-306** can provide an ejection force to a plug connector **14.1-320** inserted into the recess **14.1-310** of the receptacle connector **14.1-300**. For example, the bumpers **14.1-307** and the face seal **14.1-306** can provide resistance to the plug connector **14.1-320** as the plug connector **14.1-320** is inserted into the recess **14.1-310** and can help to eject the plug connector **14.1-320** from the recess **14.1-310** when the latches **14.1-312** are rotated to the un-engaged configuration. This can allow the plug connector **14.1-320** to be ejected from the recess by actuating the push block **14.1-314** with a tool, and without requiring a user to otherwise pull on the plug connector **14.1-320** or the like. The bumpers **14.1-307** and the face seal **14.1-306** can also retain the electrical contacts of the plug connector **14.1-320** in a desired position relative to the electrical contacts of the receptacle connector **14.1-300**.

[3546] The cover member **14.1-302** can include an upper portion **14.1-302A**, a lower portion **14.1-302B**, front portions **14.1-302C**, and a rear portion **14.1-302D**. Each of the portions of the cover member **14.1-302** can be coupled to the receptacle housing **14.1-304**. For example, the portions of the cover member **14.1-302** can adhered (such as by a polymer-based adhesive tape), fastened, welded, or otherwise affixed to the receptacle housing **14.1-304**. The cover member **14.1-302** can provide shielding and a support structure for the receptacle housing **14.1-304**.

[3547] FIGS. **14.1-5A** and **14.1-5B** illustrate front views of the receptacle connector **14.1-300**. As illustrated in FIGS. **14.1-5A**, the face seal **14.1-306** can encircle the recess **14.1-310** of the receptacle housing **14.1-304**. The bumpers **14.1-307** can be disposed at a rear surface of the receptacle housing **14.1-304** adjacent the recess **14.1-310**. When the plug connector **14.1-320** is inserted into the recess **14.1-310**, the bumpers **14.1-307** can be disposed between the plug connector **14.1-320** and a rear inner surface of the receptacle housing **14.1-304**. The bumpers **14.1-307** and the face seal **14.1-306** can each provide an ejection force to the plug connector **14.1-320**, as discussed previously.

[3548] As illustrated in FIGS. **14.1-5A**, the receptacle housing **14.1-304** can include one or more guide rails **14.1-313**. Each guide rail **14.1-313** can be disposed on a respective sidewall of the receptacle housing **14.1-304** and can extend into the recess **14.1-310**. The guide rails **14.1-313** can be molded, co-molded, or otherwise formed with the receptacle housing **14.1-304** as a singular and unitary structure. The guide rails **14.1-313** can be received within guide channels **14.1-328** of the plug connector **14.1-320** to provide a rigid connection between a support including the plug connector **14.1-320** and a display portion including the receptacle connector **14.1-300**. In other words, the guide rails **14.1-313** and the guide channels **14.1-328** can limit motion or free play between the support and the display portion. Additionally, or alternatively, the guide rails **14.1-313** can limit or prevent the plug connector **14.1-320** from being inserted into the receptacle connector **14.1-300** in an undesirable orientation and/or configuration.

[3549] FIGS. **14.1-5B** illustrates orientations of the electrical contacts (e.g., the electrical contacts **14.1-308A**, **14.1-308B**, **14.1-308C**, **14.1-309A**, **14.1-309B**, and **14.1-309C**) of the receptacle connector **14.1-300** relative to the receptacle housing **14.1-304**. As illustrated in FIGS. **14.1-5B**, each of the electrical contacts can be disposed at an angle oblique to a lateral axis and a vertical axis of the receptacle housing **14.1-304**. The electrical contacts can be disposed such that surfaces of the electrical contacts configured to contact a plug connector **14.1-320** are normal to surfaces of the plug connector **14.1-320** those electrical contacts are configured to contact. This provides for improved contact between the electrical contacts of the receptacle connector **14.1-300** and electrical contacts of the plug connector **14.1-320**, as surfaces of the electrical contacts of the receptacle connector **14.1-300** rather than just edges or corners are configured to contact the plug connector **14.1-320**. Each of the electrical contacts of the receptacle connector **14.1-300** can also be disposed at an angle oblique to adjacent electrical contacts.

[3550] FIGS. **14.1-6A** and **14.1-6B** illustrate side sectional views of receptacle connectors. In FIGS. **14.1-6A**, a receptacle connector **14.1-400A** includes an electrical contact **14.1-402A** vertically stitched to a receptacle housing **14.1-404A**. The electrical contact **14.1-402A** can have a longitudinal axis extending in a horizontal direction, and the electrical contact **14.1-402A** can be stitched to the receptacle housing **14.1-404A** in a vertical direction. The electrical contact **14.1-402A** can be stitched to the receptacle housing **14.1-404A** in a direction perpendicular to an insertion direction of a plug connector. The electrical contact **14.1-402A** is configured to contact the plug connector when the plug connector is inserted into a recess **14.1-410** of the receptacle housing **14.1-404A**.

[3551] A cover member **14.1-406** can be provided around the receptacle housing **14.1-404A** to provide support for and shielding of the receptacle housing **14.1-404A**. A face seal **14.1-408** can be provided along an inner surface of the receptacle housing **14.1-404A** adjacent the recess **14.1-410** into which a plug connector is to be inserted. A bumper **14.1-412** can be provided along an inner surface of the receptacle housing **14.1-404A** adjacent the recess **14.1-410**, at a back surface of the recess **14.1-410** relative to an opening of the recess **14.1-410** through which the plug connector is to be inserted. The bumper **14.1-412** provides a stop when the plug connector is inserted into the recess **14.1-410**, and the bumper **14.1-412** and the face seal **14.1-408** can provide ejection forces to the plug connector when the plug connector is un-latched from the receptacle connector **14.1-400A**. The bumper **14.1-412** and the face seal **14.1-408** are configured to be disposed between the plug connector and the receptacle housing **14.1-404A** when the plug connector is inserted into the recess **14.1-410**.

[3552] In FIGS. **14.1-6B**, a receptacle connector **14.1-400B** includes an electrical contact **14.1-402B** horizontally stitched to a receptacle housing **14.1-404B**. The electrical contact **14.1-402B** can have a longitudinal axis extending in a horizontal direction, and the electrical contact **14.1-402B** can be stitched to the receptacle housing **14.1-404B** in the horizontal direction. The electrical contact **14.1-402B** can be stitched to the receptacle housing **14.1-404B** in a direction parallel to an insertion direction of a plug connector. The electrical contact **14.1-402B** is configured to contact the plug connector when the plug connector is inserted into a recess **14.1-410** of the receptacle housing **14.1-404B**.

[3553] A cover member **14.1-406** can be provided around the receptacle housing **14.1-404B** to provide support for and shielding of the receptacle housing **14.1-404B**. A face seal **14.1-408** can be provided along an inner surface of the receptacle housing **14.1-404B** adjacent the recess **14.1-410** into which a plug connector is to be inserted. A bumper **14.1-412** can be provided along an inner surface of the receptacle housing **14.1-404B** adjacent the recess **14.1-410**, at a back surface of the recess **14.1-410** relative to an opening of the recess **14.1-410** through which the plug connector is to be inserted. The bumper **14.1-412** provides a stop when the plug connector is inserted into the recess **14.1-410**, and the bumper **14.1-412** and the face seal **14.1-408** can provide ejection forces to the plug connector when the plug connector is un-latched from the receptacle connector **14.1-400B**. The bumper **14.1-412** and the face seal **14.1-408** are configured to be disposed between the plug connector and the receptacle housing **14.1-404B** when the plug connector is inserted into the recess **14.1-410**.

[3554] FIGS. **14.1-7A** and **14.1-7B** illustrate perspective views of a receptacle connector **14.1-500** and a plug connector **14.1-520**, respectively. The receptacle connector **14.1-500** includes a receptacle housing **14.1-504** defining a recess **14.1-505** and a cover member **14.1-502** coupled to the receptacle housing **14.1-504**. The recess **14.1-505** can be configured to receive the plug connector **14.1-520**. The cover member **14.1-502** can be configured to provide support and shielding for the receptacle housing **14.1-504**.

[3555] A face seal **14.1-510** can be coupled to the receptacle housing **14.1-504**, such as adjacent to the recess **14.1-505**. The face seal **14.1-510** can surround or encircle the recess **14.1-505** in the receptacle housing **14.1-504**. The face seal **14.1-510** can provide sealing between the receptacle

housing **14.1-504** and the plug connector **14.1-520** when the plug connector is received in the recess **14.1-505**. A bumper **14.1-512** can be coupled to the receptacle housing **14.1-504**. The bumper **14.1-512** can be coupled to a surface of the receptacle housing **14.1-504** at a back of the recess **14.1-505**. The bumper **14.1-512** can be configured to be disposed between the receptacle housing **14.1-504** and the plug connector **14.1-520** when the plug connector **14.1-520** is inserted into the recess **14.1-505**. The bumper **14.1-512** and/or the face seal **14.1-510** can be configured to provide an ejection force to the plug connector **14.1-520**. For example, the bumper **14.1-512** and/or the face seal **14.1-510** can resist movement of the plug connector **14.1-520** as the plug connector **14.1-520** is inserted into the recess **14.1-505**. After the plug connector **14.1-520** is latched in the recess **14.1-505**, the bumper **14.1-512** and/or the face seal **14.1-510** can provide the ejection force to the plug connector **14.1-520** such that the plug connector **14.1-520** is ejected from the recess **14.1-505** when the plug connector is un-latched from the receptacle housing **14.1-504**.

[3556] The receptacle housing **14.1-504** can further include a latch **14.1-506** rotatably or pivotally coupled to the receptacle housing **14.1-504**. The latch **14.1-506** can correspond to a retention channel **14.1-526** formed in a body portion **14.1-522** of the plug connector **14.1-520**. The latch **14.1-506** can be pivotable between an engaged configuration and an un-engaged configuration. When the latch **14.1-506** is in the engaged configuration, the latch **14.1-506** extends into the recess **14.1-505** of the receptacle housing **14.1-504**, as illustrated in FIG. **14.1-7A**. Although the latch **14.1-506** is illustrated as extending into a side portion of the recess **14.1-505**, the latch **14.1-506** can be provided in a top portion, a bottom portion, or the like of the recess **14.1-505**. When the latch **14.1-506** is in the un-engaged configuration, the latch **14.1-506** can be disposed outside of the recess **14.1-505**, or less of the latch **14.1-506** can be disposed in the recess **14.1-505**. The latch **14.1-506** can be biased in the engaged configuration by a spring or the like, and can be un-latched by a user, such as with a tool actuation.

[3557] The retention channel **14.1-526** can be disposed in a sidewall of the plug connector **14.1-520**. The retention channel **14.1-526** can be etched, drilled, molded, co-molded, or otherwise formed with the plug connector **14.1-520** as a singular and unitary structure. As illustrated in FIGS. **14.1-7B**, the retention channel **14.1-526** can be disposed in the body portion **14.1-522** at an angle perpendicular to an insertion direction **14.1-528** in which the plug connector **14.1-520** is to be inserted into the recess **14.1-505** of the receptacle connector **14.1-500**, which can aid in the retention of the plug connector **14.1-520** by the latch **14.1-506**. The latch **14.1-506** can be received within the retention channel **14.1-526** of the plug connector **14.1-520** to retain the plug connector **14.1-520** within the recess **14.1-505** of the receptacle connector **14.1-500**, when the latch **14.1-506** is in the engaged configuration. The latch **14.1-506** can then be transitioned into the un-engaged configuration to release the plug connector **14.1-520** from the recess **14.1-505** of the receptacle connector **14.1-500**.

[3558] The receptacle housing **14.1-504** can further include a guide rail **14.1-508**, which can correspond to a guide channel **14.1-524** formed in a body portion **14.1-522** of the plug connector **14.1-520**. The guide rail **14.1-508** can be disposed on a sidewall of the receptacle housing **14.1-504** and can extend into the recess **14.1-505**. The guide rail **14.1-508** can be molded, co-molded, or otherwise formed with the receptacle housing **14.1-504** as a singular and unitary structure. The guide channel **14.1-524** can be disposed in a sidewall of the plug connector **14.1-520**. The guide channel **14.1-524** can be etched, drilled, molded, co-molded, or otherwise formed with the plug connector **14.1-520** as a singular and unitary structure. As illustrated in FIGS. **14.1-7B**, the guide channel **14.1-524** can be disposed in the body portion **14.1-522** at an angle parallel to an insertion direction **14.1-528** in which the plug connector **14.1-520** is to be inserted into the recess **14.1-505** of the receptacle connector **14.1-500**. The guide rail **14.1-508** can be received within the guide channel **14.1-524** of the plug connector **14.1-520** to provide a rigid connection between a support including the plug connector **14.1-520** and a display portion including the receptacle connector **14.1-500**. In other words, the guide rail **14.1-508** and the guide channel **14.1-524** can limit motion

or free play between the support and the display portion. Additionally, or alternatively, the guide rail **14.1-508** can limit or prevent the plug connector **14.1-520** from being inserted into the receptacle connector **14.1-500** in an undesirable orientation and/or configuration.

[3559] As illustrated in FIGS. **14.1-7A**, a fastener **14.1-514** can extend through holes formed in the receptacle housing **14.1-504** and the cover member **14.1-502**. The fastener **14.1-514** can be used to attach the receptacle connector to an enclosure or housing of the display portion.

[3560] FIGS. **14.1-8A** illustrates a sectional view of a support **14.1-620** including a plug connector **14.1-624** retained in a receptacle connector **14.1-600**. FIGS. **14.1-8B** illustrates a detailed view of a region **14.1-610** of FIGS. **14.1-8A**. The receptacle connector **14.1-600** can include a receptacle housing **14.1-602**, latches **14.1-604** extending through the receptacle housing **14.1-602** and configured to retain the plug connector **14.1-622**, a face seal **14.1-606** coupled to the receptacle housing **14.1-602**, and bumper seals **14.1-608** coupled to the receptacle housing **14.1-602**. The support **14.1-620** can include the plug connector **14.1-624**, an enclosure **14.1-622**, and retention channels **14.1-626** formed in the plug connector **14.1-624**.

[3561] The latches **14.1-604** can be configured to retain the plug connector **14.1-624** in a recess of the receptacle connector **14.1-600**. As illustrated in FIGS. **14.1-8B**, the latches **14.1-604** can extend through a cover member **14.1-612** or the like, which can be coupled to the receptacle housing **14.1-602**. The latches **14.1-604** can have trapezoidal shapes, and the retention channels **14.1-626** in the plug connector **14.1-624** can have corresponding trapezoidal shapes. In some examples, the latches **14.1-604** and the retention channels can have rounded shapes, U-shapes, V-shapes, triangular shapes, or any other suitable shapes. The latches **14.1-604** can have widths less than widths of the retention channels **14.1-626**, which can provide for deeper engagement of the latches **14.1-604** in the retention channels **14.1-626**. In some examples, the latches **14.1-604** can have widths equal to or greater than widths of the retention channels **14.1-626**, such that only portions of the latches **14.1-604** extend into the retention channels **14.1-626**. This can result in shallower engagement between the latches **14.1-604** and the retention channels **14.1-626** but can also provide a tighter connection between the latches **14.1-604** and the retention channels **14.1-626** and can reduce wiggle or other movement between the plug connector **14.1-624** and the receptacle connector **14.1-600**.

[3562] The face seal **14.1-606** can be coupled to the receptacle housing **14.1-602** adjacent the recess configured to accept the plug connector **14.1-624**. The face seal **14.1-606** can provide sealing between the plug connector **14.1-624** and the receptacle housing **14.1-602**. The bumpers **14.1-608** can be provided on a surface of the receptacle housing **14.1-602** at a back of the recess, such that the bumpers **14.1-608** are between the plug connector **14.1-624** and the receptacle housing **14.1-602** when the plug connector **14.1-624** is disposed in the recess. The face seal **14.1-606** and the bumpers **14.1-608** can each be configured to provide an ejection force to the plug connector **14.1-624**. For example, when the latches **14.1-604** are in an un-engaged configuration, the face seal **14.1-606** and/or the bumpers **14.1-608** can eject the plug connector **14.1-624** from the recess in the receptacle housing **14.1-602**, or aid in the ejection of the plug connector **14.1-624** from the recess in the receptacle housing **14.1-602**.

[3563] FIGS. **14.1-9A** through **14.1-9F** illustrate various configurations of latches and retention channels that can be included in receptacle connectors and plug connectors. FIGS. **14.1-9A**, **14.1-9C**, and **14.1-9E** illustrate sectional views, and FIGS. **14.1-9B**, **14.1-9D**, and **14.1-9F** illustrate detail views of regions **14.1-710A**, **14.1-710B**, and **14.1-710C** of FIGS. **14.1-9A**, **14.1-9C**, and **14.1-9E**, respectively.

[3564] In FIGS. **14.1-9A** and **14.1-9B**, a latch **14.1-706A** of a receptacle connector **14.1-700A** extends through a receptacle housing **14.1-702** and a cover member **14.1-704** adjacent a face seal **14.1-702**. The latch **14.1-706A** extends into a retention channel **14.1-722A** formed in a plug connector **14.1-720**. The retention channel **14.1-722A** and the latch **14.1-706A** can have corresponding trapezoidal cross-sectional shapes; however, any suitable corresponding shapes can

be used. The latch **14.1-706A** extends into the retention channel **14.1-722A** a distance **14.1-L1**, which can be in a range from about 0 mm to about 10 mm, from about 1 mm to about 2 mm, from about 2 mm to about 5 mm, from about 0 mm to about 2 mm, or the like. In some examples, side surfaces of the latch **14.1-706A** and the retention channel **14.1-722A** can taper at an angle in a range from about 0° to about 10°. In the example of FIGS. **14.1-9A** and **14.1-9B**, tolerances between the latch **14.1-706A** and the retention channel **14.1-722A** can be relatively large, which results in the latch **14.1-706A** extending further into the retention channel **14.1-722A**. For example, a ratio of a width of the retention channel **14.1-722A** to a width of the latch **14.1-706A** can be in a range from about 0.8 to about 1.2, from about 1.0 to about 1.2, from about 0.5 to about 2.0, from about 1.0 to about 1.5, or the like.

[3565] In FIGS. **14.1-9C** and **14.1-9D**, a latch **14.1-706B** of a receptacle connector **14.1-700B** extends through a receptacle housing **14.1-702** and a cover member **14.1-704** adjacent a face seal **14.1-702**. The latch **14.1-706B** extends into a retention channel **14.1-722B** formed in a plug connector **14.1-720**. The retention channel **14.1-722B** and the latch **14.1-706B** can have corresponding trapezoidal cross-sectional shapes; however, any suitable corresponding shapes can be used. The latch **14.1-706B** extends into the retention channel **14.1-722B** a distance **14.1-L2**, which can be in a range from about 0 mm to about 10 mm, from about 1 mm to about 2 mm, from about 2 mm to about 5 mm, from about 0 mm to about 2 mm, or the like. In some examples, side surfaces of the latch **14.1-706B** and the retention channel **14.1-722B** can taper at an angle in a range from about 0° to about 10°. In the example of FIGS. **14.1-9C** and **14.1-9D**, tolerances between the latch **14.1-706B** and the retention channel **14.1-722B** can be intermediate, which results in the latch **14.1-706B** extending less far into the retention channel **14.1-722B** than in the example of FIGS. **14.1-9A** and **14.1-9B**. For example, a ratio of a width of the retention channel **14.1-722B** to a width of the latch **14.1-706B** can be in a range from about 0.8 to about 1.2, from about 1.0 to about 1.2, from about 0.5 to about 2.0, from about 1.0 to about 1.5, or the like.

[3566] In FIGS. **14.1-9E** and **14.1-9F**, a latch **14.1-706C** of a receptacle connector **14.1-700C** extends through a receptacle housing **14.1-702** and a cover member **14.1-704** adjacent a face seal **14.1-702**. The latch **14.1-706C** extends into a retention channel **14.1-722C** formed in a plug connector **14.1-720**. The retention channel **14.1-722C** and the latch **14.1-706C** can have corresponding trapezoidal cross-sectional shapes; however, any suitable corresponding shapes can be used. The latch **14.1-706C** extends into the retention channel **14.1-722C** a distance **14.1-L3**, which can be in a range from about 0 mm to about 10 mm, from about 1 mm to about 2 mm, from about 2 mm to about 5 mm, from about 0 mm to about 2 mm, or the like. In some examples, side surfaces of the latch **14.1-706C** and the retention channel **14.1-722C** can taper at an angle in a range from about 0° to about 10°. In the example of FIGS. **14.1-9E** and **14.1-9F**, tolerances between the latch **14.1-706C** and the retention channel **14.1-722C** can be tight, which results in the latch **14.1-706C** extending less far into the retention channel **14.1-722C** than in the examples of FIGS. **14.1-9A** through **14.1-9D**. For example, a ratio of a width of the retention channel **14.1-722C** to a width of the latch **14.1-706C** can be in a range from about 0.8 to about 1.2, from about 1.0 to about 1.2, from about 0.5 to about 2.0, from about 1.0 to about 1.5, or the like.

[3567] Examples with larger tolerances between the latch and the retention channel can result in the latch extending deeper into the retention channel, but can result in wiggle or other movement between the latch and the retention channel, and between the plug connector and the receptacle connector. On the other hand, examples with tighter tolerances can result in the latch extending a shallow distance into the retention channel, but can result in less wiggle or other movement between the latch and the retention channel, and between the plug connector and the receptacle connector. Forming the latch and the retention channel with too tight of tolerances, or with the latch having too great a width relative to a width of the retention channel can result in the latch not being capable of engaging with the retention channel.

[3568] FIGS. **14.1-10** illustrates actuation of a push block **14.1-806** of a receptacle connector **14.1-**

800 with a tool **14.1-840**, which can be used to release a plug connector **14.1-820** from the receptacle connector **14.1-800**. The receptacle connector **14.1-800** can include a receptacle housing **14.1-802** and a cover member **14.1-804** coupled to the receptacle housing **14.1-802**. A fastening mechanism of the receptacle connector **14.1-800** can include the push block **14.1-806**, a spring **14.1-808**, and a guide rail **14.1-810**. Although not separately illustrated in FIGS. **14.1-10**, the push block **14.1-806** can be coupled to latches that interact with retention channels of the plug connector **14.1-820** to retain the plug connector **14.1-820** within the receptacle connector **14.1-800**. The spring **14.1-808** retains the push block **14.1-806** in an initial position in which the latches are in an engaged configuration. The spring **14.1-808** also resists a force supplied to the push block **14.1-806** by the tool **14.1-840**. When the push block **14.1-806** is pushed by the force of the tool **14.1-840**, the push block **14.1-806** slides or translates along the guide rail **14.1-810**. As the push block **14.1-806** moves along the guide rail **14.1-810**, the push block **14.1-806** pivots the latches from the engaged configuration to an un-engaged configuration in which the plug connector **14.1-820** can be removed from the receptacle connector **14.1-800**. Once the force is no longer supplied by the tool **14.1-840**, the spring **14.1-808** returns the push block **14.1-806** to the initial position and the latches pivot from the un-engaged configuration back to the engaged configuration.

[3569] The receptacle connector **14.1-800** further includes a face seal **14.1-814** and bumpers **14.1-812**. The face seal **14.1-814** and the bumpers **14.1-812** can be disposed between the plug connector **14.1-820** and the receptacle housing **14.1-802** in a direction parallel to an insertion/extraction direction of the plug connector **14.1-820**. This helps the face seal **14.1-814** and/or the bumpers **14.1-812** to provide an ejection force to the plug connector **14.1-820**. For example, when the latches are in the un-engaged configuration, the face seal **14.1-814** and/or the bumpers **14.1-812** can provide the ejection force to the plug connector **14.1-820** to push the plug connector **14.1-820** out of the receptacle connector **14.1-800**. This allows a user to remove the plug connector **14.1-820** by actuating the push block **14.1-806** with the tool **14.1-840**, without having to apply a separate force to the plug connector **14.1-820**. In some examples, the ejection force of the face seal **14.1-814** and the bumpers **14.1-812** can be relatively small, and a pull force on the plug connector **14.1-820** can be used in addition to the ejection force to eject the plug connector **14.1-820** from the receptacle connector **14.1-800**. In some examples, the bumpers **14.1-812** and the face seal **14.1-814** do not provide the ejection force. For example, the bumpers **14.1-812** can be provided to ensure proper alignment of electrical contacts of the plug connector **14.1-820** with electrical contacts of the receptacle connector **14.1-800**. The face seal **14.1-814** can provide sealing between the plug connector **14.1-820** and the receptacle connector **14.1-800**.

[3570] The receptacle connector **14.1-800** can be mounted to an enclosure **14.1-830** (also referred to as a housing) of a display. In some examples, the receptacle connector **14.1-800** can be mounted to the enclosure **14.1-830** using fasteners; however, any suitable mounting means can be used. The enclosure **14.1-830** can include an opening through which the plug connector **14.1-820** can be inserted into a recess of the receptacle connector **14.1-800**. The enclosure **14.1-830** can further include an opening through which the tool **14.1-840** is inserted to interact with the push block **14.1-806**.

[3571] FIGS. **14.1-11A** through **14.1-11D** illustrate various configurations of tools that can be used to release a plug connector from a receptacle connector. In FIGS. **14.1-11A** through **14.1-11D**, a display **14.1-904** is provided in an enclosure **14.1-902** (also referred to as a housing), and a support **14.1-906** is coupled to the enclosure **14.1-902**. The support **14.1-906** can include a plug connector, which can be retained in a receptacle connector provided in the enclosure **14.1-902**. In FIGS. **14.1-11A**, a tool **14.1-900A** can include a stadium-shaped handle with a stadium-shaped cutout therein, and a cylindrical-shaped protrusion. The cylinder-shaped protrusion of the tool **14.1-900A** can be inserted into a hole in the enclosure **14.1-902** and can actuate a push block or the like to release the plug connector of the support **14.1-906** from the receptacle connector of the enclosure **14.1-902**.

[3572] In FIGS. **14.1-11B**, a tool **14.1-900B** can include a stadium-shaped handle and a cylindrical-

shaped protrusion. The cylinder-shaped protrusion of the tool **14.1-900B** can be inserted into a hole in the enclosure **14.1-902** and can actuate a push block or the like to release the plug connector of the support **14.1-906** from the receptacle connector of the enclosure **14.1-902**.

[3573] In FIGS. **14.1-11C**, a tool **14.1-900C** can include a cylinder-shaped handle and a cylindrical-shaped protrusion. The cylinder-shaped protrusion of the tool **14.1-900C** can be inserted into a hole in the enclosure **14.1-902** and can actuate a push block or the like to release the plug connector of the support **14.1-906** from the receptacle connector of the enclosure **14.1-902**.

[3574] In FIGS. **14.1-11D**, a tool **14.1-900D** can include a handle that interfaces with the support **14.1-906** and a cylindrical-shaped protrusion. The handle of the tool **14.1-900D** can be slidably attached to the support **14.1-906** and can be used to align the cylinder-shaped protrusion with a hole in the enclosure. The cylinder-shaped protrusion of the tool **14.1-900D** can be inserted into the hole in the enclosure **14.1-902** and can actuate a push block or the like to release the plug connector of the support **14.1-906** from the receptacle connector of the enclosure **14.1-902**.

[3575] FIGS. **14.1-12A** and **14.1-12B** illustrate a configuration of a guide rail **14.1-1006** of a receptacle connector **14.1-1000** and a guide channel **14.1-1014** of a plug connector **14.1-1010**. The receptacle connector **14.1-1000** can include a receptacle housing **14.1-1002** having a recess **14.1-1004** formed therein. The recess **14.1-1004** can be configured to receive and retain the plug connector **14.1-1010**. The guide rail **14.1-1006** can be formed in the receptacle housing **14.1-1002** and can extend into a portion of the recess **14.1-1004**. The plug connector **14.1-1012** can include a main body portion **14.1-1012** having the guide channel **14.1-1014** formed therein. As illustrated in FIGS. **14.1-12A** and **14.1-12B**, the guide rail **14.1-1006** and the guide channel **14.1-1014** can be formed in longitudinally central portions of the receptacle housing **14.1-1002** and the main body **14.1-1012** of the plug connector **14.1-1010**, respectively. The guide rail **14.1-1006** and the guide channel **14.1-1014** can be disposed on corresponding portions of the receptacle housing **14.1-1002** and the main body **14.1-1012** and can have shapes corresponding to one another. The guide rail **14.1-1006** can be received within the guide channel **14.1-1014** of the plug connector **14.1-1010** to provide a rigid connection between a support including the plug connector **14.1-1010** and a display portion including the receptacle connector **14.1-1000**. In other words, the guide rail **14.1-1006** and the guide channel **14.1-1014** can limit motion or free play between the support and the display portion. Additionally, or alternatively, the guide rail **14.1-1006** can limit or prevent the plug connector **14.1-1010** from being inserted into the receptacle connector **14.1-1000** in an undesirable orientation and/or configuration.

[3576] FIGS. **14.1-13A** through **14.1-13D** illustrate various configurations of face seals that can be included in a receptacle connector **14.1-1100**. FIGS. **14.1-13B** through **14.1-13D** illustrate detail views of region **14.1-1108** illustrated in FIGS. **14.1-13A**. As illustrated in FIGS. **14.1-13A**, the receptacle connector **14.1-1100** can include a receptacle housing **14.1-1102**, which defines a recess **14.1-1104**. A plug connector can be received in the recess **14.1-1104**. A face seal **14.1-1106** can be coupled to the receptacle housing **14.1-1102** adjacent the recess **14.1-1104** and can be used to provide sealing between the receptacle housing **14.1-1102** and the plug connector. The face seal **14.1-1106** can further provide an ejection force to the plug connector to assist in the removal of the plug connector from the recess **14.1-1104** of the receptacle connector **14.1-1100**.

[3577] In FIGS. **14.1-13B**, a face seal **14.1-1106A** coupled to the receptacle housing **14.1-1102** in a region **14.1-1108A** extends from the receptacle housing **14.1-1102** at an angle perpendicular to an insertion/extraction direction of the plug connector with respect to the recess **14.1-1104**. The face seal **14.1-1106A** is then ramped down and extends distally past the receptacle housing **14.1-1102**.

[3578] In FIGS. **14.1-13C**, a face seal **14.1-1106B** coupled to the receptacle housing **14.1-1102** in a region **14.1-1108B** extends from the receptacle housing **14.1-1102** at an angle perpendicular to an insertion/extraction direction of the plug connector with respect to the recess **14.1-1104**.

[3579] The face seal **14.1-1106B** is then ramped down and extends distally past the receptacle housing **14.1-1102**. The face seal **14.1-1106B** can extend a smaller distance from the receptacle

housing **14.1-1102** relative to the face seal **14.1-1106A** and can be ramped from a proximal edge to a distal edge to a lesser degree relative to the face seal **14.1-1106A**. This provides less resistance to insertion and extraction of the plug connector, while still providing good sealing between the plug connector and the receptacle housing **14.1-1102**.

[3580] In FIGS. **14.1-13D**, a face seal **14.1-1106C** coupled to the receptacle housing **14.1-1102** in a region **14.1-1108C** extends from flush with the receptacle housing **14.1-1102**, ramps out from the receptacle housing **14.1-1102**, and then ramps back towards the receptacle housing **14.1-1102**. The face seal **14.1-1106C** extends distally past the receptacle housing **14.1-1102**. The face seal **14.1-1106C** can extend a smaller distance from the receptacle housing **14.1-1102** relative to the face seals **14.1-1106A** and **14.1-1106B** and can be ramped to lesser degrees relative to the face seals **14.1-1106A** and **14.1-1106B**. This provides less resistance to insertion and extraction of the plug connector, while still providing good sealing between the plug connector and the receptacle housing **14.1-1102**.

[3581] FIGS. **14.1-14A** and **14.1-14B** illustrate a perspective view and a sectional view of a receptacle connector **14.1-1200**. FIGS. **14.1-14A** and **14.1-14B** illustrate gaskets **14.1-1208** that can be provided with fasteners **14.1-1206** to fasten a receptacle housing **14.1-1202** of the receptacle connector **14.1-1200** to an enclosure or housing of a display portion or the like. The receptacle housing **14.1-1202** can include a cover member **14.1-1204** coupled thereto, which provides structural support and shielding for the receptacle housing **14.1-1202**. Holes can be provided in the receptacle housing **14.1-1202** and the cover member **14.1-1204**, and the gaskets **14.1-1208** and the fasteners **14.1-1206** can be provided in the holes extending through the receptacle housing **14.1-1202** and the cover member **14.1-1204**. The fasteners **14.1-1206** can be used to fasten the receptacle housing **14.1-1202** and the cover member **14.1-1204** to the enclosure of the display portion. The gaskets **14.1-1208** can be provided between the fasteners **14.1-1206** and the cover member **14.1-1204**/receptacle housing **14.1-1202** to provide give or flexibility between the receptacle connector **14.1-1200** and the enclosure. The gaskets **14.1-1208** can be formed of a relatively soft material, such as a rubber, a polymer, an elastomer, or the like. Providing the gaskets **14.1-1208** allows the receptacle connector **14.1-1200** and a plug connector inserted into the receptacle connector **14.1-1200** to move together, and to move relative to the enclosure. Large forces can be transferred between the plug connector and the receptacle connector **14.1-1200** and providing the gaskets **14.1-1208** limits any negative impacts of these forces from being transferred between the receptacle connector **14.1-1200** and the enclosure.

[3582] A plurality of electrical contacts **14.1-1210A**, **14.1-1210B**, **14.1-1210C**, **14.1-1211A**, **14.1-1211B**, and **14.1-1211B** can be coupled to the receptacle housing **14.1-1202**. As illustrated in FIGS. **14.1-14A**, the electrical contacts can be coupled to the receptacle housing **14.1-1202** along a radius of curvature, with each electrical contact being provided at an oblique angle relative to other electrical contacts, such as adjacent electrical contacts. This allows for surfaces of the electrical contacts that are to be in contact with a plug connector inserted into the receptacle connector to be disposed at angles normal to surfaces of the plug connector that the electrical contacts are to contact. This provides for improved electrical contact between the electrical contacts of the receptacle connector **14.1-1200** and electrical contacts of the plug connector.

[3583] A fastening mechanism can be provided on the receptacle connector **14.1-1200**. The fastening mechanism can include a primary push block **14.1-1212**, a spring **14.1-1214**, latches **14.1-1220**, a secondary push block **14.1-1222**, and a guide rail **14.1-1224**. The fastening mechanism can be provided to transition the latches **14.1-1220** between an engaged configuration and an un-engaged configuration, which can be used to retain and release the plug connector in the receptacle connector **14.1-1200**. The spring **14.1-1214** can be coupled to the cover member **14.1-1204** and/or the receptacle housing **14.1-1202**. The primary push block **14.1-1212** can be pivotally coupled to the spring **14.1-1214** and the cover member **14.1-1204** and/or the receptacle housing **14.1-1202**. The primary push block **14.1-1212** can be configured to interact with a tool in order to

transition the latches **14.1-1220** between the engaged configuration and the un-engaged configuration. The spring **14.1-1214** retains the primary push block **14.1-1212** in an initial position in which the latches **14.1-1220** are in the engaged configuration. The spring **14.1-1214** also resists a force supplied to the primary push block **14.1-1212** by the tool.

[3584] The secondary push block **14.1-1222** transfers movement between the primary push block **14.1-1212** and the latches **14.1-1220**. The secondary push block **14.1-1222** can be configured to translate along the guide rail **14.1-1224**, such that the secondary push block **14.1-1222** moves back and forth in a single direction. Using the tool to actuate the primary push block **14.1-1212** moves the secondary push block **14.1-1222** along the guide rail **14.1-1224**, and the secondary push block **14.1-1222** pivots the latches **14.1-1220** from the engaged configuration to the un-engaged configuration. After force from the tool is removed from the primary push block **14.1-1212**, the secondary push block **14.1-1222** returns to its initial position and the latches **14.1-1220** return to the engaged configuration. Thus, the fastening mechanism is used to retain and release a plug connector with respect to the receptacle connector **14.1-1200**.

[3585] FIGS. **14.1-15** illustrates an example in which a support **14.1-1320** is fastened to a receptacle connector **14.1-1300** and/or an enclosure **14.1-1330** using fasteners **14.1-1338**. The fastening mechanism of FIGS. **14.1-15** can be used in addition to or alternative to the fastening mechanisms of FIGS. **14.1-3A** through **14.1-14B**. The receptacle connector **14.1-1300** can include a receptacle housing **14.1-1302** having a recess **14.1-1304** formed therein. The recess **14.1-1304** can be configured to receive and retain the support **14.1-1320**. Fasteners **14.1-1306** can extend through holes in the receptacle housing **14.1-1302** and can be used to fasten the receptacle connector **14.1-1300** to the enclosure **14.1-1330**.

[3586] The enclosure **14.1-1330** (also referred to as a housing) can be an enclosure of a display portion or the like. The enclosure **14.1-1330** can include a housing body **14.1-1332**, openings **14.1-1334**, and openings **14.1-1336**. The openings **14.1-1336** can be configured to receive the fasteners **14.1-1306** and can be used to mount the receptacle connector **14.1-1300** to the enclosure **14.1-1330**. As will be discussed in greater detail below, the support **14.1-1320** can be fastened to the enclosure **14.1-1330** through the openings **14.1-1334**.

[3587] The support **14.1-1320** includes a plug connector **14.1-1322** and an enclosure **14.1-1324**. Openings **14.1-1326** can be included in the plug connector **14.1-1322** and/or the enclosure **14.1-1324**. The plug connector **14.1-1322** can include electrical contacts **14.1-1328A**, **14.1-1328B**, and **14.1-1328C**. Fasteners **14.1-1338** can be provided and can extend through the openings **14.1-1326** in the support **14.1-1320** and into the openings **14.1-1334** in the enclosure **14.1-1330**. In some examples, the openings **14.1-1334** can be provided on the receptacle connector **14.1-1300** rather than the enclosure **14.1-1330**. Fastening the support **14.1-1320** to the enclosure **14.1-1330** and/or the receptacle connector **14.1-1300** prevents unwanted disconnects of the support **14.1-1320** from the receptacle connector **14.1-1300** and/or the enclosure **14.1-1330**. The fasteners **14.1-1338** can be contained within the enclosure **14.1-1330**, unseen to a user, and thus accidental removal by a user can be avoided. Fastening the support **14.1-1320** directly to the enclosure **14.1-1330** with the fasteners **14.1-1338** further ensures the support **14.1-1320** does not move relative to the chassis **14.1-1330**.

[3588] FIGS. **14.1-16A** through **14.1-25B** illustrate various examples of receptacle connectors and plug connectors that can be used as the receptacle connectors **14.1-124** of the supports **14.1-120** and the plug connectors **14.1-142** of the cable assembly **14.1-140** illustrated in FIGS. **14.1-1A** and **14.1-1B**.

[3589] FIGS. **14.1-16A** illustrates a side section view of a receptacle connector **14.1-1400** and a cable assembly **14.1-1420** including a plug connector **14.1-1422** inserted into and retained by the receptacle connector **14.1-1400**. FIGS. **14.1-16B** illustrates an electrical contact of the plug connector **14.1-1422** and a raised portion of the receptacle connector **14.1-1400**. The cable assembly **14.1-1420** includes a shielded cable **14.1-1426** coupled to the plug connector **14.1-1422**.

The plug connector **14.1-1422** can include a boot **14.1-1424** and a central member **14.1-1430**. In some examples, the boot **14.1-1424** can have a circular or semi-circular cross-sectional shape. The plug connector **14.1-1422** can be centered or substantially centered about a central axis of the circular cross-sectional shape of the boot **14.1-1424**. The boot **14.1-1424** can form a cavity or volume and at least a portion of the central member **14.1-1430** and/or a portion of the shielded cable **14.1-1426** can be disposed within the volume.

[3590] The central member **14.1-1430** can extend from the boot **14.1-1424**. In some examples, the central member **14.1-1430** can have a cylindrical shape that enables rotation of the central member **14.1-1430** within the receptacle connector **14.1-1400**. The central member **14.1-1430** can form an external surface and an internal surface. In some examples, the central member **14.1-1430** can include one or more protrusions **14.1-1428** extending laterally from, or substantially perpendicular to, the external surface. In some examples, the central member **14.1-1430** can form a cavity or volume. For example, the internal surface can form a cavity or volume. One or more electrical contacts can be disposed on the internal surface, such as an electrical contact **14.1-1425** illustrated in FIGS. **14.1-16B**. The one or more electrical contacts can extend into the volume of the central member **14.1-1430** from the internal surface. The one or more electrical contacts can define an electrical grounding path, provide electrical power, and/or provide one or more control signals to the receptacle connector **14.1-1400**.

[3591] The shielded cable **14.1-1426** can be one or more wires (e.g., metallic wires capable of transferring electrical signals and/or electrical power). The one or more wires can be individually shielded or collectively shielded. The shielding can prevent each of the one or more wires from contacting one another. Additionally, or alternatively, the shielding can prevent or limit the influence of electromagnetic waves on each of the one or more wires.

[3592] The receptacle connector **14.1-1400** can be disposed on an enclosure of an electronic device (e.g., the supports **14.1-120** of the first electronic device **14.1-100**). In some examples, the receptacle connector **14.1-1400** can be disposed on or within the enclosure, such that a trim ring **14.1-1402** of the receptacle connector **14.1-1400** is at least partially disposed within the enclosure. In some examples, the trim ring **14.1-1402** can be disposed on or within the enclosure such that the trim ring **14.1-1402** is flush with an exterior surface of the enclosure.

[3593] In some examples, the receptacle connector **14.1-1400** can include the trim ring **14.1-1402**, a base, one or more detents **14.1-1406**, and a raised portion **14.1-1404**. The trim ring **14.1-1402** can be coupled to the base to form a recess and one or more undercut regions at a periphery of the recess. Each of the undercut regions can house or retain a respective detent **14.1-1406**. When the central member **14.1-1430** of the cable assembly **14.1-1420** is disposed within the recess, the protrusion **14.1-1428** can extend into a channel defined by a respective detent **14.1-1406**. In some examples, each of the detents **14.1-1406** can be biased toward the raised portion **14.1-1404** by a biasing element. In other words, each detent **14.1-1406** can radially translate within their respective undercut regions toward and away from the raised portion **14.1-1404**.

[3594] In some examples, the raised portion **14.1-1404** can be disposed in the center of the recess, such that the recess forms a ring within the receptacle connector **14.1-1400**. The raised portion **14.1-1404** can be a distinct component of the receptacle connector **14.1-1400**. For example, the raised portion **14.1-1404** can be adhered, fastened, interlocked, welded, or otherwise coupled to the base. In some examples, one or more electrical contacts **14.1-1407A** and **14.1-1407B** can be disposed on the raised portion **14.1-1404** and can be used to provide connections between the raised portion **14.1-1404** and the plug connector **14.1-1422**, such as the electrical contact **14.1-1425** of the plug connector **14.1-1422**. Each of the one or more electrical contacts **14.1-1407A** and **14.1-1407B** can physically contact one or more electrical contacts of the plug connector **14.1-1422** (e.g., the electrical contact **14.1-1425**) when the plug connector **14.1-1422** is coupled to the receptacle connector **14.1-1400**. The raised portion **14.1-1404** can further include one or more electrical contacts **14.1-1409A** and **14.1-1409B** that provide connections between the raised portion **14.1-**

1404 and electronic components contained within the enclosure (e.g., processors, electrical wires, digital logic circuitry, digital processing circuitry, or the like).

[3595] In some examples, the raised portion **14.1-1404** can include a body **14.1-1414** and a cap **14.1-1412**. In some examples, the body **14.1-1414** can be manufactured from a material that electrically insulates the one or more electrical contacts **14.1-1407A**, **14.1-1407B**, **14.1-1409A**, and **14.1-1409B**, such that each of the one or more electrical contacts are electrically isolated from one another. For example, each of the one or more electrical contacts can be at least partially disposed within a respective through-hole defined within the body **14.1-1414**. The cap **14.1-1412** can be coupled to the body **14.1-1414**.

[3596] The raised portion **14.1-1404** can include a seal **14.1-1408** extending around a periphery of the body **14.1-1414**. As illustrate in FIGS. **14.1-16B**, the seal **14.1-1408** can encircle the body **14.1-1414** of the raised portion **14.1-1404**. The seal **14.1-1408** can act as a seal which prevents or limits ingress of contaminants (e.g., liquid, dust, lint, debris, or the like) into the receptacle connector **14.1-1400** through the recess, such as when the plug connector **14.1-1422** is inserted into the recess of the receptacle connector **14.1-1400**.

[3597] FIGS. **14.1-17A** and **14.1-17B** show respective side and bottom views of a cable assembly **14.1-1500** including a plug connector **14.1-1502** and a shielded cable **14.1-1504**. The plug connector **14.1-1502** can include a boot **14.1-1506** and a central member **14.1-1508**. In some examples, the boot **14.1-1506** can have a circular or semi-circular cross-sectional shape.

[3598] The plug connector **14.1-1502** can be centered or substantially centered about a central axis of the circular cross-sectional shape of the boot **14.1-1506**. For example, as shown in FIG. **14.1-17A**, the central member **14.1-1508** can be centered about an axis A that extends through a center of the boot **14.1-1506**. The boot **14.1-1506** can form a cavity or volume and at least a portion of the central member **14.1-1508** and/or a portion of the shielded cable **14.1-1504** can be disposed within the volume.

[3599] The central member **14.1-1508** can extend from the boot **14.1-1506**. In some examples, the central member **14.1-1508** can have a cylindrical shape that enables rotation of the central member **14.1-1508** within a receptacle connector (e.g., the receptacle connector **14.1-1400** of FIGS. **14.1-16A**). The central member **14.1-1508** can form an external surface **14.1-1510** and an internal surface **14.1-1512**. In some examples, the central member **14.1-1508** can include one or more protrusions **14.1-1514** extending laterally from, or substantially perpendicular to, the external surface **14.1-1510**. In some examples, the central member **14.1-1508** can form a cavity or volume, for example, the internal surface **14.1-1512** can form a cavity or volume **14.1-1516**. One or more electrical contacts **14.1-1518** can be disposed on the internal surface **14.1-1512**, such that the one or more electrical contacts **14.1-1518** extend into the volume **14.1-1516** from the internal surface **14.1-1512**. The one or more electrical contacts **14.1-1518** can define an electrical grounding path, provide electrical power, and/or provide one or more control signals to a receptacle connector.

[3600] The shielded cable **14.1-1504** can be one or more wires (e.g., metallic wires capable of transferring electrical signals and/or electrical power). The one or more wires can be individually shielded or collectively shielded. The shielding can prevent each of the one or more wires from contacting one another. Additionally, or alternatively, the shielding can prevent or limit the influence of electromagnetic waves on each of the one or more wires.

[3601] FIGS. **14.1-18A** and **14.1-18B** show a perspective view and a top view of a receptacle connector **14.1-1600** disposed on an enclosure **14.1-1602** of an electronic device (e.g., the supports **14.1-120** of the first electronic device **14.1-100** of FIGS. **14.1-1A** and **14.1-1B**). In some examples, the receptacle connector **14.1-1600** can be disposed on or within the enclosure **14.1-1602**, such that a trim ring **14.1-1604** of the receptacle connector **14.1-1600** is at least partially disposed within the enclosure **14.1-1602**. Alternatively, the trim ring **14.1-1604** can be disposed on or within the enclosure **14.1-1602** such that the trim ring **14.1-1604** is flush with an exterior surface **14.1-1606** of the enclosure **14.1-1602**.

[3602] In some examples, the receptacle connector **14.1-1600** can include the trim ring **14.1-1604**, a base **14.1-1608**, one or more detents **14.1-1610**, and a raised portion **14.1-1612**. The trim ring **14.1-1604** can couple to the base **14.1-1608** to form a recess **14.1-1614** and one or more undercut regions at a periphery of the recess **14.1-1614**. Each of the one or more undercut regions can house or retain a respective detent **14.1-1610**. While a central member (e.g., the central member **14.1-1508** of FIGS. **14.1-17A** and **14.1-17B**) is disposed within the recess **14.1-1614**, a protrusion (e.g., the protrusions **14.1-1514** of FIGS. **14.1-17A** and **14.1-17B**) can extend into a channel defined by each of the detents **14.1-1610**. In some examples, each of the detents **14.1-1610** can be biased toward the raised portion **14.1-1612** by a biasing element. In other words, each detent **14.1-1610** can radially translate within their respective undercut regions toward and away from the raised portion **14.1-1612**.

[3603] In some examples, the raised portion **14.1-1612** can be disposed in the center of the recess **14.1-1614**, such that the recess **14.1-1614** forms a ring within the receptacle connector **14.1-1600**. The raised portion **14.1-1612** can be a distinct component of the receptacle connector **14.1-1600**. For example, the raised portion **14.1-1612** can be adhered, fastened, interlocked, welded, or otherwise coupled to the base **14.1-1608**. In some examples, one or more electrical contacts **14.1-1618** can be disposed on the raised portion **14.1-1612**. Each of the one or more electrical contacts **14.1-1618** can physically contact one or more electrical contacts of the plug connector (e.g., the electrical contacts **14.1-1518** of FIGS. **14.1-17A** and **14.1-17B**) when the plug connector is coupled to the receptacle connector **14.1-1600**.

[3604] FIGS. **14.1-18C** illustrates a sectional view of the receptacle connector of FIGS. **14.1-18A** and **14.1-18B**. As illustrated in FIGS. **14.1-18C**, biasing elements **14.1-1611** can be housed within the trim ring **14.1-1604** between the detents **14.1-1610** and the trim ring **14.1-1604**. The biasing elements **14.1-1611** can be configured to bias the detents **14.1-1610** radially towards protrusions of the plug connector when the plug connector is inserted into and retained in the recess **14.1-1614** of the receptacle connector **14.1-1600**. In other words, each of the detents **14.1-1610** can radially translate within their respective undercut regions toward and away from the raised portion **14.1-1612** to engage and disengage with respective protrusions of the plug connector. The biasing elements **14.1-1611** can provide forces to the detents **14.1-1610** to retain the detents **14.1-1610** in the radially extended position to engage the protrusions of the plug connector, and to resist the detents being disengages from the protrusions of the plug connector. The biasing elements **14.1-1611** are illustrated as a canted coil; however various biasing elements **14.1-1611** can be included, such as a coiled spring, an elastic foam, leaf springs, or the like.

[3605] The raised portion **14.1-1612** can include a body **14.1-1626** and a cap **14.1-1628**. In some examples, the body **14.1-1626** can be manufactured from a material that electrically insulates one or more electrical contacts of the raised portion **14.1-1612**, such that each of the one or more electrical contacts are electrically isolated from one another. For example, each of the one or more electrical contacts can be at least partially disposed within a respective through-hole defined within the body **14.1-1626**. The cap **14.1-1628** can be coupled to the body **14.1-1626**.

[3606] The raised portion **14.1-1612** can include a seal **14.1-1632** extending around a periphery of the body **14.1-1626**. As illustrate in FIGS. **14.1-18C**, the seal **14.1-1632** can encircle the body **14.1-1626** of the raised portion **14.1-1612**. The seal **14.1-1632** can act as a seal which prevents or limits ingress of contaminants (e.g., liquid, dust, lint, debris, or the like) into the receptacle connector **14.1-1600** through the recess **14.1-1614**, such as when the plug connector is inserted into the recess **14.1-1614** of the receptacle connector **14.1-1600**. In the example of FIGS. **14.1-18C**, the seal **14.1-1632** is disposed between the body **14.1-1626** of the raised portion **14.1-1612** and the base **14.1-1608**. The seal **14.1-1632** can further prevent the ingress of contaminants between the raised portion **14.1-1612** and the base **14.1-1608**.

[3607] FIGS. **14.1-18D** shows the cable assembly **14.1-1500**, including the plug connector **14.1-1502** and the shielded cable **14.1-1504** of FIGS. **14.1-17A** and **14.1-17B**, being pulled away from

the receptacle connector **14.1-1600** of FIGS. **14.1-18A** through **14.1-18C**. The plug connector **14.1-1502** can be unintentionally and abruptly extracted from the receptacle connector **14.1-1600**, for example, when the shielded cable **14.1-1504** snags on an object and pulls the plug connector **14.1-1502** in a direction away from the receptacle connector **14.1-1600** (as represented by the arrow shown in FIGS. **14.1-18D**). When the plug connector **14.1-1502** is abruptly and/or undesirably extracted from the receptacle connector **14.1-1600**, the electrical contacts **14.1-1518A** and **14.1-1518B** of the plug connector **14.1-1502** can be pulled out of contact with the correlating electrical contacts **14.1-1618A** and **14.1-1618B** of the receptacle connector **14.1-1600** causing a loss of: electrical power, control signals, an electrical ground, or a combination thereof. An abrupt loss of electrical power, electrical ground, or one or more control signals can damage the electronic device and/or compromise or corrupt information stored on the electronic device.

[3608] In some examples, at least one of the electrical contacts (e.g., electrical contact **14.1-1618A**) within the receptacle connector **14.1-1600** can detect when the plug connector **14.1-1502** is being removed from the receptacle connector **14.1-1600**. For example, the electronic device can detect an undesirable extraction of the plug connector **14.1-1502** from the receptacle connector **14.1-1600** when the electrical contact **14.1-1518A** of the plug connector **14.1-1502** is pulled out of contact with the electrical contact **14.1-1618A** of the receptacle connector **14.1-1600**. An undesirable extraction of the plug connector **14.1-1502** from the receptacle connector **14.1-1600** can occur when the plug connector **14.1-1502** is pulled away from the receptacle connector **14.1-1600** from the side of the plug connector **14.1-1502** where the shielded cable **14.1-1504** enters the boot **14.1-1506**. This undesirable extraction scenario can be most probable as the shielded cable **14.1-1504** can get snagged on an object and can pull the plug connector **14.1-1502** away from the receptacle connector **14.1-1600** (as illustrated by the arrow).

[3609] In some examples, the electrical contacts **14.1-1618** can be disposed about the raised portion **14.1-1612** such that the electrical contact **14.1-1518A** moves out of contact from the electrical contact **14.1-1618A** before the electrical contacts **14.1-1518B** of the plug connector **14.1-1502** move out of contact with the electrical contact **14.1-1618B** of the receptacle connector **14.1-1600**. While the electrical contact **14.1-1518A** is out of contact from the electrical contact **14.1-1618A** but before the electrical contacts **14.1-1518B** of the plug connector **14.1-1502** move out of contact with the electrical contact **14.1-1618B** of the receptacle connector **14.1-1600**, a duration of time can pass. The duration of time can be sufficient to enable the electronic device to mitigate any destructive effects of unintentionally extracting the plug connector **14.1-1502** from the receptacle connector **14.1-1600**. The size, shape, and position of each of the electrical contacts **14.1-1518**, **14.1-1618** can impact the duration of time when the electrical contact **14.1-1518A** is out of contact from the electrical contact **14.1-1618A** but before the electrical contacts **14.1-1518B** move out of contact with the electrical contact **14.1-1618B**.

[3610] In some examples, a contact area defined by the electrical contact **14.1-1618A** (e.g., a surface area of the electrical contact **14.1-1618** that can be engaged by one or more of the electrical contacts **14.1-1518**) can be different from a contact area defined by the electrical contact **14.1-1618B**. In other words, each electrical contact **14.1-1618A**, **14.1-1618B** can be sized or shaped such that one or more correlating electrical contacts **14.1-1518** can physically contact the contact area defined by each respective electrical contact **14.1-1618**. As shown in FIGS. **14.1-18D**, the contact area of the electrical contact **14.1-1618B** can be larger than the contact area of the electrical contact **14.1-1618A**. As such, the electrical contact **14.1-1618B** can interface with more than one corresponding electrical contacts **14.1-1518B** while the plug connector **14.1-1502** is inserted into the receptacle connector **14.1-1600**. The size, shape, and position of the contact area of each electrical contact **14.1-1518**, **14.1-1618** can impact the duration of time when the electrical contact **14.1-1518A** is out of contact from the electrical contact **14.1-1618A** but before the electrical contacts **14.1-1518B** move out of contact with the electrical contact **14.1-1618B**.

[3611] A loss of the electrical signal passed between the electrical contacts **14.1-1518A**, **14.1-**

1618A can alert the electronic device (e.g., a processor of the electronic device) that an unexpected or undesirable extraction of the plug connector **14.1-1502** is occurring. Accordingly, the electronic device can react to the loss of signal by implementing one or more actions to safeguard the electrical components and information of the electronic device. For example, a processor within the electronic device can: withdraw electrical power from one or more electronic components, cause data to be saved, or perform another preventative task or action. While FIGS. **14.1-18D** shows the plug connector **14.1-1502** being extracted from the receptacle connector **14.1-1600** from the side of the plug connector **14.1-1502** adjacent the shielded cable **14.1-1504** (e.g., rocking out of the receptacle connector **14.1-1600** starting from the side of the plug connector **14.1-1502** adjacent the shielded cable **14.1-1504**), the plug connector **14.1-1502** can be undesirably detached from another side of the plug connector **14.1-1502** in some examples.

[3612] FIGS. **14.1-18E** illustrates various configurations of the raised portion **14.1-1612** that can be used to alter the force required to insert and extract the plug connector **14.1-1502** into and out of the receptacle connector **14.1-1600**. As illustrated in FIGS. **14.1-18E**, side surfaces of the raised portion **14.1-1612** can be disposed at different angles with respect to the base **14.1-1608**. In some examples, an angle **14.1- θ .sub.1** between side surfaces of the raised portion **14.1-1612** and the base **14.1-1608** can be in a range from about 95° to about 105°, from about 90° to about 180°, from about 90° to about 120°, from about 95° to about 135°, from about 92° to about 110°, or the like. In some examples, an angle **14.1- θ .sub.2** between side surfaces of the raised portion **14.1-1612** and the base **14.1-1608** can be relatively larger than the angle **14.1- θ .sub.1** and can be in a range from about 95° to about 105°, from about 90° to about 180°, from about 90° to about 120°, from about 95° to about 135°, from about 92° to about 110°, or the like. Providing the angle **14.1- θ .sub.2** between the raised portion **14.1-1612** and the base **14.1-1608** allows the plug connector **14.1-1502** to be inserted into and extracted from the receptacle connector **14.1-1600** with a relatively small force. Providing the angle **14.1- θ .sub.1** between the raised portion **14.1-1612** and the base **14.1-1608** allows the plug connector **14.1-1502** to be inserted into and extracted from the receptacle connector **14.1-1600** with a relatively large force. Thus, the angle between the raised portion **14.1-1612** and the base **14.1-1608** can be selected based on the forces desired to insert the plug connector **14.1-1502** into the receptacle connector **14.1-1600** and to extract the plug connector **14.1-1502** from the receptacle connector **14.1-1600**.

[3613] FIGS. **14.1-19A** through **14.1-19M** illustrate various configurations of detents **14.1-1710** that can be included in a receptacle connector **14.1-1700**. The receptacle connector **14.1-1700** can include a trim ring **14.1-1704**, a base **14.1-1708**, a raised portion **14.1-1712**, and a recess between the trim ring **14.1-1704** and the raised portion **14.1-1712**. The detents **14.1-1710** can be disposed within a cavity formed in the trim ring **14.1-1704**.

[3614] FIGS. **14.1-19A** through **14.1-19C** illustrate a detent **14.1-1710A** that includes a central region **14.1-1711** and peripheral regions **14.1-1713** on either side of the central region **14.1-1711**. The central region **14.1-1711** can be notched to form a channel **14.1-1733** that can receive a protrusion **14.1-1724** of a plug connector **14.1-1720**. For example, a lower portion of the central region **14.1-1711** can be notched to form an undercut portion of the detent **14.1-1710A** in the central region **14.1-1711**. The notches can be omitted from the peripheral regions **14.1-1713**, which can prevent, limit, or reduce rotation of the plug connector **14.1-1720**. Including the un-notched peripheral regions **14.1-1713** can also increase mating forces and un-mating forces between the plug connector **14.1-1720** and the receptacle connector **14.1-1700**.

[3615] FIGS. **14.1-19C** shows a cross-sectional view of the protrusion **14.1-1724** of the plug connector **14.1-1720** disposed within the channel **14.1-1733** and interfacing the detent **14.1-1710A** taken along reference line A-A' of FIGS. **14.1-19A**. In some examples, the detent **14.1-1710A** can define the channel **14.1-1733** such that a first surface **14.1-1731** of the channel **14.1-1733** is angled relative to a second surface **14.1-1732** of the channel **14.1-1733** to form an angle **14.1- θ .sub.3**. The protrusion **14.1-1714** can engage or contact the first surface **14.1-1730** while disposed within the

channel **14.1-1733**. The angle **14.1-θ.sub.3** can directly correlate to a force required to overcome a biasing element coupled to the detent **14.1-1710A** to cause the detent **14.1-1710A** to move away from the central member **14.1-1712** and enable extraction of the protrusion **14.1-1724** from the channel **14.1-1733** (e.g., enabling extraction of the plug connector **14.1-1720** from the receptacle connector **14.1-1700**). For example, a relatively smaller angle **14.1-θ.sub.3** can require a relatively greater force to extract the plug connector **14.1-1720** from the receptacle connector **14.1-1700**. Alternatively, a relatively larger angle **14.1-θ.sub.3** can require relatively smaller force to extract the plug connector **14.1-1720** from the receptacle connector **14.1-1700**. The angle **14.1-θ.sub.3** can be about 30 degrees, between about 30 degrees and about 60 degrees, between about 60 degrees and about 90 degrees, between about 90 degrees and about 120 degrees, between about 120 degrees and about 150 degrees, between about 150 degrees and about 180 degrees, or less than 180 degrees. A third surface **14.1-1730** adjacent the first surface **14.1-1731** can extend in a direction parallel to the second surface **14.1-1732**.

[3616] FIGS. **14.1-19D** through **14.1-19F** illustrate a detent **14.1-1710B** that includes a central region **14.1-1711** and peripheral regions **14.1-1713** on either side of the central region **14.1-1711**. The central region **14.1-1711** can be notched to form a channel **14.1-1733** that can receive a protrusion **14.1-1724** of a plug connector **14.1-1720**. For example, a lower portion of the central region **14.1-1711** can be notched to form an undercut portion of the detent **14.1-1710B** in the central region **14.1-1711**. Further, an upper portion of the central region **14.1-1711** can be notched to form a sloped upper surface of the detent **14.1-1710B** in the central region **14.1-1711**. The notches can be omitted from the peripheral regions **14.1-1713**, which can prevent, limit, or reduce rotation of the plug connector **14.1-1720**. Including the un-notched peripheral regions **14.1-1713** can also increase mating forces and un-mating forces between the plug connector **14.1-1720** and the receptacle connector **14.1-1700**. In the example of FIGS. **14.1-19D** through **14.1-19F**, one of the peripheral regions **14.1-1713** can be un-notched, and the other peripheral region **14.1-1713** can be notched, which reduces breakaway forces (e.g., reduces un-mating forces). The notched upper surface of the detent **14.1-1710B** can reduce a mating force for mating the plug connector **14.1-1720** in the receptacle connector **14.1-1700**. In other words the notched upper surface of the detent **14.1-1710B** can reduce the force required to move the protrusions **14.1-1724** past the detent **14.1-1710B** into the channel **14.1-1733**. Further, the peripheral regions **14.1-1713** can extend radially inward relative to the central region **14.1-1711** and can include protuberances or the like. In the example of FIGS. **14.1-19D** through **14.1-19F**, protuberances can be included in both of the peripheral regions **14.1-1713**. This can increase mating and/or un-mating forces of the plug connector **14.1-1720** relative to the receptacle connector **14.1-1700**.

[3617] FIGS. **14.1-19F** shows a cross-sectional view of the protrusion **14.1-1724** of the plug connector **14.1-1720** disposed within the channel **14.1-1733** and interfacing the detent **14.1-1710B** taken along reference line A-A' of FIGS. **14.1-19D**. In some examples, the detent **14.1-1710B** can define the channel **14.1-1733** such that a first surface **14.1-1737** of the channel **14.1-1733** is angled relative to a second surface **14.1-1738** of the channel **14.1-1733** to form an angle **14.1-θ.sub.4**. The protrusion **14.1-1714** can engage or contact the first surface **14.1-1737** while disposed within the channel **14.1-1733**. The angle **14.1-θ.sub.4** can directly correlate to a force required to overcome a biasing element coupled to the detent **14.1-1710B** to cause the detent **14.1-1710B** to move away from the central member **14.1-1712** and enable extraction of the protrusion **14.1-1724** from the channel **14.1-1733** (e.g., enabling extraction of the plug connector **14.1-1720** from the receptacle connector **14.1-1700**). For example, a relatively smaller angle **14.1-θ.sub.4** can require a relatively greater force to extract the plug connector **14.1-1720** from the receptacle connector **14.1-1700**. Alternatively, a relatively larger angle **14.1-θ.sub.4** can require relatively smaller force to extract the plug connector **14.1-1720** from the receptacle connector **14.1-1700**. The angle **14.1-θ.sub.4** can be about 30 degrees, between about 30 degrees and about 60 degrees, between about 60 degrees and about 90 degrees, between about 90 degrees and about 120 degrees, between about 120 degrees

and about 150 degrees, between about 150 degrees and about 180 degrees, or less than 180 degrees. A third surface **14.1-1736** adjacent the first surface **14.1-1737** can extend in a direction parallel to the second surface **14.1-1738**.

[3618] In some examples, a top surface of the detent **14.1-1710B** is also notched. As such, a fourth surface **14.1-1734** of the detent **14.1-1710B** is angled relative to the third surface **14.1-1736** of the detent **14.1-1710B** to form an angle. The protrusion **14.1-1714** can engage or contact the fourth surface **14.1-1734** as the protrusion **14.1-1714** is inserted into the channel **14.1-1733**. The angle can directly correlate to a force required to overcome a biasing element coupled to the detent **14.1-1710B** to cause the detent **14.1-1710B** to move away from the central member **14.1-1712** and enable insertion of the protrusion **14.1-1724** into the channel **14.1-1733** (e.g., enabling insertion of the plug connector **14.1-1720** from the receptacle connector **14.1-1700**). For example, a relatively smaller angle can require a relatively greater force to extract the plug connector **14.1-1720** from the receptacle connector **14.1-1700**. Alternatively, a relatively larger angle can require relatively smaller force to extract the plug connector **14.1-1720** from the receptacle connector **14.1-1700**. The angle can be about 30 degrees, between about 30 degrees and about 60 degrees, between about 60 degrees and about 90 degrees, between about 90 degrees and about 120 degrees, between about 120 degrees and about 150 degrees, between about 150 degrees and about 180 degrees, or less than 180 degrees.

[3619] FIGS. **14.1-19G** and **14.1-19H** illustrate a detent **14.1-1710C** that includes a central region **14.1-1711** and a peripheral region **14.1-1713** on one side of the central region **14.1-1711**. The central region **14.1-1711** can be notched to form a channel **14.1-1733** that can receive a protrusion **14.1-1724** of a plug connector **14.1-1720**. The central region **14.1-1711** can also be notched to form a notched top surface. For example, a lower portion of the central region **14.1-1711** can be notched to form an undercut portion of the detent **14.1-1710C** in the central region **14.1-1711**. Further, an upper portion of the central region **14.1-1711** can be notched to form a sloped upper surface of the detent **14.1-1710C** in the central region **14.1-1711**. The notches can be omitted from the peripheral region **14.1-1713**, which can prevent, limit, or reduce rotation of the plug connector **14.1-1720**. Including the un-notched peripheral region **14.1-1713** can also increase mating forces and un-mating forces between the plug connector **14.1-1720** and the receptacle connector **14.1-1700**. In the example of FIGS. **14.1-19G** and **14.1-19H**, the peripheral region **14.1-1713** can be un-notched, which can increase mating and/or un-mating forces between the plug connector **14.1-1720** and the receptacle connector **14.1-1700**. The notched upper surface of the detent **14.1-1710C** can reduce a mating force for mating the plug connector **14.1-1720** in the receptacle connector **14.1-1700**. In other words, the notched upper surface of the detent **14.1-1710C** can reduce the force required to move the protrusions **14.1-1724** past the detent **14.1-1710C** into the channel **14.1-1733**. Further, the peripheral region **14.1-1713** can extend radially inward relative to the central region **14.1-1711** and can include a protuberance or the like. In the example of FIGS. **14.1-19G** and **14.1-19H**, a protuberance can be included in the un-notched peripheral region **14.1-1713** of the detent **14.1-1710C**. This can increase mating and/or un-mating forces of the plug connector **14.1-1720** relative to the receptacle connector **14.1-1700**. A cross-sectional view of the detent **14.1-1710C** taken along cross-section A-A' of FIGS. **14.1-19G** can be the same as or similar to the cross-sectional view of FIGS. **14.1-19F**, discussed above.

[3620] FIGS. **14.1-19I** through **14.1-19K** illustrate a detent **14.1-1710D** that includes a central region **14.1-1711** and a peripheral region **14.1-1713** on one side of the central region **14.1-1711**. The central region **14.1-1711** can be notched to form a channel **14.1-1733** that can receive a protrusion **14.1-1724** of a plug connector **14.1-1720**. For example, a lower portion of the central region **14.1-1711** can be notched to form an undercut portion of the detent **14.1-1710D** in the central region **14.1-1711**. Further, an upper portion of the central region **14.1-1711** and the peripheral region **14.1-1713** can be notched to form a sloped upper surface of the detent **14.1-1710D** in the central region **14.1-1711** and the peripheral region **14.1-1713**. In the example of

FIGS. 14.1-19I through 14.1-19K, the notched portions of the detent 14.1-1710D can reduce mating and/or un-mating forces of the plug connector 14.1-1720 relative to the receptacle connector 14.1-1700. The peripheral region 14.1-1713 can extend radially inward relative to the central region 14.1-1711 and can include a protuberance or the like. In the example of FIGS. 14.1-19I through 14.1-19K, a protuberance can be included in the peripheral region 14.1-1713 with the un-notched lower portion and the notched upper portion. This can increase mating and/or un-mating forces of the plug connector 14.1-1720 relative to the receptacle connector 14.1-1700. [3621] FIGS. 14.1-19K shows a cross-sectional view of the protrusion 14.1-1724 of the plug connector 14.1-1720 disposed within the channel 14.1-1733 and interfacing the detent 14.1-1710D taken along reference line A-A' of FIGS. 14.1-19I. In some examples, the detent 14.1-1710D can define the channel 14.1-1733 such that a first surface 14.1-1744 of the channel 14.1-1733 is angled relative to a second surface 14.1-1746 of the channel 14.1-1733 to form an angle 14.1- θ .sub.5. The protrusion 14.1-1714 can engage or contact the first surface 14.1-1744 while disposed within the channel 14.1-1733. The angle 14.1- θ .sub.5 can directly correlate to a force required to overcome a biasing element coupled to the detent 14.1-1710D to cause the detent 14.1-1710D to move away from the central member 14.1-1712 and enable extraction of the protrusion 14.1-1724 from the channel 14.1-1733 (e.g., enabling extraction of the plug connector 14.1-1720 from the receptacle connector 14.1-1700). For example, a relatively smaller angle 14.1- θ .sub.5 can require a relatively greater force to extract the plug connector 14.1-1720 from the receptacle connector 14.1-1700. Alternatively, a relatively larger angle 14.1- θ .sub.5 can require relatively smaller force to extract the plug connector 14.1-1720 from the receptacle connector 14.1-1700. The angle 14.1- θ .sub.5 can be about 30 degrees, between about 30 degrees and about 60 degrees, between about 60 degrees and about 90 degrees, between about 90 degrees and about 120 degrees, between about 120 degrees and about 150 degrees, between about 150 degrees and about 180 degrees, or less than 180 degrees. A third surface 14.1-1742 adjacent the first surface 14.1-1744 can extend in a direction parallel to the second surface 14.1-1746.

[3622] In some examples, a top surface of the detent 14.1-1710D is also notched. As such, a fourth surface 14.1-1740 of the detent 14.1-1710D is angled relative to the third surface 14.1-1742 of the detent 14.1-1710D to form an angle 14.1- θ .sub.6. The protrusion 14.1-1714 can engage or contact the fourth surface 14.1-1740 as the protrusion 14.1-1714 is inserted into the channel 14.1-1733. The angle 14.1- θ .sub.6 can directly correlate to a force required to overcome a biasing element coupled to the detent 14.1-1710D to cause the detent 14.1-1710D to move away from the central member 14.1-1712 and enable insertion of the protrusion 14.1-1724 into the channel 14.1-1733 (e.g., enabling insertion of the plug connector 14.1-1720 from the receptacle connector 14.1-1700). For example, a relatively smaller angle 14.1- θ .sub.6 can require a relatively greater force to extract the plug connector 14.1-1720 from the receptacle connector 14.1-1700. Alternatively, a relatively larger angle 14.1- θ .sub.6 can require relatively smaller force to extract the plug connector 14.1-1720 from the receptacle connector 14.1-1700. The angle 14.1- θ .sub.6 can be about 30 degrees, between about 30 degrees and about 60 degrees, between about 60 degrees and about 90 degrees, between about 90 degrees and about 120 degrees, between about 120 degrees and about 150 degrees, between about 150 degrees and about 180 degrees, or less than 180 degrees.

[3623] FIGS. 14.1-19L and 14.1-19M illustrate a detent 14.1-1710E that includes a central region 14.1-1711 and a peripheral region 14.1-1713 on one side of the central region 14.1-1711. The central region 14.1-1711 can be notched to form a channel 14.1-1733 that can receive a protrusion 14.1-1724 of a plug connector 14.1-1720. For example, a lower portion of the central region 14.1-1711 can be notched to form an undercut portion of the detent 14.1-1710E in the central region 14.1-1711. The notches can be omitted from lower portion of the peripheral regions 14.1-1713, which can prevent, limit, or reduce rotation of the plug connector 14.1-1720. Including the un-notched peripheral regions 14.1-1713 can also increase mating forces and un-mating forces between the plug connector 14.1-1720 and the receptacle connector 14.1-1700. The central region

14.1-1711 and the peripheral region **14.1-1713** can be notched to form a notched top surface. The notched upper surface of the detent **14.1-1710E** can reduce a mating force for mating the plug connector **14.1-1720** in the receptacle connector **14.1-1700**. In other words, the notched upper surface of the detent **14.1-1710E** can reduce the force required to move the protrusions **14.1-1724** past the detent **14.1-1710E** into the channel **14.1-1733**. A cross-sectional view of the detent **14.1-1710E** taken along cross-section A-A' of FIGS. **14.1-19L** can be the same as or similar to the cross-sectional view of FIGS. **14.1-19F**, discussed above.

[3624] FIGS. **14.1-20A** and **14.1-20B** illustrate different configurations of a cable assembly **14.1-1800** relative to a receptacle connector **14.1-1802**. FIGS. **14.1-20A** and **14.1-20B** show a portion of the cable assembly **14.1-1800** inserted into the receptacle connector **14.1-1802**. The cable assembly **14.1-1800** can be substantially similar to, and can include some or all of, the features of any of the previously described cable assemblies. For example, the cable assembly **14.1-1800** can include a shielded cable **14.1-1803** and a plug connector **14.1-1804** having a boot **14.1-1806** and a central member **14.1-1808A/14.1-1808B**. The central member **14.1-1808** can form an external surface **14.1-1810** and an internal surface **14.1-1812**. In some examples, the central member **14.1-1808** can include one or more protrusions **14.1-1814** extending laterally from, or substantially perpendicular to, the external surface **14.1-1810**. In some examples, the central member **14.1-1808** can form a cavity or volume, for example, the internal surface **14.1-1812** can form a cavity or volume.

[3625] The receptacle connector **14.1-1802** can be substantially similar to, and can include some or all of, the features of the previously described receptacle connectors. For example, the receptacle connector **14.1-1802** can include a trim ring **14.1-1816**, one or more detents **14.1-1818A/14.1-1818B/14.1-1818C** (collectively referred to as detents **14.1-1818**), and a raised portion **14.1-1820**. The trim ring **14.1-1816** can form one or more undercut regions which house or retain the detents **14.1-1818**. While a central member **14.1-1808** is disposed within the receptacle connector **14.1-1802**, the protrusions **14.1-1814** can extend into respective channels defined by the detents **14.1-1818**. In some examples, each detent **14.1-1818** can be biased toward the raised portion **14.1-1820** by a biasing element. In other words, each detent **14.1-1818** can radially translate within their respective undercut regions toward and away from the raised portion **14.1-1820** to engage and disengage respective protrusions **14.1-1814**.

[3626] Each of the detents **14.1-1818** can include different configurations and different biasing elements to customize mating and un-mating forces for each of the detents **14.1-1818**. Specific configurations of the detents **14.1-1818** and the shielded cable **14.1-1803** relative to the detents **14.1-1818** can be used to provide different mating and un-mating forces, and to provide different breakaway forces. In the example of FIGS. **14.1-20A**, the detents **14.1-1818A** and **14.1-1818C** are offset from the shielded cable **14.1-1803** (e.g., the shielded cable **14.1-1803** splits the detents **14.1-1818A** and **14.1-1818B**) distal the shielded cable **14.1-1803**, and the detent **14.1-1818B** is aligned with the shielded cable **14.1-1803** proximal the shielded cable **14.1-1803**. Protrusions **14.1-1814** distal the shielded cable **14.1-1803** are offset from the shielded cable **14.1-1803** (e.g., the shielded cable **14.1-1803** splits the protrusions **14.1-1814**), and the protrusion **14.1-1814** proximal the shielded cable **14.1-1803** is aligned with the shielded cable **14.1-1803**. In the example of FIGS. **14.1-20B**, the detents **14.1-1818A** and **14.1-1818C** are offset from the shielded cable **14.1-1803** (e.g., the shielded cable **14.1-1803** splits the detents **14.1-1818A** and **14.1-1818B**) proximal the shielded cable **14.1-1803**, and the detent **14.1-1818B** is aligned with the shielded cable **14.1-1803** distal the shielded cable **14.1-1803**. Protrusions **14.1-1814** proximal the shielded cable **14.1-1803** are offset from the shielded cable **14.1-1803** (e.g., the shielded cable **14.1-1803** splits the protrusions **14.1-1814**), and the protrusion **14.1-1814** distal the shielded cable **14.1-1803** is aligned with the shielded cable **14.1-1803**. The detents **14.1-1818** can include any of the previously described detents, such as the detents described with respect to FIGS. **14.1-19A** through **14.1-19M**, and the biasing elements can include any of the previously described biasing elements. Any combination of detents **14.1-1818** and biasing elements can be provided.

[3627] FIGS. 14.1-21A and 14.1-21B illustrate examples in which the receptacle connector 14.1-1802 includes protrusions 14.1-1830 and the plug connector 14.1-1804 includes openings 14.1-1831 configured to receive the protrusions. The protrusions 14.1-1830 and the openings 14.1-1831 can be configured to ensure that the plug connector 14.1-1804 is properly aligned with and installed in a proper orientation relative to the receptacle connector 14.1-1802. FIGS. 14.1-21A and 14.1-21B further illustrate protrusions 14.1-1832 in dashed lines. In some examples, the receptacle connector 14.1-1802 includes a single protrusion 14.1-1830; however, the receptacle connector 14.1-1802 can include the protrusions 14.1-1832 in addition to or in place of the protrusion 14.1-1830, and the plug connector 14.1-1804 can include any number of the openings 14.1-1831 configured to receive the protrusion 14.1-1830 and/or the protrusions 14.1-1832. In some examples, the receptacle connector 14.1-1802 can include one protrusion 14.1-1830; one protrusion 14.1-1830 and one protrusion 14.1-1832; one protrusion 14.1-1830 and two protrusions 14.1-1832; or a greater number of protrusions.

[3628] FIGS. 14.1-21A and 14.1-21B illustrate alternative angular spacings of the protrusions 14.1-1830/14.1-1832 of the receptacle connector 14.1-1802. In FIGS. 14.1-21A, the protrusions 14.1-1830/14.1-1832 are evenly spaced relative to one another, with angles 14.1- θ .sub.7, 14.1- θ .sub.8, and 14.1- θ .sub.9 being about 60°. A protrusion 14.1-1830 and a protrusion 14.1-1832 distal the shielded cable 14.1-1803 can be offset from the shielded cable 14.1-1803 (e.g., the shielded cable 14.1-1803 splits the protrusion 14.1-1830 and the protrusion 14.1-1832), and a protrusion 14.1-1832 proximal the shielded cable 14.1-1803 can be aligned with the shielded cable 14.1-1803. Any rotational configuration of the protrusions 14.1-1830/14.1-1832 relative to the shielded cable 14.1-1803 can be provided. In FIGS. 14.1-21B, the protrusions 14.1-1830/14.1-1832 are spaced at different intervals relative to one another. Angles 14.1- θ .sub.10, 14.1- θ .sub.11, and 14.1- θ .sub.12 between adjacent protrusions 14.1-1830/14.1-1832 can be about 30 degrees, between about 30 degrees and about 60 degrees, between about 60 degrees and about 90 degrees, between about 90 degrees and about 120 degrees, between about 120 degrees and about 150 degrees, between about 150 degrees and about 180 degrees, or less than 180 degrees. A protrusion 14.1-1830 and a protrusion 14.1-1832 distal the shielded cable 14.1-1803 can be offset from the shielded cable 14.1-1803 (e.g., the shielded cable 14.1-1803 splits the protrusion 14.1-1830 and the protrusion 14.1-1832), and a protrusion 14.1-1832 proximal the shielded cable 14.1-1803 can also be offset from the shielded cable 14.1-1803. Any rotational configuration of the protrusions 14.1-1830/14.1-1832 relative to the shielded cable 14.1-1803 can be provided. Various configurations of the protrusions 14.1-1830/14.1-1832 can be used to ensure proper alignment of the plug connector 14.1-1804 with the receptacle connector 14.1-1802 and can also be used to ensure that incompatible plug connectors are not inadvertently installed in the receptacle connector 14.1-1802.

[3629] FIGS. 14.1-22A through 14.1-22E show cross-sectional side views of a plug connector 14.1-1904 of a cable assembly 14.1-1900 disposed within a receptacle connector 14.1-1902. The plug connector 14.1-1904 includes a central member 14.1-1908 with a protrusion 14.1-1914. The protrusion 14.1-1914 engages a channel of a detent 14.1-1918. The detent 14.1-1918 is biased towards a raised portion 14.1-1920 of the receptacle connector 14.1-1902 by a biasing element 14.1-1924. The biasing element 14.1-1924 and the detent 14.1-1918 are disposed within a trim ring 14.1-1916, which is coupled to a base 14.1-1930 of the receptacle connector 14.1-1902. An external surface 14.1-1910 of the central member 14.1-1908 includes the protrusion 14.1-1914 and is configured to engage the detent 14.1-1918. An internal surface 14.1-1912 of the central member 14.1-1908 includes an electrical contact 14.1-1919 and is electrically coupled to the raised portion 14.1-1920 of the receptacle connector 14.1-1902.

[3630] FIGS. 14.1-22A through 14.1-22E show various seals 14.1-1932 disposed between the central member 14.1-1908 and the receptacle connector 14.1-1902 at various locations to prevent ingress of contaminants into the receptacle connector 14.1-1902, such as, fluids, dust, liquids, and other materials that can degrade or impede the functionality of the plug connector 14.1-1904 and

the receptacle connector **14.1-1902**. The seal **14.1-1932** can be affixed to any surface of the plug connector **14.1-1904** and/or the receptacle connector **14.1-1902**. For example, the seal **14.1-1932** can be molded, adhered, welded, fastened, or otherwise secured to a surface of the plug connector **14.1-1904** and/or the receptacle connector **14.1-1902**.

[3631] FIGS. **14.1-22A** shows a seal **14.1-1932A** disposed between the base **14.1-1930** and the central member **14.1-1908**. As illustrated in FIGS. **14.1-22A**, an opening **14.1-1934A** can be provided in a bottom portion of the central member **14.1-1908**, and the seal **14.1-1932A** can be disposed in the opening **14.1-1934A** in the central member **14.1-1908**.

[3632] FIGS. **14.1-22B** shows a seal **14.1-1932B** disposed between the base **14.1-1930** and the central member **14.1-1908**. As illustrated in FIGS. **14.1-22B**, an opening **14.1-1934B** can be provided in a portion of the base **14.1-1930** adjacent a bottom portion of the central member **14.1-1908**, and the seal **14.1-1932B** can be disposed in the opening **14.1-1934B** in the base **14.1-1930**.

[3633] FIGS. **14.1-22C** shows a seal **14.1-1932C** disposed between the raised portion **14.1-1920** and the central member **14.1-1908**. As illustrated in FIGS. **14.1-22C**, an opening **14.1-1934C** can be provided in a portion of the raised portion **14.1-1920** adjacent a bottom portion of the central member **14.1-1908**, and the seal **14.1-1932C** can be disposed in the opening **14.1-1934C** in the raised portion **14.1-1920**.

[3634] FIGS. **14.1-22D** shows a seal **14.1-1932D** disposed between the trim ring **14.1-1916** and the central member **14.1-1908**. As illustrated in FIGS. **14.1-22D**, the seal **14.1-1932D** can be coupled to the central member **14.1-1908** between the central member **14.1-1908** and the trim ring **14.1-1916**. The seal **14.1-1932D** can include a protrusion that interacts with the trim ring **14.1-1916** to provide sealing between the central member **14.1-1908** and the trim ring **14.1-1916**.

[3635] FIGS. **14.1-22E** shows a seal **14.1-1932E** disposed between the trim ring **14.1-1916** and the central member **14.1-1908**. As illustrated in FIGS. **14.1-22E**, the seal **14.1-1932E** can be coupled to the central member **14.1-1908** between the central member **14.1-1908** and the trim ring **14.1-1916**. The seal **14.1-1932E** can include an opening, which gives flexibility to the seal **14.1-1932E** as the seal **14.1-1932E** interacts with the trim ring **14.1-1916** to provide sealing between the central member **14.1-1908** and the trim ring **14.1-1916**. While FIGS. **14.1-22A** through **14.1-22E** only depict a single seal **14.1-1932** in each example disposed between the central member **14.1-1908** and the receptacle connector **14.1-1902**, other examples can include multiple seals **14.1-1932**, each disposed at one of a plurality of positions within the receptacle connector **14.1-1902** and/or the central member **14.1-1908**.

[3636] FIGS. **14.1-23A** through **14.1-23G** illustrate various configurations of seals **14.1-2002** that can be included between a raised portion **14.1-2000** and a base **14.1-2006** of a receptacle connector and a central member **14.1-2008** of a plug connector. The seals **14.1-2002** can be disposed at various locations and can include different sizes and shapes to prevent ingress of contaminants into the receptacle connector, such as, fluids, dust, liquids, and other materials that can degrade or impede the functionality of the plug connector and the receptacle connector. The seals **14.1-2002** can be affixed to any surface of the plug connector and/or the receptacle connector. For example, the seals **14.1-2002** can be molded, adhered, welded, fastened, or otherwise secured to a surface of the plug connector and/or the receptacle connector.

[3637] FIGS. **14.1-23A** shows a seal **14.1-2002A** disposed vertically between the base **14.1-2006** and the raised portion **14.1-2000** and horizontally between the raised portion **14.1-2000** and the central member **14.1-2008**. The seal **14.1-2002A** can include a triangular portion that interfaces with the central member **14.1-2008**. An opening **14.1-2004A** can be provided in the raised portion **14.1-2000** and the seal **14.1-2002A** can be disposed in the opening **14.1-2004A**. The seal **14.1-2002A** can include a protrusion **14.1-2012A** that interfaces with an opening **14.1-2010A** of the base **14.1-2006**. The protrusion **14.1-2012A** can be generally rectangular. The seal **14.1-2002A** can be attached to the raised portion **14.1-2000**. The seal **14.1-2002A** can further be attached to the base **14.1-2006** through the protrusion **14.1-2012A**.

[3638] FIGS. **14.1-23B** shows a seal **14.1-2002B** disposed vertically between the base **14.1-2006** and the raised portion **14.1-2000** and horizontally between the raised portion **14.1-2000** and the central member **14.1-2008**. The seal **14.1-2002B** can be generally L-shaped, with a rectangular portion interfacing with the central member **14.1-2008**. An opening **14.1-2004B** can be provided in the raised portion **14.1-2000** and the seal **14.1-2002B** can be disposed in the opening **14.1-2004B**. The seal **14.1-2002B** can be attached to the raised portion **14.1-2000**.

[3639] FIGS. **14.1-23C** shows a seal **14.1-2002C** disposed vertically between the base **14.1-2006** and the raised portion **14.1-2000** and horizontally between the raised portion **14.1-2000** and the central member **14.1-2008**. The seal **14.1-2002C** can include a rounded portion that interfaces with the central member **14.1-2008**. An opening **14.1-2004C** can be provided in the raised portion **14.1-2000** and the seal **14.1-2002C** can be disposed in the opening **14.1-2004C**.

[3640] The seal **14.1-2002C** can include a protrusion **14.1-2012C** that interfaces with an opening **14.1-2010C** of the base **14.1-2006**. The protrusion **14.1-2012C** can be generally rounded. The seal **14.1-2002C** can be generally D-shaped. The seal **14.1-2002C** can be attached to the raised portion **14.1-2000**. The seal **14.1-2002C** can further be attached to the base **14.1-2006** through the protrusion **14.1-2012C**.

[3641] FIGS. **14.1-23D** shows a seal **14.1-2002D** disposed vertically between the base **14.1-2006** and the raised portion **14.1-2000** and horizontally between the raised portion **14.1-2000** and the central member **14.1-2008**. The seal **14.1-2002D** can further extend vertically below a top surface of the base **14.1-2006** and laterally adjacent the base **14.1-2006** in a direction opposite the central member **14.1-2008**. The seal **14.1-2002D** can include a rounded/generally rectangular portion that interfaces with the central member **14.1-2008**. An opening **14.1-2004D** can be provided in the raised portion **14.1-2000** and the seal **14.1-2002D** can be disposed in the opening **14.1-2004A**. The seal **14.1-2002A** can include a protrusion **14.1-2012D** that interfaces with an opening **14.1-2010D** of the base **14.1-2006**. The opening **14.1-2010D** of the base can be laterally opposite the central member **14.1-2008**. The seal **14.1-2002D** can further include an opening **14.1-2014D** disposed therein, which can be generally U-shaped. The opening **14.1-2014D** can provide additional flexibility to the seal **14.1-2002D** and can improve sealing between the central member **14.1-2008** and the receptacle connector. The seal **14.1-2002D** can be attached to the base **14.1-2006** through the protrusion **14.1-2012D** in the opening **14.1-2010D**. The seal **14.1-2002D** can further be attached to the raised portion **14.1-2000**.

[3642] FIGS. **14.1-23E** shows a seal **14.1-2002E** disposed vertically between the base **14.1-2006** and the raised portion **14.1-2000** and horizontally between the raised portion **14.1-2000** and the central member **14.1-2008**. The seal **14.1-2002E** can include a rounded portion that interfaces with the central member **14.1-2008**. An opening **14.1-2004E** can be provided in the raised portion **14.1-2000** and the seal **14.1-2002E** can be disposed in the opening **14.1-2004E**. The seal **14.1-2002E** can be generally stadium-shaped, or rectangular with rounded corners. The seal **14.1-2002E** can be attached to the raised portion **14.1-2000**. The seal **14.1-2002E** can further be attached to the base **14.1-2006**.

[3643] FIGS. **14.1-23F** shows a seal **14.1-2002F** disposed vertically between the base **14.1-2006** and the raised portion **14.1-2000** and horizontally between the raised portion **14.1-2000** and the central member **14.1-2008**. The seal **14.1-2002F** can include a rounded portion that interfaces with the central member **14.1-2008**. An opening **14.1-2004F** can be provided in the raised portion **14.1-2000** and the seal **14.1-2002F** can be disposed in the opening **14.1-2004F**. The seal **14.1-2002F** can be generally stadium-shaped, or rectangular with rounded corners. In the example of FIGS. **14.1-23F**, corners of the seal **14.1-2002F** proximal the central member **14.1-2008** are more rounded relative to corners of the seal **14.1-2002F** distal the central member **14.1-2008**. The seal **14.1-2002F** can be attached to the raised portion **14.1-2000**. The seal **14.1-2002F** can further be attached to the base **14.1-2006**.

[3644] FIGS. **14.1-23G** shows a seal **14.1-2002G** disposed vertically between the base **14.1-2006**

and the raised portion **14.1-2000** and horizontally between the raised portion **14.1-2000** and the central member **14.1-2008**. The seal **14.1-2002G** can include a rounded portion that interfaces with the central member **14.1-2008**. An opening **14.1-2004G** can be provided in the raised portion **14.1-2000** and the seal **14.1-2002G** can be disposed in the opening **14.1-2004G**. The seal **14.1-2002G** can be generally stadium-shaped, or rectangular with rounded corners. In the example of FIGS. **14.1-23G**, the seal **14.1-2002E** further includes an opening **14.1-2014G** in a bottom surface thereof, adjacent to the base **14.1-2006**. The opening **14.1-2014G** can provide improved flexibility to the seal **14.1-2002G**. The seal **14.1-2002G** can be attached to the raised portion **14.1-2000**. The seal **14.1-2002G** can further be attached to the base **14.1-2006**.

[3645] While FIGS. **14.1-23A** through **14.1-23G** only depict a single seal **14.1-2002** in each example disposed between the central member **14.1-2008** and the raised portion **14.1-2000** and the base **14.1-2006** of a receptacle connector, other examples can include multiple seals, each disposed at one of a plurality of positions within the receptacle connector and/or the central member. The seals **14.1-2002** of FIGS. **14.1-23A** through **14.1-23G** can be used in addition to or in place of any of the seals discussed above, such as the seals discussed with respect to FIGS. **14.1-22A** through **14.1-22E**.

[3646] FIGS. **14.1-24A** and **14.1-24B** show exploded views of a receptacle connector **14.1-2100**. The receptacle connector **14.1-2100** includes a base **14.1-2102**, a raised portion **14.1-2110**, mounting clips **14.1-2114**, a lower mounting ring **14.1-2116**, and an upper mounting ring **14.1-2122**. The mounting clips **14.1-2114**, the lower mounting ring **14.1-2116**, and the upper mounting ring **14.1-2122** can affix the raised portion **14.1-2110** to the base **14.1-2102**. For example, the upper mounting ring **14.1-2122** and the lower mounting ring **14.1-2116** can be coupled to the base **14.1-2102**. The base **14.1-2102** can include protrusions **14.1-2106** and a raised electronic component **14.1-2104**. The upper mounting ring **14.1-2122** can include openings **14.1-2124** configured to receive the protrusions **14.1-2106** of the base **14.1-2102**. The lower mounting ring **14.1-2116** can include openings **14.1-2120** configured to receive the protrusions **14.1-2106** of the base **14.1-2102**. The openings **14.1-2124** and the openings **14.1-2120** can have shapes corresponding to shapes of the protrusions **14.1-2106** (e.g., capsule shapes, rounded shapes, rectangular shapes, or the like). The lower mounting ring can further include an opening **14.1-2118** configured to receive the raised electronic component **14.1-2104** of the base **14.1-2102**. The opening **14.1-2118** can have a shape corresponding to a shape of the raised electronic component **14.1-2104** (e.g., a capsule shape, a rounded shape, a rectangular shape, or the like).

[3647] The mounting clips **14.1-2114** can be configured to receive a raised component **14.1-2112** of the raised portion **14.1-2110**. Each of the mounting clips **14.1-2114** can be generally L-shaped, with an outer surface of the mounting clips **14.1-2114** being rounded. Central portions of the mounting clips **14.1-2114** can receive the raised component **14.1-2112**. The lower mounting ring **14.1-2116** and the upper mounting ring **14.1-2122** can be affixed to the base **14.1-2102**, and the mounting clips **14.1-2114** can affix the raised component **14.1-2112** of the raised portion **14.1-2110** to the lower mounting ring **14.1-2116**, the upper mounting ring **14.1-2122**, and the base **14.1-2102**. The upper mounting ring **14.1-2122** and the lower mounting ring **14.1-2116** can be affixed to the base **14.1-2102**, the mounting clips **14.1-2114** can be affixed to the raised component **14.1-2112**, and the raised portion **14.1-2110** can be affixed to the base **14.1-2102** by adhesives, adhesive tapes, fasteners, welding, a combination thereof, or any other mechanism from permanently or temporarily securing the various components to one another.

[3648] In the example of FIGS. **14.1-24B**, the mounting clips **14.1-2114** are replaced by mounting clips **14.1-2126**. The mounting clips **14.1-2126** interface with openings **14.1-2128** in the lower mounting ring **14.1-2116** to attach the raised portion **14.1-2110** to the base **14.1-2102** through the lower mounting ring **14.1-2116**.

[3649] FIGS. **14.1-25A** and **14.1-25B** show exploded views of a receptacle connector **14.1-2200**. The receptacle connector **14.1-2200** includes a base **14.1-2210** and a raised portion that includes a

body **14.1-2204** and a cap **14.1-2202**. The base includes a contact tab **14.1-2212** with contacts **14.1-2214** disposed in an opening **14.1-2218** in a raised portion **14.1-2216** of the base **14.1-2210**. In the example of FIGS. **14.1-25A**, the contact tab **14.1-2212** is relatively wider and includes a greater number of contacts, such as six contacts. In the example of FIGS. **14.1-25B**, the contact tab **14.1-2212** is relatively narrower and includes a lower number of contacts, such as three contacts. Any number of contacts and appropriate width of the contact tab **14.1-2212** can be provided. Providing the contact tab **14.1-2212** with a greater width and a greater number of contacts improves the strength of the connection between the body **14.1-2204** and the base **14.1-2210** and provides for more electrical connections between the receptacle connector **14.1-2200** and other electronic components, such as a plug connector.

[3650] FIGS. **14.1-25A** and **14.1-25B** further illustrate different configurations of contacts **14.1-2206** included in an opening **14.1-2208** in the body **14.1-2204**. In the example of FIGS. **14.1-25A**, four contacts **14.1-2206** are included in the opening **14.1-2208** in the body **14.1-2204**. In the example of FIGS. **14.1-25B**, two contacts **14.1-2206** are included in the opening **14.1-2208** in the body **14.1-2204**. Any number of contacts **14.1-2206** can be provided. Providing a greater number of the contacts **14.1-2206** in the body **14.1-2204** provides for more electrical connections between the receptacle connector **14.1-2200** and other electronic components, such as a plug connector.

[3651] FIGS. **14.1-26A** through **14.1-35B** illustrate various examples of receptacle connectors and plug connectors that can be used as the plug connectors **14.1-144** of the cable assembly **14.1-140** and the receptacle connectors **14.1-132** of the second electronic device **14.1-130** illustrated in FIGS. **14.1-1A** and **14.1-1B**.

[3652] FIGS. **14.1-26A** shows an electronic device **14.1-2300** having a housing **14.1-2302** forming an external surface **14.1-2304** of the electronic device **14.1-2300**. The housing **14.1-2302** can form or define one or more apertures **14.1-2306**, **14.1-2308**. Each of the one or more apertures **14.1-2306**, **14.1-2308** can receive a connector of an electrical cord or cable (see cable **14.1-2310** of FIGS. **14.1-26B**) to electrically couple the electronic device **14.1-2300** to another electronic device (not shown) and/or a power source (e.g., a wall-mounted electrical outlet).

[3653] The electronic device **14.1-2300** can be a smart phone, a laptop, a tablet computing device, a smart watch, a head-mounted display (HMD), a pair of headphones, a combination thereof, or any other electronic device. Alternatively, or additionally, the electronic device **14.1-2300** can be a sub-component or an accessory to one or more of the electronic devices previously listed above, such as, an external power supply or electrical power source, a display, a user interface, an audio component or speaker, or a combination thereof. As such, the electronic device **14.1-2300** can be electrically and/or communicatively coupled to one or more other electronic devices (not shown) via electrical receptacle connectors or ports disposed within the one or more apertures **14.1-2306**, **14.1-2308**.

[3654] The housing **14.1-2302** can define a cavity or volume wherein one or more electronic components can be disposed. For example, the one or more electrical components disposed within the housing **14.1-2302** can be a battery (e.g., a lithium battery pack), a processor, a wireless communication module, one or more antennas, or any other component or assembly housed within portable electronic devices. In some examples, the one or more electronic components can be electrically coupled to a receptacle connector (see receptacle connector **2424** in FIGS. **14.1-27A**) accessible by a cable or cord inserted into one of the apertures **14.1-2306**, **14.1-2308**. For example, the one or more electronic components can provide electrical power to the receptacle connector and the cable or cord can transfer the electrical power to another electronic device (i.e., the electronic device **14.1-2300** can be a power source or external battery electrically coupled to a portable electronic device). The housing **14.1-2302** can be molded, machined, cast, stamped, or otherwise assembled from a metal, polymer, ceramic, or combination thereof.

[3655] The aperture **14.1-2306** can be formed within or defined by the external surface **14.1-2304** of the housing **14.1-2302**. The aperture **14.1-2306** can be sized and shaped to receive and retain a

plug connector of a cable (see cable **14.1-2310** in FIGS. **14.1-26B**). In some examples, a cable operably coupled to a receptacle connector within the aperture **14.1-2306** can transfer electrical power to the electronic device **14.1-2300** while the plug connector is disposed within the aperture **14.1-2306**. In other words, a receptacle connector within the aperture **14.1-2306** can be configured to receive electrical power (e.g., from a wall-mounted electrical outlet) and deliver the electrical power to the one or more components within the housing **14.1-2302** (e.g., charging a battery disposed within the housing **14.1-2302**).

[3656] The aperture **14.1-2308** can be formed within or defined by the external surface **14.1-2304** of the housing **14.1-2302**. The aperture **14.1-2308** can be sized and shaped to receive a plug connector of a cable. In some examples, a cable operably coupled to a receptacle connector within the aperture **14.1-2308** can transfer electrical power from the electronic device **14.1-2300** while the plug connector is disposed within the aperture **14.1-2308**. In other words, the receptacle connector within the aperture **14.1-2308** can be configured to deliver or provide electrical power from the one or more components within the housing **14.1-2302** to a portable electronic device electrically coupled to the cable.

[3657] While the apertures **14.1-2306**, **14.1-2308** are shown as being positioned on a particular side of the housing **14.1-2302**, the apertures **14.1-2306**, **14.1-2308** can be formed or otherwise positioned at any location on the external surface **14.1-2304** of the housing **14.1-2302**. Similarly, while the apertures **14.1-2306**, **14.1-2308** are shown as being disposed adjacent or substantially adjacent to one another, in some examples, the apertures **14.1-2306**, **14.1-2308** can be formed or otherwise positioned at nonadjacent locations on the external surface **14.1-2304** of the housing **14.1-2302** (e.g., on opposite sides of the housing **14.1-2302**). While the apertures **14.1-2306**, **14.1-2308** are depicted as having different dimensions (i.e., the aperture **14.1-2306** is illustrated as being relatively smaller than the aperture **14.1-2308**), the apertures **14.1-2306**, **14.1-2308** can have the same dimensions or substantially similar dimensions to one another in some examples. While two apertures (e.g., apertures **14.1-2306**, **14.1-2308**) are shown in FIGS. **14.1-26A** and **14.1-26B**, some examples can have more or fewer apertures. For example, the housing **14.1-2302** can form a single aperture which is operable to provide electrical power and/or signals to a portable electronic device and also operable to receive electrical power and/or signals (e.g., from a portable electronic device and/or a wall-mounted electrical outlet). In some examples, the housing **14.1-2302** can form two apertures, such as a first aperture operable to provide electrical power and/or signals to a portable electronic device and also operable to receive electrical power and/or signals (e.g., from a portable electronic device and/or a wall-mounted electrical outlet), and a second aperture operable to receive a tool that can be used to remove a cable from the first aperture.

[3658] FIGS. **14.1-26B** shows the electronic device **14.1-2300** and a cable **14.1-2310** receivable within one or more of the apertures **14.1-2306**, **14.1-2308**. The cable **14.1-2310** can have a plug connector **14.1-2312** having a plug **14.1-2314** and a boot **14.1-2316**. The plug connector **14.1-2312** can be electrically coupled to a length of shielded wire **14.1-2318**. The shielded wire **14.1-2318** can be one or more wires (e.g., metallic wires capable of transferring electrical signals and/or electrical power). The one or more wires can be individually shielded or collectively shielded. The shielding can prevent each of the one or more wires from contacting one another. Additionally, or alternatively, the shielding can prevent or limit the influence of electromagnetic waves on each of the one or more wires. The plug connector **14.1-2312** can be received within one of the apertures **14.1-2306**, **14.1-2308** to electrically couple the cable **14.1-2310** with the electronic device **14.1-2300**. For example, the plug **14.1-2314** can be inserted within a receptacle connector (e.g., receptacle connector **14.1-2424** of FIGS. **14.1-27A**) disposed within the housing **14.1-2302** of the electronic device **14.1-2300** to enable transfer of electrical power and/or signals relative to the electronic device **14.1-2300**. The boot **14.1-2316** can be inserted within a trim ring (e.g., trim ring **14.1-2422** of FIGS. **14.1-27A**) disposed within the housing **14.1-2302** of the electronic device **14.1-2300**, between the receptacle connector and the housing **14.1-2302**. One or more of the plug

14.1-2314, the boot **14.1-2316**, the apertures **14.1-2306**, **14.1-2308**, the receptacle connector, and the trim ring can include one or more engagement features that retain the plug connector **14.1-2312** to the electronic device **14.1-2300**. The mechanisms and components capable of inter coupling the cable **14.1-2310** and the electronic device **14.1-2300** will be discussed in greater detail below with reference to FIGS. **14.1-27A** through **14.1-35B**.

[3659] FIGS. **14.1-27A** shows an electronic device **14.1-2400** and a cable **14.1-2410**. The electronic device **14.1-2400** can be substantially similar to, and can include some or all of, the features of the electronic device **14.1-2300**. For example, the electronic device **14.1-2400** can include a housing **14.1-2402** forming a cavity or volume **14.1-2420**. One or more electronic components can be disposed within the volume **14.1-2420**. For example, the one or more electronic components disposed within the volume **14.1-2420** can include a trim ring **14.1-2422** and a receptacle connector **14.1-2424** operably coupled to a battery **14.1-2426**, a processor **14.1-2428**, a combination thereof, or any other electronic component. The trim ring **14.1-2422** and the receptacle connector **14.1-2424** can form an electrical port disposed within the volume **14.1-2420**. In some examples, an external surface **14.1-2404** of the housing **14.1-2402** can form an aperture **14.1-2408** capable of receiving a plug connector **14.1-2412** of the cable **14.1-2410**. The trim ring **14.1-2422** can be positioned adjacent the aperture **14.1-2408** such that a boot **14.1-2416** of the plug connector **14.1-2412** is disposed within the trim ring **14.1-2422**. The receptacle connector **14.1-2424** can be positioned adjacent the trim ring **14.1-2422** such that a plug of the plug connector **14.1-2412** is disposed within the receptacle connector **14.1-2424**. The trim ring **14.1-2422** and the receptacle connector **14.1-2424** will be discussed in detail below with reference to FIGS. **14.1-27B**.

[3660] FIGS. **14.1-27B** shows a partially exploded view of the cable **14.1-2410**, the trim ring **14.1-2422**, and the receptacle connector **14.1-2424**. The cable **14.1-2410** can be substantially similar to, and can include some or all of, the features of the cable **14.1-2310**. For example, the cable **14.1-2410** can include the plug connector **14.1-2412** having a plug **14.1-2414** and a boot **14.1-2416**. The plug connector **14.1-2412** can be electrically coupled to a length of shielded wire **14.1-2418**.

[3661] In some examples, the boot **14.1-2416** can be sized or otherwise shaped to be at least partially receivable within a cavity **14.1-2430** formed within the trim ring **14.1-2422**. In some examples, the boot **14.1-2416** can be sized or otherwise shaped to be at least partially receivable within the cavity **14.1-2430** of the trim ring **14.1-2422** and partially receivable within the aperture **14.1-2408** of the housing **14.1-2402**. The cavity **14.1-2430** can be shaped as cubic, cylindrical, triangular, or any other geometric shape. The boot **14.1-2416** can form a corresponding geometric shape, which can be inserted or otherwise positioned within the cavity **14.1-2430**. In some examples, the boot **14.1-2416** can be partially disposed within the cavity **14.1-2430** such that a portion of the boot **14.1-2416** protrudes from the cavity **14.1-2430**. In some examples, the boot **14.1-2416** can be entirely surrounded or substantially surrounded by the trim ring **14.1-2422** and within the cavity **14.1-2430**, such that a wire-side surface **14.1-2434** of the boot **14.1-2416** adjacent the shielded wire **14.1-2418** is flush or substantially flush with a surface **14.1-2436** of the trim ring **14.1-2422** adjacent the cavity **14.1-2430**. In some examples, the surface **14.1-2432** of the boot **14.1-2416** can be disposed within the cavity **14.1-2430** of the trim ring **14.1-2422** such that the wire-side surface **14.1-2434** is flush or substantially flush with the external surface **14.1-2404** of the housing **14.1-2402** and the boot **14.1-2416** is substantially concealed or hidden within the cavity **14.1-2430** of the trim ring **14.1-2422**. In some examples, while the boot **14.1-2416** is disposed within the cavity **14.1-2430**, the boot **14.1-2416** can be substantially covered by the trim ring **14.1-2422**. This prevents the boot **14.1-2416** from being gripped (e.g., by a user) and pulled from the trim ring **14.1-2416**. In other words, a substantial portion of the boot **14.1-2416** can be disposed within the cavity **14.1-2430** such that the boot **14.1-2416** is inaccessible to grasp to pull the plug connector **14.1-2412** from the trim ring **14.1-2422** and the receptacle connector **14.1-2424**.

[3662] The plug **14.1-2414** can include one or more electrical contacts **14.1-2438A**, **14.1-2438B**, and **14.1-2438C**, which can provide electrical signals, electrical power, a grounding path, another

electrical communication, or a combination thereof between the shielded wire **14.1-2418** and the receptacle connector **14.1-2424**.

[3663] The trim ring **14.1-2422** can be molded, machined, cast, stamped, or otherwise manufactured from a metal, polymer, ceramic, or combination thereof. For example, the trim ring **14.1-2422** can be molded or co-molded from a thermoplastic or other polymer. The cavity **14.1-2430** formed within the trim ring **14.1-2422** can be at least partially defined by internal walls **14.1-2440**. In some examples, the boot **14.1-2416** can contact one or more of the internal walls **14.1-2440** while the boot **14.1-2416** is positioned within the cavity **14.1-2430** such that the boot **14.1-2416** is at least partially retained within the cavity **14.1-2430** by a frictional engagement between the boot **14.1-2416** and the internal walls **14.1-2440**.

[3664] In some examples, the receptacle connector **14.1-2424** can be adhered, molded, welded, fastened, or otherwise affixed to the trim ring **14.1-2422**. For example, the receptacle connector **14.1-2424** can define one or more receptacle apertures **14.1-2444A**, **14.1-2444B**, which correspond to and align with respective trim ring apertures **14.1-2442A**, **14.1-2442B** of the trim ring **14.1-2422**. A respective fastener (not shown) can be disposed within the receptacle aperture **14.1-2444A** and the trim ring aperture **14.1-2442A** to couple the receptacle connector **14.1-2424** to the trim ring **14.1-2422**. Similarly, a respective fastener (not shown) can be disposed within the receptacle aperture **14.1-2444B** and the trim ring aperture **14.1-2442B** to couple the receptacle connector **14.1-2424** to the trim ring **14.1-2422**. The fastener can be a pin, rivet, screw, bolt, a combination thereof, or any other fastener.

[3665] The boot **14.1-2416** can be molded, machined, cast, stamped, or otherwise manufactured from a metal, polymer, ceramic, or combination thereof. As illustrated in FIGS. **14.1-27B**, the boot **14.1-2416** can include recesses **14.1-2462A** and **14.1-2462B** formed in opposite sides of the boot **14.1-2416**. The recesses **14.1-2462A** and **14.1-2462B** can be machined, molded, etched, or otherwise recessed into an exterior surface of the boot **14.1-2416**. In the example of FIGS. **14.1-27B**, the recesses **14.1-2462A** and **14.1-2462B** extend along only a portion of the periphery of the boot **14.1-2416**. For example, the recesses **14.1-2462A** and **14.1-2462B** can extend along one or more sides of the boot **14.1-2416** (e.g., two sides in the example of FIGS. **14.1-27B**). In some examples, recesses can extend along an entire periphery of the boot **14.1-2416**. In other words, the recesses can form a channel encircling the boot **14.1-2416**. The recesses **14.1-2462A** and **14.1-2462B** can be formed between a plug-side surface and a wire-side surface of the boot **14.1-2416**. In some examples, the recesses **14.1-2462A** and **14.1-2462B** can extend parallel to and between the plug-side surface and the wire-side surface of the boot **14.1-2416**. The recesses **14.1-2462A** and **14.1-2462B** can be included to interface with structures of the trim ring **14.1-2422** in order to retain the boot **14.1-2416** and the plug connector **14.1-2412** within the channel **14.1-2430**. The trim ring **14.1-2422** and the boot **14.1-2416** will be discussed in greater detail below with reference to FIGS. **14.1-28A** through **14.1-28F**.

[3666] In some examples, the receptacle connector **14.1-2424** can include an exterior portion **14.1-2446** and an interior portion **14.1-2448**. For example, the exterior portion **14.1-2446** can at least partially encompass the interior portion **14.1-2448**. In other words, the exterior portion **14.1-2446** can act as a sleeve or shell that at least partially surrounds the interior portion **14.1-2448**. The exterior portion **14.1-2446** and interior portion **14.1-2448** can be interlocked or affixed to one another. For example, the exterior portion **14.1-2446** can include a through-hole **14.1-2450** that receives a protrusion **14.1-2452** of the interior portion **14.1-2448** when the interior portion **14.1-2448** is affixed within the exterior portion **14.1-2446**. In some examples, the receptacle connector **14.1-2424** can be formed from a singular or monolithic structure instead of the external portion **14.1-2446** and the internal portion **14.1-2448**. The receptacle connector **14.1-2424** can form a slot **14.1-2454** capable of receiving and retaining the plug **14.1-2414**.

[3667] FIGS. **14.1-28A** shows sectional views of a cable assembly **14.1-2500** including a plug connector **14.1-2502** received within an opening **14.1-2522** of a trim ring **14.1-2520** and a slot

14.1-2554 of a receptacle connector **14.1-2550**. In some examples, one or more internal contacts **14.1-2512A**, **14.1-2512B**, **14.1-2512C** are disposed within the slot **14.1-2554** of the receptacle connector **14.1-2550** such that the contacts **14.1-2512A**, **14.1-2512B**, **14.1-2512C** of the plug connector **14.1-2502** touch or otherwise engage internal contacts **14.1-2552A**, **14.1-2552B**, **14.1-2552C** positioned within the slot **14.1-2554**. In some examples, the internal contacts **14.1-2552A**, **14.1-2552B**, **14.1-2552C** disposed within the slot **14.1-2554** can be one or more metal prongs biased to engage the contacts **14.1-2512A**, **14.1-2512B**, **14.1-2512C** while the plug connector **14.1-2502** is disposed within the slot **14.1-2554**. The internal contacts **14.1-2552A**, **14.1-2552B**, **14.1-2552C** can be electrically connected to a printed circuit board (PCB), a processor, one or more wires, a combination thereof, or another component positioned within the housing (e.g., the housing **14.1-2302**, **14.1-2402**, discussed above with respect to FIGS. **14.1-26A** through **14.1-27B**). [3668] The cable assembly **14.1-2500** includes the plug connector **14.1-2502** and a shielded cable **14.1-2508**. The plug connector **14.1-2502** includes a boot **14.1-2506** having openings **14.1-2510** formed in the side surfaces thereof and a plug **14.1-2504** including the contacts **14.1-2512A**, **14.1-2512B**, and **14.1-2512C**.

[3669] In the example of FIGS. **14.1-28A** through **14.1-28F**, the trim ring **14.1-2520** includes a detent element **14.1-2530** that can be configured to interface the recesses **14.1-2510** of the boot **14.1-2506** in order to retain the plug connector **14.1-2502** within the cavity **14.1-2522**. Although a single detent element **14.1-2530** is illustrated in FIGS. **14.1-28A** through **14.1-28F**, some examples can include multiple detent elements **14.1-2530**. For example, in some examples, the trim ring **14.1-2520** can include two detent elements **14.1-2530** disposed on opposite sides of the plug connector **14.1-2502** when the plug connector **14.1-2502** is inserted into the cavity **14.1-2522**. While the plug connector **14.1-2502** is disposed within the cavity **14.1-2522**, the detent element **14.1-2530** can be inserted into the recesses **14.1-2510** defined in the boot **14.1-2506**.

[3670] FIGS. **14.1-28A** illustrates the plug connector **14.1-2502** and the detent element **14.1-2530** as the plug connector **14.1-2502** is inserted into the cavity **14.1-2522**. The detent element **14.1-2530** includes a protrusion **14.1-2532**, which is used couple the detent element **14.1-2530** to a lever arm **14.1-2540**. The lever arm **14.1-2540** can be included in the trim ring **14.1-2520** to apply a force, such as a spring force, to the detent element **14.1-2530**. The lever arm **14.1-2540** can be rotatable around a pivot **14.1-2542**. A spring (not illustrated in FIGS. **14.1-28A**) can be wrapped around the pivot **14.1-2542** to apply a force to the lever arm **14.1-2540** in a direction **14.1-D.sub.1**. The lever arm **14.1-2540** includes protrusions **14.1-2544**, which interface with the protrusion **14.1-2532** of the detent element **14.1-2530**. The detent element **14.1-2530** is physically coupled to the lever arm **14.1-2540** through the protrusion **14.1-2532** and the protrusions **14.1-2544**. The lever arm **14.1-2540** can apply the force to the detent element **14.1-2530** through the protrusions **14.1-2544** and the protrusion **14.1-2532** in a direction **14.1-D.sub.2** in order to retain the detent element **14.1-2530** in an initial position (illustrated in FIGS. **14.1-28A**) prior to the plug connector **14.1-2502** being inserted into the cavity **14.1-2522**. When the plug connector **14.1-2502** is further inserted into the cavity **14.1-2522** (illustrated in FIGS. **14.1-28C**), the force applied by the lever arm **14.1-2540** can cause friction between the detent element **14.1-2530** and the boot **14.1-2506**. Once the plug connector **14.1-2502** is fully inserted into the cavity **14.1-2522**, the force applied by the lever arm **14.1-2540** causes the detent element **14.1-2530** to slide into one of the recesses **14.1-2510** (illustrated in FIGS. **14.1-28D**).

[3671] The lever arm **14.1-2540** can be rotatable around the pivot **14.1-2542** in the direction **14.1-D.sub.1** and a direction **14.1-D.sub.3** (illustrated in FIGS. **14.1-28C** and **14.1-28E**) opposite the direction **14.1-D.sub.1**. The lever arm **14.1-2540** can include the spring, which exerts a force on the lever arm **14.1-2540** in the direction **14.1-D.sub.1**. The lever arm **14.1-2540** can include a protrusion **14.1-2546**, which can be used to rotate the lever arm **14.1-2540** in the direction **14.1-D.sub.3**. The detent element **14.1-2530** can translate or slide in the direction **14.1-D.sub.2** and a direction **14.1-D.sub.4** (illustrated in FIGS. **14.1-28C** and **14.1-28E**) opposite the direction **14.1-**

D.sub.2.

[3672] The detent element **14.1-2530** and the boot **14.1-2506** of the plug connector **14.1-2502** can include sloped portions **14.1-2534** and **14.1-2511**, respectively. The sloped portions **14.1-2534** and **14.1-2511** are provided in the detent element **14.1-2530** and the boot **14.1-2506** to allow the boot **14.1-2506** and the detent element **14.1-2530** to slide over one another as the plug connector **14.1-2502** is inserted into the cavity **14.1-2522**. This reduces the amount of force required to translate the detent element **14.1-2530** in the direction **14.1-D.sub.4** as the plug connector **14.1-2502** is inserted into the cavity **14.1-2522** and reduces the force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522**. The sloped portions **14.1-2534** and **14.1-2511** of the detent element **14.1-2530** and the boot **14.1-2506** will be described in detail below with reference to FIGS. **14.1-28B**.

[3673] FIGS. **14.1-28B** shows a detail view of the detent element **14.1-2530** and the boot **14.1-2506**. Specifically, FIGS. **14.1-28B** illustrates details of the sloped portions **14.1-2534** and **14.1-2511** of the detent element **14.1-2530** and the boot **14.1-2506**. FIGS. **14.1-28B** also illustrates details of a sloped portion **14.1-2514** of the boot **14.1-2506** adjacent the recess **14.1-2510**.

[3674] In the example of FIGS. **14.1-28B**, the detent element **14.1-2530** includes the sloped portion **14.1-2534** adjacent a linear portion **14.1-2536**. In some examples, the sloped portion **14.1-2534** can be curved, rather than sloped. The linear portion **14.1-2536** and the sloped portion **14.1-2534** can form a first angle **14.1- θ .sub.13**. The first angle **14.1- θ .sub.13** can be less than 90 degrees, such as, between about 89 degrees and about 60 degrees, between about 60 degrees and about 30 degrees, between about 30 degrees and about 5 degrees, or greater than about 5 degrees.

[3675] The boot **14.1-2506** includes the sloped portion **14.1-2511** adjacent a linear portion **14.1-2513**. In some examples, the sloped portion **14.1-2511** can be curved, rather than sloped. The linear portion **14.1-2513** and the sloped portion **14.1-2511** can form a second angle **14.1- θ .sub.14**. The second angle **14.1- θ .sub.14** can be less than 90 degrees, such as, between about 89 degrees and about 60 degrees, between about 60 degrees and about 30 degrees, between about 30 degrees and about 5 degrees, or greater than about 5 degrees. In some examples, the second angle **14.1- θ .sub.14** can be the same as the first angle **14.1- θ .sub.13**, or the second angle **14.1- θ .sub.14** can be different from the first angle **14.1- θ .sub.13**.

[3676] In some examples, a force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522** can correlate to the first angle **14.1- θ .sub.13** and/or the second angle **14.1- θ .sub.14**. For example, a relatively smaller first angle **14.1- θ .sub.13** (e.g., between about 5 degrees and 45 degrees) can cause the force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522** to be relatively greater. Similarly, a relatively smaller second angle **14.1- θ .sub.14** (e.g., between about 5 degrees and 45 degrees) can cause the force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522** to be relatively greater. Alternatively, a relatively larger first angle **14.1- θ .sub.13** (e.g., between about 45 degrees and 89 degrees) can cause the force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522** to be relatively smaller. A relatively larger second angle **14.1- θ .sub.14** (e.g., between about 45 degrees and 89 degrees) can cause the force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522** to be relatively smaller. In other words, sizes of the first angle **14.1- θ .sub.13** and the second angle **14.1- θ .sub.14** can be chosen to generate a desired force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522** (i.e., move the detent element **14.1-2530** out of the path of the boot **14.1-2506** as the plug connector **14.1-2502** is inserted into the cavity **14.1-2522**). The force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522** can be in a range from about 1 kgf to about 1.2 kgf, in a range from about 1 kgf to about 2 kgf, in a range from about 0.5 kgf to about 5 kgf, in a range from about 0 kgf to about 1 kgf, in a range from about 1 kgf to about 10 kgf, or the like.

[3677] In the example of FIGS. **14.1-28B**, the boot **14.1-2506** includes the sloped portion **14.1-2514** adjacent a linear portion **14.1-2512**. The sloped portion **14.1-2514** defines a side surface of the recess **14.1-2510**. In some examples, the sloped portion **14.1-2514** can be curved, rather than

sloped. The linear portion **14.1-2512** and the sloped portion **14.1-2514** can form a third angle **14.1-θ.sub.15**. The third angle **14.1-θ.sub.15** can be less than 90 degrees, such as, between about 89 degrees and about 60 degrees, between about 60 degrees and about 30 degrees, between about 30 degrees and about 5 degrees, or greater than about 5 degrees.

[3678] In some examples, a force required to remove the plug connector **14.1-2502** from the cavity **14.1-2522** can correlate to the third angle **14.1-θ.sub.15**. For example, a relatively smaller third angle **14.1-θ.sub.15** (e.g., between about 5 degrees and 45 degrees) can cause the force required to remove the plug connector **14.1-2502** from the cavity **14.1-2522** to be relatively smaller.

Alternatively, a relatively larger third angle **14.1-θ.sub.15** (e.g., between about 45 degrees and 89 degrees) can cause the force required to remove the plug connector **14.1-2502** from the cavity **14.1-2522** to be relatively greater. In other words, a size of the third angle **14.1-θ.sub.15** can be chosen to generate a desired force required to remove the plug connector **14.1-2502** from the cavity **14.1-2522** (i.e., move the detent element **14.1-2530** out of the path of the boot **14.1-2506** as the plug connector **14.1-2502** is removed from the cavity **14.1-2522**). The force required to remove the plug connector **14.1-2502** from the cavity **14.1-2522** can be in a range from about 5 kgf to about 7 kgf, in a range from about 1 kgf to about 12 kgf, in a range from about 10 kgf to about 20 kgf, in a range from about 5 kgf to about 50 kgf, in a range from about 1 kgf to about 10 kgf, in a range from about 10 kgf to about 100 kgf, or the like.

[3679] FIGS. **14.1-28C** illustrates the plug connector **14.1-2502** and the detent element **14.1-2530** as the plug connector **14.1-2502** is being inserted into the cavity **14.1-2522**. As illustrated in FIGS. **14.1-28C**, inserting the boot **14.1-2506** into the cavity **14.1-2522** causes the detent element **14.1-2470** to slide/translate in the direction **14.1-D.sub.4**. The movement of the detent element **14.1-2530** causes the lever arm **14.1-2540** to rotate around the pivot **14.1-2542** in the direction **14.1-D.sub.3** through the protrusions **14.1-2544** and the protrusion **14.1-2532**. The spring of the lever arm **14.1-2540** causes a force to be applied to the detent element **14.1-2530** in the direction **14.1-D.sub.2**, which results in friction between the detent element **14.1-2530** and the boot **14.1-2506** as the plug connector **14.1-2502** is inserted into the cavity **14.1-2522**. The force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522** can be in a range from about 1 kgf to about 1.2 kgf, in a range from about 1 kgf to about 2 kgf, in a range from about 0.5 kgf to about 5 kgf, in a range from about 0 kgf to about 1 kgf, in a range from about 1 kgf to about 10 kgf, or the like.

[3680] FIGS. **14.1-28D** illustrates the plug connector **14.1-2502** and the detent element **14.1-2530** after the plug connector **14.1-2502** has been fully inserted into the cavity **14.1-2522**. As illustrated in FIGS. **14.1-28D**, the spring of the lever arm **14.1-2540** causes a force to be applied to the detent element **14.1-2530**. This rotates the lever arm **14.1-2540** in the direction **14.1-D.sub.1** and translates the detent element **14.1-2530** in the direction **14.1-D.sub.2** such that the detent element **14.1-2530** slides into the recess **14.1-2510**. The detent element **14.1-2530** can contact a side surface of the boot **14.1-2506** when the plug connector **14.1-2502** is fully inserted into the cavity **14.1-2522** (e.g., the detent element **14.1-2530** is fully inserted into the recess **14.1-2510**). For example, the detent element **14.1-2530** can contact a side surface of the boot **14.1-2506** adjacent the recess **14.1-2510** and the sloped portion **14.1-2514**.

[3681] As described previously, the sloped portion **14.1-2514** of the boot **14.1-2506** can be configured such that a force required to remove the plug connector **14.1-2502** from the cavity **14.1-2522** can be in a range from about 5 kgf to about 7 kgf, in a range from about 1 kgf to about 12 kgf, in a range from about 10 kgf to about 20 kgf, in a range from about 5 kgf to about 50 kgf, in a range from about 1 kgf to about 10 kgf, in a range from about 10 kgf to about 100 kgf, or the like. A ratio of the force required to remove the plug connector **14.1-2502** from the cavity **14.1-2522** to the force required to insert the plug connector **14.1-2502** into the cavity **14.1-2522** can be in a range from about 1 to about 10, from about 1.5 to about 3, from about 2 to about 4, from about 3 to about 5, from about 1.5 to about 5, or the like. In some examples, the detent element **14.1-2530** can also include a sloped portion, which can be configured to interface with the sloped portion **14.1-**

2514 and can reduce the force required to remove the plug connector **14.1-2502** from the cavity **14.1-2522**. As illustrated in FIGS. **14.1-28D**, in some examples, the boot **14.1-2506** can be flush with the trim ring **14.1-2520** (i.e., an external-facing surface of the boot **14.1-2506** can be flush with an external-facing surface of the trim ring **14.1-2520**) when the plug connector **14.1-2502** is disposed within the trim ring **14.1-2520**.

[3682] FIGS. **14.1-28E** illustrates the detent element **14.1-2530** and the lever arm **14.1-2540** after a tool **14.1-2560** is used to release the plug connector **14.1-2502**. In some examples, the trim ring **14.1-2520** can include an aperture **14.1-2524** that can receive the tool **14.1-2560**. The tool **14.1-2560** can be shaped as cubic, cylindrical, triangular, or any other geometric shape. The aperture **14.1-2524** can form a corresponding geometric shape, into which the tool **14.1-2560** can be inserted or otherwise positioned. A user can use the tool **14.1-2560** to rotate the lever arm **14.1-2540** in the direction **14.1-D.sub.3**, which moves the detent element **14.1-2530** in the direction **14.1-D.sub.4**. This slides the detent element **14.1-2530** out of the recess **14.1-2510** and allows the plug connector **14.1-2502** to be removed from the cavity **14.1-2522**. Once the detent element **14.1-2530** is moved out of the recess **14.1-2510**, a force required to remove the plug connector **14.1-2502** from the cavity **14.1-2522** can be in a range from about 1 kgf to about 1.2 kgf, in a range from about 1 kgf to about 2 kgf, in a range from about 0.5 kgf to about 5 kgf, in a range from about 0 kgf to about 1 kgf, in a range from about 1 kgf to about 10 kgf, or the like. A force required to use the tool **14.1-2560** to remove the detent element **14.1-2530** from the recess **14.1-2519** can be in a range from about 1 kgf to about 1.2 kgf, in a range from about 1 kgf to about 2 kgf, in a range from about 0.5 kgf to about 5 kgf, in a range from about 0 kgf to about 1 kgf, in a range from about 1 kgf to about 10 kgf, or the like.

[3683] FIGS. **14.1-28F** illustrates a perspective view of the lever arm **14.1-2540**. As illustrated in FIGS. **14.1-28F**, the lever arm **14.1-2540** includes a body portion that is rotatable around a pivot **14.1-2542**. The pivot **14.1-2542** can be a pin, rivet, screw, bolt, a combination thereof, or any other fastener. A spring **14.1-2541** is disposed in the body portion and wrapped around the pivot **14.1-2542**. The spring **14.1-2541** is configured to apply a spring force to the body portion of the lever arm **14.1-2540** in a direction **14.1-D.sub.1**. The body portion of the lever arm **14.1-2540** includes protrusions **14.1-2544** and a protrusion **14.1-2546**. The protrusions **14.1-2544** can be used to physically interact with a detent element, as discussed previously. The protrusion **14.1-2546** can be used to interact with a tool, as discussed previously, and can be used to aid in removing a plug connector from a trim ring.

[3684] Providing the detent element **14.1-2530** and the lever arm **14.1-2540** to retain the plug connector **14.1-2502** in the cavity **14.1-2522** allows the plug connector **14.1-2502** to be inserted into the cavity **14.1-2522** with a relatively small force and requires a relatively large force to remove the plug connector **14.1-2502** from the cavity **14.1-2522**. This prevents the plug connector **14.1-2502** from unwantedly or accidentally being disconnected from an electronic device. Moreover, by providing the aperture **14.1-2524** and the tool **14.1-2560**, the plug connector **14.1-2502** can be easily removed when desired.

[3685] FIGS. **14.1-29A** through **14.1-29E** illustrate seals **14.1-2626** that can be included in a trim ring **14.1-2620** for providing sealing between the trim ring **14.1-2620** and a boot **14.1-2606** of a plug connector **14.1-2602** of a cable assembly **14.1-2600**. The cable assembly **14.1-2600** includes the plug connector **14.1-2602** and a shielded cable **14.1-2608** coupled to the plug connector **14.1-2602**. The plug connector **14.1-2602** includes a plug **14.1-2604** and the boot **14.1-2606** coupled to the plug **14.1-2604** and the shielded cable **14.1-2608**.

[3686] The plug connector **14.1-2602** can be received in a cavity **14.1-2622** of an enclosure **14.1-2610** and the trim ring **14.1-2620**. The trim ring **14.1-2620** can be coupled to the enclosure **14.1-2610**, and a receptacle connector **14.1-2650** can be coupled to the trim ring **14.1-2620**. The trim ring **14.1-2620** can include a detent element **14.1-2630** and a lever arm **14.1-2640**. The detent element **14.1-2630** can include a protrusion **14.1-2632** configured to interface with the lever arm

14.1-2640. The detent element **14.1-2630** can be configured to interface with openings **14.1-2609** in the boot **14.1-2606** to fasten the plug connector **14.1-2602** in the cavity **14.1-2622**. The lever arm **14.1-2640** can be rotatable around a pivot **14.1-2642** and can include protrusions **14.1-2644** configured to interface with the protrusion **14.1-2632** of the detent element **14.1-2632**. The lever arm **14.1-2640** can further include a protrusion **14.1-2646** configured to interface with a tool **14.1-2660** in order to un-fasten the plug connector **14.1-2602** from the cavity **14.1-2622**. An aperture **14.1-2624** can be provided in the trim ring **14.1-2620** and the enclosure **14.1-2610** in order to provide access to the tool **14.1-2660**. As such, the tool **14.1-2660** can extend through the enclosure **14.1-2610** and the trim ring **14.1-2620** to interface with the protrusion **14.1-2646**.

[3687] FIGS. **14.1-29A** shows a sectional view of the cable assembly **14.1-2600**, the enclosure **14.1-2610**, the trim ring **14.1-2620**, and the receptacle connector **14.1-2650**. As shown in FIGS. **14.1-29A**, the seal **14.1-2626** can be L-shaped in a cross-sectional view. The seal **14.1-2626** can be provided on the trim ring **14.1-2620**, between the trim ring **14.1-2620** and the enclosure **14.1-2610**. This provides sealing between the trim ring **14.1-2620** and the enclosure **14.1-2610**. The seal **14.1-2626** is also provided between a body of the trim ring **14.1-2620** and the cavity **14.1-2622** of the trim ring **14.1-2620**. This provides sealing between the trim ring **14.1-2620** and the boot **14.1-2606** of the plug connector **14.1-2602**. Placing the seal **14.1-2626** in this configuration provides sealing between the trim ring **14.1-2620** and both the boot **14.1-2606** and the enclosure **14.1-2610**. The seal **14.1-2626** can include an elastic material that can be stretched around a lip of the trim ring **14.1-2620** and can be secured to the trim ring **14.1-2620** by interference with the lip of the trim ring **14.1-2620**.

[3688] FIGS. **14.1-29B** shows a sectional view of the cable assembly **14.1-2600**, the enclosure **14.1-2610**, the trim ring **14.1-2620**, and the receptacle connector **14.1-2650**. FIGS. **14.1-29E** shows a perspective view of the lever arm **14.1-2640**. The lever arm **14.1-2640** includes a spring **14.1-2641** wrapped around the pivot **14.1-2642** and configured to apply a spring force to a body portion of the lever arm **14.1-2640**. A seal **14.1-2647** is coupled to the protrusion **14.1-2646** of the lever arm **14.1-2640**. The spring **14.1-2641** can apply a spring force to the lever arm **14.1-2640** that presses the seal **14.1-2647** against the enclosure **14.1-2610**. This provides sealing between the trim ring **14.1-2620** and an exterior of an electronic device. The tool **14.1-2660** can be configured to pass through an opening **14.1-2643** in the seal **14.1-2647** when the tool **14.1-2660** is used to actuate the protrusion **14.1-2646**. The seal **14.1-2647** can be a soft gasket or the like.

[3689] FIGS. **14.1-29C** shows a perspective view of the plug connector **14.1-2602** and the trim ring **14.1-2620** and FIGS. **14.1-29D** illustrates a partial sectional view of the trim ring **14.1-2620**. In some examples, the boot **14.1-2606**, the plug **14.1-2604**, and the cavity **14.1-2622** can be stadium-shaped in a cross-sectional view. In some examples, the boot **14.1-2606**, the plug **14.1-2604**, and the cavity **14.1-2622** can be rectangular, triangular, rounded, or can have any other shape in a cross-sectional view. The trim ring **14.1-2620** can include a first cavity portion **14.1-2622A** configured to receive the boot **14.1-2606** and a second cavity portion **14.1-2622B** configured to receive the plug **14.1-2604**. The trim ring **14.1-2620** can include a third cavity portion **14.1-2622C** configured to receive the detent element **14.1-2630**. The third cavity portion **14.1-2622C** can be adjacent the first cavity portion **14.1-2622A**, and can align the **14.1-2630** with openings **14.1-2609** in the side surfaces of the boot **14.1-2606**.

[3690] FIGS. **14.1-29C** and **14.1-29D** further illustrate details of the seal **14.1-2626**. As illustrated in FIGS. **14.1-29D**, the seal **14.1-2626** can have an L-shape in a cross-sectional view. The seal **14.1-2626** can be wrapped around at least a portion of a body of the trim ring **14.1-2620**, such as a lip of the trim ring **14.1-2620**. The seal **14.1-2626** can be formed of an elastic material, and an elastic force of the elastic material can retain the seal **14.1-2626** on the lip of the trim ring **14.1-2620**. In some examples, an adhesive or the like can be provided between the seal **14.1-2626** and the body of the trim ring **14.1-2620** to secure the seal **14.1-2626** to the trim ring **14.1-2620**.

[3691] The seal **14.1-2626** can act as a seal which prevents or limits ingress of contaminants (e.g.,

liquid, dust, lint, debris, or the like) into the trim ring **14.1-2620** and the receptacle connector **14.1-2650** through the cavity **14.1-2622**, such as when the plug connector **14.1-2602** is inserted into the cavity **14.1-2622**. The seal **14.1-2626** can provide sealing between the trim ring **14.1-2620** and the plug connector **14.1-2602** and between the trim ring **14.1-2620** and the enclosure **14.1-2610**. The seal **14.1-2647** can act as a seal which prevents or limits ingress of contaminants (e.g., liquid, dust, lint, debris, or the like) into the trim ring **14.1-2620** and the through the aperture **14.1-2624**.

[3692] FIGS. **14.1-30A** through **14.1-30C** illustrate a seal **14.1-2726** included on a trim ring **14.1-2720** and a seal **14.1-2705** included on a plug connector **14.1-2702** of a cable assembly **14.1-2700**. The seal **14.1-2726** can provide sealing between an enclosure **14.1-2710** and the trim ring **14.1-2720** and between the trim ring **14.1-2720** and the plug connector **14.1-2702**. The seal **14.1-2705** can provide sealing between the plug connector **14.1-2702** and the enclosure **14.1-2710**. The plug assembly **14.1-2700** includes the plug connector **14.1-2702** coupled to a shielded cable **14.1-2708**. The plug connector **14.1-2702** includes a plug **14.1-2704** and the boot **14.1-2706** coupled to the plug **14.1-2704** and the shielded cable **14.1-2708**.

[3693] The plug connector **14.1-2702** can be received in a cavity **14.1-2722** of an enclosure **14.1-2710** and the trim ring **14.1-2720**. The trim ring **14.1-2720** can be coupled to the enclosure **14.1-2710**, and a receptacle connector **14.1-2750** can be coupled to the trim ring **14.1-2720**. The trim ring **2620** can include a detent element **14.1-2730** and a lever arm **14.1-2740**. The detent element **14.1-2730** can be configured to interface with openings **14.1-2709** in the boot **14.1-2706** to fasten the plug connector **14.1-2702** in the cavity **14.1-2722**. The lever arm **14.1-2740** can be rotatable around a pivot. The lever arm **14.1-2740** can include a protrusion **14.1-2746** configured to interface with a tool in order to un-fasten the plug connector **14.1-2702** from the cavity **14.1-2722**.

[3694] The seal **14.1-2705** can be coupled to the boot **14.1-2706** between the shielded cable **14.1-2708** and the openings **14.1-2709**. In the example illustrated in FIGS. **14.1-30B**, the seal **14.1-2705** can be provided adjacent the enclosure **14.1-2710**. Providing the seal **14.1-2705** in this position provides sealing between the boot **14.1-2706** and the enclosure **14.1-2710**, which also seals the trim ring **14.1-2720** from an external environment. In the example illustrated in FIGS. **14.1-30C**, the seal **14.1-2705** can be provided adjacent the enclosure **14.1-2710** and the trim ring **14.1-2720**. Providing the seal **14.1-2705** in this position can provide sealing between the boot **14.1-2706** and each of the enclosure **14.1-2710** and the trim ring **14.1-2720** and can also provide sealing between the enclosure **14.1-2710** and the trim ring **14.1-2720**. Although the seal **14.1-2705** is illustrated as being between the seal **14.1-2726** and the shielded cable **14.1-2708**, the seal **14.1-2705** can be positioned at any point along the boot **14.1-2706**, such as between the seal **14.1-2726** and the plug **14.1-2704**, between the seal **14.1-2726** and the openings **14.1-2709**, between the openings **14.1-2704** and the plug **14.1-2704**, or the like. Moreover, the seal **14.1-2705** can be provided in place of, or in addition to the seal **14.1-2726**.

[3695] FIGS. **14.1-30B** shows a sectional view of the cable assembly **14.1-2700**, the enclosure **14.1-2710**, the trim ring **14.1-2720**, and the receptacle connector **14.1-2750**. As shown in FIGS. **14.1-30B**, the seal **14.1-2726** can be L-shaped in a cross-sectional view. The seal **14.1-2726** can be provided on the trim ring **14.1-2720**, between the trim ring **14.1-2720** and the enclosure **14.1-2710**. This provides sealing between the trim ring **14.1-2720** and the enclosure **14.1-2710**. The seal **14.1-2726** is also provided between a body of the trim ring **14.1-2720** and the cavity **14.1-2722** of the trim ring **14.1-2720**. This provides sealing between the trim ring **14.1-2720** and the boot **14.1-2706** of the plug connector **14.1-2702**. Placing the seal **14.1-2726** in this configuration provides sealing between the trim ring **14.1-2720** and both the boot **14.1-2706** and the enclosure **14.1-2710**. The seal **14.1-2726** can include an elastic material that can be stretched around a lip of the trim ring **14.1-2720** and can be secured to the trim ring **14.1-2720** by interference with the lip of the trim ring **14.1-2720**.

[3696] FIGS. **14.1-30C** shows a sectional view of the cable assembly **14.1-2700**, the enclosure **14.1-2710**, the trim ring **14.1-2720**, and the receptacle connector **14.1-2750**. As shown in FIGS.

14.1-30C, the seal **14.1-2726** can be rectangular in a cross-sectional view. Other shapes can be used for the seal **14.1-2726**, such as rounded, triangular, or the like. The seal **14.1-2726** can be provided between the body of the trim ring **14.1-2720** and the boot **14.1-2706**, between an outer surface of the trim ring **14.1-2720** facing the enclosure **14.1-2710** and an inner surface of the trim ring **14.1-2720** facing the receptacle connector **14.1-2750**. The seal **14.1-2726** can provide sealing between the trim ring **14.1-2720** and the boot **14.1-2706**.

[3697] The seal **14.1-2726** can act as a seal which prevents or limits ingress of contaminants (e.g., liquid, dust, lint, debris, or the like) into the trim ring **14.1-2720** and the receptacle connector **14.1-2750** through the cavity **14.1-2722**, such as when the plug connector **14.1-2702** is inserted into the cavity **14.1-2722**. The seal **14.1-2726** can provide sealing between the trim ring **14.1-2720** and the plug connector **14.1-2702** and between the trim ring **14.1-2720** and the enclosure **14.1-2710**. The seal **14.1-2705** can act as a seal which prevents or limits ingress of contaminants (e.g., liquid, dust, lint, debris, or the like) into the trim ring **14.1-2720** and the receptacle connector **14.1-2750** through the cavity **14.1-2722**, such as when the plug connector **14.1-2702** is inserted into the cavity **14.1-2722**. The seal **14.1-2705** can provide sealing between the plug connector **14.1-2702** and the enclosure **14.1-2710**, between the plug connector **14.1-2702** and both of the trim ring **14.1-2720** and the enclosure **14.1-2710**, or the like.

[3698] FIGS. **14.1-31A** and **14.1-31B** show sectional views of a plug connector **14.1-2800** inserted into a receptacle connector **14.1-2820** and fastened by a detent element **14.1-2830**. The plug connector **14.1-2800** includes a boot **14.1-2804** and a plug **14.1-2802** coupled to the boot **14.1-2804**. The boot includes recesses **14.1-2806** formed in side surfaces thereof. The detent element **14.1-2830** can interface with the recesses **14.1-2806** to retain the boot **14.1-2806** and the plug connector **14.1-2800** in a cavity of the receptacle connector **14.1-2820** and a trim ring.

[3699] The detent element **14.1-2830** includes a protrusion **14.1-2832** configured to interface with a lever arm **14.1-2810**. The detent element **14.1-2830** and the lever arm **14.1-2810** can be included in a trim ring, as discussed in examples previously. The lever arm **14.1-2810** includes protrusions **14.1-2814** configured to interface with the protrusion **14.1-2832** of the detent element **14.1-2830**. The lever arm **14.1-2810** can include a spring and can be configured to supply a spring force to the detent element **14.1-2830** through the protrusions **14.1-2814** and the protrusion **14.1-2832** to retain the detent element **14.1-2830** in an engaged configuration (illustrated in FIGS. **14.1-31A**). The lever arm **14.1-2810** can include a protrusion **14.1-2816** configured to interact with a tool **14.1-2840** in order to transition the lever arm **14.1-2810** and the detent element **14.1-2830** from the engaged configuration to an un-engaged configuration (illustrated in FIGS. **14.1-31B**). As such, the lever arm **14.1-2810** can be used to retain the detent element **14.1-2830** in an engaged configuration and transition the detent element **14.1-2830** to an un-engaged configuration. The detent element **14.1-2830** can also include fasteners **14.1-2834**, which can couple the detent element **14.1-2830** to a body of a trim ring. The detent element **14.1-2830** can translate relative to the fasteners **14.1-2834** as the detent element **14.1-2830** moves from the engaged configuration to the un-engaged configuration.

[3700] The lever arm **14.1-2810** can include a protrusion **14.1-2811** configured to interact with a spring **14.1-2819**. The spring **14.1-2819** can interact with the protrusion **14.1-2811** to resist movement of the lever arm **14.1-2810**. In some examples, the spring **14.1-2819** can provide a desired feel to a user using the tool **14.1-2840** to un-latch the detent element **14.1-2830** and the lever arm **14.1-2810**. For example, as illustrated in FIGS. **14.1-31A** and **14.1-31B**, the spring **14.1-2819** can move over the protrusion **14.1-2811** as the lever arm **14.1-2810** rotates from the engaged configuration (shown in FIGS. **14.1-31A**) to the un-engaged configuration (shown in FIGS. **14.1-31B**). This can provide a physical click through the tool **14.1-2840**, which informs the user that the lever arm **14.1-2810** and the detent element **14.1-2830** are un-latched, and the plug connector **14.1-2800** can be removed from the trim ring and the receptacle connector **14.1-2820**. Positions of the spring **14.1-2819** and the protrusion **14.1-2811**, as well as lengths of the spring **14.1-2819**, angles

of the spring **14.1-2819**, shapes of the protrusion **14.1-2811**, and the like can be altered to alter the force required to rotate the lever arm **14.1-2810**, as well as the feel to a user of moving the lever arm **14.1-2810**.

[3701] In some examples, the receptacle connector **14.1-2820** can include an exterior portion **14.1-2824** and an interior portion **14.1-2822**. For example, the exterior portion **14.1-2824** can at least partially encompass the interior portion **14.1-2822**. In other words, the exterior portion **14.1-2824** can act as a sleeve or shell that at least partially surrounds the interior portion **14.1-2822**. The exterior portion **14.1-2824** and interior portion **14.1-2822** can be interlocked or affixed to one another. In some examples, the receptacle connector **14.1-2820** can be formed from a singular or monolithic structure instead of the external portion **14.1-2824** and the internal portion **14.1-2822**. The receptacle connector **14.1-2820** can form a slot capable of receiving and retaining the plug connector **14.1-2800**.

[3702] FIGS. **14.1-32A** through **14.1-32C** illustrate a method of assembling a trim ring **14.1-2930** and a plug connector **14.1-2940** in an enclosure **14.1-2900**. In FIGS. **14.1-32A**, a frame **14.1-2910** is attached to the enclosure **14.1-2900**. The enclosure can include a side surface **14.1-2902**, an opening **14.1-2904** for receiving a plug connector, a bottom surface **14.1-2906**, and a top surface **14.1-2908**. The frame **14.1-2910** can be attached to structures of the bottom surface **14.1-2906** and the side surface **14.1-2902** through fasteners **14.1-2920**. The fasteners **14.1-2920** can extend longitudinally in a direction from the top surface **14.1-2908** towards the bottom surface **14.1-2906** and can pull the frame **14.1-2910** down towards the bottom surface **14.1-2906**. Although three fasteners **14.1-2920** are illustrated for attaching the frame **14.1-2910** to the enclosure **14.1-2900**, any number of the fasteners **14.1-2920** can be included.

[3703] In FIGS. **14.1-32B**, the trim ring **14.1-2930** is attached to the frame **14.1-2910**. The trim ring **14.1-2930** can be attached to the frame **14.1-2910** through fasteners **14.1-2922**. The fasteners **14.1-2922** can extend into openings **14.1-2912** of the frame **14.1-2910**, illustrated in FIGS. **14.1-32A**. The fasteners **14.1-2922** can extend longitudinally in a direction parallel to the bottom surface **14.1-2906**, and can pull the trim ring **14.1-2930** towards the frame **14.1-2910** and the side surface **14.1-2902** of the enclosure **14.1-2900**. A dummy trim ring can be inserted through the opening **14.1-2904** in the enclosure **14.1-2900** and into an opening **14.1-2932** in the trim ring **14.1-2930** in order to align the trim ring **14.1-2930** with the enclosure **14.1-2900**. Although three fasteners **14.1-2922** are illustrated for attaching the trim ring **14.1-2930** to the frame **14.1-2910**, any number of the fasteners **14.1-2922** can be included.

[3704] In FIGS. **14.1-32C**, the receptacle connector **14.1-2940** is attached to the trim ring **14.1-2930**. The receptacle connector **14.1-2940** can be attached to the trim ring **14.1-2930** through fasteners **14.1-2924**. The fasteners **14.1-2924** can extend into openings **14.1-2934** of the trim ring **14.1-2930**, illustrated in FIGS. **14.1-32A**. The fasteners **14.1-2924** can extend longitudinally in a direction parallel to the bottom surface **14.1-2906** and can pull the receptacle connector **14.1-2940** towards the trim ring **14.1-2930**, the frame **14.1-2910**, and the side surface **14.1-2902** of the enclosure **14.1-2900**. The dummy trim ring can be inserted through the opening **14.1-2904** in the enclosure **14.1-2900**, the opening **14.1-2932** in the trim ring **14.1-2930**, and into an opening in the receptacle connector **14.1-2940** in order to align the receptacle connector **14.1-2940** with the trim ring **14.1-2930** and the enclosure **14.1-2900**. Although two fasteners **14.1-2924** are illustrated for attaching the receptacle connector **14.1-2940** to the trim ring **14.1-2930**, any number of the fasteners **14.1-2924** can be included.

[3705] FIGS. **14.1-33A** through **14.1-33F** illustrate an example of a plug connector **14.1-3000** inserted into a receptacle connector **14.1-3030** and fastened in the receptacle connector **14.1-3030** by a lever arm **14.1-3010**. The plug connector **14.1-3000** includes a boot **14.1-3002** and a plug **14.1-3004** coupled to the boot **14.1-3002**. The boot **14.1-3002** includes recesses **14.1-3003**.

[3706] The lever arm **14.1-3010** can be included in a trim ring coupled to the receptacle connector **14.1-3030**. The lever arm **14.1-3010** is rotatable around a pivot **14.1-3012**. The lever arm **14.1-**

3010 includes a protrusion **14.1-3014** configured to interact with the recesses **14.1-3003** of the boot **14.1-3002** to retain the plug connector **14.1-3000** in the trim ring and the receptacle connector **14.1-3030**. The lever arm **14.1-3010** can be rotatable between an engaged configuration, with the protrusion **14.1-3014** disposed in a respective recess **14.1-3003**, and an un-engaged configuration, with the protrusion **14.1-3014** retracted from the recess **14.1-3003**.

[3707] The lever arm **14.1-3010** further includes a protrusion **14.1-3016**. The protrusion **14.1-3016** can be configured to interact with a tool **14.1-3060** to transition the lever arm **14.1-3010** from the engaged configuration to the un-engaged configuration. The lever arm **14.1-3010** includes a protrusion **14.1-3018** configured to interact with a spring mechanism **14.1-3020**. The spring mechanism **14.1-3020** can interact with the protrusion **14.1-3018** to retain the lever arm **14.1-3010** in the engaged configuration and can resist movement or forces of the lever arm **14.1-3010** transitioning to the un-engaged configuration.

[3708] The spring mechanism **14.1-3020** can include a central member **14.1-3022** extending into a spring **14.1-3024** and a stop **14.1-3026**. The protrusion **14.1-3018** interacts with the central member **14.1-3022** to compress the spring **14.1-3024** into the stop **14.1-3026**. This compression of the spring **14.1-3024** exerts a spring force on the protrusion **14.1-3018** through the central member **14.1-3022** to retain the lever arm **14.1-3010** in the engaged configuration, and to resist the lever arm **14.1-3010** from transitioning to the un-engaged configuration.

[3709] In some examples, the receptacle connector **14.1-3030** can include an exterior portion **14.1-3034** and an interior portion **14.1-3032**. For example, the exterior portion **14.1-3034** can at least partially encompass the interior portion **14.1-3032**. In other words, the exterior portion **14.1-3034** can act as a sleeve or shell that at least partially surrounds the interior portion **14.1-3032**. The exterior portion **14.1-3034** and interior portion **14.1-3032** can be interlocked or affixed to one another. In some examples, the receptacle connector **14.1-3030** can be formed from a singular or monolithic structure instead of the external portion **14.1-3034** and the internal portion **14.1-3032**. The receptacle connector **14.1-3030** can form a slot capable of receiving and retaining the plug connector **14.1-3000**.

[3710] FIGS. **14.1-33B** is a detailed view of a region **14.1-3050** of FIGS. **14.1-33A**. As illustrated in FIGS. **14.1-33A** and **14.1-33B**, as the plug connector **14.1-3000** is inserted into a trim ring and the receptacle connector **14.1-3030**, the protrusion **14.1-3014** extends along a side surface of the boot **14.1-3002**. Once the plug connector **14.1-3000** is fully inserted into the trim ring and the receptacle connector **14.1-3030**, the protrusion **14.1-3014** can extend into the recess **14.1-3003** and can retain the plug connector **14.1-3000** in the trim ring and the receptacle connector **14.1-3030**. The protrusion **14.1-3014** can extend through a body portion **14.1-3046** of the trim ring. The spring **14.1-3024** of the spring mechanism **14.1-3020** is in a compressed state and pushes the protrusion **14.1-3014** of the lever arm **14.1-3010** against the side surface of the boot **14.1-3002** through the interaction between the central member **14.1-3022** and the protrusion **14.1-3018**. The lever arm **14.1-3010** is in the engaged configuration, such that the lever arm **14.1-3010** applies a force against the boot **14.1-3002** in a clockwise direction.

[3711] FIGS. **14.1-33C** and **14.1-33D** illustrate the lever arm **14.1-3010** after the plug connector **14.1-3000** is fully inserted into the trim ring and the receptacle connector **14.1-3030**, such that the protrusion **14.1-3014** extend into the recess **14.1-3003**. Spring force from the spring **14.1-3024** of the spring mechanism **14.1-3020** through the central member **14.1-3022** and the protrusion **14.1-3018** causes the lever arm **14.1-3010** to rotate in a counterclockwise direction. The protrusion **14.1-3014** extends through the body portion **14.1-3046** of the trim ring into the recess **14.1-3003** of the boot **14.1-3002**. This is the engaged configuration in which the protrusion **14.1-3014** secures and fastens the boot **14.1-3002** into the trim ring and the receptacle connector **14.1-3030**. The spring **14.1-3024** of the spring mechanism **14.1-3020** is in a partially extended state and pushes the protrusion **14.1-3014** of the lever arm **14.1-3010** against a side surface of the boot **14.1-3002** adjacent the recess **14.1-3003** through the interaction between the central member **14.1-3022** and

the protrusion **14.1-3018**. The lever arm **14.1-3010** is in the engaged configuration, such that the lever arm **14.1-3010** applies a force against the boot **14.1-3002** in a clockwise direction.

[3712] FIGS. **14.1-33E** and **14.1-33F** illustrate the lever arm **14.1-3010** after the tool **14.1-3060** is used to actuate the lever arm **14.1-3010**, transitioning the lever arm **14.1-3010** from the engaged configuration to the un-engaged configuration. Force from the tool **14.1-3060** overcomes a spring force of the spring **14.1-3024** of the spring mechanism **14.1-3020** and causes the lever arm **14.1-3010** to rotate in a clockwise direction. The protrusion **14.1-3014** retracts from the recess **14.1-3003** through the body portion **14.1-3046** of the trim ring such that the plug connector **14.1-3000** can be removed from the trim ring and the receptacle connector **14.1-3030**.

[3713] This is the un-engaged configuration in which the protrusion **14.1-3014** no longer secures and fastens the boot **14.1-3002** into the trim ring and the receptacle connector **14.1-3030**. The spring **14.1-3024** of the spring mechanism **14.1-3020** is in a compressed state and resists the force supplied by the tool **14.1-3060**. The lever arm **14.1-3010** is in the un-engaged configuration, such that the lever arm does not apply a force against the boot **14.1-3002**.

[3714] The spring mechanism **14.1-3020** can be part of the trim ring, or part of the receptacle connector **14.1-3030** and fastened to the receptacle connector **14.1-3030**. The lever arm **14.1-3010** and the spring mechanism **14.1-3020** can overlap portions of the plug connector **14.1-3000** and the receptacle **14.1-3030**, which can enable the fastener of FIGS. **14.1-33A** through **14.1-33F** to be formed in a smaller package, providing more space within an enclosure for other components, or providing for a smaller enclosure.

[3715] FIGS. **14.1-34A** and **14.1-34B** illustrate perspective views of a cable assembly **14.1-3006** inserted into a trim ring **14.1-3040** and a receptacle connector **14.1-3030**. The cable assembly **14.1-3006** includes a shielded cable **14.1-3007** and a plug connector **14.1-3000**. The plug connector **14.1-3002** includes a boot **14.1-3002**. The boot includes recesses **14.1-3003** formed in the side surfaces thereof.

[3716] The trim ring **14.1-3040** includes a lever arm **14.1-3010** and a spring mechanism **14.1-3020**. The lever arm **14.1-3010** is provided for fastening the plug connector **14.1-3030** in the trim ring **14.1-3040** and the receptacle connector **14.1-3030**. The lever arm **14.1-3010** includes a protrusion **14.1-3014** configured to be inserted into a respective recess **14.1-3003** of the boot **14.1-3002** (such as by rotation around a pivot **14.1-3012**) and to retain the plug connector **14.1-3002** within the trim ring **14.1-3040** and the receptacle connector **14.1-3030**. The lever arm **14.1-3010** includes a protrusion **14.1-3016** configured to interact with a tool in order to transition the lever arm **14.1-3010** to an un-engaged configuration and to release the plug connector **14.1-3000** from the trim ring **14.1-3040** and the receptacle connector **14.1-3030**. The lever arm **14.1-3010** includes a protrusion **14.1-3018** configured to interact with the spring mechanism **14.1-3020**, to retain the lever arm in the engaged configuration, and to resist force supplied by the tool through the protrusion **14.1-3016**.

[3717] The spring mechanism **14.1-3020** is provided for retaining the lever arm **14.1-3010** in an engaged configuration. The spring mechanism **14.1-3020** includes a central member **14.1-3022**, a spring **14.1-3024**, and a stop **14.1-3026**. The central member **14.1-3022** is provided for interacting with the protrusion **14.1-3018** of the lever arm **14.1-3010**. The spring **14.1-3024** is provided for providing a spring force to the central member **14.1-3022** to retain the lever arm **14.1-3010** in the engaged configuration through the protrusion **14.1-3018**. The stop **14.1-3026** is provided for compressing the spring **14.1-3024**, thereby supplying the spring force to the central member **14.1-3022**.

[3718] As illustrated in FIGS. **14.1-34A** and **14.1-34B**, the lever arm **14.1-3010** can be disposed in a side surface of the trim ring **14.1-3040**, and the spring mechanism **14.1-3020** can be disposed on a top surface of the trim ring **14.1-3040** and/or the receptacle connector **14.1-3030**. The lever arm **14.1-3010** can include components extending through a portion of a sidewall of the trim ring **14.1-3040**, which allows for the trim ring **14.1-3040** and the lever arm **14.1-3010** to take up less space in

an enclosure. The spring mechanism **14.1-3020** can be disposed in different position to maximize space use within the enclosure.

[3719] In some examples, the receptacle connector **14.1-3030** can include an exterior portion **14.1-3034** and an interior portion **14.1-3032**. For example, the exterior portion **2824** can at least partially encompass the interior portion **14.1-3032**. In other words, the exterior portion **14.1-3034** can act as a sleeve or shell that at least partially surrounds the interior portion **14.1-3032**. The exterior portion **14.1-3034** and interior portion **14.1-3032** can be interlocked or affixed to one another. In some examples, the receptacle connector **14.1-3030** can be formed from a singular or monolithic structure instead of the external portion **14.1-3034** and the internal portion **14.1-3032**. The receptacle connector **14.1-3030** can form a slot capable of receiving and retaining the plug connector **14.1-3000**. The receptacle connector **14.1-3030** can further include a printed circuit board (PCB) **14.1-3036**. The PCB **14.1-3036** can be coupled to the plug of the plug connector **14.1-3000** through the interior portion **14.1-3032**. The receptacle connector **14.1-3030** can be coupled to the trim ring **14.1-3040** through a portion of the interior portion **14.1-3032**.

[3720] FIGS. **14.1-35A** and **14.1-35B** illustrate an exploded view and a side sectional view of a receptacle connector **14.1-3100**. The receptacle connector **14.1-3100** includes an interior portion **14.1-3102**, electrical contacts **14.1-3110**, a contact block **14.1-3112**, an anchor **14.1-3114**, and an exterior portion **14.1-3120**. The interior portion **14.1-3102** includes an opening **14.1-3104** for receiving a plug of a plug connector, openings **14.1-3106A**, **14.1-3106B**, and **14.1-3106C** for receiving the electrical contacts **14.1-3110** (e.g., the electrical contacts **14.1-3110A**, **14.1-3110B**, and **14.1-3110C**, respectively), and openings **14.1-3108** for fastening the interior portion **14.1-3102** to a trim ring or the like.

[3721] The electrical contacts **14.1-3110** are provided in the openings **14.1-3106** of the interior portion **14.1-3102**. The contact block **14.1-3112** can be used to isolate the electrical contacts **14.1-3110** from one another, and to aid in positioning the electrical contacts **14.1-3110** relative to the openings **14.1-3106** in the interior portion **14.1-3102**. The electrical contacts **14.1-3110** can be stitched to the anchor **14.1-3114**, which can be stitched to the interior portion **14.1-3102** to couple the electrical contacts **14.1-3110** to the interior portion **14.1-3102** of the receptacle connector **14.1-3100**. In some examples, the electrical contacts **14.1-3110** can be stitched to the contact block **14.1-3112** and/or the anchor **14.1-3114**.

[3722] The exterior portion **14.1-3120** can surround at least a portion of the interior portion **14.1-3102**. The exterior portion can include openings **14.1-3122** for fastening the exterior portion **14.1-3120** to the trim ring or the like. The openings **14.1-3122** can also fasten the exterior portion **14.1-3120** to the interior portion **14.1-3102**.

[3723] The following description is directed to certain example connectors (including example method steps of manufacturing these connectors) shown in FIGS. **14.1-36A** through **14.1-41B**. In particular, the below description will include examples of electrical connectors with a protective overmold onto a flexible circuit board at a wire termination region. In some examples, the structural features of these electrical connectors can help mitigate potential issues (e.g., solder crack, electrical discontinuity, loading stresses, heat stresses, etc.) involved with over-molding elements with lower mechanical stiffness (e.g., a flex circuit board and wires). In particular implementations, and as discussed below, a stiffener element and a wrap element can allow such sensitive components to be overmolded in an effective manner. For instance, at least one of the stiffener element or the wrap element can help decrease high-temperature exposure to the flexible circuit board and wires during the overmolding process.

[3724] FIGS. **14.1-36A** and **14.1-36B** respectively illustrate a semi-transparent view and a solid view of an electrical connector portion **14.1-3600** in accordance with one or more examples of the present disclosure. In FIGS. **14.1-36A**, a connector assembly **14.1-3612** is hidden (or made partially transparent) for illustration purposes; whereas in FIGS. **14.1-36B**, the connector assembly **14.1-3612** is depicted for viewing. The connector assembly **14.1-3612** can include an assembled (or

partially assembled) harness with prongs, sleeves, frame members, etc. The connector assembly **14.1-3612** can thus be sized and shaped to receive electrical components inside a receptacle **14.1-3616** defined by the internal walls of the connector assembly **14.1-3612**. In particular implementations, the connector assembly **14.1-3612** is sized and shaped such that one or more electrical components can be withdrawn from inside the receptacle **14.1-3616** to expose at least a portion of one or more such components for manufacturing (as will be discussed below, and as shown in FIGS. **14.1-36A**). Once these manufacturing steps have been performed, the electrical components can be inserted (e.g., pushed) back into the receptacle **14.1-3616** (as will also be discussed below, and as shown in FIGS. **14.1-36B**).

[3725] FIGS. **14.1-36A** and **14.1-36B** illustrate the electrical connector portion **14.1-3600** including a wrap **14.1-3602**. The wrap **14.1-3602** can include one or more elements that can help support or protect wires **14.1-3606** leading into a termination region. In certain examples, the wrap **14.1-3602** can help stiffen the wires **14.1-3606** (e.g., so as to decrease bending or flex at the wire termination region). Additionally or alternatively, the wrap **14.1-3602** help keep the wires **14.1-3606** from fraying under pressure from the overmold tooling. In these or other examples, the wrap **14.1-3602** includes a jacket, coating, or tape to at least partially wrap around the wires **14.1-3606** and provide the foregoing functionality. To do so, the wrap **14.1-3602** can be made of a variety of materials. In particular examples, however, the wrap **14.1-3602** includes at least one of a silicone jacket or an acetate tape.

[3726] The electrical connector portion **14.1-3600** further includes an overmold **14.1-3608** positioned adjacent to (e.g., over, around, at least partially encompassing, etc.) a wire termination region. The overmold **14.1-3608** can be injection molded, form molded, etc. The overmold can include a variety of materials. In some examples, the overmold **14.1-3608** includes a polymer material. In particular examples, the overmold **14.1-3608** includes a liquid crystal polymer material.

[3727] In addition, the electrical connector portion **14.1-3600** can include a stiffener **14.1-3610** and **14.1-3614**. The stiffener **14.1-3610** can include stiffener surfaces that can be used to glue and weld to portions of the connector assembly **14.1-3612**. Additionally or alternatively, the stiffener **14.1-3610** can include stiffener surfaces that can be used to glue or welded to portions of a flex circuit **14.1-3604**. In these or other examples, the attachment regions at the stiffener surfaces of the stiffener **14.1-3610** can help prevent assembly loading stresses from transferring to electrical components (e.g., the flex circuit **14.1-3604**). Accordingly, the stiffener **14.1-2610** can help increase the likelihood of electrical continuity after the overmold and product assembly process.

[3728] The stiffener **14.1-3610** also includes surfaces that function as contact points for overmold tooling components (e.g., heated tooling surfaces). In some examples, surfaces of the stiffener **14.1-3610** can also include datum surfaces (e.g., to guide or center overmolded parts with respect to the tool structure, thereby reducing scalding risk).

[3729] In these or other examples, the stiffener **14.1-3610** includes one or more surfaces for hotbar termination of the flex circuit **14.1-3604** to an interfacing connector (shown in FIG. **14.1-40**). In these or other examples, one or more surfaces of the stiffener **14.1-3610** can be exposed (e.g., withdrawn from inside the receptacle **14.1-3616**) to perform the hotbar termination, which reduces manufacturing complexity of needing to make the hotbar termination inside the connector assembly **14.1-3612**.

[3730] The stiffener **14.1-3610** can include a variety of materials. In some examples, the stiffener **14.1-3610** comprises a rigid material. In some examples, the stiffener **14.1-3610** comprises a conductive material. In some examples, the stiffener **14.1-3610** comprises a metal material. In particular examples, the stiffener **14.1-3610** comprises a steel stiffener.

[3731] The following description provides more detail at least with respect to one or more stiffener surfaces (e.g., datum surfaces). FIGS. **14.1-37** through **14.1-38** illustrate respective top and bottom views of the electrical connector portion **14.1-3600** discussed above. As mentioned, the stiffener

14.1-3610 can include datum surfaces for contacting the overmold tooling. Examples of these datum surfaces are depicted as datum surfaces **14.1-3702**. The datum surfaces **14.1-3702** can include extensions, wings, jut-outs, arms, flaps, flat spots, faces, curved surfaces, etc. that are designed to contact the overmold tooling at specific locations. In these or other examples, the datum surfaces **14.1-3702** can be specially designed for custom tooling. Alternatively, the datum surfaces **14.1-3702** can be designed for many different overmold tooling.

[3732] The following description provides example method steps for manufacturing the electrical connector portion **14.1-3600** discussed above. Indeed, FIGS. **14.1-39** illustrates example method steps 1-6 for manufacturing the electrical connector portion **14.1-3600** in accordance with one or more examples of the present disclosure. It will be appreciated that the following steps can be modified, omitted, or added thereto (as may be desired).

[3733] At step 1, the flex circuit **14.1-3604** is provided. At step 2, at least a portion of the stiffener **14.1-3610** is attached to the flex circuit **14.1-3604**. For example, the stiffener **14.1-3610** is adhered (e.g., via pressure sensitive adhesive) or fastened via screws or the like.

[3734] At step 3, the wires **14.1-3606** (wrapped by the wrap **14.1-3602**) are terminated onto the flex circuit **14.1-3604**. At this step, the stiffener **14.1-3610** acts as support during the termination process. At step 4, the overmold **14.1-3608** is applied to the termination region. At step 5, the wire harness portion assembled thus far can be inserted into the receptacle **14.1-3616** of the connector assembly **14.1-3612**. At step 6, an end region of the electrical connector portion **14.1-3600** is left exposed for hot-bar termination (now discussed below in relation to FIGS. **14.1-40**).

[3735] FIGS. **14.1-40** depicts example method steps of providing an interface connector to the electrical connector portion **14.1-3600** discussed above in accordance with one or more examples of the present disclosure. At step 1, the electrical connector portion **14.1-3600** is provided (e.g., as discussed above in connection with FIGS. **14.1-39**). At step 2, the end region **14.1-3900** of the electrical connector portion **14.1-3600** is exposed.

[3736] At step 3, a connection **14.1-4002** is formed between the stiffener **14.1-3610** and an interface connector **14.1-4000** (e.g., a plug-in connection, prong connection, magnetic connection, etc.). In these or other examples, the connection **14.1-4002** includes a hot-bar connection, welding connection, solder connection, metal bond connection, etc. In some examples, the stiffener **14-3610** is used to transfer loading stresses from the interface connector **14.1-4000**, thereby bypassing stresses to the sensitive electrical components (e.g., the flex circuit board). At step 4, the interface connector **14.1-4000** is pushed back into (e.g., flush with) the electrical connector portion **3600**.

[3737] FIGS. **14.1-41A** and **14.1-41B** show side schematic views of the assembling the interface connector **14.1-4000** to the electrical connector portion **14.1-3600** in accordance with one or more examples of the present disclosure. In particular, FIGS. **14.1-41A** illustrates the end region **14.1-3900** of the electrical connector portion **3600** exposed for attaching the interface connector **14.1-4000**. Then, FIGS. **14.1-41B** illustrates the end region **14.1-3900** of the electrical connector portion **3600** pushed inside the receptacle of the connector assembly (with the interface connector **14.1-4000** flush against the electrical connector portion **14.1-3600**).

[3738] As mentioned above in relation to FIGS. **14.1-40**, the stiffener **14.1-3610** can prevent the transfer of stress between the interface connector. To do so, the stiffener **14.1-3610** can include one or more attachment regions to the various components. For example, the stiffener **14.1-3610** can attach to the flex circuit **14.1-3604** at the attachment region **14.1-4104**, which is located adjacent to the interface connector **14.1-4000** at an underside of the stiffener **14.1-3610**. Additionally or alternatively, a top side of a stiffener **14.1-4110** can attach to the flex circuit **14.1-3604** at an attachment region **14.1-4108** positioned adjacent to the overmold **14.1-3608**. The underside of the stiffener **14.1-4110** can be attached to the connector assembly **14.1-3612** (e.g., a frame member of the connector assembly).

[3739] Additionally shown, an underside of the stiffener **14.1-3610** can attach to the topside of the interface connector **14.1-4000** at an attachment region **14.1-4102**. Further, a top side of the flex

circuit **14.1-3604** can attach to the bottom side of the interface connector **14.1-4000**. In addition, the wires **14.1-3606** can attach to the flex circuit **14.1-3604** at an attachment region **14.1-4100**. [3740] With the foregoing attachments at the attachment regions discussed, the stiffener **14.1-3610** can avoid the transfer of loading stress from the interface connector **14.1-4000** to either of the flex circuit **14.1-3604** (at associated attachment regions) or the wires (and associated attachment regions). The foregoing configuration can also enable the flex circuit **14.1-3604** to be rolled up or folded (e.g., into a serpentine pattern) for positioning/storage of the extended portions of the flex circuit **14.1-3604**.

[3741] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **14.1-36** through **14.1-41B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **14.1-36** through **14.1-41B**.

14.2: Modular Components for Wearable Electronic Devices

[3742] As virtual reality (VR) and mixed reality (MR) become more ubiquitous, the demand for user friendly head-mounted displays with quality components increases. Traditionally, these VR/MR systems have been devices that include a wearable display component, often referred to as a head-mounted display (HMD) and a supplemental unit. The supplemental unit can be part of the HMD, or can be a separate component that is permanently connected to the HMD through a cable, wires, or other conductors. The supplemental unit can provide power and/or added processing functionality to the system. The components of these systems are not interchangeable and thus only have a single configuration. VR/MR systems are increasingly being used in a variety of different environments and scenarios. Thus, it can be desirable for a VR/MR system to have different properties depending on the desired use or uses.

[3743] For example, certain components can maximize comfort levels and audio quality, while other components can reduce the weight of the wearable and maximize securement of the device. The modular nature of a wearable VR/MR system and its associated components can allow particular components to be selected for incorporation into the device. For example, the components that maximize comfort and audio quality can be selected for scenarios where a user may wear a device for long periods of time, while the components that maximize weight reduction and securement may be selected for scenarios where a user desired to be highly mobile.

[3744] Further, traditional VR/MR systems, including HMD systems and devices, have packaged most or all of the components of the device in a housing that is worn on the user's face. Even in systems that contain a separate processing unit, almost all of the electronic components of the HMD portion are packaged in a single housing that is mounted to the user's head. In addition to reducing the customizability of the HMD device, this type of design can concentrate the devices weight in a single location, potentially reducing comfort for the user. In contrast, the HMD devices and systems described herein can include multiple modular and connectable parts that can distribute the electronic and functional components of the device in more than one location, increasing both modularity and comfort for a user.

[3745] These and other embodiments are discussed below with reference to FIGS. **14.2-1** through **14.2-13**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature including at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the

first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[3746] FIGS. **14.2-1A** shows a wearable electronic device **14.2-100** being worn on the head **14.2-101** of a user. The wearable electronic device **14.2-100** can include a number of modular components. For example, the wearable device **14.2-100** can include a head-mounted display (HMD) **14.2-110** including a housing **14.2-105** and a display **14.2-107** attached to the housing for displaying images to a user (shown in FIGS. **14.2-1B**), one or more intermediate members, flexible straps, or connector straps **14.2-130**, a retention band **14.2-120** that can be a flexible band or a band that is moldable to a user's head, and an optional supplemental unit **14.2-140** such as an external power supply, memory component, and/or processor. Although referred to as a wearable electronic device **14.2-100**, it should be understood that the device **14.2-100** can include multiple modular components or devices and can be interchangeably referred to as a wearable electronic device, wearable electronic device system, and/or wearable electronic system. Additionally, although the particular component **14.2-110** can be referred to as an HMD, it should be understood that the terms HMD, HMD device, and/or HMD system can be used to refer to the wearable device **14.2-100** as a whole.

[3747] In some examples, and as shown, the device **14.2-100** can be worn on the user's head **14.2-101** such that the HMD **14.2-110** is worn on the user's face and disposed over one or both of their eyes. The HMD can be removably and/or releasably connected to the connector strap or straps **14.2-130** as described further herein. In some examples, the connector straps **14.2-130** can be positioned against the side of a user's head **14.2-101** and in contact therewith. In some examples, the connector straps **14.2-130** can be positioned above the user's ear **14.2-150** or ears, as shown. In some examples, the connector straps **14.2-130** can be positioned adjacent to the user's ear or ears. The connector straps **14.2-130** can be removably connected to the retention band **14.2-120**, which can extend around the user's head **14.2-101** and removably connect to the other of the connector straps **14.2-130** (not shown). In this way, the HMD, connector straps **14.2-130**, and retention band **14.2-120** can form a loop that can retain the wearable electronic device **14.2-100** on the user's head **14.2-101**. As shown in FIGS. **14.2-1A**, the connector straps **14.2-130** can connect to the HMD **14.2-110**, both mechanically and electrically, at an HMD connection location **14.2-154** that can include an electrical input or electrical connector that is attached to the housing **14.2-105** and electrically connected to the display **14.2-107** (as illustrated in FIG. **14.2-2**). This location can be identified as a temple area “Y” that can be defined as an area near a user's temple adjacent to the user's eye, and can span from in front of the user's eye to approximately 1-1.5 inches past the outer corner of a user's eye, along the side of the user's head. Similarly, the connector straps **14.2-130** can connect to the retention band **14.2-120** at a retention band connection location **14.2-152** identified as an ear area “X” that can span to include the area above the user's ear **14.2-150** or within 0.5 inches of the outer edge of the ear on either side. In this manner, the connector straps **14.2-130** are able to provide structural support between the HMD **14.2-110** and the user's ear **14.2-150**, while securely connecting the retention band **14.2-120** and translating the retention forces of the retention band through the wearable electronic device **14.2-100**. It should be understood, however, that this configuration is just one example of how the components of a modular wearable electronic device **14.2-100** can be arranged, and that in some, a different number of connector straps and/or retention bands can be included.

[3748] FIGS. **14.2-1B** shows a top view of the wearable electronic device **14.2-100**, demonstrating how the components of the device **14.2-100** can be mechanically and/or electrically connected to one another in some examples. In the example shown in FIGS. **14.2-1B**, a first connector strap **14.2-130** and a second connector strap **14.2-131** can each be removably connected or coupled to either side of the HMD **14.2-110**. In some examples, the connector straps **14.2-130**, **14.2-131** can both be electrically and mechanically coupled or connected to the HMD and can operate as an intermediate member located between the HMD and the retention band **14.2-120**. In some

examples, one of the connector straps **14.2-130**, **14.2-131** can be mechanically connected, but may not be electrically connected to the HMD. The connector straps **14.2-130**, **14.2-131** can each be connected to the retention band **14.2-120** at a different location on the connector strap **14.2-130**, **14.2-131** than the connection to the HMD. For example, the HMD can be connected to a first end of each connector strap **14.2-130**, **14.2-131**, and the retention band **14.2-120** can be connected to a second, opposite end of each connector strap **14.2-130**, **14.2-131**.

[3749] In some examples, and as shown, at least one of the connector straps **14.2-130** can also be connected to a supplemental unit **14.2-140**. The supplemental unit **14.2-140** can contain one or more processors, batteries or power supplies, antennas, and any other electrical and/or processing components, as desired. In some examples, power and/or data from the supplemental unit can be provided to the HMD through the connector strap **14.2-130** via a connection between the supplemental unit **14.2-140** and the connector strap **14.2-130**, as described herein. In the present example, a conductor or cable **14.2-142** can electrically connect the supplemental unit **14.2-140** to the connector strap **14.2-130**, through which the supplemental unit can provide power and/or communicate with the HMD **14.2-110**, second connector strap **14.2-131**, and/or retention band **14.2-120**.

[3750] Although the supplemental unit **14.2-140** is shown as being connected to the connector strap **14.2-130** through a wired connection **14.2-142**, it should be understood that in some examples the supplemental unit **14.2-140** may wirelessly connect or communicate data and/or power with the connector strap **14.2-130** and/or HMD **14.2-110** by any desired method or technology. Additionally, in some examples, the supplemental unit **14.2-140** can be incorporated into, or can be an integral part of, one or more of the other components of the device **14.2-100**, including the retention band **14.2-120**, the connector straps **14.2-130**, **14.2-131**, and/or the HMD **14.2-110**. Further, although the components of the wearable electronic device **14.2-100** are shown as being connected to one another at certain locations, it should be understood that any of the components of the device **14.2-100** can be electrically and/or mechanically connected to one or more of any of the other components of the device **14.2-100**, in any manner and location, as desired.

[3751] FIGS. **14.2-1C** shows an exploded view of the wearable electronic device **14.2-100**, including the locations of connectors on the various components, as described herein. As can be seen, the connector strap **14.2-130** can include a first connector **14.2-132**, also referred to as an HMD connector, to provide electrical communication between the connector strap **14.2-130** and the HMD **14.2-110** by removably or releasably attaching to a corresponding connector **14.2-112** on the HMD **14.2-110**. As used herein, providing electrical communication between two elements can include providing any type of electrical signal including power and/or data. Additionally, in some examples, an intermediate component can include conductive elements that are in electrical communication with two or more components and can transmit an electrical signal from one component to another. In other words, multiple components can be in electrical communication with one another, including components that conduct electrical signals, and not just components that are transmitting or receiving the electrical signals. The first connector **14.2-132** can be positioned at a first end of the connector strap **14.2-130**, and/or the flexible housing thereof. The connector strap **14.2-130** can also include a second connector **14.2-134** positioned at a second end, such as an opposite end, of the connector strap **14.2-130**. The second connector **14.2-134** can be referred to as a retention band connector or a flexible band connector, and can be removably connectable to the retention band **14.2-120** by attaching to a corresponding connector **14.2-124** on the retention band **14.2-120**. The second connector **14.2-134** can at least partially define an external surface of the connector strap **14.2-130**.

[3752] The connector strap **14.2-130** can further include a third connector **14.2-136**, also referred to as a supplemental unit connector, that can provide electrical communication between the connector strap **14.2-130** and the supplemental unit **14.2-140**, such as through a corresponding connector **14.2-141** that can be coupled to the conductor **14.2-142**. In some examples, the supplemental unit

connector **14.2-136** can be positioned at or near an end of the flexible housing of the connector strap **14.2-130**, opposite the first end. In some examples, the connector **14.2-136** can include a mechanical connector surrounding an electrical connector, and the connector **14.2-141** can be configured to rotate relative to the supplemental unit connector **14.2-136** when connected thereto to facilitate placement and relative freedom for the positioning of the supplemental unit **14.2-140**. [3753] Similar to the connector strap **14.2-130**, the second connector strap **14.2-131** can include a first connector **14.2-133**, also referred to as an HMD connector or a display connector, and a second connector **14.2-135**, also referred to as a retention band connector. These connectors **14.2-133**, **14.2-135** can be positioned at or near opposite ends of the housing of the connector strap **14.2-131**. The connectors **14.2-133**, **14.2-135**, can have substantially the same or similar functionalities as any of the connectors **14.2-132**, **14.2-134**, **14.2-136**. In some examples, the connector **14.2-133** can be releasably attached to a corresponding connector of the HMD (not shown), and the connector **14.2-135** can be releasably attached to a corresponding connector **14.2-125** of the retention band **14.2-120**.

[3754] Any number or variety of components in any of the configurations described herein can be included in a wearable electronic device, such as the HMD devices and/or HMD systems described herein. The components can include any combination of the features described herein, and can be arranged in any of the various configurations described herein. The structure and arrangement of components of a device, as well as the concepts regarding their use can apply not only to the specific examples discussed herein, but to any number of embodiments in any combination. Various examples of electronic devices and electronic device components including some having various features in various arrangements are described below, with reference to FIGS. **14.2-2A** through **14.2-4**.

[3755] FIGS. **14.2-2A** shows an exploded view of a wearable electronic device **14.2-200** that can include modular components which can be selectively and removably or releasably attached or coupled to one another as described herein. In some examples, the device **14.2-200** can be substantially similar to, or can include some or all of the features of the other wearable electronic devices described herein. As with the electronic device **14.2-100**, the device **14.2-200** can include an HMD **14.2-210**, a retention band **14.2-220**, and one or more connector straps **14.2-230**, **14.2-231**. The device **14.2-200** can also include a supplemental unit (not shown) that can be connected to one or more of the components of the device **14.2-200** to provide additional functionality, power, processing speed, and the like.

[3756] Similar to the connector strap **14.2-130**, the connector strap **14.2-230** can include an HMD connector **14.2-232** that can releasably couple to a corresponding connector **14.2-212**, a retention band connector **14.2-234** that can releasably couple to a corresponding connector **14.2-224**, and a supplemental unit connector **14.2-236** that can releasably couple to a supplemental unit. In some examples, the supplemental unit connector **14.2-236** includes a battery input and/or a data input, and can alternatively be referred to as a power connector and/or a data connector. In some examples, the HMD connected **14.2-232** can be positioned at an edge of the connector strap **14.2-230**, while the retention band connector **14.2-234** and/or the supplemental unit connector **14.2-236** can be positioned at a major surface(s) of the connector strap **14.2-230**, such as a side surface(s).

[3757] In some examples, and as shown, the retention band connector **14.2-234** and its associated connector **14.2-224** on the retention band **14.2-220**, can include multiple connection points. In some examples, these multiple connection points can be aligned in a single line and can thus rotationally fix the connector strap **14.2-230** and the retention band **14.2-220** to one another. In some examples, any other type of connection can be implemented that can provide rotational fixation between the connector strap **14.2-230** and the retention band **14.2-220**. In some examples, where the device **14.2-200** includes a second connector strap **14.2-231**, that connector strap **14.2-231** can also include an HMD connector **14.2-233** and a retention band connector **14.2-235** that can be releasably coupled to a corresponding connector **14.2-225** so as to be rotationally fixed thereto.

[3758] In some examples, and as described further herein, one or more of the connector straps **14.2-230**, **14.2-231** can include one or more operational elements or components **14.2-237**, **14.2-238** disposed at least partially in an internal volume defined by the housing of the connector strap **14.2-230**, **14.2-231**. The operational element, component, or components **14.2-237**, **14.2-238** can include any number of functional and/or electrical components, including output components, such as an audio module or component, display module, or data output component, input components, such as a touch sensor or touch sensitive element or component, button, toggle, switch, fingerprint sensor, camera, or other form of sensor, or any other functional and/or operational component desired. Further details of the connector straps **14.2-230**, **14.2-231** are described below with respect to FIGS. **14.2-2B** through **14.2-2D**.

[3759] FIGS. **14.2-2B** shows a side view of a connector strap **14.2-231** of the electronic device **14.2-200** shown in FIGS. **14.2-2A**, including an HMD connector **14.2-233** disposed at an end of the strap **14.2-231**, and a retention band connector **14.2-235** including multiple connection locations disposed at an opposite end of the strap **14.2-231**. As can be seen, an operational component **14.2-237** can be disposed at least partially in the internal volume of the strap **14.2-231** and between the HMD connector **14.2-233** and the retention band connector **14.2-235**. In some examples, and as described further herein, the housing of the strap **14.2-231** can be raised or offset at the location of the operational component **14.2-237**, as desired. In some examples, the operational component **14.2-237** itself can at least partially define an exterior surface of the strap **14.2-231**. Additionally, and as described with respect to FIGS. **14.2-2D**, the operational component **14.2-237** can be electrically connected to at least one of the first connector **14.2-233** or the second connector **14.2-235**.

[3760] FIGS. **14.2-2C** shows a side view of the other connector strap **14.2-230** of the wearable electronic device **14.2-200**. Whereas the connector strap **14.2-231** of FIGS. **14.2-2B** included an HMD connector **14.2-233** and a retention strap connector **14.2-235**, but may not have included a supplemental unit connector, the connector strap **14.2-230** of FIGS. **14.2-2C** includes an HMD connector **14.2-232**, a retention band connector (not shown), and a supplemental unit connector **14.2-236**. As with the connector strap **14.2-231**, the strap **14.2-230** can include an operational component **14.2-238** that can be disposed between the HMD connector **14.2-232** and the supplemental unit connector **14.2-236**, and at least partially within an internal volume defined by the housing **14.2-239** of the strap **14.2-230**.

[3761] FIGS. **14.2-2D** shows a cross-sectional or partial cut-away view of the connector strap **14.2-230** shown in FIGS. **14.2-2A** and **14.2-2C**. In this view, a portion of the housing **14.2-239**, as well as the operational component **14.2-238**, are not shown in order to illustrate the internal volume of the strap **14.2-230**. As can be seen, in some examples, the HMD connector **14.2-232** can be electrically connected to the supplemental unit connector **14.2-236**. In this particular example, a transmission array or transmission component(s) **14.2-251**, **14.2-253**, such as wires or other conductive elements, can be disposed in the internal volume to electrically connect the HMD connector **14.2-232** and the supplemental unit connector **14.2-236**, such as to provide data and/or power therebetween. In some examples, the transmission array can include multiple conductors, cables, wires, or other transmission lines **14.2-251**, **14.2-253** for transmitting data and/or power between the HMD connector **14.2-232** and the supplemental unit connector **14.2-236**. In some examples, the lines **14.2-251**, **14.2-253** can be in communication to the connectors through connection points **14.2-250**, as shown. Additionally, in some examples, the supplemental unit connector **14.2-236** can include a printed circuit board **14.2-256** that can be at least partially disposed in the internal volume. In some examples, the connector strap **14.2-230** can include a processor or a processing unit **14.2-257**. The conductors **14.2-251**, **14.2-253** of the transmission array can be coupled to one or more locations on this printed circuit board **14.2-256**, and/or can be directly connected to the supplemental unit connector **14.2-236**.

[3762] In some examples, the housing **14.2-239** of the strap **14.2-230** can include a relatively

flexible material, such as polymeric material, fabric, and/or any desired flexible material or materials. In some examples, the strap **14.2-230** can further include a stiffening component or a stiffener **14.2-254** disposed at least partially in the internal volume defined by the housing **14.2-239**. In some examples, the stiffener **14.2-254** can be positioned at any desired location or locations in the internal volume, and can extend along any portion of the length of the strap **14.2-230**. In some examples, the stiffener **14.2-254** can be disposed adjacent to the internal components associated with one or both of the HMD connector **14.2-232**, the supplemental unit connector **14.2-236**, and/or the retention band connector (not shown).

[3763] In some examples, such as where the housing **14.2-239** of the strap **14.2-230** is relatively flexible, the stiffener **14.2-254** can serve to provide structural support to the strap **14.2-230** in one or more directions. For example, the stiffener **14.2-254** can be flexible along a first axis, and rigid along at least one axis that is perpendicular to the first axis. In some examples, the stiffener **14.2-254** can be flexible along a first axis and can be rigid along two axes that are perpendicular to the first axis and to each other. That is, the stiffener **14.2-254** can be flexible along a first bend direction, such as a direction extending into and out of the page of FIG. **14.2-2D**, while being rigid in one or more other bend directions, such as one or more directions parallel to the plane of the page of FIGS. **14.2-2D**. In some examples, the stiffener **14.2-254** can include a polymer material, metallic material, and/or combinations thereof. In some examples, the stiffener **14.2-254** can have a higher stiffness than the material including the housing **14.2-239** in at least one bend direction or axis. In some examples, the stiffener **14.2-254** can include metal in the form of a metallic sheet. Further details regarding the connector straps of a modular wearable electronic device, including details regarding the operational components thereof are provided with respect to FIGS. **14.2-3** and **14.2-4**.

[3764] FIGS. **14.2-3** shows a side view of a component **14.2-330** that can be used as part of a module wearable electronic device as described herein. In some examples, the component **14.2-330** can be a connector strap **14.2-330** and can be substantially similar to, or can include some or all of the features of the other connector straps and/or electronic device described herein. The example shown in FIGS. **14.2-3** can be substantially similar to the connector strap **14.2-230** shown in FIGS. **14.2-2C**, and can include an HMD connector **14.2-332**, a supplemental unit connector **14.2-336**, and a retention band connector (not shown) disposed on an opposite side of the supplemental unit connector **14.2-336**.

[3765] In the example shown in FIGS. **14.2-3**, the operational component **14.2-338** of the strap **14.2-330** can include a data output component. That is, the operational component **14.2-338** can be configured to transmit data from the strap **14.2-330** and/or any of the components of a wearable device in communication with the strap, such as an HMD and/or supplemental unit, to another electronic device. In some examples, the output device **14.2-338** can output data at a relatively high bitrate. In some examples, the output component **14.2-338** can have a direct electrical connection to a secondary device through a conductor or cable **14.2-339** as shown. In some examples, however, the output component **14.2-338** can include one or more antennas to wireless transmit data to a secondary device as desired.

[3766] In some examples, a user can include the strap **14.2-330** including the output component **14.2-338** in a module electronic device in place of, for example the strap **14.2-230** shown in FIGS. **14.2-2A**, **14.2-2C**, and **14.2-2D** in situations where high throughput transfer of data from the wearable device to a secondary device is desired. In other scenarios where this functionality may not be needed or desired, however, the strap **14.2-230** can be included in the wearable device instead.

[3767] FIGS. **14.2-4** shows a side view of a component **14.2-430** that can be used as part of a module wearable electronic device, as described herein. In some examples, the component **14.2-430** can be a connector strap **14.2-430** and can be substantially similar to, or can include some or all of the features of the other connector straps and/or electronic device described herein. The

example shown in FIGS. **14.2-4** can be substantially similar to the connector strap **230** shown in FIGS. **14.2-2C**, and can include an HMD connector **14.2-432**, a supplemental unit connector **14.2-436**, and a retention band connector (not shown) disposed on an opposite side of the supplemental unit connector **14.2-436**.

[3768] The strap **14.2-430** can also include an operational component **14.2-439** including a visual or display component. In the example shown in FIGS. **14.2-4**, the operational component **14.2-439** can take the form of one or more LEDs that can be selectively illuminated in order to visually display information. In some examples, the operational component **14.2-439** can include any other type of display or display technology, or combinations thereof. In some examples, the display **14.2-439** can be used to present visual information to persons other than the user when the wearable electronic device of which the strap **14.2-430** is part. In some examples, the display **14.2-439** can be configured to present visual information to a user when the wearable device is not being worn. For example, the display **14.2-439** can communicate a battery level of the device before a user puts the device on. Further, as shown, the strap **14.2-430** can include a second additional operational component **14.2-438** in addition to the display **14.2-439**.

[3769] Any number or variety of components in any of the configurations described herein can be included in a wearable electronic device, such as the HMD devices and/or HMD systems described herein. The components can include any combination of the features described herein, and can be arranged in any of the various configurations described herein. The structure and arrangement of components of a device, as well as the concepts regarding their use can apply not only to the specific examples discussed herein, but to any number of embodiments in any combination. Various examples of electronic devices and electronic device components including some having various features in various arrangements are described below, with reference to FIGS. **14.2-5A** through **14.2-10C**.

[3770] FIGS. **14.2-5A** shows a top view of a component **14.2-530** that can be part of, or can be used with, a wearable electronic device, as described herein. In some examples, the component **14.2-530** can be a connector strap **14.2-530** and can be substantially similar to, or can include some or all of the features of the other connector straps and/or electronic device described herein. As with other connector straps described herein, the connector strap **14.2-530** can include a first connector **14.2-532** that can be an HMD connector **14.2-532**, a second connector **14.2-536** that can be a supplemental unit connector, and a third connector **14.2-534** that can be a retention band connector. As described herein, in some examples, the retention band connector **14.2-534** can include multiple connection portions. In the example shown in FIGS. **14.2-5A**, the retention band connector **14.2-534** can include a first connection portion and a second connection portion **14.2-534'** spaced apart from the first connection portion on the same side of the connector strap **14.2-530**.

[3771] The connector strap **14.2-530** can have a housing **14.2-539** that can at least partially define an internal volume and an offset surface that is proud of a major surface of the housing **14.2-539**. That is, the housing can define a protrusion or protruding portion **14.2-538** and one or more operational components can be positioned in or at this protrusion **14.2-538**. Additionally, in some examples, one or more connectors of the connector strap **14.2-530** can also be positioned at or in this protrusion **14.2-538**. For example, the supplemental unit connector **14.2-536** can be positioned at least partially at an external surface of the housing **14.2-539** defined by the protrusion **14.2-538**. In some examples, the external surface of the housing can include the outer surface of the housing. In some additional examples, the external surface can further include an outer surface of the housing that is oriented away from a user's head when worn by the user.

[3772] FIGS. **14.2-5B** shows a side view of the connector strap **14.2-530**. In addition to defining an internal volume and external surface of the connector strap **14.2-530**, the housing **14.2-539** of the connector strap **14.2-530** can define one or more openings or ports **14.2-533**. In some examples, the housing **14.2-539** can define one or more audio ports **14.2-533** that are acoustically connected to an

audio module or speaker and can allow an audio module or a speaker positioned in the internal volume defined by the housing **14.2-539** to communicate with the ambient environment. In some examples, the housing **14.2-539** can define two opposing major surfaces and the audio port **14.2-533** can be positioned at an edge or a minor surface joining the two opposing surfaces. Additionally, while one of the major surfaces, such as the major surface shown in FIGS. **14.2-5B** can be interrupted by a protrusion **14.2-538**, at least one of the major surfaces (not shown) can extend an entire length of the connector strap **14.2-530** and/or an entire length of the internal volume of the strap **14.2-530**.

[3773] FIGS. **14.2-5C** shows a cross-sectional view of the connector strap **14.2-530** in the position shown in FIGS. **14.2-5B**, but with a major surface of the housing **14.2-539** not shown in order to illustrate the positioning of components in the internal volume. In the present example, the strap **14.2-530** can include two or more operational components **14.2-557**, **14.2-558**, at least one or which can be positioned at the protrusion **14.2-538**. In some examples, a first operational component **14.2-557** can include a first audio module disposed in the internal volume and can be configured to output a first range of audio frequencies, while a second operational component **14.2-558** can include a second audio module disposed in the internal volume and configured to output a second range of audio frequencies different than the first range. Additionally drivers of the first and/or second audio modules **14.2-557**, **14.2-558** can be in communication with the audio port **14.2-533** in order to output sound therefrom. In some examples, one or more of the audio modules **14.2-557**, **14.2-558** can have open-box or ported speaker construction or closed-box or sealed speaker construction. In some examples, one audio module **14.2-557** can function as a tweeter, while the second audio module **14.2-558** can function as a woofer.

[3774] As with other straps **14.2-530** described herein, the strap **14.2-530** can include a stiffener **14.2-552** that is flexible in a first direction, and rigid in at least one direction that is perpendicular to the first direction. One or more of the operational components **14.2-557**, **14.2-558** and/or connectors **14.2-536** can be positioned on, or can be connected to a printed circuit board **14.2-556** positioned in the internal volume. The printed circuit board **14.2-556** can also include components or electronic elements, including memory and processors positioned thereon. In some examples, the strap **14.2-530** can also include a rigid support member **14.2-553** positioned between the printed circuit board **14.2-556** and the stiffener **14.2-552**, the rigid support member **14.2-553** being more rigid than the stiffener in the first direction.

[3775] In this particular example, a transmission array or transmission component(s) **14.2-551** can be disposed in the internal volume to electrically connect the HMD connector **14.2-532** to the printed circuit board **14.2-556**, the supplemental unit connector **14.2-536** and/or one or more operational components **14.2-557**, **14.2-558**. In some examples, the transmission component **14.2-551** can extend along a center-line of the housing **14.2-539**. In some examples including multiple conductors or transmission components, a first conductor can be positioned or disposed adjacent to one internal surface of the housing of the strap **14.2-530**, and a second conductor can be positioned or disposed adjacent to another different internal surface of the housing of the strap. In some examples, different conductors may be positioned on opposite sides of internal components, such as the printed circuit board **14.2-556**, operational components **14.2-557**, **14.2-558**, stiffeners, and the like. In some examples, the components of the strap **14.2-530** can be maintained in their desired locations by internal fixtures and/or attachment features defined by, or attached to the housing **14.2-539**. In some examples, such as where the housing **14.2-539** includes a relatively flexible material, such as a polymeric material, the housing **14.2-539** be at least partially overmolded around at least some of the components of the strap **14.2-530**, including the printed circuit board **14.2-556**, operational components **14.2-557**, **14.2-558**, and/or connectors **14.2-532**, **14.2-536**. Further details of connector straps for wearable electronic devices are described with respect to FIGS. **14.2-6A** through **14.2-6C**.

[3776] FIGS. **14.2-6A** shows a top view of a component **14.2-630** that can be part of, or can be

used with a wearable electronic device, as described herein. In some examples, the component **14.2-630** can be a connector strap **14.2-630** and can be substantially similar to, or can include some or all of the features of the other connector straps and/or electronic device described herein. As with other connector straps described herein, the connector strap **14.2-630** can include a first connector **14.2-632** that can be an HMD connector **14.2-632**, a second connector **14.2-636** that can be a supplemental unit connector, and a third connector **14.2-634** that can be a retention band connector. As described herein, in some examples, the retention band connector **14.2-634** can include multiple connection portions. The connector strap **14.2-630** can have a housing **14.2-639** that can at least partially define an internal volume and an offset surface that is proud of a major surface of the housing **14.2-639**. That is, the housing can define a protrusion or protruding portion **14.2-638** and one or more operational components can be positioned in or at this protrusion **14.2-638**. The protruding portion **14.2-638** can be a portion of the housing that is wider than another portion, in any direction.

[3777] FIGS. **14.2-6B** shows a side view of the connector strap **14.2-630**. In addition to defining an internal volume and external surface of the connector strap **14.2-630**, the housing **14.2-639** of the connector strap **14.2-630** can define one or more openings or ports **14.2-633**. In some examples, the housing **14.2-639** can define one or more audio ports **14.2-633** that can allow an audio module or speaker positioned in the internal volume defined by the housing **14.2-639** to communicate with the ambient environment. Additionally, as shown, the connector strap **14.2-630** may not have a constant height or thickness along its entire length. In some examples, a portion of the housing can be enlarged relative to other portions of the strap **14.2-630**, such as to provide room for operational components, and/or to position operational components nearer to their desired location relative to a user.

[3778] FIGS. **14.2-6C** shows a cross-sectional view of the connector strap **14.2-630** in the position shown in FIGS. **14.2-6B**, but with a major surface of the housing **14.2-639** not shown in order to illustrate the positioning of components in the internal volume. In the present example, the strap **14.2-630** can include two or more operational components **14.2-657**, **14.2-658**, at least one or which can be positioned at the protrusion **14.2-638**. In some examples, a first operational component **14.2-657** can include a first audio module disposed in the internal volume and configured to output a first range of audio frequencies, while a second operational component **14.2-658** can include a second audio module disposed in the internal volume and configured to output a second range of audio frequencies different than the first range. Additionally drivers of the first and/or second audio modules **14.2-657**, **14.2-658** can be in communication with the audio port **14.2-633** in order to output sound therefrom. In some examples, one or more of the audio modules **14.2-657**, **14.2-658** can have open-box speaker construction or closed-box speaker construction. In some examples, one audio module **14.2-657** may function as a tweeter, while the second audio module **14.2-658** can function as a woofer. In some examples, the strap **14.2-630** can include one or more sensors, for example a sensor that can detect a location of a user's ear relative to the strap and can aid in directing sound from the audio port **14.2-633** to the user's ear.

[3779] As with other straps **14.2-630** described herein, the strap **14.2-630** can include a stiffener **14.2-652** that is flexible in a first direction and rigid in at least one direction that is perpendicular to the first direction. One or more of the operational components **14.2-657**, **14.2-658** and/or connectors **14.2-636** can be positioned on, or connected to, a printed circuit board **14.2-656** positioned in the internal volume. The printed circuit board **14.2-656** can also include components, including memory and processors positioned thereon. In some examples, the strap **14.2-630** can also include a support member **14.2-653** positioned between the printed circuit board **14.2-656** and the stiffener **14.2-652**, the support member **14.2-653** being more rigid than the stiffener in the first direction.

[3780] In this particular example, a transmission array or transmission component(s) **14.2-650**, **14.2-651** can be disposed in the internal volume to electrically connect the HMD connector **14.2-**

632 to the printed circuit board **14.2-656**, the supplemental unit connector **14.2-636** and/or one or more operational components **14.2-657**, **14.2-658**. In some examples, the transmission components **14.2-650**, **14.2-651** can extend along a center-line “C” of the housing **14.2-639**. In some examples, the components of the strap **14.2-630** can be maintained in their desired locations by internal fixtures and/or attachment features defined by, or attached to the housing **14.2-639**. In some examples, such as where the housing **14.2-639** includes a relatively flexible material, such as a polymeric material, the housing **14.2-639** be at least partially overmolded around at least some of the components of the strap **14.2-630**, including the printed circuit board **14.2-656**, operational components **14.2-657**, **14.2-658**, and/or connectors **14.2-632**, **14.2-636**.

[3781] In some examples, the housing **14.2-639** can include a first portion, such as the portion containing the stiffener **14.2-652**, and a second portion, such as the portion containing the first audio module **14.2-657** and the second audio module **14.2-658**. In some examples, this second portion can have a greater height than the first portion, as shown. Additionally, because this second portion can also be where the protrusion **14.2-638** is located, the second portion can also have a thickness or width that is greater than a thickness or width of the first portion. Further details of connector straps for wearable electronic devices are described with respect to FIGS. **14.2-7A** through **14.2-7C**.

[3782] FIGS. **14.2-7A** shows a top view of a component **14.2-730** that can be part of, or used with a wearable electronic device, as described herein. In some examples, the component **14.2-730** can be a connector strap **14.2-730** and can be substantially similar to, or can include some or all of the features of the other connector straps and/or electronic device described herein. As with other connector straps described herein, the connector strap **14.2-730** can include a first connector **14.2-732** that can be an HMD connector **14.2-732**, a second connector **14.2-736** that can be a supplemental unit connector, and a third connector **14.2-734** that can be a retention band connector. As described herein, in some examples, the retention band connector **14.2-734** can include multiple connection portions. The connector strap **14.2-730** can have a housing **14.2-739** that can at least partially define an internal volume and an offset surface that is proud of a major surface of the housing **14.2-739**. That is, the housing can define a protrusion or protruding portion **14.2-738** and one or more operational components can be positioned in or at this protrusion **14.2-738**.

[3783] FIGS. **14.2-7B** shows a side view of the strap **14.2-730**. As can be seen in FIGS. **14.2-7A** and **14.2-7B**, in some examples, the protrusion **14.2-738** defined by the housing **14.2-739** of the strap **14.2-730** can be positioned at any desired location on the strap **14.2-730**. Additionally, the protrusion **14.2-738** can be spaced apart from an audio port **14.2-733** defined by the housing **14.2-739**. Further, as can be seen in FIGS. **14.2-7C**, this means that an operational component, such as an audio module **14.2-758** can be positioned in the internal volume of the strap **14.2-730** away from the audio port **14.2-733**. It can still be desirable, however, for the audio module **14.2-758** and/or a driver thereof to be in communication with the audio port **14.2-733** (shown with dashed lines). In order to facilitate this communication, the strap **14.2-730** can include an acoustic guide **14.2-759** that can include one or more channels in communication with the audio module **14.2-758** and the audio port **14.2-733**. In this way, the audio module **14.2-758** can be positioned away from the audio port **14.2-733**, but can still direct sound to a user from the audio port **14.2-733**. In some examples, the acoustic guide **14.2-759** can include any material desired, including metals or polymers. In some examples, the acoustic guide **14.2-759** can be flexible and can bend or flex with the strap **14.2-730** while still providing acoustic communication. In some examples, the acoustic guide **14.2-759** may not be a separate or discrete component and may be at least partially defined by the housing **14.2-739** itself.

[3784] The strap **14.2-730** can also include some or all of the other components and features of the other straps described herein. For example, while the audio module **14.2-758** can be positioned away from the audio port **14.2-733** and communicate with the audio port **14.2-733** through an acoustic guide **14.2-759**, the strap **14.2-730** can include another audio module **14.2-757** that can be

positioned near to the audio port **14.2-733** to communicate therewith. The strap can also include a printed circuit board that can be connected through various transmission components **14.2-751** to other connectors **14.2-732** and/or components **14.2-757**, **14.2-758** of the strap **14.2-730**. Further details of connector straps for wearable electronic devices are described with respect to FIGS. **14.2-8A** through **14.2-8C**.

[3785] FIGS. **14.2-8A** shows a top view of a component **14.2-830** that can be part of, or can be used with a wearable electronic device, as described herein. In some examples, the component **14.2-830** can be a connector strap **14.2-830** and can be substantially similar to, or can include some or all of the features of the other connector straps and/or electronic device described herein. As with other connector straps described herein, the connector strap **14.2-830** can include a first connector **14.2-832** that can be an HMD connector **14.2-832**, a second connector **14.2-836** that can be a supplemental unit connector, and a third connector **14.2-834** that can be a retention band connector. As described herein, in some examples, the retention band connector **14.2-834** can include multiple connection portions. The connector strap **14.2-830** can have a housing **14.2-839** that can at least partially define an internal volume and an offset surface that is proud of a major surface of the housing **14.2-839**. That is, the housing can define a protrusion or protruding portion **14.2-838** and one or more operational components can be positioned in or at this protrusion **14.2-838**.

[3786] As seen in the side view of FIGS. **14.2-8B**, the strap **14.2-830** can have a similar configuration as other straps described herein, such as straps **14.2-530**, **14.2-630**, where the protrusion **14.2-838** is positioned at an end of the strap that includes the supplemental unit connector **14.2-836**. In addition to the audio port **14.2-833**, the housing **14.2-839** can further define a vent, an exit, or a back port **14.2-835** that can be positioned at an opposite side of the strap from the audio port **14.2-833**. In some examples, the back port **14.2-835** can also be positioned at an opposite end of the strap **14.2-830** from the audio port **14.2-833**. In some examples, it can be desirable to position the back port **14.2-835** at a location on the strap **14.2-830** that is as far away from the audio port as possible or feasible.

[3787] FIGS. **14.2-8C** shows a cross-sectional view of the strap **14.2-830**, including first and second audio modules **14.2-857**, **14.2-858**, printed circuit board **14.2-856**, and transmission component **14.2-851** including a conductor such as cables or wires. In the present example, the strap **14.2-830** also includes an acoustic guide **14.2-859** that can be in communication with the audio module **14.2-858** or a driver thereof. Whereas the acoustic guide **14.2-759** shown in FIGS. **14.2-7C** is configured to direct sound generated by the audio module **14.2-758** to the audio port **14.2-733**, the acoustic guide **14.2-859** can be configured to direct the back wave generated by the driver of the audio module **14.2-858** to the back port, and thus, away from the user's ear. Accordingly, the audio module **14.2-858** can include an open-box or open-back speaker and the acoustic guide **14.2-859** can be in communication with the back or rear surface of the speaker's diaphragm so as to release backpressure and/or direct negative sound waves generated by the speaker away from the user's ear. This construction can thus allow for the audio modules **14.2-857**, **14.2-858** to generate high quality audio in a relatively small package. Further details of connector straps for wearable electronic devices are described with respect to FIGS. **14.2-9A** through **14.2-9C**.

[3788] FIGS. **14.2-9A** shows a top view of a component **14.2-930** that can be part of, or used with a wearable electronic device, as described herein. In some examples, the component **14.2-930** can be a connector strap **14.2-930** and can be substantially similar to, or can include some or all of the features of the other connector straps and/or electronic device described herein. As with other connector straps described herein, the connector strap **14.2-930** can include a first connector **14.2-932** that can be an HMD connector **14.2-932**, a second connector **14.2-936** that can be a supplemental unit connector, and a third connector **14.2-934** that can be a retention band connector. As described herein, in some examples, the retention band connector **14.2-934** can include multiple

connection portions. The connector strap **14.2-930** can have a housing **14.2-939** that can at least partially define an internal volume and an offset surface that is proud of a major surface of the housing **14.2-939**. That is, the housing can define a protrusion or protruding portion **14.2-938** and one or more operational components can be positioned in or at this protrusion **14.2-938**.

[3789] FIGS. **14.2-9B** shows a side view of the strap **14.2-930**. As can be seen in FIGS. **14.2-9A** and **14.2-9B**, in some examples, the protrusion **14.2-938** defined by the housing **14.2-939** of the strap **14.2-930** can be positioned at any desired location on the strap **14.2-930**. Additionally, the protrusion **14.2-938** can be spaced apart from an audio port **14.2-933** defined by the housing **14.2-939**. Further, as can be seen in FIGS. **14.2-9C**, this means that an operational component, such as an audio module **14.2-958** can be positioned in the internal volume of the strap **14.2-930** away from the audio port **14.2-933**. In some examples, the strap **14.2-930** can include an acoustic guide **14.2-959** that can include one or more channels in communication with the audio module **14.2-958** and the audio port **14.2-933**. In some examples, the acoustic guide **14.2-959** can define two or more, three or more, four or more, or even more channels. In some examples, the acoustic guide **14.2-959** can define three channels. In some examples, the acoustic guide **14.2-959** can also function as a stiffener for the strap **14.2-930**. That is, the acoustic guide **14.2-959** can be flexible along a first bend direction, and rigid along a second, different bend direction. In some examples, the acoustic guide **14.2-959** can thus be considered a flexible acoustic guide, or a semi-rigid acoustic guide.

[3790] The strap **14.2-930** can also include some or all of the other components and features of the other straps described herein. For example, while the audio module **14.2-958** can be positioned away from the audio port **14.2-933** and communicate with the audio port **14.2-933** through an acoustic guide **14.2-959**, the strap **14.2-930** can include another audio module **14.2-957** that can be positioned near to the audio port **14.2-933** to communicate therewith. The strap can also include a printed circuit board that can be connected through various transmission components **14.2-951** to other connectors **14.2-932** and/or components **14.2-957**, **14.2-958** of the strap **14.2-930**. Further details of connector straps for wearable electronic devices are described with respect to FIGS. **14.2-10A** through **14.2-10C**.

[3791] FIGS. **14.2-10A** shows a top view of a component **14.2-1030** that can be part of, or can be used with, a wearable electronic device, as described herein. In some examples, the component **14.2-1030** can be a connector strap **14.2-1030** and can be substantially similar to, or can include some or all of the features of the other connector straps and/or electronic device described herein. As with other connector straps described herein, the connector strap **14.2-1030** can include a first connector **14.2-1032** that can be an HMD connector **14.2-1032**, a second connector **14.2-1036** that can be a supplemental unit connector, and a third connector **14.2-1034** that can be a retention band connector. As described herein, in some examples, the retention band connector **14.2-1034** can include multiple connection portions. The connector strap **14.2-1030** can have a housing **14.2-1039** that can at least partially define an internal volume and an offset surface that is proud of a major surface of the housing **14.2-1039**. That is, the housing can define a protrusion or protruding portion **14.2-1038** and one or more operational components can be positioned in or at this protrusion **14.2-1038**.

[3792] FIGS. **14.2-10B** shows a side view of the strap **14.2-1030**. As can be seen in FIGS. **14.2-10A** and **14.2-10B**, in some examples, the protrusion **14.2-1038** defined by the housing **14.2-1039** of the strap **14.2-1030** can be positioned at any desired location on the strap **14.2-1030**. Additionally, the protrusion **14.2-1038** can be spaced apart from an audio port **14.2-1033** defined by the housing **14.2-1039**. Further, as can be seen in FIGS. **14.2-10C**, this means that an operational component, such as an audio module **14.2-1058** can be positioned in the internal volume of the strap **14.2-1030** away from the audio port **14.2-1033**. In some examples, the strap **14.2-1030** can include an acoustic guide **14.2-1059** that can include one or more channels in communication with the audio module **14.2-1058** and the audio port **14.2-1033**. In some examples,

the acoustic guide **14.2-1059** can include multiple channels. In some examples, the multiple channels of the acoustic guide **14.2-1059** can have a same size and cross-sectional area, or different sizes and/or cross-sectional areas. In some examples, one or more of the channels defined by the acoustic guide **14.2-1059** can have a rectangular cross-sectional shape, a round or circular cross-sectional shape, or any cross-sectional shape or shapes as desired. In some examples, the number of channels and their cross-sectional area can allow for the acoustic guide **14.2-1059** to be relatively flexible in at least one direction, while being relatively resistance to compression or collapse along other directions.

[3793] The strap **14.2-1030** can also include some or all of the other components and features of the other straps described herein. For example, while the audio module **14.2-1058** can be positioned away from the audio port **14.2-1033** and communicate with the audio port **14.2-1033** through an acoustic guide **14.2-1059**, the strap **14.2-1030** can include another audio module **14.2-1057** that can be positioned near to the audio port **14.2-1033** to communicate therewith. The strap can also include a printed circuit board that can be connected through various transmission components **14.2-1051** to other connectors **14.2-1032** and/or components **14.2-1057**, **14.2-1058** of the strap **14.2-1030**.

[3794] Any number or variety of components in any of the configurations described herein can be included in a wearable electronic device, such as the HMD devices and/or HMD systems described herein. The components can include any combination of the features described herein, and can be arranged in any of the various configurations described herein. The structure and arrangement of components of a device, as well as the concepts regarding their use can apply not only to the specific examples discussed herein, but to any number of embodiments in any combination. Various examples of electronic devices and electronic device components including some having various features in various arrangements are described below, with reference to FIGS. **14.2-11** through **14.2-13**.

[3795] FIGS. **14.2-11** shows an exploded view of a wearable electronic device **14.2-1100** that can include modular components which can be selectively and removably or releasably attached or coupled to one another as described herein. In some examples, the device **14.2-1100** can be substantially similar to, or can include some or all of the features of the other wearable electronic devices described herein. As with the electronic device **14.2-200**, the device **14.2-1100** can include an HMD **14.2-1110**, a retention band **14.2-1120**, and one or more connector straps **14.2-1130**, **14.2-1131**.

[3796] The connector straps **14.2-1130**, **14.2-1131** can include a number of connectors, such as HMD connectors **14.2-1132**, **14.2-1133** that can releasably couple to corresponding connectors on the HMD, such as connector **14.2-1112**. While the straps **14.2-230**, **14.2-231** of FIGS. **14.2-2A** included separate retention band connectors and supplemental unit connectors, the device **14.2-1100** can include a retention band **14.2-1120** that includes an integrated supplemental unit **14.2-1122**. Accordingly, the straps **14.2-1130**, **14.2-1131** can include a combined retention band and supplemental unit connector **14.2-1136**, **14.2-1137** that can both mechanically couple the straps **14.2-1130**, **14.2-1131** to the retention band **14.2-1120** and electrically communicate data and/or power with the supplemental unit **14.2-1122** of the retention band **14.2-1120**. The retention band **14.2-1120** itself can include corresponding connectors **14.2-1126**, **14.2-1127** that can be releasably coupled to the retention band connectors **14.2-1136**, **14.2-1137**.

[3797] Additionally, as noted above, the connector straps **14.2-1130**, **14.2-1131** can include a stiffener **14.2-1152** disposed in the flexible housing, the stiffener **14.2-1152** can be flexible along a first bend direction, such as a direction extending into and out of the page of FIGS. **14.2-11**, represented as the “Z” direction. Additionally, the stiffener **14.2-1152** can be rigid in one or more other bend directions, such as one or more directions parallel to the plane of the page of FIGS. **14.2-11**, represented as the “X” and “Y” directions, which are each substantially perpendicular to the “Z” direction. Stiffness in the “X” and “Y” directions allow the connector straps **14.2-1130**,

14.2-1131 to support both the HMD **14.2-1110** and the retention band **14.2-1120** during operation, counteracting the moment or rotational force caused by the HMD **14.2-1110** and the retention band **14.2-1120**. In this manner, the connector straps **14.2-1130**, **14.2-1131** provide sufficient structural rigidity to maintain the relative position of each component in the system during use. Simultaneously, the flexibility provided in the “Z” direction provides a snug and comfortable fit to the side of a user's head during use, and can continue and enhance the secured fit provided by the retention band **14.2-1120**. In some examples, the stiffener **14.2-1152** can include a polymer material, a metallic material, and/or combinations thereof. In some examples, the stiffener **14.2-1152** can have a higher stiffness than the material forming a majority of the connector straps **14.2-1130**, **14.2-1131** in at least one bend direction or axis. In some examples, the stiffener **14.2-1152** can include metal in the form of a metallic sheet.

[3798] According to one example, the shape and material of the connector straps **14.2-1130**, **14.2-1131** provide the desired combination of flexibility in the “Z” direction and rigidity in the “X” and “Y” directions even without the stiffener **14.2-1152**. According to this example, the approximately rectangular cross-sectional profile of the connector straps **14.2-1130**, **14.2-1131** having a height that is approximately between 5 and 10, between 10 and 20, between 20 and 50, or between 50 and 100 times that of the width facilitates bending or low-force buckling of the connector strap in the “Z” direction, while resisting buckling in the “X” and “Y” directions. In other words, buckling the connector straps **14.2-1130**, **14.2-1131** in the “X” and “Y” directions can require 10, 20, 30, 50, or more times the force that is needed to buckle the connector straps in the “Z” direction. Additional modifications, such as profile curves or directional stiffeners, can be added to the connector straps **14.2-1130**, **14.2-1131** to further strengthen the connector straps in desired directions.

[3799] In some examples, the retention band **14.2-1120** can include other operational or functional components in addition, or in alternative, to the supplemental unit **14.2-1122**. For example, the component **14.2-1122** can include a battery module that can be in electrical communication with the straps **14.2-1130**, **14.2-1131**. Additionally, the device **14.2-1100** may include a separate processing unit (not shown), such as the supplemental unit **14.2-140** shown in FIGS. **14.2-1C** that can be connected to the strap **14.2-1130**, while the retention band **14.2-1120** includes a second additional processing unit **14.2-1122**. Further details regarding various configurations of wearable electronic devices are discussed with respect to FIGS. **14.2-12**.

[3800] FIGS. **14.2-12** shows an exploded view of a wearable electronic device **14.2-1200** that can include modular components which can be selectively and removably or releasably attached or coupled to one another, as described herein. In some examples, the device **14.2-1200** can be substantially similar to, or can include some or all of the features of the other wearable electronic devices described herein. As with the electronic device **14.2-100**, the device **14.2-1200** can include an HMD **14.2-1210**, a retention band **14.2-1220**, and one or more connector straps **14.2-1230**, **14.2-1231**, and a supplemental unit **14.2-1240**.

[3801] The connector straps **14.2-1230**, **14.2-1231** can include a number of connectors, such as HMD connectors **14.2-1232**, **14.2-1233**, that can releasably couple to corresponding connectors on the HMD, such as connector **14.2-1212**. The connector straps **14.2-1230**, **14.2-1231** can also include retention band connectors **14.2-1234**, **14.2-1235** that can releasably couple to corresponding connectors **14.2-1224**, **14.2-1225** on the retention band **14.2-1220**. The connector strap **14.2-1230** can include a supplemental unit connector **14.2-1236** that can be releasably coupled to the supplemental unit **14.2-1240** through a conductor or cable **14.2-1242**. In some examples, the retention band **14.2-1220** can include a band **14.2-1222** designed to secure the device **14.2-1200** to a user, and a pocket or compartment **14.2-1226** that can be attached to the band **14.2-1222** and which can be configured to hold or retain the supplemental unit **14.2-1240**. In this way, the supplemental unit **14.2-1240** can be carried entirely on the device **14.2-1200** without the need to attach the supplemental unit **14.2-1240** to the user at another location.

[3802] FIGS. **14.2-13** shows an exploded view of a wearable electronic device **14.2-1300** that can

include modular components which can be selectively and removably or releasably attached or coupled to one another as described herein. In some examples, the device **14.2-1300** can be substantially similar to, or can include some or all of the features of the other wearable electronic devices described herein. As with the electronic device **14.2-100**, the device **14.2-1300** can include an HMD **14.2-1310**, a retention band **14.2-1320**, and one or more connector straps **14.2-1330**, **14.2-1331**.

[3803] The connector straps **14.2-1330**, **14.2-1331** can include a number of connectors, such as HMD connectors **14.2-1332**, **14.2-1333** that can releasably couple to corresponding connectors on the HMD, such as connector **14.2-1312**. The connector straps **14.2-1330**, **14.2-1331** can also include retention band connectors **14.2-1334**, **14.2-1335** that can releasably couple to corresponding connectors **14.2-1324**, **14.2-1325** on the retention band **14.2-1320**. The connector strap **14.2-1330** can include a supplemental unit connector **14.2-1336** that can be releasably coupled to the supplemental unit **14.2-1340** through a conductor or cable **14.2-1342**. Whereas some examples described herein can include two or more modular or separate connector straps, in some examples, the connector straps **14.2-1330**, **14.2-1331** can be releasably or permanently joined to one another, such as by a third strap or a head strap **14.2-1339**. In some examples, this head strap **14.2-1339** can extend over a user's head when the wearable electronic device **14.2-1300** is worn by a user. The head strap **14.2-1339** can thus provide a secured and comfortable fit to the user. Further, in examples where the strap **14.2-1339** is removably or releasably attached to the straps **14.2-1330**, **14.2-1331**, the user can opt whether or not to use the straps **14.2-1339** depending on the particular use scenario of the device **14.2-1300**. In some examples, the strap **14.2-1339** can include any material as desired, including polymeric materials, fabric materials, and any of the materials described with respect to connector straps and/or retention bands.

14.3: Modular Strap for an Electronic Device

[3804] As virtual reality (VR) and mixed reality (MR) become more ubiquitous, the demand for user friendly head-mounted displays with quality components increases. Traditionally, these VR/MR systems have been devices that include a wearable display component, often referred to as a head-mounted display (HMD) and a supplemental unit. The supplemental unit can be part of the HMD, or can be a separate component that is permanently or removably connected to the HMD through a cable, wires, or other conductors. The supplemental unit can provide power and/or added processing functionality to the system.

[3805] Devices of the present disclosure include one or more additional dongles for improved connectivity to such supplemental components. For example, an HMD system of the present disclosure includes a first dongle for attaching a power supply to a first removable strap (e.g., a modular band of the HMD system). In addition, the HMD system includes a second dongle for attaching an external computing device or display to a second removable strap (e.g., another modular band of the HMD system). With separate, designated dongles (e.g., one for power and the other for data transmission), the HMD system of the present disclosure can significantly improve throughput. Advantageously, this can facilitate developers or programmers to rapidly upload and download data from the HMD system, while separately powering the device.

[3806] In addition, the present disclosure relates to various thermal ergonomic improvements for a removable strap. Removable straps can include a variety of electrical components, including a battery, integrated circuit, and a speaker (to name a few). Such electrical components can generate large amounts of thermal energy, particularly for longer use durations and heavy computing applications. Thus, given the location of a removable strap (and its associated electrical components) being next to a user's head, managing thermal output is important.

[3807] To reduce or mitigate hotspots along the removable straps, the present disclosure includes electrical pods positioned along the removable straps. These electrical pods can include a thermal shell and other thermal elements that provide a combination of heat shielding and heat dissipation. In so doing, the removable straps can provide increased comfort to a user. In addition, the

removable straps can reduce (or more efficiently spread) a thermal load.

[3808] These and other embodiments are discussed below with reference to FIGS. **14.3-1** through **14.3-8**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[3809] FIGS. **14.3-1** shows a top view of a wearable electronic device **14.3-100** being worn on the head **14.3-101** of a user. The wearable electronic device **14.3-100**, as well as other wearable electronic devices disclosed herein, can also be referred to as HMD systems, electronic devices, or simply as devices. The device **14.3-100** can include a number of modular components. For example, the device **14.3-100** can include a head-mounted display (HMD) **14.3-102** including a housing **14.3-104** and a display **14.3-106** attached to the housing **14.3-104** for displaying images to a user.

[3810] The HMD can also be referred to as a display portion or display module having the display **14.3-106**. The display portion can include the housing **14.3-104** and display **14.3-106** that at least partially constitutes the HMD. In one or more examples, including the example shown in FIGS. **14.3-1** and other examples shown in other figures, the HMD **14.3-102** can also be referred to as an output component, output module. Such output components, modules, or portions can include one or more outputs other than visual outputs from a display. For example, an output module similar to the HMD **14.3-102** can include a speaker that outputs sound instead of or in addition to the display **14.3-106** shown in FIGS. **14.3-1**.

[3811] In addition, one example of the device **14.3-100** can include a modular securement assembly **14.3-108** that secures the HMD **14.3-102** to the user's head **14.3-101**. The modular securement assembly **14.3-108** includes removable straps **14.3-110**, **14.3-112**. The modular securement assembly **14.3-108** further includes a retention band **14.3-114**. The modular securement assembly **14.3-108** is therefore configured to removably secure the wearable electronic device **14.3-100**, including the HMD **14.3-102**, to the head **14.3-101** of the user when the removable straps **14.3-110**, **14.3-112** and retention band **14.3-114** are connected as shown in FIGS. **14.3-1**.

[3812] In particular examples, the removable straps **14.3-110**, **14.3-112** removably (e.g., detachably) connect to both of the HMD **14.3-102** and the retention band **14.3-114**. For example, each of the removable straps **14.3-110**, **14.3-112** can be removably connected to the HMD **14.3-102** (or the housing or the display portion thereof) and the retention band **14.3-114** at opposing ends of each removable strap **14.3-110**, **14.3-112** as shown in FIGS. **14.3-1**. In such an example, the modular securement assembly **14.3-108** is modular in that each of the removable straps **14.3-110**, **14.3-112**, and the retention band **14.3-114** can be connected and detached, as may be desired. For instance, each of the removable straps **14.3-110**, **14.3-112** can be removed from the modular securement assembly **14.3-108** and swapped out for one or more other modules, straps, or electronic components.

[3813] One or more other examples of the device **14.3-100** can include alternative configurations of the removable straps **14.3-110**, **14.3-112** shown in FIGS. **14.3-1**. For example, the device **14.3-100** can be positioned elsewhere along the modular securement assembly **14.3-108** (differently than what is shown in FIGS. **14.3-1**). In addition, one or more examples of wearable electronic devices described herein can include one or more intermediate members, flexible straps, or other optional supplemental components and electronic modules such as external power supplies, memory components, and/or processors.

[3814] In the example shown in FIGS. **14.3-1**, when the device **14.3-100** is worn on the head of the user, the removable strap **14.3-110** is positioned on the left side of the user's head and the removable strap **14.3-112** is positioned on the right side of the user's head. The retention band **14.3-114** can span between the removable straps **14.3-110**, **14.3-112** to wrap around the back of the user's head **14.3-101**, as shown.

[3815] In some examples, and as shown, the device **14.3-100** can be worn on the user's head **14.3-101** such that the HMD **14.3-102** is worn on the user's face and disposed over one or both of their eyes. The HMD **14.3-102** can be removably and/or releasably connected to one or more of the removable straps **14.3-110**, **14.3-112** as mentioned above. In some examples, the removable straps **14.3-110**, **14.3-112** can be positioned against the side of a user's head **14.3-101** and in contact therewith. In some examples, the removable straps **14.3-110**, **14.3-112** can be positioned above the user's ear or ears. In some examples, the removable straps **14.3-110**, **14.3-112** can be positioned adjacent to the user's ear or ears. The removable straps **14.3-110**, **14.3-112** can be removably connected to the retention band **14.3-114**, which can extend around the user's head **14.3-101** and removably connect to the other of the removable straps **14.3-110**, **14.3-112**. In this way, the HMD **14.3-102**, removable straps **14.3-110**, **14.3-112**, and retention band **14.3-114** can form a loop that can retain the wearable electronic device **14.3-100** on the user's head **14.3-101**.

[3816] As shown in FIGS. **14.3-1**, the removable straps **14.3-110**, **14.3-112** can connect to the HMD **14.3-102**, both mechanically and electrically. In particular examples, the removable straps **14.3-110**, **14.3-112** can receive and/or relay at least one of data or power via such connections. In these or other examples, the removable straps **14.3-110**, **14.3-112** can connect to the HMD **14.3-102** at an HMD connection location that can include an electrical input or electrical connector that is attached to the housing **14.3-104** and electrically connected to the display **14.3-106**. This location can be identified as a temple area that can be defined as an area near a user's temple adjacent to the user's eye, and can span from in front of the user's eye to approximately 1-1.5 inches past the outer corner of a user's eye, along the side of the user's head **14.3-101**.

[3817] Similarly, the removable straps **14.3-110**, **14.3-112** can connect to the retention band **14.3-114** at a retention band connection location identified as an area that can span to include the area above the user's ear or within 0.5 inches of the outer edge of the ear on either side. In this manner, the removable straps **14.3-110**, **14.3-112** are able to provide structural support between the HMD **14.3-102** and the user's ear, while securely connecting the retention band **14.3-114** and transferring the retention forces of the retention band **14.3-114** through the device **14.3-100**. It should be understood, however, that this configuration is just one example of how the components of a modular wearable electronic device **14.3-100** can be arranged, and that in some, a different number of removable straps and/or retention bands can be included.

[3818] While a user wearing an HMD **14.3-102** on his or her head **14.3-101** is shown as one example of a wearable electronic device, the modular components, features, and advantages of various examples of electronic devices disclosed herein can also apply to other wearable electronic devices having securement mechanisms, including but not limited to wearable smart watches, fitness trackers, smart glasses, medical monitor devices, and so forth. For example, the housing **14.3-104** and display **14.3-106** of HMD **14.3-102** shown in FIGS. **14.3-1** can also be configured as a housing and display for a smart watch module secured to the user's arm or wrist via a modular securement mechanism or assembly similar to the securement assembly **14.3-108** shown in FIGS. **14.3-1**. Although referred to as a wearable electronic device **14.3-100**, it should be understood that the device **14.3-100** can include multiple modular components or devices and can be interchangeably referred to as a wearable electronic device, wearable electronic device system, and/or wearable electronic system. Additionally, although the particular component **14.3-102** can be referred to as an HMD, it should be understood that the terms HMD, HMD device, and/or HMD system can be used to refer to the wearable device **14.3-100** as a whole.

[3819] In particular examples, the device **14.3-100** (or HMD system) includes one or more dongles

14.3-116, 14.3-118 that are connectable to external components or devices, as will now be discussed. As used herein, the term “dongle” can refer to a piece of electrical hardware such as a connection, adapter, or coupling between an external device/component and one or more components of the device **14.3-100** to provide additional functionality, or enable a pass-through to another device that adds functionality. A dongle can include electrical connections for data and/or power transmission through the dongle. In particular examples, a dongle can be unidirectional or bidirectional in terms of data and/or a power transmission.

[3820] As shown in FIGS. **14.3-1**, the dongle **14.3-116** connects the removable strap **14.3-110** to a power supply **14.3-120**. As used herein, the term “power supply” refers to any power source that supplies power to one or more components of the device **14.3-100** (e.g., to charge a battery or power a processor such as a microcontroller within the removable strap **14.3-110**). For example, a power supply can include fuel cells, battery cells, generators, alternators, solar power converters, motion-based converters (e.g., that convert vibrations or oscillations into power), etc. In particular implementations, a power supply can convert alternating current to direct current (or vice-versa) for charging or recharging components of the device **14.3-100**. Some particular examples of a power supply can include a switched mode power supply, an uninterruptible power supply, an alternating current power supply, a direct current power supply, a regulated power supply, a programmable power supply, a computer power supply, and a linear power supply.

[3821] Additionally shown in FIGS. **14.3-1**, the dongle **14.3-118** connects the removable strap **14.3-112** to an external device or display (DoD) **14.3-122**. An external device can include a smartphone, a tablet, a smart television, a desktop computer, a laptop computer, a server/network device, a virtual reality device, an augmented reality device, a smart watch, a sound/speaker device, a camera device, or other computing device. Further, an external display can include a monitor, device screen (e.g., a display screen for a laptop, television, phone, tablet, etc.), projector, or other suitable visual medium.

[3822] In at least some examples, the external DoD **14.3-122** includes a developer computer (e.g., for receiving and/or transmitting data from one or more components of the device **14.3-100**). In particular examples, the external DoD **14.3-122** uploads computer-executable instructions to one or more components of the device **14.3-100** (e.g., the removable strap **14.3-112**) through the dongle **14.3-118**. Additionally or alternatively, the external DoD **14.3-122** can also provide data to and/or receive data from certain component(s) of the device **14.3-100** (e.g., the removable strap **14.3-110**) through other dongle configurations, such as through the dongle **14.3-116** or a split-dongle configuration (described below in relation to FIGS. **14.3-6**).

[3823] In some examples, the external DoD **14.3-122** receives performance data (e.g., thermal data, power data, data throughput data, etc.). In certain implementations, the external DoD **14.3-122** provides power to one or more components of the device **14.3-100**. For example, the external DoD **14.3-122** can power or charge components of the removable strap **14.3-112** by providing power through the dongle **14.3-118**. In other examples, the external DoD **14.3-122** provides no power, in which case the power supply **14.3-120** can provide power to components of both the removable strap **14.3-110** and the removable strap **14.3-112**. For instance, the power supply **14.3-120** can relay power to the removable strap **14.3-112** via electrical connections between the power supply **14.3-120**, the removable strap **14.3-110**, the HMD **14.3-102**, and the removable strap **14.3-112**. Alternatively, the power supply **14.3-120** can relay power to the removable strap **14.3-112** via alternative dongle configurations (as described below in relation to FIGS. **14.3-6**).

[3824] Additionally or alternatively, the external DoD **14.3-122** includes a display that outputs visual information. For example, the external DoD **14.3-122** includes a monitor that outputs the same visual content that is shown at the display **14.3-106**. Thus, advantageously, the external DoD **14.3-122** can provide visual feedback (e.g., to developers or programmers) without needing to wear the device **14.3-100**. Similarly, the external DoD **14.3-122** can provide visual feedback (e.g., on a developer computer screen) while a user is simultaneously wearing the device **14.3-100**.

[3825] In one or more examples of the present disclosure, the dongles **14.3-116**, **14.3-118** can include a variety of bandwidth capacities. In certain examples, the dongle **14.3-116** includes a first bandwidth, and the dongle **14.3-118** includes a second bandwidth greater than the first bandwidth. For example, the dongle **14.3-118** can provide (relative to the dongle **14.3-116**) increased throughput of at least one of data or power. In this manner, the dongle **14.3-118** can facilitate larger and/or faster data uploads and downloads (for developer work or testing on the device **14.3-100**).

[3826] In some examples, the dongles **14.3-116**, **14.3-118** can also include variable bandwidths that can be throttled (e.g., reduced) or increased, as may be desired. For example, the dongles **14.3-116**, **14.3-118** can include adjusted bandwidths based on one or more performance metrics (e.g., throughput speeds, visual content quality/resolution, thermal output, etc.). Indeed, as will be discussed more below in relation to FIGS. **14.3-4** and **14.3-5**, an electronics pod on one or both of the removable straps **14.3-110**, **14.3-112** can generate a thermal output that is spread out or dissipated into the ambient environment (e.g., to reduce heat transfer to a user). To help lessen the thermal output of the electronics pod (and/or an amount of thermal output to dissipate), one or more components of the device **14.3-100** can vary (e.g., decrease) the bandwidth through the dongles **14.3-116**, **14.3-118**. As used herein, the term “bandwidth” refers to a rate of power transfer (e.g., power bandwidth) or a rate of data transfer (e.g., data bandwidth) through electrical wiring and connections, or else wirelessly. In some cases, the term bandwidth can include a maximum rate of transfer across a given path.

[3827] In these or other examples, at least one of data bandwidth or power bandwidth through the dongles **14.3-116**, **14.3-118** can be reduced when the thermal output achieves a threshold thermal output (e.g., a predetermined temperature reading). For instance, a micro controller (discussed below) can throttle bandwidth through at least one of the dongles **14.3-116**, **14.3-118** in response to a temperature sensor in the removable straps **14.3-110**, **14.3-112** indicating a temperature in excess of a certain temperature, such as one hundred degrees Fahrenheit.

[3828] Any of the features, components, parts, including the arrangements and configurations thereof shown in FIGS. **14.3-1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **14.3-1**.

[3829] FIGS. **14.3-2** shows a perspective view of an example of a wearable electronic device **14.3-200** that includes a display portion **14.3-202** in the form of an HMD with a housing **14.3-204**. The display portion **14.3-202** can be secured to the head of a user via a modular securement assembly **14.3-208** that can include a first removable strap **14.3-210**, a second removable strap **14.3-212**, and a retention band **14.3-214**. The first removable strap **14.3-210** can be removably connected to the display portion **14.3-202**, or the housing **14.3-204** thereof, at a first end **14.3-216** of the first removable strap **14.3-210**. A second end **14.3-218** of the first removable strap **14.3-210** can be removably connected to a first end **14.3-220** of the retention band **14.3-214**. Likewise, the second removable strap **14.3-212** can be removably connected to the display portion **14.3-202**, or the housing **14.3-204** thereof, at a first end **14.3-222** of the second removable strap **14.3-212**. A second end **14.3-224** of the second removable strap **14.3-212** can be removably connected to a second end **14.3-226** of the retention band **14.3-214**.

[3830] FIGS. **14.3-3** shows an exploded perspective view of an example of a wearable electronic device **14.3-300**, similar to the device **14.3-200** shown in FIGS. **14.3-2**. The device **14.3-300** of FIGS. **14.3-3** includes a display portion **14.3-302** in the form of an HMD with a housing **14.3-304**. The display portion **14.3-302** can be secured to the head of a user via a modular securement assembly **14.3-308** that can include a first removable strap **14.3-310**, a second removable strap **14.3-312**, and a retention band **14.3-314**. The first removable strap **14.3-310** can be removably connected to the display portion **14.3-302**, or the housing **14.3-304** thereof, at a first end **14.3-316**

of the first removable strap **14.3-310**. A second end **14.3-318** of the first removable strap **14.3-310** can be removably connected to a first end **14.3-320** of the retention band **14.3-314**. Likewise, the second removable strap **14.3-312** can be removably connected to the display portion **14.3-302**, or the housing **14.3-304** thereof, at a first end **14.3-322** of the second removable strap **14.3-312**. A second end **14.3-324** of the second removable strap **14.3-312** can be removably connected to a second end **14.3-326** of the retention band **14.3-314**.

[3831] The example of the device **14.3-300** in FIGS. **14.3-3** is shown in an exploded view to illustrate the modularity of the securement assembly **14.3-308**, where each of the first and second removable straps **14.3-310**, **14.3-312** can be individually removed or replaced and removably connected with the retention band **14.3-314** and display portion **14.3-302** or other display component or other output component such as a speaker component or module.

[3832] Any of the features, components, parts, including the arrangements and configurations thereof shown in FIGS. **14.3-2** and **14.3-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **14.3-2** and **14.3-3**. For example, at least one of the dongles **14.3-116**, **14.3-118** shown in FIGS. **14.3-1** above can be correspondingly implemented with the removable straps **14.3-210**, **14.3-212** and/or the removable straps **14.3-310**, **14.3-312** of FIGS. **14.3-2** and **14.3-3**.

[3833] FIGS. **14.3-4** illustrates a side profile view of an example removable strap of an HMD system in accordance with one or more examples of the present disclosure. In particular, FIGS. **14.3-4** illustrates a removable strap **14.3-412** that includes an electronics pod **14.3-400**, a thermal spreader **14.3-402**, and a thermal spreader **14.3-404**. Although a single removable strap is shown, it will be appreciated that both removable straps of an HMD system of the present disclosure can include the same or similar structure shown and described in relation to FIGS. **14.3-4**.

[3834] As used herein, the term “electronics pod” refers to a subassembly, an enclosure, or a shell dedicated for housing certain electronics. In certain implementations, electrical components of the electronics pod **14.3-400** are communicatively coupled to the HMD **14.3-102** (e.g., via wire(s) **14.3-534** shown in FIGS. **14.3-5**). Additionally or alternatively, the electronics pod **14.3-400** can be coupled to one or more dongles (e.g., the dongles **14.3-116**, **14.3-118** discussed above).

[3835] Furthermore, the electronics pod **14.3-400** can generate thermal energy or heat as a result of operation. In some examples, it is desirable to dissipate this thermal energy or heat away from a user (e.g., towards an ambient environment and away from a user's head). In some examples, it is also desirable to spread this thermal energy or heat away from the electronics pod **14.3-400** (e.g., by utilizing the thermal spreaders **14.3-402**, **14.3-404**). In particular examples, the thermal spreader **14.3-404** dissipates thermal energy from inside the electronics pod **14.3-400**. Optionally, the thermal spreader **14.3-402** also dissipates thermal energy from inside electronics pod **14.3-400**.

[3836] As will be shown below, a thermal spreaders **14.3-402**, **14.3-404** are in thermal communication with the electronics pod **14.3-400**. For example, the thermal spreaders **14.3-402**, **14.3-404** contact conductive portions of the electronics pod **14.3-400**. In particular examples, the thermal spreaders **14.3-402**, **14.3-404** are positioned beyond the electronics pod **14.3-400**. In these or other examples, the term “beyond” refers to an exterior or outside positioning relative to a boundary/perimeter. For instance, the thermal spreaders **14.3-402**, **14.3-404** are at least partially positioned beyond the electronics pod **14.3-400** (e.g., outside of the exterior shell of the electronics pod **14.3-400**).

[3837] As used herein, the term “thermal spreader” refers to a material that transfers or distributes heat from a hotter source to a colder source. In some examples, a thermal spreader is a thermally conductive element that disperses or spreads out thermal energy from the electronics pod **14.3-400**. To illustrate, a thermal spreader can include conductive wires, foil, strips, sheets, layers, contacts,

etc. Thermal spreaders can also include a variety of materials. For example, thermal spreaders can include copper, graphite, diamond, silver, gold, silken carbide, aluminum, tungsten, zinc, carbon fiber (e.g., pitched carbon fiber), etc. In the example of pitched carbon fiber, it will be appreciated that the fibers can be oriented in one or more predetermined directions for directional heat transfer. [3838] FIGS. **14.3-5** illustrates a top cross-sectional profile view of an example electronics pod in accordance with one or more examples of the present disclosure. As shown, the electronics pod **14.3-400** includes a thermal shell **14.3-500** and a cover **14.3-502**. Together, the thermal shell **14.3-500** and the cover **14.3-502** define an internal volume **14.3-504**. Multiple electrical components can be housed within the internal volume **14.3-504**. These electrical components can also generate thermal energy during operation.

[3839] In addition, the thermal shell **14.3-500** and the cover **14.3-502** can form an at least partially enclosed bowl or U-shaped cross-section. In particular, the example thermal shell **14.3-500** includes a plateaued surface. In some examples the plateaued surface can be oriented towards a user's head. In this manner, audio can be transmitted from the electronics pod **14.3-400** in a directional fashion towards a user's ear. In alternative examples, the plateaued surface of the electronics pod **14.3-400** can contact a user skin (e.g., for vibrational sound via bone conduction). [3840] In certain examples, the cover **14.3-502** includes one or more of a variety of materials. In some examples, the cover **14.3-502** includes at least one of a foam portion or an aluminum portion. In particular examples, the cover **14.3-502** includes a stainless steel portion. Further in some examples, the cover **14.3-502** forms a complete enclosure over the internal volume **14.3-504**. In other examples, the cover **14.3-502** forms a partial enclosure over the internal volume **14.3-504**. [3841] In these or other examples, a bottom surface of the cover **14.3-502** can attach to a removable strap. Additionally or alternatively, the cover **14.3-502** can be integrated into a removable strap itself. Further, in some examples, the thermal shell **14.3-500** includes a variety of layers of material. In some examples, the thermal shell **14.3-500** includes a clad (e.g., an overmolded clad). The clad can include a first metal layer **14.3-508**, a thermal interface material **14.3-510**, and a second metal layer **14.3-512**.

[3842] The metal layers **14.3-508**, **14.3-512** can include one or more of a variety of metal materials. In some examples, the metal layers **14.3-508**, **14.3-512** include stainless steel. In other examples, the metal layers **14.3-508**, **14.3-512** include chrome, titanium, etc. It will be appreciated, however, that the metal layers **14.3-508**, **14.3-512** need not be metal. For instance, the layers **14.3-508**, **14.3-512** can be a variety of insulative materials, such as foam, plastic, rubber, ceramic, etc. It will also be appreciated that the layers **14.3-508**, **14.3-512** can be different materials, as may be desired.

[3843] In some examples, the thermal interface material **14.3-510** can include a variety of thermally conductive materials. As used herein, the term “thermal interface material” refers to a material inserted between components or surfaces to enhance a thermal coupling therebetween and/or provide a thermal pathway. Examples of a thermal interface material include a thermal epoxy, thermal paste, thermal adhesive, thermal gap filler, thermally conductive pad, thermal tape, phase-change materials, or thermally conductive coatings (e.g., metallic coatings, diamond coatings, etc.). A thermal interface material can thus have high thermal conductivity. In specific examples, the thermal interface material **14.3-510** includes one or more copper layers. In one or more examples, the thermal interface material **14.3-510** can move, mold, or conform to the shape of the electronics pod **14.3-400**. Moreover, as will be discussed below, the thermal interface material **14.3-510** can transfer or dissipate thermal energy out of the internal volume **14.3-504**.

[3844] An overmold **14.3-506** can be positioned over the thermal shell **14.3-500**. The overmold **14.3-506** can include a variety of materials. To illustrate, the overmold **14.3-506** can include softer materials, textured materials, fabric materials, etc. In particular examples, the overmold **14.3-506** includes an elastomer material. For instance, the overmold **14.3-506** includes a fluorocarbon rubber (e.g., Viton® rubber). In these or other examples, the overmold **14.3-506** can conform to a shape of the thermal shell **14.3-500**. Additionally or alternatively, the overmold **14.3-506** can provide

rigidity or support to the thermal shell **14.3-500**.

[3845] As mentioned, the electronics pod **400** can house a variety of electrical components. Shown in FIGS. **14.3-5**, the electrical components can include a printed circuit board (PCB) **14.3-514**, a battery **14.3-516**, a processor such as a microcontroller **14.3-518**, a system-on-chip (SoC) **14.3-520**, and a speaker **14.3-522**.

[3846] The terms “PCB” or “printed circuit board” refer to a logic assembly that includes electronic components. The PCB **14.3-514** includes electrical connections and circuitry for mounting various components, including the SoC **14.3-520**. The PCB **14.3-514** can also relay power to mounted electrical components from a power supply (e.g., a battery, not separately illustrated). In certain examples, the PCB **14.3-514** is a main logic board. The PCB **14.3-514** can be a rigid board (e.g., composed of glass-epoxy compounds). In some examples, the PCB **14.3-514** is a multi-layer PCB (e.g., a laminated sandwich structure of conductive and insulating layers). In some examples, the PCB **14.3-514** is flexible (e.g., with flexible circuitry made with polyimide). In certain examples, the PCB **14.3-514** includes stiffeners added via lamination or pressure sensitive adhesive.

[3847] The battery **14.3-516** can include a myriad of different types of batteries or power sources. Indeed, the battery **14.3-516** can include one or more electrochemical cells with external connections for powering electrical devices or electrical components. For example, in some implementations, the battery **14.3-516** includes a lithium ion battery, alkaline battery, etc. In a specific example, the battery **14.3-516** includes a rechargeable battery. In other words, the battery **14.3-516** can be dispensable or rechargeable, as may be desired.

[3848] The microcontroller **14.3-518** can direct operation of one or more other components (e.g., within the electronics pod **14.3-400** the display **14.3-106**, etc.). As used herein, the term “microcontroller” refers to a device capable of generating an electrical (or digital) signal responsive to sensor data from a sensor. In one or more embodiments, a microcontroller includes any of a variety of processors (e.g., a system-on-chip, integrated circuit, driver, application processor, crossover processor, etc.). In some embodiments, a microcontroller further includes one or more memory devices (e.g., individual nonvolatile memory, processor-embedded nonvolatile memory, random access memory, memory integrated circuits, DRAM chips, stacked memory modules, storage devices, memory partitions, etc.). In particular embodiments, a microcontroller further includes one or more of input/output ports, counters, timers, etc. It will be appreciated that such a microcontroller can be mounted on a printed circuit board (e.g., a rigid circuit board or a flexible printed circuit).

[3849] The SoC **14.3-520** can include an integrated circuit that integrates a combination of electrical components for operation of the HMD system. The terms “SoC,” and “system-on-chip” refer to an electronic chip that generates thermal energy during operation. One or more of the SoC **14.3-520**, the microcontroller **14.3-518**, or the battery **14.3-516** can be a “heat source,” which term generally refers to component(s) that can generate thermal energy. In some examples, the SoC **14.3-520** can include a microchip to generate (e.g., drive) visualizations presented at the display **14.3-106** (shown in FIGS. **14.3-1**). It will be appreciated that the thermal energy or heat generated by the SoC **14.3-520** can become greater over a duration of use. Similarly, the thermal energy or heat generated by the SoC **14.3-520** can become greater when executing more data-intensive operations (e.g., for more complex or high fidelity visualizations).

[3850] The speaker **14.3-522** can include one or more audio-generating components. Example components of the speaker **14.3-522** include a transducer, magnet, coil, diaphragm/driver, etc.

[3851] Myriad other components in the electronics pod **14.3-400** can be included (albeit not expressly shown in FIGS. **14.3-5**). For example, the electronics pod **14.3-400** can include a heat sink, microphone, antenna, one or more sensors (e.g., temperature sensors, pressure sensor, accelerometer, magnetometer, etc.), or other suitable electronic elements.

[3852] In one or more examples, components of the electronics pod **14.3-400** can be electrically

coupled to a power supply positioned outside of the electronics pod **14.3-400**. For example, wire(s) **14.3-534** can extend through a wire opening **14.3-536** defined by the cover **14.3-502**. The wire(s) **14.3-534** can connect to the PCB **14.3-514** on one end, and the wire(s) **14.3-534** can connect to a power supply (not shown) on the other end. In this manner, the wire(s) **14.3-534** can provide power to each component electrically coupled to the PCB **14.3-514**.

[3853] Components of the electronics pod **14.3-400** can also be thermally coupled to at least one of the ambient environment outside the electronics pod **14.3-400** or various thermal elements. To illustrate, the electronics pod **14.3-400** comprises a thermal conduit. As used herein, the term “thermal conduit” refers to a thermal path for heat flow. A thermal conduit may be conceptualized as linear or direct (e.g., along wire paths). However, a thermal conduit is not limited to such a flow path. Indeed, thermal paths can be linear or curved (e.g., non-linear), as a function of thermal properties for a given medium (whether solid, liquid, or gaseous). Further, thermal conduits are not limited to two-dimensional space. It will be appreciated that thermal conduits of the present disclosure include three-dimensional thermal paths through a given volume. In at least some examples, however, a thermal conduit of the present disclosure includes at least one of a thermal connector **14.3-524** or a thermal connector **14.3-526**, both of which can be composed of one or more wires (e.g., copper wires) for conductive heat transfer.

[3854] In some examples, the thermal conduit includes a thermal connector **14.3-524**. The thermal connector **14.3-524** is positioned inside the internal volume **14.3-504** and connects the PCB **14.3-514** and each associated component to the thermal interface material **14.3-510** at a connection point **14.3-528**. Alternatively, the thermal connector **14.3-524** connects a specific component of the electronics pod **14.3-400** (e.g., the SoC **14.3-520**) to the thermal interface material **14.3-510**.

[3855] Thermal energy can be conductively transferred from components of the electronics pod **14.3-400** through the thermal connector **14.3-524**. In turn, the thermal energy can be spread or dissipated along the thermal interface material **14.3-510**. In at least some examples, a thermal conduit includes a thermal connector **14.3-526**. The thermal connector **14.3-526** is attached to the thermal interface material **14.3-510** at the connection point **14.3-528**. The thermal connector **14.3-526** then extends between clad layers of the thermal shell **14.3-500** (or over the thermal shell **14.3-500** underneath the overmold **14.3-506** as shown). The thermal connector **14.3-526** further extends to a connection **14.3-538** beyond the electronics pod **14.3-400**, where the thermal connector **14.3-526** and the thermal interface material **14.3-510** thermally couple to the thermal spreader **14.3-404**. By coupling the thermal connector **14.3-526** and the thermal interface material **14.3-510** to the thermal spreader **14.3-404**, thermal energy from the electronics pod **14.3-400** can be even further dissipated or spread along the removable strap. In specific examples, the thermal spreader **14.3-404** directs this thermal energy towards a second end **14.3-424** that opposes a first end **14.3-422** of the removable strap **14.3-412**. In doing so, the thermal energy is directed away from a user's temple.

[3856] However, as indicated via dashed lines, the removable strap **14.3-412** can include the thermal spreader **14.3-402** and associated heat-dissipation elements. The thermal spreader **14.3-402**, like the thermal spreader **14.3-404**, can connect to the thermal interface material **14.3-510** and a thermal connector, in a similar manner as just discussed above. In this way, thermal energy from the electronics pod **14.3-400** can be directed along the thermal spreader **14.3-402** towards the first end **14.3-422** of the removable strap **14.3-412**, as may be desired (e.g., to decrease a thermal load through the thermal connector **14.3-524**, the thermal connector **14.3-526**, and/or the thermal spreader **14.3-404**).

[3857] To accommodate these and/or other thermal couplings of the present disclosure, the thermal shell **14.3-500** includes certain etched portions that expose segments of the thermal interface material **14.3-510**. Indeed, the thermal shell **14.3-500** includes an etched opening **14.3-530** and an etched opening **14.3-532**. The etched openings **14.3-530**, **14.3-532** can be formed in a variety of ways. In some examples, chemical etching may be used to form the etched openings **14.3-530**, **14.3-532**. In other examples, laser etching may be used to form the etched openings **14.3-530**, **14.3-**

532. In such examples, the etched opening **14.3-530** is defined by a removed portion of the metal layer **14.3-512**. Similarly, the etched opening **14.3-532** is defined by a removed portion of the metal layer **14.3-508**. Through the etched openings **14.3-530**, **14.3-532**, the thermal interface material **14.3-510** is exposed for one or more conductive connections (e.g., for conductive heat transfer). [3858] Additionally or alternatively to the thermal connector **14.3-524**, heat can be transferred from electrical components of the electronics pod **14.3-400** to the thermal interface material **14.3-510** via forced convection or non-force convection. For example, thermal energy can convectively move through the etched opening **14.3-530** and impinge upon the thermal interface material **14.3-510**. As used herein, the terms “conduction” or “natural conduction” refer to the heat transfer via conductively connected components. By contrast, the term “non-force convection” refers to free or natural convection, where air motion is caused by natural, buoyancy forces that result from the density variations due to variations of thermal temperature in the air. Non-forced convection differs from forced convection, where a fluid (e.g., air) is forced to flow by an internal source such as fans, by stirring, or pumps to create an artificially induced convection current.

[3859] FIGS. **14.3-6** shows a top view of a wearable electronic device **14.3-600** being worn on the head **14.3-101** of a user. The wearable electronic device **14.3-600** is the same as or similar to the wearable electronic device **100** discussed above in relation to FIGS. **14.3-1**. Differently however, the wearable electronic device **14.3-600** includes a dongle **14.3-618**. As shown, the dongle **14.3-618** is a split dongle. In particular, the dongle **14.3-618** includes a first portion **14.3-619** connecting the removable strap **14.3-112** and the external DoD **14.3-122**. In addition, the dongle **14.3-618** includes a second portion **14.3-621** connecting the removable strap **14.3-112** to at least one of the power supply **14.3-120** or the removable strap **14.3-110**.

[3860] Advantageously, the dongle **14.3-618** can provide a variety of use-case applications. For instance, the power supply **14.3-120** can additionally provide power to the removable strap **14.3-112** via the second portion **14.3-621**. In another example, the external DoD **14.3-122** can provide data to both of the removable straps **14.3-110**, **14.3-112** via the dongle **14.3-618**. In at least some examples, the dongle **14.3-618** can improve bandwidth through the wearable electronic device **14.3-600** (e.g., by circumventing the HMD **14.3-102** for transmission of data around the wearable electronic device **14.3-600**). By transmitting data to the removable strap **14.3-110** through the dongle **14.3-618**, and in particular through the second portion **14.3-621**, the wearable electronic device **14.3-600** can also include reduced thermal loads in the bypassed components (e.g., through the removable strap **14.3-112** and the HMD **14.3-102**).

[3861] FIGS. **14.3-7** and **14.3-8** illustrate schematic views of example cable management mechanisms for HMD systems in accordance with one or more examples of the present disclosure. In particular, FIGS. **14.3-7** shows a dongle **14.3-718** connected to the removable strap **14.3-112**. The dongle **14.3-718** can be the same as or similar to the dongles **14.3-118**, **14.3-618** discussed above. The dongle **14.3-718** also includes slack **14.3-702** (e.g., a cable portion for routing to the external DoD **14.3-122**). To help prevent unwanted positioning or tangling, the retention band **14.3-114** can include a cable management mechanism **14.3-700**.

[3862] The cable management mechanism **14.3-700** can route, position, tension, or otherwise manage the slack **14.3-702** for the dongle **14.3-718**. In some examples, the cable management mechanism **14.3-700** includes an external ring, clip, button snap, Velcro® strip, tie, etc. attached to the retention band **14.3-114**. In other examples, the cable management mechanism **14.3-700** includes an aperture, slot, or recess within the retention band **14.3-114**.

[3863] Similarly, FIGS. **14.3-8** shows a dongle **14.3-818** connected to the removable strap **14.3-112**. The dongle **14.3-818** can be the same as or similar to the dongles **14.3-118**, **14.3-618** discussed above. Like the dongle **14.3-718**, the dongle **14.3-818** includes slack **14.3-802** (e.g., for routing to the external DoD **14.3-122**). A cable management mechanism **14.3-800** is configured to efficiently handle the slack **14.3-802**, including one or more of a length or tension of the slack **14.3-802**.

[3864] In some examples, the cable management mechanism **14.3-800** includes a retractable spool.

Via the retractable spool, the slack **14.3-802** can be pulled out or lengthened with manual interaction. In some examples, the slack **14.3-802** can be pulled out to a desired length and locked in place such that the retractable spool maintains an amount or tension of the slack **14.3-802**. In certain examples, the cable management mechanism **14.3-800** can automatically wind the slack **14.3-802**. In other examples, the cable management mechanism **14.3-800** winds the slack **14.3-802** in response to user interaction with at least one of the slack **14.3-802** or the cable management mechanism **14.3-800**. For instance, the cable management mechanism **14.3-800** includes a button release to initiate winding of the slack **14.3-802**. In other instances, the cable management mechanism **14.3-800** can wind the slack **14.3-802** in response to a pull (e.g., a directional tug) of the slack **14.3-802**.

14.4: Devices with Detachable Headbands

[3865] Head-mounted devices include head-mounted support structures that allow the devices to be worn on the heads of users. The head-mounted support structures may include device housings that enclose components such as displays. The displays may be used for presenting a user with visual content. The head-mounted support structures for a head-mounted device may also include headbands and other structures that help hold a device housing on the face of a user. The headband of a head-mounted device may be removable. This allows users to swap different headbands into use to accommodate different head sizes and/or to update the style of headband being used.

[3866] FIGS. **14.4-1** is a side view of a head-mounted electronic device with a detachable headband. As shown in FIGS. **14.4-1**, head-mounted device **14.4-10** may include head-mounted housing **14.4-12** (sometimes referred to as a main housing, main housing unit, head-mounted support structure, main housing portion, etc.). Housing **14.4-12** may have walls or other structures that separate an interior housing region from an exterior region surrounding housing **14.4-12**. For example, housing **14.4-12** may have walls formed from polymer, glass, metal, and/or other materials. Electrical and optical components may be mounted in housing **14.4-12**. These components may include components such as integrated circuits, sensors, control circuitry, input-output devices, etc.

[3867] To present a user with images for viewing from eye boxes (e.g., eye boxes in which the user's eyes are located when device **14.4-10** is being worn on the users' head), device **14.4-10** may include displays and lenses. These components may be mounted in optical modules **14.4-20** that face towards rear R of device **14.4-10** or may be mounted in other supporting structures in housing **14.4-12** to form respective left and right optical systems. There may be, for example, a left display for presenting an image through a left lens to a user's left eye in a left eye box and a right display for presenting an image to a user's right eye in a right eye box.

[3868] If desired, housing **14.4-12** may have forward-facing components such as cameras and other sensors on front F for gathering sensor measurements and other input and may have a soft cushion on opposing rear R. Rear R may have openings that allow the user to view images from left and right optical modules **14.4-20** (e.g., when rear R is resting on the front of the user's head).

[3869] Device **14.4-10** may have a strap such as headband **14.4-26** and, if desired, may have other structures (e.g., an optional over-the-head strap) to help hold housing **14.4-12** on the user's head. Headband **14.4-26** may have a fixed length or may be adjustable. Headband **14.4-26** may have first and second ends coupled, respectively, to the left and right sides of housing **14.4-12**. In the example of FIGS. **14.4-1**, coupling members **14.4-24**, which serve as extensions of housing **14.4-12**, are provided on the left and right sides of housing **14.4-12**. Members **14.4-24** may be formed from rigid materials such as rigid polymer and/or other materials and may contain sensors, buttons, speakers, and other electrical components. Hinges and/or other mechanisms may be used to couple members **14.4-24** to housing **14.4-12** or members **14.4-24** may be formed as integral portions of a main housing unit. The ends of headband **14.4-26** may have coupling mechanisms such as openings configured to receive posts **14.4-30** (pins) or other protrusions on members **14.4-24** or other housing structures. In some examples, these posts face inwardly towards the user's head and are not

visible to people in the vicinity of device **14.4-10** when device **14.4-10** is being worn by the user. Releasable latch mechanisms can be used to help secure the ends of headband **14.4-26** to member **14.4-24**. For example, a first detachable latch may be used to removably couple the left end of headband **14.4-26** to a left post in a left member **14.4-24** on a left side of housing **14.4-12** and a second detachable latch may be used to removably couple the right end of headband **14.4-26** to a right post in a right member **14.4-24** on a right side of housing **14.4-12**. If desired, a user may flip the headband **14.4-26** over so that the first detachable latch removable couples the end of headband **14.4-26** that was previously coupled to the left post to the right post and so that the second detachable latch removably couples the end of headband **14.4-26** that was previously coupled to the right post to the left post (e.g., the user may flip the left and right sides of the band without flipping the band inside out). A user may open and close the latches when housing **14.4-12** is being worn or when housing **14.4-12** is not being worn.

[3870] The use of latch-based coupling mechanisms in device **14.4-10** may help allow a user to removably attach headband **14.4-26** to members **14.4-24** and thereby removably attach headband **14.4-26** to housing **14.4-12**. Members **14.4-24** may have elongated shapes of the type shown in FIGS. **14.4-1** and/or other suitable shapes and may sometimes be referred to as rigid straps, rigid coupling members, power straps head-mounted device housing structures, elongated head-mounted device housing members, elongated housing structures, elongated housing members, or head-mounted device housing members (as examples).

[3871] Headband **14.4-26** may have soft flexible portions and/or rigid portions. As an example, a central portion of headband **14.4-26** may be formed from stretchable fabric. Left and right end portions of headband **14.4-26** may be coupled to opposing ends of this central portion. The left and right end portions may, as an example, have stiffening structures (e.g., the left and right end portions may be stiffer than the central stretchable portion). Other types of configuration may be used for headband **14.4-26**, if desired (e.g., arrangements with adjustable tensioning cables, etc.).

[3872] FIGS. **14.4-2** is a diagram of an end portion of a headband **14.4-26** and a corresponding end portion of a member **14.4-24**. Headband **14.4-26** and housing structures such as members **14.4-24** may sometime be collectively referred to as a forming a head-mounted device headband system (headband system **14.4-22**). As shown in the example of FIGS. **14.4-2**, member **14.4-24** may have a protruding post such as post **14.4-30** (e.g., a post that protrudes out of member **14.4-24** towards a user head while device **14.4-10** is being worn by a user). Headband **14.4-26** may have a corresponding opening **14.4-32** that is configured to receive post **14.4-30**.

[3873] Opening **14.4-32** may be a through-hole opening with a shape that matches the outline of post **14.4-30**. In the present example, post **14.4-30** and opening **14.4-32** have elongated shapes when viewed end-on (e.g., rectangular shapes with rounded corners). These elongated shapes may help resist rotational motion between longitudinal axis **14.4-34** of headband **14.4-26** and longitudinal axis **14.4-36** of member **14.4-24**. This helps prevent headband **14.4-26** from slipping up or down along the rear surface of a user's head during use. In general, post **14.4-30** and/or mating opening **14.4-32** may have any suitable shapes (e.g., the shape of post **14.4-30** and/or opening **14.4-32** may be circular, oval, rectangular, triangular, may be a shape with curved edges and/or straight edges, may be a shape with drafted edges to help with alignment and/or insertion, etc.). The use of rectangular shapes with rounded corners and/or other shapes that are elongated (e.g., along respective longitudinal axes **14.4-40** and **14.4-38** of FIGS. **14.4-2**, respectively), is merely for illustrative purposes.

[3874] FIGS. **14.4-3** is a cross-sectional side view of system **14.4-22**. As shown in FIGS. **14.4-3**, member **14.4-24** may have main portion **14.4-24M**. Main portion **14.4-24M** may serve as a support for post **14.4-30** and may be formed from rigid polymer, metal, fabric, carbon-fiber composite materials, and/or other materials. Post **14.4-30** may be attached to main portion **14.4-24M** and may protrude inwardly (or, in some examples, outwardly) from portion **14.4-24M**. Post **14.4-30** may be formed from metal, rigid polymer, other materials, and/or combinations of these materials. Post

14.4-30 may have main portion **14.4-30C**, which protrudes away from the surface of portion **14.4-24M**. Post **14.4-30** may also have a peripheral protruding portion such as peripheral protruding portion **14.4-30P**. Portion **14.4-30P** may, as an example, run along the periphery of post **14.4-30** and may protrude radially outward from main portion **14.4-30C**. In the example of FIGS. **14.4-3**, protruding portion **14.4-30P** has bevels to facilitate entry of post **14.4-30** into opening **14.4-32** of headband **14.4-26** in direction **14.4-54** and to facilitate removal of post **14.4-30** from opening **14.4-32** in the opposite direction when detaching headband **14.4-26** from member **14.4-24**.

[3875] Headband **14.4-26** has a main strap portion **14.4-44**. Strap portion **14.4-44**, which may sometimes be referred to as a strap or headband member, may have internal stiffening members, external fabric coverings and or other covering layers, strips of strengthening fabric, stretchable fabric portions (e.g., stretchable knit fabric), cosmetic coverings, and/or other headband structures. Through-hole opening **14.4-32** may be formed by cutting or otherwise forming an opening in portion **14.4-44**. The periphery of opening **14.4-32** may be strengthened using a mating pair of ring members **14.4-40**. Members **14.4-40**, which may sometimes be referred to as a cap and socket, may have ring shapes and may capture portions of strap **14.4-44**. Members **14.4-40** may be attached to each other using laser welding and/or other attachment mechanisms. Adhesive may optionally be used to help secure members **14.4-40** to strap **14.4-44**.

[3876] When members **14.4-50** are attached to each other, a ring shaped recess such as recess **14.4-52** is formed. Spring **14.4-42** (e.g., a spring formed from metal, foam, stretchy rubber gasket material or other elastomeric material, and/or other spring structures) may have a ring shape surrounding opening **14.4-32** and may be received within recess **14.4-52**. Gap **14.4-46** may be formed on one side of spring **14.4-42**. As shown in the top view of spring **14.4-42** of FIGS. **14.4-4**, the presence of gap **14.4-46** allows spring **14.4-42** to expand outwardly in direction **14.4-56** when post **14.4-30** is inserted within opening **14.4-32** and the bevels on protruding portion **14.4-30P** press spring **14.4-42** radially outward. Once post **14.4-30** has been inserted within opening **14.4-32** sufficiently for protruding portion **14.4-30P** to have passed spring **14.4-42**, spring **14.4-42** may contract inwardly against main portion **14.4-30C**, thereby retaining post **14.4-30** within opening **14.4-32** and securing headband **14.4-26** to member **14.4-24**. When it is desired to remove headband **14.4-26**, headband **14.4-26** may be pulled away from member **14.4-24**. During headband removal, the bevels on portion **14.4-30P** press against spring **14.4-42** and expand spring **14.4-42** radially, thereby allowing post **14.4-30** to be removed from opening **14.4-32**.

[3877] The illustrative spring-based headband retention arrangement of FIGS. **14.4-3** and **14.4-4** allows headband **14.4-26** to be attached and detached from member **14.4-24**. If desired a latch-based mechanism maybe used to secure and release headband **14.4-26**. This type of arrangement is shown in FIGS. **14.4-5**. As shown in the example of FIGS. **14.4-5**, system **14.4-22** may include a latching mechanism such as latch **14.4-62**. Latch **14.4-62**, which may sometimes be referred to as a latch mechanism or latch structures, may be opened and closed using magnets, springs, sliding members, toggling members, rotating members such as knobs, buttons, and/or other latch structures that may be manipulated by a user (e.g., a user's fingers). In the example of FIGS. **14.4-5**, the latch of headband **14.4-26** has a movable latch member that engages post **14.4-30** when post **14.4-30** is within opening **14.4-32** and has an associated release mechanism such as release tab **14.4-62T**. Tab **14.4-62T**, which may be formed from a flexible strip of material (e.g., fabric, polymer, and/or other material) may be pulled in direction **14.4-64** by a user (e.g., when a user grasps tab **14.4-62T** between the user's fingers), thereby moving the movable latch member out of engagement with post **14.4-30**. This releases post **14.4-30** and allows post **14.4-30** to be removed from opening **14.4-32**. Magnets, spring structures, and/or other biasing structures may be used in closing the latch. In some examples, member **14.4-24** and headband **14.4-26** may have magnets that facilitate attachment of headband **14.4-26** and member **14.4-24**. As shown in FIGS. **14.4-5**, for example, post **14.4-30** may have one or more magnets such as magnet **14.4-60**. Magnet **14.4-60** may be used to attract and/or repel corresponding magnets in strap **14.4-44**, which can assist in attaching headband

14.4-26 to member **14.4-24** and/or can assist in closing latch **14.4-62**.

[3878] FIGS. **14.4-6** is a side view of system **14.4-22** in a configuration in which latch **14.4-62** has a release tab. Tab **14.4-62T** may be attached to slidable (movable) latch member **14.4-62M**. Latch member **14.4-62M** may be moved laterally towards post **14.4-30** in response to magnetic force, spring force, or other biasing force from a latch member biasing mechanism so that the tip of the member **14.4-62M** protrudes under a lip portion (portion **14.4-30P**) of post **14.4-30**. In this way, post **14.4-30** is retained within opening **14.4-32** until latch **14.4-62** is released (e.g., by pulling on tab **14.4-62T** in direction **14.4-64**).

[3879] FIGS. **14.4-7** is a top view of system **14.4-22** in an example in which post **14.4-30** of member **14.4-24** and headband **14.4-26** have magnets. Post **14.4-30** has magnet **14.4-60**. Strap **14.4-44** of headband **14.4-26** has magnets **14.4-66**, **14.4-68**, and **14.4-70**. When post **14.4-30** is inserted within opening **14.4-32** of headband **14.4-26**, magnet **14.4-60** lies between magnets **14.4-66** and **14.4-68**. In this configuration, the south pole of magnet **14.4-60** attracts the north pole of magnet **14.4-68**. Magnet **14.4-68** is attached to slidable latch member **14.4-62M**. The magnetic attraction between magnet **14.4-60** and magnet **14.4-68** pulls latch member **14.4-62M** towards post **14.4-30** until the tip of latch member **14.4-62M** is received under protruding portion (lip) **14.4-30P** of post **14.4-30** (as shown in FIGS. **14.4-6**). In this position, latch **14.4-62** is closed and post **14.4-30** is retained within opening **14.4-32** (e.g., headband **14.4-26** is attached to member **14.4-24**).

[3880] When post **14.4-30** is not inserted in opening **14.4-32**, the south poles of magnets **14.4-70** attract the north poles of magnet **14.4-68** and vice versa, so that magnet **14.4-68** aligns with magnet **14.4-70** when post **14.4-30** is not present. Magnets **14.4-70** are attached to strap **14.4-44**, whereas magnet **14.4-68** is attached to slidable latch member **14.4-62M**. Due to the magnetic attraction between magnets **14.4-70** and magnet **14.4-68**, **14.4-70**, member **14.4-62M** is moved to open latch position **14.4-62M'**. This ensures that the tip of member **14.4-62M** that faces opening **14.4-32** will not protrude into opening **14.4-32** in the absence of post **14.4-30** and will therefore not be visible within opening **14.4-32**. The magnetic retraction of latch member **14.4-62M** thereby helps enhance the visual appearance of headband **14.4-26** when headband **14.4-26** is not attached to member **14.4-24** and latch **14.4-62** is open.

[3881] When a user desires to attach headband **14.4-26** to member **14.4-24**, the user places opening **14.4-32** of headband **14.4-26** adjacent to post **14.4-30**. As opening **14.4-32** moves towards post **14.4-30**, the south pole of magnet **14.4-66** attracts the north pole of magnet **14.4-60** while the north pole of magnet **14.4-68** attracts the south pole of magnet **14.4-60** in an almost perfectly symmetrical fashion, so that the force exhibited on post **14.4-30** feels balanced. This tends to align post **14.4-30** with opening **14.4-32** and pull post **14.4-30** into opening **14.4-32**, thereby reducing the need for the user to accurately align post **14.4-30** with opening **14.4-32**.

[3882] The illustrative example of FIGS. **14.4-7** uses magnetic force for three different functions. Firstly, while headband **14.4-26** and member **14.4-24** are detached, magnetic force is used to automatically retract latch member **14.4-62M** into position **14.4-62M'**. This automatic magnetic latch opening mechanism helps maintain latch member **14.4-62M** out of opening **14.4-32** when not in use. Secondly, when a user is first attaching headband **14.4-26** to member **14.4-24**, magnetic attraction is used to guide post **14.4-30** into opening **14.4-32** without excessive user attention. Thirdly, after post **14.4-30** has been received within opening **14.4-32**, magnetic attraction is used to pull latch member **14.4-62M** towards post **14.4-30** under protruding (lip) portion **14.4-30P**, thereby magnetically closing latch **14.4-62**.

[3883] Other biasing mechanisms may be used in addition to or in the alternative to magnetic biasing arrangements. For example, FIGS. **14.4-8**, **14.4-9**, **14.4-10**, and **14.4-11** and their related description disclose spring-based biasing mechanisms that can be used in addition to or in the alternative to magnetic biasing arrangements.

[3884] As shown in FIGS. **14.4-8**, latch member **14.4-26M** may be biased using biasing member **14.4-80**. Biasing member **14.4-80** may be a spring (e.g., a compression spring formed from metal

or other springy material, etc.), may be a spring formed from elastomeric material (e.g., polymer foam), and/or may be formed from other spring structures that exhibit spring force. Member **14.4-80** of FIGS. **14.4-8** may be used, for example, to push away from strap **14.4-44** and thereby push latch member **14.4-62M** into engagement with post **14.4-30** when post **14.4-30** has been received within opening **14.4-32**. Compression springs or other expanding biasing members may be used to move any suitable movable structures associated with latch **14.4-62** (e.g., member **14.4-62M** and/or other movable members).

[3885] As shown in FIGS. **14.4-9**, biasing members **14.4-80** may be used in tension (e.g., members **14.4-80** may be tension springs). In the illustrative configuration of FIGS. **14.4-9**, members **14.4-80** are pulling member **14.4-62M** towards opening **14.4-32**. Tension springs or other tensioned biasing members may, in general, pull on any suitable latch structures in latch **14.4-62**.

[3886] FIGS. **14.4-10** and **14.4-11** illustrate the use of a spring formed from an elastomeric band. As shown in the top view of FIGS. **14.4-10**, for example, spring member **14.4-80** (e.g., an elastomeric band having a ring shape) may extend between a portion of headband **14.4-26** (e.g., strap **14.4-44**) and latch member **14.4-62M**. When member **14.4-62M** is moved away from opening **14.4-32**, member **14.4-80** is stretched. This creates an opposing restoring spring force that tends to pull member **14.4-62M** towards opening **14.4-32**, allowing member **14.4-62M** to engage post **14.4-30** in opening **14.4-32**. This type of elastomeric band spring biasing mechanism is further illustrated in the cross-sectional side view of system **14.4-22** of FIGS. **14.4-11**. As shown in FIGS. **14.4-11**, biasing member **14.4-80** (e.g., a spring formed from an elastomeric band) may have a ring shape that extends (on the left) through a portion of strap **14.4-44** and (on the right) through a grooved portion of latch member **14.4-62M**. Post **14.4-30** may have a protruding portion such as portion **14.4-30P** that creates a post recess (recess **14.4-82**) into which latch member **14.4-62M** can be received when latch **14.4-62** is closed. When it is desired to open latch **14.4-62**, a release tab or other release mechanism that is coupled to member **14.4-62M** may pull the tip of latch member **14.4-62M** out of recess **14.4-82**. This releases protruding portion **14.4-30P** from the tip of latch member **14.4-62M** and thereby unlocks latch **14.4-62** so that post **14.4-30** can be removed from opening **14.4-32**. At the same time, moving latch member **14.4-62M** away from opening **14.4-32** tensions member **14.4-80**. When it is desired to attach headband **14.4-26** to member **14.4-24**, this tension can be used to move member **14.4-62M** into recess **14.4-82**, so that member **14.4-62M** engages post **14.4-30**.

[3887] In the latch arrangement of FIGS. **14.4-11**, latch member **14.4-62M** has a cam surface such as angled surface **14.4-84**. This surface may interact with protruding portion **14.4-30P**. For example, as post **14.4-30** is being inserted into opening **14.4-32**, portion **14.4-30P** may bear against surface **14.4-84** and thereby force latch member **14.4-62M** away from post **14.4-30** to open latch **14.4-62**. Once portion **14.4-30P** has moved across all of surface **14.4-84**, latch member **14.4-62M** may move into recess **14.4-82** under the latch-closing biasing force of biasing member **14.4-80**.

[3888] FIGS. **14.4-12** shows how latch member **14.4-62M** may have both outwardly and inwardly facing cam surfaces **14.4-88**. Protruding portion **14.4-30P** may have corresponding outwardly and inwardly cam surfaces **14.4-86**. With this type of arrangement, the outwardly facing cam surface of protrusion **14.4-30P** may bear against the inwardly facing cam surface of latch member **14.4-62M** when post **14.4-30** is being inserted into opening **14.4-32**, thereby moving latch member **14.4-62M** away from post **14.4-30** to place latch **14.4-62** in its open state. After inserting post **14.4-30** in opening **14.4-32**, the biasing mechanism that is being used to close latch **14.4-62** (e.g., magnetic biasing, spring biasing, etc.) pushes latch member **14.4-62M** toward post **14.4-30** and engages protrusion **14.4-30P** to hold post **14.4-30** in place in opening **14.4-32**. When a user desires to remove headband **14.4-26** from member **14.4-24**, the user may pull headband **14.4-26** and member **14.4-24** apart, causing the inwardly facing cam surface **14.4-86** of protruding portion **14.4-30P** to bear against the outwardly facing cam surface **14.4-88** of latch member **14.4-62M** and thereby push latch member **14.4-62M** into its open position. Because a pulling force from a user is used to

disengage member **14.4-24** and headband **14.4-26** in systems such as system **14.4-22** of FIGS. **14.4-12**, these systems may sometimes be referred to as pull-to-release systems (e.g., latch **14.4-62** of FIGS. **14.4-12** may be referred to as a pull-to-release latch). Any suitable latch closing mechanism may be used with this type of latch (e.g., member **14.4-62M** may be biased towards post **14.4-30** using magnetic biasing, metal spring biasing, elastomeric band spring biasing, etc. [3889] In addition to or instead of using a pull-to-release mechanism to release latch **14.4-62**, latch **14.4-62** may be provided with one or more of the latch release mechanisms of FIG. **14.4-13**. As shown in FIGS. **14.4-13**, latch **14.4-62** may have a movable latch member such as latch member **14.4-62M** that has a cam surface such as cam surface **14.4-94**. A biasing mechanism (magnetic, spring-based, etc.) may be used to push latch member **14.4-62M** into its closed position in which latch member **14.4-62M** engages protruding portion **14.4-30P** of post **14.4-30** and holds member **14.4-24** and headband **14.4-26** together.

[3890] In some examples, latch **14.4-62** uses a push-button release mechanism. With this arrangement, latch **14.4-62** is provided with a button. The button may have a movable button member such as button member **14.4-90** that is coupled to post **14.4-30**. When a user desires to release latch **14.4-62**, the user may press inwardly on button member **14.4-90**, to move button **14.4-90** inwardly and cause button member cam surface **14.4-92** to bear against cam surface **14.4-94** of latch member **14.4-62M**. This pushes latch member **14.4-62M** away from post **14.4-30** and thereby opens the latch so that post **14.4-30** can be removed from opening **14.4-32**.

[3891] In some examples, latch **14.4-62** is provided with a toggle release mechanism. As shown in FIGS. **14.4-13**, for example, toggle lever **14.4-96** may be coupled to latch member **14.4-62M** by pivoting connection **14.4-100** and may be coupled to structures in strap **14.4-44** of headband **14.4-26** by pivoting connection **14.4-98**. This allows a user to move latch member **14.4-62M** to its open position (disengaged from post **14.4-30**) by flipping lever **14.4-96**.

[3892] In some examples, a slider mechanism may be used for latch release operations. As shown in FIGS. **14.4-13**, for example, slider member **14.4-102** may be attached to latch member **14.4-62M**. With this arrangement, a user may open latch **14.4-62** by sliding member **14.4-102** (and thereby sliding latch member **14.4-62M**) away from post **14.4-30**.

[3893] Additional latch release arrangements may be used, if desired (e.g., release mechanisms based on pull tabs, rotating knobs, push buttons, pull buttons, sliders, toggle switches, other release structures and/or combinations of these structures). These latch release arrangements may be used with any suitable latch-biasing scheme (e.g., arrangements in which latch member **14.4-62M** is biased towards its closed position magnetically, using a metal spring, using a tension spring, using a compression spring, using a coil spring, using a leaf spring, using a spring based on a stretchable band such as an elastomeric ring, and/or other spring, using a biasing device based on compressible foam, etc.). There may be a single latch member **14.4-62M** in latch **14.4-62** or multiple latch members **14.4-62M** may be used in latch **14.4-62**.

[3894] FIGS. **14.4-14** is a perspective view of headband **14.4-26** in an example in which strap **14.4-44** has latch member recesses **14.4-104**. Recesses **14.4-104** may be formed on opposing sides of opening **14.4-32** and may be configured to receive respective latch members that are located on opposing sides of post **14.4-30**. With this type of arrangement, the latch members are retracted into post **14.4-30** to open latch **14.4-62** and are extended out of post **14.4-30** into openings **14.4-104** of headband strap **14.4-44** to close latch **14.4-62**.

[3895] FIGS. **14.4-15** is a top view of a post for member **14.4-24** that has movable latch members **14.4-62M** for forming latch **14.4-62**. In this example, post **14.4-30** has a main portion **14.4-30MP** to which latch members **14.4-26M** are slidably coupled. When it is desired to release headband **14.4-26** from member **14.4-24**, movable post latch members **14.4-26M** are placed in the retracted position shown in FIGS. **14.4-15**. After inserting post **14.4-30** into opening **14.4-32** of strap **14.4-44** of headband **14.4-26**, latch members **14.4-26M** may be moved outwardly in directions **14.4-106**. This causes latch members **14.4-62M** to move into openings **14.4-104** of strap **14.4-44** of FIGS.

14.4-14, thereby closing the latch and securing headband **14.4-26** to member **14.4-24**.

[3896] A cross-sectional side view of a system based on a strap of the type shown in FIGS. **14.4-14** and a post with latch members of the type shown in FIGS. **14.4-15** is shown in FIG. **14.4-16**. As shown in FIGS. **14.4-16**, latch **14.4-62** may have movable latch members **14.4-26M** that are slidably coupled to post **14.4-30** (e.g., in slots in main post portion **14.4-30MP**). Each latch member **14.4-26M** may have a biasing mechanism (magnetic, spring-based, etc.) that pushes that member outwardly into a corresponding strap recess **14.4-104** in strap **14.4-44** of headband **14.4-26** when post **14.4-30** is inserted into opening **14.4-32**. This closes latch **14.4-62** and secures headband **14.4-26** to member **14.4-24**.

[3897] Latch **14.4-62** may have a button release or other release mechanism. As shown in FIGS. **14.4-16**, for example, button member **14.4-110** may be coupled to post **14.4-30** and may move in and out with respect to main post portion **14.4-30MP**. When the biasing mechanisms coupled to latch members **14.4-26M** are pushing members **14.4-26M** to their greatest outward extent (away from each other), cam surfaces **14.4-116** of members **14.4-62M** bear against structures **14.4-108** (e.g., a hollow tube or other structure coupled to member **14.4-24**). This forces latch members **14.4-62M** in upward direction **14.4-120**. When it is desired to release latch **14.4-62**, a user may press on button member **14.4-130**. This causes button member **14.4-130** to move inwardly in direction **14.4-122**, so that button member surface **14.4-112** bears against surfaces **14.4-114** of latch members **14.4-62M**, causing cam surfaces **14.4-116** of latch members **14.4-62M** to bear against structures **14.4-108**. As cam surfaces **14.4-116** interact with structures **14.4-108** under pressure in direction **14.4-122** from button member **14.4-110**, latch members **14.4-62M** are forced inwardly towards each other (overcoming their outward biases). Once latch members **14.4-62M** have been fully retracted into post **14.4-30**, latch members **14.4-62M** will become disengaged with recesses **14.4-104** of strap **14.4-44**, thereby allowing strap **14.4-44** to be removed from post **14.4-30**.

[3898] In the example of FIGS. **14.4-16**, recesses **14.4-104** are located in strap **14.4-44**. If desired, recesses for receiving latch members **14.4-62M** may be located in post **14.4-30**. Consider, as an example, the arrangement shown in the cross-sectional side view of system **14.4-22** of FIGS. **14.4-17**. As shown in FIGS. **14.4-17**, latch members **14.4-26M** may be slidably coupled to strap **14.4-44** of headband **14.4-26** (e.g., latch members **14.4-26M** may slide back and forth laterally within recesses in strap **14.4-44**). Biasing mechanisms (magnetic biasing structures, spring-based biasing structures, etc.) may be used to bias latch members **14.4-26M** towards post **14.4-30**. Post **14.4-30** may have one or more recesses **14.4-124** that are configured to receive latch members **14.4-62M**. When latch members **14.4-26M** are biased into recesses **14.4-124**, post **14.4-30** will be retained within opening **14.4-32** and headband **14.4-26** will therefore be secured to member **14.4-24**. When it is desired to release latch **14.4-62**, members **14.4-26M** may be moved out of post recesses **14.4-120** (e.g., using a release mechanism based on a press or pull button, a sliding button, a toggling button, a pull tab, etc.). Recess **14.4-124** may extend around the entire periphery of post **14.4-30** or there may be one or more recesses **14.4-124** located at different locations about the periphery of post **14.4-30**, each of which is aligned with a respective member **14.4-26M**.

[3899] Although sometimes described herein in the context of examples where posts **14.4-30** are formed on members **14.4-24** and openings **14.4-32** are formed in headband **14.4-26**, posts **14.4-30** may, if desired, be formed as part of headband **14.4-26** and openings **14.4-32** may be formed in members **14.4-24**.

14.5: Cable Tensioning System and Dial

[3900] Dial-based cable tensioning systems are used in an increasing number of applications, including clothing, such as lacing systems for footwear, tensioning systems for headwear, and the like. Dial-based tensioning systems can feature various improvements over conventional tensioning systems, such as low friction and equilibrated tensioning pressure across contact surfaces (e.g., in the case of headwear, equal pressure is applied around a user's head). However, current dial-based tensioning systems are often noisy when users manipulate the dial, which can be problematic or

irritating. In addition, current dial-based tensioning systems can fail to maintain the desired pressure across the contact surface over time. Improvements and advances to dial-based tensioning systems can be desirable to provide additional functionality in a variety of situations and environments.

[3901] Portable electronic devices, such as smart phones, laptops, tablet computing devices, smart watches, head-mountable displays (HMDs), and headphones, have become commonplace for persons undertaking daily activities (travel, communication, education, entertainment, employment, etc.). Indeed, portable electronic devices can provide assistance in completing daily tasks and errands, such as, watching an instructional video or monitoring progress during and after an exercise routine. Some electronic devices, such as HMDs, require mounting systems for mounting the electronic devices on a user's body, such as on a user's head. As an example, an HMD can be mounted on a user's head using an adjustable headband, which can include a dial-based tensioning system. Tensioning systems described herein can be silent and non-back-drivable, for example tensioning systems for HMDs, so as to not cause unwanted sounds near a user's ear and to maintain the HMD on a user's head. The tensioning systems described herein can be applicable to other applications, such as lacing systems for footwear, tensioning systems for other headwear, tensioning systems for other clothing, tensioning systems for bindings, and the like.

[3902] Various examples disclosed herein relate to improved tensioning systems, such as improved dial-based tensioning systems, which can be used for adjustable headbands in head-mountable display devices along with any other applications. The improved tensioning systems include a clutch that interacts with a spool on which a cable can be wound and a wheel that can be used by a user to wind the spool. The clutch is a roller-and-wedge system, which includes rollers in roller openings of a cam wheel. When the rollers are wedged between the cam wheel and a housing of the tensioning system, the cam wheel is locked. When the rollers are not wedged between the cam wheel and the housing, the cam wheel is unlocked. The roller-and-wedge system is non-back-drivable (cable can be pushed into the tensioning system, but not pulled out of the tensioning system without rotating the wheel) and silent. The tensioning system can also include various angular restraint systems, which prevent the tensioning system from over-spooling (winding too much cable into the spool and damaging the tensioning system) and operating in a backwards direction.

[3903] HMDs described herein can include head-mountable support structures that allow the devices to be worn on the heads of users. The head-mountable support structures can include device housings for housing components, such as displays that are used for presenting a user with visual content. The head-mountable support structures for an HMD can also include headbands and other structures that help hold a device housing on the face of a user. The headband of an HMD device can be adjustable, and can include a tensioning system or the like for adjusting the headband.

[3904] These and other examples are discussed below with reference to FIGS. **14.5-1** through **14.5-9C**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[3905] FIGS. **14.5-1** is a side view of a head-mountable display (HMD) **14.5-110** with an adjustable headband **14.5-126** (sometimes referred to herein as a “band”), in accordance with some examples. As illustrated in FIGS. **14.5-1**, the HMD **14.5-110** can include a head-mountable housing

14.5-112. The head-mountable housing **14.5-112** can include a main housing, a main housing unit, a head-mountable support structure, or the like. The head-mountable housing **14.5-112** can have walls or other structures that separate an interior housing region from an exterior housing region of the head-mountable housing **14.5-112**. In some examples, the walls of the head-mountable housing **14.5-112** can be formed from a material such as polymer, glass, metal, combinations thereof, or the like. Electrical and optical components can be mounted in the head-mountable housing **14.5-112**. These components can include components such as integrated circuits, sensors, control circuitry, input-output devices, and the like.

[3906] The HMD **14.5-110** can be used to present images to a user for viewing from eye boxes (e.g., eye boxes in which the user's eyes are located when the HMD **14.5-110** is being worn on the users' head, as representatively illustrated by a user's head **14.5-122** in FIGS. **14.5-1**). Specifically, the HMD **14.5-110** can include displays, lenses, and the like in the eye boxes. These components can be mounted in optical modules or other supporting structure in the head-mountable housing **14.5-112** to form respective left and right optical systems. A left display can be included for presenting an image through a left lens to a user's left eye in a left eye box and a right display can be included for presenting an image to a user's right eye in a right eye box.

[3907] In some examples, the head-mountable housing **14.5-112** can include forward-facing components, such as cameras, other sensors, and the like on a front-side F of the head-mountable housing **14.5-112**. The forward-facing components can be used for gathering sensor measurements and other inputs. A soft cushion can be disposed on an opposing rear side of the head-mountable housing **14.5-112**. The rear side of the head-mountable housing **14.5-112** can include openings that allow the user to view images from the left and right optical systems (e.g., when the rear of the head-mountable housing **14.5-112** is resting on a front surface **14.5-120F** of the user's head **14.5-122**).

[3908] The HMD **14.5-110** can include an adjustable band, such as an adjustable headband **14.5-126** or band. In some examples, the HMD **14.5-110** can include other structures (e.g., an optional over-the-head band or the like) to help hold the head-mountable housing **14.5-112** on the user's head **14.5-122**. The adjustable headband **14.5-126** can have first and second ends that are physically coupled to left and right sides of the head-mountable housing **14.5-112**, respectively. In the example illustrated in FIGS. **14.5-1**, coupling members **14.5-124** or straps, which serve as extensions of the head-mountable housing **14.5-112**, are provided on the left and right sides of the head-mountable housing **14.5-112** to physically couple the adjustable headband **14.5-126** to the head-mountable housing **14.5-112**. The coupling members **14.5-124** or straps can be formed from rigid materials, such as rigid polymers and/or other materials. The coupling members **14.5-124** can contain sensors, buttons, speakers, and/or other electrical components. Hinges and/or other mechanisms can be used to couple the coupling members **14.5-124** to the head-mountable housing **14.5-112**. In some examples, the coupling members **14.5-124** or straps can be formed integrally as portions of the head-mountable housing **14.5-112**.

[3909] The first and second ends of the adjustable headband **14.5-126** can have coupling mechanisms, such as openings, that are configured to receive posts or other protrusions **14.5-124P** of the coupling members **14.5-124** or straps (or other coupling structures of the head-mountable housing **14.5-112**). The adjustable headband **14.5-126** can be coupled to the coupling members **14.5-124** or straps such that a user removably attaches the adjustable headband **14.5-126** to the coupling members **14.5-124** or straps. As such, the user can remove the adjustable headband **14.5-126** from the head-mountable housing **14.5-112**. The coupling members **14.5-124** or straps can have elongated shapes, as illustrated in FIGS. **14.5-1**, and/or other suitable shapes. Examples of the coupling members **14.5-124** or straps can include rigid straps, rigid coupling members, power straps, or the like.

[3910] The adjustable headband **14.5-126** can include a soft flexible portion, such as a central portion **14.5-130**. The central portion **14.5-130** can be formed between two stiffer portions, such as

end portions **14.5-128** on the left and right ends of the adjustable headband **14.5-126** (e.g., the first and second ends of the adjustable headband **14.5-126**). The end portions **14.5-128** can be stiffened using embedded polymer stiffeners (e.g., single-layer or multi-layer polymer stiffening strips) and/or other stiffening members. The central portion **14.5-130** can be formed from a stretchable material such as stretchy fabric. In some examples, the central portion **14.5-130** can be formed from a band of flat knit fabric that includes stretchable strands of material (e.g., elastomeric strands) and/or which uses a stretchable fabric construction (e.g., a stretchable knit construction). The central portion **14.5-130** can include narrowed end portions, which can extend over stiffening members in the end portions **14.5-128**. This can ensure that the adjustable headband **14.5-126** has a uniform external appearance. Forming the central portion **14.5-130** of a stretchable material and including the central portion in the adjustable headband **14.5-126** allows the adjustable headband **14.5-126** be stretched along its length. This allows the length of the adjustable headband **14.5-126** to be temporarily increased to help a user place the adjustable headband **14.5-126** over the user's head **14.5-122** when a user is donning the HMD **14.5-110**. When the adjustable headband **14.5-126** is released, the stretchiness and the elastic nature of the central portion **14.5-130** of the adjustable headband **14.5-126** help to shorten the adjustable headband **14.5-126** and pull the adjustable headband **14.5-126** against the user's head. This allows the adjustable headband **14.5-126** to rest against a rear surface **14.5-120R** the user's head.

[3911] A tensioning mechanism (such as the tensioning mechanism **14.5-100**, discussed below with respect to FIGS. **14.5-2** through **14.5-5**) can be included to provide further adjustment of the effective length and tension of the adjustable headband **14.5-126** to secure the adjustable headband **14.5-126** and the HMD **14.5-110** on the user's head **14.5-122**. The tensioning mechanism can be provided to adjust the effective length and tension in a headband tensioning cable **14.5-132**. The tensioning mechanism can be a rotatable knob, lever, slider, or other mechanism for adjusting the headband tensioning cable **14.5-132**. When the effective length of the cable **14.5-132** is greater, or the tension lower, the adjustable headband **14.5-126** will be looser. The increased effective length and low cable tension of the headband tensioning cable **14.5-132** helps to create slack in the adjustable headband **14.5-126**, which can aid in donning and doffing of the HMD **14.5-110**. When cable tension is increased, the adjustable headband **14.5-126** is secured against the user's head.

[3912] As illustrated in FIGS. **14.5-1**, the headband tensioning cable **14.5-132** can have a loop shape and can include an upper cable segment **14.5-132T** that runs along an upper edge of the adjustable headband **14.5-126** and an opposing lower cable segment **14.5-132B** that runs along the opposing lower edge of the adjustable headband **14.5-126**. In the middle of the central portion **14.5-130** of the adjustable headband **14.5-126**, the upper and lower cable segments **14.5-132T** and **14.5-132B** can be separated by a distance BW (sometimes referred to as a cable bifurcation distance). The separation between the upper and lower cable segments **14.5-132T** and **14.5-132B** helps secure the adjustable headband **14.5-126** on the curved rear surface **14.5-120R** of the user's head **14.5-122**. When the adjustable headband **14.5-126** is worn on the rear surface **14.5-120R**, as illustrated in FIGS. **14.5-1**, the upper cable segment **14.5-132T** can apply an upward force F1 on the adjustable headband **14.5-126** that can prevent the headband **14.5-126** from slipping downwards off of the user's head **14.5-122**. Similarly, the lower cable segment **14.5-132B** can apply a downward force F2 that can prevent the adjustable headband **14.5-126** from slipping upwards off of the user's head **14.5-122**.

[3913] FIGS. **14.5-2** illustrates a plan view of an adjustable headband **14.5-126**. The adjustable headband **14.5-126** can be substantially similar to, and can include some or all of, the features of the adjustable headband **14.5-126** discussed above with respect to FIGS. **14.5-1**. As illustrated in FIGS. **14.5-2**, the adjustable headband **14.5-126** can include a central portion **14.5-130** and end portions **14.5-128**. As discussed above, the central portion **14.5-130** can be stretchable and can be formed from a stretchy knit fabric, and the end portions can be stiffened using stiffeners embedded in fabric. One or more openings **14.5-140** can be formed in the end portions **14.5-128** of the

adjustable headband **14.5-126**. The openings **14.5-140** can receive posts or other protrusions, such as the protrusions **14.5-124P** discussed above with respect to FIGS. **14.5-1**. The openings **14.5-140** can be used in conjunction with the protrusions **14.5-124P** to secure the end portions **14.5-128** of the adjustable headband **14.5-126** to coupling members **14.5-124** or straps of a head-mountable housing **14.5-112**. In some examples, a single opening **14.5-140** or other attachment mechanism can be disposed on each of the end portions **14.5-128** of the adjustable headband **14.5-126**. In some examples, each of the end portions **14.5-128** of the adjustable headband **14.5-126** can include two or more of the openings **14.5-140**. In some examples, other attachment mechanisms (e.g., magnets, snaps, hook-and-loop fasteners, screws or other fasteners, combinations thereof, or the like) can be used to attach the adjustable headband **14.5-126** to the coupling members **14.5-124** or straps, or other portions of the head-mountable housing **14.5-112** in an HMD **14.5-110**.

[3914] A headband tensioning cable **14.5-132** can extend in a loop around the perimeter of the central portion **14.5-130** of the adjustable headband **14.5-126**. In the example illustrated in FIGS. **14.5-2**, the adjustable headband **14.5-126** includes a pulley **14.5-142** coupled to a left end portion **14.5-128** of the adjustable headband **14.5-126**. The adjustable headband **14.5-126** includes a tensioning mechanism **14.5-100** coupled to a right end portion **14.5-128** of the adjustable headband **14.5-126**. In some examples, the tensioning mechanism **14.5-100** can be an adjustment knob or dial. The tensioning mechanism **14.5-100** can be a two-way dial that can be rotated in a first direction **14.5-148** about a rotational axis **14.5-146** (e.g., a clockwise direction) to shorten the length of the loop of the headband tensioning cable **14.5-132** and thereby increase tension on the headband tensioning cable **14.5-132**. The length of the loop of the headband tensioning cable **14.5-132** outside the tensioning mechanism **14.5-100** and disposed in the central portion **14.5-130** can be referred to as the “effective length” of the tensioning cable **14.5-132**. A dial of the tensioning mechanism **14.5-100** can be rotated in an opposing second direction **14.5-150** about the rotational axis **14.5-146** (e.g., a counterclockwise direction) to increase the length of the loop of the headband tensioning cable **14.5-132** and thereby decrease tension on the headband tensioning cable **14.5-132**. By tightening the headband tensioning cable **14.5-132**, tension in the adjustable headband **14.5-126** can be increased and the HMD **14.5-110** can be held more snugly on a user's head **14.5-122**. The pulley **14.5-142** can be included to allow tension in upper and lower portions of the headband tensioning cable **14.5-132** to be balanced, ensuring a consistent cupping of the adjustable headband **14.5-126** across various head shapes. By loosening the headband tensioning cable **14.5-132**, effective length can be increased and the tension in the adjustable headband **14.5-126** can be decreased.

[3915] The tensioning mechanism **14.5-100** can include a variety of improvements. In some examples, the tensioning mechanism **14.5-100** can include a cam and wedge system that allows the tensioning mechanism **14.5-100** to silently spool the headband tensioning cable **14.5-132** into the tensioning mechanism **14.5-100** and silently unspool the headband tensioning cable **14.5-132** from the tensioning mechanism **14.5-100**. The tensioning mechanism **14.5-100** can be non-back-drivable. In other words, the tensioning mechanism **14.5-100** can allow the headband tensioning cable **14.5-132** to be spooled into the tensioning mechanism **14.5-100** without rotating the tensioning mechanism **14.5-100**, but cannot allow the headband tensioning cable **14.5-132** to be unspooled from the tensioning mechanism **14.5-100** without rotating the tensioning mechanism **14.5-100**. The tensioning mechanism **14.5-100** can include an angular stop mechanism (also referred to as a hard stop, an angular stop, or the like). The angular stop mechanism prevents too much of the headband tensioning cable **14.5-132** from being spooled into the tensioning mechanism **14.5-100**, preventing damage to the tensioning mechanism **14.5-100**. The angular stop mechanism also prevents the tensioning mechanism **14.5-100** from operating in a backwards direction, which can impact the user experience. The tensioning mechanism **14.5-100** can further include a spring detent or the like. The spring detent can provide a desired sound, feel, and other user experience features to the tensioning mechanism **14.5-100**. These and other features of the

tensioning mechanism **14.5-100** will be discussed below with respect to FIGS. **14.5-3** through **14.5-9C**.

[3916] FIGS. **14.5-3** illustrates a perspective view of a tensioning mechanism or system **14.5-300** that can be included in an HMD **14.5-110** to adjust the tension in a headband tensioning cable **14.5-132** of an adjustable headband **14.5-126**, in accordance with some examples. The tensioning system **14.5-300** includes a dial cap **14.5-302** and a housing **14.5-304**. The housing **14.5-304** can be directly fixed to an adjustable headband **14.5-126** of the head-mounted device **14.5-110**, or can be fixed to the adjustable headband **14.5-126** through an interface. The housing **14.5-104** can be fixed to the adjustable headband **14.5-126** such that the housing **14.5-304** cannot rotate relative to the adjustable headband **14.5-126**.

[3917] In one example, the dial cap **14.5-302** includes a center cap **14.5-306** and a wheel **14.5-308** (also referred to as a tire or the like). The wheel **14.5-308** can be configured to be rotated by a user in order to spool the headband tensioning cable **14.5-132** into the tensioning system **14.5-300** or unspool the headband tensioning cable **14.5-132** from the tensioning system **14.5-300**. The wheel **14.5-308** rotates around a central axis of the center cap **14.5-306**. The center cap **14.5-306** and the housing **14.5-304** can be fixed to one another such that the center cap **14.5-306** does not rotate relative to the housing **14.5-304**. As such, the wheel **14.5-308** can be rotatable relative to the center cap **14.5-306** and the housing **14.5-304**.

[3918] As will be discussed in more detail below, the housing **14.5-304** can be configured to contain a spool within the housing **14.5-304**. The housing **14.5-304** includes a cable opening **14.5-310**. The headband tensioning cable **14.5-132** extends through the cable opening **14.5-310** into the housing **14.5-304** and is fixed to the spool. The headband tensioning cable **14.5-132** moves through the cable opening **14.5-310** as the headband tensioning cable **14.5-132** is spooled into or unspooled from the tensioning system **14.5-300**, such as by rotating the wheel **14.5-308** of the dial cap **14.5-302**.

[3919] Any number or variety of components in any of the configurations described herein can be included in the tensioning system and the electronic device. The components can include any combination of the features described herein and can be arranged in any of the various configurations described herein. The structure and arrangement of components of a tensioning system having a roller-and-wedge locking system described herein, can apply not only to the specific examples discussed herein, but to any number of examples in any combination. Examples of a configuration of the various components of the tensioning system relative to one another are described below, with reference to FIGS. **14.5-4**.

[3920] FIGS. **14.5-4** illustrates a partially exploded view of a tensioning system **14.5-300**. The tensioning system **14.5-300** includes a headband tensioning cable **14.5-132**, a dial cap **14.5-302**, a housing **14.5-304**, a cam wheel **14.5-312**, rollers **14.5-314**, a spool **14.5-316**, and a cap post **14.5-318**. The dial cap **14.5-302** and the housing **14.5-304** can be substantially similar to, and can include some or all of, the features of the dial cap **14.5-302** and the housing **14.5-304** discussed above with respect to FIGS. **14.5-3**.

[3921] The housing **14.5-304** can include a housing post **14.5-322**. The housing post **14.5-322** can be disposed centrally in the housing **14.5-304**. The housing **14.5-304** can further include an outer wall **14.5-320** in which a cable opening **14.5-310** is disposed.

[3922] In at least one example, the spool **14.5-316** is configured to sit in the housing **14.5-304** between the outer wall **14.5-320** and the housing post **14.5-322**. The spool **14.5-316** can include a hollow central portion **14.5-330**, which surrounds the housing post **14.5-322**. The spool **14.5-316** can be configured to rotate around the housing post **14.5-322** while the headband tensioning cable **14.5-132** is spooled into or unspooled from the tensioning system **14.5-300**. The spool **14.5-316** includes a cable opening **14.5-334**, into which the headband tensioning cable **14.5-132** extends. The headband tensioning cable **14.5-132** can be retained or fixed within the cable opening **14.5-334** of the spool **14.5-316**. The spool **14.5-316** includes a barrel **14.5-338** around which the headband

tensioning cable **14.5-132** is wound, and flanges **14.5-336**, which prevent the headband tensioning cable **14.5-132** from being wound around or into other portions of the tensioning system **14.5-300** or otherwise escaping the spool **14.5-316**. The spool **14.5-316** further includes protrusions **14.5-332**, which allow the spool **14.5-316** to interface with and be rotatably coupled to the cam wheel **14.5-312**.

[3923] The cam wheel **14.5-312** and the rollers **14.5-314** are used to transfer rotation between the dial cap **14.5-302** and the spool **14.5-316**. The cam wheel **14.5-312** can be configured to rotate the spool **14.5-316** in response to a user rotating the wheel **14.5-308** in either direction. The cam wheel **14.5-312** and the rollers **14.5-314** can be configured to allow the headband tensioning cable **14.5-132** to be spooled into the tensioning system **14.5-300** by pushing the headband tensioning cable **14.5-132** into the housing **14.5-304** (e.g., without a user rotating the wheel **14.5-308**). The cam wheel **14.5-312** and the rollers **14.5-314** can be configured not to allow the cable to be unspooled from of the tensioning system **14.5-300** when the headband tensioning cable **14.5-132** is pulled. The cam wheel **14.5-312** and the rollers **14.5-314** are configured to resist torque in a first direction (e.g., a direction in which the headband tensioning cable **14.5-132** is unspooled from the tensioning system **14.5-300**) and allow torque in a second direction (e.g., a direction in which the headband tensioning cable **14.5-132** is spooled into the tensioning system **14.5-300**). Details of the cam wheel **14.5-312** and the rollers **14.5-314** are discussed below with respect to FIGS. **14.5-5**. The cam wheel **14.5-312** can include protrusions **14.5-340**, which interact with the protrusions **14.5-332** of the spool **14.5-316** to transfer rotation between the spool **14.5-316** and the cam wheel **14.5-312**. The cam wheel **14.5-312** includes an opening **14.5-342** and is configured to sit on top of the spool **14.5-316** between the spool **14.5-316** and the dial cap **14.5-302**.

[3924] The cap post **14.5-318** can be disposed in a post opening **14.5-324** in the housing **14.5-304**. The cap post **14.5-318** can be fixed to the housing **14.5-304**, such that the cap post **14.5-318** is in a fixed position. In some examples, the cap post **14.5-318** can be fixed to the housing **14.5-304** through an adhesive, a fastener, or the like. In some examples, the cap post **14.5-318** can have a semi-circular cross-sectional shape. The cap post **14.5-318** can be configured to interface with a center cap post **14.5-307** of the center cap **14.5-306** to hold the center cap **14.5-306** in place in the housing **14.5-304**. The cap post **14.5-318** can prevent the center cap **14.5-306** from rotating relative to the housing **14.5-304**. The center cap post **14.5-307** can have a complementary cross-sectional shape to the cap post **14.5-318** (e.g., the center cap post **14.5-307** and the cap post **14.5-318** can each have semi-circular cross-sectional shapes). In some examples, the center cap post **14.5-307** can have a key-shape, and the cap post **14.5-318** can have a complementary key-hole shape.

[3925] Any number or variety of components in any of the configurations described herein can be included in the tensioning system and the electronic device. The components can include any combination of the features described herein and can be arranged in any of the various configurations described herein. The structure and arrangement of components of a tensioning system having a roller-and-wedge locking system described herein, can apply not only to the specific examples discussed herein, but to any number of examples in any combination. Examples of an operation of the tensioning system are described below, with reference to FIGS. **14.5-5**.

[3926] FIGS. **14.5-5** illustrates a partial cross-sectional view of a portion of a tensioning system **14.5-300**. FIGS. **14.5-5** illustrates the operation of a cam wheel **14.5-312** and rollers **14.5-314** included in the tensioning system **14.5-300**, in accordance with some examples. The tensioning system **14.5-300** includes a housing **14.5-304**, a spool **14.5-316** in the housing **14.5-304**, and a wheel **14.5-308**. The wheel **14.5-308** can include an external portion **14.5-308A** and an internal portion **14.5-308B**. The external portion **14.5-308A** is configured to be interacted with by a user. The external portion **14.5-308A** can include ridges, bumps, grooves, or other features, which can be used by a user to grip the external portion. The external portion **14.5-308A** can be formed of a relatively soft material, such as a rubber, a polymer, or the like. The internal portion **14.5-308B** can include protrusions **14.5-308B.i**, which are configured to interact with the rollers **14.5-314** and the

cam wheel **14.5-312**. The internal portion **14.5-308B** can also be used to connect the external portion **14.5-308A** to other structures of the tensioning system **14.5-300**. In some examples, the internal portion **14.5-308B** can include posts, and the external portion **14.5-308A** can include corresponding holes in which the posts are inserted to fix the internal portion **14.5-308B** and the external portion **14.5-308A** to one another. In some examples, an adhesive can be disposed between the internal portion **14.5-308B** and the external portion **14.5-308A** to fix the internal portion **14.5-308B** and the external portion **14.5-308A** to one another.

[3927] The wheel **14.5-312** can include a protrusion **14.5-342**, which is configured to interface with the spool **14.5-316**. For example, as illustrated in FIGS. **14.5-5**, the spool **14.5-316** can include an opening **14.5-343**, which has a complementary shape to the protrusion **14.5-342**. This allows for the spool **14.5-316** and the wheel **14.5-312** to be rotationally coupled to one another. A cap post **14.5-318** is disposed in an opening in the housing **14.5-304**. The cap post **14.5-318** can be adhered to the housing **14.5-304**. The housing **14.5-304** can be physically coupled to a center cap post **14.5-307** of a center cap by a protrusion **14.5-319** of the center cap post **14.5-307**. The center cap post **14.5-307**, including the protrusion **14.5-319**, and the cap post **14.5-318** can have complementary shapes, such that the center cap post **14.5-307** is physically coupled to (e.g., rotationally coupled to) the cap post **14.5-318** by the protrusion **14.5-319**. For example, the protrusion **14.5-319** and the center cap post **14.5-307** can have a key shape, and the cap post **14.5-318** can have a complementary key-hole shape, as illustrated in FIGS. **14.5-5**.

[3928] The housing **14.5-304** can include a main body portion **14.5-304A** and an inner ring **14.5-304B**. The inner ring **14.5-304B** can be configured to interact with the rollers **14.5-314**, and can be formed of a relatively hard material, such as metals (e.g., stainless steel, such as SAE 304 stainless steel, 420 stainless steel, or the like; aluminum, brass, or any other suitable metals), high-hardness polymers, other polymers, plastics, or the like. The main body portion **14.5-304A** can be formed of a relatively soft material, such as polymers, plastics, or the like. The spool **14.5-316** can be formed of a relatively soft material, such as polymers, plastics, or the like. The inner ring **14.5-304B** can be fixed to the main body portion **14.5-304A** through an adhesive, fasteners, or the like.

[3929] The cam wheel **14.5-312** can include roller openings **14.5-313**, in which the rollers **14.5-314** and the protrusions **14.5-308B.i** are disposed. The roller openings **14.5-313** are configured to interact with the rollers **14.5-314** and the protrusions **14.5-308B.i** to lock and unlock the cam wheel **14.5-312** and the housing **14.5-304** together. Specifically, the cam wheel **14.5-312** and the housing **14.5-304** can be locked together when the rollers **14.5-314** are wedged between the cam wheel **14.5-312** and the housing **14.5-304**. This system can be referred to as a roller-and-wedge system or a roller-and-wedge clutch. The roller openings **14.5-313** include ramped surfaces **14.5-340** that are inclined relative to the inner ring **14.5-304B** of the housing **14.5-304** at a cam angle **14.5- θ** . Specifically, the cam angle **14.5- θ** for each ramped surface **14.5-340** defining a respective roller opening **14.5-313** can be measured between the respective ramped surface **14.5-340** and a line tangential to a point on the inner ring **14.5-304B** of the housing **14.5-304** in contact with a roller **14.5-314** in the respective roller opening **14.5-313**. The cam angle **14.5- θ** can be in a range from about 5° to about 30°.

[3930] In at least one example, tension from the cable can supply a force to the cam wheel **14.5-312** through the spool **14.5-316**, which causes the cam wheel to rotate in a third direction **14.5-354** (e.g., a counterclockwise direction). When the cam wheel **14.5-312** rotates in third direction **14.5-354** as a result of the tension in the cable **14.5-132**, the rollers **14.5-314** move in a fourth direction **14.5-356** (e.g., a clockwise direction) up the ramped surfaces **14.5-340**, and are wedged between the protrusions **14.5-308B.i** of the internal portion **14.5-308B** of the wheel **14.5-308** and the ramped surfaces **14.5-344** of the cam wheel **14.5-312** defining the roller openings **14.5-313**. The wedged rollers **14.5-314** lock the cam wheel **14.5-312** and the housing **14.5-304** together due to friction being present between the cam wheel **14.5-312**, the inner ring **14.5-304B** of the housing **14.5-304**, and the wedged rollers **14.5-314**, and prevents the cable **14.5-132** from rotating the spool

14.5-316 in the third direction **14.5-354**. When the cable is pushed into the tensioning system **14.5-300**, the cam wheel **14.5-312** and the housing **14.5-304** are unlocked from one another. Specifically, when the cable **14.5-132** is pushed into the tensioning system **14.5-300**, the cam wheel **14.5-312** rotates in a direction opposite the third direction **14.5-354**, the rollers **14.5-314** are unwedged and roll down the ramped surfaces **14.5-344** in a direction opposite the fourth direction **14.5-356**, and the spool **14.5-316** is free to rotate such that the cable **14.5-132** wraps around the spool **14.5-316**.

[3931] In some examples, the cam wheel **14.5-312**, the rollers **14.5-314**, and the inner ring **14.5-304B** of the housing **14.5-304** can be formed of materials such as metals (e.g., stainless steel, such as SAE 304 stainless steel, 420 stainless steel, or the like; aluminum, brass, or any other suitable metals), high-hardness polymers, other polymers, plastics, or the like. In some examples, one or more of the cam wheel **14.5-312**, the rollers **14.5-314**, and the inner ring **14.5-304B** can be coated or plated with a material to increase the hardness of the cam wheel **14.5-312**, the rollers **14.5-314**, and/or the inner ring **14.5-304B**. In some examples, the rollers **14.5-314** can be formed of materials having a higher hardness than materials of the cam wheel **14.5-312** and the inner ring **14.5-304B**. For example, the cam wheel **14.5-312**, the rollers **14.5-314**, and the inner ring **14.5-304B** can be formed of **14.5-304** stainless steel, with a coating being applied to the rollers **14.5-314** to increase the hardness of the rollers **14.5-314** relative to the cam wheel **14.5-312** and the inner ring **14.5-304B**.

[3932] The cam angle **14.5-θ** can be selected based on friction resulting from the materials selected for the cam wheel **14.5-312**, the inner ring **14.5-304B** of the housing **14.5-304**, and the rollers **14.5-314**; the scale of the tensioning system **14.5-300**; and clearances and tolerances required for the tensioning system; among other factors. Providing a smaller cam angle **14.5-θ** allows the cam wheel **14.5-312** and the rollers **14.5-314** to lock with less friction being present between the cam wheel **14.5-312**, the inner ring **14.5-304B** of the housing **14.5-304**, and the rollers **14.5-314**. Providing too large of a cam angle **14.5-θ** can allow for slipping between the cam wheel **14.5-312** and the housing **14.5-304**, such that the cam wheel **14.5-312** and the housing **14.5-304** are not sufficiently locked. Providing a smaller cam angle **14.5-θ** requires more space for the rollers **14.5-314** to move in order to lock the cam wheel **14.5-312** and the housing **14.5-304**, which requires larger tolerances and clearances. This results in a larger dead zone between movement of the cable **14.5-132** before the cam wheel **14.5-312** and the housing **14.5-304** lock, and in movement of the wheel **14.5-308** before the spool **14.5-316** spools or unspools the cable. The dead zone (also referred to as a roller travel) can be a distance or an angular distance **14.5-D1** between a center of the rollers **14.5-314** in the locked position and the unlocked position, and can be in a range from about 0.5 mm to about 8 mm, or in a range from about 5° and about 30°, respectively. A smaller cam angle **14.5-θ** results in higher normal forces, which requires more force to be used to remove the rollers **14.5-314** from between the cam wheel **14.5-312** and the inner ring **14.5-304B** when the cam wheel **14.5-312** and the housing **14.5-304** are locked. Providing the cam angle **14.5-θ** in the prescribed range allows for the cam wheel **14.5-312** and the housing **14.5-304** to be locked without slipping, while a minimal dead zone is present and minimal force is required to rotate the wheel **14.5-308**.

[3933] Although six protrusions **14.5-308B.i**, six rollers **14.5-314**, and six roller openings **14.5-313** are illustrated in FIGS. **14.5-5**, any number of the protrusions **14.5-308B.i**, the rollers **14.5-314**, and the roller openings **14.5-313** can be provided. Providing larger numbers of the protrusions **14.5-308B.i**, the rollers **14.5-314**, and the roller openings **14.5-313** can increase strength, while also increasing complexity, size, and cost. Providing smaller numbers of the protrusions **14.5-308B.i**, the rollers **14.5-314**, and the roller openings **14.5-313** can decrease strength, complexity, size, and cost.

[3934] FIGS. **14.5-5** illustrates all of the protrusions **14.5-308B.i**, the rollers **14.5-314**, and the roller openings **14.5-313** facing in the same direction with respect to an axis of rotation. In some

examples, half or another percentage of the protrusions **14.5-308B.i**, the rollers **14.5-314**, and the roller openings **14.5-313** can face in opposite directions with respect to the axis of rotation. This allows for the tensioning system **14.5-300** to be locked in both directions, rather than a single direction. In examples in which the protrusions **14.5-308B.i**, the rollers **14.5-314**, and the roller openings **14.5-313** face opposite directions with respect to the axis of rotation, the wheel **14.5-308** can be used to spool the cable into the spool **14.5-316** and out of the spool **14.5-316**.

[3935] The wheel **14.5-308** can be rotated in a first direction **14.5-350** (e.g., a clockwise direction) around a center cap post **14.5-307** of a center cap **14.5-306** (illustrated in FIGS. **14.5-4**) and the housing **14.5-304** and a second direction **14.5-352** (e.g., a counterclockwise direction) opposite the first direction **14.5-350** around the center cap post **14.5-307** and the housing **14.5-304**. Rotating the wheel **14.5-308** in the first direction **14.5-350** can spool a cable onto the spool **14.5-316**, adding tension to the cable. In other words, rotating the wheel **14.5-308** in the first direction **14.5-350** wraps the cable onto the spool **14.5-316** in order to tighten the cable. Rotating the wheel **14.5-308** in the second direction **14.5-352** can unspool the cable from the spool **14.5-316**, reducing tension in the cable. In other words, rotating the wheel **14.5-308** in the second direction **14.5-352** unwraps the cable from the spool **14.5-316** in order to loosen the cable.

[3936] Rotating the wheel **14.5-308** in the first direction **14.5-350** or the second direction **14.5-352** unlocks the cam wheel **14.5-312** and the housing **14.5-304**, and allows the cable to be spooled into the tensioning system **14.5-300** or out of the tensioning system **14.5-300**. For example, when the wheel **14.5-308** is rotated in the first direction **14.5-350**, the rollers **14.5-314** roll down the ramped surfaces **14.5-344** and the tensioning system **14.5-300** is unlocked. When the wheel **14.5-308** is rotated in the second direction **14.5-352**, the protrusions **14.5-308B.i** push the rollers **14.5-314** down the ramped surfaces and the tensioning system **14.5-300** is unlocked. This allows the wheel **14.5-308** to rotate in either direction in order to increase or decrease tension on the cable.

[3937] The tensioning system **14.5-300**, which includes the rollers **14.5-314** and the cam wheel **14.5-312**, can be locked in either direction or both directions. In contrast to conventional systems, which can use spring-driven pawls, ratcheting systems, or the like, the tensioning system **14.5-300** uses the rollers **14.5-314**, which roll within the roller openings **14.5-313** to lock and unlock the tensioning system **14.5-300**. This is a silent or relatively silent system. The tensioning system **14.5-300** can be locked in one or both directions, and is thus non-back-drivable. In other words, the cable can be pushed into the tensioning system **14.5-300**, but cannot be pulled from the tensioning system **14.5-300** without using the wheel **14.5-308**. This allows for the cable to be spooled into the tensioning system **14.5-300** and for the tensioning system **14.5-300** to hold tension in the cable.

[3938] Any number or variety of components in any of the configurations described herein can be included in the tensioning system and the electronic device. The components can include any combination of the features described herein and can be arranged in any of the various configurations described herein. The structure and arrangement of components of a tensioning system having a roller-and-wedge locking system described herein, can apply not only to the specific examples discussed herein, but to any number of examples in any combination. Examples of a configuration of the various components of a dial cap included in the tensioning system relative to one another, and a configuration of an angular restraint are described below, with reference to FIGS. **14.5-6**.

[3939] FIGS. **14.5-6** illustrates a partially exploded view of dial cap **14.5-302** and a perspective view of a cam wheel **14.5-312**. The dial cap **14.5-302** includes a center cap **14.5-306**, a center cap post **14.5-307**, a wheel **14.5-308** including an external portion **14.5-308A** and an internal portion **14.5-308B**, a wheel adhesive **14.5-360**, and a clamp ring **14.5-364**. The center cap **14.5-306**, the center cap post **14.5-307**, the wheel **14.5-308**, and the cam wheel **14.5-312** can be substantially similar to, and can include some or all of, the features of the center cap **14.5-306**, the center cap post **14.5-307**, the wheel **14.5-308**, and the cam wheel **14.5-312**, respectively, discussed above with respect to FIGS. **14.5-3** through **14.5-5**.

[3940] As discussed previously, a user can interact with the external portion **14.5-308A** of the wheel **14.5-308** to turn the wheel **14.5-308** and wind or unwind a cable in a tensioning system to tighten or loosen the cable. The internal portion **14.5-308B** of the wheel **14.5-308** can include protrusions **14.5-308B.i**. The protrusions **14.5-308B.i** can be used to interact with rollers and the cam wheel **14.5-312** to lock and unlock the cam wheel **14.5-312** and a housing of the tensioning system. The wheel adhesive **14.5-360** and the clamp ring **14.5-364** can be used to fasten the inner portion **14.5-308B** to the external portion **14.5-308A** of the wheel **14.5-308**.

[3941] The example of FIGS. **14.5-6** includes a disc-type angular stop mechanism. Specifically, in the example of FIGS. **14.5-6**, a first disc **14.5-351** and a second disc **14.5-355** are included, which provide angular stops for the dial cap **14.5-302**. The first disc **14.5-351** and the second disc **14.5-355** can be mounted on the center cap post **14.5-307** by a clip **14.5-358**. The first disc **14.5-351** and the second disc **14.5-355** can be fastened on the center cap post **14.5-307** by any method, which allows the first disc **14.5-351** and the second disc **14.5-355** to rotate around the center cap post **14.5-307**. For example, the first disc **14.5-351** and the second disc **14.5-355** can be fastened on the center cap post **14.5-307** by a pin, a clip, another fastener, or the like. The center cap **14.5-306** can include a protrusion **14.5-309**, which provides an angular stop for the first disc **14.5-351**.

[3942] The first disc **14.5-351** can include a first protrusion **14.5-352A**, which interacts with a first side of the protrusion **14.5-309** to stop angular rotation of the first disc **14.5-351** in a first direction, or interacts with a second side of the protrusion **14.5-309** opposite the first side to stop angular rotation of the first disc **14.5-351** in a second direction opposite the first direction. The first disc **14.5-351** includes a second protrusion (such as the second protrusion **14.5-350B**, discussed below with respect to FIGS. **14.5-7A** and **14.5-7B**) opposite the first protrusion, which interacts with the second disc **14.5-355**. The second disc **14.5-355** interacts with a protrusion **14.5-359** on the cam wheel **14.5-312**. The cam wheel **14.5-312** can rotate freely until the protrusion **14.5-359** interacts with a first protrusion **14.5-356A** on the second disc **14.5-355**. The cam wheel **14.5-312** and the second disc **14.5-355** can then rotate freely until a second protrusion (such as the second protrusion **14.5-356B**, discussed below with respect to FIGS. **14.5-7B**) of the second disc **14.5-355** interacts with the second protrusion of the first disc **14.5-351**. The cam wheel **14.5-312**, the second disc **14.5-355**, and the first disc **14.5-351** can then rotate freely until the first protrusion **14.5-352A** of the first disc **14.5-351** interacts with the protrusion **14.5-309** of the center cap **14.5-306**. When the first protrusion **14.5-352A** of the first disc **14.5-351** interacts with the protrusion **14.5-309** of the center cap **14.5-306**, the protrusion **14.5-309** provides an angular stop, which prevents the first disc **14.5-351**, the second disc **14.5-355**, and the cam wheel **14.5-312** from turning in one direction, but allows the first disc **14.5-351**, the second disc **14.5-355**, and the cam wheel **14.5-312** to turn in the opposite direction. This prevents a user from over-spooling a tensioning system (e.g., spooling more cable into a housing of the tensioning system than the housing is configured to hold), or from loosening the tensioning system too much such that the tensioning system winds the cable into the housing in the wrong direction.

[3943] Each of the first disc **14.5-351** and the second disc **14.5-355** can allow for roughly 360 degrees of rotation and the cam wheel **14.5-312** can allow for an additional roughly 360 degrees of rotation. Specifically, each of the first disc **14.5-351** and the second disc **14.5-355** can allow for 360 degrees of rotation minus the angular width of the protrusions included in the first disc **14.5-351**, the second disc **14.5-355**, the center cap **14.5-306**, and the cam wheel **14.5-312**. Including a larger number of discs in the angular stop increases the angular rotation allowed by the angular stop, while decreasing the number of discs in the angular stop decreases the angular rotation allowed by the angular stop. The first disc **14.5-351** and the second disc **14.5-355** can be formed of a relatively hard material, such as metals (e.g., stainless steel, such as SAE 304 stainless steel, 420 stainless steel, or the like; aluminum, brass, or any other suitable metals), high-hardness polymers, other polymers, plastics, or the like. Including the first disc **14.5-351** and the second disc **14.5-355** in the angular stop and forming the first disc **14.5-351** and the second disc **14.5-355** of relatively hard

materials provides a strong, robust angular stop, which is difficult for a user to damage.

[3944] Any number or variety of components in any of the configurations described herein can be included in the tensioning system and the electronic device. The components can include any combination of the features described herein and can be arranged in any of the various configurations described herein. The structure and arrangement of components of a tensioning system having a roller-and-wedge locking system and an angular restraint described herein, can apply not only to the specific examples discussed herein, but to any number of examples in any combination. Examples of a configuration and operation of a disc-type angular restraint included in the tensioning system are described below, with reference to FIGS. **14.5-7A** and **14.5-7B**.

[3945] FIGS. **14.5-7A** and **14.5-7B** illustrate plan views of a center cap **14.5-306**, a first disc **14.5-351**, a second disc **14.5-355**, and a cam wheel **14.5-312**. FIGS. **14.5-7A** and **14.5-7B** illustrate additional details of the disc-type angular stop mechanism of the example illustrated in FIGS. **14.5-6**. The center cap **14.5-306**, the first disc **14.5-351**, the second disc **14.5-355**, and the cam wheel **14.5-312** can be substantially similar to, and can include some or all of, the features of the center cap **14.5-306**, the first disc **14.5-351**, the second disc **14.5-355**, and the cam wheel **14.5-312**, respectively, discussed above with respect to FIGS. **14.5-6**.

[3946] As illustrated in FIGS. **14.5-7A**, the first disc **14.5-351** can sit in a channel **14.5-311** of the center cap **14.5-306**. The first disc **14.5-351** is free to rotate in the channel **14.5-311**, but is stopped by the first protrusion **14.5-352A** of the first disc **14.5-351** contacting either side of the protrusion **14.5-309** of the center cap **14.5-306**. The first disc **14.5-351** can rotate 360 degrees relative to the center cap **14.5-306**, minus the angular widths of the protrusion **14.5-309** and the first protrusion **14.5-352A**. The first disc **14.5-351** further includes a second protrusion **14.5-352B** opposite the first protrusion **14.5-352A**, which interacts with the second disc **14.5-355**, illustrated in FIGS. **14.5-7B**.

[3947] As illustrated in FIGS. **14.5-7B**, the second disc **14.5-355** sits on the first disc **14.5-351**. The second protrusion **14.5-352B** of the first disc **14.5-351** can sit in a channel **14.5-353** of the second disc **14.5-355**, which includes a second protrusion **14.5-356B** opposite the first protrusion **14.5-356A**. The second disc **14.5-355** is free to rotate, and begins to rotate the first disc **14.5-351** when the second protrusion **14.5-356B** of the second disc **14.5-355** contacts either side of the second protrusion **14.5-352B** of the first disc **14.5-351**. The second disc **14.5-355** can rotate 360 degrees relative to the first disc **14.5-351**, minus the angular widths of the second protrusion **14.5-352B** and the second protrusion **14.5-356B**.

[3948] In at least one example, the cam wheel **14.5-312** sits on the second disc **14.5-355**. The cam wheel **14.5-312** can include a channel in which the second disc **14.5-355** is disposed, and the protrusion **14.5-359** of the cam wheel **14.5-312** and the first protrusion **14.5-356A** of the second disc **14.5-355** can be disposed in the channel of the cam wheel **14.5-312**. The cam wheel **14.5-312** is free to rotate, and begins to rotate the second disc **14.5-355** when the protrusion **14.5-359** of the cam wheel **14.5-312** contacts either side of the first protrusion **14.5-356A** of the second disc **14.5-355**. The cam wheel **14.5-312** can rotate 360 degrees relative to the second disc **14.5-355**, minus the angular widths of the protrusion **14.5-359** and the first protrusion **14.5-356A**. Although the protrusion **14.5-359** has been discussed as being disposed on the cam wheel **14.5-312**, the protrusion **14.5-359** can be located on any of the parts that rotate relative to the housing and the center cap **14.5-306**, such as on the spool, the wheel, or the like.

[3949] Any number or variety of components in any of the configurations described herein can be included in the tensioning system and the electronic device. The components can include any combination of the features described herein and can be arranged in any of the various configurations described herein. The structure and arrangement of components of a tensioning system having a roller-and-wedge locking system and an angular restraint described herein, can apply not only to the specific examples discussed herein, but to any number of examples in any combination. Examples of a configuration and operation of a dial cap including a ball detent

mechanism that can be included in the tensioning system are described below, with reference to FIGS. 14.5-8.

[3950] FIGS. 14.5-8 illustrates a partial cross-sectional view of a dial cap 14.5-302. The dial cap 14.5-302 includes a center cap 14.5-306, a wheel 14.5-308, and a ball detent mechanism. The center cap 14.5-306 and the wheel 14.5-308 can be substantially similar to, and can include some or all of, the features of the center cap 14.5-306 and the wheel 14.5-308, respectively, discussed above with respect to FIGS. 14.5-3 through 14.5-6. The ball detent mechanism includes a ball 14.5-370, a detent spring 14.5-372, springs 14.5-374, and a grooved surface 14.5-376 in the wheel 14.5-308. The ball 14.5-370, the detent spring 14.5-372, and the springs 14.5-374 can be contained within protrusions 14.5-315 of the center cap 14.5-306. As described previously, the wheel 14.5-308 can rotate relative to the center cap 14.5-306. As the wheel 14.5-308 rotates, the springs 14.5-374 push the detent spring 14.5-372, which pushes the ball 14.5-370 into the grooves of the grooved surface 14.5-376 in the wheel 14.5-308. The ball detent mechanism generates both a clicking sound and a clicking feel as wheel 14.5-308 turns and the ball 14.5-370 moves between adjacent grooves of the grooved surface 14.5-376. Because the locking mechanism provided by the rollers and the cam wheel is silent, the ball detent mechanism can be included in a tensioning system in order to provide a desired user experience that includes a clicking feel and sound when the wheel 14.5-308 is rotated.

[3951] Any number or variety of components in any of the configurations described herein can be included in the tensioning system and the electronic device. The components can include any combination of the features described herein and can be arranged in any of the various configurations described herein. The structure and arrangement of components of a tensioning system having a roller-and-wedge locking system and an angular restraint described herein, can apply not only to the specific examples discussed herein, but to any number of examples in any combination. Examples of configurations and operations of various angular restraints that can be included in the tensioning system are described below, with reference to FIGS. 14.5-9A through 14.5-9C.

[3952] FIGS. 14.5-9A through 14.5-9C illustrate various angular stop mechanisms, which can be used in place of or in addition to the disc-type angular stop mechanism of FIGS. 14.5-6 through 14.5-7B. In the example of FIGS. 14.5-9A, a cable-type angular stop mechanism is illustrated. A cable 14.5-380 can be provided and attached to a center cap 14.5-306 and a cam wheel 14.5-312. The cable 14.5-380 can be attached to the center cap 14.5-306 and the cam wheel 14.5-312 using an adhesive or the like. The cable 14.5-380 can be provided such that the cable 14.5-380 wraps around a center cap post 14.5-307 of the center cap 14.5-306 as the cam wheel 14.5-312 is rotated relative to the center cap 14.5-306. Once the cable 14.5-380 is wrapped tightly around the center cap post 14.5-307, the cable 14.5-380 prevents the cam wheel 14.5-312 from rotating relative to the center cap 14.5-306, while allowing the cam wheel 14.5-312 to rotate in the opposite direction. The cable 14.5-380 can be a cheap method for providing an angular stop mechanism for the tensioning system. This prevents a user from over-spooling a tensioning system (e.g., spooling more cable into a housing of the tensioning system than the housing is configured to hold), or from loosening the tensioning system too much such that the tensioning system winds cable into the housing in the wrong direction. Although the cable 14.5-380 has been discussed as being attached to the center cap 14.5-306 and the cam wheel 14.5-312, the cable 14.5-380 can be attached to any components of the tensioning system that are desired to include an angular stop. For example, the cable 14.5-380 can be attached to the center cap 14.5-306 and a spool of the tensioning system, a wheel of the tensioning system, or the like. In some examples, the cable 14.5-380 can be attached to a housing of the tensioning system and the cam wheel 14.5-312, the housing and the spool, the housing and the wheel, or the like.

[3953] In the example of FIGS. 14.5-9B, a screw-type angular stop mechanism is illustrated. A helical ridge 14.5-382 is formed on a center cap post 14.5-307 of a center cap 14.5-306 and a

corresponding helical groove **14.5-384** is formed on a cam wheel **14.5-312**. As the cam wheel **14.5-312** rotates relative to the center cap **14.5-306**, the cam wheel **14.5-312** moves up or down the center post **14.5-307** of the center cap **14.5-306**. Vertical stops can be included on the center cap post **14.5-307**, which prevent the cam wheel **14.5-312** from rotating too far in either direction. This prevents a user from over-spooling a tensioning system (e.g., spooling more cable into a housing of the tensioning system than the housing is configured to hold), or from loosening the tensioning system too much such that the tensioning system winds cable into the housing in the wrong direction. Although the helical ridge **14.5-382** has been discussed as being disposed on the center cap post **14.5-307** of the center cap **14.5-306**, the helical ridge **14.5-382** can be disposed on a portion of a housing of a tensioning system or the like. The helical groove **14.5-384** can be disposed on a spool of the tensioning system or the like.

[3954] In the example of FIGS. **14.5-9C**, a spiral-type angular stop mechanism is illustrated. A spiral **14.5-386** is formed in a center cap **14.5-306**. A center cap post **14.5-307** of the center cap **14.5-306** can be formed adjacent the spiral **14.5-386**. The spiral **14.5-386** can be a spiral-shaped groove formed in the center cap **14.5-306**. The spiral **14.5-386** can interact with a protrusion **14.5-388** of a cam wheel **14.5-312**. The protrusion **14.5-388** can be attached to the cam wheel **14.5-312** such that the protrusion can slide relative to the cam wheel **14.5-312**. For example, the protrusion **14.5-388** can slide in a direction along a radius of the cam wheel **14.5-312**. As the cam wheel **14.5-312** rotates relative to the center cap **14.5-306**, the protrusion **14.5-388** slides relative to the cam wheel **14.5-312** along the radius of the cam wheel **14.5-312** closer to or further from a center of the cam wheel **14.5-312** depending on which direction the cam wheel **14.5-312** turns. When the protrusion **14.5-388** reaches either end of the spiral **14.5-386**, the protrusion **14.5-388** and the spiral **14.5-386** prevent the cam wheel **14.5-312** from rotating relative to the center cap **14.5-306**, while allowing the cam wheel **14.5-312** to rotate in the opposite direction. This prevents a user from over-spooling a tensioning system (e.g., spooling more cable into a housing of the tensioning system than the housing is configured to hold), or from loosening the tensioning system too much such that the tensioning system winds cable into the housing in the wrong direction. Although the spiral **14.5-386** has been discussed as being disposed on the center cap **14.5-306**, the spiral **14.5-386** can be disposed on a portion of a housing of a tensioning system or the like. The protrusion **14.5-388** can be disposed on a spool of the tensioning system, a wheel of the tensioning system, or the like.

14.6: Two-Part Speaker System

[3955] The following disclosure relates to speaker assemblies for electronic devices. More specifically, the following disclosure describes a two-part speaker assembly having a single outlet port. Electronic devices typically include a housing that surrounds internal system components, such as audio speaker assemblies, circuitry, processing units, display elements, and other electronic components. However, as electronic devices become smaller and smaller the available space for such components is reduced. Further, it may be desirable to reduce the number of openings or ports visible on the electronic device. The speaker assembly described herein uniquely addresses the challenges that arise from reductions in available space, as well as aesthetic and performance requirements.

[3956] In a particular embodiment, an electronic device, such as a head-mounted device (“HMD”), can include a housing defining an internal volume. The housing can also define an opening that is in fluid communication with the internal volume of the housing and with the ambient environment. A speaker assembly or unit can be positioned within the internal volume of the electronic device. In some examples, the speaker assembly is positioned within a support of an HMD, such as within a support arm or support band positioned along a user's head. The speaker assembly can be positioned proximate a user's ear and can be oriented to direct sound in the direction of the user's ear.

[3957] In some examples, the speaker assembly can include a speaker pod or housing that is positioned within the electronic device. The speaker pod can house various components of the

speaker assembly. The speaker pod can further define a port or outlet that is aligned with the opening of the electronic device. Thus, the port can be in fluid communication with an interior volume of the speaker pod and with the outside environment. The speaker assembly can include multiple audio transducers or speakers.

[3958] The speaker can include a diaphragm coupled to a magnetic component. The magnetic component can include gaps or grooves within which is disposed a wound coil, such as a copper coil or any conductive coil that is capable of being influenced by an electromagnetic field. In some examples, the coil is capable of generating or being influenced by a desired electromagnetic field. The diaphragm can be affixed to the coil. As pulses of electricity pass through the coil, the direction of its magnetic field is rapidly changed, resulting in alternating attraction and repulsion to the magnetic component, causing vibrations back and forth. The coil can be attached to the diaphragm which amplifies these vibrations, pumping sound waves into the surrounding air. Notably, despite having multiple audio transducers, the speaker assembly may include only one outlet or port.

[3959] In one example, the speaker assembly includes a first speaker, such as a woofer, that is positioned within the pod and is in fluid communication with a first channel. The woofer channel can include a winding, tortuous path to the outlet port. A second speaker, such as a tweeter, can also be positioned within the pod and can be in fluid communication with a second channel or tweeter channel. As described in greater detail below, the woofer channel and the tweeter channel can merge, converge, or combine into a shared channel that leads to the port.

[3960] In some examples, the tweeter can at least partially overlap the woofer, thereby reducing the footprint of the assembly along at least one major dimension. This can enable the speaker pod to better fit within an electronic device with space constraints. In some examples, the speaker assembly can include a port barrier positioned at the port. The port barrier can cover the port. The port barrier can include a woven mesh, a metallic screen defining an exterior of the electronic device, and a support frame to reinforce the woven mesh. Likewise, the speaker assembly can include a woofer mesh positioned in a channel leading to the woofer. The woofer mesh can have a higher density than the mesh of the port barrier. In some examples, the woofer mesh can have a lower density than the port barrier mesh. Thus, the speaker assembly can include multiple meshes.

[3961] A challenge that arises from a shared port system as described herein is potential interference of the tweeter from the acoustic waves or sound generated by the woofer. The woofer diaphragm is larger than the tweeter diaphragm, and can therefore negatively impact operation of the tweeter. Accordingly, in some examples, the speaker assembly can include a wall positioned between the tweeter and the woofer channel. The wall can shield the tweeter from the acoustic waves generated by the woofer. In some examples, the shielding wall can at least partially define the one or more of the channels of the speaker assembly and can be shaped to direct the flow of the sound.

[3962] These and other embodiments are discussed below with reference to FIGS. **14.6-1** through **14.6-4**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature comprising at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[3963] It will further be understood that various directional indicators used herein (e.g., up, down, sideways, vertical, horizontal, top, bottom, above, below, etc.) are made in relation to the orientation of the figure at issue. For example, a vertical axis is to be understood as an axis

extending from a top of the page to a bottom of the page.

[3964] FIGS. **14.6-1A** through **14.6-1D** illustrate various electronic devices that are capable of being used in conjunction with the speaker assembly described herein. The electronic devices can correspond to any form of a wearable electronic device, a portable media player, a media storage device, a portable digital assistant (“PDA”), a tablet computer, a computer, a mobile communication device, a GPS unit, a remote-control device, or any other electronic device. Specific examples of such devices are described below.

[3965] FIGS. **14.6-1A** illustrates an electronic device, such as a head-mounted device **14.6-100a** positioned on a user's head **14.6-120**. The head-mounted device can include a display device used for virtual or augmented reality simulations. The head-mounted device **14.6-100a** can further relate to any form of computer glasses or smart glasses. The head-mounted device **14.6-100a** can include a display **14.6-104** and a support **14.6-108**, such as a headband or support arms. In some examples, the display **14.6-104** includes an opaque, translucent, transparent, or semi-transparent screen, including any number lenses, for presenting visual data.

[3966] The head-mounted device **14.6-100a** can be worn on the user's head **14.6-120** such that the display **14.6-104** is positioned over the user's face and disposed over one or both of the user's eyes. The display **14.6-104** can be connected to the support **14.6-108**. In some examples, the support **14.6-108** can be positioned against the side of a user's head **14.6-120** and in contact therewith. In some examples, the support **14.6-108** can be at least partially positioned above the user's ear or ears **14.6-124**. In some examples, the support **14.6-108** can be positioned adjacent to the user's ear or ears **14.6-124**. The support **14.6-108** can extend around the user's head **14.6-120**. In this way, the display **14.6-104** and the support **14.6-108** can retain the head-mounted device **14.6-100a** on the user's head **14.6-120**. It should be understood, however, that this configuration is just one example of how the components of a modular head-mounted device **14.6-100a** can be arranged, and that in some examples, a different number of connector straps and/or retention bands can be included.

[3967] The head-mounted device **14.6-100a** can include a frame or housing that houses various electrical components. Various components of the display **14.6-104** can be housed within the housing. For example, the hardware and electronics which allow functionality of the HMD can be housed within the housing. The housing can be defined by any portion of the support **14.6-108** and/or the display **14.6-104**.

[3968] In some examples, the head-mounted device **14.6-100a** can include a speaker assembly **14.6-112**. The speaker assembly **14.6-112** can be positioned in a housing of the head-mounted device **14.6-100a**. For example, the speaker assembly **14.6-112** can be positioned in or on the support **14.6-108**. The support **14.6-108** can define an opening **14.6-116** that is positioned proximate the speaker assembly **14.6-112** to permit passage of acoustic waves generated by the speaker assembly **14.6-112** to emit into the outside environment. In some examples, when the head-mounted device **14.6-100a** is worn on the user's head **14.6-120**, the speaker assembly **14.6-112** and the opening **14.6-116** are positioned proximate the user's ear **14.6-124**. The support **14.6-108** and/or the opening **14.6-116** can be oriented such that sound can be directed toward the user's ear **14.6-124** when the user **14.6-120** is wearing the head-mounted device **14.6-100a**. Further details regarding electronic devices capable of implementing a single port speaker system are described below with reference to FIGS. **14.6-1B**.

[3969] FIGS. **14.6-1B** illustrates an example in which the electronic device is a wearable device **14.6-100b**. The wearable device **14.6-100b** can be a watch, such as a smartwatch. The wearable device **14.6-100b** of FIGS. **14.6-1B** is merely one representative example of a device that can be used in conjunction with the components and methods disclosed herein. In other words, the wearable device **14.6-100b** is one non-limiting example of a device that can include a single port speaker assembly as described herein.

[3970] The wearable device **14.6-100b** can include an enclosure or housing **14.6-103**. The housing **14.6-103** can be connected to a front display **14.6-105** and can have a strap **14.6-109** designed to

attach the wearable device **14.6-100b** to a user, or to provide wearable functionality. The housing **14.6-103** can define a port **14.6-116** that is in fluid communication with a speaker assembly disposed within the housing **14.6-103**. A number of input elements, such as a rotatable crown and/or a button **14.6-107** can be attached to and can protrude from the housing **14.6-103**. The housing **14.6-103** can define a single speaker port **14.6-116** that can be integrally formed in and defined by the housing **14.6-103**.

[3971] In some examples, the wearable device **14.6-100b** can include a speaker assembly (not shown in FIGS. **14.6-1B**). The speaker assembly can be positioned in the housing **14.6-103** of the wearable device **14.6-100b**. The housing **14.6-103** can define the port **14.6-116** that is positioned in relation to the speaker assembly **14.6-112** to permit passage of acoustic waves generated by the speaker assembly to emit into the outside environment. In some examples, when the wearable device **14.6-100b** is worn by a user, the port **14.6-116** is positioned or oriented to be directed toward the user's ear **14.6-124**. The port **14.6-116** can be positioned such that sound can be directed toward the user's ear when the user is wearing the wearable device **14.6-100b**. Further details regarding electronic devices capable of implementing a single port speaker system are described below with reference to FIGS. **14.6-1C**.

[3972] FIGS. **14.6-1C** illustrates an example in which the electronic device is a computer **14.6-100c**, such as a laptop or desktop. The computer **14.6-100c** of FIGS. **14.6-1C** is merely one representative example of a device that can be used in conjunction with the components and methods disclosed herein. Specifically, the computer **14.6-100c** is one representative example of a device that can include a single port speaker assembly as described herein.

[3973] The computer **14.6-100c** can include an enclosure or housing **14.6-111**. The housing **14.6-111** can be connected to a display **14.6-113**. The housing **14.6-111** can define a port **14.6-116** that is in fluid communication with a speaker assembly housed within the housing **14.6-111**. A number of input members **14.6-115**, such as a keyboard, can be attached to and can protrude from the housing **14.6-111**.

[3974] The housing **14.6-111** can substantially define an internal volume and at least a portion of an exterior surface of the computer **14.6-100c**. The display **14.6-113** can include a touch sensitive surface, such as a touchscreen. The display **14.6-113** can define an exterior surface of the computer **14.6-100c**.

[3975] The housing **14.6-111** can also include features, such as button apertures or charging port apertures. In some examples, the housing **14.6-111** defines only a single speaker port **14.6-116** by the process described herein. The speaker port **14.6-116** can be integrally formed in and defined by the housing **14.6-111**.

[3976] In some examples, the computer **14.6-100c** can include a speaker assembly (not shown in FIGS. **14.6-1C**). The speaker assembly can be positioned in the internal volume of housing **14.6-111** of the computer **14.6-100c**. The housing **14.6-111** can define the port **14.6-116** that is positioned in relation to the speaker assembly to permit passage of acoustic waves generated by the speaker assembly to emit into the outside environment. The port **14.6-116** can be positioned such that sound can be directed toward the user's ear when the user is using the computer **14.6-100c**. Further details regarding electronic devices capable of implementing a single port speaker system are described below with reference to FIGS. **14.6-1D**.

[3977] FIGS. **14.6-1D** illustrates an example in which the electronic device is a mobile device **14.6-100d**, such as a smartphone or tablet computer. The mobile device **14.6-100d** of FIG. **14.6-1D** is merely one representative example of a device that can be used in conjunction with the components and methods disclosed herein. Specifically, the mobile device **14.6-100d** is merely one representative example of a device that can include a single port speaker assembly as described herein.

[3978] The mobile device **14.6-100d** can include an enclosure or housing **14.6-117**. The housing **14.6-117** can be connected to a display **14.6-119**. The housing **14.6-117** can define a port **14.6-116**

that is in fluid communication with a speaker assembly disposed within the housing **14.6-117**.

[3979] The housing **14.6-117** can substantially define at least a portion of an exterior surface of the mobile device **14.6-100d**. The display **14.6-119** can include a touch sensitive surface, such as a touchscreen. The display **14.6-119** can define an exterior surface of the mobile device **14.6-100d**.

[3980] The housing **14.6-117** can also include features, such as button apertures or charging port apertures. In some examples, the housing **14.6-117** defines only a single speaker port **14.6-116** by the process described herein. The speaker port **14.6-116** can be integrally formed in the housing **14.6-117**.

[3981] In some examples, the mobile device **14.6-100d** can include a speaker assembly (not shown in FIGS. **14.6-1C**). The speaker assembly can be positioned in the housing **14.6-117** of the mobile device **14.6-100d**. The housing **14.6-117** can define the port **14.6-116** that is positioned in relation to the speaker assembly **14.6-112** to permit passage of acoustic waves generated by the speaker assembly to emit into the outside environment. The port **14.6-116** can be positioned such that sound can be directed toward the user's ear when the user is using the mobile device **14.6-100d**. Further details regarding speaker assemblies are described below with reference to FIGS. **14.6-2**.

[3982] FIGS. **14.6-2** shows a schematic view of an electronic device **14.6-200** that includes a speaker assembly **14.6-212**. The electronic device **14.6-200** can be substantially similar to, and can include some or all of the features of the electronic devices described herein, such as the head-mounted device **14.6-100a**, the wearable device **14.6-100b**, the computer **14.6-100c**, and the mobile device **14.6-100d**.

[3983] The electronic device **14.6-200** can define an interior volume **14.6-201** to house electronic components. The speaker assembly **14.6-212** can be positioned within the internal volume **14.6-201**. In some examples, an entirety of the speaker assembly **14.6-212** is positioned within the internal volume **14.6-201**. The electronic device **14.6-200** can define an aperture or opening **14.6-216a** that is in fluid communication with the ambient environment and the speaker assembly **14.6-212**.

[3984] The speaker assembly **14.6-212** can include a pod **14.6-213**, which can otherwise be referred to as a speaker assembly housing or enclosure, which defines an internal volume **14.6-215** of the speaker assembly. The pod **14.6-213** can house various electrical and acoustic components. The speaker assembly **14.6-212** can include various acoustic components, such as audio transducers, woofers, tweeters, midrange drivers and any other types of drivers or speaker components.

[3985] In some examples, the speaker pod **14.6-213** includes a woofer **14.6-230** and a tweeter **14.6-234**. The woofer **14.6-230** (also known as a bass speaker or a loudspeaker driver) can be designed to produce low frequency sounds. The woofer **14.6-230** can be an electrodynamic driver. In some examples, the woofer **14.6-230** can produce sound from about 50 Hz to about 2000 Hz.

[3986] The speaker assembly **14.6-212** can include a tweeter **14.6-234**. The tweeter **14.6-234** can also be referred to as a treble speaker or loudspeaker. The tweeter **14.6-234** can be a speaker driver that produces acoustic output at a high frequency range or a range of frequencies higher than the range of frequencies output by the woofer. In some examples, the tweeter **14.6-234** produces sound from about 2 kHz to about 20 kHz. In some examples, the tweeter **14.6-234** can produce sound up to about 100 kHz. The tweeter **14.6-234** can be positioned to be proximate an opening, outlet, or port **14.6-216b** in the speaker pod **14.6-213** that leads to the ambient environment. The port **14.6-216b** can be aligned with or in fluid communication with the opening **14.6-216a** defined by the housing of the electronic device **14.6-200**.

[3987] Notably, in some examples, the speaker assembly **14.6-212** includes only one port **14.6-216b** that vents sounds into the ambient environment. The tweeter **14.6-234** and the woofer **14.6-230** can be stacked vertically. For example, the tweeter **14.6-234** can be positioned over or on top (in relation to the view of FIGS. **14.6-2**) of the woofer **14.6-230**. In some examples, the tweeter **14.6-234** overlaps the woofer **14.6-230** along a horizontal axis such that the woofer **14.6-230** and

tweeter are stacked along a vertical axis. The stacking of the woofer **14.6-230** and the tweeter **14.6-234** can allow the speaker pod **14.6-213** to have a reduced dimension (e.g., along the horizontal axis). As shown, the tweeter **14.6-234** can be positioned closer to the port **14.6-216b** than the woofer **14.6-230**. In other words, a distance between the port **14.6-216b** and the woofer **14.6-230** is greater than a distance between the port **14.6-216b** and the tweeter **14.6-234**.

[3988] The speaker assembly **14.6-212** can include internal walls or barrier components that define one or more channels, passages, or tunnels that are designed to guide sound within the speaker assembly **14.6-212**. In some examples, a woofer channel **14.6-250** can be in fluid communication with an interior of the woofer **14.6-230** to receive sound produced by the woofer **14.6-230** and guide the sound, at least partially, to the port **14.6-216b**. Likewise, a tweeter channel **14.6-251** can be in fluid communication with the tweeter **14.6-234** to receive sound produced by the tweeter **14.6-234** and guide the sound to the port **14.6-216b**.

[3989] As described herein, the pod **14.6-213** can define a single outlet port **14.6-216b**. Thus, in some configurations the woofer channel **14.6-250** and the tweeter channel **14.6-251** converge into a shared channel **14.6-252** leading to the port **14.6-216b**. In at least one example, the woofer channel **14.6-250**, tweeter channel **14.6-251**, and the shared channel **14.6-252** can be formed as a single channel leading to a single port **14.6-216b**. In other words, the woofer channel **14.6-250** can direct primarily low frequency sound generated by the woofer **14.6-230** and the tweeter channel **14.6-251** can direct primarily high frequency sound generated by the tweeter **14.6-234**. Accordingly, the shared channel **14.6-252** can include both high and low frequency acoustic waves produced by both the woofer **14.6-230** and the tweeter **14.6-234**. In some examples, the woofer channel **14.6-250** and the tweeter channel **14.6-251** terminate at the port **14.6-216b**.

[3990] In at least one example, the shared channel **14.6-252** can be removed or reduced in size with the woofer channel **14.6-250** and the tweeter channel **14.6-252** extending to the port **14.6-216b**. In such an example, the woofer channel **14.6-250** can terminate at the tweeter channel **14.6-251** and the tweeter channel can terminate at the port **14.6-216**. In this way, in at least one example, the woofer channel **14.6-250** and the tweeter channel **14.6-251** can join to form a single channel that terminates at a single port **14.6-216b**. As shown in FIGS. **14.6-2**, the port **14.6-216b** and opening **14.6-216** can align to form a single opening or port defined at least in part by an external housing of the device **14.6-200** and visible from the outside environment.

[3991] In some examples, a mesh **14.6-258** can be positioned within or across the woofer channel **14.6-250**. The mesh **14.6-258** can include cloth or fabric (also referred to as grille cloth, acoustic cloth, or speaker mesh) that is designed to allow for sound transmission through the material. The mesh **14.6-258** can aid in acoustic dampening, distortion control, and flow noise control.

[3992] The speaker mesh **14.6-258** can include a predetermined sound transmissibility suitable for sound generated by the woofer **14.6-230**. The woofer **14.6-230** can have a frequency range of about 20 Hz to about 2 kHz. The tweeter **14.6-234** can have a frequency range of about 200 Hz to about 20 kHz. In some examples, the woofer **14.6-230** can have a frequency range as low as about 20 Hz to about 50 Hz, and as high as about 200 Hz to about 2 kHz. The tweeter **14.6-234** can have a frequency range as low as about 200 Hz to about 2 kHz, and as high as about 20 kHz. In some examples, the mesh **14.6-258** can allow for passage of frequencies lower than frequencies produced by the tweeter **14.6-234**. For example, the mesh **14.6-258** can substantially allow sound generated by the woofer **14.6-230** below 2000 Hz to pass there through but substantially prevent or reduce the transmission of frequencies produced by the tweeter **14.6-234** from transferring therethrough. For example, the mesh **14.6-258** can substantially block or prevent sound frequencies above about 2000 Hz from transferring from the tweeter channel **14.6-252** backward into the woofer channel **14.6-250**. The mesh **14.6-258** can at least partially block or prevent sound from the tweeter **14.6-234** from traveling into the woofer channel **14.6-250**. The mesh **14.6-258** can be made from synthetic materials or threads in a woven pattern, or can include a foam. The mesh **14.6-258** can be an air-permeable component.

[3993] The mesh **14.6-258** can be positioned at or near a terminus or outlet **14.6-249** of the woofer channel **14.6-250**. The mesh **14.6-258** can occupy substantially an entire volume defined by the woofer channel **14.6-250**. In some other examples, however, the mesh **14.6-258** may only occupy a portion of the woofer channel **14.6-250**. In use, the mesh **14.6-258** can serve to reduce the velocity of air flowing through the woofer channel **14.6-250**, thereby reducing undesirable flow noise and providing for a clearer acoustic signal to be heard by the user. In some examples, the mesh **14.6-258** can block or prevent particle ingress through the woofer channel **14.6-250**.

[3994] The speaker assembly **14.6-212** can include a port mesh **14.6-260** that is positioned within or across the opening **14.6-216a**, the port **14.6-216b**, and/or the shared channel **14.6-252**. In some examples, the port mesh **14.6-260** is positioned across the tweeter channel **14.6-251**. The port mesh can be substantially similar to the mesh **14.6-258**. However, in at least one example, the port mesh **14.6-260** can have a lower density than the mesh **14.6-258**. The port mesh **14.6-260** can include cloth or fabric that is designed to allow for easy sound transmission through the material. The port mesh **14.6-260** can include a predetermined sound transmissibility suitable for sound generated by the tweeter **14.6-234**. The port mesh **14.6-260** can supplement the mesh **14.6-258** in further dampening the sound. In some examples, the port mesh **14.6-260** can allow for passage of frequencies as high as 20 kHz. The port mesh **14.6-260** can be made from synthetic materials or threads in a woven pattern. As described in greater detail with regard to FIGS. **14.6-4**, a cosmetic steel screen can be combined with the port mesh **14.6-260** to define an exterior of the electronic device **14.6-200**. Further, a frame defined by a series of support ribs can support the port mesh **14.6-260** to prevent damage to the port mesh **14.6-260**.

[3995] The port mesh **14.6-260** can be an air-permeable component. In some examples, the port mesh **14.6-260** can occupy substantially an entire volume defined by at least one of the shared channel **14.6-252**, the port **14.6-216b**, and the opening **14.6-216a**. In some other examples, however, the port mesh **14.6-260** may only occupy a portion of at least one of the shared channel **14.6-252**, the port **14.6-216b**, and the opening **14.6-216a**. In use, the port mesh **14.6-260** can serve to reduce the velocity of air flowing through the shared channel **14.6-252**, thereby reducing undesirable flow noise and providing for a clearer acoustic signal to be heard by a user. In some examples, the port mesh **14.6-260** can block or prevent particle ingress through the shared channel **14.6-252**.

[3996] As noted above and shown in FIGS. **14.6-2**, the mesh **14.6-258** can be disposed between the woofer channel **14.6-250** and the tweeter channel **14.6-252** such that the mesh **14.6-258** aid in acoustic dampening, distortion control, and flow noise control for specific frequencies output by the woofer **14.6-230** without attenuating or otherwise interfering with the passage of sound frequencies passing through the tweeter channel **14.6-252** and out the port mesh **14.6-260**. In this way, the denser mesh **14.6-258** provides the advantages tuned for the sound output by the woofer **14.6-230** and the port mesh is configured to optimally provide dampening, distortion control, and flow noise control of the higher frequencies output by the tweeter **14.6-234** without those same high frequencies having to first passing through the mesh **14.6-258**. As noted above, the port mesh **14.6-260** is also configured to allow the passage of lower frequencies from the woofer **14.6-230** to pass through the port **14.6-216b** and the opening **14.6-216a** without substantially or noticeably attenuating or otherwise interfering with the woofer sound. In this way, with both the mesh **14.6-258** and port mesh **14.6-260** positioned as shown in FIGS. **14.6-2**, both the lower woofer frequencies and the higher tweeter frequencies produced by the speaker assembly **14.6-212** can be optimally controlled for quality as the sound passes through a single shared outlet or aperture, such as the opening **14.6-216a**. Further details regarding speaker assemblies are described below with reference to FIGS. **14.6-3A** through **14.6-3E**.

[3997] FIGS. **14.6-3A** illustrates a top perspective view of an example of a speaker assembly **14.6-312**. The speaker assembly **14.6-312** can be substantially similar to, and can include some or all of the features of the speaker assemblies described herein, such a speaker assembly **14.6-112** and

14.6-212. FIGS. **14.6-3A** is merely one illustrative example of how the speaker assembly **14.6-312** can be designed and is not limited to the depicted design or componentry. As illustrated, the speaker assembly **14.6-312** can include a housing or pod **14.6-313** that houses various components of the speaker assembly **14.6-312**, such as audio transducers. The pod **14.6-313** can include stainless steel. In some examples, the pod **14.6-313** can include a bottom portion (shown in FIGS. **14.6-3A**) and a top portion (not shown). The top and bottom portions of the pod **14.6-313** can substantially surround or encapsulate the components of the speaker assembly **14.6-312**. The bottom portion of the pod **14.6-313** can include a curved or non-planar surface that defines an internal volume, such as internal volume **14.6-215** of FIGS. **14.6-2**. The speaker assembly **14.6-312** can include an elastomeric gasket (not shown in FIGS. **14.6-3A**) positioned around a perimeter of the pod **14.6-313**. The elastomeric gasket can provide a seal between the top portion and bottom portion of the pod **14.6-313**. In some examples, the pod **14.6-313** does not include a top portion, but is instead sealed directly to an interior surface of the housing of the electronic device.

[3998] As illustrated in FIGS. **14.6-3A**, the pod **14.6-313** can house a woofer **14.6-330** and a tweeter **14.6-334**. The tweeter **14.6-334** can be positioned at least partially adjacent to the woofer **14.6-330**. As described herein, the tweeter **14.6-334** can at least partially stack or overlap the woofer **14.6-330** in order to accommodate for a reduced dimension of the pod **14.6-313**.

[3999] The pod **14.6-313** can define the port **14.6-316**, out of which sound from both the woofer **14.6-330** and tweeter **14.6-334** are emitted into the ambient environment. As discussed herein, the pod **14.6-313** can include or define a single outlet port **14.6-316**. The single outlet port **14.6-316** can improve an aesthetic appearance of the electronic device. Further, the single outlet port **14.6-316** can provide the advantage of having fewer entry points for dust, particles, or debris, thereby reducing the likelihood of foreign objects entering the electronic device. The pod **14.6-313** can define grooves **14.6-338** that are shaped to allow for electrical wires to pass into the internal volume of the pod **14.6-313** even when the top portion is secured to the bottom portion. The electrical wires can be in electrical communication with the woofer **14.6-330** and the tweeter **14.6-334**. In some examples, a printed circuit board or PCB (not shown in FIGS. **14.6-3A**) can be secured to a top of the woofer **14.6-330**. The PCB can be secured, at least partially, by protrusions **14.6-342** that anchor the PCB in place above the woofer **14.6-330**. The protrusions **14.6-342** can be helical coils. A heat sink can be positioned on top of or above the PCB. In some examples, the speaker assembly **14.6-312** can include a vent **14.6-346**. The vent **14.6-346** can be a barometric vent that allows for air equalization within the speaker assembly **14.6-312**.

[4000] FIGS. **14.6-3B** shows a cross-sectional side view of the speaker assembly **14.6-312** according to one embodiment. Similarly, FIGS. **14.6-3C** illustrates a perspective cross-sectional view of the speaker assembly **14.6-312** according to one embodiment. The woofer **14.6-330** can include a diaphragm **14.6-332** that oscillates along the vertical axis when an electrical current is provided to the woofer **14.6-330**. Sound produced by the woofer diaphragm **14.6-332** can travel downward in a woofer volume **14.6-349** and can then be guided laterally or horizontally through the woofer channel **14.6-350**. In some examples, at least a portion of the woofer volume **14.6-349** and the woofer channel **14.6-350** can be defined by an interior surface of a bottom wall of the pod **14.6-313**.

[4001] In some examples, the sound produced by the woofer **14.6-330** can take a winding or tortuous path as it travels to the port **14.6-316**. For example, the sound from the woofer **14.6-330** may need to change directions several times in order to reach the port **14.6-316**. A long tortuous path can impact acoustic modes of that path, consequently impacting the output. As used herein, a tortuous path can refer to a winding or snaking path (e.g., non-linear). The tortuous path described herein can be a result of a horizontal interface **14.6-357** between the woofer **14.6-330** and tweeter **14.6-334** with the tweeter stacked or overlapping the woofer **14.6-330** as shown.

[4002] As illustrated in FIGS. **14.6-3B**, the pod **14.6-313** can include an overhang or lip **14.6-315**. A perimeter or footprint of the lip **14.6-315** can be smaller than a perimeter or footprint of the

combined audio components (e.g., woofer **14.6-330**, tweeter **14.6-334**, etc.) housed within the pod **14.6-313**. Accordingly, the system may need to be assembled in a certain order and fashion due to the lip **14.6-315**. For example, the dimensions of the pod **14.6-313** may prevent the woofer **14.6-330** and tweeter **14.6-334** from being simply attached or assembled before being placed into the pod **14.6-313**. Instead, it may be necessary to first position the tweeter **14.6-334** or woofer **14.6-330** within the pod **14.6-313** and then place the remaining components into position. The components of the speaker assembly **14.6-312** may need to be angled or slid horizontally to be set into the correct position within the pod **14.6-313**. The lip **14.6-315** can overlap or overhang a portion of the tweeter **14.6-334**.

[4003] Further, as described herein, modern electronic devices often demand reduced dimensions that require novel solutions to fit the electronics within the housings or enclosures. For example, the electronic devices described herein can include integrated speaker assemblies that maintain a broad frequency range and desirable acoustic performance levels, despite residing in smaller, more compact housings. In some examples, this is achieved by overlapping or stacking the tweeter **14.6-334** and the woofer **14.6-330**.

[4004] The overlapping tweeter **14.6-334** and woofer **14.6-330** configuration increases the compactness of the speaker assembly **14.6-312** and allows for assembly tolerances in the horizontal direction when assembling the speaker assembly **14.6-312**. In addition, in at least one example, the overlapping configuration of the tweeter **14.6-334** and woofer **14.6-330** can result in the woofer channel **14.6-350** and/or tweeter channel **14.6-351** to change directions in a tortuous manner, for example from horizontal to vertical and/or other channel angles. The stacked nature of the speaker assembly **14.6-312** in combination with a single port **14.6-316** may result in a tortuous and winding path for the sound waves. For example, the horizontal interface **14.6-357** allows for horizontal assembly tolerances while causing the woofer channel **14.6-350** to direct upward in order to accommodate for the horizontal interface **14.6-357**. In some examples, the tortuous path can be defined by smooth or curved walls to reduce or eliminate turbulence.

[4005] As shown in FIGS. **14.6-3B**, the horizontal interface **14.6-357** can include contacting or at least overlapping portions of the tweeter **14.6-334** and woofer **14.6-330**, or the contacting or overlapping of certain walls or other components of the tweeter **14.6-334** and woofer **14.6-330**, due to the stacked nature of the tweeter **14.6-334** and the woofer **14.6-330**. Also, as used herein and as viewed in the orientation of FIGS. **14.6-3B**, the term “horizontal” can refer to a plane generally parallel to the planes in which the tweeter diaphragm **14.6-335** and the woofer diaphragm **14.6-332** are disposed. In one or more other examples, the stacked configuration/positioning of the tweeter **14.6-334** and woofer **14.6-330** can form an interface at an angle from the referenced “horizontal” orientation of the diaphragms **14.6-335**, **14.6-332**, such as any angle between horizontal and vertical (with “vertical” referencing an angle, plane, or orientation perpendicular to the diaphragms **14.6-335**, **14.6-332** shown in FIGS. **14.6-3B**). Accordingly, in at least one example, the interface **14.6-357** included surfaces or components of the tweeter **14.6-334** and the woofer **14.6-330** coming together at an angle other than vertical, for example at a 45-degree angle from horizontal or any other angle from horizontal or vertical directions. As noted above, this angled or horizontal interface **14.6-357** can cause the path of sound generated by the woofer **14.6-330** and/or tweeter **14.6-334** to be tortuous or winding as shown in FIGS. **14.6-3B**.

[4006] A sloped or curved ramp **14.6-354** can be positioned and shaped to direct sound from the woofer volume **14.6-349** to the shared volume **14.6-352**. For example, as the sound travels laterally, the curved ramp **14.6-354** guides the sound upward. The curved ramp **14.6-354** can include a smooth transition surface to reduce turbulence. The curved ramp **14.6-354** can include a radius of curvature of about 0.2 cm to 0.3 cm. It will be understood that the ramp **14.6-354** is optional and that in some examples, the sound produced by the woofer **14.6-330** travels in a primarily lateral or horizontal direction to the port **14.6-316**.

[4007] In some examples, a mesh **14.6-358** (similar to the mesh **14.6-258**) can be positioned at a

terminus of the woofer channel **14.6-350**. The mesh **14.6-358** can be positioned at an interface between the woofer **14.6-330** and the tweeter **14.6-334**.

[4008] The speaker assembly **14.6-312** can include a wall **14.6-353**. The wall **14.6-353** can be made from plastic and can be positioned below the tweeter **14.6-334** (e.g., between the tweeter **14.6-334** and the woofer channel **14.6-350**). The wall **14.6-353** can be positioned to shield the tweeter **14.6-334** from the sound produced by the woofer **14.6-330**. In some examples, the wall **14.6-353** can be curved to provide a smooth transition surface to direct upward directed sound laterally toward the port **14.6-316**. In some examples, the wall **14.6-353** is positioned and/or shaped to direct sound from the tweeter **14.6-334** toward the port **14.6-316**. The wall **14.6-353** can at least partially prevent sound from the tweeter **14.6-334** from passing into the woofer channel **14.6-350**. The wall **14.6-353** can at least partially define the tweeter channel **14.6-351** and/or the woofer channel **14.6-350**. In some examples, the wall **14.6-353** can extend substantially across the tweeter **14.6-334**. The wall **14.6-353** can extend to the port mesh **14.6-360**. Thus, the shared channel **14.6-352** can be omitted due to the wall **14.6-353** maintaining a separation of the woofer channel **14.6-350** and the tweeter channel **14.6-351**.

[4009] The tweeter **14.6-334** can include a diaphragm **14.6-335**. The diaphragm **14.6-335** can oscillate or vibrate along a vertical axis. The diaphragm **14.6-335** can oscillate in the same direction(s) as the diaphragm **14.6-332**. The woofer diaphragm **14.6-332** can include a surface area that is approximately 10 times larger than that of the tweeter diaphragm **14.6-335**.

[4010] In some examples, the tweeter **14.6-334** can be positioned closer to the port **14.6-316** than the woofer **14.6-330**. The path taken by sound produced by the tweeter **14.6-334** can be substantially linear, or at least substantially less tortuous than the path taken by sound from the woofer **14.6-330**. In some examples, sound generated by the tweeter **14.6-334** is initially directed in a downward direction but is guided substantially laterally, toward the port **14.6-316**.

[4011] The speaker assembly **14.6-312** can include several interfaces (e.g., horizontal interface **14.6-357**) between the woofer **14.6-330** and the tweeter **14.6-334** or between the speaker components and the pod **14.6-313**. Foams can be positioned between the interfaces of the components and the speaker housing **14.6-313**. Specifically, FIGS. **14.6-3B** shows a foam **14.6-363a** positioned between a sidewall **14.6-364** of the pod **14.6-313** and the tweeter **14.6-334**. Further, a foam **14.6-363b** can be positioned between the interior surface of the pod **14.6-313** and a channel wall **14.6-366**.

[4012] The foams **14.6-363a**, **14.6-363b** can include re-workable properties such as flexibility and malleability to allow free play within the speaker assembly **14.6-312**. The foams **14.6-363a**, **14.6-363b** can be used in addition to or as a replacement for pressure sensitive adhesive (PSA). The foams **14.6-363a**, **14.6-363b** beneficially may not be permanently set, allowing for adjustment and repositioning of the components of the assembly. In some examples, the foams **14.6-363a**, **14.6-363b** can be used in conjunction with an adhesive. Advantageously, the foams between various interfaces of the components and the pod **14.6-313** can take up tolerances within the system. For example, a minimum sealing tolerance can be about 300 microns. The position of the foams **14.6-363a**, **14.6-363b** are merely example locations that foam can reside. It will be understood that foam can be located at various positions or interfaces not shown in FIGS. **14.6-3B** and **14.6-3C**.

[4013] In some examples, the port **14.6-316** is about 30 mm.sup.2 across its cross-sectional face. The size of the port **14.6-316** can depend on the capability of the woofer **14.6-330**. The size of the port **14.6-316** can be driven by a desired particle velocity. For example, particle velocity can be a volume velocity (e.g., the volume of sound waves or air passing an area per unit time) divided by a cross sectional area of the port, where the volume velocity is dictated by the maximum output of the system at low frequencies. The particle velocity can be up to 50 meters per second. In some examples, the particle velocity is 10 meters per second. The port **14.6-316** can include a sleeve **14.6-319** positioned within the port **14.6-316** and extending around a perimeter of the port **14.6-316**. The sleeve **14.6-319** can be made from plastic. The sleeve **14.6-319** can extend through the

port opening **14.6-316** in the pod **14.6-313** and can also extend through the opening in the housing of the electronic device (e.g., opening **14.6-216a** in FIGS. **14.6-2**).

[4014] FIGS. **14.6-3D** shows a top perspective view of a speaker module **14.6-317**. The speaker module **14.6-317** can represent the certain electronics removed from the pod **14.6-313** described above. The speaker module **14.6-317** can include the woofer **14.6-330** and the tweeter **14.6-334**. In some examples, the speaker module **14.6-317** can include a frame **14.6-347**. The frame **14.6-347** can be secured to an interior surface of the pod **14.6-313**. FIGS. **14.6-3E** shows a bottom perspective view of a speaker module **14.6-317**. An underside surface of the frame **14.6-347** can be sealed to a bottom of the pod **14.6-313**. In some examples, the speaker module **14.6-317** is sealed to the pod using an adhesive and/or foam, such as the foam **14.6-363** discussed above with regard to FIGS. **14.6-3B**. The woofer diaphragm **14.6-332**, woofer channel **14.6-350**, and mesh **14.6-358** are also visible from the underside of the speaker module **14.6-317** shown in FIGS. **14.6-3E**. In some examples, a barometric vent **14.6-346** can be positioned as illustrated between the mesh **14.6-358** and the woofer diaphragm **14.6-332**.

[4015] FIGS. **14.6-4** shows a perspective view of a port barrier **14.6-460**. The port barrier **14.6-460** can be substantially similar to, including some or all of the features of, the meshes described herein, such as mesh **14.6-260** and **14.6-360**. The port barrier **14.6-460** can be positioned within or across the opening of the electronic device, the port of the speaker assembly, and/or a shared channel (e.g., shared channel **14.6-252**). The port barrier **14.6-460** can include an external screen **14.6-460a**, a woven mesh **14.6-460b**, and a frame **14.6-460c**.

[4016] The external screen **14.6-460a** can be a metallic structure that defines at least a portion of an exterior surface of the electronic device. In other words, the external screen **14.6-460a** can be a cosmetic feature, visible to a user. In some examples, the external screen **14.6-460a** can provide structural support to the woven mesh **14.6-460b**. The external screen **14.6-460a** can act as a particle or debris barrier to prevent particle ingress into the speaker assembly.

[4017] As discussed in greater detail above, with regard to FIGS. **14.6-3B**, the woven mesh **14.6-460b** can include a cloth or fabric that is designed to allow for easy sound transmission through the material. The speaker port barrier **14.6-460** can include a predetermined sound transmissibility suitable for sound generated by the tweeter **14.6-234**. The port barrier **14.6-460** can partially dampen the sound. In some examples, the woven mesh **14.6-460b** can allow passage of frequencies from 20 Hz to 20 kHz. In some examples, the woven mesh **14.6-460b** can allow passage of frequencies from 2 kHz to 20 kHz. The woven mesh **14.6-460b** can be made from synthetic materials or threads in a woven pattern.

[4018] The port barrier **14.6-460** can be air-permeable. The port barrier **14.6-460** can occupy substantially an entire volume defined by at least one of the shared channel, the port (e.g., port **14.6-216b**, **14.6-316**), and the opening (e.g., opening **14.6-116**, **14.6-216a**). In some other examples, however, the port barrier **14.6-460** may only occupy a portion of at least one of the shared channel **14.6-252**, the port **14.6-216b**, and the opening **14.6-216a**. In use, the port barrier **14.6-460** can serve to reduce the velocity of air flowing through the shared channel **14.6-252**, thereby reducing undesirable flow noise and providing for a clearer acoustic signal to be heard by a user. In some examples, the port barrier **14.6-460** can block or prevent particle ingress through the shared channel **14.6-252**. The port barrier **14.6-460** can have a lower density than the mesh **14.6-258**.

[4019] The frame **14.6-460c** can be positioned interior to the external screen **14.6-460a** and woven mesh **14.6-460b**. In some examples, the woven mesh **14.6-460b** is positioned between or sandwiched between the external screen **14.6-460a** and the frame **14.6-460c**. The frame **14.6-460c** can include a series of support ribs **14.6-461** to strengthen, shield, or provide support the woven mesh **14.6-460b**. In some examples, the frame **14.6-460c** is positioned external to the woven mesh **14.6-460b** (e.g., between the external screen **14.6-460a** and the woven mesh **14.6-460b**). The frame **14.6-460c** can be made from any suitable material, such as plastic or metal.

14.7: Bifurcated Band

[4020] Head-mounted devices include head-mounted support structures that allow the devices to be worn on the heads of users. The head-mounted support structures may include device housings for housing components such as displays that are used for presenting a user with visual content. The head-mounted support structures for a head-mounted device may also include headbands and other structures that help hold a device housing on the face of a user. The headband of a head-mounted device may be adjustable.

[4021] In some embodiments, it may be desirable to incorporate a bifurcated headband that contacts the user's head in multiple locations while the head-mounted device is worn. The headband may include different woven textures on opposite sides of the headband (e.g., on the side in contact with the user's head vs. facing away from the user's head), and may include hair-safe hook and loop fasteners. The headband may have ends with seamless curved webbings. To prevent the headband from slipping with respect to the user's head, stiffeners and/or cords may be included on and/or in the headband. If cords are included in the headband, the headband may have adaptive curvature based on the location of the cords within the headband. As a result, the headband may conform to the user's head.

[4022] FIGS. **14.7-1** is a side view of an illustrative head-mounted electronic device with an adjustable headband. As shown in FIGS. **14.7-1**, head-mounted device **14.7-10** may include head-mounted housing **14.7-12** (sometimes referred to as a main housing, main housing unit, head-mounted support structure, etc.). Housing **14.7-12** may have walls or other structures that separate an interior housing region from an exterior region surrounding housing **14.7-12**. For example, housing **14.7-12** may have walls formed from polymer, glass, metal, and/or other materials. Electrical and optical components may be mounted in housing **14.7-12**. These components may include components such as integrated circuits, sensors, control circuitry, input-output devices, etc.

[4023] To present a user with images for viewing from eye boxes (e.g., eye boxes in which the user's eyes are located when device **14.7-10** is being worn on the users' head such as head **14.7-22** of FIGS. **14.7-1**), device **14.7-10** may include displays and lenses. These components may be mounted in optical modules or other supporting structure in housing **14.7-12** to form respective left and right optical systems. There may be, for example, a left display for presenting an image through a left lens to a user's left eye in a left eye box and a right display for presenting an image to a user's right eye in a right eye box.

[4024] If desired, housing **14.7-12** may have forward-facing components such as cameras, other sensors, and/or a display on front **F** for gathering sensor measurements/other input and/or display information on front **F**. Housing **14.7-12** and may have a soft cushion on an opposing rear side of housing **14.7-12**. The rear of housing **14.7-12** may have openings that allow the user to view images from the left and right optical systems (e.g., when the rear of housing **14.7-12** is resting on front surface **14.7-20F** of the user's head **14.7-22**).

[4025] Device **14.7-10** may have an adjustable strap such as adjustable headband **14.7-26** and, if desired, may have other structures (e.g., an optional over-the-head strap) to help hold housing **14.7-12** on head **14.7-22**. Headband **14.7-26** may have first and second ends coupled, respectively, to the left and right sides of housing **14.7-12**. In the example of FIGS. **14.7-1**, coupling members **14.7-24**, which serve as extensions of housing **14.7-12** (e.g., extend from housing **14.7-12** directly or are attached directly to housing **14.7-12**), are provided on the left and right sides of housing **14.7-12**. Members **14.7-24** may be formed from rigid materials such as rigid polymer and/or other materials and may contain sensors, buttons, speakers, and other electrical components. Hinges and/or other mechanisms may be used to couple members **14.7-24** to housing **14.7-12** or members **14.7-24** may be formed as integral portions of a main housing unit. The ends of headband **14.7-26** may have coupling mechanisms such as openings configured to receive posts or other protrusions **14.7-24P** on members **14.7-24** or other housing structures. These coupling mechanisms allow a user to removably attach headband **14.7-26** to members **14.7-24** and thereby removable attach headband

14.7-26 to housing **14.7-12**. Members **14.7-24** may have elongated shapes of the type shown in FIGS. **14.7-1** and/or other suitable shapes and may sometimes be referred to as rigid straps, rigid coupling members, or power straps.

[4026] Headband **14.7-26** may have a soft flexible portion such as central portion **14.7-30**. Portion **14.7-30** may be formed between two stiffer portions such as end portions **14.7-28** on the left and right ends of headband **14.7-26**. Portions **14.7-28** may be stiffened using embedded polymer stiffeners (e.g., single-layer or multilayer polymer stiffening strips) and/or other stiffening members.

[4027] Portion **14.7-30** may be formed from a stretchable material such as stretchy fabric. Portion **14.7-30** may, as an example, be formed from a band of flat knit fabric that includes stretchable strands of material (e.g., elastomeric strands) and/or which uses a stretchable fabric construction (e.g., a stretchable knit construction). Alternatively, portion **14.7-30** may be formed from a band of woven fabric, which may include stretchable strands of material and/or may use a stretchable fabric construction. Narrowed end portions of the band of knit fabric may, if desired, extend over stiffening members in end portions **14.7-28** (e.g., to ensure that headband **14.7-26** has a uniform external appearance).

[4028] Portion **14.7-30** may include two bifurcated headband portions **14.7-32** and **14.7-34**. In particular, bifurcated headband portions **14.7-32** and **14.7-34** may form two portions of the same headband **14.7-26** (e.g., headband portions **14.7-32** and **14.7-34** may be connected to the same points in portions **14.7-28**). In the illustrative embodiment of FIGS. **14.7-1**, portion **14.7-32** may be attached to protrusions **14.7-24P**, and portion **14.7-34** may extend from portion **14.7-32** at a fixed angle. In other embodiments, headband portions **14.7-32** and **14.7-34** may both be attached to protrusions **14.7-24P**, thereby being able to pivot about protrusions **14.7-24P** individually.

[4029] Headband portions **14.7-32** and **14.7-34** may both contact rear portions of head **14.7-22**. In particular, headband portion **14.7-32** may contact an upper rear portion of head **14.7-22** (e.g., in the upper half of the rear of head **14.7-22**), while headband portion **14.7-34** may contact a lower rear portion of head **14.7-22** (e.g., in the lower half of the rear of head **14.7-22**). By contacting the rear of head **14.7-22** in two different locations, headband portions **14.7-32** and **14.7-34** may secure device **14.7-10** to head **14.7-22** while mitigating the amount of stress applied to any portion of head **14.7-22**.

[4030] The stretchability of headband portion **14.7-30** (and therefore headband portions **14.7-32** and **14.7-34** of headband **14.7-26**) allows headband **14.7-26** be stretched along its length. This allows the length of headband **14.7-26** to be temporarily increased to help a user place headband **14.7-26** over the user's head when a user is donning device **14.7-10**. When headband **14.7-26** is released, the stretchiness and elastic nature of portion **14.7-30** of headband **14.7-26** will help shorten headband **14.7-26** and pull headband **14.7-26** against the user's head so that headband **14.7-26** rests against rear surface **14.7-20R** the user's head.

[4031] Further adjustment of the tension of headband **14.7-26** to secure headband **14.7-26** and device **14.7-10** on the user's head may be provided by tightening doubled-back portions **14.7-33** and **14.7-35** of headband **14.7-26**. In particular, doubled-back portions **14.7-33** and **14.7-35** may pass through adjustment loops **14.7-36** and **14.7-38**, respectively. Doubled-back portions **14.7-33** and **14.7-35** may have hook-and-loop fasteners on an inner surface, allowing doubled-back portions **14.7-33** and **14.7-35** to be secured to headband portions **14.7-32** and **14.7-34**. In this way, headband portions **14.7-32** and **14.7-34** may be tightened or loosened as desired by a user of device **14.7-10**.

[4032] FIGS. **14.7-2A** is an illustrative side view of an outer surface (e.g., the surface that faces away from head **14.7-22** when worn) of headband **14.7-26** in a configuration in which headband **14.7-26** is not attached to housing **14.7-12**. As shown in FIGS. **14.7-2A**, headband **14.7-26** may have a stretchable central portion **14.7-30** that includes portions **14.7-32** and **14.7-34** formed from stretchy woven or knit fabric and may have ends **14.7-28** (e.g., end portions that are stiffened using stiffeners embedded in the fabric)—one of ends **14.7-28** is shown in the side view of FIGS. **14.7-**

2A. One or more openings **14.7-40** may be formed in the ends of headband **14.7-26** in portion **14.7-31**. Openings **14.7-40** may receive posts or other protrusions such as protrusion **14.7-24P** of FIGS. **14.7-1** to secure the left and right ends of headband **14.7-26** to the left and right members **14.7-24** of device **14.7-10**. There may be a single opening **14.7-40** or other attachment mechanism located on each end of headband **14.7-26** or each end of headband **14.7-26** may have two or more openings **14.7-40**. If desired, other attachment mechanism (e.g., magnets, snaps, latches, hook-and-loop fasteners, screws or other fasteners, etc.) may be used in attaching headband **14.7-26** to members **14.7-24** or other portions of the housing of device **14.7-10**.

[4033] As shown in FIGS. **14.7-2A**, portions **14.7-32** and **14.7-34** of headband **14.7-26** may both extend from portion **14.7-31** of headband **14.7-26**. In other words, portions **14.7-32** and **14.7-34** may extend from portion **14.7-31** at a fixed angle to one another. Portion **14.7-31** may be more rigid (e.g., less stretchy) than portions **14.7-32** and **14.7-34**, if desired. For example, portion **14.7-31** may include one or more rigid or semi-rigid layers, such as polymer layers or metal layers.

[4034] By coupling (e.g., weaving, knitting, or otherwise intertwining) portions **14.7-32** and **14.7-34** with portion **14.7-31**, portions **14.7-32** and **14.7-34** may be kept at a constant angle with respect to one another. In other words, portions **14.7-32** and **14.7-34** may remain separated by a fixed angle when worn on head **14.7-22**.

[4035] Headband portions **14.7-32** and **14.7-34** may respectively include hook-and-loop fasteners **14.7-42** and **14.7-44**, which may secure the ends of portions **14.7-32** and **14.7-34** when the end portions are doubled back to tighten headband **14.7-26** (as shown in FIGS. **14.7-1**). Hook-and-loop fasteners **14.7-42** may be, for example, hair-safe hook-and-loop fasteners that prevent hair from being caught in fasteners **14.7-42**.

[4036] The outer surface of headband **14.7-26** may be woven. As shown in FIGS. **14.7-2A**, on the outer side of headband **14.7-26**, headband **14.7-26** may have solid loop weave **14.7-46**. Solid loop weave **14.7-46** may be tightly woven and have a uniform appearance across headband **14.7-26**. The loops in solid loop weave **14.7-46** may provide the loops for hook-and-loop fasteners **14.7-42**.

[4037] FIGS. **14.7-2B** is an illustrative side view of an inner surface (e.g., the surface in contact with head **14.7-22** when worn) of headband **14.7-22** in a configuration in which headband **14.7-26** is not attached to housing **14.7-12**.

[4038] Headband **14.7-26** may have a tab, such as tab **14.7-46**. Tab **14.7-46** may be attached to a latch (or other suitable attachment mechanism) in opening **14.7-40** that attaches headband **14.7-26** to post **14.7-24P**. By pulling on tab **14.7-46**, a user may release the latch from post **14.7-24P**, and headband **14.7-26** may be removed from member **14.7-24**. However, this attachment mechanism of headband **14.7-26** to member **14.7-24** is merely illustrative. In general, headband **14.7-26** may attach to member **14.7-24** in any suitable manner.

[4039] The inner surface of headband **14.7-26** may be woven. As shown in FIGS. **14.7-2B**, on the inner side of headband **14.7-26**, headband **14.7-26** may have ribbed loop weave **14.7-47**. Ribbed loop weave **14.7-47** may provide sufficient friction for headband **14.7-26** to remain in place on head **14.7-22**, while being comfortable on head **14.7-22** (e.g., on a user's hair). Ribbed loop weave **14.7-47** may include loops **14.7-49** and rib portions **14.7-48**. By having ribbed loop weave **14.7-47**, the inner surface of headband **14.7-26** may have less tightly-packed loops than solid loop weave **14.7-46** on the outer surface of headband **14.7-26**, thereby improving the user's comfort.

[4040] Although FIGS. **14.7-2A** and **14.7-2B** show headband **14.7-26** as being woven, this is merely illustrative. In general, headband **14.7-26** may be formed from fabric including strands that are intertwined using knitting, weaving, braiding, and/or other strand intertwining techniques.

[4041] In some embodiments, it may be desirable to create a curved edge that appears seamless at the end of headband **14.7-26**. An illustrative headband strap with a curved edge with a seam that is invisible to the naked eye is shown in FIGS. **14.7-3**.

[4042] As shown in FIGS. **14.7-3**, strap **14.7-50** may be formed from inner woven portion **14.7-52** and webbing **14.7-54**. In particular, webbing **14.7-54** may be sewn, woven, or otherwise coupled to

woven portion **14.7-52** Woven portion **14.7-52** may have a solid loop weave (e.g., solid loop weave **14.7-45** of FIGS. **14.7-2A**) or a ribbed loop weave (e.g., ribbed loop weave **14.7-47** of FIGS. **14.7-2B**), as examples.

[4043] Although portion **14.7-52** has been described as being woven, this is merely illustrative. In general, portion **14.7-52** may be knitted, woven, braided, and/or formed using other strand intertwining techniques.

[4044] Webbing **14.7-54** may have portion **14.7-56** that wraps from one side of inner woven portion **14.7-52** (e.g., the left side of inner woven portion **14.7-52**), across edge **14.7-55** of strap **14.7-50**, to the opposite side of inner woven portion **14.7-52** (e.g., the right side of inner woven portion **14.7-52**).

[4045] A stiffener may be inserted into webbing **14.7-54**, if desired. In the illustrative example of FIGS. **14.7-3**, stiffener **14.7-58** may be inserted into webbing **14.7-54**, such as one side of webbing **14.7-54**. In particular, webbing **14.7-54** may have multiple layers, and stiffener **14.7-58** may be inserted between the multiple layers. Alternatively, webbing **14.7-54** may be formed as a single piece using a flat knitting technique and includes a built-in channel (sometimes referred to as a pocket or cavity), and stiffener **14.7-58** may be inserted into the built-in channel. In general, however, stiffener **14.7-58** may be inserted into webbing **14.7-54** in any desired manner.

[4046] Stiffener **14.7-58** may be formed from a cord, such as a braided cord, or a flexible strip of polymer (e.g., an elastomer such as thermoplastic polyurethane). Stiffener **14.7-58** may be sufficiently flexible to permit the headband to bend and twist, but may not stretch substantially along its length and may therefore sometimes be referred to as a non-stretchable stiffener, non-stretchable member, non-stretchable stiffening structure, etc. Stiffener **14.7-58** may significantly less stretchy and soft than the fabric of strap **14.7-50** and may serve to increase the stiffness and decrease (or eliminate) stretchiness at desired portions along strap **14.7-50**. At the same time, the flexibility of stiffener **14.7-58** may allow strap **14.7-50** to bend around the curvature of a user's head. Stiffener **14.7-58** may be inserted into selected portions of strap **14.7-50** to selectively stiffen strap **14.7-50** at desired portions along its length, if desired.

[4047] Headband **14.7-26** may have rounded corners **14.7-62** of webbing **14.7-54** and may not have visible seams. An illustrative side view of headband **14.7-26** is shown in FIGS. **14.7-4**.

[4048] As shown in FIGS. **14.7-4**, headband **14.7-26** may include layers **14.7-64** and loops **14.7-66** (e.g., in webbing **14.7-54** of FIGS. **14.7-3**). Webbing from one side of headband **14.7-26** (e.g., portion **14.7-56** of FIGS. **14.7-3**) may be bonded to the other side of the webbing at point **14.7-71** (e.g., headband **14.7-26** may have single seam at point **14.7-71** on one side of headband **14.7-26**). As shown in FIGS. **14.7-5**, there may be no visible seam at point **14.7-71**. In this way, a one-sided seam that is not visible to the naked eye may be formed in headband **14.7-26**.

[4049] Although FIGS. **14.7-2A** and **14.7-2B** show portions **14.7-32** and **14.7-34** of headband **14.7-26** extending from common portion **14.7-31**, this is merely illustrative. In general, portions **14.7-32** and **14.7-34** may be coupled and attached to a head-mounted device in any suitable manner, such as attaching portions **14.7-32** and **14.7-34** separately to a common post or wrapping a single band through a ring to double back on itself. An illustrative embodiment of a single band wrapped through a ring is shown in FIGS. **14.7-5**.

[4050] As shown in FIGS. **14.7-5**, headband **14.7-26** may be wrapped through ring **14.7-70** to form portions **14.7-32** and **14.7-34**. Ring **14.7-70** may be a metal ring, a rigid polymer ring, or a ring of other suitable material. To ensure that portions **14.7-32** and **14.7-34** remain at a desired angle with respect to one another when worn by a user, stiffeners **14.7-72** may be added to headband **14.7-26**. Stiffeners **14.7-72** may be formed from rigid polymer, metal, or other suitable material.

[4051] Stiffeners **14.7-72** may locally increase the stiffness of band **14.7-26** in the area around ring **14.7-70**. As a result, portions **14.7-32** and **14.7-34** may maintain the angle between each other about ring **14.7-70** when band **14.7-26** is worn on a user's head. Moreover, stiffeners **14.7-72** may create detents to provide a user of headband **14.7-26** with feedback regarding how tight headband

14.7-26 is on the user's head. In other words, because stiffeners **14.7-72** have to pass through ring **14.7-70** when a user tightens or loosens headband **14.7-26**, stiffeners **14.7-72** may indicate to the user how tight the headband is (e.g., by how many stiffeners the user feels pass through ring **14.7-70**).

[4052] The stiffeners shown in FIGS. **14.7-5** are merely illustrative. In general, stiffeners may be added to headband **14.7-26** in any suitable manner. For example, as shown in illustrative FIGS. **14.7-6A** and **14.7-6B**, stiffeners **14.7-72** may be added with different sizes, pitches, and/or geometries to adjust the stiffness of headband **14.7-26** about ring **14.7-70**. Stiffeners **14.7-72** may be proud of the surface of headband **14.7-26**, or may be formed flush or sub-flush with the surface of headband **14.7-26**. As another illustrative example, as shown in FIGS. **14.7-6C**, cuts **14.7-74** may be made in headband **14.7-26**. Cuts **14.7-74** may selectively modify the stiffness of headband **14.7-26** in different regions, and cuts **14.7-74** may be made so that headband **14.7-26** has more stiffness in the region adjacent to ring **14.7-70** than in other regions. In general, stiffeners in headband **14.7-26** may have any suitable geometries and may help maintain the angle between portions **14.7-32** and **14.7-34** about ring **14.7-70**.

[4053] Although stiffeners **14.7-72** are shown in FIGS. **14.7-5** as being used when headband **14.7-26** extends through ring **14.7-70**, this is merely illustrative. Stiffeners **14.7-72** may be included on a surface of headband **14.7-26** if the portions of headband **14.7-26** are in a fixed position (as shown in FIGS. **14.7-1**, as an example).

[4054] In addition to, or instead of, stiffeners **14.7-72**, headband **14.7-26** may have stiffeners that run along a length of headband **14.7-26**. An illustrative example of a headband with stiffeners of this type is shown in FIGS. **14.7-7**.

[4055] As shown in FIGS. **14.7-7**, portions **14.7-32** and **14.7-34** of headband **14.7-26** may include stiffeners **14.7-76**. Stiffeners **14.7-76** may be the same as stiffener **14.7-58** of FIGS. **14.7-3**, as an example. Stiffeners **14.7-76** may be formed from a cord, such as a braided cord, or a flexible strip of polymer (e.g., an elastomer such as thermoplastic polyurethane). Stiffeners **14.7-76** may be more rigid (e.g., less flexible/stretchy) than the other portions of headband **14.7-26**.

[4056] Stiffeners **14.7-76** may be inserted along one side of portions **14.7-32** and **14.7-34**, as shown in FIGS. **14.7-7**. In particular, because stiffeners **14.7-76** stretch less than the rest of portions **14.7-32** and **14.7-34**, portions **14.7-32** and **14.7-34** may conform to a user's head (e.g., head **14.7-22** of FIGS. **14.7-1**) and support the head-mounted device against the user's face more effectively.

[4057] Stiffeners **14.7-76** may extend entirely along the lengths of portions **14.7-32** and **14.7-34**, or may extend across parts of the lengths of portions **14.7-32** and **14.7-34**. In general, stiffeners **14.7-76** may be coupled to suitable parts of portions **14.7-32** and **14.7-34** to adjust the stiffness of portions **14.7-32** and **14.7-34** across their length as needed.

[4058] Stiffeners **14.7-76** may be inserted into channels within headband **14.7-26**. As shown in illustrative FIGS. **14.7-8**, headband **14.7-26** may include strap **14.7-78** (which may correspond to strap **14.7-50** of FIGS. **14.7-3**) having surfaces **14.7-80** and **14.7-82**. Between surfaces **14.7-80** and **14.7-82**, strap **14.7-78** may have channel **14.7-85** with opening **14.7-84** at an edge of strap **14.7-78**. Stiffener **14.7-76** may be mounted within channel **14.7-85**. In other words, stiffener **14.7-76** may be embedded within strap **14.7-78** (e.g., within headband **14.7-26**).

[4059] Stiffeners **14.7-76** may adjust the stiffness and curvature of headband **14.7-26**. In particular, by adding stiffeners **14.7-76** to different regions of headband **14.7-26**, headband **14.7-26** may curve differently in different regions. An illustrative example of a headband having stiffeners in different regions to create regions with different curvatures is shown in FIGS. **14.7-9A** and **14.7-9B**.

[4060] As shown in FIGS. **14.7-9A**, headband **14.7-26** may be straight when not under tension. In other words, when headband **14.7-26** is not on a user's head or otherwise under tension, headband **14.7-26** may return to a straight configuration. However, when headband **14.7-26** is pulled in opposite directions, headband may curve differently based on the locations of stiffeners **14.7-76**.

[4061] As shown in FIGS. **14.7-9B**, portions **14.7-90** of headband **14.7-26** may curve more than

portions **14.7-92** in response to headband **14.7-26** being pulled in directions **14.7-88** and **14.7-86**. In particular, because stiffeners **14.7-76** are formed in portions **14.7-92**, portions **14.7-92** may be stiffer than regions **14.7-90**, which do not have stiffeners **14.7-76** (in other words, stiffeners **14.7-76** are stiffer than fabric **14.7-78**). As a result, in response to pulling both ends of headband **14.7-26** in directions **14.7-88** and **14.7-86**, headband **14.7-26** may extend in a serpentine pattern.

[4062] In some embodiments, headband **14.7-26** may be straight when not under tension (e.g., FIGS. **14.7-9A**) and may bend to conform to compound curvature (e.g., a user's head) when under tension (e.g., FIGS. **14.7-9B**). However, this is merely illustrative. In general, stiffeners **14.7-76** may be included in any suitable portion of headband **14.7-26** so that headband **14.7-26** bends as desired when under tension.

[4063] Returning to FIGS. **14.7-7**, by forming stiffener **14.7-76** on the bottom edge of portion **14.7-32** and stiffener **14.7-76** on the upper edge of portion **14.7-34**, headband **14.7-26** may conform to a user's head when worn by the user. In other words, headband **14.7-26** may sit flat against the user's head across different curvatures of the head (e.g., headband **14.7-26** may distribute pressure evenly across a non-uniform surface). However, this is merely illustrative. If desired, stiffeners **14.7-76** may be formed on the upper edge of portion **14.7-32** or on the lower edge of portion **14.7-34**, depending on the shape to which headband **14.7-26** needs to conform.

[4064] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

14.8: Over the Head Strap

[4065] Head-mounted devices include head-mounted support structures that allow the devices to be worn on the heads of users. The head-mounted support structures may include device housings that enclose components such as displays. The displays may be used for presenting a user with visual content. The head-mounted support structures for a head-mounted device may also include headbands and other structures that help hold a device housing on the face of a user. The headbands of a head-mounted device may be removable. This allows users to swap different headbands into use to accommodate different head sizes and/or to update the style of headband being used.

[4066] In some embodiments, it may be desirable to attach multiple headbands to a head-mounted device. For example, one detachable headband may extend from the head-mounted support structures around the back of a user's head when worn, and another detachable headband may extend over the user's head. The user of two (or more) headbands may hold the head-mounted device against the user's face and prevent excessive pressure on the face when the device is worn. Both headbands may attach and detach from the same portion of the head-mounted support structures.

[4067] FIGS. **14.8-1** is a side view of an illustrative head-mounted electronic device with multiple detectable headbands. As shown in FIGS. **14.8-1**, head-mounted device **14.8-10** may include head-mounted housing **14.8-12** (sometimes referred to as a main housing, main housing unit, head-mounted support structure, main housing portion, etc.). Housing **14.8-12** may have walls or other structures that separate an interior housing region from an exterior region surrounding housing **14.8-12**. For example, housing **14.8-12** may have walls formed from polymer, glass, metal, and/or other materials. Electrical and optical components may be mounted in housing **14.8-12**. These components may include components such as integrated circuits, sensors, control circuitry, input-output devices, etc.

[4068] To present a user with images for viewing from eye boxes (e.g., eye boxes in which the user's eyes are located when device **14.8-10** is being worn on the users' head), device **14.8-10** may include displays and lenses. These components may be mounted in optical modules **14.8-20** that face towards rear R of device **14.8-10** or may be mounted in other supporting structures in housing **14.8-12** to form respective left and right optical systems. There may be, for example, a left display for presenting an image through a left lens to a user's left eye in a left eye box and a right display

for presenting an image to a user's right eye in a right eye box.

[4069] If desired, housing **14.8-12** may have forward-facing components such as cameras and other sensors on front F for gathering sensor measurements and other input and may have a soft cushion on opposing rear R. Rear R may have openings that allow the user to view images from left and right optical modules **14.8-20** (e.g., when rear R is resting on the front of the user's head).

[4070] Device **14.8-10** may have strap such as headband **14.8-26** and over-the-head headband **14.8-27** to help hold housing **14.8-12** on the user's head. Headbands **14.8-26** and **14.8-27** may have a fixed length or may be adjustable. Headbands **14.8-26** and **14.8-27** may have first and second ends coupled, respectively, to the left and right sides of housing **14.8-12**. In the example of FIGS. **14.8-1**, coupling members **14.8-24**, which serve as extensions of housing **14.8-12**, are provided on the left and right sides of housing **14.8-12**. Members **14.8-24** may be formed from rigid materials such as rigid polymer and/or other materials and may contain sensors, buttons, speakers, and other electrical components. Hinges and/or other mechanisms may be used to couple members **14.8-24** to housing **14.8-12** or members **14.8-24** may be formed as integral portions of a main housing unit.

[4071] The ends of headbands **14.8-26** and **14.8-27** may have coupling mechanisms such as openings, magnets, second posts, or other structures configured to attach to posts **14.8-30** (pins) or other protrusions on members **14.8-24** or other housing structures. In an illustrative configuration, these posts face inwardly towards the user's head and are not visible to people in the vicinity of device **14.8-10** when device **14.8-10** is being worn by the user. Releasable latch mechanisms can be used to help secure the ends of headband **14.8-26** and/or headband **14.8-27** to member **14.8-24**. For example, a first detachable latch may be used to removably couple the left end of headband **14.8-26** to a left post in a left member **14.8-24** on a left side of housing **14.8-12** and a second detachable latch may be used to removably couple the right end of headband **14.8-26** to a right post in a right member **14.8-24** on a right side of housing **14.8-12**. Third and fourth detachable latches may couple respective left and right ends of headband **14.8-27** to the left and right posts. Alternatively or additionally, headband **14.8-27** may be attached to the posts between the head-mounted support structures **14.8-24** and headband **14.8-26** to keep headband **14.8-27** attached to structures **14.8-24** when headband **14.8-26** is attached, and/or may use another structures, such as a magnetic structure or additional post, to attach to posts **14.8-30**.

[4072] If desired, a user may flip the headband over so that the first detachable latch removable couples the end of headband **14.8-26** that was previously coupled to the left post to the right post and so that the second detachable latch removably couples the end of headband **14.8-26** that was previously coupled to the right post to the left post (e.g., the user may flip the left and right sides of the band without flipping the band inside out). Headband **14.8-27** also may have reversible ends, if desired. A user may open and close the latches when housing **14.8-12** is being worn or, in an illustrative configuration that is sometimes described herein as an example, a user may open and close the latches when housing **14.8-12** is not being worn.

[4073] The use of latch-based coupling mechanisms, magnetic structures, and/or or other mechanisms in device **14.8-10** may help allow a user to removably attach headbands **14.8-26** and **14.8-27** to members **14.8-24** and thereby removably attach headbands **14.8-26** and **14.8-27** to housing **14.8-12**. Members **14.8-24** may have elongated shapes of the type shown in FIGS. **14.8-1** and/or other suitable shapes and may sometimes be referred to as rigid straps, rigid coupling members, power straps, head-mounted device housing structures, elongated head-mounted device housing members, elongated housing structures, elongated housing members, or head-mounted device housing members (as examples).

[4074] Headbands **14.8-26** and **14.8-27** may be straps with soft flexible portions and/or rigid portions. As an example, a central portion of headband **14.8-26** and/or headband **14.8-27** may be formed from stretchable fabric. The central portions of headband **14.8-26** and/or headband **14.8-27** may have internal stiffening members, external fabric coverings and or other covering layers, strips of strengthening fabric, stretchable fabric portions (e.g., stretchable knit fabric), cosmetic

coverings, and/or other headband structures.

[4075] Left and right end portions of headband **14.8-26** and/or headband **14.8-27** may be coupled to opposing ends of this central portion. The left and right end portions may, as an example, have stiffening structures (e.g., the left and right end portions may be stiffer than the central stretchable portion). Other types of configuration may be used for headband **14.8-26**, if desired (e.g., arrangements with adjustable tensioning cables, etc.).

[4076] FIGS. **14.8-2** is an illustrative diagram of an end portion of an illustrative headband **14.8-26**, an end portion of an illustrative headband **14.8-27**, and a corresponding end portion of an illustrative member **14.8-24**. Headband **14.8-26**, headband **14.8-27**, and housing structures such as members **14.8-24** may sometimes be collectively referred to as a forming a head-mounted device headband system (headband system) **14.8-22**. As shown in the example of FIGS. **14.8-2**, member **14.8-24** may have a protruding post such as post **14.8-30** (e.g., a post that protrudes out of member **14.8-24** towards a user's head while device **14.8-10** is being worn by the user). Headband **14.8-26** may have a corresponding opening **14.8-32** that is configured to receive post **14.8-30**.

[4077] Post **14.8-30** may protrude inwardly (or, in some embodiments, outwardly) from member **14.8-24**. Post **14.8-30** may be formed from metal, rigid polymer, other materials, and/or combinations of these materials. Member **14.8-24** may have a rigid portion to which post **14.8-30** is attached. This rigid portion may be formed from rigid polymer, metal, fabric, carbon-fiber composite materials, and/or other materials.

[4078] In the illustrative embodiment of FIGS. **14.8-2**, to attach headband **14.8-27** to post **14.8-30**, headband **14.8-27** may include post **14.8-31**. Post **14.8-31** may be formed from metal, rigid polymer, other materials, and/or combinations of these materials. In the present example, post **14.8-31** and post **14.8-30** have elongated shapes when viewed end-on (e.g., rectangular shapes with rounded corners). These elongated shapes may help resist rotational motion between longitudinal axis **14.8-34** of headband **14.8-26** and longitudinal axis **14.8-36** of member **14.8-24**. This helps prevent headband **14.8-26** from slipping up or down along the rear surface of a user's head during use. In general, post **14.8-30** and/or mating opening **14.8-32** may have any suitable shapes (e.g., the shape of post **14.8-30** and/or opening **14.8-32** may be circular, oval, rectangular, triangular, may be a shape with curved edges and/or straight edges, may be a shape with drafted edges to help with alignment and/or insertion, etc.). The use of rectangular shapes with rounded corners and/or other shapes that are elongated is illustrative.

[4079] Post **14.8-31** may be slightly larger than post **14.8-30** to fit on top of post **14.8-30**. If desired, post **14.8-31** may snugly fit onto post **14.8-30** to attach headband **14.8-27** to member **14.8-24**. Additionally or alternatively, post **14.8-31** may include a latch, such as a structure biased with a spring or other suitable latch that engages an opening in the side of post **14.8-30**, a magnet, or another suitable attachment to attach headband **14.8-27** to member **14.8-24**. If a latch is used to attach headband **14.8-27** to post **14.8-30**, tab **14.8-61T** may be attached to the latch, and may be pulled to disengage the latch so that headband **14.8-27** may be released from member **14.8-24**.

[4080] Opening **14.8-32** in headband **14.8-26** may be a through-hole opening with a shape that matches the outline of post **14.8-31**. Through-hole opening **14.8-32** may be formed by cutting or otherwise forming an opening in headband **14.8-26**. The periphery of opening **14.8-32** may be strengthened using a mating pair of ring members, which may sometimes be referred to as a cap and socket and may capture portions of headband **14.8-26**.

[4081] In the present example, post **14.8-31** and opening **14.8-32** have elongated shapes when viewed end-on (e.g., rectangular shapes with rounded corners). Headband **14.8-26** may have a latch such as a structure biased with a spring or other suitable latch, in opening **14.8-32** that engages an opening in the side of post **14.8-31** to attach headband **14.8-26** to member headband **14.8-27**. Tab **14.8-62T** may be attached to the latch, and may be pulled to disengage the latch so that headband **14.8-26** may be released from member **14.8-27**.

[4082] By including post **14.8-31** on headband **14.8-27** to overlap and attach to post **14.8-30**, both

headband **14.8-27** and band **14.8-26** may be attached to member **14.8-24**. An illustrative cross-sectional side view of member **14.8-24**, headband **14.8-26**, and headband **14.8-27** attached together is shown in FIGS. **14.8-3**.

[4083] As shown in FIGS. **14.8-3**, post **14.8-30** on member **14.8-24** may extend into post **14.8-31** of headband **14.8-27**. For example, post **14.8-31** may be a small amount larger than post **14.8-30** (e.g., less than 1 mm larger, less than 2 mm larger, or other suitable size difference) to allow post **14.8-31** to press onto post **14.8-30** and remain attached to post **14.8-30**. Post **14.8-31** may then be inserted into opening **14.8-32** of headband **14.8-26**, thereby attaching headband **14.8-26** to headband **14.8-27** and member **14.8-24**.

[4084] In some embodiments, post **14.8-31** may be attached to post **14.8-30** with a latch, and/or headband **14.8-26** may be attached to post **14.8-31** with a latch. A cross-sectional side view of an illustrative latch that may be used to attach posts **14.8-30** and **14.8-31**, and/or headband **14.8-26** and post **14.8-31** is shown in FIGS. **14.8-4**.

[4085] As shown in FIGS. **14.8-4** system **14.8-22** may include a latching mechanism such as latch **14.8-62**. Latch **14.8-62**, which may sometimes be referred to as a latch mechanism or latch structures, may be opened and closed using magnets, springs, sliding members, toggling members, rotating members such as knobs, buttons, and/or other latch structures that may be manipulated by a user (e.g., a user's fingers). In the example of FIGS. **14.8-4**, the latch of headband **14.8-26** has a movable latch member that engages post **14.8-31** when post **14.8-31** is within opening **14.8-32** and has an associated release mechanism such as release tab **14.8-62T**. Tab **14.8-62T**, which may be formed from a flexible strip of material (e.g., fabric, polymer, and/or other material) may be pulled in direction **14.8-64** by a user (e.g., when a user grasps tab **14.8-62T** between the user's fingers), thereby moving the movable latch member out of engagement with post **14.8-31**. This releases post **14.8-31** and allows post **14.8-31** to be removed from opening **14.8-32**. Magnets, spring structures, and/or other biasing structures may be used in closing the latch. In some embodiments, headband **14.8-27** and headband **14.8-26** may have magnets that facilitate attachment of headband **14.8-27** and headband **14.8-26**. As shown in FIGS. **14.8-4**, for example, post **14.8-31** may have one or more magnets such as magnet **14.8-60**. Magnet **14.8-60** may be used to attract and/or repel corresponding magnets in headband **14.8-26**, which can assist in attaching headband **14.8-26** to headband **14.8-27** and/or can assist in closing latch **14.8-62**.

[4086] Although FIGS. **14.8-4** shows latch **14.8-62** attaching headband **14.8-26** to headband **14.8-27**, latch **14.8-62** may additionally or alternatively be used to attach headband **14.8-27** to member **14.8-24**. In particular, latch **14.8-62** may be formed in post **14.8-31** of headband **14.8-27**, and may be moved in and out of a recess within post **14.8-30** of member **14.8-24** using tab **14.8-61T**. In this way, headband **14.8-27** may be removably coupled to member **14.8-24** with a latch, such as latch **14.8-62**.

[4087] In addition to, or instead of, using a post on headband **14.8-27** to attach headband **14.8-27** to member **14.8-24**, magnets may be used for this attachment. An illustrative embodiment in which headband **14.8-27** is attached to member **14.8-24** using magnets is shown in FIGS. **14.8-5**.

[4088] As shown in FIGS. **14.8-5**, headband **14.8-26** may attach to post **14.8-30** at opening **14.8-32**. For example, headband **14.8-26** may attach to post **14.8-30** in the same manner as headband **14.8-26** attaches to post **14.8-31** in FIGS. **14.8-3**. In particular, a latch in opening **14.8-32** (e.g., latch **14.8-62** of FIGS. **14.8-4**) may move into an opening within post **14.8-30**, removably attaching headband **14.8-26** to member **14.8-24**.

[4089] Headband **14.8-27** may be magnetically attached to post **14.8-30**. In particular, magnet **14.8-33** may be inserted within post **14.8-30**, and magnet **14.8-35** of headband **14.8-27** may be magnetically attached to magnet **14.8-33** of post **14.8-30**. Because post **14.8-30** passes through opening **14.8-32** of headband **14.8-26**, headband **14.8-27** may contact the upper surface of post **14.8-30**, as magnets **14.8-33** and **14.8-35** attract post **14.8-30** and headband **14.8-27** together. An illustrative cross-sectional view of a magnetic attachment of headband **14.8-27** and member **14.8-**

24 is shown in FIGS. **14.8-6**.

[4090] As shown in FIGS. **14.8-6**, member **14.8-24** may have recess **14.8-37** formed in post **14.8-30**. Magnet **14.8-33** may be coupled to post **14.8-30** within recess **14.8-37**. Similarly, headband **14.8-27** may include recess **14.8-39**, and magnet **14.8-35** may be coupled to headband **14.8-27** within recess **14.8-39**. Magnets **14.8-33** and **14.8-35** may attach headband **14.8-27** to member **14.8-24**.

[4091] Headband **14.8-26** may be between member **14.8-24** and headband **14.8-27** when headband **14.8-27** is magnetically attached to member **14.8-24**. In particular, opening **14.8-32** of headband **14.8-26** may surround post **14.8-30**. Headband **14.8-26** may be removably attached to member **14.8-24** with a latch in opening **14.8-32** (as shown in FIGS. **14.8-5**), and/or headband **14.8-26** may remain attached to member **14.8-24** due to headband **14.8-27** being magnetically attached to member **14.8-24**.

[4092] The presence of recesses **14.8-37** and **14.8-39** to accommodate magnets **14.8-33** and **14.8-35** are merely illustrative. In general, one or more magnets may be attached to each of member **14.8-24** and headband **14.8-27** in any suitable manner.

[4093] Instead of, or in addition to, having recesses to accommodate magnets in member **14.8-24** and/or headband **14.8-27**, recesses and protrusions may help attach and align headband **14.8-27** to member **14.8-24**. An illustrative embodiment in which a recess and a protrusion are used for aligning and attaching member **14.8-24** and headband **14.8-27** is shown in FIGS. **14.8-7**.

[4094] As shown in FIGS. **14.8-7**, in addition to having magnets **14.8-35** and **14.8-37** coupling headband **14.8-27** to member **14.8-24**, member **14.8-24** may have recess **14.8-40** in post **14.8-30**, and headband **14.8-27** may have protrusion **14.8-42**. Protrusion **14.8-42** may be inserted within recess **14.8-40** when headband **14.8-27** and member **14.8-24** are attached, which may improve the alignment and/or the attachment of headband **14.8-27** and member **14.8-24**.

[4095] As in the embodiment of FIGS. **14.8-6**, headband **14.8-26** may be between member **14.8-24** and headband **14.8-27** when headband **14.8-27** is magnetically attached to member **14.8-24**. In particular, opening **14.8-32** of headband **14.8-26** may surround post **14.8-30**. Headband **14.8-26** may be removably attached to member **14.8-24** with a latch in opening **14.8-32** (as shown in FIGS. **14.8-5**), and/or headband **14.8-26** may remain attached to member **14.8-24** due to headband **14.8-27** being magnetically attached to member **14.8-24**.

[4096] Although member **14.8-24** is shown as having recess **14.8-40**, and headband **14.8-27** is shown as having protrusion **14.8-42**, this is merely illustrative. Headband **14.8-27** may have a recess in which a protrusion of member **14.8-24** is inserted, if desired.

[4097] In other embodiments, headband **14.8-27** may be attached to member **14.8-24** directly, rather than at post **14.8-30**, such as by wrapping around member **14.8-24** or clipping onto member **14.8-24**. An illustrative embodiment in which headband **14.8-27** is attached to member **14.8-24** directly is shown in FIGS. **14.8-8**.

[4098] As shown in FIGS. **14.8-8**, headband **14.8-27** may be attached to member **14.8-24** with portions **14.8-44** and **14.8-46** that wrap around member **14.8-24**. In particular, portions **14.8-44** and **14.8-46** may extend from the rest of headband **14.8-27**, and may be on either side of post **14.8-30** of member **14.8-24**. In some embodiments, headband **14.8-27** may be slid onto member **14.8-24** at end **14.8-25**, and portions **14.8-44** and **14.8-46** may attach headband **14.8-27** to member **14.8-24**.

[4099] Headband **14.8-26** may attach to post **14.8-30** directly, as opening **14.8-32** of headband **14.8-26** may surround post **14.8-30**. Headband **14.8-26** may be removably attached to member **14.8-24** with a latch in opening **14.8-32** (as shown in FIGS. **14.8-5**) or another mechanism.

[4100] Although FIGS. **14.8-8** shows portions **14.8-44** and **14.8-46** wrapping entirely around member **14.8-24**, this is merely illustrative. If desired, headband **14.8-27** may have a single or multiple portions that clip onto (e.g., extend partially around) edges **14.8-21** and **14.8-23** of member **14.8-24**. In this arrangement, headband **14.8-27** may be detached from member **14.8-24** (e.g., by unclipping the clip-on portions) without removing headband **14.8-26** from post **14.8-31**.

Alternatively or additionally, portion **14.8-46** may extend around and cover end **14.8-25** of member **14.8-24**.

[4101] Instead of having portions of headband **14.8-27** that surround member **14.8-24** or clip onto member **14.8-24**, headband **14.8-27** may include lugs that fit into openings in member **14.8-24** (e.g., headband **14.8-27** and member **14.8-24** may have a lug and socket system to attach headband **14.8-27** to member **14.8-24**). An illustrative example of headband **14.8-27** attached to member **14.8-24** with a lug and socket system is shown in FIGS. **14.8-9**.

[4102] As shown in FIGS. **14.8-9**, headband **14.8-27** may have lugs **14.8-48** that are received within sockets **14.8-50** of member **14.8-24**. Lugs **14.8-48** may be coupled by one or more springs or otherwise biased outward (e.g., into sockets **14.8-50**). To release headband **14.8-27** from member **14.8-24**, sliding member **14.8-52** may be slid within opening **14.8-54** of headband **14.8-27** to retract one of lugs **14.8-50**. Headband **14.8-27** may then be removed from member **14.8-24**. Headband **14.8-27** may be reattached to member **14.8-24** by inserting lugs **14.8-48** into sockets **14.8-50**, which may include retracting one of the lugs by sliding member **14.8-52** within opening **14.8-54**. In this way, headband **14.8-27** may be attached to member **14.8-24** using a lug and socket system.

[4103] Headband **14.8-26** may attach to post **14.8-30** directly, as opening **14.8-32** of headband **14.8-26** may surround post **14.8-30**. Headband **14.8-26** may be removably attached to member **14.8-24** with a latch in opening **14.8-32** (as shown in FIGS. **14.8-5**) or another mechanism.

[4104] In some embodiments, post **14.8-30** of member **14.8-24** may be modified to accommodate both headband **14.8-26** and headband **14.8-27**. For example, headband **14.8-27** may have an additional latch that connects headband **14.8-27** to member **14.8-24**. An illustrative example of attaching headband **14.8-27** to member **14.8-24** with an additional latch is shown in FIGS. **14.8-10**.

[4105] As shown in FIGS. **14.8-10**, headband **14.8-27** may include latch **14.8-56**, and member **14.8-24** may have post **14.8-30** with recess **14.8-66**. Latch **14.8-56** may be formed from metal, rigid polymer, or other material. Latch **14.8-56** may include protruding portion **14.8-68** that attaches to post **14.8-31** within recess **14.8-66**. For example, latch **14.8-56** may be biased toward recess **14.8-66** by a spring, magnet, or other component. If it is desired to remove headband **14.8-27** from member **14.8-24**, a user may push on portion **14.8-60** of latch **14.8-56** so that latch **14.8-56** rotates about hinge **14.8-58**. Protruding portion **14.8-68** may slide out of recess **14.8-66**, and headband **14.8-27** may be removed from member **14.8-24**. To reattach headband **14.8-27**, the user may use portion **14.8-60** to move protruding portion, which may move into position when released by the user. In some embodiments, post **14.8-30** and/or protruding portion **14.8-68** may be beveled so that user may insert headband **14.8-27** into post **14.8-30** without engaging portion **14.8-60** of latch **14.8-56**.

[4106] Headband **14.8-26** may attach to post **14.8-30** directly, as opening **14.8-32** of headband **14.8-26** may surround post **14.8-30**. Headband **14.8-26** may be removably attached to member **14.8-24** with a slidable latch member **14.8-62M** that may be moved with tab **14.8-62T**. As shown in FIGS. **14.8-10**, headband **14.8-26** may be attached to member **14.8-24** between member **14.8-24** and headband **14.8-27**.

[4107] Instead of using latch **14.8-56** to attach headband **14.8-27** to member **14.8-24**, an internal structure may be used. An illustrative example in which headband **14.8-27** is attached to member **14.8-24** using an internal structure is shown in FIGS. **14.8-11**.

[4108] As shown in FIGS. **14.8-11**, headband **14.8-27** may have protruding portion **14.8-70** that is inserted into opening **14.8-72** of post **14.8-30**. Protruding portion **14.8-70** be formed from metal, rigid plastic, or other material. In some embodiments, protruding portion **14.8-70** may be slightly smaller (e.g., less than 1 mm smaller, less than 2 mm smaller, or other suitable difference) than opening **14.8-72**. By inserting protruding portion **14.8-70** into opening **14.8-72**, headband **14.8-27** may remain attached to member **14.8-24**.

[4109] To remove headband **14.8-27** from, or attach headband **14.8-27** to, member **14.8-24**, a user

would merely have to apply enough force to overcome the friction between protruding portion **14.8-70** and opening **14.8-72**.

[4110] If desired, additional material, such as material **14.8-74**, may be added to protruding portion **14.8-70** to further secure headband **14.8-27** to member **14.8-24**. Material **14.8-74** may be, for example, a polymer material, a gasket, or another suitable material. Post **14.8-30** may optionally have recessed portion **14.8-76** that matches the shape of material **14.8-74** to hold protruding portion **14.8-70** in place within opening **14.8-72**.

[4111] To remove headband **14.8-27** from, or attach headband **14.8-27** to, member **14.8-24**, a user would merely have to apply enough force to overcome the friction between material **14.8-74** and the sides of opening **14.8-72** (or the recess **14.8-76** once inserted).

[4112] Another attachment mechanism for coupling headband **14.8-27** to member **14.8-24** is a twist-to-lock system. An illustrative example of a twist-to-lock system is shown in FIGS. **14.8-12**

[4113] As shown in FIGS. **14.8-12**, headband **14.8-27** may have protrusion **14.8-78**. Member **14.8-24** may have post **14.8-30** with opening **14.8-80**. Protrusion **14.8-78** may fit through opening **14.8-80** perpendicularly, and may be twisted into proper position **14.8-82** once protrusion **14.8-78** passes through opening **14.8-80**. Post **14.8-30** may have an at least partially hollow interior to allow protrusion **14.8-78** to swivel within post **14.8-30** into position **14.8-82**. Once protrusion **14.8-78** is in position **14.8-82** within post **14.8-30**, it may be locked into place. A user may detach headband **14.8-27** from member **14.8-24** by twisting headband **14.8-27** so that protrusion **14.8-78** is aligned with opening **14.8-82**. In this way, a twist-to-lock system may attach headband **14.8-27** to member **14.8-24**.

[4114] Although not shown in FIGS. **14.8-12**, headband **14.8-27** and post **14.8-30** may include magnets to further attach headband **14.8-27** to member **14.8-24**. These magnets may be similar to those shown in FIGS. **14.8-5-14.8-7**.

[4115] Moreover, although not shown in FIGS. **14.8-12**, headband **14.8-26** may be between member **14.8-24** and headband **14.8-27** prior to attaching headband **14.8-27** to member **14.8-24** with the twist-to-lock system. In particular, opening **14.8-32** of headband **14.8-26** may surround post **14.8-30**. Headband **14.8-26** may be attached with a latch, such as the latch of FIGS. **14.8-5**, and/or may be held in place by headband **14.8-27** once it is attached to post **14.8-30** with the twist-to-lock system.

[4116] In some embodiments, headband **14.8-27** may be attached to member **14.8-24** and held in place by headband **14.8-26** attaching to member **14.8-24** with a latch. An illustrative example of such an embodiment is shown in FIGS. **14.8-13**.

[4117] As shown in FIGS. **14.8-13**, member **14.8-24** may have post **14.8-30** with shoulder **14.8-84**. Member **14.8-24** may also have pins **14.8-86**, although pins **14.8-86** may be omitted if desired. Headband **14.8-27** may have opening **14.8-88** that is slightly larger (e.g., less than 1 mm larger, less than 2 mm larger, or other suitable difference) than shoulder **14.8-84** of post **14.8-30**. Headband **14.8-27** may be attached to member **14.8-24** by passing post **14.8-30** through opening **14.8-88**, so that headband **14.8-27** surrounds shoulder **14.8-84**.

[4118] To prevent headband **14.8-27** from detaching from member **14.8-24**, headband **14.8-26** may be attached to post **14.8-30**. In particular, opening **14.8-32** of headband **14.8-26** may surround post **14.8-30**, and headband **14.8-26** may be removably attached to member **14.8-24** with a latch in opening **14.8-32** (as shown in FIGS. **14.8-5**) or another mechanism. When the latch is engaged and headband **14.8-26** is attached to member **14.8-24**, headband **14.8-26** will be unable to detach from member **14.8-24**.

[4119] Optional pins **14.8-86** may pass through optional holes **14.8-90** in headband **14.8-27**. Pins **14.8-26** may prevent headband **14.8-27** from moving/rotating with respect to member **14.8-24**. In this way, pins **14.8-26** may maintain headband **14.8-27** in a desired position on the user's head when worn.

[4120] Elastomer **14.8-92** may surround opening **14.8-88**. Elastomer **14.8-92** may be formed from

any desired elastomeric material, and may help prevent headband **14.8-27** from moving with respect to shoulder **14.8-84** or detaching from post **14.8-30**. However, elastomer **14.8-92** may be omitted, if desired.

[4121] Regardless of whether elastomer **14.8-92** or pins **14.8-86** are included in the embodiment of FIGS. **14.8-13**, a cross-sectional side view is shown in FIGS. **14.8-14**. As shown in FIG. **14.8-14**, headband **14.8-27** may surround shoulder portions **14.8-84** of post **14.8-30**. Headband **14.8-26** may then rest on top of shoulders **14.8-84** of post **14.8-30**. In this way, headband **14.8-27** may be held in place against member **14.8-24** by headband **14.8-26**.

[4122] Although not shown in FIGS. **14.8-13** and **14.8-14**, because headband **14.8-27** will be directly on member **14.8-24**, material may be included on member **14.8-24** to keep headband **14.8-27** from moving (e.g., swiveling) when the head-mounted device is worn. For example, a gasket, such as a compressible gasket, elastomeric material, or other material may be formed on the surface of member **14.8-24** that is contacted by headband **14.8-27** when headband **14.8-27** is attached to member **14.8-24**.

[4123] In some embodiments, headband **14.8-27** may be attached to member **14.8-24** with extendable magnets. Illustrative embodiments in which headband **14.8-27** is attached to member **14.8-24** with extendable magnets are shown in FIGS. **14.8-15A-14.8-17B**.

[4124] As shown in FIGS. **14.8-15A**, post **14.8-30** may have magnet **14.8-94** in a stowed (e.g., non-extended) position. Post **14.8-30** may also include opening **14.8-96**, which may accommodate a latch mechanism to attach headband **14.8-26** (e.g., the latch mechanism of FIGS. **14.8-5**). As shown in FIGS. **14.8-15B**, magnet **14.8-94** may extend into an extended position. Magnet **14.8-94** may include optional recess **14.8-98**, which may be used to attach headband **14.8-27** to member **14.8-24**.

[4125] As shown in FIGS. **14.8-16**, headband **14.8-27** may have opening **14.8-100**. Opening **14.8-100** may receive magnet **14.8-94** when it is in the expanded position. Illustrative examples of magnet **14.8-94** expanding and attaching headband **14.8-27** is shown in FIGS. **14.8-17A** and **14.8-17B**.

[4126] As shown in FIGS. **14.8-17A**, headband **14.8-27** may be brought into contact with post **14.8-30** (e.g., with member **14.8-24**). Headband **14.8-27** may have stationary magnet **14.8-102** within opening **14.8-100**. As headband **14.8-27** is moved in direction **14.8-101**, magnet **14.8-102** may attract magnet **14.8-94** in post **14.8-30**. Eventually, as shown in FIGS. **14.8-17B**, magnet **14.8-94** may be moved into opening **14.8-100** of headband **14.8-27** due to the attractive magnetic force of magnet **14.8-102**. Recess **14.8-98** of magnet **14.8-94** may be coupled to portion **14.8-104** of opening **14.8-100**, thereby attaching headband **14.8-27** to post **14.8-30** with magnet **14.8-94**. In this way, an extendable magnet may attach headband **14.8-27** to post **14.8-30** (and to member **14.8-24**).

[4127] Although the embodiments of FIGS. **14.8-2-17B** have described attaching headband **14.8-27** to member **14.8-24** in a variety of ways, this is merely illustrative. In general, headband **14.8-26** or any other headband may be attached to member **14.8-24** using any of the attachments of FIGS. **14.8-2-17B**. Alternatively or additionally, although sometimes described herein in the context of examples where posts **14.8-30** are formed on members **14.8-24** and openings **14.8-32** are formed in headband **14.8-26**, posts **14.8-30** may, if desired, be formed as part of headband **14.8-26** and openings **14.8-32** may be formed in members **14.8-24**.

[4128] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

XV: User Interface

[4129] FIGS. **1-1A** to **14.6-4** illustrate various examples of computer systems that can be used to perform the methods and provide audio, visual and/or haptic feedback as part of user interfaces described herein. In some embodiments, the exemplary computer system includes one or more display generation components (e.g., first and second display assemblies **1-120a**, **1-120b** and/or first and second optical modules **11.1.1-104a** and **11.1.1-104b**) for displaying virtual elements

and/or a representation of a physical environment to a user of the computer system, optionally generated based on detected events and/or user inputs detected by the computer system. User interfaces generated by the computer system are optionally corrected by one or more corrective lenses **11.3.2-216** that are optionally removably attached to one or more of the optical modules to enable the user interfaces to be more easily viewed by users who would otherwise use glasses or contacts to correct their vision. While many user interfaces illustrated herein show a single view of a user interface, user interfaces in an HMD are optionally displayed using two optical modules (e.g., first and second display assemblies **1-120a**, **1-120b** and/or first and second optical modules **11.1.1-104a** and **11.1.1-104b**), one for a user's right eye and a different one for a user's left eye, and slightly different images are presented to the two different eyes to generate the illusion of stereoscopic depth, the single view of the user interface would typically be either a right-eye or left-eye view and the depth effect is explained in the text or using other schematic charts or views. In some embodiments, the computer system includes one or more external displays (e.g., display assembly **1-108**) for displaying status information for the computer system to the user of the computer system (when the computer system is not being worn) and/or to other people who are near the computer system, optionally generated based on detected events and/or user inputs detected by the computer system. In some embodiments, the computer system includes one or more audio output components (e.g., electronic component **1-112**) for generating audio feedback, optionally generated based on detected events and/or user inputs detected by the computer system. In some embodiments, the computer system includes one or more input devices for detecting input such as one or more sensors (e.g., one or more sensors in sensor assembly **1-356**, and/or FIG. **11**) for detecting information about a physical environment of the device which can be used (optionally in conjunction with one or more illuminators such as the illuminators describe in FIG. **61**) to generate a digital passthrough image, capture visual media corresponding to the physical environment (e.g., photos and/or video), or determine a pose (e.g., position and/or orientation) of physical objects and/or surfaces in the physical environment so that virtual objects can be placed based on a detected pose of physical objects and/or surfaces. In at least one example, the virtual objects can represent structural objects, and can be referred to as virtual structural objects, with which the user can interact via detected hand gestures and other body movement. In one example, the virtual structural objects can include virtual buttons, dials, crowns, keyboards, or other physical input mechanisms and devices represented virtually and manipulatable by the user. In some embodiments, the computer system includes one or more input devices for detecting input such as one or more sensors for detecting hand position and/or movement (e.g., one or more sensors in sensor assembly **1-356**, and/or FIG. **11**) that can be used (optionally in conjunction with one or more illuminators such as the illuminators **6-124** describe in FIG. **61**) to determine when one or more air gestures have been performed. In some embodiments, the computer system includes one or more input devices for detecting an input such as one or more sensors for detecting eye movement (e.g., eye tracking and gaze tracking sensors in FIG. **11**) which can be used (optionally in conjunction with one or more lights such as lights **11.3.2-110** in FIG. **10**) to determine attention or gaze position and/or gaze movement, which can optionally be used to detect gaze-only inputs based on gaze movement and/or dwell. A combination of the various sensors described above can be used to determine user facial expressions and/or hand movements for use in generating an avatar or representation of the user such as an anthropomorphic avatar or representation for use in a real-time communication session where the avatar has facial expressions, hand movements, and/or body movements that are based on or similar to detected facial expressions, hand movements, and/or body movements of a user of the device. Gaze and/or attention information is, optionally, combined with hand tracking information to determine interactions between the user and one or more user interfaces based on direct and/or indirect inputs such as air gestures or inputs that use one or more hardware input devices such as one or more buttons (e.g., first button **1-128**, button **11.1.1-114**, second button **1-132**, and or dial or button **1-328**), knobs (e.g., first button **1-128**, button **11.1.1-**

114, and/or dial or button **1-328**), digital crowns (e.g., first button **1-128** which is depressible and twistable or rotatable, button **11.1.1-114**, and/or dial or button **1-328**), trackpads, touch screens, keyboards, mice and/or other input devices. One or more buttons (e.g., first button **1-128**, button **11.1.1-114**, second button **1-132**, and or dial or button **1-328**) are optionally used to perform system operations such as recentering content in a three-dimensional environment that is visible to a user of the device, displaying a home user interface for launching applications, starting real-time communication sessions, or initiating display of virtual three-dimensional backgrounds. Knobs or digital crowns (e.g., first button **1-128** which is depressible and twistable or rotatable, button **11.1.1-114**, and/or dial or button **1-328**) are optionally rotatable to adjust parameters of the visual content such as a level of immersion of a virtual three-dimensional environment (e.g., a degree to which virtual-content occupies the viewport of the user into the three-dimensional environment) or other parameters associated with the three-dimensional environment and the virtual content that is displayed via the optical modules (e.g., first and second display assemblies **1-120a**, **1-120b** and/or first and second optical modules **11.1.1-104a** and **11.1.1-104b**).

Viewpoint/View

[4130] In an augmented reality, mixed reality, or virtual reality environment, a view of a three-dimensional environment is visible to a user. The view of the three-dimensional environment is typically visible to the user via one or more display generation components (e.g., a display or a pair of display modules that provide stereoscopic content to different eyes of the same user) through a virtual viewport that has a viewport boundary that defines an extent of the three-dimensional environment that is visible to the user via the one or more display generation components. In some examples, the region defined by the viewport boundary is smaller than a range of vision of the user in one or more dimensions (e.g., based on the range of vision of the user, size, optical properties or other physical characteristics of the one or more display generation components, and/or the location and/or orientation of the one or more display generation components relative to the eyes of the user). In some embodiments, the region defined by the viewport boundary is larger than a range of vision of the user in one or more dimensions (e.g., based on the range of vision of the user, size, optical properties or other physical characteristics of the one or more display generation components, and/or the location and/or orientation of the one or more display generation components relative to the eyes of the user). The viewport and viewport boundary typically move as the one or more display generation components move (e.g., moving with a head of the user for a head mounted device or moving with a hand of a user for a handheld device such as a tablet or smartphone). A viewpoint of a user determines what content is visible in the viewport, a viewpoint generally specifies a location and a direction relative to the three-dimensional environment, and as the viewpoint shifts, the view of the three-dimensional environment will also shift in the viewport. For a head mounted device, a viewpoint is typically based on a location and direction of the head, face, and/or eyes of a user to provide a view of the three-dimensional environment that is perceptually accurate and provides an immersive experience when the user is using the head-mounted device. For a handheld or stationed device, the viewpoint shifts as the handheld or stationed device is moved, and/or as a position of a user relative to the handheld or stationed device changes (e.g., a user moving toward, away from, up, down, to the right, and/or to the left of the device). For devices that include display generation components with virtual passthrough, portions of the physical environment that are visible (e.g., displayed, and/or projected) via the one or more display generation components are based on a field of view of one or more cameras in communication with the display generation components which typically move with the display generation components (e.g., moving with a head of the user for a head mounted device or moving with a hand of a user for a handheld device such as a tablet or smartphone) because the viewpoint of the user moves as the field of view of the one or more cameras moves (and the appearance of one or more virtual objects displayed via the one or more display generation components is updated based on the viewpoint of the user (e.g., displayed positions and poses of the virtual objects are

updated based on the movement of the viewpoint of the user)). For display generation components with optical passthrough, portions of the physical environment that are visible (e.g., optically visible through one or more partially or fully transparent portions of the display generation component) via the one or more display generation components are based on a field of view of a user through the partially or fully transparent portion(s) of the display generation component (e.g., moving with a head of the user for a head mounted device or moving with a hand of a user for a handheld device such as a tablet or smartphone) because the viewpoint of the user moves as the field of view of the user through the partially or fully transparent portions of the display generation components moves (and the appearance of one or more virtual objects is updated based on the viewpoint of the user).

Air Gestures

[4131] In some embodiments, a gesture includes an air gesture. An air gesture is a gesture that is detected without the user touching (or independently of) an input element that is part of a device (e.g., computer system **101**, one or more input device **125**, and/or hand tracking device **140**) and is based on detected motion of a portion (e.g., the head, one or more arms, one or more hands, one or more fingers, and/or one or more legs) of the user's body through the air including motion of the user's body relative to an absolute reference (e.g., an angle of the user's arm relative to the ground or a distance of the user's hand relative to the ground), relative to another portion of the user's body (e.g., movement of a hand of the user relative to a shoulder of the user, movement of one hand of the user relative to another hand of the user, and/or movement of a finger of the user relative to another finger or portion of a hand of the user), and/or absolute motion of a portion of the user's body (e.g., a tap gesture that includes movement of a hand in a predetermined pose by a predetermined amount and/or speed, or a shake gesture that includes a predetermined speed or amount of rotation of a portion of the user's body).

[4132] In some embodiments, input gestures used in the various examples and embodiments described herein include air gestures performed by movement of the user's finger(s) relative to other finger(s) or part(s) of the user's hand) for interacting with an XR environment (e.g., a virtual or mixed-reality environment), in accordance with some embodiments. In some embodiments, an air gesture is a gesture that is detected without the user touching an input element that is part of the device (or independently of an input element that is a part of the device) and is based on detected motion of a portion of the user's body through the air including motion of the user's body relative to an absolute reference (e.g., an angle of the user's arm relative to the ground or a distance of the user's hand relative to the ground), relative to another portion of the user's body (e.g., movement of a hand of the user relative to a shoulder of the user, movement of one hand of the user relative to another hand of the user, and/or movement of a finger of the user relative to another finger or portion of a hand of the user), and/or absolute motion of a portion of the user's body (e.g., a tap gesture that includes movement of a hand in a predetermined pose by a predetermined amount and/or speed, or a shake gesture that includes a predetermined speed or amount of rotation of a portion of the user's body).

[4133] In some embodiments in which the input gesture is an air gesture (e.g., in the absence of physical contact with an input device that provides the computer system with information about which user interface element is the target of the user input, such as contact with a user interface element displayed on a touchscreen, or contact with a mouse or trackpad to move a cursor to the user interface element), the gesture takes into account the user's attention (e.g., gaze) to determine the target of the user input (e.g., for direct inputs, as described below). Thus, in implementations involving air gestures, the input gesture is, for example, detected attention (e.g., gaze) toward the user interface element in combination (e.g., concurrent) with movement of a user's finger(s) and/or hands to perform a pinch and/or tap input, as described in more detail below.

Air Tap and Air Pinch

[4134] In some embodiments, input gestures that are directed to a user interface object are

performed directly or indirectly with reference to a user interface object. For example, a user input is performed directly on the user interface object in accordance with performing the input gesture with the user's hand at a position that corresponds to the position of the user interface object in the three-dimensional environment (e.g., as determined based on a current viewpoint of the user). In some embodiments, the input gesture is performed indirectly on the user interface object in accordance with the user performing the input gesture while a position of the user's hand is not at the position that corresponds to the position of the user interface object in the three-dimensional environment while detecting the user's attention (e.g., gaze) on the user interface object. For example, for direct input gesture, the user is enabled to direct the user's input to the user interface object by initiating the gesture at, or near, a position corresponding to the displayed position of the user interface object (e.g., within 0.5 cm, 1 cm, 5 cm, or a distance between 0-5 cm, as measured from an outer edge of the option or a center portion of the option). For an indirect input gesture, the user is enabled to direct the user's input to the user interface object by paying attention to the user interface object (e.g., by gazing at the user interface object) and, while paying attention to the option, the user initiates the input gesture (e.g., at any position that is detectable by the computer system) (e.g., at a position that does not correspond to the displayed position of the user interface object).

[4135] In some embodiments, input gestures (e.g., air gestures) used in the various examples and embodiments described herein include pinch inputs and tap inputs, for interacting with a virtual or mixed-reality environment, in accordance with some embodiments. For example, the pinch inputs and tap inputs described below are performed as air gestures.

[4136] In some embodiments, a pinch input is part of an air gesture that includes one or more of: a pinch gesture, a long pinch gesture, a pinch and drag gesture, or a double pinch gesture. For example, a pinch gesture that is an air gesture includes movement of two or more fingers of a hand to make contact with one another, that is, optionally, followed by an immediate (e.g., within 0-1 seconds) break in contact from each other. A long pinch gesture that is an air gesture includes movement of two or more fingers of a hand to make contact with one another for at least a threshold amount of time (e.g., at least 1 second), before detecting a break in contact with one another. For example, a long pinch gesture includes the user holding a pinch gesture (e.g., with the two or more fingers making contact), and the long pinch gesture continues until a break in contact between the two or more fingers is detected. In some embodiments, a double pinch gesture that is an air gesture comprises two (e.g., or more) pinch inputs (e.g., performed by the same hand) detected in immediate (e.g., within a predefined time period) succession of each other. For example, the user performs a first pinch input (e.g., a pinch input or a long pinch input), releases the first pinch input (e.g., breaks contact between the two or more fingers), and performs a second pinch input within a predefined time period (e.g., within 1 second or within 2 seconds) after releasing the first pinch input.

[4137] In some embodiments, a pinch and drag gesture that is an air gesture includes a pinch gesture (e.g., a pinch gesture or a long pinch gesture) performed in conjunction with (e.g., followed by) a drag input that changes a position of the user's hand from a first position (e.g., a start position of the drag) to a second position (e.g., an end position of the drag). In some embodiments, the user maintains the pinch gesture while performing the drag input, and releases the pinch gesture (e.g., opens their two or more fingers) to end the drag gesture (e.g., at the second position). In some embodiments, the pinch input and the drag input are performed by the same hand (e.g., the user pinches two or more fingers to make contact with one another and moves the same hand to the second position in the air with the drag gesture). In some embodiments, the pinch input is performed by a first hand of the user and the drag input is performed by the second hand of the user (e.g., the user's second hand moves from the first position to the second position in the air while the user continues the pinch input with the user's first hand. In some embodiments, an input gesture that is an air gesture includes inputs (e.g., pinch and/or tap inputs) performed using both of the

user's two hands. For example, the input gesture includes two (e.g., or more) pinch inputs performed in conjunction with (e.g., concurrently with, or within a predefined time period of) each other. For example, a first pinch gesture performed using a first hand of the user (e.g., a pinch input, a long pinch input, or a pinch and drag input), and, in conjunction with performing the pinch input using the first hand, performing a second pinch input using the other hand (e.g., the second hand of the user's two hands).

[4138] In some embodiments, a tap input (e.g., directed to a user interface element) performed as an air gesture includes movement of a user's finger(s) toward the user interface element, movement of the user's hand toward the user interface element optionally with the user's finger(s) extended toward the user interface element, a downward motion of a user's finger (e.g., mimicking a mouse click motion or a tap on a touchscreen), or other predefined movement of the user's hand. In some embodiments a tap input that is performed as an air gesture is detected based on movement characteristics of the finger or hand performing the tap gesture movement of a finger or hand away from the viewpoint of the user and/or toward an object that is the target of the tap input followed by an end of the movement. In some embodiments the end of the movement is detected based on a change in movement characteristics of the finger or hand performing the tap gesture (e.g., an end of movement away from the viewpoint of the user and/or toward the object that is the target of the tap input, a reversal of direction of movement of the finger or hand, and/or a reversal of a direction of acceleration of movement of the finger or hand).

Attention

[4139] In some embodiments, attention of a user is determined to be directed to a portion of the three-dimensional environment based on detection of gaze directed to the portion of the three-dimensional environment (optionally, without requiring other conditions). In some embodiments, attention of a user is determined to be directed to a portion of the three-dimensional environment based on detection of gaze directed to the portion of the three-dimensional environment with one or more additional conditions such as requiring that gaze is directed to the portion of the three-dimensional environment for at least a threshold duration (e.g., a dwell duration) and/or requiring that the gaze is directed to the portion of the three-dimensional environment while the viewpoint of the user is within a distance threshold from the portion of the three-dimensional environment in order for the device to determine that attention of the user is directed to the portion of the three-dimensional environment, where if one of the additional conditions is not met, the device determines that attention is not directed to the portion of the three-dimensional environment toward which gaze is directed (e.g., until the one or more additional conditions are met). Further details of the user interface and user experience are provided below with reference to FIG. 15-1.

[4140] In FIG. 15-1, electronic device 15.1-700 detects that an interpupillary distance (IPD) setting of electronic device 15.1-700 does not match an interpupillary distance of a user of electronic device 15.1-700 (e.g., as detected, estimated, and/or determined by electronic device 15.1-700). For example, in some embodiments, electronic device 15.1-700 is a head-mounted system that includes two or more optical components (e.g., two or more optical lenses and/or two or more display generation components such as, for example, two or more transparent display generation components), with a first optical component positioned in front of a first eye of the user and a second optical component positioned in front of a second eye of the user. In some embodiments, the IPD setting of electronic device 15.1-700 corresponds to the distance between the two optical components, and the IPD setting is adjustable to move the two optical components further apart or closer together so that they are positioned correctly relative to the two eyes of the user (e.g., the IPD setting is adjustable to move the two optical components further apart or closer together until the IPD setting of electronic device 15.1-700 matches the interpupillary distance of the user). At FIG. 15-1, electronic device 15.1-700 detects the interpupillary distance of the user (e.g., via sensors 15.1-707), and determines that the IPD setting of electronic device 15.1-700 does not match the interpupillary distance of the user (e.g., the two optical components should be moved

closer together or further apart). In FIG. 15-1, in response to detecting that the IPD setting of electronic device 15.1-700 does not match the interpupillary distance of the user (and, optionally, in response to determining that electronic device 15.1-700 does not satisfy the one or more error conditions), electronic device 15.1-700 displays user interface 15.1-712. In some embodiments, when electronic device 15.1-700 detects that the IPD setting of electronic device 15.1-700 matches the interpupillary distance of the user (e.g., the two optical components of electronic device 15.1-700 are positioned in a predetermined position, or within a predetermined range of positions, relative to the two eyes of the user), electronic device 15.1-700 forgoes display of user interface 15.1-712 (and, optionally, displays a different user interface).

[4141] In FIG. 15-1, user interface 15.1-712 includes object 15.1-714, which is representative of electronic device 15.1-700, and objects 15.1-716a-15.1-716b, which are representative of buttons 15.1-706a and 15.1-706b, respectively. User interface 15.1-712 also includes objects 15.1-718a-15.1-718c. In some embodiments, one or more of objects 15.1-718a-15.1-718c are representative of physical components of electronic device 15.1-700. For example, in some embodiments, object 15.1-718a is representative of a first optical component (e.g., a first optical lens and/or a first display generation component), and object 15.1-718b is representative of a second optical component (e.g., a second optical lens and/or a second display generation component). As will be described and shown in greater detail below, in some embodiments, object 15.1-718a and object 15.1-718b move relative to one another on display 15.1-702 to indicate movement of the corresponding physical components of electronic device 15.1-700. User interface 15.1-712 also includes arrow 15.1-720 and prompt 15.1-722, which instruct the user to hold down button 15.1-706b (represented in user interface 15.1-712 by object 15.1-716b) in order to align one or more physical components of electronic device 15.1-700 (e.g., in order to adjust the IPD setting of electronic device 15.1-700 and/or to move one or more optical components of electronic device 15.1-700). In FIG. 15-1, while the IPD setting of electronic device 15.1-700 is not being adjusted (e.g., while a user does not provide user input to button 15.1-706a and/or button 15.1-706b), electronic device 15.1-700 outputs audio output 15.1-711a. In some embodiments, audio output 15.1-711a is an ambient audio output that indicates that the IPD setting of the electronic device 15.1-700 is not currently being modified and/or changed. At FIG. 15-1, electronic device 15.1-700 detects user input 15.1-724, which is a button press of button 15.1-706b. In FIG. 15-1, user input 15.1-724 is a button press of button 15.1-706b. However, in some embodiments, user input 15.1-724 is a different type of user input (e.g., a gesture, an air gesture, a touch input, a rotation input, a gaze-based input, and/or any combination of the foregoing). In addition, the device 15.1-700 can include other buttons, including button 15.1-704, which can also be pressed, turned, or otherwise actuated to perform similar or different functions to those described above with reference to buttons 15.1-706a-b. Buttons 15.1-706a-b and 15.1-704 can include mechanically depressible buttons, touch sensitive buttons such as capacitive sensing buttons, twistable/rotatable dials such as the crowns or buttons described elsewhere herein, or any other type of button.

[4142] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 15-1 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown elsewhere and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to figures shown elsewhere can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 15-1.

[4143] As illustrated in FIG. 15-2A, while displaying mode control affordance 15.2-746 having photo mode affordance 15.2-746A and video mode affordance 15.2-746B, with video mode affordance 15.2-746B highlighted and shutter affordance 15.2-718 with the color fill, computer system 15.2-700 detects a sixth potential media capture input, such as button press input 15.2-748A of hardware button 15.2-706, air gesture input 15.2-748B, and/or tap input 15.2-748C while gaze

15.2-732 is directed at shutter affordance **15.2-718** and accordingly initiates media capture. As the video media capture mode is currently selected, computer system **15.2-700** begins capturing video media. The computer system **15.2-700** can include an electronic device **15.2-702**, which in one example can be a part of a head-mountable device or other wearable electronic device including a display screen displaying images. The device **15.2-700** can include a border **15.2-714** of a housing or other structural frame, a user interface **15.2-710** displayed on a display **15.2-708**. The device **15.2-702** can be configured to detect eye gaze and hand **15.2-726** gestures for controlling the user interface **15.2-710**. The user interface **15.2-710** of the display **15.2-708** can include indicators **15.2-722A** and **15.2-722B** as well as darkened area **15.2-716**. The display **15.2-708** can be a user facing display of the HMDs described herein. Finger **15.2-724** can be used by the user to depress any number of buttons of the HMD to input commands to alter the user interface **15.2-710** as described herein.

[4144] As illustrated in FIG. **15-2B**, while capturing video media, computer system **15.2-700** ceases display of mode control affordance **15.2-746** and instead displays video status affordance **15.2-750** in the lower region of camera viewfinder **15.2-712**. Video status affordance **15.2-750** indicates that video media is currently being captured and includes the currently elapsed time of the video capture. Upon beginning the video capture, computer system **15.2-700** initially displays video status affordance **15.2-750** opaquely. After a threshold amount of time (e.g., a few seconds) without detecting gaze **15.2-732** directed at video status affordance **15.2-750**, computer system **15.2-700** increases the translucency (e.g., reduces the opacity) of video status affordance **15.2-750** (e.g., as described above with respect to changing the visual prominence of captured media icon **15.2-738** and/or mode control affordance **15.2-746**). In response to detecting gaze **15.2-732** directed at video status affordance **15.2-750**, computer system **15.2-700** decreases the translucency (e.g., increases the opacity) of video status affordance **15.2-750**.

[4145] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **15-2A** and FIG. **15-2B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown elsewhere and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to figures shown elsewhere can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **15-2A** and FIG. **15-2B**.

[4146] FIG. **15-3A** and FIG. **15-3B** show an example of a computer system **15.3-101**, which can include a device such as an HMD. The HMD can include rearward facing displays or display screens, similar to the display showing the three-dimensional environment **15.3-702** illustrated in FIG. **15-3A** and **15-3B**. The device can include one or more image sensors **314**, similar to other sensors and cameras of HMDs shown and described elsewhere herein. The display can show various virtual objects **15.3-a-15.3-h**, **15.3-706**, **15.3-708**, **15.3-732a** and **15.3-732c**, as well indicators for thresholds **15.3-734a-d** and **15.3-738a-b**. The display can be configured to project light toward a user's eyes when donning the HMD and forward facing cameras and sensors can capture the environment external to the device and the display can display those images, as well as various virtual objects, to the user through the display projecting the various virtual objects noted herein and shown in the figures.

[4147] In some embodiments, if the attention input indicates that the attention of the user has changed away from the gaze virtual object (or corresponding virtual object) and subsequently moved back to the gaze virtual object (or corresponding virtual object), the computer system **15.3-101** updates the visual indication of progress of the attention of the user directed towards satisfying the one or more criteria for performing an operation associated with a corresponding virtual object based on whether the attention input moved back to the object within a threshold time (e.g., 0.03, 0.5, 0.07, 0.09, 0.1, 0.15, 0.2, 0.25, 0.3, 0.5, 1, 3, 5, 10, 15, 20, or 30 seconds). For example, in FIG. **15-3A**, the visual indication of progress in the gaze target for object **15.3-704a** is maintained

(e.g., continued from the amount of fill prior to the attention of the user changed away) because the attention of the user (e.g., attention input **15.3-716**) changed from being directed away from the gaze virtual object **15.3-704a'** of virtual object **15.3-704a** to being directed to the gaze virtual object **15.3-704a'** of virtual object **15.3-704a** within the time threshold. In some embodiments, the computer system **15.3-101** updates the visual indication of progress, different from maintaining the indication of progress as will be described in the following figures.

[4148] In some embodiments, if the attention input includes an activation input, the computer system **15.3-101** performs the operation associated with the virtual object. For example, in FIG. **15-3A**, the computer system **15.3-101** detects an activation input (e.g., finger of hand **15.3-710** touching trackpad **15.3-746** and/or an air pinch gesture from hand **15.3-710**) before attention **15.3-716** satisfies the one or more criteria with respect to the gaze virtual object **15.3-704a'** for object **15.3-704a** for performing the operation associated with the virtual object; for example, before the duration of the attention input **15.3-716** directed towards the gaze virtual object **15.3-704a'** associated with virtual object **15.3-704a** reaches threshold **15.3-734d** as shown by timer **15.3-734** in FIG. **15-3A**. In response to detecting the activation input before satisfying the one or more criteria (e.g., reaching threshold **15.3-734d**), the computer system **15.3-101** optionally performs the operation associated with the virtual object **15.3-704a** such as updating the right side of virtual object **15.3-704** to include content associated with virtual object **15.3-704a** as shown in FIG. **15-3B**.

[4149] In some embodiments, in response to detecting that recent interactions with computer system include attention-only inputs or non-attention inputs, the computer system **15.3-101** optionally changes (e.g., shortens or lengthens) the threshold requirement to display gaze virtual objects. For example, after detecting that the attention input includes attention-only inputs (e.g., does not include non-attention inputs, such as inputs from hand **15.3-710**), the computer system **15.3-101** shortens the time threshold for displaying a gaze virtual object for a virtual object to **15.3-732b'**, as shown in FIG. **15-3A**. In another example, after detecting that recent interaction with computer system **15.3-101** includes non-attention inputs, the computer system **15.3-101** optionally lengthens the time threshold for displaying a gaze virtual object for a virtual object to **738b**, as shown in FIG. **15-3B**.

[4150] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **15-3A** and **15-3B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown elsewhere and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to figures shown elsewhere can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **15-3A** and **15-3B**.

[4151] FIG. **15-4A** and FIG. **15-4B** show first and second display generation components **15.4-7100** and **15.4-7102**, which can include cameras or sensors **15.4-7104** and **15.4-7106**, respectively. The display generation components **15.4-7100** and **15.4-7102** be or include head mountable display devices operated in different locations **15.4-7000-a** and **15.4-7000b**, as shown in FIGS. **15-4A** and **15-4B**. In at least one example, the HMDs can also include one or more buttons **15.4-7302** to control the displayed content or other functionalities of the device, as described elsewhere herein. The button **15.4-7302** can include a dial. In at least one example, the display generation components **15.4-7100** and **15.4-7102** can display mixed/virtual reality (MR, VR, or XR) content **15.4-7002**. Various virtual objects, including a progress bar **15.4-7004** and overlay **15.4-7008** can be displayed as shown.

[4152] In some embodiments, the computer system detects whether criteria for changing the current level of immersion are met, while displaying the XR content via the first display generation component **15.4-7100** with a respective level of immersion. In response to detecting that the criteria for changing the current level of immersion are met, the computer system changes the level of

immersion with which the XR content is displayed via the first display generation component **15.4-7100**, and changes the appearance of the graphical elements that represent the status associated with the user shown via the second display generation component **15.4-7102**. For example, as shown in FIGS. **15-4A-15-4B**, in some embodiments, the computer system determines that the criteria for changing the level of immersion are met in accordance with detection of actions of the second user **15.4-7204** that is present in the physical environment (e.g., in the location B **15.4-7000-b** in FIG. **15-4A**). For example, in some embodiments, detecting the actions of the second user **15.4-7204** that cause the computer system to change the current level of immersion (e.g., when the current level of immersion is a fully immersive level, or more than a threshold level of immersion) includes detecting that the second user **15.4-7204** has moved into a region in front of the second display generation component **15.4-7102** (e.g., also the region in front of the first user **15.4-7202** if the first user is wearing the HMD including the first display generation component **15.4-7100** and the second display generation component **15.4-7102**). In some embodiments, detecting the actions of the second user **15.4-7204** that cause the computer system to change the current level of immersion (e.g., when the current level of immersion is a fully immersive level, or more than a threshold level of immersion) includes detecting that the second user **15.4-7204** is making a gesture (e.g., waving to the first user **15.4-7202**, tapping on the shoulder or arm of the first user **15.4-7202**, or other gestures that calls for the attention of the first user **15.4-7202**) to the first user **15.4-7202**. As shown in FIG. **15-4A**, the second user **15.4-7204** has moved to a region in front of the second display generation component **15.4-7102**, is waving to the first user **15.4-7202**, and/or is calling out the name of the first user **15.4-7202**; and upon detecting one or more of the above, the computer system determines that the criteria for changing the current level of immersion are met, in accordance with some embodiments. In Figure M, in response to detecting that the criteria for changing the current level of immersion are met, the computer system changes the way that XR content is displayed via the first display generation component in accordance with the change of the current level of immersion, in accordance with some embodiments. In this example scenario shown in FIGS. **15-4A-15-4B**, the computer system decreases the level of immersion from a fully immersive level to an intermediate level of immersion, and displays a representation of the physical environment (e.g., including a representation **15.4-7312** of the physical object **15.4-7314** in the physical environment (e.g., in location B **15.4-7000-b**), and a representation **15.4-7010** of the second user **15.4-7204**) in front of the first user (e.g., in location B **15.4-7000-b**) among the XR content displayed via the first display generation component **15.4-7100**. In addition, when the computer system decreases the level of immersion from a fully immersive level to an intermediate level of immersion, the computer system also displays the representation **15.4-7006** of the portion of the body of the first user **15.4-7202** via the second display generation component **15.4-7102**, in accordance with some embodiments. The representation **15.4-7006** of the portion of the body of the first user **15.4-7202** are updated in accordance with the changes in appearance of the first user **15.4-7202**, and the graphical elements that indicate the status of the XR content and the current level of immersion are updated in accordance with the changes in the status of the XR content and the level of immersion.

[4153] In at least one example, the computer system can vary a level of engagement between the user and the user interface, including changing the level at which a user can engage with (e.g., manipulate, interact with, move . . . etc.) a virtual object displayed to the user. In at least one example, the computer system can vary a level of engagement between the user and the user interface, including changing the level at which a user can engage with (e.g., manipulate, interact with, move . . . etc.) a 3-D space or 3-D environment, including virtual 3-D spaces displayed to the user. In one example, the levels of engagement between the user and the virtual objects and/or 3-D spaces can be automatically changed based on detected facial features, hand gestures, gaze direction, and so forth. In one example, the levels of engagement between the user and the virtual objects and/or 3-D spaces can be manually adjusted using the buttons, dials, crowns, and other

physical interactions between the user and the device having the computer system described herein. [4154] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. 15-4A and 15-4B can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown elsewhere and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to figures shown elsewhere can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 15-4A and 15-4B.

[4155] FIGS. 15-5A-15-5C illustrate examples of setting levels of immersion for operating system user interfaces as compared with application user interfaces. FIG. 15-5A illustrates an electronic device 15.5-101 displaying, via a display generation component (e.g., a display screen or generation component of a head mountable device), a three-dimensional environment 15.5-901 in a user interface. The electronic device 15.5-101 optionally includes a display generation component (e.g., a touch screen) and a plurality of image sensors (e.g., image sensors including cameras of an HMD described herein). The image sensors optionally include one or more of a visible light camera, an infrared camera, a depth sensor, or any other sensor the electronic device 15.5-101 would be able to use to capture one or more images of a user or a part of the user while the user interacts with the electronic device 15.5-101. In some embodiments, the user interfaces shown below could also be implemented on a head-mounted display that includes a display generation component that displays the user interface to the user and sensors to detect the physical environment and/or movements of the user's hands (e.g., external sensors facing outwards from the user), and/or gaze of the user (e.g., internal sensors facing inwards towards the face of the user). [4156] As shown in FIG. 15-5A, device 15.5-101 captures one or more images of the physical environment 15.5-900 around device 15.5-101, including one or more objects in the physical environment 15.5-900 around device 15.5-101. In some embodiments, device 15.5-101 displays representations of the physical environment in three-dimensional environment 15.5-901. For example, three-dimensional environment 15.5-901 includes a representation 15.5-902b of a tree (e.g., corresponding to tree 15.5-902a in physical environment 15.5-900) and a representation 15.5-904b of a person (e.g., corresponding to person 15.5-904a in physical environment 15.5-900). In some embodiments, device 15.5-101 also displays one or more virtual objects (e.g., objects that are not in the physical environment 15.5-900) in the three-dimensional environment 15.5-901. For example, in FIG. 15-5A, device 15.5-101 is displaying virtual object 15.5-910 (e.g., an ornament) on representation 15.5-902b of a tree, virtual object 15.5-906 (e.g., a hat) on representation 15.5-904b of a person, and representation 15.5-908 of the sun. In some embodiments, representation 15.5-908 of the sun is displayed and treated as a light source in the three-dimensional environment 15.5-901, and the lighting effects resulting from representation 15.5-908 of the sun are applied, by device 15.5-101, to representations of physical objects and/or virtual objects displayed in three-dimensional environment 15.5-901.

[4157] In FIG. 15-5A, device 15.5-101 is also displaying a user interface 15.5-912 of the operating system of the device 15.5-101. User interface 15.5-912 is optionally an application browsing user interface of the operating system of the device 15.5-101 from which one or more applications that are accessible on device 15.5-101 can be displayed/launched. In some embodiments, user interface 15.5-912 is any other user interface of the operating system of the device 15.5-101 (e.g., as compared with being a user interface of an application on device 15.5-101). In FIG. 15-5A, user interface 15.5-912 includes icons corresponding to different applications that are accessible on device 15.5-101 (e.g., an icon for App A, an icon for App B, an icon for App C, and an icon for App D), which are selectable to display the respective selected application via the display generation component 15.5-120 of device 15.5-101. In at least one example, the display generation component 15.5-120 can be a display module or screen of a head mountable device configured to project light toward a user's eyes when the user dons the head mountable device. The button 15.5-920 can be

similar to or the same as the crowns, dials, and other buttons shown and described on other head mountable devices described herein and shown in other figures.

[4158] In some embodiments, user interface **15.5-912** is displayed at a location within three-dimensional environment **15.5-901** such that it overlays one or more representations of physical objects and/or virtual objects in three-dimensional environment **15.5-901** (e.g., overlaying and in front of part of representation **15.5-904b** of the person, corresponding to person **15.5-904a** in physical environment **15.5-900**). In some embodiments, user interface **15.5-912** is located within three-dimensional environment **15.5-901** such that one or more virtual objects are in front of and overlay part of user interface **15.5-912** and/or one or more representations of physical objects are in front of and overlay part of user interface **15.5-912**.

[4159] In some embodiments, device **15.5-101** displays three-dimensional environment **15.5-901** and/or user interfaces displayed via display generation component **15.5-120** at a particular level of immersion. In FIG. **15-5A**, device **15.5-101** is displaying three-dimensional environment **15.5-901** at a particular level of immersion **15.5-918** (e.g., indicated on immersion scale **15.5-916**, the left-most side of which corresponds to no immersion, and the right-mode side of which corresponds to maximum immersion) at which representations **15.5-902b** and **15.5-904b** of physical objects are visible via display generation component **15.5-120**. In some embodiments, a level of immersion includes an associated degree to which the content displayed by the electronic device obscures background content (e.g., content other than the user interface **15.5-912**) around/behind the user interface **15.5-912**, optionally including the number of items of background content displayed and the visual characteristics (e.g., colors, contrast, opacity) with which the background content is displayed. In some embodiments, the background content is included in a background over which the user interface **15.5-912** is displayed. In some embodiments, the background content includes additional user interfaces (e.g., user interfaces generated by the device **15.5-101** corresponding to applications, system user interfaces, etc.), virtual objects (e.g., files, representations of other users, etc. generated by the device **15.5-101**) not associated with or included in the user interface **15.5-912**, and/or real objects (e.g., pass-through objects representing real objects in the physical environment of the electronic device **15.5-101** that are displayed by the device such that they are visible via the display generation component). In some embodiments, at a first (e.g., low) level of immersion, the background, virtual and/or real objects are displayed in an unobscured manner. For example, a respective user interface with a low level of immersion is optionally displayed concurrently with the background content, which is optionally displayed with full brightness, color, and/or translucency. In some embodiments, at a second (e.g., higher) level of immersion, the background, virtual and/or real objects are displayed in an obscured manner (e.g., dimmed, blurred, removed from display, etc.). For example, a respective user interface with a high level of immersion is displayed without concurrently displaying the background content (e.g., in a full screen or fully immersive mode). As another example, a user interface displayed with a medium level of immersion is displayed concurrently with darkened, blurred, or otherwise de-emphasized background content. In some embodiments, the visual characteristics of the background objects vary among the background objects. For example, at a particular immersion level, one or more first background objects are visually de-emphasized (e.g., dimmed, blurred, displayed with increased transparency) more than one or more second background objects, and one or more third background objects cease to be displayed.

[4160] In FIG. **15-5A**, an input is received on input element **15.5-920** of device **15.5-101**. Input element **15.5-920** is optionally a mechanical input element that can be manipulated to change the level of immersion of the user interface(s) currently-displayed by device **15.5-101**, such as a slider element or a rotational input element, where the amount and direction (e.g., increase or decrease) of the change in immersion is based on the magnitude and direction of the manipulation of the input element **15.5-920** (e.g., the direction and/or magnitude of the movement of input element **15.5-920**, or the direction and/or magnitude of the rotation of a rotatable input element). In some

embodiments, device **15.5-101** and/or input element **15.5-920** generate tactile feedback as the level of immersion is changes (e.g., minor tactile outputs for each increases or decreases stage of immersion, and different major tactile outputs upon reaching minimum or maximum immersion). [4161] In FIG. **15-5A**, the input received on input element **15.5-920** is an input to slide the input element **15.5-920** upward to increase the level of immersion of the currently displayed user interface(s) by an amount based on the amount of movement of element **15.5-920** upward. In response to the input in FIG. **15-5A**, device **15.5-101** increases the level of immersion at which operating system user interface **15.5-912** is displayed to an intermediate level of immersion, as shown in FIG. **15-5B** and immersion scale **15.5-916**. In particular, in some embodiments, increasing the level of immersion at which user interface **15.5-912** is displayed causes device **15.5-101** to display user interface at a larger size via display generation component **15.5-120**, as shown in FIG. **15-5B**. Further, as described previously, additionally or alternatively, the content displayed surrounding and/or behind user interface **15.5-912** (e.g., representations of physical objects **15.5-902b** and **15.5-904b**, and virtual objects **15.5-910** and **15.5-906**) is darkened and/or blurred in FIG. **15-5B**, while user interface **15.5-912** is not darkened or blurred. In some embodiments, increasing the level of immersion at which the currently-displayed user interface(s) is displayed additionally or alternatively increases atmospheric visual and/or audio effects associated with virtual or physical objects displayed via display generation component **15.5-120**. For example, in FIG. **15-5B**, the rays of light being emitted from representation **15.5-908** of the sun have been lengthened (e.g., corresponding to more virtual light being emitted from representation **15.5-908**), and ornament **15.5-910** on representation **15.5-902b** of the tree has begun to shine as a result. Additional or alternative changes in atmospheric lighting, visual, audio, etc. effects in conjunction with changes in levels of immersion are similarly contemplated.

[4162] As previously mentioned, in some embodiments, the levels of immersion at which different user interfaces are displayed are optionally set independently from one another. For example, changes in the level of immersion at which operating system user interfaces are displayed optionally do not change the level of immersion at which application user interfaces are displayed, and in some embodiments, changes in the level of immersion at which the user interface of a first application is displayed do not change the level of immersion at which the user interface of a second application is displayed. Thus, in some embodiments, in response to an input to switch from displaying one user interface to displaying another user interface, device **15.5-101** optionally displays the switched-to user interface at the level of immersion at which that user interface was last displayed, independent of any changes of immersion that were applied to the currently-displayed user interface.

[4163] For example, in FIG. **15-5B**, device **15.5-101** detects an input selecting the icon corresponding to App C in user interface **15.5-912**. In response, device **15.5-101** displays user interface **15.5-934** of App C, as shown in FIG. **15-5C**. App C is optionally an application that is accessible via (e.g., installed on) device **15.5-101**, and is optionally any type of application (e.g., a content viewing application, a word processing application, etc.). App C is optionally different from the operating system of device **15.5-101**, and thus user interface **15.5-934** is optionally not a user interface of the operating system of device **15.5-101**, but rather a user interface of App C.

[4164] As shown in FIG. **15-5C**, in response to the input to display App C, device **15.5-101** is displaying user interface **15.5-934** at a level of immersion **15.5-932** that is higher than the level of immersion **15.5-918** at which device **15.5-101** was displaying user interface **15.5-912** in FIG. **15-5B** (e.g., as indicated by immersion scale **15.5-930** for App C). This is optionally because, as previously described, the levels of immersion set for applications are optionally independent of the level of immersion set for the operating system of device **15.5-101**. Therefore, it is optionally the case that the higher level of immersion at which device **15.5-101** is displaying user interface **15.5-934** of App C in FIG. **15-5C** is the level of immersion at which user interface **15.5-934** was last displayed by device **15.5-101**. If user interface **15.5-934** had last been displayed at a lower level of

immersion than the level of immersion **15.5-918** for user interface **15.5-912** shown in FIG. **15-5B**, device **15.5-101** would have optionally displayed user interface **15.5-934** in FIG. **15-5C** at that lower level of immersion rather than at the higher level of immersion shown in FIG. **15-5C**. Also shown in FIG. **15-5C** is the increased size of user interface **15.5-934** on display generation component **15.5-120** (as compared with the size of user interface **15.5-912** in FIG. **15-5B**) and the increased darkening and/or blurring of the content outside of/surrounding user interface **15.5-934** (as compared with the darkening and/or blurring of the content outside of/surrounding user interface **15.5-912** in FIG. **15-5B**)—these changes in display characteristics are optionally some of the changes that occur in response to increased immersion, as previously described. Additional or alternative changes in display characteristics, such as those previously described, are also contemplated.

[4165] As previously described, changes in immersion at which a user interface in one application is displayed optionally do not change the level of immersion at which user interfaces of other applications and/or user interfaces of the operating system are displayed. For example, if device **15.5-101** detects an input to change the level of immersion of user interface **15.5-934** of App C in FIG. **15-5C** (e.g., via input element **15.5-920**), device **15.5-101** would change the level of immersion at which user interface **15.5-934** is displayed in accordance with that input, as previously described. After changing the level of immersion at which user interface **15.5-934** is displayed, in response to an input to redisplay user interface **15.5-912** (e.g., an input to cease display of user interface **15.5-934**), device would optionally return to the display of FIG. **15-5B** in which user interface **15.5-912** is displayed at level of immersion **15.5-918**, which is the level of immersion at which user interface **15.5-912** was last displayed, and is independent from the changed level of immersion at which user interface **15.5-934** was displayed when the input to redisplay user interface **15.5-912** was detected.

[4166] In some embodiments, applications (e.g., App C) include controls in their user interfaces for changing the level of immersion at which those applications are displayed. For example, in some embodiments, user interface **15.5-934** of App C includes controls for increasing or decreasing the level of immersion at which user interface **15.5-934** is displayed, and the immersion at which App C is displayed is changed by device **15.5-101** in response to interaction with those controls additionally or alternatively to interaction with input element **15.5-920**. Further, in some embodiments, immersion above a threshold (e.g. maximum immersion) for an operating system user interface is reachable in response to inputs detected at input element **15.5-920**. However, in some embodiments, immersion above that threshold for an application user interface is not reachable in response to inputs detected at input element **15.5-920**—in some embodiments, the immersion at which an application user interface is displayed can only reach the threshold immersion using the input element, and once at that threshold, a different type of input is required to increase the immersion at which the application user interface is displayed beyond that threshold. For example, in FIG. **15-5C**, the level of immersion **15.5-932** at which user interface **15.5-934** is displayed is optionally the highest immersion that can be reached for user interface **15.5-934** via inputs at input element **15.5-920**. To increase immersion past the level of immersion **15.5-932**, device **15.5-101** optionally requires an input selecting a user interface element shown in user interface **15.5-934**. For example, in FIG. **15-5C**, user interface **15.5-934** includes user interface element **15.5-950** that is selectable to increase the immersion at which user interface **15.5-934** is displayed past level of immersion **15.5-932** (e.g., a user input selecting user interface element **15.5-950** optionally results in device **15.5-101** displaying user interface **15.5-934** at maximum immersion, or full screen, where no content surrounding user interface **15.5-934** is displayed by device **15.5-101**). In some embodiments, user interface **15.5-934** only includes element **15.5-950** once the level of immersion at which user interface **15.5-934** is displayed has reached the threshold level of immersion previously described. In some embodiments, user interface **15.5-934** includes element **15.5-950** irrespective of whether the level of immersion at which user interface **15.5-934** is

displayed has reached the threshold level of immersion previously described.

[4167] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **15-5A-15-5C** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown elsewhere and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to figures shown elsewhere can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **15-5A-15-5C**.

[4168] FIG. **15-6A** shows application user interface **15.6-11002** displayed in three-dimensional environment **15.6-11000**, **15.6-7128** that includes representation **15.6-7014'** of physical object **15.6-7014** (e.g., a physical table), representations **15.6-7004'** and **15.6-7006'** of physical walls **15.6-7004** and **15.6-7006**, respectively, and representation **15.6-7008'** of physical floor **15.6-7008**. In some embodiments, both computer-generated virtual objects that do not correspond to any specific objects in the physical environment (e.g., box **15.6-7016**) and representations of objects in the physical environment are displayed to a user.

[4169] In some embodiments, as shown in FIG. **15-6A**, application user interface **15.6-11002** provides media content to a user. For example, application user interface **15.6-11002** is a video player that includes a display of playback controls. In some embodiments, physical environment is a spatially constrained space (e.g., indoor space) surrounded by walls, physical objects, or other obstructions. Features of physical environment can be visible in the three-dimensional environment **15.6-11000** (e.g., an XR environment) displayed to a user, as shown in FIG. **15-6A**, by the presence of representations **15.6-7004'** and **15.6-7006'** of physical walls **15.6-7004** and **15.6-7006**, representation **15.6-7008'** of physical floor **15.6-7008**, and representation **15.6-7014'** of physical object **15.6-7014** (e.g., a table). In some embodiments, computer generated virtual content, such as box **15.6-7016**, which has no correspondence in the physical environment is also displayed to a user by computer system.

[4170] As shown in FIG. **15-6A**, a first portion of the three-dimensional environment **15.6-11000** includes computer-generated virtual objects that are not present in the physical environment, and a second portion of the virtual three-dimensional environment includes representations of objects in physical environment displayed as a part of three-dimensional environment **15.6-11000**. FIG. **15-6A** shows a first level of immersion generated by the display generation component of computer system in which a display of the XR environment concurrently includes a passthrough portion of a physical environment of computer system, virtual content from the application user interface **15.6-11002**, and computer-generated virtual content, such as box **15.6-7016**, that is distinct from the content provided by application user interface **15.6-11002**.

[4171] In some embodiments, when a level of immersion is increased, an amount of input signals from the physical environment that user receives is reduced. For example, reducing inputs from the physical environment by not displaying representations **15.6-7004'**, **15.6-7006'**, and **15.6-7008'** to a user, and/or displaying computer generated virtual content that provide visual inputs to a user to simulate a more spacious environment (e.g., open fields, outdoor spaces, or outer space) than that offered by physical environment, increases a level of immersion that is provided to a user in the three-dimensional environment **15.6-11000**. In some embodiments, increasing a level of immersion that is provided to a user decreases an influence the physical environment has on user.

[4172] User may wish to be more fully immersed in the content of the media that is provided through application user interface **15.6-11002**. Without exiting the application user interface **15.6-11002**, user provides rotational inputs to rotational input element **15.6-7108** (e.g., by turning rotational input element **15.6-7108** or **15.6-7508**). The input elements **15.6-7108** and **15.6-7508** can be similar to the buttons, crowns, dials, and other input mechanisms of head mountable display devices described herein. In some embodiments, user provides a first rotational input to rotational input element **15.6-7108** by turning in a first rotational direction (e.g., a clockwise rotation, or a

counter-clockwise rotation). The first rotational direction changes a level of immersion presented by the display generation component (e.g., increases a level of immersion, or decreases a level of immersion). For example, by turning in a first rotational direction that increases the level of immersion, passthrough portions of physical environment are displayed with lower fidelity and/or fewer passthrough portions of physical environment are displayed (e.g., some passthrough portions of the experience cease to be displayed) than before the first rotational input. For example, representation **15.6-7014** ceases to be displayed, as shown in FIG. **15-6B**. Instead, computer generated content, shown by dash lines, is presented as virtual content **15.6-11004** to a user.

[4173] In addition to increasing a level of immersion during a media consumption experience (e.g., watching a video, or listening to music), the level of immersion can also be increased to help user's concentration and focus while working. For example, increasing the level of immersion reduces noises from the ambient physical environment, and/or content from a smaller number of applications (e.g., a single application, a word processing application) is presented to a user (e.g., notifications from other applications are blocked while user is using a particular application at a high level of immersion).

[4174] Increasing a proportion of virtual (e.g., computer generated) content that is not a representation of the physical environment, increases a level of immersion presented to a user. Virtual content refers generally to content that is distinct from representations of the physical world (e.g., physical environment). For example, presenting a larger number of computer-generated virtual content, such as box **15.6-7016**, which has no correspondence in the physical environment, increases a level of immersion represented to a user. In some embodiments, as the immersion level increases, computer generated content previously presented at a lower immersion level continues to be displayed (e.g., maintaining display of computer-generated content as level of immersion increases).

[4175] In some embodiments, a level of immersion presented to a user is related to an associated size (e.g., a magnitude) of a spatial extent (e.g., an angular range) of a field of view within which computer-generated content is displayed. At a lower level of immersion, computer generated virtual content is displayed within a smaller field of view. At a higher level of immersion, computer generated virtual content is displayed within and covers a larger field of view.

[4176] For example, in FIG. **15-6A**, although box **15.6-7016** is computer-generated virtual content, it appears in a peripheral portion of a field of view of user. In contrast, as shown in FIG. **15-6B**, in response to detecting a rotational input to the rotatable input element **15.6-7108** along the first direction, the passthrough portion showing representation **15.6-7014** ceases to be displayed, and a first portion (e.g., central portion) of the user's field of view is replaced with computer-generated virtual content **15.6-11004**. In some embodiments, the computer-generated virtual content occupies an extended portion of user's field of view, starting in the central portion of user's field of view.

[4177] In some embodiments, the central portion of the user's field of view coincides with a middle region of the application user interface **15.6-11002**. In some embodiments, a level of immersion has an associated angle of view, which is an angular size of a view cone of the field of view within which computer-generated virtual content is displayed. A higher level of immersion has a larger associated angle of view, and a lower level of immersion has a smaller associated angle of view.

[4178] In some embodiments, for the level of immersion shown in FIG. **15-6B**, the angle of view is about $\pm 10^\circ$. For example, for an angle of view of about $\pm 10^\circ$, computer generated virtual content is displayed, from a perspective of user, starting at an angle of about 10° to the left, to about 10° to the right, and from about 10° above, to about 10° below an axis of user (e.g., an axis lying in a sagittal plane of user, for example, the axis of user projects forward, perpendicularly from a front surface plane of user's body). In some embodiments, computer generated virtual content occupies (e.g., completely occupies) the field of view of user within the angle of view. In some embodiments, computer generated virtual content partially occupies a portion of the field of view of user within the angle of view, but no input from the physical environment of computer system is

provided, within the angle of view, in the field of view of user.

[4179] In some embodiments, decreasing a level of immersion involves changing a level of immersion from an initial virtual reality (VR) environment in which no passthrough portion of a physical environment of the computer system is displayed to a first immersion level that includes a display of the XR environment. In some embodiments, a highest level of immersion for a three-dimensional environment is a virtual reality environment in which no passthrough portion of the physical environment is provided.

[4180] Level of immersion influences a user's perception experience by changing properties of a mixed reality three-dimensional environment. Changing a level of immersion changes a relative prominence of virtual content to content from the physical world (visual and/or audio). For example, for audio components, increasing an immersion level includes, for example, increasing noise cancellation, increasing a spatiality of spatial audio associated with the XR environment (e.g., by moving audio sources to more points around the user or increasing a number and/or volume of point sources of audio), and/or by increasing a volume of audio associated with the virtual environment. In some embodiments, increasing a level of immersion changes a degree to which the mixed-reality environment reduces (or eliminates) signals from the physical world that are presented to the user (e.g., audio and/or visual passthrough of a portion of the physical environment of the computer system). For example, increasing an immersion level includes increasing a proportion of the visual field of view that displays the virtual content, or decreasing a prominence of a representation of the real world (e.g., physical environment) by dimming, fading, or reducing an amount of the representation of the real world that is displayed to the user.

[4181] Changing a level of immersion can also include changing a visual presentation of the mixed-reality environment, including an extent of a field of view and a degree to which visibility of the external physical environment is reduced. Changing a level of immersion can include varying a number or extent of sensory modalities that a user can use to interact with the mixed-reality three-dimensional environment (e.g., interacting through user's voice, gaze, and body motion). Changing a level of immersion can also include changing an extent of a fidelity and a resolution with which the mixed-reality environment simulates a desired environment. Changing a level of immersion can also include modifying an extent to which a viewpoint of the mixed-reality environment is modified to match a user's viewpoint or perspective, e.g., through capture of the user's motion and timely adjustment of portions of the three-dimensional environment that lie within a field of view.

[4182] At the level of immersion shown in FIG. 15-6B, some visual inputs from the physical environment are still provided to a user. In some embodiments, the representations **15.6-7004'**, **15.6-7006'**, and **15.6-7008'** are provided at a lower fidelity (e.g., less focused, with less contrast, or more monochromatic) in the level of immersion shown in FIG. 15-6B compared to the level of immersion shown in FIG. 15-6A. Thus, while inputs from the physical environment are still provided to a user, the visibility of the external physical environment is reduced at a higher level of immersion compared to a lower level of immersion.

[4183] Rotatable input mechanism **15.6-7108**, in addition to changing a level of immersion presented to a user, is also able to receive one or more press inputs that cause the computer system (e.g., a wearable device) to perform various operations, as described in Table 3 below.

[4184] The use of a single input device (e.g., rotatable input mechanism **15.6-7108**) that accepts two (or more) different types of input (e.g., rotational inputs as a first type of input, and/or press inputs as a second type of input) reduces the number of distinct input devices that have to be provided to request or instruct performance of different functionalities. Reducing the number of input devices that have to be provided reduces manufacturing costs of the computer system, and reduces the number of components in the computer system that can fail. Reducing the number of components reduces the cost and complexity of manufacturing the computer system, and increases the reliability of the computer system. Reducing the number of input devices also reduces physical clutter on the computer system, freeing up more physical space on the computer system and helps

to prevent accidental inputs from inadvertent contacts.

[4185] Table 15-1 below describes the behavior of the computer system in response to different operations on button **15.6-7108** (e.g., a hardware button, a solid-state button), in accordance with some embodiments.

TABLE-US-00001 TABLE 15-1 Number Device Device On Device of presses On (Worn) (Not Worn) Off 1 press Show home UI Standby N/A or passthrough (temporary or exit full screen tracking state) 2 presses Force quit menu N/A N/A 3 presses Accessibility mode N/A N/A Turn Change immersion N/A N/A Press and Re-center N/A N/A hold (fade out and fade in)

[4186] For a computer system that is a wearable device (e.g., a head-mounted device, a strapped-on device, or a watch), while the wearable device is turned on and worn on the body of user, in response to detecting a rotational input to rotatable input element **15.6-7108** (e.g., a bidirectional rotatable input element), a level of immersion presented by the wearable device changes in the manner described above in reference to FIGS. **15-6A** and **15-6B**. For example, a level of immersion increases when a rotational input in the first rotational direction is provided to rotatable input element **15.6-7108**. In another example, when the wearable device is turned on but not worn on the body of a user (e.g., user), in response to detecting a rotational input to rotatable input element **15.6-7108**, no operation is triggered (e.g., the wearable device remains in a sleep state).

[4187] For a computer system that is a wearable device (e.g., a head-mounted device, a strapped-on device, or a watch), while the wearable device is turned on and worn on the body of user, in response to detecting a single press input to rotatable input element **15.6-7108**, a home menu user interface is presented to a user, or an application user interface that was in an immersive or full-screen display mode when the single press input was detected exits the full-screen or immersive display mode. Thus, the wearable device is configured to perform different operations depending on a type of user input provided to the rotatable input element **15.6-7108** (e.g., a press input, or a rotational input).

[4188] For a computer system that is a wearable device (e.g., a head-mounted device, a strapped-on device, or a watch), while the wearable device is turned on but not worn on the body of user, in response to detecting a single press input to rotatable input element **15.6-7108**, the wearable device transitions from a sleep state into a standby state.

[4189] For a computer system that is a wearable device (e.g., a head-mounted device, a strapped-on device, or a watch), while the wearable device is turned on and worn on the body of user, in response to detecting two press inputs within a preset time interval (e.g., less than 3 seconds, less than 2 seconds, less than 1 second) to rotatable input element **15.6-7108**, an operating system menu (e.g., a force quit menu) is presented to a user.

[4190] For a computer system that is a wearable device (e.g., a head-mounted device, a strapped-on device, or a watch), while the wearable device is turned on and worn on the body of user, in response to detecting three press inputs within a preset time interval (e.g., less than 5 seconds, less than 3 seconds, less than 2 second) to rotatable input element **15.6-7108**, an option to enter an accessibility mode is presented to a user.

[4191] In some embodiments, the rotatable input mechanism **15.6-7108** is also used to re-center a field of view of user. For example, instead of having a central portion of the view of field of view of user be aligned with a middle portion of the application user interface **15.6-11002**, user recenters a central portion of his field of view to a different location in the three-dimensional environment **15.6-11000**. For example, the new center of user's field of view corresponds to a point along an intersection of representations **15.6-7004'** and **15.6-7006'**. In some embodiments, recentering a field of view includes a display of computer generated virtual content fading out of the user's field of view at the previous center location of the user's field of view and then fading in, at the newly defined center location of the field of view of user, computer generated virtual content in that region. Optionally, the virtual content is presented with a higher fidelity or displayed with a higher contrast than prior to the recentering of the field of view of the user. While the wearable device is

turned on and worn on the body of user, a press input on the rotatable input mechanism **15.6-7108** that is held (e.g., persists) for a preset time duration (e.g., about 2 second, or about 5 second) causes the wearable device to begin the recentering operation described above. While the press input is held, user can rotate or move her head so a central portion of her field of view is re-positioned to a new location. Upon selecting the new location, releasing the press input recenters the central portion of user's field of view to the new location.

[4192] As shown in Table 3, different numbers of inputs (e.g., individual or sequential inputs) of a type distinct from rotational inputs (e.g., press inputs) cause different operations to be performed by computer system. For example, in some embodiments, for a single press input: (1) a home menu user interface is displayed, (2) a passthrough portion of the physical environment is provided, or (3) the application exits a full-screen mode. For two press inputs provided in close succession (e.g., within 3 seconds, within 2 seconds, within 1 second), a force quit menu is displayed. For three press inputs provided in close succession (e.g., within 5 seconds, within 3 seconds, within 2 seconds), an accessibility mode is activated, or an option to activate the accessibility mode is provided.

[4193] Using the number of press inputs to determine which operation(s) to perform reduces the number of distinct input devices that have to be provided to accomplish different. Reducing the number of input devices that have to be provided reduces physical clutter on the device, freeing up more physical space on the device and helps to prevent accidental inputs from inadvertent contacts. Reducing the number of input devices also reduces the need to provide additional hardware wiring within the device, and instead, the processor can be programmed to interpret more types of inputs (e.g., based on a number of press inputs) from a particular input device.

[4194] The use of a rotational input mechanism allows the user to easily provide a range of inputs, which may be a continuous range or a range that encompasses a sequence of discrete steps or values, and bidirectionality of the rotational input mechanism allows the input to be easily and intuitively varied, in either direction, without having to display additional controls to the user. The same rotational input mechanism **15.6-7108** is able to receive a second type of input (e.g., a press input) that requests or instructs performance of discrete functions (e.g., dismiss or display a user interface object). Reducing the number of input devices that have to be provided reduces physical clutter on the device, freeing up more physical space on the device and helps to prevent accidental inputs from inadvertent contacts. The use of the rotational input mechanism provides direct access to changes in immersion levels and the performance of different operations, reducing the amount of time needed to effect particular outcomes (e.g., the user does not have to navigate through menus or visually displayed control elements to make a selection for performing the operation and/or changing an immersion level), thereby improving operational efficiency of the computer system.

[4195] With respect to immersion level, increasing an immersion level can help to lessen constraints imposed by the physical environment of the computer system. For example, a more spacious virtual or XR environment can be realistically simulated by blocking out sensory out inputs from the physical environment (e.g., blocking visual input of a small/confined room, removing (audio) echoes from a small physical space) to provide a virtual environment that is more conducive for the user to interact with an application in the three-dimensional environment **15.6-11000**.

[4196] User further increases a level of immersion from the level of immersion shown in FIG. **15-6B** by applying a further rotational input in the same direction (e.g., the first rotational direction) as that applied to increase the level of immersion from that shown in FIG. **15-6A** to FIG. **15-6B**. In response to a second rotational input, the immersion level associated with display of the XR environment generated by the display generation component shown in FIG. **15-6B**, increases to a second immersion level by displaying additional virtual content to a user.

[4197] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **15-6A** and **15-6B** can be included, either alone or in any

combination, in any of the other examples of devices, features, components, and parts shown elsewhere and described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to figures shown elsewhere can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. 15-6A and 15-6B.

[4198] While a number of user interface features and experiences that can be achieved with the system described in FIGS. 1-1A to 14.6-4 are provided above, additional user input systems and methods can operate in conjunction with the exemplary systems and methods in a variety of ways. According to one exemplary embodiment, the user can dynamically change the level of immersion by actuating an input. For example, a user can manipulate a input to gradually increases the level of immersion of the AR experience as an input progresses. This gradual change in the immersion can include, but is in no way limited to, a switch between a plurality of levels of immersion, a smooth animation between levels of immersion, the display of a window from AR space to VR space, where light from VR space is cast onto objects in AR space. Additionally, any number of motions can be detected and cause the initiation of a designated action, including, but in no way limited to, a user swipe to change a view, a swipe to change lighting, a horizontal swipe to change location or time of day while a vertical swipe changes level of immersion (and vice-versa), or a detection of a progression of swipes, such as a first swipe to expand windows, a second swipe that removes a wall, and a third swipe removes multiple walls and a roof in the displayed environment, as an example. Other detected input variations can also be used to modify the user's experience with the HMD, including, but in no way limited to, detecting repeated inputs to change levels of immersion, detecting a single continuous input to change through multiple levels of immersion, detecting a relatively long press to get navigation menu (geographic location, time of day, level of immersion). Similarly, as the AR/VR environment is being adjusted or presented, different combinations of removal and immersion can be performed, such as the floor remains even when the walls disappear, some furniture remains in place even when walls disappear, and/or a detected air swipe over by the user can change content, while a detected air swipe up or down changes a level of immersion. Additional input motions and resulting system outputs can be incorporated by the exemplary system.

[4199] Additionally, as noted above, some examples of the HMD include a set of power straps that include a speaker system, or can be in communication with earphones or another separate audio device. According to some examples, the audio output style can change or otherwise be modified to correspond to a level of immersion. According to one example, the power strap can be designed to produce spatial audio. As the dial or button 1-328 is actuated (either by pushing, pulling, rotating, etc.), the actuation can cause a dedicated audio change or setting. By way of example, if the user is watching a movie on the HMD, the system can provide stereo audio when movie is displayed in frame, and can provide surround sound audio when movie is displayed in an immersive environment. Furthermore, in some examples the audio level corresponds to a change in a level of immersion (e.g. window slide down, walls slide away, etc.).

[4200] In other examples, the effect of the level of immersion can be related to the operating system only. For example, the system can use dedicated control to adjust a level of immersion for the operating system, and according to this example, applications have their own immersion controls (while the operating system is in a first level of immersion, a user can enter an application, adjust the immersion in the application, and then return to the operating system that is at the first level of immersion). According to one example of this configuration, the Digital Crown can be a dedicated control wherein the immersion varies based on a direction and/or magnitude of rotation of the Digital Crown. In some examples, major haptics can be disposed at either end of the crown, and minor haptics can be located throughout the dial of the crown to provide desired haptic effects. In some examples, the system can gradually change the level of immersion in the first experience in accordance with a user input (e.g., based on magnitude and/or direction of the user input).

Additionally, application controls can be different for different applications, and can be displayed in the user interface. Furthermore, in some examples, the level of immersion can include directing amount of passthrough content that is replaced. The level of immersion include an amount of de-emphasis of passthrough content or an amount of display area occupied with the corresponding experience. In some examples the level of immersion also controls or corresponds to atmospheric effects, and/or audio effects. In some examples, the controls enable a 360 degree experience. The video content provided to the user can be in either a VR or an AR environment.

[4201] In some examples, the user will leave a first application and enter a second application, each having a different level of immersion. According to one example, when the user toggles between the applications, an adjustment to a level of immersion in a first application, will not affect the second application or the system as a whole. If the user returns to the first application (or the second application) after exiting to the operating system, the system will return to the level of immersion set in the application previously (e.g., without regard to any changes in level of immersion for the operating system). Similarly, the system can remember the user's level of immersion when toggling between AR/VR modes. For example, while displaying a first experience with a first level of immersion, the system can detect an input to adjust the level of immersion, increase to a second level of immersion selected based on user input, and detect a request to reduce to a low level of immersion. Upon resuming the experience, the user can resume at the second level of immersion.

[4202] Additionally, the present system includes various levels of immersion and passthrough. According to one example, a low level of immersion is essentially a passthrough. In one example, a level of immersion can be adjusted by rotating a digital crown (e.g., based on direction and/or magnitude of rotation). In some examples, a request to reduce to a low level of immersion can be a press of the digital crown (e.g., in a direction perpendicular to the direction of rotation). Similarly, in some examples, rotating the crown can resume the experience from the second level of immersion, or can resume the experience gradually increasing the level of immersion from the low level of immersion (e.g., not from the second level of immersion). In addition to the crown rotation, any number of gestures can be used to trigger the system to resume an experience. Similarly, a long press that exceeds a threshold amount of time can cause the system to initiate a passthrough setting, and a release of the long press can cause the system to go back out of passthrough mode.

[4203] In some examples, when displaying a simulated environment, the system can use a first transition if the physical environment meets a first criteria, and can use a second transition if the physical environment meets a second criteria. According to this example, the first criteria can be that a portion of the physical environment in front of the user is more than a threshold distance away from a viewpoint of the user. The second criteria can be that a portion of the physical environment in front of the user is less than the threshold distance away from the viewpoint of the user. According to some embodiments, the environment can replace at least a portion of a view of the physical environment, and the transition can be gradual based on user input, such as a rotation of the crown. In some examples, the beginning and end states can be the same, but the intermediate states can be different. In some examples, the first transition can take into account depth information about a respective portion of the environment and the second transition does not take into account depth information about the respective portion of the environment. The first transition can be a cross dissolve starting with a portion of the simulated environment that is furthest away from a viewpoint of the user, and the second transition can be a transition that gradually encompasses portions of the physical environment based on a distance from the viewpoint of the user to the portions of the physical environment. According to these examples, portions of the room (e.g., architecture or furniture) that are “in front of” the portion of the physical environment that has been taken over by the simulated environment can be shown in front of the simulated environment (e.g., occlude portions of the simulated environment) and portions of the room (e.g., architecture or furniture) that are “behind” the portion of the physical environment that has been

taken over by the simulated environment can be hidden by the simulated environment (e.g., are occluded by portions of the simulated environment). In some examples, the environment includes light sources that shine onto portions of the physical environment, the environment can take into account the geometry of the room, the system can take into account the architecture of the room (location of walls/windows/doors, etc.), the environment can include volumetric effects that fill space within the physical environment, can include lighting effects, and/or particle effects.

[4204] In some examples, the present system includes a spatial effect (environments/atmosphere) immersion adjustment. According to this example, when initiating a spatial effect, the spatial effect can be in an intermediate immersive state, detect an input, and if the input is a first input, increase immersion in spatial effect if the input is a second input, decrease immersion in spatial effect. In some examples, the spatial effect can be an environment that replaces at least a portion of the view of the physical space. In some examples, the system can detect if the user looks to the side and can, in response to the detected motion, fade the audio component of the environment. The user can also, in some examples, choose extent of visuals for maximum level of immersion (e.g., between 180 degree immersion and 360 degree immersion as maximum level of immersion). Alternatively, the application can designate an extent of visuals for maximum level of immersion (e.g., between 180 degree immersion and 360 degree immersion as maximum level of immersion).

[4205] In some examples, the spatial effect is an atmosphere that modifies at least a portion of the view of the physical space (e.g., with lighting or particle effects). In some examples, the system can detect the user look to the side to fade the atmosphere effect. In some examples, there are more levels of immersion for environments than atmospheres. Additionally, displaying the atmosphere can include replacing something outside of the room (e.g., ceiling, imagery outside of windows), can include increasing an amount of light in the room, and/or can include reducing the amount of light in the room.

[4206] Any number of visual indicators can be used to communicate the level of immersion, including, but in no way limited to a graphical indicator. According to this example, major indicators can indicate major levels of immersion, and minor indicators can indicate minor levels of immersion. In some examples, the first major level of immersion can be audio only, the second major level of immersion can be audio and some visuals, and the third major level can include audio and expanded visuals.

[4207] In some examples, a threshold distance is established by the system such that if a user moves more than a threshold distance while displaying an environment, the system can cease to display the environment. According to this example, the environment gradually fades out as the user approaches the threshold distance. In some examples, if the user moves more than a threshold distance while displaying an environment, the system can cease to display the environment/if displaying an atmosphere, maintain display of the atmosphere.

[4208] In one example, the system can detect the motion of the user and adjust the user interface accordingly. According to one example, when the user looks to the side or back, the system can add duck audio for the environment. In response to a detected movement, the system can reduce spatial audio, reduce noise cancellation, and or reduce complexity of audio (e.g., number of tracks). Similarly, when the user looks to the side or back, the system can fade out some of the visuals of the environment. For example, the system can feather cutout around user's feet.

[4209] In some examples, the system can include an environment breakthrough where the system will reveal a portion of the physical environment during use. According to one example, the environment appears to approach a user as the user increases immersion. Further, in some examples, the environment rotates based on body movement, but not based on head movement. For example, a virtual environment is viewable from a particular region, and when a user shifts position toward a boundary of the region, the system can reveal a portion of the physical environment to the user. Alternatively, the system can reduce a level of immersion in the environment when revealing the portion of the physical environment; can reveal the portion of the physical environment without

regard to the presence or absence of obstacles; and/or if user moves away from the boundary, cease to display the portion of the physical environment. In some examples, the amount of the physical environment that is revealed can be based on the distance from the boundary, the speed of movement toward the boundary, and/or the acceleration of the user. In some examples, if the user continues moving toward a boundary, the system can cease to display the virtual environment. In some examples, the physical environment is not displayed, such as if the user is just looking around, or if the user moves toward the boundary but is not within a threshold distance of the boundary. In some examples, the system uses spatial transition to reveal physical space. In some examples, the breakthrough is not applied to atmospheres. Similarly, if a user leaves a virtual environment when an application is in a virtual environment, the system can cease to display the application. If, however, the user leaves a virtual environment when the application is outside the virtual environment, the system can continue to display the application on the system.

Alternatively, if the user leaves a virtual environment when the application is in the virtual environment, the system can reduce the size of the application and move into physical space.

[4210] Additionally, the present system can, in some examples, set time and day for the system, to be used with different applications. In one example, when a user requests to display an environment, the system can pick a time of day for the environment based on a system setting (e.g., if time of day has a first setting, environment is displayed with a first time of day, if time of day has a second setting, environment is displayed with second time of day). In some examples the user can select the time of day manually or automatically by the device. In some examples, the selection can be based on a current time of the day, or alternatively, can be selected based on a lighting level of the room. In some examples, the system can provide the user with access to a control center that includes setting inputs for setting the time of day, using any number of system controls. According to this example, the control center can fade from display after the time of day is set. Additionally, according to one example, when displaying an environment picker user interface, the system can update displayed icons to represent current time of day (e.g., daytime or nighttime); can display different simulated times of day for different nighttime environments (e.g., dusk, midnight, or predawn); and can display different simulated times of day for different daytime environments (e.g., morning, afternoon, or early evening). According to one embodiment, the system can include a night mode that darkens passthrough for portions of the field of view of the user that are showing passthrough content. According to one example, if an atmosphere is currently selected, the system can disable time of day picker. Similarly, the time of day setting can be ignored for some applications/experiences (e.g., mindfulness and other contemplative experiences). In some examples, changing the time of day setting can affect UI outside of the app/experience. Further variations can also include, but are in no way limited to, setting different degrees of passthrough dimming for different times of day (e.g., more dimming for nighttime, less dimming for daytime); changing immersion in environment with a physical control or a control center setting; allowing access to settings for environment even if environment is hidden (e.g., at 0% immersion); and/or displaying the environment volume control.

[4211] In at least one example, when joining a communication session while in a simulated environment with a user in a different environment, the devices described herein can maintain a simulated environment. The first user can see a representation of the second user in the first user's environment and the second user can see a representation of the first user in the second user's environment. In one example, when predetermined criteria are met, the environment(s) can be shared between users. In such an example, the environment can automatically switch when one user switches, the switch can occur when a user accepts a prompt to switch environments, the switch can occur after the environment is downloaded after a user accepts a prompt to switch environments, and/or the switch can occur after the new environment has been fully downloaded.

[4212] In one example, the current environment can be shared from an information panel. The system/user-interface of the HMD can prompt the user to share the environment with other users if

the user switches while in a communication session. An environment picker can indicate which environment is currently being shared with the user. The environment picker can indicate which environments have been downloaded. The user-interface can present an option to switch for all users or just switch for themselves. Switching environments can be based on a user request that switches the system environment even when a communication session ends, switching the environment based on a prompt to share other user's environment may not switch the system environment. In one example, if a user switches the environment while not in a communication session, the user may not be prompted to share with others.

[4213] In one example, the user-interface can prompt a user to join an environment of another user and the other user has the option to decline or accept. In one example, an even can be based on a user switching the environment. In one example, an even can be based on a user starting to share content. In one example, if a user declines to switch environments, that user can leave the sharing session. In one example, when content sharing sessions end, the user-interface can revert back to the original environment. In one example, when users are in a shared environment, the users can have shared spatial truth. In one example, the users can be seeing different portions of the environment when there is a shared environment. In one example, there is no spatial truth for even if both users have selected the same environment if the users are only in a shared environment and if they are not in a shared environment.

[4214] In one example, when switching to a shared environment, the current immersion level can be maintained. In one example, when switching to a shared environment, if the level of immersion is below a threshold, the current immersion level may not be maintained. In one example, different users can set different immersion levels. In one example, different users can set different volume levels. In one example, different users can set different immersion levels while not in full screen viewing mode, but if the content is changed to full screen viewing mode, all the user can have immersion set to at least a threshold level (e.g., the same level or a level that is a "high" level). In one example, and if users are sharing an environment, the time of day is synchronized, if users aren't sharing an environment, the time of day is not synchronized. In one example, any user can change the time of day. In one example, other users can receive a notice of the change of time of day.

[4215] In one example, while participating in a communication session and the environment changes, and if a first set of conditions is met, the user-interface can change the environment for multiple users in the communication session. If the first set of conditions are not met, the environment for different users remain different from each other. The first set of conditions can include a user confirmation that the environment should be shared. In one example, when a user displays an environment, the user is displayed at a predefined orientation. In one example, if the user joins a communication session with a shared environment, the user can be placed at a different orientation. In one example, the different orientation can be based on the positions of other participants in the environment. The different orientation can be based on the number of other participants in the environment. In one example, when an activity of the communication session changes, the participants can be shifted based on the selected activity. In one example, the participants can be shifted to face the same direction for shared visual experience (e.g., video). In one example, the participants can be shifted to face different directions (e.g., toward each other) for a participatory experience (e.g., a game).

[4216] In at least one example, the immersion state can be maintained for different participants and a notification can be displayed when the shared environment is changed. The notification can indicate who changed the environment. In one example, other users may need download the environment before displaying the environment. In one example, multiple participants (e.g., all participants) can be placed in a center of the environment, and in some examples facing different directions. In one example, one user can see other users at locations that are different from where those users see themselves. In one example, the time of day for the environment can be

synchronized between different users. In one example, when one user changes the time of day, the time of day can change for multiple or all users.

[4217] In one example, the user can pick a new environment from session UI for communication session (e.g., with audio/video start/stop option, call end option, and/or user status information). In one example, the user can pick a new environment from home UI (e.g., with multiple environment options, option to display available apps, and/or option to display available people for a communication session). The UI can highlight around a currently selected environment. Different color highlights can be used for shared environments vs. non-shared environments. In one example, the home UI can be displayed over representations of one or more other participants. In one example, new participants can be added to the conversation.

[4218] In one example, while the user is in a first viewing angle viewing an environment from a first angle, the system can detect a predetermined event in accordance with a determination that the perspective of the user meets respective criteria, change a viewing angle of the environment in accordance with a determination that the perspective of the user doesn't meet the respective criteria, and maintain the viewing angle of the environment. In one example, the angle of the environment can determine a location of the portal into the environment. In one example, the predetermined event can be dragging a window. In one example, the predetermined event can be detecting a change in an orientation of the user. In one example, the change in orientation can be a change in orientation of the user's body. In one example, the change in orientation can be a change in orientation of the user's head or a change in orientation of a portion of the user relative to the horizon of the environment or relative to the horizon of the physical world. In one example, the viewing angle for a different type of change in perspective of the user is not shifted (e.g., turning left/right as opposed to tilting with respect to the horizon). In one example, the predetermined event includes detecting a request to change a position of content. In one example, the predetermined event includes detecting a request to re-center content. These can be carried out by pressing a hardware button or by activating a software button. The software button can be displayed in response to a determination that the perspective of the user has changed.

[4219] In one example, a horizon of the environment remains consistent as the viewing angle of the environment changes. In one example, a degree of immersion of the environment remains consistent as the viewing angle of the environment changes. In one example, the environment can be viewed through a portal in the real world and the location of the portal can shift. In one example, when the environment is shifted, content that was at a first distance into the environment is shifted in distance relative to the users, for example shifted closer or content was docked at a predefined location and is turned into a window that can be repositioned at a user defined position. In one example, if a first kind of content was displayed with the environment, the first kind of content can shift to maintain the same or similar angle relative to the perspective of the user and if a second kind of content was displayed with the environment (e.g., user-positionable windows), that content can remain at a world locked position.

[4220] In one example, another predetermined event can be detected after the perspective of the user shifts again and in response, the viewing angle of the environment can be shifted. In one example, if the perspective of the user shifts back, the viewing angle of the environment can shift back. If the perspective of the user shifted to a different perspective, the viewing angle of the environment can shift to a different viewing angle. If a docked window was undocked when the viewing angle shifted the first time, the window can move back to the docked location.

[4221] In one example, a mechanical or software button of the HMD can be pressed once to go to home UI and again to hide home UI. In one example, while displaying an application, a button can be pressed to replace at least a portion of the application with a home UI. When pressed again, the home UI can be hidden. In one example, home UI can include representations of people, environments, and apps. Hiding home UI can include replacing home UI with passthrough display. In one example, hiding home UI can include ceasing to display a virtual environment in which the

home UI is display. The home UI can be used to switch which virtual environment is displayed. IN one example, if an application is a media application with media playing, the UI can display a min-player, including video PiP, audio player, including representation of video or audio content, playback controls. In one example, the mini-player can persist after home UI is hidden.

[4222] In one example, when the home UI is displayed, the user can scroll through home UI and the user can navigate through sections of the UI (e.g., environments, people, and/or apps), In one example, a current section can be maintained when the user leaves and returns to home UI. In one example, the current selection can be maintained if the user returns to home UI within a time threshold and resets to a predetermined section (e.g., first page of applications) if the user returns to home UI after the time threshold (e.g., the next day, next session, or one hour). In one example, a quick look object (e.g., an object dragged out of an application) can remain when the application is dismissed. In one example, the quick look object is an object pulled out of the application before it was replaced with the home UI. In one example, the quick look objects hide when home UI is hidden (e.g., on a second button press). In one example, a second application can be launched from home UI while the quick look object is displayed. In one example, the quick look object can be dragged by the user in the UI into a second application to perform an operation in the second application based on the quick look object (e.g., added to a message or document).

[4223] In one example, the user can press the home button again to re-display the home UI. In one example, the button is a hardware button. In one example, the button is a solid state button. In one example, the system UI can be treated as the same as was the application (e.g., a button press while they are displayed can cause the system UI to be replaced with the home UI). In one example, a quick double press of the button can be a command to get an application management UI (e.g., force quit menu or multitasking UI).

[4224] In one example, the system can detect a press input on a button. If the device is displaying an immersive experience of an application, the UI can display the application in a non-immersive mode. If the device is displaying a non-immersive experience, the UI can replace the non-immersive experience with the home UI. In one example, the button can be pressed again during a non-immersive experience to cease to display the home UI. In one example, the non-immersive experience can be displayed in a virtual environment that continues to be displayed after the home button is pressed. In one example, virtual environment continues to be displayed while the home UI is displayed. In one example, the home UI can be used to switch which virtual environment is displayed.

[4225] In one example, the rotatable input mechanism of the HMD can be used to perform a UI immersion level change operation when a first input type is detected and a different operation when a second input type is detected. For example, the change in immersion can be based on a rotation of the mechanism, a direction of rotation, an amount of rotation, and so forth. In one example, the first input type is a rotation and the second input type is a depression of the mechanism. In one example, the number of presses can change what operation is performed. In one example, the duration of the press can change what operation is performed. In one example, the second operation can be a home display operation. In one or more other examples, the second operation can include an application dismiss operation, a virtual object dismiss operation, an application manager UI display operation, an accessibility mode operation, reentering virtual UI elements in the AR/VR environment, and/or an operation that can be used in accordance with another input device to perform a third operation. In one or more other examples, the third operation can depend on the duration and/or pattern of inputs and can include taking a screenshot, powering off the device, restarting the device, and/or entering a hardware reset mode.

[4226] As described above, one aspect of the present technology is the gathering and use of information such as information from input-output devices. The present disclosure contemplates that in some instances, data may be gathered that includes personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can

include demographic data, location-based data, telephone numbers, email addresses, twitter ID's, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, username, password, biometric information, or any other identifying or personal information.

[4227] The present disclosure recognizes that the use of such personal information, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to deliver targeted content that is of greater interest to the user. Accordingly, use of such personal information data enables users to calculated control of the delivered content. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

[4228] The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the United States, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA), whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

[4229] Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide certain types of user data. In yet another example, users can select to limit the length of time user-specific data is maintained. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an application (“app”) that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

[4230] Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when

appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data at a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods. [4231] Therefore, although the present disclosure broadly covers use of information that may include personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data.

[4232] Physical environment: A physical environment refers to a physical world that people can sense and/or interact with without aid of electronic systems. Physical environments, such as a physical park, include physical articles, such as physical trees, physical buildings, and physical people. People can directly sense and/or interact with the physical environment, such as through sight, touch, hearing, taste, and smell.

[4233] Computer-generated reality: in contrast, a computer-generated reality (CGR) environment refers to a wholly or partially simulated environment that people sense and/or interact with via an electronic system. In CGR, a subset of a person's physical motions, or representations thereof, are tracked, and, in response, one or more characteristics of one or more virtual objects simulated in the CGR environment are adjusted in a manner that comports with at least one law of physics. For example, a CGR system may detect a person's head turning and, in response, adjust graphical content and an acoustic field presented to the person in a manner similar to how such views and sounds would change in a physical environment. In some situations (e.g., for accessibility reasons), adjustments to characteristic(s) of virtual object(s) in a CGR environment may be made in response to representations of physical motions (e.g., vocal commands). A person may sense and/or interact with a CGR object using any one of their senses, including sight, sound, touch, taste, and smell. For example, a person may sense and/or interact with audio objects that create 3D or spatial audio environment that provides the perception of point audio sources in 3D space. In another example, audio objects may enable audio transparency, which selectively incorporates ambient sounds from the physical environment with or without computer-generated audio. In some CGR environments, a person may sense and/or interact only with audio objects. Examples of CGR include virtual reality and mixed reality.

[4234] Virtual reality: A virtual reality (VR) environment refers to a simulated environment that is designed to be based entirely on computer-generated sensory inputs for one or more senses. A VR environment comprises a plurality of virtual objects with which a person may sense and/or interact. For example, computer-generated imagery of trees, buildings, and avatars representing people are examples of virtual objects. A person may sense and/or interact with virtual objects in the VR environment through a simulation of the person's presence within the computer-generated environment, and/or through a simulation of a subset of the person's physical movements within the computer-generated environment.

[4235] Mixed reality: In contrast to a VR environment, which is designed to be based entirely on computer-generated sensory inputs, a mixed reality (MR) environment refers to a simulated environment that is designed to incorporate sensory inputs from the physical environment, or a representation thereof, in addition to including computer-generated sensory inputs (e.g., virtual objects). On a virtuality continuum, a mixed reality environment is anywhere between, but not including, a wholly physical environment at one end and virtual reality environment at the other end. In some MR environments, computer-generated sensory inputs may respond to changes in sensory inputs from the physical environment. Also, some electronic systems for presenting an MR environment may track location and/or orientation with respect to the physical environment to enable virtual objects to interact with real objects (that is, physical articles from the physical environment or representations thereof). For example, a system may account for movements so that

a virtual tree appears stationary with respect to the physical ground. Examples of mixed realities include augmented reality and augmented virtuality.

[4236] Augmented reality: an augmented reality (AR) environment refers to a simulated environment in which one or more virtual objects are superimposed over a physical environment, or a representation thereof. For example, an electronic system for presenting an AR environment may have a transparent or translucent display through which a person may directly view the physical environment. The system may be configured to present virtual objects on the transparent or translucent display, so that a person, using the system, perceives the virtual objects superimposed over the physical environment. Alternatively, a system may have an opaque display and one or more imaging sensors that capture images or video of the physical environment, which are representations of the physical environment. The system composites the images or video with virtual objects, and presents the composition on the opaque display. A person, using the system, indirectly views the physical environment by way of the images or video of the physical environment, and perceives the virtual objects superimposed over the physical environment. As used herein, a video of the physical environment shown on an opaque display is called “pass-through video,” meaning a system uses one or more image sensor(s) to capture images of the physical environment, and uses those images in presenting the AR environment on the opaque display. Further alternatively, a system may have a projection system that projects virtual objects into the physical environment, for example, as a hologram or on a physical surface, so that a person, using the system, perceives the virtual objects superimposed over the physical environment. An augmented reality environment also refers to a simulated environment in which a representation of a physical environment is transformed by computer-generated sensory information. For example, in providing pass-through video, a system may transform one or more sensor images to impose a select perspective (e.g., viewpoint) different than the perspective captured by the imaging sensors. As another example, a representation of a physical environment may be transformed by graphically modifying (e.g., enlarging) portions thereof, such that the modified portion may be representative but not photorealistic versions of the originally captured images. As a further example, a representation of a physical environment may be transformed by graphically eliminating or obfuscating portions thereof.

[4237] Augmented virtuality: an augmented virtuality (AV) environment refers to a simulated environment in which a virtual or computer generated environment incorporates one or more sensory inputs from the physical environment. The sensory inputs may be representations of one or more characteristics of the physical environment. For example, an AV park may have virtual trees and virtual buildings, but people with faces photorealistically reproduced from images taken of physical people. As another example, a virtual object may adopt a shape or color of a physical article imaged by one or more imaging sensors. As a further example, a virtual object may adopt shadows consistent with the position of the sun in the physical environment.

[4238] Hardware: there are many different types of electronic systems that enable a person to sense and/or interact with various CGR environments. Examples include head mounted systems, projection-based systems, heads-up displays (HUDs), vehicle windshields having integrated display capability, windows having integrated display capability, displays formed as lenses designed to be placed on a person's eyes (e.g., similar to contact lenses), headphones/earphones, speaker arrays, input systems (e.g., wearable or handheld controllers with or without haptic feedback), smartphones, tablets, and desktop/laptop computers. A head mounted system may have one or more speaker(s) and an integrated opaque display. Alternatively, a head mounted system may be configured to accept an external opaque display (e.g., a smartphone). The head mounted system may incorporate one or more imaging sensors to capture images or video of the physical environment, and/or one or more microphones to capture audio of the physical environment. Rather than an opaque display, a head mounted system may have a transparent or translucent display. The transparent or translucent display may have a medium through which light representative of images

is directed to a person's eyes. The display may utilize digital light projection, OLEDs, LEDs, uLEDs, liquid crystal on silicon, laser scanning light sources, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In one embodiment, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface.

[4239] The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. A wearable electronic device, comprising: a housing; an input; a first camera configured to capture an environment external to the housing, the camera facing a first direction; an optical module comprising a display facing a second direction different than the first direction; and a processor electrically coupled to the display, the first camera, and the input; wherein the processor is configured to cause the display to present augmented reality content including an amount of virtual content superimposed over video passthrough content captured by the first camera, the amount based on a manipulation of the input.
2. The wearable electronic device of claim 1, further comprising a second camera oriented to face a third direction different from the first direction and the second direction, the second camera configured to detect a hand gesture of a user donning the wearable electronic device.
3. The wearable electronic device of claim 2, wherein: the input comprises the second camera; and the amount of virtual content is based on the hand gesture.
4. The wearable electronic device of claim 1, wherein: the optical module further comprises: a chassis; and a second camera adjacent the display and facing the second direction; the display and the second camera are secured to the chassis; and the chassis is secured to the housing via a frame.
5. The wearable electronic device of claim 4, wherein the second camera is configured to detect a facial feature of a user donning the wearable electronic device.
6. The wearable electronic device of claim 5, wherein the facial feature comprises a gaze direction of the user.
7. The wearable electronic device of claim 1, wherein: the input comprises a capacitive touch sensor; and the manipulation comprises a contact with the touch sensor.
8. A wearable display, comprising: a housing defining an external surface; a dial; an outward facing camera secured within the housing; and an optical module secured within the housing, the optical module comprising: a display configured to project light toward an eye of a user donning the wearable display; wherein: the display is configured to display augmented reality content including a first immersion level of passthrough content captured by the outward facing camera and a second immersion level of virtual content; and a manipulation of the dial changes the first immersion level and the second immersion level.
9. The wearable display of claim 8, wherein: the outward facing camera is oriented to capture the passthrough content in front of the user when donning the wearable display; and the wearable display further comprises an inward facing camera oriented to face toward the eyes of the user when donning the wearable display.
10. The wearable display of claim 9, wherein: the inward facing camera is a first inward facing camera; and the wearable display further comprises a second inward facing camera.
11. The wearable display of claim 10, wherein the second inward facing camera is configured to detect a facial feature.

- 12.** The wearable display of claim 11, wherein the facial feature includes a gaze direction.
- 13.** The wearable display of claim 8, wherein: the outward facing camera is a first outward facing camera; and the wearable display further comprises a second outward facing camera configured to capture the passthrough content.
- 14.** The wearable display of claim 13, wherein the first immersion level and the second immersion level are based on a facial feature.
- 15.** The wearable display of claim 8, wherein: the dial is rotatable; and the manipulation comprises a rotation of the dial.
- 16.** The wearable display of claim 8, wherein: the dial is depressible; and the manipulation includes a depression of the dial.
- 17.** A head-mountable display device, comprising: a housing; a button; a first camera configured to capture passthrough content, the first camera facing a first direction; an optical module within the housing, the optical module comprising: a display facing a second direction different than the first direction; and a second camera facing the second direction; wherein the display is configured to project mixed reality content including a level of virtual content superimposed over the passthrough content, the level based on a manipulation of the button.
- 18.** The head-mountable display device of claim 17, further comprising a frame secured to the housing, wherein the optical module is secured to the frame.
- 19.** The head-mountable display device of claim 18, wherein: the head-mountable display device further comprises a dial disposed on a first side of the housing; the button is disposed on a second side of the housing; and the optical module is disposed between the dial and the button.
- 20.** The head-mountable display device of claim 17, wherein: the display comprises a pixelated screen; and the second camera is disposed adjacent the pixelated screen.
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